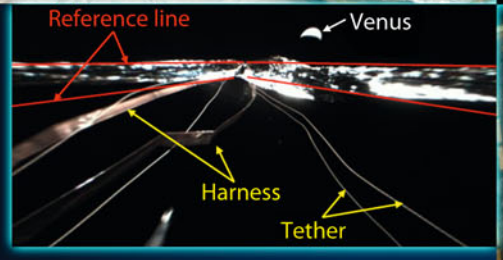
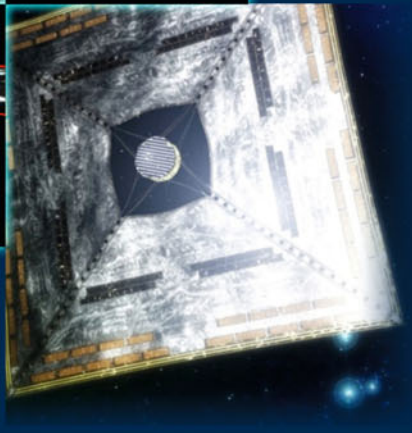
Advances in Solar Sailing

Malcolm Macdonald
Editor





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Malcolm Macdonald
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Advances in Solar Sailing

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Preface

Hosted by the Advanced Space Concepts Laboratory within the department of Mechanical and Aerospace Engineering of the University of Strathclyde, the third International Symposium on Solar Sailing was held in McCance Building at 16 Richmond Street, Glasgow, between 11 and 13 June 2013.

The symposium attracted over 90 delegates from 19 different counties, bringing together international experts from across the globe to discuss funded solar sail flight programs alongside on-going technology development and testing programs. The symposium also provided a forum for the discussion of enabling technologies, new application concepts, materials and structural concepts, space environmental effects, dynamics, navigation, control, and much more. This volume contains the unabridged symposium proceedings, in the gathered experts own words.

As symposium chair, I thank our partners at Scottish Enterprise and L'Garde, Inc., the symposium's gold sponsor, for their support in realising this symposium.

Malcolm Macdonald
Chair of the Third International
Symposium on Solar Sailing

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Part I

Flight Programs

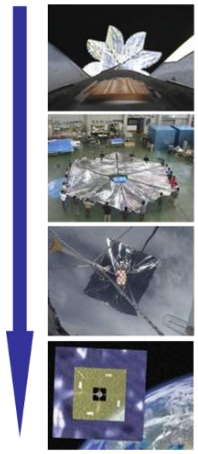
An overview of solar sail related activities at JAXA

Junichiro Kawaguchi[1]
Japan Aerospace Exploration Agency (JAXA), .

Japan Aerospace Exploration Agency (JAXA) made world's first Solar Sail mission, Ikaros in 2010 successfully. This mission was performed as a precursor and technology verification mission toward our goal of having Solar Power Sail mission to Trojan asteroids rendezvous. This paper overviews what Ikaros demonstrated and presents the future mission currently studied at JAXA. It includes some introduction on technology development programs.

I. Introduction

THE solar sail spacecraft has been studied many years before at JAXA since solar yacht race to the moon was discussed internationally late 80th celebrating 500 years from Columbus voyage. The significance in flight by solar radiation was quite clear, while legible application was not clearly seen with the burden of transportation cost. Besides deploying large membrane was still a high hurdle for realization, since space activity costs high and always requests verification on the ground. This is contradictory for solar sails. JAXA was also at the tough position similar to the circumstances in the world. JAXA anyhow started the flight experiments via sounding rocket, balloons and piggy back payload attached to the launch vehicles.



Aug. in 2004
Sounding Rocket (S310-34)
Dynamic Membrane Deployment (10m)
Spinner

Feb. 2006
Subpayload on M-V#8
Quasi-Static Membrane Deployment (10m)
Deployment resulted in Partial Success.

Aug. 2006
Balloon Experiment on B200-7
Quasi-Static Membrane Deployment (20m)
Deployment resulted in Success.

Sept. 2006
Piggy-back ultra-small satellite on M-V#7
Thin Film Solar Power Generation
Film Deployed. Functioned for 6 hrs using Battery.
Resulted in Failure.

Fig. 1 History in Solar Sail activity at JAXA.

[1] Senior Fellow, ISAS, Japan Aerospace Exploration Agency (JAXA), 3-1-1 Yoshinodai, Chuo-ku, Sagamihara, Kanagawa 252-5210, Japan

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Fig. 1 summarizes what we did before Ikaros. This paper describes Ikaros, how it was born and what it demonstrated first, and secondly what is planned beyond Ikaros, especially about the mission to Trojan asteroids rendezvous, and associated technology developments performed at JAXA finally.

II. Ikaros Mission – how it was born and what was demonstrated

While we had made a proposal of putting a Solar Power Sail to the Jupiter, the proposal hardly passed through due to its high technology risks. (Agency is always sensitive to the risks.) The study team (WG: working group) then proposed a technology demonstrator to show if the deployment of large membrane is feasible. The proposal consisted of a small spacecraft carried by a smaller independent dedicated vehicle. At that time, the demonstration was assumed done on Earth orbit. However, the proposal still did not go through, just because the cost is still high enough and also in view of the risks.

When JAXA's Akatsuki (PLANET-C) spacecraft to Venus was decided launched via our H-IIA vehicle, large amount of payload surplus was identified found , while the launch needed a dummy mass damper on the vehicle. I, at that instance, raised my hand to propose a further low-cost solar sail demonstration by taking the advantage of this opportunity. The development period was just 2.5 years and the development was not easy. But we tried and the mission was approved started. It was defined as a technology demonstrator. The mission outline was shown below in Fig.2.

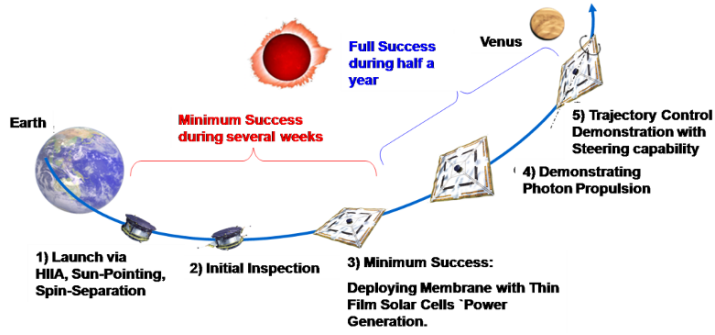

Fig. 2 Ikaros mission at a glance.

The membrane of it is spanned by Spin, Centrifugal force without use of any spar structural member. Exclusion of spar leads to significant mass reduction. This is the advantage in spin-spanned sail craft. However, there is a drawback in adopting spinning configuration. It is hard to control huge angular momentum and in other words, to steer the sail. But we found a good solution for it. It is the use of Liquid Crystal device to control and steer the sail direction via reflectivity control. The demonstration of the device is also a part of missions.

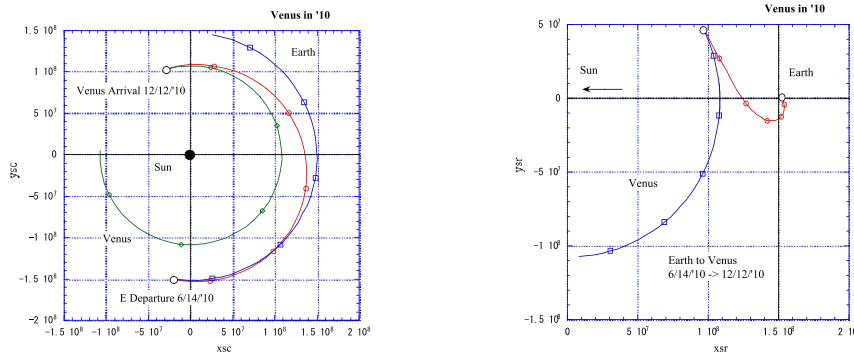

Fig. 3 Ikaros Trajectory.

The trajectory of Ikaros is shown in Fig. 3. The probe was injected toward Venus along with PLANET-C spacecraft on May 21st of 2010. The spacecraft was built and it is shown in Fig. 4 below. The sail membrane was wound around the hub rotator, and extracted by rotation bar as the hub rotates. Deployment consists of two phases. First phase extracts the sail and made it in cross configuration, and it was followed by second phase in which the cross shaped sail is released out to the fully expanded shape. These steps are depicted in Fig. 5 and 6.

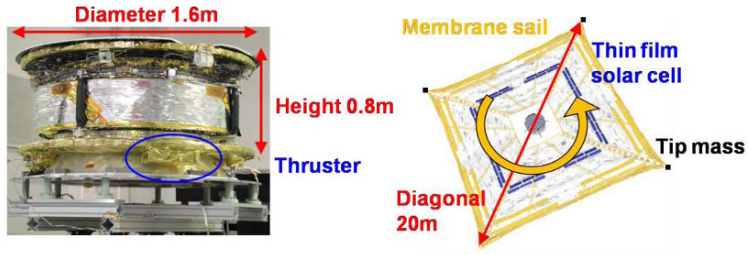

Fig. 4 Ikaros Spacecraft.

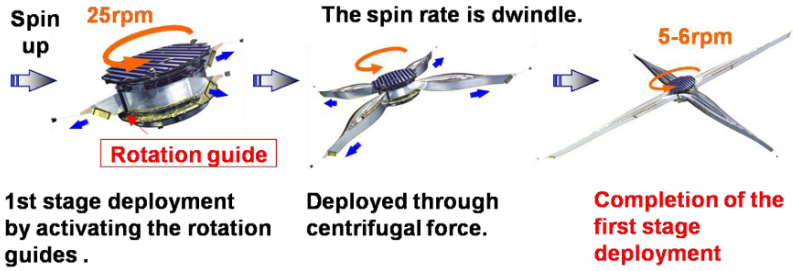

Fig. 5 First Deployment.

One of the most characteristic point introduced in Ikaros is the use of detached cameras aboard to photograph the deployed shape of the spacecraft on orbit. This attempt resulted in very good success and they indicated the membrane was fully deployed as expected and planned.

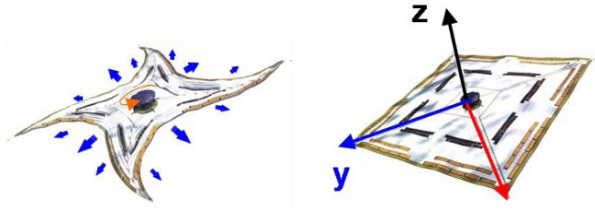

Fig. 6 Second Deployment.

The photos taken are given in Fig. 7. As the previous reports showed, the spacecraft made an attempt to steer the sail via Liquid Cristal device, what we call Reflection Control Device (RCD) . The photo taken by a detached camera well captured LCD operation actually.

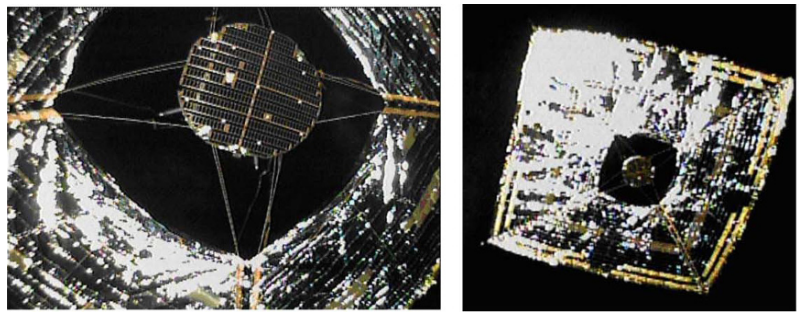

Fig. 7 Spacecraft Entire View photographed on orbit.

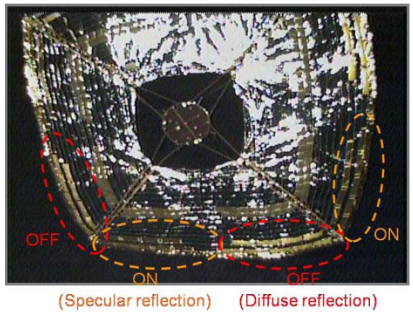

Fig. 8 RCD Operation Pictured.

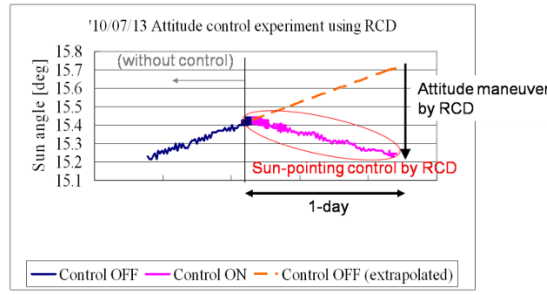

Fig. 9 Effective Reorientation via RCD.

Fig. 8 and 9 show switching RCD on and off during the flight enabled the reorientation effectively in space. This provides the steering solution to spinning solar sail.

Fig. 10 Venus photographed from Ikaros at flyby.

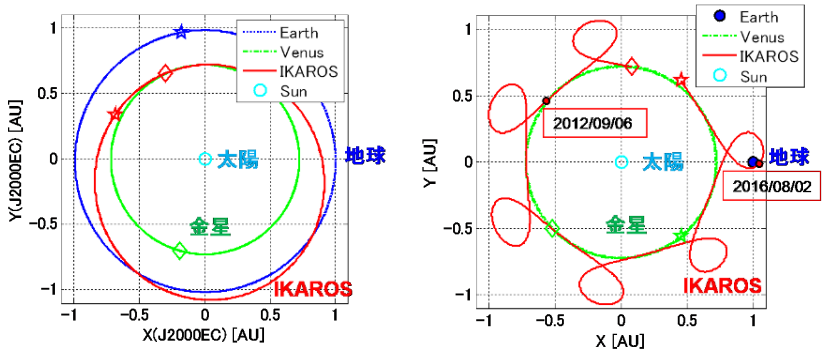

Fig. 11 Extended Operation Plan.

The Ikaros made Venus flyby in December of 2010, when the close-up view of the planet was taken by Ikaros camera. (Fig. 10) Since no attitude control fuel is available, operation of Ikaros is restricted and intermittent. However, the project team continues to track the spacecraft to obtain further information until the spacecraft approaches to the Earth in 2016. (Fig. 11)

III. Solar Power Sail to Trojan Asteroids

The mission at JAXA following the Ikaros is the Solar Power Sail spacecraft to Trojan asteroids. The scientific aspects in the missions to Trojan asteroids are not mentioned here. It is obvious. To have a fruitful mission there requests the electric propulsion system driven at the distance of Jupiter. Since the power availability at that distance is significantly degraded to 4% with respect to that at Earth distance, driving the electric propulsion engine inevitably requests huge power availability.

Sooner or later, Nuclear Propulsion represented by Fission / Fusion Reactors will prevail for Large and Heavy Interplanetary Transportation in 21st century. However, Nuclear Propulsion weighs heavy and not

efficient for smaller spacecraft, and Solar powered electric propulsion continues to be sought for the time being. Contemporary Ion Engines performance characterized in terms of specific impulse will shift for higher region dramatically. Ultra-High specific impulse era does come.

Pure Solar Photon propulsion may be useless for rendezvous and orbiter missions. And the applications of pure photon solar sails will be confined only to smaller flyby probes. This Prospect infers : Solar powered high-specific-impulse Ion Engines combined with Solar photon propulsion will be utilized for medium-to-smaller missions in near future. (Hybrid Propulsion) This is a Solar Power Sail.

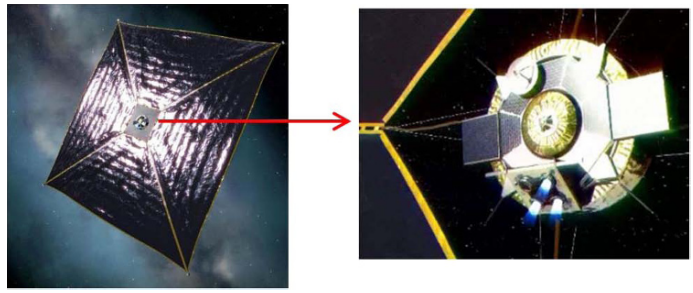

Fig. 12 Solar Power Sail uses Hybrid Propulsion, equipped with Electric Propulsion Engines

A Solar Sail is a space yacht that gathers energy for propulsion from sunlight pressure. (Photon Sail) A Solar Power Sail is a Japanese original concept that gets electricity from thin film solar cells in addition to acceleration by solar radiation. (Hybrid Propulsion) The Japan Aerospace Exploration Agency (JAXA) developed the world's first Solar Power Sail craft demonstrator for both its photon propulsion and thin film solar power generation. This is the Ikaros spacecraft and is the technology step-up toward the goal of Solar Power Sail to Trojan asteroids.

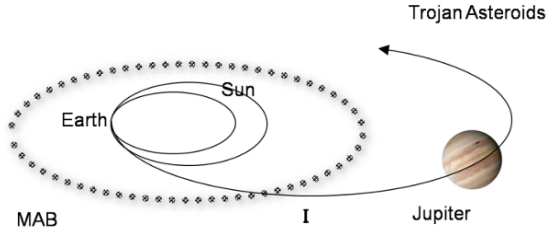

Fig. 13 Flight to Trojan Asteroids by way of Jupiter Swingby

The spacecraft may fly directly for Trojan asteroids. However, the flight period around aphelion point is not long enough for the low thrust propulsion to exhibit rendezvous delta-V due to the low acceleration. The flight require relatively long stay at the distance of Jupiter. The solution is to make the spacecraft to have a swing-by

Jupiter that can retain the Jupiter's distance for sufficient duration. Fig. 13 illustrates typical flight sequence to Trojan asteroids by way of Jupiter swing-by. How to get to Jupiter also depends on whether the electric propulsion is made use of before the arrival at Jupiter. In case the flight utilizes the electric propulsion most, the combination of multiple EDVEGA (Electric Delta-VEGA Earth Gravity Assist) sequences will be the most efficient way. In that case, the spacecraft uses Double Earth swingbys to Jupiter as shown in Fig. 14a. It shall be commented that there is a direct flight option still exists if the flight period is requested to shorten entire mission duration.

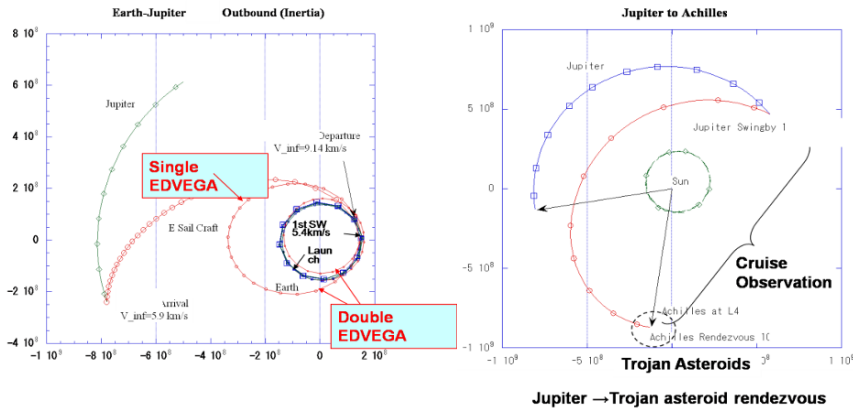

Fig. 14a, b Trans-Jupiter & Trans-Trojan Trajectories (L4)

There are two kinds of Trojan asteroids; around L4 or L5 points. Fig. 14b presents typical flight sequence to Achilles located around L4 point. This sequence is advantageous over the flight to L5 points. The flight to L5 flies farther than that to L4 and solar power availability is degraded.

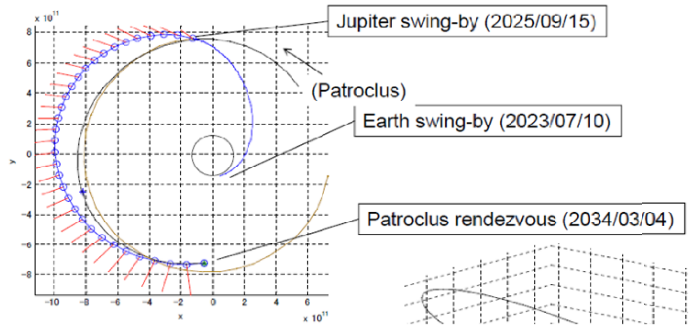

Fig. 14c Trans-Trojan Trajectories (L5)

IV. Preparation Activities for Solar Power Sail mission

While JAXA succeeded in performing Ikaros mission, it is much smaller than real missions to Trojan asteroids. And JAXA still identifies some key technology to be matured before the actual missions start.

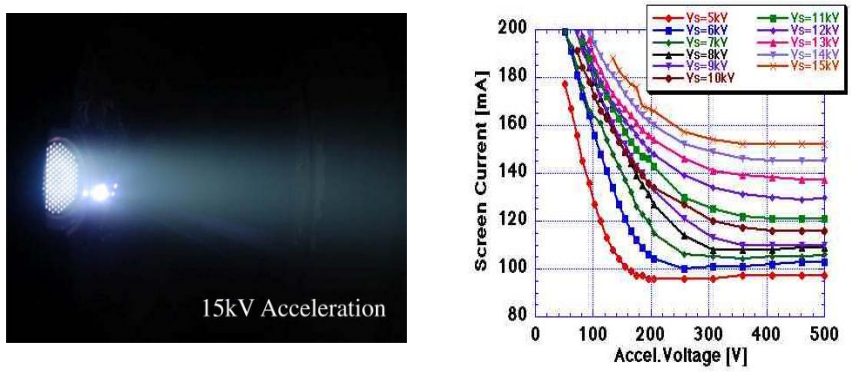

Fig. 15 Ultra High Specific Impulse Ion Engines Test

Fig. 15 presents the ion engine test performed at JAXA. The flight to rendezvous with Trojan asteroids requires ultra high specific impulse engines. The mission aims at developing 15kV ion engine for it. JAXA has already completed preliminary test for it. Ikaros adopted adhesion to make films mated. But as for huge solar sails, adhesion shall be replaced self-melt-seal technology. JAXA made an assessment of the technology by fabricating part of the membrane for verification as in Fig. 16.The power sail will also utilize thin film solar cell as well as RCD made of Liquid Crystal Device as in Ikaros.

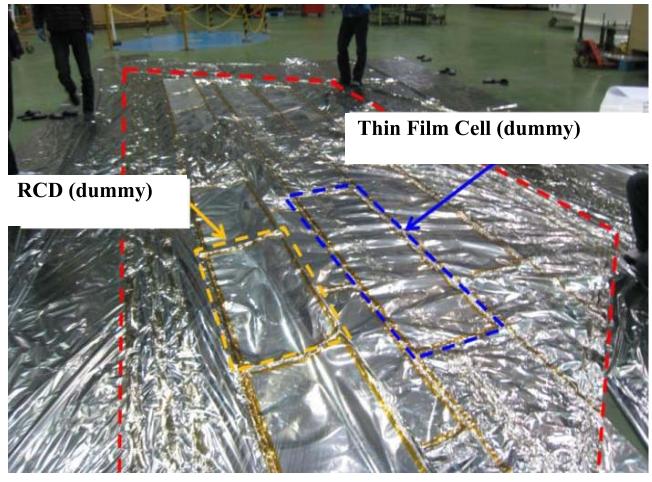

Fig. 16 Fabrication Test for New Membrane

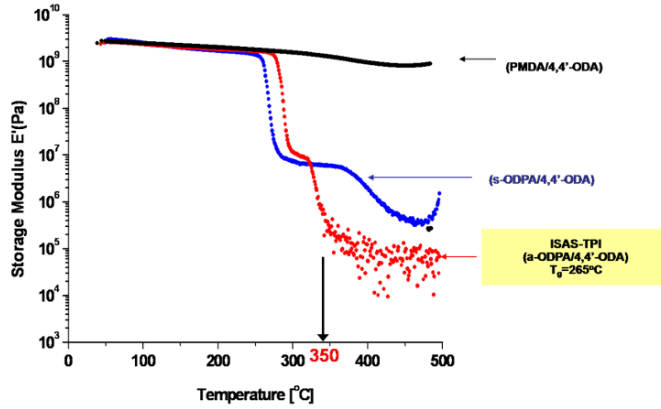

Fig. 17 New Polyimide Film for Heat Seal

In order that the self-heat-seal capability be available. JAXA developed its own membrane material. From the radiation resistance point of view, current space technology has only Polyimide as usable plastic. However, polyimide has a significant drawback. That is, it is too hard to be heat sealed. It is characterized by glass transition temperature. JAXA has developed new material that allows heat seal characteristics at the same time has high radiation resistance. (Fig. 17)

Fig. 18 Spiral Folding

Ikaros adopted petal-separated membranes and tool double deployment sequences. JAXA still thinks the method is nominal even for Solar Power Sail mission to Jupiter and Trojan asteroids. However, JAXA has been investigating another various kinds of folding not only for the mission to Trojan asteroids but also for other practical applications. This is not a single method but the researchers at JAXA looks at spiral folding that can be folded down to a single cylinder as shown in Fig. 18.

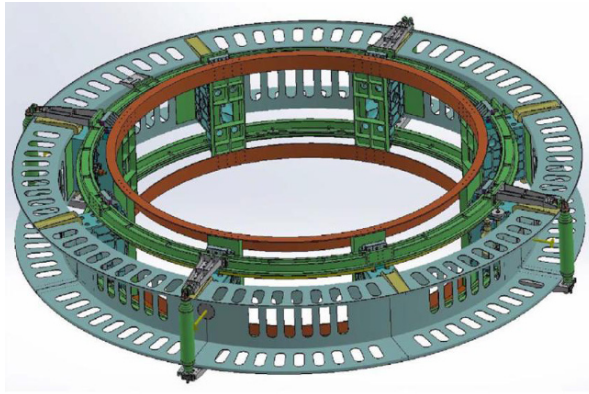

Fig. 19 Slip Ring for Double Spinner Configuration

Since the acceleration direction by ion engine is fixed with respect to the inertial frame, there is a configuration idea about how to have ion engine thrust frozen to inertial space. It is the use of double spinner configuration, in which the membrane is spanned via centrifugal force, but the ion engines are mounted on the stator that is free from spin motion. As the huge power must be provided from solar cell on the sail membrane, there must be a slip ring incorporated on the spacecraft. Fig. 19 shows an idea about it. JAXA has fabricated the model for it for system verification. It shall be noted that the use of double spinner configuration has not been confirmed and may resort to a single spin configuration for simplicity.

V. Conclusion

While JAXA successfully accomplished the Ikaros mission as a step for the real Trojan asteroids mission, we think we should have a new era, Another Age of Discovery. Humans will have a Deep Space Port at L1 or L2 points, from where spaceships have round trip cruises for returning resources from asteroids or small objects. Our activity including solar sails or solar power sails all lead to the realization of the age. (Fig. 20)

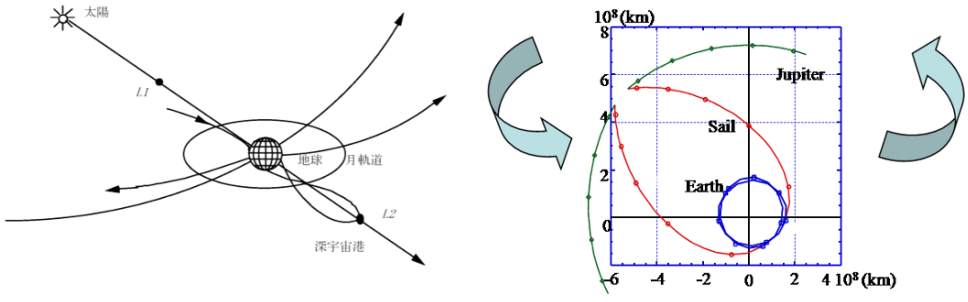

Fig. 20 Deep Space Port and Round Trip Cruise for Resource Utilization

We have had some important lessons as for solar sails. They are, 1) Folding, Extraction of Membrane, 2) Still puzzled to understand Deployment, 3) Convex or Concave, Torsion Deformation, etc., 4) Anti-Environmental Liquid Crystal Device, and 5)Thin Coating for Thermal and Structural measure for This Film Cells. We think we shall continue to secure all of these technology issues.

The paper has presented the following subjects and here are some remarks as to it.

A Solar Power Sail taps for a new era, consisting of : *Thin-Film Solar Cells in Large Area Membrane deployment technology*, A Baseline : *Solar Power, Hybrid propulsion spacecraft, aiming at Trojan asteroids rendezvous*, is provided. And , the paper presents *World's First & First Class Scientific output and Technology demonstration , Ikaros Demonstrator Mission* was successfully performed. It flew by Venus at the beginning of December 2010. It demonstrated Gravity Assist is feasible via Solar Sails.

JAXA thinks we are ready to host joint Work Shops with International Partners, such as the Planetary Society.

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An Overview of Solar Sail Propulsion within NASA

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Solar Sail Propulsion (SSP) is a high-priority new technology within The National Aeronautics and Space Administration (NASA), and several potential future space missions have been identified that will require SSP. Small and mid-sized technology demonstration missions using solar sails have flown or will soon fly in space. Multiple mission concept studies have been performed to determine the system level SSP requirements for their implementation and, subsequently, to drive the content of relevant technology programs. The status of SSP technology and potential future mission implementation within the United States (US) will be described.

Nomenclature

ΔV	=	delta-v (change in velocity)
AU	=	astronomical unit
g	=	gravitational (measurement of acceleration felt as weight; a force per unit mass)
GeV	=	giga-electron-volts
kg	=	kilogram
L1	=	Lagrange point 1
m^2	=	meters squared

I. Introduction

Solar Sail Propulsion (SSP) was identified as a high-priority new technology by the Space Studies Board of the National Research Council in their 2012 Heliophysics Decadal Survey [1], and several potential future space

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missions have been proposed that will require SSP to be achieved. Missions identified include those that would support enhanced space weather monitoring and prediction by placing an instrumented satellite at Sun/Earth L1 (often called *Heliostorm*), high inclination solar orbiting science satellites (*Solar Polar Imager*), and an ambitious short-travel time mission to beyond the Heliopause (also known as *The Interstellar Probe*).

This renewed interest in solar sails is manifested by several small and mid-sized technology demonstration missions within the US that use solar sails, ranging in size from cubesat-scale sails designed for primary operation in Earth orbit to 40-meter class sails capable of operating anywhere in Earth-Sun space. These include the NASA Marshall Space Flight Center's (MSFC) already-flown *NanoSail-D*, the *CubeSail* being developed by The University of Illinois at Urban/Champaign, and the L'Garde/NASA *Sunjammer*.

Several mission concept studies have also been performed to determine the system level SSP requirements for their implementation and, subsequently, to drive the content of NASA's technology programs. NASA's MSFC recently studied the use of SSP in a Mars sample return architecture, providing a potential cost reduction to the overall mission by eliminating the need for one of the three launch vehicles required in the baseline chemical propulsion approach [2]. Another MSFC-led study examined the use of SSP to enable asteroid survey missions to perform low-cost reconnaissance of candidate Near Earth Asteroids (NEAs) for future human exploration [3]. The NASA Innovative Advanced Concepts (NIAC) program is funding innovative advanced sail technology studies, such as the "Steering of Solar Sails Using Optical Lift Force" [4] and "Solar System Escape Architecture for Revolutionary Science (SSEARS)" [5] studies. Various academic studies have been conducted looking at very advanced sails and architectures, including those that make use of new materials like graphene for enabling interstellar missions [6] and flotillas of sails to act as sunshades for mitigating the effects of global warming by reducing the incident solar flux [7].

II. Solar Sails In Perspective: 1990's to the Present

While SSP systems have been studied since the early twentieth century, it wasn't until the 1990's that serious engineering development began. In 1993, the Russian Space Agency conducted a successful solar sail experiment, called *Znamya*, which was deployed from an unmanned Progress vehicle after it departed from their Mir Space Station. A 20-meter circular sail-like reflector was successfully deployed; a follow-up experiment in 1999 collided with a deployed spacecraft antenna and was destroyed [8].

The first free-flying and autonomous solar sail spacecraft built for flight was The Planetary Society's privately funded *Cosmos 1*. Once launched, the sail was to have deployed eight triangular 'heliogyro' blades in

LEO and demonstrated controllable solar sailing [9]. Unfortunately, the sailcraft never reached orbit due to a launch vehicle failure.

In 1999, Germany's Deutschen Zentrum fur Luft-und Raumfahrt (DLR) performed a series of ground system demonstration tests of a 20-meter 3-axis stabilized solar sail system. The system was not flown in space, but it clearly showed that the technology was mature enough to be considered for near-term space mission applications [10].

Partially in response to these significant international milestones, in the early 2000's NASA made substantial progress in the development of its own solar sail propulsion systems. Two different 20-meter solar sail systems were produced. The two design configurations successfully completed functional vacuum testing in the Glenn Research Center's Space Power Facility at Plum Brook Station, Ohio. The sails were designed and developed by Alliant Techsystems Space Systems and L'Garde, respectively. Life and space environmental effects testing of sail and component materials were also conducted [11]. NASA planned to fly a 40-meter class solar sail in 1999, but the project was canceled due to funding constraints.

JAXA began a series of deployment test flights in 2004, leading to the successful flight of the Interplanetary Kite-craft Accelerated by Radiation Of the Sun (*IKAROS*) in 2010. The *IKAROS* is the first deep-space demonstration of solar sailing. The *IKAROS* verified solar radiation pressure effects on the sail and performing in-flight guidance and navigation techniques using the solar sail [12].

Using sail material left over from its ground system demonstrations, NASA MSFC developed *NanoSail-D*, a cubesat-based solar sail deployment verification experiment. It was not intended for actual solar sailing, but rather as a demonstration of a sail to act as an atmospheric drag device for deorbit. The initial launch failed in 2008. A second attempt was made in 2010 using the flight spare. The second launch attempt was succeeded, the sail deployed, and the spacecraft reentered the atmosphere several months later [13].

III. Potential Future Missions and Concepts

As SSP technology matures, entirely new classes of high ΔV deep space missions will be enabled. Many of the scientific questions these missions address have been unanswered for decades because the technology to fly spacecraft to the destinations required to address them was not in reach. With the successes of *NanoSail-D* and *IKAROS*, that is no longer the case, and the scientific community is starting to seriously plan for using SSP-propelled spacecraft to answer them.

A. Solar Polar Imager

Taking advantage of a solar sail's ability to provide nearly unlimited total thrust, an SSP spacecraft can fly into a highly-inclined orbit around the Sun in order to study the Sun's poles. Current spacecraft observations of the Sun are limited to being in or very nearly in the ecliptic plane, limiting solar observations to the solar equator or, at best, mid-latitude regions. A 150-meter class solar sail could slowly raise its inclination from the ecliptic plane and reach a solar polar orbit within just a few years, easily reaching an orientation only previously possible with multiple gravity assist maneuvers and considerable flight time [14].

B. Heliostorm

Space between the Earth and the Sun is not empty. Events on the Sun, such as Coronal Mass Ejections (CMEs) send massive clouds of plasma into space with energies as high as 1 GeV. The radiation in these plasma clouds is significant and can cause damage to unprotected spacecraft and humans operating beyond the protection of the Earth's magnetosphere. The events that occur in this ever-changing space environment are called "space weather."

The *Heliostorm* mission would use a solar-sail propelled spacecraft to station a space weather detection and characterization payload sunwards of the classical L1 point along to the Sun-Earth line, potentially placing it twice as close to the sun's natural L1 point at 0.993 AU. The sail is used to maintain an artificial equilibrium position displaced both sunward of and slightly off the Sun-Earth line. Onboard instruments will detect solar CMEs much earlier than is now possible and potentially double the warning time of impending terrestrial space weather events [15].

C. Interstellar Probe

Humanity's first intentional step into nearby interstellar space may occur with the launch of the *Interstellar Probe* (*ISP*) [16]. The *ISP* would use a 160,000 m^2 solar sail with an areal density of less than 1 g/m^2 to achieve a flight time of 20 – 25 years to 250 AU. The velocity required for such a short flight time would be achieved by having the sail make a very close solar flyby, taking advantage of the increased solar flux and the resultant increased solar photon pressure, to accelerate the sailcraft. The *ISP* would be instrumented to study the heliopause and the interstellar medium and make a significant penetration into nearby interstellar space, with a minimum goal of reaching 250 AU and a desire to reach as far as 400 AU before the spacecraft systems cease functioning. Sails this large are significantly beyond the state-of-the-art, and additional research and development will be required in sail materials, deployment, and attitude control systems to make it viable.

Additional missions to ever-increasing interstellar distances would follow, culminating in one traveling to about 1000 AU, aptly named the Thousand AU, (TAU) Mission.

D. Near Earth Asteroid Rendezvous

Crewed missions to NEAs are planned for early in the 2020's. To reduce the inherent risk associated with visiting any solar system body for the first time, an assessment of it must be performed robotically, so the crew systems appropriate for the NEAs unique environment can be developed. For example, operations on or near an asteroid composed of a single large piece of rock will be very different from those required to support the exploration of a loosely gravitationally-bound "rubble pile" asteroid.

Conventional rockets, and even unconventional ones using solar electric propulsion, have been considered for this class of precursor mission, and none can match the ability of a solar sail to visit multiple targets sequentially in a relatively short period of time.

NASA MSFC studied various options for implementing an SSP NEA precursor mission and found that an 80-meter 3-axis stabilized solar sail spacecraft launched by a small rocket can easily rendezvous with asteroids 1999 AO10, Apophis, and 2001 QJ142 – all of which are on the most recent listing of "Human Accessible NEAs" – within six years of launch. The notional spacecraft science payload was selected to be simple and lightweight, consisting of only the multispectral imager flown on the *NEAR* Mission to 433 Eros and 253 Mathilde. Most non-sail spacecraft systems baselined were used in the *Messenger* and other flight-proven mission spacecraft.

E. Mars Sample Return

Scenarios for robotic missions to return samples from other planets are inherently complex and will require the sequential launch of many spacecraft, landers, and sample return stages to implement. For this type of mission, as well as any multi-staged space system, it is clear that by minimizing the mass of the final stage, the initial system mass will be reduced by simply reducing the propellant required in earlier stages. Therefore, the best way to minimize the launch mass of the *Mars Sample Return* (MSR) mission is to reduce the mass of the Earth return stage (ERS).

Using chemical or even solar electric propulsion, the Mars ERS will require a significant amount of fuel to leave Mars orbit and travel back to Earth; hence, the earlier mission stages will be required to transport this mass to Mars, in turn requiring more propellant in these mission stages. In fact, under current designs, nearly half of the orbiter mass is dedicated to returning samples to Earth and stopping the Earth-return propellant at Mars. By

replacing the chemical propulsion system of the ERS with a solar sail, most, if not all, of this fuel is eliminated. This reduces the fuel mass of the earlier stages and can result in smaller spacecraft, fewer required launches, or both.

The current MSR orbiter is planned to arrive two years ahead of the Mars ascent vehicle (MAV) and lander. The orbiter will provide communication links to the lander and monitor the lander's entry, descent, and lander phases. The orbiter is also required to be the active partner in the rendezvous and capture of the sample canister once the MAV launches it from the surface of Mars.

NASA MSFC studied the use of a solar sail to perform the Earth return portion of the MSR Mission. All other required propulsive functions would still be performed with traditional storable liquid propellant thrusters. Upon retrieval of the sample canister, the solar sail will deploy and begin the low-thrust spiral out of the Martian gravity well. As a means to save mass and reduce the required size of the solar sail, it is proposed that the liquid propulsion system be jettisoned in Mars orbit, potentially reducing the mass of the return system by an additional 10-20 percent.

MSFC also studied the use of a solar sail to perform the Earth-Mars transfer, as well as the return leg. While this would entirely eliminate the orbiter propulsion system and further reduce the system's mass, it would increase the overall trip time beyond what is required for the all-chemical mission approach. It also may be possible, given the large surface area of a solar sail, to use the sail to perform an aerocapture maneuver through the extremely thin upper atmosphere of Mars. .

IV. Technology Development and Demonstrations

Within NASA, solar sail propulsion is moving forward to mission implementation on both the large (100 kg class spacecraft payloads) and small (cubesat) scales. From Sunjammer's planned deep space flight validation to the Earth-orbital CubeSail, solar sail capability is advancing rapidly.

A. Sunjammer

The L'Garde Sunjammer Mission will demonstrate large solar sails (~1200 m^2) for future NASA mission use. The Sunjammer design relies on L'Garde inflatable rigidizable technology for the sail's structure. The sail structure consists of four conically-stowed, deployable, cold-rigidized booms. These four booms are connected at the center of the sail area to the sailcraft bus. Shortly after launch, the solar sail will be deployed to full size and separated from the carrier. Within a few days of launch, the sail will be calibrated and trimmed, and for the next several weeks, the sail will demonstrate attitude control, guidance, navigation, and control by flying in deep

space to near the Earth-Sun L1 and finally to sub-L1. Along the way, an onboard magnetometer will be measuring magnetic field and comparing the results with The National Oceanic and Atmospheric administration (NOAA's) Advanced Composition Explorer. It is hoped that these measurements will foster infusion of this technology into the space weather monitoring community. Launch is anticipated in 2014 [17].

B. Steering of Solar Sails Using Optical Lift Force

Scientists at the Rochester Institute of Technology (RIT) found that transparent refractive objects may settle into a position where they feel a force that is perpendicular to the incoming light direction. Because this transverse force is analogous to the lift experience by an airplane wing, it is called "optical lift." Dr. Swartzlander and his team at RIT completed a NIAC Phase I study in 2001 looking at the physics, technology, and mission applications of optical lift with an emphasis on their application to solar sailing.

Optical wing structures were theoretically and numerically analyzed, and prototype arrays of wings called optical flying carpets, were fabricated with solar sail material clear polyimide (CP1). Various optical wing sizes and shapes were analyzed to develop design strategies for thrust and torque applications. The developed ray-tracing model has undergone continual advancement and stands as an effective tool for modeling most types of solar sails.

The distributed forces and torques have been reduced to a simple theoretical, whereby the fundamental mechanics may be understood in terms of the numerically-determined center of pressure offset from the center of mass. This research has established a foundation for understanding the force and torque afforded by optical wings. The study began by considering transparent wings and ended by considering wings having a reflecting face. The latter was found to afford the advantages of high thrust and both intrinsic and extrinsic torque. The discovery of the intrinsic torque on optical wings (meaning that a moment arm is not required) has no analogy for a flat reflective solar sail, and therefore provides an extra degree of control that may be useful for sailcraft attitude and navigation purposes.

C. Solar System Escape Architecture for Revolutionary Science (SSEARS)

NASA's Jet Propulsion Laboratory (JPL) is exploring the feasibility of using 10 or more sail-propelled probes to explore the heliopause in multiple orthogonal locations in an integrated and synergistic approach, providing a three dimensional map of the solar system's boundaries.

V. Conclusion

For many years, solar sail propulsion was an interesting idea with many mission benefits, but within NASA, the technology was without mission pull. In other words, no official future missions were being planned that would require their use. Since the success of *IKAROS* and the selection and funding of the *Sunjammer* demonstration mission, the reality of solar sails has manifested in the Agency's planning, and there are now numerous mission and project teams considering their use.

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Overview of IKAROS Mission

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The Japan Aerospace Exploration Agency (JAXA) makes the world's first solar power sail demonstration of photon propulsion and thin film solar power generation during its interplanetary cruise by IKAROS (Interplanetary Kite-craft Accelerated by Radiation Of the Sun). It deployed and spans a membrane of 20 meters in diameter taking the advantage of the spin centrifugal force. It accelerates and controls the orbit using solar radiation pressure successfully. This is the first actual solar sail flying an interplanetary voyage. This paper presents the summary of development and operation of IKAROS.

I. Introduction

A solar sail can move forward without consuming propellant as long as it can generate enough energy from sunlight. This idea of a solar sail was born about 100 years ago and we often find it in science fiction novels. The solar sail missions are studied in the world [1, 2].

A Solar Power Sail is a Japanese original concept that gets electricity from thin film solar cells on the membrane in addition to acceleration by solar radiation. A solar power sail craft can save the fuel using a solar sail and it can also gain the necessary electric power using a vast area of thin film solar cells on the membrane even when it is away from the sun. It can be a hybrid propulsion system with a solar sail by activating the ultra-

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high specific impulse ion engines with the power generated by thin film solar cells. Thus solar power sails are suitable for outer planetary exploration.

The authors have studied an Advanced Solar Power Sail mission toward Jupiter and Trojan asteroids via hybrid electric photon propulsion as shown in Fig. 1 [3, 4]. The mission proposal passed the Mission Definition Review and now is eligible to go into the pre-project phase (Phase-A).

However solar sail and solar power sail were not realized. It was considered as high risk mission. We applied first for the small technology demonstrator mission, IKARO as a Front-Loading of new key technical issues of advanced solar power sail craft [5]. The proposal was endorsed in fall, 2007. IKAROS (Interplanetary Kite-craft Accelerated by Radiation Of the Sun) demonstrated the membrane deployment and thin film power generation. It is also the world's first actual solar sail flying an interplanetary voyage as shown in Fig. 2. IKAROS is not hybrid propulsion system, because ion propulsion engine is not mounted. This paper presents the summary of the development and operation of IKAROS.

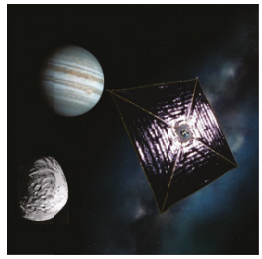


Fig. 1 Advanced solar power sail mission.

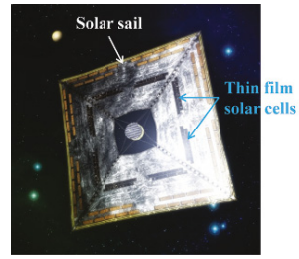


Fig. 2 IKAROS mission.

II. IKAROS Mission

The success criteria of the IKAROS mission are summarized as follows;

(1) Deployment of Large Membrane Sail

-Deployment and Expansion of a Large Membrane in space using similar mechanical device and procedures to those in Advanced Solar Power Sail craft.

-Obtaining a number of data indicating the expansion status of the membrane.

(2) Generating Electricity by Thin Film Solar Cells

-Demonstrating Solar Power from Thin Film Solar Cells

-Evaluating performance of thin film solar cells on the membrane in space.

These two items belong to the Minimum Success Criteria.

(3) Demonstrating Photon Propulsion

-Verification and Comparison of Reflectance with Specular & Diffuse Property.

-Measurement of overall Reflectance with a rigorous relation examination of the temperature and surface status.

(4) Demonstrating Guidance, Navigation Control Skills for Solar Sail Propulsion

-Navigation / Orbit Determination under continuous and small acceleration.

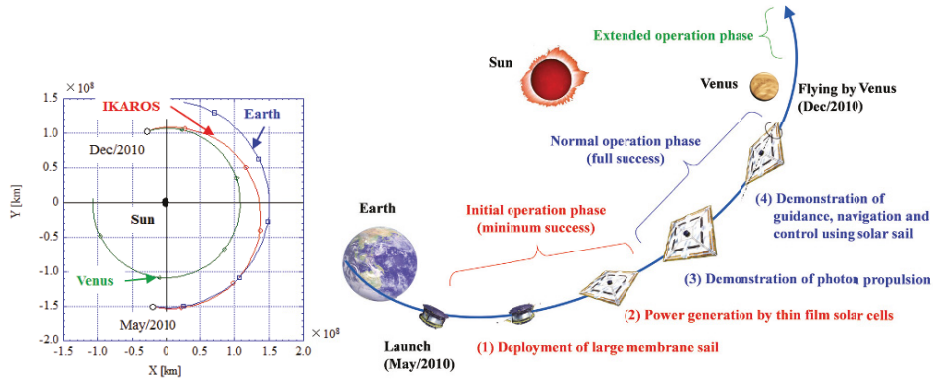
-Acceleration Direction Control by Steering via appropriate attitude control means.

These latter two items belong to the Full Success Criteria.

IKAROS was launched together with the Venus Climate Orbiter, AKATSUKI in May 2010 as shown in Fig. 3. The orbit and mission sequence was designed as shown in the Fig. 4. After sun-pointing and spin separation from the rocket, the initial operation check including the link establishment was performed. IKAROS span up to 25 rpm and deployed the large membrane using the centrifugal force. The power generation by thin film solar cells was confirmed after the deployment. It took several weeks to achieve the minimum success. The photon propulsion was verified and the guidance, navigation and control by solar sail were demonstrated for full success mission in the next six months. IKAROS flew by Venus at the beginning of December 2010.

IKAROS entered its extended operation phase at the beginning of 2011. In the extended operation, the spin rate was decreased to observe the deformation of the sail under low centrifugal force environment. On Oct. 18, 2011, IKAROS was transferred to the reverse spin to enhance the knowledge about the effect of stiffness of membrane against the solar radiation pressure. At the end of 2011, IKAROS moved to hibernation mode because the Sun angle was increased. We searched for IKAROS considering the attitude and orbital motion during hibernation successfully.

Fig. 3 Launch of IKAROS by H-IIA launch vehicle.



a) Orbit. **b) Mission sequence.**

Fig. 4 Orbit and mission sequence.

III. Development of IKAROS Spacecraft

A. IKAROS System

The major characteristic of IKAROS is shown in Table 1. The spacecraft spans a square membrane sail whose tip-to-tip length is 20 m long, and weights about 310 kg. The sail, spanned using the its spin centrifugal force, includes thin film solar cells which occupy approximately 5 % of the total area. The main body is a spinner and its shape is cylindrical. It carries a drum around which a membrane is wound to be unfolded via a special mechanics aboard.

Where sail membrane is wound is the mission portion that is independent of the bus hub portion as shown in Fig. 5. The spacecraft design started from fall of 2007, just 2.5 years prior to the launch and the design process intentionally adopted No-EM (Engineering Model) development strategy taking the advantage of relatively sufficient mass margin indicated from the launch vehicle. Besides, the existing hardware surplus as well as reproduction of existing design was fully exploited to shorten the development period and to cut cost.

Table 1 Major characteristics.

Size (m)	Main body	Diameter 1.6, Height 0.8 (Cylinder)
	Sail	Diagonal 20 (Square)
Mass (kg)	Wet mass	310 (including sail mass)
	Dry mass	290 (including sail mass)
	Sail mass	15 (including 4 tip masses)

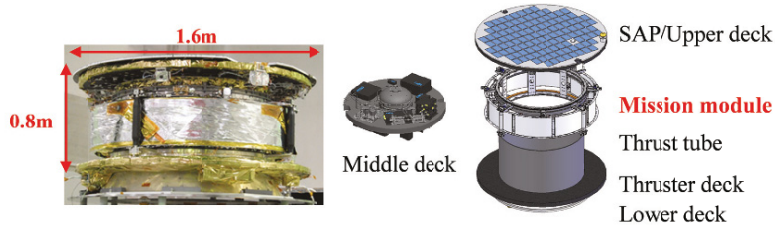


Fig. 5 IKAROS appearance.

B. Sail

A solar sail requires a thin film mirror that is extremely light and strong enough to hold a vast area. The sail of IKAROS is made of ultra-thin polyimide resin whose thickness is 7.5 μm (about 1/10 of a human hair.) It has a high tolerance for space environment requirement. Figure 6 shows the sail shape and equipment layout.

Membrane [5]: The shape of the membrane is a square whose diagonal distance is 20 m. The direction of folded lines is perpendicular to the direction of the centrifugal force. It is made of two kinds of polyimide resin as shown in Fig. 7. Polyimide-1 is an existing film and it needs to be connected to each other by adhesive. On the other hand, polyimide-2 is thermoplastic polyimide which is developed and manufactured by JAXA newly for the advanced solar power sail mission. It can be fused with each other by heater control without adhesive. If the membrane is made of thermoplastic polyimide, the membrane becomes lighter and stronger and can decrease outgassing. Aluminium is vapor deposited on front side of the membrane as shown in Fig. 8, in order to reflect sunlight more efficiently. In addition, the film is reinforced in such a way as to prevent it from splitting all the way if it's ripped. If it is ripped, the solar sail performance declines slightly, but it can still continue its space travels.

Thin film solar cells [6]: a-Si (amorphous silicon) solar cells as shown in Fig. 9 are attached to certain areas of the membrane. They generate almost 300 W. The area ratio is 5 %.

Reflection control device [7]: Orbit control requires changing the angle (facing direction) of the membrane to the sun. Liquid crystal films are loaded near the tips of the membrane. They can be switch specular and diffuse reflection by power ON and OFF according to the spin rate to control the sun angle as shown in Fig. 10.

ALADDIN: PVDF (PolyVinylidene Di-Fluoride) film is attached on the back side as dust counter. It is the optional equipment which is not for the mission of solar power sail demonstration.

Tip mass: A 0.5 kg weight is connected to each tip of the membrane. It supports the deployment of the membrane.

Tether and harness: Tether connects the membrane with the main body mechanically and harness connects them electrically.

In addition, electric charge-measuring patches and thermometers are equipped on the membrane.

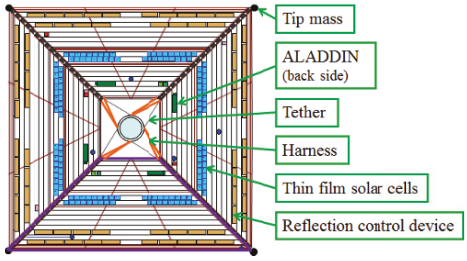


Fig. 6 Sail shape and layout.

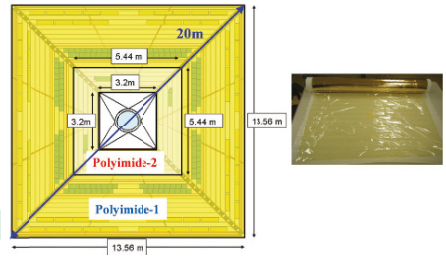


Fig. 7 Membrane size and polyimide resin.

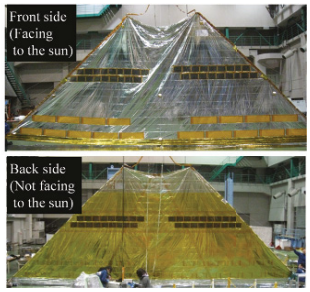


Fig. 8 Quarter of the square sail.

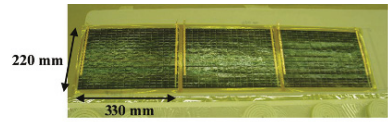


Fig. 9 Thin film solar cells.

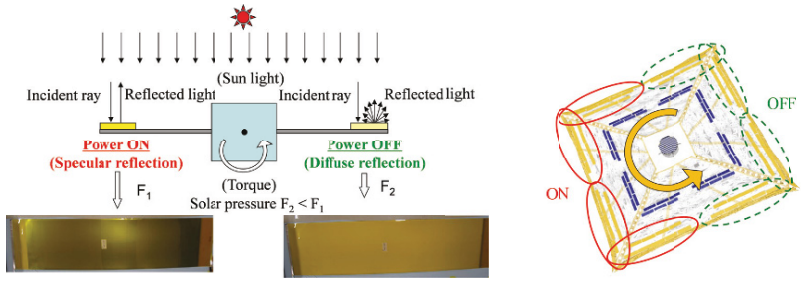


Fig. 10 Attitude control by reflection control device.

C. Development System

The most difficult mission of IKAROS is the deployment of membrane sail. There are two types of solar sail. First is the mast type which uses some rigid support structure to deploy and maintain the sail [8, 9]. The other is the spin type which uses spinning centrifugal force. The deployment motion and attitude control of mast type are simpler than those of spin type. On the other hand, spin type can be accomplished with lighter-weight mechanisms than mast type because it does not require rigid structural elements. Thus spin type should be

selected in case of a large membrane. IKAROS demonstrated a spin type solar power sail whose area was 200 m^2 for advanced solar power sail whose area is 3000 m^2. The feasibility was verified by a lot of experiments and numerical simulations [10, 11].

The proposed deployment method of the sail is shown in Fig. 11 [12]. It consists of three processes: tip mass separation, first stage deployment and second stage deployment.

(1) Tip mass separation

In the folded configuration, each quarter of the membrane is line-shaped and rolled up around the main body. After the separation from the launch vehicle with the slow spin at 5 rpm, IKAROS spins down to 2 rpm using gas jet injections in order to decrease the centrifugal force of four tip masses. They are separated at the same time by activating the tip mass separation mechanism.

(2) First stage deployment

IKAROS spins up to 25 rpm using gas jet injections to increase the centrifugal force. In the first stage deployment, each quarter of the membrane is extracted like a Yo-Yo despinner and the membrane forms a cross shape. If the deployment is performed dynamically, each quarter is twisted around the main body again just after the deployment. Therefore the first stage deployment is performed quasi statically by activating the rotation guides that hold the membrane through the relative rotation mechanism (motor drive). The membrane is deployed slowly and gradually through centrifugal force and the spin rate dwindles as the relative rotation mechanism activates. After the completion of the first stage deployment, the cross shape is maintained by rotation guides.

(3) Second stage deployment

The second stage deployment is performed dynamically by activating the rotation guides and releasing the hold of the membrane. The membrane expands quickly to form a square shape. The spin rate is decreased in the first and second stages of the deployment, because the inertial momentum of the sail is increased.

The motion of the main body and the membrane is checked by a sun sensor, a three-axis gyro, range rate, four monitor cameras, two separation cameras [13] and so on as shown in Fig. 12.

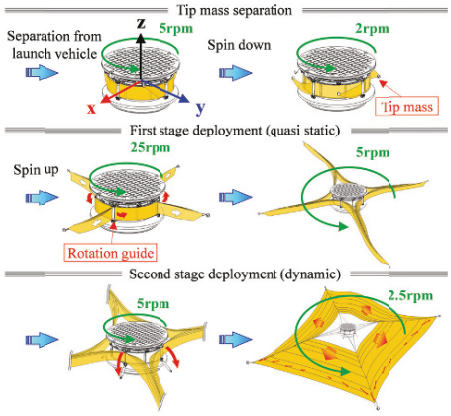


Fig. 11 Deployment method.

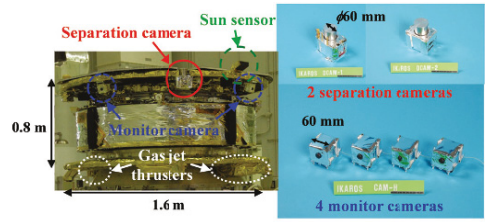


Fig. 12 Sensors and gas jet thruster.

IV. IKAROS Operation

IKAROS was launched together with AKATSUKI aboard H-IIA F17 at 6:58:22 am on May 21, 2010 (Japan Standard Time) from the Tanegashima Space Center. The initial operation was performed to achieve the minimum success until June 29, 2010. The normal operation was conducted for full success. IKAROS entered its extended operation phase at the beginning of 2011.

A. Initial Operation

Summary of the initial operation is as follows.

May 21, 2010: Launch, Spin separation (=> 5 rpm)

May 23-25, 2010: Spin down (=> 2 rpm)

May 26, 2010: Tip mass separation

May 27-31, 2010: Spin up (=> 25 rpm)

June 2-8, 2010: First stage deployment (=> 5 rpm)

June 9, 2010: Second stage deployment (=> 2.5 rpm)

June 10, 2010: Solar power generation through thin film solar cells

June 14, 2010: 1st separation camera experiment

June 16-18, 2010: Spin down (=> 1 rpm)

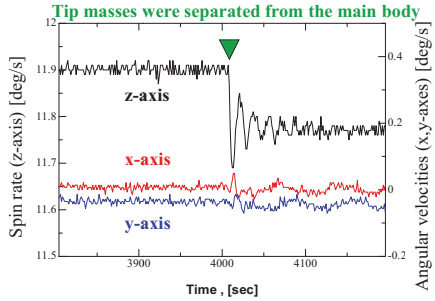
June 19: 2nd separation camera experiment

June 21-25, 2010: Start up of optional equipments

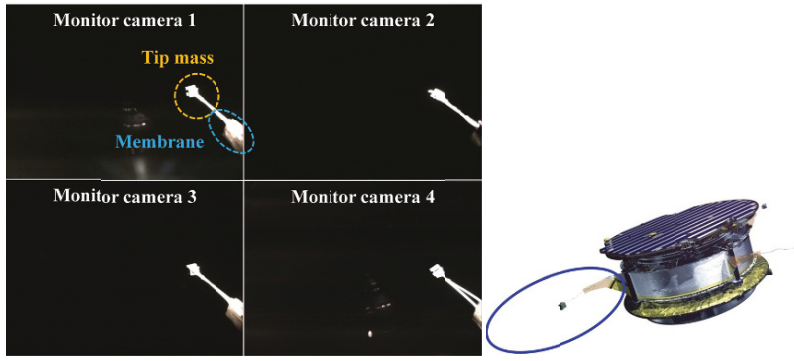
June 29, 2010: Transition to normal operation

From May 26 to June 9, 2010, the membrane was deployed. Figure 13 shows the data of the tip mass separation. Due to conservation of angular momentum, the spin rate was decreased, when these tip masses were separated from the main body. The angular velocities of x, y-axes were zero before and after this event. If one of tip mass does not be separated, the nutation motion of the main body was occurred. Thus four tip masses were separated at the same time, despite a lot of locks for flight security. The pictures by four monitor cameras also confirm that four tip masses were separated successfully.

Figure 14 shows the data of the first stage deployment. The rotation guides hold the membrane. By rotating them relatively, the membrane was deployed gradually and similarly. The pictures by four monitor cameras confirm that the membrane was deployed smoothly and formed a cross shape after the first stage deployment.

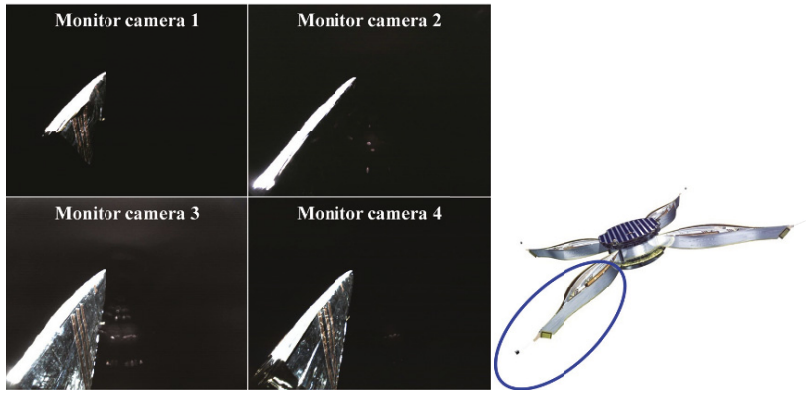


a) Spin rate and angular velocity of main body (May 26, 2010).

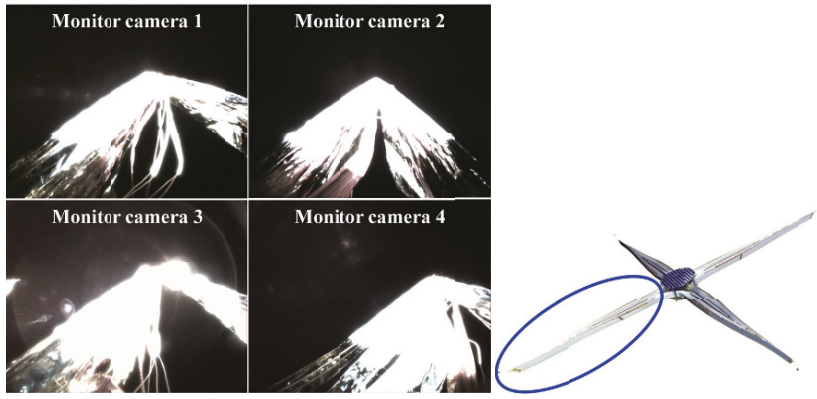


b) Pictures taken after tip mass separation by monitor camera 1 (May 29, 2010).

Fig. 13 Tip mass separation.



a) Pictures taken in the middle of first stage deployment by monitor camera 1 (June 4, 2010).



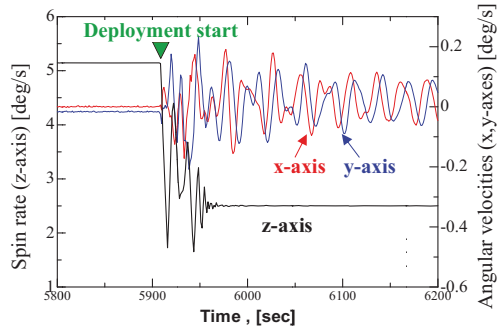
b) Pictures taken after first stage deployment by monitor camera 1 (June 8, 2010).

Fig. 14 First stage deployment.

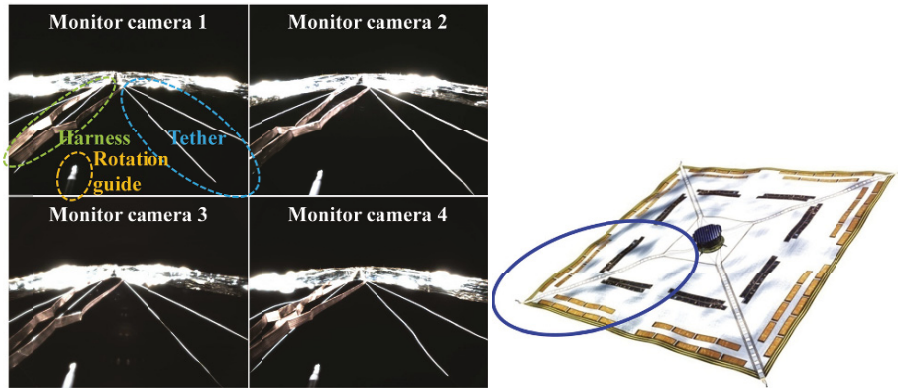
Figure 15 shows the data of the second stage deployment. Just after deployment start, the vibrations of angular velocities were occurred. However they were dumped gradually. The spin rate of z-axis was decreased due to conservation of angular momentum. Thus the membrane was deployed dynamically and symmetrically. The pictures by four monitor cameras confirm that the membrane, tethers and harnesses were not gotten hung up on the released rotation guides.

On June 10, 2010, the membrane deployment and solar power generation by thin film solar cells were confirmed. We gathered V-I characteristics and several related data constantly as shown in Fig. 16.

On June 14 and 19, 2010, two separation cameras successfully took impressive images as shown in Fig. 17. The membrane was expanded to form a square shape without damages. It was shining in the dark space. The second separation camera made sure that the reflection control device was working properly. How ON and OFF reflection control devices looks like in orbit was confirmed.



a) Pictures taken in the middle of first stage deployment by monitor camera 1 (June 9, 2010).



b) Pictures taken after first stage deployment by monitor camera 1 (June 10, 2010).

Fig. 15 Second stage deployment.

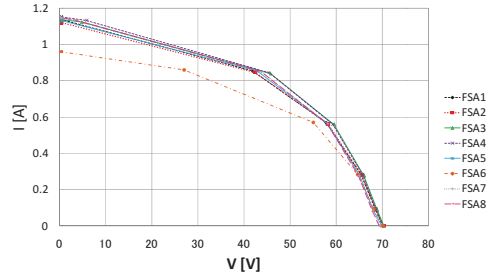
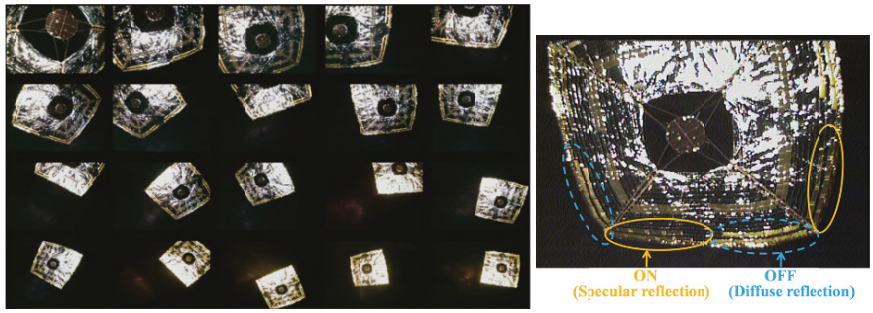


Fig. 16 Observed power generation by thin film solar cells (June 10, 2010).



a) First separation camera (June 14, 2010). b)Second separation camera (June 19, 2010).

Fig. 17 Pictures shot by separation cameras.

B. Normal Operation

After the initial operation, the normal operation was conducted for full success. The photon acceleration was confirmed in the course of determining its precise orbit after its sail deployment. Figure 18 shows the difference between calculated value (without photon acceleration) and observed value (result of range rate measurement) of IKAROS's line-of-sight velocity on June 9, 2010. For about one hour around 9:36, when the second stage deployment was in operation, data was lost; however, you can clearly see that the graph line starts inclining after the second-stage deployment compared to the flat line prior to the event. The calculated value does not incorporate photon acceleration, thus, the velocity difference becomes a flat line if there is no photon acceleration. It is not equal to zero due to the error of estimated orbit motion. With photon acceleration, the line inclines. The thrust force is calculated as 1.1 mN and it is nearly equal to the designed value. This is the world's first data of solar sailing.

IKAROS continued to be accelerated by solar sail. The delta-V of 100 m/s was achieved within half a year.

The attitude control performance using the reflection control device was successfully accomplished. Figure 19 shows the evaluation result of the attitude control experiment by the reflection control device on July 13, 2010 based on sun angle. The solar angle was gradually increasing without control, while it was decreasing when the reflection control device was used to control the attitude toward the sun direction. Very smooth attitude maneuver was achieved. The reflection control enabled IKAROS to change its attitude for orbit control.

IKAROS flew by Venus on December 8, 2010 as shown in Fig. 20. The distance at closest point to Venus was 80800 km. IKAROS was arrived at Venus one day later than AKATSUKI, because IKAROS was decelerated by solar sail.

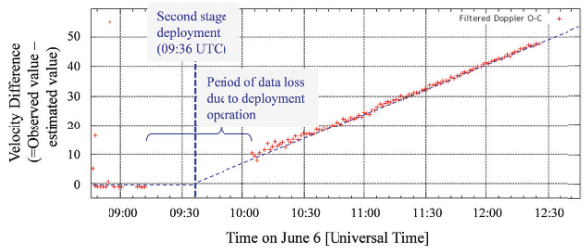


Fig. 18 Confirmation of photon propulsion by range rate measurement (June 6, 2010).

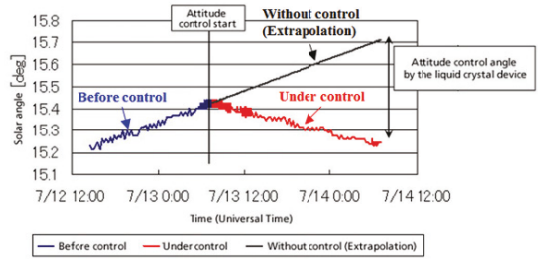


Fig. 19 Attitude control performance using reflection control device (July 12-14, 2010).

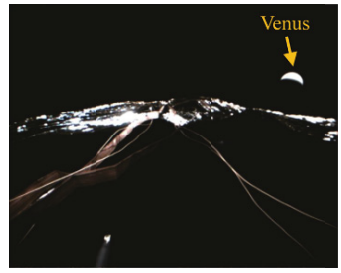


Fig. 20 Flying by Venus (Dec. 8, 2010).

C. Extended Operation

(1) Slow Spin Mission

To expand and maintain the shape of the sail membrane against the solar radiation pressure, the spin rate of IKAROS had been kept higher than 1 rpm for about one year from the completion of the sail deployment on June 9, 2010.

The slow spin mission was started in June 2011 in the extended operation. This mission aimed to see the effect of stiffness of membrane against the solar radiation pressure under low centrifugal force environment.

The spin rate of IKAROS had decreased naturally by solar radiation pressure torque (windmill behavior). In the normal operation, the spin rate had been kept by periodical thrust control of gas jet injections. By suspending the spin control, it went down to 0.055 rpm in the slow spin mission.

Figure 21 shows the angular velocity of the main body at 0.055 rpm. ω_z indicates the angular velocity around the z-axis, which is equivalent to the spin rate. This denotes that the oscillations of out of plane angular velocities ω_x and ω_y are nearly equal to the spin rate. This nutation motion can be occurred in a case of the inertia tensor of the system is close to the one of thin plate. From this result, it is considered that the main body had rotated combined with the sail membrane.

In the slow spin mission, some images of the sail were taken by monitor cameras. Figure 22 shows the sail images shot by monitor camera and the sail images expected by numerical simulations at 2.5 rpm and 0.055 rpm [14]. It had been expected by numerical simulations that the decrease of the centrifugal force would cause the deflection of the sail by solar radiation pressure, however the images shot by monitor camera denotes that the sail shape at 0.055 rpm is not so different from the one at 2.5 rpm.

From these results, we concluded that the sail membrane has a larger bending stiffness on an average than expected. By reflecting these results, we modified numerical models of the sail membrane, which can simulate flight data appropriately.

After the slow spin mission, we planned a reverse spin mission. The control sequence of the thrust on this mission was designed based on the result of simulation using the modified high bending stiffness model.

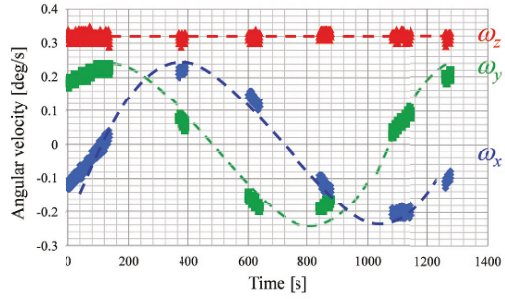


Fig. 21 Angular velocity of the main body during slow spin mission (July 5, 2011).

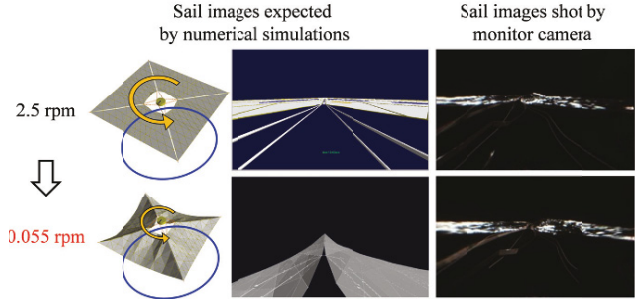


Fig. 22 Sail images at 2.5 rpm and at 0.055 rpm.

(2) Reverse Spin Mission

On Oct. 18, 2011, the reverse spin mission was performed to enhance the knowledge about the effect of stiffness of membrane against the solar radiation pressure and to investigate the attitude dynamics under the different spin direction.

This mission was conducted by gas jet thrust to shift spin direction to the reverse way. The gas jet injections were executed 118 times with 0.4 sec pulse width and at intervals of 10 seconds as planned in advance.

As a result of this mission, the spin rate was turned down from 0.15 rpm to -0.24 rpm, and the sail membrane was successfully spun in the reverse course without being entangled.

Figure 23 shows the angular velocity of the main body during the reverse spin mission. This figure denotes that the spin rate of the main body shifts to the reverse way with the gas jet injections. The spin rate of the main body is pulled back after the gas jet injections by the in-plane oscillation between the sail membrane of large inertia moment, however it shifts totally to the reverse way gradually.

Figure 24 shows the sail images shot by monitor cameras at 0 rpm. These images denote that the shape of the sail was not so deflected even at 0 rpm. From these results, it is considered that the sail membrane of IKAROS has large bending stiffness as expected after the slow spin mission.

Figure 25 shows the attitude motion in the case of reverse spin. The attitude motion of spinning solar sail is governed by the solar radiation pressure. Viewed in the Sun fixed coordinates, The spin axis draws a spiral motion with a certain equilibrium point as the center (spiral behavior). The spin rate varies (windmill behavior). They are caused by the deflection and torsion of the sail. However, the spin direction tended to be inertia oriented according to the increase of absolute value of spin rate. Because the electricity were less than power consumption due to the increase of Sun angle, IKAROS moved to hibernation mode (shutdown) at the end of 2011.

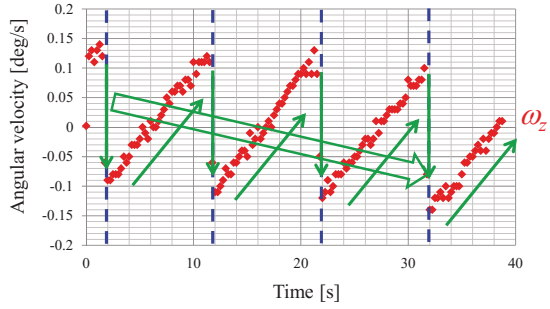


Fig. 23 Angular velocity of the main body during reverse spin mission (July 5, 2011).

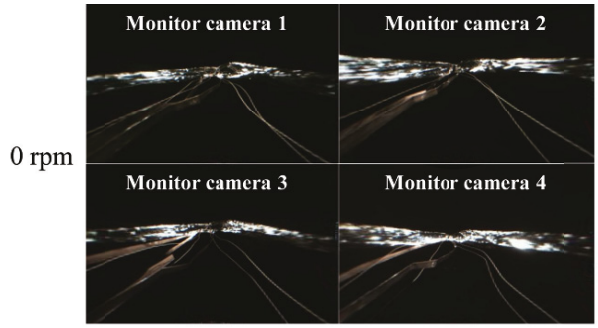
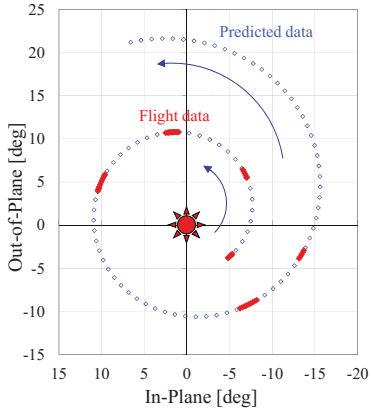
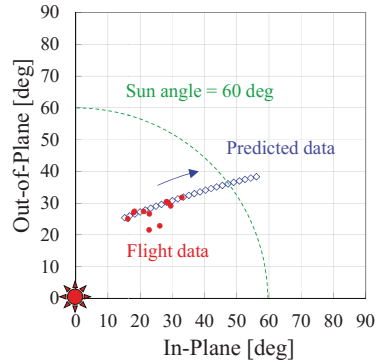


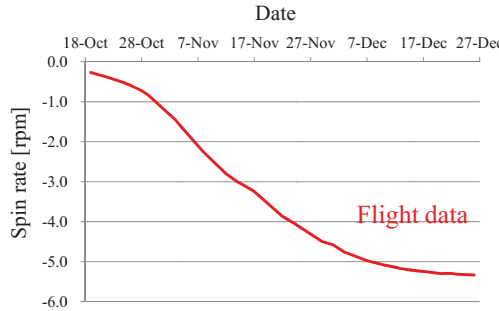
Fig. 24 Sail images at 0 rpm (Oct. 18, 2011).



a) Spin axis history (Oct. 18-29, 2011).



b) Spin axis history (Nov. 29 - Dec. 24, 2011).



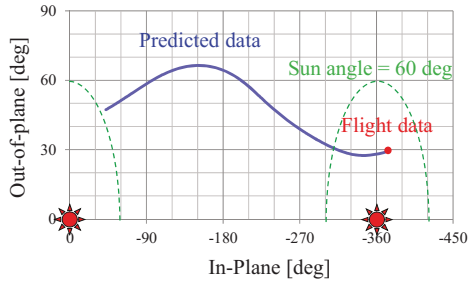
c) Spin rate history (Oct. 18 - Dec. 26, 2011).

Fig. 25 Attitude motion in the case of reverse spin.

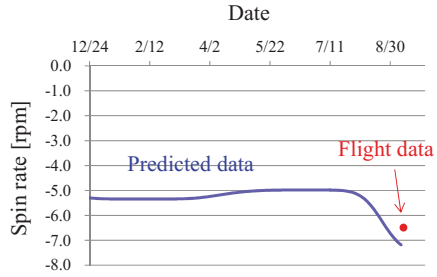
(3) Search Mission

Because IKAROS is accelerated by solar sail, we need to calculate the orbit considering the attitude motion in detail during hibernation, in order to search the IKAROS. Figure 26 shows the predicted attitude motion after

moving to hibernation. We succeeded in tracking IKAROS on Sep. 6, 2012. IKAROS came out of hibernation in latter half of July, 2012 according to the predicted attitude motion.



a) Spin axis history (Dec. 24, 2011 - Sep. 6, 2012).



b) Spin rate history (Dec. 24, 2011 - Sep. 6, 2012).

Fig. 26 Attitude motion after moving to hibernation.

V. Conclusion

This paper presents summary of the development and operation of IKAROS. IKAROS was launched on May 21, 2010. In the initial operation, it deployed the large membrane using the centrifugal force. The power generation by thin film solar cells was confirmed after the deployment. In the normal operation, the photon propulsion was verified and the guidance, navigation and control by solar sail were demonstrated. IKAROS flew by Venus at the beginning of December 2010. In the extended operation, the slow spin and reverse spin missions were performed to observe the deformation of the sail under low centrifugal force environment and to enhance the knowledge about the effect of stiffness of membrane against the solar radiation pressure. At the end of 2011, IKAROS moved to hibernation mode because the Sun angle was increased. We searched for IKAROS considering the attitude and orbital motion during hibernation successfully. IKAROS continues the first actual solar sail flying. The advanced solar power sail mission toward Jupiter and Trojan asteroids is planning based on the results of IKAROS mission.

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How IKAROS Shape is Designed -Attitude Stability of Spinning Solar Sail

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This paper discusses the attitude stability criteria for spinning spacecraft equipped with large flexible membrane, such as solar sail. One actual realization of such a configuration is IKAROS, which is the world-first interplanetary solar sail spacecraft launched by JAXA in 2010. The authors previously derived stability criteria for spinning sail craft. This paper revisits the stability criteria of spinning solar sail spacecraft, and shows how it has been applied to the practical design of the IKAROS spacecraft configuration. The validity of the stability criteria is then evaluated by some numerical simulations and flight data of IKAROS.

I. Introduction

THIS paper discusses the attitude stability criteria for spinning spacecraft equipped with large flexible membrane, such as solar sail, flexible solar array and large aperture light-weight antenna, etc. One actual realization of such a spacecraft configuration is IKAROS (Interplanetary Kite-craft Accelerated by Radiation Of the Sun)[1], which is the world-first interplanetary solar sail spacecraft launched by JAXA in 2010.

IKAROS is a spin-stabilized spacecraft equipped with 14m×14m rectangular solar sail (Fig.1). The shape of the sail is kept by centrifugal force generated by spinning of a whole spacecraft, and thus it does not require any rigid structural members (such as deployable booms) to hold the shape of the sail, which contributes to the realization of light-weight and simple mechanism-solar sail system.

The spin stability of rigid bodies in space is one of the most fundamental knowledge, and appears in many elementary textbooks of space technology. There are also many studies dealing with spinner spacecraft equipped with flexible structural members, such as boom, wire antenna and sloshing tank. For example, McIntyre/Miyagi[2], Agrawal[3] use Lyapnov function to derive the spin stability criteria of spinning spacecraft with liquid-filled tank. Hughes/Fung[4] provides stability criteria for spinning spacecraft equipped with radial

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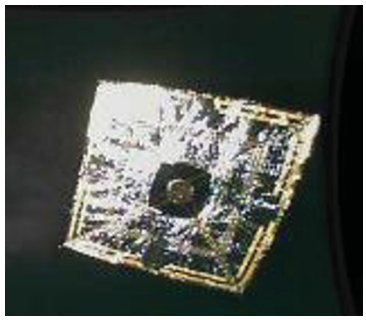

Fig.1 IKAROS with solar sail fully extended.
This picture was taken by the deployable camera (DCAM) on June 14, five days after the deployment of the solar sail.

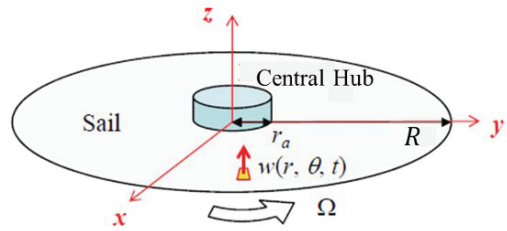

Fig.2 Axisymmetric spinning sail mode for FMM

flexible appendages, introducing a structural mode function to apply to the Lyapnov analysis. Hinada et al.[5] developed stability criteria for spinning spacecraft equipped with radial wire antennas and liquid-filled tanks.

In contrast with those papers, Nakano, et al.[6] and Tsuda[7] have derived an analytical solution for the stability criteria for spinning sail-craft. They provided a fundamental nature of spinning spacecraft with axisymmetric sail. The stability of such spacecraft is worse than rigid spacecraft, and thus one must pay special attention when designing such spacecraft.

This paper revisits the stability criteria of spinning solar sail spacecraft, and shows how it has been applied to the practical design of the IKAROS spacecraft configuration. The validity of the stability condition is then evaluated by numerical simulations and IKAROS flight data.

II. Derivation of Attitude Stability for Spinning Solar Sail

A. Stability Based on First-order Mode Model

One of the special characteristics of solar sails is that, because they are made of thin flexible membrane, their bending rigidity is usually considered negligible in structural analysis. This property is highlighted when considering spinning sail-craft, such as IKAROS, as they extend the sails only by centrifugal force generated by spinning of the whole sail (with spacecraft body), and do not have stems to structurally support the extension of the sail membrane. In such configurations, the out-of-plane oscillation of the sail is directly coupled with central hub (spacecraft main body) dynamics. Hence it is important to understand the attitude dynamics and the stability of the spinning sail especially focusing on the coupling between the out-of-plane oscillation and the central hub dynamics.

The in-plane oscillation can also be coupled with central hub dynamics through buckling of the sail, which is also important to understand for designing spinning sail-craft. However, buckling is a complicated and nonlinear effect which should be analyzed through different type of thorough analysis with extensive numerical simulations. The in-plane dynamics of the sail is out of scope of the present paper.

In order to derive the attitude dynamics taking into account the out-of-plane oscillation modes of the spinning sail, we consider a simple configuration which consists of a cylindrical main body and a circular sail[6]. They are connected to each other and rotate around the z-axis at a rate of Ω as shown in Fig. 2. In this paper, we adopt the First-order Mode Model (FMM) which was proposed by Nakano et al. The FMM extends the Euler's equations and provides the coupling dynamics between first-order steady oscillation of the sail and the rigid central hub. The equations of motion given through the FMM derivation is

$$\begin{aligned} I_2\ddot{\phi} + I_1\Omega^2\phi + I_1\left(\Omega\omega_y + \dot{\omega}_x\right) &= 0 \\ I_2\ddot{\psi} + I_1\Omega^2\psi + I_1\left(-\Omega\omega_x + \dot{\omega}_y\right) &= 0 \\ I\dot{\omega}_x + \left(J - I\right)\Omega\omega_y + \frac{1}{2}I_2\left(\ddot{\phi} + \Omega^2\phi\right) &= 0 \\ I\dot{\omega}_y - \left(J - I\right)\Omega\omega_x + \frac{1}{2}I_2\left(\ddot{\psi} + \Omega^2\psi\right) &= 0 \end{aligned} \tag{1}$$

where I_1 is the moment of inertia (MOI) of the sail defined as

$$I_1 = 2\pi\int_{r_a}^{R}\mu(r)r^3dr \tag{2}$$

(R : sail radius, r_a : central hub radius, μ : sail area density.) I is the MOI of overall spacecraft about x- and y-axes (transverse axes), J is the MOI of overall spacecraft about z-axis (spin axis). I and J satisfy $I = I_T + \frac{1}{2}I_1$, $J = I_S + I_1$, where I_T , I_S are the MOIs of the central hub about transverse axis and spin axis, respectively. The angular velocity about x, y and z with respect to the body frame (central-hub fixed coordinates) are denoted by ω_x, ω_y, and Ω, respectively. $\phi(t)$ and $\psi(t)$ constitute the first order mode of the out-of-plane sail deformation about x- and y-axes. I_2 is a parameter having the dimension of MOI. By putting the first-order mode of the out-of-plane deformation of the sail w as

$$w(r,\theta,t) = k(r - r_a)\phi(t)\sin\theta - k(r - r_a)\psi(t)\cos\theta \tag{3}$$

with an arbitrary amplitude k, we get

$$I_2 = 2\pi\int_{r_a}^{R}\mu(r)r^2(r - r_a)dr \tag{4}$$

The roots of the characteristic equation of Eq.(1) can be obtained in the form $s = \pm B_0\Omega, \pm B_1\Omega, \pm B_2\Omega$, with

$$
\begin{aligned}
B_0{}^2 &= \frac{2\beta + N_{sc}^2 - 1}{2} - \frac{N_{sc} - 1}{2}\sqrt{\left(N_{sc}+1\right)^2 + 4\left(\beta - 1\right)} \\
B_1{}^2 &= 1 \\
B_3{}^2 &= \frac{2\beta + N_{sc}^2 - 1}{2} + \frac{N_{sc} - 1}{2}\sqrt{\left(N_{sc}+1\right)^2 + 4\left(\beta - 1\right)}
\end{aligned}
\tag{5}
$$

where

$$
N_{sc} = \frac{I_S}{I_T} - 1,\ \beta = \frac{I_1}{I_2}\frac{I_T + \frac{1}{2}\left(I_1 - I_2\right)}{I_T} \tag{6}
$$

The stability of the system can be obtained by imposing that all the coefficients of the characteristic equation are positive. Consequently, the stability condition becomes

$$
J - I > \frac{1}{2}I_2 \tag{7}
$$

which is equivalent to

$$
I_S - I_T > -\frac{1}{2}I_3 \tag{8}
$$

where

$$
I_3 = I_1 - I_2 = 2\pi\int_{r_a}^{R}\mu(r)r^2 r_a\,dr \tag{9}
$$

Compared with the stability condition of a rigid body, i.e. $J - I > 0$, the stability is degraded by $\frac{1}{2}I_3$ due to flexibility of the sail. On the other hand, compared with the stability condition of the central hub alone, i.e. $I_S - I_T > 0$, the stability is increased by $\frac{1}{2}I_2$ due to adding the sail around the central hub.

B. Stability Based on Energy-Angular Momentum Analysis

More general form of the stability condition is investigated by Tsuda[7], employing relations between energy and angular momentum of a rotational body with a flexible sail. The analysis focuses on the first order out-of-plane oscillation as shown in Fig.3 (represented by radial unit vector of the sail $\mathbf{e}'$). The paper investigated the energy and angular momentum changes due to first-order mode deformation of the sail, and derived the

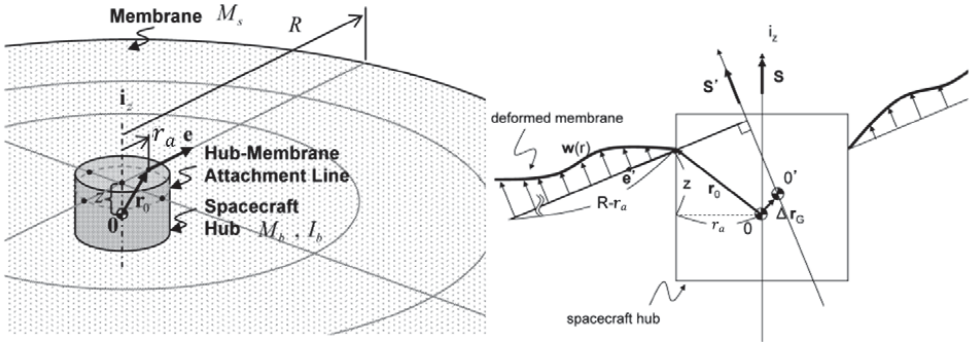

Fig.3 Modeling of Spinning Sail for Energy-Angular Momentum Analysis

"effective MOI" of spinning sail-craft, with which one can derive the spin stability exactly in the same way as that of rigid bodies.

The effective MOI takes the form

$$I_{eff} = \begin{bmatrix} b_{11} & b_{12} & b_{13} \\ b_{12} & b_{22} & b_{23} \\ b_{13} & b_{23} & b_{33} \end{bmatrix} \tag{10}$$

with which the stability condition about z-axis is written as

$$b_{33} - \frac{b_{11}+b_{22}}{2} > \sqrt{\left(\frac{b_{11}-b_{22}}{2}\right)^2 + b_{12}^2} \tag{11}$$

By concretely calculating each element of Eq.(10) and substituting it into Eq.(11), the following stability condition is obtained

$$I_{zz} - \frac{I_{xx}+I_{yy}}{2} > \sqrt{\left(\frac{I_{xx}-I_{yy}}{2}\right)^2 - I_{xy}^2} - I_3 + \frac{(M_b+M_s)q_0 z^2}{M_b+M_s+q_0} \tag{12}$$

where I_{xx}, I_{yy}, I_{zz}, I_{xy} are the MOI and product of inertia (POI) of the central hub, M_b, M_s are the mass of the central hub and the sail, respectively, z is the attachment offset of the sail from the spacecraft center of mass along z-axis. q_0 is a parameter having the dimension of mass, and defined as

$$q_0 = \pi \int_{r_a}^{R} \mu(r) \frac{r}{r_a}(r - r_a)\, dr = \frac{1}{2}\left(\frac{I_3}{r_a^2} - M_S\right) \tag{13}$$

The stability condition Eq. (12) is a generalized form of Eq.(8). Indeed, Eq.(8) can be obtained from Eq.(12) by substituting $I_T = I_{xx} = I_{yy}$, $I_S = I_{zz}$, $I_{xy} = 0$, and $z = 0$ into Eq.(12). Thus two derivations reach the

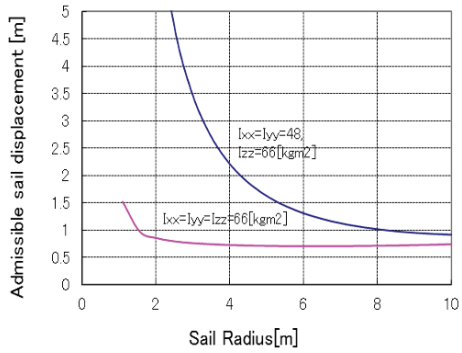

Fig.4 Admissible sail axial displacement
This graph uses the following parameters; M_b=300kg, M_s=16kg, r_a=0.63m, R=10m

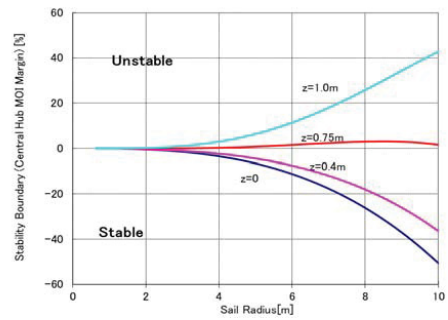

Fig.5 MOI margin of the central hub for different sail axial displacement
This graph uses the following parameters; I_S=I_T=66kgm^2, M_b=300kg, M_s=16kg, r_a=0.63m, R=10m

identical result. However, the former derivation can be connected to modal analysis of sail deformation, and the latter can deal with asymmetric central hub with sails displaced from spacecraft center of mass. Note that for axisymmetric sail-craft, Eq.(12) is simplified as

$$I_S - I_T > -\tilde{M}_s r_a^2 + \frac{M_0 \tilde{M}_s}{M_0 + \tilde{M}_s} z^2 \tag{14}$$

where we define $\tilde{M}_S = I_3 / (2r_a^2)$, $M_0 = M_b + M_s$.

III. Designing Spinning Sail Spacecraft Configuration

This section discusses several useful criteria for designing spinning sail-craft derived from the stability condition Eq.(8) , Eq.(12) or Eq.(14).

A. Effect of Sail Attachment Displacement

As discussed in Section II.A, spinning sail-craft with no displacement in the sail attachment (i.e. the sail plane exactly aligns with the spacecraft center of mass) gain more stability than the central hub alone, but are far less stable than the central hub with rigid plate.

Eq.(14) also reveals effect of displacement of sail attachment on the stability. Since the last term of the right hand side of Eq.(14) is positive semidefinite, the sail displacement always lowers the stability. In the special case when the central hub is spherical (i.e. $I_S = I_T$), the system is stable only when the sail attachment displacement is within the following range.

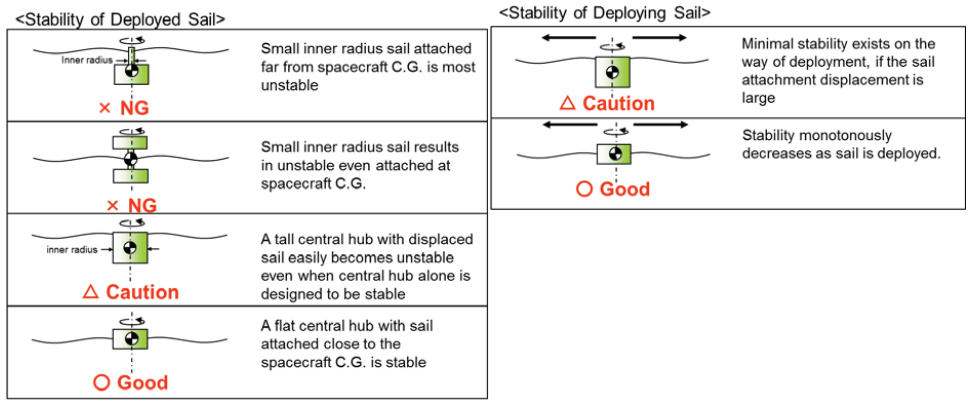

Fig.6 Stable and unstable configurations of spinning sail

$$-\sqrt{1+\frac{\tilde{M}_s}{M_0}}r_a < z < \sqrt{1+\frac{\tilde{M}_s}{M_0}}r_a \tag{15}$$

Fig.4 shows admissible sail displacement as a function of sail radius, which is obtained by solving Eq.(14) for z . The parameters used to draw this graph adopt specifications of IKAROS with two different MOI configurations, as is noted in the caption of Fig.4.

If the central hub is spherical, the sail displacement should be 0.7m or lower. Fig.4 indicates that this limitation is relaxed for more flat central hub with spin axis aligned with maximum principle moment of inertia.

B. Stability on Sail Deployment Process

The stability condition Eq.(14) is an indirect function of sail radius R via $\tilde{M}_s$. Let us consider then that the sail is gradually deployed such that $\tilde{M}_s$ monotonously increases. The right hand side of Eq.(14) is maximal when

$$\tilde{M}_s = \left(\frac{z}{r_a} - 1\right) M_0 \tag{16}$$

Eq.(16) suggests the followings; (i) if $z > r_a$,

Fig.5 shows stability boundaries as a function of sail radius for four different sail attachment displacement case. The central hub "MOI margin" in the vertical axis is calculated by $(I_S - I_T)/I_S$ [%]. In this case, the whole deployment sequence is stable for $z = 0, 0.4[\mathrm{m}]$ as long as the central hub is a stable shape (i.e.

$I_S > I_T$). If $z = 0.75[\mathrm{m}]$, there is a critical radius on the way of deployment process, and the central hub must satisfy $I_S - I_T > M_0 \left(r_a - z\right)^2$ to secure stability at any moment of the deployment process. If $z = 1.0[\mathrm{m}]$, the stability gets worse monotonously as the deployment is proceeded.

Fig.6 illustratively summarizes relations between the attitude stability and the sail-central hub configuration. Among many conceptual designs arose in the design work of IKAROS, the present appearance of IKAROS adopts the configuration indicated as "Good" in Fig.6, which reflects the stability discussions in the previous sections; the sail attachment displacement z is attempted to be minimized, and resulted in $z = 0.04[\mathrm{m}]$. Meanwhile the sail attachment radius r_a and the MOI margin of the central hub are attempted to be maximized, resulted in $r_a = 0.63[\mathrm{m}]$ and MOI margin=30%, by placing the sail along the outmost envelope of the central body, and shaping the central hub as flat as possible.

IV. Numerical and In-Flight Evaluation of Stability

A. Stability Evaluation of Displaced Sail

Numerical simulations were conducted to validate the stability criteria Eq.(14). The Multi-Particle Model (MPM), which has been developed, validated and extensively used in the IKAROS development, was employed to calculate attitude behavior of a spinning sail-craft. The MPM models the sail as multiple particles connected by springs and dampers (Fig.7). By neglecting the bending rigidity, it realizes fast and reasonably accurate way of solving attitude and sail deformation behavior of the spinning sail-craft.

The MOI of the central hub is set as $I_S = I_T$, and the attitude motion is calculated with different sail attachment displacement z. The critical displacement in this case is $z = 0.44[\mathrm{m}]$. Fig.8 shows the time history of the nutation angle of the central hub for the case $z = 0.3[\mathrm{m}]$ and $z = 0.5[\mathrm{m}]$, starting from 1Hz spinning with a small transverse angular velocity disturbance. As expected, $z = 0.3[\mathrm{m}]$ shows a stable result, whereas $z = 0.5[\mathrm{m}]$ shows an unstable result.

Fig.9 (left) shows the nutation angle at $t = 1000[\mathrm{sec}]$ as a function of sail attachment displacement, calculated using the MPM. Two cases of initial conditions are evaluated; $[\omega_x \ \omega_y \ \Omega] = [0.01 \ 0 \ 1.0][\mathrm{Hz}]$ (case1) and $[0.1 \ 0 \ 1.0][\mathrm{Hz}]$ (case2). Fig.9 (right) shows the amplitude ratio (the ratio between nutation

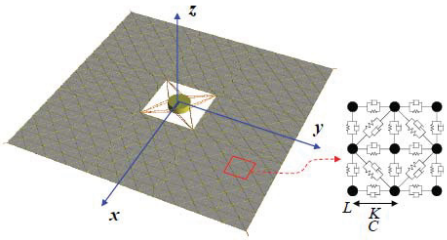

Fig.7 Modeling of Spinning Sail-craft for Multi-Particle Model (MPM)

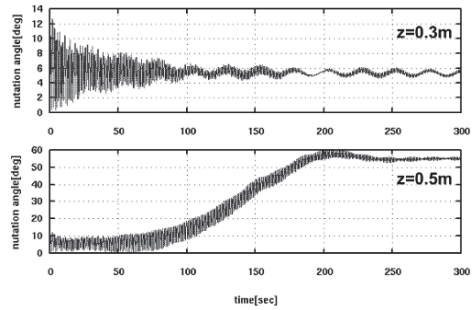

Fig. 8 MPM simulation for 1-Hz spin with initial transverse angular velocity error

The simulation uses the following parameters; $I_S=I_T=500kgm^2$, $M_b=500kg$, $M_s=53.8kg$, $r_a=0.25m$, $R=15.8m$, sail rigidity=10E+7Pa, damping=10E-6Ns/m.

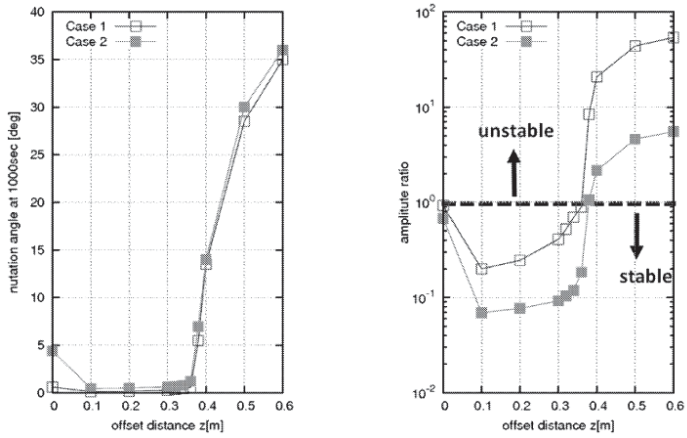

Fig.9 Nutation angle (Left) and Amplification ratio of nutation angle (Right) as functions of sail attachment displacement

amplitudes at $t = 1000[\mathrm{sec}]$ and $t = 0[\mathrm{sec}]$). Seen from the figure is that, regardless of initial conditions, the critical sail displacement (i.e. when the nutation amplitude ratio crosses 1) is slightly lower than the analytical result, but almost consistent with the theoretical value of $z = 0.44[\mathrm{m}]$. The small inconsistency is considered to be due to an effect of the higher mode of the sail deformation and a specific configuration of the sail attachment part modeled for the numerical simulation.

B. IKAROS Flight Result

The JAXA's solar sail demonstration spacecraft IKAROS was launched on May 21, 2010. It successfully completed deploying the 20m-span solar sail on June 9, 2010[1]. The final phase of the deployment process comes with drastic change of the sail in a few seconds, reducing the spin rate from 5.2rpm to 2.5rpm due to the

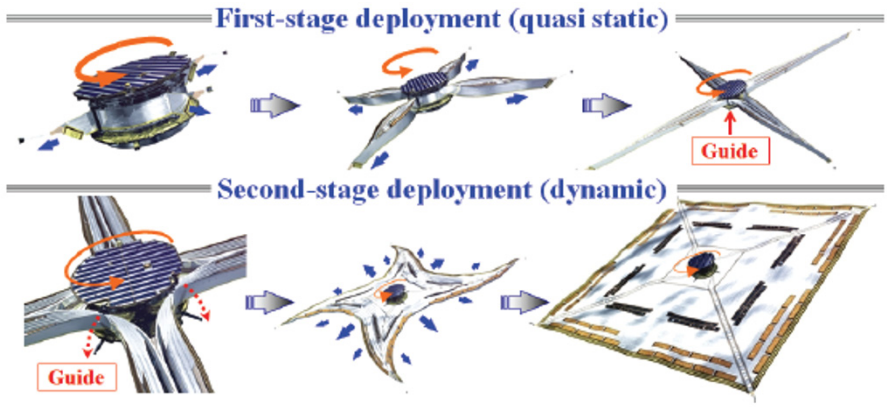

Fig.10 Sail deployment procedure of IKAROS

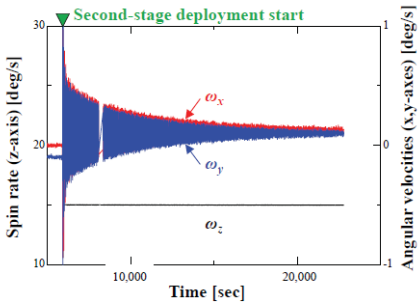

Fig.11 Transverse gyros output right after completion of the sail deployment[8]

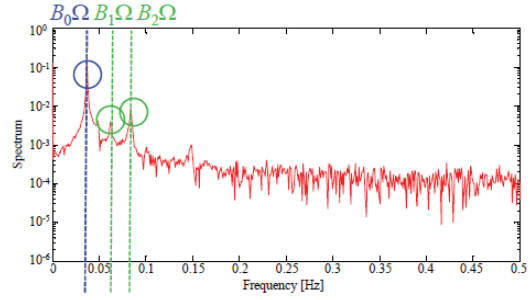

Fig.12 FFT result of the gyro data[8]
Three dashed vertical lines indicate analytical values of mode frequency based on FMM (Eq.(5)).

law of angular momentum conservation(Fig.10). This induces oscillation on the central hub and the sail, which was observed by three-axis gyros equipped on the central hub. As this phase did not use active attitude control, the gyro data right after the completion of the sail deployment can be used for a validation of the FMM (Eq.(1)).

Fig.11 shows the gyro telemetry right after the completion of the sail deployment. The transverse angular velocity is converged to zero, showing that the attitude is stable around the nominal spin axis. The converging characteristic is thought to be provided mainly by structural damping of the sail. Although the central hub is equipped with a nutation damper, it affects little after the sail is extended.

Fig.12 shows the comparison of oscillation modes obtained by Eq.(5) and the flight data. A FFT result of the x-axis angular velocity is used as the flight data in this graph. As is seen from the figure, the three major modes accurately coincide with the peaks of the spectrum produced from the flight data.

Thus the FMM can be said to be a useful reduced model for reproducing the attitude dynamics of spinning sail-craft. This provides a firm collateral evidence that Eq.(8), Eq.(12) and Eq.(14) are valid stability criteria for spinning sail-craft.

V. Conclusion

This paper discussed attitude stability criteria for spinning spacecraft equipped with large flexible membrane, such as solar sail. The paper reviewed the authors' previous work on stability criteria for spinning sail craft, and showed how it has been exploited for actual design of IKAROS, the JAXA's interplanetary solar sail demonstration spacecraft launched in 2010. The validity of the stability criteria was evaluated by some numerical simulations and flight data of IKAROS. IKAROS showed excellent attitude stability as expected by pre-launch analyses, and the oscillation mode of the sail, which is related tightly with the stability criteria, was observed to be in good agreement between flight data and the theory.

Acknowledgments

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Evaluation of Sail Mechanics of IKAROS on its Slow-spin and Reverse-spin Operation

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In the post operational phase of spin type solar sail "IKAROS", a slow-spin operation and a reverse-spin operation were conducted to acquire basic knowledge of mechanics of sail membrane. The flight data indicates that the sail membrane kept its shape against the solar radiation pressure even with low centrifugal force. From these results, it is concluded that the sail has a much higher bending stiffness than expected. In this paper, the flight results of these operations are reported precisely and compared to the numerical simulation. The causes of unexpected stiffness of the sail are also discussed.

Nomenclature

d = cross section diameter of tether

E = Young's modulus of membrane

E_t = Young's modulus of tether

h = thickness of membrane

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F = inter-particle force

I_C = second moment of area of a curled beam

(Ixx, Iyy, Izz) = moments of inertia of main body

(Ixy, Iyz, Izx) = products of inertia of main body

K = spring constant of membrane

L = distance between two particles

L_0 = natural length of spring

M = weight of main body

M_b = weight of membrane

M_t = weight of tip mass

(x,y,z) = coordinate frame fixed to main body

(x_g, y_g, z_g) = center of mass of main body

α = coefficient of compression stiffness

η = coefficient of dumping

θ = bending angle of components

θ_0 = natural angle of components

ν = Poisson's ratio of membrane

ρ = density of membrane

$(\omega_x, \omega_y, \omega_z)$ = angular velocity vector of (x,y,z) frame with respect to reference frame

I. Introduction

THE Japan Exploration Agency (JAXA) is studying the spin type solar sail whose sail membrane is deployed and maintained flat by the centrifugal force. This method is expected to be realized with simpler and lighter-weight mechanism than other ways, because it does not require rigid structural elements. To demonstrate the deployment and expansion of a 20m-class membrane, JAXA launched a powered solar sail “Interplanetary Kite-craft Accelerated by Radiation Of the Sun (IKAROS)” on May 21, 2010. IKAROS performed a sail deployment sequence and have confirmed that the sail was successfully expanded [1] (Fig.1).

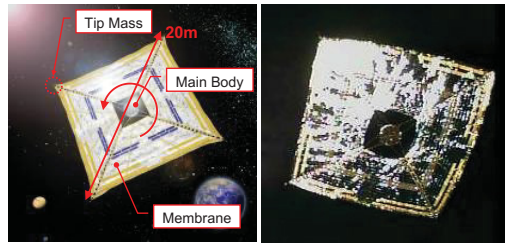

Fig. 1 IKAROS, left: illustration, right: image taken by a separation camera.

From the results of analysis using separation cameras' images and attitude data, it was estimated that the membrane was deformed toward to the sun direction[2][3][4]. This was different from the results obtained from the numerical simulation which had predicted that the membrane would be deformed in the opposite direction by the solar radiation pressure and we had come to consider that the sail has a higher bending stiffness than expected[5].

To investigate the behavior of the sail membrane in a low centrifugal force environment, a slow-spin operation and a reverse-spin operation were planned in the post operational phase of IKAROS. In this paper, the flight results of these operations are reported and compared to the numerical simulation. The causes of unexpected stiffness of the sail are also discussed.

II. Configuration of the Sail

Figure 2 shows the shape and equipment layout of the sail of IKAROS. The sail is made of polyimide base membrane whose thickness is 7.5μm. The direction of folded lines is perpendicular to the direction of the centrifugal force. The base membrane mounts following components:

1) Tether: The membrane should not be contact on the main body after the deployment. They are connected by tethers.

2) Tip mass: Four masses are connected to the tip of membrane by the tethers. Each mass is about 0.5kg. It adjusts the centrifugal force and the inertial momentum of the membrane.

3) Solar cells: Thin flexible solar arrays are attached on the membrane partly. The area ratio is about 5%.

4) Reflection Control Device (RCD): Variable reflectance elements are loaded near the tips of the membrane. The control of the spin direction and rate can be performed using them.

5) Dust counter: PVDF detectors are attached on the membrane.

6) Harness: Several elements on the membrane are connected electrically to a processing unit on the main satellite through harnesses.

7) Reinforcing tape: The sail is reinforced by polyimide tape to prevent the extension of cracks.

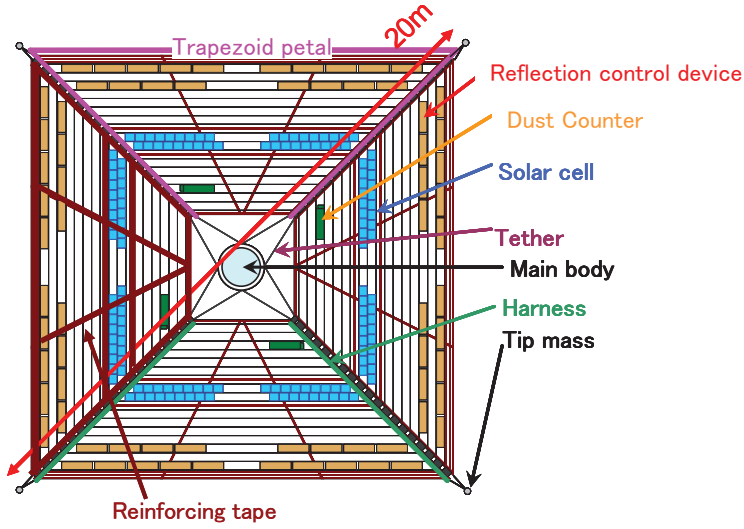

Fig. 2 Sail shape and device layout of IKAROS.

In this study, we focus on the structural effect of solar cells, reflection control device (RCD) and reinforcing tape. These components are much thicker than base membrane and have a relatively large area. Figure 3 shows images and layer constructions of these components.

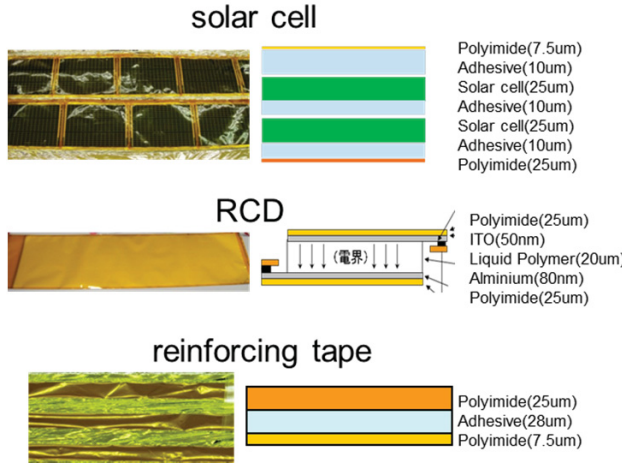

Fig. 3 Images and layer constructions of components.

III. Slow-spin Operation

To expand and maintain the shape of the sail membrane against the solar radiation pressure, the spin-rate of IKAROS had been kept higher than 1 rpm for about one year from the completion of the sail deployment on June 9, 2010.

In contrast, in the extended operations started after the completion of the regular missions on January 2011, the attitude of IKAROS was changed actively to enhance the knowledge about the statics and dynamics of the sail membrane and to construct a precise numerical mode of IKAROS.

The "slow-spin operation" was started in June 2011 as a part of the extended operation. This operation aimed to see the effect of stiffness of membrane against the solar radiation pressure under low centrifugal force environment.

During the regular operation, the spin rate of IKAROS had decreased naturally by solar radiation pressure torque and it had been kept by periodical thrust control of gas jet injection. By suspending the spin control, it went down to 0.055 rpm in the slow-spin operation.

This section shows the flight results of the operation compared to the results obtained from numerical simulations.

A. Operation results of the attitude trend

Figure 4 shows the history of the spin-rate and the sun angle of the main body of IKAROS. The sun angle is equivalent to the angle between the sun direction and the spin axis. This figure shows that the sun angle increased form 3deg to 20deg for about 2.5 days with the decrease of the spin-rate from 0.2 rpm to 0.055 rpm. It is considered that the attitude of the main body became unstable by the solar radiation pressure torque. This is also associated with the decrease of the rotation stiffness.

Figure 5 shows flight data of the rate gyro mounted on the main body at 0.055rpm. The angular velocity around the z-axis ω_z is equivalent to the spin-rate. This data denotes that the oscillations of out of plane angular velocities ω_x and ω_y are nearly equal to the spin-rate. This nutation motion can be occurred in a case of the inertia tensor of the system is close to the one of thin plate structure like a sail. From this result, it is considered that the main body had rotated combined with the sail membrane.

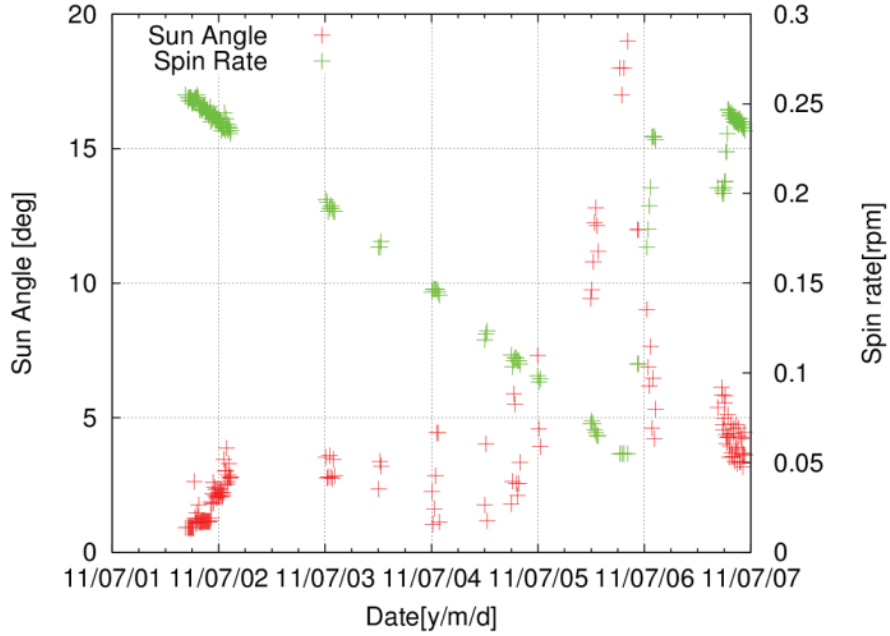

Fig. 4 History of the spin-rate and the sun angle during slow-spin operation.

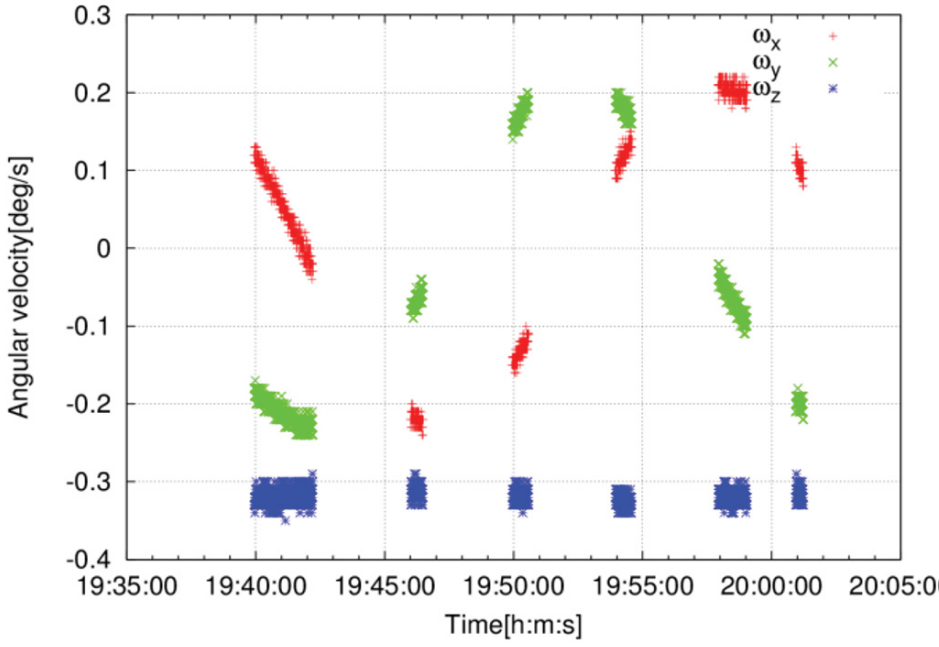

Fig. 5 Flight data of the rate gyro at 0.055rpm.

B. Operation results of the shape of the sail

In the slow-spin operation, some images of the sail were taken by side monitor cameras. Figure 6 shows its field of view. It had been expected by numerical simulations that the decrease of the centrifugal force would cause the deflection of the sail by solar radiation pressure as shown in Fig. 7, however the image denotes that the shape of the sail is not so different from the one taken at 2.1 rpm. Figures 8 and 9 show images of the sail taken by these cameras at 2.1rpm and 0.055rpm. The red lines denote the reference line of the edge of the sail membrane on the assumption that the sail is deployed completely flat. The edge line of the sail is approximately-same with the reference line regardless of its spin rate.

From these results, we concluded that the sail membrane has a larger bending stiffness on an average than expected and tried to modify the numerical models of the sail membrane to simulate flight data appropriately as described in the next chapter.

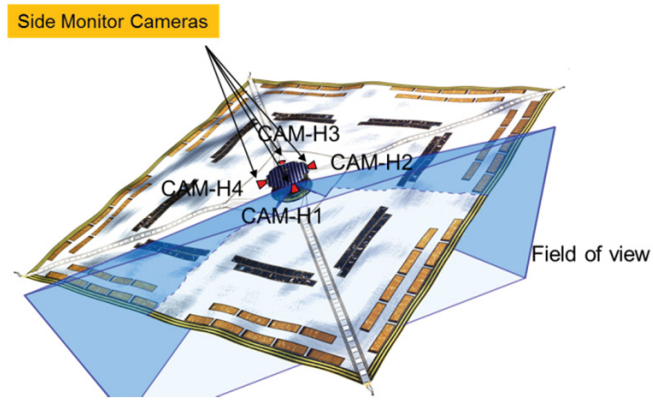

Fig. 6 Field of view of the side monitor camera.

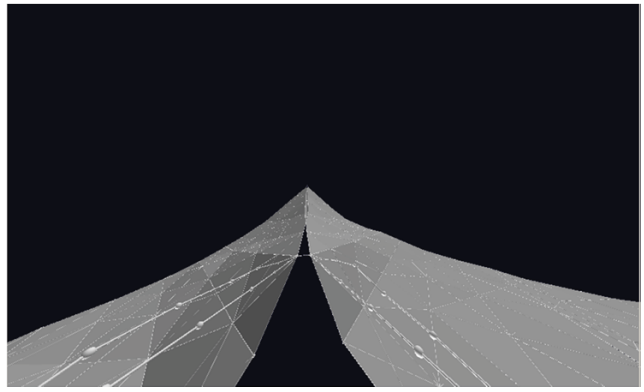

Fig. 7 Simulation of the Image of the sail at 0.055rpm.

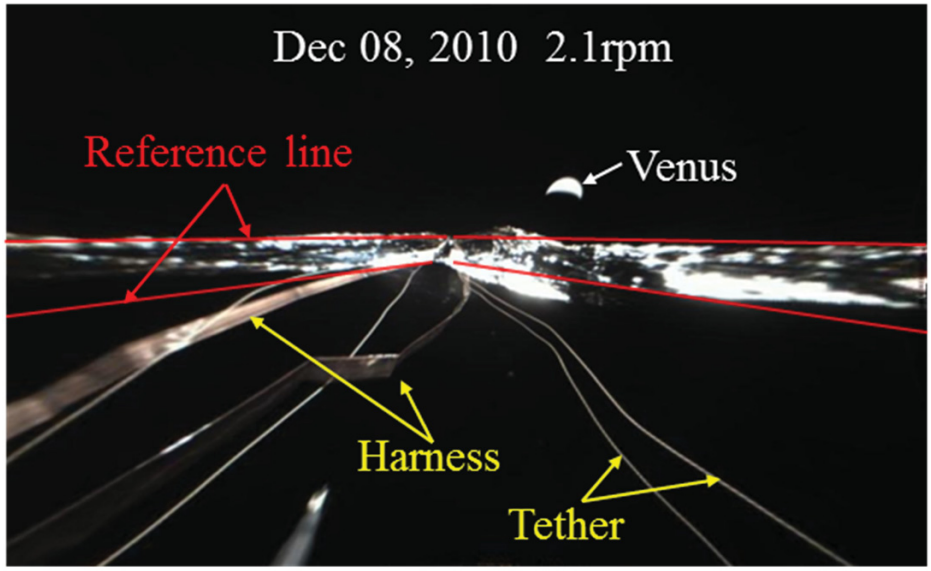

Fig. 8 Image of the sail at 2.1 rpm.

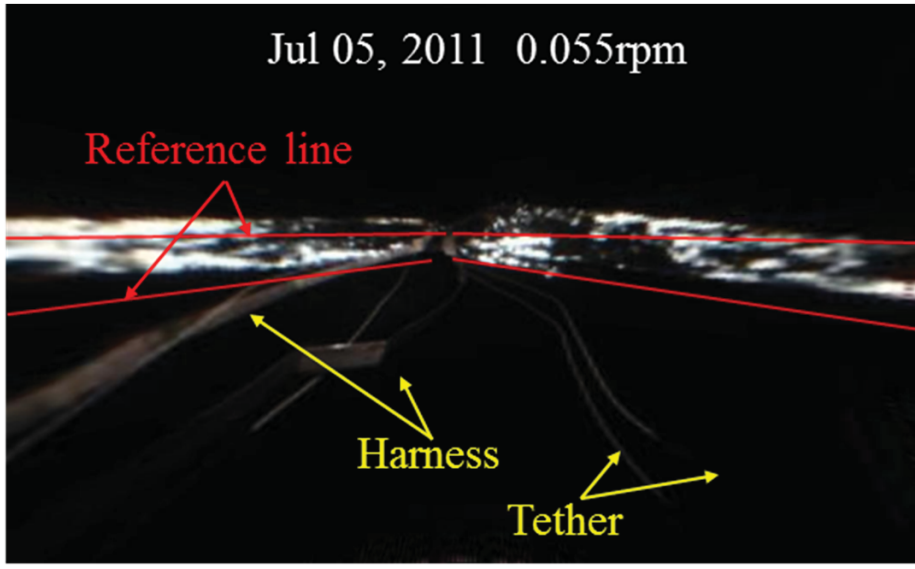

Fig. 9 Image of the sail at 0.055rpm.

IV. Numerical Analysis of the Shape of the Sail

A. Numerical model

In order to analyze the dynamic motion of membrane, a multi-particle model is introduced. In this model, each elements of the membrane is substituted by particles connected by springs and dampers. This approximation facilitates the construction of model and lowers the computational cost.

The spring constant K are determined by applying the principle of virtual work on an element. The inter-particle force F can be described as following form:

$$F = \begin{cases} K(L - L_0) + \eta K\dot{L} & (L \geq L_0) \\ K\alpha(L - L_0) + \eta\alpha K\dot{L} & (L < L_0) \end{cases}$$

where L, L_0, α, η denotes inter-particle distance, a natural length of spring, a compressive ratio and a dumping ratio, respectively.

In addition, a bending stiffness model is introduced by implementing rotational spring [6]. Figure 10 denotes the concept of bending moment generated by translational force in multi particle model. The force act on each particle is described as:

$$f_A = \frac{Eh^3}{l_A^3} \cdot \vec{AA'} \cdot \vec{BC}(\theta - \theta_0)$$

$$f_D = \frac{Eh^3}{l_D^3} \cdot \vec{DD'} \cdot \vec{CB}(\theta - \theta_0)$$

$$f_{AB} = -f_A \cdot \frac{A'C}{BC},\ f_{AC} = -f_A \cdot \frac{A'B}{BC}$$

$$f_{DB} = -f_D \cdot \frac{D'C}{BC},\ f_{DC} = -f_D \cdot \frac{D'B}{BC}$$

where E and h denote Young's modulus and thickness, respectively. θ_0 denotes the natural bending angle.

Table 1 shows simulation parameters for normal case. For simplicity, it is assumed that each parameters of solar cell, RCD, dust counter and reinforcing tape are the same as the ones of base membrane, except the thickness. Since the stiffness of adhesive is very small, the thickness parameters are calculated by laminate thickness with the exclusion of the one of adhesive described in Fig.3.

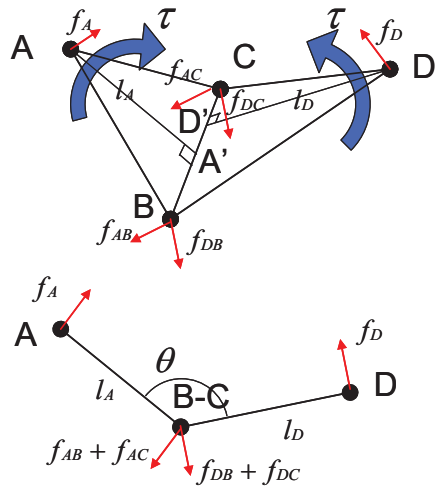

Fig. 10 Bending force model.

Table 1 Simulation parameters for normal case.

Parameter name	Symbol	Value	Unit
Angular velocity	$(\omega_x, \omega_y, \omega_z)$	(0.0,0.0,15.3)	deg/s
Main body's weight	M	302.11	kg
Main body's iniertia moment			
moments of inertia	(Ixx, Iyy, Izz)	(47.91,47.46,66.44)	kgm^2
products of inertia	(Ixy, Iyz, Izx)	(-6.3e-4,4.6e-5,1.6e-3)	kgm^2
Center of mass	(x_g, y_g, z_g)	(-1.6e-4,2.6e-4,-2.7e-2)	m
Membrane's weight	M_b	14.253	kg
Tip mass's weight	M_t	2.035	kg
Membrane			
Young's modulus	E	3.0	GPa
Poisson's ratio	ν	0.4	-
Density	ρ	1420.0	kg/m^3
Thickness	h	7.5	um
Dumping coefficient	η	3.9e-5	sec
Compressive ratio	α	1.0e-7	-
Tether			
Young's modulus	E_t	100.0	GPa
Cross section diameter	d	0.73	mm
Dumping coefficient	η	2.0e-5	sec
Solar cell			
Thickness	h	82.5	um
RCD			
Thickness	h	50.0	um
Dust counter			
Thickness	h	40.0	um
Reinforcing tape			
Thickness	h	32.5	um

B. Result of simulations

The dynamic numerical simulations are performed employing the bending stiffness model by implementing rotational springs. In this simulation, the force of solar radiation pressure is taken into account. The results are evaluated in terms of the shape of the sail.

In a previous study, we had considered that the bending force based on the design value was too small compared to the centrifugal force. In addition, we had studied that the local bending deformation would not induce significant displacement of other part of the sail or unless it is connected continuously to propagate the bending moment[5]. From these view points, we introduce a high bending stiffness model to reinforcing tapes on the sail by increasing their thickness in the simulation model.

Figure 11 shows the shape of membrane simulated on the condition that θ_0=180deg and h =650um (20 times that of normal case) for reinforcing tapes. This result shows that the deflection of the sail is almost same with the flight data as compared to Fig.9.

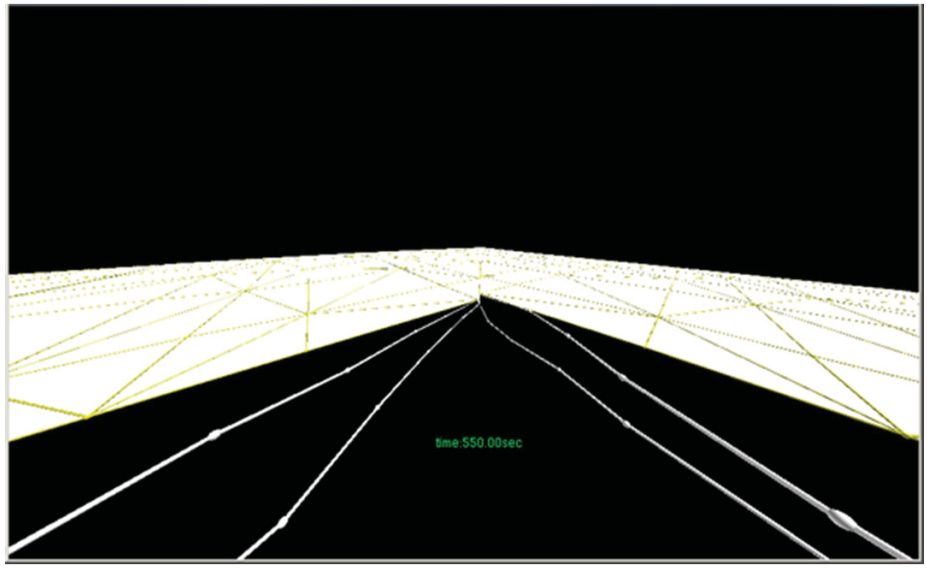

Fig. 11 Simulation of the Image of the sail at 0.055rpm with high bending stiffness of reinforcing tape.

By using these parameters of bending stiffness model, we designed a thrust control sequence for a reverse-spin operation.

V. Reverse-spin Operation

On Oct. 18, 2011 (JST), the "reverse-spin operation" was performed to enhance the knowledge about the effect of stiffness of membrane against the solar radiation pressure and to investigate the attitude dynamics under the different spin direction.

This operation was conducted by gas jet thrust to shift spin direction to the reverse way. The gas jet injection was executed 118 times with 0.4 sec pulse width and at intervals of 10 seconds as planned in advance. As a result of this operation, the spin-rate was turned down from 0.15rpm to -0.24rpm, and the sail membrane was successfully spun in the reverse direction without being entangled.

According to the result of the attitude determination, it was confirmed that the spin axis of the main body tilted about 18 degrees during the operation. This was within a predictable range.

A. Operation results of the attitude trend

Figure 12 shows flight data of the rate gyro during the reverse-spin thrust control. Pink and light-blue lines indicate the timing of gas jet injections and camera shots, respectively. This figure denotes that the spin-rate of the main body shifts to the reverse way with the gas jet injections. The spin-rate of the main body is pulled back after the gas jet injections by the in-plane oscillation motion between the sail membrane of large inertia moment, however it shifts totally to the reverse way gradually.

Figure 13 shows the simulation result of the spin rate of the main body and the sail membrane on the same configuration with the flight operation. In this figure, the angular velocity of the main body (red points) shows the same oscillation as the flight data. The trend of the angular velocity of the sail membrane (green points) denotes that the spin rate of the sail is almost 0rpm and is free from the effect of the centrifugal force about 250sec in this figure.

B. Operation results of the shape of the sail

Figure 12 also denotes that the eight times camera shots were performed at near-zero spin. Figures 14 and 15 show the images taken at the time correspond to 247sec (shot 4) and 252sec (shot 5) in Fig.12, respectively. These images denote that the shape of the sail is not so deflected even at 0rpm. In addition, no significant deformation is observed between Figs 14 and 15 though the sail membrane is subjected to external force between two shot times by the gas jet injection of pulse 47 as shown in Fig. 12. The prediction of the shape of the sail by the modified numerical simulation is coincides with these flight data. From these results, it is concluded that the sail membrane of IKAROS has large bending stiffness as expected after the slow-spin operation.

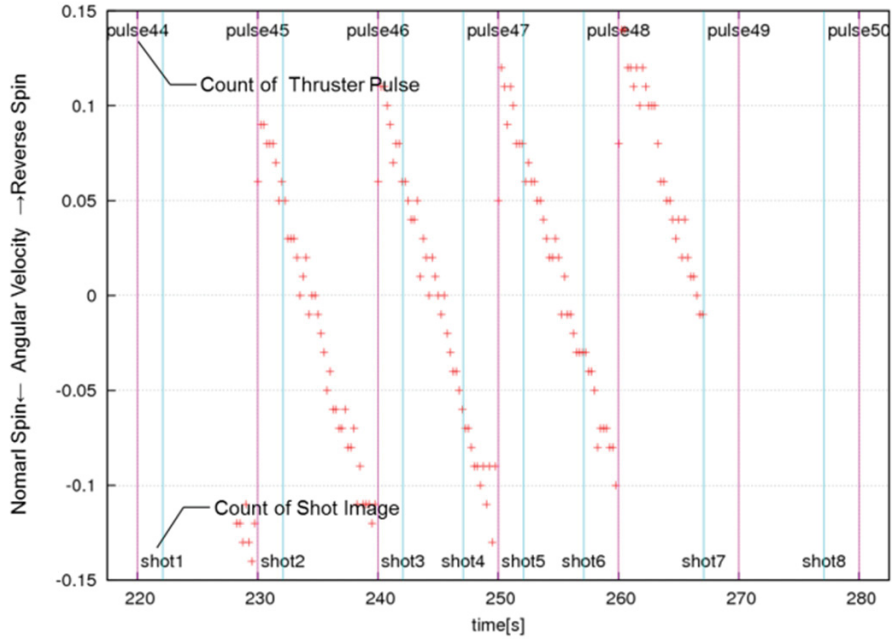

Fig. 12 Flight data of the rate gyro during reverse-spin operation.

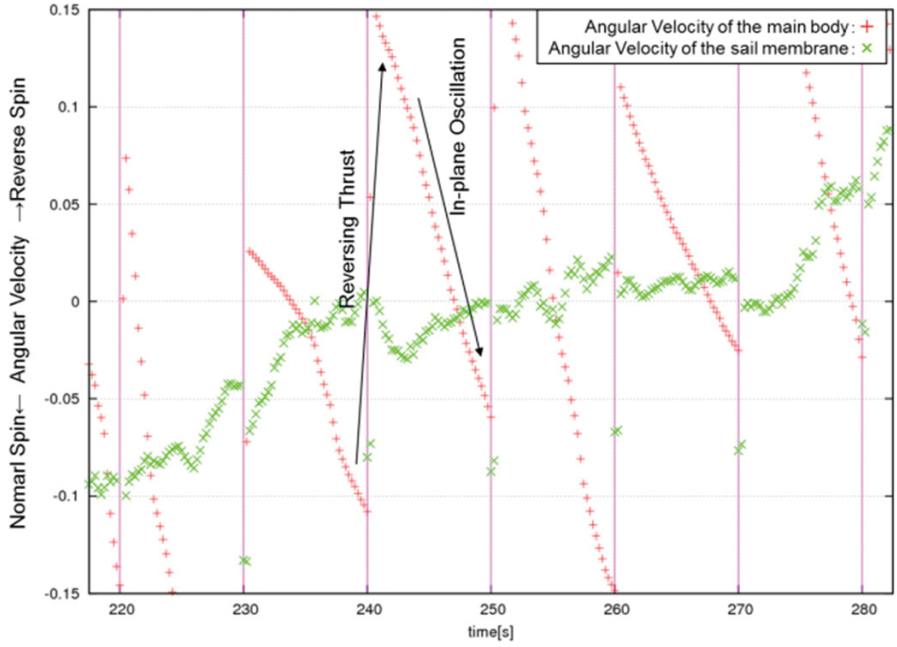

Fig. 13 Simulation result of the rate gyro during reverse-spin operation.

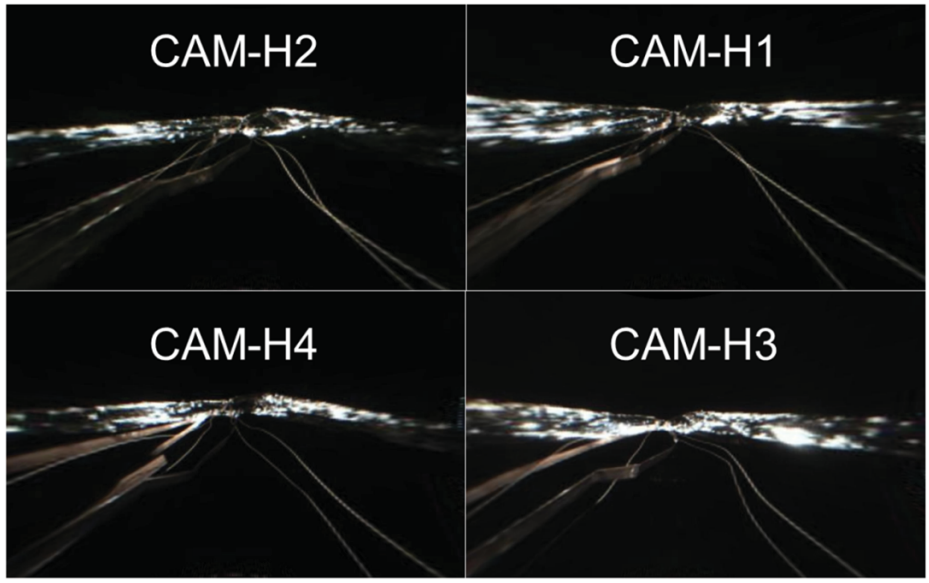

Fig. 14 Images of the sail at 0.0rpm (shot 4 on Fig.12).

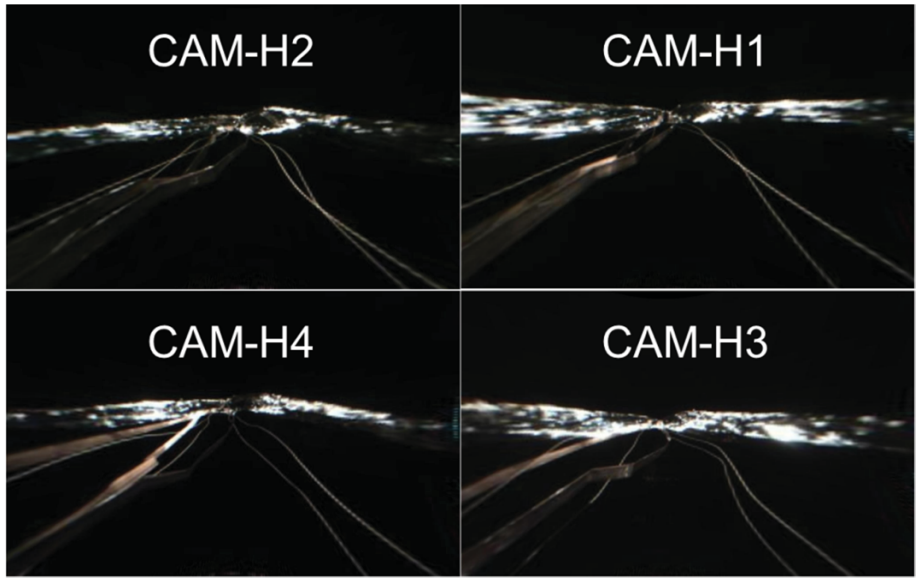

Fig. 15 Images of the sail at 0.0rpm (shot 5 on Fig.12).

VI. Discussion

In this chapter, the factor affecting to keep the shape of sail against the solar radiation pressure is considered.

First, the bending stiffness of the components on the sail is investigated by simple experiments. Figure 16 shows test pieces of the components and the configuration of the experimental setup. The test piece is simply supported as shown in Fig.17, and the deflection under its own weight is measured. By regarding the test piece as a beam, the bending stiffness EI is calculated from its deflection. The theoretical values of EI are also calculated from the extensional stiffness and the cross-section shape for each layer of component. The results are summarized in Table 2. From these results, it found that the bending stiffness obtained by the experiment is nearly equal to the theoretical ones except the one of reinforcing tape.

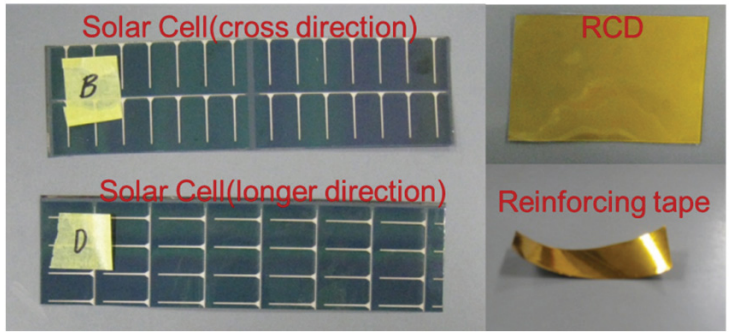

Fig. 16 Test pieces of the components.

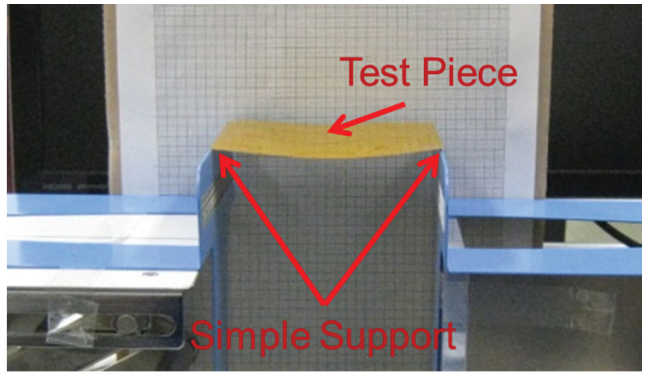

Fig. 17 Experiment setup.

Table 2 Bending stiffness of each component.

Components	EI[Nm^2] Theoretical	EI[Nm^2] Experimental
Solar cell(cross direction)	1.0x10^{-5}	1.8x10^{-5}
Solar cell(longer direction)	1.0x10^{-5}	1.5x10^{-5}
RCD	3.4x10^{-6}	2.0x10^{-6}
Reinforcing Tape	1.9x10^{-6}	-

As shown in Fig.16, we can see that the reinforcing tape have a curvature by its adhesive stress. This curl shape makes it difficult to measure the deflection by simple support, so the value of EI is not derived in Table.2. This curled shape of the cross-section suggests that the bending stiffness along the axis normal to the curled section would rise drastically because its second moment area is increased.

Figure 18 illustrates a curled cross-section view of a beam. In this figure, the cross section area S and the distance L_G between the center of curvature O and barycenter G are calculated as follows:

$$S = rh\theta$$

$$L_G = \left(\frac{h^2 + 12r^2}{6r}\right)\frac{\sin(\theta/2)}{\theta}$$

The second moment of area of the beam I_C around G is calculated as follows:

$$I_C = \frac{1}{8}(\theta + \sin\theta)(4r^3h + rh^3) - L_G^2 S$$

By measuring the test piece of reinforcing tape in Fig.16, we have $r\theta = 0.052$ m and $G\theta = 40$ deg , then the total bending stiffness of each layer is obtained as EI_C =8.1 x 10^{-3} Nm^2. This denotes that the bending stiffness of becomes about 4300 times larger than the one of flat beams and the curvature cross-section shape achieves equivalent effect of 16 times thicker model, which is the same order supposed in chapter IV. From these result, it is considered that the curvature cross-section shape of each component would be one of the factor of the increase of equivalent bending stiffness to keep the shape of sail against the solar radiation pressure.

This conclusion indicates that other curved structures of the sail also increase the equivalent bending stiffness of the sail. Such curved structures may include curved thin film solar cells, curved RCDs, crease and wrinkle of the base film. To simulate the behavior of the sail under a weak effect of its tension, these factors of the equivalent bending stiffness of the sail should be considered in numerical models. A detailed validation by numerical simulations which implement this factor is still under consideration.

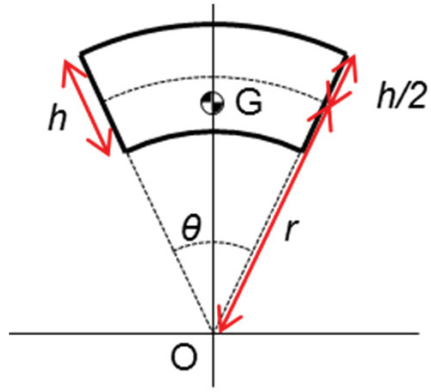

Fig. 18 Curled cross-section of a beam.

VII. Conclusion

This paper reported a slow-spin operation and a reverse-spin operation in the post operational phase of IKAROS. In the slow-spin operation, it was observed that the shape of sail was kept nearly flat against the solar radiation pressure. This can be explained by introducing the bending stiffness model on the reinforcing tapes of 20 times thicker than the designed value to have high out-of-plane stiffness. After the slow-spin operation, reverse-spin operation was conducted in Oct 18, 2011. The control sequence of this operation was designed based on the result of simulation using the high bending stiffness model. In this operation, the spin direction was successfully reversed and the images taken at near 0rpm were obtained. The flight data was coincides with the prediction by the modified numerical simulation.

The bending stiffness of components was also investigated by using test pieces, and the effect of its curvature cross-section shape was considered. This would be one of possible factors to have an unexpectedly-strong out-of-plane stiffness. Further detailed research should be performed to understand the dynamics of the sail membrane.

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Attitude and Orbit Prediction of IKAROS in Actual Flight Operation

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The world's first solar sail IKAROS (Interplanetary Kite-craft Accelerated by Radiation Of the Sun) which is operated by Japan Aerospace Exploration Agency (JAXA) lost communication with the ground station due to the power short-age on December 24, 2011. In order to acquire IKAROS again after the power comes back, we immediately initiated to predict the attitude and orbit for the spacecraft. As the result of the effort for the prediction, finally we acquire IKAROS after 9 months. This paper presents that the attitude and orbit prediction technique, while IKAROS was lost in space.

Nomenclature

A = effective area of the sail

B_f = Lambertian coefficient

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c = light speed

C_{spe} = specular reflectivity

C_{dif} = diffusive reflectivity

C_{abs} = absorptivity

G = gravitational constant

h = offset between the center of the SRP and the center of mass

I_S = moment of inertia for the spin-axis

I_T = moment of inertia for the orthogonal axis to the spin-axis

l = radius of the sail

m = mass of the spacecraft

M_1 = mass of the Sun

$\boldsymbol{n}$ = normal direction vector to the sail membrane

r = solar distance of the spacecraft

$\boldsymbol{r}$ = position vector of the spacecraft

R = the solar distance of the Earth

$\boldsymbol{s}$ = Sun direction vector

S_0 = Solar Constant

η = deflection angle of the sail

θ_s = Sun cone angle

κ = thermal emissivity

μ = gravitational parameter equal to $G(M_1 + m)$

ξ = distortion angle of the sail

ϕ = out-of-orbit-plane Sun angle

ϕ_{eq} = out-of-orbit-plane Sun angle of the attitude equilibrium point in Sun-pointing

ψ = in-orbit-plane Sun angle

ψ_{eq} = in-orbit-plane Sun angle of the attitude equilibrium point in Sun-pointing

ω_s = the angular rate of the Sun direction with respect to the spacecraft (equals to the orbit motion)

ω_x = angular rate about x axis in spin-free frame

ω_y = angular rate about y axis in spin-free frame

Ω = spin rate of the spacecraft

I. Introduction

THE Japan Aerospace Exploration Agency (JAXA) launched the world's first operational solar sail IKAROS (Interplanetary Kite-craft Accelerated by Radiation Of the Sun) on May, 2010 [1]. IKAROS has a 14 m x 14 m squared-shape type solar sail as shown in Fig. 1, and the wet mass is 307 kg. IKAROS has been continued to operate after its nominal missions, and succeeded many additional engineering and scientific missions. However, the propellant for the attitude control was eventually almost run out, and IKAROS no longer has a sufficient capability to control its attitude to a desired orientation. As a result, the attitude of IKAROS deviated from the Sun direction due to the interaction of the orbit motion and the solar radiation pressure (SRP) torque. Consequently, we lost the communication with IKAROS in December 24, 2011, due to the lack of solar power generation.

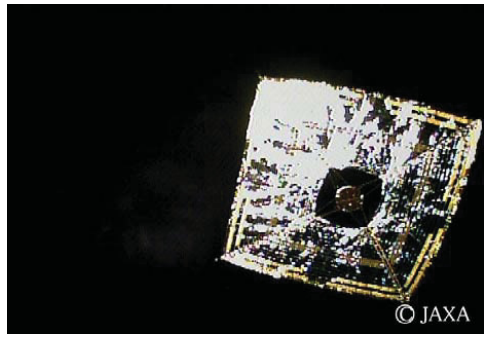

Fig. 1 Actual picture of spinning solar power sail IKAROS taken by deployable camera.

II. Characteristics of IKAROS

The solar sail demonstration spacecraft IKAROS investigated by JAXA has the cylindrical body and the square sail membrane as shown in Fig. 1. This spacecraft was launched on May 21, 2010 as the sub-payload of the PLANET-C; the Venus climate orbiter (which is also developed by JAXA). On 9th June 2010, IKAROS finally succeeded to deploy its sail membrane in deep space, and became the world's first operational solar sail. At present it is in the optional operation phase after nominal operation phase.

The main specifications of IKAROS are summarized in Table 1. The sail membrane of IKAROS consists of 7.5 μm Polyimide and it is evaporated by aluminum in 80 nm thickness [2]. A polyimide has the high thermal, mechanical and chemical resistance properties and is very light. Although the specular reflectivity of the membrane itself is relatively high due to the reflective properties of aluminum, the averaged specular reflectivity is degraded by several devices on the membrane. The main devices are the Flexible Solar Array (FSA) [3] and

the Reflectivity Control Device (RCD) [4]. The FSA is a thin power collector. If the large power collection system using this FSA can be constructed, it would be possible to load the high power ion engine on a future solar sail spacecraft [5]. The RCD is mounted on the edge of the sail membrane, and it can generate a torque by changing the induced force on each small element's surface. By using this RCD, the spacecraft would not need any propellant to control the attitude. Although the RCD has the capability to become a feasible attitude actuator in the future, it is one of the experimental instruments at the current stage. Therefore, IKAROS uses the gas-liquid equilibrium thrusters [6] as the nominal attitude actuator.

There are four antennas mounted on IKAROS. Two of them are Low Gain Antennas (LGA's) which are used for normal operation, and one is mounted on +Z panel of the body, and the other is mounted on –Z panel of the body. The Middle Gain Antenna (MGA) is mounted on +Z panel of the body. Although this MGA is supposed to be used in the case that the distance from the Earth is large, we cannot use it in the nominal operation phase due to the Earth aspect angle constraint. The other antenna is for the VLBI (Very Long Baseline Interferometry) operation.

Table 1 Main Specifications of IKAROS

Parameter	Specification
Size of Membrane	14 m x 14 m
Material of Membrane	Polyimide
Total Mass	< 307 kg
Moment of Inertia	(Ix, Iy, Iz) = (434, 434, 868) kgm^2
Specular Reflectivity	0.719
Diffusive Reflectivity	0.117
Absorptivity	0.162
Antenna Directionality	LGA-1: +Z of Body; LGA-2: -Z of Body MGA: +Z of Body; VLBI: +Z of Body
Attitude Control	Spin
Attitude Actuator	Gas-liquid equilibrium thrusters

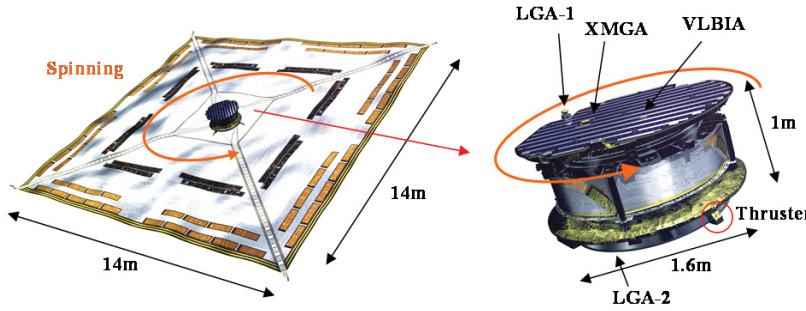

Fig. 2 Configuration of IKAROS.

III. Search Operation

After we lost the communication with IKAROS, the prediction of the IKAROS trajectory was immediately initiated in order to re-establish the communication. The complexity is that the IKAROS trajectory is influenced by its attitude, since IKAROS is solar sailing. Therefore, the attitude must also be predicted for the precise prediction of the trajectory of the solar sail. Firstly, we introduce the detail of the search operation for IKAROS.

A. Condition of Communication Recovery

The condition of communication recovery with IKAROS is defined as follows:

1) Sun cone angle condition:

In order to obtain the power, the solar arrays mounted on the top of the body must be pointed to the Sun direction. Because the spin-axis direction and a normal vector direction almost correspond, the generated power is large when the Sun cone angle is small. Nevertheless, it cannot be known exactly that in how small Sun angle case IKAROS can wake up, because there are several factor for the power generation, i.e., Sun cone angle, solar distance, aging of the solar array, and temperature of the solar array. According to the actual flight results, the latest observed Sun cone angle was about 60 deg. Therefore, we define the criterion for the Sun cone angle is to be smaller than 60 deg.

2) Earth cone angle condition:

In order to communicate with IKAROS, the antenna mounted on IKAROS should be pointed to the Earth direction. Although there are four antennas on IKAROS introduced in the previous section, one of two LGA's is used in this search operation. Same as the condition of the Sun cone angle, there are several factors to be caught by the ground station; Earth cone angle, distance from the Earth, radio wave interference of the sail membrane. Thus, we define the criterion for the Earth cone angle is to be smaller than 80 deg in accordance with the actual flight result.

3) Pointing accuracy of the antenna on ground

Even if IKAROS normally wakes up and the antenna mounted on IKAROS points to the Earth direction, we cannot communicate with IKAROS in the case that the antenna of the ground station does not point to spacecraft direction. Therefore, we have to predict the spacecraft position precisely to point the spacecraft direction by the antenna on ground.

B. Outline of Search Operation

In three conditions introduced above, the satisfactions for conditions 1) and 2) depend on the natural motion of the attitude. Only for the condition 3), we can make an effort to satisfy. In order to satisfy the condition 3), main objectives are defined as follows.

- To predict the IKAROS attitude with adequate accuracy
- To predict the IKAROS orbit with adequate accuracy

By means of these predictions, the direction of antenna on ground station can be predicted in the operation. This predicted antenna direction is used to change the orientation of the antenna of the ground station.

IV. Attitude Prediction

During the period from when the propellant run out to when we lost the communication, the attitude of IKAROS drifted away from the Sun direction mainly due to the SRP torque. This attitude drift motion has been observed after the sail was deployed. It has been clear that this motion is occurred by the balance of the SRP torque and the angular momentum of the solar sail. Thus, we succeeded to model this attitude drift motion. This dynamics model can be defined as follows:

A. Attitude Dynamics of IKAROS

First, we introduce the mechanism of the attitude photon-drift motion. After the deployment of the sail membrane on June 9, 2010, the attitude of IKAROS changed without any attitude control, which had not been seen before the sail deployment. This attitude motion is clearly caused by the SRP torque induced by the large sail. During the first few days after deployment of the sail, this motion was expected to be the same as in Hayabusa's case [7] (circular motion). However, the determined attitude did not obey the attitude dynamics model which was established based on Hayabusa's flight results. Therefore, the IKAROS project team remodeled the attitude dynamics by using the flight result of IKAROS. Here, we introduce the attitude dynamics modeling for the solar sail under the SRP torque.

The attitude drift motion in this frame is somehow different from the Hayabusa's. Although the attitude drift motion is circular around the equilibrium point in the Hayabusa case, IKAROS attitude motion is more spiral toward the equilibrium point. This spiral attitude motion can be explained from deformation of the sail. For the flat surface like the solar array panel of Hayabusa, the SRP torque direction induced by the absorption and the diffusive reflection of the solar arrays surfaces is orthogonal to both the Sun direction and the spin axis direction (i.e., normal direction of the effective surface area). This spiral motion is found to be due to the collaboration

between the deflection and distortion of the sail and the variation of the optical properties of the devices on the membrane [8]. The SRP torque induced by this deformed sail is defined as follows:

$$\begin{bmatrix} T_x \\ T_y \\ T_z \end{bmatrix} = \begin{bmatrix} A & -B & 0 \\ B & A & 0 \\ 0 & 0 & C \end{bmatrix} \begin{bmatrix} \sin\psi \\ \sin\phi\cos\psi \\ 1 \end{bmatrix} \tag{1}$$

where the subscript x, y and z are the components of the spin-free coordinate system. The torque parameters A and B are defined as follows:

$$A = \frac{\pi l^3}{3}(2p_{n1} + p_{n2} + p_s)\xi\,, \quad B = \frac{\pi l^3}{3}\left\{(2p_{n1} + p_{n2})\eta + p_s\left(\eta + \frac{3h}{R}\right)\right\} \tag{2}$$

with the parameters consisted by the optical parameters:

$$p_{n1} = \frac{S_0}{c}\left(\frac{R}{r}\right)^2 (2C_{spe})\,, \quad p_{n2} = \frac{S_0}{c}\left(\frac{R}{r}\right)^2 (B_f C_{dif} + \kappa C_{abs})\,, \quad p_s = \frac{S_0}{c}\left(\frac{R}{r}\right)^2 (C_{abs} + C_{dif}) \tag{3}$$

On the other hand, z component of the torque parameter C is derived from the actual flight data of the spin rate shown in Fig. 3.

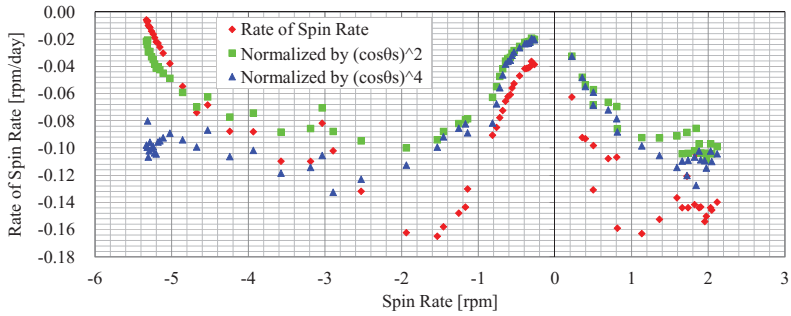

Fig. 3 Rate of spin rate with respect to the spin rate.

The red diamond in Fig. 3 is the rate of spin rate in the unite rpm/day, which is almost raw data; the green square is the normalized rate of spin rate by the $\cos 2\theta_s$; the blue triangle is the normalized rate of spin rate by the $\cos 4\theta_s$. In Fig. 3, the blue triangle seems to be the easiest trend to model as the quadratic function of the spin rate.

$$C = \frac{S_0}{c}\left(\frac{R}{r}\right)^2 \cos^4\theta_s (a\Omega^2 + b\Omega + c) \tag{4}$$

where a, b and c are the coefficients of spin rate. Equation (5) means that the z component of torque varies with the fourth power of cosine of the Sun cone angle and the quadratic function of the spin rate. If the sail membrane is flat plate, there is no torque for z component. In the real world, however, the sail is deformed and the z component of torque exists.

By using Eq. (1), the dynamics equation and kinematics equation for the x and y components can be derived under the assumption that the moment of inertia of the orthogonal axes to the spin-axis is equal:

$$\frac{d}{dt}\begin{bmatrix}\phi\\ \psi\\ \omega_x\\ \omega_y\\ \Omega\end{bmatrix}=\begin{bmatrix}\omega_y/\cos\phi\\ -\omega_x\\ (-I_S\Omega\omega_y+T_x)/I_T\\ (I_S\Omega\omega_x+T_y)/I_T\\ T_z/I_S\end{bmatrix} \tag{5}$$

By the algebraic manipulation, the attitude solution of Eq. (1) is obtained as follows:

$$\begin{bmatrix}\phi\\ \psi\\ \Omega\end{bmatrix}=\begin{bmatrix}\phi_{eq}+D\exp\left(\frac{A}{I_S\Omega}t\right)\sin\left(\frac{B}{I_S\Omega}t+\varphi\right)\\ \psi_{eq}+D\exp\left(\frac{A}{I_S\Omega}t\right)\cos\left(\frac{B}{I_S\Omega}t+\varphi\right)\\ \Omega_0+\frac{C}{I_S}t\end{bmatrix} \tag{6}$$

with the equilibrium points of ϕ and ψ:

$$\begin{bmatrix}\phi_{eq}\\ \psi_{eq}\end{bmatrix}=\frac{I_S\Omega}{A^2+B^2}\begin{bmatrix}B\omega_s\\ A\omega_s\end{bmatrix} \tag{7}$$

where φ is the initial phase angle of the attitude in the Sun-pointing reference frame (refer to Fig. 4); D is the angular length of the attitude from the equilibrium point in the Sun-pointing reference frame.

From Eq. (6), the behavior of the attitude drift motion can be predicted. The convergence of the attitude motion is determined by the exponential term, i.e., the value and sign of A/Ω. The rotation direction of the spin axis is determined by the trigonometric function, i.e., the value and sign of B/Ω. The spin up/down is determined by the value and sign of C. The feature due to each parameter can be summarized as follows;

- $A/\Omega > 0$: Divergence, $A/\Omega < 0$: Convergence
- $B/\Omega > 0$: Anti-Clockwise, $B/\Omega < 0$: Clockwise
- $C > 0$: Spin-up, $C < 0$: Spin-down

As we can see from Eqs. (2) and (3), the sign of A and B are determined by coupling of the optical properties and deformation of the sail membrane. Similarly, C is also known to be determined by the optical properties and the deformation of the sail in the equation, and the sign of the value of C is always minus as shown in Fig. 3 in IKAROS case.

B. Actual flight results

The feature of the attitude drift motion is observed in the actual operation. At first, we introduce how to see the figures of attitude. Figure 4 shows the attitude orientation in the attitude sphere in the Sun-pointing reference frame. In the figure of attitude, the origin is the Sun direction in the Sun-pointing reference frame, the horizontal axis denotes the in-plane Sun angle ψ, the vertical axis denotes the out-of-plane Sun angle ϕ (plane means orbit plane here). In this figure, the Earth direction is moved depending on the season.

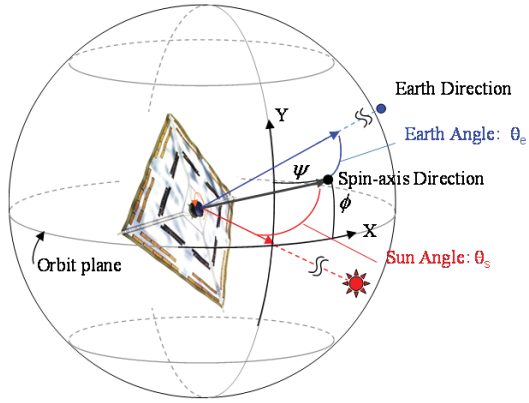

Fig. 4 Attitude definition in the attitude unit sphere.

As the results of the actual flight operation, we introduce two case attitude drift motion. Case-1 is the attitude result of the dynamics attitude control operation which was conducted during the period from September to October, 2011. This motion is shown in Fig. 5, and this is the representative case for the normal spinning case. In this case, the spin rate is minus; torque parameters A is minus, B is plus, C is minus; and the total motion is convergent and anti-clockwise drift motion as the result of the combination of each parameter. Case-2 is the attitude result after the inverse-spin operation executed on October 18, 2011. This motion is shown in Fig. 6, and this is the representative case for the inverse-spinning case. In this case, the spin rate is plus; torque parameters A is minus, B is plus, C is minus; and the total motion is divergent and clockwise drift motion. The features of drift motion for these two cases are summarized in Table 2.

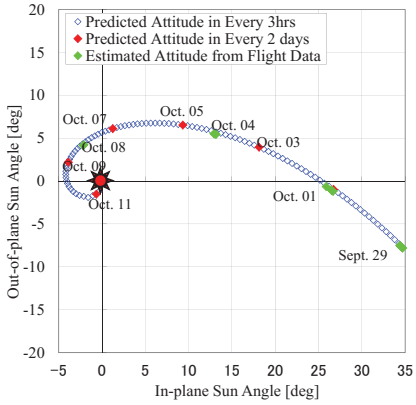

Fig. 5 Estimated and predicted attitude in dynamic attitude control operation

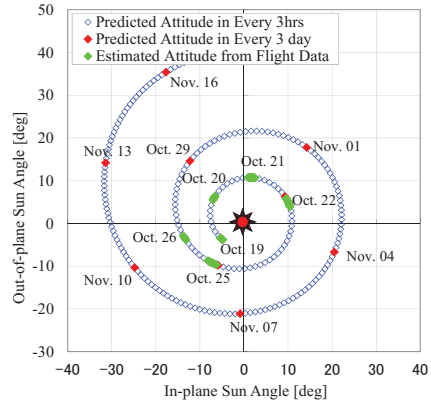

Fig. 6 Estimated and predicted attitude after inverse-spin operation

Table 2 Features of attitude drift motion.

Case No.	Ω	A	B	C	Convergence/ Divergence	Rotation Direction	Spin up/down
Case-1	Plus	Minus	Plus	Minus	Convergence	Anti-Clockwise	Spin-down
Case-2	Minus	Minus	Plus	Minus	Divergence	Clockwise	Spin-down

As shown in Figs. 5 and 6, the attitude solution expressed in Eq. (6) is verified in the both case that the spin rate is plus/minus. In both period shown in Figs. 5 and 6, the torque parameters are estimated based on the actual attitude data.

As case-3, Fig. 7 shows the estimated and predicted attitude motion just before IKAROS lost the communication with the ground station. In this period, the fuel for the attitude control is already almost run out, and the attitude move completely naturally. In this case, the attitude should be drifted away spirally from Sun direction and the rotation direction is a clockwise, because the conditions of the torque parameters are almost same as case-2 (Fig. 6).

Nevertheless, as shown in Fig. 7, the attitude drifts away from the Sun direction in the shape like a logarithm function, and completely different from case-2. The main differences between case-2 and case-3 are the amplitudes of the Sun cone angle and the spin rate. Both parameters of case-3 is larger than case-2. This fact indicates that the influence of the SRP torque in the case-3 is weaker than case-2, because the attitude drift motion is determined by the balance of the solar radiation torque and the angular momentum. If the Sun cone angle is large, the magnitude of the SRP decreases with the cosine of the Sun cone angle. Similarly, if the

absolute value of the spin rate increases, the influence of the SRP relatively decreases due to the balance of the angular momentum. As the result of these differences of the parameters, the attitude behaves more like “inertial-fixed motion” (if the attitude completely points to inertial-fixed direction, the spin-axis direction in the Sun-pointing frame flows to the minus direction of the horizontal axis).

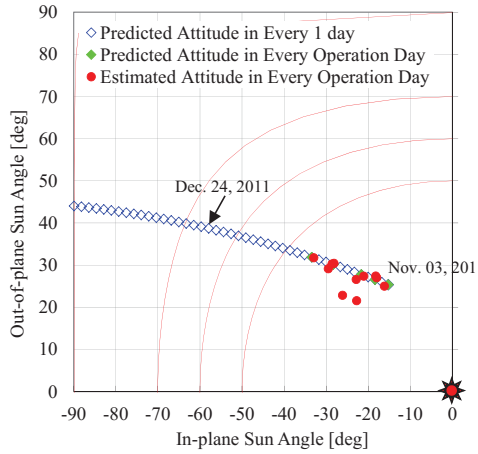

Fig. 7 Predicted and estimated attitude after November 3, 2011.

The attitude is predicted based on the SRP torque model defined as Eq. (1) and the estimated torque parameters. The estimated parameters from the actually observed attitude data in case-1, 2 and 3 are summarized in Table 3.

Table 3 Estimated torque parameters.

Case No.	Date in 2011	Solar Distance [AU]	Spin Rate [rpm]	A	B	C/I_S [rpm/day]
Case-1	Sept.28 – Oct. 7	0.73	1.17	-1.63e-4	1.81e-4	-0.146
Case-2	Oct. 24 – Oct. 29	0.72	-0.633	-1.12e-5	2.93e-5	-0.0689
Case-3	Nov. 29 – Dec. 10	0.83	-4.7	-9.51e-5	3.11e-5	-0.05

C. Attitude Prediction in Search Operation

The attitude drift motion is modeled correctly, and the torque parameters estimated by means of the actual flight data in the case that the spin rate is plus/minus and minus large. It seems to have enough information to predict the future attitude. For the attitude prediction in the search operation, however, there is the crucially insufficient information, i.e., the behavior of the drift motion when the front of the sail (aluminum evaporated

side) is oriented to point the anti-Sun direction (the case of $\theta_s > 90$ deg and $\theta_s < 180$ deg). It is impossible to observe the attitude motion in the case when the sail orients to the anti-Sun direction, because the solar arrays are mounted on the +Z panel of the body and it is same side as the front side of the sail membrane. Therefore, the attitude motion in the rear side Sun-pointing case is involuntarily assumption.

The assumption for the attitude prediction in the rear side Sun-pointing case is as follows:

- The sail deformation does not change.
- The difference of the optical properties between the front and rear side in the torque parameters is not taken into account.

From these assumptions and also the observed fact introduced in the previous sub-sub-sections, the relationships of signs of the torque parameters and spin rate, and the attitude motion can be sorted out as Table 4:

Table 4 Features of attitude drift motion.

Sun-pointing Side of Sail	Ω	A	B	C	Convergence/ Divergence	Rotation Direction	Spin up/down
Front	Minus	Minus	Plus	Minus	Convergence	Anti-Clockwise	Down
Front	Plus	Minus	Plus	Minus	Divergence	Clockwise	Down
Front	Plus/Minus Large Value	Minus Small	Plus Small	Minus Small	Inertial-fixed	---	Down
Rear	Minus	Plus	Minus	Plus	Divergence	Clockwise	Up
Rear	Plus	Plus	Minus	Plus	Convergence	Anti-Clockwise	Up
Rear	Plus/Minus Large Value	Plus Small	Minus Small	Plus Small	Inertial-fixed	---	Up

From this table, we can interpret the almost all attitude drift motion of the sail. Under this knowledge about the attitude drift motion, we conduct the simulation to find out IKAROS signal. Once we estimate torque parameters A, B and C based on the actual flight data, the attitude can be predicted in accordance with the dynamics equation defined in Eq. (5) in principle. Nevertheless, these torque parameters are strongly influenced by the shape (deflection and distortion) of the sail and the attitude determination accuracy. Thus, A, B and C are considered to have large errors, and we assumed that the errors are 20 % (1σ) of the estimated value, when we propagate the attitude based on the estimated values.

Using the dynamics defined as Eq. (5), the attitude is predicted by means of the Monte Carlo simulation with 20 % (1σ) error of the estimated torque parameters. Although there are over a thousand simulation results, here, we show only typical attitude profile result, by which we managed to find IKAROS. Figure 8 shows the simulation results for the actual search operation. As shown in Fig. 8, there are roughly three categories as follows:

I. Inertial-fixed flow motion firstly; Sun-cone angle is larger than 90 deg; start to spin-up; converge to the anti-Sun direction; the spin rate is changed from negative to positive and spin-up; start to diverge to the positive horizontal direction; Sun-cone angle is less than 90 deg; start to spin-down; converged to the Sun-direction.

II. Inertial-fixed flow motion firstly; Sun-cone angle is larger than 90 deg; start to spin-up; converge to the anti-Sun direction; the spin rate is changed from negative to positive and spin-up; start to diverge to the negative horizontal direction; Sun-cone angle is less than 90 deg; start to spin-down; converged to the Sun-direction.

III. Inertial-fixed flow motion firstly; Sun-cone angle is larger than 90 deg; start to spin-up; the influence of the SRP is too small and attitude flow like inertial-fixed to negative horizontal direction; Sun-cone angle is less than 90 deg; start to spin-down; converged to the Sun-direction.

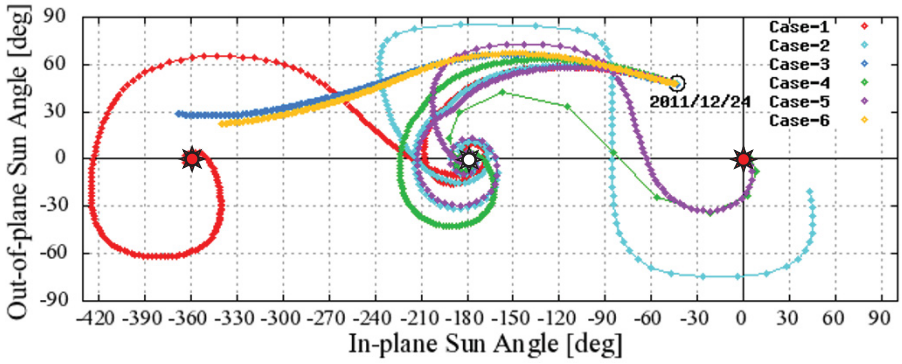

Fig. 8 Attitude prediction in the period of search operation.

V. Orbit Prediction

As introduced in the section I, the orbit of IKAROS is changed by the attitude direction, because IKAROS is solar sailing. Based on the attitude profile obtained as the previous section, the trajectory during the dis-communication period can be predicted.

A. Orbit Dynamics of IKAROS

The SRP acceleration depends on the sail shape, and the value of the SRP acceleration can be changed by the deformation of the sail similar to the attitude. However, this acceleration difference due to the sail deformation is too small to be considered in the orbit dynamics. Thus, the sail shape of IKAROS is assumed to be a flat plate in

the orbit estimation and prediction. That is why, the acceleration model for IKAROS can be expressed in the simple formulation.

The SRP acceleration is defined as follows:

$$\mathbf{a}_{SRP} = -\frac{S_0 A}{mc}\left(\frac{R}{r}\right)^2 \cos\theta_s \left\{(C_{abs} + C_{dif})\mathbf{s} + \left(\frac{2}{3}C_{dif} + \kappa C_{abs} + 2C_{spe}\cos\theta_s\right)\mathbf{n}\right\} \tag{8}$$

where A is the effective area of the sail; m is the mass; S_0 is the solar constant; c is the light speed; r is the solar distance of the spacecraft; R is the solar distance of the Earth; θ_s is the Sun-cone angle; κ is the thermal emissivity; $\mathbf{s}$ is the Sun direction vector; $\mathbf{n}$ is the normal direction vector to the sail membrane. For the orbit calculation, several non-SRP accelerations also should be considered in order to precisely estimate or predict, unlike with the attitude dynamics. Therefore, the orbit dynamics equation can be defined as follows:

$$\ddot{\mathbf{r}} = -\frac{\mu \mathbf{r}}{r^3} + \mathbf{a}_{NB} + \mathbf{a}_{rel} + \mathbf{a}_{SRP} + \mathbf{a}_{Other} \tag{9}$$

where $\mathbf{a}_{NB}$ is the acceleration due to the N-Body gravity; $\mathbf{a}_{rel}$ is the acceleration due to the relativistic effect; $\mathbf{a}_{\mathrm{Other}}$ is the acceleration induced by the other factors. As shown in Eq. (9), the orbit dynamics of IKAROS is general formulation.

B. Orbit Prediction in Search Operation

The orbit for the search operation is predicted based on Eq. (9). Figures 9 - 10 show that the predicted trajectory and acceleration direction in the relative frame (X is the radial direction; Y is the tangential direction of the orbit; Z is the normal to orbit plane) with respect to the ballistic orbit which is calculated by without SRP acceleration. The case numbers correspond to the attitude case numbers in Fig. 8. Thus, case-1 and -2 are cases that the SRP torque is relatively strong, and case-3 is that the attitude is more inertial-fixed. Compared to three cases, the case-1 and -2 are almost 500,000 km away from the ballistic orbit, and case-3 is almost 200,000 km away from ballistic orbit. This means that IKAROS could not be found in the cases that the orbit is propagated by using the attitude of cases-1 and -2, or ballistic without SRP acceleration. Indeed, we could not find the IKAROS signal in the actual operation with the predicted antenna direction based on the case-1 and -2, and also ballistic case.

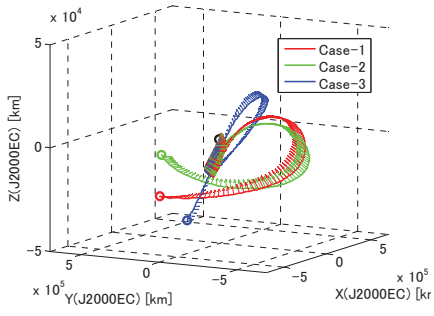

Fig. 9 Predicted position and acceleration relative to ballistic orbit in 3D.

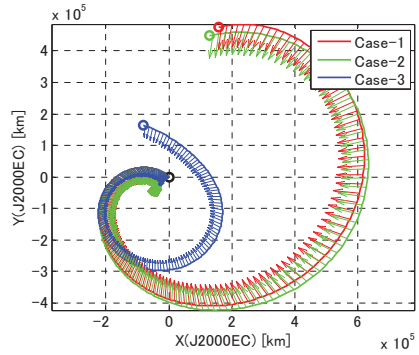

Fig. 10 Predicted position and acceleration relative to ballistic orbit in X-Y plane.

VI. Results of Search Operation

A. Attitude during Hibernation Mode

Although we could not know any observed information about the attitude during hibernation mode, the most probable attitude can be predicted after we caught the IKAROS's signal. Since the attitude was determined by means of the Sun cone angle and the Earth cone angle obtained in the operation after re-acquisition, we tried to predict the attitude more precisely to get close to the determined attitude. Figure 11 shows the attitude improved to get close to the actual data and Fig. 12 shows the spin rate in the same period. The red dot denotes the predicted attitude at September 8, 2012, and the black dot is the estimated attitude by means of the actual flight data. Therefore, the difference between the red and black dot means the prediction error, and there is still few degrees offset in Fig. 11, and few rpm offset in Fig. 12. This prediction error is mainly from the uncertainty of the attitude drift motion model in the rear side Sun-pointing case. Thus, we plan to analyze more detail of the dynamics of the rear side of the sail under several assumptions in the future.

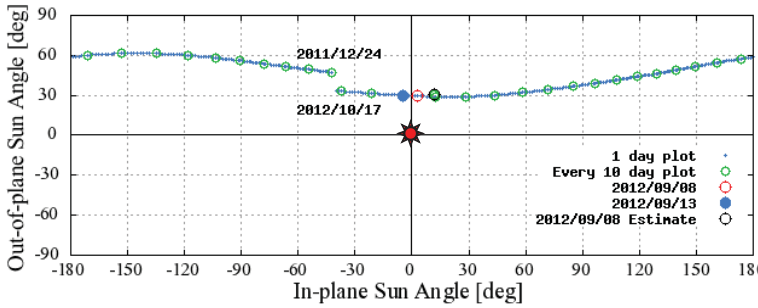

Fig. 11 Improved attitude profile from December 24, 2011.

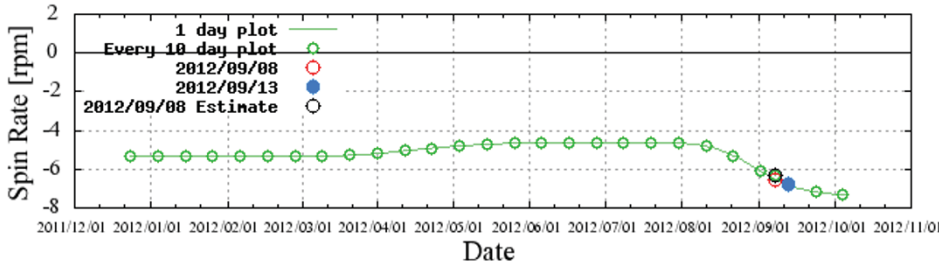

Fig. 12 Improved spin rate profile from December 24, 2011.

B. Trajectory during Hibernation Mode

The orbit obtained by the propagation based on the predicted attitude is illustrated in Figs. 13 and 14. The diamond means the position at epoch (October 17, 2011), and the star means the position at September 6, 2012. Although it cannot be seen clearly, the trajectory is changed from the initial orbit planning. Figure 15 shows that the relative orbit motion with respect to the ballistic orbit case calculated without consideration of the SRP. As shown in this figure, the position is actually varies from the ballistic case.

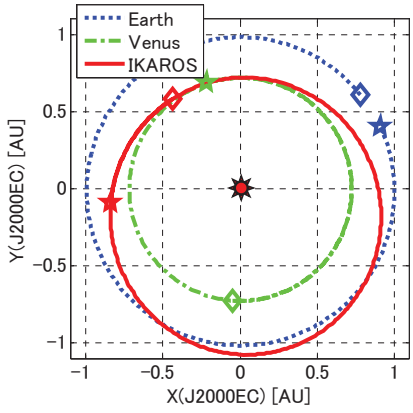

Fig. 13 Predicted trajectory in J2000EC frame.

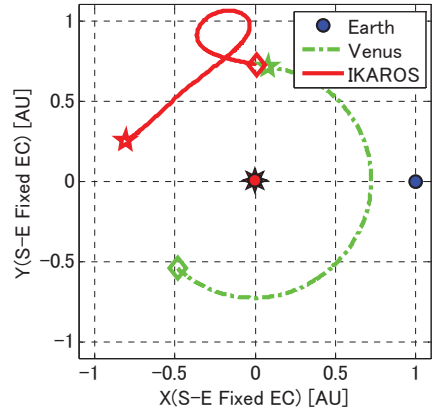

Fig. 14 Predicted trajectory in Sun-Earth fixed frame.

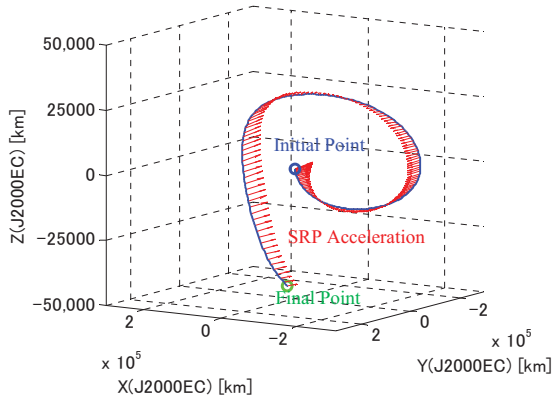

Fig. 15 Relative orbit motion from December 24, 2011.

VII. Analysis to Improve Trajectory Prediction

In order to improve prediction accuracy and find the signal more quickly at the next search operation, we try to modify the predicted trajectory. The most uncertain part of this prediction for both attitude and orbit is the dynamics in the period when the rear side of the sail orients to the Sun direction. The predicted attitude is gradually getting close to the observed attitude by means of the parameter survey of the torque parameters. As the effort of the trajectory prediction, we try to estimate the optical properties of the rear side of the membrane with the O-C (Observed minus Calculated) data of the Doppler. Figure 16 shows the residual data of the Doppler. The blue diamond is the improved case of the estimation of the optical parameters of the rear side, and the red cross is the original residual calculated from the nominal optical properties of the rear side. It can be seen that the amplitude of the residual of the improved case is reduced. However, there is still systematic trend even in this improved one. Therefore, we try to estimate also the initial state of this trajectory which is fixed in this analysis. The estimated SRP torque parameters and the optical properties of the rear side of the sail are summarized in Table 5. Although the optical properties of the total value of sail rear side are slightly larger than front side, the orbit fit well. On the other hand, the absolute values of torque parameters are minimums in the estimated values so far. This indicates that the influence of the SRP torque is relatively weak compared to the angular momentum of the spacecraft.

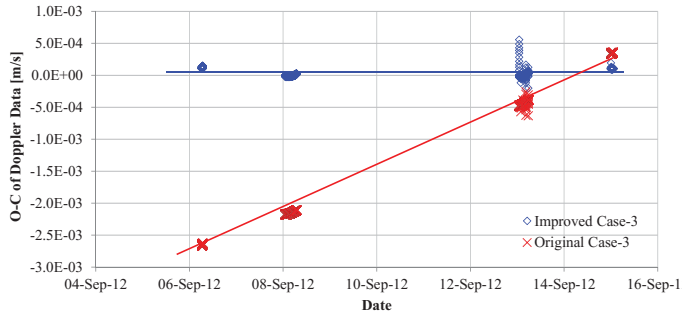

Fig. 16 Predicted attitude profile from December 24, 2011.

Table 5 Summary of estimated parameters

Estimated Parameters	Symbol	Value	Unit
Specular Reflectivity of Sail Rear Side	C_{spe}	0.545	---
Diffusive Reflectivity of Sail Rear Side	C_{dif}	0.466	---
Torque Parameter during Hibernation	A	-1.81e-5	Nm
Torque Parameter during Hibernation	B	2.31e-5	Nm

VIII. Summary

During the period from December 24, 2011 to September 6, 2012, we conducted the search operation for IKAROS which shift to the hibernation mode passively due to the shortage of the solar power. In order to re-acquire IKAROS, we must know the orbit precisely to make the antenna of the ground station point to the IKAROS direction. The precise orbit prediction is based on the precise attitude prediction, because IKAROS is solar sailing. In the both predictions of the orbit and attitude, the prediction during the period when the rear side of the sail is Sun-pointing is the difficult part, because we do not have any information about the dynamics in that period.

Under the reasonable assumptions, however, we model the dynamics even during the sail-rear-side Sun-pointing period. By means of this attitude drift motion model, the attitude of IKAROS is predicted, and the orbit is propagated based on the obtained attitude profile. Moreover the antenna direction is predicted by using the predicted position of the spacecraft, and finally we could find the signal of IKAROS which was orbiting in the deep space. This means it is verified that the attitude and orbit of IKAROS was predicted within enough accuracy to catch the signal.

As the future work, we plan to modify the model of the rear-side of the sail, and improve the prediction accuracy of the orbit, in order to find the IKAROS signal at the next timing she wakes up again.

Acknowledgments

Authors sincerely acknowledge to all IKAROS-team members and all supporters for their support to find the lost IKAROS.

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NanoSail – D Orbital and Attitude Dynamics

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NanoSail-D unfurled January 20th, 2011 and successfully demonstrated the deployment and deorbit capability of a solar sail in low Earth orbit. The orbit was strongly perturbed by solar radiation pressure, aerodynamic drag, and oblate gravity which were modeled using STK HPOP. A comparison of the ballistic coefficient history to the orbit parameters exhibits a strong relationship between orbital lighting, the decay rate of the mean semi-major axis and mean eccentricity. A similar comparison of mean solar area using the STK HPOP solar radiation pressure model exhibits a strong correlation of solar radiation pressure to mean eccentricity and mean argument of perigee. NanoSail-D was not actively controlled and had no capability on-board for attitude or orbit determination. To estimate attitude dynamics we created a 3-DOF attitude dynamics simulation that incorporated highly realistic estimates of perturbing forces into NanoSail-D torque models. By comparing the results of this simulation to the orbital behavior and ground observations of NanoSail-D, we conclude that there is a coupling between the orbit and attitude dynamics as well as establish approximate limits on the location of the NanoSail-D solar center of pressure. Both of these observations contribute valuable data for future solar sail designs and missions.

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Nomenclature

β	=	angle between sun-earth line and orbit plane
e	=	eccentricity
AOP	=	Argument of Perigee
I	=	inertia tensor
α	=	angular attitude [deg]
ω	=	angular velocity [deg s^{-1}]
τ	=	torque [N m]
ρ	=	density [kg m^3]
v	=	velocity [m s^{-1}]
Cd	=	drag co-efficient
A	=	sail drag area [m^2]
S	=	total sail area [m^2]
We	=	solar constant [J s^{-1} m^{-2}]
c	=	speed of light in vacuum [m s^{-1}]
r_e	=	distance of earth from sun [km]
r_s	=	distance of satellite from sun [km]
k	=	gravitational constant
R	=	semi-major axis [km]
I_{xx}	=	inertial moment about x-axis
I_{yy}	=	inertial moment about y-axis
I_{zz}	=	inertial moment about z-axis
a_x	=	direction cosine of x-axis in body frame
a_y	=	direction cosine of x-axis in body frame
a_z	=	direction cosine of x-axis in body frame

I. Introduction

NanoSail-D was released from the NASA FASTSAT-HSV01 satellite on January 17, 2011 and unfurled three days later on January 20 to become the first American solar sail deployed in space [1]. NanoSail-D (NSD) successfully demonstrated deployment of a solar sail in Low Earth Orbit (LEO) and stayed on orbit nearly nine months, re-entering on September 17, 2011. NSD had no active control capability and no onboard instrumentation to determine attitude or position, so ground observations were the sole source of flight data. These observations included orbit estimates in the form of Two Line Element (TLE) sets as well as a few ground-based visual observations that provided rough estimates of spin rate and limited insight into the attitude. Our goal in this research is to use the limited set of flight data to construct a time history of the ballistic coefficient based on orbit decay, attempt to model the effect of orbital lighting, and seek insight into the attitude dynamics of the NSD with an emphasis on bounding the location of the Center of Pressure (CP) of the solar force on the sail. In addition to the demonstration of solar sail technology, NSD also had a flight objective to demonstrate its utility as a de-orbit device [2] and this objective was clearly demonstrated.

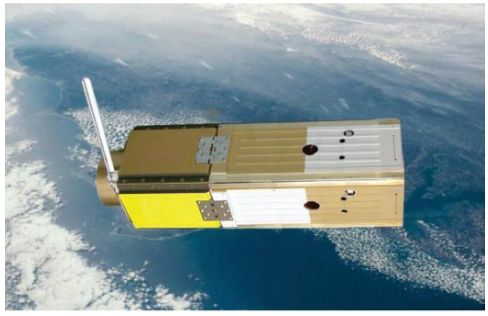

Fig. 1 NanoSail-D Cubesat Prior to Deploy

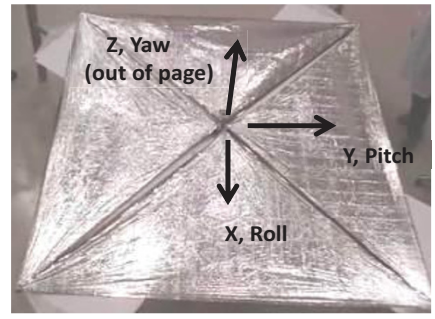

Fig. 2 NanoSail-D Deployed in Ground Test

II. Background

NSD was a 10 square meter sail deployed from a 3U Cubesat as depicted in Fig. 1. Fig. 2 shows the deployed sail during a ground test. Note in Fig. 2 that the X and Y axes are symmetric and located in the sail plane. From this, we can observe that the sail is axisymmetric about the Z axis which implies a potential for spin-stability about

that axis. The sail structure consisted of four thin triangular sheets of aluminized CP-1 material attached to four Triangular Rollable and Collapsible (TRAC) booms as depicted in Fig. 3. Each triangle of the sail is mounted to two TRAC boom ends, each 2.2 m. long, and the bus. The TRAC booms depicted in Fig. 3 were collapsed inside the cube satellite before launch and deployed the sail in approximately five seconds in space.

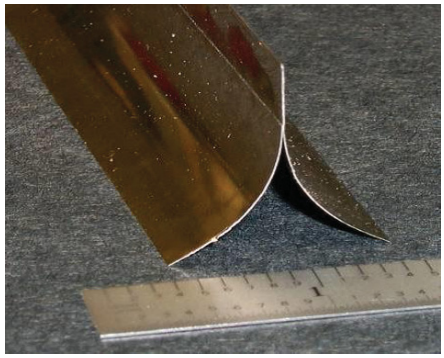

Fig. 3 TRAC Boom

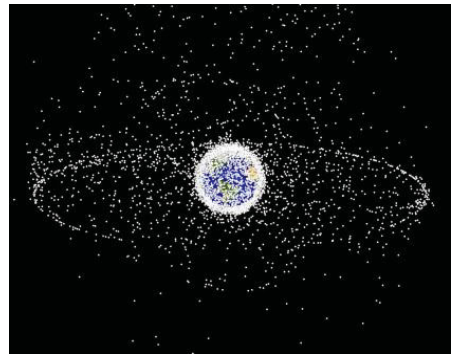

Fig. 4 Space Junk Orbiting Earth

NSD was inserted into a 71.9 degree inclination orbit at an altitude of 654 km. At this high inclination, full sun occurred periodically during the 9 months the sail was on orbit. The orbit decayed swiftly in spite of the high initial altitude due to the high area/mass ratio of the NSD.

NASA developed the Solar Sail Spaceflight Simulation Software (S5) [3,4] to model all aspects of interplanetary solar sail missions, but since Low Earth Orbit (LEO) is not generally considered a good mission fit for solar sails, S5 does not currently include atmospheric models. Thus for the orbit decay analysis we used Systems Tool Kit (STK) High Precision Orbit Propagator (HPOP) to determine the time history of the ballistic coefficient. The default Solar Radiation Pressure (SRP) model in STK HPOP was also used to model the SRP effect on the orbit and was compared to higher fidelity models in S5 over relatively short simulation time periods. Since the flight data available for inclusion into the various simulations was rather limited in accuracy, the relative lack of fidelity in the SRP model in STK HPOP was not an important issue in the analysis and STK HPOP was the primary simulation for the orbit dynamics. To model NSD attitude dynamics we created a new Matlab 3-Degree-Of-Freedom (3-DOF) rigid body dynamics simulation.

One of the objectives of NSD was to demonstrate that a sail can be used to de-orbit space junk. More than 521,000 human-made defunct devices greater than 1 cm in diameter now orbit Earth at velocities up to 8 km/s (see Fig. 4). Many pose an impact danger to operational satellites and the International Space Station. Thus the demonstration of the potential of a sail-like device for rapid de-orbit of space junk is strong ancillary benefit of the NSD mission.

III. Analysis

Given the lack of detailed mission data, the challenge was to extract as much useful information as possible from what was available. We started by identifying what was well known and proceeding to incorporate the lower-fidelity data sets. The best-characterized set of data was the orbit history consisting of Two Line Element (TLE) state vectors. Although the TLE vectors have a known limit of accuracy, the frequency with which they were updated allowed an orbit determination more than sufficient to characterize the time history of the ballistic coefficient of the NSD.

Solar radiation perturbations in LEO tend to affect eccentricity and argument of perigee more than the other orbital elements [5]. Thus we modeled the orbit with and without a Solar Radiation Pressure (SRP) model to determine whether the SRP effects on these two parameters could be isolated. For this analysis we used the baseline SRP model in STK, which is the spherical model. The spherical SRP model aligns the solar force completely along sun-spacecraft line, which means that it was essentially like a flat plate pointed directly at the Sun with no attitude variations possible. However, since from ground observations we know that NSD was rotating much more rapidly than the orbital period, an assumption can be made that NSD over the course of an orbit points an “average solar area” at the sun that is a flat plate equivalent. In the same way that the ballistic coefficient can be varied, one can then use the spherical SRP model in STK as a “solar ballistic coefficient” area. The SRP model can then be fitted as an average solar area as the orbital geometry (and spin axis of the NSD) evolves over time.

The analysis also included the creation of a 3-Degree-Of-Freedom attitude dynamics simulation called Tester. This model used the orbit history generated in the ballistic coefficient study to drive an attitude dynamics simulation. Since STK HPOP has high fidelity drag models available, density was also available as an input to Tester. Using the STK results, it was possible to recreate to a high degree of accuracy the perturbing forces that create environmental torques on NSD. This includes the gravity gradient, aerodynamic, and solar perturbing forces.

By contrast, the attitudes and attitude rates of the NSD are unavailable. A spin rate can be estimated at times from optical ground observations but attitude data is limited to approximations based on engineering judgment and sailcraft geometry. Studies have been conducted that combine estimates of starting spin rate and attitude with high-fidelity models of perturbing torques to attempt to establish limits on the offset between the Center of Pressure (CP) and Center of Gravity (CG) of the NSD and match the average solar area derived from the orbit simulation.

IV. Models

To model the aerodynamic and solar perturbation forces, we used STK HPOP. The drag model used the default STK drag coefficient (CD) of 2.20 along with the NRL MSISE atmospheric model and daily solar activity values obtained from the STK website. The SRP model in HPOP used 1.88 as an input for the pressure coefficient. In both cases, the area/mass ratio was varied to match the observed orbit behavior. The analysis was performed in 10-day segments, with a few smaller time increments analyzed for periods of interest such as full sun. A 10-day increment was a sufficient time increment to match the orbit within a prescribed tolerance. The as-flown TLE vectors were available on a roughly daily basis. The 10-day increments were times in which the solar and aero area/mass ratios were varied in STK HPOP.

We also developed a 3-DOF attitude dynamics simulation in Matlab called Tester. The 3-DOF rigid body dynamics model solved the equation:

$$I \cdot \alpha + \omega \times (I \cdot \omega) = \tau \tag{1}$$

In Eq. (1) I is the moment of inertia tensor, α represents the rotation acceleration vector; ω is the angular velocity vector; and τ is the sum of the disturbance torque vectors. Aerodynamic drag, solar radiation pressure, and gravity torques were modeled. Drag force is calculated through the equation:

$$f_{drag} = \frac{1}{2} \rho \, v^2 \, Cd \, A \tag{2}$$

Eq. (2) illustrates aerodynamic drag's dependency on density ρ, satellite velocity v, and A – the reference drag area. Cd represents the Co-efficient of drag. Solar force is calculated through the equation:

$$f_{solar} = c_r \frac{2\,S\,We}{c}\left[\left(\frac{r_e}{r_s}\right)^2\right] \tag{3}$$

where c_r is coefficient of reflectivity, r_e is the average distance of earth from the sun, r_s is the distance of the sailcraft from the sun, S is the total sail area, We is the solar constant equal to 1368 $\frac{J}{sm^2}$, and c is the speed of light. STK HPOP uses c_r in the spherical SRP model that was used in this research. The 3-DOF attitude dynamics simulation used the more complex formulation of optical coefficients in McInnes [6].

The x, y, and z components of the gravity gradient torque are calculated through the equations [7]:

$$\begin{aligned} L_x &= \frac{3k}{R^3}\left[\left(I_{zz} - I_{yy}\right)a_y a_z\right] \\ L_y &= \frac{3k}{R^3}\left[\left(I_{xx} - I_{zz}\right)a_z a_x\right] \\ L_z &= \frac{3k}{R^3}\left[\left(I_{yy} - I_{xx}\right)a_x a_y\right] \end{aligned} \tag{4}$$

The variable k is the Earth gravitational parameter (the universal gravitational constant times Earth's mass), R is the orbital radius vector; I_{xx}, I_{yy}, I_{zz} are the inertial moments along the x, y, and z axes respectively, and a_x, a_y, and a_z are the direction cosines of R with respect to the body frame of the sail. L_x, L_y, and L_z are the gravitational torques in the body axis frame of the sailcraft.

All these torques were combined into Tester and Matlab ODE 45 was used to integrate the equations of motion as depicted in Eq. (1). The simulation was driven by orbit state vector data fit to the raw TLE data in STK HPOP.

V. Results

A. Drag Model Results

The results of the orbit decay study in STK HPOP are shown in Fig. 5. The STK predictions in blue align well with the TLE data in red. In HPOP we selected a detailed atmospheric model and daily values for solar activity, thus ensuring an accurate density. Referring to Eq. (2) above, the velocity was also well known from the orbit and the drag coefficient was always assumed to be 2.20, which is a reasonable approximation for a flat plate. Since optical ground observations established that the sailcraft was spinning at a rate far greater than the orbit rate, the drag area solved for in Eq. (2) and used with the area/mass ratio required by STK HPOP as an input can be

considered an average area over the orbit as the sailcraft spun. If it were possible to know the precise spin axis and rate of the sailcraft as a function of time, the drag could be modeled more accurately. However, Fig. 5 demonstrates that the assumption of an average cross-section for the sailcraft was more than sufficient to model the orbit decay.

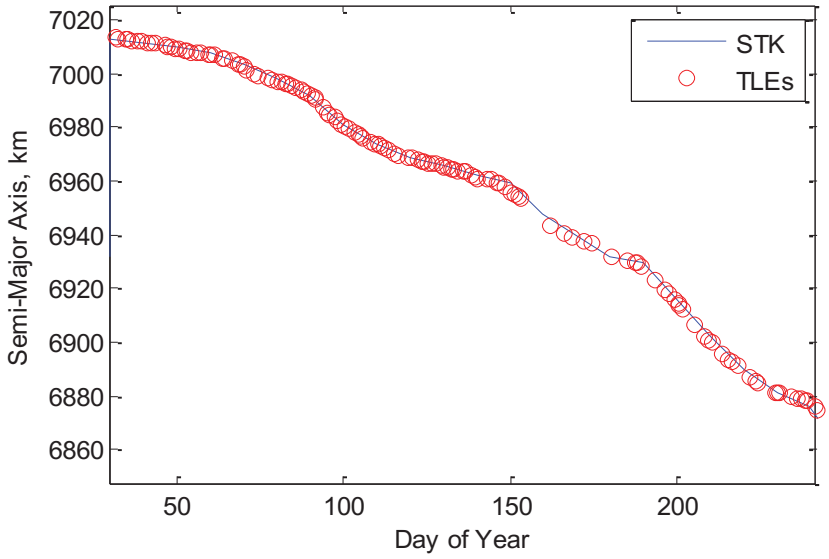

Fig. 5 Observed (TLEs) and Simulated (STK) Mean Semi-Major Axis

Over the course of the mission, TLE vectors were only available on a roughly daily basis with some gaps in coverage (a total of 167 in 240 days). In order to model the orbit decay, multiple observed state vectors are required for curve fitting of the mean Semi-Major Axis (SMA). Engineering judgment and trial and error led us to select a period of 10 days for each average cross-sectional drag area.

Thus the estimate of the drag area is based on an average fitted to TLEs over a period of 10 days and includes the rotation of the sailcraft relative to the inertial frame. Imagine that the sail is spinning and for a given orientation of the spin axis, the average area in the direction of the velocity vector can be estimated for purposes of calculating drag forces. That the area stayed more or less constant for periods of roughly 10 days could perhaps imply that the spin axis of the sailcraft for that time period was relatively constant.

In fact, the working hypothesis throughout this research was that the sailcraft may have approached a spin – stabilized condition at times. This is an assumption, but note that the sailcraft is axisymmetric about the Z axis (as

depicted in Fig. 2), and that it is well known from classical dynamics that an axisymmetric rigid body can be readily spin-stabilized [8].

To further explain the orbit geometry and disturbance torque environment during full sun conditions, Figures 6 and 7 show the orbit geometry for the full sun condition and the minimum sun condition.

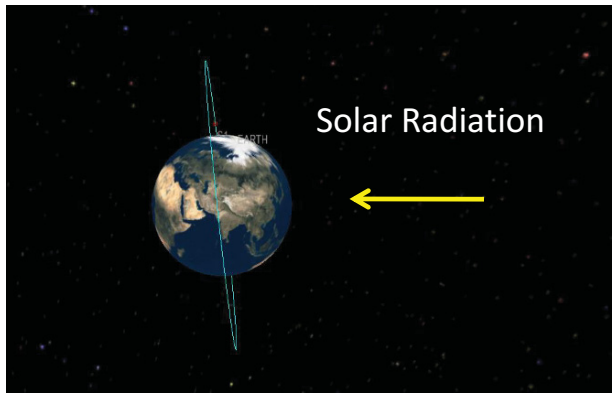

Fig. 6 Full Sun Orbital Geometry

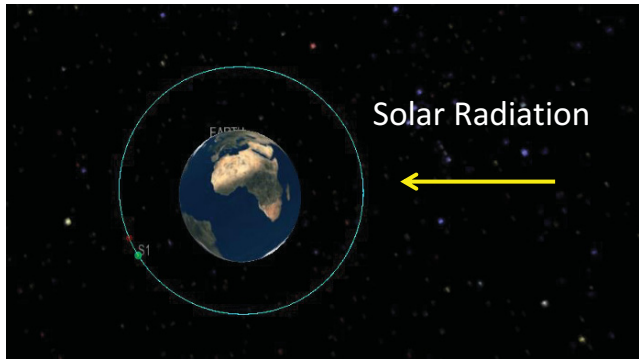

Fig. 7 Minimum Sun Orbital Geometry

From Fig. 6, one can deduce that the solar force will be roughly constant in the direction of the orbit angular momentum vector during the full sun condition. On the contrary, the orbit geometry in Fig. 7, where the solar direction is in the orbit plane, shows that in this case the solar disturbing force will constantly change direction over the course of an orbit. On the left and right of the plot in Fig. 7, the solar force will roughly be coincident with the positive and negative radius vector of the orbit. At the top and bottom the solar force is roughly coincident with the

negative and positive velocity vector of the orbit (assuming a clockwise rotation in the orbit plane of the sailcraft in its orbit). Thus it can be seen at a glance that the disturbance force environment in Fig. 6 is more consistent over an orbit than that in Fig. 7.

Other factors causing a difference between full sun and minimum sun are the presence of a shadow cone during the minimum sun period and the magnitude of the solar density over an orbit. For the shadow time, observe that on the left side of Fig. 7 there is a period when there is no solar force at all. Atmospheric density is also a strong function of orbital lighting [5]. Spot checking of STK HPOP density data showed that atmospheric density was a factor of 4-5 times higher between orbit midnight and orbit noon. But notice in Fig. 6 that since the orbit plane is almost perpendicular to the solar direction and the orbit is completely lighted, solar heating along the orbit is closer to constant and thus the density is also. Thus the drag force is less variable in Fig. 6 than it is in Fig. 7 over the course of an orbit.

Taking all these considerations into account, it is clear that the disturbance force direction and magnitude with respect to the orbit geometry is much more consistent over an orbit for the full sun condition as compared to the solar minimum. Thus, we might expect the sailcraft to tumble less, perhaps, in the full sun condition and so present a smaller effective drag area to the velocity vector.

To test this hypothesis, we closely examined the derived average drag area as a function of orbital lighting. Table 1 gives the result of this study by comparing β to average drag area at different times. β is used as a measure of sunlight exposure over the course of an orbit and is defined as the angle between the sun-earth line and the orbit plane. When the absolute value of β is near zero (Fig. 7) the sail orbit experiences the maximum shadow and minimum sun exposure. Conversely, when the absolute value of β is a maximum (Fig. 6), the sail can experience full sun for the duration of the orbit.

Table 1 shows a comparison of average drag over 10-day periods near both solar minimum and full sun conditions. There is a clear pattern of smaller average drag areas near solar maximum and vice versa. A smaller drag area could potentially imply a smaller spin precession axis during the full sun periods, or perhaps less tumbling of the sailcraft. The first ~ 40 days of the mission gave less consistent results, perhaps due to an initial “settling down” period for the sailcraft, lower aero forces or a combination of the two. Note also that due to the sun being in the northern hemisphere while orbit noon was in the southern hemisphere, the absolute value of β Day 160 was only 49 degrees, so that full sun did not occur, thus this period is called maximum sun.

Table 1 Average Drag Area in Full Sun and Minimum Sun Conditions

Days After Deploy	Condition	β (deg)	Area (meters^2)
76	Min Sun	0	5.34
105	Full Sun	88	2.04
135	Min Sun	0	5.77
160	Max Sun	-49	2.11
184	Min Sun	0	5.38
214	Full Sun	84	0.69

B. Solar Radiation Model Results

The average Solar Radiation Pressure (SRP) area can be determined in similar way. To model the solar radiation pressure, we examined argument of perigee and eccentricity, since those two parameters are the most sensitive to solar radiation pressure perturbation force. [5] Fig. 8 presents a plot of eccentricity over the course of the entire mission with respect to β.

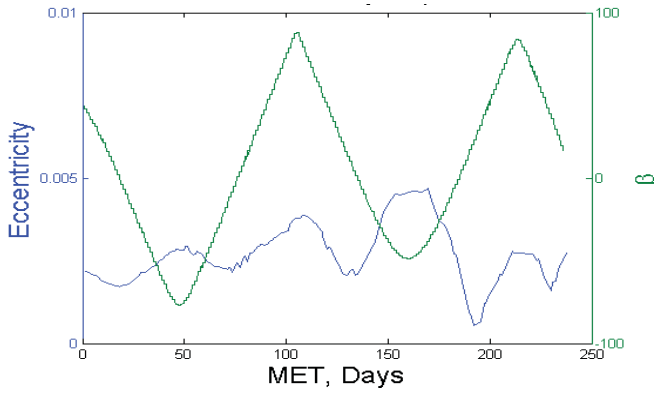

Fig. 8 Eccentricity and Beta Angle Comparison

Fig. 8 shows a clear correlation of eccentricity to β. Given the knowledge that argument of perigee and eccentricity are strongly affected by solar radiation pressure, we used these two elements to iterate for correct values

of average projected solar area over 10-day periods in the same way that drag area was derived using the observed mean SMA history.

In order to help determine the significance of solar radiation pressure on the NSD orbit, STK HPOP simulated the trajectory with and without the SRP model. The results of this simulation are presented in Fig. 9 for Argument of Perigee (AOP). In Fig. 9 the green line is the STK HPOP without SRP modeled, the red line is the same simulation with SRP modeled, and the blue dots are the AOP from the TLE data which was based on observations.

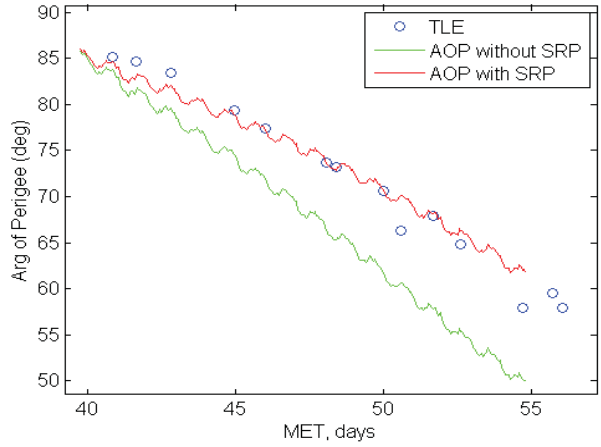

Fig. 9 Simulated AOP With and Without SRP Modeled

Some conclusions can be drawn from Fig. 9. First, the spherical model of SRP in STK HPOP is a good approximation of a spinning sailcraft provided the model is supplied with the correct reference area and radiation coefficient. For this study we used 1.88 for the radiation pressure coefficient in HPOP based on a simple average of the measured radiation pressure coefficients of the front (1.92) and back (1.83) of the NSD sail material. Note that the front and back coefficients are based on laboratory optical testing of the actual sail material that flew in space on NSD, since this material was carefully measured during the NASA In-Space Solar Sail Project [9]. Second, Fig. 9 demonstrates conclusively that the SRP effect on the orbit was significant and that the orbit could not be modeled accurately without an SRP model. Thus, NSD definitely demonstrated solar thrust that seemed to consistently be in a single direction roughly along the sail-sun line, even though it was spinning rapidly.

Table 2 SRP Area in Full Sun and Minimum Sun Conditions

Days After Deploy	Condition	β (deg)	Area (meters^2)
76	Min Sun	0	1.92
105	Full Sun	88	9.09
135	Min Sun	0	1.42
160	Max Sun	-49	4.61
184	Min Sun	0	4.68
214	Full Sun	84	6.91

Table 2 presents the results from STK HPOP that determined a best fit to SRP area for 10-day periods near solar minimum and full sun conditions. These values were selected by matching the simulated and observed eccentricity and AOP of the NSD orbit. The task was more difficult than matching the SMA decay rate. This difficulty was caused in part by the fact that eccentricity and AOP tend to be less well defined and noisier for low values of eccentricity and at times the eccentricity of NSD was below 0.01. Another reason that it was more difficult to match eccentricity and AOP compared to SMA is that although the spherical model of SRP in STK HPOP did a remarkably good job of modeling the optical properties of a spinning sailcraft, the limitation that the solar force acts only along the sailcraft-sun line undoubtedly still introduced some error.

Nevertheless, the values in Table 2 did provide a good fit to the eccentricity and AOP of the NSD orbit, as can be seen in Fig. 9. Keeping in mind that the goal of the research presented in this paper was to do a general characterization of the orbit and attitude dynamics of the NSD based on a limited set of flight data, the fit to the observed orbit parameters is surprisingly good considering the relatively sparse flight data available.

Based on heuristic insight, one might expect the effective or average solar area to increase during full sun conditions, and Table 2 displays exactly this pattern. As discussed above the local minimum of β was only -49 on Day 160 and full sun did not occur. It can be seen that the average solar area for the 10-day period around Day 160 was significantly lower than that of Day 105, but given the difference in the absolute value of the respective βs this is to be expected (this logic also applies to the drag area in Table 1, but that does not seem to have been affected as it is still comparable to the full sun drag areas). As expected, the solar area for solar minimums is generally much

smaller, although the solar area on Day 184 was larger than that of the two earlier solar minimum periods, perhaps because the atmospheric density was beginning to be the dominant disturbance force at this point in time.

The estimates of solar area give valuable information about the NSD and provide confirmation that the sail produced a significant thrust while spinning. Additionally the data in Table 2 also can assist with studies of the attitude dynamics of the NSD. To see this, refer to Fig. 10, which presents the definition of the Local Vertical Local Horizontal (LVLH) coordinate frame that is the one in general used by NASA. In this system of reference, the Y axis is aligned in the direction opposite the angular momentum vector of the orbit, the Z axis is in the opposite direction of the radius vector, and Y cross Z completes the triad in the general direction of the velocity vector.

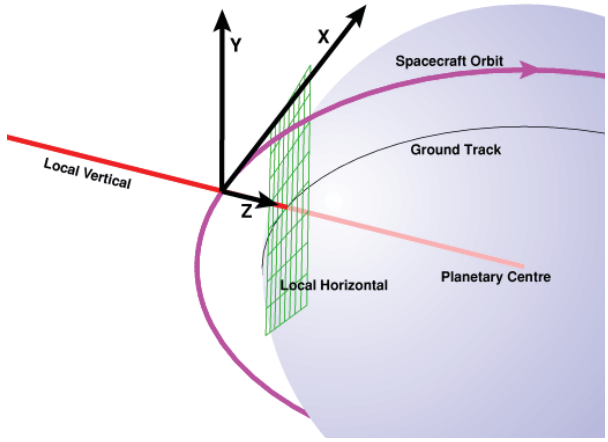

Fig. 10 LVLH Coordinate Frame

Consider the disturbance forces in the LVLH frame during a full sun condition. Since the orbit plane is nearly perpendicular to the sun, the solar pressure direction in the LVLH frame is then aligned mostly along the +/- momentum vector of the orbit (the +/- Y axis in Fig. 10) for the duration of the entire orbit. This is in contrast to the minimum sun condition when the solar disturbance force rotates between the +/- radial and +/- velocity directions (the +/-X and +/-Z axes in Fig. 10.) Given that in a full sun condition the solar force remains aligned with the angular momentum vector of the orbit, intuitively one might expect the sail to be “pushed into the orbit plane” by the solar torque. If this were the case, then one would also expect the sail to present a flatter aspect to the velocity

vector. The combination of these two effects would be to maximize solar area and minimize drag area when the sail is in a full sun condition. Table 1 and 2 confirm this heuristic reasoning very well, as they do indicate that the average 10-day solar areas and drag areas behave as expected during full sun. The gravity gradient and drag disturbance forces will always respectively be in the direction of the LVLH Z (or - radius) and LVLH X (or + velocity) and so will be consistent between full sun and minimum sun conditions.

The fact that the average solar and drag areas of the sail are somewhat predictable during full sun presents an opportunity to estimate a spin axis orientation (and thus a rough estimate of the attitude history) for the NSD during the time near full sun. Another piece of information that is relevant is that the NSD was axisymmetric and so the effect of the gravity gradient torque would tend to be to align the larger inertia axis (the axis with the bus or the body Z axis in Fig. 2) with the negative radius vector. Finally and fortuitously, there exist a few estimates of the spin rate of the NSD from ground observations, and two independent observations just days from the Day 160 maximum sun event.

C. Spin Rate Estimate from Optical Ground Measurements

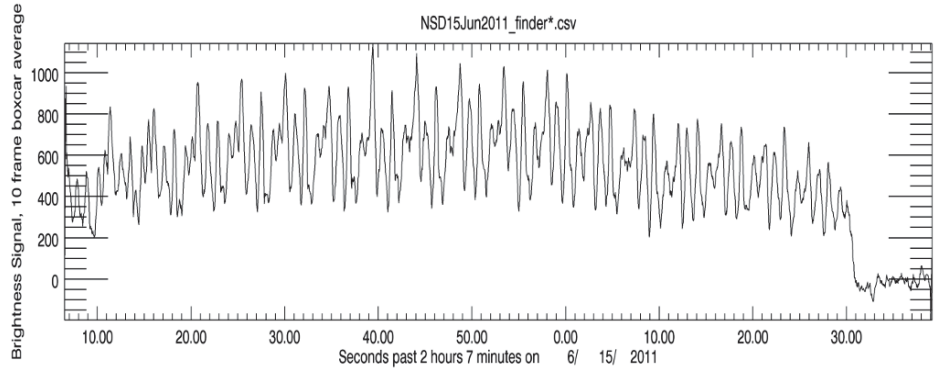

Fig. 11 Optical Brightness Measurement of NSD

Fig. 11 presents the results of an optical observation using a ground-based telescope at the Marshall Space Flight Center (MSFC) in Huntsville, Alabama by Dr. Robert Suggs. The information presented in Fig. 11 consists of

brightness measurements recorded during a pass of NSD over MSFC on June 15, 2011 (Day 165). Dr. Suggs also measured the brightness of NSD on June 13, 2011 (Day 163) and Ron Dantowitz of the Clayton Observatory Center in Boston, Massachusetts provided similar and independent data from measurements taken on June 8, 2011 (Day 158). Analysis of all three measurements by Dr. Suggs provided an estimate of approximately one rotation every 5 seconds which is equivalent to a spin rate of roughly 1.3 rad/sec. A close examination of Fig. 11 reveals smaller periodicities of four peaks per major cycle. The sub-peaks in amplitude clearly correspond to the four quadrants of the NSD, thus helping to confirm the estimate.

Optical observations were also achieved in mid-February and mid-August. The February observations estimated a rotation rate of once every 8 seconds or 0.8 rad/sec while the August observation estimated a rotation once every 3 seconds or 2.1 rad/sec. The spin rate of the NSD appeared to be increasing with time, perhaps due to the increasing atmospheric density as the altitude decreased.

D. Attitude Dynamics Simulation Results

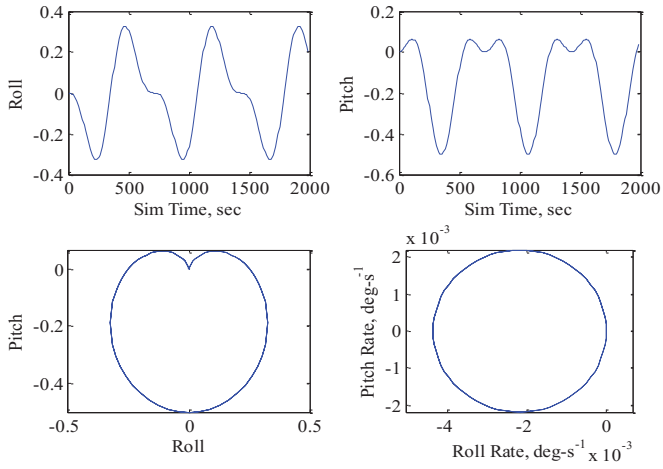

Fig. 12 Replication of Result from Wie

As discussed above, it's possible to derive a reasonable estimate of the NSD attitude around the time of the Day 160 solar maximum by combining STK-derived drag area and SRP area with the estimate of spin rate from the preceding section. This estimate can then be simulated in the 3-DOF attitude dynamics simulation Tester, which

takes the Best Estimate Trajectory (BET) from the STK results to provide high fidelity models of disturbance forces for use in environmental torque calculations.

Tester was developed in a modular fashion and tested as work proceeded. Each torque model was verified independently using idealized test cases to confirm the models were correct. The final verification was a comparison to a result in Wie [10]. The result of this simulation appears in Fig. 12 and is identical to Fig. 10 in Wie. It is worth noting that Wie's result was computed analytically while Tester solved the rigid body dynamics equations of motion numerically.

Using the average SRP and average drag areas from Tables 1 and 2, initial conditions to match those areas for the maximum sun condition of Day 160 and the full sun condition of Day 232 were estimated. For both of these days, estimates of spin rate based on ground observations were also available and these completed the initial conditions for Tester. Some selected results of these simulations are presented in Table 3.

Table 3 Attitude Dynamics Simulation Results

Days After Deploy	Mean Solar Area (meters^2)	Center of Pressure Offset (meters)
160	4.78	0.02
214	6.21	0.02

In Table 3, the mean solar area is the area averaged over one orbit of the spinning NSD that is perpendicular to the Sun-sail vector. The results show that the mean solar area was matched well compared to the results in Table 2. Once that result was matched with the values in Table 2, for both of the test cases a Center of Pressure (CP) offset from the geometric center of the sail was then varied. When the mean solar area predicted by the attitude dynamics simulation varied by a significant amount from the mean solar area in Table 3, then the value of CP was considered a rough upper limit on the size of a bias in the CP-CM offset on the sailcraft. Put another way, column 2 of Table 3 is the mean solar area for the simulation with no CP offset, and column 3 is an estimate of where the mean solar area deviates significantly from that in column 2. We also tracked the average drag area of the simulation, but it was not a primary focus.

For the results in Table 3, the matching of the average solar area is an indication that the estimate of spin rate and initial attitudes had some value. We do not claim to have replicated the precise attitude nor the exact spin axis of the

NSD, but the initial conditions were probably representative of the actual attitude and spin axis orientation on Days 160 and 214. The CP offset estimate should be considered more of a rough upper limit to the size of this parameter for NSD. The results were not completely unambiguous, and the time simulated and initial spin rates were varied from what was observed to aid a general understanding of the problem.

VI. Conclusions

The orbital and attitude dynamics of the NSD have been modeled based on a limited set of flight data. The simulations have produced an accurate model of the sail's orbit. The results of the orbit simulations produced estimates of mean solar area and drag area for 10-day increments of the mission. These estimates in combination with spin rate measurements from ground observations allowed initial conditions to be derived for an attitude dynamics simulation that correctly predicted the general observed behavior of the NSD mean solar area over one orbit during periods of full sun. The attitude simulations also led to a rough estimate of the CP bias from the geometric center of the sail, which is an important parameter for future sail missions.

In addition to the specific conclusions on the orbit and attitude dynamics of the NSD, simulation tools have been developed and honed that will aid future missions. The results of this research will directly aid verification work currently underway at NASA in support of the Sunjammer mission, the first controlled and instrumented American solar sail that is currently scheduled to be launched in November, 2014.

NSD also proved that a thin membrane of sail-like material with a large area/mass ratio is an excellent candidate for de-orbiting LEO satellites at the end of their lifetimes, which was one of its flight objectives. The attitude dynamics simulation results aid understanding the interplay between the solar and aerodynamic perturbing forces in LEO that will assist future mission designers who choose to use a sail-like device for de-orbiting satellites.

NSD achieved the first successful deployment of a solar sail for America, and the lessons learned from it are helping light the way to a bright future for solar sails.

Acknowledgements

We would like to acknowledge Dr. Robert Suggs and Rod Dantowitz for the optical measurements of the NSD and NASA for sponsoring the internships of Chelsea Katan and Brent Faller. We also acknowledge the NSD Principal Investigator, Dean Alhorn for his support and assistance during the NSD mission.

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Sunjammer: A Solar Sail Demonstration

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NASA's newly minted Space Technology Mission Directorate (STMD) is supporting the fabrication of a sail of this design in preparation of a planned 2014 flight. This mission, dubbed Sunjammer, will further advance the potential of propellantless solar sails. The Sunjammer mission is being led by the private company L'Garde Inc. of Tustin, CA. Sunjammer is named after a short story written by Sir Arthur C. Clarke. This ambitious project aims to prove the efficacy of a versatile and scalable solar sail design. The Sunjammer mission is being designed with the following objectives as guidance:

1. Demonstrate segmented deployment of a 1200m^2 solar sail.
2. Demonstrate attitude control plus passive stability and trim using tip vanes.
3. Execute a navigation sequence with mission-capable accuracy.
4. Fly to and maintain position at a sub-L1 and/or pole sitter positions.

NASA's Space Technology Mission Directorate is providing the necessary leadership and support to sail this flight-ready technology to the launch pad.

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Sunjammer Mission Overview

Figure 1. Sunjammer (Artist Rendering)

The Sunjammer mission is built around the desire to fly a sail beyond the bounds of an Earth orbit. Also, the mission is targeted to be low cost. These two constraints drive much of the design of the entire spacecraft. First, in order to achieve the low-cost constraint, this mission is planned as a secondary payload. As a secondary payload, the mission is forced to conform to several regulations and requirements. Specifically, Sunjammer is being designed to meet EELV Secondary Payload Adapter (ESPA) standards. This conformance, among other things, puts volume and mass constraints on the spacecraft.

Launch is scheduled for late 2014 onboard a SpaceX Falcon 9. The insertion trajectory will be sunward, but with slightly less than Earth escape velocity. This necessitates an onboard hydrazine propulsion system to impart enough delta-V to assure Sunjammer does not return to Earth.

Approximately two hours after this burn, sail deployment will begin. This process will take up to two more hours. The Sunjammer spacecraft is split into two distinct portions: the sailcraft and the carrier. The sailcraft is the portion of the spacecraft that will perform the solar sail demonstration after being boosted to earth escape. The carrier is the portion of the spacecraft that is expendable, and so will be jettisoned after deployment of the sailcraft, more than doubling the characteristic acceleration. Figure 2 shows the assembled spacecraft and the separated sailcraft and carrier, as well as the jettison simulation.

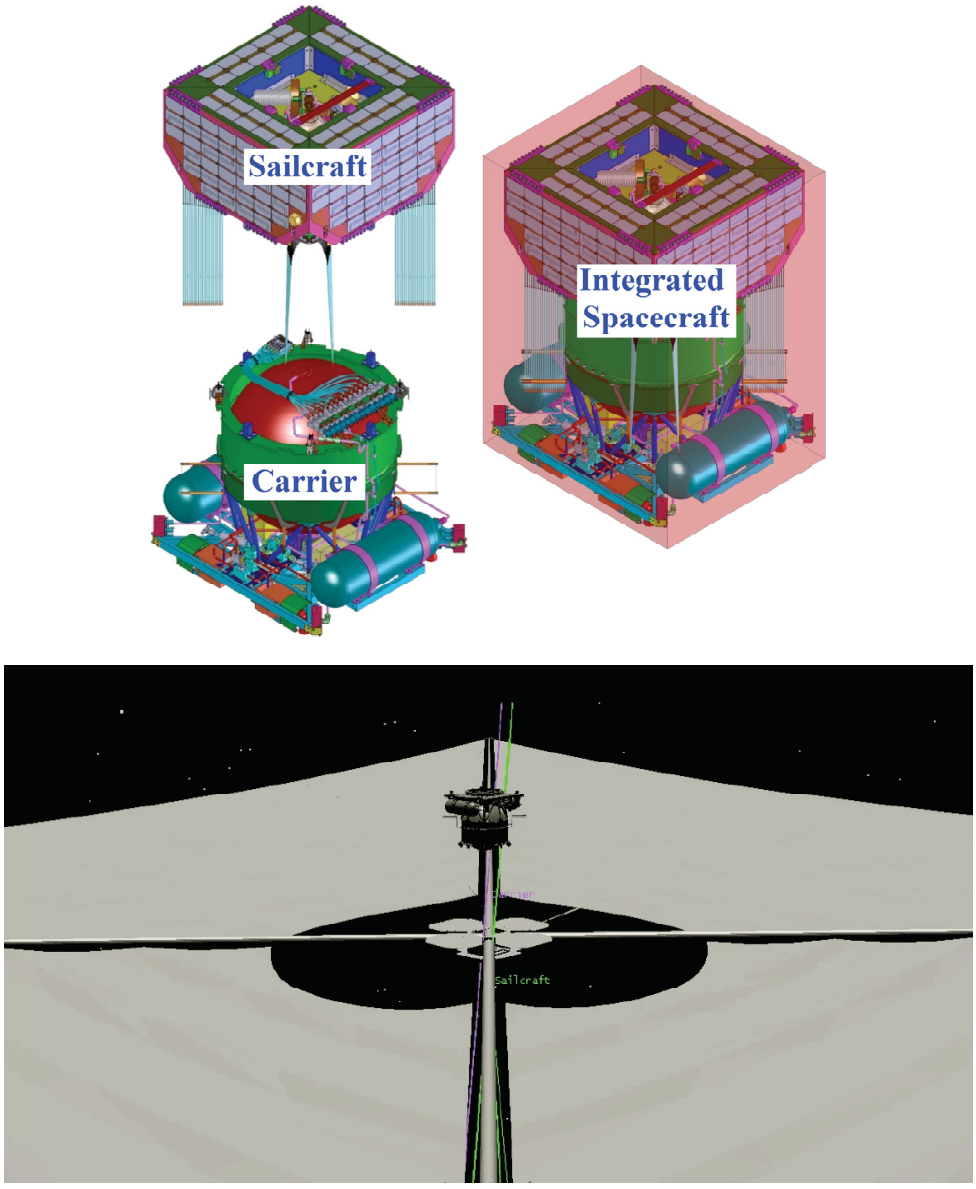

Figure 2. Packaged Sailcraft and Carrier, and Jettison Simulation

The operational sailcraft will be 1200 m^2 with a characteristic acceleration of approximately 0.25 mm/s^2. Deployment to a functional state will satisfy NASA's first requirement. An deployed overview is shown in Figure 3.

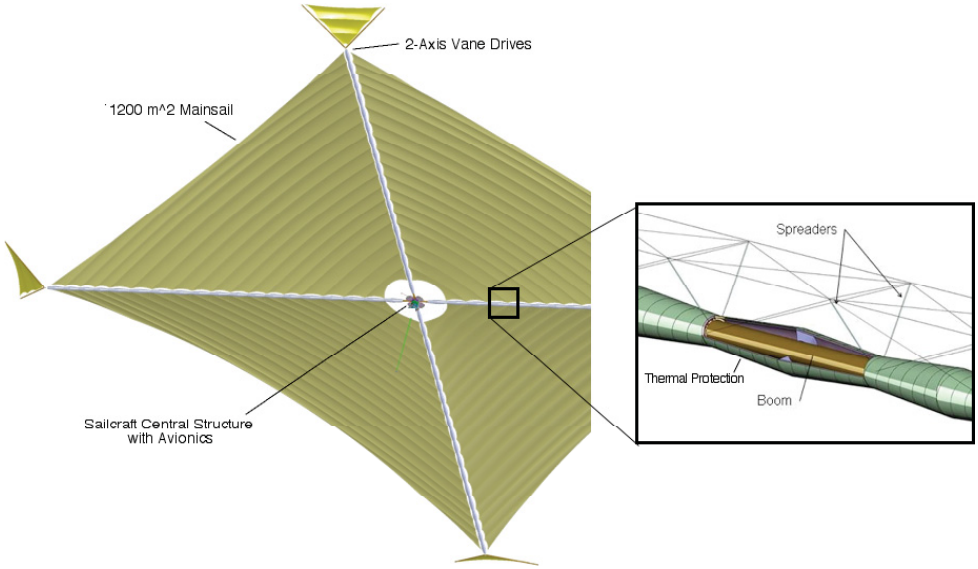

Figure 3. Major Components of the Deployed Sailcraft

Within six days of launch the sail will be calibrated for the moments that affect attitude control, then the thrust that affects navigation. The ability to retrim to new passively stable attitudes off-normal to the sunline will have been demonstrated as part of fulfilling NASA's attitude control requirements.

For the next 30 days the sail will navigate to demonstrate and fulfill the navigation requirements specified by NASA.

At this point, L'Garde desires to continue the mission to fly the sail near L1 and finally to a sub-L1 location, to stationkeep.

Along the way, the onboard magnetometer and plasma detector will be measuring particles from the Sun and comparing the results with NOAA's Advanced Composition Explorer. These important instruments are being provided by the UK Space Agency and will be instrumental in demonstrating the usefulness of solar sails in this particular application.

After six months, the sail will be recalibrated to demonstrate that it can still be controlled and navigated in the presence of degradation.

Deployment

A camera boom will be deployed first, then the vanes, then the mainsail.

All deployments are driven by L'Garde's telescopic boom deployment technology. The four mainsail booms, for example, are tapered such that they can be folded in on themselves. When pressurant is introduced, the boom deploys linearly, base-first, in a controlled and structurally stable manner, folded segment by folded segment, repeatably. The boom folding also allows rings fixed to the boom to be exposed such that the sail membrane quadrants may

be attached to the boom periodically along the boom length, so the booms take the sail along for the ride. There are line management devices for the spreader system lines, and membrane management devices to keep the quadrant deploying orderly, stripe-by-stripe. A deployment in progress showing the booms pulling out the managed sail quadrants is seen in Figure 4.

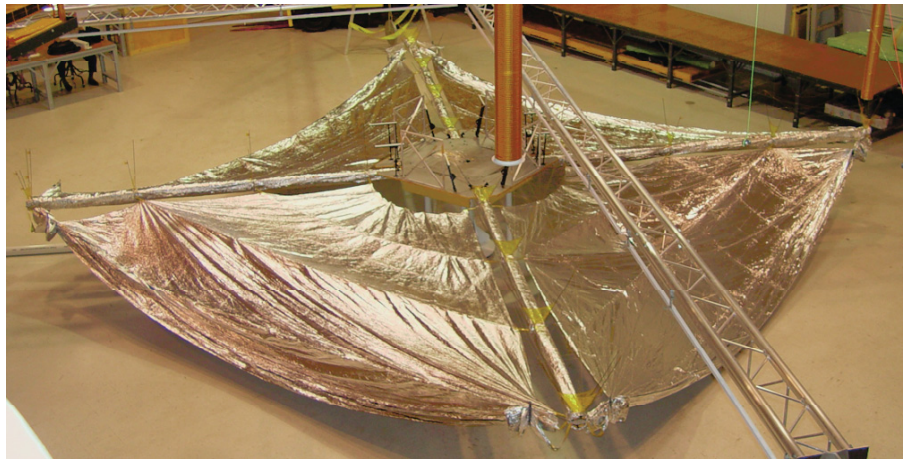

Figure 4. 10m ISP Deployment in Progress

The Sunjammer design relies on L'Garde's inflatable rigidizable technology in the sail structure. The booms are warmed before deployment to make them flexible enough to deploy. Once fully extended, the heaters are turned off and the booms allowed to cool below the glass-transition temperature of their composite matrix, under the protection of the thermal protection system. Now rigid, the booms will be intentionally vented to prevent gas-jet moments due to micrometeoroid strikes. The combination of this boom with its sunside spreader system makes an efficient scalable beam.

The sail area is built from 5µm Kapton. During previous efforts Kapton of this gauge was unavailable. However, the promise of projects like this has incited DuPont to invest in developing Kapton of sufficiently low thickness as to be credible for use in large solar sail projects. They are now marketing this film for other applications, a technology spinoff.

This design and deployment technique was matured by L'Garde through the execution of several programs including the NASA In-Space Propulsion (ISP) efforts. The major thrust of that program was scalability, both in deployment and the resulting structure. The "stripe-net" sail suspension (Greschik, Ref. 5) imparts the lowest loads possible to the beams, and together with the resulting periodic follower loading on the beam makes for a scalable structural architecture. A scaling of approximately four times was demonstrated in ISP, to the largest size that can fit in the largest vacuum chamber in the world, and Sunjammer will demonstrate another factor of four in space (Figure 5).

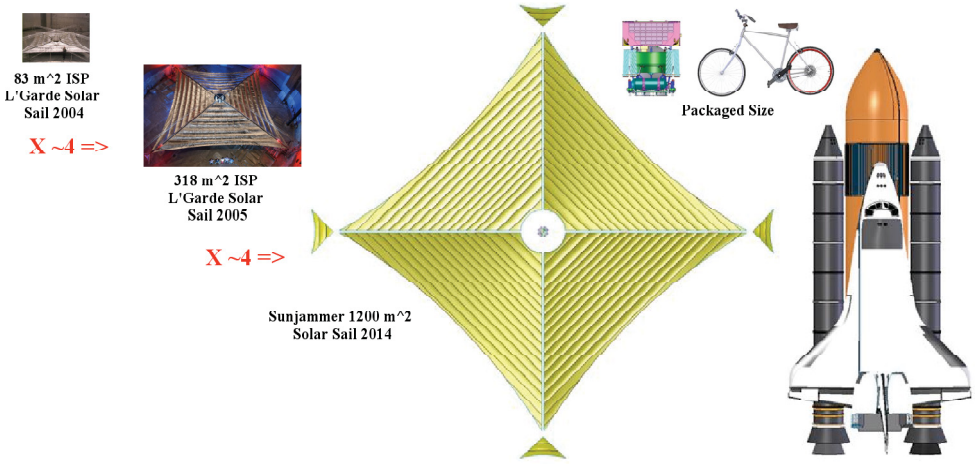

Figure 5. Scaling from ISP Deployments

The deployment will be viewed via cameras on a six-meter boom deployed sunward of the sail. There may be sponsorship logos painted on the sail surface. Carrier jettison will also be viewed, possibly from a camera on the fly-away carrier itself, which would give spectacular views of the deployed sail and Earth behind from a standoff distance.

Attitude Calibration

Sunjammer will be equipped with four vanes at the tip of the mainsail beams, each with two axes of control, "Twirl" and "Cant." With eight degrees of freedom, there are several possible ways to control attitude, but L'Garde will use a simple and safe method, and demonstrate basic backup modes.

The vanes will rely on the same technology as the sail area. The use of vanes makes the Sunjammer design truly propellantless. This will be an important feature to future mission planners who envision ultra-long solar sail flights.

All sails possess passive static stability about the two axes normal to the sunline. In addition to the familiar "drag" stabilization due to the center of mass (CM) being forward of the "center" of pressure (CP), the effectively shuttlecocked shape of the sail due to sail billow also provides stability. This can be greatly enhanced by canting the vanes antisunward, as illustrated in Figure 6.

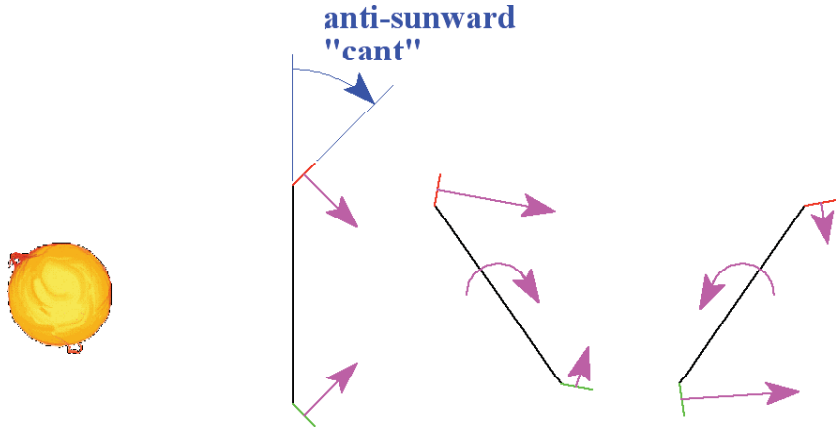

Figure 6. Restoring Moments due to Vane Cant or Mainsail Shuttlecock (Vanes Set to Trim the Sail at Sunline-Normal)

By changing the vane cant angles, the sail can be retrimmed to an attitude off-normal to the sunline, as shown in Figure 7. This is similar to retrimming aircraft pitch using elevator trim tabs.

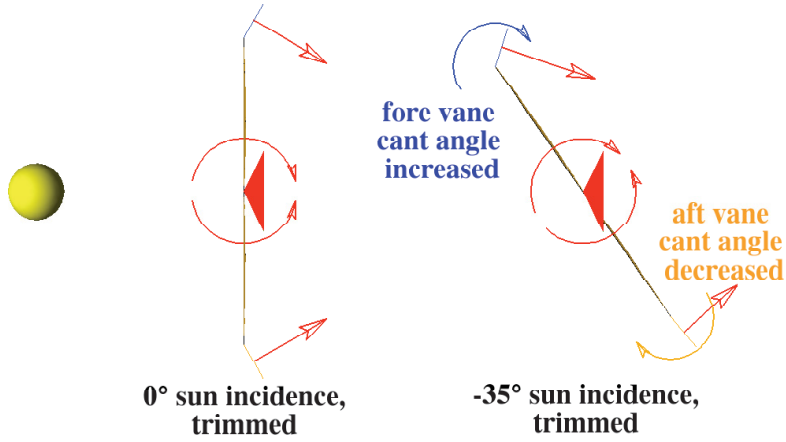

Figure 7. Retrim to Passively Stable 35° Sun Incidence Using Vane Cant Angles

There can be no passive stability about the sunline, so more spacecraft-like methods must be used to maneuver in “Top” and actively stabilize that angle. Two opposing vanes are twirled about the mainsail boom axes to produce a “windmill” torque, as shown in Figure 8.

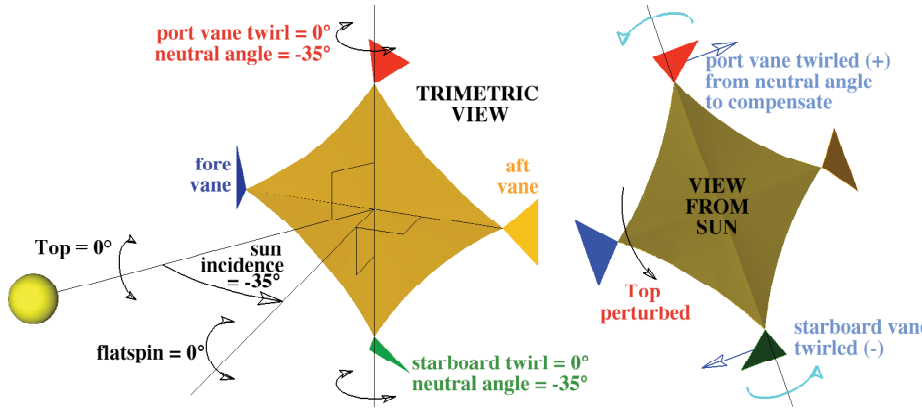

Figure 8. Differential Vane Twirl to Windmill the Sail about the Sunline

In order to perform these maneuvers, the restoring moments on the mainsail at the desired attitude must be known so that vane angles can be calculated to overcome them. The sail shape is gridded, each little element a flat plate to which is applied an elementary solar force model for its particular angle to the sun (Figure 9). The forces from all these elements are integrated to determine the total forces and moments on the mainsail.

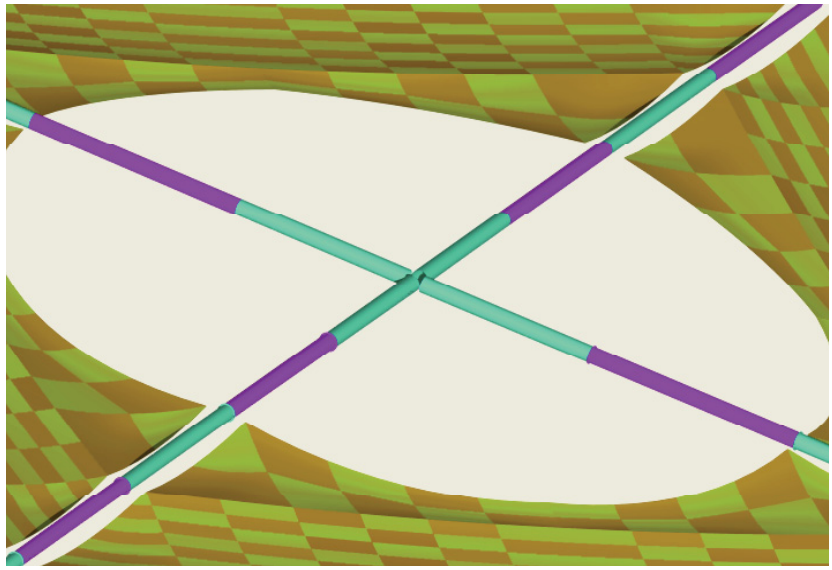

Figure 9. Sail Grid Consisting of Elemental Flat Plates

If the mainsail were simply modeled as a perfectly reflecting flat plate, there would be no restoring moments at all. If the "drag" force due to an imperfectly reflecting sail is introduced, the moment due to CM-CP offset would appear, but this may be minor compared to the effect of mainsail billow. The sail shape must therefore be captured, and as this shape is affected by bending of this gossamer structure, structural analysis must be included as well. One other significant shape effect to consider is "photoflexibility." Because the forces on an

elemental piece of sail material are a function of solar attitude, as the sail flexes in response to pressure, the forces change, so the flexing must be recalculated, iteratively.

L'Garde has calculated these moments, but recognizes that as this is a gossamer structure, there will be bias moments due to imperfect balancing and shimming, as well as residual deployment effects. Sunjammer's vanes are oversized to overcome both mainsail restoring moments and estimated bias moments, with margin. Post-deployment, the sail will be calibrated using onboard attitude determination to determine the real moments vs. attitude, providing vane trim tables for the practical onboard control of the sailcraft. This attitude calibration will demonstrate the attitude accuracy required by NASA.

Thrust Calibration

Thrust calibration will follow, using ground ranging measurements. While shape effects have a major impact on attitude control, they have far less effect on thrust and navigation. What is most important is surface quality – wrinkles. This effect is captured in the elemental solar force model used by L'Garde, which differs from the frequently seen optical model.

The optics model of solar force due to reflection is flawed for two reasons:

1. The optics model separates reflected light into specular and diffuse components, and adds the resulting solar forces as scalars. However, the definition of what is "specular" in optics is actually arbitrary and restrictive (+/- 6° or +/-10° of the specular reflection line), and the resulting forces from rays reflected all about the specular reflection line, all useful to propulsion, add as vectors, not scalars.

2. The diffuse reflectance is symmetric about the specular reflection line; not about the surface normal, as is assumed in optics. In fact, there is a little more scattered forward of the specular reflection line than backward. Visibly, wrinkling simply facets the material (Figure 10). It doesn't make it diffuse. This is evidenced by the fact that total reflectivity doesn't change much even for heavily wrinkled samples.

It is possible to use the optics model on a flat plate to do trajectory simulations by assuming a mirror-high value of specularity, but when actual data for wrinkled sail membranes under pragmatically low stress is used, the optics model predicts a poorly performing sail, while actually adding force vectors from the same data shows quite good performance.

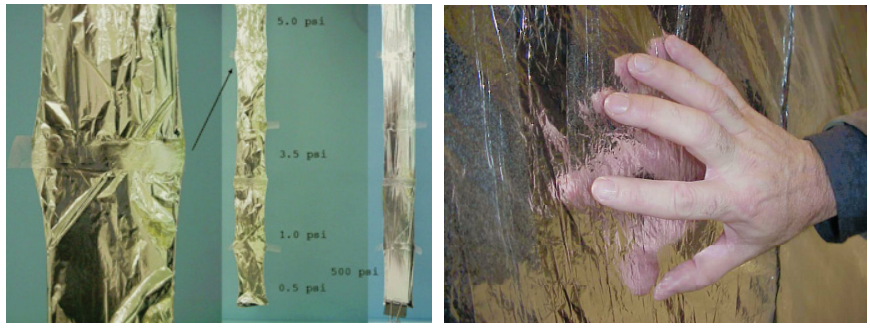

Figure 10. Wrinkles at Various Stress Levels

Seeking a better measure of the propulsive performance of wrinkled films, L'Garde in 2001 had material samples, some more wrinkled than others, tested for bi-directional reflectance (BDR) under zero stress. Forces from each individual ray reflected into the hemisphere above the sample were added as vectors. L'Garde developed a mathematical technique to boil the hundreds of data points at any solar incidence angle down to just two values, "Propulsive Reflectivity" (Rp) and "Propulsive Zenith" (Θp) (Figure 11). The resulting data indicates that despite the wrinkled appearance of the material, very good propulsive performance is attained.

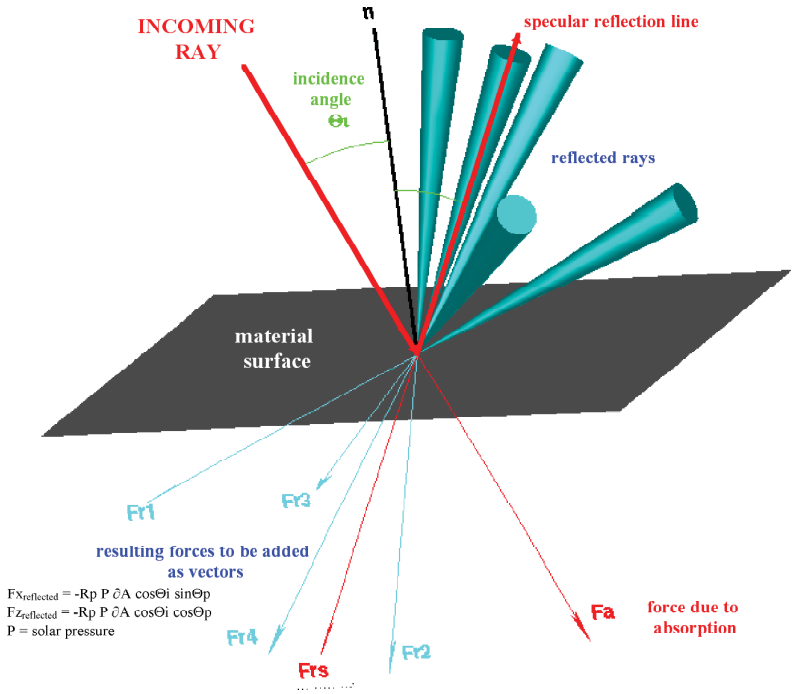

Figure 11. L'Garde's Reflected Solar Force Model is a Vector Summation of BDR Data

L'Garde theorizes that propulsive reflectivity Rp is fairly constant with incidence angle, and that propulsive zenith Θp, which must be zero at zero incidence, varies from actual incidence Θi only mildly with increasing incidence angle, due to forward scattering. We call on interested experimenters to take bi-directional reflectance data on sail films at other incidence angles.

Total forces are summed from the elemental flat plates in the gridded model of the shaped sail. Because there will be coating and wrinkle variations throughout the sail which cannot be fully evaluated, the sail thrust must be calibrated on-orbit. This will allow better navigation solutions, starting with the 30 day NASA-required navigation sequence immediately following calibration.

There is also a requirement to recalibrate the sail after six months. The intention is not to isolate the reason for any degradation, but rather to demonstrate that the sail can still be flown in the presence of degradation. This is true as well of the initial attitude and thrust

calibrations. There is not instrumentation to determine why the sail is different than predicted, nor is there a requirement to validate our predictive models, but calibration will allow us to better demonstrate attitude control and navigation despite these differences.

Flyout and Other Extended Operations

L'Garde intends to operate the spacecraft for 1 year. After the completion of this one year flight, the ultimate infusion goal is to hand over operation of the spacecraft to an interested party. The spacecraft can then be used directly by another organization to demonstrate its capabilities. NOAA is one infusion partner with L'Garde. Possible activities include extended stationkeeping, polesitting, and transfers.

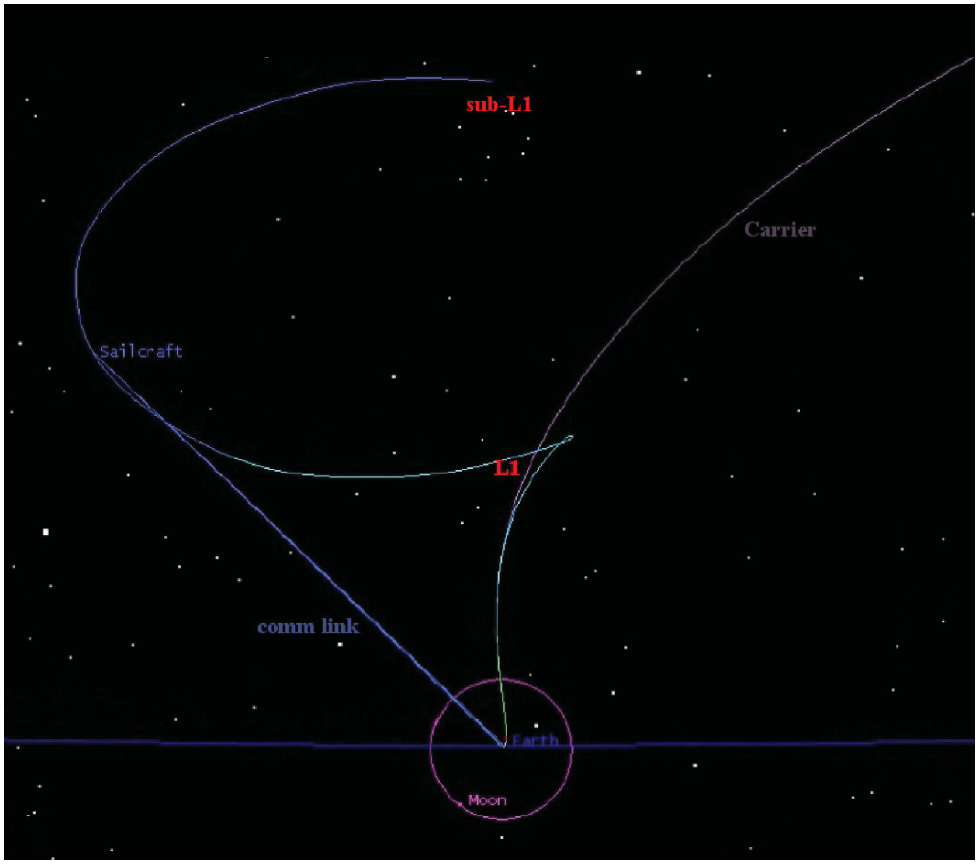

Figure 12. Flyout to Sub-L1

Conclusion

The L'Garde built Sunjammer, with its planned 2014 launch, will fully demonstrate the propellantless propulsion potential of solar sails. Upon completion of this Technology Demonstration Mission (TDM) flight, mission planners will have a flight-proven scalable solar sail design with which to build operations around. The technologies required for future

larger sail missions will be demonstrated in this single mission. This relatively inexpensive mission will greatly advance the solar sail technology that NASA has already invested in. It will provide NASA and the USA with a solar sailing technology and capability that is second to none.

NASA's Space Technology Mission Directorate is providing the necessary leadership and support to sail this flight-ready technology to the launch pad.

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Commercial Solar Sail Applications: Overview and Update on NASA's Sunjammer Mission

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Space Services Holdings, Inc. (SSHI) is the commercial infusion partner for the NASA Technology Demonstration Mission Sunjammer. In partnership with LGarde, Inc. and the U. S. National Oceanic and Atmospheric Administration (NOAA), SSHI will demonstrate the commercial market for solar sail missions, augment the capabilities of the Sunjammer mission, and produce innovative content about the mission for a global audience. This paper will explore commercial infusion activities associated with Sunjammer and outline the potential for follow on commercial solar sail missions enabled by the Sunjammer mission.

I. About Sunjammer

Sunjammer is a NASA-funded Technology Demonstration Mission that will launch, deploy, and navigate the largest solar sail to date. The Sunjammer team is led by L'Garde, Inc. and includes Micro Aerospace Solutions, as well as infusion partners the National Oceanic and Atmospheric Administration and SSHI. NASA oversight is provided by the Marshall Spaceflight Center.[1]

Sunjammer, named, with the estate's approval, to honor Sir Arthur C. Clarke, is scheduled to launch in Q4, 2014 from Cape Canaveral, FL as a secondary payload aboard a Falcon 9 rocket carrying the Deep Space Climate Observer (DSCOVR). Sunjammer will be deployed from Falcon near the Sun-Earth L1 point and begin the sail deployment procedure. Once the sail is fully deployed, Sunjammer will execute a series of maneuvers to demonstrate a full range of navigation functions.

The primary objective of the Sunjammer mission is demonstration of sail deployment and subsequent navigation capabilities. Successful completion of the Sunjammer mission should result in the removal of any

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final barriers to the adoption of sails by future space mission planners for use when they are deemed cost-effective.

Sunjammer has three secondary mission objectives, to be performed after the primary NASA mission objectives are completed.

First, Sunjammer will perform magnetometer calibration tests seeking to assess any potential interference issues for the instrument in the presence of a sail. By comparing in situ readings between Sunjammer and the Advanced Composition Explorer (ACE) spacecraft currently on operational service for NOAA at L1, mission personnel should be able to ascertain what, if any, effects the sail's presence may induce in magnetometer measurements. Magnetometer output is a vital component of operational space weather requirements, hence any interference issues uniquely raised by employing a sail for propulsion must be identified and understood before committing to sails for operational space weather missions.

Second, Sunjammer will sail from L1 to a location significantly (2 million kilometers) "upstream" to demonstrate the ability to cost-effectively deliver substantially earlier solar storm warnings from a sail-enabled orbit nearer to the Sun. If successful, this demonstration should be a "game-changer" as it defines the standard for future space systems and provides 4-6 hours of warning for solar storms destined for Earth.

Third, Sunjammer will carry the Sunjammer Cosmic Archive, including Celestis Memorial Spaceflight capsules containing a symbolic portion of the cremated remains of participants from many countries, as well as Celestis MindFiles and BioFiles from the global public. MindFiles and BioFiles allow people to include their photos, writings, music, and even their dna aboard the Archive, creating a new "virtual planet" orbiting the sun with representations of human history, culture, and societies aboard.

II. About SSHI

Space Services Holdings, Inc. is a Houston, TX commercial space company whose heritage reaches back more than thirty years. In 1982, then Space Services Inc. of America (SSIA) - under the direction of retired Mercury astronaut Donald K. "Deke" Slayton - conducted the first privately funded launch of a rocket and payload into space. Conestoga 1 was launched from the world's first privately owned, developed, licensed, and operated spaceport - Matagorda Island, Texas on the Gulf Coast. The legal and regulatory regime that today governs all private launches from the US was established for the Conestoga 1 launch, and subsequently embodied in legislation by the 1984 Commercial Space Launch Act.[2]

SSHI today incubates and operates commercial space businesses including Celestis, Inc. - the pioneer of Memorial Spaceflight Services; Sentinel Satellite, Inc - developer of commercial space weather services; and, Spaceport Launch Systems - a partnership with UP Aerospace, Inc.

SSHI's business model focuses on deploying market-driven space technologies to serve traditional and non-traditional customers - often from the same platform. The addition of Internet, social media, and sponsor activities to these missions further enhances revenue, broadens investor appeal, and reduces customer mission costs through virtual "ride share" techniques. Sunjammer is the largest platform to encompass the SSHI business model to date.

III. Precursor Private Solar Sail Efforts

Solar sail technology has been seen as an enabler of a host of privately funded missions since at least the late 1970s, with the formation of the World Space Foundation. Subsequent efforts include the proposed multi-nation race to Mars proposed to the Commission celebrating the 500th anniversary of Columbus' voyages to the New World,[3] a late 1990s effort titled "Forever Bound," that sought to send human dna on a journey beyond the solar system, and several attempts by the Planetary Society to fund and launch a solar sail.

In the late 1990s, members of the present SSHI management team were part of an effort – called Team Encounter – to develop a privately funded solar sail mission that would send a spacecraft beyond the solar system. While ultimately not successful, Team Encounter funded and led critical sail technology advancements, entering into a data purchase arrangement with NOAA and a Space Act Agreement with the NASA Langley Space Center focused on the mutual development of sail technology.[4]

IV. SSHI and Sunjammer

SSHI was selected as the exclusive commercial infusion partner for Sunjammer as part of the NASA competition that awarded the mission to L'Garde. As a condition of award, NASA required infusion partners to participate by making significant contributions to the mission, with the expectation that the infusion partners will benefit from, and subsequently utilize, sails for their own purposes.

SSHI has three roles in the Sunjammer Mission. First, SSHI and L'Garde will team to provide NOAA with further insights into the potential for a public-private partnership to meet NOAA's mid to long-term space weather data needs. The insights provided to NOAA will be developed as a direct result of Sunjammer mission activities, including financial performance results from SSHI's implementation of its business model around the Sunjammer mission.

Second, SSHI's subsidiary, Celestis, Inc., has agreed to purchase secondary capacity aboard the Sunjammer sail for Memorial Spaceflight capsules including cremated remains, MindFils, and BioFiles. The funds provided by Celestis will be utilized for mission assurance.

Third, SSHI will conduct an extensive Education and Public Outreach (EPO) activity – in conjunction with NASA and global partners – designed to maximize awareness of the Sunjammer mission and to create multiple learning opportunities for students interested in space and solar sailing. SSHI is conducting a two-year series of events and activities surrounding the Sunjammer mission.

V. Post Sunjammer Commercial Infusion Activities and Opportunities

Successful demonstration of solar sail technology via Sunjammer should enable several follow on commercial space mission opportunities. Three will be discussed in this paper.

A. Space Weather

Sunjammer should demonstrate that sails permit the significantly earlier warning times required by an increasingly wired planet. Locating space weather sensors "upstream" at a sub-L1 position – a mission made economical because of solar sails – permits improvements in solar storm warnings by as much as 4-6 hours, critical lead times when addressing electric power grid, airline, satellite, and other system responses to solar storms threatening Earth.

Over the course of several competitively awarded contracts from NOAA, SSHI has defined a sail based spacecraft that could provide enhanced solar storm warning services at a significantly reduced spacecraft mass (and hence significantly reduced cost). The SSHI business model, envisioning a data purchase by NOAA serving as the "anchor tenant," supplemented by other revenue streams from US Federal agencies and private companies, created an opportunity that achieves both a low price to the government and attractive returns to potential investors.[5]

NOAA is attempting to service an increasing number of space weather customers (from less than 3,000 in 2005 to nearly 27,000 in 2012), and is increasingly looking to partner with other nations and commercial providers to meet its evolving needs. Dr. Kathryn Sullivan, Assistant Secretary of Commerce for Environmental Observation & Prediction has labeled the Sunjammer mission a "Potential game-changer!!" in that regard.[6]

Improved quality of space weather data may also help to drive the development of a space weather "value added" industry akin to the value added industry spawned by commercial land remote sensing systems. A 2001 study of the European commercial space weather market noted that: "commercial opportunities are currently

limited although the availability of more and better data, development of better SW (space weather) forecast models and a conducive commercial environment could see this figure increase substantially."[7]

B. Telecommunications

In 2005, the National Science Foundation issued its annual Facility Plan which reviews and projects needs for all NSF facilities for the near, medium, and long term. In 2004, NSF provided funds to NOAA to task SSHI to assess the possibility of using a commercial solar sail platform to provide "modern, broadband digital communications" services to its South Pole Station facility. The South Pole facility is beyond the view of geosynchronous communications satellites, creating unique coverage issues for the facility.

NSF reviewed several alternative scenarios for meeting their requirements and concluded that for its long term requirements, a "strong possibility" was to "develop 'pole sitter' satellites positioned in solar (not Earth) orbit so as to provide continuous line-of-sight to South Pole Station. " The report envisions a co-development approach to share costs with "NOAA, the Department of Defense (DOD) and private developers." [8]

The same study by SSHI also addressed possible architectures for sail provided telecommunications services for lunar and deep space applications. Taken together, these potential applications make a strong business case for solar sail enabled telecommunications services.

C. Exploration/Entertainment Missions

A final near term class of commercial missions enabled by sail technology includes missions conducted primarily for exploration and entertainment. Historically, organizations such as the National Geographic Society have provided private funding for a host of Earth exploration missions with significant entertainment components.

Indeed, most of the early proposed solar sail missions were in this class, including, for example, the World Space Foundation's effort to visit Comet Halley and the Team Encounter "Humanity's First Starship" mission.[9]

Privately funded exploration and entertainment solar sail missions are appealing because of the highly visible and interesting nature of the sail; by the events and/or explorations enabled; and by the sponsorship, participation, and advertising opportunities created by such missions. SSHI is exploring and considering several candidate opportunities in this area to pursue upon successful completion of the Sunjammer mission.

VI. Conclusion

In March 2006, the publication *Business 2.0*, projected a global market for solar sail propulsion at $350 million by 2014.[10] With the projected launch of Sunjammer in Q4, 2014, the market for commercial applications of solar sail technology could exceed that projection in the near future. Commercial solar sail missions in the areas of space weather, telecommunications, and exploration/entertainment have significant potential, promising a follow-on wave of new applications and business opportunities.

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The Preliminary Design of the GOSSAMER-1 Solar Sail Membrane and Manufacturing Strategies

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The aim of the GOSSAMER-1 mission is an in-orbit deployment of versatile ultra-lightweight deployable sail hardware as a feasibility demonstration. The presented work focuses on the design and manufacturing of the sail membrane for GOSSAMER-1. The work has reached a preliminary design status which shall be presented in this paper. It includes mechanical analysis as well as a description of manufacturing strategies at the department of System Conditioning at the DLR Institute of Space Systems in Bremen.

Nomenclature

A_{cut} = cut area at the sail edges [m^2]

A_{eff} = effective area for solar pressure [m^2]

A_{sail} = sail area [m^2]

α = force angle [rad]

α_{s} = solar absorption coefficient

d_{Al} = thickness of Aluminum [m]

$d_{\mathrm{cu,CIGS}}$ = thickness of copper and CIGS photovoltaic [m]

d_{P} = thickness of Polyimid [m]

dl = differential length [m]

$d\varphi$ = differential angle [rad]

dN = differential normal force [N]

dR = differential friction force [N]

E = Young modulus [Pa]

E_{Al} = Aluminum Young modulus [Pa]

$E_{\mathrm{AO,kin}}$ = kinetic energy of atomic oxygen [eV]

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$E_{\mathrm{cu,CIGS}}$ = Young modulus of copper and CIGS [Pa]

E_{P} = Polyimid Young modulus [Pa]

E_{PV} = Young modulus of photovoltaic [Pa]

e = efficiency factor

e_{l} = efficiency factor describing the wrinkling effect of load introduction

e_{s} = efficiency factor describing the wrinkling effect due to storage

ε_{h} = angle between hypotenuses sail edge and force line [rad]

ε_{k} = angle between cathetus sail edge and force line [rad]

ϵ_{Al} = Aluminum strain

ϵ_{P} = Polyimide strain

ϵ_{IR} = infrared emissivity coefficient

ϵ = strain

F = force [N]

F_0 = force introduced to the rigging (equal to F_{k} or F_{h}) [N]

$F_{100\%}$ = solar pressure force (perfect reflective, no wrinkling) [N]

F_{Al} = force acting on Aluminum [N]

F_{h} = force acting along the hypotenuses [N]

F_{k} = force acting along the cathetus [N]

F_{P} = force acting on Polyimid [N]

$F_{\mathrm{max}}^{232°\mathrm{C}}$ = maximum force carried at 232°C [N]

F_{red} = solar pressure force reduced due to wrinkling [N]

f = view factor

k_{B} = Stefan-Bolzmann constant [$\mathrm{Wm^{-2}K^{-4}}$]

k_{n} = unit vector perpendicular to the sail surface

k_{s} = unit vector directed to the sun

$m_{\mathrm{Atom}}^{\mathrm{AO}}$ = atom mass of Aluminum [kg]

N = normal force [N]

n^{φ} = force distribution per angle [N/rad]

φ_{h} = angle of the hypotenuses cut out [rad]

φ_{k} = angle of the cathetus cut out [rad]

Q_E = Earth energy flux [Wm^{-2}]

R = friction force [N]

S_0 = solar constant [Wm^{-2}]

σ = stress [Pa]

σ_{Al} = stress in Aluminum [Pa]

$\sigma_{Al0.2}$ = yield stress of Aluminum [Pa]

σ_P = stress in Polyimid [Pa]

T_{max} = highest temperature [°C]

T_{min} = lowest temperature [°C]

$\tau_{max,a}$ = maximum shear stress of adhesive [Pa]

v = velocity [m/s]

ν = poison ratio

ν_P = Polyimid poison ratio

ν_{Al} = Aluminum poison ratio

I. Introduction

GOSSAMER-1 is part of an intended three steps solar sail technology development described in [1]. Within this project, scalable solar sail technologies including sails, booms, photovoltaic and mechanisms will be developed. Scalable means that GOSSAMER-1 is a technology demonstrator that will deploy a 5m x 5m sail using technology that is suited to build GOSSAMER-2 with a 25m x 25m sail and GOSSAMER-3 with a 50m x 50m sail. The mission objective is the demonstration of a successful and reliable deployment, not yet its use as a solar sail. Therefore neither a realistic sail assembly load nor navigation of a solar sail S/C are in the scope of this demonstrator mission. Correspondingly, it is sufficient to demonstrate the deployment in Low Earth Orbit (LEO), and it is not required to launch into altitudes where solar photonic pressure indeed dominates over atmospheric drag effects.

The GOSSAMER-1 is based on a crossed boom configuration with four sail segments as shown in Fig. 1. Four boom and sail deployment units (BSDUs) will deploy the booms and sail segments. During the deployment process, the BSDUs will move away from a central bus unit by uncoiling the booms stored in these mechanisms while at the same time deploying the sail segments between the booms.

Fig. 1: The GOSSAMER-1 System, four sail segments deployed in a crossed boom configuration

The presented work focusses on the development of the solar sail membrane, including the load introduction and integration of thin film photovoltaic, from a mechanical point of view. First the storage and deployment strategy will be described, followed by a specification of the materials used and a mechanical analysis of the load introduction and the tensioned state of the sail. The load introduction for the sail tensioning will be accomplished using ropes – the rigging of the sailcraft. Specifically, the edges of the sail are built with a curvature, and they are folded over and glued to create a pocket in which the rigging can move. This allows a more homogeneous tensioning of the sail membrane compared to a load introduction only at the corners of the sail. Each sail segment is frog-leg folded and coiled. The fourth and fifth fold, counting from the inner mounting point, is holding the thin film photovoltaic. The overall geometry is shown in Fig. 2. In the deployed state, the sail will be tensioned by the rigging with a maximum force of 2N introduced to the rigging. The maximum force is chosen according to the boom specifications.

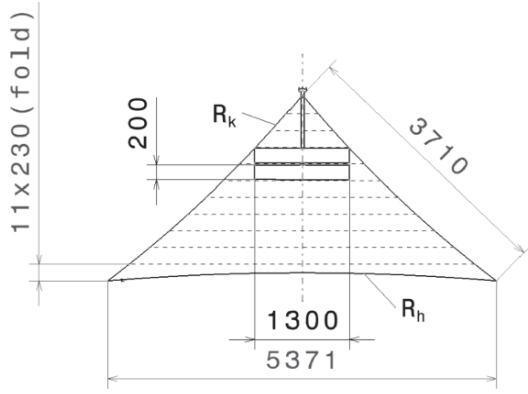

Fig. 2: Sail Geometry

II. Sail Storage

The folding strategy is illustrated in Fig. 3. Each sail segment is frog-leg folded ten times. The folded segment is then coiled on two spools from the outer corners to the middle of the sail. It is important to fold the sail such that the two solar cell areas are laid on one another as this is an electrical requirement to ensure electrical insulation of the photovoltaic.

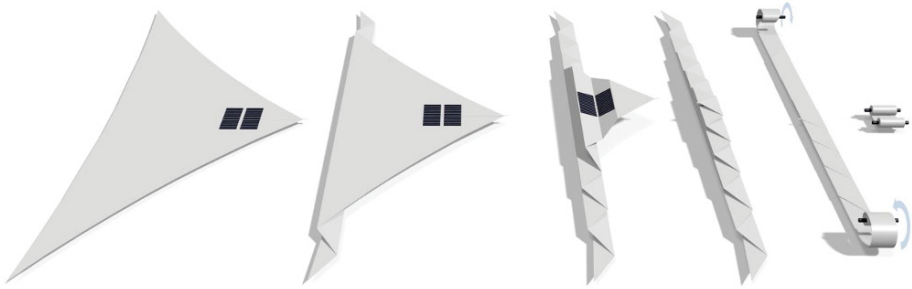

Fig. 3: Sail Storage: Frog-leg folding of each sail segment and coiling on two spools

With respect to the two coiling directions, two possible configurations are possible as shown in Fig. 4. Coiling the sail such that the opening of the folds is directed to the outside (Fig. 4, right) has the advantage that the configuration is very compact.

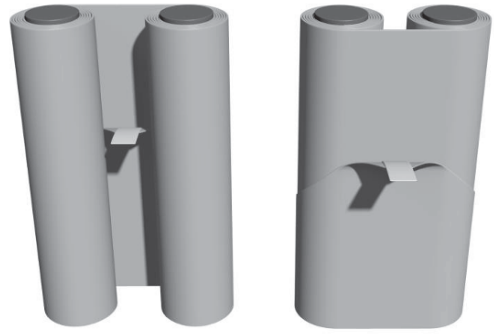

Fig. 4: Influence of the coiling directions on the sail package, option one on the right and option two on the left

The advantage of the other configuration (Fig. 4, left) is that it reduces the bending moments on the booms. This is because the load transfer is along the cathetus, meaning that the boom loading is mostly along the boom axis and bending moments are small. Considering the given sail attachments point a load reduction along the

hypotenuses can be reached if the uncoiling point is parallel shifted to the boom axis. During the deployment the sail spool with its uncoiling point is moving along a certain path according to Fig. 5.

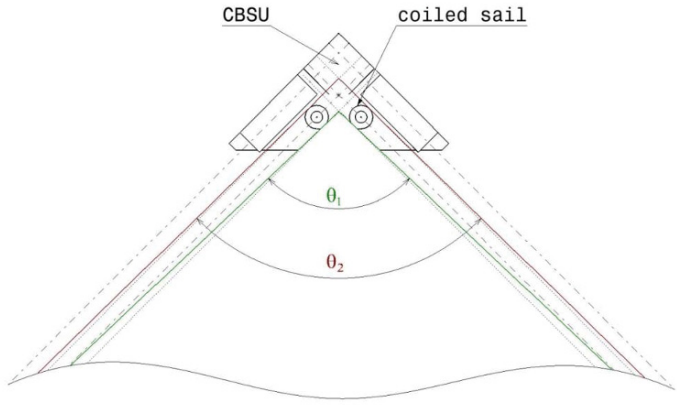

Fig. 5: Opening angles of the sail according to the two coiling options

Because the booms are arranged perpendicular to one another, the paths of the uncoiling points are perpendicular. After the deployment, the stretched sail is attached to the booms at a certain mounting point. Due to the parallel shift of the uncoiling path and the path of the fixation point, the stretched sail needs to have an opening angle that is bigger than 90° although the opening angle during the deployment is 90°. This geometry exactly reaches the goal of not transferring loads through the hypotenuses, as it is not stretched during the deployment. While for the first coiling option the angle Θ_1 is 90.92° (which is only slightly bigger than 90°), the angle Θ_2 for the second coiling options is 92.75°.

In order to ensure a proper load introduction to the booms the second coiling option was chosen for GOSSAMER-1. Using this coiling option with the opening of the folds directed to the inside of the coiled package also seems to be a little more stable with respect to ventilation and launch loads.

III. Material Selection

The material selection is mostly driven by temperature limits and Atomic Oxygen (ATOX) resistance. While the latter would be of no concern for a solar sail flying in interplanetary space, the GOSSAMER-1 technology demonstrator needs to withstand the low earth orbit environment. The highest temperatures are calculated as equilibrium temperature of the sail in case one side faces the Sun while the other side is directed to the cold space. The cold case is calculated when the sail is in the Earth's shadow with a medium view factor of 0.56. In

Fig. 6 the ATOX flux in LEO is given. In an orbit with 300km height the flux is about $15\mathrm{cm}^{-2}\mathrm{s}^{-1}$. This would degrade plastic materials in a very fast rate.

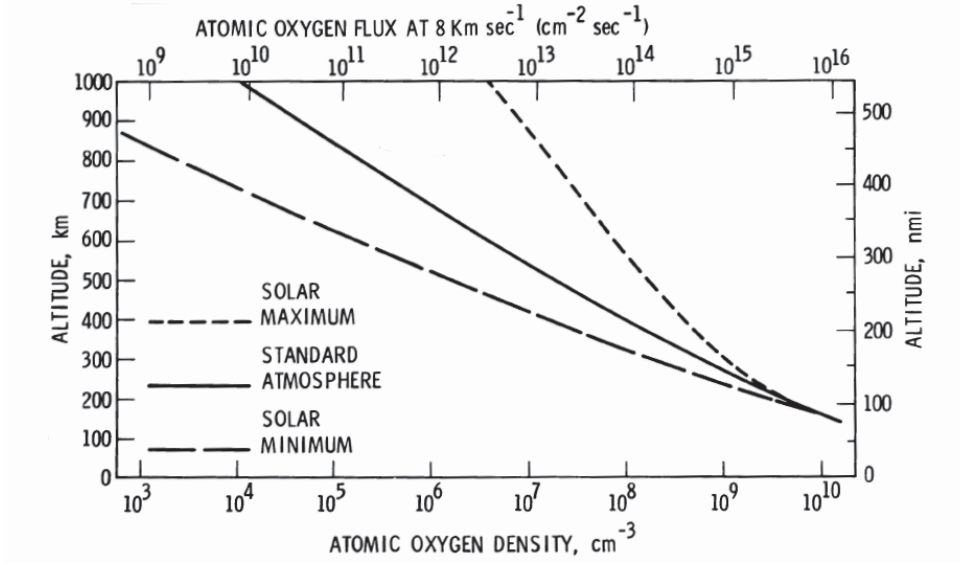

Fig. 6: ATOX Flux in low earth orbits [2]

The ATOX in low earth orbit has a low thermal energy of approximately 0.1eV. When the satellite is traveling through such medium with the speed of roughly $8\,\mathrm{kms}^{-1}$ the impact energy of a single oxygen atom with a mass of 15.999u ($2.66 \cdot 10^{-23}\mathrm{kg}$) and the satellite is approx.

$$\begin{aligned} E_{\mathrm{AO,kin}} &= \tfrac{1}{2} \cdot m_{Atom}^{AO} \cdot v^2 \\ &= \tfrac{1}{2} \cdot 2.66 \cdot 10^{-26}\mathrm{kg} \cdot 8000\ \mathrm{ms}^{-1} \\ &= 8.5 \cdot 10^{-19}\mathrm{J} \\ &= 5.31\mathrm{eV} \end{aligned} \tag{1}$$

The total energy (thermal and kinetic) of the ATOX is hereby about 5.4eV. To remove atoms from a surface the binding energy of the atoms and the displacement energy must be overcome. For metals this energy is much higher than 5.31eV (e.g. 28.36eV for Aluminum [6]). Csequently, it can be assumed that the metal materials and coatings withstand the ATOX.

A. Polyimid Film

The GOSSAMER-1 sail will be made from polyimide foil. The foil is 7µm thick and both sides are coated with 100nm vacuum deposited aluminium (VDA). The coated aluminium has an effect on the mechanical material parameters. Therefore the material will be treated as a composite material. It is assumed that the

material properties are the same for both directions in the plane of the foil and that the poison ratio and strain for each material is equal

$$\epsilon = \epsilon_P = \epsilon_{Al} \tag{2}$$

$$\nu = \nu_P = \nu_{Al} \tag{3}$$

The force acting in the material is spitted accordingly:

$$F = F_P + F_{Al} = \sigma_P d_P dl + \sigma_{Al} d_{Al} dl \tag{4}$$

Including the material law

$$\sigma = E\epsilon \tag{5}$$

and combining it with the force and stress expressions this leads finally to:

$$\frac{E_P \epsilon d_P + E_{Al} \epsilon d_{Al}}{d_P + d_{Al}} = E\epsilon \tag{6}$$

$$=> E = \frac{E_P d_P + E_{Al} d_{Al}}{d_P + d_{Al}}$$

Table 1 provides the material data for Polyimide and Aluminum and the composite foil. For polyimide the material data from UBEs Upilex S foil is used. The data is used for the calculations given in section IV.

Table 1: Material Properties

Material	**Young's Modulus [GPa]**		**Poison Ratio**	**Material Thickness [µm]**
	250°C	**25°C**		
Polyimide Upilex®-S	5.886	8.829	0.34	7.5
Aluminum	70	70	0.34	$2 \cdot 0.1 = 0.2$
Composite foil	7.551	9.893	0.34	7.7

As material stress limit for Polyimide the yield point at 3% strain is used. This correlates with yield point of Aluminium. The values decrease with the temperature, for 200°C it is:

$$\sigma_{Al0.2} \approx \sigma_{P0.3}^{200°C} = 41\text{MPa} \tag{7}$$

The connected thin film solar cells are a composite of many different materials. The 4µm thick solar cell material consisting of Copper, Indium, Gallium, Sulfur, and Seleniu (CIGS) is built up on 25µm Polyimide foil and they are mounted and electrical contacted on a 25µm Polyimide foil with a 3µm copper. The Young Modulus of this composite is estimated with:

$$E_{PV} = \frac{E_P d_P + E_{Cu,Cigs} d_{Cu,Cigs}}{d_P + d_{cu,cigs}} = \frac{50\mu m \cdot 8.829\text{GPa} + (3\mu m + 4\mu m) \cdot 70\text{GPa}}{50\mu m + 3\mu m + 4\mu m} = 16.34\text{GPa} \tag{8}$$

Because the Polyimide foil could be used in a wide temperature range up to several hundred degrees the foil itself is not considered to be critical with respect to the thermal requirements. On the other hand the thermal properties of the foil are important because other components, e.g. adhesives, also need to withstand the temperatures arising in the Polyimide foil. The Aluminum coated Upilex-S foil is specified with a solar absorbance coefficient α_s of 0.12 and an infrared emission coefficient ϵ_{IR} of 0.03. The equilibrium temperature in the hot case when the sail is directed to the Sun is calculated by

$$\mathrm{T_{max}} = \sqrt[4]{\frac{\alpha_S \; S_0}{\epsilon_{IR} \; k_B} \cdot \frac{1}{2}} = 468°\mathrm{K} = +195{,}44°\mathrm{C} \tag{9}$$

In the equation a solar constant of 1367 W/m^2 was considered and k_B is the Stefan–Boltzmann constant ($5.670400 \cdot 10^{-8} \frac{\mathrm{W}}{\mathrm{m^2 K^4}}$). For the cold case in the Earth's shadow a view factor f of 0.56 was considered which means the sail is slightly directed to the Earth. The energy flux emitted from the Earth Q_E is 250 W/m^2. With these assumptions the minimum temperature is

$$\mathrm{T_{min}} = \sqrt[4]{\frac{\mathrm{Q_E \; f}}{\mathrm{k_B}}} = 222.91°\mathrm{K} = -50.24°\mathrm{C} \tag{10}$$

Here it should be mentioned that especially with respect to the thin film solar cells it is considered to locally use foils with a different coating like Germanium, Indium Tin Oxide and Silicon Oxide to enhance the thermal properties. For a solar sail with a proper attitude control system that flies in interplanetary space it is worth considering different coatings for the sail sides. The side directed to the Sun should be optimized with respect to a high specular reflectivity. The back side instead should be high-emitting in the infrared spectrum while the reflective properties are of no concern. In case of the GOSSAMER-1 demonstrator the attitude with respect to the Sun is more uncertain. Consequently both sides need to have the same coating so that it does not matter which side points to the sun. Considering a Silicon Oxide coating as offered by the company Sheldahl an $\alpha_s/\epsilon_{\mathrm{IR}}$ ratio of 1 seems possible. This would lead to a minimum temperature of approximately -50°C and a maximum temperature of approximately 58°C. A Silicon Oxide coating is transparent. Anyway its influence on the solar pressure force is not yet known. In order to demonstrate the deployment technology with GOSSAMER-1 it is not necessary to optimize the coatings with respect to a high specular reflectivity for solar sailing.

B. Transfer adhesive

To reinforce the edges of the sail and for creating the rigging pockets as well as integrating the solar cells on the foil a transfer tape 966 of the company 3M™ with adhesive 100 will be used. According to the datasheet the adhesive can be used in a temperature range between -40°C to 232°C. As these are already limits in which the

adhesive can be used it is recommended to perform additional tests for the specific use case. For low temperatures down to -142°C this was made in the precursor project Geosail [7]. The hot case was only tested up to 98°C. In the proceeding of the GOSSAMER-1 project a test with temperatures up to 200°C will be made. In case the adhesive cannot be used in this temperature range, it is still an option to use additional coatings. The allowed static shear decreases at high temperature. For the sail design the value at 232°C is used. The data sheet provides the maximum shear force for an 1x1 inch test sample that carried the load for 10,000 minutes.

$$F_{\max}^{232°C} = 3.9\text{N} \tag{11}$$

$$\tau_{\max,a} = \frac{F_{\max}}{2.54\times10^{-2}\ \text{m}\cdot 2.54\times10^{-2}\ \text{m}} = 6{,}045\,\frac{\text{N}}{\text{m}^2} \tag{12}$$

C. Rigging

For the rigging the following materials were considered

- Dyneema®,
- Polyimide fibres,
- Glass fibres,
- Steel ropes

The materials are shown in Fig. 7. At some point when considering degradations due to ATOX or UV-radiation, especially during long term missions, all plastic materials are affected more than metals. One could consider a coating with Aluminum like it is used for the sails. First attempts with a Polyimide fibre coating did not lead to adequate results, for example the coating was peeling off. The Glass fibres are too brittle to be used as rigging.

Fig. 7: Investigated materials for the rigging: Dyneema®, polyimide, glass fiber, stainless steel rope (200x magnified)

Very flexible steel ropes made of several filaments which themselves are made of twisted wires are state of the art material. They are available in various configurations with diameters from 0.1mm to the size of bridge ropes. Fig. 7 (right) shows a 0.4mm rope. A rope with a diameter of 0.1mm can withstand loads up to 10N and steel could be used in wide temperature field, from far below cero degree up to several 100°C. As these values all

exceed the sails requirements it seems to be the perfect candidate. Different thermal expansion is not a problem as the rigging can move freely in the edge pockets.

IV. Membrane Design

As previously described, the membrane design for gossamer one will use a rigging that is guided along the edges of the sail. As shown in Fig. 8 the edge pocket with these ropes is building a reinforced frame around the sail segments.

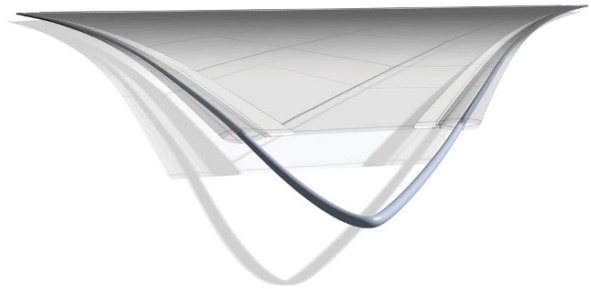

Fig. 8: Rigging guided in pockets along the sail edges

Furthermore each sail segment is made of three sheets with a width of one meter (see section IV). If micrometeoroids or space debris hit the sail it is likely that it is punctured only locally. In any case a rip would be stopped by the frame or the bonding between the sheets. Furthermore the rigging can be used to enhance the load introduction

A. Load introduction

First the forces transmitted into the rigging needs to be calculated. The geometry is given in Fig. 9.

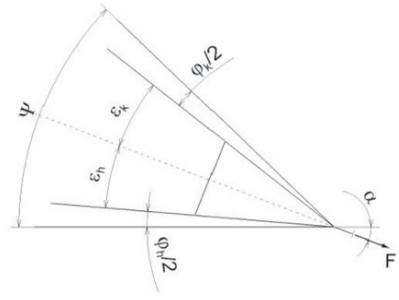

Fig. 9: Load introduction geometry at sail edge with rigging

Following equations describes the equilibrium:

$$F = F_k \cdot \cos(\varepsilon_k) + F_h \cdot \cos(\varepsilon) \tag{13}$$

$$F_k \cdot \sin(\varepsilon_k) + F_h \cdot \sin(\varepsilon_h) = 0 \tag{14}$$

The load introduction angle is

$$\alpha = \frac{\varphi_h}{2} + \varepsilon_h \tag{15}$$

In order to introduce tensioning forces, it should apply:

$$\frac{\varphi_h}{2} < \alpha < (\Psi - \frac{\varphi_k}{2}) \tag{16}$$

Furthermore the Capstan equation is employed to describe the load transmission through the rigging in means of belt friction according to Fig. 10:

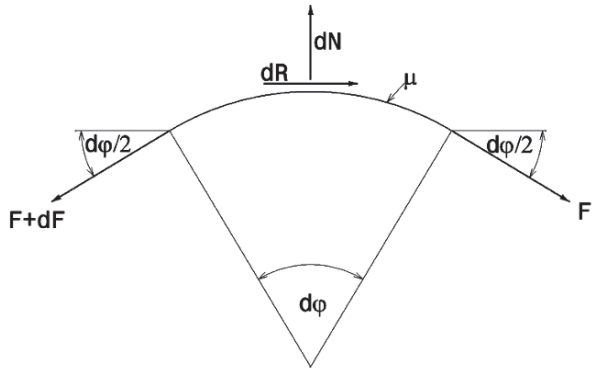

Fig. 10: Rigging loads according to the Capstan equation

$$dN = 2 \cdot F \cdot \sin\left(\frac{d\varphi}{2}\right) \tag{17}$$

$$= F d\varphi$$

Thus, the load distribution is given by

$$n^{\varphi} = \frac{dN}{d\varphi} = F = F_0 \cdot e^{-\mu\varphi} \tag{18}$$

The friction load is given by

$$dR = \mu dN \tag{19}$$

$$\Rightarrow \quad R = \mu F_0 \cdot e^{-\mu\varphi}$$

Here F_0 is the Force (F_h or F_k) introduced to the specific rigging at the introduction point. Fig. 11 gives an impression of the force distribution along the edges of a sail segment. It was taken into consideration that the rigging and the foil are bonded at the central mounting point. Due to that reason the normal force and the friction force at the catheti are decreasing from the introduction point at the outer corners to central mounting point. Along the hypotenuses the force decreases from the outer edges to the middle of the hypotenuses. The minimum is reached at the middle because, in contrast to the load introduction at the catheti, the force is introduced at both outer edges

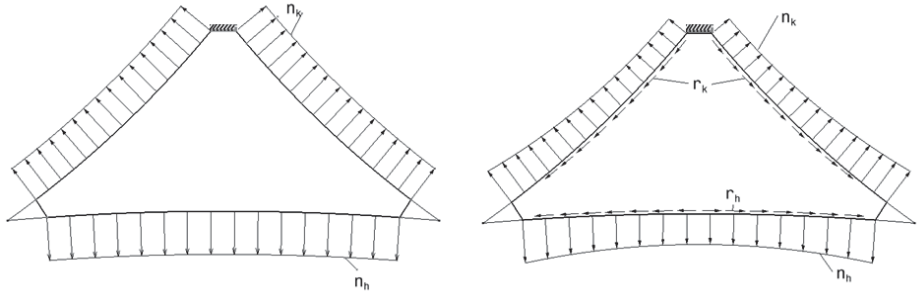

Fig. 11: Load distribution on the edges of a sail segment without friction on the left and considering friction on the right

B. FEM Model

In order to analyse the effect of changes of curvature respectively the radii R_h and R_k of the sail segments and the load introduction angle α a membrane model was implemented in Ansys. According to the load introduction described in the last paragraph the main influence of those factors on the sail stresses are given in Table 2. In first measurements it was found that the friction coefficient μ is approximately 0.5.

Table 2: Geometry influence on the sail stresses

	σ_x	σ_y
R_h ↑	-	↓
R_k ↑	↓	-
α ↑	↓	↑

The sail segment geometry was meshed with triangular Shell181 elements using the Membrane Only option. The forces are applied to the membrane model using force tables that provide the force components in x and y

direction for every node at the sail corners. In an automated script that finally provides an Ansys input file the following steps are performed:

- Normal force and friction force for each edge is calculated in a polar coordinate system with its origin in the centre of the circle of the edges radii (as described in the previous section).
- The forces are transferred in a global ANSYS Cartesian coordinate system splitting them into x and y components.
- The forces are discretised with respect to the number of nodes at the sail edges and written in force tables (one table per edge and axis).

First parameters were identified that are suitable to evaluate how well the sail membrane is stretched. The sail should be tensioned such that wrinkling due to the force introduction is minimized and wrinkling due to the folding process is reduced.

Wrinkling due to the load introduction can be roughly analysed by calculating the smallest principle stress. Such analysis is also presented in [3]. While the presented model does not cover a detailed wrinkling analysis it can be used to identify areas in which wrinkling likely occurs by identifying areas in which the second principle stress is zero (or negative). The geometry for the load introduction with a rigging is chosen such that those areas are not present.

The sail also contains wrinkles due to the folding process, that also reduce the solar pressure force by a certain amount. These wrinkles can be reduced by increasing the tensioning stress parallel to the symmetry axis of the sail (y-direction). In [4] a minimum stress of 1.5 psi (10300Pa) is specified to sufficiently reduce the packaging wrinkling. The load introduction with the rigging is designed such that this minimum stress is reached.

Fig. 12 shows the smallest principle stress in a sail segment that is tensioned by only introducing the load in the corners of the segment. The red areas marking areas in which the model produces compressive loads, these areas tend to wrinkle.

Fig. 13 to Fig. 16 show a comparison of the load introduction strategies. On the left the stress distribution according to the load introduction at the corners of the sail is presented while the right hand side illustrates the stress distribution when employing the rigging. With a force angle α of 25.8°, a hypotenuses radius R_h of 28m and a cathetus radius R_k of 30m the model produce a minimum stress in the y-direction in the order of the mentioned 10300Pa while no wrinkling areas appear. The figures also show that the sail is tensioned more uniformly when using the rigging. The deformation is shown highly exaggerated.

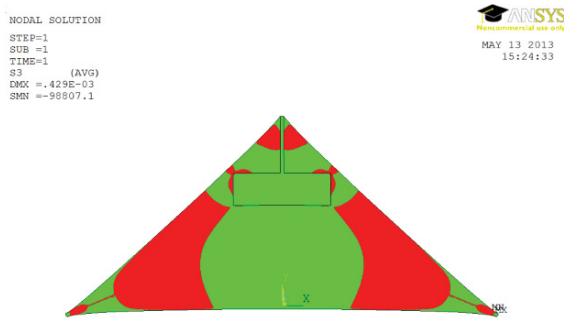

Fig. 12: Sail only tensioned in the corners ($\alpha = 21.81°$): red marked areas have a smallest principle stress equal or smaller as zero and therefore tend to wrinkle

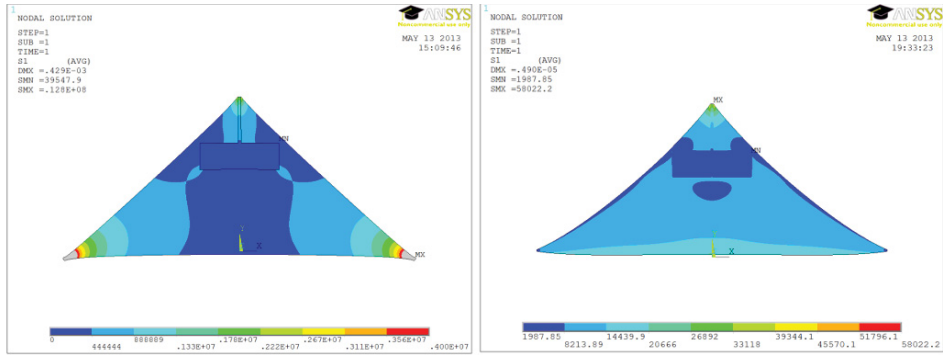

Fig. 13: First principle stress, load introduction in the sail's corners on the left and load introduction with a rigging on the right, stress in Nm^{-2}

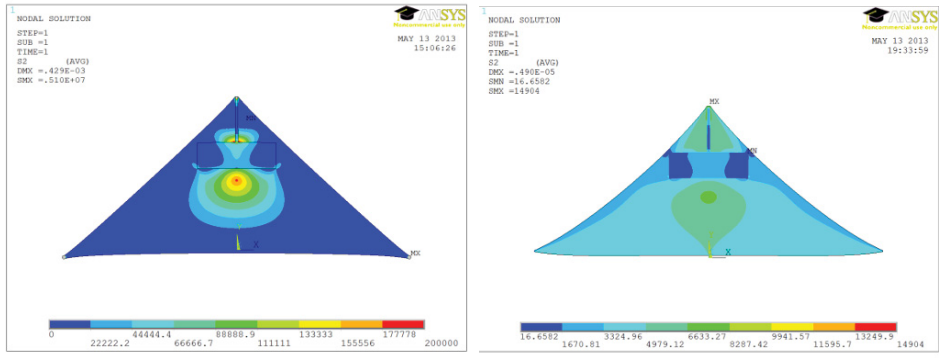

Fig. 14: Second principle stress, load introduction in the sail's corners on the left and load introduction with a rigging on the right, stress in Nm^{-2}

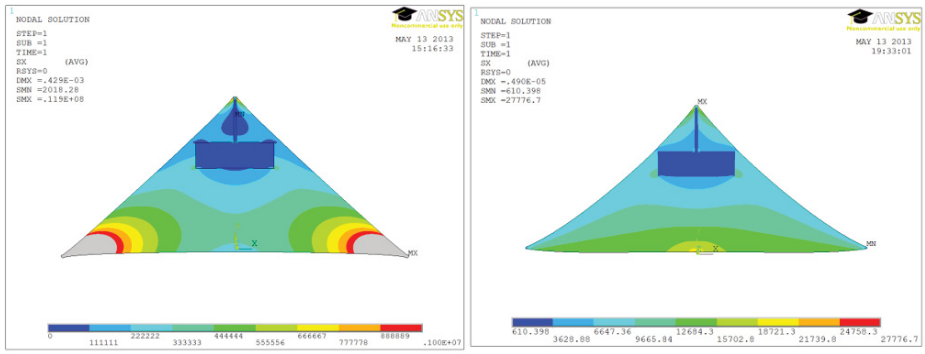

Fig. 15: Stress parallel to the hypotenuses (x-direction), load introduction in the sail's corners on the left and load introduction with a rigging on the right, stress in Nm^{-2}

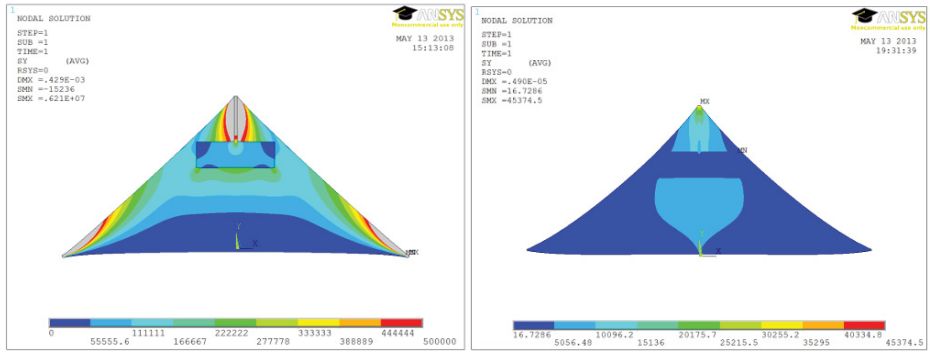

Fig. 16: Stress parallel to the symmetry axis (y-direction), load introduction in the sail's corners on the left and load introduction with a rigging on the right, stress in Nm^{-2}

With respect to the structural integrity the loads in the sail are low compared to the maximum loads described in section III. For the presented ANSYS model the following preliminary results for the Eigenfrequency were found:

- 1^{st} : 0.15 Hz
- 2^{nd}: 0.38 Hz
- 3^{rd}: 0.43Hz

It is worth mention that the better tensioning was achieved by the expense of sail area. While the red marked areas in Fig. 12 only reduce the solar pressure force about a certain percentage, the cut outs on the sails edges are completely lost. In order to optimize the sail the influence of the wrinkling on the solar pressure force must be evaluated. Considering efficiency coefficients for the effect of the wrinkling on the solar pressure force one could formulate an optimization function. With the efficiency coefficient for the wrinkling due to load

introduction e_l and the coefficient for the wrinkling due to the storage of the sail e_s the effective sail area is described by

$$A_{eff} = e_l \cdot e_s \cdot A_{sail} - A_{cut} \tag{20}$$

For the load introduction in the sail corners A_{cut} is zero. Consequently the curved sail edges make only sense if the effective sail area increases at least by this amount.

At this point the wrinkling is neither measured nor simulated. It is intended to measure the complete sail surface with a laser measurement device. These measurements are currently prepared. With the geometry of the wrinkled surface of the sail it should be possible to integrate the solar pressure force over the surface. As an reference case a perfect reflecting sail that is directed to the sun is analysed (also this might not be the case during a solar sail mission). For a flat sail the pressure force is:

$$F_{100\%} = 2 \cdot \frac{W}{c} \cdot A \tag{21}$$

Taking into account a wrinkled surface (multiple reflections are neglected) the pressure force is:

$$F_{red} = \int 2 \cdot \frac{W}{c} \cdot \left(\vec{k}_s \cdot \vec{k}_A\right)^2 dA \tag{22}$$

The efficiency factor for such a reference case is thus given by:

$$e = \frac{F_{100\%}}{F_{red}} \tag{23}$$

These calculations are left to future work and the sail geometry will be optimized accordingly in the course of the GOSSAMER-1 development. When considering the various optical parameters and attitudes of the sail the description would be more complex. In [5] such analysis is presented. Wrinkling could have of course also other negative effects. For example the attitude of the sail could be influenced or hot spots in which the solar radiation is focused could destroy the foil. These aspects must also be taken into account.

V. Sail Membrane Manufacturing

The sail membranes are manufactured in facilities of the DLR in Bremen, Germany. A vacuum table allows a precise positioning and fixation of the foil during the manufacturing. The manufacturing is shown in Fig. 17 and Fig. 18.

First three foil sheets with a width of one meter are adhesively bonded. This provides a sufficient size to cut out the desired area. Furthermore, building the sail segments out of three foil sheets provides more robustness in case the sail is hit by space debris or micro meteorites because rips are stopped at the bonding the sheets. A pattern printed in ratio 1:1 is used to cut out the desired sail geometry. The curved rigging pocket at the edges is built by dividing the curvature into several straight segments. At the end of the cuts for these segments are

reinforcements made of adhesive and foil and a circular stamp avoid tearing of the cut. After cutting out the segments the transfer adhesive is applied to the sail edges. Finally the pocked is formed around the rigging.

Thin thin-film solar cells will be integrated on the sail. On the test sail the solar cells were replaced by thin flex PCB dummies which later would be used for the electrical contacting of the solar cells. The flex PCB as well as the solar cells are adhesively bonded to the sail.

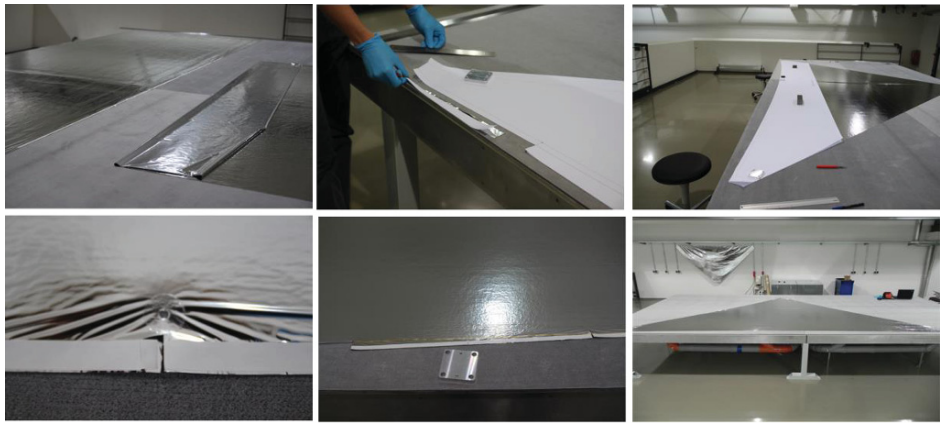

Fig. 17: Sail manufacturing, bonding film stripes as bases from which the sail geometry is cut out using cutting patterns, cuts are reinforced and the edge pockets are formed around the rigging

Fig. 18: Thin film solar cell (Flex PCB dummy) integration, folding and testing of sail segments

Using the materials described in section III to build a sail according to the design given in section IV a sail was built with the mass given in Table 3.

Table 3: Sail Mass

Component	Weight [g]
sail foil	330 (11.8g/m^2)
rigging	5
adhesive	32
photovoltaic	407
total	774

VI. Conclusion

For the GOSSAMER-1 system a concept for the sail storage was developed. The materials for the sail were chosen according to the environmental requirements. Furthermore, tools were developed to understand and analyze the load introduction to the sail membrane. Using these first models different designs are presented. While an analysis of the wrinkling effect on the solar pressure force is not yet implemented the strategy for an optimization of the sail geometry is presented. For a first design manufacturing strategies were developed and first test sails were built.

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Demonstrator Flight Missions at the Surrey Space Centre involving Gossamer Sails

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This paper presents an overview of the different gossamer sail flight projects being undertaken at the Surrey Space Centre. The missions consist of a 25 m^2 solar sail to be launched in Q1 2014 (CubeSail), a gossamer deorbiter for future European space assets (DGOSS), a scalable sailcraft that will demonstrate satellite deorbiting in Low Earth Orbit (DeorbitSail), and a drag sail that uses inflatable and rigidizable technology to be flown as part of the QB50 mission (InflateSail). The key technologies currently being developed for each project will be summarized and the most relevant scientific results presented.

I. Introduction

In the last decade there has been an extensive research focus in the field of solar sails. Many unique missions have been proposed for them and we have seen their potential in the successful missions by JAXA (IKAROS) and NASA (NanoSail-D2). With three new upcoming launches: the Planetary Society's LightSail-1[1], DLR's Gossamer-1[2], and the Surrey Space Centre's CubeSail[3], solar sail technology could soon reach the level of maturity and heritage required for more demanding missions and critical applications to be attempted.

One of the future applications of these sails is as ultra-light deorbiters of satellites and launch vehicle upper stages. Whether they are utilized as drag augmentation devices at altitudes below 700 km, or at higher altitudes in a pure solar sail mode, these gossamer sails can have significant mass savings with respect to the more traditional technologies considered for deorbiting [4].

The Surrey Space Centre has been working for the past few years on several demonstrator missions involving gossamer sails for pure solar sailing and sail-based deorbiting applications. Some of these programs are international collaborative projects that have created and reinforced links amongst some of the major players involved in solar sail technology research such as NASA, ESA, DLR and EADS Astrium.

Section I provides an introduction to the paper. Sections II through V review the Surrey Space Centre's flight projects independently and the show the key technologies developed for them. Section VI concludes the paper.

II. CubeSail

CubeSail is an EADS Astrium funded flight project developed at the Surrey Space Centre (SSC) that is currently in the qualification testing stage. CubeSail is a low-cost 5 m by 5 m nano-solar sail to be fitted inside a 3U CubeSat platform. The primary objective of the mission is educational, and over the past few years the

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project has played an important role within the Surrey Space Centre as an educational platform, with its developments forming the core of several PhD research topics.

Another important CubeSail objective is to advance the state of the art of CubeSat, solar sail propulsion, and deorbiting technology. The proposed mission will demonstrate several key technologies:

- deployment of a 25 m^2 solar sail, the largest structure deployed from a CubeSat to date;
- first demonstration of solar sailing in Low Earth Orbit (LEO), by utilizing the propulsive effect of solar radiation pressure (SRP) to attain a change in orbit inclination;
- first demonstration of a three-axis controlled solar sail, where the attitude control is enhanced by means of a novel center of mass/center of pressure (CM/CP) offset technique;
- demonstration of satellite deorbiting using a large gossamer structure to increase drag.

In the pursuit of these mission objectives, the Technology Readiness Level (TRL) of several components will be raised to flight-demonstration level:

1) motor-controlled deployment of novel metal tape-spring booms;
2) sail membrane with integrated rip-stops;
3) strain-energy driven deployment of bistable carbon composite booms;
4) two-axis translation stage between the satellite bus and sail structure, which provides a CM/CP offset used to provide 2-axis attitude control of the solar sail;
5) control algorithms for the 3-axis attitude control system using a range of actuators, including CM/CP offset, magnetorquers and a momentum wheel.

Mission phases

The CubeSail mission will consist of three stages:

- Commissioning and sail deployment (2 months);
- Solar sailing demonstration (9 months);
- Satellite deorbiting using drag (dependent on orbit and system parameters).

During the solar sailing demonstration phase of the CubeSail mission, the sail is oriented parallel to the direction of travel of the satellite (edge onto the velocity direction) in order to minimize aerodynamic drag; see Figure 1. In this configuration the resultant solar force will be mainly aligned with the sail normal vector and will be parallel to the orbit fixed direction $\overline{\mathbf{Y}}$. As there will be no solar force in the eclipse, an average torque in the $\overline{\mathbf{Z}}$ direction is generated. Since the orbit angular momentum direction is aligned with $\overline{\mathbf{Y}}$ the orbit precession direction is towards $\overline{\mathbf{X}}$ leading to an inclination change over time. The solar radiation pressure (SRP) induced inclination drift will continue until the orbit is fully sunlit, and there is no longer an effective solar torque.

Figure 1 illustrates the change in orbit inclination for a 3U CubeSat with and without a 25 m^2 solar sail; starting from an initial sun-synchronous orbit of 800 km, the simulations indicate that an inclination change of 1.9° over a period of 260 days can be attained.

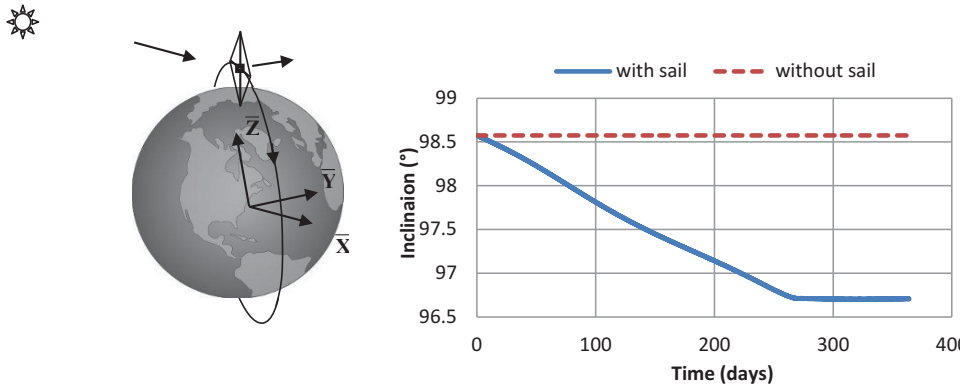


Figure 1: Change in orbit inclination for a 3U CubeSat of 3 kg with and without a 25 m^2 sail attached.

To achieve efficient drag deorbiting during the last phase of the mission, the sail must be reoriented to maximize the drag force at all times. This is accomplished when the sail normal vector points along the velocity vector of the satellite and therefore the sail will be first flipped 90° (face onto the velocity direction). Afterwards, to maximize drag, a rotation relative to the inertial reference frame of 360° per orbit revolution will be imposed, as shown in Figure 2. The requirement to align the sail normal with the velocity vector implies that control of the sail attitude is necessary. The Y-axis momentum wheel on board will be used for this maneuver.

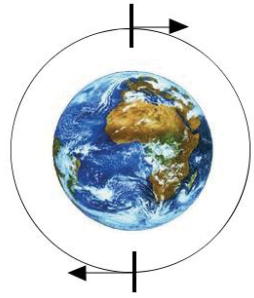


Figure 2: Sail orientation during drag deorbiting phase

Initial altitude (km)	Deorbit time <u>without</u> sail (years)	Deorbit time <u>with</u> sail (days)
600	26.5	39
650	48.1	75
700	88.5	132
750	160.5	190

Table 1: simulated deorbiting times for a 3 kg satellite with 0.03m^2 frontal area and with 25m^2 sail.

Table 1 shows the simulated deorbiting times for a 3 kg CubeSat with and without a 25 m^2 sail at different initial altitudes. Therefore, in order to guarantee sufficient time in orbit to demonstrate solar sailing, the initial orbit altitude for CubeSail will have to be above 700 km. The orbit will also need to be sun-synchronous. Fortunately, the majority of Earth observation satellites launched in LEO use this type of orbit and altitude and therefore the "piggy-back" option for CubeSail is feasible.

CubeSail layout

The satellite layout is divided into two main sections connected by a translation stage (3 cm x 10 cm x 10 cm). There is a 1U (10 cm x 10 cm x 10 cm) volume for the bus and a 2U (20 cm x 10 cm x 10 cm) volume for the sail payload. CubeSail uses the ISIPOD CubeSat deployer that has an additional cylindrical volume on the top side. This will be used to house a momentum wheel, a Sun sensor and a self-deployable bistable CFRP tape spring that will position the magnetometer 40 cm from the main bus electronics. The four 3U solar panels are spring-hinged and will deploy 90° into a four-petal cross configuration. This will reveal the packed sail to the space environment and upon activation sail deployment will commence. Figure 3 shows the different deployables of CubeSail and their deployment sequence.

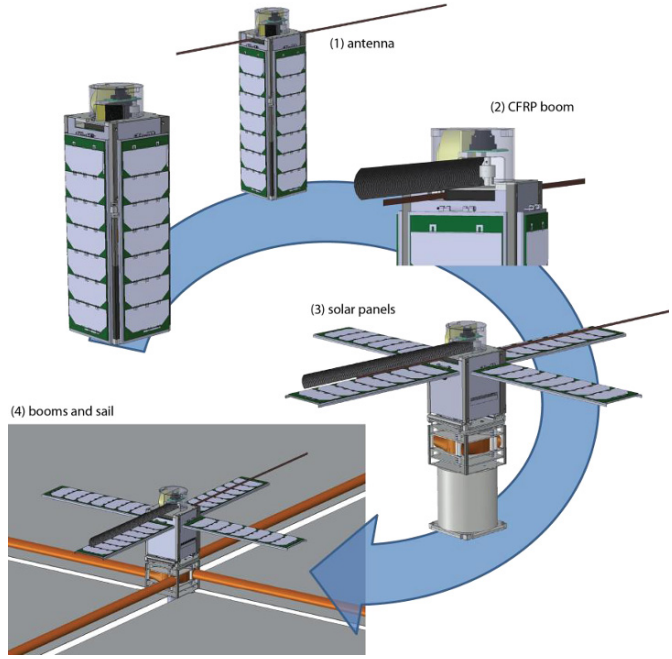


Figure 3: CubeSail deployment stages: (1) timed release of antennas to enable communications, (2) CFRP boom with magnetometer for attitude determination (3) release of the solar panels for power generation, and (4) deployment of the sail support booms and the sail membrane payload.

The square solar sail will have a four-quadrant configuration with four rigid booms used for deployment and support. Each of the four sail triangles will be supported at three points: at the satellite hub and at the tip of its two neighboring booms. The sail membrane is a 7-µm-thick double-aluminized Kapton film with a grid of Kapton tape rip-stops spaced at 62.5 cm. A low-cost manufacturing technique was developed that could be applied to larger size sails. The different sail segments are accurately cut to size using a large vacuum table and a CNC laser cutter. The sail segments are taped together on an overlaid joint using 3M's Y966 PSA transfer tape. Black "E" Kapton/966 PSA electrically conductive tape is used to ground the sail segments at the top and bottom sides of sail at its three edges.

A new type of metallic boom, and a boom deployer, have been developed in-house. The booms are assembled by sliding two beryllium-copper (BeCu) tape springs (of 16 mm radius of curvature) into a cylindrical Kapton sheath with the convex faces of the tape springs facing one another. This produces a "closed"

section lenticular-shaped structure, which increases the overall torsional stiffness. An aluminized Kapton version of the sheath will be on the flight model for better thermal control. The boom can be elastically coiled in an efficient manner as a standard tape spring holding a large amount of strain energy. The boom deployer will use a commercial off-the-shelf (COTS) pin-puller as a hold-down release mechanism, a motor to control the deployment rate of the boom, a set of eight radial spring-loaded rollers to manage the stored strain energy of the packed boom, and two fixed rollers at each corner of the deployer to guide the booms out linearly. More details of the boom and mechanism can be seen in Section III.

The spacecraft will have a compact 3-axis stabilized attitude control system. The sensor suite will consist of a COTS magnetometer, and Sun and Nadir sensors developed by the Stellenbosch University. The actuators will include three magnetorquers aligned with the spacecraft principal axis, a Y-axis momentum wheel, and a novel two-dimensional translation stage separating the spacecraft bus from the sail. The translation stage will be used to move the bus with respect to the sail in a plane, and therefore control the lateral offset between the center of pressure of the sail, and the center of mass to produce the desired control torques on the sailcraft. Two custom peripheral boards will be used to control the ADCS subsystem and the deployment actuation electronics such as the boom motor, the NiChrome burn wires and the pin-puller. The rest of the satellite bus (EPS, Battery, OBC, Communications) will be constructed using COTS components.

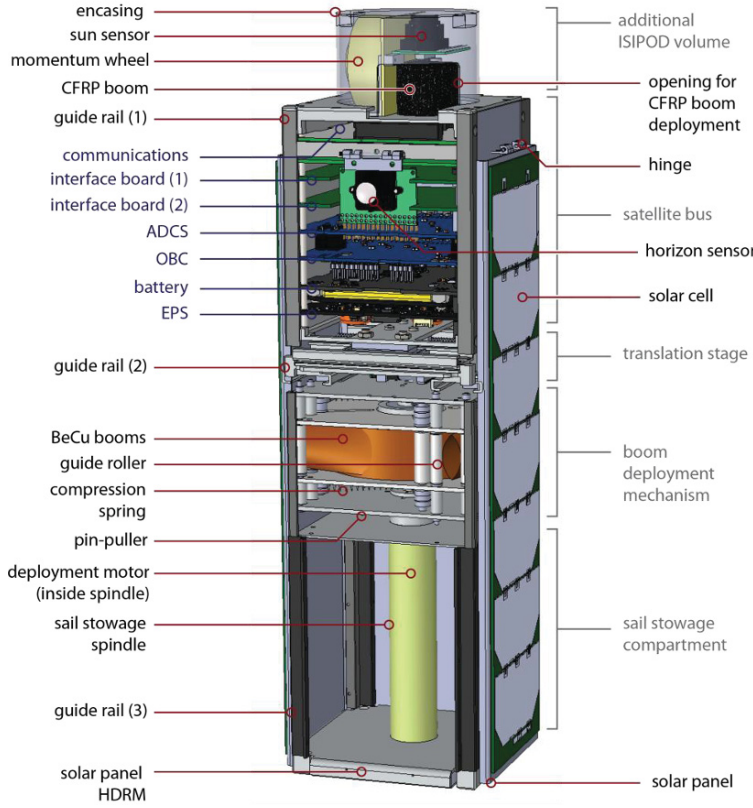


Figure 4: Overview of the CubeSail satellite configuration for launch; to note that one solar panel and the packed sail have been removed to reveal the internal components.

III. DGOSS - Deployable Gossamer Sail for Deorbiting

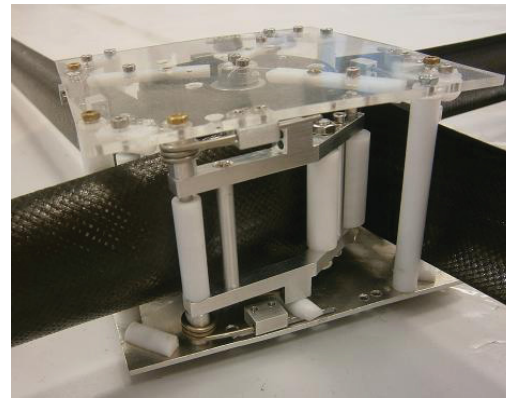
Through its ARTES 5.1 program, ESA has funded a project called the "Deployable Gossamer Sail for Deorbiting," or DGOSS. It comprises the study and advance to a Technology Readiness Level of 5 of a deorbiting concept based on gossamer technology.

The objective of the Gossamer Deorbiter is to be able to deorbit space assets in LEO within 25 years, after a period of up to 15 years of storage using only solar radiation pressure and drag. When the host asset loses functionality, the deorbiter will autonomously activate and the deployment of the sail will commence. It also needs to provide an overall 75% mass reduction from a traditional on-board chemical propulsion system. For this to be the case, the total mass of the Gossamer Deorbiter needs to be less than 3.5 kg for a large range of satellites and orbits. The volume envelope of the system when stored is 15 cm x 15 cm x 26.5 cm. It is fully enclosed by an external system that also offsets the sail plane 60 cm from the host satellite prior to deployment of the sail in order to clear any possible external appendages of the host. The size of the square solar/drag sail is 5 m by 5 m and it makes use of four 3.6 m booms for deployment and structural support.

The Gossamer Deorbiter has two main sections: the Sail Deployer System (SDS), which consists of the boom deployer, the booms and the sail; and the Telescopic Deployer System (TDS), which serves as the enclosure for the SDS, attachment point to the host satellite/rocket upper stage, and the system that displaces the sail away from the host.

Two alternative Sail Deployment Systems, initially developed for CubeSail, are being qualified as part of this project [5]. Both occupy a 2U volume (20 cm x 10 cm x 10 cm) and fit inside the inner box of the Telescopic Deployer. One concept utilizes the BeCu lenticular booms, and the other novel bistable CFRP scalable booms [6]. The two boom deployers can be seen in Figure 5. To constrain the booms and avoid blossoming problems the BeCu version uses eight spring loaded rollers that move radially inwards as the booms are extended out of the deployer. Since the CFRP booms are completely bistable and therefore the coiled state is an energy minimum, the boom deployer mechanism is simpler. It uses four torsion spring loaded lever arms that prevent the boom from transitioning from the coiled to extended curvature under the compressive forces required to unfurl the sail. This mechanism is simpler, lighter and more compact.

(a) BeCu version of the SDS without the sail.



(b) Boom deployer for the CFRP concept.

Figure 5: The DGOSS Sail Deployer System.

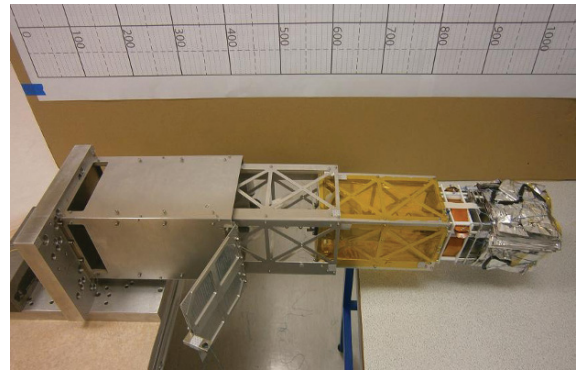
The Telescopic Deployer System (TDS) consists of three box-shaped skeletonized structures open on one end, a top enclosure door locked by a pin-puller HRDM, and the spring system. The boxes of different sizes are designed so that the smaller ones can be stored inside the bigger ones, resembling the design of a standard telescopic boom. Low friction guides located at the four corners of each box provide the sliding capability required during extension.

The extension of the successive telescopic stages is actuated by the force of three large compression springs. These springs are compressed during stowage, storing elastic strain energy that will be released after the top door is freed by the pin-puller activation. The choice of this extension actuation method was made based on its simplicity, long term storage reliability and minimal moving parts. Special care was taken to design springs and spring supports that prevent buckling during compression/extension. The locking mechanism of each telescopic stage consists of two ball plungers per stage that lock on holes and two hard stoppers made from Aluminum blocks used for redundancy in case the plungers fail to lock.

The complete deployment sequence of the Gossamer Deorbiter can be seen in Figure 6. To test the complete sequential deployment on ground a 6 m x 6 m table with a low-friction and anti-static coating of Sentrex film has been manufactured. The table is 0.9 m high and has a square hatch in the center so that the TDS can extend vertically from underneath, levelling the boom deployer with the table's top surface before the sail deployment commences. Polytetrafluoroethylene (PTFE) stands attached to the tip and mid-length points of the booms were used to reduce friction between the booms and the table. Two gravityoff-loading techniques for the booms have also been tried. The first involves helium balloons, and the second involves a cross-shaped gantry made with rails and low-friction rolling carts. Both methods suspend the booms from two points: at the tip and at the mid-length point after half of the boom has been deployed.

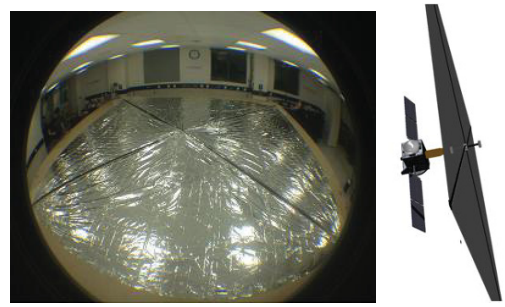
DGOSS has undergone an extensive qualification testing campaign that includes: boom characterization tests; vibration tests; functional tests in ambient, at temperature extremes and under vacuum condition of a scaled-down version of the sail; thermal-vacuum (TVAC) tests on the real scale system in a stored configuration with subsequent deployments; and survivability tests for the sail (micrometeorite impacts) and booms (loading and creep during storage). The SSC and ESA are negotiating a follow up project that will involve using DGOSS on a commercial satellite.

(a) Gossamer Deorbiter in stored configuration.



(b) Top door of Telescopic Deployer System (TDS) opens by a pin puller release and spring loaded hinges. TDS self-extends and displaces the stored Sail Deployer System (SDS) away from the host satellite.

(c) Second pin-puller is used to unlock the boom spindle. After this motor-driven sail deployment will start.



(d) Sail deployed and final configuration during host deorbiting.

Figure 6: Deployment sequence of DGOSS.

IV. DeorbitSail

Deorbitsail is a nanosatellite demonstration mission, expected to launch in 2014 and funded by the European Commission's Seventh Framework Programme (FP7). The project objectives are to provide research in the field of deorbiting, provide a demonstrated and verified design for deorbiting of satellites and debris using a deployable gossamer sail, and provide effective and efficient propulsion technologies based on sails.

The deployable sail consists of a 5-metre-by-5-metre membrane of 12.5 μm-thick Kapton, divided into four quadrants. A small version of DLR's ultra-light CFRP booms has been developed in order to meet the volume requirements of the mission. The four 3.6 m booms will extend from the boom deployer using a motor.

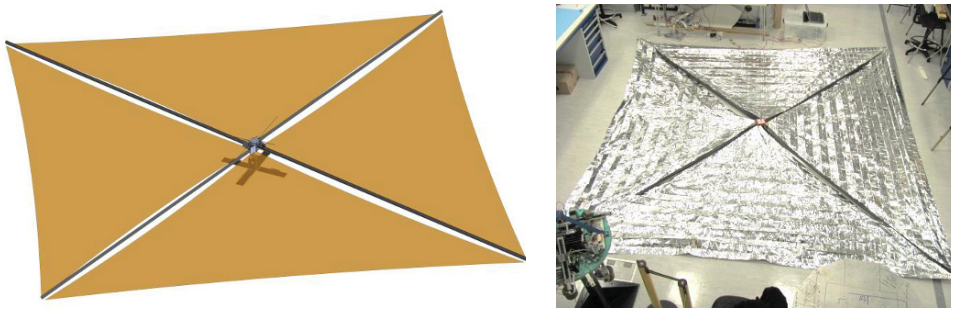


Figure 7: CAD image of Deorbitsail (left) and the 4m x 4m prototype in the lab (right).

Nine organizations are partners in the Deorbitsail project. The Surrey Space Centre is responsible for coordination and testing, and the other partners are:

- Athena SPU, Greece. *Software, bus subsystems, and website*
- Deutsches Zentrum für Luft- und Raumfahrt (DLR), *Germany. Boom and deployer*
- EADS Astrium SAS, France. *Sail membrane*
- Innovative Solutions in Space (ISIS), Netherlands. *Launch services and power*
- Middle East Technical University (METU), Turkey. *Boom damping analysis*
- Surrey Satellite Technology Ltd (SSTL), UK. *Ground operations*
- University of Patras, Greece. *CubeSat chassis*

- Stellenbosch University, South Africa. *Attitude determination and control system.*

In addition to the analysis necessary for the project, which is shared between many partners, METU has completed a study on active and passive damping system compatible with the Deorbitsail system, the University of Patras has characterized the carbon fiber chassis with FEA, and SSC has developed orbital analysis software for use with deorbiting and ADCS systems.

Deorbitsail mission requirements

The driving limitation on Deorbitsail is volume. The available volume is defined by the Cubesat deployer that ejects the satellite from the launch vehicle. An ISIPOD 3U Cubesat deployer[3] will be used to eject Deorbitsail. This deployer has a mass of 2.0 kg and allows a payload mass of up to 6.0 kg. For a satellite with a standard 10-by-10-cm CubeSat cross-section, the ISIPOD can accommodate lengths up to 339.5 mm, and it accepts some additional protrusions beyond the standard CubeSat square profile.

In order to convincingly demonstrate drag deorbiting, Deorbitsail carries two payloads: the sail system and the attitude determination and control system (ADCS). Alone, the sail payload will demonstrate a robust deployment system for a 5-by-5-metre sail in an approximately 2U package. The ADCS will make it possible to align the sail perpendicular to the direction of motion, maximizing drag and demonstrating optimal deorbiting performance for the system.

Volume is divided into five regions, for a total length of 34 cm: the main bus, consisting of the power system, radio, and onboard computer (5.5 cm); the ADCS (8.5 cm); the sail membrane storage area (9.5 cm); the booms and deployer (8.5 cm); and the hold-down/release mechanism for the deployable solar panels and antenna (1.5 cm). Therefore, the sail payload occupies approximately the same volume as a 2U CubeSat.

Property	Value
Boom length	3.53 m
Sail membrane areal density	18 g/m^2
Boom linear density	20 g/m
Required packing efficiency of sail	45%
Experimental load required to compress folded sail to stowed volume	9 N
Total mass of booms (no margin)	280 g
Total mass of sail membrane segments (no margin)	460 g
Total mass of sail system with deployer (20% margin)	1630 g
Sail area per unit mass for full system	15 m^2/kg or 65 g/m^2

Table 2: Deorbitsail vital statistics

Bus components are COTS products available from European suppliers. These include an ISIS[3] antenna and UHF/VHF radio and a GomSpace[4] Nanopower P31U electrical power system.

The Deorbitsail ADCS

Stellenbosch University (SU) is responsible for the Deorbitsail ADCS, which was developed from a modified version of SU's CubeSense attitude determination system. As a payload, it will demonstrate a very

[3] www.cubesatshop.com
[4] gomspace.com

compact three-axis attitude determination and three-axis control system, including a translation stage for low-power steering of sail spacecraft. Three magnetorquers and a single momentum wheel will be used for detumbling and orientation of the satellite; after the sail is deployed, a translation stage between the sail and the spacecraft bus will be able to shift the relative position of the center of mass and center of pressure by ±3 cm along the plane of the sail.

A number of accommodations have been required to integrate the ADCS with Deorbitsail, beyond the compacted ADCS stack. The initial phase of detumbling uses a magnetometer and a set of six coarse sun sensors to determine orientation. Traditionally, the magnetometer should be placed far from motors and magnetorquers, and the coarse sun sensors would ideally be on flat surfaces with reliable relationships to the satellite body. On Deorbitsail, the magnetometer will be placed at the base of the spacecraft, as far as possible from the ADCS system motors. Because all four side panels deploy, each of these panels will have two coarse sun sensors mounted to it. When the panels change orientation, the active sensor will also be switched.

The ADCS can be divided into boards, determination peripherals, and control peripherals:

- Boards: CubeSense board and CubeAIM board.
- Determination system: Coarse sun sensors (12), magnetometer, Sun camera, and Nadir camera.
- Control system: magnetorquers (3), momentum wheel (1), two-axis translation stage.

The Deorbitsail sail system

The sail system has been developed from DLR's longstanding heritage in deployable booms [7]. The sail membrane will be 12.5 μm-thick Kapton HN. Transparent yellow film, without the reflective coating that would be used for a solar sail, will be used for the membrane. This will improve the field of view of the attitude determination cameras. Similar to existing DLR heritage booms, the deployable booms will be made of carbon-fiber-reinforced polymer and have a collapsible closed cross-section.

An innovative system that increases the booms' extension force while preventing buckling during the sail deployment phase has been produced. The deployable doors also provide a stiffer root section at the end [8].

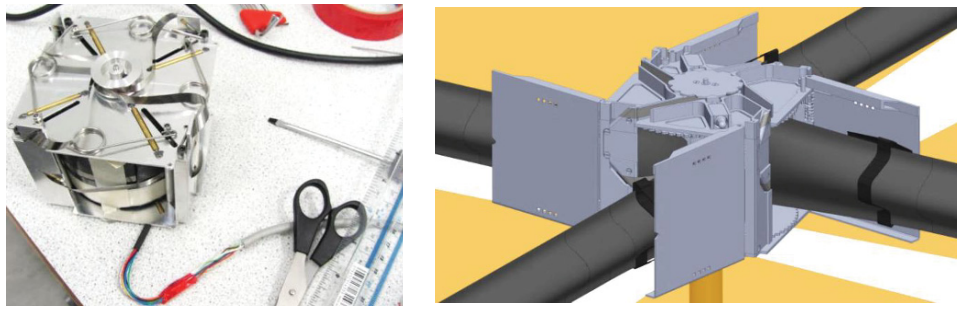


Figure 8: The prototype boom deployer in the lab (left) and the updated design (right) (DLR).

The Deorbitsail project will continue into Q2 2014. Current efforts are focused on finalizing the flight design.

V. InflateSail

InflateSail is an inflatable gossamer structure for use as a deorbiting drag brake. It will be tested in orbit on a 3U CubeSat platform as a technology demonstrator on the QB50 mission [9]. It is funded by the European Commission as part of an FP7 collaborative project named DeployTech [10].

The primary mission objective for InflateSail is the flight demonstration of an inflatable rigidizable sail structure, with an additional objective to demonstrate drag-augmented deorbiting. The expected initial orbit altitude for InflateSail on the QB50 launch opportunity will be 320 km, with the successful deployment of a drag sail expected to return the satellite to Earth within several days. The QB50 launch is scheduled for Q2 2015.

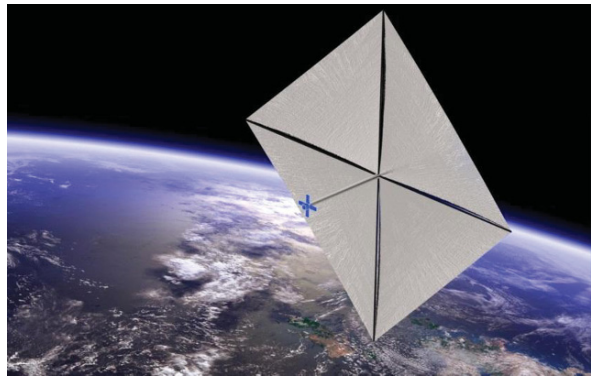
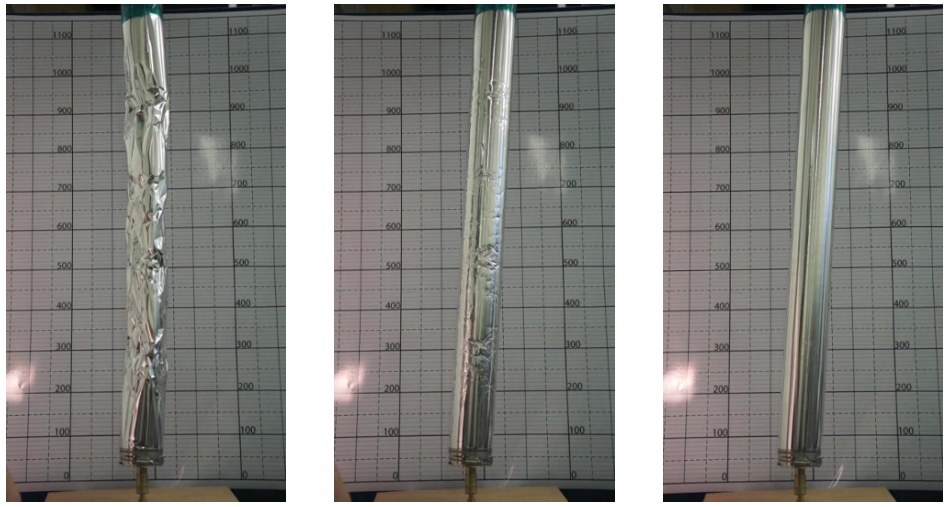


Figure 9: InflateSail in the deployed configuration.

InflateSail consists of a 1 m long inflatable rigidizable cylindrical boom, which supports a 10 m^2 sail that is deployed and held taut using four co-coiled deployable bistable open section tubular composite that will self-unravel as proposed by Fernandez [11]. It will be housed within a 3U CubeSat with approximately 1U reserved for avionics and the remainder dedicated to the payload. The goal is not only to demonstrate a drag deorbiting device, but also to demonstrate a rigidizable inflatable structural member which could be used for other purposes. The main structural member, the inflatable boom, serves to offset the sail from the body of the satellite to which it is attached. This reduces the likelihood of the unfurling sail encountering protruding parts of its host satellite during deployment, and also places the ‘center of pressure’ of the drag sail well aft of the center of mass of the entire system. The latter increases the longitudinal stability of the system, and allows it to remain aligned with the velocity vector, thus generating the maximum amount of drag.

Despite a long history of experimentation and development, few in-space rigidization missions have flown to date. Examples include the strain-rigidization of metal-polymer laminates used for the Explorer IX in 1961 [12], Explorer XIX in 1963, and the larger Echo II in 1964 [13-15], and wire meshes in photolyzable skins [16]. More recently, the Cibola Flight Experiment demonstrated Sub-Tg rigidisation of L'Garde deployable conical booms [17]. In 2009 the Astrium/Lavochkin Inflatable and Rigidizable Structure (IRIS) was launched, although little information about this mission is available. Also launched in 2009 was The Aerospace Corporation's AeroCube 3, a 1U CubeSat carrying a non-rigidisable balloon which was intended to be inflated using a miniaturized inflation system. If successful, InflateSail will be the first inflatable boom deployed from a CubeSat, and also

the first load bearing inflatable laminate structure flown. A major design challenge has been to adapt the inflatable structure concept to the scale of a CubeSat.



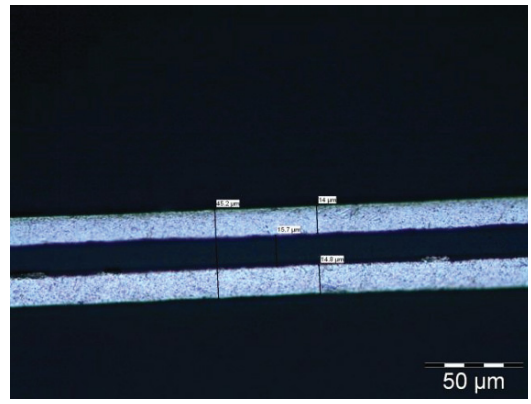
a) Initial wrinkled state b) During inflation c) Imperfections removed by slight yielding of skin

Figure 10: Strain rigidization process of a metal-polymer laminate boom.

Strain-rigidization of metal laminates has been selected as the rigidization method for InflateSail because of its extensive space heritage, simplicity of design, inherent reliability and insensitivity to in-orbit conditions. A slight over-inflation of the inflatable boom causes the metal in the laminate skin to strain just beyond its yield point. This results in a structure which gains rigidity locally through the strain hardening of the metal foil, and globally through the removal of creases and imperfections (Figure 10). The inflatable boom will be folded using an origami inspired fold pattern (Figure 11). Sub-T_g resins were also considered, but eventually rejected because of the difficulties associated with keeping the resin below its glass transition point at all times in orbit.

A key part of the InflateSail technology demonstration is the inflation and in-orbit rigidization using Cool Gas Generators (CGGs). CGGs have already demonstrated reliable operation after a number of years in the space environment on the PROBA-2 satellite [18], making them a good candidate for use in end-of-service-life deorbiting missions. CGGs also outperform pressurized gas systems in terms of total system mass and volume. TNO and its spin-off company CGG Technologies are developing a new design of Nitrogen CGG for InflateSail. The gas generator will contain approximately five grams of N_2 stored in a stable state until an electrical signal triggers a decomposition reaction, which releases the gas. The firing of the CGG will drive an inflation-driven deployment, followed by a slight over-pressurization of the deployed structure to rigidize the metal laminate skin. A miniaturised CGG is currently under development as part of the InflateSail project.

(a) Origami folded metal-polymer laminate cylinder (95 mm diameter) in stowed configuration



(b) Microscope image of Al-Mylar-Al laminate

Figure 11: Metal laminates in the construction of the InflateSail main mast

The sail material for InflateSail is expected to be a thin (less than 10 µm) biaxially-oriented polyethylene terephthalate (BoPET, or Mylar) film. The short duration of the InflateSail flight experiment means that more durable but expensive polyimide films are not required. For deorbiting from higher altitudes, a more durable metalized film will be required.

The sail will be deployed following the inflation of the main metal-laminate mast. It will consist of a single large square sheet of film co-wrapped with four CFRP bistable booms. The strain energy stored in the booms will be used to drive the unwrapping of the sail-boom structure. The sail fold pattern will be based either on the inextensional wrapping pattern proposed by Guest [19], or the spiral wrapping pattern proposed by Banik [20]. The bistable carbon booms will be specially constructed such that their coiled radius changes along their length, preventing local buckling effects when stored [6].

VI. Conclusions

Over the last fifteen years there has been an extensive research effort in the field of solar sails with many unique missions proposed for them. One of the recent applications considered for them is the use of solar sails as ultralight deorbiters of satellites and launch vehicle upper stages. At altitudes above 700km as pure solar sails and below that region as drag sails, these gossamer sails can have significant mass savings with respect to the more traditional technologies considered for active deorbiting. However many technologies to be used on these solar/drag sails are not at the necessary TRL for an actual flight mission.

This paper has presented the four upcoming flight missions (CubeSail, DGOSS, DeorbitSail and InflateSail) involving gossamer sails towards which the Surrey Space Centre is working. These low-cost demonstrator missions will serve as test-beds for some of the key scalable technologies developed for them, providing flight heritage to these new systems. Hence novel technology in the field of slender booms, membranes and attitude control actuators for gossamer structures will be given a chance to demonstrate their feasibility for larger and more demanding missions. For example, the CFRP lenticular booms to be flown on DeorbitSail are a scaled-down version of the booms that DLR will use on their 3-step Gossamer program. DeorbitSail will also fly a similar membrane using the low-cost technique developed during CubeSail and DGOSS as well as the X-Y

translation stage for attitude control. In addition, the single 1m inflatable origami boom with CGG Technology's cool gas generator that will fly aboard InflateSail could be potentially scaled up as the booms and deployment actuator of a larger ultralight sail on a bigger platform.

Acknowledgments

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ISS-based solar sail deployment experiment "BMSTU-Sail"

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The paper is about heliogyro solar sail deployment experiment called BMSTU-Sail which is developed by Bauman Moscow State Technical University (BMSTU) students and this experiment's main aim is the validation of heliogyro deployment technology. Small prototype of two bladed solar sail will be deployed from picosatellite which will be launched by cosmonaut during extravehicular activity on International Space Station. BMSTU-Sail experiment was included into Russian segment experiment list as an educational experiment in December 2012 and it is financed by Russian Federal Space Agency "Roscosmos". It is planned to be performed in the end of 2014.

I. Introduction

A. Two-blade solar sail

TWO blade solar sail is a type of frameless spin-stabilized solar sail which was proposed in 2010 by student enthusiasts from Bauman Moscow State Technical University as a prospective picosatellite mission [1]. Schematic drawing of spacecraft with two blade solar sail is shown on fig. 1.

This conception is similar to R. MacNeal Heliogyro solar sail [2] but has two major differences: it has two solar sail blades and different attitude control mechanism for blades and central body. This method is based on phenomena of spinning instability of such bodies and was patented by authors [3].

Advantages of this solar sail type are:

- Simplicity of solar sail folding;
- Attitude control.

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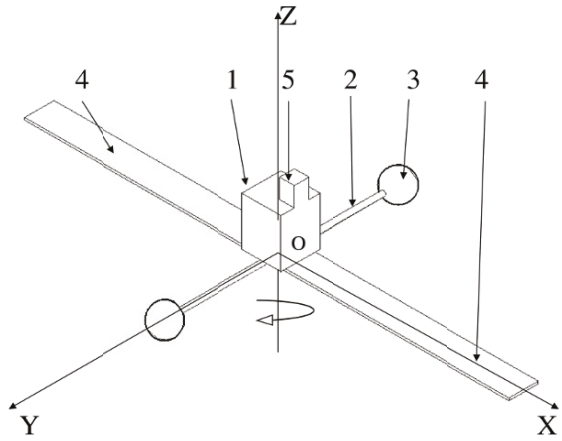

Fig. 1 Spin-stabilized spacecraft with two blade solar sail
1- spacecraft, 2 - sliding rod, 3 – tip mass, 4 – solar sail blade, 5 - payload
Z – axis spinning

The main disadvantages of two-blade solar sail are:

- Kinematic momentum is higher relative to other spin-stabilized solar sail with equal area (see fig. 2);
- Spacecraft motion induces disturbance of solar sail (e.g. waves on solar sail blades).

B. BMSTU-Sail as a student space mission

BMSTU-Sail is a space experiment developed by students and professors of Bauman Moscow State Technical University. The main goal of this experiment is developing and testing of deployment technology for two blade solar sail onboard of picosatellite. Testing of attitude changing system is not included in this experiment and will be implemented in future experiments.

The International Space Station (ISS) we selected as a base platform to launch mission picosatellite during extravehicular activity (EVA). Any frameless structure launched from ISS cannot operate as a solar sail – it can operate only as an atmospheric drag sail. ISS safety regulations for manned space vehicles and for delivered scientific equipment are another challenge with which we currently work. Now there are several successful missions launched during EVA from Russian Segment of ISS: Radioskaf[††] program, TNS[‡‡] and Sphera[§§] satellites. No doubt, these difficult and ambitious projects promote STEM education.

[††] http://radioskaf.ru/en/ [cited 06 May 2013]
[‡‡] https://directory.eoportal.org/web/eoportal/satellite-missions/t/tns [cited 06 May 2013]
[§§] http://www.spaceref.com/news/viewsr.html?pid=41766 [cited 06 May 2013]

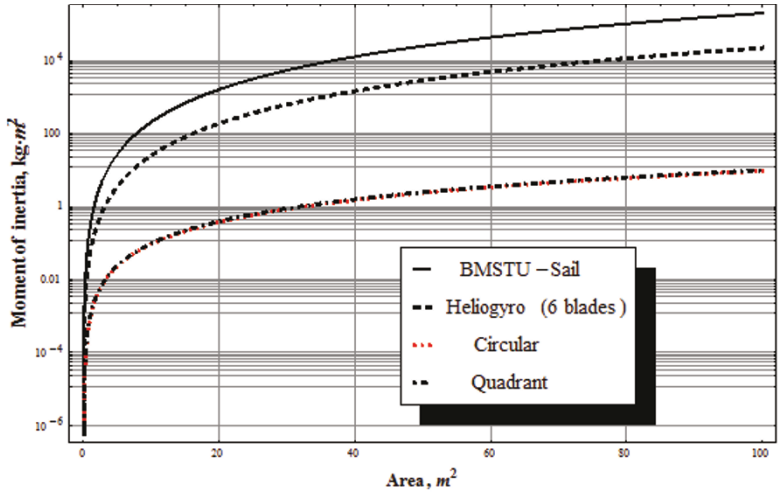

Fig. 2 Comparison of spin-stabilized solar sails

II. Mission design

BMSTU-Sail mission consists of three stages:

1) Preliminary stage – preparations inside and outside of pressurized volume of ISS;

2) Deployment experiment with logging of experimental data about deployment process;

3) Transmission of experimental data into ground stations.

A. Preliminary stage and launch

Preliminary stage begins in pressurized volume onboard of Progress automated transport vehicle where BMSTU-Sail scientific equipment (picosatellite and launch device) is stowed with another cargo for ISS. This scientific equipment can be stored inside pressurized volume of ISS up to 6 months waiting for EVA for launch. As far as picosatellite has electric batteries inside, it should be stowed inside special sealed container that should be opened only prior to launch. After unsealing, cosmonaut should implement testing power on of picosatellite to check its operability and to charge batteries.

From 2013 to 2015, ISS should incorporate some new modules for Russian segment[***]. Integrationof these new modules will be done during numerous EVAs. We propose to extend one of these EVA’s program to include launch of our picosatellite. One cosmonaut will implement launch sequence, which is described below:

[***] http://spaceexp.tumblr.com/post/48041932057/artists-impression-showing-the-european-robotic [cited 06 May 2013]

1 At the first, he should attach picosatellite to its picosatellite launch device (PLD) which looks similar to firing gun.
2 Then, by toggling switch No1 on outer shell of picosatellite, he initiates starting program of picosatellite onboard software.
3 To start reaction wheel cosmonaut should toggle the second switch.
4 After this, he should point PLD in safe launch direction and, after appearance of LED No2 indicator, initiate spring pusher.
5 After separation, he should take pictures of frameless thin film structure during deployment by camera integrated into Orlan space suit.

Total duration of this sequence is about 10 minutes of EVA time of one cosmonaut.

After switching on picosatellite is performing the following programs step by steps (see fig. 3):

1 Starting initialization and self-diagnosis.
2 Spinning up of reaction wheel and waiting for launch.
3 Deployment of two-blade solar sail prototype and collecting of data.
4 Transmitting of collected data.

Self-diagnosis program is necessary to test onboard software after long time of storage onboard of ISS. In case of failure, which was detected by self-diagnosis program, the onboard software can be restored inside the station by software installed on cosmonaut's laptop by USB interface. Batteries should also be charged by USB interface.

Spinning up of reaction wheel started briefly after toggling switch No2, cosmonaut should anchor to external structures of ISS. After reaching the appropriate velocity of reaction wheel, onboard software signalizing by LED No2 to the cosmonaut that picosatellite is ready for launch.

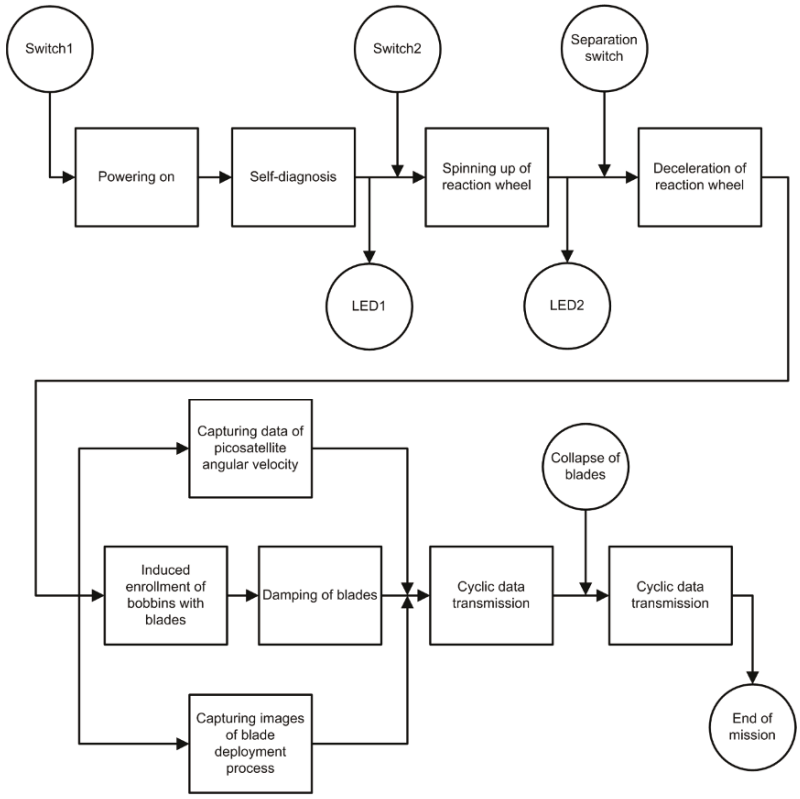

Fig. 3 BMSTU-Sail picosatellite flow diagram of flight program

B. Deployment process

After separation of picosatellite from PLD, separation switch turns electric motor of reaction wheel into braking mode. This procedure is necessary to translate kinematic momentum from reaction wheel to picosatellite structure. Deceleration process stops when spinning velocity of reaction wheel relative to picosatellite becomes zero.

After reaching of appropriate kinematic momentum the deployment process starts. To provide deployment, onboard software controls angular velocity of bobbin’s unrolment. Considering experiment parameters given in table 1, the control law for angular velocity is shown on fig. 4.

Table 1 Main parameters of experiment

Parameter	Dimension	Value	Description
J_{zz}	$\mathrm{kg}\cdot\mathrm{m}^2$	0.005	Inertia momentum of picosatellite
J_{RW}	$\mathrm{kg}\cdot\mathrm{m}^2$	$3.3\cdot10^{-5}$	Inertia momentum of reaction wheel
b	m	0.05	Width of each blade
l	m	5	Length of each blade
h	m	10^{-5}	Thickness of solar sail film
ρ	$\mathrm{kg/m}^3$	1470	Density of solar sail material (PET)
m_{tip}	kg	$5\cdot10^{-4}$	Tip mass weight
ω_{end}	*rpm*	2.06	Required spinning velocity of picosatellite after deployment
ω_{RW}	*rpm*	5550	Required spinning velocity of reaction wheel before launch

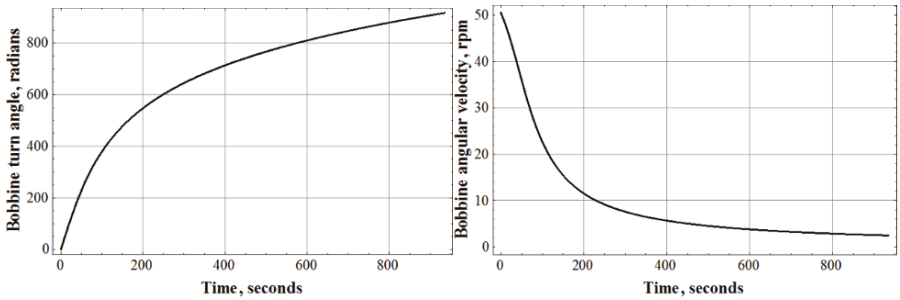

Fig. 4 Control law for angular position (left) and velocity (right) of bobbin depending on time from beginning of deployment

Following this law, spinning velocity of picosatellite and enrolled length of blades will cange as it showed on fig. 5.

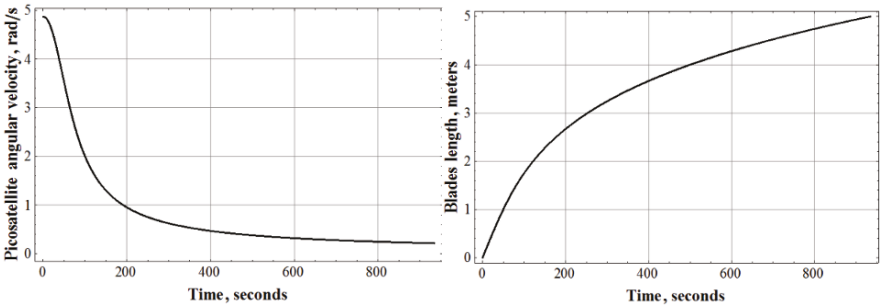

Fig. 5 Control law for picosatellite angular velocity (left) and blade enrolled length (right) depending on time from beginning of deployment

III. BMSTU-Sail picosatellite and launch device

Two-blade thin film solar sail prototype, which length is 5 m for each blade and width is 0.05m, will be deployed from picosatellite launched from PLD. Both picosatellite and PLD were developed by BMSTU students using their experience in other small satellite projects of BMSTU [4].

A. Picosatellite

Main differences of BMSTU-Sail picosatellite from other picosatellites are:

- small lifetime – few days (no less than 3 days);
- high power consumption during experiment – up to 1.5W;
- method of launch which consists of launch from Earth onboard of Progress transporting vehicle, and storing in pressurized volume of ISS where it can be manipulated by cosmonauts.

The exterior view of picosatellite is represented on fig. 6 and 7.

Picosatellite consists of following parts:

- Picosatellite frame with solar cells panel,
- Rechargeable batteries with radiators and heaters,
- Motored bobbins with solar sail tape,
- Reaction wheel,
- Electronic motherboard,
- Photo camera,
- Downlink antennas, Control switches and LEDs.

BMSTU-Sail picosatellite is spin-stabilized satellite; its structure should meet special requirements:

- Spinning axis of reaction wheel should be its principal central axis of inertia;
- Spinning axis of reaction wheel should be collinear with principal central axis of inertia of picosatellite;
- Blade’s plane should intersect mass center of picosatellite to minimize centrifugal moments of inertia;
- Principal central moment of inertia around spinning axis should be much greater than other central momenta of inertia.

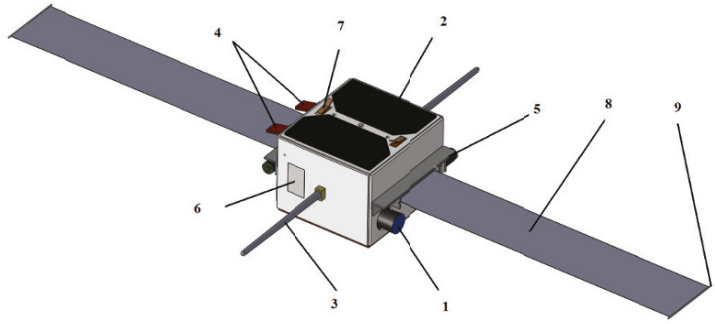

Fig. 6 BMSTU-Sail picosatellite (MLI is not shown); 1 – camera, 2 – solar cells, 3 – antenna, 4 – switches, 5 – stepper motor with bobbin, 6 – radiation surface, 7 – separation switch, 8 –blade, 9 – tip mass

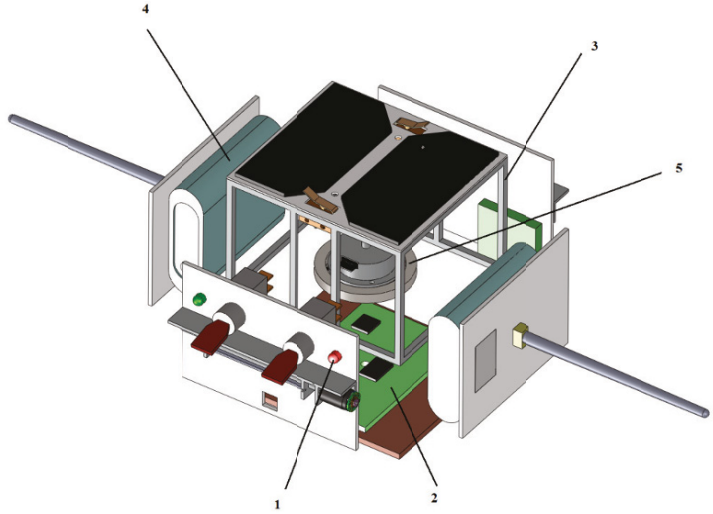

Fig. 7 BMSTU-Sail picosatellite (MLI is not shown); 1 – LED, 2 – electronic motherboard, 3 – frame, 4 – rechargeable battery, 5 –reaction wheel

Special assembly process was developed to meet these requirements. At the first, picosatellite and reaction wheel should be balanced separately, dynamically and statically. Than both part should be integrated to unite structure by with special seating fit hole.

Rechargeable batteries and electronic board is thermally insulated from other parts (bobbins with solar sail, reaction wheel and camera) because as far as experiment finishes, they don't need to be contained in thermal state. Batteries and electronic board is installed on panels, which shouldn't be oriented to the sun. Downlink antennas should be attached to pins prior before launch by cosmonaut.

Experiment sequence is implemented by onboard electronic hardware and software. Onboard electronics is built around small motherboard with 8-bit RISK microcontroller (see fig. 8). It controls temperature onboard,

makes the satellite follow the flight program by controlling of two stepper motors of bobbins and brushless motor for reaction wheel, collects experimental data, and prepares it for sending to ground stations. Another task of this microcontroller is power saving options, because energy capacity of batteries is small.

Experimental data collected from three-axis digital MEMS gyroscope and camera is stored on external EEPROM as well as time stamps for each measurement. Time stamps are provided by timer, which should be synchronized on Earth before launch or, as an option, onboard of ISS.

Information is transmitted from the Space to Earth by a radio line using amateur frequency 435 MHz FSK modulation. The bit rate is 9600 baud. Micro radio transmitter module is plugged into the picosatellite’s motherboard. Transmitter output is amplified by amplifier module; output power can be increased up to 7W however, it was calculated that output power of about 2W is enough to ensure ground data capturing, minimize data loss, and decrease requirement to ground stations.

Currently we assembled and tested engineering model of electronic motherboard (fig. 9) and performed development tests of deployment process of thin film structure on engineering model of picosatellite (fig. 10). In these test cables simulate blades.

The picosatellite mass is no more than 1 kg. Size is no more than 305x130x75 mm (with antenna)

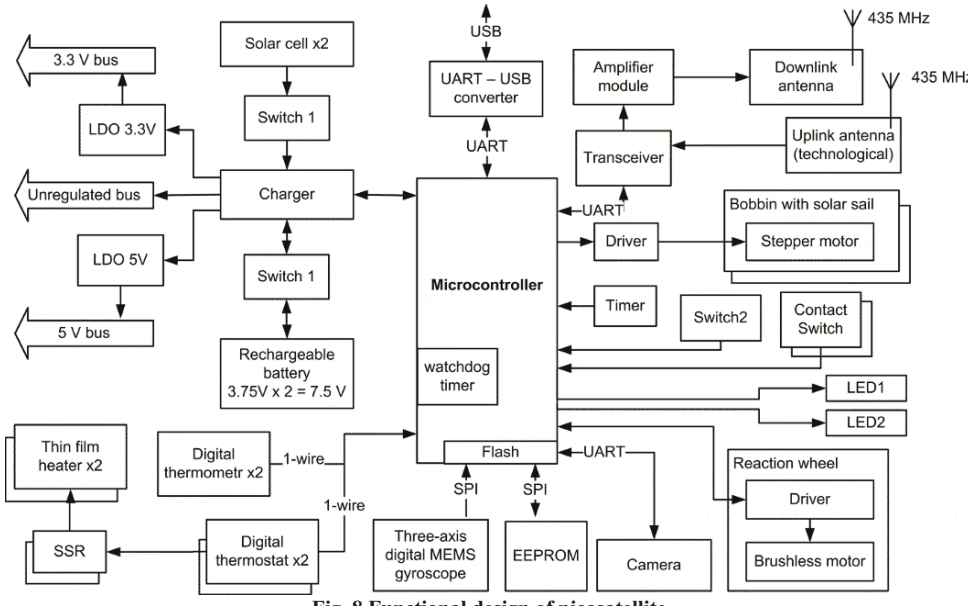

Fig. 8 Functional design of picosatellite

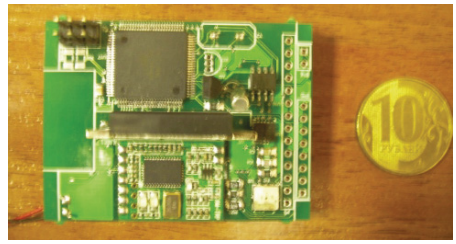

Fig. 9 Engineering model of electronic motherboard

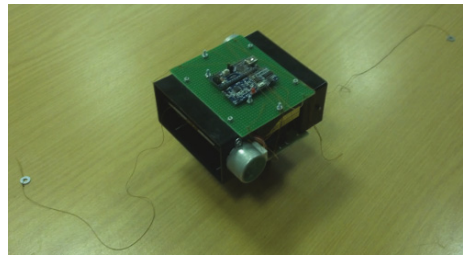

Fig. 10 Engineering model of picosatellite

B. Launch device

Picosatellite launch device designed to minimize parasite angular velocities which can appear during separation. Conception of this device is shown on fig. 11.

After arming, picosatellite should be attached to the pusher plate 5 and fixed by permanent magnet 1 on fixed part of launch device. Trigger activation frees the spring which is pushing picosatellite through hummer. Hummer travel is limited by plain bearer 6. After this, picosatellite separates from pusher plate. Spring tension can be changed by spring tensioner.

Launch device has two handles to minimize cosmonaut hands movements. To prevent accidental triggering the special safety lock is implemented. Launch device should also be tethered to space suit's belt; it is also should be tethered to picosatellite before arming. Technical specifications of picosatellite are given below:

- Weight: up to 2.5 kg;
- Maximum dimensions 400x130x40 mm;
- Hummer travel length 50 mm;
- Spring tensioner travel length 30 mm;
- Length of spring which is corresponding to minimum tension: 95 mm.

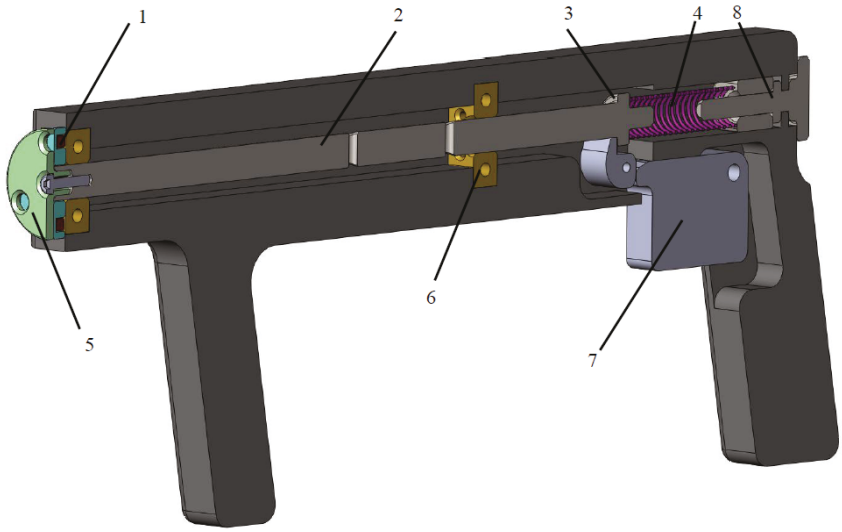

Fig. 11 Picosatellite launch device
1 – permanent magnet, 2 – hummer, 3 – stopper, 4 – spring, 5 – pusher plate, 6 – plain bearer, 7 – trigger, 8 - spring tensioner.

IV. Data acquisition

Data obtained during space experiment will be collected from two independent channels: from external cameras and from internal sensors. Possible amount of data from external cameras lie in the range of allowable limits of data that can be transmitted to Earth directly or by memory stick with ISS crew member.

Data from onboard sensors (images from camera and angular velocity data) should be transmitted to ground station. Downlink telemetry consists of three parts: vector of current state of picosatellite, scientific data from angular velocity sensors, images from onboard camera. Vector of current state of picosatellite includes current onboard absolute time, temperature data from sensors on PCB and in other parts of picosatellite, current angular velocity vector.

Scientific data is represented as array of measurements absolute time-tagged angular velocity vector (1.5 Kbytes total). Each black-and-white JPEG image from camera takes about 10 Kbytes, 20 images total. The total volume of data of all types is about 200 Kbytes.

To provide the maximum access time for picosatellite we choose packet protocol AX.25 [5] as data link layer. In order to transmit long messages the special open source transport protocol was proposed.

Each envelope of transport protocol consists of full vector of state of picosatellite, full array of scientific data and one of the images from camera. The length of such envelope is about 11.6 Kbytes. Then this envelope should be divided into small unnumbered information frames of AX.25 with data length of 256 bytes (276 bytes

including AX.25 header and footer). All AX.25 frames of one transport envelope should be transmitted in sequence without any pause. Transmission time of one transport envelope is 9.9 seconds considering that downlink speed is 9600 baud. After finishing of transmission picosatellite waits for about 50 seconds and then it transmits another envelope. After sending of 20 envelopes (with 20 different images) picosatellite should transmit another envelope with new image from onboard camera (not from experiment). First 20 envelopes should be transmitted in random order to increase the reliability in case of accidental reboot of onboard controller. Total transmission time of all envelopes is 21 minutes. Transmission pause between different envelopes can vary according to thermal regulation system and/or for economy of batteries power.

To determine the number of ground station we analyzed the picosatellite access time for different possible stations:

1) BMSTU, Moscow, Russia;
2) Possible point in Krasnoyarsk, Russia;
3) Possible point in South France;
4) Possible point in Texas, USA;
5) Possible point in Japan;
6) Possible point in Vietnam.

Calculation was performed for mission period of 3 days. Results shown in fig. 12. Parameter “number of ground station” is the number of stations from list above, which were included in calculation.

Summarizing calculation data, the minimum number of ground station should be two in order to receive all experimental data several times. However, it is important to have more ground stations to minimize error rate, dependence of accidental situations. In addition, the more ground stations we will have, the more additional images (envelope number 21) we will receive.

The use of a communication protocol AX.25 allows utilization of standard radio amateur hardware and software for receiving of data frames from picosatellite. Received frames will be collected in mission control center of BMSTU by e-mail or other communication channel. Collected frames will be incorporated into envelopes; then we will extract data and images from envelopes and share this information in our website[†††]. To promote volunteers from all over the world we will send signed certificates of receiving for all radio amateurs.

[†††] This information will be available in website http://parus.bmstu.ru

Our estimates show that the moment of two bladed construction collapse because of atmospheric drug will be after 3 days of mission. The total satellite lifetime before burn out is 60 days (Jacchia 1970 model, solar activity 150 sfu).

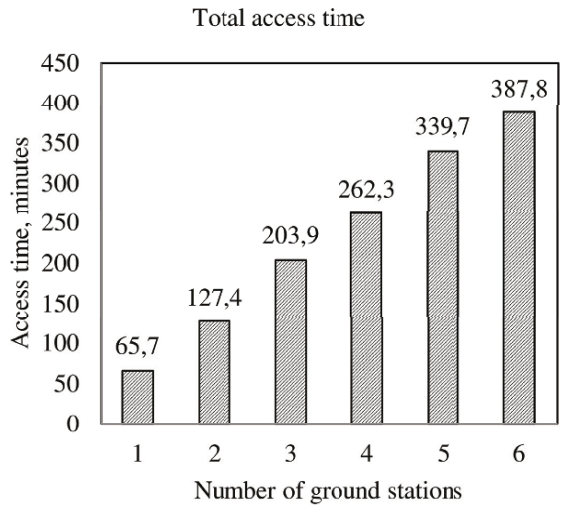

Fig 12 Total access time depending on number of ground station

V. Conclusion

In this paper we considered one of the types of frameless solar sail – two blades solar sail, provided a brief introduction to BMSTU-Sail project, described experiment sequence, structure of picosatellite and launch device, and the process of collecting and transmitting of experimental data. Proposed way of space experiments onboard picosatellites launched from ISS as well as technical solutions can be implemented in other space experiment with gossamer space structures.

Acknowledgments

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Part II

Mission Applications

Magnetic field measurements from a solar sail platform with space weather applications

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Structure in the solar wind magnetic field can cause space weather (e.g. geomagnetic storms) at Earth. The magnetic field must be measured in situ and to improve warning times, monitoring platforms operating further upstream of the Earth than existing satellites are ultimately needed. This can be accomplished by solar sail missions, but any instrumentation must be extremely light-weight. Here we describe the development of a highly miniaturized magnetometer (MAGIC), which as part of the Sunjammer payload will conduct scientific studies of the solar wind magnetic field and its evolution to better understand the fundamental sources and causes of space weather.

I. Motivation: An Introduction to the Near Earth Space Environment and Space Weather

The solar atmosphere is not in static equilibrium and continuously expands at supersonic speeds into space. This flow is known as the solar wind and the region of space filled by the solar wind is known as the heliosphere

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[1]. The first order properties of the solar wind are described quite well by ideal Magnetohydrodynamics (MHD), which treats the plasma as a single fluid with an effectively infinite conductivity [e.g. 2]. In particular, the solar magnetic field behaves as if it is `frozen' to the solar wind plasma and is convected out into space, forming the Interplanetary Magnetic Field (IMF). The interaction of the solar wind with the Earth's magnetic field leads to the formation of the magnetosphere, as shown in Fig. 1. Here the Geocentric Solar Magnetospheric (GSM) coordinate system is used: **x** points Sunward, the **x**-**z** plane contains the dipole moment (**z** points northward), and **x**-**y**-**z** is a right handed triple.

The solar wind and the magnetosphere are separated by the magnetopause; on the sunward side the magnetopause sits where the solar wind ram pressure is balanced by the magnetic pressure of the confined dipole whereas on the nightside the terrestrial magnetic field extends hundreds of Earth radii downstream, forming the magnetotail. The magnetosphere itself is highly structured (see e.g. [1] for a more detailed review).

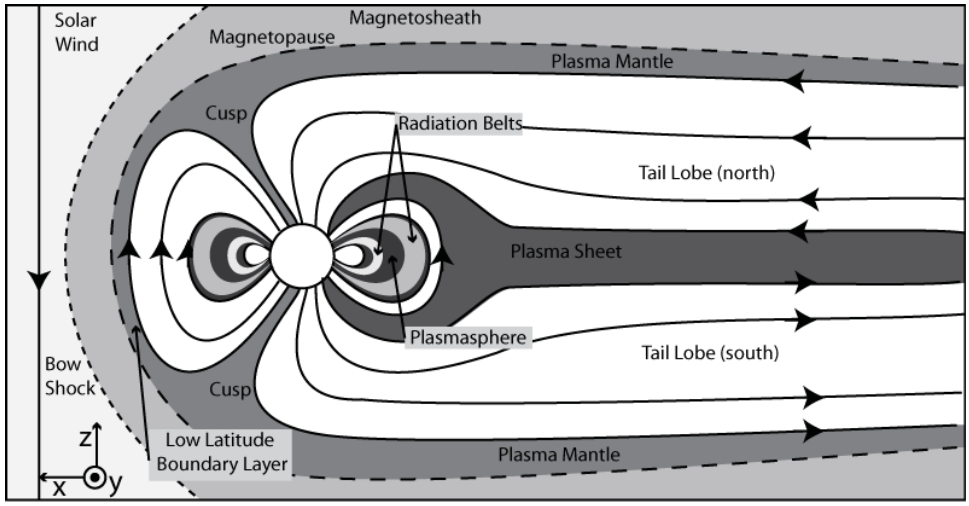

Fig. 1 The Earth's magnetosphere

The dynamics of the Earth's magnetosphere are mainly controlled by the solar wind and magnetic reconnection is the principal mechanism controlling the flow of energy into the magnetosphere [3]. If the IMF points southward (i.e. points in the $-\mathbf{z}_{GSM}$ direction), magnetic reconnection occurs between the IMF and the oppositely directed dipole field at the nose of the magnetosphere, resulting in open field lines (which are connected at one end to the Earth with the other end in the solar wind). These field lines, frozen to the plasma, are convected through the cusps and over the poles by the solar wind flow. Ultimately flux is added to the tail

lobes, resulting in the steady increase of energy stored there in the magnetic field. The stored energy is explosively released in a magnetospheric substorm [4], where magnetic reconnection across the magnetotail current sheet results in dynamic auroral displays, significant perturbations to ground based magnetometers resulting from auroral electrojet currents, and the ejection of plasma out of the magnetotail [e.g. 5-6]. Under intense and continuous solar wind driving with the IMF pointing southward for a prolonged period of time, substorm activity can develop into a geomagnetic storm, with intense auroras at low latitudes, a noticeable depression in the Earth's equatorial magnetic field strength caused by an enhanced particle ring current in the inner magnetosphere and enhanced radiation belt fluxes [7]. The relationship between substorms and storms is discussed in more detail by [8].

Such processes are one manifestation of space weather [9]. Geomagnetic storms and substorms can impact human activity both in space and on the ground and can have significant effects on terrestrial and space-based technologies such as power distribution networks, aviation, radio, satellites and global positioning systems. These effects, collectively known as “space weather,” have major economic impact and indeed space weather is now an entry on the UK Government Risk Register. Here we briefly mention three impacts which are specific to geomagnetic storms. There are many other space weather phenomena: for a comprehensive review see [10].

A. Power distribution networks

Power grids are susceptible to space weather effects [11-13]. Induced electric fields can be generated over large regions of the Earth’s surface in response to time variation in the Earth’s magnetic field during geomagnetic storms [14]. These electric fields, modulated by variations in the crustal resistivity, generate electric currents through conducting networks, in particular power grids. Ground Induced Currents (GICs) in the power transformer windings saturate the core of the transformer, which can eventually result in the failure of the transformer and potentially significant economic consequences [10].

B. Global Navigation Satellite Systems (GNSS)

GNSS receivers use data from several satellites simultaneously to accurately determine location in space and time, but operations can be degraded by irregularities in the ionosphere that cause receivers to lose lock and/or generate ranging errors. At mid-latitudes, GPS scintillations occur more often near solar maximum and in association with geomagnetic storms [15]. At high latitudes, so called polar cap patches can cause scintillations and auroral arcs have also been observed to cause GPS receivers to lose lock [16]. The refractive effects of

scintillations can be removed by using two frequencies. The new GALILEO system will have two public signals, as will GPS after 2014. However, this increases sensitivity to diffractive effects.

C. Satellite anomalies

Space weather can cause a number of different types of satellite anomaly [17]. Energetic particles that penetrate satellite electronics, generated as a consequence of geomagnetic storms and substorms, are just one example of this. Fluxes of 10 - 100keV electrons during substorms can cause spacecraft surface charging. Electrons above this energy trapped in the outer radiation belt just inside geostationary orbit (so-called killer-electrons) can cause deep dielectric charging as well as background counts in sensors. Protons between 0.1 and 1 MeV damage the surface of satellites, 1 - 10 MeV protons degrade the solar panels, and more energetic protons (SEPs) increase ionization, solar panel damage, and single event effects.

II. Drivers of Geomagnetic Activity

The degree to which solar wind plasma, momentum and energy enters the magnetosphere, leading to substorms and storms, is characterized by so-called solar wind coupling functions [18-19]. Physically, magnetic reconnection at the dayside magnetopause is enhanced if there is a strong southward IMF component, supplemented by fast solar wind, for an extended period of time. Whilst intervals of southward IMF occur naturally in the solar wind, and so substorms occur on a daily basis [20], strong driving and the largest storms tends to occur in response to coherent solar wind structure: Coronal Mass Ejections and Corotating Interaction Regions (21). We now describe these in more detail.

A. Coronal Mass Ejections

Coronal Mass Ejections (CMEs) are transient eruptions of coronal material into space [22]. CMEs propagate at super-magnetosonic speeds relative to the ambient solar wind, leading to the formation of shock waves at their leading edge [e.g 8]. CMEs play a particularly important role in the dynamics of the Earth's magnetic field because they can contain long intervals of southward IMF [e.g. 21]. In general, the largest geomagnetic disturbances are associated with CMEs, with the level of activity being directly related to the flow speed, the field strength and the southward component of the magnetic field [23-24]. In particular, the solar wind is more frequently disrupted by CMEs during solar maximum and the declining phase of the solar cycle [25].

B. Corotating Interaction Regions

During solar minimum the dipole is well defined and there are few sunspots. At this time, the heliosphere is structured in such a way that fast solar wind (~ 750 kms^{-1}) is observed to emanate from coronal holes over the poles and slow solar wind (~ 450 kms^{-1}) is observed closer to the equator [26]. However, since the dipole is tilted with respect to the rotation axis, and since the boundary between fast and slow wind is not at a uniform magnetic latitude, fast solar wind can find itself running into slow solar wind further ahead. This leads to the formation of Corotating Interaction Regions (CIRs), compressed regions of plasma following a spiral pattern in the heliosphere [27]. CIRs, exhibit enhanced densities and magnetic field strengths at their central interface. The trailing high speed solar wind contains large-amplitude Alfven waves, which can have long duration intervals of southward IMF [28]. Intervals of high speed, southward IMF solar wind in the trailing part of CIRs are also a significant source of magnetic storms, although they tend to be weaker than the ones generated by CMEs [29].

III. Measurement of the Interplanetary Magnetic Field upstream of the Earth

The ability of the solar wind, and discrete structures such as CMEs or CIRs, to generate geomagnetic storms and substorms (referred to as 'geo-effectiveness') depends on the orientation of the IMF as it reaches the nose of the magnetosphere. This cannot be detected remotely, and must be measured in situ in the upstream solar wind. Such measurements are made by satellites at the Sun – Earth L1 Lagrange point (1.5×10^6 km/ ~235 Re (Earth radius) upstream of the Earth), first by the ISEE-3 spacecraft [30] from 1978 – 1982, and then by the ACE (1997 – date) [31] and Wind (continuously since 2004) [32] spacecraft. Accurate timing, together with an accurate understanding of how the solar wind evolves between the upstream measurement point and the Earth is important to understand what specifically triggers geomagnetic activity.

Whilst there is a significant societal need to develop space weather monitoring platforms that can operate for prolonged periods upstream of the Earth closer to the Sun than the L1 point, this introduces technical and scientific challenges, in particular to understand how the solar wind evolves as it flows from the upstream measurement point to the Earth.

Early work simply considered the correlation between measurements made at the upstream point and in the solar wind near the Earth [e.g. 33] and the simplest convection models assume that the solar wind conditions at Earth can be predicted by time lagging the upstream data according to

$$\Delta t = t_2 - t_1 = \frac{(x_1 - x_2)}{\langle v_{sw} \rangle} \quad (1)$$

where x_1 and x_2 are the relative GSM-x locations of the upstream and downstream monitors, and $<v_{sw}>$ is the average solar wind speed. [e.g. 34]. This essentially assumes that the solar wind is one dimensional. For a typical solar wind speed of 400 km/s, the travel time from L1 to the nose of the bow shock (x ~ 13 Re) is ~59 minutes.

However, the arrival time of solar wind structure at Earth does not simply depend on the speed of the solar wind and the separation of the upstream and downstream monitor. Figure 2 illustrates the problem, where one may imagine that a solar wind 'phase front' (for example a discrete change in the density or magnetic field strength) may be oriented obliquely to the solar wind flow [e.g. 35].

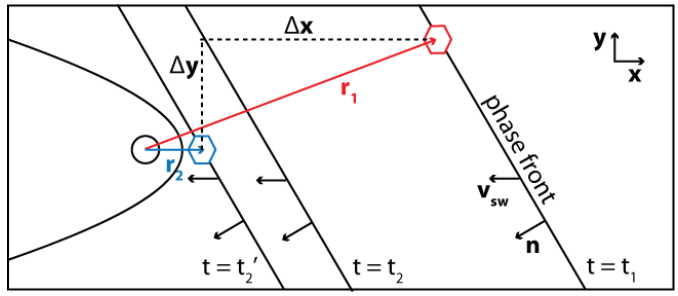

Figure 2 Geometry of phase front normals in the solar wind. (Here we assume that the upstream and downstream measurement points are at the same z location.)

Since no upstream monitor lies exactly on the Sun-Earth line, oblique solar wind phase fronts introduce error into the simple predictions of arrival time at Earth described above. A more accurate arrival time can be modelled according to

$$\Delta t' = t_2' - t_0 = \frac{\mathbf{n}\cdot(\mathbf{r}_1 - \mathbf{r}_2)}{\mathbf{n}\cdot<\mathbf{v}_{sw}>} \quad (2)$$

where **n** is the phase front normal (PFN). Evidently this requires knowledge of the PFN orientation which can be established by analysing the magnetic field data: for example, calculation of the normal vector can be derived from magnetic coplanarity calculations [36]. In the past few years, algorithms have been developed which take oblique PFNs into account (e.g. [37] and references therein) and this is used for example to produce the OMNI dataset (http://omniweb.gsfc.nasa.gov), an estimated timeseries of the solar wind conditions at the nose of the Earth's magnetosphere, used for magnetospheric studies.

Although accounting for PFN orientation improves the prediction time, the evolution of the solar wind is still not well understood. For example, in a recent study comparing different propagation methods, it was found that although on a statistical level the performance of different techniques is similar, in individual events the difference in predicted arrival time can be of the order of 30 minutes [38]. This could be due to a variety of reasons:

- Propagation methods often assume time invariant solar wind structure, which is expected to be overly-simplistic, particularly when there are large gradients in solar wind pressure. Rapid evolution of the solar wind structure can occur between L1 and the Earth [39].
- Even on relatively short spatial scales, it is possible for there to be significant variation in the solar wind perpendicular to the Sun-Earth line [40].
- Solar wind phase fronts are unlikely to be planar on large scales. Establishing the curvature of phase fronts requires the use of multi-point measurements in the solar wind [41].

More generally, if the correlation length in the solar wind is of the order of 100 Re [34], then as the separation between the upstream monitor and the Earth increases, understanding how the solar wind evolves as it flows away from the Sun becomes progressively more important.

The weaknesses of the different techniques used to propagate data from the upstream measurement point to the Earth will ultimately be revealed only by increasing the separation, providing a new dataset that provokes new approaches to the problem. Increasing the spatial domain over which propagation techniques can be tested and developed will thus establish an improved understanding of how the solar wind evolves, and more precise knowledge of how variation in the solar wind controls magnetospheric and geomagnetic activity.

IV. The Sunjammer solar sail mission

It has been shown that theoretically, solar sails can be used to maneuver a spacecraft into a highly non-Keplerian orbit and station-keep upstream of the Earth, closer to the Sun than the L1 point (e.g. [42] and references therein). This has been more specifically studied and defined by the GeoStorm mission concept (e.g. [43-44]. In this context, NASA has approved a solar sail technology demonstration mission, Sunjammer, due to launch in 2014 (http://www.nasa.gov/mission_pages/tdm/solarsail/solarsail_overview.html). The mission profile sees Sunjammer launch as a secondary payload, using its own propulsion to escape Earth orbit. The sail will then be deployed, and non-required propulsion systems will be jettisoned. Sunjammer's trajectory will ultimately take it into interplanetary space, beyond L1 where it will station-keep near the Sun-Earth line.

Sunjammer's prime contractor, L'Garde Inc, has extensive experience in the development of solar sail technology [45].

Being beyond the first Lagrangian point offers Sunjammer the prospect of providing advanced warning of interplanetary phenomena such as shocks and CMEs. For science purposes, the availability of multiple observation points on a 100 Re scale in the solar wind further opens up the field of solar wind studies (e.g. phase front curvature) using multi-point satellite measurements (Sunjammer, ACE, Wind, DSCOVR, Artemis and Earth orbiting satellites such as THEMIS and MMS). In particular, Sunjammer will substantially increase the spatial domain over which such multi-point studies can be performed and this data is crucial to testing different existing solar wind propagation techniques to their limit and developing new and improved algorithms for understanding solar wind evolution.

NOAA is an institutional partner in the Sunjammer project, providing additional funding and spacecraft communication support. The presence of an institutional partner places a requirement on the project to perform relevant activities satisfying the partner's participation. In this case, the activity is a demonstration of technologies needed to improve space weather monitoring and in particular to demonstrate the feasibility or otherwise of making space-weather quality magnetic field and solar wind measurements from a solar sail platform beyond L1.

V. MAGIC: MAGnetometer from Imperial College

The engineering constraints associated with solar sail missions mean that the payload must be very light-weight. Imperial College has developed a miniaturised DC (0-20Hz) vector magnetometer (MAGIC) and in the context of the UK-US Memorandum of Understanding on space weather, MAGIC will form part of the Sunjammer payload. MAGIC comfortably satisfies the NOAA performance requirements for an upstream monitor magnetometer. The main objectives of the Sunjammer MAGIC instrument are to:

- investigate the technical capability or otherwise of a solar sail mission to measure the interplanetary magnetic field and establish the influence of the sail on magnetic field measurements.
- conduct scientific studies of solar wind evolution and propagation, in the context of space weather monitoring from a location closer to the Sun than the L1 point.

A. Magnetoresistive sensor technology

MAGIC is based on solid state magnetoresistive (MR) sensors which offer two orders of magnitude mass and volume reduction compared to the more conventional fluxgate [46]. MR sensors are thin film solid state

devices whose operation is based on the variation in resistance of some magnetic materials in the presence of an applied field. The advantage of these devices over more traditional technologies such as fluxgates is their extremely low mass (<5 g for a three axis sensor design) and volume (~2 cm^3) compared to high performance fluxgates (~100 g and > 100 cm^3). In addition they can be packaged as ceramic micro-circuits which means they can be mounted directly onto a PCB (i.e. without dedicated sensor mechanical support). In fact Anisotropic MR (AMR) sensors are identified as the best candidate MR technology due to their superior detectivity at low frequencies (0 – 10 Hz).

In order to improve its sensitivity and thermal stability, the sensor is implemented as hybrid design: Imperial has developed a hybrid micro-circuit combining three orthogonal single axis AMR sensors together with a common drive circuit into a single package. The AMR sensors and associated drive circuitry are implemented as a thick film on a Low Temperature Co-Fired Ceramic (LTCC) substrate. This permits the use of an ultra-light weight connecting harness of several meters and in fact the concept of co-locating the sensor drive circuitry came from the requirement to complement the ultra light MR sensor with a matching ultra-low mass harness in order to facilitate longer distances between the sensor and the FEE (Front End Electronics) on CubeSat platforms. The sensor runs in feedback mode controlled by a bus mounted electronics card communicating via SPI. A detailed description of the design principle is given in [46]. Combining the MR triad and drive circuit in this way results in a very efficient sensing element as it combines the physical resources savings (mass/power/volume) intrinsic to the MR sensor together with the detectivity enhancement realized through operation in a driven mode, into a single device that can be packaged for space environments.

B. Flight Heritage: CINEMA

CINEMA is a CubeSat built by UC Berkeley under the auspices of the NSF CubeSat programme. Designed to operate as a spin-stabilized platform in low-Earth-orbit, CINEMA carries two instruments – STEIN and MAGIC – to study magnetospheric dynamics. The STEIN instrument is derived from the STE instrument on STEREO and uses solid-state detectors in combination with an electrostatic deflection system to measure the local ion and electron population, as well as detect energetic neutral atoms (ENAs). ENAs are created by charge exchange between exospheric neutrals and energetic ions in the magnetosphere. Significant fluxes of ENAs are generated by charge exchange in the radiation belts, and so they can be used to remotely detect and image changes in the radiation belts [47]. The MAGIC instrument has a two-fold purpose. The first is to provide magnetic field data required for attitude control of CINEMA, and the second is to conduct scientific studies of magnetospheric ultra-low-frequency waves observed as a consequence of solar wind-magnetosphere coupling.

Two further identical CINEMA CubeSats will be built by Kyung Hee University to ultimately form the TRIO constellation, with a fourth CINEMA CubeSat also planned.

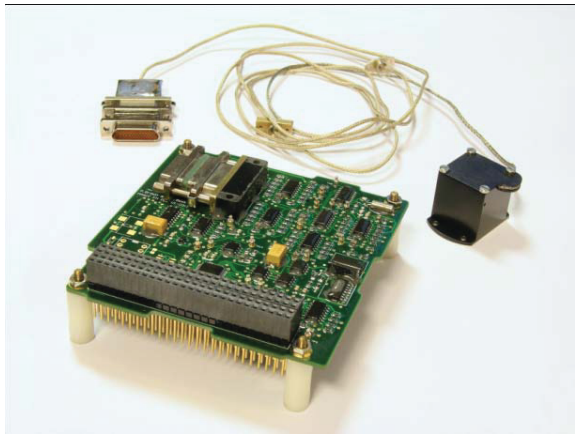

Fig. 3 CINEMA MAGIC flight instrument

On CINEMA, MAGIC uses a dual sensor design consisting of an electronics card, outboard and card-mounted sensor heads and an interconnecting harness, as shown in Fig. 3. The harness, providing a link to the outboard sensor, is designed for a deployable boom length of up to 1m. The outboard is a hybrid sensor, whereas the card-mounted sensor is composed of a discrete MR triad offering lower performance, but also lower power requirements. The mass of the electronics card is 71 g and the mass of the outboard sensor head is 33 g. The maximum power requirement of MAGIC onboard CINEMA is 560 mW. The MAGIC instrument on CINEMA has configurable data rates up to a maximum of 8 Hz per sensor axis, a sensitivity of better than 2 nT and a Noise Spectral Density (NSD) of less than 150 pT/√Hz above 1 Hz. It can operate in two modes - 'science' and 'attitude': in science mode the instrument operates at a higher sampling rate and higher resolution, whereas in attitude mode the instrument power and data rate is reduced whilst still providing sufficient data for the purposes of spacecraft control.

CINEMA launched as a secondary payload on 13 September 2012 into a 770 km × 480 km orbit at an inclination of 64º. Although operations on CINEMA have been impeded by communication difficulties, thought to be due to non-deployment of an antenna, the attitude mode data returned thus far appear to show that MAGIC is operating nominally. An example of magnetic field data recorded by MAGIC on CINEMA is shown in Fig. 4. The data, collected in attitude mode, are shown in the spacecraft coordinate system; the oscillation in the time series is due to the fact that the satellite is rotating in the Earth's magnetic field. These and similar

measurements have been used to determine the attitude and orientation of CINEMA (M. Archer, private communication).

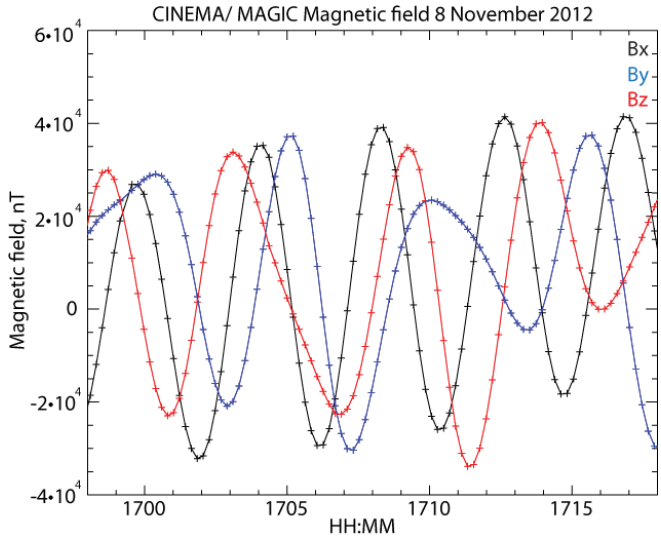

Fig. 4 Time series plot of the three components of the MAGIC inboard magnetometer sensor on board the CINEMA CubeSat. Data are shown in the rotating spacecraft coordinate system. The oscillation is due to the rotation of the CubeSat in the Earth's magnetic field, and can be used to determine the attitude of the CubeSat.

C. Implementation of MAGIC on Sunjammer

The Sunjammer magnetometer will be derived from the CINEMA design and will use the same very light-weight MR sensor technology as described above. The very low mass and power requirements of the modified design are ideal for the tight constraints of a solar sail mission and can easily be accommodated on the spacecraft within the estimated available mass and power.

CINEMA operates in the high-field environment of low-Earth orbit, whereas Sunjammer will operate in the solar wind and so the Sunjammer implementation of MAGIC will feature significant developments beyond that of CINEMA. Development work has made it possible to remove the original card-mounted sensor, duplicate the outboard sensor and electronics, and thus produce an instrument with two boom mounted sensors (placed mid-way and at the end of the boom). For Sunjammer, MAGIC will therefore utilise two remotely deployed hybrid sensors instead of a single outboard sensor and this will allow MAGIC to run in a so-called gradiometer mode, which will characterize the magnetic field environment of the spacecraft more precisely. As a result, this will improve the calibration of the magnetic field measurements and their potential usefulness for space weather purposes. The card delivers the three components of the magnetic field and the temperature at the sensor

(necessary for calibration) from each of two remotely accommodated hybrid sensors as 12-bit values over SPI. The use of a 12-bit radiation hardened Analogue-Digital Converter (ADC) (as opposed to the 24-bit commercial ADC employed in the CINEMA design) means it will be necessary for the instrument to have two ranges, a high resolution range for operation in the solar wind and a lower resolution range for operation closer to the Earth. Whilst not yet finalized, the range scales are expected to be of the order ± 512 nT and ± 10,000 nT respectively. The instrument sensitivity is better than 1 nT averaged over 60 seconds. Power consumption will be less that 1.0 W and the total instrument mass is less than 125 g. The design will be modified to use space qualified parts where possible. A full environmental test and qualification campaign as well as radiation testing of the hybrid magnetoresistive sensor are also planned.

A key goal of the MAGIC experiment in addition to conducting scientific studies of the solar wind and its evolution is to understand the basic magnetic signatures generated by a solar sail through its interaction with the solar wind. For example, it is expected that eddy currents, in response to the changing solar wind magnetic field, and spacecraft charging currents, in response to differential illumination, will be generated. These (and other) effects must be understood and characterized in order to remove their signature from the measured data.

VI. Conclusion

The development of the ultra lightweight MAGIC magnetometer will enable magnetic field measurements to be made by Sunjammer and other solar sail missions in the future. Operation of MAGIC on board Sunjammer provides the opportunity to test the feasibility of using solar sail missions to study the solar wind magnetic field and its evolution to better understand the fundamental sources and causes of space weather. Successful operations may potentially result in a significant increase in the warning time for geo-effective CMEs, which will improve the ability of infrastructure operators (both in space and on the ground) to mitigate the effects of space weather.

Acknowledgments

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The Solar Wind Analyser (SWAN) for the Sunjammer solar sail mission

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The Solar Wind Analyser (SWAN) is a miniaturised analyser to be delivered for NASA's Sunjammer solar sail mission. The aim of the Sunjammer mission is to demonstrate the technology readiness of solar sails and the ability to carry out good measurements of the solar wind with in-situ instrumentation in the presence of a large solar sail. Solar wind data from upstream of the Earth is extremely valuable, particularly when available in real-time, for advanced space-weather warning systems, alerting satellite operators and utilities on Earth of potential hazards from geomagnetic storms caused by coronal mass ejections from the sun. SWAN combines an electrostatic analyser with an energetic particle detection system to provide a comprehensive measurement of the low to medium energy solar wind plasma. The electrostatic analyser, based on the Charged Particle Spectrometer (ChaPS) built for the TechDemoSat mission, is designed to measure energy distribution functions of proton and alpha particles and provide real-time data for future space weather operational missions. The energetic particle detection system on the other hand will extend the energy range to enable measurement of solar energetic particles. This paper will present instrument requirements for a solar wind space weather monitor, present technical details of ChaPS and SWAN and discuss the challenges of making plasma measurements in the presence of a large solar sail.

I. Introduction

With the recognition of the increasing vulnerability of space and ground based assets to space weather events, there is considerable activity worldwide both to understand the science behind the phenomena that cause the disruption as well as to develop technologies for monitoring such phenomena. In this context, solar sails are particularly attractive for space weather operational missions as the technology can potentially be used to place an observatory upstream of L1,to provide more warning time than an L1 observatory, and use the solar sails for

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station keeping. The aim of NASA's Sunjammer mission is to demonstrate the technology readiness of solar sails and the ability to 'sail' the spacecraft. 'Sailing' to L1 and then subsequently to sub-L1 heliocentric distances and possibly out of the ecliptic are further goals of the mission.

A key requirement for a future operational mission however is instrumentation that can carry out reliable and continuous in-situ measurements of the environment.Sunjammer will hence carry two in-situ instrumentson board, a solar wind ion instrument and a magnetometer and the mission also aims to demonstrate the ability to carry out good measurements of the solar wind in the presence of a large area solar sail. Besides calling out a subset of these observations for continuous real-time telemetry for space weather purposes, the instruments will contribute towards furthering our understanding of the science of space weather. This is still a relatively young field and significant work is required to fully understand the physics of these space-based events.This is needed in order to develop operational processes that deliver useable space weather measurements and forecasts.The mission brings together a combination of novel technology developments, providing an opportunity to raisetheir technology readiness level for future space weather operational missions.

For solar wind ion measurements, there are currently a number of scientific instruments operating in a variety of orbits providing data in the energy range < 40 keV. These are largely top-hat type analysers, typically weighing 1 kg - >3.5 kg, designed either for electrons or ions. A number of these are often a part of instrument suites providing both electron and ion analysers in a resource efficient package. The SWEPAM [1] instrument on the ACE satellite routinely monitors the low energy space plasma environment at the L1 Lagrangian point. The instrument consists of two sensors for measurement of electron and ions. Another example of an investigation in the solar wind is the IMPACT [2] instrumentsuite onboard the Stereo mission combining both low and high energy detection capabilities. The Supra-Thermal Electron telescope (STE) [3]and Solar Electron and Proton Telescope(SEPT) [4] instruments in the suite cover electrons in the energy range from ~2 keV to 400 keV and protons from 20 keV to 7 MeV. Amongst instruments currently under development for solar wind measurements is the SWA sensor suite [5] for the Solar Orbiter mission, consisting of an Electron Analyser System (EAS), a Proton and Alpha Spectrometer (PAS) and a Heavy Ion Spectrometer (HIS). Relevant to this study is the PAS sensor, providing full 3-D distributions and moments at a high time resolution of 4s.

The instruments described above are all primarily scientific instruments aimed at detailed solar wind science studies. While they provide comprehensive measurements of the solar wind with suites of instruments optimized for different plasma populations, they require relatively large mass and power resources. For space weather monitoring purposes, low resource instrumentation providing continuous monitoring of key parameters is

desirable. The Solar Wind Analyser (SWAN) described in the following sections is a miniaturizedinstrument designed to carry out measurements of the low to medium energy solar wind plasma, providing real-time data for advanced space-weather warning systems.

II. Measurement requirements

The key requirements for a solar wind space weather monitor are described below.

1. Continuous data flow, 24 hours per day, seven days a week. Minimum requirement is 96% coverage per day with no data gap exceeding 15 minutes
2. Each measurement is available for use in operations within 5 minutes after completion of the measurement
3. Plasma Ion Measurement
    a. At least one measurement of the solar wind velocity vector (Vx, Vy, Vz), average ion temperature, and ion density moments every minute
    b. Must deliver data in GSM coordinates in real time
    c. Velocity range 200 to 2000-km/sec with 5% relative accuracy
    d. Temperature range: 40,000 to 2,000,000 K with 20% relative accuracy
    e. Density range: 1 to 100 cm-3, with 20% absolute accuracy
4. Characterization of low energy ionparticle population with at least one set complete set of measurements every 5 minutes
5. Necessary housekeeping data for each instrument sent in real time
6. Carry out solar wind measurements close to the Sun-Earth line

III. Instrument technical details

A. Charged Particle Spectrometer

The design of SWAN is largely based on the Charged Particle Spectrometer (ChaPS), built at MSSL for the UK's TechDemoSat mission, with the sensor geometry altered for SWAN to address the measurement requirements of solar wind plasma, i.e., ion densities and flow velocities. Characteristics of ChaPS are presented here. The key aim of ChaPS is to raise the TRL level of the miniaturized analysers being developed at MSSL. The spectrometeris designed to make measurements of the electron and ion populations in Low Earth Orbit (LEO). The measurement requirements for the different populations can be significantly different. For example, to make measurements of accelerated auroral electrons, typically few to several keV energy, an energy resolution

of the order of 30% is sufficient. On the other hand, in order to study ionospheric photoelectron populations in LEO, the energies are considerably lower but the energy resolution required is considerably higher. Suchmeasurements aretypically carried out by multiple instruments, each optimized for the particle population of interest.ChaPS consists of a number of miniaturized analysers integrated into a single unit and will attempt to make these measurements with a single instrument. The design and accommodation of each of the analysers is optimized for a specific measurement goal. The three targeted measurement goals are to study the upward-downward electron flows in the Earth's auroral region, study cold ions in the ionosphere in the spacecraft ram direction and measure the spacecraft potential. The analysers usevariants of the Bessel box [6] geometrywith performance tuned to the three measurement goals.Two variants are used on ChaPS, one with an analyser constant of 2.5, shown in Figure 1a, to study the lower energy particle populations below 100 eV and the other with an analyser constant of 120 for particles above 100 eV. The response is further refined by choice of collimator/aperture geometry and the geometry of the extraction region.

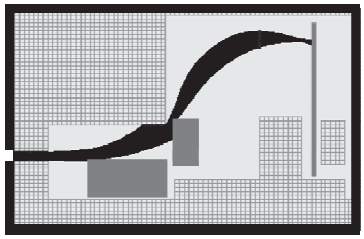

Figure 1a

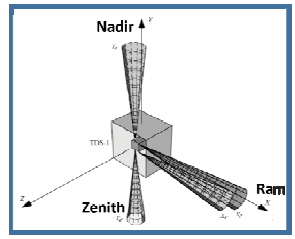

Figure 1b

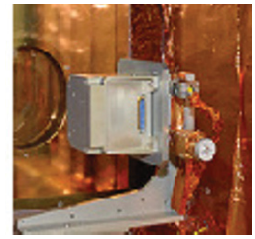

Figure 1c

Figure 1a: Schematic of the analyser geometry created with the SIMION software, with particle trajectories shown in black. The electrodes shown solid are biased to the required voltages whereas the rest of the structure is held at spacecraft ground. Figure 1b is a schematic showing the accommodation of the four analysers and Figure 1c is a photo of ChaPS in the electron calibration chamber

Figure 1b shows a schematic of the accommodation of the sensors. The two auroral analysers are accommodated to point in the zenith and nadir directions to sample upward and downward travelling electrons in the Earth's polarregions whereas the ionospheric ion and photoelectron analysers are accommodated to face the spacecraft ram direction.Figure 1c is a photo of ChaPS in the MSSL calibration system. Figure 2 shows simulation data for the low energy sensor shown in Fig. 1a. The calculated k-factor is 2.5 and the energy resolution is 4.9%.

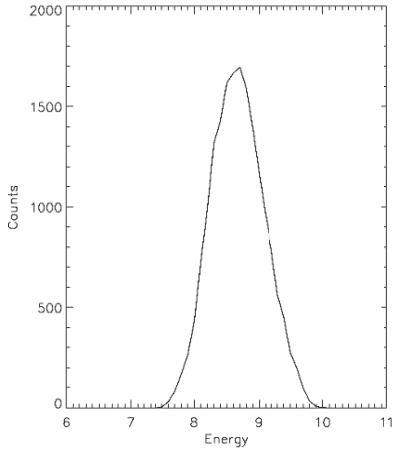

Figure 2: Simulated Energy response of the ChaPS low energy ion sensor

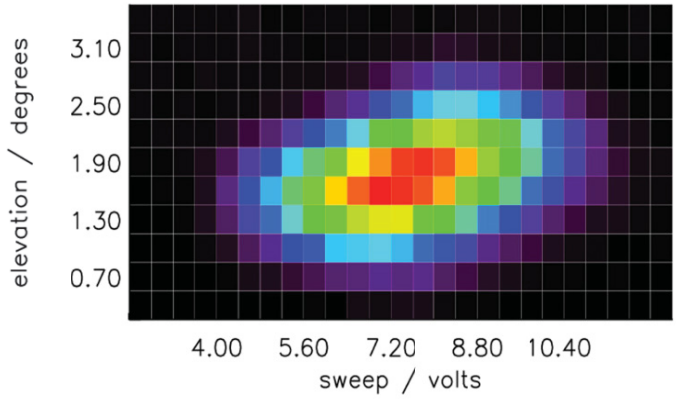

Figure 3: Energy Elevation scan for the ChaPS Nadir electron sensor at 1 keV

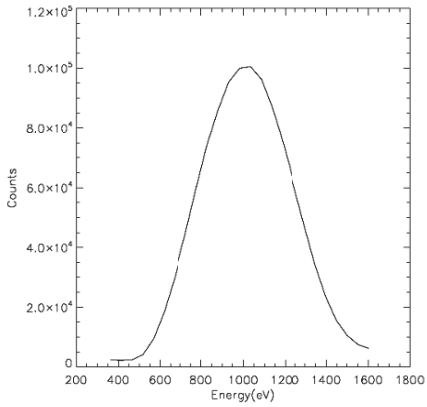

Figure 4: Calibrated energy response of the ChaPS nadir electron sensor

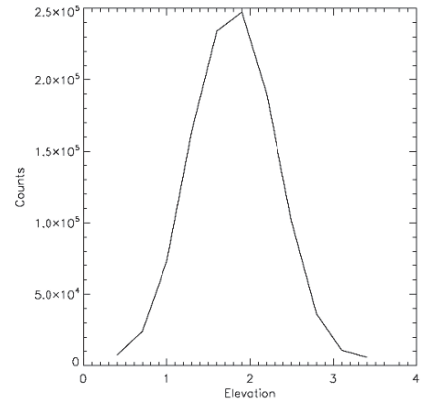

Figure 5: Calibrated elevation response of the ChaPS nadir electron sensor

Figure 3- 5 show calibration data from ChaPS for the Nadir sensor for a 1 keV electron beam incident on the analyser at a single azimuth. Figure 3 is the elevation-voltage response of the analyser. Converting the analyser voltage to its equivalent energy response and summing over all elevations provides the energy response of the analyser, shown in Figure 4, giving an analyser constant (k-factor)of 127 and an energy resolution of 50%. Figure 5 is the elevation response, obtained by summing over all voltages and gives an angular acceptance of 1.3°. ChaPS has a symmetric design in azimuth with an acceptance of 20°.

The design chosen for ChaPS provides a good scheme to carry out measurements of the plasma populations in LEO as well as demonstrate the analysis technique, the Bessel box geometry and the ability to "tune" performance to specific applications.

B. SWAN designoptimization

The baseline design for SWAN is that of the nadir/zenith analyser as a number of the requirements for SWAN are already met by the design subject to the necessary changesto measure solar wind ions instead of auroral electrons. The key additional features and design optimizations required to meet the measurement goals of the mission are described in the following sections.

1. Fly-through geometry

As can be seen in Figure 1a, ChaPS does not have a fly-through geometry. As SWAN is a sun-facing sensor, a fly-through geometry is highly desirable to minimize secondary contamination from reflected UV and to minimize flow of current from secondary electrons generated off exposed electrodes. Design modifications are being incorporated to include a fly-through geometry as well as UV rejection baffles. The electrodes will also be

put through the Ebanol-C [7] process to provide the blacked surfaces typical of most low energy plasma analysers, to further enhance UV rejection.

2. Field of view

A key additional feature that is required for SWAN is an increased field of view, both in azimuth and in elevation.The elevation-azimuth acceptance of ChaPS is 1.3° x 20° whereas the ideal requirement for SWAN is±22.5° x 60°. As ChaPS has a symmetric design in azimuth, extending to 60° is relatively simple. However, to extend the field of view in elevation, additional electrodes will be required in the design. A technique to achieve this has been used quite effectively with top hat analysers (5) and is currently also being developed for the Solar Orbiter EAS instrument at MSSL. A variant of theSolar Orbiter electrode designis being incorporated in SWAN.

3. Solar Wind ionmeasurements

Two key aspects need to be addressed for solar wind ion measurements, 1) the changes required to operate the analyser for ions as opposed to electrons and 2) the optimization of the analyser performance parameters, in particular, the energy resolution and geometric factor. The modifications required for the first are relatively simple, primarily requiring changes to the power supplies to obtain the correct voltage polarities and magnitudes, both on the analyser as well as the detector. The second aspect requires the analyser geometry to be tuned. Measurement of auroral electrons does not require very high energy resolution and ChaPS is designed for an energy resolution of ~40%. Thedesired energy resolution for SWAN is ~5%, closer to the resolution of the ionospheric ram sensor on ChaPS. At 5% resolution, it should be feasible to separate out the dominant ions, H and He, in the solar wind through energy measurements only and avoid a more complex ion mass identification technique. The design of SWAN is being optimized to obtain 5% resolution while retaining the analyser constant of 125 and a geometric factor of $1x10^{-6}$ cm^2 sr eV/eV.

4. Energetic particle measurements

The electrostatic analyser on SWAN is currently being designed for 20 keV. However, the capability to measure solar energetic particles in the range 30 keV to 5 MeV would provide a significant performance enhancement for a space weather operational mission. Detection of particles in this energy range with electrostatic analysers, particularly on miniaturised instrumentation, is significantly challenging and requires considerably more mass, power and volume. Direct detection, e.g., using Silicon detectors, are instead conventionally used for this energy range. SWAN will include an energetic particle detector based sensor system with a design that will deliver the performance parameters give in Table 1.

Parameter	Analyser	Energetic particle detector
Particle Species	Proton and Alpha particles	Energetic electron and ions
Energy Range	100 eV – 10 keV	Electrons <200 keV, Protons <5 Mev, Alphas < 20 MeV
Energy resolution	8%	10%
Analyser constant	100	-
Polar Angle	±22.5°	±22.5°
Resolution	11.25°	-
Azimuthal Angle	-15° - +45°	±22.5°
Resolution	Unresolved	-
Geometric factor	1×10^{-6}	0.1
Time resolution	30s	30s

Table 1: Performance parameters required for SWAN

The complete instrument to include the detector and the sensor electronics is estimated to weigh less than 500 gms and draw 1 W.

C. SWAN development, next steps

This section briefly describes some of the key technical issues to be addressed for the development of SWAN.

1. Electromagnetic Cleanliness and spacecraft charging simulations

In making clean measurement of the solar wind plasma, magnetic and electrostatic cleanlinessplays a key role. The impact of this can potentially be exacerbated by the presence of Sunjammer's large solar sail. Although the primary aim of SWAN is to demonstrate its ability to provide real-time data as an operational instrument, reporting the ambient conditions of the solar wind plasma environment, quality data along with a good understand of instrument or spacecraft related effects will be crucial to interpreting the data returned from the instrument. Due to the characteristic features of the solar wind ions, with typical energies of a few 100 to several 1000 eV, static magnetic and electrostatic fields are not expected to affect the ion measurements significantly under most conditions, particularly for space weather monitoring purposes. However, to ensure a good understanding of the electro-magnetic environment of the spacecraft under static and dynamic conditions, e.g., during a CME, and its impact on the measurements, future work will aim to model ambient plasma conditions and associated spacecraft charging simulations.

2. Thermal and Radiation environment

Further work being carried out on SWAN includes thermal and radiation modelling. The impact of these on ChaPS is relatively lower as TechDemoSat is in a LEO environment. Due to this, ChaPS uses a number of commercial-off-the-shelf components which may not be suitable for Sunjammer. As SWAN will be a sun-facing instrument, thermal modelling will also be required to ensure the analyser components and the detectors remain within their operational temperature ranges. While changes required to the electronics are already under way to address the thermal and radiation requirements, increased background in the detectors, particularly during CME conditions, will need to be assessed through the radiation simulations.

IV. Conclusions

SWAN is designed to address the measurement requirements of solar wind plasma for the Sunjammer mission. The design of SWAN is based on one of the analysers on the ChaPS instrument on TechDemoSat. Regarding ChaPS as a prototype for SWAN, simulation and performance results from the instrument are presented. Further, design modifications, performance enhancements and optimization of instrument response being pursued for SWAN are also discussed. In particular, the addition of an energetic particle element is presented. This aspect will provide a significant capability enhancement for a space weather monitor. Finally, some of the crucial aspects that may affect the measurements and future work associated with these aspects have also been presented. SWAN will not only demonstrate the ability to carry out good measurements of the solar wind in the presence of a large solar sail but potentially also provide real-time data for advanced space-weather warning systems, alerting satellite operators and utilities on Earth of potential hazards from geomagnetic storms caused by coronal mass ejections from the sun.The complete instrument to include the detector and the sensor electronics is estimated to weigh less than 500 gms and draw 1 W.

Acknowledgments

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Gossamer Roadmap Technology Reference Study for a Multiple NEO Rendezvous Mission

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A technology reference study for a multiple near-Earth object (NEO) rendezvous mission with solar sailcraft is currently carried out by the authors of this paper. The investigated mission builds on previous concepts, but adopts a strong micro-spacecraft philosophy based on the DLR/ESA Gossamer technology. The main scientific objective of the mission is to explore the diversity of NEOs. After direct interplanetary insertion, the solar sailcraft should – within less than 10 years – rendezvous three NEOs that

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are not only scientifically interesting, but also from the point of human spaceflight and planetary defense. In this paper, the objectives of the study are outlined and a preliminary potential mission profile is presented.

I. Introduction

Small bodies – asteroids and comets – are in some sense the fossils of the solar system, which hold key information for understanding its origin and evolution. Small bodies vary highly in size, surface properties, composition, and probably origin (see Fig. 1*).

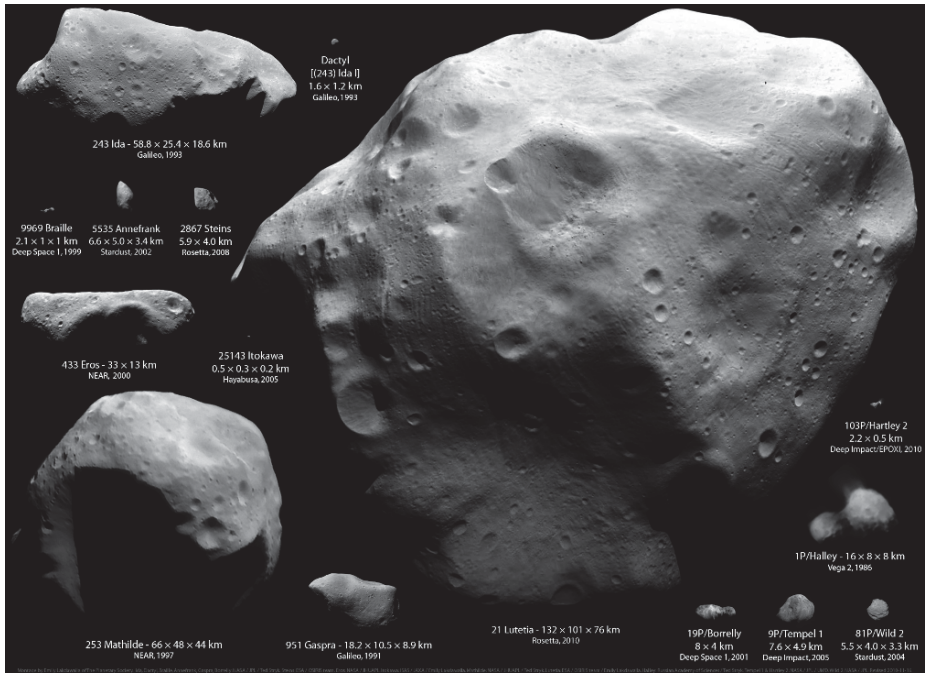

Figure 1. All asteroids and comets visited by spacecraft as of November 2010

Although especially comets and the most primitive undifferentiated asteroids, which have undergone little physical and chemical alteration, represent most closely the properties of the primordial solar nebula, all types of small bodies are of scientific importance and it is necessary to understand their diversity and the relations between the different types and classes. Most asteroids are located in the main asteroid belt, where they are not easily accessible by spacecraft, but a significant number of those objects populate the near-Earth space and represent a sample of the main asteroid belt, which is much easier to access [1]. Near-Earth objects (NEOs) comprise thousands of asteroids (and some comets) with orbits that intersect or pass

*Image credits: Montage by Emily Lakdawalla. Ida, Dactyl, Braille, Annefrank, Gaspra, Borrelly: NASA / JPL / Ted Stryk. Steins: ESA / OSIRIS team. Eros: NASA / JHUAPL. Itokawa: ISAS / JAXA / Emily Lakdawalla. Mathilde: NASA / JHUAPL / Ted Stryk. Lutetia: ESA / OSIRIS team / Emily Lakdawalla. Halley: Russian Academy of Sciences / Ted Stryk. Tempel 1, Hartley 2: NASA / JPL / UMD. Wild 2: NASA / JPL.

near the orbit of Earth (perihelion ≤ 1.3 AU). 9808 objects have been discovered so far, 1075 of them being larger than ≈ 1 km in diameter[†]. All NEOs with an Earth Minimum Orbit Intersection Distance (MOID) ≤ 0.05 AU and an absolute magnitude H ≤ 22 (diameter $\gtrsim 200$ m) are termed Potentially Hazardous Objects (PHOs) or Potentially Hazardous Asteroids (PHAs). According to the latest discovery statistics, there are currently 1396 known PHOs, 195 of them with H ≤ 18 (diameter $\gtrsim 1$ km), and 7 of them with H ≤ 15 (diameter $\gtrsim 5$ km)[‡]. This population poses a significant hazard to human civilization and to life on Earth. Today it is widely accepted that NEO impacts have caused at least one mass extinction, 65 million years ago at the Cretaceous/Tertiary boundary, and they are suspected to have caused several global catastrophes before [2, 3]. A 2-km object is capable of causing catastrophic alteration of the global ecosystem [4]. Ocean impacts of even smaller objects are of some concern because the destructive potential caused by the resulting tsunamis may be above that from a land impact [4, 5]. Even objects that do not intersect Earth's orbit may evolve into Earth-crossers because their orbits are chaotic, having a relatively short dynamical lifetime in the order of $10^7 - 10^8$ years [6, 7, 8]. One day, it will become necessary to prevent a specific object from impacting Earth and causing a global catastrophe by nudging it out of its orbit. Depending on their physical properties, however, different objects may require different deflection strategies, so that a successful and efficient deflection of such an object requires a good understanding of the diversity of NEOs. Many NEOs have considerable inclinations and large synodic periods, which makes it difficult to rendezvous them by using conventional chemical propulsion systems. To explore their diversity, a multiple rendezvous mission would be very advantageous. The high ΔVs that are associated with such missions, however, are prohibitive for conventional spacecraft and even very challenging for solar electric propulsion systems [9]. Solar sails, in contrast, offer the advantage of (in principle) unlimited ΔV-capability for such missions and the number of NEOs that one single solar sailcraft can rendezvous is only limited by its lifetime.

In this paper, the current results of a technology reference study for a multiple NEO rendezvous mission with solar sailcraft are presented. This mission builds on previous concepts and studies [10, 11], as they are described in the next section, but adopts a strong micro-spacecraft philosophy, based on the DLR/ESA Gossamer technology demonstration missions Gossamer 1–3 [12] (see Section IV).

II. Previous Work on Solar Sail Missions to NEOs

The present study is not the first study that investigates the use of solar sails for NEO exploration. Already in August 2000, a dedicated mission for the exploration of NEAs with solar sailcraft (ENEAS) was proposed by DLR in cooperation with the University of Münster (Germany) as a candidate within the German small satellite program for space sciences. Using a $(50\,\mathrm{m})^2$ square sail with a specific mass

[†]As obtained on 08 May 2013 from the European Asteroid Research Node (EARN) website (http://earn.dlr.de).
[‡]As obtained on 08 May 2013 from the European Asteroid Research Node (EARN) website (http://earn.dlr.de).

(or sailcraft loading) of $30\,\mathrm{g/m^2}$, ENEAS was intended to rendezvous the near-Earth asteroid (NEA) $1996\,\mathrm{FG_3}$ with a *sail* payload (i.e., the total system except the sail assembly) of 75 kg within less than 5 years from launch [10, 13]. The 5-kg *scientific* payload comprised a CCD camera, an IR spectrometer, and a magnetometer. The ENEAS proposal, however, was not selected for funding because the technology readiness level (TRL) of solar sails was not considered to be high enough. The study of solar sail missions to NEOs, however, continued at DLR. In addition to ENEAS, also a multiple NEO rendezvous mission (ENEAS+), a NEO sample return mission (ENEAS-SR), and even a multiple NEO sample return mission (ENEAS+SR) was studied [11]. The most important mission parameters of those mission studies are summarized in Table 1.

Table 1. Comparison of the most important mission parameters for the four different ENEAS mission options (adapted from Ref. [11])

Mission	Objective	NEO target bodies	Mission duration [yr]
ENEAS	1 × rendezvous	$1996\,\mathrm{FG_3}$	4.2
ENEAS-SR	1 × sample return	$1996\,\mathrm{FG_3}$	10.0
ENEAS+	3 × rendezvous	$2000\,\mathrm{AG_6}$, $1989\,\mathrm{UQ}$, $1999\,\mathrm{AO_{10}}$	7.6
ENEAS+SR	3 × sample return	$2000\,\mathrm{AG_6}$, $1989\,\mathrm{UQ}$, $1999\,\mathrm{AO_{10}}$	10.1

Mission	Payload [kg]	Sail size [m]	Sail assembly loading $[\mathrm{g/m^2}]$	Char. acc. $[\mathrm{mm/s^2}]$	Launch mass [kg]
ENEAS	75	50	30.0	0.140	148
ENEAS-SR	295	70	22.7	0.100	406
ENEAS+	75	70	22.7	0.218	186
ENEAS+SR	295	139	22.7	0.218	734

Also at NASA MSFC's Advanced Concepts Office, multiple NEO rendezvous with solar sails have been studied recently [14]. In their study, a solar sail is used to rendezvous three NEOs ($1999\,\mathrm{AO_{10}}$, (99942) Apophis, and $2001\,\mathrm{QJ_{142}}$) consecutively. With a total mass of 243 kg, the $(80\,\mathrm{m})^2$ solar sailcraft achieves a characteristic acceleration of $0.35\,\mathrm{mm/s^2}$. Performing proximity operations of about 30 days at each object, it accomplishes the triple NEO rendezvous within a total mission duration of about 6 years.

III. Study and Mission Objectives

Given a renewed interest in solar sailing at DLR and ESA, and the development of a Gossamer roadmap that includes three consecutive in-orbit solar sail technology demonstration missions (Gossamer 1–3), the goal of the Gossamer Roadmap Technology Reference Study for a multiple NEO rendezvous mission is to take a fresh look at multiple NEO rendezvous missions and assess their suitability for a first scientific solar sailing mission that is based on the Gossamer technology. It is one of three studies that are currently carried out by international teams to identify scientific missions for which solar sailing is an enabling technology. The other two studies investigate a sub-L_1 space weather mission [15] and a solar polar mission [16].

The main mission objective of the Gossamer Roadmap Technology Reference Study for a multiple NEO rendezvous mission is to explore – during rendezvous and/or slow flybys – the diversity of NEOs in terms of bulk density, mineralogical composition, surface topography, surface texture, thermal properties, surface cratering age, and size-dependency of their features and structures. To keep the total mission duration and the engineering requirements within reasonable limits, three NEO rendezvous should be performed within a maximum mission duration of 10 years. If possible, also fast flybys of other NEOs should be made between two rendezvous. Such intermediate flybys, however, would probably be associated with longer flight times between the respective rendezvous.

The scientific mission requirements that the study team has defined are listed in Table 2. The selected scientific model payloads, which are based on state-of-the-art instruments (already flying or soon to be flying), are listed in Table 3. Their total mass is about 12 kg.

Table 2. Scientific requirements for the multiple NEO rendezvous mission

S1:	Rendezvous at least 3 NEOs with a stay time in the order of a few days (at least one rotation period) within a distance of less than 10 object diameters (rendezvous or very slow flyby)
S2:	Support several intermediate NEO flybys, if possible (including fast flybys)
S3:	Support a scientific payload of ≈ 12 kg (based on the state of the art of currently flying instruments)
S4:	Support observations at a low to moderate solar phase angle (requires significant agility, attitude and trajectory control near the target)
S5:	Provide a 3-axis stabilized (not spinning) platform to allow for pointing of body-mounted scientific payload
S6:	The mission lifetime should be ≈ 10 years
S7:	Support a data generation rate of ≈ 100 Mbyte/day

Table 3. Scientific model payloads for the multiple NEO rendezvous mission

#	Scientific model payload	Mass
MP1:	Multispectral imager (7 filters + clear), 100 mrad FOV, $1\,\mathrm{k}^2$ detector or larger: for visual imaging, similar to Dawn frame camera	≤ 5 kg
MP2:	Point spectrometer in the Vis-NIR range: for near-infrared mineralogy, similar to Dawn VIR mapping spectrometer	≤ 3 kg
MP3:	Infrared radiometer: for determination of thermal surface characteristics, similar to MASCOT MARA	< 1 kg
MP4:	3 small drop-probes: to measure free-fall descent velocity profile and target marking (based on the CubeSat standard)	< 1 kg each

At the target bodies, the sailcraft has to provide a 3-axis stabilized platform that allows for the pointing of the body-mounted scientific payloads to observe the targets at low to moderate solar phase angles. This requires sufficient agility for attitude and trajectory control near the target bodies.

No clear priority for the selection of NEO target bodies is given by the scientific community. All of them are generally considered as different and equally interesting. Therefore, priority was given according to the their relevance as targets for human NEO missions and for planetary defense. After direct interplanetary insertion with an hyperbolic excess energy $C3 \geq 0\,\mathrm{km}^2/\mathrm{s}^2$, the solar sailcraft should consecutively rendezvous three NEOs that comply with the following objectives:

1. At least one of the selected NEO target bodies should be a potentially hazardous object, on which a small transponder should be placed to determine its trajectory very accurately and to measure the non-gravitational forces that are acting onto the body in order to improve the knowledge of its future trajectory.

2. At least one of the selected NEO target bodies should be among the targets for a potential future human mission (see Table 4) to study its suitability for such a mission.

3. The last NEO target body should be a very small object (less than $\approx 20 - 50$ m in diameter) to place a transponder onto its surface, to demonstrate the feasibility of a solar sail as a gravity tractor [17, 18], and to study the controllability of the solar sailcraft and the gradual deflection of the object. This phase may take as long as the solar sail is operational.

Table 4. NASA list of potential NEO target bodies for future human missions

Designation	Orbital type
(251732) 1998 HG_{49}	Amor
1999 AO_{10}	Aten
2000 SG_{344}	Aten
2001 QJ_{142}	Apollo
2003 LN_6	Aten
(99942) Apophis	Aten
(341843) 2008 EV_5	Aten
2009 CV	Apollo
2009 HC	Apollo

An additional advantage of a solar sail for such a mission with respect to a conventional propulsion system is seen in the mission's post-launch flexibility. If, for example, a very interesting object is detected after the mission was launched – e.g., an object with a high short-term impact probability – it is still possible to change the mission profile and fly directly to the newly discovered object. Such a diversion would be difficult for an electric propulsion system and probably not be possible at all for a chemical propulsion system because of the likely high (unplanned) ΔV.

IV. Solar Sailcraft Design

The solar sailcraft design for this Gossamer Roadmap Technology Reference Study is based on the Gossamer technology that is currently developed at DLR in cooperation with ESA [12]. It adopts a strong micro-spacecraft philosophy, based on the DLR/ESA Gossamer technology demonstration missions (Gossamer 1-3) [12]. This ensures a small and lightweight spacecraft bus, so that only a modest deployed sail area is required for the mission, commensurate with the solar sail performance that can be achieved with the Gossamer technology. Off-the-shelf subsystems are used for the platform to provide a functional and capable deep space micro-spacecraft mission. More up-to-date information on the Gossamer technology can be found in several other papers that are presented at the 3rd International Symposium on Solar Sailing. In this paper, only a very preliminary design analysis is done. A more elaborate design has to be postponed for a later publication.

For the current Gossamer technology, the specific mass of the Upilex® sail film is $\sigma_{film} = 11.8\,\mathrm{g/m^2}$ and the specific mass of the CFRP booms is $\sigma_{boom} = 35\,\mathrm{g/m}$. The mass of the booms scales with the side length ℓ as $m_{boom} = 2\sqrt{2}\ell \cdot \sigma_{boom}$. The mass of the sail film scales with the sail area[§] $A = \ell^2$ as

[§]We neglect here that the real "propulsive" solar sail area is smaller than the squared side length of the solar sail.

$m_{film} = \ell^2 \cdot \sigma_{film}$. Assuming that all other solar sailcraft mechanisms, components, and subsystems do not depend to a great extent on the sail size, they can be embodied in the mass of the spacecraft bus m_{bus} (note that in this definition the spacecraft bus contains everything but the booms and the sail film, i.e., also those parts of the deployment modules which remain attached, and the scientific payload). Note that the mechanisms that are jettisoned after the boom and sail deployment are not to be taken into account. The total mass of a sailcraft with side length ℓ is then

$$m = m_{bus} + 2\sqrt{2}\ell \cdot \sigma_{boom} + \ell^2 \cdot \sigma_{film} \tag{1}$$

The sailcraft loading is

$$\sigma = \frac{m}{A} = \frac{m_{bus}}{\ell^2} + \frac{2\sqrt{2}}{\ell} \cdot \sigma_{boom} + \sigma_{film} \tag{2}$$

Therefore, for a given σ_{film} and σ_{boom}, the sail size depends on the required sailcraft loading σ and the bus mass m_{bus}. Solving eq. (2) for ℓ yields

$$\ell = \frac{\sqrt{2}\sigma_{boom} + \sqrt{2\sigma_{boom}^2 + m_{bus}(\sigma - \sigma_{film})}}{\sigma - \sigma_{film}} \tag{3}$$

Alternatively, eq. (3) can be written in terms of the characteristic acceleration $a_c = P_s/\sigma$, where P_s is the effective solar radiation pressure. In this case, the sail size depends on the required characteristic acceleration and the bus mass as

$$\ell = \frac{\sqrt{2}\sigma_{boom} + \sqrt{2\sigma_{boom}^2 + m_{bus}\left(\frac{P_s}{a_c} - \sigma_{film}\right)}}{\frac{P_s}{a_c} - \sigma_{film}} \tag{4}$$

Figure 2 shows the required sail side length over bus mass for different values of the characteristic acceleration and Fig. 3 shows the required sail side length over the characteristic acceleration for different values of the bus mass. In both figures, $P_s = 2\eta P_0 = 8.286\,\mu\mathrm{N/m^2}$ was used for the effective solar radiation pressure, where $\eta = 0.908$ is the sail efficiency parameter and $P_0 = 4.563\,\mu\mathrm{N/m^2}$ is the solar radiation pressure at Earth distance from the Sun [19].

V. Preliminary Potential Mission Scenario

In this section, the results for one potential mission scenario are given and the operations at the NEO target bodies are detailed. Preliminary trajectory optimization runs with the low-thrust trajectory optimization code InTrance [9, 20, 21] have shown that the following combination of NEO traget bodies would be feasible for a triple-rendezvous – using a solar sailcraft with $a_c = 0.3\,\mathrm{mm/s^2}$¶: Earth $\rightarrow 2004\,\mathrm{GU}_9$

¶The ideal SRP force model was used to reduce the time required for the preliminary trajectory optimization runs.

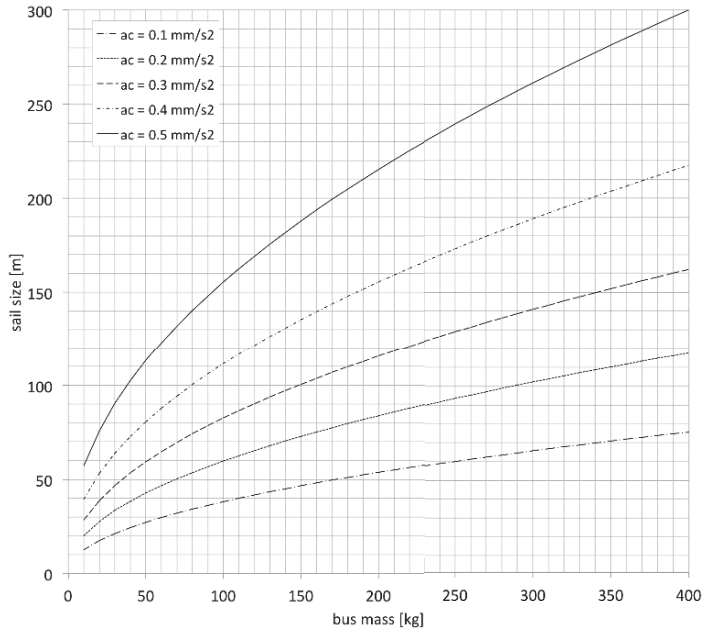

Figure 2. Sail side length over bus mass for different values of the characteristic acceleration

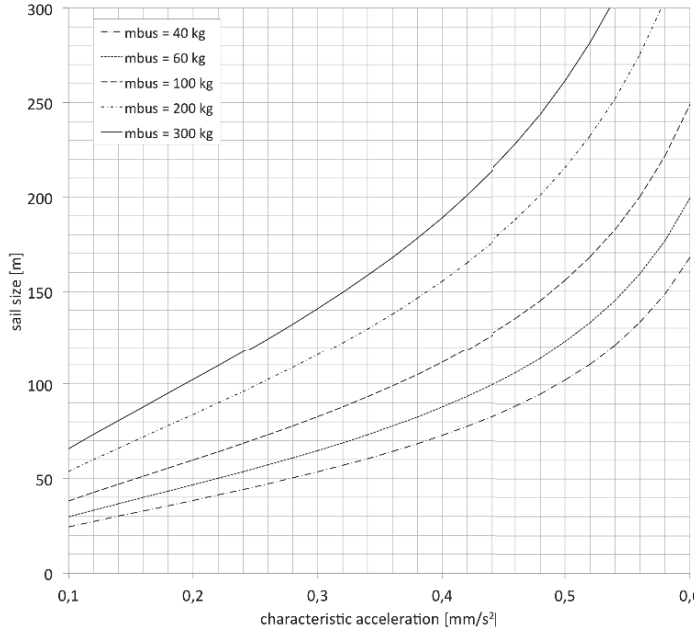

Figure 3. Sail side length over characteristic acceleration for different values of the bus mass

$\rightarrow 2001\,\mathrm{QJ}_{142} \rightarrow 2006\,\mathrm{QQ}_{56}$. The properties of those NEOs are summarized in Table 5[‖] . The first two targets, $2004\,\mathrm{GU}_9$ and $2001\,\mathrm{QJ}_{142}$, are potentially hazardous objects (PHOs), $2001\,\mathrm{QJ}_{142}$ being also a member of the list of potential NEO target bodies for future human missions (Table 4). The third target, $2006\,\mathrm{QQ}_{56}$, is very small, so that a change of its orbit by a solar sailcraft hovering close to it should be measurable.

Table 5. Properties of the selected NEO targets.

	$2004\,\mathrm{GU}_9$	$2001\,\mathrm{QJ}_{142}$	$2006\,\mathrm{QQ}_{56}$
Orbital type	Apollo	Apollo	Aten
Taxonomic type	?	?	?
Semi-major axis [AU]	1.00119	1.06183	0.984896
Eccentricity	0.136319	0.086133	0.045581
Inclination [deg]	13.65	3.103	2.796
Absolute magnitude [mag]	21.1 – 21.2	23.5 – 23.7	25.9
Estimated size [m]	170 – 380	50 – 120	10 – 40
Estimated rotation period [h]	?	0.16	?
Earth MOID [AU]	0.00056	0.0094	0.01519
Potentially hazardous object	yes	yes	no

For such a mission, a bus mass of $40 - 60\,\mathrm{kg}$ may be realistic, if compared to the more elaborate mass breakdown of a sub-L_1 space weather mission that is also based on the Gossamer technology [15]. A complete mass breakdown, however, especially with respect to the communication system, still has to be done. According to eq. (3), a characteristic acceleration of $0.3\,\mathrm{mm/s^2}$ yields sail sizes between $(54\,\mathrm{m})^2$ and $(65\,\mathrm{m})^2$, which are also realistic for a near-term solar sailing mission. This scenario, however, is still preliminary and has to be re-assessed and optimized with respect to the following criteria:

- Reduction of total mission duration
- Reduction of required characteristic acceleration (and therefore sail size)
- Prioritization with respect to the target objects' hazard potential

The final goal of our investigations is to obtain a larger set of NEO target combinations and required characteristic accelerations. This, however, requires much more trajectory analysis and optimization than it was done until the writing of this paper. The final goal is to achieve a triple rendezvous with

[‖]The data in Table 5 were obtained on 08 May 2013 from the European Asteroid Research Node (EARN) website (http://earn.dlr.de), the Near Earth Objects Dynamic Site (NEODyS) website (http://newton.dm.unipi.it/neodys/) and the Minor Planet Center (MPC) website (http://www.minorplanetcenter.org).

a characteristic acceleration of about $0.2\,\mathrm{mm/s^2}$, which would result in sail sizes between about $(39\,\mathrm{m})^2$ and about $(48\,\mathrm{m})^2$, depending on the bus mass.

A. Potential Combination for a Triple NEO Rendezvous

Because it is difficult and time consuming for solar sailcraft with a low characteristic acceleration to gain orbital energy in the Earth's gravitational field, the launcher inserts the sailcraft directly into an interplanetary trajectory ($C_3 = 0\,\mathrm{km^2/s^2}$). After injection and deployment, the sail is oriented to follow the pre-calculated attitude profile that leads to an optimal interplanetary transfer trajectory to the first target body, $2004\,\mathrm{GU_9}$ (transfer leg 1, see Fig. 4). During the transfer, the sailcraft runs almost autonomously, so that ground monitoring is carried out on a weekly basis only [13]. At the end of leg 1, the solar sailcraft makes a rendezvous with $2004\,\mathrm{GU_9}$, after a flight time of 1111 days (3.04 years). The proximity operations at the target objects are described in the next subsection. Table 6 collects the most relevant mission parameters.

After a stay time of 113 days to investigate the object in detail, the solar sailcraft flies to the next object, $2001\,\mathrm{QJ_{142}}$ (transfer leg 2, flight time of 1396 days (3.82 years), see Fig. 5), where it stays for 90 days. Afterwards, the solar sailcraft flies to the last NEO, $2006\,\mathrm{QQ_{56}}$, which is a very small body with a diameter of only about 10–40 meters. There, it will demonstrate the feasibility of a solar sail as a gravity tractor [17, 18]. Hovering next to the NEO, the gravitational acceleration that the solar sailcraft exerts onto the NEO will change the orbit of the NEO. After some time, this orbit change will be measurable, if it is larger than the non-gravitational accelerations that are acting onto the asteroid (an issue that is still to be investigated). The acceleration that the NEO exerts onto the solar sail is compensated by the solar radiation pressure, as it is described in the next subsection. The total mission duration from launch until arrival at the final target body is 9.16 years.

Table 6. Mission parameters for triple NEO rendezvous

	Start	End	Duration [days]
Transfer leg 1	28 Nov 2019	13 Dec 2022	1111
Stay at $2004\,\mathrm{GU_9}$	13 Dec 2022	05 Apr 2023	113
Transfer leg 2	05 Apr 2023	30 Jan 2027	1396
Stay at $2001\,\mathrm{QJ_{142}}$	30 Jan 2027	30 Apr 2027	90
Transfer leg 3	30 Apr 2027	24 Jan 2029	635
Arrival at $2006\,\mathrm{QQ_{56}}$	24 Jan 2029		

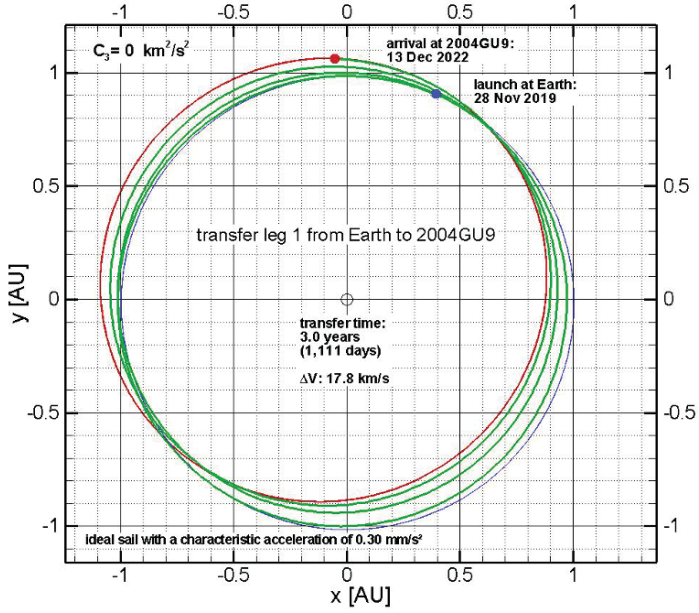

Figure 4. Transfer leg 1 from Earth to $2004\,\mathrm{GU}_9$

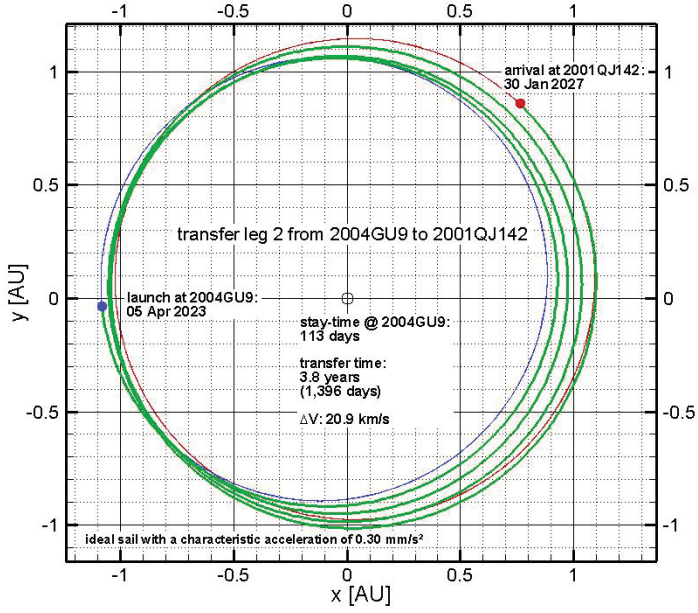

Figure 5. Transfer leg 2 from $2004\,\mathrm{GU}_9$ **to** $2001\,\mathrm{QJ}_{142}$

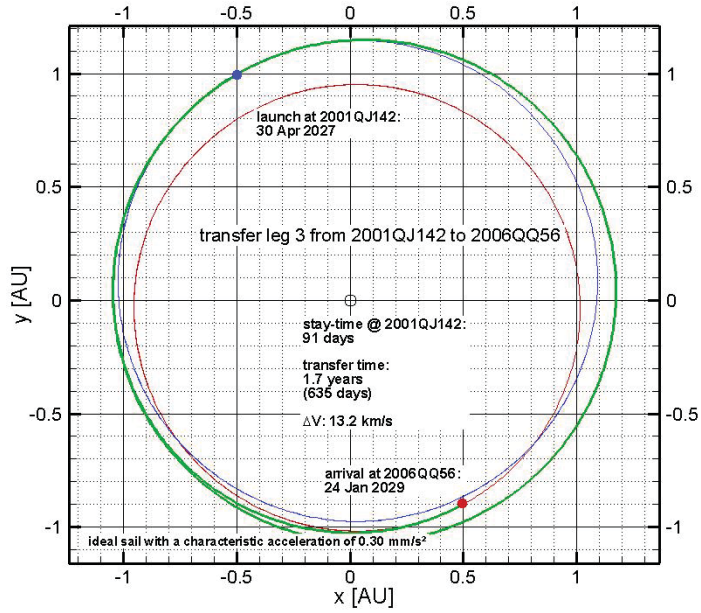

Figure 6. Transfer leg 3 from $2001\,QJ_{142}$ to $2006\,QQ_{56}$

B. Operations at the NEO Target Bodies

Even in the near-field of the NEO target bodies, the solar radiation pressure acceleration is larger than the gravitational acceleration, so that the sailcraft can be placed along a continuum of hovering positions in the hemisphere that is opposite to the sun [22, 23]. On those quasi-stationary hovering positions, the SRP acceleration cancels the gravitational acceleration of the sun and the NEO, as well as the small centrifugal acceleration that is due to the slightly larger distance from the sun (Fig. 7). Those positions are unstable in the Lyapunov sense but can be stabilized using a feedback attitude control loop, so that they can be maintained for extended periods of time. Hovering near the NEO, the (likely complex) gravitational field is studied, so that a coarse gravitational field model can be determined.

Additional to the observation of the NEO and the determination of its gravitational field, a small transponder will be placed onto the surface of the NEO, so that its trajectory can be determined with a very high accuracy. This would allow the measurement of the non-gravitational forces that are action onto the object.

VI. Conclusions and Future Work

Building on previous concepts and studies, the Gossamer Roadmap Technology Reference Study for a multiple NEO rendezvous mission adopts a strong micro-spacecraft philosophy, which is based on

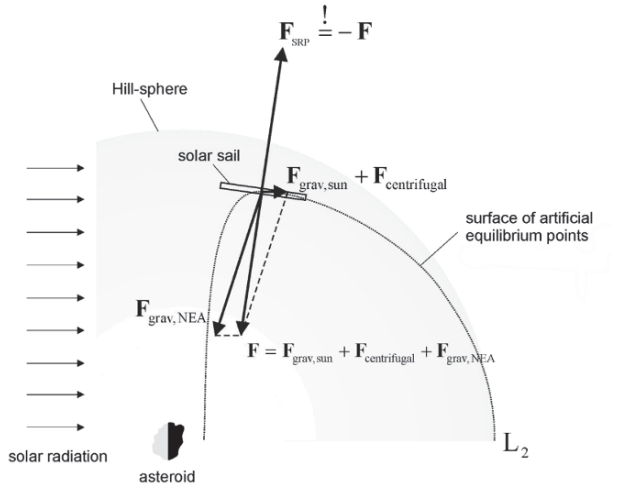

Figure 7. Hovering at a NEO target body

the DLR/ESA Gossamer technology demonstration missions Gossamer 1–3, which are currently under development. It is one of three studies that are currently carried out by international teams to identify scientific missions for which solar sailing is an enabling technology. The objective of the multiple NEO rendezvous mission is to explore the diversity of NEOs. A preliminary mission analysis has shown that a solar sail with a characteristic acceleration of $0.3\,\mathrm{mm/s^2}$ is able to rendezvous 3 NEOs within less than 10 years, but we expect that better combinations of NEO target bodies can be found that lower the required characteristic acceleration for a 10-year mission to about $0.2\,\mathrm{mm/s^2}$, which would allow for a smaller solar sail and/or a heavier sailcraft bus. Further investigations in this direction will give a relation between sailcraft performance and mission duration. Before this study is finished, it is also necessary to elaborate – based on the required characteristic acceleration for this mission – a more detailed solar sailcraft design. The proximity operations of solar sails at small bodies with an irregular gravitational field is a wide research discipline in its own that requires more research before such a mission can be undertaken. This is also important because it is a key driver for the sailcraft's attitude control design.

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Gossamer Roadmap Technology Reference Study for a Sub-L_1 Space Weather Mission

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A technology reference study for a displaced Lagrange point space weather mission is presented. The mission builds on previous concepts, but adopts a strong micro-spacecraft philosophy to deliver a low mass platform and payload which can be accommodated on the DLR/ESA Gossamer-3 technology demonstration mission. A direct escape from

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Geostationary Transfer Orbit is assumed with the sail deployed after the escape burn. The use of a miniaturized, low mass platform and payload then allows the Gossamer-3 solar sail to potentially double the warning time of space weather events. The mission profile and mass budgets will be presented to achieve these ambitious goals.

Nomenclature

a = sail acceleration vector

n = sail normal vector

r = sail position vector relative to barycenter (rotating frame)

$\mathbf{r}_1$ = sail position vector relative to Sun (rotating frame)

$\mathbf{r}_2$ = sail position vector to Earth (rotating frame)

U = augmented 3-body potential field

α = sail pitch angle

β = sail lightness number

η = sail reflectivity

θ = sail cone angle

μ = three-body mass ratio

ϕ = sail center-line angle

$\boldsymbol{\omega}$ = angular velocity of rotating frame of reference

I. Introduction

SOLAR sailing has long been considered as an enabling technology for enhanced space weather missions, exploiting solar radiation pressure to deliver artificial equilibria in the restricted three-body problem [1-3]. Using a modest solar sail, rich new families of artificial equilibria appear which allow the sail to be located sunwards of the classical L_1 point [4-6]. From this new vantage point space weather events are detected earlier than for conventional space weather missions at L_1, thereby significantly increasing warning times. This is of operational importance as the growing global digital economy becomes more reliant on satellite applications, such as accurate satellite navigation services which can be severely degraded by space weather events.

The mission concept proposed in this paper adopts previous mission architectures for displaced-L_1 space weather missions, such as Geostorm and Sunjammer [1-3], but uses a micro-spacecraft philosophy in an attempt to reduce the mission mass budget to fit within the envelope of the DLR Gossamer-3 solar sail [7]. The

subsequent displaced-L_1 mission concept (DL1) then provides an interface between the DLR Gossamer-1 and 2 technology demonstration missions in Earth orbit, and a useful first operational solar sail mission using DLR gossamer technology. As will be shown, if significant miniaturization of both the science payload and platform is possible, a space weather mission with a 65 x 65 m sail and a launch mass of order 110 kg appears possible, using existing DLR Gossamer-3 technology.

II. Mission Requirements

The top-level mission requirement is that the DL1 mission shall double the warning time of impending space weather events. This requirement will drive the required sail characteristic acceleration and hence the deployed sail area. A doubling of the current warning time (typically 12 min for fast 2000 kms^{-1} solar wind to 60 minutes for slower 350-400 kms^{-1} solar wind) is selected as the top-level mission requirement in order to provide a significant enhancement over conventional space weather services from the classical L_1 position.

In order to meet this top-level objective, and provide a first operational mission after the Gossamer-1 and 2 technology demonstrators, a set of other mission requirements are defined, as listed in Table 1. The requirement for a doubling of the warning time of impending space weather events defines the location of the DL1 position (requirement M2), while the need to avoid the solar radio disk defines a small offset of the DL1 position from the Sun-Earth line (requirement M3). M2 therefore drives a requirement for a sail characteristic accelerator of 0.3 mms^{-2}, assuming a conservative effective sail reflectivity of 0.85.

To limit the demands on the solar sail assembly, a kick-stage is assumed to deliver the stowed solar sail to Earth escape, avoiding the need for long Earth escape spirals and rapid sail turning rates (requirement M1). Similarly, to ensure continuity, the sail assembly shall be a direct evolution from the Gossamer-1 and 2 demonstrators (requirement M5), using thin 2 μm film to provide an improved sail characteristic acceleration. Finally, in order to provide a space weather service of duration comparable to missions using the classical L_1 point (such as ACE), a mission duration of 10 years shall be required.

Table 1 Top level mission requirements

	Requirement	Comment
M1	Sail shall be deployed after Earth escape	*Kick-stage to escape from GTO*
M2	Sail shall station-keep at DL1 (double L_1 distance)	*Sail acceleration of 0.3 mms^{-2}*
M3	Sail will station-keep away from solar radio disk	*Require active control of sail*
M4	Mission will provide space weather data for 10 years	*To meet ACE performance*
M5	Design will use DLR Gossamer-3 technology	*Require 2 μm film with 65 m sail*

In order to provide enhanced space weather services, a set of mission science requirements are defined, as listed in Table 2. First, the requirement that the mission will double space weather event warning times (S1) flows directly from mission requirement M1. In order to deliver this requirement a set of measurements are required, specifically 3-axis magnetic field along with solar wind speed and density (S2), as noted in Table 2. Accuracy requirements are then set (S3) along with knowledge of the spacecraft attitude (S4). Importantly, it will be assumed that the science instruments are free from interference from the sail membrane through a central cut-out, or the impact of the membrane on the instruments is known, allowing measurements to be de-convolved.

Table 2 Mission science requirements

	Requirement	Comment
S1	DL1 mission shall double event warning time	*Top level mission goal*
S2	Measure 3-axis B-field, solar wind speed and density	*All 3 measurements are essential*
S3	Min. 1 nT with 0.1 nT (1 Hz) goal, wind speed 10 km/s	-
S4	Knowledge of spacecraft attitude better than 1 deg	*From ACS system*
S5	Cut-out in center of sail assembly for instruments	*c.f. Geostorm/Sunjammer*

III. Mission Architecture

The DL1 mission requires the use of a solar sail to enable the mission science payload to remain in equilibrium, sunward of the classical L_1 Lagrange point. The underlying dynamics are shown schematically in Fig. 1. At the classical L_1 point, 1.5 M km sunward of Earth, the outwards force due to the gravitational attraction of the Earth, and the centripetal force due to orbital motion about the Sun, balance the inwards forced due to the gravitational attraction of the Sun. At the DL1 position, 3M km sunward of Earth, it can be seen that an additional outwards force is required for equilibrium, due to the increase in inwards force due to the gravitational attraction of the Sun, and decreased gravitational attraction of the Earth and centripetal force. This additional force which is necessary for equilibrium is provided by the solar sail. Since the effective 3-body potential is relatively flat close to the classical L_1 point, only a modest acceleration is required from the solar sail to displace the equilibrium point sunwards. This is the key advantage of the DL1 mission concept as an early operational solar sail mission. A detailed analysis of the requirements for artificial equilibria in the Sun-Earth restricted 3-body problem is provided in Appendix A, both for an ideal and a partially reflecting solar sail [4-5]. This analysis can be used to determine both the sail assembly loading and the sail cone angle required for an artificial equilibrium position in the vicinity of the classical L_1 Lagrange point.

It should be noted that the operational DL1 position is also displaced from the Sun-Earth line. A spacecraft stationed directly on the Sun-Earth line will be at the center of the solar radio disk as viewed from Earth, with

significant impacts on telecommunications. In order to displace the DL1 position away from the Sun-Earth-line, a small pitch bias is required to generate a transverse force to offset the artificial equilibrium position.

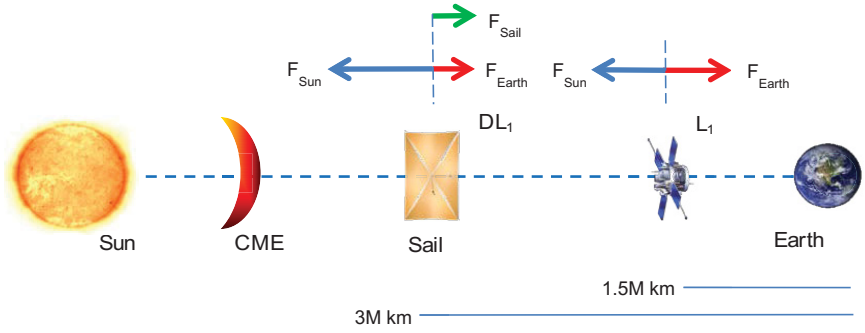

Fig. 1 Schematic DL1 mission concept.

In order to deliver the solar sail to the DL1 artificial equilibrium position, a direct transfer strategy will be adopted, as shown schematically in Fig. 2. The stowed solar sail and kick-stage will be delivered as a secondary payload on a midnight launch to GTO. The kick-stage will then be used to transfer the stowed solar sail to a ballistic trajectory towards the DL1 position, with the sail deployed after Earth escape. This transfer strategy avoids using the solar sail for a slow Earth escape spiral which would require rapid turn rates. Such rapid turn rates impose severe drivers on the sail attitude control system to ensure that sufficient slew rates can be achieved, particular as the sail passes through perigee. In addition, the low perigee of GTO leads to air drag and dynamic pressure loads on the sail assembly.

Once on-station at the DL1 position, the sail is used to provide the additional force required to maintain the artificial equilibrium position, with only modest slew rates required for station-keeping. Active station-keeping is required since the artificial equilibria are unstable, although are fully controllable using state feedback to the sail attitude [4]. Again, it is noted that the sail requires a small pitch bias to generate a side force to ensure that the artificial equilibrium position is displaced from the Sun-Earth line.

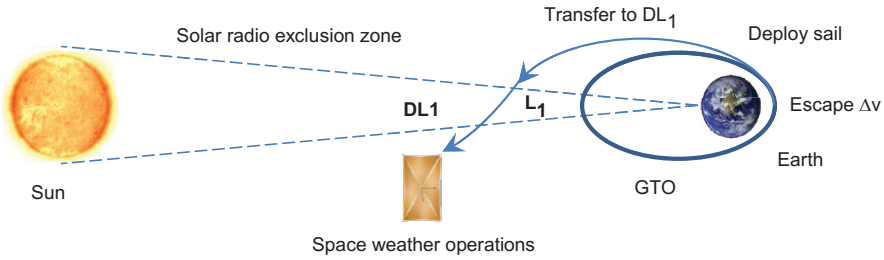

Fig. 2 Schematic DL1 transfer concept.

IV. Spacecraft Sizing

A. Science payload

In keeping with the micro-spacecraft philosophy used in this paper, a suite of miniaturized instruments are selected, based on existing CubeSat heritage. These include the MAGIC magnetometer and the ChaPS plasma analyzer [8]. Such instrumentation has been developed for missions such as QB50 and provide the means to ensure that science payload and platform can fit within the envelope of Gossmaer-3 technology. The instrument mass budget, shown in Table 3, has a total mass of order 1 kg, including a deployable boom for the magnetometer.

Table 3 Science instrument mass budget

Instrument	Mass (kg)	Comment
Magnetometer	0.17	*Imperial College MAGIC instrument*
Boom	0.12	*1 m deployable boom and harness*
Plasma analyzer	0.5	*MSSL ChaPS instrument*
Langmuir probe	0.2	*SWARM instrument*
Total	**1**	*QB50/TechDemoat instrument heritage*

B. Platform sizing

As noted earlier, in order to provide a first operational solar sail mission using Gossamer-3 technology, a micro-spacecraft philosophy is required. This will ensure that a small spacecraft bus, and so modest deployed sail area are required for the mission, commensurate with the Gossamer-3 solar sail assembly. In order to provide a representative mass budget, an existing deep space micro-spacecraft concept will be considered. The platform uses off-the-shelf subsystems to provide a functional deep space micro-spacecraft mission with a low platform mass. The mass budget for the spacecraft platform is listed in Table 4. Allowing a mass margin of 30%, a total

platform mass of 14 kg is required. Including the 1 kg instrument suite, a total mass of order 15 kg is therefore defined as useful payload for the solar sail.

C. Sail assembly sizing

The total mass budget for the DL1 mission can now be determined from the science payload and platform mass, as listed in Table 5. A total sail payload of 15 kg is required (1 kg science payload, 14 kg platform (inc. 30% margin)) with a required solar sail characteristic acceleration of 0.3 mms^{-2} (assuming an effective sail efficiency of 0.85) to deliver the top-level mission goal of doubling the warning time of impending space weather events. Using the DLR gossamer-3 sail assembly with 2 micron film a sail assembly loading 22.2 g m^{-2} is provided on a 65 x 65 m sail with a total sail assembly mass of 95 kg [7]. This provides the required 15 kg of solar sail payload mass. The total spacecraft mass is therefore 110 kg, which is delivered to GTO with a kick-stage for Earth escape. The schematic spacecraft platform and solar sail concept is shown in Fig. 3.

Table 4 Platform mass budget

Item	No.	Unit mass (kg)	Total mass (kg)
Structure	1	2.6	2.6
Sun sensors	6	0.035	0.21
Star trackers	2	0.375	0.75
Flight computer	1	0.75	0.75
Solar arrays	6	0.5	1.5
Batteries	16	0.0445	0.7
Power electronics	1	0.8	0.8
S-band radio	1	0.7	0.7
Diplex	1	0.25	0.25
Omni-antenna	2	1.0	2.0
Unallocated	-	-	4 (30% margin)
Total			**14**

Table 5 Spacecraft mass budget

Item	Mass (kg)	Comment
Science payload	1	*3-axis B-field, plasma speed and density*
Platform	14	*Micro-spacecraft*
Subtotal	**15**	*Includes 4 kg unallocated margin*
Sail assembly	95	*65 x 65 m DLR sail with 2 μm film*
Total	**110**	*Total mas to be delivered to C_3=0*

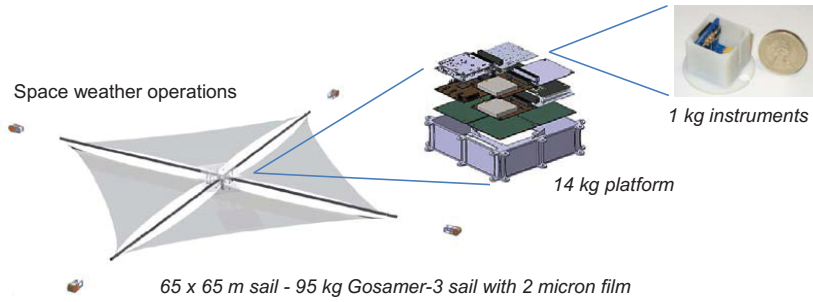

Fig. 3 Schematic DL1 spacecraft concept.

V. Trajectory Analysis

In order to generate an optimal transfer from GTO to the DL1 position, the optimal control problem to be solved can be described as follows: find the state history, and control history in the Sun-Earth three-body problem that minimizes the cost function $J = \Delta V_{GTO} + w\left(t_f - t_0\right)$, which is a trade-off between the impulse required for Earth escape ΔV_{GTO} and the time of flight to DL1 through a weighting factor w. The optimal control problem defined is solved with PSOPT, a particular implementation of a direct pseudo-spectral method. In the analyses for this paper, Legendre polynomials are used to approximate and interpolate the dependent variables at the nodes and IPOPT (Interior Point OPTimizer) is used as the large-scale NLP solver [9].

Note that the optimal control problem only spans the solar sail phase and that the ballistic phase only comes into play when computing ΔV_{GTO} in the objective function. The ballistic phase is thus not part of the state or control history.

In order to initialize the optimization, PSOPT requires an initial guess. Here the initial guess is assumed to be a Hohmann transfer from perigee of GTO to the DL1 position. Clearly, this initial guess has errors mainly due to neglecting third body perturbations from the Sun, but appears to provide a stable optimization and quick convergence within PSOPT [10].

The optimized transfer is shown in Fig. 4, with the ballistic arc represented in grey, while the solar sail arc is presented in blue. The projection of the transfer on the *x*-*z* plane shows that the transfer is almost entirely located in the ecliptic plane. The total time of flight is 201 days, comprising 25 days for a ballistic arc following escape from GTO and 176 days for the solar sail arc to the DL1 position. The required Δv for the kick-stage to deliver the stowed sail from GTO to escape is 749.5 ms^{-1}.

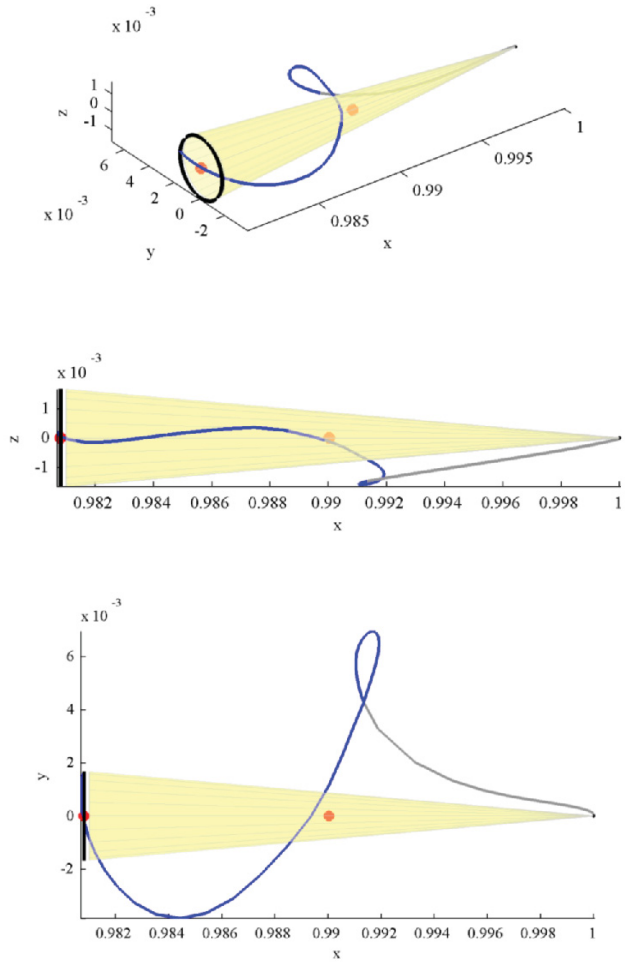

Fig. 4 Minimum-time trajectory from Earth scape to sub-L1 position on edge of 5° solar radio disk (barycentric coordinates (AU), grey – coast arc, blue – solar sail arc).

VI. Propulsion System Comparison

In order to provide a comparison with chemical and SEP propulsion technologies, the mass fraction required for the DL1 mission can be found. For a fixed mission duration and assumed specific impulse, the mass fraction remaining can be determined, as shown in Table. 6. It can be seen that chemical propulsion is entirely unsuitable for the DL1 mission, while SEP is competitive, but only for mission durations of a few years. Assuming that the

DL1 mission reaches the goal of 10 years of operations (comparable with conventional missions such as ACE), then solar sailing is clearly the enabling propulsion technology, and hence DL1 is an attractive first operational solar sail mission for Gossamer-3 sail technology.

Table 6 Comparison of propulsion options

Mission duration (yrs)	Effective Δv (ms^{-1})	Chemical mass ratio	SEP mass ratio
1	9,467	0.05	0.72
2	18,935	0.002	0.53
5	47,336	-	0.20
10	94,673	-	**0.04**

VII. Conclusions

A micro-spacecraft philosophy has been used to configure a space weather solar sail mission using Gossamer-3 sail technology. Using a miniaturized instrument suite being developed for CubeSat-class missions, and existing micro-spacecraft platform subsystems, a 15 kg platform can be stationed at a displaced L_1 point (DL1), sunward of the classical L_1 point to double the warning times of impending space weather events. A 65 x 65 m square solar sail using Gossamer-3 technology and 2 μm film is required. The stowed sail is injected to Earth escape from GTO, after which the sail deploys and transfers to the DL1 position. For a mission duration of up to 10 years, solar sailing is a truly enabling propulsion technology, demonstrating that the concept represents an attractive first operational solar sail mission using Gossamer-3 sail technology.

Appendix A

1. Equilibrium solutions for an ideal solar sail

Firstly, equilibrium solutions for an idealised, perfectly reflecting solar sail will be derived. The ideal sail will be considered in a frame of reference co-rotating with two primary masses m_1 (Sun) and m_2 (Earth) at constant angular velocity $\boldsymbol{\omega}$, as shown in Fig. A1. The sail attitude is defined by a unit vector $\mathbf{n}$ normal to the sail surface, fixed in the rotating frame of reference. In addition, the ratio of the solar radiation pressure force to the solar gravitational force exerted on the sail is defined by the sail lightness number β. Since both forces have an inverse square variation with solar distance, the sail lightness number is a constant. It can be shown that the sail lightness number is related to the total solar sail mass per unit area by $\sigma[\text{g m}^{-2}]=1.53/\beta$. The units of the problem will be chosen such that the gravitational constant, the distance between the two primary masses and the sum of the primary masses are all taken to be unity.

The vector equation of motion for a solar sail in this rotating frame of reference may be written in standard form as

$$\frac{d^2\mathbf{r}}{dt^2} + 2\boldsymbol{\omega} \times \frac{d\mathbf{r}}{dt} + \nabla U = \mathbf{a} \tag{1}$$

with the three-body gravitational potential U and the solar radiation pressure acceleration **a** defined by

$$U = -\left[\frac{1}{2}\left(x^2 + y^2\right) + \frac{1-\mu}{r_1} + \frac{\mu}{r_2}\right] \tag{2a}$$

$$\mathbf{a} = \beta \frac{1-\mu}{r_1^{\,2}} \left(\hat{\mathbf{r}}_1 . \mathbf{n}\right)^2 \mathbf{n} \tag{2b}$$

where $\mu = m_2 / \left(m_1 + m_2\right)$ is the mass ratio of the system and the sail position vectors are defined as $\mathbf{r}_1 = \left(x + \mu, y, z\right)$ and $\mathbf{r}_2 = \left(x - \left(1 - \mu\right), y, z\right)$.

Equilibrium solutions are now required in the rotating frame of reference so that the first two terms of Eq. (1) vanish. The five classical Lagrange points are then obtained as the solutions to $\nabla U = 0$ with $\hat{\mathbf{r}}_1 . \mathbf{n} = 0$ and so **a**=0. However, for $\hat{\mathbf{r}}_1 . \mathbf{n} \succ 0$ there is an additional acceleration **a**, which is a function of the lightness number β and attitude **n**, so that new artificial equilibrium solutions may be generated. Since the vector **a** is oriented in direction **n**, taking the vector product of **n** with Eq. (1) it follows that

$$\nabla U \times \mathbf{n} = 0 \;\; \Rightarrow \;\; \mathbf{n} = \lambda \nabla U \tag{3}$$

where λ is an arbitrary scalar multiplier. Using a normalisation condition of the sail normal vector, λ is identified as $|\nabla U|^{-1}$ so that the required sail attitude is defined by

$$\mathbf{n} = \frac{\nabla U}{|\nabla U|} \tag{4}$$

which can be used to obtain the sail pitch angle α. The required sail lightness number may also be obtained by taking a scalar product of Eq. (1) with **n**. Again requiring an equilibrium solution it is found that

$$\beta = \frac{r_1^{\,2}}{\left(1-\mu\right)} \frac{\nabla U . \mathbf{n}}{\left(\hat{\mathbf{r}}_1 . \mathbf{n}\right)^2} \tag{5}$$

Since the sail lightness number and attitude can be selected, the set of five classical Lagrange points will be replaced by an infinite set of artificially generated equilibrium surfaces.

The regions in which these new solutions may exist are defined by the constraint $\hat{\mathbf{r}}_1.\nabla U \geq 0$ with a boundary surface defined by an equality. This constraint may be understood physically since the solar radiation pressure acceleration vector **a**, and so the sail attitude vector **n**, can never be directed Sunward. The boundary surface has two topologically disconnected surfaces S_1 and S_2 which define the region of existence of equilibrium solutions near the m_2, as shown in Fig. A2. The classical equilibrium solutions lie on either S_1 or S_2 since they are the solutions to $\nabla U = 0$. Surfaces of constant sail lightness number generated from Eq. (5) for the Earth-Sun system are also shown in Fig. A2. In general, the surfaces of constant sail lightness number approach the boundaries asymptotically with $\beta \to \infty$ when $\hat{\mathbf{r}}_1.\nabla U \to 0$ as is clear from Eq. (5). It can be seen that as the sail lightness number increases larger volumes of space are accessible for artificial equilibrium positions.

2. Equilibrium solutions for a partially reflecting solar sail

A realistic solar sail force model which includes absorption will now be considered. To allow a closed-form solution, the solar sail will be assumed to have perfect specular reflectivity and no thermal re-emission but will still have an overall reflectivity η less than unity. Then, the radiation pressure acceleration will act in direction **m** and may be written as the sum of components normal **n** and transverse **t** to the sail surface such that

$$a\mathbf{m} = \frac{1}{2}\beta\frac{1-\mu}{r_1^2}(1+\eta)(\hat{\mathbf{r}}_1.\mathbf{n})^2\mathbf{n} + \frac{1}{2}\beta\frac{1-\mu}{r_1^2}(1-\eta)(\hat{\mathbf{r}}_1.\mathbf{n})(\hat{\mathbf{r}}_1.\mathbf{t})\mathbf{t} \tag{6}$$

It can be seen that the main effect of the non-perfect reflectivity of the sail is to reduce the acceleration magnitude and to introduce an off-set in the direction of the radiation pressure acceleration. The acceleration now acts in direction **m** rather than normal to the sail surface in direction **n**. This off-set is defined by the centre-line angle ϕ with the actual radiation pressure force direction defined by the cone angle θ, as shown in Fig. A1.

The analysis presented in the previous section will be repeated using the sail force model defined by Eq. (6) so that the equation of motion may now be written as

$$\frac{d^2\mathbf{r}}{dt^2} + 2\boldsymbol{\omega}\times\frac{d\mathbf{r}}{dt} + \nabla U = a\mathbf{m} \tag{7}$$

For an equilibrium solution the first two terms of Eq. (7) will again vanish so that the sail attitude must be chosen as

$$\mathbf{m} = \frac{\nabla U}{|\nabla U|} \tag{8}$$

The unit vector **m** can now be defined by the cone angle θ between the radial direction $\hat{\mathbf{r}}_1$ and **m** as

$$\tan\theta = \frac{\left|\hat{\mathbf{r}}_1 \times \nabla U\right|}{\hat{\mathbf{r}}_1 . \nabla U} \tag{9}$$

In addition, using Eq. (6) the centre-line angle can be obtained from the ratio of the transverse and normal accelerations as

$$\tan\phi = \frac{1-\eta}{1+\eta}\tan\alpha \tag{10}$$

where the sail pitch angle $\alpha=\theta+\phi$. Noting that **n.t**=0 and taking a scalar product of Eq. (7) with the unit vector **n** gives the required sail lightness number as

$$\beta = \frac{2r_1^{\,2}}{1-\mu}\frac{\nabla U.\mathbf{n}}{(1+\eta)(\hat{\mathbf{r}}_1.\mathbf{n})^2} \tag{11}$$

The centre-line angle may be obtained explicitly by again noting that $\alpha=\theta+\phi$. Then, after some reduction, Eq. (10) yields the centre-line angle directly from the cone angle as

$$\tan\phi = \frac{\eta}{(1+\eta)\tan\theta}\left[1-\left[1-\frac{1-\eta^2}{\eta^2}\tan^2\theta\right]^{1/2}\right] \tag{12}$$

Lastly, using Eq. (11) it is found that the required sail lightness number may be obtained in terms of the lightness number for an ideal solar sail $\tilde{\beta}$ as

$$\beta = \frac{2}{(1+\eta)}\frac{\sqrt{1+\tan^2\phi}}{(1-\tan\theta\tan\phi)^2}\tilde{\beta} \tag{13}$$

where $\tilde{\beta}$ is defined by Eq. (5). Therefore, using Eqs. (9), (12) and (13) the sail orientation and sail lightness number required for an artificial equilibrium position can be obtained.

The effect of a non-ideal solar sail is shown in Fig. A3 for a reflectivity of 0.8. Firstly, it can be seen that the volume of space available for equilibrium solutions about L_2 is significantly reduced. This is due to the centre-line angle which limits the direction in which the solar radiation pressure force vector can be oriented. For solutions near L_1 the main effect of the non-ideal sail is to displace the equilibrium solutions towards the Earth. This is due to the reduction in the

magnitude of the solar radiation pressure force, rather than the centre-line angle. In general then, equilibrium solutions sunward of L_1 are not greatly affected by a realistic sail while equilibrium solutions about L_2 are severely constrained.

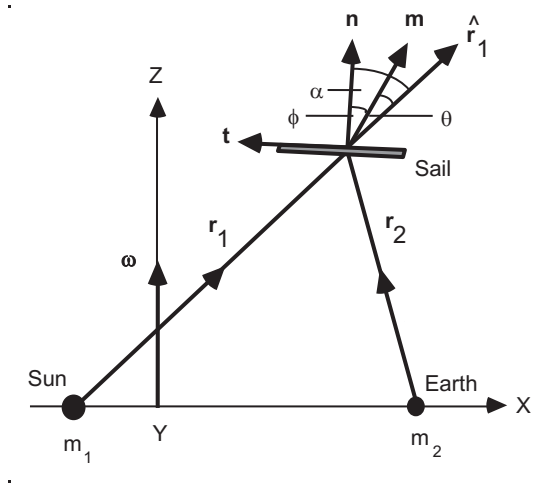

Fig. A1 Artificial equilibria in the restricted three-body problem for a non-ideal solar sail.

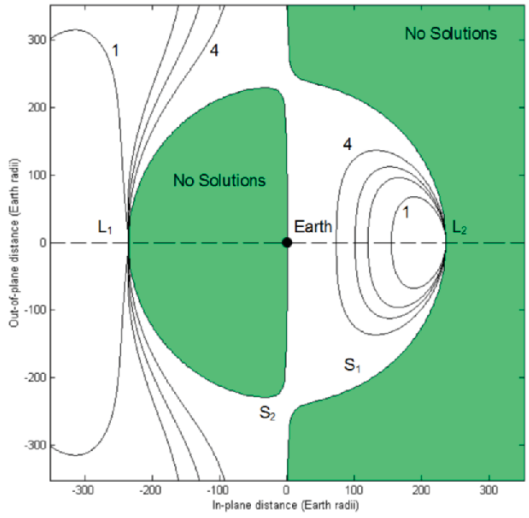

Fig. A2 Contours of sail loading in the x-z plane with reflectivity η=1. Contours (gm^{-2}): [1] 30 [2] 15 [3] 10 [4] 5.

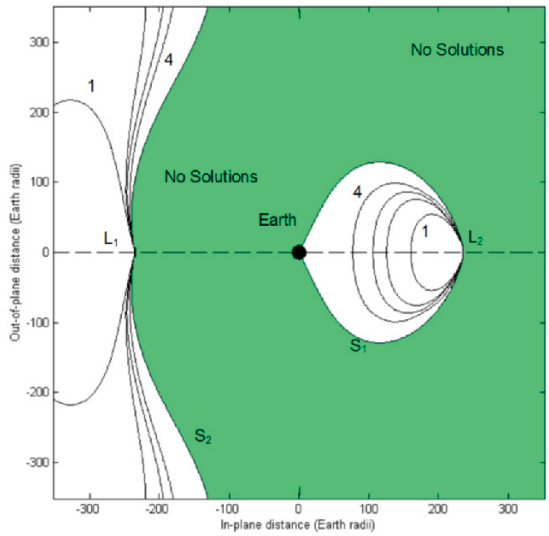

Fig. A3 Contours of sail loading in the x-z plane with reflectivity η=0.8. Contours (gm^{-2}): [1] 30 [2] 15 [3] 10 [4] 5.

Acknowledgments

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Gossamer Roadmap Technology Reference Study for a Solar Polar Mission

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A technology reference study for a solar polar mission is presented. The study uses novel analytical methods to quantify the mission design space including the required sail performance to achieve a given solar polar observation angle within a given timeframe and thus to derive mass allocations for the remaining spacecraft sub-systems, that is excluding the solar sail sub-system. A

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parametric, bottom-up, system mass budget analysis is then used to establish the required sail technology to deliver a range of science payloads, and to establish where such payloads can be delivered to within a given timeframe. It is found that a solar polar mission requires a solar sail of side-length 100 – 125 m to deliver a 'sufficient value' minimum science payload, and that a 2. 5μm sail film substrate is typically required, however the design is much less sensitive to the boom specific mass.

I. Introduction

TERRESTRIAL observations of the Sun are restricted to within the ecliptic plane, and hence ±7.25 deg., of the solar equator, whilst space-based observations have largely been similarly restricted. The Ulysses spacecraft used a Jupiter gravity assist to pass over the solar poles, obtaining field and particle measurements but no images of the poles. Furthermore, the Ulysses orbit is highly elliptical, with a pole revisit time of approximately 6 years. It is desired that future solar analysis be performed much closer to the Sun, as well as from an out-of-ecliptic perspective. ESA's Cosmic Vision 2015 – 2025 Programme mission Solar Orbiter intends to deliver a science suite of order 180 kg to a maximum inclination of order 35 degrees with respect to the solar equator and to a minimum solar approach radius of 0.22 au [1].

The inability of the Solar Orbiter mission to attain a solar polar orbit highlights the difficulty of such a goal with conventional chemical or electric propulsion. Solar sailing has previously been identified as a key enabling technology to place a spacecraft into an high-inclination, or polar orbit over the Sun [2]; such a mission concept has been extensively studied since the nineteen seventies [3 – 6]. This paper presents an analysis of the mission design trade-space to inform future technology development within the DLR-ESA Gossamer Technology Roadmap [7, 8], whilst seeking to inform analysis of the 'sufficient value' minimum science payload [9].

II. Mission Architecture & Analysis

In many previous solar polar mission concepts [3 – 6, 10 –13] the mission science goals required an Earth resonant orbit, placing the spacecraft near to the solar limb as seen from Earth to allow observation of the corona along the Sun-Earth line. An Earth resonant orbit limits the orbit radius to $N^{-2/3}$ au, where N is an integer representing the resonant value. If this science requirement is not present then the only other reason to adopt a resonant orbit would be to allow continuous spacecraft visibility from Earth for ease of spacecraft operations. It is of note that in mission concepts that do not require a resonant orbit, a resonant orbit radius is still typically chosen [9, 14].

The peak-duration of communication blackouts due to solar conjunctions against a range of orbit radius for both X-band and K-band communications is shown in Fig. 1 assuming perpendicular relative motion. Note that a 4-degree field-of-view exclusion is assumed for X-band communications, while a 1.5-degree field-of-view exclusion is assumed for K-band communications. It is seen from Fig. 1 that the likely peak-duration of communication blackouts, on a solar polar orbit, due to solar conjunctions is around two-days for X-band systems and less than one-day for K-band systems. It is concluded that a resonant orbit is not required for spacecraft operations.

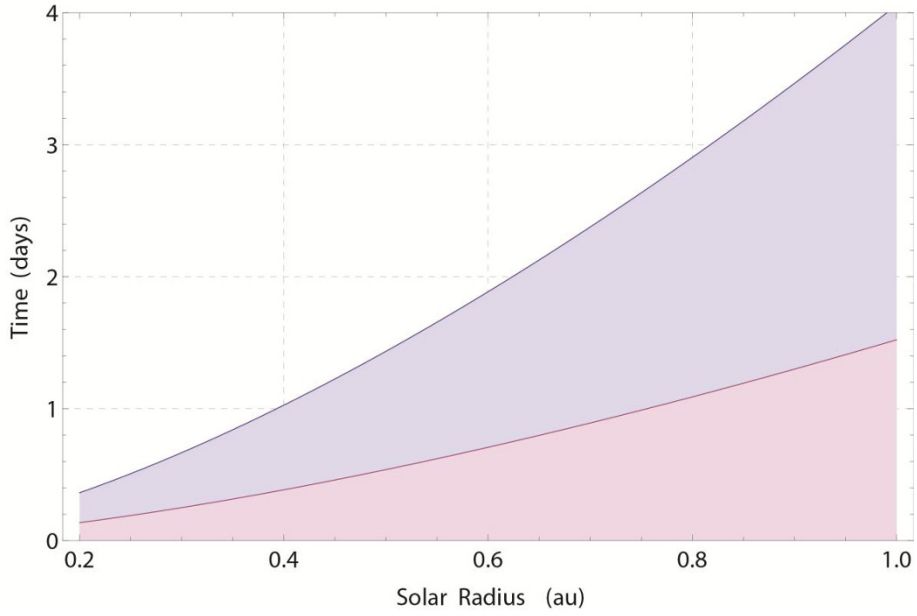

Fig. 1 Peak-duration of communication blackout due to solar conjunction against orbit radius for X-band (top) and K-band (bottom) communication, assuming perpendicular motion.

A. Solar Pole Visibility

The visibility of the solar pole is illustrated in Fig. 2 for a given observer location, determined by the observers distance (orbit radius, assumed equal to the orbit semi-major axis) and the observers elevation above the pole (the orbit inclination measured from the Solar equator, equal to the inclination from the Ecliptic plane plus 7.25 degrees).

Using the cosine rule the observers Slant Range, *SR*, to the solar pole can be determined as,

$$SR^2 = R_S^2 + r^2 - 2R_S r \sin i_{se} \qquad (1)$$

where $i_{se} > i_{se,min}$ and R_S is the mean volumetric radius of the Sun. Knowing the *SR* to the solar pole, the observers elevation from the solar pole can be determined using the sine rule as,

$$\epsilon = i_{se} - \sin^{-1}\left[\frac{R_S \cos i_{se}}{\sqrt{R_S^2 + r^2 - 2R_S r \sin i_{se}}}\right] \tag{2}$$

where $i_{se} > i_{se,min}$. The Observer-to-Solar pole zenith angle, OZA, is thus simply,

$$OZA = \frac{\pi}{2} - \epsilon \tag{3}$$

Note that Eq. (1) and (2) can be converted into the standard ecliptic plane reference plane by addition of 7.25 degrees to the given inclination, i.e. $i = i_{se} + 7.25°$. Assuming a circular orbit, the spacecraft mean motion can be used to determine the time that the observer will be able to maintain an Observer-to-Solar pole zenith angle above a given defined minimum.

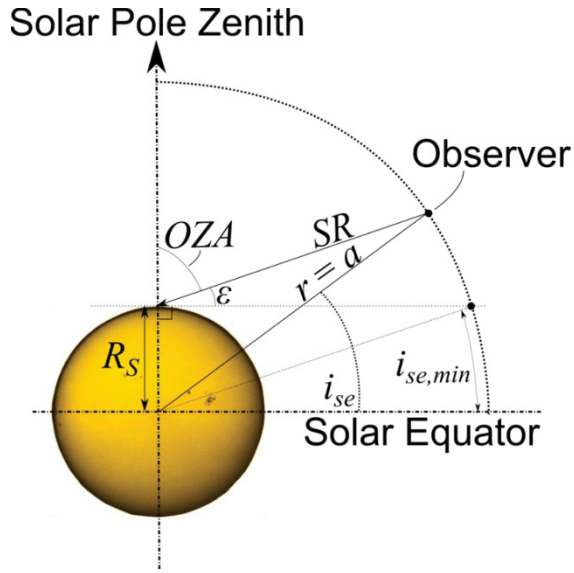

Fig. 2 Solar pole visibility; note that the solar equator is offset from the ecliptic plane by 7.25 degrees

The time below an Observer-to-Solar pole zenith angle of 30 degrees and 50 degrees over a range of semi-major axis and inclinations, with inclination measured from the ecliptic plane, is shown in Fig. 3. It is seen that the minimum required Observer-to-Solar pole zenith angle critically defines the time that the solar pole is visible and hence critically defines the minimum orbit radius at which a complete rotation of the Sun would be visible during a single orbit pass over the solar pole. Adopting the sidereal rotation period of the Sun as 609.12-hr (25.38 days), based on a solar latitude of 16 degrees, it is found that at a orbit radius of 0.4 au a solar polar orbit (orbit inclination 82.75 degrees) can maintain an Observer-to-Solar pole zenith angle below 50 degrees for a complete rotation of the Sun. Increasing the orbit radius of 0.5 au, it is found that the Observer-to-Solar pole

zenith angle can be maintained below 40 degrees for the same duration at an orbit inclination of 78 degrees, but that the Observer-to-Solar pole zenith angle cannot be maintained below 30 degrees at any inclination for this duration. Finally, increasing the orbit radius to 0.6 au, the Observer-to-Solar pole zenith angle can be maintained below 40 degrees for this duration at an orbit inclination of 70 degrees, and can be maintained below 30 degrees for the required duration at an orbit inclination of 80 degrees.

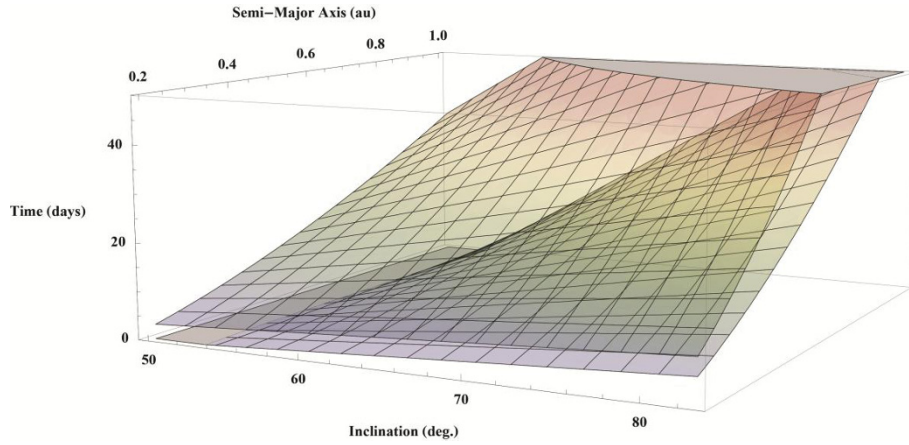

Fig. 3 Time below an Observer-to-Solar pole zenith angle of 30 deg. (bottom) and 50 deg. (top) over a range of semi-major axis and inclinations.

B. Transfer Trajectory

It has been seen previously that for solar sail transfers to high-inclination, circular orbits the trajectory can be divided into multiple phases [3 – 6, 10 – 13]. Phase 1 is the reduction of orbit radius to below the target radius, whilst perhaps performing some modest plane change maneuver at the same time. Phase 2 is the rapid increase of orbit inclination while maintaining a near-constant orbit radius and Phase 3 is increasing the orbit radius to the target value, whilst perhaps performing some modest plane change maneuver at the same time. In [3 – 6, 10 – 13] the optimal level of plane-change in each phase was simply defined using empirical evidence based on the use of either, local and/or global numerical optimization techniques, or semi-analytical approximations. However, in [15] orbit averaging techniques are used to develop a general perturbations solution of circle-to-circle low-thrust trajectory transfers with plane-change about the Sun that allow the optimal level of plane change in Phase 1 to be analytically derived. In [15] it is found that the minimum time, analytically derived, solar sail transfer is within 1% of the optimal transfer.

It has been shown previously that a three-phase transfer can be used to minimize either the transfer time or the technology requirements on the solar sail [6]. That is, for fixed transfer duration, the required sail size can be minimized by correctly defining the orbit radius in Phase 2. As this radius is reduced the spacecraft size

increases due to the thermal loading on the spacecraft, however the sail performance also increases due to the $(1/r^2)$ variation in sail performance, reducing the required sail size. In [6] it is shown that a turning point can be defined at which the mass penalty due to the thermal loadings overcomes the increase in sail performance, this turning point defines the minimum sail technology requirement design point. However, the analysis in [6] does not directly consider spacecraft or mission cost, and it is likely that the trajectory profile for a minimum cost mission would increase the Phase 2 radius above both the time optimal and sail technology optimal radius. Indeed assuming the target orbit radius is less than that of, say, Venus', it is likely that the minimum cost mission would result in the removal of phase three, with phase two occurring at the target orbit radius and hence the spacecraft thermal design load would be the target/operational orbit radius and not that during the transfer trajectory.

The presented analysis assumes a two-phase trajectory using [15], whereby the orbit radius is restricted to greater than or equal to the target orbit radius in an attempt to minimize the technology requirements across the complete spacecraft system. Considering a solar sail propelled spacecraft transfer from a circular 1 au orbit within the ecliptic plane to a circular orbit of semi-major axis a_1 inclined at a target inclination of i_t, it is shown in [15] that the total transfer time (in seconds) can be written in the form,

$$t = \frac{1}{\pi\beta}\left[\frac{1}{\cos\tilde{\gamma}}\left(\mathfrak{A} - \mathfrak{B}\, a_1^{\frac{3}{2}}\right) + a_1^{\frac{3}{2}}(\mathfrak{C} i_t + \tan\tilde{\gamma}(\mathfrak{B}\,\mathrm{Log}[a_1] - \mathfrak{D}))\right] \tag{4}$$

where, $\mathfrak{A}, \mathfrak{B}, \mathfrak{C}$ and $\mathfrak{D}$ are all constants determined by the orbit parameters, the value of which is not relevant to this analysis, β is the sail lightness number [16], and $\tilde{\gamma}$ switches between $\{-\gamma, +\gamma\}$and is an augmented form of the sail clock angle, δ, [16], measured from the orbit transverse axis towards the normal axis. As such, $\gamma = (\pi/2 - \delta)$, meaning that the augmented clock angle, $\tilde{\gamma}$, will switch sign at true anomaly ($^{\pi}/_{2}$) and ($^{3\pi}/_{2}$) when considering the change of orbit inclination. Analysis of Eq. (4) shows that the trip time is minimized when the out-of-plane thrust angle is $0 < \gamma < (\pi/2)$ and that a turning point can be identified when the trip time is minimized.

Equation (4) cannot be generalized for the optimal case due to the variation in the optimal out of plane thrust angle, $\tilde{\gamma}$, with target orbit radius. However, in [15] the optimal out of plane thrust angle is approximated using a third order polynomial, allowing Eq. (4) to be re-written as,

$$t = \frac{1}{\pi\beta}\left[\frac{\left(\mathfrak{A} + \mathfrak{B}a_1^{\frac{3}{2}}\right)}{\cos[\mathfrak{C}(\mathfrak{D} + a_1)(\mathfrak{E} + \mathfrak{F}a_1 + a_1^2)]}\right.$$

$$\left. + a_1^{\frac{3}{2}}\left(\mathfrak{G}i_t + \left(\mathfrak{H} + \mathfrak{I}\,\mathrm{Log}\left[a_1^{\frac{3}{2}}\right]\right)\tan[\mathfrak{J} + \mathfrak{K}a_1 + \mathfrak{L}a_1^2 + \mathfrak{M}a_1^3]\right)\right] \text{ years} \quad (5)$$

where $\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \mathfrak{D}, \mathfrak{E}, \mathfrak{F}, \mathfrak{G}, \mathfrak{H}, \mathfrak{I}, \mathfrak{J}, \mathfrak{K}, \mathfrak{L}$ and $\mathfrak{M}$ are constants determined by the orbit parameters, distinct from those used in Eq. (4), and are defined in Table 1, and where a_1 is measured in meters and t in seconds.

Table 1 Constants for Eq. (5)

Constant	Value
$\mathfrak{A}$	0.43303013661188056
$\mathfrak{B}$	$-7.483934949353608\times10^{-18}$
$\mathfrak{C}$	$3.4425263133185114\times10^{-34}$
$\mathfrak{D}$	$-8.392144253966071\times10^{+9}$
$\mathfrak{E}$	$3.5784751639989546\times10^{+22}$
$\mathfrak{F}$	$-1.995309183700735\times10^{+11}$
$\mathfrak{G}$	$3.5267212585249795\times10^{-17}$
$\mathfrak{H}$	$2.8885612757886467\times10^{-16}$
$\mathfrak{I}$	$-7.483934949353608\times10^{-18}$
$\mathfrak{J}$	0.103382782379
$\mathfrak{K}$	$-1.2895443276860758\times10^{-11}$
$\mathfrak{L}$	$7.157806142290288\times10^{-23}$
$\mathfrak{M}$	$-3.4425263133185114\times10^{-34}$

Using Eq. (5) the attainable orbit region within five years is shown in Fig. 4 for a range of sail characteristic accelerations [16], a metric of solar sail performance; Fig. 4 also shows the orbit region that is required to be obtained to enable 25-day observation of the poles at a range of maximum observer-to-solar pole zenith angles. It is seen in Fig. 4 that with a sail characteristic acceleration of 0.2 mm/s^2 a peak solar latitude of 60 degrees cannot be reached in 5-years. However, with a sail characteristic acceleration of 0.3 and 0.4 mm/s^2 a peak solar latitude of 60 degrees at orbit radius 0.43 and 0.62 au, respectively, can be reached in a similar timeframe. It is also seen in Fig. 4 that restricting sail characteristic acceleration to below 0.5 mm/s^2, it is not possible using a two-phase 5-year transfer trajectory to attain a solar orbit that maintains the observer-to-solar pole zenith angle below 30 degrees for 25 days. It is seen however that a sail characteristic acceleration below 0.5 mm/s^2 can, using a two phase 5-year transfer trajectory, attain a solar orbit that maintains the observer-to-solar pole zenith angle below 40 degrees for 25 days.

From Fig. 4 it is apparent that a range of two-phase transfer trajectories are possible within any set of mission requirements. Assuming the spacecraft is required to reach an orbit inclination of 52.75 degrees in 5-

years, equivalent to a peak solar latitude of 60 degrees, and that the final observer-to-solar pole zenith angle should be less than 40 degrees, and ideally should be less than 30 degrees, during a sidereal rotation period of the Sun, the range of two-phase transfer trajectories are defined by the observer-to-solar pole zenith angle 30 and 40 degree contour lines, giving a target solar radii between 0.550 au and 0.447 au. Table 2 summaries a range of possible mission architectures, with an increasing level of technology requirement on the solar sail, ranging from a characteristic acceleration of 0.28 mm s^{-2} up to 0.53 mm s^{-2}.

III. System Analysis

A. Generic Gossamer Structure Model

A generic gossamer structure / solar sail model is used to allow general top-level technology capabilities to be analyzed and quantified. It should be noted that this generic gossamer structure / solar sail model is not intended to be a complete or final model, but rather is intended to allow general top-level technology capabilities to be analyzed and quantified.

Noting that the possible films typically considered range in density from Polyethylene terephthalate (PET) at 1350 kg m^{-3} to LaRCT-CP1 at 1430 kg m^{-3}, the generic model is similarly based on Kapton film as this is one of the heaviest films likely to be used. In addition to the film, an allocation of 0.42 g m^{-2} is given for adhesives and edge enhancement, but without swivels and without constant force springs. Sail film thickness can be arbitrarily selected, however 7.5 μm and 2.5 μm films will be considered.

Assuming a 3-axis stabilized gossamer structure, as opposed to a spin-stabilized structure, the support booms for the structure are a primary technology requirement. The impact of boom technology can hence also be quantified within the generic gossamer structure / solar sail model as shown in Fig. 5, which presents the range of possible sail assembly loadings over sail length sizes from 35 – 125 m. Note within Fig. 5 the DLR boom technology is described within [17, 18], whilst the CoilAble booms are described in [19]. It should also be noted that the inflatable booms developed by L'Garde [20 – 22] for use in the NASA-funded Sunjammer mission have a lower specific mass than the booms considered in Fig. 5; however, this data is not in the public domain. The boom technology specific mass data points impose a quasi-limit on the gossamer structure / solar sail model to side lengths of less than or equal to 125 m.

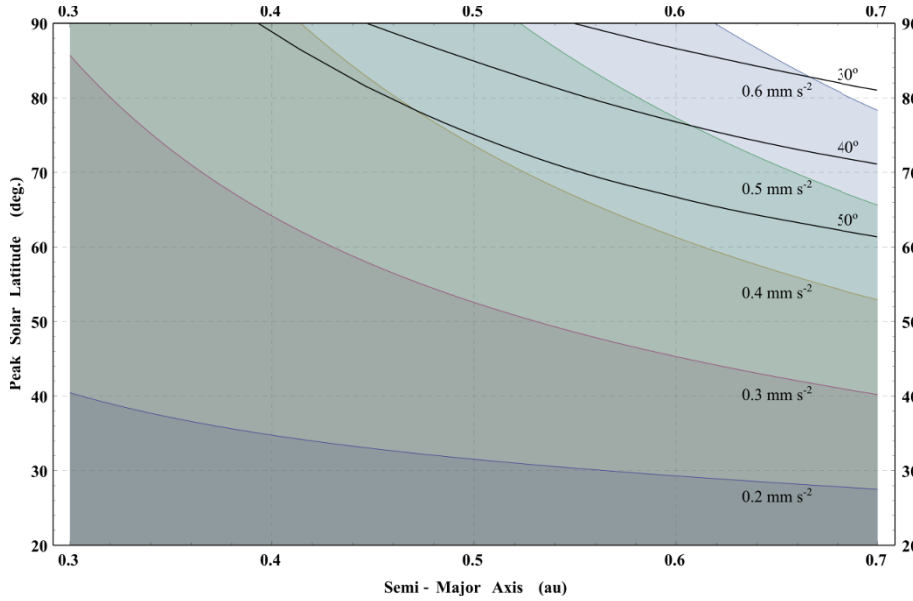

Fig. 4 Attainable peak solar latitude and semi-major axis within 5-years for a sail characteristic accelerations of 0.2 mm s^{-2} (bottom); 0.3 mm s^{-2}; 0.4 mm s^{-2}; 0.5 mm s^{-2}; 0.6 mm s^{-2} (top); 25-day maximum observer-to-solar pole zenith angle contours also shown.

Table 2 Summary and comparison of solar sail mission architecture options and requirements; superscript number indicates number of trajectory phases

	A^2	$A+^2$	B^2	$B+^2$	C^2	$C+^2$
Solar pole maximum OZA in one-sidereal rotation period	50 deg.	50 deg.	40 deg.	40 deg.	30 deg.	30 deg.
Target orbit radius	0.397 au	0.397 au	0.461 au	0.461 au	0.559 au	0.559 au
Required sail characteristic acceleration	0.29 mm/s^2	0.39 mm/s^2	0.32 mm/s^2	0.44 mm/s^2	0.37 mm/s^2	0.54 mm/s^2
Phase 1 duration	2.4 years	1.8 years	2.1 years	1.5 years	1.7 years	1.1 years
Inclination at end of phase 1	8.8°	8.8°	8.8°	8.8°	8.5°	8.5°
Time to 60° latitude	5.0 years	3.7 years	5.0 years	3.6 years	5.0 years	3.4 years
Time to 90° latitude	6.8 years	5.0 years	7.0 years	5.0 years	7.3 years	5.0 years
Time to 50° OZA view pole over a sidereal rotation	6.8 years	5.0 years	6.3 years	4.5 years	5.8 years	4.0 years
Time to 40° OZA view pole over a sidereal rotation	n/a	n/a	7.0 years	5.0 years	6.5 years	4.5 years
Time to 30° OZA view pole over a sidereal rotation	n/a	n/a	n/a	n/a	7.3 years	5.0 years

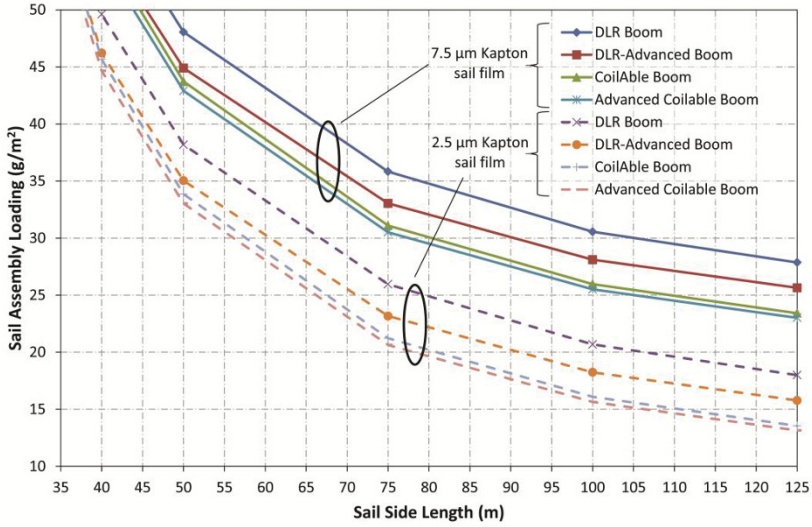

Fig. 5 Sail assembly loading using generic gossamer structure / solar sail model over a range of sail lengths and technology levels.

B. System Mass Budget Allowances

For each mission architecture option detailed in Table 2 the generic gossamer structure / solar sail model illustrated in Fig. 5 can be used to determine a platform mass allowance for of the required performance level of that mission architecture as shown in Table 3 – Table 8. Note that the platform mass allowance is given for the two film thicknesses considered and as a range over the boom technologies considered as in Fig. 5.

Table 3 Summary of mission architecture option A[2] mass allowance estimates, including ROM payload allowance estimate.

Architecture option A[2]				
GO Side Length (m)	**GO Mass Range with 7.5 µm film (kg)**	**GO Mass Range with 2.5 µm film (kg)**	**Platform Mass range with 7.5 µm film (kg)**	**Platform Mass range with 2.5 µm film (kg)**
50	*120 – 107*	*95 – 83*	*negative*	*negative*
75	*202* – 172	146 – 116	< 12	38 – 67
100	306 – 255	207 – 156	21 – 71	120 – 170
125	435 – 360	281 – 205	75 – 150	229 – 305

Table 4 Summary of mission architecture option A+[2] mass allowance estimates, including ROM payload allowance estimate.

Architecture option A+[2]				
GO Side Length (m)	**GO Mass Range with 7.5 µm film (kg)**	**GO Mass Range with 2.5 µm film (kg)**	**Platform Mass range with 7.5 µm film (kg)**	**Platform Mass range with 2.5 µm film (kg)**
50	*120 – 107*	*95 – 83*	*negative*	*negative*
75	*202* – 172	146 – 116	*negative*	< 20
100	306 – 255	207 – 156	*negative*	34 – 85
125	435 – 360	281 – 205	< 17	96 – 172

Table 5 Summary of mission architecture option B^2 mass allowance estimates, including ROM payload allowance estimate.

Architecture option B^2				
GO Side Length (m)	**GO Mass Range with 7.5 µm film (kg)**	**GO Mass Range with 2.5 µm film (kg)**	**Platform Mass range with 7.5 µm film (kg)**	**Platform Mass range with 2.5 µm film (kg)**
50	*120 – 107*	*95 – 83*	*negative*	*negative*
75	*202 – 172*	146 – 116	*negative*	20 – 49
100	*306* – 255	207 – 156	< 39	87 – 138
125	435 – 360	281 – 205	24 – 100	179 – 254

Table 6 Summary of mission architecture option $B+^2$ mass allowance estimates, including ROM payload allowance estimate.

Architecture option $B+^2$				
GO Side Length (m)	**GO Mass Range with 7.5 µm film (kg)**	**GO Mass Range with 2.5 µm film (kg)**	**Platform Mass range with 7.5 µm film (kg)**	**Platform Mass range with 2.5 µm film (kg)**
50	*120 – 107*	*95 – 83*	*negative*	*negative*
75	*202 – 172*	*146* – 116	*negative*	< 2
100	*306 – 255*	207 – 156	*negative*	4 – 54
125	*435 – 360*	281 – 205	*negative*	48 – 124

Table 7 Summary of mission architecture option C^2 mass allowance estimates, including ROM payload allowance estimate.

Architecture option C^2				
GO Side Length (m)	**GO Mass Range with 7.5 µm film (kg)**	**GO Mass Range with 2.5 µm film (kg)**	**Platform Mass range with 7.5 µm film (kg)**	**Platform Mass range with 2.5 µm film (kg)**
50	*120 – 107*	*95 – 83*	*negative*	*negative*
75	*202 – 172*	*146* – 116	*negative*	< 26
100	*306 – 255*	207 – 156	*negative*	45 – 96
125	*435* – 360	281 – 205	< 34	113 – 188

Table 8 Summary of mission architecture option $C+^2$ mass allowance estimates, including ROM payload allowance estimate.

Architecture option $C+^2$				
GO Side Length (m)	**GO Mass Range with 7.5 µm film (kg)**	**GO Mass Range with 2.5 µm film (kg)**	**Platform Mass range with 7.5 µm film (kg)**	**Platform Mass range with 2.5 µm film (kg)**
50	*120 – 107*	*95 – 83*	*negative*	*negative*
75	*202 – 172*	*146 – 116*	*negative*	*negative*
100	*306 – 255*	*207* – 156	*negative*	< 17
125	*435 – 360*	*281* – 205	*negative*	< 65

C. Payload Budget

Three science priorities have been identified [9]. The first two require long-duration observations at high-latitudes. The first is latitudinal variations of solar irradiance and requires a payload mass allocation of 5 kg to carry a Total Solar Irradiance instrument. The second priority considers the internal solar structure and dynamo, and increases the payload mass allocation requirement to 15 kg with the addition of Doppler velocity and magnetic field instruments. The third scenario addresses Heliophysics and dynamo issues and requires a payload

of remote and in-situ instruments including Doppler and Stokes Imager, Heliospheric imager (coronagraph), Magnetometer, Solar Wind Analyzer (Suprathermal elec.) and Energetic Particle Detector. The third scenario requires a payload mass of 40 kg.

D. System Mass Budget Analysis

A parametric, bottom-up, system mass budget analysis was developed to allow the platform mass, without the solar sail, to be evaluated across a range of payload masses and minimum orbit radius. The analysis is restricted to technology assessed as TRL 6 or greater, and is based on worst-case assumptions to ensure a suitably conservative mass budget estimate. The results from the parametric, bottom-up, system mass budget analysis are shown in Fig. 6; note that this analysis, as before, is intended to inform analysis of the 'sufficient value' minimum science payload and hence is intended only as an indicative mass estimate.

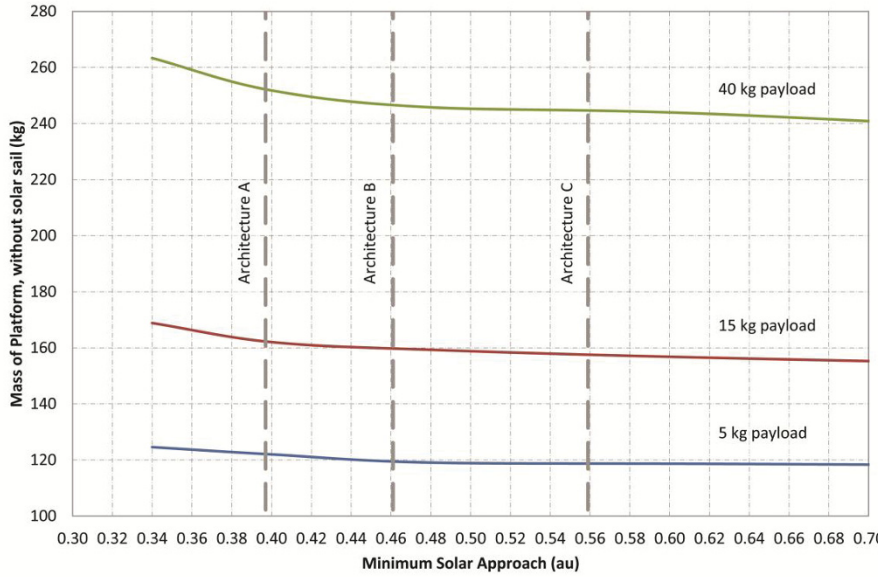

Fig. 6 Platform mass budget analysis across a range of minimum solar approaches and payload masses

From Fig. 6 and Table 3 – Table 8 the required sail size and technology for each mission architecture and payload mass can be established.

1. Architecture A

It is found that Architecture A+[2] is restricted to a payload mass of less than about 15 kg, assuming both 2.5μm film and a boom specific mass of order 70 g/m on a sail of side length 125m. Meanwhile, Architecture A[2] can deliver a 40 kg payload with a 2.5μm film and a boom specific mass of order 145 g/m on a sail of side length

125m. Alternatively, a reduced payload of 15 kg is enabled with a reduced boom specific mass to order 70 g/m on a sail of side length 100m or a payload of 5 kg with an increased boom specific mass to order 190 g/m. A 5 kg payload can also be delivered with a 7.5μm film and a boom specific mass of order 100 g/m on a sail of side length 125m.

2. *Architecture B*

It is found that Architecture B+[2] is restricted to a payload mass of less than about 5 kg, assuming both 2.5μm film and a boom specific mass of order 70 g/m on a sail of side length 125m. Meanwhile, Architecture B[2] is can deliver a 40 kg payload with a 2.5μm film and a boom specific mass of order 70 g/m on a sail of side length 125m. Alternatively, a reduced payload of 15 kg is possible with an increased boom specific mass to order 240 g/m on a similar size side or a payload of 5 kg with a boom specific mass to order 90 g/m on a sail of side length 100m.

3. *Architecture C*

It is found that Architecture C+[2] is not feasible with a meaningful payload mass, while Architecture C[2] is restricted to a sail of side length 125 m, with 2.5μm film with a boom specific mass to order 120 g/m (15 kg payload) too 200 g/m (5 kg payload).

4. *Analysis*

It is of note that in all architectures, except one, the 7.5μm film does not provide a sufficiently meaningful payload mass allocation, and in the one architecture where it does the payload mass is limited to around 5 kg on a 125 m sail side length. Meanwhile the feasible payload mass appears much less sensitive to the considered variation in boom specific mass.

IV. Conclusions

Analysis of a Solar Polar Mission design trade-space to inform required future technology development, whilst seeking to inform analysis of the 'sufficient value' minimum science payload, showed that a 40 kg instrument payload can only feasibly be delivered to a maximum solar pole observer-to-zenith angle in one solar sidereal rotation period of 40 degrees or more. Further, a sail of side length 125 m would be required to achieve this, along with 2.5μm film substrate and a boom specific mass of order 70 – 145 g/m, corresponding to a 40 – 50 degree maximum solar pole observer-to-zenith angle in one solar sidereal rotation period after 7 – 6.8 years, respectively. It was found that a 15 kg instrument payload could feasibly be delivered to a maximum solar pole observer-to-zenith angle in one solar sidereal rotation period of 30 degrees or more, and that a 2.5μm film was again required. This lower observation angle can be achieved in 7.3 years by maintaining a sail side-length of 125 m and a boom specific mass of order 120 g/m. Alternatively, the boom technology requirements, or the transfer time, can be reduced for an increased observation angle. For example, maintaining a sail side-length of

125 m the boom specific mass can be increased to around 240 g/m, corresponding to a 40-degree maximum solar pole observer-to-zenith angle in one solar sidereal rotation period after 7-years. On the other hand, the sail size can be reduced to a side-length of 100 m, with a boom specific mass to order 70 g/m, corresponding to a 50-degree maximum solar pole observer-to-zenith angle in one solar sidereal rotation period after 6.8-years. As expected, a 5 kg payload offers the most flexibility on solar pole observer-to-zenith angle, sail technology and transfer time. It was found that only a 5 kg payload could be considered with a 7.5μm film substrate, corresponding to a 50-degree maximum solar pole observer-to-zenith angle in one solar sidereal rotation period after 6.8 years with a boom specific mass of order 100 g/m.

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SOLARIS: Solar sail investigation of the Sun

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In this paper, we detail the scientific objectives and outline a strawman payload of the SOLAR sail Investigation of the Sun (SOLARIS). The science objectives are to study the 3D structure of the solar magnetic and velocity field, the variation of total solar irradiance with latitude, and the structure of the corona. We show how we can meet these science objective using solar-sail technologies currently under development. We provide a tentative mission profile considering several trade-off approaches. We also provide a tentative mass budget breakdown and a perspective for a programmatic implementation.

I. Introduction

Understanding the origins of solar magnetic activity has been at the forefront of solar and stellar physics since the discovery of the 11-year sunspot cycle nearly two centuries ago. Unraveling this mystery has broad

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implications not only for promoting a deeper knowledge of the Sun itself but also for understanding the Sun's influence on the heliosphere, the geospace environment, and potentially the Earth's climate system. Such influences regulate space weather, with increasing economic impacts on our technological society as our reliance on telecommunications systems, power grids, and airline travel continues to grow. As a readily observable example of an astrophysical magneto-hydrodynamic (MHD) dynamo, the Sun offers unique insights into the generation of magnetic fields by turbulent plasma flows throughout the universe, from planetary and stellar interiors to stellar and galactic accretion disks to interstellar clouds.

The global magnetic polarity of the Sun reverses during each 11-year sunspot cycle so that the overall period of the solar magnetic activity cycle is 22 years. It is a formidable challenge to understand how such remarkable regularity arises from the highly turbulent conditions of the solar convection zone and how magnetic flux emerges from the solar interior to energize the solar atmosphere and to drive solar variability. Large-scale flows (differential rotation and meridional circulations) established by turbulent convection, plasma instabilities, and nonlinear feedbacks all play an important role, spanning many orders of magnitude in spatial and temporal scales.

Modern solar observations coupled with sophisticated theoretical and numerical models have yielded important insights into many aspects of solar magnetism but the basic physical mechanisms responsible for generating these fields are still not understood. To achieve scientific progress in our understanding the Sun and the fundamental problems of cosmic magnetism we need to redouble both theoretical and observational efforts. On the observational side, measuring solar internal flows is of the greatest importance. For this task helioseismology has proven to be a powerful tool.

Helioseismic measurements are based on surface wavefield data, normally and preferably temporal series of photospheric Dopplergrams, which are then analyzed to probe the solar interior structure and flows. With the so-called global methods, the wavefield data are used to measure (mainly acoustic) eigenfrequencies of the Sun. The eigenfrequencies are then analyzed, often by way of inverse methods, to probe the solar interior for thermal and dynamical structure of high degrees of symmetry, such as the spherically symmetric distribution of sound speed, or differential rotation as the axisymmetric component of flows. With new local methods, wavefield data are used to measure local resonant properties or wave propagation time for a given pair of points, by cross-correlating local wavefields. These travel-time data are then analyzed to probe the interior for local and/or asymmetric structures, such as meridional flow, convection and flows around active regions [5].

Differential rotation and meridional flows have already been measured by such helioseismology techniques, up to about 60° latitude with a typical uncertainty of the order of a few m s^{-1} [10]. It is essential to extend this measurement to the polar region, partly because these measurements will validate our understanding of dynamics of the Sun as a whole, and partly because the polar region is where the magnetic flux reverses and the meridional flow, which plays an important role in carrying the magnetic flux, should turn in towards the solar interior.

Another related mystery is the total solar irradiance variation over the solar cycle. The total irradiance of other solar-like stars with similar activity level, varies around 0.3 per cent, on average, over their activity cycles. On the other hand, the solar irradiance varies only by 0.1 per cent. There is a well-founded suspicion that the solar irradiance depends on latitude, thereby creating a great interest in measuring the solar irradiance from high heliographic latitudes.

Because of the orbit characteristics of the vast majority of spacecraft, the solar flux has predominantly been measured at Earth or generally in the plane of the ecliptic. Therefore, the existing data cannot directly demonstrate whether the latitudinal distribution of the extreme-ultraviolet (EUV) solar flux is anisotropic. Indeed, the non-uniform distribution of very contrasted EUV bright features (i.e., active regions) and dark features (i.e., coronal holes) at the surface of the Sun produces both the obvious rotational (or longitudinal) modulation of the flux and also a strong latitudinal anisotropy. This has been demonstrated by [4], but no direct measurements exist. Although largely ignored up to now, the latitudinal anisotropy has several important implications. A first consequence is that the total EUV output of the Sun is currently overestimated, which affects comparisons with other stars. The EUV latitudinal anisotropies also affect the photo-ionization rates of helium that are used do derive the helium abundance in the local interstellar medium (LISM) from measurements made in-situ in the heliosphere (e.g. by GAS/Ulysses). The helium abundance in the LISM is a major tracer of galactic evolution. Measuring the solar flux in at least two EUV wavebands will enable to test and to complete the models of the irradiance anisotropies derived from in-Ecliptic observations.

Many facts point to a weak coupling between the magnetic fields in the northern and southern hemispheres of the Sun. The reversals of the magnetic field polarity at the poles do not occur simultaneously, with one reversal delayed by as much as two years relative to the other. The Emergence of the magnetic field in the photosphere was organized in a persistent large-scale pattern with different rotation periods in the northern and southern hemispheres, in cycles 20 and 21, pointing to a weak interdependence of the field systems originating in the two hemispheres [1]. Galactic cosmic rays modulated by solar activity are affected by a north-south gradient that reverses at the reversal of the solar magnetic dipole, indicating a different level of activity, and thus magnetic

flux, in the two hemispheres. A significant north-south asymmetry in the EUV solar flux was discovered in solar cycle 23. Measurements of solar irradiance at the poles have not yet been possible and therefore we cannot assess whether this quantity is the same at the north and the south pole. Thus observations suggest that the magnetic field systems generated in the two hemispheres can indeed evolve independently, although they tend to exchange characteristics at the polarity reversal, following the magnetic cycle.

Recent results show that the corona itself is site of a vast variety of fluctuations. The coronal images, obtained with STEREO, show the existence of quasi-periodic non-stationary density variations characterized by a wide range of temporal and spatial scales and strongly confined by the magnetic topology. In closed field line structures, at mid and low latitudes, the density variations might be interpreted as due to slow standing magneto-acoustic waves excited by the convective super-granular motions [12]. In the outer corona, observed in the ultraviolet line HI Ly α with UVCS-SOHO, density fluctuations are coherent and persistent in the slow coronal wind, at mid and low latitudes, whilst in the fast wind, at high latitudes, they are stochastic and non-correlated [11]. The analysis and interpretation of this kind of coronal phenomena is still in its infancy and a complete scenario which links the coronal fluctuations back to their magnetic roots in the photosphere and which helps to assess their role in the energetics of the solar corona and the solar wind, is still to be developed.

Finally, several aspects of the large-scale magnetic configuration of the corona and the interplay between explosive activity and background magnetic field remain either elusive or ambiguous. Questions such as: *what is the 3D structure of the global corona? What is the longitudinal extent of coronal mass ejections (CMEs)? What is the magnetic connectivity between the outgoing CME and low corona? Are post-CME rays the plasma envelopes of the post-CME current sheet predicted by theories of eruptions?* remain open issues despite several years of multi-viewpoint observations from STEREO. For example, both STEREO spacecraft lie on the ecliptic and hence can provide information for the density distribution of the background corona and CMEs only normal to the ecliptic. For this reason, inversion techniques cannot be used reliably unless out-the ecliptic information on the longitudinal spread of coronal structures is available. Such observations provided by a wide angle EUV imager will probe directly the longitudinal evolution of eruptions from the very low corona (< 70,000 km) to the outer corona (~15 R_s) and verify whether there exists a super-radial expansion phase during the early part of the event, as postulated by [9], whether the changes in the CME angular width are due simply to projection effects (i.e. rotation) or due to interaction with the ambient coronal structures, whether CMEs can be deflected longitudinally by nearby coronal holes [8], and whether CMEs can have angular momentum coupling with the Sun as proposed by [7]. All these effects can have important consequences for understanding the development

of CMEs and eventually predicting their geoeffectiveness. Wide-angle EUV observations of post-CME structures, from high latitudes, will finally uncover the natures of these features. They will show both the extent and connectivity of these structures between the outgoing CME and post-eruption flaring arcade. Although the upcoming Solar Orbiter mission will make the first inroads towards addressing these problems, its modest maximum latitude of 34^{o} does not result in a significantly different view of coronal structures in regards to ecliptic observations. Besides, those extreme latitudes will last for only a few days making it difficult to study the intermittent solar eruptive activity from that platform. The best science return can be achieved from long-term synoptic observations from high latitudes as envisioned in the SOLARIS white paper proposed in May 2013 to the European Space Agency in response to its call for L2/L3 mission ideas.

II. Scientific Objectives

The outstanding issues confronting our current understanding of the solar dynamo and its manifestation in the corona may be summarized through several key scientific questions:

- How is the global, cyclic, solar magnetic field generated?
- What is the nature of flows in the polar regions of the Sun and how do they interact with magnetism?
- Which is the degree of coupling between the magnetic field generated in the northern and southern hemispheres of the Sun?
- How does the radiative energy output of the Sun depend on latitude?
- How does the solar dynamo work and drive connections between the Sun and the heliosphere?
- What is the 3D structure of the global eruptive and quiescent corona?

Progress on these scientific questions requires detailed observations of the solar polar regions, where data is currently scarce and where much of the subtle interplay between plasma flows and magnetic fields that gives rise to cyclic polarity reversals is thought to occur. The out-of-ecliptic observations of the Sun, for the first time, will provide an opportunity for detailed investigations of the magnetic structure and dynamics of the polar region. High-latitude photospheric observations will also provide an unprecedented vantage point for helioseismic imaging that can be used to probe flows and fields in the deep convection zone and tachocline where solar activity is ultimately thought to originate.

In addition to measurements at the photospheric level, the structures of the outer solar atmosphere in polar region and the heliospheric structures merit observations from outside the ecliptic. The poles of sun undergo dramatic change during the 11-year solar cycle, driven by the dynamo action in the solar convection zone. The

polar vantage point gives unique opportunities for understanding the origin of the fast solar wind spectroscopically and for stereo viewing of surface vector magnetic fields, coronal structures, and Earth-directed CMEs in coordination with observatories near the Earth. The unique inclination for the SOLARIS observatory will also permit unprecedented measurements of the total solar irradiance. This may help resolve the discrepancy of the cycle variation of the solar irradiance of ~0.1% while solar analogues vary, on average, by 0.3%.

With this in mind, we propose the following prime measurement targets for the SOLARIS mission:

T1) Photospheric magnetic flux distribution and evolution in the polar region

T2) Dynamical coupling between magnetic fields and flows

T3) High-precision measurement of total solar irradiance

T4) Dynamics of the background and transient solar winds

T5) Latitudinal variations of the EUV irradiance and coupling with the magnetic activity

III. Instrument and Mission Requirements

SOLARIS from its highly inclined orbit around the Sun aims to combine helioseismic and magnetic observations, solar irradiance measurements and EUV images at various latitudes. A total mass of 35 to 50 kg is envisaged for the following three instruments.

The highest priority science objective of SOLARIS is measuring subsurface flow at high-latitude regions, by local helioseismology techniques, based on Dopplergrams acquired by an HMI-MDI type instrument. From this measurement we will derive differential rotation, meridional circulations and convective flows in the upper convection zone. These flow-field measurements will then be cross-correlated with surface magnetic field measurement, to reveal how magnetic flux is transported to the polar region, and how the polarity reversals take place as interplay between plasma flows and magnetic fields. The predominantly vertical kG-field patches that Hinode has found in the polar regions are large enough to be observed. A serious attempt will also be made for stereoscopic helioseismology, for investigating deeper layers, including the best ever shot of the solar tachocline region. Space heritage on this type of instrument is strong if we include also the lighter development done for the Polarimetric and Helioseismic Imager of Solar Orbiter.

Solar irradiance measurements at various latitudes will, for the first time, enable us to measure the anisotropy of the total solar irradiance. The unexplained low photometric variability of the sun may be explained by higher variability at higher latitudes, likely caused by faculae. If it is not the case, then we must conclude that the Sun

is a rather atypical star, which will lead to more fundamental questions in astrophysics. Space heritage is also strong given the various irradiance monitor flown on VIRGO, PICARD and other missions.

The wide-field EUV telescope must be able to produce images of the corona in at least two EUV wavelengths (e.g. 30.4 nm and 17.4 nm) and cover a field of view (FOV) of 6 -10 degrees (~10-15 R_s from 0.4 AU) centered on the Sun. Over the past 10 years, the quality of the EUV optics has improved to such a degree that we can seriously contemplate using EUV imagers, that are lighter, more compact and easier to build, to replace low corona coronagraphs whose design remains challenging due to the intrinsically high contrast between the solar disk and the corona at visible wavelengths. Indeed the most recent developments in EUV technologies make it possible to design and build a wide field of view (5 degrees or more), compact (shoebox size) and lightweight (5kg) multi-band EUV telescope. The instrument will be based on the compact design of the Full Sun Imager (FSI) on board Solar Orbiter [3, 6]. FSI includes a movable occulting disk to limit the level of instrumental stray-light for observations above $3R_s$. The 30.4 nm and 17.4 nm bands are dominated by spectral lines formed by resonant scattering above 2 to 3 R_s, which produces a signal proportional to the electron density, as in white light coronagraphs. Several technical options, such as using a smaller pitch detector, will be studied to make the design even more compact than on Solar Orbiter

The main mission requirements are as follows:

- Data collection for the primary science goals shall begin within five years of launch
- Data collection for the primary science goals shall begin when the payload can observe above ±60 degrees of solar latitude
- In an orbit passing directly over the solar poles, the pole shall be continuously visible at an observation-zenith angle of less than 30 degrees (TBC) for one full rotation period
- The orbit shall have a very low eccentricity < 0.2 (close to a circular orbit).
- EUV observations will be performed at all latitudes with primary data collection at latitudes ±45 degrees of solar latitude.
- At most two spacecraft for having a 3D view of the corona.

These requirements are much more severe than those put on the Solar Orbiter mission which is not suited for long helioseismic observation and solar irradiance measurements mainly due to the lower inclination to the highly eccentric orbit (>0.8).

IV. Solar-sail mission design

In the framework of a study led by DLR and ESA, the Gossamer working group (GWG) was set up in order to study how missions based on the use of solar sail could be feasible. The GWG is led by M. Macdonald (Chair and engineer PI) and by T. Appourchaux (Science PI) assisted of several European engineers and scientists comprising L. Gizon (Max Planck Institute for Solar System Research) and T. Sekii (NAOJ). The activities started in November 2011 and will continue beyond the present workshop on Solar Sail.

Table 1: Summary and comparison of solar sail mission architecture options and requirements. Superscript number indicates number of trajectory phases.

	Architecture option			
	A^2	B^2	C^2	D^2
Solar pole maximum OZA in one-sidereal rotation period (in deg)	50	40	30	30
Target solar radius (in AU)	0.393	0.447	0.550	0.550
Target orbit period (in years)	0.25	0.30	0.41	0.41
Required sail characteristic acceleration (in mm s^{-2})	0.2843	0.3103	0.3655	0.5300
Time to target solar radius, i.e. phase 1 duration (in years)	2.44	2.16	1.68	1.16
Phase 2 duration, i.e. time to reach 60 deg. solar latitude (in years)	2.56	2.84	3.32	2.29
Time to 60 deg. solar latitude (in years)	5.00	5.00	5.00	3.45
Time to 90 deg. solar latitude (in years)	6.75	6.94	7.25	5.00

The preliminary mission analysis assumes that the orbit is reached using a two-phase approach: Phase 1 during which a solar sail is deployed and the orbit radius is reduced; Phase 2 during which the orbit is cranked to 90 degrees. Table 1 provides the results for different options. Table 2 provides the spacecraft and payload characteristics only for option A^2. With a solar sail of 100 m, a payload mass of 35 kg is enabled within a total launch mass of about 320 kg. Increasing the sail size to 125 m, a payload mass of 60 kg is enabled within a total launch mass of about 500 kg.

Table 2: Summary of mission architecture option A^2 mass allowance estimates, including ROM payload allowance estimate.

Architecture option A^2						
Sail Side Length (m)	**S/C Mass Range with 7.5 µm film (kg)**	**S/C Mass Range with 2.5 µm film (kg)**	**Platform Mass range with 7.5 µm film (kg)**	**Platform Mass range with 2.5 µm film (kg)**	**Payload Mass with 7.5 µm film (kg)**	**Payload Mass with 2.5 µm film (kg)**
50	*120 – 107*	*95 – 83*	*negative*	*negative*	*negative*	*negative*
75	*202* – 172	146 – 116	< 13	39 – 69	< 5	10 – 15
100	306 – 255	207 – 156	23 – 73	122 – 172	5 – 15	25 – 35
125	435 – 360	281 – 205	78 – 154	232 – 308	15 – 30	45 – 60

V. Interest of the scientific community

The French community interested in this project is located in the Paris area (Institut d'Astrophysique Spatiale, Orsay; Laboratoire d'Etudes Spatiales et d'Instrumentation en Astrophysique, Meudon; Institut de Recherche sur les lois Fondamentales de l'Univers, Saclay). These laboratories are involved in the Solar Orbiter mission and were involved in the study of the Plan A of Solar-C. Additional laboratories that are interested are the Observatoire de la Côte d'Azur and Institut de Recherche en Astrophysique et Planétologie in Toulouse.

The international community interested is the same as the one that proposed several out-of-ecliptic concepts such as Solar Polar Investigation (SPI), POLARIS or the Plan-A of Solar-C. These are namely the Jet Propulsion Laboratory (USA), Stanford University (USA), the Naval Research Laboratory (NRL), the Max-Planck institute for Solar System Research, the World Radiation Center (Switzerland) and the NAOJ (Japan), the INAF-Astrophysical Observatory of Turin (Italy). China is also potentially interested since they are considering the SPORT mission that is also an out-of-ecliptic mission.

VI. Conclusion

The POLARIS mission was submitted in 2007 as L-class mission to the Cosmic Vision programme of ESA [2]. It was rejected based on the low technology readiness of the solar sails. The Gossamer road map established by the Working Groups of DLR/ESA aims at the successful deployment of a solar sail system of 50 m square by 2018 (Gossamer-3). Therefore we can reasonably envisage a demonstrated solar-sail technology by 2020 for a potential mission in the framework 2025-2030. The *Solaris* concept developed in this paper was submitted both to the Centre National d'Etudes Spatiales in January 2013 and to the European Space Agency in May 2013. Such a mission would require a large international collaboration involving several agencies (ESA, JAXA, NASA, China). A mission of the L-class allowing a payload of about 50 kg with solar sails of 100 m to 150 m seems realistic. Nevertheless, at the time of writing, there is not yet any programmatic window either in Europe or in the US, but this could change in the Fall of 2013 if a mission using solar sail is selected by ESA.

The prospect of having a potential dual-spacecraft solar mission out of ecliptic using solar sail is an exciting opportunity for great scientific return with large discovery-like potential. This opportunity and challenge shall not be missed.

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A Fractionated Space Weather Base at L_5 using CubeSats and Solar Sails

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The Sun-Earth L_5 Lagrange point is an ideal location for an operational space weather forecasting mission to provide early warning of Earth-directed solar storms (coronal mass ejections, shocks and associated solar energetic particles). Such storms can cause damage to power grids, spacecraft, communications systems and astronauts, but these effects can be mitigated if early warning is received. Space weather missions at L_5 have been proposed using conventional spacecraft and chemical propulsion at costs of hundreds of millions of dollars. Here we describe a mission concept that could accomplish many of the goals at a much lower cost by dividing the payload among a cluster of interplanetary CubeSats that reach orbits around L_5 using solar sails.

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I. Introduction

The ascendancy of CubeSats has brought renewed interest in solar sail propulsion for the simple reason that commercially available sails are large enough to propel low mass CubeSats over interplanetary distances in reasonable amounts of time. Here, we present a concept for a space weather forecasting base to provide early warning of approaching solar storms utilizing a loose cluster of interplanetary CubeSats that reach orbits around the Sun-Earth L_5 Lagrange point (Fig. 1) using solar sails. The most destructive solar storms result from fast coronal mass ejections (CMEs) - enormous clouds of plasma and magnetic field explosively ejected from the corona at speeds reaching 2000 km/s. The release of these CMEs is often accompanied by a large solar flare that sends relativistic particles streaming into space. The fast CMEs drive shock waves that accelerate large fluxes of energetic particles. These solar energetic particles are hazardous to spacecraft and humans in space. CMEs impacting Earth's magnetosphere can cause geomagnetic storms that can damage power grids and communications systems. By placing appropriately instrumented spacecraft at L_5, solar storms heading towards Earth can be detected in time to give early warning so the effects can be mitigated. The major advantages of the L5 position for forecasting are that it provides a view of the inner heliosphere to image CMEs one to five days before they reach Earth and it allows viewing of solar active regions behind the solar limb that are rotating Earthward. Both provide greater warning time than simply having sensors at Sun-Earth L1. The insert in Fig. 1 shows an Earth-directed CME imaged from one of NASA's Solar TErrestrial RElations Observatory (STEREO) twin spacecraft.

The CubeSat mission presented here draws heavily on a NASA Innovative Advanced Concepts (NIAC) study [1] that developed a concept for a small spacecraft for interplanetary missions with a target volume of 10 cm x 20 cm x 30 cm (6U in CubeSat parlance, where 1 U = 10 cm x 10 cm x 10 cm). This study allocated 2U for a solar sail; the sail system was based on the Planetary Society's LightSail-1™ architecture (www.planetary.org/explore/projects/lightsail-solar-sailing/). The concept also draws on two other activities: (1) the unsuccessful SolWISE (Sailing On Light With Interplanetary Science and Exploration) 6U CubeSat proposal, led by A. Klesh, submitted to NASA's Edison call in 2012; and (2) the proposed 3U INSPIRE (Interplanetary NanoSpacecraft Pathfinder In a Relevant Environment) interplanetary CubeSat mission, also led by A. Klesh, now under development at JPL, which has been selected by NASA's CubeSat Launch Initiative for an upcoming launch, possibly as soon as 2014.

The "fractionated" L_5 mission concept was developed at a 2012 workshop entitled "Small Satellites: A Revolution in Space Science," hosted by the Keck Institute for Space Studies at the California Institute of

Technology (http://www.kiss.caltech.edu/study/smallsat/). The concept would enable a permanent space weather forecasting base at L_5 that could accomplish, at a much reduced cost, the goals of a conventional single-spacecraft L_5 mission, as described in the recent heliophysics decadal report [2]. Rather than using a single conventional spacecraft, multiple small satellites would be used with the scientific payload divided among them; thus the name fractionated Space Weather Base (SWB-L5). Key to the concept is that only one of the CubeSats would carry a high-gain antenna and other hardware necessary for sending high-rate science data to Earth (~1 AU from L_5). The other spacecraft would carry a much smaller communication system to send the science data to the communication hub and low-rate engineering data to Earth.

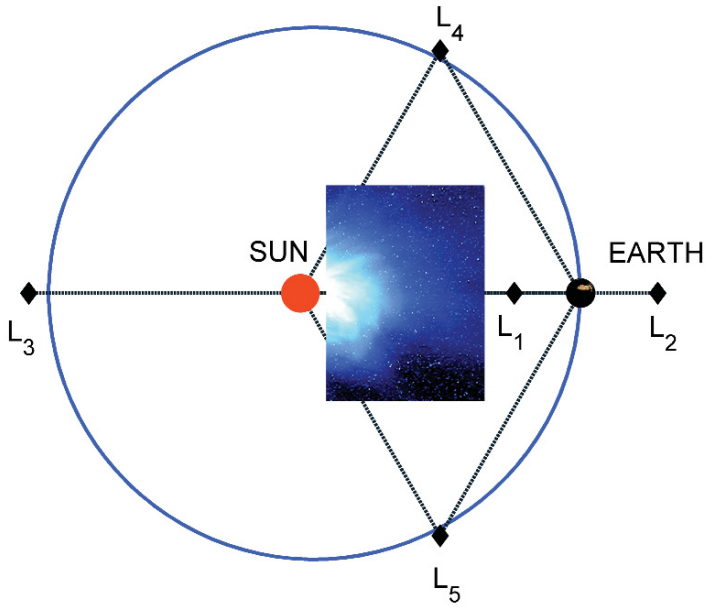

Fig. 1 Schematic showing Earth – L5 - Sun relative locations. Insert shows a white light image of a CME taken from an Heliospheric Imager on one of NASA's STEREO spacecraft.

For the first phase of operation, a loose (~1000 km maximum separation) constellation of five 6U CubeSats would be placed at L_5. Each of the ~10 kg CubeSat allocates ~2U for the solar sail, ~2U for the engineering subsystems common to all five constellation members (attitude control, avionics, etc.) and ~2U for each constellation member's unique payload. This allocation of space is based on the architecture developed for the SolWISE proposal, shown in Figs. 2 and 3 (deployment sequence). The SWB-L5 mission could later be

expanded incrementally to add new instruments and new objectives by sending additional small spacecraft to the L_5 base. The mission concept described in this paper represents a potential beginning for a permanent space weather forecasting base at L_5.

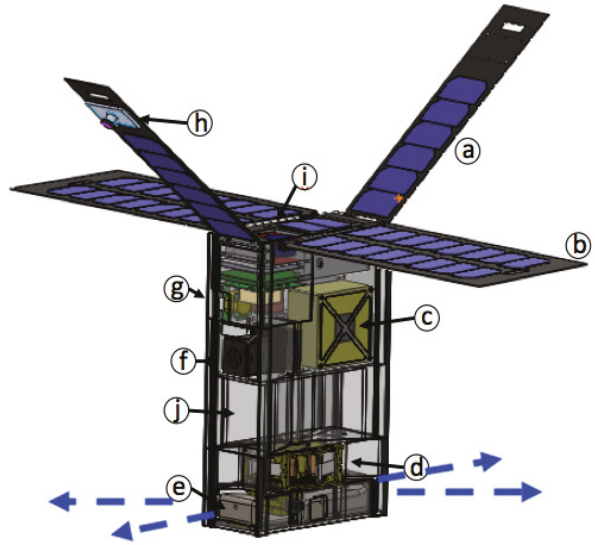

Fig. 2 The 6U SolWISE CubeSat shown after sail deployment (with sail, booms and sides hidden). The booms deploy in the direction of the blue arrows. (a) Fixed solar panels, (b) Rotatable solar arrays, (c) Attitude determination and control system, (d) Sail deployer, (e) Cold-gas system, (f) Star tracker, (g) Magnetometer (stowed), (h) Camera (to observe sail deployment), (i) Solar wind instrument, (j) Sail cavity. The SWB-L5 spacecraft could have a similar architecture.

While detailed engineering studies of the fractionated mission concept have not yet been done, the approach has several obvious advantages and cost savings: (1) Existing solar sails are sufficient for propulsion: the trajectory calculation described in Section II shows that a ~6U spacecraft with a sail approximately twice the Lightsail-1^{TM} size (using thinner sail material and stronger material in the sail boom support structure, at somewhat higher cost than the lowest-possible-cost Lightsail-1^{TM} implementation) could reach an orbit around L_5 in less than 3 years; (2) Spacecraft requirements are eased when the *in situ* instruments are not on the same spacecraft as the imaging instruments: fields and particle instruments prefer spinning spacecraft whereas imaging instruments require spacecraft with 3-axis stabilization, and often instruments are susceptible to interference from each other; (3) Integration and testing is much easier and cheaper for several simple small spacecraft than one large spacecraft with many instruments with conflicting requirements; (4) The cluster could be built up incrementally; (5) Fractionation allows incremental replacement of degrading or failed spacecraft at dramatically lower cost than if a single spacecraft is utilized to provide full capability; (6) Different agencies or institutions

could contribute their own CubeSat; (7) Individual CubeSats could be replaced to upgrade capabilities; and (8) The Space Weather Base could be expanded later by adding other (perhaps larger) spacecraft with new instruments (solar coronagraph, solar EUV and X-ray imagers and spectrometers) to address additional science goals.

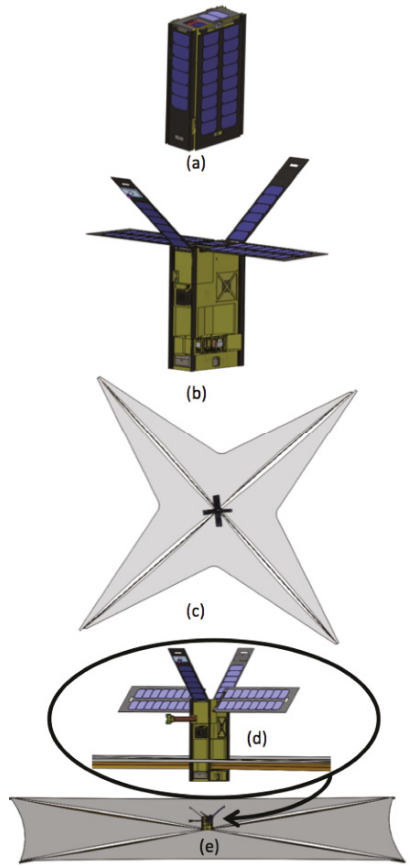

Fig. 3 SolWISE deployment sequence with precisely controllable and monitored elements: a) stowed spacecraft, b) solar panel deployment, c) boom and sail deployment, d) magnetometer deployment, e) fully deployed spacecraft. The SWB-L5 spacecraft could have a similar deployment sequence.

In Section II, calculation of the trajectory to L_5 and the constellation formation are discussed. Section III contains an overview of the fractionated mission, including the science goals, the division of the payload among the five constellation members and potential expansion of the concept beyond the first phase. In Section IV, the science and engineering subsystems of the spacecraft are described and the technology challenges discussed. Section V contains a summary and discussion.

II. Solar Sail Trajectory Calculation

Since a ~10 kg 6U CubeSat cannot carry enough propellant to reach L_5 using conventional propulsion, a solar sail is used. Each of the constellation members would transfer directly from an Earth-escape orbit into an orbit around L_5 using its own solar sail. The spacecraft may be launched on a single or multiple launches as secondary payloads at very low cost. The sail system itself is described in Section III. For this mission, it was desired that the final orbit about L_5 be relatively small (~1000 km maximum separation of constellation members) to ease the requirements on the inter-CubeSat communication system. The SWB-L5 trajectory results are generated for a solar sail design with characteristic acceleration (the acceleration at 1 AU with sail facing the Sun) a_0=0.068 mm/s^2 and 100% efficiency, equivalent to a perfectly reflecting 8.7 m square sail with total mass of 10 kg. This assumption is actually somewhat conservative given that the supplier has stated their ability to provide a sail up to 10 m on a side within the allocated stowage and deployer volume as soon as 2015.[13]

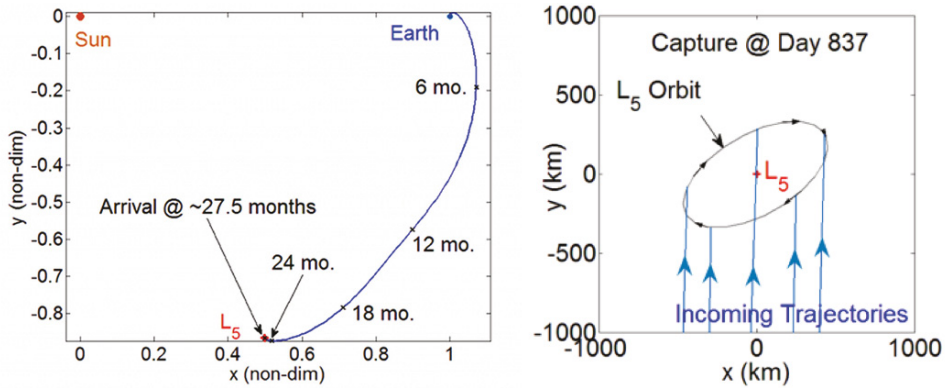

Fig. 4 (a) Overview of solar sail trajectory from an Earth-escape orbit to the ~1000 km diameter orbit around L_5 orbit, shown in SEMB PCR3BP rotating frame (see text). Travel time to L_5 is 837 days. (b) Close up of the L_5 orbit, showing points of insertion of the five trajectories. Orbits were calculated to place five spacecraft with equal temporal spacing in final orbit to show ease of assembly of the constellation.

In order to transfer to the ~1000 km diameter L_5 orbit directly starting from an Earth-escape orbit (characteristic energy C_3≈0 km^2/s^2) using solar sail propulsion, a backward integration scheme is used. The initial state condition of the backward integration is set to coincide with a chosen location on the periodic L_5 orbit so as to constrain the transfer trajectory to end in the desired state around L_5. Thereafter a simple single-parameter control law, described below, is used to propagate the trajectory backward from L_5 and, by varying the one parameter, find a ~200 km altitude flyby of Earth with C_3≈0. One such trajectory is shown in Fig. 4a. The reason for starting from an Earth Escape trajectory is because the Earth's gravity well is so deep, that our current

[13] Tomas Svitek, CEO, Stellar Exploration Inc. personal communication to Robert L. Staehle, 2013 April 16.

sail design would require over two years to escape the Earth starting from geostationary Earth orbit (GEO). To eliminate this long escape trajectory, we assumed the launch vehicle would deliver our CubeSats into an Earth-escape orbit with a perigee altitude of ~200 km and $C_3 \approx 0$. Upper stage disposal to escape is not unusual, having been used most recently on the Landsat 8 launch in 2013 February. There is often excess payload capability within which one or more CubeSats can be accommodated. Deployment using excess payload capability from GEO satellites[14] remains a backup option. Such secondary launch opportunities may be available for costs on the order of $1 M for a 6U CubeSat.

The underlying dynamical equations are the Planar Circular Restricted 3 Body Problem (PCR3BP) with the Sun (S) and Earth-Moon Barycenter (EMB) as primary and secondary bodies respectively, referred to as SEMB PCR3BP hereafter [3]. Acceleration due to solar radiation pressure is also modeled, where a perfectly reflective solar sail is assumed. The equations of motion are given in [4]. The control is the angle of the sail normal with respect to the Sun-Spacecraft line. The control law chosen to achieve the transfer is based on the idea that an Earth to L_5 transfer is similar to a heliocentric co-orbital rendezvous to a point trailing Earth by approximately 60°. Thus, one must first increase the orbital specific energy to slow down with respect to Earth and thereafter decrease the energy to speed up and re-achieve orbital velocity to remain at the target once it is reached. An approximation of a locally optimal sail steering law was used for maximizing and minimizing the orbital energy rate of change. For a sail in a circular heliocentric orbit in the 2-Body Problem, the sail angle that maximizes the rate of heliocentric orbital energy increase is approximately 35°, derived by McInnes[5]. Conversely, the sail angle that maximizes the rate of subtracting energy from the heliocentric orbit is approximately -35°. The parameterized control law is set up to use the maximizing sail angle while leaving Earth and switches over to the minimizing sail angle once a specified "mean anomaly" is achieved, where the mean anomaly is measured by the angle between the X-axis of the SEMB PCR3BP coordinate system and the spacecraft position vector (see Fig. 4a, which shows a trajectory in a frame rotating with the center of mass of the Sun-Earth-Moon system (SEMB PCR3BP rotating frame)**.** This parameter, the switching mean anomaly, is the single control parameter that was manually modified by trial and error to achieve an Earth flyby at a desired ~200 km altitude with $C_3 \approx 0$. The selection of a 200 km perigee for this example is merely a convenience used in order to estimate approximate flight time; in reality a likely starting orbit would have a higher perigee but still have $C_3 \approx 0$. Flight time from Earth's vicinity to L_5 is relatively insensitive to this perigee parameter.

[14] David Lackner, Space Systems Loral, private communication to Robert Staehle, 2011 June 23. Published costs are also available from Spaceflight Services online (www.spaceflightservices.com).

The initial guess for the switching mean anomaly was -30° from Earth, since this is halfway to the destination. Thus the spacecraft is accelerating for the first half and decelerating for the second half. Five individual trajectories were generated from an ~200 km altitude flyby to the same (periodic) L_5 orbit at five positions, equally spaced in time on the periodic L_5 orbit (See Fig. 4b). This was done to illustrate that a constellation could be placed in orbit at L_5 with arbitrary spacing requirements through use of solar sail propulsion. Since the concept is to incrementally build the constellation and to continually add to the constellation, the number of constellation members could vary substantially. For this paper, we used five spacecraft to demonstrate the feasibility and ease with which such constellations may be assembled and controlled at L_5. All resulting trajectories are slightly different but have an approximate transfer time of 837 days. The Earth flyby altitudes vary in the range 143-270 km. At 90% efficiency, the transfer time is 961 days. These results are summarized in Fig. 4. Fig. 4a shows a trajectory from Earth to L_5 and Fig. 4b shows the final periodic orbit around L_5 and the locations of the five trajectories with equal temporal spacing; both are in the SEMB PCR3BP rotating frame. Orbits about L_5 are operationally stable, so very little station keeping should be required.

In the above calculations, it was assumed that the spacecraft had a mass of 10 kg, but there is no requirement that the mass of each constellation member be exactly the same. Fig. 5 illustrates the relationship between spacecraft mass and sail size for this specific sail performance (characteristic acceleration a_0=0.068 mm/s^2 and 100% efficiency, where a_0= 8.22 / σ and σ is the sail areal density in g/m^2, which includes the support structure**).**

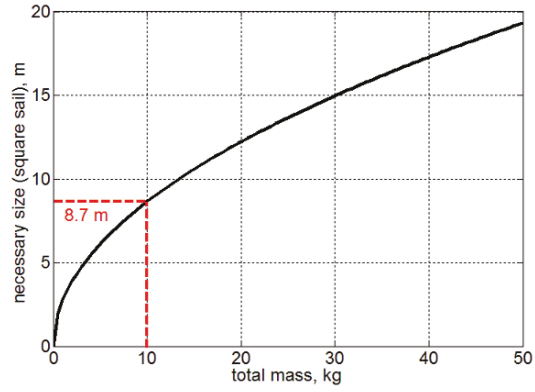

Fig. 5 Sail size vs. mass relationship for a sailcraft with a characteristic acceleration of a_0=0.068 mm/s^2. A perfectly reflecting square sail is assumed.

III. Mission and Spacecraft Overview

The L_5 point has been identified in the NRC Solar and Heliospheric Panel report as a prime location for a space weather forecasting mission for several reasons: (1) it provides global coverage of the inner heliosphere with the potential for observing CMEs and other transient disturbances one to five days before they reach Earth; (2) it allows viewing of solar active regions behind the solar limb rotating Earthward, and (3) it allows *in situ* sampling of solar wind structure at a heliographic longitude distinct Earth's and rotating Earthward, giving ~5 days warning. An L_5 mission would build upon experience using NASA's STEREO mission that had twin spacecraft, carrying both *in situ* and remote sensing instruments, drifting slowly away from Earth at 1 AU. The twin STEREO spacecraft did not have adequate propulsion to stop their drift away from Earth at L4 or L5. This illuminates one advantage of using propellantless solar sails; they can trade time for a velocity change ΔV, instead of trading limited mass for limited ΔV.

The minimal instrumentation necessary to accomplish the above goals is (1) a heliospheric imager to observe and predict trajectories of CMEs that might impact Earth; (2) a magnetograph to measure magnetic active regions on the Sun behind the solar limb which are rotating Earthward (3) an *in situ* solar wind package with a magnetometer to measure the solar wind magnetic field and instruments to measure the solar wind plasma temperature, velocity, density and composition; and (4) an energetic particle instrument to warn of solar energetic particle conditions in the solar wind rotating Earthward. This is the payload that would be carried in the first phase of SWB-L5 mission, divided among the five 6U CubeSats of this fractionated space weather forecasting base. Each of the constellation members would carry a unique payload, fitting in ~2U of the 6U CubeSat, as well as the required engineering subsystems that would be nearly identical for the five CubeSats.

A. Common Engineering Subsystems (~4U)

Each spacecraft would carry a solar sail system requiring two units (2U) of the 6U sailcraft. The remaining basic engineering subsystems, also common to all five, would require an additional 2U. These are (1) a communications system for inter-spacecraft communication and for sending very low rate engineering data to Earth, (2) an attitude determination and control system which would include a star tracker, cold gas thrusters, low-quality gyroscope, and basic Sun sensors, (3) avionics which would include the primary computer, power regulator, battery, onboard memory and a fault tolerant watchdog system, and (4) solar panel system (solar panels and deployment mechanisms). These systems and their current level of technology development for CubeSats are described in the next section.

B. Unique Payloads (~2U)

The unique payloads of the five sailcraft, each require the remaining 2U, are listed here. These subsystems and their current level of technology development are also described in the next section.

Spacecraft 1: High Gain Antenna (HGA) and other hardware necessary to collect science data (from the other constellation members) and communicate the data to Earth. This spacecraft is referred to as the communications hub (3-axis stabilized).

Spacecraft 2: A white light telescope (Heliospheric Imager) to image coronal mass ejections (3-axis stabilized).

Spacecraft 3: A magnetograph to measure the magnetic fields on the surface of the Sun (3-axis stabilized).

Spacecraft 4: A solar wind plasma instrument and magnetometer to measure the interplanetary magnetic fields (spin stabilized, after arriving on-station in L5 orbit).

Spacecraft 5: An energetic particles instrument and magnetometer to characterize the solar energetic particles (spin stabilized, after arriving on-station in L5 orbit).

The L_5 mission envisioned in [2] included additional instruments, to image the Sun and corona in white light, EUV and X-rays, that are unlikely to fit in the available 2U payload volume and that also require larger data rates than the instruments listed above. These instruments increase our understanding of the link between solar magnetic activity and its consequences in the heliosphere and would help develop the capability for even longer term space weather forecasting, e.g. the ability to predict which active regions are about to spawn a CME or flare. These other instruments could be included on larger spacecraft which would travel to L_5 by more traditional methods, or efforts could be undertaken to reduce the size and power requirements of these instruments to fit within the available CubeSat volume, if possible.

IV. Engineering and Science Subsystems

CubeSat capabilities have advanced rapidly as the community has matured over the last ten years. Today, attitude control systems are commercially available providing degree and sub-degree accuracy, power systems can provide 30+W of power, and onboard processing systems include FPGAs (Field-Programmable Gate Array), ARM [Advanced RISC (Reduced Instruction Set Computing) Machine] and other microcontroller architectures. A survey of available technologies can be found in [6,7]. But a unique aspect of CubeSats is that they are constrained by volume due to the launch system. Thus, though many individual capabilities are available through subsystems, not all the subsystems at the “bleeding edge” of capability, can fit within the same spacecraft. Yet

thinking “inside the box" has led to major improvements in design that has continued to increase the overall spacecraft functionality without increasing size. 1U, 3U and 6U platforms are now standardly available, the shapes being dictated primarily by the launch availability.

Below is an overview of each subsystem needed for the SWB-L5 mission and its current state of the art:

A. Engineering Subsystems

Solar Sail Subsystem: The SWB-L5 solar sail system would be similar to that developed for the SolWISE proposal. For SolWISE, the solar sail deployment mechanism, booms and sail were scaled up versions of the existing LightSail-1™ design, built by Stellar Exploration (see http://www.stellar-exploration.com/#!lightsail). These booms are ~3 m longer with upgraded material and the deployer spindle is wider to accommodate them. The full spacecraft areal density would be 134 g/m^2. The sail material would be 5.5 μm Kapton™-E, which is available from DuPont. Kapton™ is also the material used by the Japanese solar sail mission IKAROS for its multi-year mission. Kapton™ is well known for its strong performance in space applications: testing at NASA MSFC showed it vastly out-endures Mylar™, the material used by NASA Ames and Marshall to construct NanoSail-D [8]. The sail would be covered by a thin layer of Al (nominal reflectivity ~90%) and reinforced with a 0.5" x 0.5" grid of 67-denier polyester yarn, providing a strong rip-stop material in case of damage. A reinforced, doubled-over edge, along with grommets at the corners, would assure sail strength during deployment. The sail is grounded to the spacecraft through the metallic booms, minimizing the potential for charge build-up. Due to the high-efficiency Z-fold packing, tested with both Kapton™ and Mylar™, the sail can significantly scale in size, limited only by the packed volume (~2U equivalent volume for SolWISE; somewhat less for LightSail™-1) and boom length. The sail deployment was shown in Fig. 3.

Avionics: Today, CubeSat avionics are generally commercially available radiation-soft hardware that is commonly found in hobbyist hardware or smart-phones. Simple, very cheap microcontrollers, advanced RISC Machine (ARM) processors, and even FPGA’s have served as the central processing core of CubeSats. As a common theme, each system is designed to expect resets; many systems have cascaded watchdog controllers to monitor and reset the system if latch-up occurs. Other systems include hardware timers that will reset the spacecraft automatically after a set period (typically a few days). These approaches allow latch-up robustness to be added to commercial processors. As the community matures, more information will become available regarding the performance of these controllers on orbit, and particularly, how well they have survive ionizing radiation (whose rate is higher in low Earth orbit than in the deep space environment of L_5). The mass of an avionics processor is typically ~100g and a single 10x10 cm printed **c**ircuit board in size. More advanced

avionics boards often include memory storage, real-time clocks, telemetry sensors, and basic attitude determination devices (such as a micro-electro-machined gyroscope). INSPIRE, which may launch as early as 2014, would use a slightly modified version of the Radio Aurora Explorer (RAX-2) avionics boards, developed at the University of Michigan (http://rax.engin.umich.edu/). For the longer duration L_5 mission, it would be desirable to have more radiation tolerant boards. One example is JPL's CubeSat Onboard Processor Validation Experiment (COVE) flown aboard the University of Michigan's M-Cubed spacecraft [9], with COVE-2 for slated flight later in 2013. COVE has at its heart the rad-hard by design Xylinx Virtex-5QV FPGA. This is a rapidly evolving area and it can be expected that more fault tolerant boards will be available in a few years.

Electrical Power System and Battery: The electrical power system can be viewed as the core of the spacecraft. As CubeSats have limited surface area for solar cells, regulators are often needed to buck or boost the voltage to appropriate levels for distribution to systems, and for battery charging. Input regulators are often simple direct-energy circuits, but have been advanced to include single-set-point controllers, or even maximum power point trackers. Onboard batteries are typically lithium-ion cells, with a total energy around 5-10 A-hrs (for a 6U). The batteries are often the densest equipment aboard the spacecraft, and the overall system has a mass of around 500 g and is approximately 10x10x4cm in size. INSPIRE would use the RAX-2 system, which would be sufficient for the L5 mission as well.

Navigation/Communication: While the SolWISE proposal made use of an S-Band radio, the INSPIRE mission would use a JPL X-band navigation/communication radio called Iris that is capable of coherent transmissions for radiometric tracking of the spacecraft. This radio allows for 1000 bps at 1.5 Mkm distances to a Deep Space Network 34 m dish using INSPIRE's onboard omnidirectional patch antenna at 5W of radiated power. At 0.5 kg and 10x10x4 cm, the Iris radio has heritage from the Low-Mass Radio-Science Transponder (LMRST) and NASA's Electra radio. For this mission to L_5, the same Iris-based communication/navigation system, but using a ~0.5m–diameter dish antenna, would be sufficient for inter-spacecraft communication, sending engineering data directly to Earth, and for radiometric tracking of each constellation member. The inter-spacecraft communication rate would roughly 2Mbps, assuming a 5W transmitter and 0.5m antenna. One such 0.5m deployable dish antenna is currently flying on the University of Southern California's *Aeneas* CubeSat; it deploys much like an umbrella (see http://www.isi.edu/projects/serc/aeneas).

Spacecraft 1, the communication hub, would need a larger antenna and more radiated power. Based on 70+W solar panels now being offered for CubeSats and the continuous solar illumination for this mission, the transmitter dc input power available would be >20W continuous or >50W in burst mode. It should be possible to

fit an ~1m deployable antenna and the more powerful communication systems in the ~2U additional space allocated for this purpose in Spacecraft 1. CubeSat navigation/communications subsystems are evolving rapidly and the L_5 mission could take advantages of these advances.

Attitude Determination and Control System (ADCS): Today's attitude determination and control systems often make use of magnetometers and magneto-torquers to orient the spacecraft in LEO. Both of these systems become inoperable in deep space due to the lack of a strong enough magnetic field. Instead, for the L5 mission, a reaction wheel system would be needed to torque the solar sail into the correct position, and the solar sail itself and adjustable solar panels (as in the SolWISE concept, see Fig. 2) used to de-saturate the reaction wheels (using calibrated and controlled asymmetries in solar pressure). A system using cold gas thrusters could also be used as a backup for de-saturating the reaction wheels, for the initial despin of the spacecraft after deployment, and for performing any necessary small impulsive maneuvers. A three-dimensional printed cold-gas system, developed at the University of Texas for the Bevo-2 CubeSat (http://lightsey.ae.utexas.edu/research/bevo-2/), would be sufficient. The thruster system remains at approximately 10x10x5cm in size (~½ U), and weighs approximately 0.5kg. Integrated into this system, a small star tracker provides attitude determination accurate to less than 0.05°. A low-quality Micro-electromechanical systems (MEMS) gyroscope, along with basic photodiodes, allow for initial despin (by nulling rates and orienting toward the Sun). This system is based upon the INSPIRE attitude determination and control system, but expands its capability through the addition of reaction wheels per the SolWISE proposal. All the pieces needed for the SWB-L5 mission ADCS system exist and, after INSPIRE's flight, would have been tested in space, but they would not have been used together with the solar sail.

B. Science Instruments

The L_5 Heliospheric Imager: The required field-of-view for the L_5 heliospheric imager (L_5HI) is ~60° in order to image solar wind transients near the Sun-Earth line from around 3 solar radii out from the Sun (where most CMEs have formed well defined fronts) to Earth (see Fig. 1). A cadence of 30 mins to one hour would be sufficient to follow those fronts to Earth. The resolution and image size could be adjusted to fit the telemetry capabilities of a CubeSat (<1 kbps at present). The requirements are based on the actual operations of the Heliosphieric Imager (HI) instrument aboard the STEREO mission which has 1 kbps telemetry allocated to it. Images at much lower bit rate (~100 bps) are transmitted daily and used for operational purposes through a so-called space weather beacon (of 500 bps total downlink capability).

Thanks to the STEREO mission, the L_5HI has a strong design heritage. It derives from the designs of the heliospheric imagers on the Solar Orbiter (SoloHI) and Solar Probe Plus (WISPR) missions currently under

development. WISPR, in particular, measures about 30cm x 14cm x 58cm and weighs 6 kg and is a smaller version of STEREO's HI instrument (55cm x 26cm x 84cm, and weighing 15 kg) currently in operation on the STEREO mission. WISPR is an optimal starting point for the L_5HI because (1) its focal plane is actually 1U, (2) it could easily accommodate a 60° telephoto lens, and (3) the instrument electronics meet our requirements in a ½ U package.

The most important factor for an HI telescope is successful suppression of the stray light entering the detector assembly. This requires baffling which defines the instrument size and can be quite complicated depending on the location of stray light sources on the spacecraft. In the case of a CubeSats, where L_5HI is the only instrument, the necessity for baffling is reduced, and the plane of its aperture is placed entirely sunward of the solar sail. Solar photovoltaic panels would need to be placed amidships, rather than on the sunward end of the spacecraft as shown for SolWISE, but this is a reasonably straightforward change. The width and height of the instrument are then driven by the size of the detector focal plane. The length of the instrument, on the other hand, is driven by solar occultation considerations. There needs to be a certain distance between the forward occulters and the lens depending on the desired minimum elongation angle from the solar limb. The longer the distance, the higher resolution and single-to-noise ratio we could achieve close to the solar limb.

To remain within the 2U volume, we insert the WISPR focal plane that is conveniently around 1U (10cm x 10cm) in the 2U box (see Fig. 6). To gain some extra distance to the occulters, we incorporate them on the top lid of the instrument, making it a deployable structure. This novelty would result to an effective 40cm distance when the lid deploys. This allows us to set 3-5 occulters along the first 10cm and thus minimize the solar stray light sufficiently to image starting as close as 3 solar radii from the Sun. We estimate the instrument mass at about 3 kg.

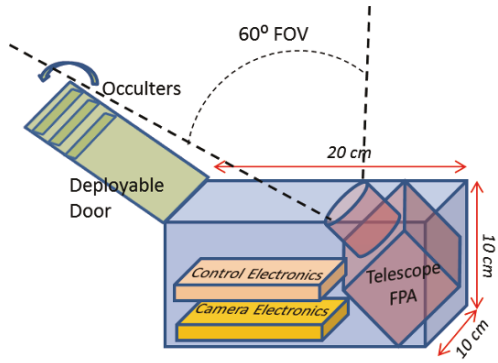

Fig. 6 Strawman concept for a 2U Heliospheric Imager based on the WISPR instrument on the Solar Probe Plus mission.

The telescope concept is based on technology at Technology Readiness Level (TRL) 6 and above. The deployable lid is a new concept. The structure must satisfy an alignment requirement between the last occulter and the lens of about <100 micrometers and must maintain the positioning of the occulters to the same accuracy. However, these requirements would not difficult to meet with properly designed and tested mechanical structures. Another open issue is the pointing stability of the spacecraft. Our concept of operations for the L5HI baselines a 2k x 2k CMOS/Active Pixel Sensor, developed for SoloHI, binned 2x2 to increase the signal-to-noise ratio. The effective spatial resolution is about 7 arcmin. This is beyond presently-demonstrated CubeSat pointing capabilities (of the order of one arcmin), and the instrument needs to maintain stable pointing for the duration of the exposure, which can be one minute or more. One approach being pursued for other investigations involves an MIT/JPL piezoelectric stage using guide stars, as described for the ExoplanetSat concept [10]. While development of pointing control mechanisms/algorithms to achieve arcmin-level pointing stability for long durations is required, at least one plausible path has been shown as referenced, and there are many applications driving the desire for CubeSat-compatible implementations.

The L_5 Magnetometer: Solar wind *in-situ* magnetic fields would be measured using a compact vector helium magnetometer (CVHM), which would occupy ~½ U volume at launch, and would have a boom-deployed sensor to reduce the effects of residual spacecraft magnetic fields. The CVHM is the latest in a long heritage of helium magnetometers, and is chosen because of its high stability, which facilitates accurate measurements of field differences between spacecraft. The CVHM is a laser-pumped helium instrument, where the traditional helium lamp is replaced with a diode laser. Laser pumping increases the potential sensitivity of the sensor significantly, with this sensitivity being traded for a reduction in volume in the CVHM. The elimination of the lamp reduces further the sensor size. The laser is fiber coupled to the sensor, allowing the laser diode and its electronics to be

housed with the instrument electronics. The most recent heritage for the CVHM is the Dynamo sounding rocket experiment, which validated the fiber-coupled CVHM instrument. Further development is underway for JPL's INSPIRE CubeSat mission, with an anticipated launch as soon as 2014. The CVHM sensitivity would be < 10pT, with a frequency range of DC - 10 Hz. For the SWB-L5 mission, no further development is needed beyond what is being done now for INSPIRE. The data rate is variable with a peak at about 500 bps for10 samples/sec and 2 b/s at the low end for 2 samples/min; both rates assume a factor of two compression.

Solar Wind Instrument: The solar wind measurements required at L_5, the ambient velocity, density and temperature of the solar wind, can be measured by a Faraday Cup (FC) instrument. This could be packaged in 1U by taking advantage of modern electronics miniaturization. The single FC instrument derives its heritage from the instruments on NASA's WIND (Solar Wind Experiment-SWE) and on Triana [11, 12]. The instrument has a field of view of ~60° and provides a reduced velocity distribution of the solar wind along its mounting direction. It utilizes the spacecraft spin to determine the angular variation of the distribution in the spin phase direction, and a split pre-amplification and detection system from which the direction of the solar wind in the other direction (spin elevation). By on-board processing, moments of the distribution are thus obtained. The data rate would be approximately 50 bps, assuming a factor of two compression.

Energetic Particle Detector: The energetic particle detector is required at L_5 to determine, using the measured arrival times of protons and electrons accelerated at shocks, the shock distance at the time of the acceleration [13]. The arrival times of these particles, when correlated with X-ray fluxes as measured by NOAA Geostationary Operational Environmental Satellites (GOES) provide verification of the presence of reconnection or shock-acceleration at emerging shocks at the base of the corona. The approximate field line length as determined by modeling of local magnetic field measurements and remote observations of the photospheric field, could be used to ascertain the length estimates from energetic particle measurements. The requirements are energy range from 30keV to 500keV for ions and electrons, with an energy resolution dE/E ~ 50%, and an angular resolution of 45x45°. These could be easily met using the principles of solid state detector (SSD) systems flown on NASA's WIND, STEREO and Time History of Events and Macroscale Interactions during Substorms (THEMIS) missions, using standard electronics miniaturization technologies. The SSD instrument to be flown on Spacecraft 5, takes advantage of the spacecraft spin; it uses four detectors for each species with a combined 180° field of view along the elevation angle direction. This approach provides a full 4π steradian distribution once per spin with the aforementioned energy range and resolution, using a volume of 1U. The data rate would be approximately 500 bps, assuming a factor of two compression.

Magnetograph: The L_5 magnetograph (L_5MI) would provide line-of-sight magnetograms (magnetic field at the surface of the Sun) from the perspective of L_5, to allow early warning of the appearance of solar active regions before they become visible from Earth. Used together with magnetograms from Earth, observation from the L5 vantage point also enhances the fidelity of heliospheric solar wind models by providing a wider angular view of the photospheric magnetic field and thus a more accurate boundary condition on the magnetic field at the Sun. L_5MI would be a filter-based magnetograph, based on a magneto-optical filter (MOF) [14]. The MOF has extensive ground-based heritage in Doppler and magnetic imagers, and a flight version of the filter has been developed to TRL 6 at JPL. L_5MI would observe the photospheric Potassium line at 770nm, with a 5cm aperture. Recent flight designs of MOF based Doppler-magnetographs have estimated masses around 10 kg, for an instrument with a 7 cm aperture: we estimate that L_5MI, with it's reduced aperture would weigh approximately 5 kg, and could be packaged in a 2U volume. The data rate is 1200 b/s for a 6 hour cadence of 1024 by 1024 pixel magnetograms.

V. Summary and Discussion

In this paper, we have presented a concept for an operational space weather forecasting mission at L_5 using a constellation of five CubeSats. The concept presented could accomplish many of the goals of L_5 missions proposed using conventional spacecraft in the last decadal review, possibly at a much reduced cost. The constellation could later be expanded by adding other spacecraft with new science instruments and new goals. The fractioned Space Weather Base concept presented here represents a beginning for a permanent space weather base at L5. This is just one of the many advantages of this fractioned approach; this and others were discussed in Section I**.**

The orbits at L_5 would be achieved using a solar sail. The trajectory calculation results in Section II demonstrate that starting from an Earth Escape trajectory, it is feasible to send a constellation of multiple CubeSats using solar sail propulsion to reach and insert into a 1000 km diameter orbit around L_5 in less than three years. Starting from GEO altitude adds ~2.5 years to the travel time.

The very rapid rate of technology advances in the CubeSat community has made it so that progenitors of most of the science and engineering subsystems needed for this mission have already flown (RAX-2) or are ready to be flown (Bev-2). INSPIRE, to be launched to interplanetary space in ~2014, would fly several of them, including the communication and navigation system, the avionics system, the cold gas system and one science

instrument, the magnetometer. It would also demonstrate inter-spacecraft communication. The other science instruments discussed in this paper also require little additional technology development.

For some of the technologies and subsystems, further technology development and/or flight testing is needed before this mission is ready to be executed. Certainly a mission is needed to test the solar sail system together with the attitude determination and control system (although all the necessary parts of this system exist). Also the existing avionics systems have not demonstrated the radiation tolerance (and thus lifetime) needed for this length of mission (at least 5 years). There is also a need to test the attitude control necessary for the heliospheric imager (spacecraft 2). As a precursor for this mission, a launch of two of the constellation members (the communication hub and spacecraft 2) to interplanetary space would give all the required flight tests. In addition, there is no design for the communication system for the constellation's communication hub, nor is there an operations concept. A more detailed design will be needed to fully encompass the mission requirements.

Even with this technology development and flight testing needed, it is clear that this mission could be launched before the next Heliophysics decadal review in ~2022. Looking further ahead, the mission concept presented here is on the path towards development of a solar sail/CubeSat-based "fractionated" Solar Polar Imager mission to explore the polar regions of the Sun [15].

Acknowledgements

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Design and Trade-offs of a Pole-Sitter Mission

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This paper provides a mission analysis and systems design of a pole-sitter mission, i.e. a spacecraft that is continuously above an Earth Pole, and can provide real-time, continuous and hemispherical coverage of the polar regions. Two different propulsion strategies are proposed: solar electric propulsion (SEP) and SEP hybridized with a solar sail. For both, minimum-propellant pole-sitter orbits and transfers are designed, assuming Soyuz and Ariane 5 launch options. A mass budget analysis allows for a trade-off between mission lifetime and payload mass capacity (up to 7 years for 100 kg), and candidate payloads for a range of applications are investigated.

I. Introduction

SPACECRAFT in geostationary orbit (GEO) have demonstrated the significant benefits offered by continuous coverage of a particular region. However, GEO platforms can only provide their services in the equatorial and temperate zones, where elevation angles are sufficiently high. At higher latitudes, similar services are provided at present by mainly two types of conventional platforms: highly-eccentric, inclined orbits, or low or medium polar orbits.

The first class includes the well-known Molniya-type orbits: they exploit the oblateness of the Earth (effect of the J_2 term in the geopotential) to maintain the argument of the pericentre constant in time [1]. In order to achieve this condition, their inclination must be fixed at the critical value of 63.4°. This value is still relatively low to obtain a satisfactory coverage not only of the polar caps, but also of the high-latitude regions [2]. Due to their high eccentricity, the orbit apocentre is usually at a distance that is comparable to GEO, and therefore offers a hemispheric view of the Earth. However, an intrinsic limitation is the impossibility of providing continuous coverage over time: despite that the spacecraft spends a significant fraction of its orbital period around the orbit apocentre, the viewing conditions of the poles change continuously, and there is an interval of time in which the

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spacecraft is at the pericentre and coverage is not available. With an orbital period of approximately 12 hours, it has been shown that three to six Molniya spacecraft are necessary to provide satisfactory continuous coverage [3]. Recent research [4] has considered changing the critical inclination of the Molniya orbit to 90°, using a continuous acceleration generated by a solar electric propulsion (SEP) system for maintaining the orbit. However, a continuous and constant polar view is not available.

The second class of orbits largely consists of Sun-Synchronous orbits. Spacecraft in these orbits are used due to the high spatial resolution that they can provide. However, only a narrow swath is imaged at each polar passage, relying on multiple passages (and/or multiple spacecraft) for full coverage. For example, Landsat 7 (altitude of 705 km at 98.2°) completes just over 14 orbits per day, covering the entire Earth between 81 degrees north and south latitude every 16 days.[*] This results in a poor temporal resolution for the entire polar region, as different areas are imaged at different times, hence missing the opportunity to have a simultaneous and continuous real-time view of the pole. At present, these images are post-processed to make a composite image, which can be used, for example, for weather forecasting and wind vector prediction. However, the data that can be extracted is neither complete nor accurate [5].

To overcome these limitations, the ideal platform would be one constantly above one of the poles, stationary with respect to the Earth, in the same way as a GEO spacecraft is stationary above one point on the equator. In this position, the footprint of the spacecraft will be constantly at the pole, in the same way as the footprint of a geostationary spacecraft is constantly at some longitude on the equator. This spacecraft is known in literature as "pole-sitter", which uses low-thrust propulsion to maintain a position along the polar axis (see Fig. 1). The pole-sitter is the only platform that can offer a truly continuous hemispheric view of one of the poles, enabling real-time imaging over the entire disc (Fig. 1b). The first study of this concept was apparently made by Driver [6] in 1980.

[*] Landsat 7 Handbook, http://landsathandbook.gsfc.nasa.gov/ [Cited 17/08/2012]

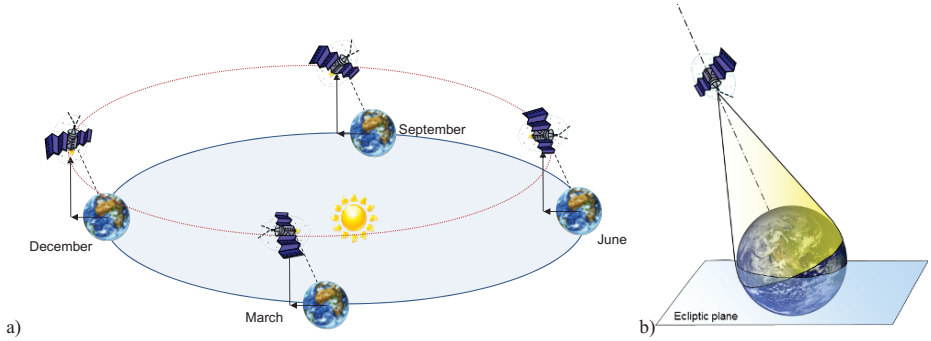

Fig. 1 Pole-sitter concept in (a) Sun-centered inertial frame and (b) Earth-centered frame.

Driver proposed the use of solar electric propulsion (SEP) to maintain the pole-sitter position counterbalancing the gravitational forces. Despite the high efficiency of SEP, the thrusting time and hence the mission duration is always limited by the mass of propellant on-board.

In order to avoid this drawback, some authors investigated the use of a solar sail [7] instead of SEP as a means to provide the continuous acceleration. Solar sailing is a propellant-less spacecraft propulsion system that exploits the solar radiation pressure due to solar photons impinging on a large, highly reflecting surface (the sail) to generate thrust. Solar sails have been recently demonstrated in space [8, 9] and due to the interesting potential of providing acceleration without propellant mass, they seem to be suitable for potentially long duration missions that require a small, but continuous, thrust, such as the pole-sitter. One of the intrinsic limitations of solar sailing is the relationship between the direction of the sail force vector and its magnitude; in particular, the force can never be directed towards the Sun, which is the reason why a sailcraft cannot maintain the pole-sitter position indefinitely [10]. Other stationary positions in the Sun-Earth system [11] and non-Keplerian orbits [12] were studied in the literature, however the spacecraft does not achieve satisfactory conditions for continuous coverage of high-latitude regions.

In order to overcome the limitations of a platform based on a pure sail, which constrains the positions (or more generally the orbits) that can be achieved and maintained, hybrid propulsion has been proposed [13]. Hybridizing SEP and solar sailing is a comparatively recent idea [13], nevertheless research is flourishing in this field, investigating its potential for novel, interesting applications [14-18]. The reason for this interest is due to the fact that in the hybrid system, at the cost of increased spacecraft and mission design complexity, the two propulsion systems complement each other, cancelling their reciprocal disadvantages and limitations. While the hybrid propulsion pole-sitter can in principle enable a mission that is not feasible using only a solar sail and can extend the mission lifetime with respect to the pure SEP scenario, there are still issues associated with the pole-

sitter mission related to the large distance from the Earth. As expected, the acceleration required increases dramatically as the spacecraft is stationed closer to the Earth, and reasonable values of acceleration are obtained only if the Earth-spacecraft distance is of the order of millions of kilometers [6, 10]. Nevertheless, the advantage of having a static or quasi-static platform is such that these concepts are worth considering and investigating in more detailed studies [19]. Although high-bandwidth telecommunications and high-resolution imagery are difficult due to the large Earth-spacecraft distance, a number of novel potential applications are enabled, both in the fields of observation and telecommunications, which will be discussed in the paper. Due to these capabilities, and with the aim of increasing mission lifetime, the authors have undertaken an extensive investigation focused on the concept of a hybrid-propulsion pole-sitter: from the generation of optimal hybrid pole-sitter orbits [10], to system mass budgets [20], and from design of optimal transfers to the pole-sitter orbit from LEO [21], to design of optimal north-to-south transfers [22].

This paper presents the full mission and systems analysis of a pole-sitter mission, covering the launch phase and the operational phase. Several different options will be proposed and assessed, including different propulsion systems for the spacecraft (SEP or hybrid sail-SEP), different mission scenarios, enabling coverage of one pole only or both poles, and different launcher options. Then a systems design will determine the lifetime of the spacecraft, and some candidate payloads will be identified to enable the potential applications of the pole-sitter.

II. Mission Description and Propulsion Options

A. Mission Phases

The complete pole-sitter mission is split into different phases, which will be described in detail and designed in the following sections. The phases are schematically represented in Fig. 2.

The mission starts with a launch and transfer phase. This phase begins with the spacecraft injected into a low Earth orbit (LEO) by the launcher; the type and size of the LEO depends on the launcher used. The launcher upper stage provides a number of impulsive maneuvers, before jettisoned; the spacecraft continues the transfer using its own low-thrust propulsion system, up to the injection point into the pole-sitter orbit.

At this point, the operational phase begins. The operational phase is the one in which the spacecraft is stationed along the polar axis of the Earth, maintaining its nominal pole-sitter orbit. This is obviously the most important phase of the mission, and its duration shall be maximized.

Since each pole is lit only 6 months per year, it is an option, especially for observations in the visible part of the spectrum, to transfer the spacecraft from a North Pole operational orbit to a symmetric orbit below the South Pole, and vice-versa, according to their lighting conditions. Therefore, an additional north-to-south transfer phase

is designed. This phase can be inserted at appropriate points along the nominal orbit to enable the transfer to the other pole, where a symmetric operational orbit can be followed.

These three phases will be described and designed sequentially, starting from the operational phase, which defines the optimal nominal orbit. Then, the transfer from Earth to this orbit will be designed. The optimization of the transfer allows the determination of the maximum mass at the pole-sitter orbit injection, and therefore the spacecraft can be sized and the lifetime assessed. Finally, the north-to-south transfers will be designed, between optimal pole-sitter orbits.

The design of the three phases requires the solution of optimal control problems, which are solved numerically using a direct method based on pseudospectral transcription, implemented in the tool PSOPT [23], and the non-linear programming problem is solved through IPOPT (interior point optimizer, [24]).

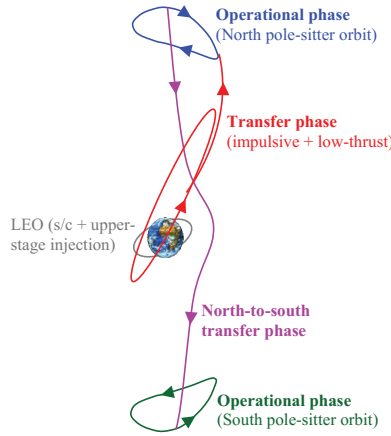

Fig. 2 Representation of the mission phases.

1. Trajectory models

For the transfer and operational phases we consider a three-body problem in which the spacecraft is subject to the gravitational attraction of both the Earth and the Sun. This choice is made since the gravity of the Earth and the Sun are both important for a practical pole-sitter. In particular, we use the well-known circular restricted three-body problem (CR3BP), which describes the motion of the spacecraft, of negligible mass, under the influence of the Sun and Earth (the *primaries*) that rotate in circular motion around each other (at a constant distance of 1 Astronomical Unit, AU). The reference frame is synodic, with its origin at the center-of-mass of the system, the x-axis passing through the Sun and the Earth, and oriented towards the latter, and the z-axis

perpendicular to the ecliptic plane, see Fig. 3. The equations of motion for the spacecraft can be found in Ref. [10].

Note that the launch phase, which occurs close to the Earth, will be modeled using a two-body approach. Details of this model will be given in Section III.B.

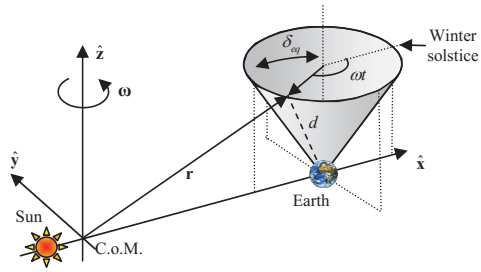

Fig. 3 Synodic reference frame of the restricted three-body problem and pole-sitter reference.

B. Spacecraft Propulsion Options

We consider two different spacecraft architectures. The first is a pure SEP spacecraft, in which solar electric propulsion is used to provide the acceleration required throughout the mission. The spacecraft complexity is relatively low in this case, due to the high TRL of this type of propulsion system. The second is a more advanced, long-term spacecraft that exploits both solar sail and solar electric propulsion on the same bus.

1. Pure SEP

The pure SEP spacecraft can be considered as a conventional spacecraft with deployable solar arrays to power the propulsion system. Usually, the solar panels can be rotated along their longitudinal axis, so as to modulate the power delivered according to the instantaneous need of the spacecraft. The thruster is assumed to be rigidly connected to the spacecraft bus, and the thrust vector is steered by changing the attitude of the spacecraft (the instruments can be mounted on a gimbal).

The key technology parameters of an SEP thruster are the maximum thrust that it can provide, usually in the order of a fraction of a Newton, and its specific impulse. In this paper, we assume that the maximum thrust is used to size the SEP system, as will be explained later, and a fixed specific impulse of 3200 s is conservatively assumed, based on current ion engine technology (existing NSTAR/DS1 [25] or EADS/Astrium RIT-XT [26]). It is foreseen that this specific impulse allows levels of thrust suitable for the spacecraft and mission under consideration.

Since the fuel mass consumption is directly related to the magnitude of the thrust, in general we will try to find trajectories that minimize propellant consumption, in order to maximize the mission lifetime, or alternatively, to maximize the payload mass for a given lifetime.

2. Hybrid Sail and SEP

In this scenario, we envisage the use of a spacecraft that combines solar sailing and SEP. As noted earlier, this adds system complexity, but it can be advantageous in terms of mission lifetime, as it will be shown in this paper.

The hybrid spacecraft has a bus from which the sail is deployed, and thus the sail is rigidly connected to it (see Fig. 4). We assume that the sail can be steered, with relatively modest angular acceleration, by using the attitude control system of the spacecraft. The SEP thruster is mounted on a gimbal system, because it is required that the SEP thrust vector can steer independently of the sail orientation, for control purposes. Furthermore, the SEP system requires electrical power in order to operate. In conventional spacecraft, this is collected through solar arrays that are hinged on the spacecraft bus and can be oriented towards the Sun when power is needed. This type of architecture would be difficult to implement due to the presence of the sail. We instead envisage a layer of thin film solar cells (TFSC) which partly occupy the sail surface, similarly to the IKAROS spacecraft [27].

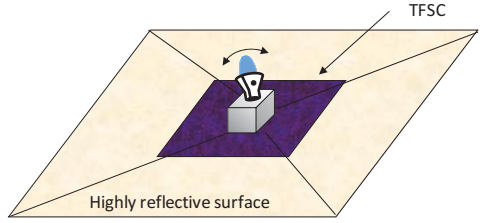

Fig. 4 Hybrid spacecraft with sail, thin film solar cells and steerable thruster.

We assume a partially reflective, partially absorbing solar sail (optical sail model) of total area A_s, such that the solar radiation pressure generates an acceleration with a component perpendicular to the sail and one parallel to it [13, 14]. The sail acceleration is proportional to the parameter β, the *system lightness number*, which is a function of the sail loading $\sigma = m/A$ of the spacecraft (spacecraft mass over total sail area):

$$\beta = \sigma^* / \sigma \tag{1}$$

with $\sigma^* \cong 1.53 \cdot 10^{-3}$ kg/m^2 the critical sail loading. However, β can also be defined as the ratio of the maximum solar radiation pressure acceleration at 1 A.U. (*sail characteristic acceleration*) to the gravitational acceleration. Note that β varies during the mission, because of the decrease in spacecraft mass m due to SEP propellant consumption. We therefore introduce the value $\beta_0 = \beta \frac{m}{m_0}$, constant, in which the subscript "0" refers to the parameters at the injection into the pole-sitter operational orbit, when the time is t_0. Note that the pure SEP spacecraft can be considered as a particular hybrid system with $\beta = \beta_0 = 0$. Values of β up to 0.05 can be assumed for a near-term hybrid system [28]. Recent solar sail demonstrators, however, had considerably lower lightness numbers: JAXA's IKAROS [29] has a 20-m-diagonal square sail and weighs 350 kg (β = 0.001), while NASA's NanoSail-D2 [9] is 4 kg for 10 m^2 (β = 0.003).

In the hybrid spacecraft, the TFSCs are assumed to cover an area $A_{TF} = 0.05A$ on the sail. This area ratio is a conservative estimation based on previous studies [14] and the IKAROS mission [29], and it is used to compute the optimal orbits. The actual value of this area depends on the spacecraft technology parameters, as well as the selected orbit, and will be computed in Section IV.

Another important technological parameter of a sailcraft is the areal density of the sail assembly, σ_s. It measures the mass of the sail per unit surface area, and it is expected that technological developments [28] should enable sails of 10 g/m^2 in the near future. Ultra-thin (around 2 μm of thickness) sails are expected in the mid- to far-term timeframe [30] and can lead, for large sails, to sail loadings of the order of 5 g/m^2.

Given these considerations, the value of β_0 for the hybrid scenario can be decided following the study in Ref. [20]. That work showed that, if a sail assembly areal density of 10 g/m^2 is considered, then the hybrid spacecraft is beneficial only if very long missions are considered (i.e. lifetime > 7 years). Instead, considerable mass saving (or extended lifetime) is expected considering a hybrid system with $\sigma_s = 5$ g/m^2 (or below). Furthermore, for this value, it is found that that $\beta_0 = 0.035$ represents the lightness number in which the spacecraft initial mass, for a given payload, is lowest over a range of mission lifetimes. For these reasons, we select for this scenario: $\beta_0 = 0.035$, $\sigma_s = 5$ g/m^2.

III. Mission Design

A. Pole-Sitter Operational Orbits

An optimal pole-sitter orbit is defined as the one that minimizes the propellant consumption of the spacecraft, while maintaining the pole-sitter condition at any time, and being one-year periodic. The design of this orbit

requires the (numerical) solution of a constrained optimal control problem, in which the control and state history over time is to be determined. Details of the optimization process are presented in Ref. [10].

In the optimization process, the distance from the Earth can vary, however a maximum is set to $d_{max} = 0.01831\,\mathrm{AU}$, i.e. 2.74 million km, in order to prevent the trajectory from going too far away from the Earth, thereby excessively decreasing the spatial resolution for instruments or the data bandwidth of the platform. In fact, it was found [10] that optimal, unconstrained trajectories move further away from the Earth in summer, which is the period in which the North Pole is lit, and therefore observations at optical wavelengths can be performed.

The result of the optimization, for the two scenarios, is presented in the following figures. Note that the figures showing the orbits (or later on in the paper, the transfers) employ a Sun-Earth synodic reference frame centered at the Earth rather than at the Sun-Earth barycenter for interpretation purposes. The optimal SEP-only path is essentially symmetric around spring and autumn, and the spacecraft is closest to the Earth at the summer and winter solstices. Instead, in the hybrid case, the spacecraft is closest to the Earth in winter and farthest in summer: the constraint on the maximum distance is active across the summer solstice for approximately 150 days. This can be seen in Fig. 5. The same figure also highlights that the SEP spacecraft distance to Earth varies between 0.01568 and 0.01831 AU, while for the hybrid case it varies between 0.01369 and 0.01831 AU.

It results also that the SEP acceleration required is less for the hybrid spacecraft than for the pure SEP spacecraft. This is expected, as the sail provides a beneficial contribution to the total acceleration, and results in propellant mass saving. For example, for a spacecraft of mass 1000 kg at injection, the maximum SEP thrust would be 170 mN for the pure SEP system, and 144 mN for the hybrid system. At this point, however, the spacecraft mass is still unknown: it will be determined by the launcher capability and the transfer phase.

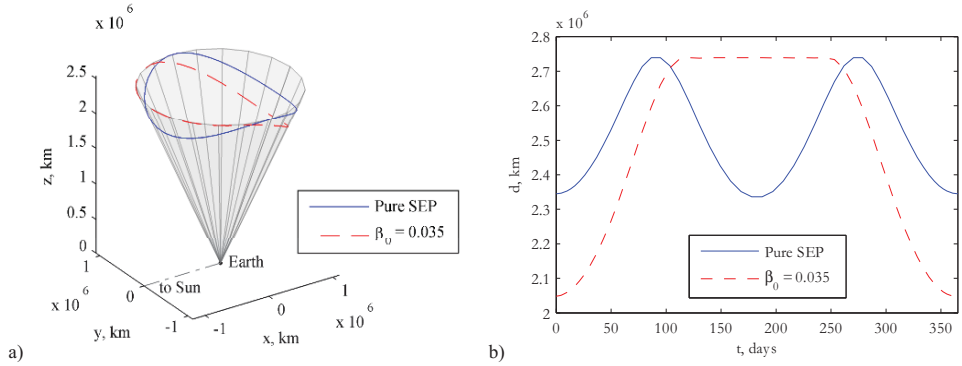

Fig. 5 Fuel-optimal pole-sitter orbits for the pure SEP case and the hybrid case. (a) Optimal trajectories in the synodic reference frame. (b) Distance from the Earth.

B. Launch and Transfer Phase

In order to find the maximum mass that can be inserted into the pole-sitter orbit, this section investigates optimum transfers from LEO up to insertion into the pole-sitter orbit. The transfer is modeled as a launch phase and a transfer phase, as shown in Fig. 2. The launch phase is designed as an impulsive, two-body Soyuz or Ariane 5 upper-stage transfer from a low Earth parking orbit up to insertion into the transfer phase. The transfer phase is modeled in the Earth-Sun three body problem using either pure SEP or hybrid propulsion, depending on the system architecture. The overall objective is to maximize the mass upon insertion into the pole-sitter orbit, assuming that the full launch vehicle capacity into the parking orbit is used. More details on this approach can be found in Ref. [21] . Table 1 provides data of the Soyuz and Ariane 5 parking orbits, as well as performances in terms of mass deliverable to these parking orbits and characteristics of their upper stages.

Table 1 Soyuz and Ariane 5 parking orbit and launch vehicle specifications

Launcher	Parking orbit			Upper stage		Adapter
	Altitude, km	Inclination, deg	Performance, kg	Mass, kg	Specific impulse, s	Mass, kg
Soyuz	200	51.8	7185	1000	330	100
Ariane 5	400	51.6	19000	4540	446	160

The resulting optimal transfers, see Fig. 6a, inject the spacecraft into the operational orbit at winter solstice, i.e. at the point closest to the Earth. More details on the optimal transfers, including the maximum thrust magnitude and mass injected into the pole-sitter orbit can be found in Table 2.

Since the SEP and hybrid pole-sitter orbits are not the same, a direct comparison of the performances of the transfers using the two propulsion techniques cannot be made. However, the mass injected into the hybrid pole-sitter orbit is larger than for the SEP case: an increase of 58 kg for a Soyuz launch and an additional 160 kg for an Ariane 5 launch. Part of this better performance will be due to the smaller Earth to pole-sitter distance at

winter solstice (i.e. upon injection) for the hybrid pole-sitter orbit. However, part will also be due to the smaller propellant consumption in the transfer, because the hybrid case allows for a much longer duration in which the SEP thruster is switched off due to the contribution of the solar sail, see Fig. 6b. However, the total transfer duration is considerably longer.

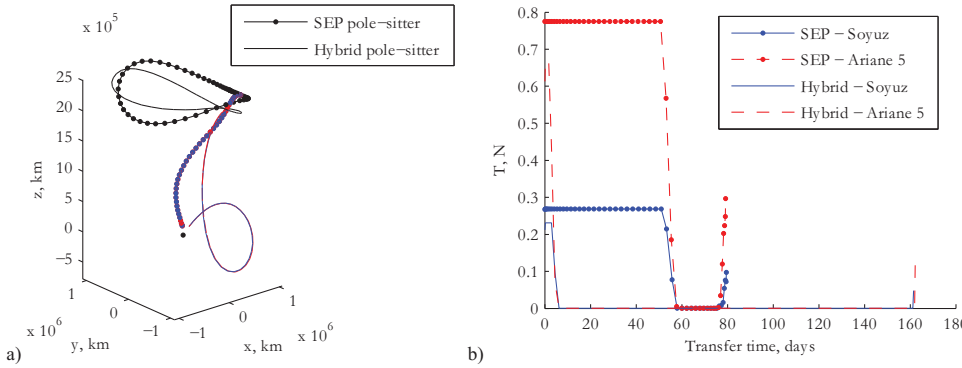

Fig. 6 Optimal transfers for SEP and hybrid propulsion with Soyuz and Ariane 5 launches. (a) Transfer in synodic reference frame. (b) Thrust profile.

Table 2 Results for the optimization of transfer to SEP and hybrid low-thrust pole-sitter orbits including the maximum SEP thrust magnitude and the mass injected into the pole-sitter orbits.

Architecture	Launcher	Transfer time, d	T_{max} , N	m_0 , kg
SEP	Soyuz	79	0.269	1537
SEP	Ariane 5	79	0.775	4439
Hybrid	Soyuz	161	0.231	1595
Hybrid	Ariane 5	161	0.667	4599

C. Transfers between North and South Poles

Due to the tilt of the Earth's polar axis with respect to the ecliptic plane, the North and South Poles are alternately situated in darkness for 6 months per year. For observations being performed in the visible part of the spectrum, this significantly constrains the mission scientific return. Therefore, an additional transfer is introduced that allows the pole-sitter spacecraft to change between pole-sitter orbits above the North and South Poles before the start of the Arctic and Antarctic winters, see Fig. 2. For that, the SEP and hybrid pole-sitter orbits shown in Fig. 5 are mirrored in the ecliptic plane. Viewed in the synodic frame, the Poles are illuminated when the spacecraft is in the Sun-ward part of the pole-sitter orbit, see Fig. 7. Ideally, this means that the pole-sitter spacecraft would follow the north pole-sitter orbit from March to September and the south pole-sitter orbit from September to March. Clearly, in reality this is not feasible since some time needs to be allowed for the

spacecraft to transfer from north to south and vice-versa. Instead, departure takes place between summer and autumn (June – September), while arrival takes place between autumn and winter (September – December), see Fig. 7, where this paper conventionally refers to the seasons in the northern hemisphere. Note that due to the symmetry of the problem, the optimal transfers from north to south can also be used to transfer from south to north.

The concept of transfers between north and south pole-sitter orbits has been introduced before [22], and is applied in this section to the optimal SEP and hybrid pole-sitter orbits of Fig. 5 (using the respective types of propulsion to perform the transfer). Also, for both propulsion strategies, both the Soyuz and Ariane 5 launch cases will be considered together with the corresponding values for the maximum thrust magnitude as provided in Table 2.

In the transfer design, a tradeoff has to be made between the observation time on the operational orbit (which shall be maximized), and the propellant consumption required for the transfer: a fast transfer guarantees long observation times, but requires more propellant.

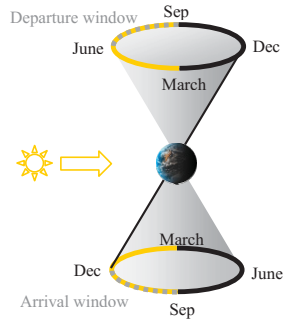

Fig. 7 Schematic of dark (black line) and light (yellow line) conditions on the North and South Poles during the year and departure and arrival window (dotted) for a north-to-south transfer.

For the pure SEP spacecraft, a transfer that guarantees about 90 days of observation time in the operational orbit per year for each Pole uses as much propellant as the operational orbit itself. If the minimum-propellant (i.e. longest) transfer is used, the observation time reduces to 54 days, however the gain in propellant mass is 279.6 kg after 5 years, with respect to staying on the same pole-sitter orbit. On the other end, observation times of up to 94 days can be achieved, by using the quickest transfer. This comes, however, at the cost of an increase in the propellant consumption (45.3 kg more are required over 5 years). Very similar results are obtained for the hybrid case. Note that in the following, the minimum-propellant transfers are used.

D. Full Mission Profile

Finally, to illustrate the full mission profile, Fig. 8 and Fig. 9 represent the trajectory of the pole-sitter spacecraft, including the launch phase, operational phases and north-to-south transfers. Fig. 8 refers to the pure SEP mission, Fig. 9 to the hybrid propulsion one. Both figures represent the Ariane 5 option; the trajectory followed by the Soyuz option is very similar. In each figure, the plot (a) is in an Earth-centered synodic reference frame (Earth and Lagrange points L_1 and L_2 are represented with dots), while the plot (b) is in an inertial, Sun-centered reference frame (the Sun represented with a dot, and the orbit of the Earth with a solid black line). Note that in plots (b) the z-direction is not to scale with x and y, in order to appreciate the out-of-plane displacement of the pole-sitter, which would otherwise be too small with respect to the orbit of the Earth.

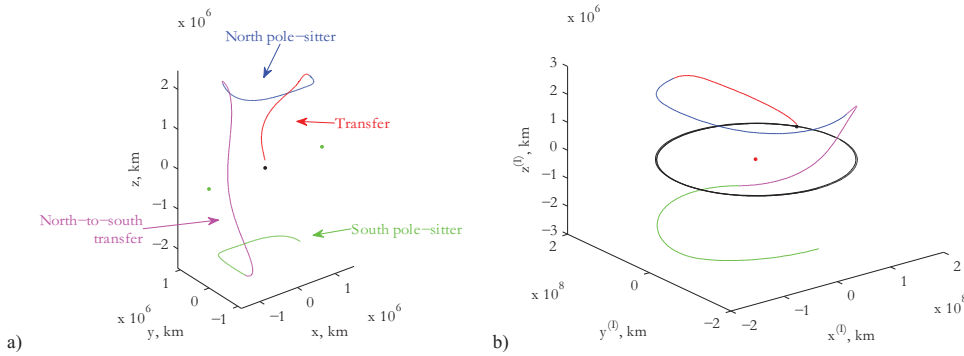

Fig. 8 Pure SEP mission trajectory (Ariane 5 launch) in synodic reference frame (a) and inertial Sun-centered reference frame (b).

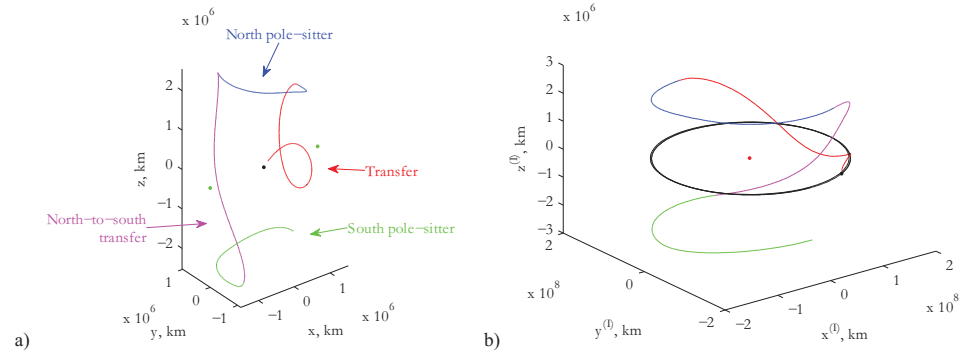

Fig. 9 Hybrid propulsion mission trajectory (Ariane 5 launch) in synodic reference frame (a) and inertial Sun-centered reference frame (b).

IV. Spacecraft Subsystems Sizing

Starting from the total mass m_0 that can be injected into the operational orbit of the pole-sitter (Table 2), a systems design provides the mission lifetime that can be achieved into this orbit, or alternatively the payload mass that can be carried for a given lifetime of the spacecraft. For a preliminary mass budget, the total mass of the pure SEP spacecraft can be split into propellant, tank, SEP thruster, solar arrays, other subsystems and payload. For the hybrid spacecraft, instead, we also consider the gimbal of the SEP thruster, the sail, and the radiators to dissipate any excess of power collected by the sail-mounted thin film solar cells.

Note that the systems mass budget proposed here differs to the one in Ref. [20] in several ways: it considers multiple SEP thrusters, but working in parallel to achieve the necessary thrust, rather than being redundant; the mass of the other subsystems (ADCS, thermal, structure, OBDH, TT&C) is taken into account explicitly, as 30% [1] of the spacecraft dry mass; the pure SEP spacecraft does not employ thin film solar cells, but solar arrays mounted on panels (higher efficiency and areal mass); the pure SEP spacecraft does not employ a gimbal system, as it is assumed that attitude maneuvers can be used to steer the thrust vector; lastly, margins are also taken into account for each subsystem [1]. In particular, the sail and the thin film solar cells are considered new technologies, and therefore a margin of 20% is used. Conversely, the other subsystems are considered to be well-proven technologies, and its margin is set to 5%. The same margin is added to the propellant mass for contingency maneuvers.

Given the initial mass of the spacecraft in the pole-sitter orbit (as found in Table 2 for a given mission scenario), the optimal 1-year trajectory is used to compute the propellant mass m_{prop} needed for a given lifetime $t_{mission}$. Once the propellant mass is found, the mass budget returns the payload mass m_{pl}. This is plotted, as a function of the lifetime, in Fig. 10. Each plot refers to one spacecraft architecture (pure SEP, hybrid).

It can be noted that the mission lifetime does not depend on the injected total mass m_0, but only on the technology that is used to build the spacecraft. The lifetime for the pure SEP system is limited to approximately 4.5 years, while this value extends to about 7 years for the hybrid architecture. This result itself should be sufficient to justify the interest in the hybrid propulsion technology for this type of mission, and in general for all those missions which require a continuous acceleration [20]. Furthermore, for nearly equal injected masses, the hybrid spacecraft can carry the same payload mass for a longer mission lifetime. Finally, for a particular spacecraft architecture, the payload mass scales with the injection mass m_0.

Considering a 100-kg payload mass, launching with Soyuz, the lifetime is 3.6 years if the spacecraft is using pure SEP technology, or 5.6 years if using hybrid propulsion. These lifetimes extend to 4.24 and 6.58 years when launching with Ariane 5 for the two architectures respectively.

The subsystem design also allows the computation of the mass of the other subsystems, some of which are reported in Table 3. The size of the total sail assembly (reflective surface and thin film solar cells) of the hybrid spacecraft is 191 m for the Soyuz launch and 324 m for the Ariane launch, assuming a square assembly.

The hybrid configuration furthermore allows a lower power budget, by reducing the thrust needed per unit mass of the spacecraft from the SEP thruster. For the pure SEP spacecraft, the maximum power required by the SEP thruster is 6 kW (Soyuz) and 17.4 kW (Ariane 5). For the hybrid spacecraft, instead, the power is 5.2 kW (Soyuz) and 15 kW (Ariane 5), despite the fact that the total injected mass of the spacecraft, m_0, is slightly larger than in the SEP case.

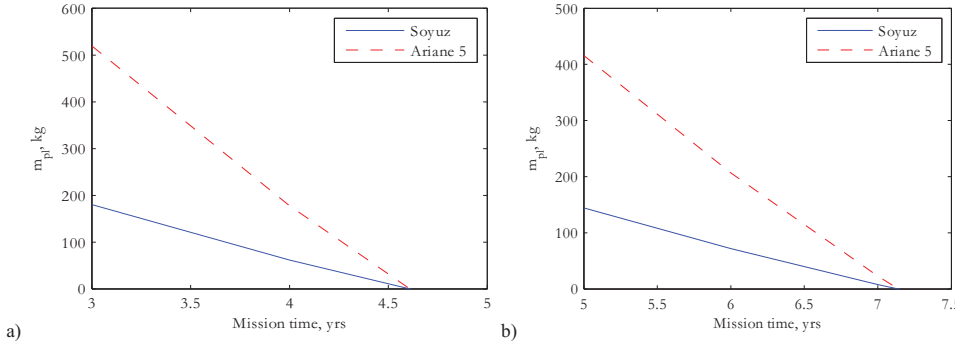

Fig. 10 Payload mass (m_{pl}) as a function of the mission lifetime, for the pure SEP spacecraft (a) and the hybrid spacecraft (b).

Table 3 Design points. Masses are without margins.

Architecture	Pure SEP		Hybrid	
Launcher	Soyuz	Ariane 5	Soyuz	Ariane 5
Lifetime, $t_{mission}$, yrs	3.6	4.24	5.6	6.58
Payload mass, m_{pl}, kg	100	100	100	100
Pole-sitter injection mass, m_0, kg	1537	4439	1595	4599
SEP mass, kg	121	348	104	299
Propellant mass, m_{prop}, kg	675	2192	698	2242
Other subsystems mass, kg	259	674	269	707
Solar array/TFSC area, m^2	44	127	121	349
Solar sail mass (reflective), kg	-	-	182	524
Total sail area (reflective + TFSC), m^2	-	-	191 × 191	324 × 324
Maximum SEP thrust, mN	269	776	231	667
Maximum SEP power, kW	6	17.4	5.2	15.0

A. Payload Selection and Applications

We now wish to investigate possible payloads that could be used on the pole-sitter spacecraft. As noted in the introduction, the pole-sitter spacecraft could serve as a platform for Earth observation and science, and a data relay for telecommunications. Concerning the former, taking into account the considerable distance of the spacecraft from the Earth, high-resolution imaging is limited to the near-visible part of the spectrum (from infrared to ultra-violet). Spatial resolution in the visible wavelength in the range 10-40 km should be available for the pole-sitter [1], which enables real-time, continuous views of dynamic phenomena and large-scale polar weather systems [19]. The creation of atmospheric motion vectors (AMV) would also make use of the stationary location of the platform, avoiding gap problems related to geolocation and intercalibration that composite images introduce [5]. Glaciology and ice-pack monitoring would also benefit from continuous, but low resolution polar observation [5]. Ultraviolet imagery of the polar night regions at 100 km resolution or better would enable real-time monitoring of rapidly-changing hot spots in the aurora that can affect high frequency communications and radar [19]. As the resolution degrades with increasing wavelength, it is unlikely that sensing in the microwave band could provide some useful information. However, radio science can detect the total flux of radiation reflected and emitted by the Earth at the poles.

Candidate instruments are found in the literature of deep space missions. In fact, optics for deep space mission typically have long focal lengths and narrow filed of views (FOVs), and they are therefore ideal for the pole-sitter. For example, the NASA mission Galileo, launched in 1989 towards Jupiter and its satellites, was designed to perform observations of Jupiter's atmosphere composition, weather phenomena and auroras. The Solid State Imaging (SSI) instrument (29.7 kg of mass) had an 800x800 pixel sensor in the near-visible. This instrument would provide a resolution of about 27 km at nadir for the pole-sitter. The Near-Infrared Mapping Spectrometer (NIMS), instead, could provide 137-km resolution with about 18 kg of mass. Both these instruments had apertures of less than 30 cm.

Another mission, the NASA Deep Space Climate Observatory (DSCOVR), was designed for Earth observation and science from the L_1 Lagrange point, which is located approximately 1.5 million km from the Earth, of the same order as the distance of the pole-sitter mission (where angular size of the Earth disc is approximately 0.5 deg). Despite that this mission was initially canceled, its payload would be similar to that which could be used on a pole-sitter spacecraft. The Earth Polychromatic Imaging Spectrometer (EPIC) was designed to study ozone, aerosols, clouds, sulfur dioxide, precipitable water vapor, volcanic ash and UV irradiance. It offered a 2048x2048 pixel image in the near visible, with an aperture of 30 cm and mass of 65 kg.

At pole-sitter distance, it would offer a resolution of about 14.5 km. A complete discussion on potential applications enabled by such instrumentation is found in Ref. [2]; suffice to underline that all these instruments are well below the 100 kg mass assumed in the pole-sitter design.

Concerning the use of the pole-sitter for telecommunications, the main application would be its use as a low-bandwidth data relay with polar regions, and may be particularly useful in the southern hemisphere, where there are key Antarctic research activities ongoing and communication capabilities are limited. McInnes and Mulligan [5] envisaged possible applications including data links for scientific experiments, links to automated weather stations, emergency airfields and telemedicine. In the same work, the authors proposed the use of polar stationary spacecraft as data relay for future NOAA polar orbiting satellites, such as the NPOESS system. In this case, the telecommunication subsystem would be the main payload of the spacecraft. However, even for Earth science, there will be a need to download a relatively large volume of data from the spacecraft to the Earth real-time (high resolution imaging, for example). Considering the distance of the pole-sitter from the Earth, a high-gain steerable antenna will be required for either or both of these tasks.

Finally, a polar-sitter type platform could be used for high latitude ship tracking and telecommunications, and to support future high-latitude oil and gas exploration, especially if northern sea routes open due to climate change.[*] We refer again to Ref. [2] for further discussion on possible applications.

V. Conclusions

In this paper a full, preliminary mission analysis and systems design of a near-term and far-term pole-sitter mission is provided, where the distinction comes from the use of either existing, solar electric propulsion (SEP) or more far-term hybrid SEP and solar sail propulsion. The platform would provide a vantage view point on either pole of the Earth, with essentially unlimited temporal resolution (real-time observations) by means of one spacecraft only. Optimal transfers from north to south and vice-versa allow observation of the pole that is lit, at no extra cost in terms of propellant consumption. Moreover, in some particular cases propellant savings can be achieved through the use of these transfers, allowing for an extension of the mission lifetime or alternatively an increase in the payload mass. Using the full potential of either a Soyuz or Ariane 5 launch vehicle, a systems mass budget demonstrated that it is potentially possible to maintain a 100-kg payload in the pole-sitter orbit for at least 4 years for the short-term SEP-only mission, and 6 years or more for the far-term hybrid mission.

[*] http://www.esa.int/esaEO/SEMT7TRTJRG_index_0.html [cited 17/08/2012].

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Reduction of Martian Sample Return Mission Launch Mass with Solar Sail Propulsion

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Solar sails have the potential to provide mass and cost savings for spacecraft traveling within the inner solar system. Companies like L'Garde have demonstrated sail manufacturability and various in-space deployment methods. The purpose of this study was to evaluate a current Mars sample return architecture and to determine how cost and mass would be reduced by incorporating a solar sail propulsion system. The team validated the design proposed by L'Garde, and scaled the design based on a trajectory analysis. Using the solar sail design reduced the required mass, eliminating one of the three launches required in the original architecture.

Acronyms

ACO	Advanced Concepts Office
ACS	Attitude Control System
AR&C	Arrival, Rendezvous & Capture
C&DH	Command & Data Handling
CP1	Colorless Polymer 1
EEV	Earth Entry Vehicle
ERS	Earth Return Stage

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GN&C	Guidance Navigation and Control
GR&A	Ground Rules and Assumptions
GSD	Ground System Demonstration
HGA	High Gain Antenna
IEM	Integrated Electronics Module
Isp	Specific Impulse
JPL	Jet Propulsion Laboratory
LEO	Low Earth Orbit
LGA	Low Gain Antenna
MEL	Master Equipment List
MIDAS	Mission Design and Analysis Software
MLI	Multi-Layer Insulation
MOI	Mars Orbit Insertion
MPDM	Multi-Purpose Data Module
MPS	Main Propulsion System
MSFC	Marshall Space Flight Center
MSR	Mars Exploration Sample Return
NASA	National Aeronautics and Space Administration
ORU	Orbital Replacement Unit
OS	Orbiting Sample
PDU	Power Distribution Unit
PMAD	Power Management and Distribution
SBC	Single Board Computer
SRO	Sample Return Orbiter
SSPA	Solid State Power Amplifier
TRL	Technology Readiness Level
UHF	Ultra High Frequency

I. Introduction

Utilizing solar sails instead of traditional chemical propulsion can provide mass and cost savings for interplanetary travel within the inner solar system [1]. L'Garde and other private ventures have proven both the manufacturability of solar sail materials and various in-space deployment methods. This study involved the incorporation of an existing design of a solar sail for a Solar Sail Technology Demonstration Mission [2] by L'Garde called Sunjammer [3]. The sail design would be applied to the propulsion system of an existing Mars Exploration Sample Return Mission Architecture [4,5,6] developed by the Jet Propulsion Laboratory (JPL). The solar sail is currently in manufacturing development phases in support of NASA's Office of the Chief Technologist Advanced In-Space Propulsion Systems [7]. The purpose of this study was to estimate the reduction of mission cost and mass for an existing architecture through the inclusion of the solar sail.

Current Mars Sample Return Project Architecture

The point of departure for this study was the Mars Exploration Sample Return (MSR) Mission Architecture [4,5,6] which requires separate launches of an exploration rover, an orbiter with an Earth return stage, and a Martian ascent vehicle with a sample return canister. The current launch manifest spans six years with operations lasting an additional four years and costs totaling billions of dollars.

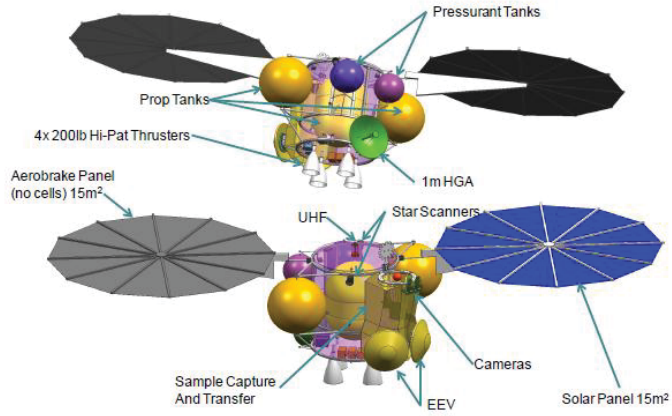

Figure 1 Sample Return Orbiter Design, Credit NASA JPL

The Sample Return Orbiter (SRO), Figure 1, consists of a spacecraft bus capable of relaying communications from Earth to the exploration rover on Mars and vice versa. A Sample Capture and Transfer canister attached to an Earth Entry Vehicle (EEV) retrieves the Orbiting Sample (OS) launched on the ascent vehicle in orbit and

returns back to Earth. The amount of chemical propellant required to send the orbiter to Mars, orbit during surface operations, rendezvous and capture the Martian soil sample, and return back to Mars eliminates the option of ridesharing with any of the other two elements. The first step in the study looked into replacing the required chemical propellant for Earth return with a solar sail propulsion unit.

II. Ground Rules and Assumptions

To reduce the amount of mass the solar sail would need to propel the SRO was split into two vehicles: Orbiter and the Earth Return Stage (ERS). The orbiter would support the primary mission of traveling to Mars, performing relay communications, and assist in rendezvous and capture of the sample. The ERS would separate from the Orbiter after the sample has been secured, deploy the sail, and spiral out from Martian orbit to travel back to Earth. The basic ground rules and assumptions are listed in Figures 2 and 3 for the Orbiter and ERS.

Orbiter	General GR&A	Notes
Orbiter Mission	1. Provide telecommunication relay operations for the MSR lander, fetch rover, and MAV 2. Perform orbital sample (OS) rendezvous and capture 3. Deploy solar sail on return vehicle and perform separation of return vehicle 4. AR&C sensors and OS capture mechanism will be contained on the ERS	Phase 3 of mission ops is all new for the solar sail return proposal.
Risk Class: Fault Tolerance	Risk Class C: Single fault tolerance for critical systems	
Orbital Debris and Disposal	Spacecraft will comply with appropriate sections of NASA-STD-8719.14	
Spacecraft Orbital and Cruise Attitude	No particular attitude assumed	
Rover and Lander Restrictions	The landing locations of each of the science elements should be where the orbiter can support them	
Number of launches	Reduce number of launches from 3 to 2	Current MSR mission architecture calls for three launches

Figure 2 Ground Rules and Assumptions for the Orbiter

Earth Return Stage	General GR&A	Notes
ERS Mission	1. Perform a Mars spiral departure and Earth direct entry 2. Sails will be inflated before separation 3. Separation will be performed by the orbiter 4. AR&C sensors and OS capture mechanism will be contained on the ERS	
Risk Class: Fault Tolerance	Risk Class C: Single fault tolerance for critical systems	
Orbital Debris and Disposal	Spacecraft will comply with appropriate sections of NASA-STD-8719.14	Controlled disposal at Earth
Mass Minimization	Minimize Earth return mass by utilizing solar sail and staging (leaving as much mass as possible in Mars orbit)	
Spacecraft Orbital and Cruise Attitude	Sun pointing	

Figure 3 Ground Rules and Assumptions for Earth Return Stage

III. Orbiter Chemical Propulsion Mission Analysis

To maintain consistency with the MSR architecture, the transit to Mars in this study was performed by a traditional bi-propellant chemical propulsion stage. MIDAS (MIssion Design and Analysis Software) developed by NASA JPL was used to optimize Earth-to-Mars trajectories over a date range of 2022 to 2035.

Earth Departure			Outbound		At Mars		
Departure Date	C3 (km^2/s^2)	Launch Declination (deg)	TOF (days)	HCA (deg)	Arrival Date	Vinf (km/s)	MOI ΔV (km/s)
08/30/2022	15.151	3.319	347.024	212.10	08/12/2023	2.604	0.877
09/30/2024	11.369	14.915	332.696	209.17	08/29/2025	2.445	0.804
10/31/2026	9.229	28.746	311.054	205.71	09/07/2027	2.571	0.861
11/23/2028	9.116	28.696	300.113	212.83	09/19/2029	2.970	1.059
12/24/2030	*10.794*	*14.994*	*283.084*	*216.15*	*10/03/2031*	*3.480*	*1.342*
04/17/2033	*9.086*	***-55.268***	*199.722*	*148.44*	*11/02/2033*	*3.317*	*1.248*
02/21/2033	*13.646*	*7.369*	*260.000*	*206.40*	*11/08/2033*	*3.934*	***1.617***
06/27/2035	10.372	8.822	201.889	148.63	01/15/2036	2.630	0.889

Figure 4 MIDAS generated optimal Earth-to-Mars Trajectories

The arrival delta-V at Mars was minimized while keeping the Earth-departure C3s between 15 – 20 km^2/s^2. At Mars, the spacecraft must perform the Mars orbit insertion (MOI) maneuver into a 1-day Sol orbit. Afterwards, the periapsis must be lowered into the Martian atmosphere, so that aerobraking can begin. Aerobraking lowers the apoapsis altitude from 33,810 km to 500 km. Then, a "walk-out" maneuver is performed to raise the periapsis out of the atmosphere and to an altitude of 500 km. Finally, the spacecraft must rendezvous with and capture the OS, which will be launched from the Martian surface.

For propellant computations, an MOI delta-V of 1.1 km/s is assumed. This value was previously used by JPL and is assumed to contain gravity losses. This assumption means that the launch dates in 2030 and 2033 must be avoided.

To obtain the propellant requirement, an initial mass of 2110 kg was assumed based on initial study team mass estimations. Removing the launch vehicle adapter from the initial mass estimate brings the mass to a starting value of 2080 kg. The propulsion system was assumed to be a dual-mode system with hydrazine and NTO propellants. Large burns are performed by the main propulsion system (MPS), which is a bipropellant system with a specific impulse (Isp) of 326s. The smaller maneuvers are performed by the smaller, monopropellant attitude control system (ACS) thrusters, which have an Isp of 209 s. The ACS engines also perform attitude control during the MPS maneuvers.

No sample return delta-V was given in the JPL design reference documentation. A value was reverse-engineered such that a launch mass of 3270 kg generated a total propellant load and burnout mass consistent with their design. This resulted in a sample rendezvous (which also contains margin) delta-V of 250 m/s.
Using these assumptions, the required MPS maneuver propellant is computed to be 440.2 kg, and the required ACS propellant is 9.3 kg.

IV. Solar Sail Mission Analysis

Our proposed solar sail has a heritage with the L'Garde 20 m on a side Ground System Demonstration (GSD) hardware that was developed and successfully vacuum deployed during the 2003-2006 timeframe at the Plumb Brook facility at NASA Glenn Research Center. This design had heritage from a JPL, NOAA and Air Force Space Technology 5 (ST-5) proposal to the NASA New Millennium Program. To demonstrate scalability, the GSD 20-meter system was the central section of a larger 100 m on a side solar sail, and the mass estimates for the larger sail were derived from the 20 m system. This hardware is essentially the 38 m on a side SunJammer flight demonstration that has been selected by the NASA Office of Chief Technologist and will launch in 2014.

Mars Spiral Out

Once the OS has been captured and the sail has deployed in Martian orbit, the ERS will spiral out from Mars before performing a Mars-to-Earth transfer. The time of flight calculated to perform the Mars Spiral Out was 435 days assuming a characteristic acceleration of 0.50 mm/s^2 starting from a 500 km Mars circular orbit. The sail area was assumed to be 18,887 m^2 propelling a payload mass of 200kg which includes the sample. The total sail mass system was 115 kg utilizing a 2.5 μm Colorless Polymer 1 (CP1).

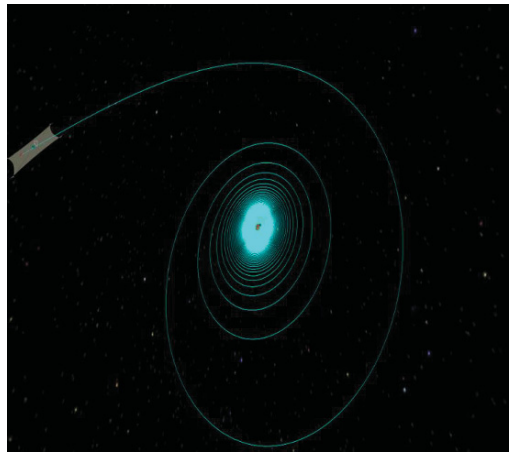

Figure 5 ERS Spiral Out from Mars

Mars-to-Earth Transfer

After an estimated time of 435 days performing the Mars spiral out maneuver, the ERS will begin the Mars-to-Earth Transfer. Assuming the 0.50 mm/s^2 characteristic acceleration, the ERS will leave from Mars perihelion to achieve an Earth entry velocity of 12.8 km/s^2 after 507.5 days.

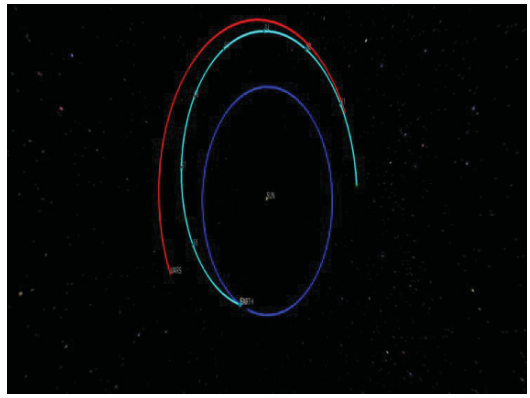

Figure 6 Mars-to-Earth Transfer of 507.5 days

V. Configurations

The Orbiter and ERS launch as one spacecraft and separate after Arrival, Rendezvous & Capture (AR&C) of the OS. The Orbiter will perform primary operations during launch, Earth-to-Mars transit, MOI, and Mars operations. The ERS will be kept in "keep alive" mode during that timeframe. The following section will outline the configurations of the Orbiter and ERS.

Orbiter

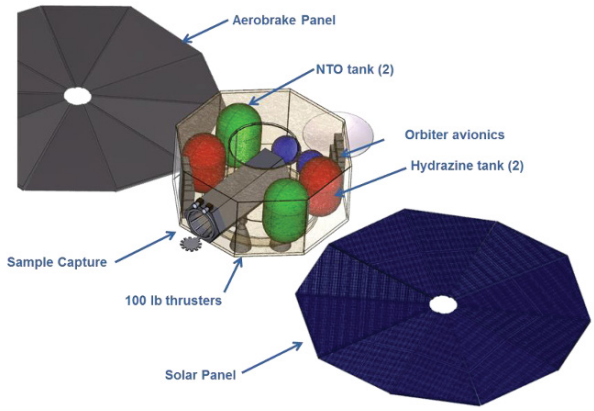

Figure 7 Orbiter Configuration

The configuration of the orbiter, Figure 7, consists of an octagonal bus. The orbiter bus holds all of the orbiter systems. It holds four bi-propellant tanks which feed the main propulsion and reaction control propulsion system. The sample capture system is integrated such that the EEV interface is in the central core. The orbiter avionics are attached to the outer structural walls which also provides the required thermal radiating area. The orbiter also has one 6 m diameter solar array panel and a dummy solar array panel that is used as an aerobrake panel.

Earth Return Stage

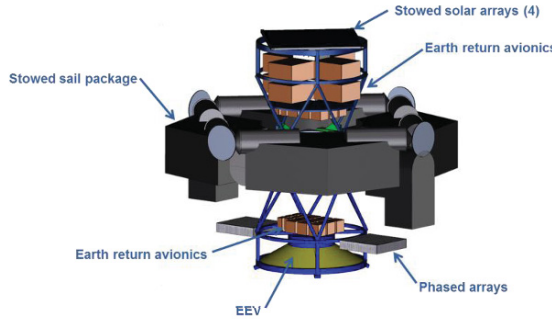

Figure 8 Earth Return Stage (ERS), Sail Stowed

The ERS is comprised of the L'Garde sail package and some new structure that holds the ERS avionics and the EEV capsule. It also has supporting solar arrays and communication phased arrays. See Figure 8. The sail/ERS mates to the top deck of the orbiter. The spacing of the two is set such that the sail package sits on the top deck just as the EEV capsule mates to the sample capture assembly. The total height of the ERS and Orbiter is about 3.3 m tall and 2.4 m across the base. See Figure 9 for details.

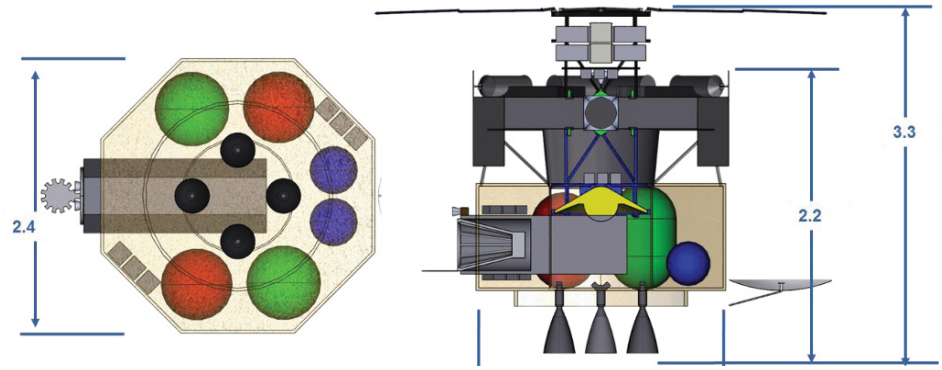

Figure 9 Orbiter & ERS launch configuration and dimensions

VI. Spacecraft Structure Analysis

Orbiter Structures

All of the sail, boom and beam mass and approximate dimensions were taken from a previous L'Garde MSR study or extrapolated from study or experimental data. These masses were placed in the FEMAP model as mass elements. The sail main support, inflation tanks and the sail retainment doors were not sized; L'Garde's mass was used. The sail retainment doors as well as thermal treatments and much of the power and avionics mass was "smeared" over some structural panels as "non-structural mass" to account for all of the predicted mass.

The FEMAP model was modeled using Al 2024 with $1/8^{th}$ in thick panels. Support beams below the sail carrier were circular Al tubing with $1/8^{th}$ in wall thickness. Hypersizer sized most panels to the minimum and would have sized them thinner, but all panel thicknesses were limited to a minimum of 0.060 in. See Figure 10.

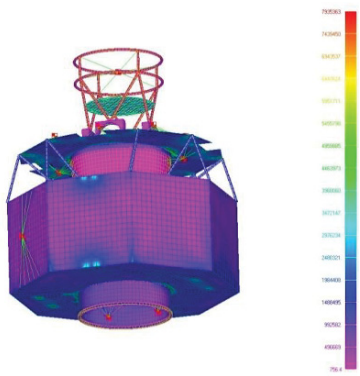

Figure 10 Orbiter FEMAP Model

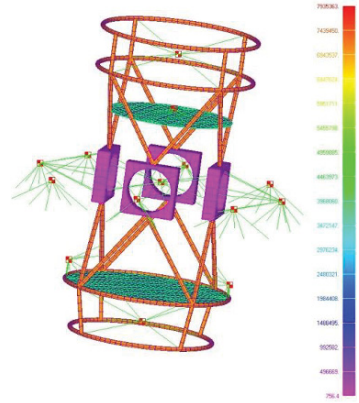

Figure 11 ERS FEMAP Model

ERS Structures

The combined vehicle configuration, the SRO and the ERS were subjected to body loads equal to 6 g's axially and 2 g's laterally at 45 degree intervals radially. The Al structure was meshed and showed a very high Margin of Safety (MOS). When the FEA results were fed into Hypersizer, panel and beam masses were reduced by over 80%. Minimum mass was limited in Hypersizer sizing to select reasonable raw material for manufacturing and provide thick enough panels to conduct heat away from mounted electronics.

FEMAP model versus Pro-E depiction of ERS un-deployed configuration. The ERS structure was completely Al tubing with a couple of 1/8th" thick round avionics panels. These small ERS panels were maintained at 1/8th" thick to facilitate thermal rejection. See Figure 11.

VII. Solar Sail Structures

Solar Sail Structures for the study breakdown into three sections: boom design, beam design, and the net/membrane sail design.

Boom Design

The booms are designed in a semi-monocoque configuration. High modulus longitudinal fibers impregnated with a Sub Tg resin are used to rigidize the structure after deployment and are oriented to absorb the compressive loads in the booms, while the lateral fibers absorb the inflation loads and stabilize the longitudinal fibers and the cross section. The external layer doubles as a bladder and shear web.

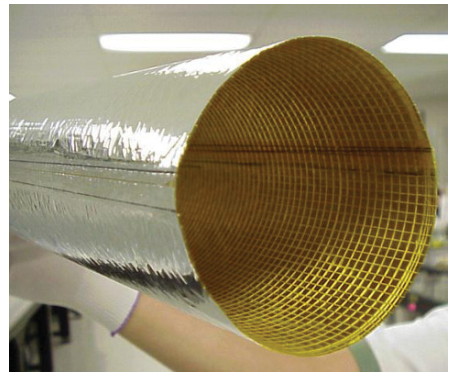

Figure 12 L'Garde Boom

Sub Tg or cold rigidization takes advantage of the increase in modulus of certain materials below their glass transition temperature. Sub Tg rigidizable structures are simple, reliable and are completely passive and in general require no heaters or vents. Multilayer insulation (MLI) will be used to mitigate the effects of on-orbit thermal gradients and to lower overall boom temperature, adding margin to sub Tg rigidization. Thin heater lines are packaged along with the boom material when stowed to keep the boom warm enough to be flexible, and double as actuator control lines during the mission. They deploy concurrently with the boom providing a heat source until the deployment sequence is complete and then will provide power and control the tip vanes at the ends of the booms.

Beam Design

The beam is composed of a boom and spreader system. The spreader system is composed of rings, spreader bars and tension lines. Low mass rigid rings are mounted periodically along the boom. They serve to stabilize the boom cross-section, provide a hard point for mounting of the spreader bars and also provide an ideal point for mounting of the net/membrane lines. Additionally the spreader system consists of a number of tension elements running along the boom. The anchor lines running from the spreader bar tips to the rings provide a shear component transmitting bending loads. The longeron lines running from the boom tip, along the spreader bar tips provides a large sectional moment of inertia to the truss to resist the bending generated by the solar flux. Finally the diagonals crossing over along alternating spreader bar tips increase the torsional stiffness of the truss while providing longitudinal redundancy in the lines in case one or more lines are severed. See Figure 13.

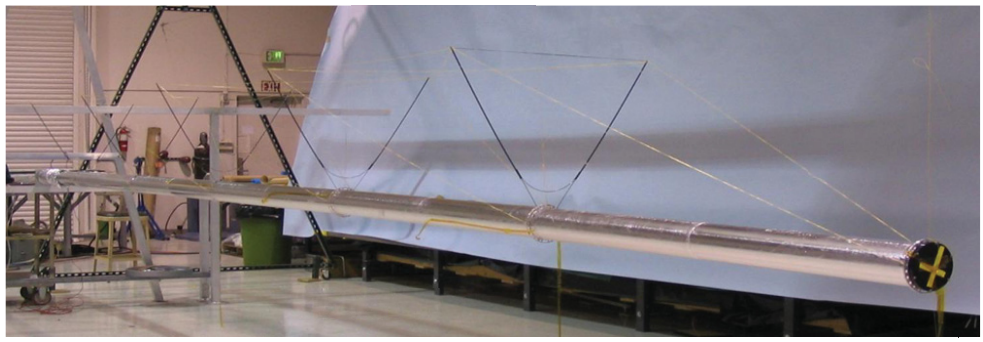

Figure 13 Deployed boom with spreader bars

Sail Net Membrane Design

The sail membranes are fabricated from easily acquired and metalized 2.5μm CP1 material. Rip stop fibers, used to halt rips across the membrane, are bonded to the sail in a 1 m grid pattern. Grounding straps are incorporated to provide a conductive path between the front and back surfaces. Also included are the net chords that define the sail shape and provide the attachment points to the rings on a quadrant beams. The wrinkles in the radial direction are formed by a small amount of extra material specifically designed to absorb any lateral deformations in the film due to thermal effects.

Sail stress loads are transferred through the sail material to the nearest net element that then transfer the load to the structure. Utilizing this method there are minimal stress concentrations in the sail and attach points. Low stress concentrations in the sail membrane have a host of benefits. Small holes and minor damage are far less likely to propagate and tear. Degradation in the film due to environmental effects such as particle radiation and UV are similarly less likely to result in a failure. See Figure14.

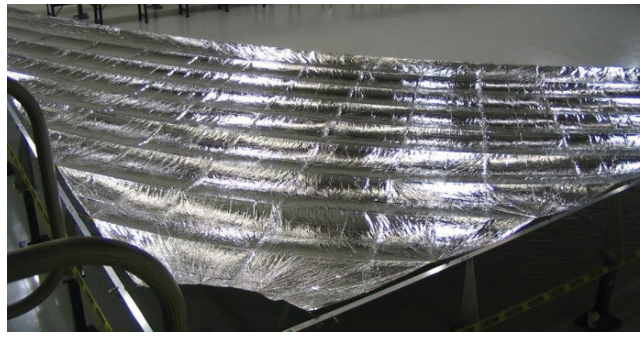

Figure 14 L'Garde solar sail membrane with netting

Figure 15 depicts the deployment of the L'Garde Solar Sail. For deployment the sail attitude will be normal to the sun-sail line. The carrier ACS will shut off for deployment so as not to impart loads on the sailcraft structure. Passive stability should keep the sail sun-normal, but in the event some bandwidth is exceeded, the carrier ACS will be used. The sequence initiates with the opening of the canister lids. Next the vanes rotate into the plane of the sail and the vane booms are now deployed bringing with them the vane membranes. The main sail deployment initiates with the deployment of the spreader bars. The main booms are then deployed bringing with them the sail quadrants. All four beams will deploy deploy simultaneously, taking the sail with them. Deployed length is measured and fed back to control relative beam deployment via pressure. After completion of the main sail deployment the canister is ejected to lighten the sailcraft. The final step in the deployment is to deploy the solar arrays for spacecraft power generation.

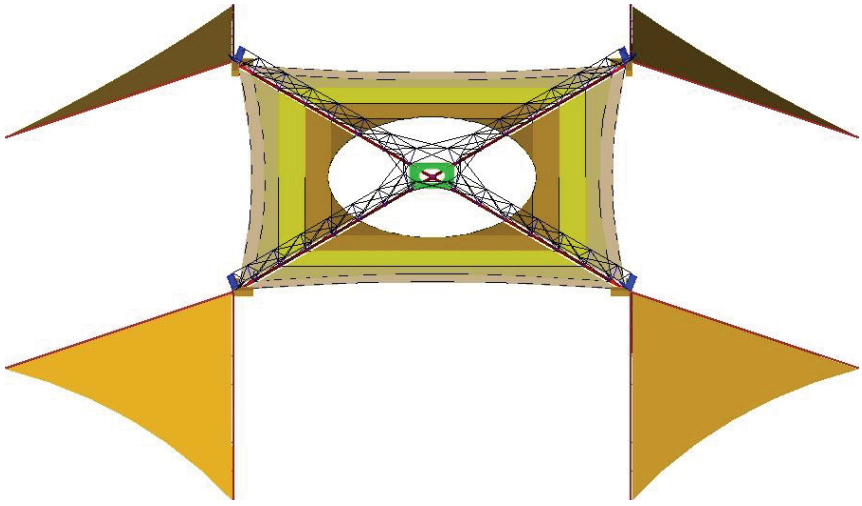

Figure 15 Deployment of the L'Garde Solar Sail and tip vanes

VIII. Power

The ERS and Orbiter will have their own solar arrays and power distribution, but the array regulation and battery charge control are common and will return with the ERS. Each stage has its own flight computer and Power Distribution Unit (PDU) which affords redundancy on the trip to Mars. The two stages are connected by a breakaway harness connector which can be separated on command. After separation, the Orbiter solar arrays will cease to generate power and the batteries will no longer charge. The batteries will then power the Orbiter during its disposal while the ERS deploys its arrays and sails back to Earth. The solar irradiance – 1361 W/m^2 is a minimum value chosen to be conservative at a 1.67 AU solar distance, which is the aphelion of Mars. The 35° incidence angle is taken from the mission analyst's estimate of the maximum angle required to support solar sail propulsion on the return trajectory.

IX. Avionics

Orbiter C&DH – Approach

The original MSR C&DH system was baselined for the Orbiter. All primary functions are contained in a single flight computer, including data acquisition, data processing, flight commands, and data storage. Also included is a communications interface board and an image processing board. A special function board is assumed to handle the AR&C operations. Two exactly alike flight computers are used for single fault tolerance.

The communication system includes a 1 meter parabolic pointing High Gain Antenna (HGA) for direct link to Earth using 34 meter DSN antennas in X-band, and a 15 watt SSPA. This system is used to relay data from the MSR lander and fetch rover back to Earth, and to uplink commands and operation modifications. This same system should be more than sufficient to support any sail deployment and ERS separation status and uplinks.

The OS may be several hundred kilometers down range from the Orbiter after the Mars ascent. The Orbiter will use the UHF beacon signal to track and maneuver to the OS using the main engines. Once within 10 km of the OS, the Optical Navigation cameras of the capture system will be used to maneuver the Orbiter to within 1 meter. At this point, it's assumed a close proximity lider system, or some similar capable system, guides the Orbiter to capture the OS within the basket without making contact. At this point, the capture door can close and the basket frame work used to dissipate the OS spin momentum.

Orbiter GN&C – Approach

The pointing accuracy required for the Orbiter is driven by the AR&C maneuver fidelity. In the MSR orbiter, 505 arcsec accuracy is specified. However, for most of the mission time this high accuracy is not required. A 2.5 degree low accuracy is assumed sufficient during low accuracy demand times. Typical solar array sun pointing is only 5 degrees. The gimbal HGA can maintain tight Earth pointing independent of the Orbiter's pointing accuracy.

ERS C&DH – Approach

For the ERS C&DH, the Messenger spacecraft Integrated Electronics Module (IEM) system was baselined. Messenger is a Mercury surveillance mission, in which images are transmitted back to earth. Although the data handling capabilities are probably more than what is needed for the ERS, it has proven heritage for reduced risk. Two IEM units are booked on the ERS for single fault tolerance.

For the ERS communications, the Messenger spacecraft communication system was baselined. It uses 11 watt solid state power amplifiers (52 w input) and phased array antennas that give about 24 dBi of gain. This system allows over 2 kbps of data to be transmitted from Mercury to Earth in X-band to 34 meter DSN antennas. Since the distance from Mars to Earth is nearly the same as Mercury to Earth (about 1.5 AU), it can be assumed that this system will work for the ERS at Mars.

Since the ERS will not be transmitting science data, it's assumed that 1 kbps is a sufficient data rate for transmitting navigation information and health and status data. Either smaller arrays or lower power Solid State Power Amplifiers (SSPAs) can be considered, but to keep the high TRL level the system was unaltered.

Fanbeam antennas are used for a 31.25 bps uplink capability anywhere within the forward looking hemisphere. LGA are included for emergency backup operations.

ERS GN&C - Approach

All the ERS sail attitude and control hardware is included in the sail structures. These components would include sun sensors on the tipvanes, along with the tipvane motors and required electronic interface units and cabling. The use of MSFC Multi-Purpose Data Module (MPDM) for a remote electronic distribution units is suggested to minimized system mass. These units are small and lightweight, about 100 grams each. It's believed one unit in each tip vane could support 3 sun sensors along with controlling and monitoring the tipvane motors. A single power cable and data bus cable would run the length of the booms to these MPDM units at the tipvanes. The power cable is thought to double as a heater cable during deployment. An appropriate switched power circuit would need to be included in the MPDMs for this operation.

The same MPDMs are suggested for use by the ERS solar arrays for monitoring those sun sensors. Low accuracy AeroAstro star trackers (0.1 deg) and Northrop LN200 IMUs are suggested for navigation of the ERS since no high accuracy sun pointing is required (+/- 2.5 deg from the 35 degree sun angle pointing accuracy required). The EEV retention and pyrotechnic release mechanism control is assumed in the secondary structures.

An observation camera is included in the ERS. This camera is for monitoring the deployment of the sail and the ERS separation from the Orbiter. It can also be used to monitor the deployment of the EEV for earth entry. The IKAROS mission used deployment cameras for visual observation.

X. Thermal Control

Orbiter Analysis

A passive thermal design concept was developed for the Orbiter and ERS. Thermal control will utilize components including multilayer insulation, high emissivity paint and coatings, heatpipes, heaters, etc. to maintain spacecraft subsystem components within acceptable temperature ranges. There are no dedicated radiators, however spacecraft structural panels act to dissipate avionics heat by conduction and also act as radiative surfaces. The Orbiter bus outer surfaces are covered in low absorptivity materials in order to cold bias the structure. Propellant tanks located in the orbiter are wrapped in MLI and propellant tank heaters are sized for a solar synchronous orientation. RCS thrusters, antennas, solar arrays and solar array mechanisms are not part of the analysis.

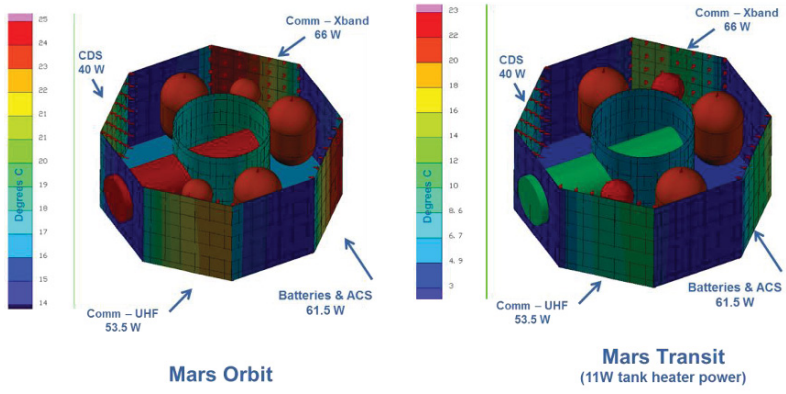

Figure 16 Steady state Thermal Desktop® model of orbiter bus developed to assess subsystems interface temperatures.

A system level thermal model of the spacecraft orbiter bus structure was developed using Cullimore and Ring Technologies' Thermal Desktop® (www.crtech.com) to assess support structure and subsystem equipment

interface temperatures. The structure is modeled as aluminum and the panel thickness is consistent with the structural design. Environmental heat loads were calculated for a Mars orbit altitude of 450 km. Mars transit environmental heat loads were calculated at 500,000 km from Mars. Steady state spacecraft structure temperatures for both Mars orbit and transit to Mars were generated in order to determine a range of temperatures to be expected during the course of the flight for the orbiter bus. Steady state temperature profiles for the Orbiter is shown in Figure 16. Heat loads were distributed on the spacecraft panels according to box locations, as shown. Structure temperatures range from 14 °C to 25 °C for the bus during orbit about Mars and 3 °C to 23 °C for the bus during transit to Mars. Additionally 11 watts of heater power is required for the propellant and pressurant tanks for the Mars transit case. Environmental heat loads are averaged about an orbital period for the Mars orbiting case and a single position at an altitude of 500,000 km from Mars was used for the Mars transit case.

	Total Heat Dissipation	Operating Temp Range
ACS (IMU Assembly)	24 W	-40 C to 85 C
Command & Data System	40 W	-40 C to 85 C
Communication System (UHF)	53.5 W	-40 C to 85 C
Communication System (X-band)	66 W	-40 C to 85 C
Batteries (3)	37.5 W	0 C to 45 C
Propellant and Pressurant Tanks (Heaters)	11 W	10 C to 30 C
Total	232 W	

Table 1 Subsystem equipment heat loads on the Orbiter

Subsystems equipment heat loads used as input to the orbiter thermal analysis are shown in Table 1. A total of 232 W of spacecraft power/heat dissipation was considered in the orbiter thermal analysis, this includes 11 watts of heater power for the propellant tanks. All heat loads are imposed directly on the structure and modeled as area averaged heat loads.

ERS Analysis

The Earth Return Stage is designed with multiple avionics shelves to contain the subsystems equipment. For purposes of the preliminary model, the equipment heat load is equally divided between the upper avionics shelf and the lower shelf near the sample return mechanism. The ERS structure is wrapped with multilayer insulation. ERS thermal analysis results show that during the operational phase of the mission the avionics shelves maintain an average temperature of between 35 and 45 degrees C. The total heat load of 277 watts for the ERS during the

operational mission phase is divided between the forward and aft avionics shelves. The operational phase is defined as the transit from Mars orbit to Low Earth Orbit. During the trip from Earth to Mars and during most of the Mars Orbit period, the ERS is maintained in a keep-alive mode. A reduction in avionics power during this period results in an average avionics shelf temperature of between -16 and -6 degrees C. A total head load of 132 watts was divided between the forward and aft avionics shelves.

A summary of the heat dissipation and operating temperature range for the subsystems avionics and power systems equipment contained on the ERS is shown in Table2 for both the operational and keep alive mission phases. The keep alive mission phase required 36 watts of heater power to maintain interface temperatures above -16 degrees C. All predicted interface temperatures are within acceptable range of -16 to 45 degrees C for the mission phases analyzed.

	Operational Heat Dissipation (Return to Earth)	Keep Alive Heat Dissipation (Transit to Mars & Mars Orbit)	Operating Temp Range
ACS (IMU Assembly)	24.0 W	-	- 40 C to 85 C
Command & Data System	52 W	26.0 W	- 40 C to 85 C
Avionics Heaters	-	36.0 W	n/a
Power Systems	70 W	70 W	- 40 C to 85 C
Communications (X-band)	131 W	-	- 20 C to 70 C
Total	277 W	132 W	

Table 2 Heat dissipation and operating temperature ranges for ERS

Total thermal control mass for the configuration is estimated at 50.65 kg which includes a 30% margin. Based on the analysis, all interface temperatures are within acceptable range for the defined orientation and mission phase. Thermal management is accomplished with typical, flight proven components and no technology development is required.

XI. Mass Properties

The Orbiter basic Dry Mass total is 495 kg and is a rollup of the following subsystem masses: Structures, Chemical Propulsion, Thermal, Avionics, and Power. The predicted Dry Mass increases to 591 kg once a 19% Contingency (96 kg) is added to the basic Dry Mass.

After adding non-propellant fluids, payload, and maneuver propellant to the basic Dry Mass, the Orbiter Total Mass increases to 1,450 kg. This total increases to 1,591 kg once contingency is added. See Table 3 for details.

		Basic Mass (kg)	Contingency (%)	Contingency (kg)	Predicted Mass (kg)
1.0	Structures	139.45	23%	32.33	171.78
2.0	Chemical Propulsion	117.67	25%	29.42	147.09
3.0	Thermal	22.14	30%	6.64	28.78
4.0	Avionics	109.41	13%	14.44	123.85
5.0	Power	106.60	12%	13.22	119.82
Dry Mass		**495.27**	**19%**	**96.04**	**591.31**
6.0	Non-Propellant Fluids	26.46			26.46
7.0	Payload	149.00	30%	44.70	193.70
Inert Mass		175.46			220.16
Total Less Propellant		670.73			811.47
8.0	Maneuver Propellant	779.20			779.20
Total Orbiter Mass		**1449.93**			**1590.67**

Table 3 Total Orbiter Launch Mass

The Earth Return Stage (ERS) basic Dry Mass total is 268 kg and is a rollup of the following subsystem masses: Structures, Solar Sail (Propulsion), Thermal, Avionics, and Power. The predicted Dry Mass consists of the basic Dry Mass total and the contingency (margin) added by each subsystem, which averages 14% (36 kg); this increases the Dry Mass total to 305 kg. The ERS basic Dry Mass increases to 320 kg once the payload (52 kg) is added. After contingency is added, the Total ERS mass increases to 357 kg. See Table 4.

		Basic Mass (kg)	Contingency (%)	Contingency (kg)	Predicted Mass (kg)
1.0	Structures	38.02	26%	10.05	48.07
2.0	Solar Sail	125.08	12%	14.51	139.59
3.0	Thermal	16.82	30%	5.05	21.87
4.0	Avionics	51.81	6%	3.12	54.94
5.0	Power	36.70	10%	3.60	40.30
Dry Mass Total		**268.43**	**14%**	**36.33**	**304.77**
6.0	Payload	52.05			52.05
Total ERS Mass		**320.48**			**356.82**

Table 4 Total ERS Mass + OS Payload

The total launch mass is 1,947.5 kg and includes the Orbiter Mass 1,591 (kg) and the ERS Mass (357 kg). See Table 5 for details.

	Basic Mass (kg)	Predicted Mass (kg)
Orbiter Mass	1449.93	1590.67
ERS Mass	320.48	356.82
Total Launch Mass	**1770.41**	**1947.49**

Table 5 Total Launch Mass of the combined Orbiter and ERS

XII. Study Conclusion

The primary objective was to reduce the baseline mission from three to two launches. The study was required to maintain the baseline mission objectives for all three elements, only redesigning the Earth return propulsion system. The Advanced Concepts Office MSR Orbiter and ERS had a launch mass too large to be packaged on the Max-C Rover or Lander launch vehicles if the design only replaced the ERS chemical propulsion system with the solar sail propulsion unit. However, the amount of mass saved was significant. The original architecture design had an Orbiter launch mass of 3270kg. The solar sail propulsion unit reduced the launch mass by 1322.5 kg to a new launch mass of 1947.5 kg. See Table 6 below.

	Predicted Launch Mass (kg)	Launch Vehicle	Launch Capability (kg)	Contingency (kg)
Max-C Rover	4457.4	Atlas V 531	4980	522.6
Lander and Ascent Vehicle	4668	Atlas V 551	5130	462
ACO MSR Orbiter and ERS	1947.49	--	--	--

Table 6 Three MSR elements to be launched include the new ACO MSR Orbiter and ERS. The first trade of replacing the ERS propulsion unit with a solar sail greatly reduced the launch mass but not enough to ride share with the Max-C Rover or Lander and Ascent Vehicle on their current launch vehicle manifest.

A second trade, called Delta, consisted of utilizing solar sail propulsion on both trips. If the orbiter used a solar sail as the main propulsion method for transit to and from Mars, the orbiter would be able to launch with the Max-C Rover eliminating the need for one of the three launches. The updated mass numbers are located in Table 7 below.

Delta	Predicted Mass (kg)
Max-C Entry System	1550.7
Descent Stage	1313.1
Pallet	327.5
Max-C Rover	364.5
ACO Delta Orb+ERS	508.8
Total	**4064.6**

Delta Launch Summary	
2028 Atlas V 551 Mass (kg)	5150
Delta Mass (kg)	4064.6
Launch Contingency:	**21%**

Table 7 Updated mass numbers from the Delta trade replacing the entire chemical propulsion unit with a solar sail propulsion unit.

In conclusion, the team was able to feasibly repackage the Orbiter and Earth Return System into a configuration that allows stowage and deployment of the 150 M solar sail system based on the scalable L'Garde design and share a dual launch manifest with the Max-C Rover. The team concluded that the addition of the solar sail for the trip to Mars and back to Earth eliminated the need for one of three planned launches therefore reducing the mission operations timeline and launch manifest costs. The team estimated that even with the reduced mass, all mission requirements would be met. Future work would involve re-evaluating launch vehicle options and further design modifications to the spacecraft to further reduce mass.

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Distributed Reflectivity Solar Sails for Extended Mission Applications

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The dynamics of solar sails with a variable surface reflectivity distribution are investigated. When changing the reflectivity across the sail film, solar radiation pressure forces and torques can be controlled without changing the attitude of the spacecraft relative to the Sun or using attitude control actuators. The reflectivity can in principle be modified using electro-chromic coatings, which are applied here as examples to counteract gravity-gradient torques in Earth orbit and to enable specific shape profiles of a flexible sail film. This 'optical reconfiguration' method introduces an adaptive solar sail as a multi-functional platform for novel mission applications.

I. Introduction

USING conventional solar sailing technology, the solar radiation pressure (SRP) force vector direction and magnitude depend strongly on the sail attitude relative to the Sun, limiting the applicability of solar sails compared to other low-thrust propulsion systems such as solar-electric propulsion. Furthermore, the SRP force magnitude follows an inverse square law with solar distance, making the sail less efficient at large distances from the Sun [1]. In order to increase the flexibility of modulating the SRP forces and torques and to increase the range of potential solar sail mission applications, we introduce the concepts of shape change and variable optical

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properties for large gossamer spacecraft [2]. Merging these concepts, potential benefits of such a system over conventional solar sails can be identified. It is envisaged to use future shape-changing solar sails as multi-functional platforms that can deliver additional key mission functionality such as power collection, sensing and communications. For example, the sail may begin at Earth escape in a flat configuration on a small-body science mission towards a designated target object. In close proximity to the target body, the sail reconfigures to a parabolic shape, using its reflective film as a remote sensing device or as a large communication antenna, before continuing again in a flat thrust mode.

To demonstrate these capabilities of solar sails, this paper investigates the use of variable film reflectivity for attitude and deflection control. First, in Section II, the attitude dynamics of a rigid, flat solar sail with a variable surface reflectivity distribution will be investigated. When changing the reflectivity coefficient across the sail film, the SRP forces and torques acting on the sail can be controlled without changing the incidence angle relative to the Sun and without using additional attitude control actuators. The reflectivity can be modified using electro-chromic coatings, which consist of an electro-active material that changes its surface reflectivity according to an applied electric potential [3]. By assigning an appropriate reflectivity function across the sail area, the center-of-pressure can be shifted optically relative to the center-of-mass (CoM), rather than mechanically, as for example through the use of moving payload masses. This enables more flexible steering of the solar sail and minimizes actuation effort.

Subsequently, in Section III, a variable reflectivity distribution is used to control the deflection of a flexible sail film due to the transverse SRP load acting on the surface. The very thin membrane-like sail film experiences relatively high deflections from the initially flat shape and requires the use of non-linear flexible bending theory [4], accounting for the non-negligible in-plane tension within the material. The deflection magnitudes and shapes due to a constant surface reflectivity are presented for various circular sail films of different radius and distances from the Sun. The nominal deflection shapes are then manipulated through the use of suitable reflectivity functions across the surface. It is shown that when a particular deflection shape is selected a priori (e.g. a parabolic one to achieve the remote sensing device and/or antenna as mentioned above), the required reflectivity distribution can be calculated by formulating an inverse problem. Resulting paraboloid-type deflection shapes and deflection magnitudes are assessed in Section IV in terms of their usefulness for novel mission applications for solar sails, using shape-changing reflective sail films.

II. Sail Attitude Control using Variable Reflectivity Distribution

As a first example, a variable reflectivity distribution is used to control the attitude of a solar sail in low Earth orbit (LEO). In particular, this steering method is applied to counteract the gravity-gradient torques acting on a large rigid sail film. It will be shown that a constant Sun-pointing attitude can be maintained along the orbit, while likewise minimizing the necessary surface reflectivity. A square solar sail is modeled as a rigid flat Kapton® film of density $\delta = 1{,}572\ \mathrm{kg/m^3}$ [5], thickness $d = 2.5\times10^{-6}\,\mathrm{m}$ and edge length $L = 50\ \mathrm{m}$, without assembly or payload masses, as shown in Fig. 1.

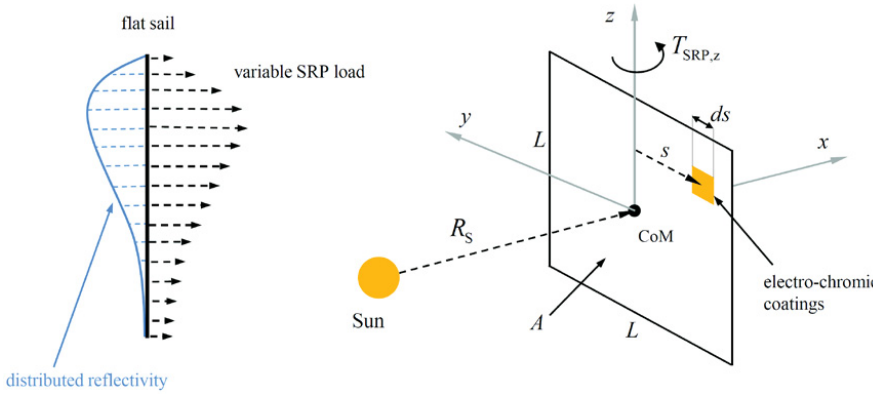


Fig. 1 Square flat solar sail with variable reflectivity distribution across the surface (left) and sail model using patched electro-chromic elements (right)

A sail-fixed Cartesian coordinate frame $(\mathbf{x},\mathbf{y},\mathbf{z})$ is used to describe the sail attitude along its orbit, where the $\mathbf{z}$-axis is constrained to be always perpendicular to the ecliptic plane. This allows only one degree of rotational freedom, while rotation about the sail $\mathbf{x}$ and $\mathbf{y}$ axes is omitted. The mass moments of inertia of the Kapton® film are $I_{xx} = 4093.8\ \mathrm{kgm^2}$ and $I_{yy} = I_{zz} = 2046.9\ \mathrm{kgm^2}$, with a total mass $m = 9.83\ \mathrm{kg}$. It is ideally assumed that the entire sail area A is covered with electro-chromic coating elements that are able to change their reflectivity. The additional mass and thickness of the elements is neglected throughout this preliminary study. When changing the reflectivity of a surface element $dA = L\cdot ds$ on the sail, where s denotes the distance of the element from the axis of rotation, the difference in SRP force creates a torque about the CoM of the sail.

The CoM of the sail is assumed to move on a circular LEO of 400 km altitude in the ecliptic plane, as shown in Fig. 2, while air drag and solar eclipses are ignored for illustration. A constant sail cone angle $\alpha = 0\ \mathrm{deg}$ between the Sun-sail line $\mathbf{R}_S$ and the sail plane normal is assumed.

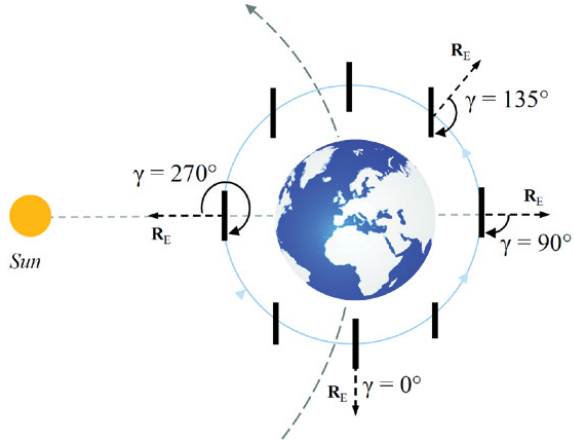


Fig. 2 Rigid flat solar sail on Earth-centered circular LEO of 400 km altitude and constant Sun-pointing attitude

The gravity-gradient torque along the orbit depends on the angle γ between the sail plane and the Earth radial direction $\mathbf{R}_\mathrm{E}$. It can be approximated using the standard relation

$$T_{\mathrm{GG,z}} = 3\frac{\mu_E}{R_{\mathrm{CoM}}^3}\sin\gamma\cos\gamma\,(I_{\mathrm{xx}} - I_{\mathrm{yy}}) \tag{1}$$

with μ_E being the gravitational parameter of the Earth and R_{CoM} the radial distance of the sail's CoM from the Earth's center [6]. For example, the maximum gravity-gradient torque acting on the sail considered here is of the order $T_{\mathrm{GG,z}} = 2\times10^{-3}$ Nm about the sail $\mathbf{z}$-axis.

A variable surface reflectivity distribution $0 \le \rho(s) \le 1$ will now be assumed to counteract this gravity gradient torque. The SRP force and torque about the sail's CoM are calculated using a simplified SRP model [1]. It assumes that the sail surface is a perfectly (specular) reflecting mirror, neglecting all other form of optical interactions between the solar photons and the sail surface such as diffuse reflection, absorption and thermal re-emission. It also does not account for wrinkles or sail billowing, and thus assumes a perfectly flat sail surface. Therefore, the SRP for reflectivity $\rho(s) = 1$ is

$$p_{\mathrm{SRP}} = 2p_0\left(\frac{R_{\mathrm{S},0}}{R_{\mathrm{S}}}\right)^2\cos^2\alpha \tag{2}$$

at a radial distance R_S from the Sun, with the pitch angle α between the Sun-sail line and the sail plane normal and $p_0 = 4.563\times10^{-6}\,\mathrm{N/m^2}$ being the solar radiation pressure at $R_{\mathrm{S},0} = 1$ AU.

When now assuming that the sail film reflectivity is no longer constant, but changes along the surface, the local SRP becomes a function of $\rho(s)$ as

$$p_{\mathrm{SRP}} = p_0\left[1+\rho(s)\right]\left(\frac{R_{S,0}}{R_S}\right)^2 \cos^2\alpha \tag{3}$$

where $\rho = 1$ represents the perfect mirror in Eq. (2) that experiences the maximum possible SRP load $p_{\max} = 2p_0$, while $\rho = 0$ reduces the effective SRP load to $p_{\max}/2 = p_0$, because only the momentum of the incoming photons is applying a force to the surface. As a consequence, the solar radiation pressure induced forces can be modified directly when changing the surface reflectivity.

According to Fig. 1, the incremental SRP force and torque acting about the CoM on a rectangular sail surface element can now be written as

$$dF_{\mathrm{SRP}} = p_0 L\left[1+\rho(s)\right]\left(\frac{R_0}{R_{\mathrm{S}}}\right)^2 \cos^2\alpha\ ds \tag{4}$$

$$dT_{\mathrm{SRP},z} = p_0 L\left[1+\rho(s)\right]\left(\frac{R_0}{R_{\mathrm{S}}}\right)^2 \cos^2\alpha\ s\,ds \tag{5}$$

Assuming a linear reflectivity function

$$\rho(s) = a_0 + a_1 \cdot s \tag{6}$$

the above equations can be integrated over the interval $\left[-L/2, L/2\right]$, while constraining the sail surface to be always perpendicular to the Sun-sail line, thus $\alpha = 0$, and assuming that the solar distance $R_{\mathrm{S}} = 1\ \mathrm{AU}$ is constant on the Earth-centered orbit. The resulting SRP force and torque are found to be

$$F_{\mathrm{SRP}} = p_0 L \int_{-L/2}^{L/2} \left[1+a_0+a_1\cdot s\right] ds\ = p_0 L^2\left[1+a_0\right] \tag{7}$$

$$T_{\mathrm{SRP},z} = p_0 L \int_{-L/2}^{L/2} \left[1+a_0+a_1\cdot s\right] s\ ds = \frac{1}{12} p_0 L^4 a_1 \tag{8}$$

It can be seen that the magnitude of the resulting SRP force is only a function of the absolute coefficient a_0. Furthermore, the SRP torque only depends on the gradient of the linear reflectivity function across the film, thus the difference of SRP force on the left and right hand side of the sail. If the forces are equal on both sides $(\mathrm{a}_1 = 0)$, the torque on the sail is clearly zero, regardless of the absolute value of the SRP force, as determined by a_0.

Equation (8) represents the SRP torque that must be generated in order to counteract the gravity-gradient torque $T_{\mathrm{GG,z}}$ along the orbit, thus $T_{\mathrm{SRP,z}}$ must always be equal $T_{\mathrm{GG,z}}$. Thus, the coefficients a_i of the linear reflectivity function $\rho(s)$ can in general be calculated using the following equations

$$a_1 = \frac{12 T_{\mathrm{GG,z}}}{p_0 L^4}, \qquad a_0 = \frac{F_{\mathrm{SRP}}}{p_0 L^2} - 1 \tag{9}$$

Although a_1 is always determined through the current $T_{\mathrm{GG,z}}$ that needs to be compensated, a_0 is in principle only constrained by the maximum possible $F_{\mathrm{SRP,\,max}} = 2 p_0 L^2$, for which $a_0 = 1$ and also $\rho(s) = 1 = \text{const.}$ across the surface. However, a reflectivity gradient $a_1 \neq 0$ is necessary to create a torque, while at the same time $\rho(s) \leq 1$ across the entire sail. Thus, the upper constraint is $a_{0,\max} < 1$. The lower constraint is determined through the additional condition $\rho(L/2) = 0$ or $\rho(-L/2) = 0$ at the edge of the sail, in order to fix the linear reflectivity curve across the film. Inserting this into the linear reflectivity function (Eq. (6)), while keeping a_1 fixed, results in $a_{0,\min} = \pm a_1 (L/2)$ as the lower limit, depending on the sign of the torque that needs to be created. Since always $\rho(s) > 0$, this second condition also defines the minimum reflectivity distribution that likewise matches the torque requirement. Fig. 3 shows the resulting minimum reflectivity distribution $\rho(s)$ along the film as a function of the sail attitude angle γ during one revolution along the orbit for the chosen sail configuration.

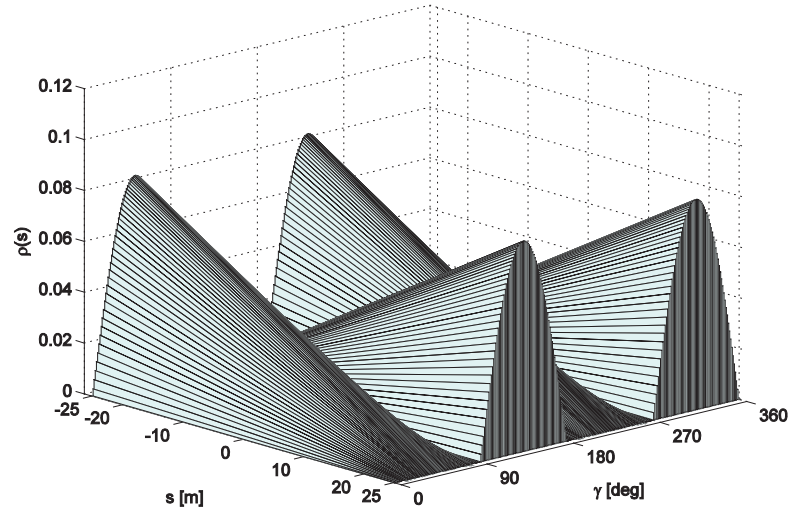


Fig. 3 Reflectivity distribution along a square rigid sail (L = 50 m) as function of sail attitude angle γ during one revolution along a 400-km LEO

III. Sail Shape Control using Variable Reflectivity Distribution

As mentioned in the introduction, a variable SRP reflectivity distribution can also be used to create desirable film deflections of a flexible circular disk sail. For that investigation, the sail is modeled with the same Kapton film used in section II, supported by a circumferential hoop structure, forming hinged-support type boundary conditions at the edges. Electro-chromic elements are again assumed to be distributed across the surface, as shown schematically in Fig. 4. For now, only the structural bending of the sail film is accounted for, ignoring time-dependent orbital and attitude dynamics of the structure due to distributed SRP forces and torques.

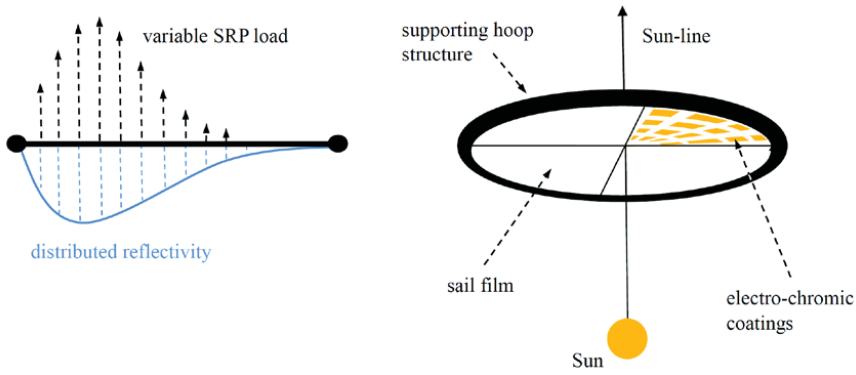


Fig. 4 Schematic of circular disk sail with electro-chromic coatings distributed across the surface (left) and supporting hoop structure (right)

In general, the SRP load distribution across the sail surface of diameter D causes an out-of-plane deflection w of the sail film from its initial flat shape. Due to the very small thickness, d, of the film material and relatively large deflections, w, (i.e. a high $W = w/d$ ratio), the non-linear theory of circular membranes needs to be used: in general, thin membrane-like structures offer a very small flexural rigidity and therefore cannot resist bending loads [7]. Furthermore, in-plane radial and transversal tension stresses are non-negligible, while for low W ratios these can be assumed negligible within the well-known linear beam theory [8]. Figure 5 shows a circular sail film of radius R and thickness d, under a uniform transverse SRP load $p_w = p_{SRP}$ perpendicular to the surface, initial radial in-plane tension $N_{r,0}$ at the edges and central deflection w_0.

The symmetrical out-of-plane deflection w can be described as a system of non-linear coupled ODEs of 2nd order. According to [9] these can be written in non-dimensional form as

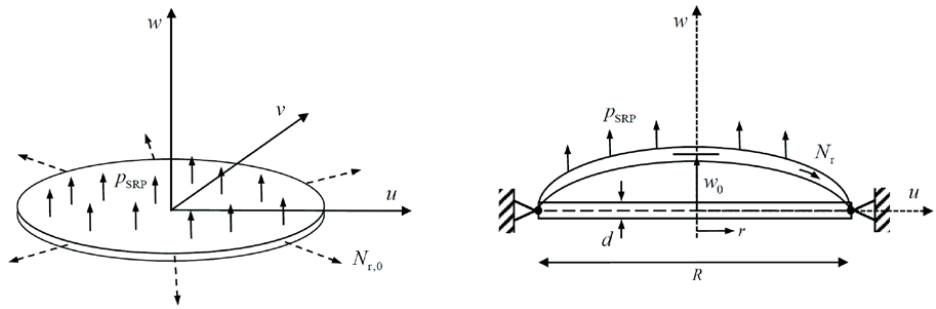


Fig. 5 Circular disk sail with transverse SRP load and initial in-plane tension (left) and sail film cross-section with hinged-edge support and deflected shape (right)

$$\xi^2\ddot{\theta}+\xi\dot{\theta}-[1+\xi^2(k^2+12(1-\upsilon^2)S_r)]\theta=6(1-\upsilon^2)P_{\mathrm{SRP}}\xi^3 \tag{10}$$

$$\xi^2\ddot{S}_r+3\xi\dot{S}_r=-\frac{1}{2}\theta^2 \tag{11}$$

where the following non-dimensional variables are used

$$\xi=\frac{r}{R},\ \dot{(\,)}=\frac{d}{d\xi},\ W=\frac{w}{d},\ \mathrm{U}=\frac{u}{d} \tag{12}$$

$$\theta=\frac{dW}{d\xi}=\frac{R}{d}\frac{dw}{dr},\ \mathrm{S}_r=\frac{N_rR^2}{Ed^3} \tag{13}$$

with r the radial distance from the center of the sail disk, the dimensional radial in-plane tension N_r and in-plane deflection u. Furthermore, the non-dimensional parameters

$$k=\sqrt{\frac{N_{\mathrm{r},0}R^2}{D}}\ \text{ and }\ P_{\mathrm{SRP}}=\frac{p_{\mathrm{SRP}}R^4}{Ed^4} \tag{14}$$

denote the tension parameter k and the loading parameter P_{SRP}, using the Young's modulus E, Poisson ratio υ and the flexural rigidity

$$D=\frac{Ed^3}{12(1-\upsilon^2)} \tag{15}$$

of the sail film material. Within the scope of this paper, no initial in-plane tension $N_{\mathrm{r},0}$ has been accounted for to maximize the deflection, so the parameter k is zero. The corresponding boundary conditions (BC) to solve the boundary value problem (BVP) for a hinged edge support are then

$$\left.\begin{aligned}\theta&=0\\ S_r&=0\end{aligned}\right\}\ \text{for } \xi=0 \quad \text{and} \quad \left.\begin{aligned}\dot{\theta}&=0\\ \dot{S}_r+(1-\upsilon)S_r&=0\end{aligned}\right\}\ \text{for } \xi=1 \tag{16}$$

This type of support was chosen to represent the real conditions in the best way, since the (almost rigid) hoop structure inhibits sail film deflections in the u direction, but allows for a non-zero slope $dW/d\xi$ at the edges.

In principle, the above BVP can be solved for any radially-symmetric load distribution $P_{\mathrm{SRP}}(\xi)$, whereas in the literature P is widely assumed to be constant across the film surface. Later in this section, variable load distributions $P(\xi)$ will be used to change the nominal deflection curves of sail films subjected to constant pressure loads P, as used, for example, in microelectronics and microfluidics [10]. After the BVP has been solved for $\theta(\xi)$, the vertical sail film deflection is obtained by integration from

$$W = \frac{w}{d} = \int \theta d\xi \tag{17}$$

The BVP is solved with the MATLAB™ bvp4c routine, using the three-stage Lobatto IIIa collocation method [11]. It is validated using the results observed in [10] for Silicon Nitride membranes with a clamped-edge support under a constant vertical load $P(\xi)$, showing that the found deflections $w(\xi)$ and non-dimensional in-plane tensions $S_r(\xi)$ can be reproduced. Furthermore, the chosen analytical approach is validated with a numerical finite-element analysis (FEM) conducted in [12] using circular Mylar® films ($\rho = 1350\,kg/m^3$, $E = 3.5\times10^9\,N/m^2$, $\upsilon = 0.38$ and $d_{\mathrm{Mylar}} = 1.0\times10^{-6}$ m) of varying radius and thickness. Figure 6 shows the relative deflections w/R obtained when solving the coupled ODE system for a constant SRP load at 1 AU and using hinged edge support.

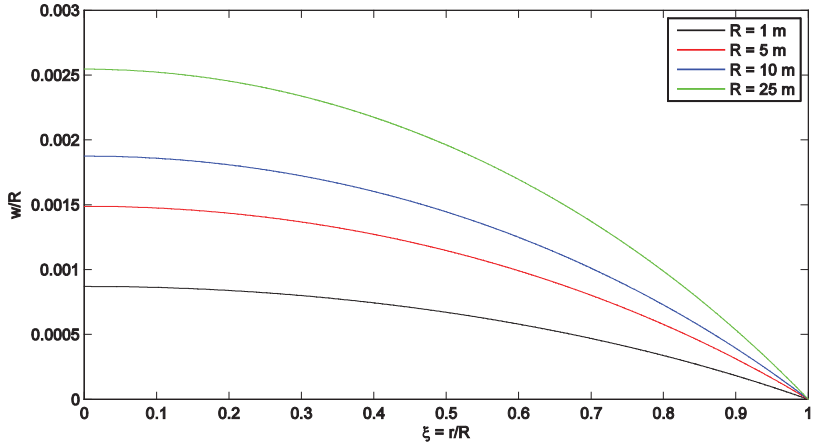


Fig. 6 Reproduction of relative out-of-plane deflections for Mylar sail films (thickness d = 1μm) at 1 AU for different sail disk radii found in [12]

The central sail film deflections are in the order of 0.2% of the sail disk radius and in good agreement with the results found in [12]. According to that analysis, the variation of load magnitude due the local deflection of the film was also taken into account. The film deflection changes the local light incidence angle, reducing the nominal SRP load. However, comparing results with [12] indicates that this effect is negligible, due to very small angular deflections of the surface.

A. Basic Sail Film Deflection for Constant Reflectivity

The material likely to be employed for future solar sails is Kapton®, due to its higher heat resistance and higher resistance to radiation compared to Mylar®. Accordingly, the deflection properties of a Kapton® film (thickness $d = 2.5\times10^{-6}\,\mathrm{m}$, density $\rho = 1572\,kg/m^3$, Young's Modulus $E = 2.48\times10^{9}\,N/m^2$ and Poisson ratio $\upsilon = 0.34$), according to [5], are investigated in the following. The sail surface is assumed to be always perpendicular to the Sun-sail line. Figures 7 and 8 show the relative sail film deflections w/R obtained for different sail disk sizes R = 1, 5, 10, 25, 50 and 100 m and for solar distances R_S = 0.5, 0.75, 1.0, 1.5, 2.5 and 3.0 AU, using a 100-m radius disk sail.

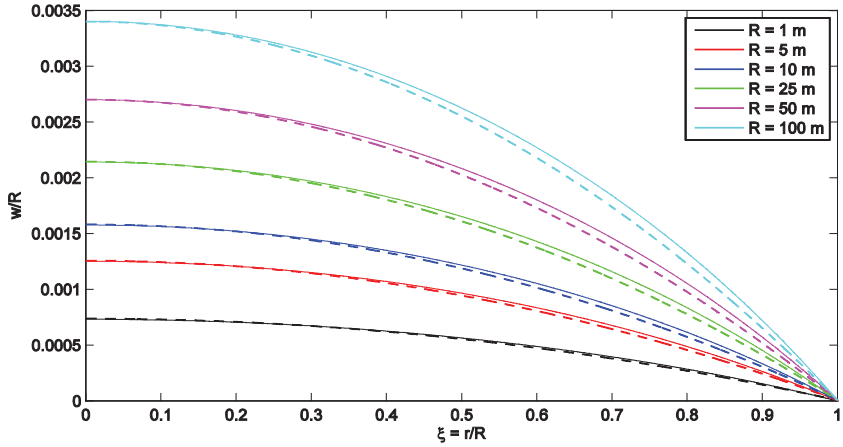


Fig. 7 Relative out-of-plane deflections for Kapton® sail film (thickness d = 2.5 µm) at 1 AU for different sail disk radii (solid lines) and hypothetic parabolic reference curves (dashed lines)

The dashed lines for each case indicate hypothetic parabolic reference curves that are satisfying the same boundary conditions and the same central deflections w_0/R. As can be seen, the deflection surfaces obtained are clearly not ideal paraboloids, as will be discussed below. The central deflections increase for larger sail disk sizes and smaller solar distances, as expected. In general, the absolute deflections stay below 0.5 m, even for

relatively large sail disks (100 m radius) and close to the Sun (0.5 AU), indicating large focal distances when using the sail film as a solar collector or antenna. The focal distances obtained will be discussed in section IV.

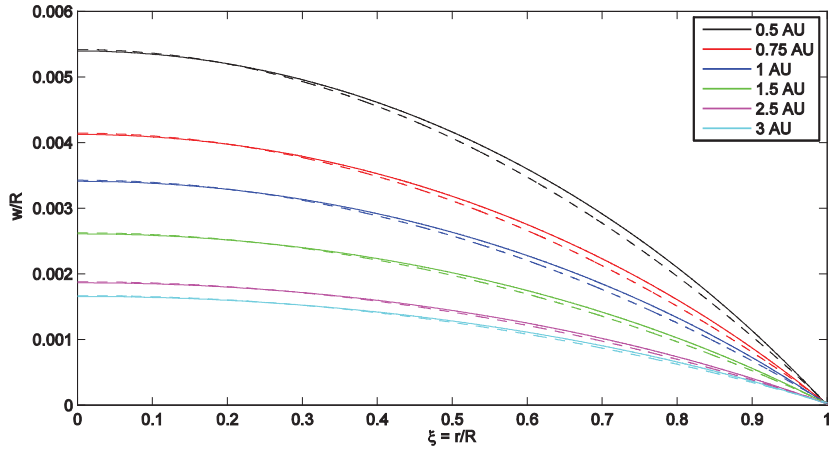


Fig. 8 Relative out-of-plane deflections for Kapton® sail film (radius R = 100 m, thickness d = 2.5 µm) at different Solar distances (solid lines) and hypothetic parabolic reference curves (dashed lines)

The trends of the relative central deflection w_0/R scale with $R^{1/3}$ for the sail disk radius and with $1/R_S^{2/3}$ for the solar distance, as can be seen in Fig. 9. The first result means that a larger disk radius is not necessarily beneficial, since the achievable sail deflections (and thus the focal length) are growing slower than the sail size. Furthermore, the deflection decreases slower than $1/R_S^2$ with solar distance, indicating moderate focal lengths are also available far from the Sun.

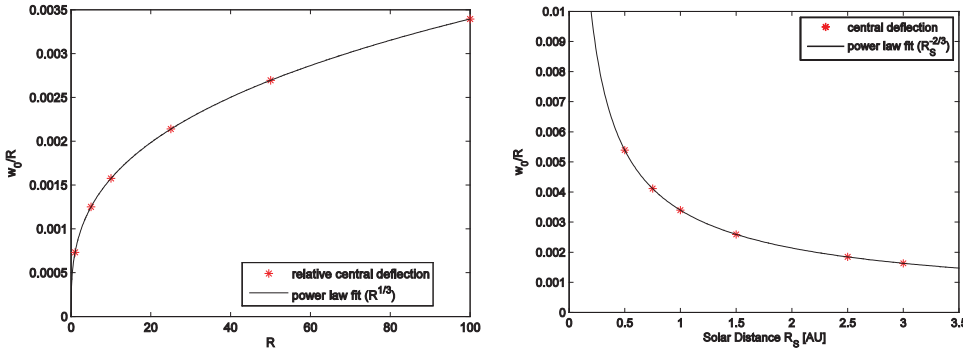


Fig. 9 Relative central sail deflections as function of sail disk radius R (left) and solar distance R_S (right), and power law fits to the data

When applying a polynomial fit to the sail deflection curves, a 3rd order (cubic) trend can be seen. This trend is shown in Fig. 10 for a 100-m radius sail disk at 1 AU, together with a parabolic fit using a 2nd order (parabolic) polynomial. Both polynomial fits are constrained to the central deflection value w_0/R.

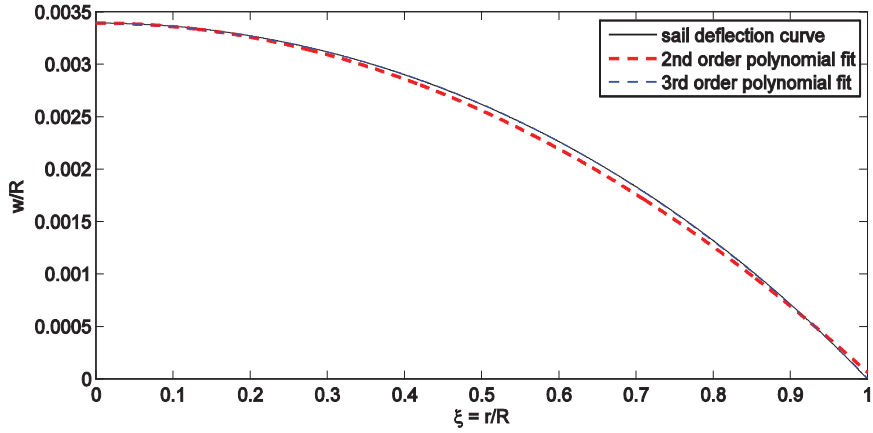


Fig. 10 Second and third order polynomial fits to sail film deflection curve, constrained to central deflection value

The cubic fit (solid blue line) is almost identical with the sail deflection curve (solid black line). Although the parabolic fit (dashed black line) does not match the BC at the edges exactly, the deviation of the sail deflection curve from the ideal parabolic shape is most visible in the mid-region of the sail disk. Here, the local gradient $dw/d\xi$ is smaller than the gradient of the parabola and vice-versa close to the edges. This indicates that the highly reflective Kapton® surface will not concentrate incoming light (or other forms of electro-magnetic radiation) into a single focal point due to aberration. In particular, the cubic surface reflects incoming light at the mid-part towards higher focal lengths and vice-versa for light impinging close to the edges. In the following, it will be shown that the cubic deflection can now be corrected to a parabolic one using a distributed reflectivity function across the sail surface.

B. Controlling Sail Film Deflection using Variable Reflectivity Distribution

In order to change the nominal (cubic) deflection shapes that are found for a constant SRP load distribution across the sail film, the surface reflectivity distribution needs to be controlled. As seen in Eq. (3), a modified SRP force model is employed throughout this study that incorporates the reflectivity $\rho(\xi)$ as a function of the position ξ along the sail surface

$$p_{\mathrm{SRP}}(\xi) = p_0\left[1+\rho(\xi)\right]\left(\frac{R_{S,0}}{R_S}\right)^2 \cos^2\alpha \tag{18}$$

When uniformly distributing electro-chromic elements across the sail disk, the reflectivity function $\rho(\xi)$ directly represents the SRP load distribution. Solving Eq. (19) for $\rho(\xi)$, while again assuming a constant pitch angle $\alpha = 0$, results as

$$\rho(\xi) = \frac{p_{\mathrm{SRP}}(\xi)}{p_0}\left(\frac{R_{S,0}}{R_S}\right)^{-2} - 1 \tag{19}$$

where an arbitrary load function $p_{\mathrm{SRP}}(\xi)$ can be used, as long as the physical constraint for $\rho(\xi) \in [0,1]$ is satisfied. We are now using this property to control the sail film shape, initially neglecting the additional mass and thickness that would be introduced to the sail film when distributing an electro-chromic coating layer on the surface.

Connecting $p_{\mathrm{SRP}}(\xi)$ with the non-dimensional load parameter P_{SRP} in Eq. (14) such that

$$P_{\mathrm{SRP}}(\xi) = \frac{p_{\mathrm{SRP}}(\xi)R^4}{Ed^4} \tag{20}$$

and substituting $P_{\mathrm{SRP}}(\xi)$ for the constant load distribution P_{SRP} within the coupled ODE system (Eq. (10)) introduces an arbitrary (radial symmetric) load function into the system that can be solved as a BVP, with corresponding boundary conditions at the center and at the edges.

Furthermore, an inverse problem can also be formulated, which is defined as calculating the necessary reflectivity function $\rho(\xi)$ to obtain a given sail deflection shape. This can, for example, be a parabolic shape in order to use the sail film as a large antenna. A parabolic deflection curve, as used already in Figs. Fig. 7 and Fig. 8, is of the general form

$$W_{\mathrm{parab}}(\xi) = -A\xi^2 + B\xi + C \tag{21}$$

where again $W = w/d$. When inserting the ideal parabolic curve into the coupled ODE system, it can be solved for $P_{\mathrm{SRP}}(\xi)$ in order to obtain the load distribution necessary to create this curve. Rearranging Eq. (10) for $P_{\mathrm{SRP}}(\xi)$ and Eq. (11) for the non-dimensional in-plane tension $S_{\mathrm{r}}(\xi)$ gives

$$P_{\mathrm{SRP}}(\xi) = \frac{1}{\upsilon^*}\left[\frac{\ddot{W}}{\xi} + \frac{\dddot{W}}{\xi^2} - [1+\xi^2(k^2+2\upsilon^* S_r)]\frac{\dot{W}}{\xi^3}\right] \tag{22}$$

$$\ddot{S}_r = -\frac{3}{\xi}\dot{S}_r - \frac{1}{2\xi^2}(\dot{W})^2 \tag{23}$$

using the Poisson parameter $\upsilon^* = 1-\upsilon^2$. Inserting the parabolic curve $W_{\text{parab}}(\xi)$ for *W*, the above equations now become

$$P_{\text{SRP}}(\xi) = -\frac{B}{\upsilon^*}\frac{1}{\xi^3} + 4AS_r - 2BS_r\frac{1}{\xi} \tag{24}$$

$$\ddot{S}_r = -\frac{3}{\xi}\dot{S}_r - \frac{B^2}{2\xi^2} - 2A^2 + 2AB \tag{25}$$

Now, Eq. (26) can be generally solved for S_r, without specifying boundary conditions

$$S_r = -\frac{1}{4}A^2\xi^2 + \frac{2}{3}AB\xi - \frac{C_1}{2\xi^2} - \frac{B^2}{4}\ln\xi + \frac{B^2}{8} + C_2 \tag{26}$$

When using again boundary conditions for hinged edge support, the above equation results in

$$S_r = -\frac{1}{4}A^2\xi^2 + \frac{2}{3}AB\xi - \frac{B^2}{4}\ln\xi + 1.0076A^2 - 1.6768AB + 0.3788B^2 \tag{27}$$

Inserting Eq. (28) into Eq. (25), the load distribution associated to a general parabolic deflection curve can now be written as

$$P_{\text{SRP}}(\xi) = -\frac{B}{\upsilon^*}\frac{1}{\xi^3} + \left(4A - \frac{2B}{\xi}\right)\left(-\frac{1}{4}A^2\xi^2 + \frac{2}{3}AB\xi - \frac{B^2}{4}\ln\xi + 1.0076A^2 - 1.6768AB + 0.3788B^2\right) \tag{28}$$

It can be seen that $P_{\text{SRP}}(\xi)$ is fully determined by the polynomial coefficients A and B.

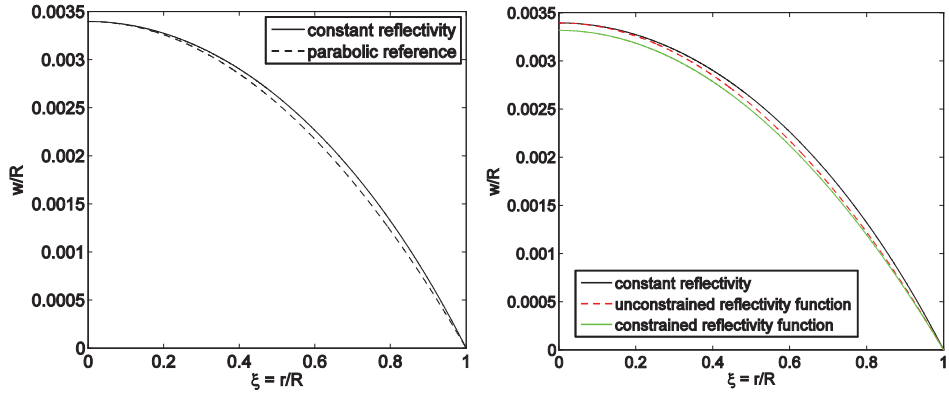
The inverse problem is now applied to create a parabolic deflection shape for a 100-m disk sail of thickness $d = 2.5\times10^{-6}$ m at 1 AU. Fig. 11a) shows the nominal cubic deflection for constant reflectivity, thus a uniform load distribution. Furthermore, a parabolic reference deflection curve (dashed black line) of the simplified form

$$W_P(\xi) = -A\xi^2 + C \tag{29}$$

is taken as input for the inverse problem. The coefficient A is the slope and C is the vertex of the parabola. The parameter *B* is zero, thus $W_P(\xi)$ has no horizontal offset from the symmetry axis. In order to match the central deflection of the nominal cubic deflection curve and the zero-deflection condition at the edges, the coefficients are chosen to be $A = C = W_{0,\text{nom}}$, where $W_{0,\text{nom}}$ represents the central deflection obtained for constant reflectivity.

Accordingly, the parabolic reference curve is now

$$W_P(\xi) = -W_{0,\text{nom}}\xi^2 + W_{0,\text{nom}} \tag{30}$$



a) **b)**

Fig. 11 a) Nominal deflection curve for constant surface reflectivity using 100-m disk sail (thickness $d = 2.5$ µm) at 1 AU (solid line) and parabolic reference curve (dashed line). b) Unconstrained parabolic deflection curve (dashed red line) and constrained parabolic curve (dashed green line)

After inserting the reference parabola $W_P(\xi)$ into Eq. (29), the $P_{\mathrm{SRP}}(\xi)$ function becomes

$$P_{\mathrm{SRP}}(\xi) = 4W_{0,\mathrm{nom}}\left(-\frac{1}{4}W_{0,\mathrm{nom}}^2\xi^2 + 1.0076\ W_{0,\mathrm{nom}}^2\right) \tag{31}$$

and simplifies to

$$P_{\mathrm{SRP}}(\xi) = W_{0,\mathrm{nom}}^3\left(4.0304 - \xi^2\right) \tag{32}$$

When introducing this $P_{\mathrm{SRP}}(\xi)$ function into the coupled ODE system, it can be solved as a regular BVP. As can be seen in Fig. 11b), the resulting deflection curve (dashed red line) exactly matches the input curve in Fig. 11a). When reformulating Eq. (33) , the absolute SRP load distribution can be written as

$$p_{\mathrm{SRP}}(\xi) = \frac{Ed^4}{R^4}W_{0,\mathrm{nom}}^3\left(4.0304 - \xi^2\right) \tag{33}$$

However, the $p_{\mathrm{SRP}}(\xi)$ distribution does not necessarily match the condition not to exceed the maximum possible $p_{\max} = 2p_0\left(R_{S,0}/R_S\right)^2$ at a certain solar distance or, equivalently, not to exceed the maximum reflectivity $\rho_{\max}(\xi) = 1$. As can be seen in Fig. 12, the normalized load distribution $p_{\mathrm{SRP}}(\xi)/p_0$ for the chosen case exceeds $p_{\max}/p_0 = 2\left(R_{S,0}/R_S\right)^2$, showing that it is not possible to achieve the same nominal central deflection when simply constraining the sail film to a parabolic deflection. After introducing the additional

constraint $p_{\mathrm{SRP}}(\xi) \le p_{\max}$ into the SRP load function, the coefficients $A_C = C_C = W_{0,C}$ for the constrained parabola can be calculated from Eq. (34) as

$$W_{0,C} = \left(\frac{2 p_0 R^4}{4.0304 E d^4} \left(\frac{R_{S,0}}{R_S} \right)^2 \right)^{\frac{1}{3}} \tag{34}$$

In here, the constrained central deflection $W_{0,C}$ is now fully determined by the sail film parameters, the sail disk size and the solar distance. The constrained load distribution is also shown in Fig. 12 (green solid curve).

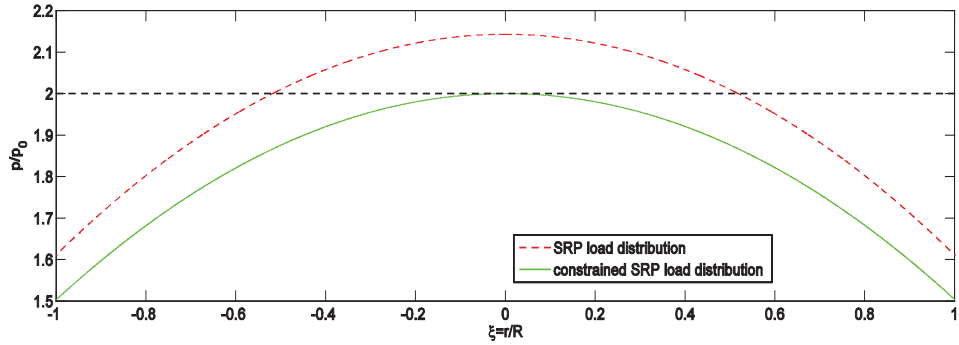


Fig. 12 Non-dimensional load distribution for unconstrained parabolic deflection curve (dashed red line) and for constrained parabolic deflection (solid green line)

The constrained parabolic deflection curve is also shown in Fig. 11b) (dashed green line); its central deflection is about 3% smaller compared to the unconstrained parabolic deflection curve (dashed red line).

The corresponding reflectivity distribution $\rho(\xi)$, according to Eq. (20), is finally found after inserting the SRP load distribution of Eq. (34)

$$\rho(\xi) = \frac{E d^4}{p_0 R^4} \left(\frac{R_S}{R_{S,0}} \right)^2 W_{0,\mathrm{C}}^3 \left(4.0304 - \xi^2 \right) - 1 \tag{35}$$

Inserting the constraint for the central deflection $W_{0,\mathrm{C}}$ into Eq. (36) results in

$$\rho(\xi) = 1 - \frac{2}{4.0304} \xi^2 \tag{36}$$

It can be seen that the reflectivity distribution in order to create a parabolic deflection shape is independent of the sail film parameters and solar distance. The reflectivity distribution is visible in Fig. 13, along with the constrained distribution that abides by $\rho \le 1$.

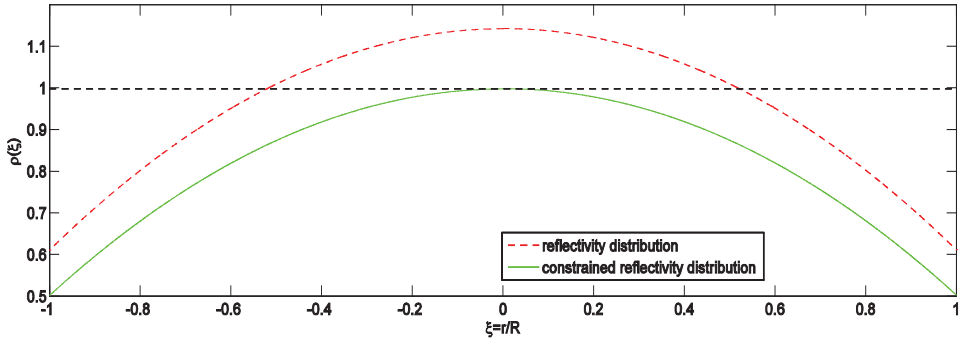


Fig. 13 Reflectivity distribution for unconstrained parabolic deflection curve (dashed red line) and distribution for constrained parabolic deflection (solid green line)

In summary, the necessary reflectivity distribution in order to create a parabolic sail film deflection is independent of the disk radius, film thickness and solar distance. The magnitude of central sail film deflection can be calculated analytically for a circular sail of radius R and film thickness d, perpendicular to the Sun at a solar distance R_S.

IV. Computation of Focal Length of Parabolic Sail Reflector

It was shown in the preceding section that the deflection curve of a sail film exposed to SRP can be controlled by changing the reflectivity distribution across the surface. Accordingly, the nominal deflection shape due to a constant reflectivity distribution, which is normally a cubic polynomial function, can be transformed into a true parabolic shape. A large, highly reflective parabolic surface has many potential applications, such as communication, sensing and power collection. In order to evaluate the usefulness of the shapes that can be generated, some properties of parabolic reflectors have to be identified.

A paraboloid concentrates the incoming electro-magnetic radiation into a single focal point, depending on the geometrical precision of the surface generated. When approximating the nominal cubic deflection shapes calculated in Section III by a parabolic curve $W_P = A\xi^2 + C = W_{0,C}\xi^2 + W_{0,C}$, the corresponding focal lengths, thus the focal distance from the vertex of the parabola, can be calculated as

$$f = \frac{1}{4A} = \frac{1}{4W_{0,C}} \tag{37}$$

using the central deflection $W_{0,C}$ of the constrained parabola, as it was shown in Eq. (35). When transforming the parabolic function W_P into dimensional form

$$w_P = W_{0,C} d \left(\frac{r}{R} \right)^2 + W_{0,C} d = \frac{W_{0,C} d}{R^2} r^2 + W_{0,C} d = a r^2 + c \tag{38}$$

where $a = W_{0,C} d / R^2 = w_{0,C} / R^2$ and $c = W_{0,C} d = w_{0,C}$, thus the focal length is now

$$f = \frac{1}{4a} = \frac{R^2}{4 w_{0,C}} \tag{39}$$

The resulting focal lengths for the deflected circular disk sails calculated in Section III are shown in Fig. 14 as a function of disk radius and solar distance. The results are in good agreement with [12].

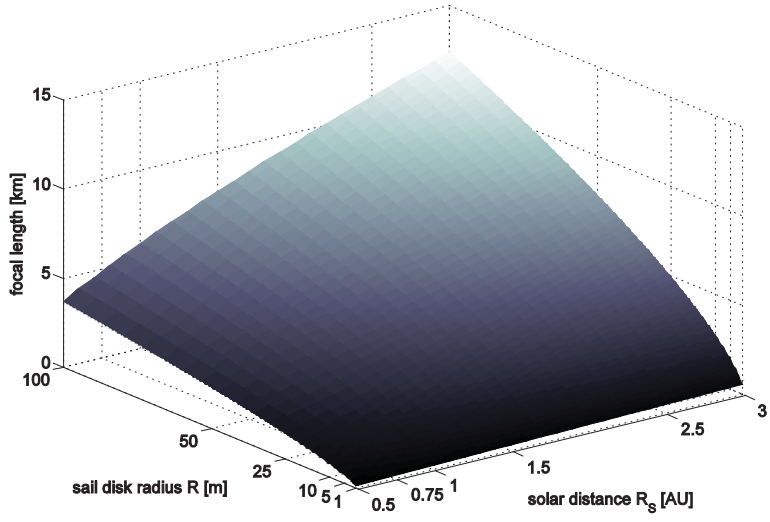


Fig. 14 Focal length of parabolic reference deflection curve of a circular solar sail disk as a function of disk radius and solar distance

For example, a circular disk sail of 100 m radius at 1 AU has a focal length $f_{100} = 5.98$ km. When positioning a detached platform at the focus, and so formation-flying with the solar sail, the platform will be at the focus of a large true antenna, or will be at the focus of a power collector.

V. Conclusion

A variable reflectivity distribution across the surface of a thin Kapton® sail film has been used to investigate potentially novel mission applications for solar sails. The surface reflectivity can be manipulated when distributing electro-chromic elements across the sail film that change their reflectivity according to an applied electric potential. First, a linear reflectivity function was employed to control the attitude of a solar sail in low

Earth orbit. In particular, the steering method has been applied to counteract the gravity-gradient torques acting on a large rigid sail film. It was shown that a constant Sun-pointing attitude can be maintained along the orbit, while simultaneously minimizing the necessary surface reflectivity variation.

Furthermore, a variable reflectivity distribution has been used to change the deflection shape of a flexible circular sail film, supported by a rigid hoop structure around the edge. The nominal sail film deflection due to a constant solar radiation pressure load distribution perpendicular to the surface has been calculated for various sail disk radii and solar distances, using non-linear theory of thin circular membranes. The results indicate that the deflected sail film surface does not concentrate incoming light (or other forms of electro-magnetic radiation) into a single focal point due to non-parabolic aberration. However, an analytical expression for the reflectivity function across the surface necessary to create a true parabolic deflection shape has been derived. This radially symmetric reflectivity function does not depend on sail size, film material or solar distance. Although limited to sail films perpendicular to the Sun-sail line, this can be used for prefabricating a fixed reflectivity distribution on the film surface. In space, this allows the solar sail to obtain a parabolic film deflection at any solar distance. Although the focal length will be changing, this can be compensated by a detached platform formation-flying in the current focus.

The focal length of the resulting parabolic reflector can be calculated using the derived analytical function. It was shown that the focal length increases with sail radius and solar distance.

Throughout this preliminary analysis, simplifying assumptions have been made such as neglecting the additional mass and thickness that would be introduced to the sail film when distributing an electro-chromic coating layer on the surface. Accordingly, extending the non-linear approach to incorporate multi-layer composite membranes will result in better estimates of the actual deflection shapes. Furthermore, no initial billowing of the sail film has been accounted for, as it is assumed to be kept flat by the hoop structure. Thus, only the flexible structural bending has been accounted for so far, which has shown to lead towards relatively small film deflections and thus long focal lengths. Incorporating an initial ‘floppiness’ of the sail film, the dynamical inflation of the film prior to flexible bending can be used to increase the overall deflection and to create shorter focal lengths and thus a higher performance of an optical sail reflector.

Acknowledgments

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Alternative Application of Solar Sail Technology

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The development of Gossamer sail structures for solar sails contributes to a large field of future space applications like thin film solar generators, membrane antennas and drag sails. The focus of this paper is the development of a drag sail based on solar sail technology that could contribute to a reduction of space debris in low Earth orbits. The drag sail design and its connections to solar sail development, a first test on a sounding rocket, as well as the ongoing integration of the drag sail into a triple CubeSat is presented

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Nomenclature

μ	=	friction coefficient
b	=	width [mm]
E	=	Young's modulus [N/mm²]
F_N	=	force normal to the surface [N]
k_c	=	local curvature [1/mm]
k_x	=	curvature in y direction [1/mm]
k_y	=	curvature in x direction [1/mm]
r_m	=	middle radius [mm]
t	=	thickness [mm]
T_f	=	friction torque [Nmm]
U	=	stored energy [J/mm²]

I. Introduction

The focus of this paper is the development and testing of a drag sail based on solar sail technology that could contribute to a reduction of space debris in low Earth orbits (LEO). In order to build solar sails the development of gossamer sail structures is a necessity. These deployable structures contribute to a large field of future space applications like thin film solar generators, membrane antenna and drag sails. Solar sail technology development as part of the described alternative applications opens the possibility to gain experience with deployable gossamer structures in less complex projects with lower needs for funding e.g. in CubeSat projects. In preceding

studies it was shown that a sail with 2 m² area is sufficient to deorbit a triple CubeSat from a circular 800 km orbit in 25 years [1], which is consistent with the code of conduct (ISO 24113). This deorbit requirement is likely to be tightened and therefore sail areas up to 10 m² were considered in the design process. The presented drag sail fits into one unit of a triple CubeSat so that it is possible to fly further payloads.

Gravity induced deflection of gossamer structures as shown in Fig. 1 prevents a proper ground qualification testing of deployment mechanisms. Accordingly, the Space Sailors team [2] has developed a micro gravity experiment to test the deployment of a drag sail in size of a CubeSat. It was launched on a sounding rocket in the REXUS 13/14 campaign (Rocket Experiments for University Students) [3]. This project was supported by various institutes of the RWTH Aachen, the Faculty of Aerospace Engineering of FH Aachen University of Applied Sciences, the German Aerospace Center (DLR) and several companies. Further development of the presented drag sail will take place at the Faculty of Aerospace Engineering of FH Aachen in their Compass 2 project.

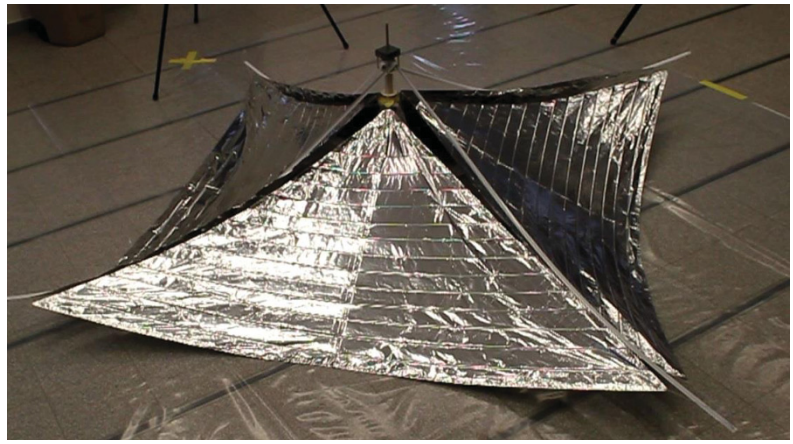

Fig. 1 Drag sail demonstrator [1].

II. System Overview

The general concept of the Space Sailors drag sail mechanism is the use of bi-stable metal leaf springs. By this the energy for the deployment is mechanically stored which makes an additional energy source superfluous. The development followed the guidelines “Systematic approach to the development and design of technical systems and products” (VDI 2221). This implied several steps of iteration to get the final experiment design as

shown in Fig. 2. The left hand figure shows the cube structure with stored booms and sail on two spools in the launch configuration. In the center the deployment mechanism is illustrated with the sail segments hidden to show details of the sail deployment mechanism. The deployed sail consisting of four segments is shown on the right.

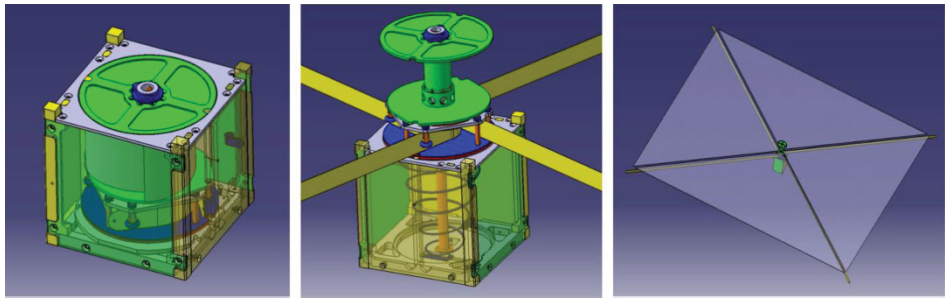

Fig. 2 Drag sail design: Stored configuration, deployment mechanism and deployed configuration.

The deployment of the sail is triggered by cutting a Dyneema® cord. Thus allows the conical coil spring to push the sail mechanic with the two spools out of their containment. They are free to rotate once they have left the guiding rails integrated in the sidewalls of the structure. While the bi-stable leaf springs are coiled on the lower of the two spools the sail foil is folded and coiled on the upper one. When the spools are free to rotate the leaf springs uncoil. Thereby they deploy the four foil segments. After the deployment the bi-stable leaf springs function as booms to keep the sail in its shape. To document the deployment, four cameras which provide 100 frames per second cover every part of the sail. Furthermore the rotation of the boom coil is measured. Due to safety reasons of the REXUS flight it is necessary to get rid of the complete sail after deployment. This is triggered by a second Dyneema® cord that is cut, allowing four small springs to jettison the complete cube with the sail. An overview of these interactions is presented in Fig. 3. The complete Space Sailors experiment mounted on the rocket structure (the black ring at the bottom) is shown in Fig. 4.

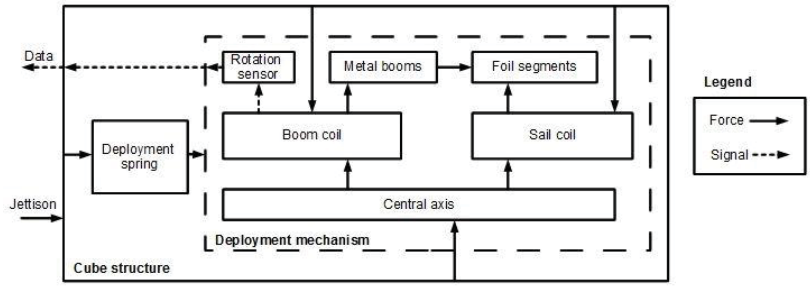

Fig. 3 Functional overview of the drag sail mechanism.

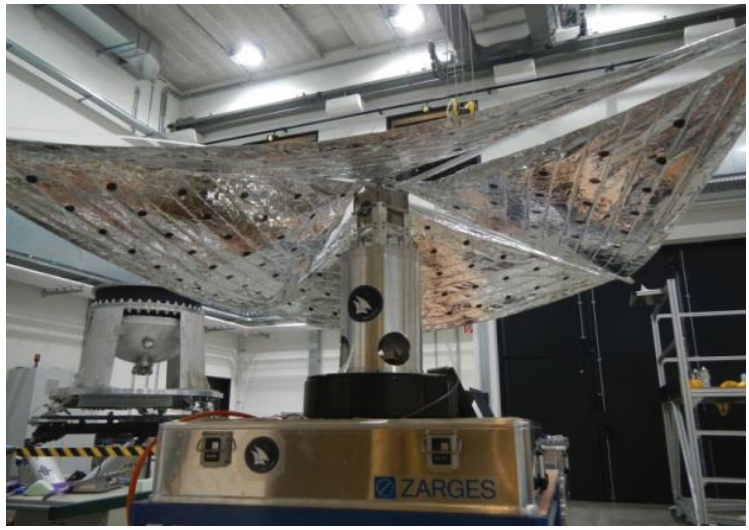

Fig. 4 The Space Sailors experiment.

III. Key Technology

A. Booms

The booms are made of bi-stable steel leaf springs. They have two mechanical stable configurations, one in an extended and one in a coiled up state. This is shown in Fig. 5. As both states are not free from internal tensions the two states contain stored strain energy. The difference in stored energy is used to open the sail. The contained tensions are a result of pre-stressing the leaf springs. The stored energy itself is as a function of curvature. This can be seen in Fig. 6 for a bi-stable leaf spring. While the global minimum of stored energy is a stable configuration of the spring the (higher) local minimum corresponds to the metastable second configuration. Is the boom in its metastable state, a small geometrical disturbance will trigger the transition of the boom into the state with the least stored energy. It is further noticeable that the curvature orientations of the stable configurations are different for k_x and $k_{y,}$ as shown at the left side of Fig. 6. Hence the curvature k_c in the diagrams in Fig. 6 is changing from k_x to k_y during the zero-crossing, where the zero-crossing is the entire flat state. This behavior and the relation between energy and curvature was investigated in “Bistable pre-stressed shell structures” [4] by E. Kebadze. There an equation to calculate the stored energy is also given. As the utilized steel springs were taken from a large-batch production with wide tolerances in geometry and shape, the usable moment of uncoiling was determined by measurements to be 265 Nmm. Their properties are given in Table 1.

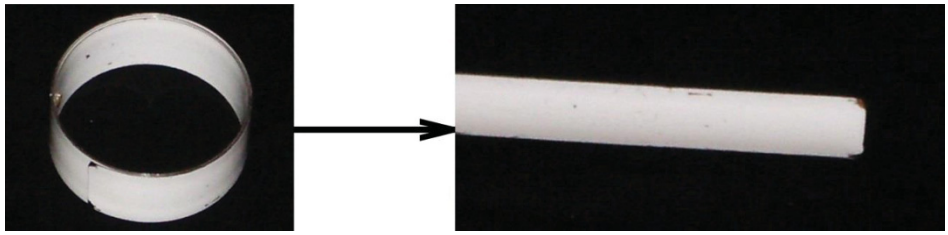

Fig. 5 The mechanically stable configurations of the pre-stressed metal leaf springs.

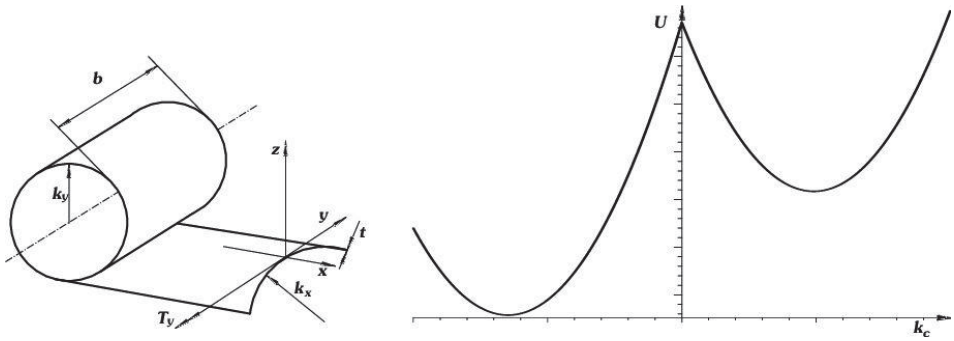

Fig. 6 The right side shows the geometric definitions and the left side the relation between the stored strain energy and the curvature for a bi-stable metal boom.

Table 1 Properties of the used bi-stable metal booms.

Properties of the used bi-stable leaf springs					
E [N/mm²]	b [mm]	t [mm]	k_x [1/mm]	k_y [1/mm]	T_y [Nmm]
$2 \cdot 10^5$	16	0.2	0.069	0.060	265

B. Sail

The technology to build the foil sail for the Space Sailors experiment was transferred from the solar sail project Gossamer-1 and its precursor project. Further the sail was built at the same facilities and with the same materials provided by DLR Bremen. It is based on a 7.5 µm thick Polyimide foil which is coated with aluminum. It consists of four triangular segments each mounted using Dyneema® strings from the company Royal DSM. The strings are adhesively bonded in a reinforced pocket along the edges of the segment. At the corners of the triangular segments the strings are fixed. The pockets were built with a space qualified transfer adhesive.

Launcher constraints leave no choice but developing sails that can be stored for launch in a compact configuration. Furthermore, in case of a drag sail it is desired only to deploy the sail at the end of a mission to increase the drag area. This makes it necessary to fold and coil the sail membranes in a way that an automatic deployment in orbit is possible. While in projects like Gossamer the spacecraft size is big enough to store every sail segment separately, a strategy was adapted to reach the miniaturization requirements of CubeSats. The strategy of frog-leg folding and coiling the segments has been modified in such a way that all four segments can be coiled on one spool as illustrated in Fig. 7. In a first step every sail segment is frog-leg folded 16 times. Afterwards it is folded over in the middle such that the folds are directed to the outside. Finally all segments are laid on top of one another and are coiled onto one spool. The Dyneema® strings at the inner corners are fixed to the spool itself.

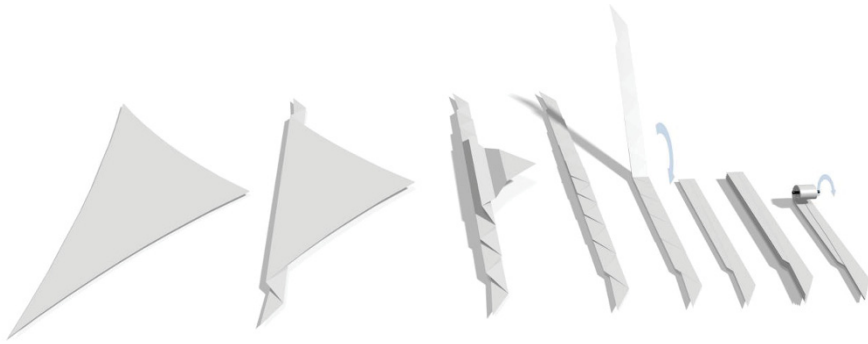

Fig. 7 Sail Storage Strategy: Every Segment is frog-leg folded, every frog-leg folded segment is then folded over in the middle, all segments are laid on top one another and are coiled onto one spool.

Once the segments are coiled on the sail spool, the Dyneema® strings at the outer corners are connected to the tips of the booms which later pull the sail of the spool. Figure 8 shows the integration procedure. In a final step the last centimeters of the booms and sail are coiled and the coils are slid into the CubeSat segment.

Some impressions of the manufacturing are given in Fig. 9. A vacuum table was used which allows a precise fixation of the foil when the desired geometry was cutted out. The edges are reinforced with pockets that are formed around the Dyneema® strings. Finally all segments are folded and coiled as described in the previous section. Figure 10 shows a drawing that specifies the geometry of one segment with the edge pockets.

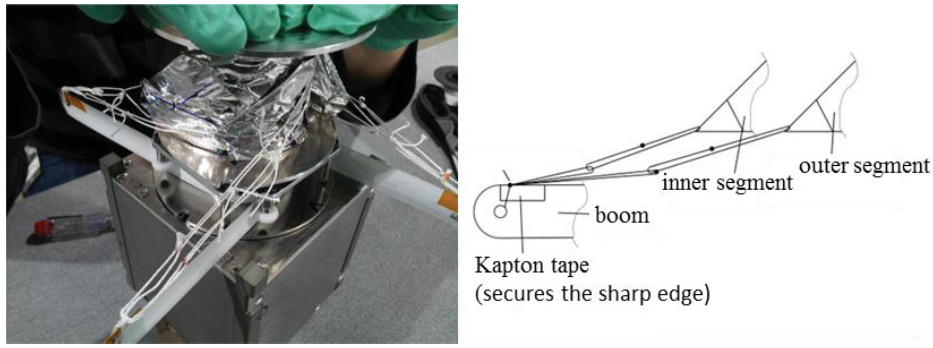

Fig. 8 Fixation of the outer sail edges to the booms.

Fig. 9 Sail manufacturing at DLR Bremen.

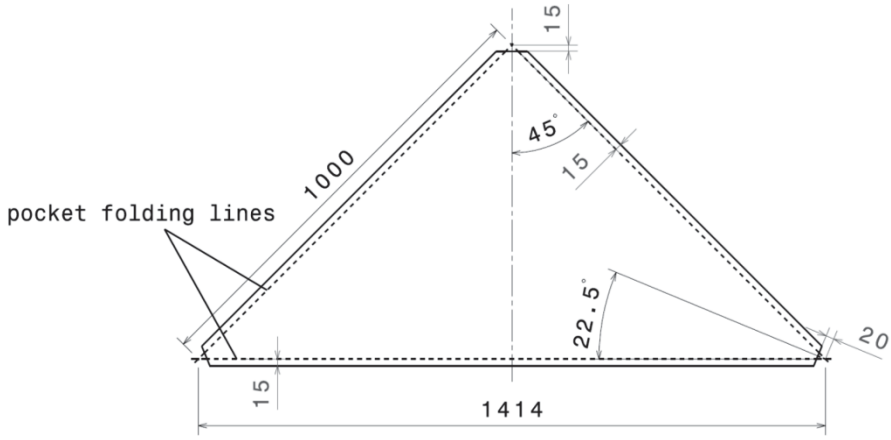

Fig. 10 Basic sail geometry with edge reinforcement pockets

IV. Design Details

For the Space Sailors experiment the sail and its deployment mechanism were integrated into a housing which fulfills the CubeSat standards [5]. The sail fits into the dimensions of one CubeSat segment with 100 x 100 x 113 mm and a mass of less than 1.34 kg. The construction maintains the possibility of expanding the structure from a single to the length of a triple CubeSat. The sidewalls were kept clear to allow a later installation of solar cells or sensors for further developments.

A. Sail scaling

The drag force on a 10 m² sail in a circular 200 km orbit is below 0.2 N [1]. On the other hand the Space Sailors experiment will take place at the apogee of a ballistic flight with different flow characteristics during the deployment. For this the altitude, the relative velocity and the attitude of the rocket is of importance. If the sail does not point into flight direction, which is likely on a ballistic flight path, the drag force is nearly zero while a maximum would arise if the sail is directed into flight direction. Between these extreme values the drag force is scaled by the sine function of the pitch angle. Figure 11 shows the force due to drag for a 5 m² sail as assumed for the worst case of influences. The calculation is based on flight data of REXUS 09 and an assumed flight to an altitude of approximately 100 km with the same velocity profile. The atmospheric density was considered according to NRLMSISE-00 for a solar maximum as provided in ECSS E-ST-10-04C [6]. The drag force is calculated according to a hypersonic Newtonian flow [7] normal to the sail. Because it is likely that the rocket reaches only 80-90 km altitude the sail area was scaled to 2 m². Nevertheless, a collapse of the sail structure shortly after deployment, due to increasing drag forces during descent, was taken into account.

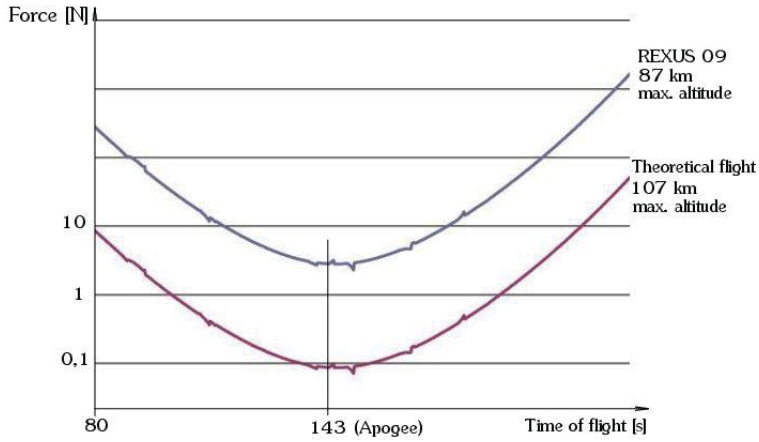

Fig. 11 Estimated drag force for a 5 m² sail around the apogee of a REXUS flight.

B. Friction consideration

For the deployment itself the involved forces and friction between interacting parts were investigated. As the stored strain energy in the bi-stable leaf springs is limited, the reduction of friction is a major design aspect. Therefore PTFE coating is used as well as a design which minimizes moving parts e.g. all four sail elements are stored on a common coil. An estimation of the maximum friction which the leaf springs have to overcome for a deployment was made with the worst case assumption of 1 g acceleration in longitudinal direction as the initial idea for an experiment on a sounding rocket was to eliminate that influence. The friction that needs to be overcome occurs in the areas shown in Fig. 12. The friction coefficients and the resulting torques are given in Table 2. The friction torque is calculated by:

$$T_f = \mu F_N r_m$$

The resulting friction torque is 114.12 Nmm. It shows that the four booms each with a moment of 265 Nmm should be strong enough for a successful deployment even under Earth bound conditions. This estimation neglects the influence of deflection of the booms and the hereby induced friction as this is virtually impossible to estimate. As this aspect prevents a proper deployment under Earth bound conditions it created the necessity for the micro gravity experiment.

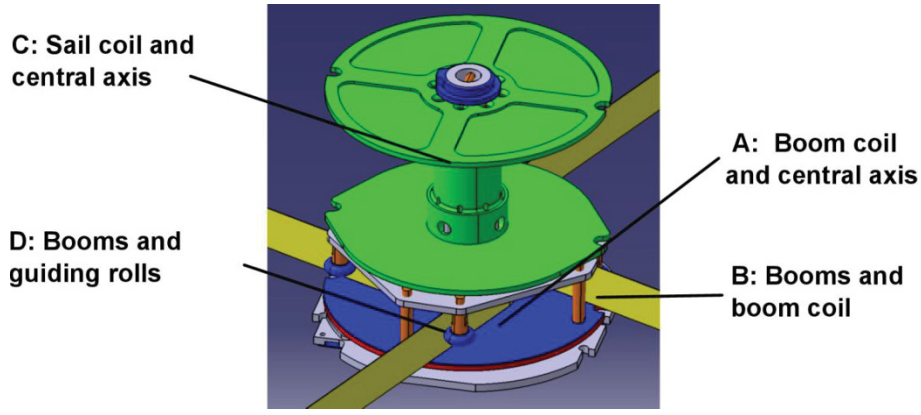

Fig. 12 Areas in which friction occurs during deployment.

Table 2 Friction coefficients and resulting friction torques involved in the deployment.

	Involved materials	Coefficient of friction μ	Friction Torque T_f [Nmm]	Quantity
A	Aluminum/PTFE coating	≤0.73	21.4	1
B	Steel /steel	0.2	21.45	4
C	Aluminum/PTFE coating	≤0.73	5.8	1
D	Steel/PTFE	0.04	0.53	4
∑			**115.12**	

C. Testing

Testing of the Space Sailors experiment gives the results that it fulfills the requirements given by the CubeSat Standards [5] and by the requirements for a flight on a REXUS rocket [8]. Vibration and vacuum tests where successfully performed with the Space Sailors experiment hardware. As the flight on a REXUS rocket generates higher loads than usual for CubeSats (e.g. 20 g of longitudinal acceleration [8]) the complete construction was designed to withstand these loads. Figure 13 shows the coiled up sail after a vacuum test. These tests demonstrated that the folded and coiled up sail keeps its shape during decompression and vibration. Tests of the deployment under gravity conditions on the ground showed the expected results. The booms deflect, thereby causing friction which prevents a full extension. Figure 14 illustrates that the deployment stops after 5.25 turns which is 2 turns short of a full deployment.

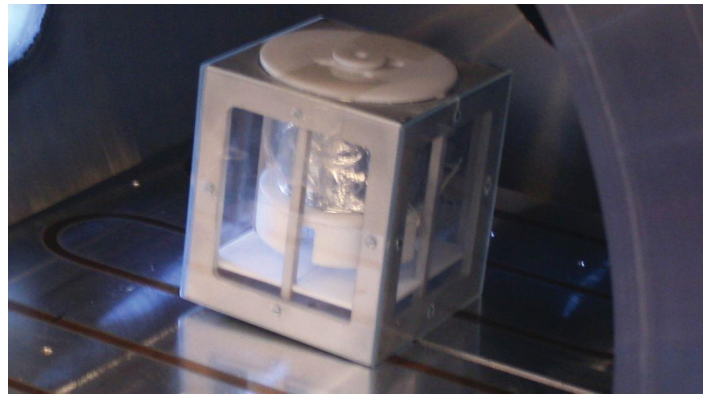

Fig. 13 Vacuum chamber test of the sail package.

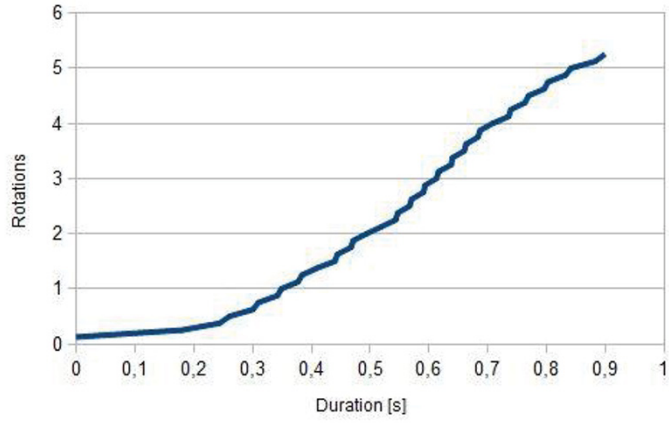

Fig. 14 Rotations of the boom coil during an Earth bound Deployment.

D. Measurements

The build mechanic (see Fig. 15) shows that a sail with up to 8 m² of sail area can be stored within a single CubeSat segment. The Space Sailors experiment masses of parts and assembly groups are summarized in Table 3. The numbers are taken from the build mechanic with the 2 m² sail and the ones in brackets were calculated for an 8 m² sail.

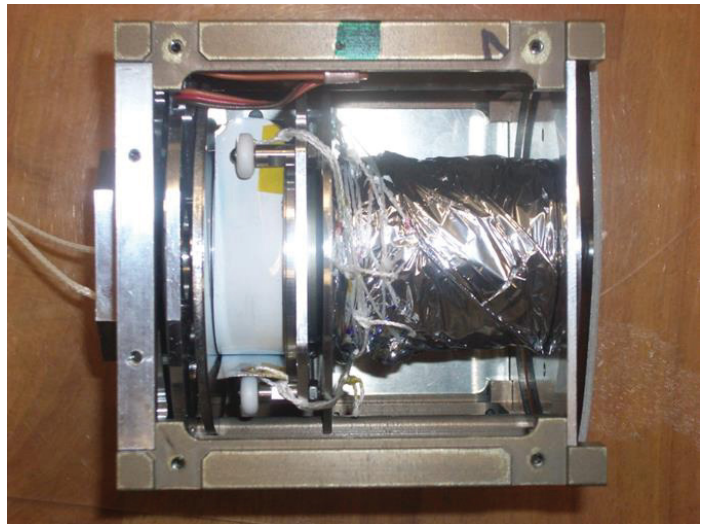

Fig. 15 The Space Sailors drag sail in its stored configuration.

Table 3 Masses of the Space Sailors experiment.

Part / Assembly group	Mass [g]	Quantity	Entire Mass [g]
Structure			
Structure	400	1	400
Deployment spring	30	1	30
Rotation sensor	20	1	20
			450
Sail Components			
Booms	60 (85)	4	240 (410)
Kaptoin foil segment	2.5 (6.5)	4	10 (40)
			250 (450)
Deployment Mechanic			
Sail spool	95	1	95
Boom spool	125	1	125
Central axis	110	1	110
Boom guidance	70	1	70
			400
Entire cube segment			**1100 (1300)**

V. Flight Results

The experimental deployment was performed near the apogee of the REXUS 14 flight at an altitude of 81 km. The deployment was successful. The drag sail extended completely without being visibly affected by friction. It took 0.65 s. which can be compared to the 0.9 s of the incomplete deployment of the Earth bound test. This emphasizes the assumed influence of gravity caused friction under ground conditions. The different stages of deployment are presented in Fig. 16 and the fully extended sail in Fig. 17. A few moments after deployment, at the beginning of the descent, the sail collapsed like expected due to increasing aerodynamic drag.

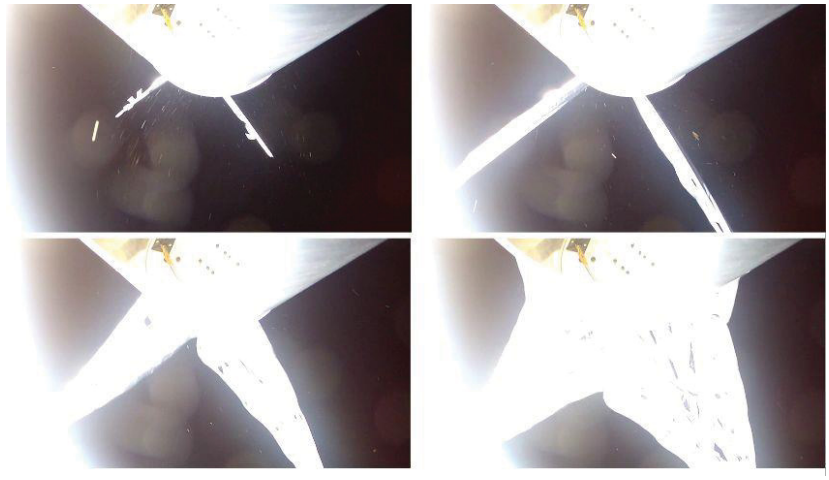

Fig. 16 Four stages of deployment of the Space Sailors drag sail.

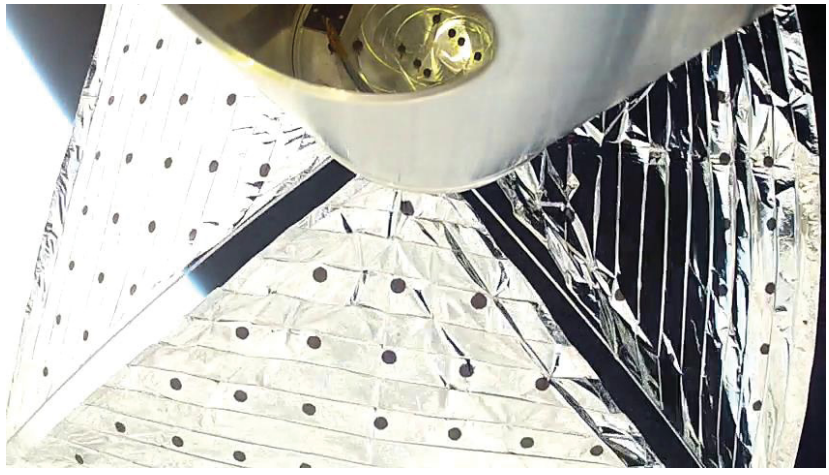

Fig. 17 The fully deployed sail at the apogee of the REXUS 14 flight.

VI. Prospect

The next step will be the integration of the drag sail on a satellite as deorbiting device. This is planned for a triple CubeSat which is in development at FH Aachen University of Applied Sciences in their Compass 2 project. The early stage of the design can be seen in Fig. 18. The planned orbit is a sun synchronous one with about 98° inclination and 350 to 400 km altitude. From that orbit the deorbiting duration is calculated to 43 to 24 hours depending on sun activity. The payloads are a drag sail as described above and thin film solar cells with a similar deployment system. The deployment of the sail will be captured by cameras and the rotation of the boom coil will be measured. Further the effects the sail has on the satellite, its attitude and the attitude control system shall be investigated. To integrate the drag sail system into the CubeSat some design improvements have to be made. Mainly the weight has to be reduced. Further it is planned to give the sail a pyramidal shape to assist the attitude control system, achieving a satellite-sail-configuration similar to a shuttlecock. An improved rigging to allow better stretching of the sail shall be considered as well.

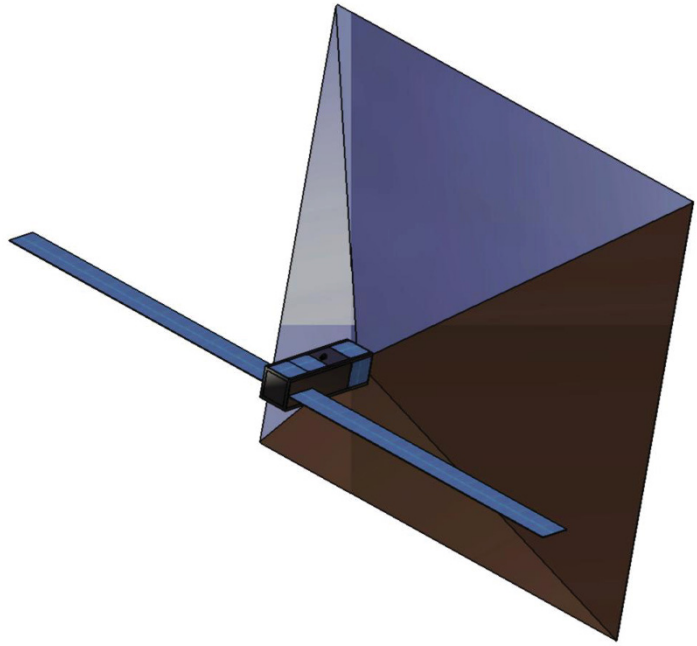

Fig. 18 Early stage of the DragSail-Cubesat.

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Earth-crossing asteroids deflection with a sailcraft

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Fast intercept trajectories are discussed to impact the dangerous Earth-crossing asteroids by using a solar sail spacecraft for the near to far-term missions. The heliocentric trajectory with a single solar photonic assist is considered to transfer the sail kinetic impactor from the Earth orbit to the collision point with the asteroid. Two options of such trajectories, i.e., the direct flyby and the angular momentum reversal trajectory, are compared via numerical simulations with respect to the transfer time and the relative collision velocity to determine which one is more suitable. The variation trend of the relative collision velocities are presented for different mission scenarios as a function of the sail lightness number. For a square flat ideally reflective sail with a given assembly loading, the determination of the impactor mass is studied to obtain the highest value of the net increment velocity of the asteroid.

I. Introduction

A FALLING meteor blazed and exploded over the Russia's Ural Mountains region on February 15, 2013 has been attracting attentions from all over the world. It is estimated that the meteor entered the Earth's atmosphere at a speed of approximately 33−40 km/s by Russian Academy of Science and NASA The energy released by the meteor is in the level of hundreds of kilotons with the force of an atomic bomb. The threat posed

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by such falling meteors and Earth-crossing asteroids (ECAs) cannot be excluded as one of the most unpredictable risks to the Earth and human beings. The effect of an impact between two bodies of non-ignorable dimensions has been clearly demonstrated by the colliding of comet Shoemaker-Levy 9 on Jupiter in 1994.

There are three typical families of near Earth asteroids, including Atens, Apollos and Amors. All of these near Earth asteroids have the potential to be perturbed into the trajectory of ECAs [1]. The ECAs or celestial bodies with diameters of 5−10 m usually explode in the upper atmosphere by approaching the Earth once per year. If the ECAs are large enough to pass through the atmosphere they may cause significant damage to the impacting region or even global effects. Recently, a number of large ECAs passing the Earth with a minimum distance of 0.05 AU have been identified as the potentially hazardous objects. Particularly, as a member of those ECAs, the asteroid 99942 Apophis, is discovered to be possible to hit the Earth in 2036 if it passes through one of several gravitational keyholes [2]. Due to these events and risks induced by the ECAs, the scientific and space communities have shown their interest on ECAs in both practical and theoretical aspects.

Some significant advances have been seen in the field of small celestial bodies in the past decades. On the practical side, Hayabusa, Deep Impact and Rosetta, have been launched by different space agencies (JAXA, ESA and NASA) to asteroids or comets to improve our knowledge about the small celestial bodies and enhance the current technology to rendezvous/impact an asteroid. On the academic side, a variety of deviation methods, that have been proposed to deflect an asteroid, can be organized into two ways. One is the impulsive strategies to instantaneously change the asteroid's orbital velocity, such as the kinetic impactor or nuclear explosion. The other is the slow-push strategies by applying a continuous low-thrust on the targeted asteroid in a long time, such as surface ablation, Yarkovsky effect and attached propulsion devices [3]. Taking the current technology into account, the kinetic impactor seems to be one of the most mature ways.

The kinetic impactor has gained considerable interest for future asteroid deflection missions. Casalino and

Simeoni [4] developed an indirect procedure for the determination of the optimal deflection mission of an asteroid by using an electric propulsion. Park and Choi [1] presented the optimal intercept and rendezvous trajectories for the near earth asteroids by using an advanced magneto plasma spacecraft. McInnes [5] proposed to impact an asteroid by the use of a solar sail spacecraft delivered into a retrograde orbit. Wei [6] and Dachwald [7] investigated the solar sail kinetic energy impactor on a heliocentric retrograde orbit for both ideal and non-ideal sail models. The transfer time for the sailcraft to complete such a mission is usually about 10 years or even longer. To reduce the transfer time, the angular momentum reversal (*H*-reversal) trajectory was adopted by Gong [8] to put a sailcraft into a retrograde orbit. It still takes about 3 years for the sailcraft to maneuver from the Earth orbit to the targeted collision point.

Until now, most studies involving the asteroid deflection trajectory by using solar sailing focus on the maximization of the relative collision velocity. There are few discussion and research about the fast intercept trajectories to the dangerous asteroids. Due to the chaotic dynamic of the ECAs and the complex perturbations of the solar system, it is very difficult to exactly predict the next impact between an asteroid and our planet. Fast intercept trajectories can transfer a sailcraft to the collision point of the asteroid with the kinetic impactor in a short time. It is a key factor to deflect or intercept with an asteroid which cannot be detected many years in advance.

The subject of this paper is to look for a type of trajectory that a sailcraft can maneuver from its starting point to the collision point in a short mission time. A time optimal control model is solved by using an indirect method to obtain the fast intercept trajectory. There are two options for the trajectories with a single solar photonic assist [9], including the direct flyby and the *H*-reversal trajectory. Comparisons are made via numerical simulations to determine which one is appropriate to be as the fast intercept trajectory. Then, a parametric study with respect to different asteroid radius and sail lightness numbers is done to show the variation trend of the relative collision

velocities between the asteroid and the sail kinetic impactor. When the area of the sail surface and its mass are given, the determination of the impactor mass is presented to obtain the highest value of the net increment velocity of the asteroid.

II. Options of Deflection Trajectories

A. Time optimal control model

The single solar photonic assist trajectories are adopted for a sailcraft to complete the fast intercept with an asteroid. Different from a rendezvous problem, the deflection mission only requires transferring the sailcraft to the same space position with the targeted asteroid where both the sailcraft and the asteroid are modeled to be mass points. The final condition can be written as

$$\Phi_f = \boldsymbol{r}_{\mathrm{s}}\left(t_f\right) - \boldsymbol{r}_{\mathrm{a}}\left(t_f\right) = 0 \tag{1}$$

where t_f is the flight time for the sailcraft from the starting point to the collision point. The vector $\boldsymbol{r}$ is the heliocentric orbital radius for the space object away from the Sun. The subscript 's' and 'a' denote the sailcraft and the asteroid, respectively. An ideally reflective flat sail model is adopted in this paper. The sail characteristic acceleration is

$$\boldsymbol{f}_{\mathrm{s}} = \beta \frac{\mu}{r^2}\left(\hat{\boldsymbol{r}} \cdot \boldsymbol{n}\right)^2 \boldsymbol{n} \tag{2}$$

where β is the sail lightness number, defined as the ratio between the sail acceleration and the solar gravitational acceleration. μ is the solar gravitational constant in a value of 1.3271244×10^{11} km^3/s^2. The scalar r is the magnitude of the orbital radius of the sailcraft and $\hat{\boldsymbol{r}}$ is the unit vector along the orbital radius. $\boldsymbol{n}$ is the unit vector along the sail normal direction which is perpendicular to sail surface. If $\hat{\boldsymbol{r}}$ and $\boldsymbol{n}$ are in the same direction, the sail achieves its maximum acceleration when β is given. For example, the maximum sail acceleration for the case of $\beta = 1$ at 1 AU (1 AU $\approx 1.496 \times 10^{11}$ m) is about 5.93 mm/s^2 which is referred to as a

high performance solar sail. As a sailcraft usually consists of a sail and a payload mass, the relation between the sail lightness number and the system of the sailcraft is

$$\beta = \frac{\sigma^*}{\sigma} \tag{3}$$

where σ^* is the critical sail loading parameter in a value of 1.53 g/m^2. σ is the area to mass ratio which is defined as a function of the total assembly loading

$$\sigma = \frac{m}{A} = \frac{m_s + m_k}{A} = \sigma_s + \frac{m_k}{A} \tag{4}$$

In the above equation, m is the system mass, including the mass of the solar sail m_s and the attached mass m_k which is the mass of the kinetic impactor. The parameter σ_s is the mass per unit area of the sail assembly, named as the sail assembly loading [10]. A is simply assumed as the sail film area. To simulate the deflection trajectory, the sailcraft equations of motion are

$$\begin{cases} \dot{\boldsymbol{r}} = \boldsymbol{v} \\ \dot{\boldsymbol{v}} = -\dfrac{\mu}{r^2}\hat{\boldsymbol{r}} + \boldsymbol{a} \end{cases} \tag{5}$$

where $\boldsymbol{r}$ and $\boldsymbol{v}$ are the position and velocity vectors of the sailcraft relative to the Sun. $\boldsymbol{a}$ is the system disturbing acceleration which can be expressed as

$$\boldsymbol{a} = \boldsymbol{f}_s + \boldsymbol{f}_o \tag{6}$$

The vector $\boldsymbol{f}_o$ represents the total gravitational perturbation forces from other celestial bodies. In this paper, the simple two body dynamic problem is considered. There are only solar gravitational force and solar radiation pressure force exerting on the sail. Hence, the second right term $\boldsymbol{f}_o$ can be neglected in this study. The single solar photonic assist trajectory is adopted to be as the heliocentric transfer trajectory for the asteroid deflection. As a preliminary study, the distance of the collision point (between the sailcraft and the asteroid) away from the Sun is considered instead of Eq.(1) for the planar case

$$\bar{\boldsymbol{\Phi}}_f = \left\| \boldsymbol{r}\left(t_f\right) \right\| - r_{\mathrm{a}} = 0 \tag{7}$$

The minimum flight time for the sailcraft from one point of the Earth orbit to the targeted distance r_{a} should be a constant value. To simulate the fast intercept trajectory, a sailcraft departs from the Earth orbit with a zero hyperbolic excess speed after the sail deployment. The Earth is assumed to maneuver on a circular orbit at 1 AU away from the Sun. Since there is only distance constraint about the collision point in the model, all points on the Earth orbit should be equivalent with respect to the mission time. For the single solar photonic assist (SPA), the sailcraft approaches the Sun to sharply increase its orbital velocity by gaining enough energy. Due to the thermal limit of the sail film, the minimum solar distance of the trajectory should be specified as

$$\Phi_1 = \begin{bmatrix} \left\| \boldsymbol{r}\left(t_1\right) \right\| - r_{\min} \\ \boldsymbol{r}^{\mathrm{T}}\left(t_1\right) \boldsymbol{v}\left(t_1\right) \end{bmatrix} = 0 \tag{8}$$

where t_1 is the perihelion time when the sailcraft passes the allowable minimum solar distance $r_{\min}$. Fast trajectories are required to intercept a suspected impacting asteroid which can provide an estimation of the mission time and allow orbit corrections to begin earlier. In order to minimize the transfer time one can maximize the following performance index

$$J = -\int_0^{t_f} dt \tag{9}$$

In this paper, an indirect method is applied to solve the time optimal control problem based on the Pontryagin's maximum principle. By introducing the adjoint vector $\boldsymbol{\lambda} = [\boldsymbol{\lambda}_r, \boldsymbol{\lambda}_v]$ related to the orbital position and velocity, the Hamiltonian of the system is formed as

$$H = -1 + \boldsymbol{\lambda}_r\left(t\right) \cdot \boldsymbol{v} + \boldsymbol{\lambda}_v\left(t\right) \cdot \left[-\frac{\mu}{r^2} \hat{\boldsymbol{r}} + \beta \frac{\mu}{r^2} \left(\hat{\boldsymbol{r}} \cdot \boldsymbol{n} \right)^2 \boldsymbol{n} \right] \tag{10}$$

The time derivatives of the adjoint variables are deduced by the Euler-Lagrange equations

$$\begin{cases} \dot{\boldsymbol{\lambda}}_r = \dfrac{\mu}{r^3}\boldsymbol{\lambda}_v - \dfrac{3\mu}{r^3}\left(\hat{\boldsymbol{r}}\cdot\boldsymbol{\lambda}_v\right)\hat{\boldsymbol{r}} - 2\beta\dfrac{\mu}{r^3}\left(\hat{\boldsymbol{r}}\cdot\boldsymbol{n}\right)\left(\boldsymbol{n}\cdot\boldsymbol{\lambda}_v\right)\left[\boldsymbol{n} - 2\left(\hat{\boldsymbol{r}}\cdot\boldsymbol{n}\right)\hat{\boldsymbol{r}}\right] \\ \dot{\boldsymbol{\lambda}}_v = -\boldsymbol{\lambda}_r \end{cases} \tag{11}$$

The optimal control law $\boldsymbol{n}(t)$ is the solution of

$$\boldsymbol{n}(t) = \arg\max_{\alpha,\delta\in\Re} H \tag{12}$$

where the subscript $\Re$ is the domain of sail feasible control angles. The sail orientation can be fully determined by the two sail attitude angles α and δ. The sail cone angle α is defined as the angle between the sail normal vector and the sunlight direction. The clock angle δ is defined as the angle between the transverse direction and the projection of the vector $\boldsymbol{n}$ on the plane perpendicular to the sunlight direction. It has been proved that the minimum time heliocentric transfer of an ideal solar sail requires the vector $\boldsymbol{n}$ to be in the plane defined by the primer vector $\boldsymbol{\lambda}_v$ and the position vector $\boldsymbol{r}$. It can be expressed as

$$\boldsymbol{n} = \begin{cases} \dfrac{\sin\left(\tilde{\alpha}-\alpha\right)}{\sin\tilde{\alpha}}\hat{\boldsymbol{r}} + \dfrac{\sin\alpha}{\sin\tilde{\alpha}}\dfrac{\boldsymbol{\lambda}_v}{\|\boldsymbol{\lambda}_v\|}, & \tilde{\alpha}\in(0,\pi) \\ \hat{\boldsymbol{r}}, & \tilde{\alpha}=0 \end{cases} \tag{13}$$

where $\tilde{\alpha}$ is cone angle of the primer vector, defined as the angle between $\boldsymbol{\lambda}_v$ and $\boldsymbol{r}$. The feasible domain of $\tilde{\alpha}$ is [0, π]. The case of $\tilde{\alpha}$ = π indicates that the value of α is π/2 corresponding to the scenario that the sail radiation pressure force is zero. Specifically, the optimal cone angle obtained from the optimal control law is given by

$$\alpha = \begin{cases} \tan^{-1}\left(\dfrac{-3+\sqrt{9+8\tan^2\tilde{\alpha}}}{4\tan\tilde{\alpha}}\right), & \tilde{\alpha}\in(0,\dfrac{\pi}{2}] \\ \tan^{-1}\left(\dfrac{-3-\sqrt{9+8\tan^2\tilde{\alpha}}}{4\tan\tilde{\alpha}}\right), & \tilde{\alpha}\in\left(\dfrac{\pi}{2},\pi\right) \\ 0, & \tilde{\alpha}=0 \\ \dfrac{\pi}{2}, & \tilde{\alpha}=\pi \end{cases} \tag{14}$$

According to the baselines of the transversality conditions, the final adjoint variables satisfy

$$\Pi_f = \begin{bmatrix} \boldsymbol{\lambda}_r\left(t_f\right) - \gamma_1 \dfrac{\boldsymbol{r}\left(t_f\right)}{\left\|\boldsymbol{r}\left(t_f\right)\right\|} \\ \boldsymbol{\lambda}_v\left(t_f\right) \end{bmatrix} = 0 \tag{15}$$

The requirement of the allowable minimum solar distance in Eq.(8) is settled as an inner constraint. Hence, the adjoint variables at t_1 should be

$$\begin{cases} \boldsymbol{\lambda}_r\left(t_1^+\right) = \boldsymbol{\lambda}_r\left(t_1^-\right) - \gamma_2 \dfrac{\boldsymbol{r}\left(t_1\right)}{\left\|\boldsymbol{r}\left(t_1\right)\right\|} - \gamma_3 \boldsymbol{v}\left(t_1\right) \\ \boldsymbol{\lambda}_v\left(t_1^+\right) = \boldsymbol{\lambda}_v\left(t_1^-\right) - \gamma_2 \boldsymbol{r}\left(t_1\right) \end{cases} \tag{16}$$

where $\gamma_i\,(i = 1, 2, 3)$ are the Lagrange multipliers corresponding to the constraints of the trajectory. The superscript + and − are the moments just after and before a position. The stationary conditions can be specified with respect to the inner constraint and the final constraint as

$$\Pi_H = \begin{bmatrix} H\left(t_1^-\right) - H\left(t_1^+\right) \\ H\left(t_f\right) \end{bmatrix} = 0 \tag{17}$$

The time optimal control problem has been transferred to be a multi-point boundary value problem (TPBVP) with the shooting function

$$\boldsymbol{S} = \left\{\bar{\boldsymbol{\Phi}}_f, \boldsymbol{\Phi}_1, \Pi_f, \Pi_H\right\} = 0 \tag{18}$$

There are totally 11 constraints in Eq.(18) where the unknown variables for the problem is also 11, including six initial adjoint variables, three Lagrange multipliers, the transfer time of the fast intercept trajectory and the time for the sailcraft passing the minimum solar distance.

After the derivation of the shooting function, the next problem is to solve the nonlinear equations by guessing the appropriate initial values of the adjoint variables. Since the sensitivity of the transfer time to small variations in the history of sail orientations, the convergence region of the time optimal control problem is very

small for the indirect method and difficult to be determined. Two techniques are adopted to reduce the sensitivity of the problem and the calculation effort. The first is the scaling techniques by normalizing the initial values of the adjoint variables [11]. One can observe that the equations of state/adjoint variables, the Hamiltonian, the performance index and the optimal sail control vectors are homogeneous with respect to the adjoint variables and the Lagrange multipliers. It indicates that a solution of the equations multiplied by a scale parameter should be also a solution. A positive constant λ_0 is introduced into the performance index which can make the Hamiltonian match the transversality conditions and enforce the initial adjoint variables on a unit sphere. Specifically, the performance index of Eq.(9) is rewritten as

$$\overline{J} = -\lambda_0 \int_0^{t_f} dt \tag{19}$$

and the Hamiltonian of Eq.(10) should be

$$H = -\lambda_0 + \boldsymbol{\lambda}_r(t) \cdot \boldsymbol{v} + \boldsymbol{\lambda}_v(t) \cdot \left[-\frac{\mu}{r^2} \hat{\boldsymbol{r}} + \beta \frac{\mu}{r^2} (\hat{\boldsymbol{r}} \cdot \boldsymbol{n})^2 \boldsymbol{n} \right] \tag{20}$$

The new Lagrange multipliers of the system can be defined as

$$(\lambda_0, \boldsymbol{\lambda}_r, \boldsymbol{\lambda}_v, \gamma_1, \gamma_2, \gamma_3) = \frac{(\lambda_0, \boldsymbol{\lambda}_r, \boldsymbol{\lambda}_v, \gamma_1, \gamma_2, \gamma_3)}{\sqrt{\lambda_0^2 + \boldsymbol{\lambda}_r^{\mathrm{T}}(t_0) \boldsymbol{\lambda}_r(t_0) + \boldsymbol{\lambda}_v^{\mathrm{T}}(t_0) \boldsymbol{\lambda}_v(t_0) + \sum_{i=1}^{3} \gamma_i^2}} \tag{21}$$

Obviously, the new Lagrange multipliers must satisfy the following equation

$$\Pi_\lambda = \sqrt{\lambda_0^2 + \boldsymbol{\lambda}_r^{\mathrm{T}}(t_0) \boldsymbol{\lambda}_r(t_0) + \boldsymbol{\lambda}_v^{\mathrm{T}}(t_0) \boldsymbol{\lambda}_v(t_0) + \sum_{i=1}^{3} \gamma_i^2} - 1 = 0 \tag{22}$$

As such, the optimal control problem consists of the performance index Eq.(19), the Hamiltonian Eq.(20), the state equations of Eq.(5), Eq.(6) and Eq.(2), the adjoint equations of Eq.(11), the trajectory constraints of Eq.(7) and Eq.(8), and the transversality conditions relevant to the Lagrange multipliers of Eq.(15), Eq.(17) and Eq.(22). Taking the new parameter λ_0 into account, there are 12 unknowns while the new 12 dimensional

shooting function for the system is

$$\boldsymbol{S} = \left\{ \bar{\boldsymbol{\Phi}}_f, \boldsymbol{\Phi}_1, \Pi_f, \Pi_H, \Pi_\lambda \right\} = 0 \tag{23}$$

The second is to nondimensionalize the dynamic equations with the following unit system. The distance is set to be 1 AU during the calculation of the orbital maneuver. The time unit is about 58.1 days to make the solar gravitational constant to be unity. In such an unit system, the Earth takes 2π time units to travel around the Sun. Since the optimal control history of an ideal solar sail is continuous, a fourth (fifth) order Runge-Kutta method with an adaptive step size should be adequate to calculate the nonlinear equations of state/adjoint variables. The absolute and relative errors during the simulation are set to be 10^{-10} while the program is stopped with a tolerance of the position and velocity errors in 10^{-8} with the dimensionless units.

B. Solar Photonic Assist Trajectories

According to the magnitude of the solar radiation pressure acceleration, solar sailing can be classified into three main groups, referred to as the low/mid/high performance solar sails. The sailcrafts under current technologies belong to the low performance solar sailing whose solar acceleration is no more than 1 mm/s^2 at 1 AU whereas the acceleration of high performance sailing is not less than 3 mm/s^2 at 1 AU [12]. The mid performance sailing has an acceleration between those two boundary values. There have been many discussions and potential mission applications ranging from the low to high performance solar sailings. Since solar sails can gain a large amount of energy by approaching the Sun, solar photonic assist (SPA) trajectories have been widely discussed in the previous studies in the area of interstellar missions [13, 14]. The low performance solar sails need to make multiple close approaches to the Sun to gain enough orbital energy whereas the high performance ones need only one encounter. Since the orbital angular momentum of those trajectories is always positive during the interplanetary transfer, i.e., the sailcraft maneuvers in a counter-clockwise direction, such trajectories are referred to as “direct solar flybys” throughout this paper. The single SPA trajectory is the subject of this paper

whereas the trajectories with multiple close approaches will be not presented. Hence, the lowest sail lightness number adopted in the simulation of this paper is 0.25 corresponding to an acceleration of 1.48 mm/s^2 at 1 AU, which can guarantee the sailcraft to complete the single SPA flyby meeting the requirement of the minimum solar distance.

When the sail accelerations are increased to be greater than 3 mm/s^2 at 1 AU, there is another type of single SPA flyby named “angular momentum reversal trajectory” abbreviated as “*H*-reversal trajectory”. The reversal trajectory was proposed by Vulpetti [15] in 1992 and it was adopted to complete the interstellar mission with one close approach to the Sun in an appropriate fixed sail orientation during the whole transfer. It will be first discussed that which type of trajectory, the direct flyby or the reversal trajectory, is more suitable to be as the fast intercept trajectory for a high performance solar sail spacecraft.

Without loss of generality, a sail lightness number of 0.75 is adopted to investigate the two types of single SPA trajectories. The value of 0.75 for β corresponds to an acceleration of 4.45 mm/s^2 and is also the highest value discussed in this paper. Fig. 1 shows the two different SPA trajectories of $\beta = 0.75$ with the same departure point and final distance of the collision points. The minimum solar distance is 0.4 AU. The departure point is arbitrarily selected as [1, 0, 0, 0, 0, 1.8] corresponding to the classical orbital elements [$a\ e\ i\ \Omega\ w\ f$]. The value of the true anomaly f is equivalent to each other for the circular 1 AU Earth orbit. The length unit for the semi-major axis is AU while the unit for the angles is radian. The fictitious dangerous asteroid orbit is partially presented with the two collision points at 1.5 AU. The optimal sail orientations during the whole transfer are shown in every 5.8 days.

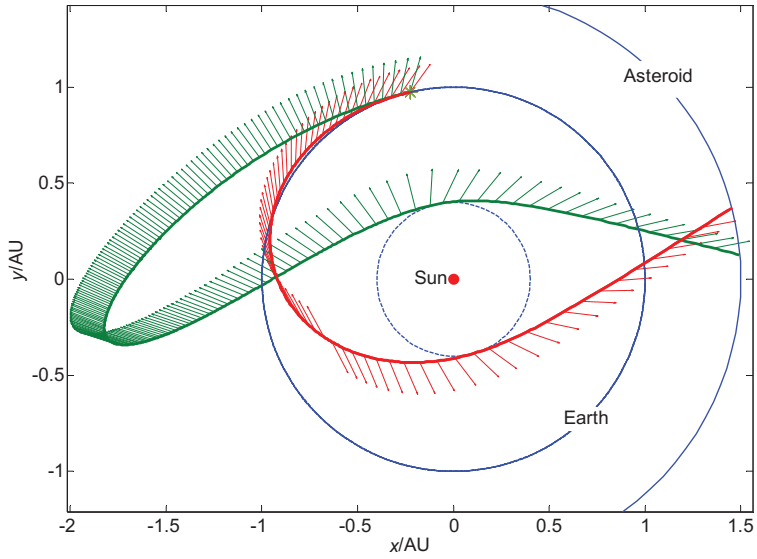

Fig.1 Asteroid deflection trajectories with a sailcraft.

For ease of comparison between the two SPA trajectories with respect to the collision effect, the orbital velocity of the fictitious asteroid at the collision point in a distance of r_a away from the Sun is assumed to be the local circular orbital velocity which can be expressed in the heliocentric inertial frame as

$$\boldsymbol{v}_a = \sqrt{\frac{\mu}{r_a}}\left[-\sin\theta \quad \cos\theta \quad 0\right]^{\mathrm{T}} \tag{24}$$

where θ is the polar angle defined as the angle from the inertial axis ox to the orbital radius of the collision point. The inertial axis ox is defined to be the direction from the Sun pointing to the zero true anomaly point of the circular Earth orbit. Axis oz is along the direction perpendicular to the ecliptic plane where axis oy completes the right-handed inertial frame. The magnitude of relative collision velocity between the sailcraft and the asteroid is

$$\Delta v = \left\| \boldsymbol{v}_{\mathrm{s}}\left(t_f\right) - \boldsymbol{v}_{\mathrm{a}}\left(t_f\right) \right\| \tag{25}$$

To compare the two single SPA trajectories, the transfer time and collision velocity for each trajectory in four different mission scenarios are listed in Table 1, i.e., the inner (r_a = 0.5 AU) and outer (r_a = 1.5 AU) cases

with different perihelion distances (r_p = 0.2 or 0.4 AU).

Table 1. Flight time and velocities for different target distances

SPA Flyby	r_p (AU)	t_{pi} (day)	t_i (day)	v_i (km/s)	Δv_i (km/s)	t_{po} (day)	t_o (day)	v_o (km/s)	Δv_o (km/s)
Direct	0.2	155.93	164.92	92.18	82.57	156.13	184.42	92.08	91.39
	0.4	150.65	160.09	59.69	34.28	151.50	193.55	61.66	58.80
Reversal	0.2	323.68	335.72	72.21	99.36	324.83	360.30	74.70	82.52
	0.4	485.17	497.53	48.22	86.74	487.17	538.95	51.63	63.97

Regardless of the trajectory mode in direct or reversal, the relative collision velocities for the case of r_p = 0.2 AU are greatly higher than the case of r_p = 0.4 AU. In order to achieve a higher velocity for the fast intercept trajectories, the minimum solar distance in the following simulations is set to be 0.2 AU which is also adopted in many previous studies [14, 16] evolving the interstellar missions. The relative collision velocities for the direct flybys and H-reversal trajectories with r_p = 0.2 AU are in the range from 82 to 99 km/s. The mission durations of H-reversal trajectories are nearly double as much as the direct flybys. Since the focus of this paper is to look for a type of fast intercept trajectory, the direct SPA flyby is selected in the following discussions.

III. Case Studies

A. Variation trend of the transfer time and relative collision velocity

In this paper the momentum of the system is assumed to be conserved during the impact between the asteroid and the sailcraft. The momentum conservation of the system at t_f can be expressed as

$$K\left(m_k \boldsymbol{v}_k + m_a \boldsymbol{v}_a^-\right) = \left(m_k + m_a\right)\boldsymbol{v}_a^+ \tag{26}$$

where m_a is the mass of the target asteroid. $\boldsymbol{v}_a^-$ and $\boldsymbol{v}_a^+$ are the orbital velocities of the asteroid before and after the collision, respectively. $\boldsymbol{v}_k$ is the collision velocity of the kinetic impactor which is the same $\boldsymbol{v}_s$ of Eq.(25). The coefficient K is used to scale the collision effect between these two objects [17]. In this study the value of K is assumed to be exactly one as a conservative estimation. Taking the inelastic impact into account, the value of K

can be much greater than one which depends on the materials of the kinetic impactor. Therefore, the variation of the asteroid's orbital velocity can be obtained as

$$\Delta \boldsymbol{v}_{\mathrm{a}} = \boldsymbol{v}_{\mathrm{a}}^{+} - \boldsymbol{v}_{\mathrm{a}}^{-} = \frac{m_{\mathrm{k}}}{m_{\mathrm{k}} + m_{\mathrm{a}}}\left(\boldsymbol{v}_{\mathrm{k}} - \boldsymbol{v}_{\mathrm{a}}^{-}\right) \tag{27}$$

The sail lightness number evaluated in the simulations is in the region of [0.25, 0.75] with a step size of 0.05 to show the variation trend of the mission time and relative collision velocities. The variation trend of the variables $\Delta \boldsymbol{v}_{\mathrm{a}}$ and flight time is presented as a function of the sail lightness number. The variable $\Delta \boldsymbol{v}_{\mathrm{a}}$ is difficult to be described explicitly. Therefore, the scalar parameter Δv defined in Eq.(25) is given along with the flight time in Fig. 2 for the outer case of $r_{\mathrm{a}} = 1.5$ AU and $r_{\mathrm{p}} = 0.2$ AU. As such, the relative collision velocity of Eq.(27) can be rewritten as

$$\Delta \boldsymbol{v}_{\mathrm{a}} = \frac{m_{\mathrm{k}} \Delta v}{m_{\mathrm{k}} + m_{\mathrm{a}}} \cdot \vec{\boldsymbol{v}} \tag{28}$$

where $\vec{\boldsymbol{v}}$ is the unit vector of the relative collision velocity. With the increasing value of β, the transfer time decreases whereas the impactor's velocity increases. The relative collision velocities are a little lower than the corresponding impactor's velocities.

A family of fast intercept trajectories is presented with all trajectories departing from the Earth orbit with a state vector of [1, 0, 0, 0, 1, 0] in the dimensionless unit. Due to the high acceleration with $\beta = 0.75$, the sailcraft maneuvers along nearly a rectilinear trajectory after the perihelion point. The relative collision velocity reaches up to 90 km/s which is consistent with previous studies [6, 7]. To accomplish near and far-term deflection missions, a group of trajectories with $r_{\mathrm{a}} = 0.5$ AU and $r_{\mathrm{p}} = 0.2$ AU referred to as the "inner case" is simulated and the variation trend of relevant parameters is given in Fig. 3.

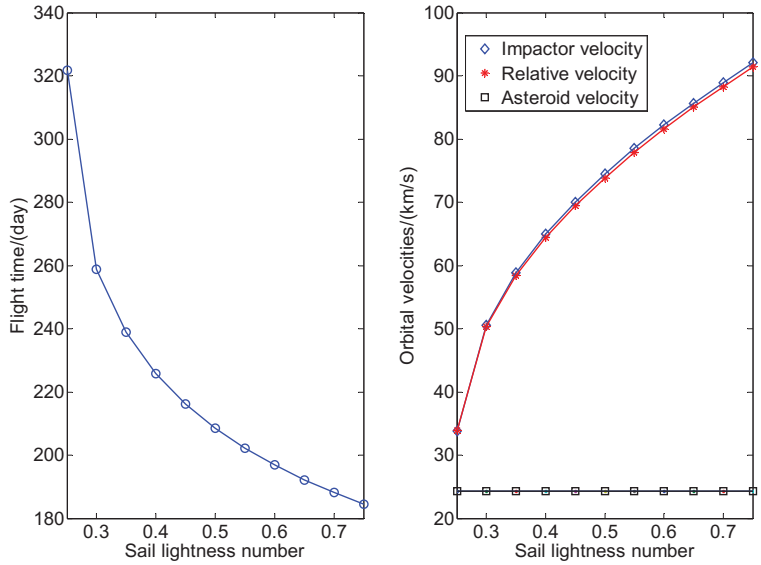

Fig.2 Flight time and collision velocity variation for the outer case.

The rule of parameters' variation for the inner case is similar to the outer case shown in Fig. 2. The gaps between the relative collision velocities and the corresponding impactor velocities are larger than the trajectories of the outer case. The magnitude of the relative collision velocities for the inner case is averagely 10 km/s lower than the impactor velocities. Taking the case of $\beta = 0.75$ as an example, the relative collision velocity for the outer case Δv_o is 91.39 km/s and the impactor velocity v_o is 92.08 km/s which can be seen from Table 1. For the inner case, the relative collision velocity Δv_i is 82.57 km/s and the impactor velocity v_i is 92.18 km/s. It seems that the collision point is better to be selected beyond the Earth orbit rather than the inner solar system. Note that such a result is obtained based on the assumption that the asteroid velocity at the collision point is the local circular orbital velocity. Additionally, the transfer time for the inner case is always shorter than the outer case with the same initial conditions.

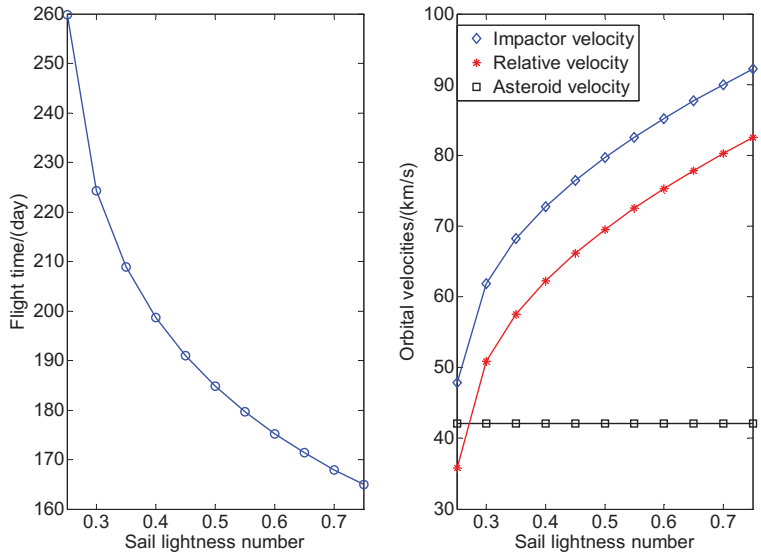

Fig.3 Flight time and collision velocity variation for the inner case.

B. Impactor mass and increment velocity

Without consideration of the impactor's material, the mass of the impactor also affects the collision effect from Eq.(28). Additionally, for a specified solar sail with a given assembly loading σ_s, the scalar relative collision velocity Δv is a function of the impactor mass. To investigate the influence of the impactor mass on the collision effect, a flat square solar sail is adopted in this section. In order to cover the region of $\beta \in$ [0.25, 0.75], the value of σ_s in Eq.(4) is assumed to be 1.53 g/m^2. It indicates that when the impactor's mass m_k is zero, the sail lightness number is exactly one. The relation between the lightness number and the mass of the impactor can be simply obtained from Eq.(3) and Eq.(4)

$$m_k = \frac{\sigma^* \cdot l_s^2}{\beta} - m_s \tag{29}$$

where l_s is the side length of the solar sail. The masses and densities of the asteroid are difficult to be determined due to their small size. Based on the previous studies, the density for the fictitious asteroid is assumed to be 2 $\times$ 10^3 kg/m^3 which is relatively a mid value for the observed asteroids [18]. It is estimated that

an asteroid with a diameter of 100 m can pass through the Earth atmosphere and cause damages to the impacting area. The mass of such an asteroid is easily obtained as 8.38×10^9 kg. If the side length l_s is 1000 m, the mass of the sail itself is 1.53×10^3 kg. Then the mass of the impactor can be obtained for different sail lightness numbers given in Table 2.

Table 2. The mass of impactor for different mission scenarios

β	0.25	0.3	0.35	0.4	0.45	0.5	0.55	0.6	0.65	0.7	0.75
m_k (10^3 kg)	4.59	3.57	2.84	2.30	1.97	1.53	1.25	1.02	0.82	0.66	0.51

With such a sail and different masses of the impactor, the net increment velocities of the asteroid for the inner and outer cases have been shown in Fig. 4. Since the mass of the asteroid is averagely 10^6 times than the masses of the impactors, the net increment velocities of the asteroid are a little low even though the relative collision velocities are in the range of 40 – 90 km/s.

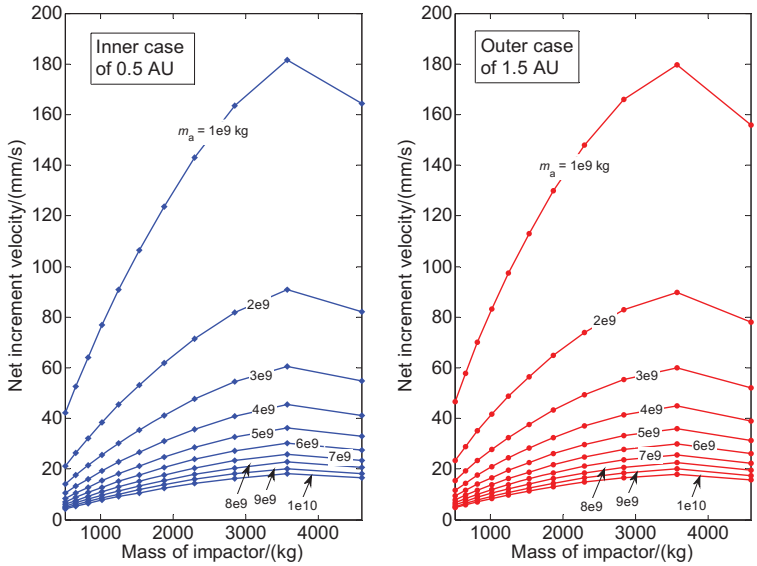

Fig.4 Net increment velocities for the inner and outer cases.

For both of these two cases there is a single ripple when $\beta = 0.3$ corresponding to the highest increment

velocity. It indicates that when the solar sail is given, the assembly loading, i.e., the mass of the impactor needs to be determined based on the mass of the asteroid and the impactor, the collision point and the transfer time. To increase the net increment velocities, two methods can be applied based on the fast intercept trajectories. One is to increase the area of the sail surface for the same lightness number which increases the mass of the impactors. The other is to impact the asteroid in a series with different collision points before its approaching to the Earth.

IV. Concluding Remarks

In this paper a type of fast intercept trajectories has been presented by using solar sailing to deflect the dangerous near Earth asteroids. Compared to the angular momentum reversal trajectories, the direct flybys with a single solar photonic assist are adopted to impact the asteroid by considering the transfer time and relative collision velocity. Two cases, referred to as the inner and outer cases, are investigated with the collision points at 0.5 AU and 1.5 AU, respectively. The variation trend of the relative collision velocities and the transfer times for the two cases are simulated by exploiting a time optimal control model. The transfer times decrease with the increasing value of the sail lightness number from 0.25 to 0.75 whereas relative collision velocities increase from 40 km/s to 90 km/s. With the same initial parameters, the relative collision velocity of the outer case is averagely a little higher than the inner case although the transfer time of the outer case is also a little longer. Taking the mass of the impactor into account, the net increment velocities of the asteroid are relatively low in a range of [2, 20] m/s when the sail lightness number is one as there is no impactor. There is a single ripple of the velocity variation for both the inner and outer cases corresponding to the highest increment velocity. For a given sail and a targeted asteroid, the mass of the impactor should be determined with the framework provided in this study to obtain the highest increment velocity.

Acknowledgments

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Solar Sails and the Search for Dark Matter

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Solar-photon sails have been proposed for decades-duration missions to the heliopause (~200AU) and the Sun's inner gravitational focus (~550 AU). A more advanced goal for space-manufactured sails capable of ~500 km/s interstellar cruise velocities is a search for WIMPS (weakly interacting massive particles), a suggested form of dark matter, within the solar vicinity. Newton's Shell Theorem is applied, in which WIMPS within the sphere defined by the solar distance are treated as at the Sun's center; those outside this sphere are ignored. A spherically symmetric near-Sun WIMP cloud will produce an anomalous spacecraft acceleration towards the Sun. Consideration of the Pioneer Anomaly demonstrated to be caused by differential spacecraft thermal emissions, reveals that WIMP mass within ~60AU is <0.2 Earth masses. Increasing the published accuracy of Pioneer 10/11 acceleration measurements by a factor of 10X allows probe trajectory measurements at ~10,000AU to confirm or falsify the existence of a ~3X star mass WIMP cloud in the Sun's galactic vicinity. Various sail configurations, sail/space-environment interactions, and mission planning are discussed.

Nomenclature

a_{dm} = dark-matter caused anomalous spacecraft acceleration, m/s^2

ε = radius of Sun-centered sphere, AU

G = Universal Gravitational Constant, 6.67×10^{-11} Nm^2/kg^2

κ = dark-matter mass/stellar mass

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M_{dm} = dark matter mass, kg

ρ_{dm} = WIMP density, kg/m^3

V_{sp} = Sun centered sphere volume, m^3

I. Introduction

Near-term solar photon sails have been proposed to propel robotic exploratory missions to various solar system destinations and the heliopause at around 200 AU from the Sun. [1,2]. In the not-too-distant future, possibly before 2050, sail technology may have advanced enough that space agencies could consider sail-launched probes to the Sun's inner gravity focus at and beyond 550 AU [3].

Some research has been conducted on hyper-thin space manufactured all- metal sails that could ultimately serve as true starships, propelling robots or humans on millennial expeditions to nearby stars [4, 5]. Although such ventures are far in the future, it would be nice if some destination intermediate between 550 AU and the nearest stars at about 240,000 AU existed, so that small-scale interstellar craft could be built and tested in the foreseeable future.

In its long range planning for space exploration in the post-2040 time frame, NASA has considered designing an Oort-cloud probe to explore the inner fringes of the Sun's Oort comet cloud, >1,000 AU from the Sun [6]. But Sedna, a member of the Oort cloud with a perihelion of about 76 AU and an aphelion of ~1,000 AU is currently at ~100 AU [7]. A near-term Earth-launched sail such as those proposed for the heliopause mission could visit this object in a human lifetime. Because of Sedna's eccentric orbit and relative current accessibility, it may be difficult to make a compelling case for a dedicated Oort-cloud probe.

But there is an alternative justification. It has been known for at least seven decades that the kinematics of stars in the Milky Way galaxy is not what one would expect from application of Newtonian gravitational theory. Instead of stars farther from the galactic center revolving more rapidly than those closer in, in the manner of planets in our solar system, stars farther out tend to move faster. Stars move around the galaxy almost as if they are attached to the spokes of an invisible wheel.

Some have attempted to explain this anomaly using modifications to the inverse-square nature of universal gravitation. The problem with this approach is that different modifications seem necessary at different astrophysical distances [8]. Others have suggested that familiar paradigms of physical thought may require substantial revision [9].

Most astrophysicists likely take the conservative approach that a mysterious form of matter dubbed "dark matter" is a possible explanation for anomalistic stellar motions [10]. This hypothetical dark matter might out-mass normal matter by a factor of three or four to one and comes in two possible forms: MACHOS (massive halo objects) and WIMPS (weakly interacting massive particles).

Advocates of MACHOS must deal with the fact that gravitational lens observations have not revealed a large number of invisible objects in the galactic halo and globular clusters in our galaxy's halo do not appear to be disrupted by large numbers of invisible, stellar-massed objects or brown dwarfs.

But WIMPS have their issues too. For WIMPS to produce the necessary stellar-motion anomalies and remain undetected, these particles must be invisible and non-reactive with ordinary matter. As evidenced by accurate and exhaustive trajectory analysis of humanity's earliest extra-solar probes, Pioneer 10 and 11, it seems unlikely that a large cloud of dark matter is in the solar system, at least to a distance of ~80 AU from the Sun.

If WIMPS do not gather in circum-stellar clouds, they may be more-or-less uniformly distributed in space. The following sections of this paper demonstrate that very accurate observations of the trajectory of a sail-launched probe to 10,000 AU might allow for the determination of an upper limit to WIMP density in the galactic vicinity of our solar system.

II. Dark Matter: Its Effect on Interstellar Probe Trajectories

The first step in this analysis of WIMP density determination is a consideration of the volume of interstellar space dominated by the Sun, The Sun's mass is 1.99 X 10^{30} kg [10] and the distance to the Sun's nearest known stellar neighbors (Alpha/Proxima Centauri) is about 4 light years. Therefore the Sun dominates a volume of 2.84 X 10^{49} m^3. The density of dark matter WIMPS in this volume is expressed:

$$\rho_{dm} \approx 7X10^{-20}\kappa \quad \text{kg/m}^3, \tag{1}$$

where κ is the local ratio of dark matter mass to stellar matter mass.

By comparison, if we assume 0.1 proton or neutral hydrogen atom per cubic centimeter in the Sun's galactic vicinity, the local interstellar proton mass density is 1.67 X 10^{-22} kg/m^3. So "normal" interstellar matter in the solar vicinity can be safely ignored in this analysis.

We next apply Newton's familiar Shell Theorem. If a spacecraft is at a distance R_{au} Astronomical Units from the Sun, all spherically symmetric matter closer to the Sun than R_{au} can be treated gravitationally as if it is centered on the Sun. All matter farther from the Sun than R_{au} can be ignored when calculating gravitational effects.

The volume of a sphere (V_{sp}) centered on our Sun with a radius ε AU is 1.4 ε^3 X 10^{34} m^3. The mass of dark matter within this Sun-centered sphere is therefore:

$$M_{dm} \approx 10^{15} \varepsilon^3 \kappa \text{ kg.} \quad (2)$$

Since this mass is assumed spherically symmetric and concentrated at the Sun's center from the point-of-view of gravity theory, the anomalous acceleration on a spacecraft towards the Sun at this solar distance is estimated:

$$a_{dm} = \frac{GM_{dm}}{\left(1.496X10^{11}\varepsilon\right)^2} \approx 3X10^{-18}\varepsilon\kappa \text{ m/s}^2, \quad (3)$$

where G is the gravitational constant.

Attention has been devoted to the trajectories of humanity's first extra-solar probes (Pioneer 10 and 11), which were launched in the early 1970's. The final signal from Pioneer 10 was received on January 22, 2003, when this spacecraft was at an approximate distance of 82 AU [11]. Contact with Pioneer 11 ended in 1995, when it was more than 40 AU from the Sun [12].

Analysis of Pioneer 10 and 11 trajectory data between 40 and 60 AU reported in 1998 revealed an initially unexplained acceleration towards the Sun of 8.09 X 10^{-10} m/s^2 for Pioneer 10 and 8.56 X 10^{-10} m/s^2 for Pioneer 11, with errors of about 0.2 X 10^{-10} m/s^2[13]. The accuracy of the acceleration estimates of these spin-stabilized space probes has been estimated as about 10^{-12} m/s^2, averaged over a period of 5 days [13].

An effort to understand this so-called "Pioneer Anomaly" was supported by The Planetary Society [14]. The published results of the study indicate that the cause of the anomalous Sunward acceleration is differential thermal emissions from spacecraft surfaces, not a new physical phenomenon [15].

It seems that the published accuracy estimate of the Pioneer 10/11 anomalous acceleration measurements, about 10^{-12} m/s^2, as the determination of an upper estimate for the WIMP density within 60 AU. Applying Eq. (3), we find that $\kappa < 5600$. From Eq. (1), the maximum average density of dark matter within 60 AU of the Sun is 4 X 10^{-16} kg/m^2. This corresponds to a maximum dark matter mass within a sphere of 60 AU radius centered on the Sun is 1.2 X 10^{24} kg. This is about equal to 0.2 Earth masses.

It seems indeed strange that the solar-system density of this allegedly ubiquitous substance that dominates the mass of the galaxy and the motion of stars is so low that dark matter seems not to be attracted by stars to form circum-stellar clumps. But an interstellar probe to ~10,000 AU may confirm or disprove the existence of WIMPS that avoid stellar vicinities.

Consider a probe to 10,000 AU and a value of κ =3.3. Applying Eq. (3), it is found that the Sun-directed acceleration caused by a spherically symmetric cloud of WIMPS within this radius is about 10^{-13} m/s^2. If the 1998-era accuracy of acceleration measurements on spin-stabilized extra-solar spacecraft increases by a factor of 10X, the existence of such a dark matter cloud in the solar vicinity can be confirmed or falsified by trajectory observations of such a spacecraft.

III. Sailcraft for a Mission to 10,000 AU

Current generation and near-term solar photon sails that could be used for missions to the heliopause or the Sun's inner gravitational focus are clearly inadequate for flights deep within the Sun's Oort comet cloud. The same is true for such missions propelled by the nuclear electric rocket.

To reach 10^4 AU in a flight time of 100 years, an interstellar cruise velocity of about 500 km/s is required. This is roughly an order of magnitude greater than the velocity required for a voyage to the heliopause or the Sun's inner gravity focus within a human lifetime.

Consider a nuclear electric rocket with an exhaust velocity of 100 km/s (Isp = 10,000 s). The mass ratio for such an exhaust velocity and interstellar cruise velocity is 148. At an average acceleration of 10^{-3} m/s^2, a 15-year acceleration period is required.

A hyper-thin space solar photon sail could achieve this velocity during a very close solar pass. As is reviewed in Refs. 4 and 5, the parachute-sail configuration with a ~30-nm thin space-manufactured metallic sail might be capable of such a performance. But further work has revealed that parachute sails

closer to the Sun than ~0.2 AU might focus sunlight on the payload, producing intolerable thermal effects [16].

To reach an interstellar cruise velocity of ~500 km/s during a close perihelion pass, accelerations will be higher than experienced in contemporary photon sail missions. This probably rules out disc and square sails that have been considered to propel the heliopause probe [17,18].

Another promising concept that can probably be ruled out for an Oort cloud mission, at least at present, is Forward's thin-film Starwisp, which was further investigsted by Landis [19, 20]. This perforated sail with a thin-film payload on the anti-Sunward face would certainly be capable of high acceleration. However, the microwave and laser-beaming infrastructure in space is in the distant future and the perforated sail optical theory presented in Ref. 5 is quite preliminary. However, it must be admitted that progress in the field of beamed-sail propulsion is advancing rapidly [21].

Another possible sail configuration is the hoop sail, where the sail is within an inflated hoop from which the payload is suspended. The analysis presented for this device's application to extra-solar travel is very preliminary and requires elaboration [22].

The best option for this mission in the not-too-distant future may be a hollow-body or inflatable "pillow" sail with hydrogen fill gas, the payload mounted on the anti-Sunward sail face and pushed by the sail. Originally suggested by Strobl [23], this concept has been further developed for a beryllium sail material [24].

The advantages of beryllium for this application include high melting point and low specific gravity. A disadvantage is the toxicity of this metal.

Initial analysis of the beryllium hollow body sail considered a 150-kg spacecraft with a 30-kg payload. The ~1-km radius sail, always oriented normal to the Sun, would be inflated in an initially parabolic solar orbit at 0.05 AU from the Sun and the peak acceleration would be less than 4 g. The version of this spacecraft initially considered, with a Sun-facing sail thickness of 20-nm and an anti-Sunward sail thickness of 10 nm could achieve and interstellar cruise velocity of about 740 km/s [24].

Further consideration of the interaction of photon sails with the near-Sun interplanetary medium and penetrating solar electromagnetic radiation revealed that a perihelion distance of 0.06-0.07 AU and a Sun-facing sail thickness of 40 nm are more realistic [25]. Interstellar cruise velocities of 500 km/s are not unreasonable for this spacecraft, especially if other trajectory options such as optimized solar aspect angle during acceleration are considered [26],

IV. Conclusions: Can We Conduct ~100-Year Extra-Solar Missions?

Exploration beyond the heliopaause and inner solar gravitational focus presents operational challenges because of the long travel times, regardless of the propulsion system. One approach is to have different science teams concentrating on different mission phases at different times.

During the perihelion approach, one science group might study solar activity or particles and fields in the inner solar system. Since the close perihelion pass occurs within the gravitationally warped space-time surrounding the Sun, the sail acceleration maneuver could be treated by a second science team as a laboratory to treat various aspects of relativity theory [27, 28].

After the conclusion of sail acceleration, probably near or beyond the orbit of Jupiter, interplanetary and near-interstellar particles and fields can once again become the science focus of the mission, until the Sun's inner gravitational focus is reached ~6 years into the mission. Observations of Kuiper Belt Objects might an additional science goal during this mission phase.

If the spacecraft has some cross-range capability, different astrophysical objects occulted by the Sun could be studied using the Sun's gravity focus. Without cross-range capability, variation in the electromagnetic emissions from an extended astrophysical object (such as the galactic center or M31 can be monitored.

During the final decades of extra-solar travel towards ~10,000 AU, the spacecraft's trajectory can be carefully monitored. This could allow astrophysicists to detect or place an upper limit on dark matter WIMPS in the solar vicinity.

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Part III

Technology Activities

Light pressure measurement at DLR Bremen

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The solar light pressure exerted on a thin foil is the quantity that determines crucially the performance of a solar sail propulsion technology. Besides the distance from the sun and the relative position of the sail to the sun, the light pressure depends on the quality of the sail surface. The surface quality, especially the reflectivity, degrades by exposing the sail to particle and electromagnetic irradiation under space conditions. This process will be simulated by a test facility at DLR Bremen. By directly measuring the light pressure after exposing the foils to a certain dose yields a kind of integral quantity of the degradation effects. To this end a light pressure measurement facility based on a high precision balance has been developed, completed and tested at DLR Bremen.

Nomenclature

A	=	surface area [m²]
A_{eff}	=	effective surface area [m²]
c	=	speed of light [m/s]
E	=	energy [N m]
ΔE	=	energy transfer [N m]
F	=	force [N]
g	=	gravitational acceleration [m/s²]

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h = Planck's constant [N m s]

m = mass [kg]

m_0 = rest mass [kg]

P = light pressure [N/m²]

p = momentum [N s]

Δp = momentum transfer [N s]

R = reflectivity

S = one solar constant [W/m²]

Δt = time step [s]

W = energy flux [W/m²]

θ = angle [°]

ν = frequency [1/s]

I. Introduction

Thin foils like solar sails experience under space conditions and irradiation the force of solar light pressure. The knowledge of the light pressure quantity is crucial not only for solar sail mission analyses. There are three parameters that influence the intensity of the light pressure:

- the magnitude of energy flux. It determines the strength of light pressure. In space it is determined by the distance from the sun. In laboratory the energy flux is adjustable by a suitable light source.
- the relative position of the foil surface to the light source. A perpendicular radiation will ensure the maximum momentum transfer.
- the reflectivity of the sail surface.

The quality of the foil degrades already since its fabrication, during the tailoring of the sails, furthermore by vibrations during launch, and finally in space by several degradation processes.

To determine the magnitude of the light pressure as a function of energy flux and quality of foil surface, a test facility, the light pressure measurement facility (LPMF), was developed at DLR - Bremen. The light pressure of one solar constant energy flux S, assuming perfect reflection, is $P = 2*S/c = 9.12*10^{-6}$ N/m^2. This relative smallness of this effect is a challenge to its measurement.

Onto an area of 1km² acts a force of only 9N. To measure such small forces, it is necessary to design a very sensible measuring system. Therefore, the core of the light pressure measurement facility is a high precision balance.

II. Theory

Light pressure is transferred by electromagnetic radiation onto surfaces; when photons hit them, they transfer their momentum to them. This momentum can be used as motive force for the solar sail propulsion technology.

According to special relativity the mass-energy equivalence is

$$E^2 = m_0^2 c^4 + p^2 c^2. \tag{1}$$

Since the rest mass of photons vanishes, Eq. (1) simplifies to

$$E = pc. \tag{2}$$

Because the energy of a photon is proportional to its frequency ν

$$E = h\nu, \tag{3}$$

the momentum transported by a single photon is

$$p = \frac{E}{c} = \frac{h\nu}{c}. \tag{4}$$

The light pressure can be expressed in terms of the energy flux W

$$W = \frac{\Delta E}{A\Delta t}. \tag{5}$$

Since the pressure is force per area and the force can be approximated by the momentum transfer per time step, the light pressure is eventually

$$P = \frac{1}{A}\frac{\Delta p}{\Delta t} = \frac{W}{c}. \tag{6}$$

This value for the light pressure doubles for a perfectly reflecting surface, i.e. R = 1. In this case and for an energy flux W = S = 1370 W/m² at earth's distance from the sun (one solar constant) the light pressure P is

$$P = (1+R)\cdot\frac{S}{c} = 9.12\cdot 10^{-6} N/m^2. \tag{7}$$

Clearly, the effective light pressure depends on the angle θ between the incident light and the surface normal. The effective sailing surface is reduced to $A_{eff} = A \sin\theta$. In the LPMF a vertical light incidence on the foil surface is realized. A factor 2 in Eq. (7) arises from the assumption that the surface is an ideal reflector with a

reflectivity of R = 1 [2]. This means that neither absorption, nor transmission, nor diffuse reflection appears. The surface consider as being perfectly plane. For realistic studies, the factor 2 has to be replaced by (1+R) [1].

To determine the light pressure by the LPMF, the following relations were applied. The weight m indicated by the balance, defines via the gravitational acceleration g the force F acting onto the foil

$$F = m \cdot g \,. \tag{8}$$

With this measured force F and the irradiated foil surface area A, the light pressure P is given by

$$P = F / A \,. \tag{9}$$

As seen in Eq. (7) the light pressure depends on energy flux W of radiation. With this relation the force can be written as

$$F = (1 + R) \cdot \frac{W}{c} \cdot A. \tag{10}$$

In the here presented test arrangement the irradiated foil surface area is A = 0.02 m². For an energy flux of one solar constant S = 1370 W/m² acting upon this test area, the force can be estimated to F = 0.187 µN. This force induces on the balance a weight of m = 0.0186 mg. This weight corresponds to the light pressure given in Eq. (7).

III. Test Facility LPMF

The LPMF established at DLR-Bremen consists of a vacuum chamber to simulate space conditions, a sun simulator for irradiation, a sample of solar sail foil, and a highly precise balance to measure the force acting by the radiation onto the foil. As shown in Fig. 1, the measuring system includes also a mirror to change the direction of radiation. The LPMF is in principle a precisions balance which weighs the photons.

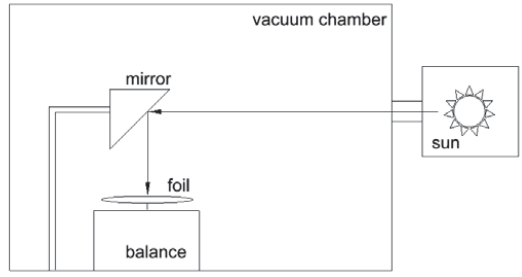

Fig. 1 Measuring system in the vacuum chamber.

In the vacuum chamber a final pressure of $p = 3*10^{-7}$ mbar is achieved. The sun simulator (1600-Watt Xenon lamp) provides an energy flux between 600 and 1200 W/m² at the test area on foil level. The energy flux is continuously adjustable. It can be directly measured in the test area. The spectrum of the radiation covers the wavelength range from 0.2 to 2.5 µm, and its beam diameter is about 180 mm. The foil sample used in the experiments was a 7.5 µm thick Kapton ® foil coated on both sides with 100 nm Aluminum. Given a diameter of 160 mm, the sample area is about 0.02 m². Since the resolution of the precision balance is in the order of 0.1 µg, forces in the range of nN can be measured. At the first glance it seems to be a very simple experimental setup. However, this test setup is highly sensitive to perturbations. The precision balance, as bought from the industry, is not designed to work under vacuum and nearly space conditions. However, a modification of the balance has been achieved, that makes it appropriate for the demanded test procedures.

IV. Results

The small forces acting onto the foil surface can easily be superimposed by other small forces. The most striking effect is the desorption of residual gas molecules. Water vapor is an always present contamination in a vacuum system. The water molecules adsorb on the surfaces in the whole vacuum system, e. g. also on the foil.

If the binding energy of the water molecules to the foil is less than the energy of the photons, the photons will remove the molecules from the surface. During the first irradiation of the foil surface the molecules desorb from the contaminated foil surface. How many molecules can be adsorbed back when the solar simulator has been switched-off, depends on the ambient pressure. At an ambient pressure lower then $5*10^{-7}$ mbar and an irradiation period of one minute desorption can be neglected. Another disturbing feature is the radiometric effect. This effect must not be considered in case of the LMPF, because it can be neglected for ambient pressures less than 10^{-3} mbar, certainly achieved in that facility.

The LPMF quantifies the light pressure in the above described way. In a first test series foil samples were irradiated at energy fluxes of $W_1 = 1100$ W/m² and $W_2 = 970$ W/m². Figure 2 shows clearly the increase of weight after switch-on of the sun simulator. For $W_1 = 1100$ W/m² the value of weight rises within a few seconds up to $m_1 = 0.0133$ mg. Then, the sun simulator radiates about one minute and the weight value stays nearly constant. After one minute of radiation the sun simulator has been switched-off and again the balance reacts immediately. Within seconds the value of weight decreases back, almost to the level of activation. During one minute without radiation the LMPF reaches a stable state. This test process is repeatable.

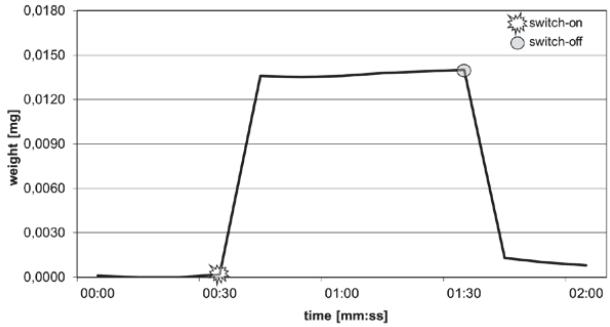

Fig. 2 Increase of weight during irradiation W_1 = 1100 W/m².

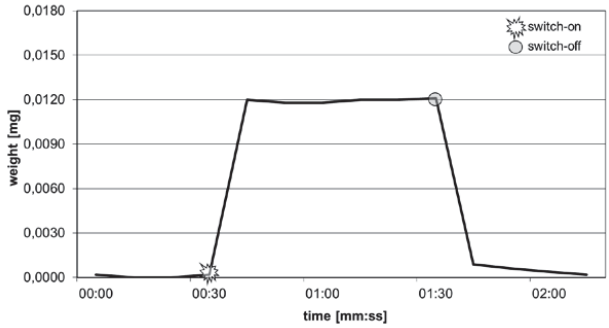

Fig. 3 Increase of weight during irradiation W_2 = 970 W/m².

Then, the energy flux was reduced from W_1 to W_2 = 970 W/m² and the test procedure has been repeated, see Fig. 2. Expectedly, the amount of weight is m_2 = 0.0113 mg, significantly less than for the previous irradiation with W_1. After the drop of energy flux it was raised again to W_1 = 1100 W/m². As seen in Table 1 the value of weight rises to the same value as in the first sequence. The mass value was measured in the switch-off reaction phase of the balance.

Table 1 Results of light pressure measurement for different irradiation

Sequence	Energy flux, W/m²	Weight measured, mg	Weight calculated, mg	Deviation	Light pressure (measured), μN/m²
1	1100	0.0133	0.0150	88%	6.5
2	970	0.0113	0.0132	86%	5.5
3	1100	0.0133	0.0150	88%	6.5

The calculations in chapter II refer to an ideal sail foil. The measured weights shown in Table 1 are between 12% and 14% less than predicted by its idealized theoretical values. That difference corresponds to the real reflectivity $R < 1$, to the roughness, and bumpiness of the foil surface.

Reflectivity measurements with three different foil samples have been performed. The reflectivity was measured in steps of one nm over the wavelength range between 300 and 1000 nm. Kapton was the supporting material of all three sail foil. The first sample was coated with Aluminum on both sides and the second one only on one side. The third sample was also coated on both sides but it was degraded during several hours of irradiation. As shown in Table 2 the average reflectivity over the whole wavelength range of all three samples is close to 0.91.

Table 2 Average reflectivity

Foil Sample	Coating	Feature	Average Reflectivity
1	both sides	fresh	91,10%
2	one side	fresh	90,51%
3	both sides	degraded	90,86%

With this value for the real reflectivity the calculated weight reduces according to Eqs. (10) and (8). Obviously, the non-ideal reflectivity causes losses of about 5%. Additionally the roughness and bumpiness of the surface causes a reduction of the acting light pressure. The foil surface gets crinkles caused by the manufacturing process and sharp kinks obtained during the packing. Furthermore, the foil is not clamped perfectly smooth on its fixture. The angles of the bumpiness are quite small, however, the sum of these effects causes the total measured losses of about 12-14%.

87 light pressure measurement sequences have been performed at the same energy flux $W_1 = 1100$ W/m² and the same foil sample. As seen in Fig. 3 the values lie in a range between 0.0125 mg and 0.0135 mg. The analysis shows, that the process is repeatable; it returns an average light pressure caused weight of 0.0130 mg. The reflectivity of the irradiated foil has been measured. It is $R = 0.91$, corresponding to a “theoretical” weight of 0.0143 mg. The difference of about 9% to the measured value is presumably caused by the roughness and bumpiness of the surface.

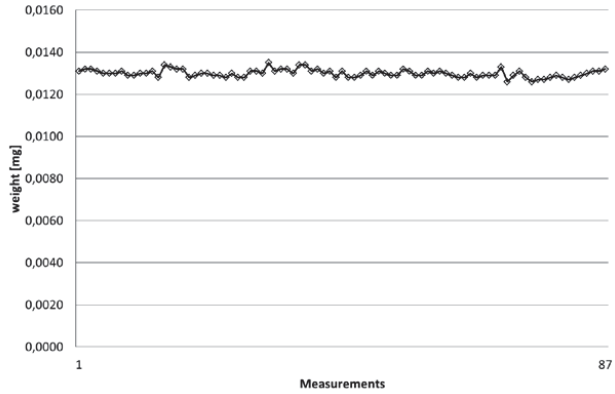

Fig. 3 87 measurements at W = 1100 W/m².

V. Conclusion

With the LPMF at DLR-Bremen, a test facility has been installed which permits the determination of both the effect of energy flux variations and of decreasing reflectivity on to the momentum transfer from solar photons to a sail with sufficient accuracy. Therefore, the efficiency of solar sail propulsion can be quantified under conditions that are similar to those in space.

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Realistic Solar Sail Thrust

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Before a solar sail will be used as a primary propulsion system for a space flight mission, many technical areas must be developed further. One of these areas concerns understanding the propulsion performance of a realistic solar sail well enough so that solar sail orbits can be confidently predicted to meet defined mission requirements. This paper identifies major contributors to solar sail thrust uncertainty, and analyzes the most significant ones to provide a better understanding of thrust generation by a "realistic" solar sail. Performance of representative "realistic" sailcraft are compared to similar "ideal" and "non-ideal" sailcraft to illustrate the differences.

I. Introduction

Severalscientifically important space flight missions have been identified that, at this time, can only be practically achieved using a solar sail propulsion system. These missions take advantage of the potentially continuous force on the sail, provided by solar radiation, to produce significant changes in the spacecraft's velocity, in both magnitude and/or direction, without the need for carrying the enormous amount of fuel that conventional propulsion systems would require to provide the same performance. However, to provide thrust levels that would support these missions requires solar sail areas in the (tens of) thousands of square meter sizes. To realize this, many technical areas must be developed further and demonstrated in space before solar sails will be accepted as a viable space mission propulsion system.

One of these areas concerns understanding the propulsion performance of a realistic solar sail well enough for mission planning. In most mission studies to date, either an "ideal" sail's performance is used for mission planning, or some top-level assumptions of certain non-ideal sail characteristics are incorporated to give a slightly better estimate of the sail performance. However, for missions using a 'realistic" solar sail, uncertainties

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in performance could propagate over the years of a mission's lifetime, potentially affecting the mission objectives and success. Without a better understanding, solar sail orbits could not be predicted well enough to meet defined mission requirements, such as rendezvous or station-keeping, and solar sail orbit optimization, such as minimizing flight time, could prove to be impossible.

Up to now, studies of realistic solar sail properties and their effect on thrust have been limited to single areas of concern with sometimes varying results. This paper identifies the major sources of solar sail thrust performance uncertainty, and analyzes the most significant ones to provide a more comprehensive understanding of thrust generation by a "realistic" solar sail. With this understanding, mission planners should be able to more confidently and accurately estimate the capabilities of such a system.

II. Ideal Solar Sail Characteristics

Those who have readMcInnes' text [1],are familiar with how solar radiation pressure and reflection from the surface of a solar sail creates a sailcraft's thrust. Figure 1illustrates the incident and reflected radiation from an ideal (flat, perfect reflector) surface, and the resulting force generated.

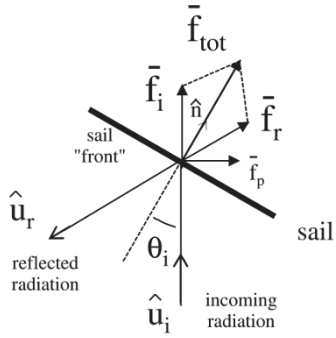


Fig. 1 Ideal Solar Sail Force

Taking into account the incidence angle θ_i the sail has with respect to the sun-line, the total force *vector* is ideally directed normally from the surface (as indicated by the unit vector **n**) at a magnitude proportional to the incident radiation pressure ($\mathbf{P_I}$, dependent on distance from the Sun) and the sail surface area A:

$$\bar{f}_{tot} = 2P_I A\left(\cos\theta_i\right)^2 \hat{n} \qquad (1)$$

Figure 2 illustrates the change in magnitude of the total force on a 100 x 100 meter perfect sail with changing sun-incidence angle. Though small (9 *micro*Newtons per square meter), the inexhaustible supply of light from the Sun can create a correspondingly small, but constant acceleration that enables the unique capabilities of a solar sailcraft.

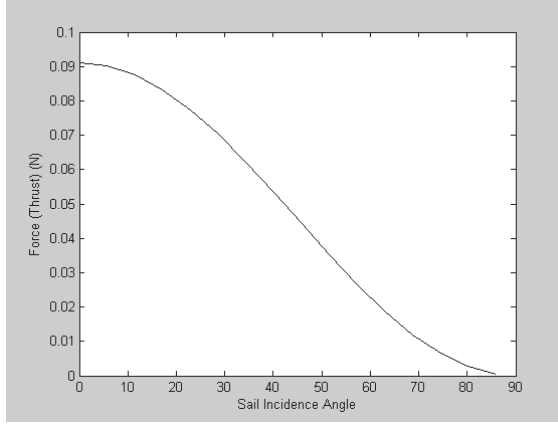


Fig. 2 Solar Sail Total Force (Thrust) Vs. Sun-Incidence Angle (For a 100 x 100 meter perfect sail @ 1 A.U.)

III. Non-ideal Sail Characteristics

For the purposes of this section, a non-ideal sail is defined as a flat surface with *non-ideal optical characteristics*. These non-ideal optical properties can be characterized by an absorption coefficient $\mathbf{a_b}$, a transmission coefficient $\mathbf{t_r}$, and a reflection coefficient $\mathbf{r_f}$, where $\mathbf{r_f} + \mathbf{a_b} + \mathbf{t_r} = 1$. In addition, there are emission coefficients for the front (reflective) and back surfaces of the sail ($\boldsymbol{\varepsilon_f}$ and $\boldsymbol{\varepsilon_b}$, respectively) and non-Lambertian coefficients ($\mathbf{B_f}$ and $\mathbf{B_b}$) which specify the uniformity of the intensity of the non-specular (or diffuse) reflected and emitted radiations (also from the front and back side surfaces). Table 1 compares the ideal values of these solar sail optical parameters to some typical non-ideal values, and Fig.3 depicts how these optical/thermal properties affect the reflection/re-radiation of the incoming solar radiation. The figure also depicts the sail thermal radiation due to absorbed energy.

Table 1. Ideal and Non-ideal Solar Sail Optical Properties

	t_r	a_b	r_f	s	ε_f	ε_b	B_f	B_b
Ideal sail	0	0	1	1	0	0	2/3	2/3
Non-ideal sail	0.02	0.1	0.88	0.94	0.05	0.55	0.79	0.55

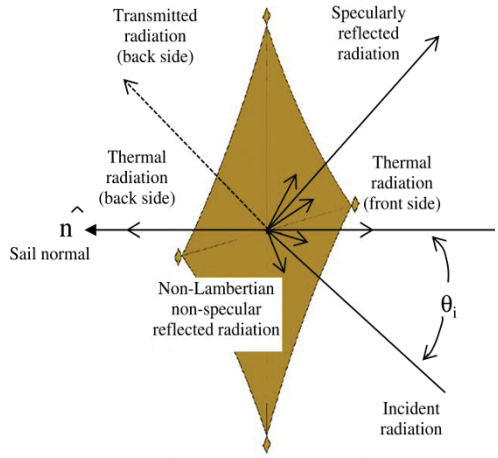


Fig. 3 Solar Sail Optical Properties

A. Non-ideal Solar Sail Force

This non-ideal reflection and emission of radiation from the solar sail surface has an effect on both the magnitude, and direction, of the resulting solar sail force, as explained in Fig. 4. Due to less than ideal optical properties of the sail, the actual solar radiation pressure force (denoted by the vector **m**) will be offset from the sail normal by an angle (ϕ), called the centerline angle. The angle between the actual force direction and the sun-line is defined as the cone angle(α). In general, the attitude of a solar sail is defined using the angle between the sail normal and the sun-line, represented by the incidence angle, and then the actual force direction determined, defining the cone angle, which is used in calculating the sail force effect on the sailcraft orbit.

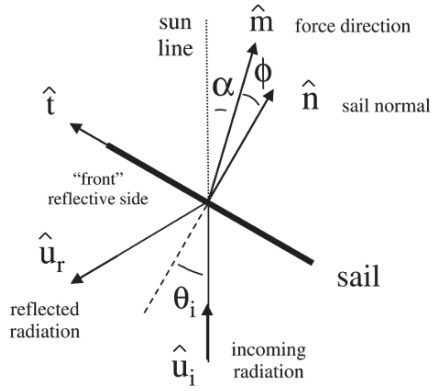


Fig. 4 Non-Ideal Solar Sail Force Direction

B. Solar Sail Force Components

Combining all the normal and tangential force components generated from the absorption, reflection and emission of the incident solar radiation, equations for these components of force on a non-ideal solar sail are obtained:

$$\overline{f}_n = P_I A\left[\left(1+r_f s\right)\cos^2(\theta_i)+B_f\left(1-s\right)r_f\cos\left(\theta_i\right)+\left(1-r_f\right)\frac{\varepsilon_f B_f-\varepsilon_b B_b}{\varepsilon_f+\varepsilon_b}\cos\left(\theta_i\right)\right]\hat{n} \tag{2}$$

and:

$$\overline{f}_t = P_I A\left(1-r_f s\right)\cos(\theta_i)\sin(\theta_i)\hat{t} \tag{3}$$

and, the total force magnitude can be found from:

$$f_{tot} = \left(f_n^2 + f_t^2\right)^{1/2} \tag{4}$$

The angle between the normal direction and the force vector was described earlier as the centerline angle, which now can be determined by:

$$\tan(\phi) = \frac{f_t}{f_n} \tag{5}$$

Figure 5 plots the total force magnitude for a non-perfectly reflecting 100 x 100 meter area square flat sail (using the Table 1 parameters) at 1 A.U., and compares this to a perfectly reflecting sail of the same area over a range of incidence angles. It can be seen that there is a noticeable decrease in the non-ideal solar sail performance compared to the ideal sail.

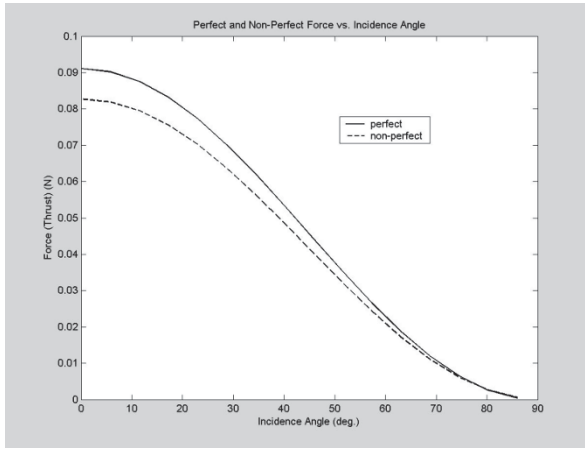


Fig. 5 Perfect and Non-perfect Solar Sail Force Vs. Sun-Incidence Angle (For a 100 x 100 meter sail @ 1 A.U.)

C. Relative Effects on Thrust

In [2], the author examines the relative effects of each of the non-ideal optical properties on overall thrust, using Eq. 2 – 5 above. Using a steady 30 degree sun-incidence angle attitude, one at a time each optical value is increased, then decreased, by 20% of its baseline value as given in Table 1[3], and the total force magnitude calculated. Figure 6 shows the percent change in the force from that which would be obtained using just the constant Table 1 values. It is clear that the sail reflectivity has the biggest influence on thrust magnitude, and that the other optical properties have much less of an effect. However, what Fig. 6 also reveals is a relationship between uncertainty in sail properties and the associated uncertainty in performance. Obviously, even a modest 10% uncertainty in actual sail reflectivity could cause appreciable uncertainty in thrust performance expectations.

Similarly, Fig. 7 illustrates the relative effect of the changes of each optical property on the overall total sail thrust *direction*. Using the same steady 30 degree sun-incidence angle attitude, and separate changes in the sail

[3]Up to a maximum of 0.99 where applicable, so as not to exceed ideal values.

optical properties values from the baseline value as given in Table 1, the figure shows the percent change in the force cone angle from that which would be obtained using just the constant Table 1 values. In this case, it can be seen that both reflectivity and specularity have relatively large influence over sail force direction, while the effect of the other sail optical properties are essentially insignificant in this regard. In addition, the graph reveals the relation between uncertainty in these key values and the associated uncertainty in force direction, and that they may or may not be additive or offsetting. As much as the force magnitude uncertainty, this could make solar sail mission planning difficult.

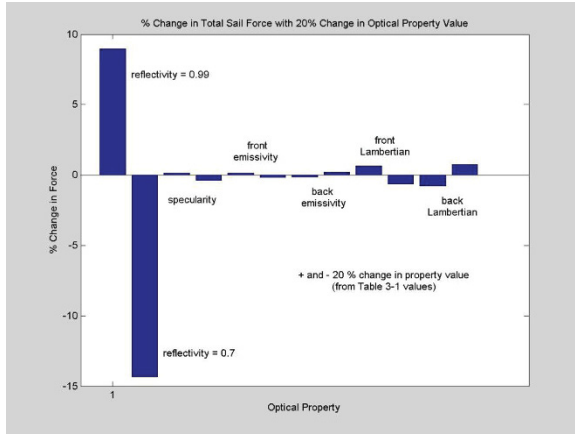


Fig. 6 Relative Effect of Optical Properties on Sail Thrust Magnitude (For a 100 x 100 meter sail @ 1 A.U.)

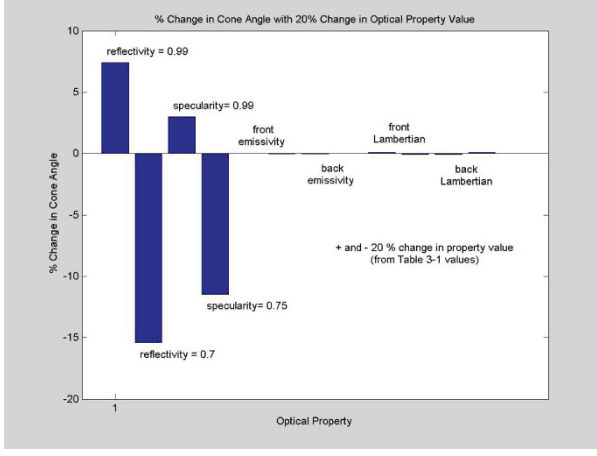


Fig. 7 Relative Effect of Optical Properties on Sail Thrust Direction (For a 100 x 100 meter sail @ 1 A.U.)

IV. Realistic Solar Sail Characteristics

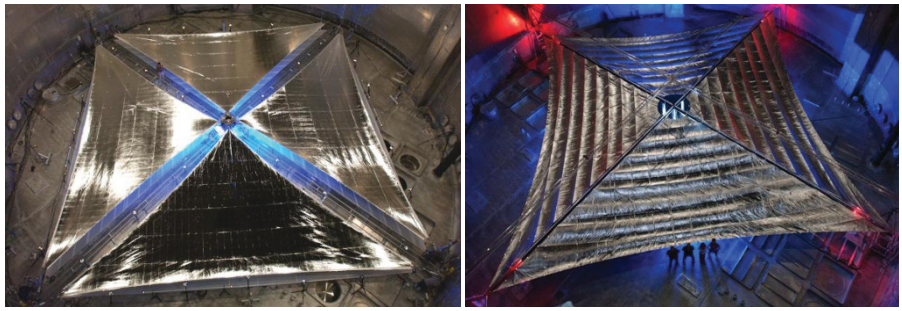
The last section introduced the solar sail optical characteristics that differentiate an ideal solar sail from a non-ideal sail. Though the non-ideal model provides the solar sail mission planner with a better approximation of the performance of a solar sail, there are a number of additional characteristics that define a more realistic solar sail. These characteristics include factors associated with actual manufacturing and use of a solar sail system, including materials, handling, testing, packaging and deployment, attitude control, and operation of the sailcraft in orbit. In this section, sail characteristics associated with many of these factors are identified.

Much of the realistic characterization of solar sails comes from results of the NASA In-Space Propulsion Technology (ISPT) development efforts, for which two different sail design configurations were built and tested [3, 4]. In addition, significant sail manufacturing and deployment testing has been accomplished in Europe [5] and in Japan [6], from which important information has been obtained. During these developments, testing provided measured values for solar sail properties and characteristics, and provided insight as to how these properties come about and change under certain conditions. What became very clear was that solar sails are not the homogeneous entities that, to this point, have been assumed, and that there will be a healthy uncertainty in predictions of the actual sail characteristics, and the subsequent thrust (and attitude control) performance of the sail in space.

Figure 8a) depicts a 20 x 20 meter square sail prototype that was built as part of the NASA ISPT effort by ABLE/ATK[4]. The sail material was manufactured by SRS Technologies and consisted of a 2 – 3 micron thick CP1 polyimide substrate with a 1000 Angstrom deposited surface of aluminum on the "front" (reflective) side and no coating on the "back" side of the sail material. The figure shows that, for this particular configuration, the sail material was placed in tension between the sail booms in an effort to produce a flat, smooth surface to minimize shape and wrinkling effects, which are discussed later.

Figure 8b) shows the 20 x 20 meter square sail manufactured by L'Garde, Inc. for the ISPT program. This sail was made from 2 – 3 micron thick Mylar material with a 900 Angstrom deposited surface of aluminum on the front side and a 200 Angstrom coating of black chromium on the back side of the sail material for thermal control. For this particular configuration, the sail material was draped over "stripes" of wires hung between the sail booms. This was in an effort to reduce the mass required for the sail booms, and so to reduce the overall mass of the sail propulsion system, thereby increasing sail performance (despite the potential decrease in performance for a more diffuse, non-flat surface).

[4] ABLE was acquired by ATK during this time.



a), CP1 Tensioned Sail (NASA/ATK) **b), Mylar Draped Sail (NASA/L'Garde)**

Fig. 8 "Realistic" Solar Sail Test Articles

These photos show someapparent differences between the two sail configurations in many of the areas that have been discussed so far. There seems to be a difference in reflectivity (at least based on the flash photography), and due to the design decisions involving applying tension to the sail booms, there is an obviousdifference in the surface shape of the sails[5]. It is these two factors, realistic optical properties and sail shape, and their effect on sail thrust, that are the subject of the rest of this paper.

A. Realistic Solar Sail Optical Properties

As we saw earlier, the total reflectivity of the sail and the percent of specular reflectivity have the greatest effect on solar sail thrust magnitude and direction (at least for a flat sail). Since this is the case, this section concentrates on these sail characteristics for a realistic sail, to determine any associated thrust effects.

1. Reflectivity

Figure 9 [7] illustrates how the average reflectivity of the aluminum-coated CP1 sail was measured over the applicable wavelengths of solar radiation. Note that this reflectivity was measured based on the spectral irradiance possible in the test set-up, also shown in the figure (dark blue line). It is clear that the reflectivity changes, in some cases significantly, with incident wavelength. Over the entire wavelength spectrum, the average total reflectance for the CP1 sail was 91.8%, as indicated, which is the value sometimes used for thrust calculation purposes.

[5] These differences, including the differences in sail materials and manufacturing methods, were actually factors in these configurations being selected for development – to allow comparison of the resulting sails in all these areas.

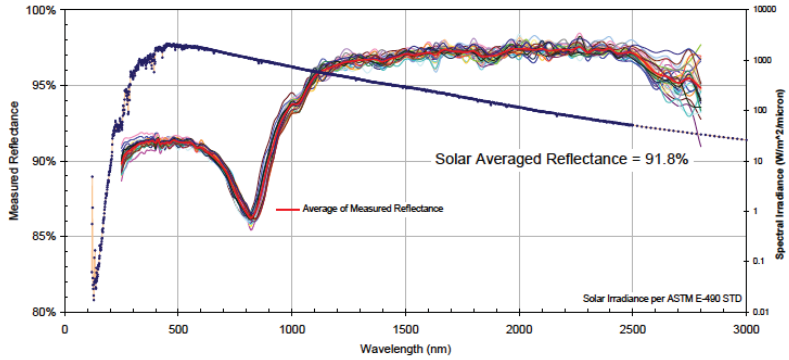


Fig. 9 Measured Solar Reflectance (ATK)

2. Specularity

The total reflection of radiation by the sail surface is made up of specular and non-specular (or diffuse) components, which were described earlier using a specular reflection coefficient (or specularity ***s***). Though also considered constant over the sail surface,Vulpetti's experiments [8] showed that this sail characteristic is dependent on the surface roughness of the sail material, which can change with the manufacturing process, mainly a result of irregularities in the deposited aluminum coating on the sailmaterial. Figure 10 [8] shows that for different surface roughness values, the specular reflectivity of an aluminum surface can vary significantly, especially at low sun-incidence angles, with the rougher surface having a lower specular reflectivity (as would be suspected).

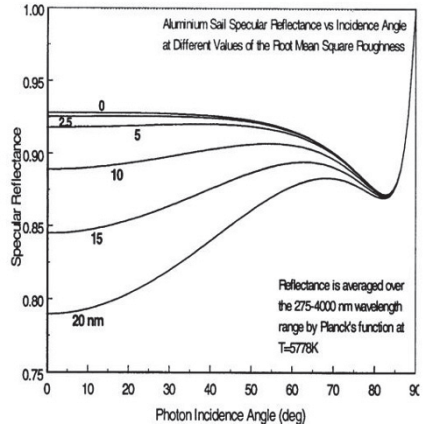


Fig. 10 Specular Reflectance vs. Sun-Incidence Angle (Vulpetti)

The figure shows that the change in the specular reflection coefficient between a perfect surface and a considerably rough surface (20 nm rmsr) is about 10% (at zero degrees sun-incidence, and less at higher incidence angles). While it was shown earlier that a similar 10% change in specularity only affects the total sailcraft thrust magnitude by about 2%, it was also shown that changes in specularity have a noticeable effect on thrust direction, especially on maximum thrust cone angle. Figure 11 illustrates the differences in cone and centerline angles over the range of specularities associated with the roughness values used in the computations to produce Fig. 10. This effect is not insignificant.

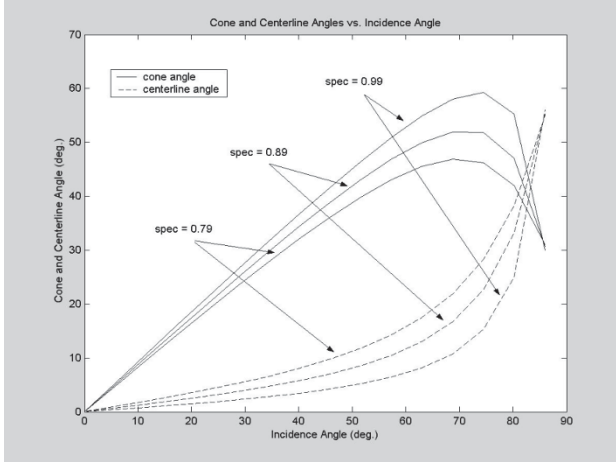


Fig. 11 Cone and Centerline Angles Vs. Sun-Incidence Angle for Different Specularities (For a 100 x 100 meter sail @ 1 A.U.)

To ameliorate this effect, surface roughness for the sail will need to be measured prior to flight, and depending on the value and desired accuracy, incidence-dependent thrust direction calculations will need to be performed. Fortunately, it is relatively easy to get decent measurements of this sail characteristic, and the values for this inherently physical characteristic would not be expected to change during a mission.

3. Wrinkles, Crinkles and Creases

Solar sails are made up of thin polyimide substrates with very thin metallic layers applied. As these materials are being fabricated, handled, folded and deployed, the unavoidable associated stresses and strains will produce temporary and permanent deformations in the form of wrinkles, crinkles and creases. Figure 13 shows sections of sail material that demonstrate the difference between these sail material characteristics.

a), Wrinkles and Creases (ATK) **b), Crinkles (L'Garde)**

Fig. 13 Wrinkles, Crinkles, and Creases

For the purpose of this discussion, *wrinkles* are defined as temporary deformations formed in the sail material from the unbalanced distribution of the strains placed on the sheet of material. This is visible in Fig. 13a) as the rolling deformations emanating from the sail sheet edges where the tension forces have been applied. These deformations are temporary, in that if the associated strains were eliminated the wrinkles would disappear. This sheet of material also shows a *crease* horizontally across the material, which could result from folding the sail for packaging before deployment. Creases are permanent deformations, much like what happens to folded paper when making a paper airplane.

Figure 13b) shows a section of sail material with many *crinkles*. Like creases, these are small permanent deformations that can form during the handling of the material, similar to balling up a piece of paper and then laying it out flat. In general, a realistic solar sail will have all three of these surface features, which have the potential for affecting the thrust produced.

Various models have been used to describe these surface effects (Murphey and Murphy [9], Zeiders[10, 11]) some of which use a sinusoid to represent the surface, as illustrated in Fig. 14.

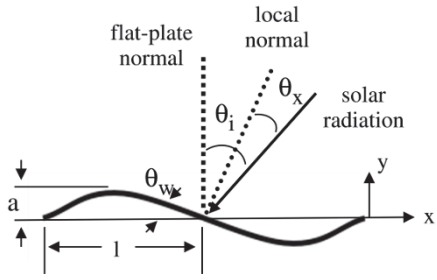


Fig. 14 Surface model

The sinusoid in this model can be described by:

$$y = a\sin\left(\pi\frac{x}{l}\right) \tag{8}$$

where **a** is the amplitude of the wrinkle and ***l*** is the length of one half of the sinusoid. A *wrinkle aspect ratio* can be defined as ***a/l***. As shown in the figure, a wrinkle in the sail surface changes the local surface normal from that of the mean (flat plate) normal, which affects both the magnitude and direction of the resulting force at each location along t6he sinusoid. These forces can be found by computing the *wrinkle angle* (θ_w):

$$\theta_w = \arctan\left(\frac{dy}{dx}\right) = \arctan\left(\frac{a\pi}{l}\cos\left(\pi\frac{x}{l}\right)\right) \tag{9}$$

andthe*local incidence angle* (θ_x), which represents the angle between the local normal along the sine-wave wrinkle and the incoming solar radiation:

$$\theta_x = \theta_i - \theta_w \tag{10}$$

where θ_i is the solar radiation incidence angle to the flat plate sail normal. Note that both the wrinkle and local incidence angles are functions of distance along the wrinkle (**x**) and of the wrinkle aspect ratio.These relationships can be used to derive expressions for the radial (in the direction of the incoming solar radiation along the non-flat wrinkle surface) and tangential (also along the non-flat wrinkle surface) solar radiation pressure forces, and thus the total force over the wrinkled surface. Figure 15[9] plots the (normalized) radial force component, over a range of wrinkle aspect ratios, for different sail sun-incidence angles.

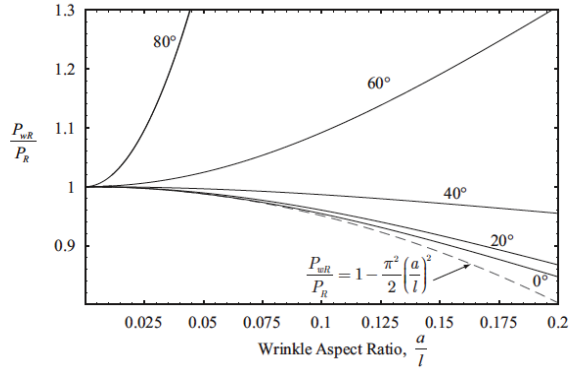


Fig. 15 Effect of Wrinkles on Solar Sail Radial Thrust (Murphey).

It is obvious that wrinkle aspect ratios much over 10% can result in a significant change in thrust than achievable with a perfectly smooth sail. The major difference between the two sail articles shown in Fig. 8 comes from the tensioning of the sail material in the ATK sail to produce a highly specular surface, versus the L'Garde sail which drapes a low tension sail over wire supports (which reduces boom stiffness requirements) creating the more wrinkled "striped" configuration shown.

As can be seen, quantification of these factors for a realistic solar sail is highly design dependent, and requires a significant amount of testing during the development process. However, incorporation of these factors into total solar sail thrust estimation, after the sail is properly characterized by the testing, may be as simple as modifying the sail optical properties or physical size to represent these effects on thrust. One approach would be to include these wrinkle-type considerations, along with the surface roughness and incidence angle dependency described earlier, into a modified realistic specularityterm $\mathbf{S}_{w}(\delta r,\theta)$ for use in realistic sail thrust calculations. Another approach could be to incorporate a "wrinkle factor" (w_f) to the sail effective area to accommodate the "knock-down" of achievable thrust from these effects.

4. Other Non-Ideal Optical Factors

Other characteristics of a realistic solar sail that could affect thrust magnitude and/or direction include changing emissivity (on both sides of the sail) with changing sail temperature, non-perfect and changing Lambertian radiation from the surface, and additional radiation pressure force from a close infrared source (like from the Earth in a LEO orbit). Like the wrinkle factor above, these factors may be incorporated into thrust calculations by using modification of some of the optical characteristics, based on pre-flight sail testing and measurements. Table 2 is an update of Table 1 thatincludes some representative values of optical properties for these realistic sails to the familiar ideal-sail values that have been used for many solar sail studies to date.

B. Realistic Solar Sail Shape Properties

Up to now, it has been assumed that the reflecting surface of the solar sail is essentially a flat plate of known dimensions for which the magnitudes and directions of reflected and emitted energies could be easily understood. Unfortunately, realistic solar sails are not so easy. Depending on the sail configuration, as well as orientation and location in space, both the actual sail dimensions as well as shape may be uncertain. The following sections discuss these factors and their effects on sailcraft thrust.

Table 2 Ideal, Non-ideal, and Realistic Solar Sail Optical Properties

		Ideal Sail	Non-ideal (Flat) Sail	Non-ideal CP1 Sail	Non-ideal Mylar Sail
Solar	t_{rs}	0	0.02	0.002	0.02
	a_{bs}	0	0.1	0.08	0.12
	r_{fs}	1	0.88	0.918	0.86
	s_s	1	0.94	0.978	0.55
Infrared	t_{rir}	0	0	0	0
	a_{bir}	0	ε	ε	ε
	r_{firf}	1	0.97	0.98	0.97
	r_{firb}	1	0.7	0.731	0.7
	S_{irf}	1	0.98	0.9898	0.97
	S_{irb}	1	0.8	0.9617	0.703
	ε_f	0	0.05	0.02	0.03
	ε_b	0	0.55	0.269	0.4
	B_f	2/3	0.79	0.79	0.89
	B_b	2/3	0.55	0.55	0.65

1. Sail Billow

Depending on the sail geometry and support configuration, the surface of the sail may take on a billowed shape when exposed to solar radiation pressure in space. Figure 16 illustrates a billowed square sail made up of four triangular panels (quadrants) using a three-point support configuration.

Due to the non-ideal optical properties, as well as the shape of the sail and other realistic factors, the resultant solar radiation pressure force on each of the sail quadrants (depicted as the arrows in the figure) may be different in both magnitude and direction. For illustration, the dotted lines in the figure represent the normal direction for the equivalent flat plate sail (quadrant). The uncertainty and magnitude of this effect on both sail thrust and attitude control may be a major factor in realistic sailcraft design and operation.

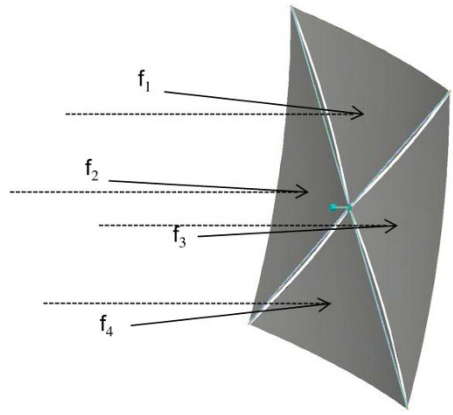


Fig. 16 Solar Sail Billow

2. Sail Droop

Another characteristic of a realistic, gossamer solar sail may be an offset of the sail quadrant below the "equivalent flat plane" by a bending of the sail booms due to the solar radiation force on the sail material. This is indicated in Fig. 17 by the angle θ_d which is called the *droop angle*.

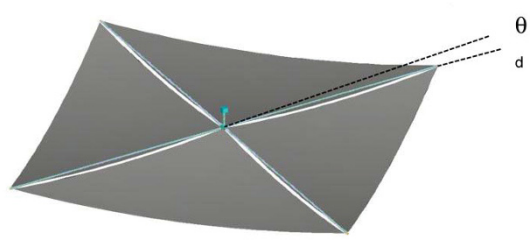


Fig. 17 Sail Droop Angle

Other realistic sail shape considerations include non-symmetry characteristics of the sail material (which might also include the aforementioned optical properties, as well as mass properties) which could include differential areas of the sail sections, as well as different billows and droops. Depending on the structural construction of the sail, all of these factors may affect both the magnitude and direction of the resulting sailcraft thrust, as well as center-of-pressure ($\mathbf{C_p}$) and center of mass ($\mathbf{C_m}$) differentials and associated attitude control uncertainties.

3. Computational Modeling

Believing that sail shape has a significant effect on thrust (and sailcraft attitude control), several approaches have been investigated for estimating the in-space shape of a realistic solar sail [12-15]. Since ground testing may be impossible for useable realistic sail sizes of 100 meters on a side and larger, sail structure and membrane models must be developed and somehow validated [16-18]. Part of the difficulty in this, is that any testing on the ground will subject the sail to gravitational forces orders of magnitude higher than solar radiation will impart. Somehow, the combination of tests under these conditions, and the associated models, must be related to the situation the sail will see in orbit.

Currently, two methods of modeling solar sail shape are predominantly used. The first is analytical, using geometric representations and relationships. In their papers [14, 19], Rios-Reyes and Scheeres use a geometric sail shape to derive a number of coefficients that can be used to analytically determine the forces (and moments). Figure 18 defines the geometry used to describe a single sail quadrant as a section of an oblique cone.

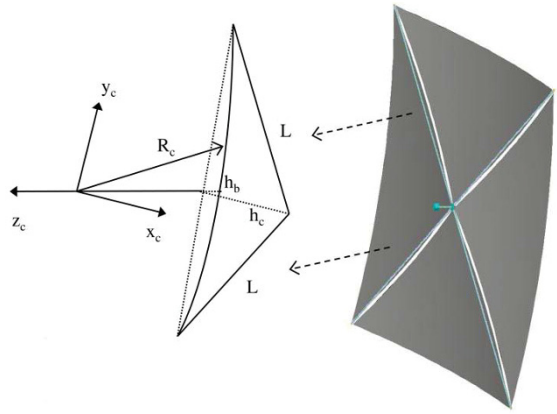


Fig. 18 Geometric Sail Shape (derived from Rios-Reyes)

Some of the pertinent characteristics illustrated include the *cone section length* (**L**, equivalent to the sail boom length), the *cone radius of curvature* ($\mathbf{R_c}$), the *cone height* ($\mathbf{h_c}$), and the *billow depth/height* ($\mathbf{h_b}$). From this basic geometry, the normals for the surface can be described and coefficients computed, and used in a general formula to determine thrust and moments for a sail of that particular shape.

The most common way of representing the effects of sail shape on resultant thrust is to use finite element modeling (FEM) techniques. In this method, the sail material is broken into many small area elements, and

analyzed for reactions to applied forces. Figure 19[16] illustrates an FEM model for one of the 10-meter test articles used in the ISPT development effort.

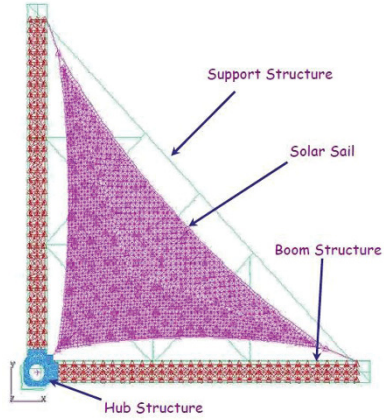


Fig. 19 FEM Sail Model (Taleghani)

Actual modeling of a solar sail is made difficult due to its thin-film membrane characteristic and inherently non-linear behavior when subjected to forces and stresses, especially when including material nonlinearities due to wrinkles and creases and such. Nonetheless, the advent of solar sails and sun-shields has advanced the field of thin-membrane and shell analysis techniques in recent years, and the method has been applied and accepted as giving useable information about solar sail shape for preliminary analysis and mission planning. Due to its large and gossamer nature, the testing of solar sails prior to flight will be quite limited, so as in other solar sail development areas, FEM modeling will still be only an estimation. Though it has the best chance to represent the sail, including many of its non-ideal characteristics such as unsymmetrical shape and non-homogeneous properties, as well as its ability to model dynamic reactions to forces, solar sail and mission designers will still have to include healthy margins to FEM predictions to offset these inherent uncertainties.

For the purposes of this paper, a mixed method of representing solar sail shape properties is used within the Princeton Satellite Systems (PSS) Solar Sail Module software (Appendix A) [20] that is used for the analysis that follows. In creating the solar sail model, a geometric sail shape can be specified, after which an FEM-like routine is used to divide the surface into discrete sections. The force (vector) for each of the discrete sections is calculated, and the discrete force elements added to calculate a resultant force vector. This is illustrated in Fig.20, which is an output taken directly from the PSS simulation software.

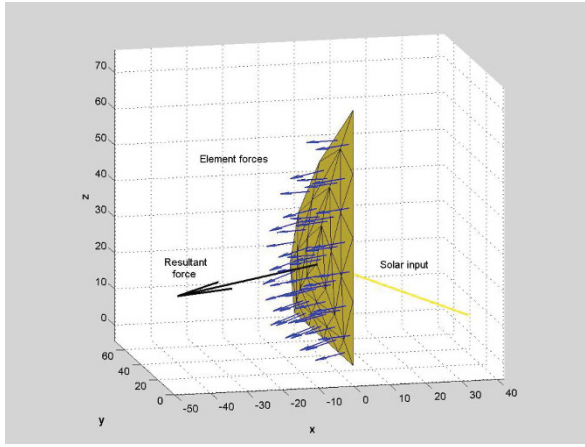


Fig. 20 PSS Model Sail Quadrant Forces

For the purposes of this paper, a simple conic section is used to represent the shape of a quadrant. To describe the amount of billow (or curvature of the conic section) the **billow height** ($\mathbf{h_b}$) is defined as the amount of out-of-plane displacement over the length of the sail quadrant (between the booms). For instance, a 0.1 billow height for a 100 x 100 meter square sail would have a maximum billow of 10 meters below the equivalent "flat plate" plane defined by straight sail booms. Figure 21 shows three realistic sails with different billows.

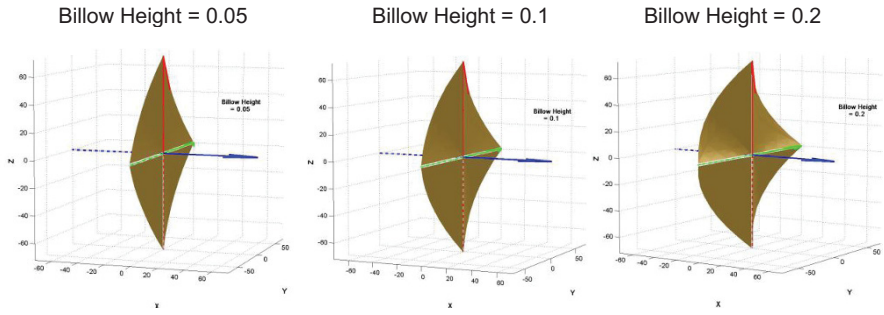


Fig. 21 Sail Billow Heights

Figure 22 shows the resulting thrust over sun-incidence angles for the three sails, along with the thrust curve for a flat, non-ideal reflector sail of the same sail area. It is clear that the magnitude of the thrust produced from a solar sail can be significantly affected by the amount of billow that is allowed.

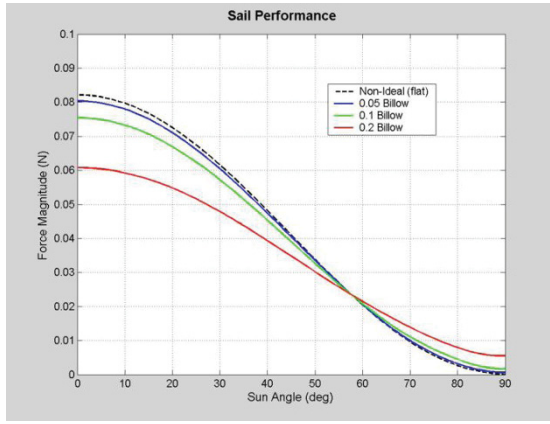


Fig. 22 Billow Thrust vs. Sun-Incidence Angle (For 10,000 m^2 sails @ 1 A.U.)

V. Solar Sailcraft Characteristic Performance Comparison

In this section, four sailcraft models are defined and their performance, in terms of thrust generated as well as ability to modify a given orbit, are shown. Table 3 provides a listing of the pertinent characteristics of the four sailcraft configurations (one ideal, one flat non-ideal, and our 2 realistic models) that are used for the following performance comparisons. Sailcraft characteristics, given in the left-hand column, include sailcraft mass ($\mathbf{m_{sc}}$), sail (propulsion system) mass ($\mathbf{m_s}$), payload mass (spacecraft bus and everything other than the sail system) ($\mathbf{m_p}$), and sail area ($\mathbf{A_s}$).

Some of the obvious differences in the model optical properties include a higher reflectivity for the billowed sail and a lower reflectivity for the striped sail, a higher specularity for the billowed sail and a lower specularity for the striped sail, increased Lambertian coefficients for the striped sail, and lower emissivities for both the shaped sails, with the billowed sail having the lowest values. Since this analysis is looking at the heliocentric case, IR optical properties are not used, since the effect on thrust at the operational location around 1 A.U. is negligible. Though the shaped sails (billowed and striped) both have the same billow and droop angles, the surface geometries are different, as explained above.

Each sailcraft has a fixed total mass of 300 kg, and a total *flat* sail area of 10,000 m^2. Projected sail area depends on sail shape and sun-incidence angle, and computation of the effective sail area in the PSS software also incorporates the effects of billow and droop. For each model, the sail surface is divided into many smaller

areas and the solar radiation pressure force for each area calculated (using average solar luminosity values) to allow the combined effects of shape and sail attitude to be realized. These force components are vectorally added to produce the total sail force and direction. The PSS software then uses these forces to continuously update the orbit of each of the test sailcraft.

Table 3 Ideal, Non-ideal, and Realistic Solar Sail Model Characteristics

	Ideal sail	Non-ideal (flat) sail	Realistic (billowed) sail	Realistic (striped) sail
m_{sc}	300 kg	300 kg	300 kg	300 kg
m_s	60 kg	60 kg	60 kg	60 kg
m_p	240 kg	240 kg	240 kg	240 kg
A_{sail}	10,000 m^2	10,000 m^2	10,000 m^2	10,000 m^2
Billow (θ_b)	0	0	10^o	10^o
Droop (θ_d)	0	0	5^o	5^o
t_r	0	0.02	0.002	0.02
a_b	0	0.1	0.08	0.12
r_f	1	0.88	0.918	0.86
s_{act}	1	0.94	0.978	0.55
ε_f	0	0.05	0.02	0.03
ε_b	0	0.55	0.269	0.4
B_f	2/3	0.79	0.79	0.89
B_b	2/3	0.55	0.55	0.65

A. Thrust Performance Comparison

Figure 23 shows the calculated thrust performance of each of the four configurations described above, over sun-incidence angles, at 1 A.U., using the PSS software. Table 4 provides the numerical values obtained from the PSS software for the sail thrust levels at zero and 35 degree incidence angles. The 35 degree values indicate the difference in thrust between the sail configurations at the incidence angle at which operation of the sails in the following simulations occur.

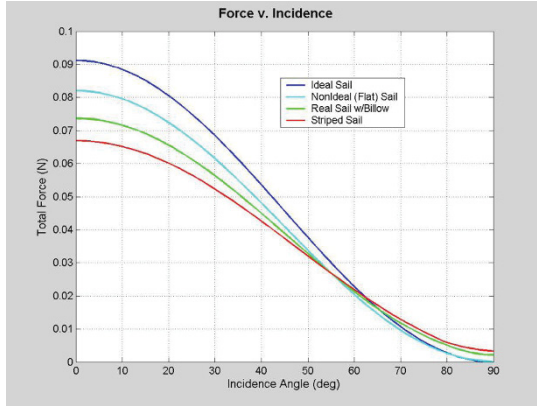


Fig. 23 Perfect, Non-perfect and Realistic Solar Sail Forces Vs. Sun-Incidence Angle (For 100 x 100 meter sails @ 1 A.U., based on PSS models)

Table 4 Ideal, Non-ideal, and Realistic Solar Sail Model Thrust Levels

	Ideal sail	Non-ideal(flat) sail	Realistic (billowed) sail	Realistic (striped) sail
0^o (θ_i)	0.091	0.082	0.074	0.067
35^o (θ_i)	0.063	0.056	0.053	0.048

Figure 24 shows the cone angle vs. incidence angle curves for each of the sailcraft models. Notice that the maximum achievable cone angle (direction of the thrust vector) is much more limited in the case of the realistic sails. It is the combination of both poor reflectivity *and* specularity that affects the cone angle performance of the striped sail model so significantly.

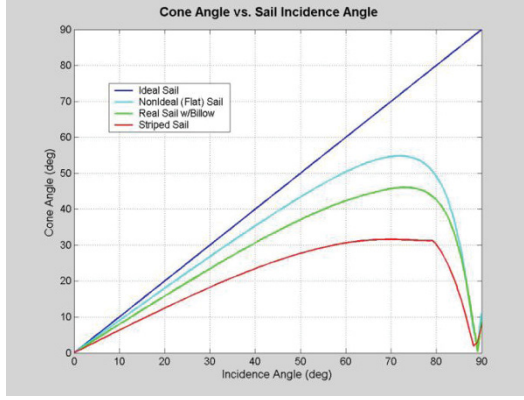


Fig. 24 Perfect, Non-perfect and Realistic Sail Cone Angles Vs. Sun-Incidence Angle (For 100 x 100 meter sails @ 1 A.U., based on PSS models)

B. Orbital Performance Comparison

To demonstrate operational performance differences for the sail models, two different simulations are run. The reason for using two different simulations becomes clear after the first one is completed.In the first simulation, each sail follows the optimum sail attitude profile to decrease the orbital radius of the 1 A.U. circular starting heliocentric orbit (basically a 35 deg incidence angle to the Sun where the force vector points away from the sailcraft velocity vector). Figure 25 shows the performance of each sail, in terms of distance travelled around the Sun and change in orbital radius, for a simulation lasting 365 days. Though expected, the ideal sail performance is seen to be significantly better than the other more realistic sail models.

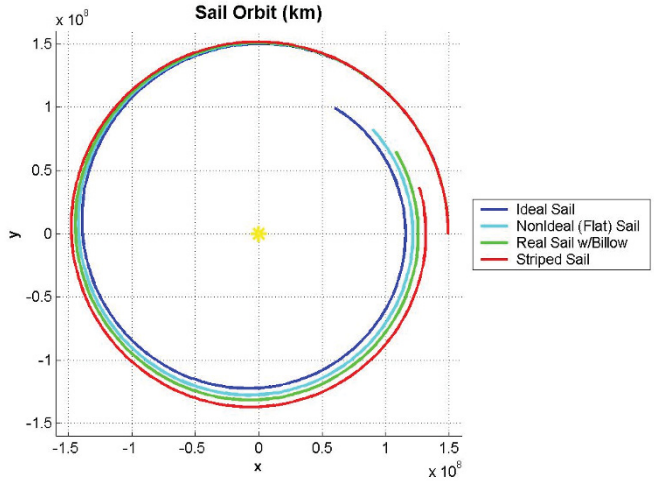


Fig. 25 Heliocentric Orbital Semi-Major Axis Decrease Performance Comparison (For 100 x 100 meter sails @ 1 A.U., based on PSS models)

Figure 26 plots the magnitude of the reduction in orbit semi-major axis for each of the sail models over the 365 day simulation. This chart clearly shows the significant difference between an ideal sail and the more realistic sail models, and also reflects the not-insignificant difference in performance due to shape effects.

Because some of the difference in performance in the above simulation can be attributed to the resulting changes in distance from the Sun, which favors the higher-performing sail models, a second simulation is run where the sailcraft maneuver while maintaining a constant distance from the Sun. In this case, each sail follows the optimum sail attitude profile to increase the inclination of the orbit (basically a 35 deg incidence angle to the Sun where the force vector now points perpendicular to the sailcraft velocity vector, flipping twice each orbit to

continue to “crank” the inclination in the same direction). To expedite the simulation results, the sailcraft start from a 0.5 A.U. circular heliocentric orbit for this case.

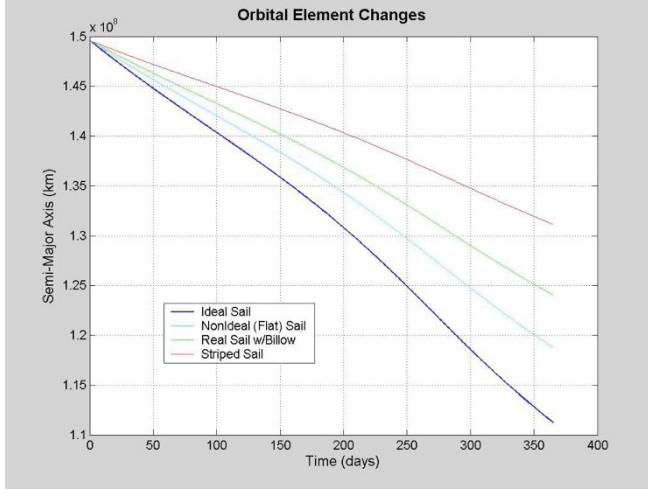


Fig. 26 Orbit Semi-Major Axis Decrease Performance Comparison (For 100 x 100 meter sails @ 1 A.U., based on PSS models)

Figure 27 shows the performance of each sail, in terms of inclination change, again for a simulation lasting 365 days. The bottom plot in Fig. 27 is a blow-up of a portion of the middle plot showing the end positions of the sailcraft after the 365 day simulation. Figure 28 plots the magnitude of this change for each of the sail models over the simulation. This chart clearly shows the significant difference between an ideal sail and the more realistic sail models, and also reflects the not-insignificant difference in performance due to shape effects.

VI. Conclusion

The major result from this study is a demonstration that a “realistic” solar sail will perform differently than an ideal or even a non-ideal sail that simply incorporates realistic optical properties. The effects of factors such as surface properties and sail shape are not insignificant and must be taken into account for solar sail development and planning purposes. Models, such as in the PSS Solar Sail Module software, must incorporate these factors as part of their simulation capability, so they can provide more accurate estimates of sailcraft performance. This is especially important when considering that many of the envisioned solar sail missions have very long operational lifetimes, during which performance differences could propagate to significance. This also demonstrates the importance of a sail validation flight, to provide the only information that can allow planners to update their models to gain confidence in their results – actual in-flight measurements.

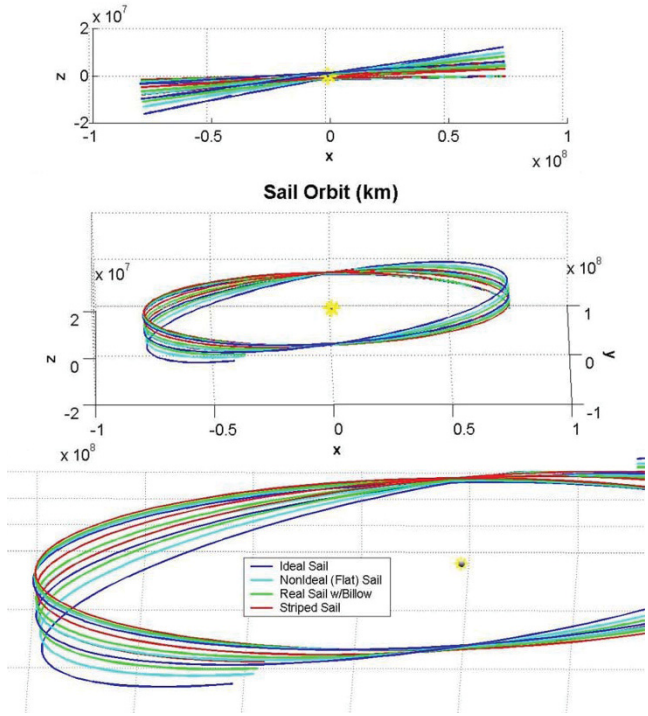


Fig. 27 Heliocentric Orbital Inclination Increase Performance Comparison (For 100 x 100 meter sails @ 0.5 A.U., based on PSS models)

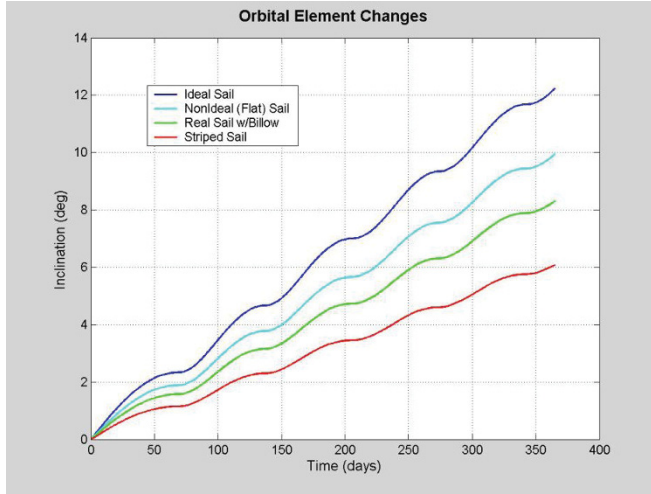


Fig. 28 Inclination Change Performance Comparison (For 100 x 100 meter sails @ 0.5 A.U., based on PSS models)

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Appendix A. The PSS Solar Sail Module

A MATLAB-based software tool, called the Solar Sail Module (SSM), was developed by Princeton Satellite Systems (PSS) for the NASA In-Space Propulsion Technology (ISPT) Solar Sail Program to provide solar sail analysis, design and simulation capabilities. SSM is based upon, and takes advantage of, the capabilities in Princeton Satellite Systems' existing Spacecraft Control Toolbox (SCT) that has been commercially available since 1995. SCT is composed of the following modules:

- **Core**: Orbit and attitude dynamics, computer-aided modeling, sensor and actuators
- **Attitude Control**: Complete attitude control design examples
- **Orbit**: Precision orbit propagator, mission planning, formationflying
- **Estimation**: Attitude and orbit estimators
- **Propulsion**: Propulsion system modeling
- **Thermal**: Thermal system modeling
- **Link**: Link and radar analysis
- **Spin Axis Attitude Determination**: Attitude determination with horizon and sun sensors
- **Formation Flying**: Control and estimation of satellite formations

The PSS Solar Sail Module is based on functions from several of these modules with the addition of numerous solar sail specific functions and scripts. The SSM is self-contained and only requires basic MATLAB 5.2 or later. Simulink is not used. It has a number of interactive Graphical User Interfaces but the core simulation capability is based on MATLAB scripts. The Solar Sail Module includes all source code (scripts) and no dll or p-code files are used. All functions follow a consistent format, have many comments, and can be easily customized by the user.

Simulations are script-based. MATLAB's ode113, a variable order, variable step size integrator, is used to propagate the kinematical and dynamical equations. The right-hand-side function called by ode113 is created by the user in a standard format script. This script can be configured to use a selected disturbance model, attitude dynamics model and orbital dynamics model. Several sail-specific attitude dynamics models are available and configurable. The controller is called from the simulation script. Controllers are implemented digitally and are not part of the simulation right-hand-side.

The software includes functions for multi-body dynamic modeling, attitude control systems design, thrust vector control design, orbit analysis, solar sail mission analysis, thermal analysis and power subsystem analysis. Sailcraft models can be created, analyzed and simulated without the need to deal with any other software tools. The software couples attitude and orbit dynamics into the same simulations. The user can choose from several different attitude dynamics models, including specially-developed multibody models for moving mass and gimbaled boom control systems, and from several orbit models including point mass, n-body and non-spherical earth.

Sailcraft models are created using a graphical computer-aided design package that is included in the software. The component information defined in the computer aided design (CAD) package is used to generate

mass properties and for all disturbance (force) calculations. Component-level data includes optical and thermal properties for the surfaces; mass, center-of-mass, and inertia; and magnetic dipoles, RF sources, etc. Additional information can also be stored with each component so that the CAD file serves as a database for all model data.

A disturbance modeling package is included which can be used independently or integrated with the solar sail simulations. Disturbances on a sailcraft include:

- Solar pressure force and torque
- Earth radiation force and torque
- Earth albedo force and torque
- Aerodynamic force and torque
- Gravity gradient torque
- Thermal emission force and torque
- Magnetic torque
- Radio frequency emission force and torque
- Outgassing force and torque

This list includes some disturbances only applicable to a sailcraft in a planetary orbit, and does not include disturbances caused by actuators on the spacecraft.

The PSS Solar Sail Module includes a number of scripts associated with developing a sailcraft model for use within the simulations. The user creates a separate script that provides the information input to the Module scripts. Within the user script, the Module script **BuildCADModel.m** sets up the sailcraft model data structure and collects all of this data into a saveable file. After initializing **BuildCADModel**, the user uses additional SSM scripts, such as **CreateBody**, and **CreateComponent** to define the parts of the sailcraft that will be incorporated into the overall model. These scripts define the properties of each component, including shape, size and mass, as well as important solar sail properties such as optical and thermal coefficients. The Module scripts include many generic shapes, such as “box”, “sphere” and “cylinder” that make inputting components easier. Figure A-1 shows the component model for a triangular sail quadrant. In this example, the quadrant includes a defined amount of billow, and the surface is broken up into a number of “triangles” that can be used as individual sub-components for computing radiation, aerodynamic, and even magnetic forces on each element.

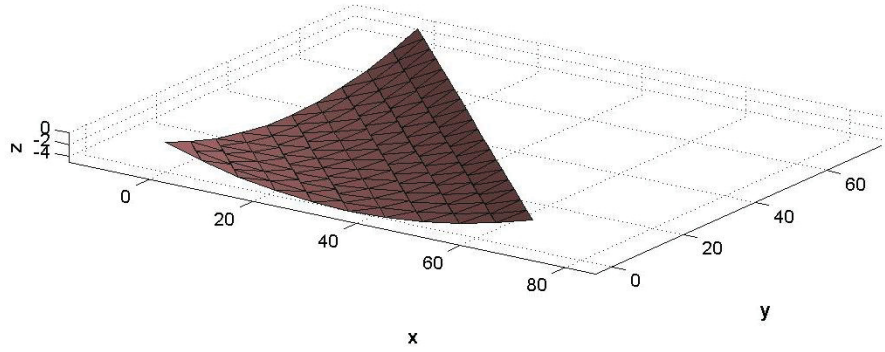

Fig. A-1 Solar Sail Quadrant Model

Of equal importance, these “create” scripts define the orientation and placement of each component, with respect to the body as a whole. When brought into the overall sailcraft model, using the “add” command, **BuildCADModel** uses this information to rotate and displace the component into its proper place. For instance, the quadrant above would need to be rotated such that the side visible in the figure (associated with the reflective side of the sail) would have the axis normal to this surface pointed in the “X” direction, and also rotated and displaced to its proper quadrant position.

At the same time it adds the component, **BuildCADModel** computes the component mass properties, and updates the overall sailcraft properties (total mass, total inertia, etc.). When completed, the user can save the overall model structure into a separate file that can be called for use in simulations, and can use several Module display and analysis scripts to analyze the final sailcraft model. Figure A-2 shows the 100 x 100 meter square, non-ideal sail with billow model used for many of the simulations conducted in this dissertation.

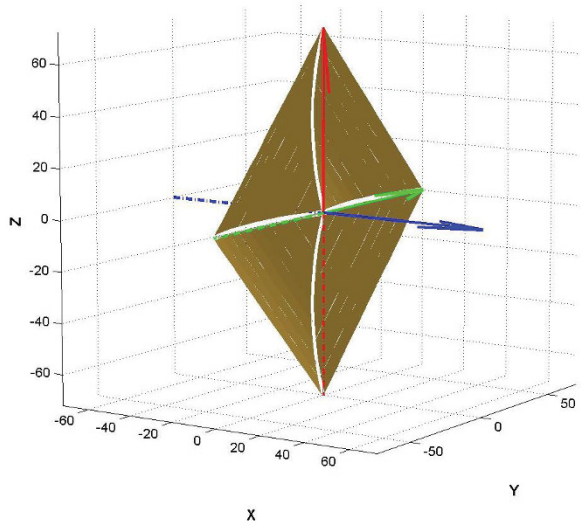

Fig. A-2 SolarSailcraft Model

Once the model is created, it can be used in simulations. Once again, the user creates a stand-alone script that calls and provides information to a number of Module scripts. The most used of these is **FSailCombined.m**. This script (function) creates the data structure used for each step in the simulation. As inputs, the user specifies the Sun and planet (if applicable) ephmerides to be used and the orbit gravity model, the environment the sail is to operate in (in terms of planetary atmospheric density, radiation, magnetic fields, etc.), the attitude dynamics and guidance for the sailcraft – including initial orbital elements, and the sail disturbances (forces) to be considered during the simulation.

As in **BuildCADModel**, several existing Solar Sail Module scripts can be used within **FSailCombined** to incorporate standard quantities easily. The most demanding for the user is specification of the sailcraft dynamics (multiple moving bodies, such as moving masses or tip vanes for attitude control can be accommodated) and specification of the sailcraft guidance. The guidance can be simply a result of the reaction of the system to the sailcraft dynamics (moving masses, etc. – set up by a user-defined control logic) based on calculations each step of the iteration, or simply a specification of the sail attitude profile over time. Of particular interest to the user, the **SailDisturbance.m** script (function) is where the forces and torques on the sailcraft are computed each iteration. When specified, and where applicable, these calculations are performed on each "triangle" of the components created in the sailcraft CAD model. The **SolarPressureForce** script (function) is called upon to calculate the combined thermal and optical forces on the sail. If in a planetary orbit (with specified characteristics) aerodynamic, albedo, and even magnetic field forces can be calculated.

Once the required information is input, the simulation can be run using an iterative differential equation integration method. One of the standard MATLAB methods is **ode113**, which calls the values and functions within **FSailCombined** to compute the new sailcraft state at each specified time step. These state parameters at each time step are saved within **FSailCombined** and can be retrieved, graphed, and/or plotted by the user-created simulation script. Many of these products are visible throughout this paper.

Direct Thrust Efficiency for the L'Garde Sail Surface with a Linear Reflectivity Model

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A measure of propulsive efficacy for the grooved L'Garde solar sail surface — the photonic thrust efficiency in the context of an equivalent smooth (i.e., not grooved) sheet — is numerically assessed for thrust in the surface normal direction, and the dependence of this metric on the illumination incidence angles (in the directions along and across the grooves) is found to have some remarkable counter-intuitive characteristics. The study is based on a simple but powerful reflectivity model which, despite being a straightforward approximation to the full optical formulation, has received little attention in the past. This model, referred to as "linear" thrust model, simplifies analysis at the cost of only a minor loss of detail shown to be insignificant in the context of other common approximations. A result of this simplification is rigorous proof that the L'Garde sail surface groove contour shapes well approximate the classic catenary curve — the hyperbolic cosine function. The insight here offered contributes both to a practical appreciation of photonic thrust models and to the better understanding of some of the thrust characteristics of the L'Garde solar sail.

Nomenclature

A	=	Parameter characterizing trough contour shape ($2 + \eta t/\eta n$).
B_f, B_b	=	Sail sheet front and back face non-Lambertian radiation coefficients.
c_1, c_2, ...	=	Integration constants.
l	=	Length, distance.
N	=	Film tension [force/length].
P, p	=	Photon flow radiation pressure (P=4.563 μ Pa at 1 AU) and thrust pressure exerted on a surface.
P_{max}	=	Maximum thrust pressure on an ideally reflecting surface; =2P=9.126 μ Pa at 1 AU.
R	=	Radius of curvature.

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r, s	=	Fraction of irradiation reflected, and the fraction thereof reflected specularly.
s	=	Trough contour arc length, from the shape apex.
t_f	=	Film thickness.
u,v,w	=	Trough contour frame of reference aligned with the direction of illumination.
$u(v)$	=	Trough contour shape.
0	=	Contour shape apex (illuminated in the normal direction).
n, t	=	Surface normal and tangent directions.
s	=	Arc length along trough contour; sagging film.
w	=	Trough width (distance between sheet support cords).

Symbols

$\ldots'$	=	"Prime:" differentiation with respect to a Cartesian variable.
α	=	Irradiation angular offset from the surface normal.
α_z	=	Contour slope angle in the trough coordinate system.
$\dot{\ldots}$	=	"Dot:" differentiation with respect to the contour arc length s.
η_n, η_t	=	Photonic thrust coefficients in the film surface normal and tangent directions.
κ	=	Curvature ($\beta = 1/R$).
σ	=	Film skin stress.
$\varepsilon_f, \varepsilon_b$	=	Front and back film surface emissivities.
ξ, ζ	=	Sail surface frame with ξ perpendicular, ζ parallel to the cords.

I. Introduction

L'Garde, Inc., of Tustin, California, the USA, is currently fabricating Sunjammer, the newest member of the family of its signature solar sails. This structure, according to the design concept proposed and structural analysis performed in 2001 and 2002 by Greschik [1, 2, 3], is one of three "crosscutting flight demonstrations" being prepared for space flight under special support by NASA[4]. Launch is expected in late 2014.

Figure 1: Grooved sail surface [5, Fig. 14].

The characteristically grooved surface of this architecture, Fig. 1, is unique. The geometric complexity resulting from the very presence of the grooves creates reflective and thrust conditions markedly different from other sails which all employ smooth surface shapes. Consequently, thrust analysis — and the control of propulsion and attitude — for the L'Garde sail is even less trivial than for alternative designs. The added effect of coupled structural deformations, numerically addressed by Greschik for sun-facing attitudes [6, 1, 2] and later termed by Derbes [7] "photoflexibility" in a context combined with attitude stability considerations, makes the analysis problem even more difficult.

A straightforward answer to this modeling challenge is to "back away" from predictive analysis: to rely on the latter only for guidance for the design of in-flight performance calibration and responsive control techniques. The emphasis of attitude control and propulsion design has to shift, as previously said [8], *"... to the development of very robust and flexible navigation and control strategies, in order to permit the progressive and adaptive correction of a variety of unforseen differences between modeling and reality."* This shift of focus is not new. For example, in response of the near-impossible challenge to reliably predict performance metrics, a mathematical formulation for in-flight sail calibration has already been given by Rios-Reyes and Scheeres [9], albeit in a structurally simplified context (without accounting for structural deformations).

However, despite the trend to reduce direct reliance on thrust and structural analysis, the latter retains a critical role in solar sail structural and mission design. Only analysis can provide detailed insight into the coupled structural, attitude stability, propulsion, and control problems that must be answered for hardware to be robust. It is in this spirit that the results described in the present paper are put forth.

We begin with a discussion of typical reflective thrust models. The approximations behind some common approaches are highlighted, and the case is made for a straightforward derivative of the full optical model which has received little attention in the past. This model is [8] named the “linear photonic thrust model” after the linear relationship between its two material parameters.

The L’Garde solar sail architecture is discussed next; its history and key mechanical features are briefly reviewed. The power of the linear photonic thrust model is illustrated by rigorously showing that the L’Garde sail groove contours approximate the hyperbolic cosine shape.

The linear thrust model is finally numerically applied to the L’Garde sail grooves to obtain on the global level (for an equivalent hypothetical smooth surface replacing adjacent grooves) a reflective efficiency coefficient for normal thrust. The dependence of this coefficient on the direction of illumination is investigated for an example scenario, and some remarkable observations are made.

II. Solar Radiation Force Modeling and the Linear Thrust Model

SOLAR sail engineering relies on more profound simplifications than the engineering of perhaps any other space structure. The assessment of sail structural response is out of the reliable reach of most engineering tools [10], and sail optical performance is grossly affected by conditions [11, 12, 13] that are rarely even explicitly highlighted, let alone accounted for, in most structural and trajectory studies in the literature. Against the backdrop of such approximations, the approach to choose a photonic thrust model primarily to streamline the calculations can be deemed acceptable.

Nevertheless, the formulation of photonic thrust is rather straightforward in the context of radiation physics and some (simplified) film properties [14] — the nominal optical force model. Most common propulsive force approximations are direct derivatives of this formulation.

A. The Nominal Optical Force Model and Some of its Derivatives

The modeling of reflectance on the local scale is generally based on the assumption of a symmetric spatial pattern for the illuminating, reflected, and absorbed-emitted light about the plane subtended by the surface normal and the direction of incidence. As a result of this assumption — which is appropriate for optically isotropic surfaces — the photonic thrust will develop in the plane of symmetry and can, therefore, be described in two dimensions within that plane. Accordingly, the thrust pressure (force divided by area) exerted on the surface can be subdivided into surface-normal and -tangential components p_n

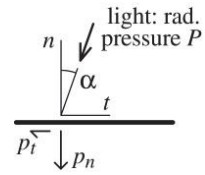


Figure 2. Surface normal and in-plane radiation pressure components p_n and p_t

and p_t as indicated in Fig. 2, where n and t denote the associated frame of reference and the illumination incidence angle is α. Thus any photonic thrust model is but a particular form of the functions

$$p_n = p_n(P,\alpha) \quad \text{Eq. 1}$$

$$p_t = p_t(P,\alpha) \quad \text{Eq. 2}$$

wherein the radiation pressure of the photon flow is denoted with P (4.563 μ N/m^2 at 1 AU from the Sun) and the functions incorporate all film reflectance parameters. In general, functions Eqs. $p_n = p_n(P,\alpha)$ Eq. 1 and 2 also depend on the optical parameters of the illumination beyond the radiation pressure (intensity) P, such as on wavelength/frequency spectrum, polarization, coherence. However, such details are mute in the context of the present study where solar illumination alone is considered, with its optical properties given.

The nominal optical force formulation derives the pressure with simple geometric assumptions from a few optical surface parameters. Often, this rule is cited in the form used by Write [15, Appendix A, Eqs. A.6, A.7] and McInnes [14, Section 2.6.1, Eqs. 2.57a, 2.57b]. Cast in the form of Eqs. 1 and 2, these expressions are

$$p_n = P\left[(1+rs)\cos^2\alpha + B_f(1-s)r\cos\alpha + (1-r)\frac{\varepsilon_f B_f - \varepsilon_b B_b}{\varepsilon_f + \varepsilon_b}\cos\alpha\right] \quad \text{Eq. 3}$$

$$p_t = P\left[(1-rs)\cos\alpha\sin\alpha\right] \quad \text{Eq. 4}$$

with r the reflection coefficient (the fraction of all incident photons reflected, as opposed to absorbed), s the fraction thereof reflected specularly, ε_f and ε_b the film front and back side emissivities, and B_f and B_b coefficients quantifying the extent to which these surfaces differ from Lambertian radiators (B_f and B_b would be 2/3 for a Lambertian surface which appears to be equally bright from any viewing direction).

This law is also used in calculations other than solar sail design. An example is the study of the movement of dust, debris, comets and other small objects in the solar system [16, 17].

An alternative form of the nominal law is

$$p_n = 2P\cos\alpha\ (a_1\cos\alpha + a_2) \quad \text{Eq. 5}$$

$$p_t = 2P\cos\alpha\ \ (a_3\sin\alpha) \quad \text{Eq. 6}$$

in which the optical parameters of Eqs. 3 and 4 have been combined in the resultant coefficients a_1 through a_3.

A recent example for the use of full optical model is an orbital study [18] by Quanta and Mengali. Another one is the work [19, 20] by Rios-Reyes and Scheeres in which a tensor formulation is developed to map irradiation to thrust in the context of an entire sailcraft — for any constant (non-deforming) shape for the latter.

Simplifying the nominal law to the extreme with $r = s = 1$, the model for perfect specular reflectance results:

$$p_n = 2P\cos^2\alpha \qquad \text{Eq. 7}$$

$$p_t = 0 \qquad \text{Eq. 8}$$

Due to its obviously simplistic nature, this formulation is rarely used. Two examples are a recent paper [21, Eq. 2] on the UltraSail architecture and a study of a spinning membrane disk under photonic pressure [22, Eqs. 12 and 13].

An approximation slightly more refined than the ideal specular model, however, is very common. This is the normal thrust approximation by which thrust in the surface normal direction only is accounted for, and the non-ideal reflective efficiency of the surface is characterized by a single parameter $0<\eta<2$.

The normal thrust model corresponds to keeping only the first term of Eq. 3, and altogether dropping the other terms there as well as Eq. $p_t = P\left[(1-rs)\cos\alpha\sin\alpha\right]$ Eq. 4.

$$p_n = P\,(1+rs)\cos^2\alpha \;=\; \eta\cos^2\alpha \qquad \text{Eq. 9}$$

$$p_t = 0 \qquad \text{Eq. 10}$$

This model is widely used. Examples include the recent study of the mm-scale “Sprite” sailcraft by Atchinson [23] (where the thrust coefficient used corresponds to $\eta = 1.7$ in the present nomenclature); Bolle’s work on sail attitude control [24] (with $\eta = 1.8$), and Circi’s study of the attitude control of regular spacecraft augmented with sail elements [25].

In the first part of his 2004 review of attitude control and dynamics Wie [26] reviews, without references, derivatives of the nominal optical formulation. In addition to the models already touched upon above, he mentions in Eq. 4 a model which can be derived from the nominal law by dropping the diffuse reflectance and thermal radiation terms. This model is herein termed the linear thrust model and is described in the following subsection.

However, before moving on, note that the photonic thrust assumptions are not always consistently used and communicated in the literature. One example is Burton’s paper [27] introducing the UltraSail architecture: four

radial ribbon arms with tip satellites spinning about a hub. In Eqs. 2 and 3 this publication cites the nominal optical force model and consequently applies it to estimate the load on a ribbon arm. However, a few paragraphs later, Section III already opens with the statement that *"... the normal of the sail points in the direction of acceleration"*, which implies the surface normal model — a model also used later in the same paper, in Eq. 9.

Another example for underdocumented approximations is Ewing's 2007 study [28] on the L'Garde solar sail thrust profile, where the type of propulsion model is not revealed. Yet another example is Derbes' paper [7] on photonic thrust itself, in which some key assumptions behind the nominal optical model are reported to fail, but enough data is provided neither to justify the alternative modeling framework there proposed, nor to enable the reader to critically review the quantitative results presented.

1. The Linear Photonic Thrust Model

Yet another derivative of the nominal optical law is obtained if, from Eqs. and the terms related to diffuse reflectance and thermal radiation are removed. The resulting formulation is

$$p_n = \eta_n \, P \cos^2 \alpha \qquad \text{Eq. 11}$$

$$p_t = \eta_t \, P \cos \alpha \sin \alpha \qquad \text{Eq. 12}$$

with

$$\eta_n = (1 + rs) \qquad \text{Eq. 13}$$

$$\eta_t = (1 - rs) \qquad \text{Eq. 14}$$

These equations capture both in-plane and normal thrust in an attractively simple manner. Indeed, the two material coefficients involved, η_n and η_t, are linearly interdependent

$$\eta_t = 2 - \eta_n \qquad \text{Eq. 15}$$

so the formulation, in fact, is a single-parameter approximation. This model can be [8] called "linear" in reference to the linearity of Eq. 15.

However, the linear model is attractive not only because of its simple form. It turns out to be also a very good quantitative approximation of the full optical formulation because the terms of the latter ignore contribute little to the photonic thrust. This can be seen by substituting realistic values for the film optical parameters in Eqs. 3 and 4, such as those presented by Wright [15, p.228] and here listed in the first data row of Table Table 1.

	Optical surface parameters						Resulting coefficients to		
	r	s	B_f	B_b	ε_f	ε_b	$\cos^2\alpha$	$\cos\alpha$	$\cos\alpha\sin\alpha$
as cited by Wright [15]	.88	.94	.79	.55	.05	.55	1.827	-0.011	0.173
Ref [15] + reduced reflectance	.70	.94	.79	.55	.05	.55	1.658	-0.098	0.342
Ref [15] + reduced spec. frac.	.88	.75	.79	.55	.05	.55	1.704	0.086	0.296
Ref [15] + reduced refl. & sp. frac.	.70	.75	.79	.55	.05	.55	1.560	-0.021	0.440
Ref [15] + degraded front s. emiss.	.88	.94	.79	.55	.10	.55	1.827	0.000	0.173
Ref [15] + all above degradations	.70	.75	.79	.55	.10	.55	1.560	0.007	0.440
						Contributing to:	p_n, Eq. 3		p_t, Eq. 4

Table 1: Coefficients to the terms of the thrust expressions according to the nominal optical thrust model (Eqs. 3 and 4), defined by alternative sets of film optical parameters.

The thrust coefficient associated with these parameters are shown in the rightmost three columns of the same table; of these coefficients the middle one (that associated with the $\cos\alpha$ term of the surface-normal thrust component) is ignored by the linear thrust model. The ignored coefficient is 0.011, a little more than one half of a percent of the dominant coefficient, 1.827. For a rather large incidence angle of α=60^o, the ratio of the actual thrust contributions associated with these terms increases to about 1% (from 0.5%), while the surface-tangent thrust component remains unaffected. This is an excellent quantitative approximation.

To explore the errors associated with the linear model for a wider range of parameters, a limited parametric study has been performed. Namely, the effects of a 20% drop of the reflectance r and of the specular reflectance fraction s, and a 100% increase of the front side emissivity ε_f have been explored via applying combinations of these degraded values in lieu of the Wright numbers. (The 20% drop of r and s is compatible with the most extreme environmental degradations predicted by Dachwald, *et. al.* in 2006 [12], while the 100% increase of the emissivity is beyond that foreseen in this cited paper.) The results of this study are summarized in the second through last rows of Table 1.

The numbers reveal generally small contributions by the $\cos\alpha$ term ignored in the linear model — in fact, when all degradations are together applied (the last row), the coefficient of this term is even less than for the unperturbed Wright numbers.

While there do exist selective combinations of the degraded values considered (rows two and three) where the surface normal thrust component predicted by the linear model for α=60^o would be somewhat more than 10% off the prediction by the full nominal model, this deviation is still deemed acceptable in light of other, more

significant approximations associated with solar sail engineering. Therefore, the linear model has been deemed to be a very attractive thrust formulation, and has been chosen for the present study.

Note, that despite the advantages highlighted above the linear model has been relatively rarely used in the past. The writer has been able to find only few examples for its application in publications [29, 30, 31]. Examples of a modified application with Eq. 15 not enforced can be found in the writer's earlier work [6, 2].

B. Better Approximations of Reality

While the nominal optical force model and its derivatives clearly dominate the literature, their typical implementation entirely ignores a number of effects that may influence thrust magnitude and direction more significantly than the selective application of optical modeling terms. Some attempts to quantify a few of these effects are touched upon in the following paragraphs.

The complex nonlinear dependence (and interdependence) of the nominal optical model's parameters on the sail temperature, in the context of other coupled effects such as surface roughness, solar irradiation angle, and solar distance have been investigated by Mengali, *et. al.* [13]. Their study reveals a very significant interplay between the considered effects. For example, at 1 AU solar distance, the incidence angle via the film temperature can produce variations of surface reflectance and specular reflectance in the $r \in [0.8731,\ 0.9224]$ and $s \in [.7853,1.0]$ ranges — variations of 5% and 21%, respectively. (This study also relies on previous work by Vulpetti [11] who studied non-predictable variations including thermo-optical dependence, and effects by light polarization and wavelengths.)

Rachwald, *et. al.* in 2006 studied [12] how material degradation in the space environment may effect optical performance. As irradiation doses increase, they chart up to 20% degradation of the reflectance and specular reflectance coefficients r and s, and front side emissivity ε_f, of the nominal optical model.

Yet another publication to highlight gross approximations in the nominal optical force model is Derbes's 2006 paper on propulsive reflectivity [7]. The paper reports that the distribution of reflected light experimentally recorded didn't match some basic assumptions inherent in the classic optical model. Accordingly, a modified semi-empirical framework for photonic thrust modeling is proposed, and results for the L'Garde sail's trough geometry (which has been recently solved [8] for the linear thrust model with rigor) are shown. However, the publication fails to disclose methodology and results to a depth that would be necessary for the reader to critically review the work or to effectively use the results. It also remains silent on the specifics of the thrust model with which the billow shapes shown have been generated.

Nevertheless, Derbes' paper well highlights that reflectance patterns on the macroscopic material scale may significantly differ from those traditionally assumed. The most profound practical impact of that work, however, remains to be the documentation of high propulsive reflectance for slack films. This observation justified the use of virtually unstressed sheets for solar sailing and, in turn, enabled Greschik to develop the architecture [3] that became the structural backbone of the L'Garde solar sails: Encounter [32] and its successors [33, 5].

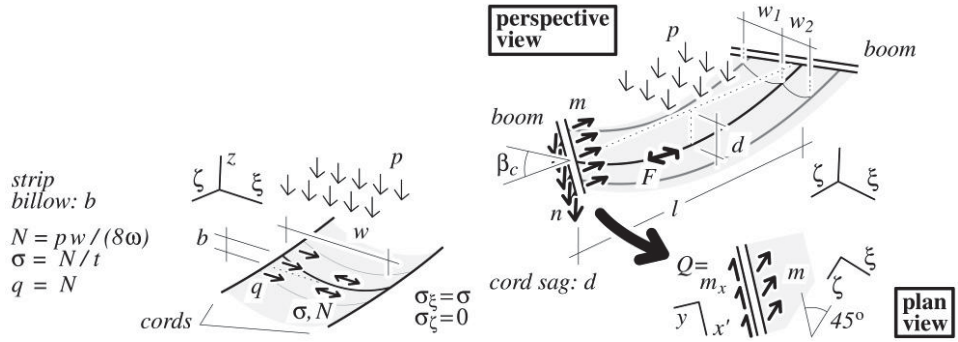
III. The L'Garde Sail Architecture and its Trough Shape

The L'Garde solar sail is distinguished by a structure like no other square sailcraft. The film surface in each quadrant is constituted by troughs defined by a array of mast-to-mast cords, to which the reflective sheet is loosely attached, Fig. 1. The film sags between the pairs of adjacent cords; the cords sag between the booms.

This architecture was proposed by Greschik [1, 2, 3] for the Encounter 2001 mission [32] as a practicable solution to approach the theoretical square sail performance limit earlier investigated [6]. With the Encounter effort abandoned, incremental adjustments to the design were made to fit the needs of subsequent NASA In-Space Propulsion Technology programs [34, 33, 5]. Most recently, the architecture was selected [4] by NASA as one of three *"crosscutting flight demonstrations" due to "their potential to provide tangible, near-term products and infuse high-impact capabilities into NASA's future space operations missions."* Within the framework of this program, the L'Garde sail, under the program name Sunjammer, will soon be flown, deployed, and maneuvered in space.

However, despite the attention this design has received, the thrust characteristics of its inherently non-smooth surface and how they impact attitude control and trajectory planning have received little public attention. One recent contribution to the understanding of the associated billow mechanics is a paper by the writer [8] in which rigorous proof is presented that the trough contour shapes in the context of the linear thrust model well approximates the classic catenary curve: the hyperbolic cosine function *cosh*. The derivation of this shape is enabled by the linear reflectivity model and it, therefore, well illustrates the practical advantages the simplicity of this model offers.

The derivation of the billow geometry in Ref. [8] is based on the two-dimensional analysis of its contour in the plane of the trough cross section. This approach is justified by the geometric proportions and hierarchical structure of the design, Fig. 1, wherein parallel and uniformly distributed cords in each sail quadrant transform the quadrant surface from a largely smooth geometry into a multitude of slender troughs.



(a) Force flow to the cords from the billow. (b) Sail quadrant and force flow from cords to boom.

Figure 3: Schematic of hierarchical force flow in L'Garde solar sail quadrant.

As indicated in the figure, the photonic pressure p billows the sheet within each trough in the $-z'$ direction and, via the catenary action of the billow in the trough cross section direction ξ, the cords are loaded. A cord tension F results which, in turn, transfers the loads in the edge-parallel $zeta$ direction to the booms. The latter end up effectively loaded with follower traction m (along with lateral loads and tip forces [3]).

With the radiation force components in the trough direction (ζ in Fig. 3) ignored as irrelevant to the contour shape, the proof is laid out in the two dimensional context of the trough cross section plane with the frames of reference and variables defined in Fig. 4. Namely, in addition to the quadrant surface coordinates xi and z (from cord to cord and rising from the surface), a trough contour Cartesian frame is also defined with its v axis pointing towards the light source and its origin **O** on the contour where the surface tangent and normal directions align with u and v, Fig. 4. As the photons arrive with a θ angular offset from the quadrant surface normal z (i.e., the u and z axes subtend a θ angle), the origin **O** will not be at the center of the contour between its endpoints **A** and **B** (the support cords), nor will the contour possess symmetry about the z axis.

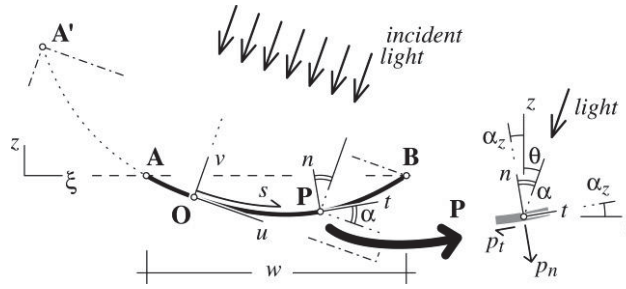


Figure 4: Trough contour geometry and variables for formulation.

As the $z(\xi)$ function defines the contour shape in the quadrant surface coordinates, the contour slope angle in the z, ξ frame at a generic point **P**, Fig. 4, is

$$\alpha_z = \arctan z' \qquad \text{Eq. 16}$$

$$z' = dz/d\xi \qquad \text{Eq. 17}$$

and the local light incidence angle there is

$$\alpha = \theta + \alpha_z \qquad \text{Eq. 18}$$

As α is the contour slope angle in the u, v frame, Eqs. $p_n = \eta_n P \cos^2\alpha$ Eq. 11 and 12 imply symmetric film loading and contour shape about the v axis. Recognizing this symmetry, refer to the origin **O** as the contour shape apex and formulate the problem in the u, v frame. (The actual trough between cords **A** and **B** will follow a section of a symmetric function of a greater domain, shown between points **A'** and **B** in Fig. 4.)

Denote membrane tension, the product of skin stress σ and film thickness t_f, along the contour with N

$$N = \sigma t_f \qquad \text{Eq. 19}$$

The equilibrium in the n and t directions of an infinitesimal contour section at location **P** implies

$$p_n = N\dot{\alpha} \qquad \text{Eq. 20}$$

$$p_t = \dot{N} \qquad \text{Eq. 21}$$

with the "dot" denoting differentiation with the arc length s, Fig. 4. Next, divide Eq. 20 with $\dot{\alpha}$, subsequently differentiate with respect to s to produce $\dot{N}$, and substitute this result into Eq. 21 to yield

$$p_n\ddot{\alpha} = \dot{p}_n\dot{\alpha} - p_t\dot{\alpha}^2 \qquad \text{Eq. 22}$$

Substituting functions Eqs. 11 and 12 into this relation finally yields

$$0 = \ddot{\alpha} + A\dot{\alpha}^2\tan\alpha \qquad \text{Eq. 23}$$

$$A = 2 + \eta_t/\eta_n \qquad \text{Eq. 24}$$

which is the governing differential equation of the contour in terms of the slope angle from the $+u$ direction expressed as a function of the arc length: $\alpha = \alpha(s)$. The sole parameter in this equation is the A material constant which can thus be named the trough contour shape parameter [8].

Dividing Eq. 23 with $\dot{\alpha}$, then integrating both sides in terms of s gives an alternative form which can be written as

$$\int \frac{1}{\cos^{A}\alpha} d\alpha = c_1 s \qquad \text{Eq. 25}$$

where another add-on integration constant is not shown on the right hand side as it is implied in the indefinite integral on the left.

A simple explicit closed form expression for Eq. 25 can be obtained when $\eta_t=0$ — when the in-plane photonic pressure component in the linear thrust model is assumed zero. In this case, $A=2$ and the left hand side of Eq. 25 becomes $(\tan\alpha+c_2)$ so

$$\alpha = \arctan(c_1 s+c_2) \qquad \text{Eq. 26}$$

The c_2 constant disappears by virtue of the symmetry of $\alpha(s)$ about $s=0$, and c_1 can be shown to be equal to the curvature at $s=0$. Accordingly,

$$\alpha = \arctan(s/R_0) \qquad \text{Eq. 27}$$

with R_0 the apex radius of curvature. Finally, symbolically obtaining the definite integrals

$$u = \int_0^s \cos\alpha(s) ds = R_0 \cosh^{-1}\frac{1}{R_0} + c_3 \qquad \text{Eq. 28}$$

$$v = \int_0^s \sin\alpha(s) ds = R_0\sqrt{1+s^2/R_0^2} + c_4 \qquad \text{Eq. 29}$$

then eliminating the constants and combining the results yields

$$v(u) = R_0\left[\cosh\frac{u}{R_0} - 1\right] \qquad \text{Eq. 30}$$

The billow shape, therefore, is the classic catenary curve bi-directionally scaled with the apex radius of curvature. The latter, in turn, can be expressed with the apex film tension and the photonic pressure there as

$$R_0 = N_0/(P\eta_n) \qquad \text{Eq. 31}$$

which can be used to express the billow shape Eq. $v(u) = R_0\left[\cosh\frac{u}{R_0} - 1\right]$ Eq. 30 with the apex film tension.

In the general case when $A>2$ (when $\eta_t \neq 0$ in the linear thrust model and, therefore, an in-plane thrust component also emerges under non-perpendicular illumination), no closed form solution with elementary functions can be derived. Instead, a series approximation can be applied which, expanded to three terms for high accuracy, gives the shape as

$$v(u) = R_0 \left[\frac{1}{2}\left(\frac{u}{R_0}\right)^2 + \left(\frac{1}{8} - \frac{A}{24}\right)\left(\frac{u}{R_0}\right)^4 + \left(\frac{A^2}{72} - \frac{A}{15} + \frac{19}{240}\right)\left(\frac{u}{R_0}\right)^6 \right] \quad \text{Eq. 32}$$

This general solution approaches Eq. 30, the special case for $A=2$, as the latter substitution transforms its polynomial terms into the first four terms of the power series expansion for the hyperbolic cosine function less 1:

$$v(u)_{a=2} = R_0 \left[\frac{1}{2}\left(\frac{u}{R_0}\right)^2 + \frac{1}{24}\left(\frac{u}{R_0}\right)^4 + \frac{1}{720}\left(\frac{u}{R_0}\right)^6 \right] \approx R_0 \left[\cosh\frac{u}{R_0} - 1 \right] \quad \text{Eq. 33}$$

These solutions have also been numerically verified by comparison with contour shapes explicitly obtained with numeric integration using the assumed photonic thrust models. A highlight from this verification has been shown in Ref. [8].

With the above results known, the calculation of the actual billow geometry becomes a numerical exercise in properly orienting and scaling the normalized contour shapes Eqs. 30 and 32 to fit the "gap" between cords **A** and **B**, Fig. 4, and to match the curved material length of the contour. This has been implemented for the present work in two independent ways: an *MS Excel* spreadsheet with extended graphical capabilities for visual feedback on the solution shapes, and a C program developed and run in a Linux environment. While both tools used the shooting method as the numerical engine of the contour tracing, the implementations differed not only to the extent necessitated by the different software platforms. The integration algorithm themselves with formulated differently, with different iterative techniques to improve the accuracy of the results. Both pieces of software were developed and tested in a modular manner. The verification of both tools has been deemed thorough and sufficient; Results by the two are identical within numerical accuracy.

Example output from the spreadsheet program is shown in Fig. 5: the contour shape normalized to the trough span for $\theta = 15^o$ illumination angle (off the normal direction to the plane between the support points), $\eta_n = 1.80$ thrust efficiency in the film surface normal direction (which, according to the linear model used, implies $\eta_t =$

0.20 for surface tangential thrust, Eq. 15), and a contour material length 2% greater than the distance between the support points.

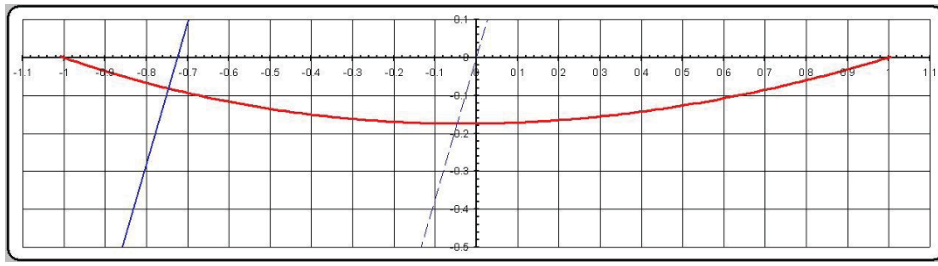


Figure 5: Contour shape for η_n = 1.80 , illumination subtending 15° with the global surface normal direction, and billowed contour 2% longer than the span. Drawn to scale, normalized to the billow span. The contour shape axis (at the contour shape apex) is shown with Continous blue lineEffective Reflective Efficiency Parameters for an Equivalent Surface

For practical insight into the thrust performance of the L'Garde solar sail surface, consider the conceptual replacement of the grooved surface with an equivalent smooth sheet which produces the same overall thrust in the context of an appropriate photonic thrust model as the real thrust on the former.

In the context of the linear photonic thrust model here chosen as preferred, this conceptual approach would call for the introduction of "effective" photonic thrust efficiency parameters $\eta_{n,eff}$ and $\eta_{t,eff}$ schematically indicated in Fig. 6. The rigorous execution of this conceptual step, however, calls for the resolution of a number of nontrivial questions related to the inherently anisotropic nature of the reflectance over the grooved surface. The extension (generalization) of the linear radiation force model to this scenario appears to be a formidable challenge because of the very basis of this model, the full nominal optical formulation, is inherently isotropic in nature. Accordingly, a general "quasi-linear" formulation for the equivalent surface is not sought in the present paper. Rather, the surface-normal thrust component is considered alone, and the equivelent $\eta_{n,eff}$ associated with this force component is examined. This is accomplished as follows.

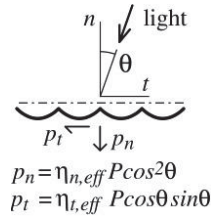


Figure 6: Linear model for a grooved surface.

First, the direction of illumination is defined in the orthotropic context of the grooved surface with the projected inclination angles θ_ξ and θ_ζ shown by the upper left inset in Fig. 7. Next, the trough geometry is numerically solved for this illumination with the film photonic thrust model of choice — for the present work, the linear model. With the geometry of and stresses in the billowed film known, the loads exerted thereby on the

supporting cords — the full thrust — is calculated. An equivalent thrust pressure on the flat strip between the two cords is calculated next. Finally, from the component of this thrust pressure in the direction perpendicular to the cord-to-cord strip and from the global illumination incidence angle θ with respect to the normal to this strip, an effective surface-normal thrust efficiency coefficient $\eta_{n,eff}$ is obtained from Eq. 11 with the globally defined θ used instead of α.

To explore how this global normal thrust coefficient $\eta_{n,eff}$ depends on the direction of illumination and on the normal thrust coefficient η_n characteristic to the film itself, a parametric study has been performed for a film contour length 2% greater than the distance between the cords. (This contour length means a rather slack film, chosen so to accentuate deformation effects.) Both projected incidence angles have been varied in steps of 2.5 from 0 (lateral illumination) to 50. Four values for the film normal thrust coefficient have been considered, in equal increments from $\eta_n = 1.2$ to $\eta_n = 1.8$ — from an unlikely low value to a value that can be deemed realistic in the context of typical solar sail studies. The results of the parametric study are shown in Fig. 7. The effective thrust coefficients are shown as contour maps. The legend, shown on the left side, is shared across all plots.

Two striking trends are apparent in the plots. One, the effective coefficient stays (nearly) constant if either one of the other incidence angle is varied alone, but it drops sharply if the two angles are combined. In other words: the coefficient is little affected by the illumination direction turned in either of the two orthogonal directions, but it is significantly reduced by skew illumination directions. The other counterintuitive trend is that the coefficient, in fact, increases slightly if the illumination direction is moved sideways in the trough cross section plane. No explanation to these trends has been found so far. Efforts to locate computational or coding errors also failed to date, and numerical verification with an independent piece of software also showd this trend. It is, therefore, likely that the phenomenon observed is a true orthogonal characteristic of photonic thrust performance. Without a conclusive explanation, the plots are reported as work-in-progress results.

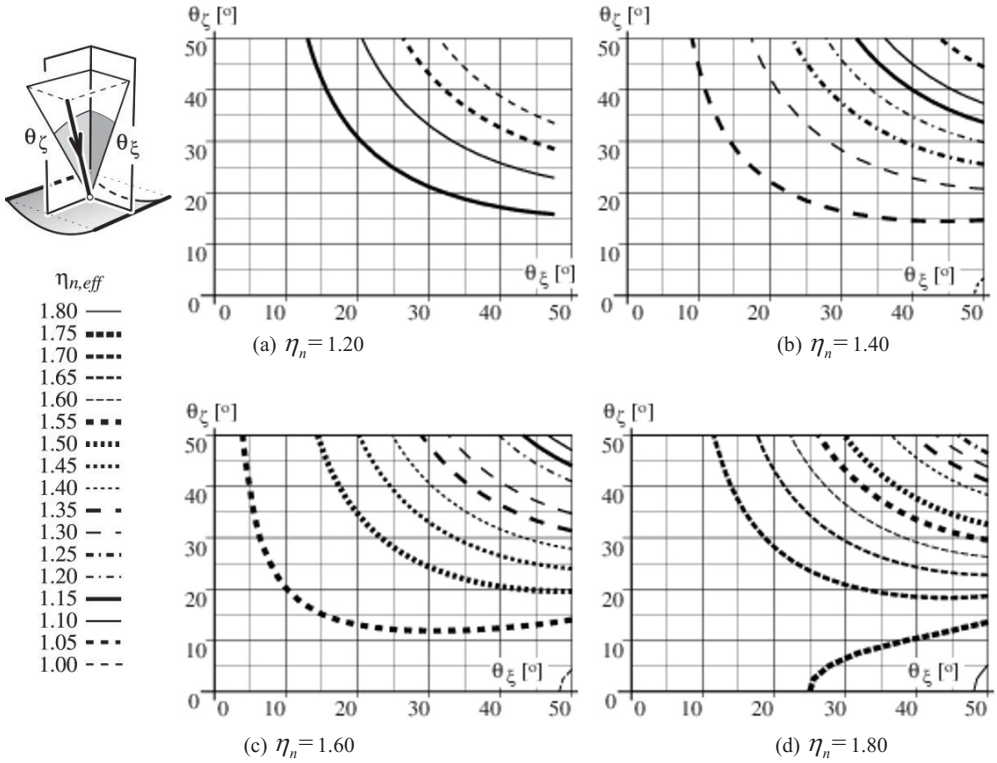


Figure 7: The dependence of effective surface normal thrust efficiency coefficients $\eta_{n,eff}$ on projected illumination incidence angles θ_ζ and θ_ξ for a billow contour length 2% longer than the cord gap, for four values of the film normal reflectance efficiency parameter η_n.

IV. Conclusion

A practical review of representative solar radiation force models has been presented, and a formulation — the linear thrust model — to strike an attractive middle ground between simplicity and theoretical fidelity has been chosen for the study of the thrust characteristics of the grooved L'Garde solar sail surface. This model has been used both to rigorously show that the sail's characteristic trough contours approximate the hyperbolic cosine function, and to numerically map the effective surface-normal thrust efficiency of the overall surface. This equivalent efficiency coefficient has been found to vary little with the change of the light incidence angle either along or across the troughs. However, incidence angles in skew directions resulted in a significant drop of the efficiency coefficient. It has also been observed that the efficiency coefficient slightly higher for

nonzero incidence angles in the cross-trough direction than under lateral illumination. No explanation for these phenomena has yet been found, rendering the discussion a work-in-progress report.

Ongoing work will explain these phenomena, and it will extend the scope of the grooved surface thrust efficiency study to also include the tangential (in-plane) thrust components. One challenge to overvom is to extend a reflectance thrust model with a characteristically isotropic philosophy into a generalized form that is capable of capturing the anisotropic characteristics of the grooved L'Garde sail surface.

IV. Acknowledgments

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Thrust Efficiency on an Idealized Deformable Sail

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A solar sail membrane that deforms under radiation pressure is numerically analyzed to determine the thrust efficiency at different attitudes. The deformable sail is compared to flat and other rigid structures. Our numerical model uses ray tracing and simulated annealing to calculate the force, torque, and steady-state contour of a loose-fitting, thin membrane adhered to a rigid support.

Nomenclature

(*Nomenclature entries should have the units identified*)

$\mathbf{F}$	=	force due to radiation pressure in vector form [N]
A	=	sun-facing area of a solar sail of arbitrary shape [m^2]
I	=	intensity of incident light in [W/m^2]
c	=	speed of light in vacuum [3x10^8 m/s]
$\hat{\mathbf{n}}$	=	unit normal of surface element
Q	=	total thrust efficiency of an arbitrary sail
Q_s	=	longitudinal thrust efficiency in direction of illumination of arbitrary sail
Q_l	=	transverse thrust efficiency to direction of illumination of arbitrary sail
α	=	angle between incoming rays and the surface normal [degrees]
$\mathbf{T}$	=	torque due to an offset of the center of mass and center of radiation pressure in vector form [Nm]
$\mathbf{r_o}$	=	offset of the center of mass and center of pressure in vector form [m]
Q_t	=	torque relative to force efficiencies
r_e	=	magnitude of CM-CP offset relative to projected area at normal incidence
s	=	degree of sag relative to projected area of sail

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r = radius of curvature of sagging sail with circular cross-section relative to projected area

κ = curvature of sagging sail with circular cross-section relative to projected area

θ = angular position along circular sagging sail [degrees]

b = maximum angle of incidence for a circular sagging sail [degrees]

x, y = coordinate frame of shaped sail structures

$x_{\min}$ = x position of lowest point on a catenary curve

a = radius of curvature of the catenary at $x_{\min}$ relative to projected area

I. Introduction

THRUST on a solar sail is dependent on many factors, but a grouping that is often overlooked involves the shape and curvature of the sail membrane. Sail curvature, defined by the degree of sag, changes both the direction and magnitude of incremental forces across the sailcraft. When summed, these incremental changes often diminish the efficiency of total thrust. The shape of a sail may be described as being flat, in the form of a pyramid, or a segment of a sphere, yet a real sail will be some arbitrary shape that changes with radiation pressure [1]. Disregarding effects from non-ideal optical properties of sail materials, there are still many aspects of real solar sails related to shape that play a large part in thrust efficiency and attitude control.

Global shape variations such as billowing [2,3], drooping [4], twisting [5], dynamic fluctuations or flapping [6,7], and deployment asymmetries [8] as well as local deformation of the sail shape [9,10] have been analyzed, assuming low degrees of sag, for their impact on attitude control through single reflection force calculations. Each of these will alter the direction and magnitude of thrust on a real sailcraft, and modeling these effects is necessary for prediction purposes to the development of accurate attitude and navigation control systems.

With the Sunjammer technology demonstration mission on the horizon, understanding the effects of sag and shape on solar sail mission controls has proximate gain. Sunjammer employs a striped architecture of the sail membrane to allow scalability of the design without significant mass penalties [8]. The chosen design lets the sail membrane sag at regular intervals of 1 m along the length of each boom. The loose membrane will affect the total thrust on the sailcraft in two ways. By increasing the curvature of each element, the total thrust efficiency will decrease. Secondly, each sagging element has the freedom to dynamically change shape under radiation pressure. This deformation may adversely impact the thrust vector on the entire sailcraft. Radiation pressure influences sail curvature similar to how gravity induces curvature of a sagging chain; however, local forces due to radiation pressure on a surface change in direction and magnitude with the curvature of the surface. Therefore,

numerical methods are required to determine the relaxed shape of the sail membrane under a radiation pressure load [2-3,6-7].

This proceeding compares the efficiency of thrust for a set of ideal (i.e. perfectly reflecting) sail shapes through rigorous ray tracing that accounts for multiple reflections off surfaces. Longitudinal and transverse force efficiencies due to radiation pressure as well as a torque due to a center of mass (CM) to center of pressure (CP) offset are first defined. The impact of increased curvature, by increasing the degree of sag, is presented both on-axis and as a function of attitude with analytical expressions derived for special cases. Sail membrane deformation is modeled as a catenary for the purpose of example to present the expected challenges that arise for deformable sails. Finally, results from a simulated annealing routine used as a low-order prediction of sail deformation under radiation pressure are compared to the catenary.

II. Force efficiency and CM-CP offset

This work focuses on changes in the thrust vector due to radiation pressure on a solar sail. The radiation pressure force is broken into two components: the scatter force and the lift force. The scatter force is defined to lie along the line of incidence, or in the direction of propagation of the incident light. The lift force is a transverse component of incident force positioned in a plane perpendicular to the direction of incidence. In order to maneuver to higher orbits further from the sun, a solar sail must be oriented such that a component of the thrust vector points in the direction of rotation to change the orbital angular momentum of the sail. The lift force gives this component of thrust in the plane of the orbital path. Some solar sail missions rely heavily on the 'lifting' ability of a sailcraft for orbital transfers [1]. In these cases, it is helpful to design the sailcraft to maximize the efficiency of lift. The combination of the lift and scatter component forces on a sail is the sail thrust vector.

The force due to radiation pressure from uniform light of intensity, I, on a flat reflective surface of area, A, is given by $\mathbf{F} = Q(IA/c)\hat{\mathbf{n}}$, where c is the speed of light in vacuum, $\hat{\mathbf{n}}$ is the surface normal of the mirror, and Q is the fraction of light effectively reflected off the mirror [11], i.e. thrust efficiency of the mirror. The Q value is determined from many physical parameters of the surface including shape, orientation, reflectivity, and emissivity. We decompose Q into its scatter (s) and lift (l) components such that $Q = \sqrt{Q_s^2 + Q_l^2}$, where Q_s has a maximum value of 2 and Q_l has a maximum value of 1 by the conservation of momentum. A perfectly reflecting, flat mirror has thrust efficiency in the scatter and lift directions given by

$$Q_s = 2\cos^2(\alpha)\left|\cos(\alpha)\right| \quad \text{and} \quad Q_l = 2\cos^2(\alpha)\sin(\alpha), \tag{1}$$

where α is the angle between incoming rays and the surface normal. The maximum scatter efficiency of the ideal flat mirror is 2 at normal incidence, $\alpha = 0^\circ$, and the maximum lift efficiency is 0.7698 at $\alpha = 35^\circ$.

The center of mass (CM) of a solar sail does not always coincide with the center of radiation pressure (CP). A CM-CP offset induces a torque on the sailcraft given by $\mathbf{T} = \mathbf{r_0} \times \mathbf{F}$, where $\mathbf{r_0}$ is the position of CP relative to CM. This torque may also be expressed in terms of a Q-factor as was done for the component forces [12], where r_ε is the magnitude of CM-CP offset relative to A.

$$Q_t = Qr_\varepsilon \tag{2}$$

In the following sections, force efficiencies at various angles of attack are given for a set of rigid and deformable sails all with perfect reflectivity to compare the effect of sail shape and curvature on thrust magnitude and direction. Numerical force efficiencies and relative CM-CP offsets are calculated from a ray tracing program developed by the authors and affiliates described in more detail in [13].

III. Impact of sail sag

In this proceeding, degree of sag is defined relative to the projected area of a sail at normal incidence, A. Degree of sag is the distance from a plane containing the fixed ends of a sail to the furthest point on the sail surface along the direction of normal incidence assuming a circular cross-section. Figure 1 diagrams this geometry. The degree of sag, s, is related to the radius of curvature, r, by Eq. 3, where $\kappa = 1/r$ is the curvature of the circular sail.

$$s = r - \sqrt{r^2 - 0.25} \tag{3}$$

A. Analytical solutions

In the special cases of a sail with a circular cross-section or catenary cross-section at normal incidence, analytical solutions for the component force efficiencies are derived. An element of a circular sail surface, $d\theta$, has component force efficiencies due to radiation pressure given by Eq. (4) and (5).

$$dQ_{s-circ} = 2\cos^3\left(\theta + \alpha\right) d\theta \tag{4}$$

$$dQ_{l-circ} = 2\cos^2\left(\theta + \alpha\right)\sin\left(\theta + \alpha\right) d\theta \tag{5}$$

The maximum value of θ at the surface end points is given by $b = \theta_{\max}$, such that $\sin(b) = 0.5\kappa$. Therefore, the total component force efficiencies on the circular sail are

$$\begin{aligned} Q_{s-circ}(\alpha) &= 2\int_{\theta=-b}^{+b} \cos^3(\theta+\alpha)\, d\theta \\ &= \frac{1}{2}\left(3\cos\alpha + \cos 3\alpha\right) - \frac{\kappa^2}{6}\cos 3\alpha \end{aligned} \quad \text{and} \tag{6}$$

$$\begin{aligned} Q_{l-circ}(\alpha) &= 2\int_{-b}^{+b} \cos^2(\theta+\alpha)\sin(\theta+\alpha)\, d\theta \\ &= \frac{1}{2}\left(\sin\alpha + \sin 3\alpha\right) - \frac{\kappa^2}{6}\sin 3\alpha \end{aligned} \tag{7}$$

.

At normal incidence, the efficiency of forward thrust, i.e. the scatter efficiency, is $Q_{s-circ} = 2 - \kappa^2/6$. At an angle of attack of $\alpha = 35^\circ$, the lift efficiency is $Q_{l-circ} = 0.7698 - \kappa^2/6.2$. These expressions provide a useful relationship between the component force efficiencies and the degree of sag.

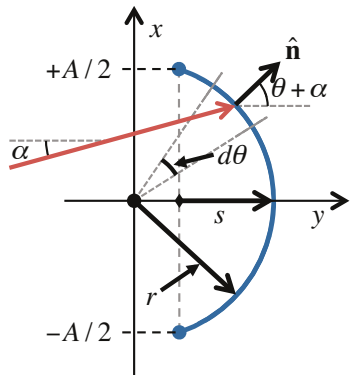

Fig. 1 Geometry of the circular cross-section defining the degree of sag, s.

A catenary is the shape of a chain under a gravitational load. It is dependent on the direction of gravity, and is used here as a representative solution for the curvature of a deformable sail membrane. The expression for a catenary is $y = a\cosh\left((x - x_{\min})/a\right) + y_o$, where a is the radius of curvature of the minimum point positioned at $x_{\min}$ and y_o is a bias in position along the y-axis. Following similar steps as above, the scatter force efficiency for the catenary at normal incidence may be written

$$Q_{s-cat} = 4a\sin\varphi_{\max} \tag{8}$$

where $\tan\varphi_{\max} = \frac{L}{2a}$ is the tangent of the angle between either end point of the symmetric catenary curve and the x-axis and L is the arc length of the entire curve. The arc length is calculated from the circular model and applied to Eq. (6) to maintain a consistent definition of the degree of sag.

These analytical models for the force due to radiation pressure account only for one reflection of the light. At high degrees of sag, the light must be traced through multiple reflections until its intensity becomes negligible for an accurate calculation of the thrust. In the next section, a ray tracing model is used to determine the force of radiation pressure on the circular and catenary sail shapes accounting for multiple reflections.

B. Ray tracing results

In this section, force efficiencies for a rigid circular sail and deformable catenary-shaped sail are numerically obtained with a ray tracing model and compared to analytical solutions derived above. Examples of the output ray tracing plots are given in Fig. 2 for the circular and catenary structures at a few angles of attack and degrees of sag at 0.05 and 0.3. Only a subset of the total number of rays is represented in the plots. In the following results, 20,000 rays are traced through eight bounces and the momentum transfer from each ray-surface interaction is summed to obtain the force data. The shape surfaces are perfectly reflecting so that the rays do not loose intensity upon each reflection.

As a function of the degree of sag, modeled scatter force efficiency at normal incidence and lift force efficiency at 35° angle of attack are each plotted in Fig. 3. Also plotted is a calculation of the lift force on the catenary sail with variable sag at 35° obtained numerically with only a single reflection. Similar to most numerical models of arbitrary solar sail shapes, this technique models each discrete element of the shape function as a flat surface. The elemental force efficiencies are calculated using Eq. (1) scaled by the element size and summed over the length of the shape function to obtain the total force efficiency in each component direction.

At normal incidence, clear disparity between the analytical model and ray tracing results is apparent for a degree of sag of 0.25 or greater due to additional opportunities for momentum transfer when multiple reflections are considered. At 35°, the lift efficiency is less impacted by multiple reflections; however, when s =0.3 and 0.5, the catenary has a small boost in efficiency due to added reflection of a relatively flat region of the shape. The rays responsible for this boost can be seen in Fig. 2, $\alpha = 35°,\ s = 0.3$ catenary ray plot.

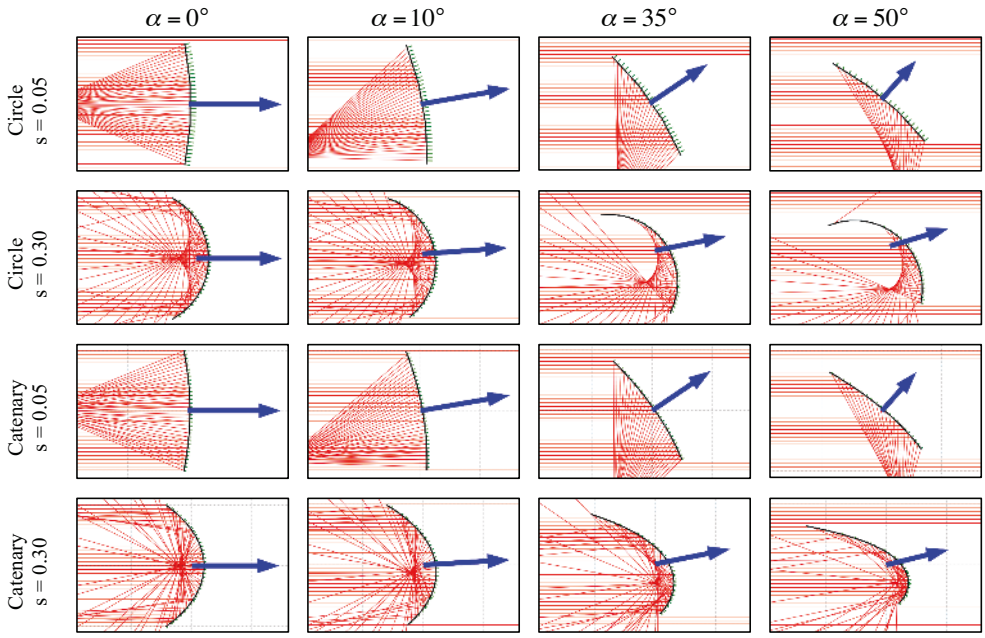

Fig. 2 Ray tracing plots. Large blue arrows are the calculated thrust vectors, and small green lines on the surface of each sail are individual forces from each ray.

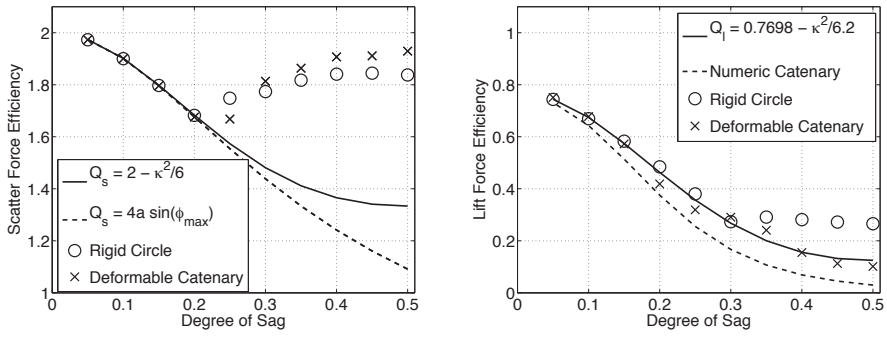

Fig. 3 Scatter force (left) and lift force (right) efficiencies at $\alpha = 0^\circ$ and $\alpha = 35^\circ$, respectfully, for a rigid circular sail and a deformable catenary sail compared to the analytical cases.

The angle 35° is selected as a representative pitch angle because the lift efficiency of a flat reflector peaks at this orientation. Navigation control systems currently plan to pitch solar sails at 35° during orbital transfers to take advantage of this maximum in lift efficiency. However, it is reasonable to assume that not all sail shapes will have maximum lift efficiency at 35°. In Fig. 4, lift efficiency is plotted as a function of angle of attack. At

very low degrees of sag, the peak lift efficiency remains at 35°, but as s increases, the peak in lift efficiency wonders to other angles of attack for both the rigid circular and the deformable catenary sails.

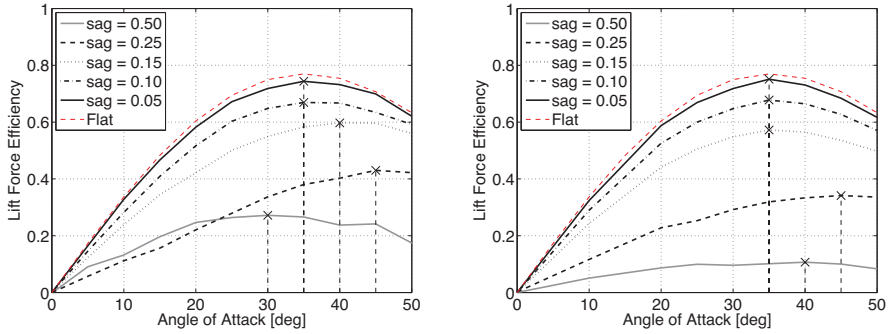

Fig. 4 Change in peak lift efficiency due to increased sag for the rigid circular sail (left) and catenary deformable sail model (right).

An output of the ray tracing program is a torque on the object about CM determined by summing discrete torques from each ray force on the surface of the sail. From this, Eq. (2) is used to determine the relative CM-CP offset for the rigid and deformable cases at a variety of sag values. These results are plotted in Fig. 5 indicating that a deformable sail has much smaller offset values.

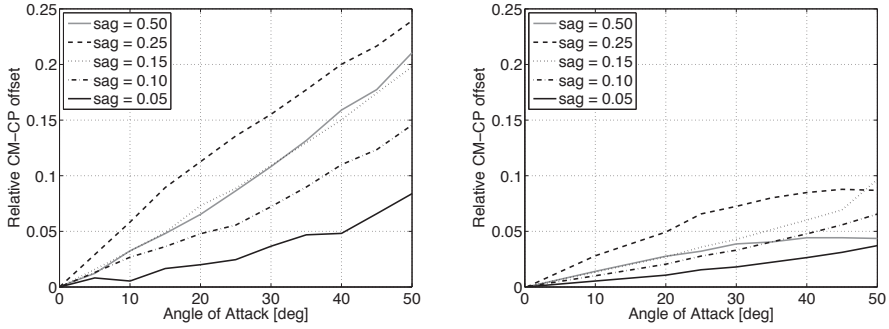

Fig. 5 CM-CP offset distances relative to projected area at normal incidence for the rigid circular sail (left) and deformable catenary sail model (right).

IV. Model of a deformable sail

To this point, the deformable sail has been modeled as a catenary. This was due to the ease in expressing the catenary shape in analytical form. Unlike gravity, radiation pressure changes with the curvature of the sail so that areas with high angles of incidence (e.g. near the fixed end points) will have small applied load from radiation

pressure. In this section, a coarse approximation of the curvature of a deformable sail under a load from radiation pressure is solved with a simulated annealing numerical model.

Simulated annealing is a widely used probabilistic optimization technique, esp. for systems with a large number of degrees of freedom [14]. The standard method of simulated annealing is a rather simple algorithm to implement, which finds the global minimum of a cost function. The algorithm mimics the mechanical process of annealing in metallurgy where temperature is slowly decreased until the system freezes. Assuming the temperature has been decreased slowly enough and the system has been given enough freedom to search parameter space, the final state of the system will be that of minimum energy [14]. The relaxed shape of a thin, deformable solar sail under radiation pressure is determined using a standard method of simulated annealing by minimizing tension in the sail membrane.

In this model, nearest-neighbor tension in the sail membrane is calculated at all points along the length of a one-dimensional curve by taking the dot product of each element unit tangent and neighboring forces. The sum of elemental tension values plus the weighted difference in total arc length of the curve from the input is used as the cost function to be minimized in the simulated annealing scheme. No material stiffness is considered in this model. Solutions are presented only for normal incidence. From Section III, we understand that low degrees of sag lead to solutions similar to the flat sail; therefore, two large values of sag are used in this section, s = 0.3 and 0.5.

A circular sail cross-section with the desired degree of sag was used as the starting shape. The program began at a temperature of 1.443, and progressed through 50 cycles, each time reducing the temperature by 0.026. The solution was taken as the current best after the 50th cycle had completed. The numerically modeled deformed sail was found to be catenary-like in curvature, see Fig. 6. In Table 1, the thrust efficiency of the deformed sail is compared to that of the catenary, and the results are nearly identical.

Table 1 Scatter force efficiency of modeled deformed sail and the catenary.

	$s = 0.3$	$s = 0.5$
Deformed Sail	$Q_s = 1.8778$	$Q_s = 1.9317$
Catenary	$Q_s = 1.8134$	$Q_s = 1.9291$

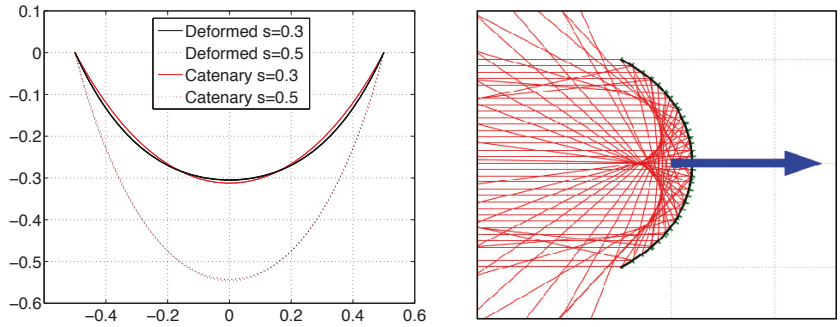

Fig. 6 (left) Shape comparison of the modeled deformed sail shape and a catenary for two degrees of sag. (right) Ray plot of the deformed sail with $s = 0.3$.

V. Conclusion

Analysis of force efficiency and CM-CP offset has been presented for a rigid circular, deformable catenary, and modeled deformable solar sail. These results indicate that a sail with a degree of sag of 0.25 or greater will have multiple reflections inside its curvature. As the degree of sag is increased, the angle of attack for maximum lift wonders from 35°, yet this effect is greater for the rigid circular structure. Since a deformable sail membrane relaxes to equalize with a radiation pressure load, it is unsurprising that the CM-CP offset is much reduced. When material elasticity and optical properties are accounted for, deformation of a real solar sail may be expected to be between a circle and a catenary. If a scaling to account for loss due to optical properties is applied, the force efficiencies presented above roughly provide bounding limits for the expected efficiencies of a real solar sail.

Acknowledgments

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The Solar Radiation Pressure Force Models for a General Sail Surface Shape

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In this study, a solar sail is modeled as smooth curved surfaces and the total force and moment models for the deformed sails are investigated, and two different forms of the models, namely the tensor expressions and the parameterized expressions, are proposed. Both forms can describe the total force and moment off-line by using the constant coefficient tensors or coefficient matrix correspondingly. Besides, some other properties as a result of deformation such as the sail mass center are also discussed. Thus this work paves the way for sail craft's precise navigation and control where exact forces are needed.

I. Introduction

SOLAR sails consist of large-area reflective films that achieve thrust by reflecting solar photons without fuel. In recent decades, solar sails arouse more and more interests in theoretical and experimental research. Since the first successful solar sail mission IKAROS was launched in May 2010 [1], it indicates that the solar sail technology has entered into the engineering stage. In fact, many space research institutes, such as NASA, ESA and DLR et al. have some projects to develop the solar sail technology [2, 3]. For actual solar sail missions, an important step is to determine the solar radiation pressure forces precisely in the sense of navigation accuracy. It is noted that most solar sails have large flexible reflective membrane, and then the elastic deformation may be large enough to render the actual radiation pressure forces much different from that derived from the ideal flat sail model. However, this deformation was often ignored as an ideal flat model in the previous studies. Actually, with the development of engineering application, a more accurate force model for the deformed non-planar solar sail is to be in urgent need. It can be expected that a modified model can help the solar sail navigation and

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control accurately. In most research papers, the deformation of the sail film is not considered in the radiation pressure force model. The flat sail force model contains two common forms. The first model form is assumed that the solar sail is a perfect reflector, and the second is based on a non-perfect solar sail with considering reflection, absorption and re-radiation by the sail film [4]. It is noted that these theoretical models are implicit assumption that the sail is flat. Rios-Reyes and Scheeres [5] developed a methodology to estimate the force and moment based on a generalized model for arbitrary shape. Yamaguchi et al. [6] introduced a spinning model to examine the deformation of solar sails based on the IKAROS mission, and the expression of the solar radiation pressure force is finally obtained by using five parameters with respect to the attitude angle; however, this model can only be used for a deformation with special symmetry. Sakamoto et al. [7] studied the finite element model of sail deformation by the numerical method, and using this method the radiation pressure force can be calculated accurately, and this method is useful for attitude control and structure designing of sail-craft; however, the finite element model is complicated and not appropriate for the orbit navigation and control.

In this study, based on the methodology in Ref. [5], a parameterized general radiation pressure force and moment models are developed under the assumption of an arbitrary shape of the sail film. Two different forms are investigated in this paper. The first has a simple and clear expression by tensor form and is convenient for theoretical analysis; the second has a parametric description in specific frame to be practical for numerical calculation and provide a possibility for solar sail on-orbit parameter estimation. Since the sail surface is not flat anymore, the mass center, inertia matrix and pressure center will vary depending on the shape of the sail surface, and then the theoretical equations of calculating these properties are also investigated. As a caution, it should be noted that the deformation and the force on the sails are definitely coupled together. To get rid of potential difficulties due to this coupling, we assume that the coupling is one-way in our preliminary study. That is, we aim to establish the model to calculate the total force by giving a specific fixed shape without considering the shape deformation varying with the force. In principle the actual shape can only be obtained after sail's deployment in space because the sail always needs to be deployed after launch onto space. Since the sail's deployment may cause wrinkles or deformation from the nominal flat shape that was expected before launch and the degradation of materials may also lead to uncertainty about the sail's shape, then the final shape of the solar sail in orbit cannot be the same as it is on ground. Therefore, in this study, a modified solar sail force model is investigated to describe the radiation pressure force for arbitrary sail shapes which is hoped to estimate the force conveniently on ground after launch.

II. Sail Surface Assumption and the Differential Force & Moment

Assume that the sail surface is an arbitrary curved surface, which can be described in the sail body-fixed frame by the general surface equation as

$$\begin{cases} x = x(u,v) \\ y = y(u,v) \\ z = z(u,v) \end{cases} \quad (u,v) \in D_{uv} \subset \mathbb{R}^2 \tag{1}$$

where u, v are the parameters in the equation. Note that the sail surface is assumed to be fixed shown in Fig.1.

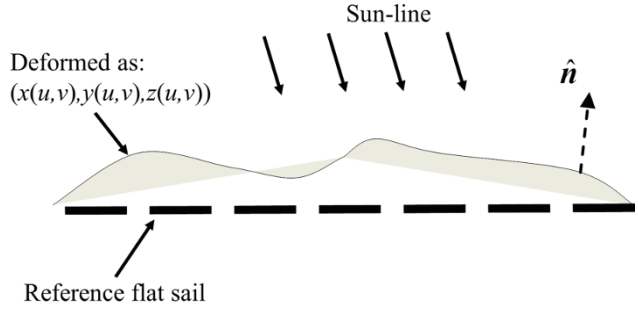

Fig. 1 The assumption of the sail surface deformation.

Then the unit normal vector for local surface can be obtained by

$$\hat{\boldsymbol{n}}(x,y,z) = \frac{\left(y_u z_v - y_v z_u, x_v z_u - x_u z_v, x_u y_v - x_v y_u\right)^{\mathrm{T}}}{\sqrt{\left(y_u z_v - y_v z_u\right)^2 + \left(x_v z_u - x_u z_v\right)^2 + \left(x_u y_v - x_v y_u\right)^2}} = \left(\hat{n}_1, \hat{n}_2, \hat{n}_3\right)^{\mathrm{T}} \tag{2}$$

where $x_u = \dfrac{\partial x}{\partial u}$ is the partial derivative of x with respect to the parameter u, and similar relationships exist for the others. $\hat{n}_1$, $\hat{n}_2$, and $\hat{n}_3$ are the components in the body-fixed frame of the unit normal vector $\hat{\boldsymbol{n}}$, respectively. Note that $\hat{\boldsymbol{n}}$ is not a constant vector but a function of the location on the surface, and it can be determined only by the shape of sail surface. It is noted that in this study, the '^' notation on a vector's top means its unit vector.

The differential area dA is defined as

$$dA = \sqrt{EG - F^2}\, du dv \tag{3}$$

where *E*, *F*, and *G* are obtained as follows:

$$
\begin{aligned}
E &= \left(\frac{\partial x}{\partial u}\right)^2 + \left(\frac{\partial y}{\partial u}\right)^2 + \left(\frac{\partial z}{\partial u}\right)^2 = x_u^2 + y_u^2 + z_u^2 \\
G &= \left(\frac{\partial x}{\partial v}\right)^2 + \left(\frac{\partial y}{\partial v}\right)^2 + \left(\frac{\partial z}{\partial v}\right)^2 = x_v^2 + y_v^2 + z_v^2 \\
F &= \frac{\partial x}{\partial u}\frac{\partial x}{\partial v} + \frac{\partial y}{\partial u}\frac{\partial y}{\partial v} + \frac{\partial z}{\partial u}\frac{\partial z}{\partial v} = x_u x_v + y_u y_v + z_u z_v
\end{aligned} \tag{4}
$$

Taking into account the reflectance, absorption and emissivity of the sail film, the differential force $d\boldsymbol{F}$ on the differential area element dA due to solar radiation pressure (SRP) may be written as

$$
d\boldsymbol{F} = P(s)\left[a_1\left((-\hat{\boldsymbol{n}})\cdot\hat{\boldsymbol{s}}\right)\hat{\boldsymbol{s}} + a_2\left((-\hat{\boldsymbol{n}})\cdot\hat{\boldsymbol{s}}\right)(-\hat{\boldsymbol{n}}) + 2a_3\left((-\hat{\boldsymbol{n}})\cdot\hat{\boldsymbol{s}}\right)^2(-\hat{\boldsymbol{n}})\right]dA \tag{5}
$$

where $\hat{\boldsymbol{s}}$ is the position unit vector of the solar sail from the Sun and there the SRP is $P(s)$, and a_1, a_2, and a_3 are the optical parameters of the sail surface shown in Appendix A.

To consider the moment due to SRP for deformed sail surface, the arm of force on the differential area element dA relative to a particular point is assumed as $\boldsymbol{\rho}$, so the differential moment is defined as

$$
\begin{aligned}
d\boldsymbol{M} &= \boldsymbol{\rho}\times d\boldsymbol{F} \\
&= P(s)\left[-a_1(\hat{\boldsymbol{n}}\cdot\hat{\boldsymbol{s}})(\boldsymbol{\rho}\times\hat{\boldsymbol{s}}) + a_2(\hat{\boldsymbol{n}}\cdot\hat{\boldsymbol{s}})(\boldsymbol{\rho}\times\hat{\boldsymbol{n}}) - 2a_3(\hat{\boldsymbol{n}}\cdot\hat{\boldsymbol{s}})^2(\boldsymbol{\rho}\times\hat{\boldsymbol{n}})\right]dA
\end{aligned} \tag{6}
$$

Since the differential force and moment on each differential element dA are given by Eq.(5) and Eq.(6), the total force and moment can be obtained by integrating over the whole sail surface area A. The results can be written as different forms, which are the vector form, tensor form and the parameterized form. Except for the sail position and the sail orientation in space, the factors that influence the force and moment also include the sail surface properties such as the shape, size and the sail film optical properties. For orbit determination and orbit propagation, if we assume that the solar sail position and orientation information can be obtained by measuring, then the influence of the sail surface properties are important factors required to be considered for deformed sail, which are expected to be independent of the sail position and orientation. Therefore, the tensor form and the parameterized form of the force & moment models are applicable.

III. The Tensor Forms of the Total Force & Moment

Integrating over the sail surface area, the total force and moment are given in the vector forms as

$$
\boldsymbol{F} = P(s)\int_A\left[-a_1(\hat{\boldsymbol{n}}\cdot\hat{\boldsymbol{s}})\hat{\boldsymbol{s}} + a_2(\hat{\boldsymbol{n}}\cdot\hat{\boldsymbol{s}})\hat{\boldsymbol{n}} - 2a_3(\hat{\boldsymbol{n}}\cdot\hat{\boldsymbol{s}})^2\hat{\boldsymbol{n}}\right]dA \tag{7}
$$

$$
\boldsymbol{M} = P(s)\int_A\left[-a_1(\hat{\boldsymbol{n}}\cdot\hat{\boldsymbol{s}})(\boldsymbol{\rho}\times\hat{\boldsymbol{s}}) + a_2(\hat{\boldsymbol{n}}\cdot\hat{\boldsymbol{s}})(\boldsymbol{\rho}\times\hat{\boldsymbol{n}}) - 2a_3(\hat{\boldsymbol{n}}\cdot\hat{\boldsymbol{s}})^2(\boldsymbol{\rho}\times\hat{\boldsymbol{n}})\right]dA \tag{8}
$$

Note that in the integral the sail position $\boldsymbol{s}$ is independent of the surface properties, and can be treated as a constant vector on the entire sail surface because the size is much smaller than the magnitude of sail position. Therefore, the total force & moment can be calculated in tensor forms [5]:

$$\boldsymbol{F} = P(s)\left[-\left(\int_A a_1\hat{\boldsymbol{n}}dA\cdot\hat{\boldsymbol{s}}\right)\hat{\boldsymbol{s}}+\int_A a_2\hat{\boldsymbol{n}}\hat{\boldsymbol{n}}dA\cdot\hat{\boldsymbol{s}}-2\hat{\boldsymbol{s}}\cdot\int_A a_3\hat{\boldsymbol{n}}\hat{\boldsymbol{n}}\hat{\boldsymbol{n}}dA\cdot\hat{\boldsymbol{s}}\right] \tag{9}$$

$$\boldsymbol{M} = P(s)\left[-\hat{\boldsymbol{s}}\cdot\left(\int_A a_1\hat{\boldsymbol{n}}\tilde{\boldsymbol{\rho}}dA\right)\cdot\hat{\boldsymbol{s}}+\left(\int_A a_2\tilde{\boldsymbol{\rho}}\cdot\hat{\boldsymbol{n}}\hat{\boldsymbol{n}}dA\right)\cdot\hat{\boldsymbol{s}}-\hat{\boldsymbol{s}}\cdot\left(\int_A 2a_3\hat{\boldsymbol{n}}\tilde{\boldsymbol{\rho}}\cdot\hat{\boldsymbol{n}}\hat{\boldsymbol{n}}dA\right)\cdot\hat{\boldsymbol{s}}\right] \tag{10}$$

where $\tilde{\boldsymbol{\rho}}$ is the 2-rank skew-symmetric tensor relative to the arm of force vector $\boldsymbol{\rho}$, so the cross product can be equated as

$$\boldsymbol{\rho}\times\boldsymbol{a} = \tilde{\boldsymbol{\rho}}\cdot\boldsymbol{a} \tag{11}$$

where $\boldsymbol{a}$ is an arbitrary vector.

Here define the tensors to describe the integral in the expressions of Eq.(9) and Eq.(10) as

$$\begin{aligned}\boldsymbol{J}^1 &= \int_A a_1\hat{\boldsymbol{n}}dA\\ \boldsymbol{J}^2 &= \int_A a_2\hat{\boldsymbol{n}}\hat{\boldsymbol{n}}dA\\ \boldsymbol{J}^3 &= \int_A a_3\hat{\boldsymbol{n}}\hat{\boldsymbol{n}}\hat{\boldsymbol{n}}dA\end{aligned} \tag{12}$$

$$\begin{aligned}\boldsymbol{K}^{31} &= \int_A a_1\hat{\boldsymbol{n}}\tilde{\boldsymbol{\rho}}dA\\ \boldsymbol{K}^2 &= \int_A a_2\tilde{\boldsymbol{\rho}}\cdot\hat{\boldsymbol{n}}\hat{\boldsymbol{n}}dA\\ \boldsymbol{K}^{32} &= \int_A 2a_3\hat{\boldsymbol{n}}\tilde{\boldsymbol{\rho}}\cdot\hat{\boldsymbol{n}}\hat{\boldsymbol{n}}dA\\ \boldsymbol{K}^3 &= \boldsymbol{K}^{31}+\boldsymbol{K}^{32} = \int_A\left(a_1\hat{\boldsymbol{n}}\tilde{\boldsymbol{\rho}}+2a_3\hat{\boldsymbol{n}}\tilde{\boldsymbol{\rho}}\cdot\hat{\boldsymbol{n}}\hat{\boldsymbol{n}}\right)dA\end{aligned} \tag{13}$$

where $\boldsymbol{J}^1$ is a 1-rank tensor namely a vector, $\boldsymbol{J}^2$ and $\boldsymbol{K}^2$ are 2-rank tensors, and $\boldsymbol{J}^3$, $\boldsymbol{K}^{31}$, $\boldsymbol{K}^{32}$, and $\boldsymbol{K}^3$ are 3-rank tensors. It can be seen that the tensors are obtained by integrating the sail's local properties over the whole surface which are only related to the optical parameters and the sail's shape. Therefore, the tensor forms of total force & moment can be obtained as

$$\boldsymbol{F} = P(s)\left[-\left(\boldsymbol{J}^1\cdot\hat{\boldsymbol{s}}\right)\hat{\boldsymbol{s}}+\boldsymbol{J}^2\cdot\hat{\boldsymbol{s}}-2\hat{\boldsymbol{s}}\cdot\boldsymbol{J}^3\cdot\hat{\boldsymbol{s}}\right] \tag{14}$$

$$\begin{aligned}\boldsymbol{M} &= P(s)\left[-\hat{\boldsymbol{s}}\cdot\boldsymbol{K}^{31}\cdot\hat{\boldsymbol{s}}+\boldsymbol{K}^2\cdot\hat{\boldsymbol{s}}-\hat{\boldsymbol{s}}\cdot\boldsymbol{K}^{32}\cdot\hat{\boldsymbol{s}}\right]\\ &= P(s)\left[\boldsymbol{K}^2\cdot\hat{\boldsymbol{s}}-\hat{\boldsymbol{s}}\cdot\boldsymbol{K}^3\cdot\hat{\boldsymbol{s}}\right]\end{aligned} \tag{15}$$

So the total force and moment expressions of Eq.(14)~(15) consist of two categories factors: the first category is the local solar radiation pressure $P(s)$ and the unit position vector $\hat{\boldsymbol{s}}$, which have the same values throughout the entire sail surface; the second category is the integral tensors, which are the overall performance of the sail's optical and shape properties. The benefits of the expressions written as Eq.(14)~(15) include two

aspects. Firstly the tensors can be used to calculate an arbitrary surface shape and non-uniform optical properties. Furthermore, since the tensors are obtained by integrating, the force and moment can be described as the general tensor expressions including any local properties on the sail surface. Secondly, when the optical properties and the shape of the solar sail are fixed, considering in the sail body-fixed frame, the tensors are all constant tensors. In addition, the tensors are independent of the solar sail position and orientation, namely the tensors vary only with the sail surface shape and optical properties, so the tensors can be called coefficient tensors which can describe the total force and moment off-line.

Although the expressions as Eq.(14)~(15) are very clear and simple, it needs pay attention to calculating the components due to the more than 2-rank tensors. According to the tensor notation, the components of the tensors are given by

$$\begin{aligned} J_i^1 &= \int_A a_1 \hat{n}_i dA \\ J_{ij}^2 &= \int_A a_2 \hat{n}_i \hat{n}_j dA \qquad i,j,k = 1,2,3 \\ J_{ijk}^3 &= \int_A a_3 \hat{n}_i \hat{n}_j \hat{n}_k dA \end{aligned} \tag{16}$$

$$\begin{aligned} K_{ijk}^{31} &= \int_A a_1 n_i \tilde{\rho}_{jk} dA \\ K_{ij}^2 &= \int_A a_2 \tilde{\rho}_{ik} n_k n_j dA \\ K_{ijk}^{32} &= \int_A 2a_3 n_i \tilde{\rho}_{jr} n_r n_k dA \\ K_{ijk}^3 &= K_{ijk}^{31} + K_{ijk}^{32} = \int_A \left(a_1 n_i \tilde{\rho}_{jk} + 2a_3 n_i \tilde{\rho}_{jr} n_r n_k \right) dA \end{aligned} \qquad i,j,k = 1,2,3 \tag{17}$$

Note that in all the expressions, a couple of same indexes means summation, i.e., the subscript 'k' in the right hand of the expression 'K_{ij}^2' appears as a couple, so $K_{ij}^2 = \int_A a_2 \sum_{k=1}^{3} \tilde{\rho}_{ik} n_k n_j dA$ and 'k' is called dummy index. Then, the components of the total force and moment can be obtained by

$$F_i = P(s)\left[-\left(J_j^1 \hat{s}_j \right) \hat{s}_i + J_{ik}^2 \hat{s}_k - 2\hat{s}_r J_{rit}^3 \hat{s}_t \right] \tag{18}$$

$$\begin{aligned} M_i &= P(s)\left[-\hat{s}_r K_{rit}^{31} \hat{s}_t + K_{ik}^2 \hat{s}_k - \hat{s}_r K_{rit}^{32} \hat{s}_t \right] \\ &= P(s)\left[K_{ik}^2 \hat{s}_k - \hat{s}_r K_{rit}^3 \hat{s}_t \right] \end{aligned} \tag{19}$$

Note that all the component forms of tensors and vectors must be defined as the projections in a particular coordinate system, which can be chosen as any different coordinate system, thus the components in different coordinate systems are different, correspondingly. As discussed above, the sail body-fixed frame chosen as the projection coordinate system can bring much convenience such as the tensors are constants. So in the later section we will discuss the component-based parameterized forms of the total force and moment in the sail body-fixed frame and the comparison with the tensor forms will also be given.

As we known, the results in the previous sections are the composition by integrating the distribution forces on the sail surface. Note that the result of total force vector is unique, but the moment vector is defined relative to the reference point, and the moments about different reference points have the relationship as

$$\boldsymbol{M}_P = \boldsymbol{M}_O + \boldsymbol{F} \times \boldsymbol{r}_{OP} \tag{20}$$

where P and O are different points, $\boldsymbol{M}_P$ and $\boldsymbol{M}_O$ are the moments about them respectively, $\boldsymbol{r}_{OP}$ is the vector from O pointing to P. Usually the reference point is chosen as the mass center of the sail. Generally, the mass center cannot find out directly for an arbitrary surface, and it will discuss in the later section.

IV. The Parameterized Forms of the Total Force & Moment

A. The Definition of the Coordinate Systems

To obtain the components, we must define the projection coordinate system above all. In this manner, all the components discussed in the particular coordinate systems are significant. According the previous discussion, the sail body-fixed frame is properly chosen as the projection coordinate system. In this study it is defined according to the following principles: The coordinate's origin is at the center point O of the reference flat sail surface. The z-axis is aligned normal to the reference flat sail surface, and both the x-axis and y-axis are in the surface plane. The body-fixed frame is obtained by the 3-1-2 Euler rotation by reference the orbit frame which is usually used for the 3-axis stabilized satellites. According to the rotation sequence the Euler angles are defined as ψ, ϕ, θ, respectively. Note different rotation sequences lead to different component forms. Besides, the body-fixed coordinate frame is fixed not on the flexible actual sail but on the reference rigid body, which means the body-fixed frame does not vary with the changing of the sail's shape.

Here the definition of the body-fixed frame is concerned with the reference orbit frame, and we define it as follows: the orbit coordinate system's origin is also at the center point O of the reference flat sail surface. The z'-axis is directed along the opposite of the Sun-line, the y'-axis is directed normal to the orbit plane, and the x'-axis is in the orbit plane completing the right-hand set and pointing to the velocity direction. So the unit vectors of Cartesian coordinate axes are written as $\hat{\boldsymbol{i}}' = \hat{\boldsymbol{j}}' \times \hat{\boldsymbol{k}}'$, $\hat{\boldsymbol{k}}' = -\dfrac{\boldsymbol{r}_s}{\|\boldsymbol{r}_s\|}$, and $\hat{\boldsymbol{j}}' = \dfrac{\boldsymbol{v}_s \times \boldsymbol{r}_s}{\|\boldsymbol{v}_s \times \boldsymbol{r}_s\|}$. The orbital coordinate is the bridge between the inertial coordinate system and the body-fixed frame. Here the inertial frame is the heliocentric ecliptic coordinate system shown in Fig. 2.

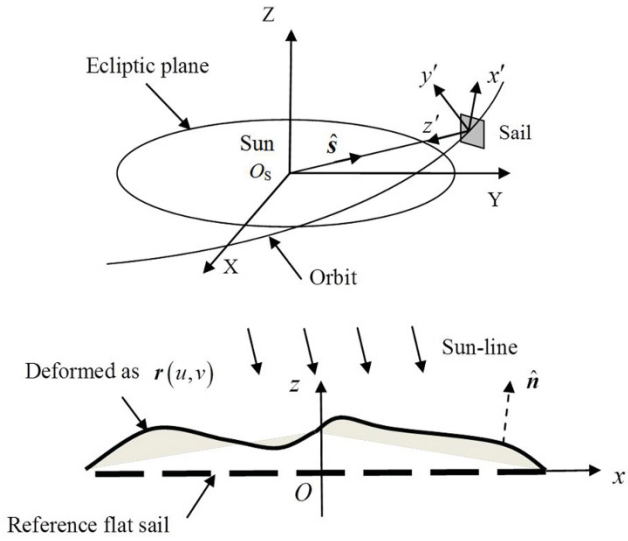

Fig. 2 The definition of coordinate systems.

To separate the components in different coordinate systems, we use $[*]_I$, $[*]_O$ and $[*]_B$ to stand for the components in the inertial frame, orbit frame and body-fixed frame, respectively. Thus, if $[\boldsymbol{u}]_I$, $[\boldsymbol{u}]_O$, and $[\boldsymbol{u}]_B$ are the column components in three different coordinate systems of the same arbitrary vector $\boldsymbol{u}$, they are related by

$$[\boldsymbol{u}]_I = R_{IO}[\boldsymbol{u}]_O = R_{IO}R_{OB}[\boldsymbol{u}]_B = R_{IB}[\boldsymbol{u}]_B \tag{21}$$

Note that

$$\begin{aligned} &R_{IO} = R_{OI}^T, R_{OB} = R_{BO}^T \\ &R_{IB} = R_{IO}R_{OB} \end{aligned} \tag{22}$$

The transformation matrices are defined as

$$\begin{aligned} R_{OI} &= \left(-\frac{(\boldsymbol{v}_s \times \boldsymbol{r}_s)\times \boldsymbol{r}_s}{\|(\boldsymbol{v}_s \times \boldsymbol{r}_s)\times \boldsymbol{r}_s\|}, \frac{\boldsymbol{v}_s \times \boldsymbol{r}_s}{\|\boldsymbol{v}_s \times \boldsymbol{r}_s\|}, -\frac{\boldsymbol{r}_s}{\|\boldsymbol{r}_s\|} \right)_I^{\mathrm{T}} \\ R_{BO} &= R_{312} = R_y(\theta)R_x(\phi)R_z(\psi) \\ &= \begin{pmatrix} \mathrm{c}\theta\mathrm{c}\psi - \mathrm{s}\phi\mathrm{s}\theta\mathrm{s}\psi & \mathrm{c}\theta\mathrm{s}\psi + \mathrm{s}\phi\mathrm{s}\theta\mathrm{c}\psi & -\mathrm{c}\phi\mathrm{s}\theta \\ -\mathrm{c}\phi\mathrm{s}\psi & \mathrm{c}\phi\mathrm{c}\psi & \mathrm{s}\phi \\ \mathrm{s}\theta\mathrm{c}\psi + \mathrm{s}\phi\mathrm{c}\theta\mathrm{s}\psi & \mathrm{s}\theta\mathrm{s}\psi - \mathrm{s}\phi\mathrm{c}\theta\mathrm{c}\psi & \mathrm{c}\phi\mathrm{c}\theta \end{pmatrix} \end{aligned} \tag{23}$$

Note that the subscript I in Eq.(23) means that the column vectors of matrixes are measured in the inertial frame, and the short hand notation 'c' means 'cos' and 's' means 'sin'.

B. The Parameterized Forms of the Total Force & Moment in the Sail Body-fixed Frame

As shown in Eq.(2), the local unit normal vector $\hat{\boldsymbol{n}}$ of the differential area element dA measured in the sail body-fixed frame is $[\hat{\boldsymbol{n}}]_B = (\hat{n}_1, \hat{n}_2, \hat{n}_3)^{\mathrm{T}}$. In addition, the components of heliocentric unit position vector $\hat{\boldsymbol{s}}$ is easily to find out in the orbit frame, the components can be written as

$$[\hat{\boldsymbol{s}}]_O = (0, 0, -1)^{\mathrm{T}} \tag{24}$$

According to the transformation matrix, the components in the body-fixed coordinate can be obtained as

$$[\hat{\boldsymbol{s}}]_B = R_{BO}[\hat{\boldsymbol{s}}]_O = (\cos\phi\sin\theta, -\sin\phi, -\cos\phi\cos\theta)^{\mathrm{T}} \tag{25}$$

Then, integrating Eq.(5) over the entire surface D_{uv} the total force due to SRP in the body-fixed frame is obtained as

$$[\boldsymbol{F}]_B = P(s)(F_1, F_2, F_3)^{\mathrm{T}} \tag{26}$$

If the moment reference point is chosen as the original point O of the body-fixed coordinate, the arm of force on the differential area element dA has the component form of

$$[\boldsymbol{\rho}]_B = (x, y, z)^{\mathrm{T}} = (x(u,v), y(u,v), z(u,v))^{\mathrm{T}} \tag{27}$$

Similarly, integrating Eq.(6) in the body-fixed frame, the components of the total moment about the original point O yields

$$[\boldsymbol{M}_O]_B = P(s)(M_1, M_2, M_3)^{T} \tag{28}$$

Here we find that the force and the moment expressions have the same form. In addition, the components in each axis are just the linear combination of the trigonometric functions of the two Euler angles ϕ and θ, furthermore the trigonometric functions in each axis involve the same nine items:

$$\begin{aligned}
&4^{\text{th}}\ \text{Order} - \cos^2\phi\sin^2\theta \quad \cos^2\phi\cos^2\theta \quad \cos^2\phi\sin\theta\cos\theta \\
&3^{\text{rd}}\ \text{Order} - \sin\phi\cos\phi\sin\theta \quad \sin\phi\cos\phi\cos\theta \\
&2^{\text{nd}}\ \text{Order} - \cos\phi\sin\theta \quad \cos\phi\cos\theta \quad \sin^2\phi \\
&1^{\text{st}}\ \text{Order} - \sin\phi
\end{aligned} \tag{29}$$

Thus the total force and moment can be written as the same form by coefficient matrixes:

$$[\boldsymbol{F}]_B = P(s)\boldsymbol{\Gamma}^F\boldsymbol{\Theta} \tag{30}$$

$$[\boldsymbol{M}_O]_B = P(s)\boldsymbol{\Gamma}^M\boldsymbol{\Theta} \tag{31}$$

where the sizes of coefficient matrixes $\boldsymbol{\Gamma}^F$ and $\boldsymbol{\Gamma}^M$ are both 3×9, the angle matrix $\boldsymbol{\Theta}$ is certainly 9×1, and they are written as

$$\boldsymbol{\Gamma}^F = \begin{pmatrix} C_1^F & C_2^F & \cdots & C_9^F \\ D_1^F & D_2^F & \cdots & D_9^F \\ E_1^F & E_2^F & \cdots & D_9^F \end{pmatrix}_{3\times 9} \tag{32}$$

$$\boldsymbol{\Gamma}^M = \begin{pmatrix} C_1^M & C_2^M & \cdots & C_9^M \\ D_1^M & D_2^M & \cdots & D_9^M \\ E_1^M & E_2^M & \cdots & D_9^M \end{pmatrix}_{3\times 9} \tag{33}$$

$$\boldsymbol{\Theta} = \begin{pmatrix} \cos^2\phi\sin^2\theta \\ \cos^2\phi\cos^2\theta \\ \vdots \\ \sin\phi \end{pmatrix}_{9\times 1} \tag{34}$$

where all the components C_i^F and C_i^M in Eq.(32) and (33) are given in Appendix B and C.

Although the total moment in Eq.(28) is defined relative to the original point *O* of the body-fixed frame, the moment about the other point can be obtained by Eq.(20). It is clear that the coefficient matrixes $\boldsymbol{\Gamma}^F$ and $\boldsymbol{\Gamma}^M$ are just related to the sail surface properties but independent to the solar position and orientation, and the angle matrix $\boldsymbol{\Theta}$ is just related to the solar sail's attitude but nothing to do with the surface properties. Therefore, the total force & moment can be described to be parameterized forms off-line by the coefficient matrixes, just like the tensor forms by using the coefficient tensors. Actually, the parameterized forms and the tensor forms are equivalent. Calculating the tensors and reflecting into the sail body-fixed frame, we can get the parameterized forms by some particular mathematical manipulations. Thus if tensor forms are also reflected into the sail body-fixed frame there is a transformational relationship between the coefficient tensors of the tensor forms and coefficient matrixes of the parameterized forms:

$$\begin{aligned}
&C_1^F = -J_1^1 - 2J_{111}^3, C_2^F = -2J_{133}^3, C_3^F = J_3^1 + 4J_{113}^3, C_4^F = J_2^1 + 4J_{112}^3, \\
&C_5^F = -4J_{123}^3, C_6^F = J_{11}^2, C_7^F = -J_{13}^2, C_8^F = -2J_{122}^3, C_9^F = -J_{12}^2, \\
&D_1^F = -2J_{211}^3, D_2^F = -2J_{233}^3, D_3^F = 4J_{213}^3, D_4^F = J_1^1 + 4J_{212}^3, \\
&D_5^F = -J_3^1 - 4J_{223}^3, D_6^F = J_{21}^2, D_7^F = -J_{23}^2, D_8^F = -J_1^2 - 2J_{222}^3, D_9^F = -J_{22}^2, \\
&E_1^F = -2J_{311}^3, E_2^F = -J_3^1 - 2J_{333}^3, E_3^F = J_1^1 + 4J_{313}^3, E_4^F = 4J_{312}^3, \\
&E_5^F = -J_2^1 - 4J_{323}^3, E_6^F = J_{31}^2, E_7^F = -J_{33}^2, E_8^F = -2J_{322}^3, E_9^F = -J_{32}^2
\end{aligned} \tag{35}$$

$$
\begin{aligned}
&C_1^M = -K_{111}^3, C_2^M = -K_{313}^3, C_3^M = K_{113}^3 + K_{311}^3, C_4^M = K_{112}^3 + K_{211}^3, \\
&C_5^M = -K_{213}^3 - K_{312}^3, C_6^M = K_{11}^2, C_7^M = -K_{13}^2, C_8^M = -K_{212}^3, C_9^M = -K_{12}^2, \\
&D_1^M = -K_{121}^3, D_2^M = -K_{323}^3, D_3^M = K_{123}^3 + K_{321}^3, D_4^M = K_{221}^3 + K_{122}^3, \\
&D_5^M = -K_{223}^3 - K_{322}^3, D_6^M = K_{21}^2, D_7^M = -K_{23}^2, D_8^M = -K_{222}^3, D_9^M = -K_{22}^2, \\
&E_1^M = -K_{131}^3, E_2^M = -K_{333}^3, E_3^M = K_{331}^3 + K_{133}^3, E_4^M = K_{231}^3 + K_{132}^3, \\
&E_5^M = -K_{332}^3 - K_{233}^3, E_6^M = K_{31}^2, E_7^M = -K_{33}^2, E_8^M = K_{232}^3, E_9^M = -K_{32}^2
\end{aligned}
\tag{36}
$$

It should be pointed out that the component forms are related to the definition principle of the body-fixed frame, and for different frame definition principles, the results and all the transformational relations above may be different. In this study, the most common and traditional frames are used and the results are all based on them. Although the two kinds of forms are equivalent, it depends on the conditions to use them properly. The tensor forms have simple forms of expression and clear physical significance to describe the total force and moment, but the 3-rank tensors are difficult to calculate, so it is adequate for theoretical analysis. However, the parameterized forms are just using the matrixes equal to the 2-rank tensors, so it is convenient for numerical calculations.

V. Some Properties for the Deformed Sail Surface

When the sail does not remain regular flat surface anymore, the mass center and the inertia matrix will be depending on the sail's shape. Besides, the distributed forces are related to the deformation condition, so the pressure center will also be affected. For the deformed sail with an arbitrary shape, the effect of SRP is acting as a form of distributed forces on the sail surface. The distributed forces can be equivalent to a single force and a moment about a particular point which is often chosen as the mass center of the solar sail, and what we get in the sections above is just the results of equivalence. For a reference flat solar sail, the forces due to SRP exerted on the whole sail surface is same, so the distributed forces on the whole sail is equivalent to a single force exerted on the mass center, with zero moment about the mass center, then the mass center is also the pressure center. However, for a deformed solar sail, since the pressure center depends on the forces exerted on it, the mass center is different from the pressure center, namely the moment about the mass center is usually not zero anymore. Here we will discuss these properties for deformed sails assumed as arbitrary shapes, including the mass center, the inertia matrix and the pressure center.

A. Mass Center

Assume that the areal density of the sail surface is $\eta(u,v)$, where the surface equation is given by Eq.(1), then the components of the mass center's relative position (the vector from original point *O* to mass center *C*) in the sail body-fixed frame is obtained by

$$[\boldsymbol{r}_{OC}]_B = (x_{OC}, y_{OC}, z_{OC})^T = \left(\frac{m_{23}}{m}, \frac{m_{31}}{m}, \frac{m_{12}}{m}\right)^{\mathrm{T}} \tag{37}$$

where the total mass of the sail surface *m* and the three static moments about the different coordinate plane m_{23}, m_{31}, m_{12} are defined by

$$\begin{aligned} m &= \int_A \eta dA, \quad & m_{23} &= \int_A x\eta dA \\ m_{31} &= \int_A y\eta dA, \quad & m_{12} &= \int_A z\eta dA \end{aligned} \tag{38}$$

Since the moment relative to the original point *O* is given by Eq.(28), according to Eq.(20), the moment relative to the mass center can be calculated by

$$\boldsymbol{M}_C = \boldsymbol{M}_O + \boldsymbol{F} \times \boldsymbol{r}_{OC} \tag{39}$$

It is important to obtain the mass center and the moment about it for attitude motion and control, especially for the deformed solar sail the mass center cannot be determined directly.

B. The Inertia Matrix

The inertia of the sail also depends on the surface shape. As discussed above, the arm of force for a differential area element *dA* is vector $\boldsymbol{\rho}$, and $\tilde{\boldsymbol{\rho}}$ the 2-rank skew-symmetric tensor relative to it. The areal density is still $\eta(u,v)$, thus the inertia tensor about the body-fixed original point is defined by

$$\boldsymbol{I}_O = \int_A \left(-\tilde{\boldsymbol{\rho}}\tilde{\boldsymbol{\rho}}^T\right)\mu dA \tag{40}$$

Actually the inertia tensors about the mass center is more useful, and it can be calculated by

$$\boldsymbol{I}_C = \boldsymbol{I}_O + m\tilde{\boldsymbol{r}}_{OC}\tilde{\boldsymbol{r}}_{OC}^T \tag{41}$$

where *m* is the sail surface total mass, and $\tilde{\boldsymbol{r}}_{OC}$ is the 2-rank skew-symmetric tensor relative to the mass center's relative position $\boldsymbol{r}_{OC}$.

Projecting $\boldsymbol{I}_O$ into the sail body-fixed frame, the inertia matrix about the origin point *O* can be obtained as

$$[\boldsymbol{I}_O]_B = \begin{pmatrix} I_{11} & -I_{12} & -I_{13} \\ -I_{21} & I_{22} & -I_{23} \\ -I_{31} & -I_{32} & I_{33} \end{pmatrix} \tag{42}$$

where all the components are obtained by

$$
\begin{aligned}
I_{11} &= \int_A \left(y^2 + z^2\right)\eta dA, \quad I_{22} = \int_A \left(x^2 + z^2\right)\eta dA \\
I_{33} &= \int_A \left(x^2 + y^2\right)\eta dA, \quad I_{12} = \int_A \left(xy\right)\eta dA = I_{21} \\
I_{23} &= \int_A \left(yz\right)\eta dA = I_{32}, \quad I_{31} = \int_A \left(xz\right)\eta dA = I_{13}
\end{aligned} \tag{43}
$$

Then inertia matrix about the mass center in the body-fixed frame can be written by Eq.(41) as

$$
\left[\boldsymbol{I}_C\right]_B = \left[\boldsymbol{I}_O\right]_B + m\begin{pmatrix} -y_C^2 - z_C^2 & x_C y_C & z_C x_C \\ x_C y_C & -z_C^2 - x_C^2 & y_C z_C \\ z_C x_C & y_C z_C & -x_C^2 - y_C^2 \end{pmatrix} \tag{44}
$$

The inertia matrix is necessary for the attitude dynamics discussing. Since the inertia matrix about the mass center in the body-fixed frame $\left[\boldsymbol{I}_C\right]_B$ is given by Eq.(44), then we can establish the principal body-fixed frame and obtain the principal inertia matrix about the mass center by calculating the eigenvalues and eigenvectors of $\left[\boldsymbol{I}_C\right]_B$, because in some cases the principal inertia matrix about the mass center is useful and can simplify the dynamic equations. It needs to note that when the sail surface shape is fixed, the mass center and the inertia matrix are also treated as a constant had no concern with the forces exerted on the sail.

C. Pressure Center

Generally the total force is impossible to be a zero vector. According to the definition of the pressure center, in some cases, the distributed forces on the solar sail are equivalent to a single force, and then the pressure center is unique existence. The unique pressure center can be obtained by

$$
\boldsymbol{r}_{OP} = \frac{\boldsymbol{F} \times \boldsymbol{M}_O}{\left\|\boldsymbol{F}\right\|^2} \tag{45}
$$

where $\boldsymbol{r}_{OP}$ is the relative position of the pressure center P (the vector from O to P), $\boldsymbol{F}$ and $\boldsymbol{M}_O$ are the total force and moment about the original point O, respectively. Note that in this case, they are orthogonal to each other, namely $\boldsymbol{F} \cdot \boldsymbol{M}_O = 0, \boldsymbol{F} \times \boldsymbol{M}_O \neq \boldsymbol{0}$. Thus, the moment about point P is

$$
\boldsymbol{M}_P = \boldsymbol{M}_O + \boldsymbol{F} \times \boldsymbol{r}_{OP} = \boldsymbol{M}_O + \boldsymbol{F} \times \left(\frac{\boldsymbol{F} \times \boldsymbol{M}_O}{\left\|\boldsymbol{F}\right\|^2}\right) = \boldsymbol{M}_O - \boldsymbol{M}_O = \boldsymbol{0} \tag{46}
$$

It can be found that the moment about point P is a zero vector which meets the traditional definition of pressure center. However, for some special cases the distributed forces are equivalent to a force screw but not a single force. In this condition, if the moment vector about a particular point is parallel to the total force, we define this point is the equivalent pressure center. For the general cases, the total force vector is usually neither parallel nor perpendicular to the moment vector about the original point O, namely $\boldsymbol{F} \cdot \boldsymbol{M}_O \neq 0, \boldsymbol{F} \times \boldsymbol{M}_O \neq \boldsymbol{0}$. In

these cases, although there exists no point about which the moment is zero, we can still obtain the equivalent pressure center by Eq.(45). Therefore, the moment about the equivalent pressure center P is

$$\begin{aligned}\boldsymbol{M}_P &= \boldsymbol{M}_O + \boldsymbol{F}\times\boldsymbol{r}_{OP} = \boldsymbol{M}_O + \boldsymbol{F}\times\left(\frac{\boldsymbol{F}\times\boldsymbol{M}_O}{\|\boldsymbol{F}\|^2}\right) = \boldsymbol{M}_O + \left(\frac{\boldsymbol{M}_O\cdot\boldsymbol{F}}{\|\boldsymbol{F}\|^2}\right)\boldsymbol{F} - \left(\frac{\boldsymbol{F}\cdot\boldsymbol{F}}{\|\boldsymbol{F}\|^2}\right)\boldsymbol{M}_O \\ &= \left(\frac{\boldsymbol{F}\cdot\boldsymbol{M}_O}{\|\boldsymbol{F}\|^2}\right)\boldsymbol{F}\end{aligned} \tag{47}$$

It can be found that the moment vector about the equivalent pressure center $\boldsymbol{M}_P$ is parallel to the total force vector $\boldsymbol{F}$, namely $\boldsymbol{M}_P\times\boldsymbol{F}=\boldsymbol{0}, \boldsymbol{M}_P\cdot\boldsymbol{F}\neq 0$.

VI. Example

Here we give an example of the solar sail with a general surface shape, and the surface equation is given by

$$\begin{cases} x = u \\ y = v \\ z = \dfrac{1}{4}\left(\sin\left(0.1u+0.2v\right)+\cos\left(0.3u-0.4v\right)\right) \end{cases} \tag{48}$$
$$u, v \in D_{uv} = \left(\left[-10,10\right],\left[-10,10\right]\right)$$

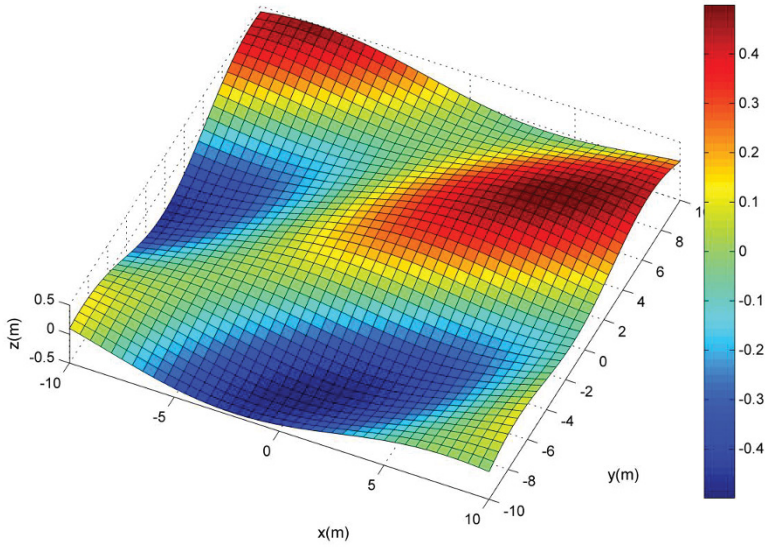

Fig. 3 The surface shape of the example.

In the example we assume that the reference flat sail is a square with a size of $20\text{m}\times 20\text{m}$, and the general curved surface shape after deformation is described by Eq.(48) and shown in Fig. 3. The sail areal density η and the optical parameters are assumed to be constants for simplify calculating. To compare the deformed shape model to the flat surface model, the parameters of the reference flat sail are the same as the deformed sail. Thus the constant parameters of the sail are given by Table 1.

Table 1 The constant parameters for the example sail

Constants	a_1	a_2	a_3	η(kg/m^2)	Area(m^2)
Reference flat sail	0.1728	-0.010888	0.8272	0.001	400
General shape sail	0.1728	-0.010888	0.8272	0.001	401.85

According to the surface shape, the mass center, pressure center and inertia matrix are given by Table 2.

Table 2 The mass center, pressure center and inertia matrix for the example sail

Paramters	$[\boldsymbol{r}_{OC}]_B$ (m)	$[\boldsymbol{r}_{OP}]_B$ (m)	$[\boldsymbol{I}_O]_B$ (kg·m^2)
Reference flat sail	$\begin{pmatrix}0\\0\\0\end{pmatrix}$	$\begin{pmatrix}0\\0\\0\end{pmatrix}$	$\begin{bmatrix}13.33 & 0 & 0\\0 & 13.33 & 0\\0 & 0 & 26.67\end{bmatrix}$
General shape sail	$\begin{pmatrix}0.50\\2.45\\-2.17\end{pmatrix}\times 10^{-3}$	Uncertain	$\begin{bmatrix}13.42 & 0.69\times 10^{-3} & -0.14\\0.69\times 10^{-3} & 13.42 & -0.37\\-0.14 & -0.37 & 26.79\end{bmatrix}$

Note that the mass center and the inertia matrix are just related to the surface shape, but the pressure center is also related to the forces and moments which vary depending on the solar sail position and attitude. For the reference flat sail, we can know in advance that the moment about the mass center namely the original point is always zero, so the pressure center is the same as the mass center. However, for the sail with general surface shape, the total forces and moments are different in different positions and attitudes, and then the pressure center will be uncertain.

Here take only one attitude for an example, namely $\phi=\theta=45^\circ$, then the total force and moment are

$$\begin{aligned}[\boldsymbol{F}]_B &= \begin{pmatrix}0.096\\-0.112\\-0.808\end{pmatrix}\times 10^{-3}[N]\\ [\boldsymbol{M}]_B &= \begin{pmatrix}-0.026\\0.221\\0.056\end{pmatrix}\times 10^{-3}[N\cdot m]\end{aligned} \tag{49}$$

The mass center has been obtained as

$$[\boldsymbol{r}_{OC}]_B = \begin{pmatrix} 0.50 \\ 2.45 \\ -2.17 \end{pmatrix} \times 10^{-3}[m] \tag{50}$$

It can be found that the mass center is quite close to the original point, and the total moment about the mass center can be calculated by Eq.(39)

$$[\boldsymbol{M}_C]_B = \begin{pmatrix} -0.024 \\ 0.221 \\ 0.056 \end{pmatrix} \times 10^{-3}[N \cdot m] \tag{51}$$

At the particular attitude of $\phi = \theta = 45^\circ$ the total force and moment are both known, and we can find out that the total force vector is usually neither parallel nor perpendicular to the moment vector about the original point *O*, namely $\boldsymbol{F} \cdot \boldsymbol{M}_O \neq 0, \boldsymbol{F} \times \boldsymbol{M}_O \neq \boldsymbol{0}$. So we can also calculate the pressure center by Eq.(45) as

$$[\boldsymbol{r}_{OP}]_B = \begin{pmatrix} 0.255 \\ 0.024 \\ 0.027 \end{pmatrix} [m] \tag{52}$$

Thus the moment about the pressure center *P* is

$$[\boldsymbol{M}_P]_B = \begin{pmatrix} -0.103 \\ 0.121 \\ 0.869 \end{pmatrix} \times 10^{-4}[N \cdot m] \tag{53}$$

Note that the point *P* is the equivalent pressure center namely $\boldsymbol{M}_P \times \boldsymbol{F} = \boldsymbol{0}, \boldsymbol{M}_P \cdot \boldsymbol{F} \neq 0$. Although the true pressure center about which the moment is a zero vector does not exist at this force condition, the moment about the equivalent pressure center is the minimum than that relative to other points on the surface.

According to the discussion above, although the total force and moment are related to the sail position and attitude, the coefficient tensors and matrixes in the force and moment expressions are independent. So we can calculate the coefficients for the different expressions. The results are given by

$$\boldsymbol{\Gamma}^F = \begin{bmatrix} 0.72 & 6.32 & 73.20 & -1.31 & 4.18 & -0.01 & -0.04 & -0.05 & -0.01 \\ -0.01 & 12.76 & -4.18 & -0.56 & -77.17 & 0.01 & -0.08 & 1.53 & 0.03 \\ -2.04 & -724.82 & -13.31 & -4.18 & 26.23 & 0.04 & 4.34 & -4.02 & -0.08 \end{bmatrix} \tag{54}$$

$$\boldsymbol{\Gamma}^M = \begin{bmatrix} 1.96 & 3.10 & -5.58 & -1.33 & -7.37 & 0.02 & -0.01 & -5.16 & 0.02 \\ -0.47 & -1.27 & 67.28 & 0.24 & 89.76 & -0.21 & 0.01 & 1.06 & -0.28 \\ -0.44 & 0 & 47.10 & -4.22 & -1.69 & -0.01 & 0.30 & 5.08 & 0.01 \end{bmatrix} \tag{55}$$

Furthermore, all of the components for the coefficient tensors in the tensor forms also can be calculated, and they are given by Eq.(35)~(36).

VII. Conclusions

This paper models the force and moment for the deformed solar sail under the assumption of the sail surface deformed as a general surface. Two different forms of the total force and moment: the tensor forms and the parameterized forms are investigated. The tensor forms have simple and clear expressions, and they are convenient for theoretical analysis but difficult for numerical calculation due to the existence of more than 2-rank tensors. However, the parameterized forms are practical for numerical calculation because they just use the matrixes, and the total force and moment have the same expression forms. It requires comment that the component forms are depending on the definition of coordinate systems, and the results may be different if the definition principles are different. In this study, the chosen coordinate systems are commonly used, so the results are practical. Both of the two forms use the coefficient tensors and coefficient matrixes discussing in the sail body-fixes frame to describe the sail surface deformation conditions including the optical properties. The benefits are possible to describe an arbitrary deformed sail shape, and separate the sail deformation independent of the sail position and orientation so discussing the force and moment off-line. Then some important and useful properties of the sail including the mass center, the pressure center and the inertia matrix are investigated. Where the mass center and the inertia matrix are just determined by the sail surface shape, but the pressure center also varies depending on the force conditions. After all, to meet the requirement of the precise solar sail navigation including orbit determination and orbit propagation, in this study the pre-flight force and moment models lay a good foundation for on-orbit parameter estimation.

Appendix

A. The definition of the optical parameters

The optical parameters in the study are defined as

$$\begin{aligned} a_1 &= 1-\rho s \\ a_2 &= B_f \rho(1-s)+(1-\rho)\frac{\varepsilon_f B_f - \varepsilon_b B_b}{\varepsilon_f + \varepsilon_b} \\ a_3 &= \rho s \end{aligned} \tag{A1}$$

where the reflection coefficient ρ means that a fraction ρ of the incident photons will be reflected, and of that fraction of photons a fraction of s will be specularly reflected, ε_f and ε_b are the differential area of the sail's front and back emissivities, respectively, and B_f and B_b are the front and back Lambertian coefficients, respectively.

B. The expressions of the force coefficients $\boldsymbol{\Gamma}^F$:

In the parameterized form of the total force, it is described by a series of coefficients in Eq.(30), and they are defined as

$$\begin{aligned}
&C_1^F=-\int_A a_1\hat{n}_1dA-\int_A a_3 2\hat{n}_1\hat{n}_1\hat{n}_1dA, C_2^F=-\int_A a_3 2\hat{n}_1\hat{n}_3\hat{n}_3dA,\\
&C_3^F=\int_A a_1\hat{n}_3dA+\int_A a_3 4\hat{n}_1\hat{n}_1\hat{n}_3dA, C_4^F=\int_A a_1\hat{n}_2dA+\int_A a_3 4\hat{n}_1\hat{n}_1\hat{n}_2dA,\\
&C_5^F=-\int_A a_3 4\hat{n}_1\hat{n}_2\hat{n}_3dA, C_6^F=\int_A a_2\hat{n}_1\hat{n}_1dA, C_7^F=-\int_A a_2\hat{n}_1\hat{n}_3dA,\\
&C_8^F=-\int_A a_3 2\hat{n}_1\hat{n}_2\hat{n}_2dA, C_9^F=-\int_A a_2\hat{n}_1\hat{n}_2dA
\end{aligned} \tag{B1}$$

$$\begin{aligned}
&D_1^F=-\int_A a_3 2\hat{n}_1\hat{n}_1\hat{n}_2dA, D_2^F=-\int_A a_3 2\hat{n}_2\hat{n}_3\hat{n}_3dA, D_3^F=\int_A a_3 4\hat{n}_1\hat{n}_2\hat{n}_3dA,\\
&D_4^F=\int_A a_1\hat{n}_1dA+\int_A a_3 4\hat{n}_1\hat{n}_2\hat{n}_2dA, D_5^F=-\int_A a_1\hat{n}_3dA-\int_A a_3 4\hat{n}_2\hat{n}_2\hat{n}_3dA,\\
&D_6^F=\int_A a_2\hat{n}_1\hat{n}_2dA, D_7^F=-\int_A a_2\hat{n}_2\hat{n}_3dA,\\
&D_8^F=-\int_A a_1\hat{n}_2dA-\int_A a_3 2\hat{n}_2\hat{n}_2\hat{n}_2dA, D_9^F=-\int_A a_2\hat{n}_2\hat{n}_2dA
\end{aligned} \tag{B2}$$

$$\begin{aligned}
&E_1^F=-\int_A a_3 2\hat{n}_1\hat{n}_1\hat{n}_3dA, E_2^F=-\int_A a_1\hat{n}_3dA-\int_A a_3 2\hat{n}_3\hat{n}_3\hat{n}_3dA,\\
&E_3^F=\int_A a_1\hat{n}_1dA+\int_A a_3 4\hat{n}_1\hat{n}_3\hat{n}_3dA, E_4^F=\int_A a_3 4\hat{n}_1\hat{n}_2\hat{n}_3dA,\\
&E_5^F=-\int_A a_1\hat{n}_2dA-\int_A a_3 4\hat{n}_2\hat{n}_3\hat{n}_3dA, E_6^F=\int_A a_2\hat{n}_1\hat{n}_3dA,\\
&E_7^F=-\int_A a_2\hat{n}_3\hat{n}_3dA, E_8^F=-\int_A a_3 2\hat{n}_2\hat{n}_2\hat{n}_3dA, E_9^F=-\int_A a_2\hat{n}_2\hat{n}_3dA
\end{aligned} \tag{B3}$$

Then, the coefficients can be written as a coefficient matrix $\boldsymbol{\Gamma}^F$ with the form of Eq.(32).

C. The expressions of the moment coefficients $\boldsymbol{\Gamma}^M$:

In the parameterized form of the total moment, it is described by a series of coefficients in Eq.(31), and they are defined as

$$\begin{aligned}
C_1^M&=\int_A 2a_3\left(-y\hat{n}_1\hat{n}_1\hat{n}_3+z\hat{n}_1\hat{n}_1\hat{n}_2\right)dA,\\
C_2^M&=\int_A a_1\left(-y\hat{n}_3\right)+2a_3\left(-y\hat{n}_3\hat{n}_3\hat{n}_3+z\hat{n}_2\hat{n}_3\hat{n}_3\right)dA\\
C_3^M&=\int_A a_1\left(y\hat{n}_1\right)+4a_3\left(y\hat{n}_1\hat{n}_3\hat{n}_3-z\hat{n}_1\hat{n}_2\hat{n}_3\right)dA,\\
C_4^M&=\int_A a_1\left(-z\hat{n}_1\right)+4a_3\left(y\hat{n}_1\hat{n}_2\hat{n}_3-z\hat{n}_1\hat{n}_2\hat{n}_2\right)dA,\\
C_5^M&=\int_A a_1\left(-y\hat{n}_2+z\hat{n}_3\right)+4a_3\left(-y\hat{n}_2\hat{n}_3\hat{n}_3+z\hat{n}_2\hat{n}_2\hat{n}_3\right)dA,\\
C_6^M&=\int_A a_2\left(y\hat{n}_1\hat{n}_3-z\hat{n}_1\hat{n}_2\right)dA,\\
C_7^M&=\int_A a_2\left(-y\hat{n}_3\hat{n}_3+z\hat{n}_2\hat{n}_3\right)dA,\\
C_8^M&=\int_A a_1\left(z\hat{n}_2\right)+2a_3\left(-y\hat{n}_2\hat{n}_2\hat{n}_3+z\hat{n}_2\hat{n}_2\hat{n}_2\right)dA\\
C_9^M&=\int_A a_2\left(-y\hat{n}_2\hat{n}_3+z\hat{n}_2\hat{n}_2\right)dA
\end{aligned} \tag{C1}$$

$$
\begin{aligned}
D_1^M &= \int_A a_1\left(-z\hat{n}_1\right)+2a_3\left(-z\hat{n}_1\hat{n}_1\hat{n}_1+x\hat{n}_1\hat{n}_1\hat{n}_3\right)dA,\\
D_2^M &= \int_A a_1\left(x\hat{n}_3\right)+2a_3\left(-z\hat{n}_1\hat{n}_3\hat{n}_3+x\hat{n}_3\hat{n}_3\hat{n}_3\right)dA,\\
D_3^M &= \int_A a_1\left(-x\hat{n}_1+z\hat{n}_3\right)+4a_3\left(z\hat{n}_1\hat{n}_1\hat{n}_3-x\hat{n}_1\hat{n}_3\hat{n}_3\right)dA,\\
D_4^M &= \int_A a_1\left(z\hat{n}_2\right)+4a_3\left(z\hat{n}_1\hat{n}_1\hat{n}_2-x\hat{n}_1\hat{n}_2\hat{n}_3\right)dA,\\
D_5^M &= \int_A a_1\left(x\hat{n}_2\right)+4a_3\left(-z\hat{n}_1\hat{n}_2\hat{n}_3+x\hat{n}_2\hat{n}_3\hat{n}_3\right)dA,\\
D_6^M &= \int_A a_2\left(z\hat{n}_1\hat{n}_1-x\hat{n}_1\hat{n}_3\right)dA,\\
D_7^M &= \int_A a_2\left(-z\hat{n}_1\hat{n}_3+x\hat{n}_3\hat{n}_3\right)dA,\\
D_8^M &= \int_A 2a_3\left(-z\hat{n}_1\hat{n}_2\hat{n}_2+x\hat{n}_2\hat{n}_2\hat{n}_3\right)dA,\\
D_9^M &= \int_A a_2\left(-z\hat{n}_1\hat{n}_2+x\hat{n}_2\hat{n}_3\right)dA
\end{aligned}
\tag{C2}
$$

$$
\begin{aligned}
E_1^M &= \int_A a_1\left(y\hat{n}_1\right)+2a_3\left(-x\hat{n}_1\hat{n}_1\hat{n}_2+y\hat{n}_1\hat{n}_1\hat{n}_1\right)dA,\\
E_2^M &= \int_A 2a_3\left(-x\hat{n}_2\hat{n}_3\hat{n}_3+y\hat{n}_1\hat{n}_3\hat{n}_3\right)dA,\\
E_3^M &= \int_A a_1\left(-y\hat{n}_3\right)+4a_3\left(x\hat{n}_1\hat{n}_2\hat{n}_3-y\hat{n}_1\hat{n}_1\hat{n}_3\right)dA,\\
E_4^M &= \int_A a_1\left(-y\hat{n}_2+x\hat{n}_1\right)+4a_3\left(x\hat{n}_1\hat{n}_2\hat{n}_2-y\hat{n}_1\hat{n}_1\hat{n}_2\right)dA,\\
E_5^M &= \int_A a_1\left(-x\hat{n}_3\right)+4a_3\left(-x\hat{n}_2\hat{n}_2\hat{n}_3+y\hat{n}_1\hat{n}_2\hat{n}_3\right)dA,\\
E_6^M &= \int_A a_2\left(x\hat{n}_1\hat{n}_2-y\hat{n}_1\hat{n}_1\right)dA,\\
E_7^M &= \int_A a_2\left(-x\hat{n}_2\hat{n}_3+y\hat{n}_1\hat{n}_3\right)dA,\\
E_8^M &= \int_A a_1\left(-x\hat{n}_2\right)+2a_3\left(-x\hat{n}_2\hat{n}_2\hat{n}_2+y\hat{n}_1\hat{n}_2\hat{n}_2\right)dA,\\
E_9^M &= \int_A a_2\left(-x\hat{n}_2\hat{n}_2+y\hat{n}_1\hat{n}_2\right)dA
\end{aligned}
\tag{C3}
$$

Then, the coefficients can be written as a coefficient matrix $\boldsymbol{\Gamma}^M$ with the form of Eq.(33).

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Applying Vector Scattering Theory to Solar-Photon Sail Thrust Modeling

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The Rayleigh-Rice Vector Scattering Theory is applied to the interaction between sunlight and sail's reflective layer, thus going beyond the scalar scattering theory already applied to solar-photon sailing. Diffuse-reflectance sailcraft acceleration is not negligible in accurate sailcraft trajectory propagation. The related values amount to about 20-110 times those ones due to the total solar irradiance fluctuations. The resultant of the diffuse-reflectance photon momenta per unit time and unit area, as function of the sail surface properties and sunlight incidence angle, exhibits many meaningful properties not previously reported in the solar-photon sailing literature.

I. Introduction

ONE of the fundamental problems in Solar-Photon Sailing (SPS) is an accurate modeling of thrust induced by (1) the solar irradiance, and (2) chiefly the diffraction of the electromagnetic (e.m.) waves impinging a real surface. There are various approaches used by different authors for considering sail's small/large curvature scales of a deployed sail. This is a geometric problem. However, the main problem is the physics of the interaction between sunlight and the sail's reflective metal layer. If one uses a very simple physical model, then the integrated thrust over an even accurately detailed surface will ultimately result in a rough approximation of the sailcraft's nominal acceleration history. Not only, SPS navigation, guidance and control, and the orbit determination process could benefit from a model appropriate to high-sensitivity missions. In addition, the analysis of the sail - viewed as a very large thin sheet under a distributed pressure - would utilize a model significantly more appropriate for trajectory design.

In the very vast literature of optical diffraction, scattering from *real* surfaces regards the distribution of the intensity/radiance[2]; much probably, there is no practical interest in photon momenta within the framework of a different-from-SPS research. In contrast, the reaction momenta to photons scattered by *non-ideal* metal surfaces, like those ones stemming from the sail making process, are meaningful components of the total thrust.

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[2] Here, radiometric quantities are meant according to the *International Commission on Illumination* (CIE in French).

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This paper regards the application of the full electromagnetic theory to the interaction between sunlight and sail material; here, we are concerned with the diffuse-reflectance thrust model. It is a further step - via Vector Scattering Theory (VST) - of a work initiated in [1], which goes beyond a scalar scattering theory (SST) applied to SPS [2]. The spectral and angular distributions of the scattered photon momenta depend on a basic quantity called the Bidirectional Reflectance Distribution Function (BRDF). In turn, in the limit of a sufficiently smooth sail surface, BRDF is an explicit function of four quantities: (a) the blue-sky term, (b) a theory-dependent geometric term, (c) the polarization factors, and (d) the surface's Power Spectral Density (PSD).

The paper is organized as follows: Sect. II reminds the reader the basic concept of BRDF; Sect. III regards the modeling of the surface of a photon sail; Sect. IV reports and comments on the polarization factors with regard to the sunlight-sail interaction. Once the above key concepts are grasped, the sail's reflectance photon momenta are dealt with in Sect. V. In the current framework, many numerical results regarding metal sails are discussed in Sect. VI, whereas the conclusions and the next steps of investigation are described in Sect. VII.

II. BRDF: a reminder

The concept of *reflectance* of a surface regards the specular-reflection *and* the diffuse reflection (often referred to as the scattered light by some authors). Both are expression of the diffraction of light[3]. Fig. 1 sketches the geometry of incident/scattered waves. An infinitesimal sail surface of area $\mathrm{d}A$ receives energy with input wave vector $\mathbf{k}_{(in)} = -k_{(in)}\,\mathbf{d}_{(source)}$ from the source direction $\mathbf{d}_{(source)} = -\mathbf{d}_{(in)}$ as observed in the local sail frame $F^{(sail)}$ (see Fig. 1). A fraction of this energy is reflected with wave vector $\mathbf{k}_{(re)} = k_{(re)}\,\mathbf{d}_{(re)}$, where $k_{(re)} = k_{(in)} = 2\pi/\lambda \equiv k_{\mathrm{e}}$ is the wave number. In the SPS case, $\mathbf{d}_{(source)} = \mathbf{d}_{\mathrm{e}}$ and $\mathbf{k}_{(in)} = -k_{\mathrm{e}}\,\mathbf{d}_{\mathrm{e}} \equiv \mathbf{k}_{\mathrm{e}}$. Considering the reflection in a general way, the definition of Bidirectional Surface-Scattering Reflectance Distribution Function (BSSRDF) states that the spectral radiance is proportional to the spectral irradiance:

$$L_{(re),\lambda} = \Upsilon \frac{\mathrm{d}\Phi_\lambda}{\mathrm{d}A} \quad \Leftrightarrow \quad \Upsilon = \frac{\mathrm{d}^2\Psi_{(re),\lambda} / \mathrm{d}\omega_{(re)}}{\cos\vartheta_{(re)}\,\mathrm{d}\Phi_\lambda}\,,\ \Upsilon = \Upsilon(\lambda,\, \mathbf{d}_{(in)},\, \mathbf{x}_{(in)},\, \mathbf{d}_{(re)},\, \mathbf{x}_{(re)},\, t,\, \{\mathrm{P}\}) \qquad (1)$$

Let us explain the relationships (1). L is the radiance induced by the spectral power $\mathrm{d}\Phi_\lambda$ impinging on the area $\mathrm{d}A$. The power reflected from this area into the solid angle $\mathrm{d}\omega_{(re)}$ about the direction $(\phi, \vartheta)_{(re)}$ (spherical

[3] From a basic viewpoint, scattering entails diffraction, and no diffraction phenomenon occurs without interference. Some authors distinguish diffraction from scattering in their specific research areas; here, we will use the terms scattering and diffraction interchangeably, even though our focus is on the diffuse reflection. Any process for measuring a scattered power (in the radiometric sense) will result in some average of the wavefront's very rapid fluctuations over a time much longer than the wave period(s).

angles) is expressed by $\mathrm{d}^3\Psi_{(re),\lambda} = L_{(re),\lambda}\cos\vartheta_{(re)}\,\mathrm{d}A\,\mathrm{d}\omega_{(re)}\,\mathrm{d}\lambda \equiv \mathrm{d}^2\Psi_{(re),\lambda}\,\mathrm{d}\lambda$ (by definition of spectralradiance). The set of parameters denoted by $\{\mathsf{P}\}$ is related to the scattering model.

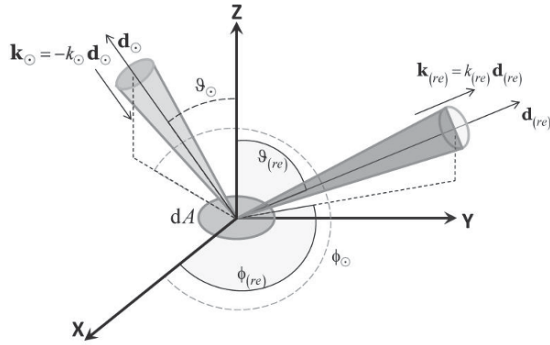

Fig. 1 Geometry for defining BRDF; $F^{(sail)}$ has the local tangent plane as the reference plane, and the normal to the surface in the same semispace of the sunlight incidence as the z-axis.

Function $\Upsilon(..)$ has units sr^{-1}; it is plain that it depends on wavelength and the incident/reflected directions. Instead, less obvious is that BSSRDF also depends on the coordinates of the impingement and exit points, which may be *different*; as a point of fact, in the surface material, sub-surface (multiple) scattering may occur so that the exiting wave may be located at some $\mathbf{x}_{(re)} = \mathbf{x}_{(out)} \neq \mathbf{x}_{(in)}$. Time dependence of Υ is ascribed to changes of the refractive index caused by radiation bombardment, chemical reaction, temperature variations, and so forth. Sails of the current generation have a metal film as the reflective layer: this is our baseline here. In addition, we will suppose that the optical quantities are time independent. Models for sail's optical degradation in the space environment are dealt with in [1]. Using a metal as reflecting layer allows one to ignore the sub-surface scattering. Thus, the BSSRDF concept turns into the Bidirectional Reflectance Distribution Function (BRDF):

$$L_{(re),\lambda} = \Lambda_\lambda \frac{\mathrm{d}\Phi_\lambda}{\mathrm{d}A} \quad \Leftrightarrow \quad \Lambda_\lambda = \frac{\mathrm{d}^2\Psi_{(re),\lambda} / \mathrm{d}\omega_{(re)}}{\cos\vartheta_{(re)}\,\mathrm{d}\Phi_\lambda}$$
$$\Lambda_\lambda = \frac{\mathrm{d}F_{(re),\lambda}}{\cos\vartheta_{(re)}\,\mathrm{d}\Phi_\lambda} \cong \frac{\Psi_{(re),\lambda}}{\omega_{(re)}\cos\vartheta_{(re)}\,\Phi_\lambda} \tag{2}$$

In Eqs. (2), BRDF is denoted by Λ_λ and depends on wavelength, the incident and diffraction directions, and $\{\mathsf{P}\}$.Note that $\mathrm{d}F_{(re),\lambda} = L_{(re),\lambda}\,\mathrm{d}A_\perp$ is the (differential) spectral radiant *intensity*that emerges from the projected area $\mathrm{d}A_\perp = \cos\vartheta_{(re)}\,\mathrm{d}A$. The rightmost expression in the second of Eq. (2) is utilized for defining BRDF experimentally [3] because it focuses on finite-area samples and finite solid-angle sensors. The spectral

bidirectional reflectance, the spectral and total directional-hemispherical reflectances follow from the BRDF [4] respectively as

$$\mathrm{d}R_\lambda = \Lambda_\lambda \cos\vartheta_{(re)}\, \mathrm{d}\omega_{(re)}\ , \qquad R_\lambda = \int_{2\pi\ \mathrm{sr}} \mathrm{d}R_\lambda$$
$$R = \int_{\lambda_{min}}^{\lambda_{max}} R_\lambda \Phi_\lambda \,\mathrm{d}\lambda \ / \int_{\lambda_{min}}^{\lambda_{max}} \Phi_\lambda \,\mathrm{d}\lambda \tag{3}$$

One should note that any practical total reflectance is (1) a *broadband* reflectance as the integral's bounds are different from zero and ∞ , and (2) relative to the spectral power actually received by $\mathrm{d}A$, which for the SPS case becomes the Spectral Solar Irradiance (SSI).

Remark 1: Some authors are used to utilize the concept of Angle-Resolved Scattering (ARS): one has easily ARS= $\Lambda^{\mathrm{j}}_\lambda \cos\vartheta_{(re)}$ where $\Lambda^{\mathrm{j}}_\lambda$ denotes the *diffuse* part of BRDF. In addition, according to ISO 13696, one should use the term *backscattering* referred to the power scattered (from a surface) into the same hemisphere of the incident beam(s). However, *here*, this might cause some confusion because a component of thrust comes from polarized waves backscattered or retro-reflected, i.e. diffused antiparallely to the incidence wave vector. We will keep this second convention in this paper.

Remark 2: What could the BRDF role be in SPS? For modeling thrust accurately, one has to know the distribution of the photon momenta in order to calculate the vector sum in the sail frame, and then to transform this vector to the reference frame where the sailcraft motion should be described. The photon momentum (per unit wavelength, unit area, and unit time) in the infinitesimal solid angle $\mathrm{d}\omega_{(re)}$ is proportional to Λ_λ . We will make BRDF explicit in subsection V-A via a vector theory of diffraction. The conditions, under which the surface-light interaction models have been developed for computer graphics, do *not* hold for the nanometric topography of a solar sail, which is irradiated by and sensitive to a bandwidth covering two orders of magnitude in wavelength. In computer graphics, geometric optics can be applied because micro-facet areas are assumed much larger than λ^2, and the incident-light band is practically the visible one, e.g. see [5] and Section 6.4 of [1].

III. Solar-Sail Surface Modeling

The role that topography plays in the interaction between photons and sail surface is particularly important. In a thrust model, one should conceptually distinguish between the (macroscopic) geometric problem and the (microscopic) physical problem. The former regards the medium/large scale of the surface (i.e. sail curvature and wrinkling, essentially), whereas the latter is strongly affected by the small-scale structure of the sail surface. Let

us consider an elementary surface of the whole sail; if such element were *ideally* smoothed, then the direction of the locally reflected photons at the interface vacuum/reflective-layer would be known. A reflectance complication would be caused since, in principle, each infinitesimal surface may have a different index of refraction coming from variable density and/or composition, or imperfections, just to cite a few. If the related scattering modes were negligible, then the problem of thrust computation would be solved well enough, and the only demanding task would be to integrate the differential thrust over the surface shape (which is an output of the mechanical sail design subsystem). However, a real-world surface is far to be geometrically ideal, and a difficult problem of diffraction arises. This could be further complicated by sub-surface non-local scattering, and obviously by the vacuum-film-substrate interface reflection if some transparency takes place.

The following assumptions/considerations about photon-sail interaction are central points in our analysis:

i. Though a general surface may exhibit a BRDF stemming from the contributions of various distributed and/or localized scatterers, we suppose that topographic features are dominant throughout the sail surface.

ii. The field scattered from a stochastically modeled surface is itself stochastic. In particular, the field supposed coming from a regular stochastic surface is regular, and so the corresponding scattered radiance[4].

iii. Sunlight wavelengths (from middle UV up) and the reflective-layer roughness of a solar sail satisfy the criterion that the ratio of the root mean square roughness on wavelength is sufficiently less than unity throughout the sunlight's chosen band. This means that the 2nd-order moment of the diffracted radiance could be considered sufficiently small in order to apply various diffraction theories. Simple polarization modes are important for getting an accurate BRDF; if sail is smooth, then one gets analytical expressions of the wave's polarization states (Section IV).

iv. Sail surface's local height $z(x,y)$, with respect to the experimental zero-level or the mean surface level [6], is described by a two-dimensional Gaussian distribution $P(Z_1,Z_2)$[5] with auto-correlation function given by

$$\tilde{C}(x,y)=\tilde{z}^2 C(x,y)=\tilde{z}^2\exp\left(-(x^2+y^2)/\tilde{l}^2\right) \tag{4}$$

[4] Regular stochastic processes have probability densities *prescribed*. They differ from *predictable* stochastic processes that are modeled via parameters described by probability densities.

[5] Z_1 and Z_2 are the random height values at any two points separated by $x_2-x_1\equiv x$ and $y_2-y_1\equiv y$.

where $\mathsf{C}(x,y)$ is the auto-correlation coefficient, and $\tilde{l}$ is named the auto-correlation length. In the framework of surface modeling, $\tilde{z}$ is known as the root-mean-square roughness (or *rms* roughness/height, for short), because $\tilde{z}^2$ is the variance of the zero-mean marginal (Gaussian) probability density of either Z_1 or Z_2.

For characterizing a surface independently of the employed measuring technique, we need concepts that focus on both the surface shape and the "probing objects", namely, e.m. waves (or, better, a photon flow). The appropriate concepts are (*a*) the spatial frequency, and (*b*) the (area) Power Spectral Density (PSD), e.g. [3, 6, 7]. Spatial frequencies are the components of the difference between the diffracted wave vector and the incident wave vector: for SPS, this frequency vector is equal to $\Delta\mathbf{k} \equiv \mathbf{k}_{(re)} - \mathbf{k}_{\mathsf{e}} = \mathbf{k}_{(re)} + k_{\mathsf{e}}\mathbf{d}_{\mathsf{e}}$, according to Fig. 1. In diffraction theories, the incidence plane is usually taken as the *xz*-plane of $F^{(sail)}$ where the *z*-axis is along the surface's local normal (in the incidence semispace), and the *x*-axis is such that $\phi_{\mathsf{e}} = \pi$. The *y*-axis completes the positive triad. The components of $\Delta\mathbf{k}$ all enter the phase difference between diffracted waves and incident waves. Particularly important here are the *xy*-components:

$$\mathbf{f} \equiv \begin{pmatrix} \sin\vartheta_{(re)}\cos\phi_{(re)} - \sin\vartheta_{\mathsf{e}} \\ \sin\vartheta_{(re)}\sin\phi_{(re)} \end{pmatrix} / \lambda \tag{5}$$

Thus $z(x,y)$ can be viewed as a spatial-evolution stochastic process (assumed wide-sense stationary here) brought about mainly by the surface manufacturing, which is part of the sail making. It is responsible for scattering features different from the well-behaved ones related to precise and clean diffraction grating. We are interested in the $z(x,y)$-induced diffraction pattern average as a function of the scattering angles. The area PSD is defined as

$$\begin{aligned} &\mathrm{PSD} \equiv \mathsf{S}_2(\mathbf{f}) = \lim_{\ell\to\infty}\frac{1}{\ell^2}\left\langle \left| \int_{-\frac{1}{2}\ell}^{\frac{1}{2}\ell} \mathrm{d}y \int_{-\frac{1}{2}\ell}^{\frac{1}{2}\ell} z(x,y)\exp\left(i\,2\pi\mathbf{f}\cdot\mathbf{r}\right)\mathrm{d}x \right|^2 \right\rangle \\ &\Delta A = \ell^2\,, \qquad \mathbf{r} = \begin{pmatrix} x \\ y \end{pmatrix} \end{aligned} \tag{6}$$

where i is the imaginary unit, $\langle\ \rangle$ denotes the averaging process, and the spatial frequencies are given by Eq. (5). If the area ΔA of the sample is sufficiently large, then the actual PSD can be considered coincident with that of an infinite area. As it is clear from the definition-(6), power spectral density is the squared modulus of the Fourier-transform of the shape $z(x,y)$, which in SPS is the interface between the vacuum and the layer reflecting sunlight. The units of area PSD are length[4].

Note 1: Definition (6) is appropriate for the $z(x,y)$ process. As for any wide-sense stationary process, power spectral density can be defined and obtained via the Fourier transform of the auto-correlation function.

Note 2: PSD is symmetric with respect to spatial frequency, and is a property of the sampled surface, not a measurement method. However, there is the problem to relate PSD *univocally* to BRDF. For a wide and authoritative discussion, see [3]; a smooth surface allows solving the Inverse Scattering Problem (ISP).

In our context of SPS, particularly important is the concept of *isotropic* surface, the PSD of which depends on the magnitude $\|\mathbf{f}\| \equiv f$ of the spatial frequency vector. What does this mean physically? One aspect lies on Eq. (4) that contains only one auto-correlation length. More important is that *all* information on its spectrum can be got via measurements of scattering in the incidence plane[6]: in other words, an isotropic surface entails that its PSD does not depend on the azimuth of the reflected wave direction. If the isotropic surface were probed by a stylus instrument, then the same roughness would be measured throughout. One should realize that measurements are always carried out in a finite range of spatial frequencies; as a result, the volume under the function $\mathsf{S}_2(f)$ in the (f_x, f_y, S_2) space is no longer equal to the variance of $P(Z_1, Z_2)$:

$$\begin{aligned} &\tilde{z}_{eff}^2 = \int_0^{2\pi} \mathrm{d}\phi_{(re)} \int_{f_{min}}^{f_{max}} \mathsf{S}_2(f)\, f\, \mathrm{d}f = 2\pi \int_{f_{min}}^{f_{max}} \mathsf{S}_2(f)\, f\, \mathrm{d}f \;<\; \tilde{z}^2 \\ &0 \le f_{min} < f_{max} < \infty \;, \qquad f \in [f_{min}\,,\, f_{max}] \end{aligned} \tag{7}$$

where $\tilde{z}_{eff}$ is the *effective* value of the *rms* roughness due to the finite bandwidth of the sampling spatial frequencies; it is not an intrinsic property of the surface, but a band-limited quantity. If we use Eq. (4) for modeling a sail surface, then

$$\mathsf{S}_2(f) = \mathrm{FT}\left[\tilde{z}^2 \exp\left(-(x^2+y^2)/\tilde{l}^2\right)\right] = \pi\left(\tilde{z}\,\tilde{l}\right)^2 \exp\left[-\left(\pi\,\tilde{l}\,f\right)^2\right] \tag{8}$$

Inserting Eq. (8) into Eq. (7), and integrating, one gets

$$\tilde{z}_{eff} = \sqrt{\exp\left[-\left(\pi\,\tilde{l}\,f_{min}\right)^2\right] - \exp\left[-\left(\pi\,\tilde{l}\,f_{max}\right)^2\right]}\;\tilde{z} \tag{9}$$

We will see later that the square root value in Eq. (9) for the (currently conceived) smooth sails is close to 1.

Note 3: When comparing measurements from different instruments, it is fundamental that the lower and upper limits of the spatial bandwidth are not modified. This shall hold even in analyzing the samples representative of the surface of the sail that will be actually launched.

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[6] In the *isotropic* case, the two-dimensional $\mathsf{S}_2(f)$ can be obtained from the one-dimensional spectral density $\mathsf{S}_1(f_x)$ via the integral transform known as the inverse Abel transform; the *vice versa* can be done via the Abel transform, e.g. [3,7].

At Eq. (4), we have pointed out that the surface height probability density contains an additional important parameter, i.e. $\tilde{l}$. The quantity $z(r)$ is a scalar Gaussian stochastic process, and is *mean square* differentiable. The scalar stochastic process $\mathsf{p} = \mathrm{d}z(r)/\mathrm{d}r$ is Gaussian as well; $\mathsf{p}(r)$ represents how the surface slope varies with the radial coordinate, and has the following statistics:

$$\langle \mathsf{p} \rangle = 0 \ , \qquad \tilde{\mathsf{p}} = \sqrt{\langle \mathsf{p}^2 \rangle} = \sqrt{2}\ \tilde{z} / \tilde{l} \equiv \tan\xi \tag{10}$$

$\tilde{\mathsf{p}}$ is named the *rms* slope. We will see the quantitative importance of $\tilde{l}$ for the SPS thrust calculation in Section VI. The here-defined slope angle ξ will be useful for arranging the numerical results.

IV. Polarization in SPS

There are many optical diffraction theories, progressively developed since the 19th century, though a first qualitative explanation of the diffraction phenomenon can be traced some centuries back to Leonardo da Vinci. Some theories are exact in their conceptual nature, namely, they quite rely on the Maxwell equations. Physical and geometric configurations enter the formulation of a theory of diffraction, according to the class of problems one wants to investigate and subsequently apply to the real-world situations [8]. In the last six decades, theories such as the (source-based) Physical Theory of Diffraction (PTD), and the Geometric Theory of Diffraction (GTD) - with its modified versions known as the Uniform Theory of Diffraction (UTD) and the Uniform Asymptotic Theory of Diffraction (UATD) - have been set up. They have been succeeding in high-tech applications (from intercommunication of cellular telephones to special aircraft, and in the diffraction from non-optical objects), and added to more classical or recent theories such as the Rayleigh-Rice Theory, the (modified) Beckmann-Kirchhoff Theory, and the (generalized) Harvey-Shack Theory. Some authors group diffraction theories in two large sets: (1) Rigorous Diffraction Theories (RDTs), and (2) Analytic Diffraction Theories (ADTs). RDTs are very few, and entail a very large amount of calculation for 2D or 3D scatterers. In addition, the ISP is not allowed via such theories. ADTs start from the Maxwell equations as well, but the explicit scattering results are obtained by some approximation(s). Such theories are not particularly complicated to implement, and many offer solution to ISP. We are remarking these items in the two sets of diffraction theories because two aspects of SPS are decisive here for selecting either an RDT or ADT for thrust modeling. They are: (*a*) the capability to be employed in *pre*-launch ISP, and (*b*) avoiding unnecessary complications in implementing the thrust equation in the fundamental *post*-launch trajectory determination system. Point-(*a*) is particularly important inasmuch as the surface parameters shall be measured on representative samples of the

(large) sail *to be flown* in order to re-compute the final sailcraft acceleration profile before the mission begins. In general, point-(b) requires a certain computation rapidness; for the SPS case, in particular, quickness is important because this propulsion mode is limited in acceleration, and many parameters have to be estimated, including those ones related to the sail's optical degradation and the variability of the solar irradiance (total and spectral).

In the SPS case, each small area of a large and thin reflective layer is a scatterer: Aluminium, nowadays, or variable-reflectance liquid crystals, and/or Carbon nanotube textures in future sailcraft. From a practical viewpoint, SST does not consider polarization of light, whereas VST does. If one is interested in the solar-sail thrust modeling, then one should include the four polarization factors $\perp\to\|$, $\perp\to\perp$, $\|\to\perp$, $\|\to\|$. Sunlight incoherence is considered via the semi-summation of these terms. The factor $\|\to\|$ from backscattering is particularly important like the radar effect on metal targets [9]. The following formulas provide the polarization factors for any smooth surface of a non-magnetic material:

$$a_s \equiv |\varepsilon-1|^2 \sin^2\phi_{(re)}\,,\quad a_c \equiv |\varepsilon-1|^2\cos^2\phi_{(re)}\,,\quad b_\odot \equiv \sqrt{\varepsilon-\sin^2\vartheta_\odot}\,,\quad b_{(re)} \equiv \sqrt{\varepsilon-\sin^2\vartheta_{(re)}}$$

$$\mathcal{Q}_{\perp\to\perp} = \frac{a_c}{\left|\left(\cos\vartheta_{(re)}+b_{(re)}\right)\left(\cos\vartheta_\odot+b_\odot\right)\right|^2}\;,\quad \mathcal{Q}_{\perp\to\|} = \frac{a_s\left|b_{(re)}\right|^2}{\left|\left(\varepsilon\cos\vartheta_{(re)}+b_{(re)}\right)\left(\cos\vartheta_\odot+b_\odot\right)\right|^2} \tag{11}$$

$$\mathcal{Q}_{\|\to\perp} = \frac{a_s\left|b_\odot\right|^2}{\left|\left(\varepsilon\cos\vartheta_\odot+b_\odot\right)\left(\cos\vartheta_{(re)}+b_{(re)}\right)\right|^2}\;,\quad \mathcal{Q}_{\|\to\|} = |\varepsilon-1|^2\left|\frac{\cos\phi_{(re)}\,b_{(re)}\,b_\odot-\sin\vartheta_{(re)}\sin\vartheta_\odot\,\varepsilon}{\left(\varepsilon\cos\vartheta_{(re)}+b_{(re)}\right)\left(\varepsilon\cos\vartheta_\odot+b_\odot\right)}\right|^2$$

where ε denotes the dielectric function, and a factorization has been accomplished for showing symmetries and reducing the number of irrational/transcendental complex operations. Each spectral bidirectional reflectance per unit solid angle, i.e. $\left(\mathrm{d}\mathcal{R}_\lambda\right)_{a\to b} / \mathrm{d}\omega_{(re)}$ (differential reflectance), is referred to the incident irradiance *corresponding* to the given polarization (either $\|$ or $\perp$). As shown by Eqs. (11), polarization is a function of the dielectric function of the scatterer, not of its topography. If the incident light is not polarized, then the sail's non-magnetic metal material may scatter it polarized partially; in the plane of incidence, specular or diffuse reflection leaves the incident polarization unchanged. If a detector receives unpolarized light and is insensitive to polarization, then the overall polarization factor (e.g. see [3, 10]) can be expressed by:

$$\mathcal{Q}_\odot = \tfrac{1}{2}\left(\mathcal{Q}_{\perp\to\|}+\mathcal{Q}_{\perp\to\perp}+\mathcal{Q}_{\|\to\perp}+\mathcal{Q}_{\|\to\|}\right) \tag{12}$$

A general solar sail is assumed to be in such conditions and Eq. (12) applies to thrust calculation. Let us see whether some $\mathcal{Q}_\odot$-approximation may be useful for calculating the photon momenta stemming from the diffuse reflectance. On the other side, if one puts (independent) cutoffs on the incidence and the scattering angle (e.g.

75°, 80°, depending on the reflective material), one can carry out a meaningful polarization simplification that does not include the ε-function. In this paper, we will follow this approach; the general one will be dealt with in a next paper (with a remarkably more complicated numerical analysis). Therefore, in the limit of a high ε, but with angular cutoff values on incidence/scattering angles not too close to $\pi/2$, is an easy task to carry out the ensuing polarization factors. We have used a prime for denoting such approximation:

$$\begin{aligned} \mathcal{Q}'_{\perp\to\perp} &= \cos^2\phi_{(re)} \quad , \qquad \mathcal{Q}'_{\perp\to\parallel} = \left(\frac{\sin\phi_{(re)}}{\cos\vartheta_{(re)}}\right)^2 \\ \mathcal{Q}'_{\parallel\to\perp} &= \left(\frac{\sin\phi_{(re)}}{\cos\vartheta_{\odot}}\right)^2 \quad , \qquad \mathcal{Q}'_{\parallel\to\parallel} = \left(\frac{\cos\phi_{(re)} - \sin\vartheta_{\odot}\,\sin\vartheta_{(re)}}{\cos\vartheta_{\odot}\,\cos\vartheta_{(re)}}\right)^2 \end{aligned} \tag{13}$$

Of course, the formula corresponding to Eq. (12) is

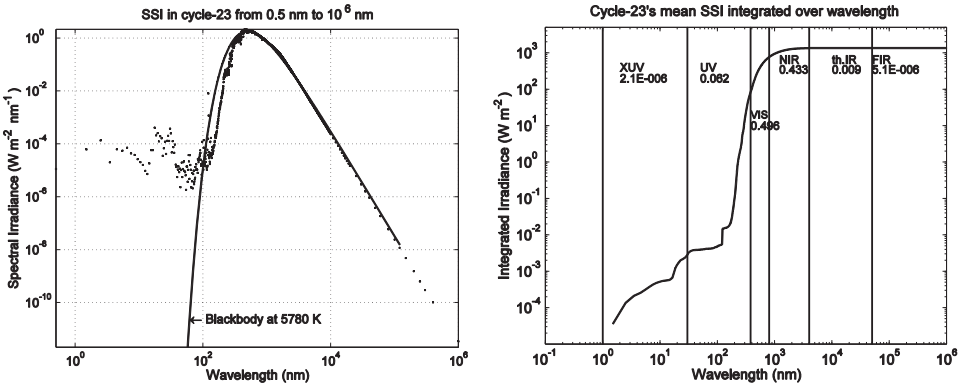
$$\mathcal{Q}'_{\odot} = \tfrac{1}{2}\left(\mathcal{Q}'_{\perp\to\parallel} + \mathcal{Q}'_{\perp\to\perp} + \mathcal{Q}'_{\parallel\to\perp} + \mathcal{Q}'_{\parallel\to\parallel}\right) \tag{14}$$

This formula will be employed below in the integrand related to the total momentum of photons diffusely reflected by a metal solar sail.

V. Sail Reflectance Photon Momenta

Sail-making technology is an advanced one in any case. The above assumption (iii) about sail smoothness regards a broadband from middle UV (MUV) to the thermal or middle IR (MIR). SSI differs from the solar blackbody spectrum best fit just in the low wavelength region, as shown on the left of Fig. 2; on its right, the integrated SSI is shown as function of wavelength. From the histogram of the measured TSI over solar cycles 21-23 (Chapter 3 of [1]), one gets that the TSI mean amounts to 1365.94 W/m^2, while the standard deviation is as low as 0.56 W/m^2, i.e. TSI fluctuations are of the order of 1-2 W/m^2. Now, depending on the trajectory sensitivity, Ref. [11] explained how such fluctuations may not allow the mission success if were neglected.

If we consider the SSI broadband, say, from 0.2 to 10 μm, we capture the quasi-totality of TSI, leaving out an amount of the order of the TSI variability. In addition, λ_{min} should be compliant with the Aluminum layer thickness, or 90-100 nm. Though (to the author knowledge) no surface parameters about IKAROS and NanoSail-D2 sails have been published hitherto, one should expect an *rms* roughness no more than 1/5 of the reflecting layer thickness, or 15-20 nm at most, for avoiding roughness-induced absorption.



a) b)

Fig. 2 (a): SSI in solar cycle-23 (dots) and the irradiance at 1AU caused by the Sun considered as a 5780K blackbody (solid line). **(b):** SSI-23 integrated over wavelength. From [1], courtesy of Springer

A. BRDF made explicit

Rayleigh-Rice theory, or the RR-theory here, is a VST that holds for smooth surfaces, and works through even large incidence/scattering angles. Another important experimental confirmation of it can be found in [12]. One should note that a sail of *rms* height of 20 nm might not be considered smooth with respect to the MUV band wavelengths (200-300 nm). However, in the NUV range (300-380 nm), the ratio $\tilde{z}/\lambda$ does not exceed 0.067. In the sequel, we will consider 10-15 nm as a realistic range for a very large surface. From the visible band up, this sail's reflective layer can be treated as a smooth surface. Consequently, considering that the RR-theory is not complicated to implement, we start with it for dealing with the current physical problem in the mentioned broadband 0.2-10 μm. Accordingly, the sunlight diffuse-reflection BRDF can be then written as

$$\Lambda_{\lambda}^{\mathrm{j}} = \frac{16\pi^2}{\lambda^4}\left(\cos\vartheta_{\mathrm{e}}\,\cos\vartheta_{(re)}\right) \mathrm{Q}\;\mathrm{S}_2(\mathbf{f}) = \frac{16\pi^2}{\lambda^4}\left(\cos\vartheta_{\mathrm{e}}\,\cos\vartheta_{(re)}\right) \mathrm{Q}'\;\mathrm{S}_2^{(iso)}(f) \tag{15}$$

The second expression in Eq. (15) refers to as an isotropic surface with high dielectric-function modulus, i.e. Q' is given by Eqs. (13)-(14), and PSD is provided by Eq. (8). Since $\mathrm{S}_2^{(iso)}$ has rotational symmetry, then we can consider only the incidence plane and get explicitly

$$\begin{aligned}\mathrm{S}_2^{(iso)} &= \pi\left(\tilde{z}\tilde{l}\right)^2 \exp\left[-\left(\frac{\pi\tilde{l}}{\lambda}\right)^2\left(\sin\vartheta_{(re)}-\sin\vartheta_{\mathrm{e}}\right)^2\right] \\ &= 2\pi\,\frac{\tilde{z}^4}{\tilde{\mathrm{p}}^2}\exp\left[-2\left(\frac{\pi\tilde{z}}{\tilde{\mathrm{p}}\lambda}\right)^2\left(\sin\vartheta_{(re)}-\sin\vartheta_{\mathrm{e}}\right)^2\right]\end{aligned} \tag{16}$$

Note that, apart from the obvious case $\tilde{z} = 0$, the other ways to strictly zeroing the isotropic-surface Gaussian covariance PSD are to have $\tilde{l} = 0$ or $\tilde{l} \to \infty$. Eq. (16) expresses the clear and important angular shift invariance; the second line visualizes easily how the current PSD varies with the *rms* height for given surface *rms* slope and spatial frequency. For a given spatial frequency f, there is one value of $\tilde{\rho}$ that maximizes PSD:

$$\left(\tilde{\rho}\,,\,\mathrm{PSD}\right)_{max} = \left(\pi\sqrt{2}\,\tilde{z}\,f\;,\;\tilde{z}^2 / e\,\pi\,f^2\right) \tag{17}$$

Property (17) is simple whenever the incident waves may be considered monochromatic. However, for a broadband beam with complicated spectral irradiance (like the sunlight on a sail), the only way to know the overall effect is to integrate numerically (and to verify experimentally, of course). This observation will be useful for understanding the results from the numerical analysis below.

B. Diffuse Reflectance Momenta

Let us turn to the total momentum of the photons diffused by such a surface type. Recalling the meaning of the above quantity $\mathrm{d}^3\Psi_{(re),\lambda}$, the momentum of photons in the infinitesimal band $[\lambda, \lambda + \mathrm{d}\lambda]$ scattered by the surface $\mathrm{d}A$ into the solid angle $\mathrm{d}\omega_{(re)}$ during the time interval $[t,\, t + \mathrm{d}t]$, is expressed by

$$\begin{aligned} \mathrm{d}^4\mathbf{Q}_{(re)} &= \frac{I_\lambda}{c}\left(\mathrm{d}A\;\mathrm{d}\lambda\;\mathrm{d}t\right)\mathrm{d}R^{\mathrm{j}}_{\lambda}\;\mathbf{d}_{(re)} = \frac{I_\lambda}{c}\left(\mathrm{d}A\;\mathrm{d}\lambda\;\mathrm{d}t\right)\left(\Lambda^{\mathrm{j}}_{\lambda}\cos\vartheta_{(re)}\,\mathrm{d}\omega_{(re)}\right)\mathbf{d}_{(re)} \\ &= \frac{\mathrm{d}A\;\mathrm{d}t}{c}\;\;I_\lambda\,\mathrm{d}\lambda\;\;\Lambda^{\mathrm{j}}_{\lambda}\cos\vartheta_{(re)}\begin{pmatrix}\sin\vartheta\cos\phi\\ \sin\vartheta\sin\phi\\ \cos\vartheta\end{pmatrix}_{(re)}\mathrm{d}\omega_{(re)} \end{aligned} \tag{18}$$

Here the symbol I_λ denotes the spectral irradiance, and again (ϕ, ϑ) are the classical spherical angles with ϑ as the colatitude. Note that $\Lambda^{\mathrm{j}}_{\lambda}$ is the *diffuse* BRDF given by Eq. (15). In order to get the corresponding thrust density, we have to integrate $\mathrm{d}^4\mathbf{Q}_{(re)}$ over solid angle and wavelength, and to divide by $\mathrm{d}A\,\mathrm{d}t$ (or viceversa). Thus, one obtains

$$\begin{aligned} &\mathbf{D} \equiv \frac{\iint_{\lambda,\omega}\mathrm{d}^4\mathbf{Q}_{(re)}}{\mathrm{d}A\;\mathrm{d}t} = \frac{\mathrm{d}^2\mathbf{Q}_{(re)}}{\mathrm{d}A\;\mathrm{d}t} = \frac{1}{c}\int_{\lambda_{min}}^{\lambda_{max}} I_\lambda \mathrm{d}\lambda \int_{\overline{\omega}_{(re)}} \Lambda^{\mathrm{j}}_{\lambda}\;\mathbf{d}_{(re)}\cos\vartheta_{(re)}\,\mathrm{d}\omega_{(re)} \\ &\mathrm{d}\omega_{(re)} = \sin\vartheta_{(re)}\,\mathrm{d}\phi_{(re)}\,\mathrm{d}\vartheta_{(re)}\;, \qquad \overline{\omega}_{(re)} = 2\pi\left(1-\cos\overline{\vartheta}_{(re)}\right), \qquad 0 \le \overline{\vartheta}_{(re)} \le \pi/2 \end{aligned} \tag{19}$$

In Eq. (19), we introduced a cutoff value $\overline{\vartheta}_{(re)}$ on the scattering angle according to the considerations about the high-ε polarization formulas usage. In order to highlight the triple integral, we re-arrange the vector **D**:

$$\mathbf{D} = \frac{1}{c}\int_{\lambda_{min}}^{\lambda_{max}} \frac{\mathrm{SSI}}{R^2}\,\mathbf{G}\,\mathrm{d}\lambda\ , \qquad \mathbf{G} = 16\pi^3\left(\tilde{z}\,\tilde{l}/\lambda^2\right)^2\,\mathbf{G}_1$$
$$\mathbf{G}_1 = \int_0^{\bar{\vartheta}_{(re)}} \sin\vartheta_{(re)}\cos^2\vartheta_{(re)}\exp\left[-\left(\pi\tilde{l}/\lambda\right)^2\left(\sin\vartheta_{(re)} - \sin\vartheta_e\right)^2\right]\mathbf{G}_2\,\mathrm{d}\vartheta_{(re)} \tag{20}$$
$$\mathbf{G}_2 = \cos^2\vartheta_e\int_0^{2\pi} Q'\,\mathbf{d}_{(re)}\,\mathrm{d}\phi_{(re)}$$

after noting that, by definition of SSI, $I_\lambda = \mathrm{SSI}\,\cos\vartheta_e\ /\ R^2$ where R is the Sun-sailcraft distance expressed in Astronomical Units (AU). Of course, if one substitutes the spectral irradiance at 1 AU due toa 5780K-blackbody for I_λ, one can compare pairs (blackbody, observed SSI) of vectors **D** as function of the surface model parameters and the solar incidence angle. Q' is the expression of polarization from the high dielectric-function approximation.Equations (20) contain nine scalar integrals with $\mathbf{G}_2$ obtained by quadrature:

$$\mathbf{G}_2 = \pi\begin{pmatrix} -\tan^2\vartheta_{(re)}\sin\vartheta_e \\ 0 \\ \sin\vartheta_{(re)}\tan\vartheta_{(re)}\ \sin^2\vartheta_e + \frac{1}{2}(\cos\vartheta_{(re)} + \sec\vartheta_{(re)})(1+\cos^2\vartheta_e) \end{pmatrix} \tag{21}$$

Equations (20) and (21) show clearly that vector **D** will not be orthogonal to the local sail surface – as implied by SST – unless the sunlight is on perfect normal incidence. The always-zero y-component of $\mathbf{G}_2$, $\mathbf{G}_1$ and **D** is due to the assumption of isotropic surface. Calculating $\mathbf{G}_1$ is not very simple. If one employs some *ad-hoc* approximating expression(s) for the integrand, then this integral may be obtained in terms of trigonometric/exponential functions, and special functions, which are available directly in modern computer algebraic systems; however, even in such cases, a large number of terms have to be computed. Searching for a strict closed-form expression of $\mathbf{G}_1$ has resulted in no success. For any realistic SSI, a robust numerical integration should be performed in any case. This is the policy followed in the next section, where the number of numeric double integrals to be carried out are therefore two.

VI. Numerical Results

The $\mathbf{G}_2$ expression has to be inserted into Eq. (20), then one has to compute a double integral numerically with respect to $\left(\vartheta_{(re)}, \lambda\right)$, if the SSI is given and the surface parameters, namely, $\left(\tilde{z}, \tilde{l}\right)$ or $\left(\tilde{z}, \tilde{\rho}\right)$ are given. Here, we will to assign the *rms* slope and rms height for a better visualization of the results. Tables of figures are built; each entry is a figure that shows how the vector **D** varies with the sunlight incidence angle, once the pair $\left(\tilde{z}, \tilde{\rho}\right)$ is prefixed together with the SSI type, the bandwidth, and the scattering and incidence cutoff angles; the

last two items determine the range of spatial frequencies in the diffraction theory here, namely, the RR-theory. Each matrix of figures is characterized by a prefixed $\tilde{z}$ and a number of chosen values of ξ. The models of SSI that we have considered are: (1) the spectral irradiance at 1 AU caused by a blackbody of 5780 K, also denoted by Sun-5780 here, and (2) the mean spectral solar irradiance (i.e. averages of daily means) in the solar cycle 23, or SSI-23 for short, as shown in Fig. 3.

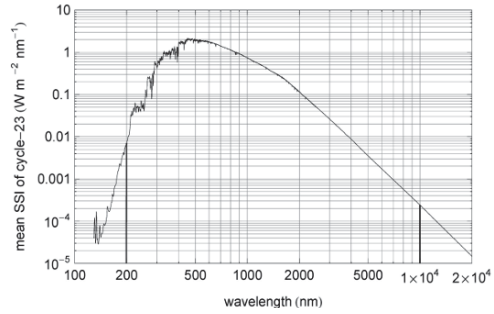
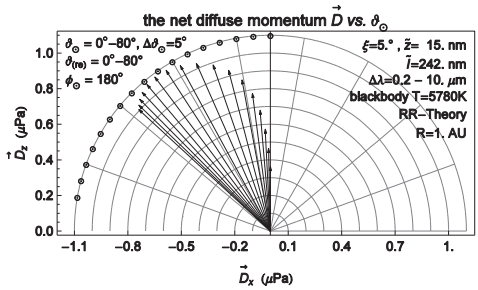
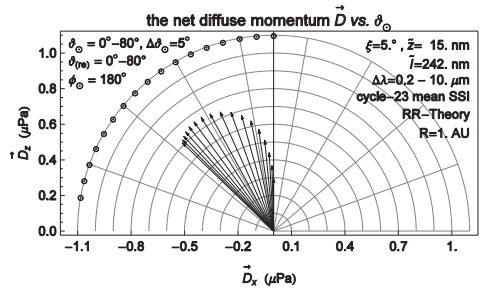


Fig. 3 Mean spectral solar irradiance (averages of daily means in the solar cycle-23). The vertical thick lines represent the bounds of the SSI-23 bandwidth chosen for computing the vector **D** (Section V).

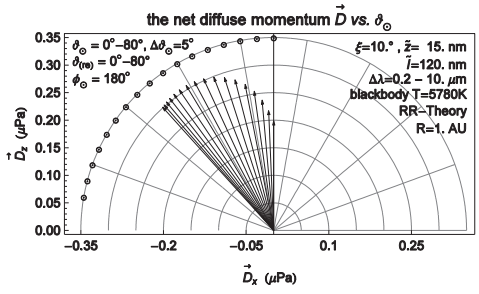
From what said in Sects. IV and V-A, and with an incidence/scattering cutoff of 80°, one gets the spatial frequency range equal to $[0\ ,\ \sin\left(\frac{4}{9}\pi\right)/\lambda_{min}] \cong [0\ ,\ 4.92]\mu\text{m}^{-1}$; it is kept fixed in all numerical examples. With regard to numerical integration, we have employed the NIntegrate facility of Mathematica™ 8. In Eq. (20), if SSI is given the function Sun-5780, then the double integral is computed by the general adaptive method. Differently, when the SSI-23 is considered, the integration procedure is carried out in two steps. The tabulated values shown in Fig. 3 are interpolated via *spline* in order to obtain an interpolating function, say, $\Im(\lambda)$, which accepts a wavelength value as argument and returns the corresponding (very well approximated) value of SSI; second, using $\text{SSI-23} = \Im(\lambda)$, the double integral is performed by the method known as the Gauss-Kronrod (here with 15 points for the integral's error estimate). Relative errors have been required/checked to be $< 10^{-7}$. The numerical results below can be figured out by summarizing the current basic points. *First*, the interaction between sunlight and sail surface is appropriately described via a vector theory of diffraction.*Second*, the $\square\rightarrow\square$ polarization mode is the highest around retro-reflection.*Third*, sunlight-based sailing requires SSI broadband contribution. *Fourth*, the sail's reflective layer is a metal with very high dielectric function. *Fifth*, the sail surface exhibits random topographic-only irregularities. *Sixth*, any piece of the sail surface has a PSD isotropic. *Seventh*, independently of its local curvature, the sail surface is sufficiently smooth even in the 0.2 – 0.3 μm range so that here we employ *one* theory here (i.e. the RR-theory) throughout the chosen band 0.2-10 μm.



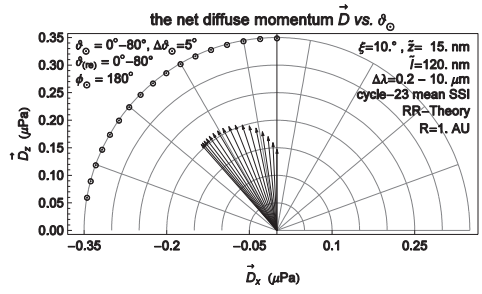
a)



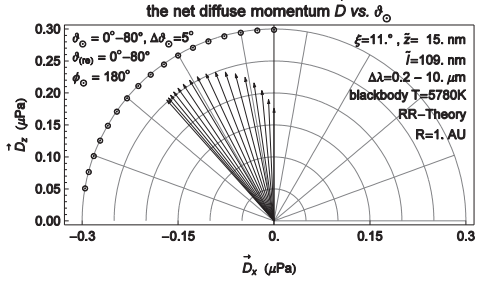
e)



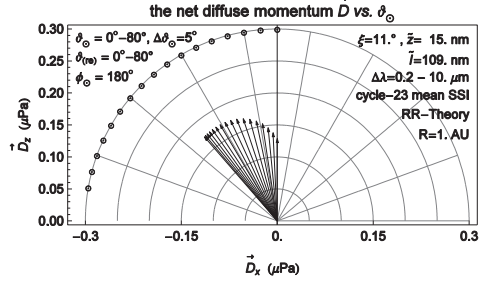
b)



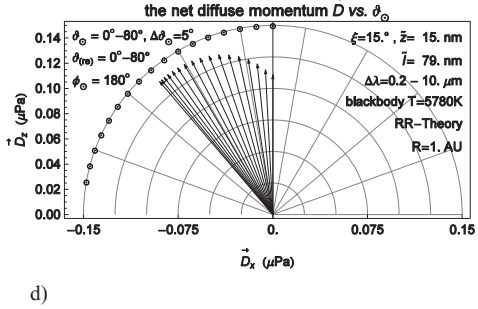
f)



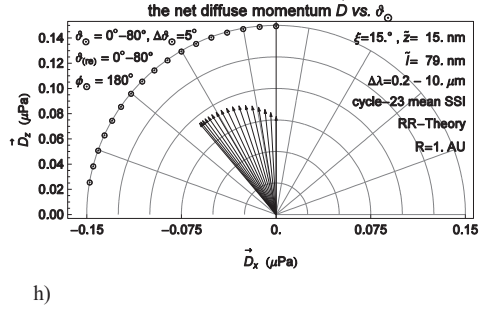
c)



g)



d)



h)

Fig. 4 Subplots showing vector **D** for high-ε surface with a given *rms* height (15 nm) and different slope angles ξ chosen equal to $(5^\circ, 10^\circ, 11^\circ, 15^\circ)$.

Fig. 4 contains a set of subplots each of which displays the vector **D** for high dielectric function material surface with a given *rms* height of 15 nm and different *rms* slopes. The first column (a-d) regards the Sun modeled as a blackbody of 5780K, whereas the second column (e-h) is relevant to the actual SSI of the solar cycle 23. Each vector pertains to a given solar incidence, namely, vector $\mathbf{D}_j$ $(j = 0\ldots16)$ corresponds to the incidence angle $\vartheta_e = \Delta\vartheta_e\, j = 5^\circ j$ marked (in each subplot's second quadrant) by $\mathbf{e}$, which points to the Sun as observed by sensors in the local sail frame (defined in Fig. 1). Such data layout is repeated for Figs. 6-7.

The first significant feature, which one can infer from a simple inspection of the subplots of Fig. 4, is that the total scattered momentum **D** has a negative component, progressively stronger as the incidence angle increases. (Note that $D_x \equiv 0$ in anSST). This has to be ascribed mainly to the $\square\rightarrow\square$ backscattering around the retro-reflection $(\vartheta_{(re)} = \vartheta_e\ ,\ \phi_{(re)} = \pi)$, which is significant for a high-ε material, especially at large incidence angles. For small slope angles, $\|\mathbf{D}\|$ increases markedly as ϑ_e increases, achieves a maximum, and then decreases slowly. As a point of fact, as the incidence angle rises from zero, more details of the surface irregularity (i.e. random "hills" and "valleys") appear reachable by the incoming waves that are thus scattered to a greater extent. At large incidence and scattering angles, masking/shadowing effects begin. Besides, **D** lags behind $\mathbf{d}_e$ at any ϑ_e, and results in $D_x \cong D_z > D_z\ (\vartheta_e = 0)$ at large incidence; in particular, if $\tilde{\beta}$ is the angle of **D** with the *z*-axis, one gets numerically $\tilde{\beta} / \vartheta_e \sim 0.5 - 0.6$. This important thrust effect can be understood by focusing on a monochromatic wave impinging on the sail; the integrand of vector $\mathbf{G}_1$, say, $d\mathbf{G}_1$ is proportional to the photon momentum scattered in $d\omega_{(re)}$. Its relative components, considered as function of the incidence and scattering angles, are shown in Fig. 5.

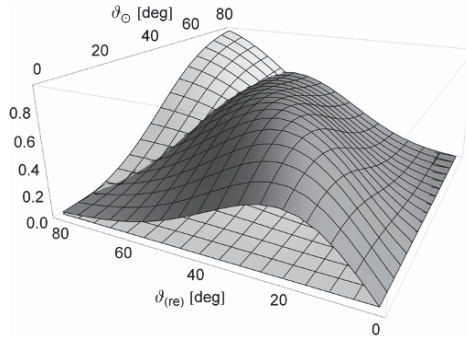
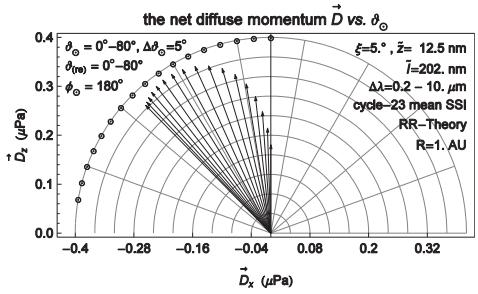


Fig. 5 The relative values of the components of the vector $d\mathbf{G}_1$; the dark gray surface is the *z*-component, whereas the light gray is the *x*-component. In this example, we set $\tilde{l} / \lambda = 0.5$.

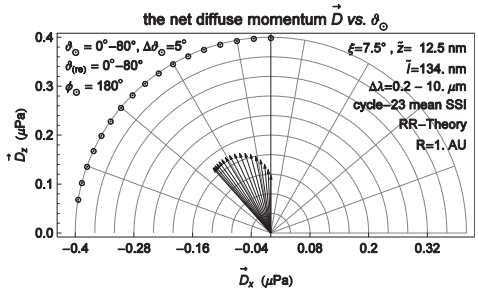
Their behaviors are due mainly to the polarization factors and the obliquity term of the RR-theory. Only at sufficiently large ϑ_e *and* $\vartheta_{(re)}$, the *x*-values of the scattered momenta prevail over the *z*-values. Therefore, once d$\mathbf{G}_1$ is integrated over $\left[0\ ,\ \overline{\vartheta}_{(re)}\right]$ to give $\mathbf{G}_1$, and hence **G**, the direction of **D** cannot turn as fast as $\mathbf{d}_e$.

The actual SSI-23 is lower than Sun-5780 in the band ~100-400 nm; thus, the second column of Fig. 4 exhibits result smaller than the corresponding blackbody SSI model. This behavior is systematic, and it will not be repeated; only the SSI-23 results will be plotted in the other figures.

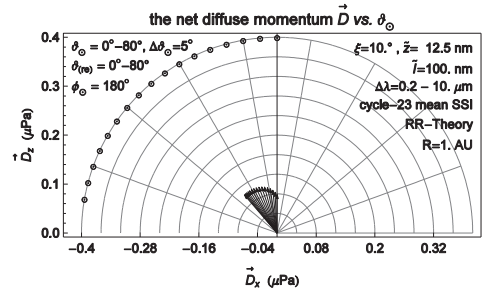


a)

Fig. 6 Subplots showing vector **D** for high-ε surface with a given *rms* height (12.5 nm) and different slope angles $(\xi = 5^\circ\ ,\ 7.5^\circ\ ,\ 10^\circ)$. As in Fig. 4, the auto-correlation length $\tilde{l}$ is derived from Eq. (10). In this set of plots, the Sun-5780 option is ignored because its corresponding behavior is similar to that of Fig. 4.



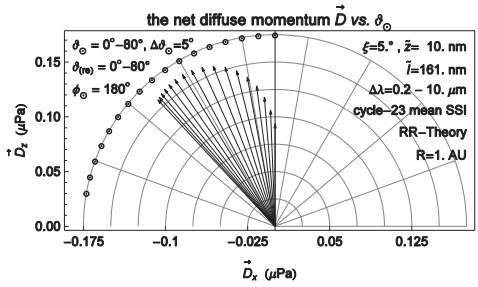
b)



c)

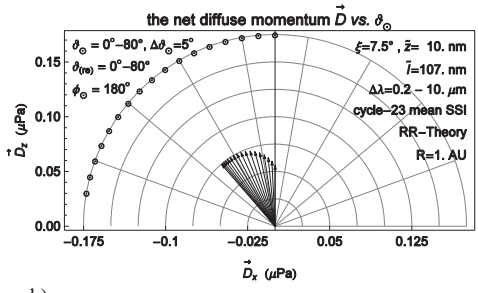
Figure 6 contains the subplots regarding $\tilde{z} = 12.5\,\text{nm}$ and three values of the slope angle, and the $\tilde{l}$ values are determined by Eq. (10). Again, decreasing the auto-correlation length induces a reduction of the diffusely reflected momentum; for instance, comparing subplot 6(a) to subplot 4(a), the behaviors are very similar, but the momentum values scales more than $(\tilde{z}\,\tilde{l})^2$ as a result of the exponential term of the current PSD, which depends on $\tilde{l}$ and not on $\tilde{z}$. This means that the current ISP is a well-posed problem under surface smooth conditions.

Finally, Fig. 7 is similar to Fig. 6 but $\tilde{z} = 10\,\text{nm}$. Note that the scale common to these three subplots has decreased from 0.4 to 0.175 μPa, i.e. notably less than $(10/12.5)^2$ x 0.4 μPa = 0.256 μPa.

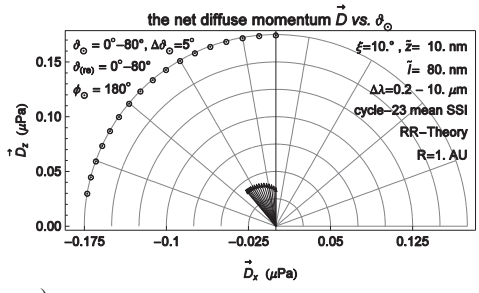


a)

Fig. 7 Subplots showing vector **D** for high-ε surface with a given *rms* height (10 nm) and different slope angles $(\xi = 5^\circ\ ,\ 7.5^\circ\ ,\ 10^\circ)$. Again, the auto-correlation length $\tilde{l}$ is derived from Eq. (10). In this set of plots, the Sun-5780 option is ignored because its corresponding behavior is similar to that of Fig. 4.



b)



c)

Case 7(c) represents the weakest example of diffuse-reflection momentum analyzed here, and might seem to be negligible; for example, $\left\|\mathbf{D}(\vartheta_e = 25^\circ)\right\| = 0.04$ μPa, namely, 21 times the standard deviation of TSI time series in the whole cycle-23. Not only such an amount is not negligible, but also the direction of **||D||** is meaningfully different from the surface normal for non-paraxial incidence.

Once the contribution to thrust induced by diffraction from a *non-ideal* sail is computed by the difference between input and output photon momenta in $F^{(sail)}$, then the word devolves on the computation code of the sailcraft trajectory, one of the main flight design blocks of the required mission analysis.

VII. Conclusion and Next Development

A general approach for computing the diffuse-reflection component of the thrust density has been proposed in this paper. It uses concepts well employed in advanced Optics, but so far unapplied to solar-photon sailing. This approach mainly consists of going beyond the scalar-scattering approach via a vector theory of diffraction, where polarization and area power spectral density are important both. To begin with, the various polarization factors, which depend on the employed material in general, have been simplified by the high dielectric function approximation because the reflective layer of a solar sail is a metal. However, a cutoff angle of incidence/scattering has been set for applying the high-ε formulas. The problem of modeling the scattering

surface has been dealt with by assuming topographic-only scattering from a bi-Gaussian surface with two parameters, i.e. the *rms* height and the auto-correlation length. Another basic assumption is that diffraction takes place from a sail surface sufficiently smooth even in the low-wavelength region of the UV band. With such hypotheses, we have proceeded with the physical problem of scattering from a solar-photon sail by employing the Rayleigh-Rice vector theory for computing the total momentum $\mathbf{D} = (D_x \, , \, 0 \, , \, D_z)^{\cdot}$ of the photons scattered by the sail. The numerical results may be summarized as follows:

1. Diffuse-reflection photon momenta increase with the *rms* roughness (of course), but also with the auto-correlation length in the range of *rms* slopes here taken as realistic for a solar sail;
2. the normal-to-sail component D_z is meaningfully different from that one computed via SST;
3. the parallel-to-sail component D_x (identically zero in SST) is always negative and may be neglected only for sunlight's paraxial incidence;
4. the length $\|\mathbf{D}\|$ increases with the incidence angle, but not always a clear maximum is present;
5. Vector $\mathbf{D}$ lags behind the Sun's direction as observed from the sailcraft, especially at large incidence angles, at least to within the cutoff values set on incidence/scattering angles.

Sail making regards the production of high-quality surfaces *huge* with respect to those ones of devices used in electronics, astronomy, laboratory and commercial optics, and so forth. Thus, a-priori we may not exclude that some part(s) of the sail may result in a moderately rough product, also depending on the very large-scale deposition method of the reflective-layer. In addition, the assumption of very high dielectric function of the sail's material will eventually become inadequate as the incidence and the scattering angles go to grazing the sail. The current solution (with a cutoff angle) may be viewed as a lower bound of the scattering momenta covering the $[0 \, , \, \pi / 2]$ range in colatitude.

Next work will focus on the following main points: (1) inserting the general expressions of the polarization factors (e.g. related to Aluminium) in the calculation of the integrals in Eqs. (20) for any incidence and scattering angles, but still using the RR-theory; (2) considering a relatively recent theory such as the generalized Harvey-Shack for smooth and medium rough sails; and (3) using different PSDs for characterizing the reflective layer. Of course, the final word is to appropriate experiments on sail materials in the large bandwidth considered in this paper. Once the physical model of sail-photon interaction is reliable for any sail, then the global/local curvature effects in the sail design and thrust acceleration optimization can be trustily evaluated significant or negligible in a given sailcraft mission.

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The Solar Sail Materials (SSM) Project – Results of Activities

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Introduced at ISSS 2010, the SSM project conducted for ESA came to its end at the beginning of 2013 with interesting and practical results. The project, relying also on past major European solar sails design projects, aimed at developing and testing future sail

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technologies suitable for large solar sailcrafts featuring low Sail Film Areal Density σ_{Film}. The proposed paper presents the project's achievements, mainly in the fields of (1) Sail film and coating materials, (2) Sail-to-Structure Interfaces (SSIF) and, (3) Correlation of ground sail tensioning strategies with 3D imaging and Finite Element Model (FEM) simulations.

Nomenclature

R = Reflectivity (-)

α_s = Solar Absorptance (-) integrated over [250-2500 nm]

ε_h = Infrared Emittance (-) integrated over [3-21 microns]

σ_{Film} = Sail Film Areal Density (g/m^2)

σ_{SC} = Sail Loading (kg/m^2)

I. Introduction

Introduced at ISSS 2010 [1], the Solar Sail Materials (SSM) project conducted for ESA came to its end at the beginning of 2013 with interesting and practical results for future European scientific/exploration missions supported by photonic propulsion.

After recalling the project objectives and work logic, this paper presents the main achievements of SSM.

II. Project Objectives and Logic

A. Project Objectives

The SSM project objectives [2] were to make progresses on sail materials, their assembly and integration for future Advanced European Solar Sail missions.

Out of all, major progresses were expected on :

- Identification of the reference film materials to consider for European projects.
- Thinning of the reference film material(s) to 3 microns or less.
- Identification of suitable film coatings (reflective and emissive) and protective overcoats
- Identification of an environmentally resistant assembly concept and related adhesive(s)
- Selection of a preferred sail-to-structures fastening concept

Although focusing on technologies necessary to assemble efficient and reliable sails, the SSM project investigated also some key system aspects relating to :

- Sail-to-Structure Interfaces (SSIF) e.g. sail tensioning devices and sail containers
- Optimal tensioning strategy, under 1g
- Scalability of results, from 1g observations and simulations of small scale mock-ups/breadboards up to large scale operational sails in orbit.

The considered European Solar Sail missions targeted both a Sail Deployment Demonstration “First Steps” mission close to Earth and, for the advanced “Full Capacity” ones, a close approach to Sun.

B. Work Logic

To achieve the project’s objectives defined here above, a work logic has been proposed/followed consisting mainly of :

- Definition and preparation of the test campaign (up to Test Readiness Review - TRR)
- Execution of the agreed test campaign and its three components :
    - Tests on materials
    - Tests at sail system level with small scale mock-ups/breadboards
    - FEM simulations and validation

All proposed activities aimed at finding practical and reliable solutions ready to be used in the “First Steps” missions framework but still considering the “Full Capacity” framework (and its environmental constraints).

Up to this TRR, the project’s activity was divided in three consecutive steps :

- Sail Systems Analysis : this first step reviewed the existing documentation in order to define/derive :
    - (1) the “frameworks” and requirements (based on the analysis of 108 solar sail missions listed in [3]) and,
    - (2) the key technical challenges relating to SSM, sail manufacture, integration and test, and SSIF (based on the analysis of European solar sailcraft designs : VSE, ODISEE, Daedalus and Libellule).
- Sail System Design : this second step proposed a sail concept to be further investigated, also by including the results of the parallel studies on its main components (films, coatings, assembly methods, SSIF and booms).
- SSM Test Planning : this third step started by defining the Overall Test Plan proposed for SSM. After approval of this document by ESA, the specific Test Plan & Procedures (TP&P) were prepared for :
    - (a) Films and Coatings tests,
    - (b) Film Thickness Reduction process and tests,

- (c) Thermo-mechanical and UV tests,
- (d) Optical means and tests, and
- (e) Sail breadboarding and test activities.

After reception of the raw materials and closure of the TRR, the SSM (test) activity was then split in three interacting tasks :

- Material (Sample) Tests : in two refining "Phases", the set of "validated" SSM was thoroughly qualified, after having been "selected" as the optimal configuration of materials :
  - Phase-1 indeed aimed at selecting by tests the optimal configuration of materials (films, coatings and adhesives) out of the ones identified previously during the bibliographic surveys. Based on test results, the "Selected SSM" configuration was approved during the first Test Review Board (TRB-1).
  - Phase-2 further qualified with additional tests the "Selected SSM" configuration and could ultimately validate it as suitable for future Advanced European Solar Sail missions.
- Sail Mock-Up Tests : various sail mock-ups have been built and tested to gain expertise at (sail) system level, with the objective to improve the knowledge on how sails are made and should be operated.
- FEM Validation : FEM simulations have been performed, based on all collected information, and validated thanks to optical 3D metrology conducted on mock-ups, to allow future scalability of results for larger operational sails.

III. Synthesis of Material Tests

This synthesis of Material tests recalls :

- The materials which were considered during SSM
- The tests performed and observed issues/results

A. Materials Considered

As a sail is composed of strips of coated film assembled by adhesives, material candidates had to be found for Films, Coatings and Adhesives.

i. Films (Tested)

Although other films were identified as possible candidates, the following were tested during SSM :

- Polyimide (PI) Upilex SN 7.5 microns
- Polyimide (PI) Kapton® 50HN 12.5 microns (the 30HN 7.5 microns was not available for SSM due to ITAR limitations)
- PolyEthylene Naphathalate (PEN) Teonex® Q72 4 microns

ii. Coatings (Tested)

The following coatings and protective overcoats were identified and tested during SSM :

- Reflective coating (SUN facing side) : 100 nm Al
- Emissive coating (Anti-SUN facing side) : 10 nm Cr
- Oxide overcoats on Aluminium : 30 nm Silica or Alumina

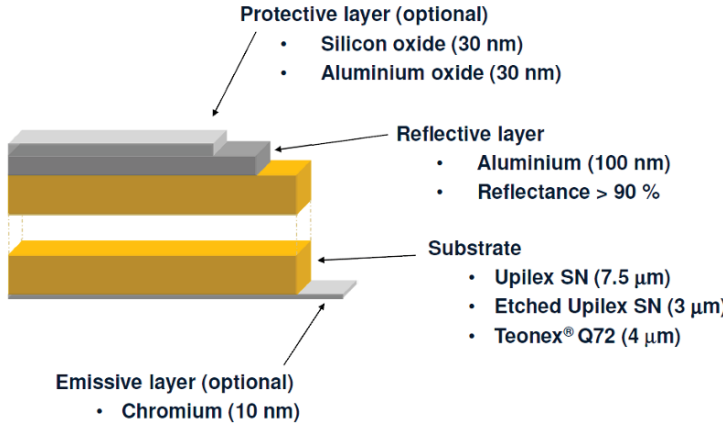

Fig. 1: Various coatings configurations investigated with SSM

iii. Adhesives (Tested)

The full range of attachment and joining techniques was considered in SSM for the assembly of coated plastic films : mechanical fastening (like sewing) and joining by either adhesive bonding or welding. About this last option, some trials were performed on Kapton® HN with negative outcome. Out of all, adhesive bonding was finally preferred for SSM.

The following adhesives were identified and tested during SSM :

- Two-component epoxide resin
- Polyimide resin
- Pressure Sensitive Adhesive (PSA) tape

B. Tests Performed and Main Issues

An overall test plan has been generated for both Phase-1 (i.e. screening of materials, leading to the “Selected SSM” configuration) and Phase-2 (i.e. for the validation of the “Selected SSM” configuration).

iv. Tests Performed

In short, films (raw and coated, thinned or not) and assembly samples have been produced and tested both functionally and environmentally :

- Both Kapton® 50HN and Upilex SN 7.5 microns have been thinned, this last down to about 1.4 microns :

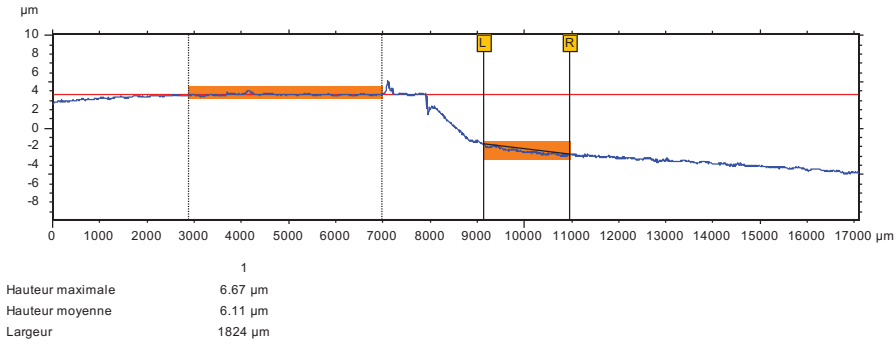

Fig. 2: Profilometry after plasma etching of Upilex – Average step obtained

- Raw films have been functionally tested wrt optical, electrical and outgassing properties, and environmentally tested by thermal cycling, immersion in liquid Helium and exposed to UV for two simulated orbit periods.
- Coatings and overcoats have been deposited on various films, tested functionally wrt optical, electrical and mechanical (adherance and abrasion) properties, and environmentally tested in the same way as the raw films.
- Assemblies have been produced with various films (raw and coated, thinned or not), tested functionally w.r.t. mechanical resistance, and environmentally tested by thermal cycling (20 cycles over [-170°C, +120°C for Teonex® and +200°C for Upilex]), immersion in liquid Helium (4K) and exposed to UV for two simulated orbit periods (i.e. 2x188 hours).

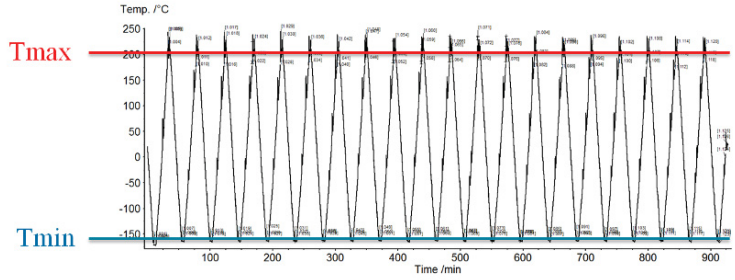

Fig. 3: Thermal cycling test used in SSM (here for Upilex, sensing on sample)

v. Main Issues

During the tests, the following issues were observed :

- Teonex® Q72 4 microns yellowed rapidly and featured a strong mechanical deformation with UVs, a sign of material degradation but with no observed effects on mechanical resistance to operational constraints. Samples were also difficult to produce and assemble after the coating process
- Strap joints have been produced with PSA tape but failed under thermal and UV aging and did not pass the cryogenic test. From this it was concluded that such a product is not suitable for the targeted missions
- Polyimide resin failed to adhere to Chromium on a double-sided (Al/Cr) coated Upilex sample
- Non protected coatings made in laboratory conditions suffered during the "strong" abrasion test performed according to ISO 9211 method 01, degree 01 : 50 strokes with a cheesecloth, bearing force of 5N. Industrial commercial products and oxide-protected samples showed increased resistance to this test :

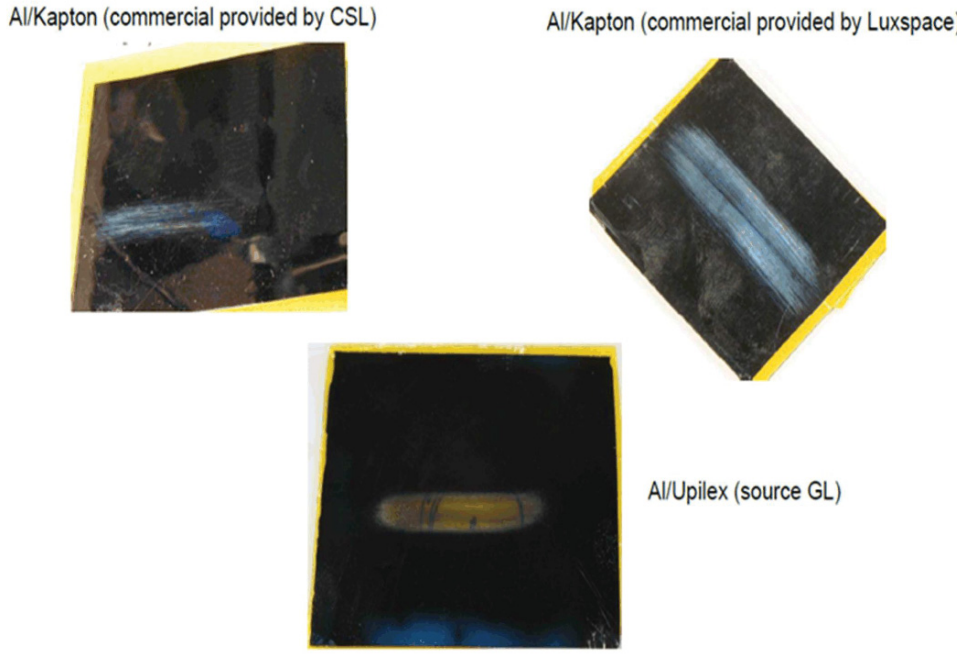

Fig. 4: Issue observed during abrasion test on SSM coatings made by CRP-GL

- Although etched for the targeted thickness of 3 microns, the thinned Upilex samples featured after delivery and coating an increased areal density probably due to process-improved water uptake and/or to uncertainties in the thickness of the raw material.

vi. Main Results for Selected SSM

The Selected SSM configuration (for future Advanced European Solar Sail missions) was identified by various trade-offs based on Phase-1 performed tests, complemented with data sheet information. It consists of :

- Polyimide (PI) Upilex SN plasma etched down to 3 microns
- Coated with 100 nm Al on the SUN facing side and with 10 nm Cr on the Anti-SUN facing side. No oxides overcoats were considered/deposited on the Selected SSM configuration
- Assembled by simple lap joint of 5 mm made with two-component epoxide resin

The Selected SSM configuration was validated by its successful Phase-2 test campaign :

- Commercial Upilex SN was first plasma-etched from 7.5 to the requested 3 microns with satisfactory surface (roughness) properties
- Etched samples have been coated with Al/Cr featuring the following optical properties :
    - On SUN facing side : R = 0,934 ; α_s = 0,094 ; ε_h (22°C) = 0,023
    - On Anti-SUN facing side : ε_h (22°C) = 0,580
- Adhesion properties of the coatings showed no degradation after environmental tests
- Assemblies survived to :
    - Durability tests : after one hour at -170°C followed by one hour at +200°C, 720.000 cycles at 100 Hz with loading varying sinusoidally between 0,1 and 0,3 N (over 5 mm sample width)
    - Loaded thermal cycling : 20 cycles over [-170°C, +200°C] loaded at 0,56 MPa with heating/cooling rate of 17±2°C/min
    - Cryogenic immersion test : within Liquid N_2 and within Liquid He

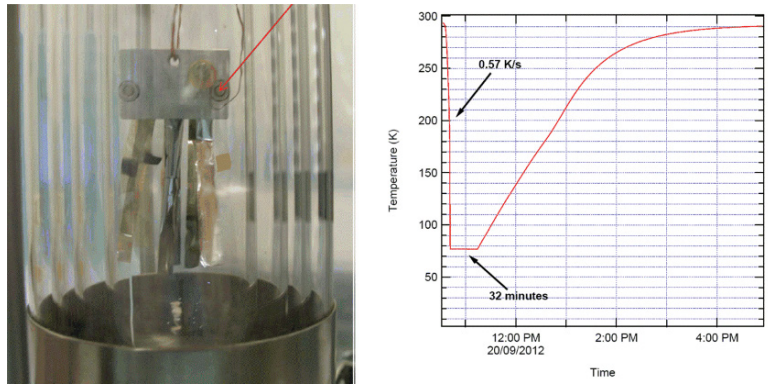

Fig. 5: Immersion of assembly samples in liquid N_2

- Dilatation tests during thermal cycling over [-170°C, +200°C] :

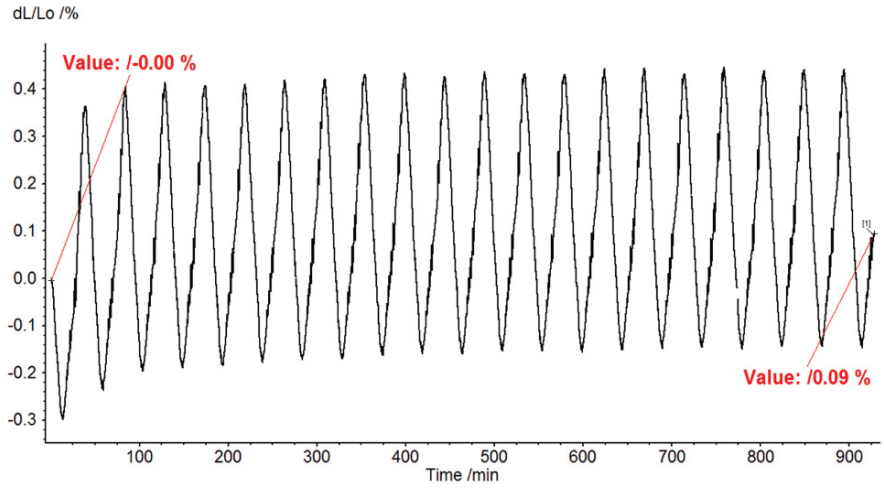

Fig. 6: Dilatation of assemblies during thermal cycling

IV. Synthesis of Breadboarding

Real breadboarding activities started after TRB-1.

In line with the project's agreements, the list of breadboards deliverables consisted of several items, each one devoted to a specific learning purposes :

- The tests with Item #3.1: one flat, unassembled small sail (~1 x 2 m) served cutting/handling practice purposes and tension effects visualisation with an unperturbed sail.
- The tests with Item #3.2: one flat, assembled (no SSIF) small sail (~1 x 2 m) served assembly practice purposes and tension effects visualisation around (PSA) tape assemblies.
- The tests with Item #3.3: one flat, unassembled (SSIF implemented) small sail served fasteners assembly practice purposes and tension effects visualization around these.
- The tests with Item #3.4: one folded, assembled small sail (~1 x 2 m) served folding/unfolding practice purposes and tension effects visualisation around folds.
- The tests with Item #3.5: one full (integrating assembly, reinforcement, SSIF and folding functionalities) "zero-g" sail (~1.5 x 3 m) were initially planned to serve as the reference sail for all tests aimed at validating the FEM calculations.

Here below, the Deployment testbench (Item #1) with the Breadboard sail Item #3.5 made of aluminised Kapton® 50HN assembled with PSA tape, both considered as representative for the scopes of breadboarding :

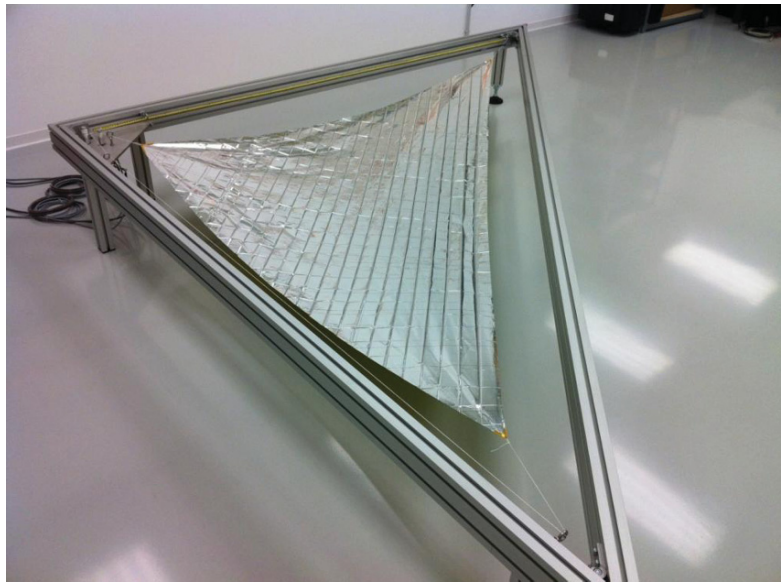

Fig. 7: Testbench and breadboard sail in LuxSpace's laboratory

Here below, the different fastener types produced and tested (grouped under Item #3.3A) :

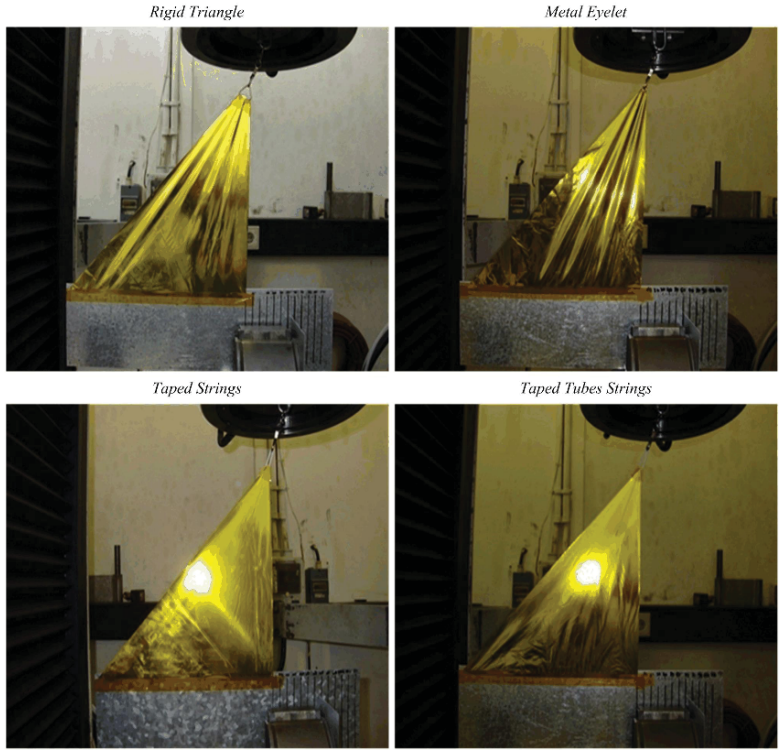

Fig. 8: Comparative results for different fasteners type with a 20° angle of tension

As can be observed in Fig. 8, the best results were obtained with the string solution either passing through rigid tubes or not, probably because it introduces its tension force components more optimally in/along the sail.

Sail Mock-up Item #3.4, depicted in the following Fig. 9 showed once more the easiness of manufacture and the reliability at deployment of the VSE folding scheme presented for the first time in [4] later renamed as “Frog Legs” pattern.

Here again, the SSM project made interesting progresses towards real mission implementation by proposing a compact and lightweight sail container compatible with this folding scheme.

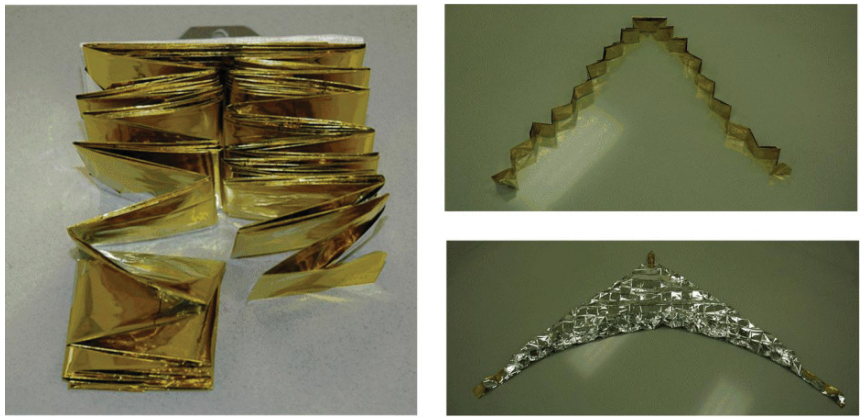

Fig. 9: Deployment sequence of the “Frog legs” folding scheme

Because of its flatness, Sail Mock-up Item #3.3 was finally chosen to perform the 3D Imagery activities at CSL. After marking and powdering the sail, four cases have been imaged :

- Case 80 (α = 25 deg with 0,5N weights) : Untensioned sail showing the nicest shape/curvature
- Case 81 (α = 19 deg with 5N weights) : Close to the identified flattest configuration under 1g
- Case 82 (α = 11 deg with 5N weights) : Lower bound aside Case 81 of about 10 deg
- Case 82 (α = 28 deg with 5N weights) : Upper bound aside Case 81 of about 10 deg

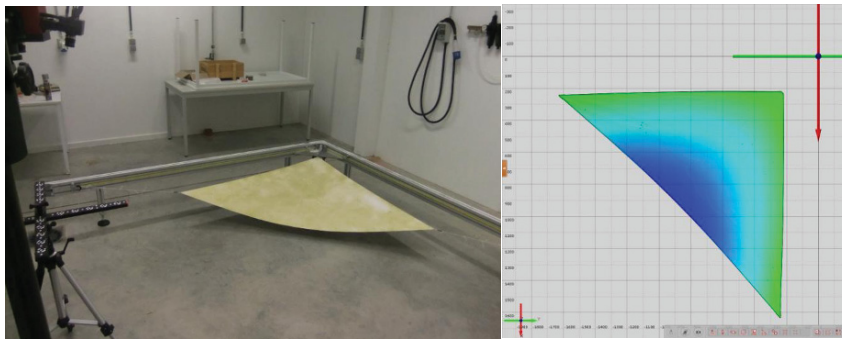

Fig. 10: 3D Imagery of the Untensioned Sail (Case 80)

V. Synthesis of FEM Activities

Since its inception, the SSM project proposed to allow future scalability of FEM simulations to larger, operational sails tensioned in space.

For this, FEM simulations were first to be performed replicating the 1g observations of small scale mock-ups/breadboards. The idea was indeed to compare the extremely accurate 3D Imaging results with the FEM calculated shape of the sail mock-up : if comparable, the FEM simulations would be validated opening thus the way to operational simulations on advanced, larger sails.

FEM simulations have been performed on 3D Imaging Case 80 showing the nicest sail curvature due to untensioned conditions. The qualitative comparison showed excellent results :

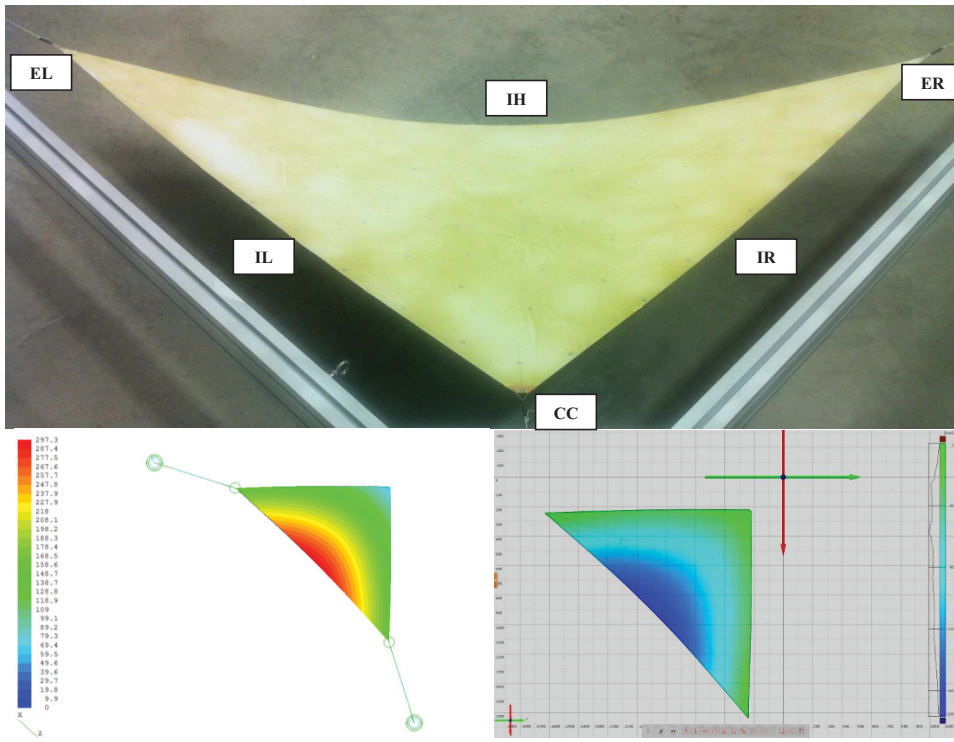

Fig. 11: Excellent qualitative validation of FEM (left) with 3D imagery (right) wrt real (above)

Quantitatively, the results of the FEM simulations could be compared with high and low precision 3D observations, performed at six key positions located around the sail perimeter (indicated on Fig. 11). Comparison (see following Table 1) allowed further positive conclusions also.

Table 1: Relative error between FEM results vs. 3D-Imaging data

Case 80	CC	IH	IR	IL	ER	EL	Average
3D Imaging Data (CSL)							
Absolute z (mm)	-63,14	-320,20	-124,50	-122,00	-147,16	-148,80	
Absolute z corrected (mm)	-63,14	-320,20	-123,25	-123,25	-147,98	-147,98	
Relative height ($z-z_{CC}$)	0,00	-257,06	-61,36	-58,86	-84,02	-85,66	
Relative height corrected ($z-z_{CC}$)	0,00	-257,06	-60,11	-60,11	-84,84	-84,84	
6-Points Data (LXS)							
Absolute z (mm)	341,50	84,50	280,50	283,00	254,00	258,00	
Relative height ($z-z_{CC}$)	0,00	-257,00	-61,00	-58,50	-87,50	-83,50	
FEM Data (GDT)							
Absolute z (mm)	-63,14	-297,29	-127,18	-127,34	-148,51	-148,60	
Relative height ($z-z_{CC}$)	0,00	-234,15	-64,04	-64,20	-85,37	-85,46	
Relative Heights							
6-Points/3D Imaging		100,0%	99,4%	99,4%	104,1%	97,5%	
Differece		0,0%	0,6%	0,6%	-4,1%	2,5%	-0,1%
FEM/3D Imaging		91,1%	104,4%	109,1%	101,6%	99,8%	
Relative error		8,9%	-4,4%	-9,1%	-1,6%	0,2%	-1,2%
FEM/3D Imaging corrected		91,1%	106,5%	106,8%	100,6%	100,7%	
Relative error		8,9%	-6,5%	-6,8%	-0,6%	-0,7%	-1,2%

With this FEM analysis and the relative (high and low precision) 3D observations of the most curved sail test case, the qualitative and quantitative validation of the simulation tool has thus been achieved : indeed, although a maximal error of about 9% could be identified at the lowest point of the sail when comparing FEM-calculated relative heights with the high-precision 3D observations, the overall (i.e. average) error at sail level was only of -1,2% and none of the errors exceeded 10%.

VI. Synthesis of System Activities

During SSM, system level activities were conducted to :

- consolidate the material test requirements with the generation of two (environmental) “Frameworks”,
- consolidate the system configuration, especially for SSIF items
- compare the project achievements with other sail(craft) development (e.g. IKAROS)

A. SSM “Frameworks”

The review of past and future solar sail missions performed during the SSM proposal phase indicated the necessity for categorising the needs related to sailcraft mission and their working environments.

Two SSM “Frameworks” have been defined :

- Near-Term (or First Steps) : for those first Solar Sail missions, planned in the near-term, aiming at Proof-of-Concepts of Solar Sailing (deployment, navigation, …). Near-Term Framework missions include only

those staying close to Earth i.e. at ~ 1 AU. These are based on sails of reduced physical dimensions and will typically have operational (in-orbit) lifetimes between less than 1 and up to 4 years.

- Mid-/Long-Term (or Full Capacity) : for those missions, planned in the mid-/long-term, aiming at taking full advantages of Photonic Propulsion. Mid-/Long-Term Framework missions include those proposed to travel in the inter-planetary medium and even beyond to other solar systems. These larger missions (also in terms of physical dimensions) with operational lifetimes of more than 4-7 years, consider typically first, a Sun approach/fly-by (as close as 0.15 AU) to maximize thrust, and lastly a sail jettisoning at 5 AU.

The SSM test specifications for mechanical, thermal and UV loads were derived from the analysis of these two "Frameworks". As described under Section III.A., the Selected SSM configuration passed succesfully the tests relating to the most demanding Mid-/Long-Term (Full Capacity) "Framework".

B. SSIF Items

System activities defined Sail-to-Structure Interfaces (SSIF) as consisting of all elements of a solar sailcraft except satellite, masts (with respective mechanisms) and sails. In this sense, the SSIF provides all the interfaces between the sails and the sailcraft structure. To achieve its mission, the SSIF provides a storage system for the folded sail up to orbit, ensures the deployment and tensioning of the sail (enabling the user to be informed about the sail status or tension) and relaxes the sail as needed. Following this definition, the composition of the SSIF can be described by the following three elements:

- The Tensioning System : it ensures the deployment of the sail and its tension state. This system can be typically a set of motors (with gearboxes and pulleys) that will act on the corners of the sail, or springs that deploy/control the sails automatically.
- The Linking Harness elements: these are the items that will ensure a physical link between the sail (fastener) and the Tensioning System. They consist typically of strings (metal, polymer …) with a linkage system (knot, hook, …) at each tips of the string.
- The Storage System : It is a bag or box that will contain the sail folded up to the orbit. This system is also referred to as container.

C. Benchmarking of SSM Outcomes

The results of the SSM project at materials and system levels were integrated taking the GeoSail mission as reference : with these, this 50m hypothenuse 3 microns sail would feature a Sail Film Areal Density σ_{Film} of 5.1 g/m^2 still dominated by the sail film contribution (i.e. 87% of total).

Although compliant with the GeoSail requirement of ≤ 6 g/m^2, this figure was also compared to those of some relevant projects :

- IKAROS (estimated) :
    - Sail Film Areal Density σ_{Film} of ~75 g/m^2 probably explained by the experiments located on the sail film
    - Sail Loading σ_{SC} falling within [1,76 – 2,07] kg/m^2, depending if sailcraft mass is considered wet or dry, and if solar actuators are reflective or not.
- NanoSail-D2 (estimated) :
    - Sail Loading σ_{SC} of 0,42 kg/m^2
- VSE (design of 1995) :
    - Sail Film Areal Density σ_{Film} of ~14 g/m^2
    - Sail Loading σ_{SC} estimated to be of ~0,10 kg/m^2
- ESA small solar sail project for the New Millennium celebration [5] :
    - Sail Loading σ_{SC} estimated to be of ~0,42 kg/m^2

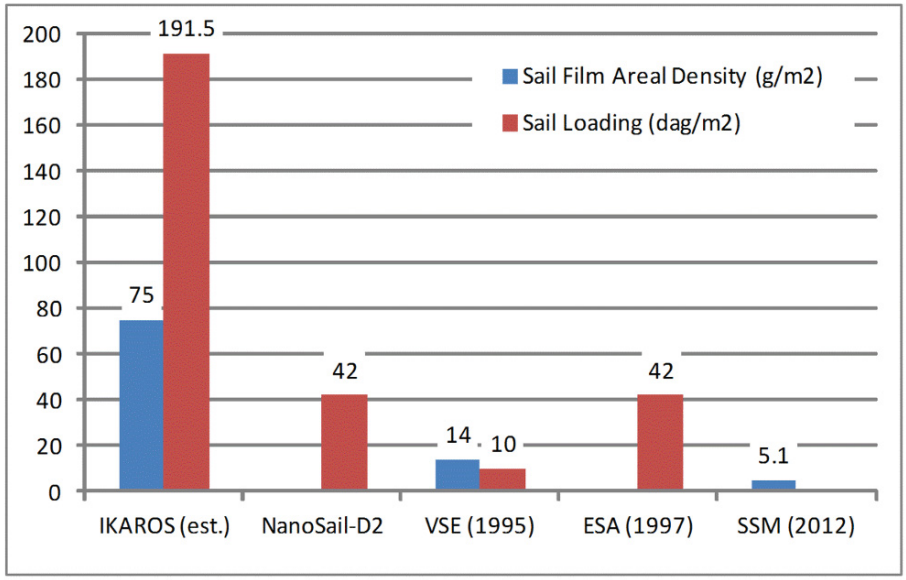

Fig. 12 : Comparison of key parameters

VII. Conclusions

The SSM project made key advancements in developping and qualifying feasible technologies and concepts applicable for future sails of advanced European solar sailcraft missions :

- For what relates to the sails, the SSM project was indeed able to define a mix of suitable (film, coatings, assembly) materials featuring an overall Sail Film Areal Density σ_{Film} compliant with the GeoSail requirement of ≤ 6 g/m^2
- According to analyses and tests performed with SSM, the best sail fastening solution is made with Taped (Tubed) Strings and Rigid Triangle fasteners, holding the sail according to Tip Holds.
- SSIF elements have been identified and each of them and first concepts have been proposed (especially for the sail container)
- For a complete 7,5 microns European sail, a TRL of [5,7/2] was estimated at the end of the SSM project. For a complete ≤ 3 microns European sail, the TRL should be [4,9/2]. The SSM SSIF's TRL is today estimated to be [5,7/2] (using the ESA TRL format [Avg/Min])
- Thanks to the validation of FEM simulations with 1g observations/measurements on small scale mock-ups/breadboards, the scalability of FEM results to large scale operational sails in orbit is now ensured.

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Selection and Manufacturing of Membrane Materials for Solar Sails

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Commercial metallized polyimide or polyester films and hand-assembly techniques are acceptable for small solar sail technology demonstrations, although scaling this approach to large sail areas is impractical. Opportunities now exist to use new polymeric materials specifically designed for solar sailing applications, and take advantage of integrated sail manufacturing to enable large-scale solar sail construction. This approach has, in part, been demonstrated on the JAXA IKAROS solar sail demonstrator, and NASA Langley Research Center is now developing capabilities to produce ultrathin membranes for solar sails by integrating resin synthesis with film forming and sail manufacturing processes. This paper will discuss the selection and development of polymer material systems for space, and these new processes for producing ultrathin high-performance solar sail membrane films.

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I. Introduction:

The first use of polymer film as the primary structure for space technology was NASA's ECHO I. Launched in 1960, ECHO I was the first passive "reflector" communications satellite used to bounce microwave transmissions beyond the horizon.[1] ECHO I was an inflatable balloon constructed of 0.5 mil thick Mylar® film, both sides metalized with aluminum. The success of the series of these balloon-type structures, ECHO I, ECHO II and Explorer IX demonstrated the utility of polymeric materials for space applications. Beyond their intended use of scientific study, these satellites showed that sunlight solar pressure acts as the largest deforming force on these structures in the zero-gravity ultra-vacuum of space.[1] Additionally, this early work outlined 3 major areas of concern for the use of polymeric materials in long duration space missions. They were thermal (-105 to 150°C), photochemical effects resulting from ultraviolet (UV) radiation imparting (~0.75 J/cm^2-min), and ionizing radiation (IR) (10-100 R/hr).[2] These effects can be additive and cause additional problems such as material loss, changes in physical properties, and accelerated degradation. These effects can be reduced if the dielectric films are protected with reflective and thermally conductive coatings, specific chemical structures, physical reinforcement, and electrostatic dissipaters. As the areal density of the laminated polymeric structure decreases, larger space structures can be developed per payload weight. Unfortunately the thinner the structural membrane, the more likely the space environmental effects will have on decreasing the lifetime of the space vehicle. The thermal effects, resulting from the a_s/e (absorptance/emittance), can increase membrane temperatures that may result in permanent distortion of the structural material leading to decreased performance and failure. When thermal effects are combined with IR and UV, additional mechanical failure modes, such as material ablation (radical fragmentation and static discharge), chemical hardening (radical crosslinking), and physical failure (cracking and tearing) will occur. Several possible solutions for these material based problems were: using a less dense polymer allowing for a thicker film, punching holes in the polymer film to lighten it and reduce solar pressure effects, developing a thermal control coating that conducts the absorbed heat energy over the entire surface of the balloon, thereby physically increasing the radiative surface area, and physically reinforcing the film membrane with battens or fibers.

Solar sail membranes are similar in construction and concepts used in the inflatable membrane structures, ECHO and Explorer. Their mission enabling performance was based on mission longevity, size, and areal density of the membrane and support structure in the space environment.[3-5] The primary performance metrics for sail materials are deployable reflective area, areal density, and operational temperature range.[6] Solar sail membranes

have been made from Polyimides, such as CP1 for Nanosail-D, ISAS-TPI, for IKAROS, and polyesters such as polyethylene terephthalate (Mylar®) for the L'Garde solar sail demonstrator.[5,6] Commercial film materials are generally capable of meeting the less demanding performance criteria for most solar sail technology demonstration missions; however film thickness is often greater than desired. For long duration missions, partially aromatic polymers such as Mylar® are more prone to degradation by space environmental factors such as temperature fluctuations, (UV) and cosmic radiation. Many other polymeric materials have flown in space for non-solar-sail applications, including thin films for use as flexible electronic cabling, solar array components, and flexible fabrics. Some of these materials could potentially serve as alternative solar sail materials.[2,7-9]

The process of solar sail manufacturing usually begins with the procurement of commercial-off-the-shelf (COTS) rolls of metalized film as stock material, followed by several basic sail assembly steps:[7,8,10]

1. Layout and seaming or bonding the films together.
2. Preparing for a secondary support structure (mast, battens, etc.).
3. Integrating secondary support structure to the sail membrane (bonding, lacing, etc).
4. Packaging the assembled sail structure (rolling, folding, etc.).

For small technology demonstrations, this process, and the use of COTS membranes and film materials that can be hand-assembled into a sail are suitable.[3,5] However, as the required sail area increases, Steps 1-4 become increasingly difficult. In particular, the addition of non-uniform, non-membrane structural features, e.g., bonded joints, stiffeners, rip-stops, etc., complicates the handling and packaging of large and delicate sail membranes, which may now no longer be easily roll processed.

For long-duration missions, and missions where large solar sail areas are required, an alternative approach is to begin with the processing of the film resin system itself. This affords several new possibilities for sail making:

- Creating ultrathin membranes on a continuous basis from most any resin system.
- Making films of nonuniform width, thickness, and mechanical orientation.
- Directly integrating rigid support structures, physical features, or functional material during film processing.
- Using “staged” reaction chemistry to tailor material process behavior.
- Metallizing films during a more advantageous point in the sail system production process.

- More efficient direct packaging of the sail assembly, thereby decreasing manufacturing floor space requirements.

JAXA's Interplanetary Kite-craft Accelerated by Radiation Of the Sun (IKAROS) sail membrane used a new material, ISAS-TPI, jointly developed by JAXA and the Kaneka Corporation, using a nonstandard synthesis process. Rather than relying on the current methodology of selecting COTS materials and fitting them into a nonstandard application, the IKAROS team developed a material and process to fit the application. This is an important departure. Here, the polymer synthesis was designed to optimize material properties for that specific technology demonstration. [11] This provides others with an impetus to include other suitable polymers, and develop manufacturing processes that revolutionize how a large sail or blade is fabricated.

At NASA's Langley Research Center (LaRC), high performance polymer development has been ongoing for over four decades. In fact, several LaRC polymeric systems have been commercially licensed. They include CP1™, CP2™, LaRC-SI™, and the PETI™ resin series. This paper will discuss the selection and development of polymer material systems for space, and a process that LaRC is developing to produce ultrathin high-performance membrane films.

II. Discussion:

The reason for using a polymeric membrane as the support structure for a solar sail is that in order to get the areal density required for mission performance, a metallic sail would be too physically frail to survive. Therefore, the most obvious solution is to find a material that is extremely thin, highly reflective and not effected by heat. However, until such a material is developed, coating one side of a durable lightweight material with a reflective material and the other side with a material that emits heat efficiently is the current best method. For use in Space, polymers suffer from three basic problems, they are dielectrics, thermal insulators, and lack broad-band reflectivity. However, they can be formed into most any shape and are inherently tough. In order to be enabling for Space applications, polymers are combined with other materials, coatings, fillers and fibers. The next process stage involves taking the polymer film and integrating structural support into the sail. The simplest process for reinforcing and building lengths of film into a planer rectangular sail or a sail blade have been studied in detail and referenced in the introduction. These studies are based using COTS metalized film as the starting point in the fabrication process. For rectangular sails, the process starts by cutting the metalized film, bonding structural support members, then taping or stitching the long edges of the film together to obtain the required sail dimension, and adding additional

structural support to the sail as required using an adhesive. This method requires starting with metalized film, because the two-dimensional sail geometry would dictate the size and complexity of the thermal evaporator required to accurately deposit a layer of different metal on both sides of the polymer membrane. Additionally, as this membrane becomes thinner, additional infrastructure and care is required in handling the increased area of the sail. From a materials and manufacturing standpoint, the utility of this class of sail geometry is physically limited by the areal density, footprint, and deployment risk. The making of bladed sails were detailed in a similar fashion except that the film was cut to length, the film strips rotated 90 degrees, then sown together edge-to-edge to make the long blade. Reinforcements were added to the blade during assembly, as required.

There are many types of commercial polymeric materials to select as potential sail membranes (Table 1). The polymers are divided into three categories, aliphatic, partially aromatic and wholly aromatic. Aliphatic polymers, also referred to as polyolefins, are synthesized from unsaturated monomers and share the characteristic of single carbon-carbon (C-C) bonds along the polymer backbone. These bonds are highly susceptible to UV, IR, and radical attack (e.g. atomic oxygen). This results in the formation of radical and ionic organic fragments that recombine, attack, and propagate throughout the polymer, generating crosslinks and volatile organic fragments. These species have the effect of changing the physical properties of the film and accelerating the degradation of the material. Furthermore, aliphatic materials tend to lack the thermal stability and mechanical integrity associated with the other two groups of polymers. The partially aromatic polymers have double bonds and double bond character and have the classical resonance structure seen in the “circle-hexagon” unit, known as a benzene ring. These polymers gain an advantage from the resonance ring because the carbon atoms share a double bond (C=C), the ring structure is able to store electric charges generated by radical fragmentation and imparts a molecular stiffness that manifests itself through an increase in Tg and modulus. If a bond near or on the aromatic ring is broken, the atoms remain in place, increasing the chances for recombination of the severed atoms than radical propagation or volatile fragmentation. Thus, the addition of aromatic units within a polymer increases its thermal and radiation stability. Therefore, for the highest performance, a wholly aromatic polymer is a more logical choice.[11] These polymers tend to be higher in modulus, may not exhibit any melt flow or creep, and have strong intermolecular dipole interactions. These polymers possess the classic benzene ring structure throughout, and also have other ring structures (e.g. 5 membered imide ring) as part of the polymeric backbone. One thing to note is that in some of these polyimides, there is a C-C bond and a ring-O-ring single bond (Aryl ether). In discussing the C-C bond, the

connecting atoms are either benzene rings or -CF_3 carbons. These molecular structures impart the ability to stabilize radical charges and, in the case of oxygen, impart a partial resonance to the single oxygen atom via the unshared pairs of electrons orbiting the oxygen atom. Thus, the single bonds are reinforced single bonds imparting additional stability to the polymer. These aromatic polymers are characteristically radiation and UV resistant. However, like the majority of organic materials, they are susceptible to atomic oxygen attack, without the use of specific monomers designed to mitigate those effects. Although some of the polymers listed on Table 1, have found utility in Space, they are not available in the form factor required for a truly high performance solar sail. The reason is that the processes to make these material formats are optimized for other applications. Thus, using these materials as purchased may place limitations on some nontraditional applications, like solar sails.[11]

Polymer films are made by several standard processes; they include extrusion through a film die followed by roll-pressing, solution casing on a heated drum or belt and melt-blow forming (Figure 1). The melt-blow forming of polymer films is used for polymers that undergo melt flow, such as polyolefins and polyesters.

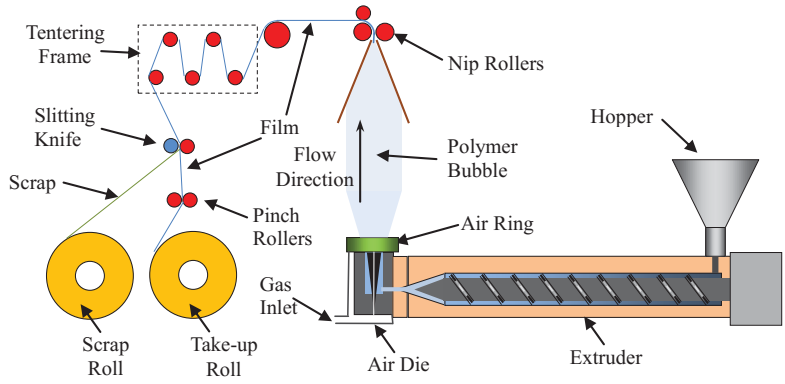


Fig. 1 Generalized Polymer Melt Blow-forming Film Process.

Here, polymer pellets are introduced into the hopper that feeds the extruder. The extruder melts and pressurizes the polymer and forces it through the air die which balances the air pressure inside the bubble controlling its size. The air ring keeps the molten film from touching, and adhering to, any external part of the die. As the bubble travels upward, it expands, becomes biaxially oriented and cools. As the polymer cools, its modulus increases and it is able to maintain its circumference. The film is then directed towards the Nip rollers where it is pinched together and fed down line into a tentering frame (optional) that further stretch orients and finishes the film. The film is then slit to

size and taken up on a roller for additional post processing as required. Thus, the Nip rollers provide a film that has a minimal film width of the bubble circumference divided by 2, with a thickness nearly double that of the bubble wall. The advantage in a blow forming process is that the polymer film is biaxially oriented from the circumferential expansion of the bubble and the axial orientation from the upward pull of the Nip rollers, and a high quality surface is produced. The solution or gel film casting process uses a drum or belt, and a gel applicator in place of an extruder and air die system. This setup is required for films that cannot be easily melt processed, or the economics do not justify such a system. In a film casting process, the extrusion system is replaced with a heated drum or belt and the film is essentially dried and cured, prior to entering a tentering frame. However, when making ultrathin films, less than 1 micron, from polymers that are not readily melt processable, the gel casting method becomes increasingly difficulty as the film is dried, chemically staged and mechanically manipulated. The casting surface must be wider than the film, increasing the requirement of even heating, and highly polished without surface contaminates that could overwhelm the physical integrity of the sub-micron film. Since the film is so thin, it is possible to create pinholes, tears and cracks as the film shrinks and is stripped from the casting surface. Thus, a system similar to the melt blow forming technique could be employed (Figure 2).

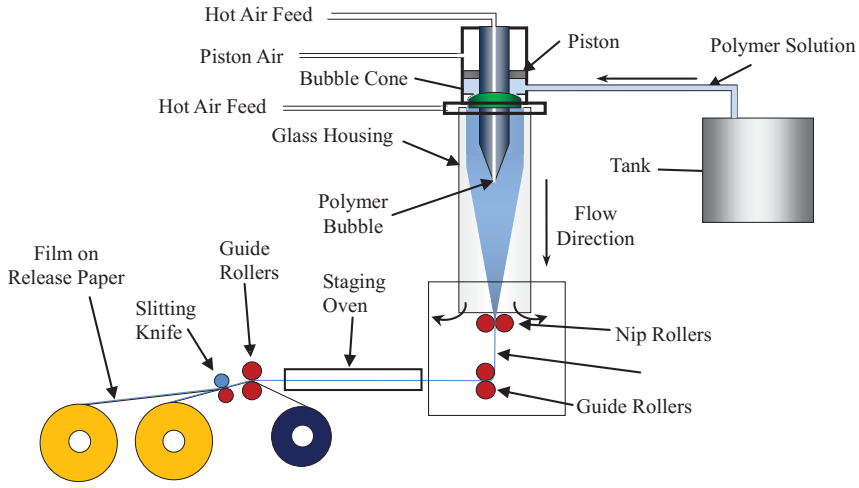
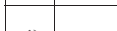
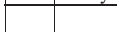
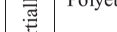
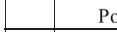


Fig. 2 Generalized Polymer Solution Blow-forming Film Process.

Table 1 Polymers Used or Considered for Use in Space as Metalized Thin Films [12].

	Unit Chemical Structure & Name	Tg / Tm (°C)	CTE (ppm/K)	Density (g/cm^3)	Tensile Strength (kPA)	Thickness (μm)	*Comment
Aliphatic	 Polyethylene (PE)	~ -100 / 110-138	180-200	0.9 - 1	5	12	Not UV stable, melt processed, can creep, poor adhesion.
	Polytetrafluoroethylene (PTFE or Teflon®)	125/327	135	2.15	21	25	An engineering polymer, not UV stable in high Vacuum, can creep, good hot-melt adhesive.
Partially Aromatic	 Polyethylene Terephthalate "PET" (Mylar®)	80/246	60-65	1.39	34	7	An engineering polymer, not UV stable, melt processed resin. Mainly used as a molding resin.
	Polyethylene Naphthalate "PEN" (Teonex®)	120/270	60-65	1.36	34	7	An engineering polymer, not UV stable, melt processed resin. Slightly higher Tg than PET
	 Polycarbonate (Lexan®, Macrofoil®)	145	75-80	1.2	60	75	An engineering polymer, formed through melt processing. A high degree of toughness, UV Stable.
Wholly Aromatic	 Polyimide (Kapton®, Apical®)	360-410	20-40	1.42 – 1.54	33.5	7.5	A gel cast film. Considered space durable, UV stable, creep resistant, excellent thermal stability, no melt flow, Tg is theoretical.
	Polyimide CP1®	263	50	1.43	14.5	2-3	A gel cast film. Considered space durable, UV stable, creep resistant, excellent thermal stability below Tg, heat sealable.
	Polyimide CP2®	209	50	1.43	17	2-3	A gel cast film. Considered space durable, UV stable, creep resistant, excellent thermal stability below Tg, heat sealable.
	Polyimide ISAS-TPI (IKAROS)	270	Not Det.	Not Det.	Not Det.	7.5	A gel cast film. Considered space durable, UV stable, creep resistant, excellent thermal stability below Tg, soluble in the Imide form, heat sealable.[11]

*Unless stated otherwise, physical properties are determined at 23°C and are approximate.

In the diagram shown in Figure 2, the film is sent downward, rather than upward (Figure 1), because the polymer gel has a minimal amount of mechanical integrity and gravity will distort the bubble, scraping the material. The downward forming of the bubble has several unique dynamics, as the bubble size increases under flowing hot air; the solvent is driven off causing the polymer gel to contract. If the polymer is a reactive precursor, usually the case for many polyimides, there are curing mechanisms that also shrink the film, evolve volatiles, and cause a non-linear change in viscosity, modulus and tensile strength. The formed film is very thin, biaxially oriented, and can be partially cured, supported and stored on release paper for additional post processing, (Figure 3). Bubble forming from solution requires additional process knowledge that involves downstream manipulation once the bubble is pressed into film at the Nip rollers. Some of the advantages of this process are that COTS resins and unique polymeric chemistries can be used that create high strength films that cannot melt-flow and are therefore thermally stable (degradation, physical properties and CTE) from cryogenic to above 300°C. The intermediate chemical staging allows for post process heat sealing of these films that possess a Tg, or have residual solvent that helps with viscous flow prior to the final processing step. The film is not in physical contact until the Nip rollers which, due to their small diameter, are easier to surface finish than a large rotating belt or drum. These Nip rollers are easier to evenly heat and have minimal contact with the film. Lastly, a roll of film, blown from either process, can be used for other applications including, separation membranes, dielectric innerlayers, hot-melt bonding adhesives, and metalized for other products. Additional applications are available when certain filler particles are added to the resin prior to blow-forming (Figure 4).

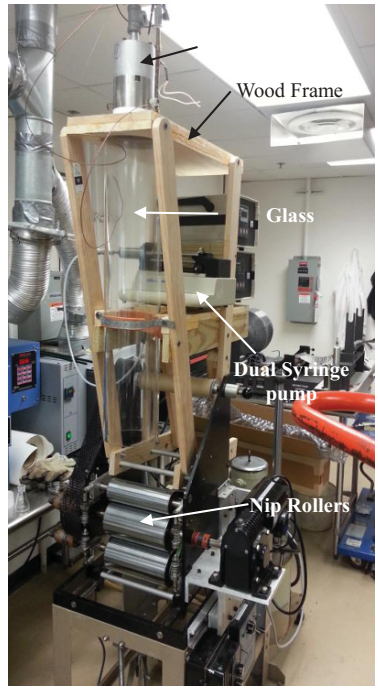


Fig. 3 NASA Langley's Prototype Polymer Solution Blow Film Caster.

At NASA's Langley Research Center, we have designed a lab scale prototype film caster that targets continuous production of 20 cm wide film (Figure 3). Pictured is the nearly complete assembly featuring the piston film forming head, glass enclosure, dual syringe pump, thermally control nip rollers and the take-up reel. The temporary wood stand allow for rapid adjustments in the configuration prior to optimization where aluminum framing elements will be used. This system will allow us to demonstrate the ability to produce ultrathin films from COTS resins, or our own chemistries. The advantage in exploring these different film chemistries is that properties can be tailored specifically for an application. For example, the polymers in metallized films rarely match the CTE of the metallic coatings over a broad temperature range. This is important if a mission requires that a solar sail need to approach the sun below 0.5 astronomical units (AU). Here, thermal build-up in the film can cause the metallic substrate to crack and peel, and the film can suffer permanent deformation or physical failure. There are two methods where this CTE matching can be achieved, by creating a metalized film that minimizes the temperature rise i.e. $a_s/e \sim 0$, or using chemistry and fillers to tailor the CTE of the substrate to the metal coating. Below is a

commercial product, developed and licensed from NASA, that uses a seamless polyimide belt with carbon nanotube (CNT) fillers. This Intermediate Transfer Belt (IBT) transfers the digital image from the 4 color dye Organic Photo Conductor (OPC) Drum onto the paper in color laser printers. This polymer film is made in-house by Kolon Industries of Korea specifically for this application that requires the right amount of electrical conductivity for image retention/renewal and thermal conductivity at elevated temperatures to avoid image distortion (Figure 4).

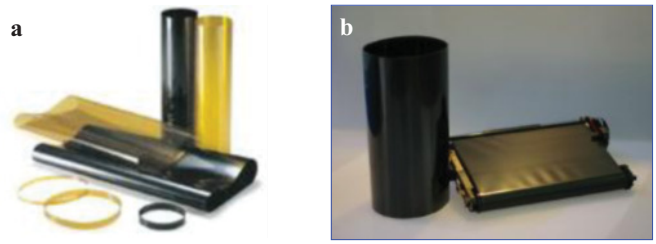


Fig. 4 (a) Filled (CNT) and neat resin polyimide belts, (b) IBT belt and IBT belt on mechanism.

A second approach would be to develop a film with the chemical structure that closely matches the CTE of the metallic coating, or using fillers that decrease the overall CTE of the polymer film towards that of the metal coating. Unlike rectangular sails, sail blades can be made by a different process starting with neat polymer film, including melt-blown or solution-blown film. Reinforcement is then added or processed into the membrane to increase rigidity prior to metallization. In order to do this, there should only be two different thicknesses, that of the film, and that of the reinforcing structure. This will assist in the degassing of the sail after folding or roll-up for pre-flight storage. The advantage of this method is that the process is scalable and therefore, sail area is theoretically unlimited.

The next process step involves taking the polymer film and integrating a structural reinforcement into the sail (Figure 5). Here the roll containing the thin polymeric film backed with release paper and fed through an embossing and cutting roller dies, that eject the scrap, and past a roll where the release paper is removed. Next, a roller die folds the film along the embossed lines and lightly heat bonds the layered film. The film is then passed through rotating folding bars that fold the film transversely while moving along with the film until the folds are clamped by heated blocks that hot press the film while the folding bars are rapidly removed, counter-rotated and moved back into position. The folded film then travels into a hot roller section where heated embossed rollers weld the folds of the film into a continuous reinforced edge with axial stiffeners and transverse battens, and remove any

residual volatiles. The sail blade then passes through cold rollers, then punch rollers that put tiny holes in the sail to assist in degassing during roll-up, and to assist in making electrical connection during metallization.

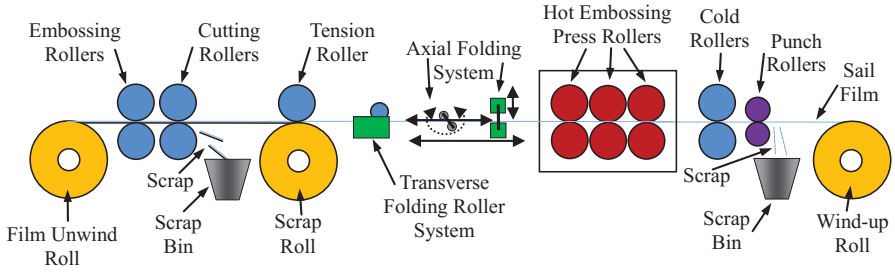


Fig. 5 Post Processing Line for Creating the Sail blade Architecture.

The details of folding the membrane to add different types of structural reinforcement on the sail blade are shown in Figures 6 and 7, and are some examples of 3 and 5 layer folded reinforcement. Table 2 contains the symbol key. Here, the punch out areas (6a and 7a) are required to fold the film, with biaxial reinforcement, into a continuous blade with only two thickness (6b,c and 7b,c). The reason for this is to avoid bumps and localized strains that occur on both roll-up and deployment. This allows the sail membrane to remain suspended between the thicker reinforced areas and to decrease static charge build-up during furling/unfurling. This also helps alleviate the effects of auto-adhesion of the membrane material to itself during storage in an evacuated environment. For this reason, it is equally important that the fold spacing be equal depth on each side of the sail membrane.

Table 2 Key to Figures 6 and 7.

Line Style	Key
————————	3 Layer Fold
– – – – – – – –	2 Layer Fold
- - - - - - - - - - - -	Fold/Edge Line
Bold Letters	Above Sail Film
Gray Line	Underneath
	Cut Out
	1 Layer Film
	3 or 5 Layer Film

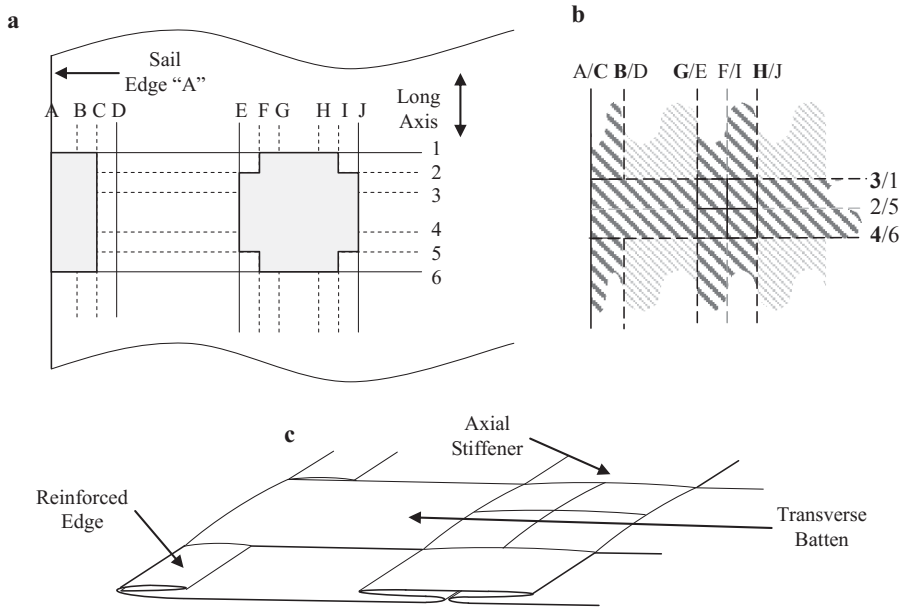


Fig. 6 3-Layer Folded Sail Membrane: (a) Film with Cutout Slots, (b) Folding Diagram, (c) Expanded Vertical View of folded sail without punch holes.

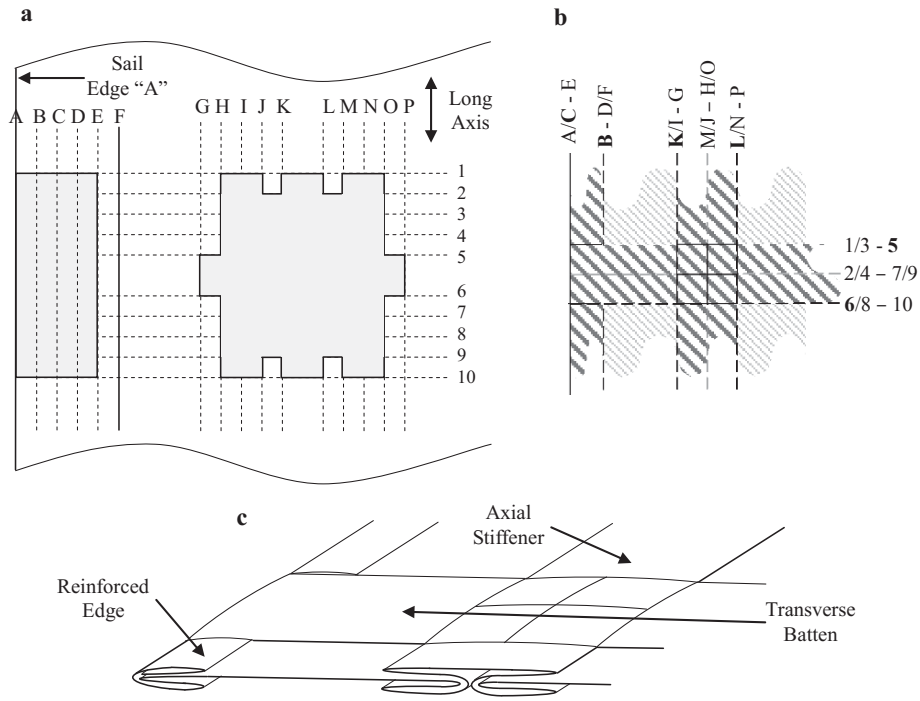


Fig. 7 5-Layer Folded Sail Membrane: (a) Film with Cutout Slots, (b) Folding Diagram, (c) Expanded Vertical View of folded sail without punch holes.

What is evident is that any internal support feature requires an odd number of folds, whereas the edge can take any thickness. The actual placement, number, width, thickness, and type(s) of the reinforcing folds comprising the structure depend on factors involving the sail parameters and the environmental requirements of the mission. These folds can be replaced by adhesive-backed tape or a hot-bonded structure, but then there are additional considerations of the effects that dissimilar materials may have on the sail over time.[11] This process requires additional fabrication hardware, especially if the film is extremely thin.

The last step involves metalizing the blade (Figure 8). Here, the folded membrane wind-up reel (Figure 5) becomes the sail membrane reel and the blade is fed into two metal deposition chambers that deposit a layer of metal on each side of the folded film.

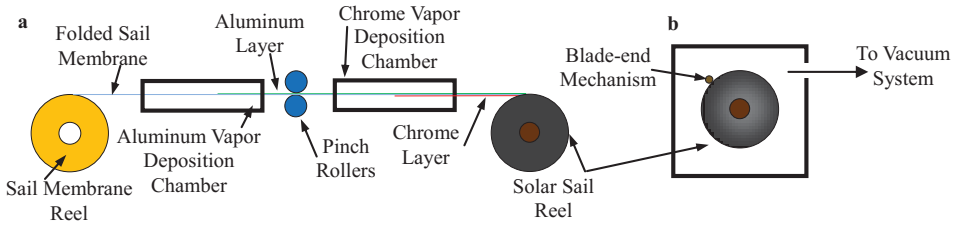


Fig. 8 (a) Continuous Two Sided Metallization of the Sail Blade Membrane, (b) Vacuum Degas and Storage.

As previously stated, the minute punched-out holes in the blade provide a conductive pathway connecting the metallic layers thereby alleviating static charge buildup across the dielectric membrane, and allows for degassing of trapped air during storage. The post-metalized sail blade is then wound onto the solar sail reel, slowly degassed and stowed until unfurling is required.

III. Conclusions:

In order for Solar Sails to be selected as mission enabling technology, the move away from COTS technology for the sail membrane, and traditional solar sail design may have to occur. This shift has already taken place with the IKAROS solar sail demonstrator. However, more progression is needed, especially in the sail material and construction. This paper has put forward construction methods, which are current manufacturing practices, which could be applied towards the advancement of the major enabling factor for solar sails, the sail material. Here, the advocacy is focused on optimizing materials and processes, rather than compromising the application with a COTS material. The details of basic chemistry, thin film manufacture, mitigating static charge build-up, building in several types of reinforcement, roll-to-roll evaporative coating, and post-storage degassing issues, have been covered to the point where subject matter experts can began to design in the further details.

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The Complex Irradiation Facility at DLR-Bremen

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The performance of solar sails is significantly affected by both the sail geometry and the physical properties of used materials which degrade under space conditions. Hence it is of utmost importance for a sail-mission to select such sail materials which degrade as little as possible in space. The Complex Irradiation Facility (CIF) is a new laboratory at DLR-Bremen to study the degradation behavior of materials under space conditions. The CIF allows the simultaneous irradiation with three light sources and with a dual beam irradiation system for the bombardment of materials with electrons and protons having energies up to 100 keV. It is eminently suitable to perform degradation investigations at solar sail foils.

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I. Introduction

THE Complex Irradiation Facility (CIF) was designed and commissioned with the aim to perform material investigations under radiation conditions as prevalent in space environment. The idea is to combine multiple radiation sources at a vacuum irradiation chamber. A driving motivation to establish the CIF at the DLR Institute of Space Systems was its participation in the DLR-ESA Gossamer project. Although the CIF can be used for a large variety of material studies, many of its features are especially designed for the study of solar sail materials. This concerns e.g. the special design of the sample holder that is well-suited for thin foils. It concerns as well the dimensioning of the electron and proton accelerator with respect to energy and intensity range. Both cover the bulk of the corresponding solar wind parameters. Last but not least it concerns the quality of the vacuum which can be achieved in the CIF. To attain a high vacuum special effort was made.

The quality of the vacuum is an important aspect to detect material degradation because molecules of the rest gas can interact with the radiation and degrade the surface of the sample. This would distort the results of degradation studies which are actually designed for conditions of the interplanetary space. Therefore the complete facility has been manufactured in UHV-technology with metal sealings and without organic compounds (rubber vacuum sealings, pump oil) to avoid self-contamination which could affect the results of the experiment.

It is known from many evaluation tests [1, 2] that particle and UV radiation can significantly degrade materials and, e.g., lead to changes in their mechanical behavior or thermo-optical properties [3]. That was taken into account for the determination of the required parameters of the CIF.

An important number that characterizes the performance of the CIF is the acceleration factor. It is defined as the ratio of the intensity of the degrading radiation applied to a material during a laboratory test to the intensity of the same degrading parameter in space environment [2]. The spectra of the light sources are shown in comparison to the extraterrestrial solar spectrum [4, 5] in Subsections III A-C. The spectral distribution of the accelerating factor of the complete electromagnetic radiation provided by all light sources is presented in the conclusion.

Another aspect, beside the irradiation performance, is the measurement engineering to detect the changes of material properties caused by irradiation. For this purpose the CIF is equipped with a mass spectrometer at the irradiation chamber to qualify the outgassing behavior of the sample. Furthermore, our department has commissioned an optical measurement system, consisting of a FTIR spectrometer and Ulbricht spheres. It captures the changes of thermo-optical properties by measuring reflectivity and emissivity. This can be done presently only ex-situ, i.e. the sample has to leave the vacuum and the influence of the atmosphere cannot be

avoided. However, it is foreseen until 2014 to achieve the development and commissioning of an in-situ measurement system. Another possibility to qualify optical properties is the measurement of the light pressure [6] before and after an irradiation test. Light pressure measurements return a kind of integral information about the state of degradation of a solar sail. Such a light pressure measurement facility will be placed outside the CIF.

In order to study the effects of degradation on to the elastic properties of the irradiated materials, a tensile testing facility will be established at the CIF. Clearly, such studies can be performed only ex-situ.

The experimental studies of degradation effects have to be accompanied by theoretical efforts. This is necessary not only to get an idea about the physical processes that cause degradation. It is also necessary for profound predictions of changes in the material properties on long-term space missions, where a simple scaling by use of the acceleration factors provided by the CIF will fail. First attempts are in progress at DLR-Bremen [7].

II. Overview of the configuration and instrumentation of the CIF

A. Geometry and technical parameters

The CIF consists of a vacuum irradiation chamber (400mm in diameter) which is connected to a lock chamber for the placement of the sample into the vacuum environment of the facility (see Fig. 1). The irradiation chamber has got four tubes with flanges for the connection with the radiation sources. They are arranged at an angle of 30 degrees to the neighboring one in the same level for the accelerator beam line, the solar simulator, and the argon-VUV-source. The Deuterium-UV-source is located above the solar simulator at an angle of 30 degrees between their axes. The axes of all radiation sources are crossing in the center of the irradiation chamber where the target station is mounted. The geometry of that arrangement, the target mounting, and the radiation sources itself are dimensioned in a way, that a square area of 80mm can be irradiated simultaneously with all sources.

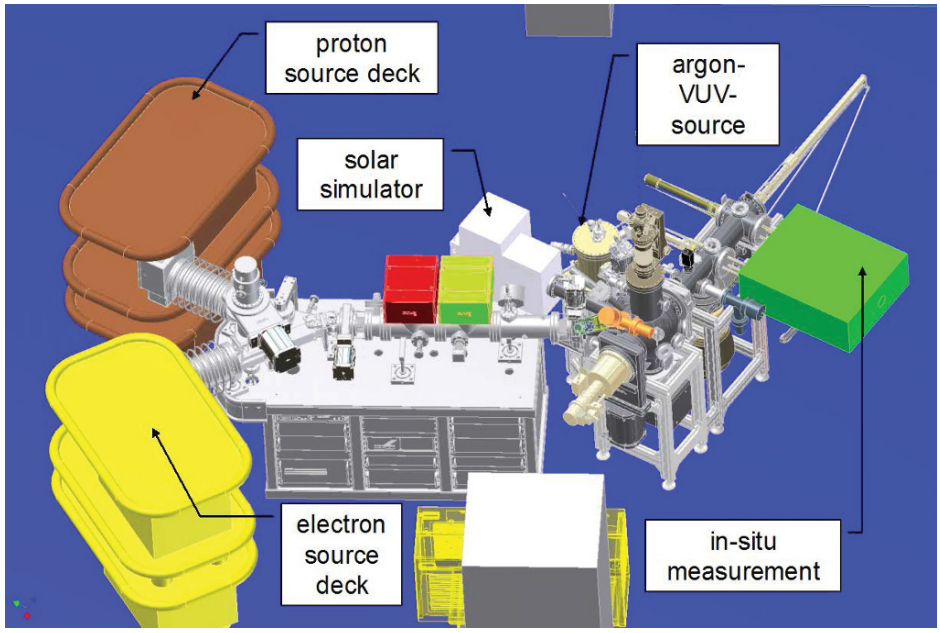

Fig. 1 Schematic overall view of the CIF configuration.

The sample mounted in a holder will be inserted into the lock chamber and transferred after vacuuming by a magnetic manipulator into the sample station of the irradiation chamber. Additionally, there are magnetic manipulators installed in cross direction at the lock chamber. They can bring the sample holder in front of the

Ulbricht spheres of the planned in-situ measurement system to measure the reflectivity before and after the irradiation without leaving the vacuum environment.

The vacuum system at the irradiation- and the lock chamber consists of a turbo molecular pump, an ion getter pump and a cryogenic pump to reach a pressure in the UHV-range. This will allow measurements with the quadrupole mass spectrometer which is installed at the irradiation chamber.

Table 1 Technical Parameters of the CIF

Description		Value
Vacuum irradiation chamber	Volume	Circa 33.5 l (400 mm diameter)
	Irradiated zone	80 mm diameter
	Vacuum pressure	$<10^{-8}$ mbar (without VUV-source) $<10^{-6}$ mbar (depending on VUV settings)
Light sources	Solar simulator	250 to 2500 nm (5000 W/m^2)
	Deuterium UV source	112 to 400 nm (1.65 W/m^2)
	Argon VUV source	40 to 410 nm (50 mW/m^2)
Proton and electron source	Current at lower energy range (1 to 10 keV)	1 to 100 nA
	Current at higher energy range (10 to 100 keV)	0.1 to 100 µA
Target thermal conditioning	Heating	Halogen radiation (600 W, 450 K)
	Cooling	Liquid nitrogen (lN_2: 80 K)

B. Target mounting

The target mounting (see Fig. 2) can be rotated about 30° into two directions. Thereby, either the VUV-radiation or the particle beam line can be perpendicularly oriented with respect to the sample surface. In the normal transfer position the sample surface is oriented perpendicular to the axis of the solar simulator.

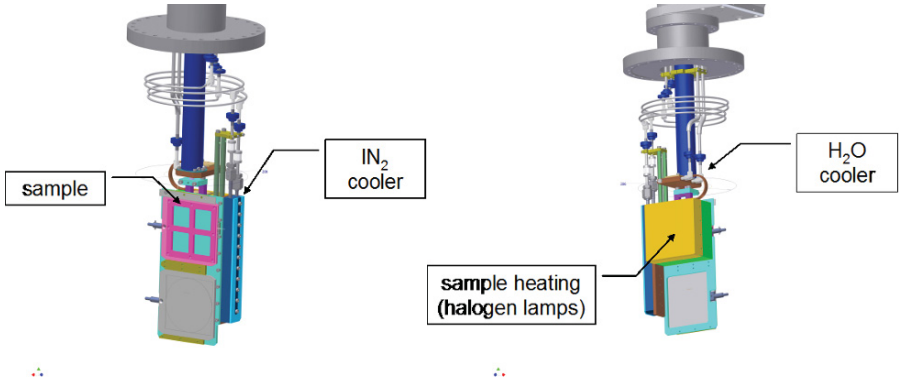

Fig. 2 Sample station in the center of the irradiation chamber (front and back side).

The sample station is vertically adjustable, so that the sample holder can be inserted at the upper or the lower position at the station. The upper position provides a heating compartment from the backside of the sample. This

can be used e.g. to simulate ambient conditions closer to the Sun, which cannot be supplied by the solar simulator. Furthermore the sample can be cooled to liquid nitrogen level to simulate ambient conditions of the deep space.

III. The radiation sources of the CIF

A. The solar simulator

The solar simulator is used to simulate the electromagnetic spectrum of the Sun in the wavelength range from 250 nm to 2500 nm. It consists of a Xenon lamp with 1600 W electrical power, an optical mirror and lens system to concentrate and homogenize the light spot. An air mass zero filter is mounted to eliminate the Xenon peaks and to fit the spectral distribution to the extraterrestrial one defined in the ASTM E-490 [5]. Figure 3 illustrates the spectral irradiance of the solar simulator as measured at DLR-Berlin with the minimum, and the maximum settings of electrical power.

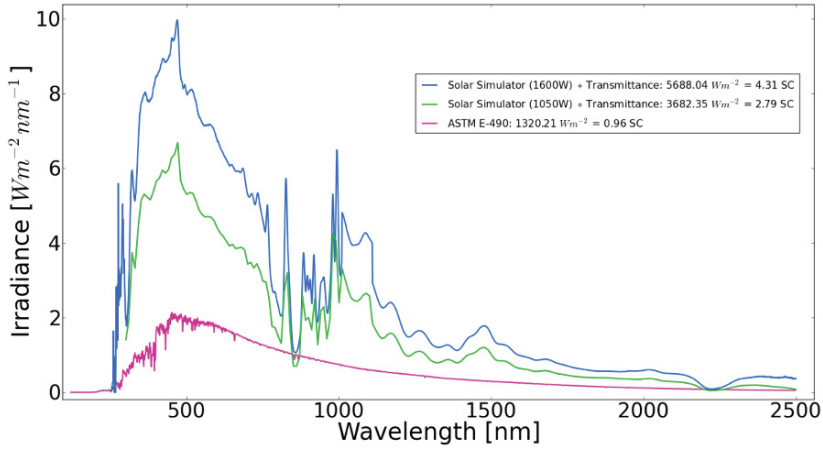

Fig. 3 Spectral irradiance of the solar simulator with the minimum and maximum setting of electrical power in comparison to [4].

The measured values are corrected with the transmittance of the quartz glass window at the irradiation chamber. Presently there are ongoing activities to enhance the intensity in the UV range below 250 nm (see Section IV).

B. The deuterium-UV-source

The deuterium-UV-source is a modified standard solution which is intended to be used as an excitation source in research and development. The modifications are the MgF_2-window that allows a transmission down to 112 nm, and an indium sealing between window and vacuum flange to avoid contamination.

The working principle of the source is the electrode excited gas discharge. The construction consists of a glass bulb, which is filled with deuterium, a cooling water jacket, and a vacuum flange (DN40CF). The electrical leads from the socket of the housing to the interior of the bulb are melt-sealed and vacuum tight. The glass bulb contains the thermionic cathode, which is heated by means of direct or alternating current, the ring-shaped anode, and an aperture of 1 mm diameter, where the discharge arc between cathode and anode is guided through. The light intensity can be controlled by the anode current between 0.8 A and 1.9 A. That corresponds to 120 W and 200 W electric power. The spectral irradiance, both at minimum and maximum setting, was calibrated by Physikalisch Technische Bundesanstalt in Berlin (PTB) (see Fig. 4).

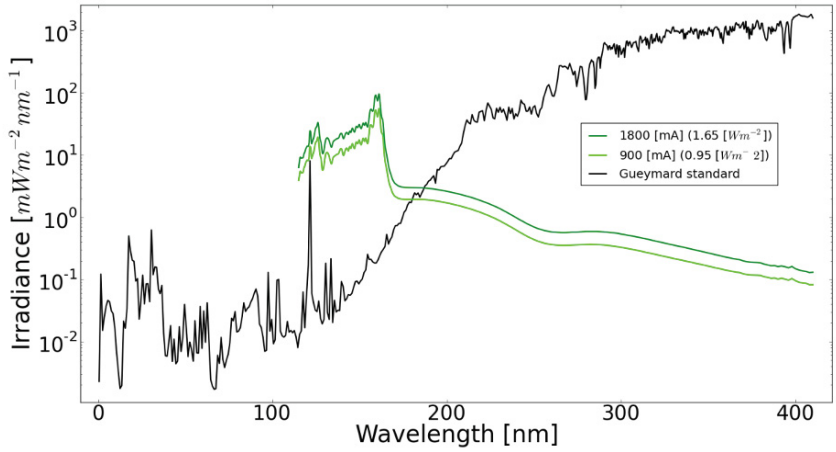

Fig. 4 Spectral irradiance of the deuterium-UV-source with the minimum and maximum setting of electric power in comparison to [5].

The stability of the source can be observed by shifting a monitoring sensor, mounted on a linear manipulator, onto the axis between source and sample.

C. The Argon-gas-jet-VUV-simulator

The following description of the VUV-simulator is an abstract of the detailed presentation, its design, and calibration (see [8]). The advantage of the VUV-simulator is the generated electromagnetic radiation in a broad

spectral range (soft X-rays, VUV, NUV) with relatively high intensities at lower wavelengths (<110 nm). This is ensured by the design that avoids windows and is based on the semi-cryogenic version of a previous simulator, developed by the Institute of Low Temperature Physics and Engineering in Charkov, Ukraine [9].

The radiation is produced by the transition of electrons belonging to excited gas atoms into their ground state. The excitation occurs by electron bombardment of a spatially limited supersonic gas jet which flows into a vacuum chamber. Semi-cryogenic design means that the vacuum is maintained by a combination of cryogenic and mechanical pumps. Figure 5 illustrates the arrangement of the electron source and the gas jet inside the VUV-chamber of the simulator. The generated VUV-light is realized at the spot side perpendicular to the figure's plane. The electrons which pass through the gas jet are caught by the collector at the opposite site of the source. The components of the vacuum system are visualized better in Fig. 6, whose plane is perpendicular to the plane of Fig. 5.

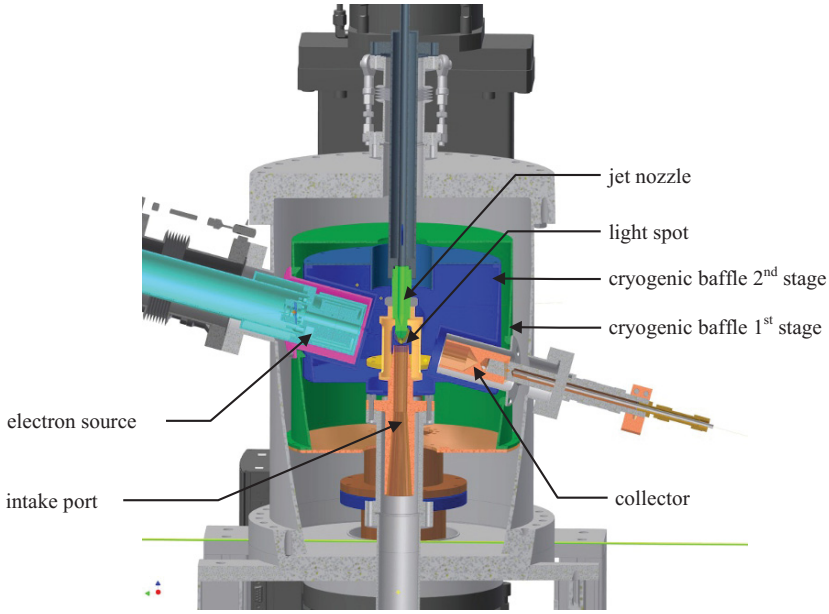

Fig. 5 Sectioning of the VUV-simulator along the electron beam.

The gas jet is injected by a nozzle from top of the VUV-chamber into the vacuum. The flow rate is stabilized by a flow controller and is adjustable by software in the range between 0 and 5000 sccm (standard cubic centimeter per minute). The bulk of the gas load is pumped out through an intake port at the bottom of the VUV-chamber by a screw pump. The rest of the gas, which is a small fraction of about 3%, is removed by cryogenic condensation at two baffles. Each of them is connected to one of the both stages of the cold head of a commercial cryogenic pump (helium cooling machine). The 1st stage reaches a bottom temperature of 15K

without gas load. It increases to about 20K under gas load by formation of Argon ice, which decreases the pumping power gradually and limits the operating time. The temperature is logged permanently as an indicator when regeneration is necessary to remove the ice. During this process the cryogenic pump is turned off while the mechanical pumps continue. The temperature inside the VUV-chamber increases and the residual gas, which was frozen on the cold baffles, is pumped out.

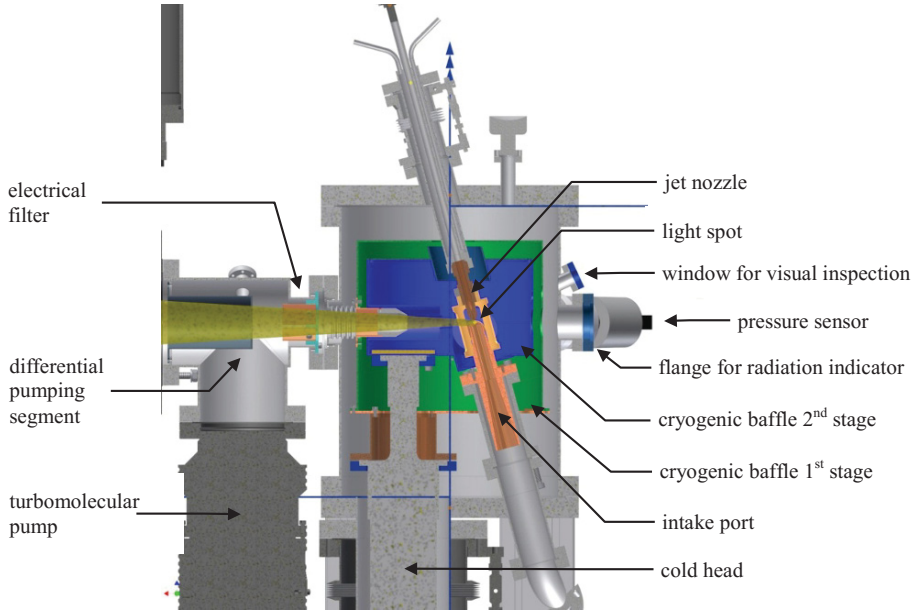

Fig. 6 Sectioning of the VUV-simulator along the light outlet (yellow).

The gas nozzle is thermally connected to a water circuit for tempering it to avoid a freeze. A differential pumping segment is installed at the light outlet consisting of a turbomolecular pump and an aperture assembly. It improves the pressure conditions inside the irradiation chamber. So tests can be performed in the chamber at pressures of 10^{-6} mbar and less. The fraction of light which is permitted to enter the irradiation chamber is colored yellow in Fig. 6. An opening angle of about 6° is defined by the apertures in the cryogenic baffles and at the differential pumping segment. It ensures the irradiation of the target area with a diameter of 80mm at a distance of about 770mm from the spot. To avoid that charged particles produced at the spot can reach the irradiated object, an electrical filter is installed at the light outlet, which deflects them beyond the radiation flow.

The electron source is realized as a Pierce-type model with a magnetic lens behind the anode. The LaB_6-cathode is heated up electrically by adjusting the cathode voltage depending on the demanded emission current. The electronic control unit of the source provides a PID algorithm which stabilizes the emission current by varying the Wehnelt voltage automatically. The beam is focused by setting the current for the magnetically lens.

Two flanges are located at the opposite site of the light outlet. Each is connected with a window for visual inspection of the luminous jet and with a radiation indicator compartment to observe the stability of the radiation (see Fig. 6).

The spectral intensity distribution of the radiation depends on the gas mixture, the flow rate of the jet, the electron current, on the alignment between beam and jet, and the focus of the electron beam. Figure 7 illustrates these relations qualitatively by showing the size and intensity of the spot with different settings for the emission current and for the gas flow. The photos were taken through the window at the opposite site of the light outlet (see Fig. 6)

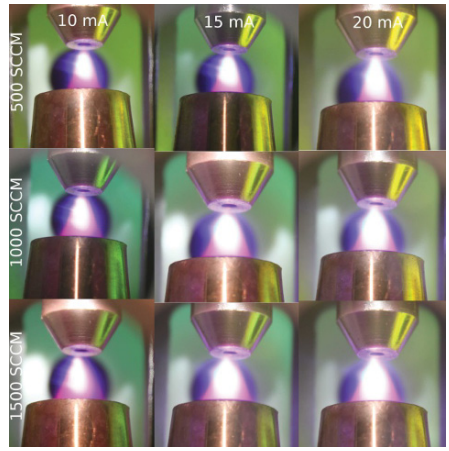

Fig. 7 Pictures of the VUV-spot with different settings of the gas flow rate (rows: 500, 1000, 1500 sccm) and the emission current (columns: 10, 15 and 20 mA).

During the calibration campaign at the radiometry laboratory of the PTB at Berlin Electron Storage Ring Society (Bessy II) for Synchrotron Radiation in Berlin-Adlershof the alignment and the focus mentioned above were optimized. The following settings were chosen for the calibration:

Table 2 Parameters of the VUV-simulator for calibrated spectra.

Parameter	Value, Description
Gas mixture	Ar (98.5 %), He (0.5 %), Kr (1 %)
Electron energy	1 keV
Electron emission current	20 mA
Current of magnetic lens	175 mA
Flow rate of the gas jet	300, 600, 1200 sccm

The results of the calibration campaign are shown in Figs. 8 and 9 in the wavelength range from 40 nm to 410 nm.

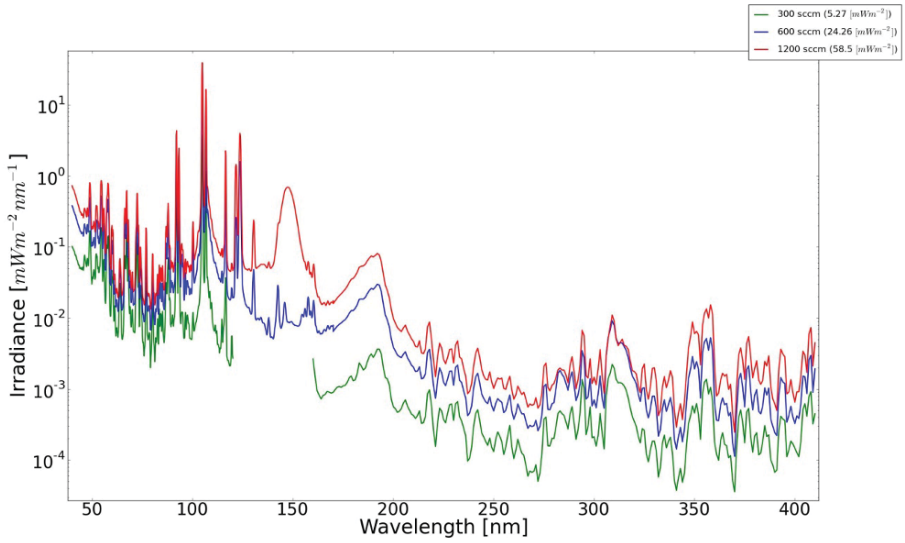

Fig. 8 Validated spectra of the VUV-simulator with an emission current of 20 mA and a gas flow of 300, 600 and 1200 sccm.

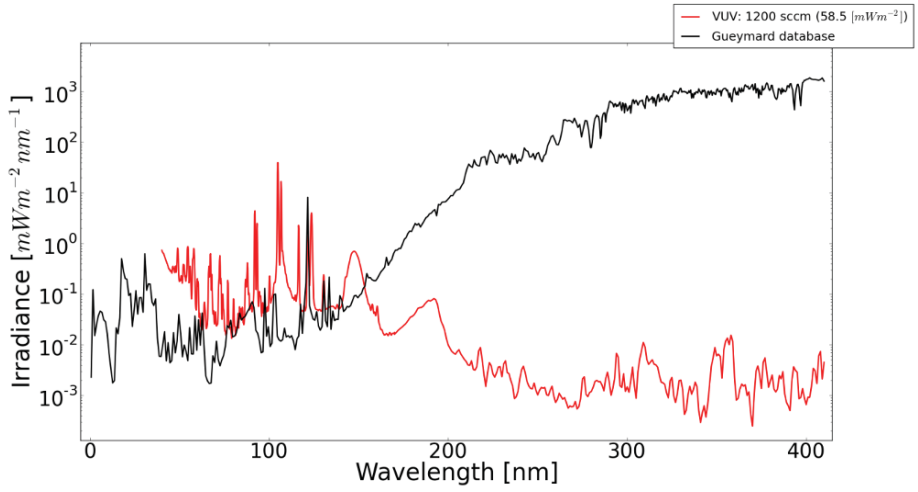

Fig. 9 Validated spectrum of the VUV-simulator with an emission current of 20 mA and a gas flow of 1200 sccm in comparison to [5].

D. The 100 keV Proton / Electron Dual Beam Irradiation System

The Dual Beam Irradiation System is designed to irradiate samples with protons or electrons independently or with both particle species simultaneously. The selected beam(s) can be scanned over the samples. All relevant parameters can be adjusted remotely via the computer control system. Figure 10 illustrates the configuration of the beam line with its main items in view from top.

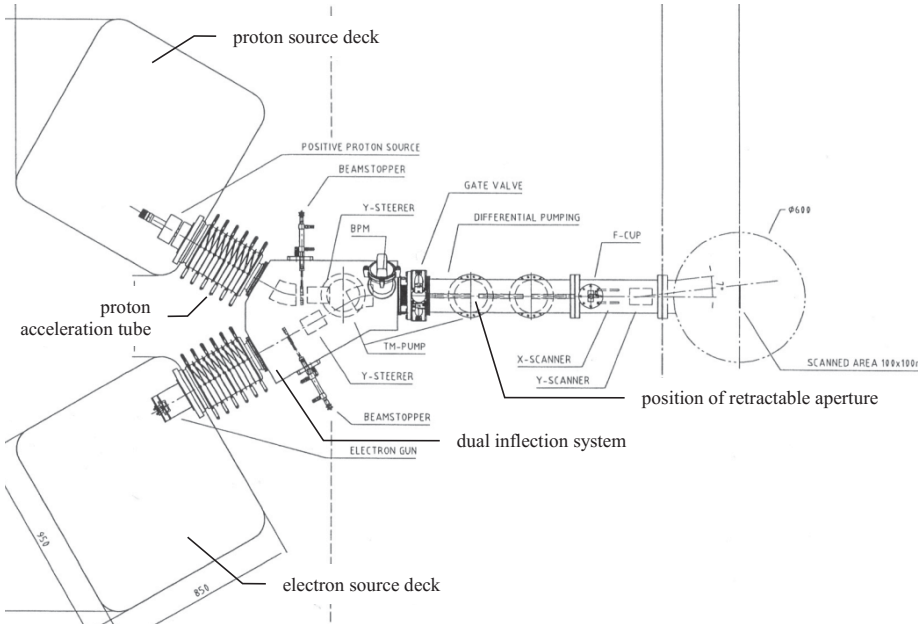

Fig. 10 Layout of the dual beam irradiation system.

The vacuum system of the beam line consists of the acceleration tubes (one for each particle species), the dual inflection system, and the differential pumping segment. The dual inflection system is equipped with a turbomolecular pump and the differential pumping segment with two ion getter pumps.

The protons are produced by ionization of hydrogen, which is stored in a lecture bottle inside the source deck. After pressure reducing the hydrogen is guided through a thermo-mechanical gas inlet valve with remote control to the ion source. The ionization takes place inside the glass bulb of the source by excitation with a radio frequency, which is capacitive coupled to the bulb. The plasma is confined and positioned by an axial permanent magnetic field. The source output is optimized by control of the source gas pressure and oscillator loading. There are not only positive ions H^+ (1 atomic mass unit) generated with hydrogen gas, but also the molecule ions H_2^+ (2 atomic mass units) and H_3^+ (3 atomic mass units). This makes a mass selector necessary, which is installed in the dual inflection system (see below).

The electrons are generated by a lanthanum hexaboride (LaB_6) cathode, which is a high performance, resistively heated, thermionic electron source. A heater current and Wehnelt voltage control the electron current.

Both particle species are accelerated in appropriate tubes by a high voltage, which corresponds to the required energy. The acceleration tubes are manufactured as a metal to ceramic brazed assembly with no organic compounds. After the acceleration the beam(s) are deflected onto a common axis in the dual inflection system. This is realized by inflection magnets. The proton inflection magnet works additionally as a mass selector for the different ion species (elemental and molecular Hydrogen ions). The electron beam is magnetically shielded from the comparatively strong magnetic fields that are applied in the proton inflection line. Both the vertical and the horizontal position of each beam are adjusted by separate magnetic steerer, which are used to compensate the influence of the electron inflection magnet to the proton beam too. The negative influence of the magnetic components of the proton beam and/or the Earth magnetic field to the electron beam is corrected with special magnetic shielding techniques, compensating magnetic fields by correction coils and advanced software tools. A separate beam stopper for each particle species can be inserted pneumatically to block the beam while the other source is running. This is useful to tune each source separately.

The Beam Profile Monitor (BPM), see Fig. 10, is used to analyze the horizontal and vertical beam size and position as well as its intensity distribution separately for each particle species. For that purpose, either the proton or the electron beam has to be blocked. The BPM is located behind the inflection segment at the beam line.

A retractable aperture can be inserted pneumatically into the beam at the differential pumping segment. It is used to reduce the current.

If the current is adjusted it can be measured with the retractable Faraday cup (F-cup, see Fig. 10). It can be inserted pneumatically in front of the scanning section. This must be done separately for each particle species, i.e. again one beam must be blocked.

The electrostatic scanning segment contains two sets of deflection plates, which deflect the beam in two directions perpendicular to each other. The triangular voltages for these plates and its crystal locked frequencies are carefully chosen to eliminate the possibility of synchronization caused dose non-uniformity. The amplitude of the deflection voltages is adjusted by use of the Faraday cups, which are mounted at the corners of the sample station. They detect if the sample area is scanned completely.

IV. Conclusion

The performance of solar sail propulsion is significantly affected by both the sail geometry and the physical properties of sail materials [3]. Hence, when planning a solar sail mission, one has to choose those materials which suffer as little as possible under space conditions. The knowledge of degradation processes of materials under space conditions must be deepened. Therefore, the DLR Institute of Space Systems in Bremen has commissioned a new facility to study the behavior of materials under complex irradiation and to investigate their degradation in a space environment. With the CIF it is possible to irradiate samples with three light sources (for the simulation of the spectrum of solar electromagnetic radiation) simultaneously with protons and electrons. The light sources are a solar simulator with a Xenon-lamp (wavelength range 250 to 2500 nm), a deuterium-UV-source (112 to 400 nm), and an Argon-gas-jet-VUV-simulator (40 to 410 nm). The latter enables the irradiation of samples with shorter wavelengths below the limitation of any window material. All sources have been calibrated by the PTB and DLR-Berlin. The availability of UV-radiation below 115 nm is a particular advantage of the CIF. For such small wavelengths ionization of metal atoms can appear. In the Figs. 11and 12 the spectral distribution in the short wavelength range is presented in detail, while Fig. 13 gives an idea about the acceleration factor of the whole electromagnetic spectrum covered by the CIF sources.

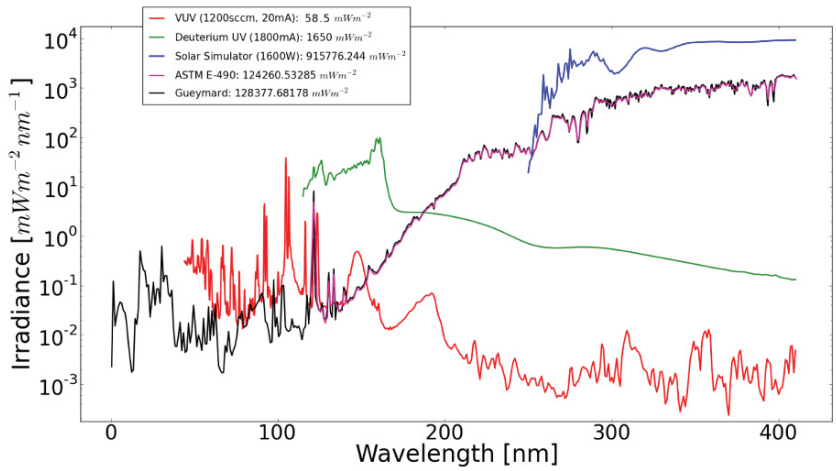

Fig. 11 The spectra of the CIF light sources compared to the solar radiation standards [4, 5]. Both standards are shown, however, the ASTM E-490 standard (pink line) reaches down to 120 nm only, while the Gueymard standard (black line) covers the range from 0.5 nm to 450 nm.

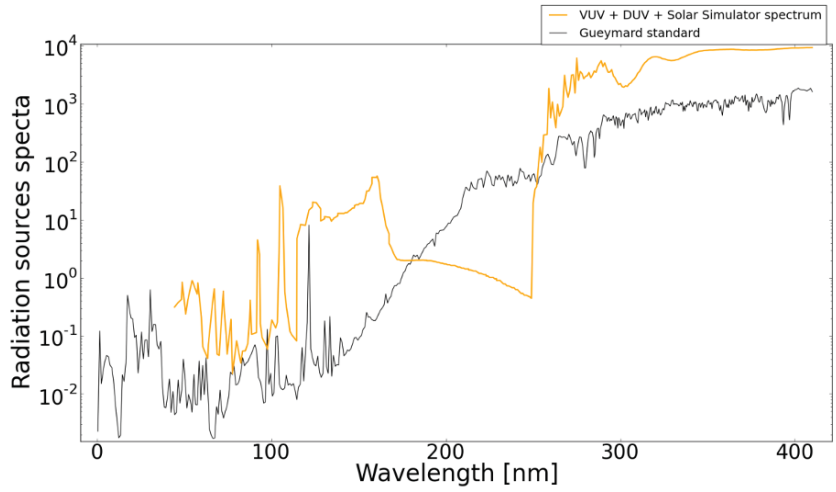

Fig. 12 Overall spectrum of electromagnetic radiation of the CIF in the wavelength range from 40 nm to 410 nm in comparison to [5].

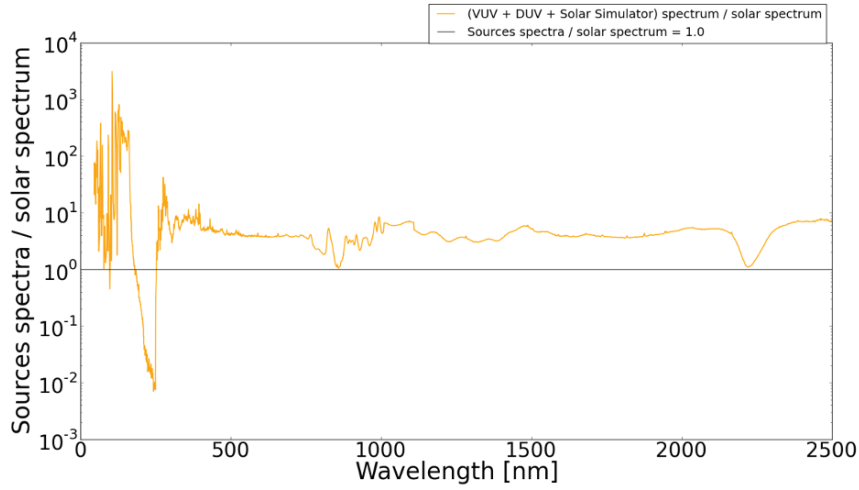

Fig. 13 Spectral distribution of the acceleration factor of electromagnetic radiation in comparison to [4,5] in the whole spectral range covered by the CIF radiation sources.

As seen in Fig. 11, 12, and 13 there is a significant gap between 180 and 250 nm in the UV-spectrum and in the corresponding acceleration factor. This will be closed in the near future by the installation of a not ozone free Xenon lamp in the solar simulator.

In addition to the different light sources, CIF provides electron and proton irradiation. The charged particles are generated in a low energy range from 1 to 10 keV with currents from 1 to 100 nA and in a higher range from 10 to 100 keV with 0.1 to 100 μA. Both particle sources can be operated simultaneously. However, another advantage of the CIF is that electrons and protons are led in a common beam onto the sample. In order to model temperature variations as appear in free space, the sample can be cooled down to liquid Nitrogen level and heated up to about 450 °C by halogen lamps behind the target.

The complete facility has been manufactured in UHV-technology with metal sealing. It is free of organic compounds to avoid self-contamination. The different pumping systems achieve a final pressure in the 10^{-10} mbar range (empty irradiation chamber). Besides the installed radiation sensors, which control the stability of the various radiation sources, and an attached mass spectrometer for analyzing the outgassing processes in the irradiation chamber, the construction of CIF allows adding other in-situ measurement systems to measure parameters that are of user's interest. We are currently planning to develop an in-situ measurement system in order to determine degradation caused changes in the thermo-optical properties of the samples without leaving the vacuum.

The future activities of the CIF facility are focused on tests of materials which will be used in the Gossamer solar sail project [10]. The Complex Irradiation Facility is a device that can be used in a wide spectrum of degradation investigations. With its light sources together with proton and electron guns one can plan, manage and execute simulations of erosion effects of materials with large acceleration factors. Thereby, the CIF allows the study of degradation effects as may appear in long-term solar sail missions.

Acknowledgments

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H_2 blister formation on metallic surfaces – a candidate for degradation processes in space

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H_2-blisters are metal bubbles filled with hydrogen molecular gas resulting from recombination processes of protons in metal lattice. Bubble formation depends on many physical parameters, for instance: proton energy, proton flux, or the temperature of an exposed sample. Up to now no metallic sample that has been exposed to conditions prevalent in the interplanetary medium has been returned to Earth. Therefore, a direct evidence that blistering appears in space is missing. However, blistering is certainly a candidate of degradation processes which may occur in space. It could play an important role in the solar sail technology, where the performance of the sail is significantly affected by both the sail geometry but especially by optical properties of sail materials. Thus, both theoretical and laboratory studies of the blistering process have to be performed.

The here presented model simulates the growth of molecular hydrogen bubbles on metallic surfaces. Additionally, it calculates the decrease of reflectivity of the by blistering degraded foils. First theoretical results show that the reflectivity of an Aluminum foil decreases by about 27% for a bubble surface density of 1500 cm^{-2} and an average bubble radius of 100 µm. Therefore, if blistering occurs, the propulsion performance of any sail-craft will be decreased by a significant factor.

Nomenclature

A	=	area of the sample
BS	=	backscatter coefficient
d_{csda}	=	Continuous Slowing Down Approximation (CSDA-range)
η	=	relation between the number of H_2 molecules and the H atoms in the metal lattice

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ϵ = binding energy of the H_2 molecule to a vacancy

ϵ' = migration energy of the H atom in the metal lattice

$\varepsilon_{\text{cell}}$ = size of a grid cell that covers the irradiated sample

E = Young module

E_{int} = Internal energy of molecules/atoms located at certain positions in the metal lattice

$F_{\text{gas,i}}$ = free energy of gas inside the i^{th} bubble

F_{H} = free energy of H atoms located outside bubbles in the metal lattice

F_{H_2} = free energy of H_2 molecules located outside bubbles in the metal lattice

$F_{\text{md,i}}$ = free energy of metal deformation caused by expanding i^{th} bubble

$F_{\text{surf,i}}$ = surface free energy of the i^{th} bubble cap

γ = Poisson coefficient

H_{i} = sum of partial derivatives

I = proton flux

k_{B} = Boltzmann constant

M_{u} = molar mass of the sample's material

N = number of iterations

N_0 = number of lattice sites

N_{B} = number of bubbles per unit area

N_{B}^{T} = total number of bubbles at the irradiated sample

N_{cell} = number of cells

N_{H}^{T} = total number of H atoms in the sample

$N_{\text{H}_2}^{\text{T}}$ = total number of H_2 molecules in the sample

$\Delta N_{\text{p}^+,\text{j}}$ = number of protons sent to the sample in a unit time period

p_{i} = pressure of the gas inside the i^{th} bubble

q = momentum

Δq_{i} = momentum transfer of a photon to the i^{th} cell of the degraded foil

$\Delta q_{\text{max,i}}$ = momentum transfer of a photon to the i^{th} cell of a perfect mirror

$\bar{r}$ = average bubble radius

$r_{\text{i,0}}$ = initial radius of the bubble

R = reflectivity

S = entropy

σ = metal strain

θ = angle between the surface normal and the path of a light ray

Δt_j = time step

T = temperature of the sample

V_i = volume of the i^{th} bubble

$V_{max,i}$ = maximum volume of the i^{th} bubble

V_{min} = minimum volume of a bubble

ξ = relation between the number of H_2 molecules inside and outside the bubbles

$\Xi_{i,j}$ = increment of increase of the i^{th} bubble radius during the j^{th} period of time

I. Introduction

Materials covered with metallic surfaces are often used in space technology. For instance, thin Polyimid foils like Kapton, covered on both sides with a thin Aluminum layer, are manufactured for solar sails. The performance of the sails is significantly affected by both the sail geometry and the optical properties of its surface materials [4]. Degradation processes will change the physical properties of these materials. Studies, both theoretical and experimental, allow to choose the best material for a given sail mission.

Degradation of structural properties of solids caused by hydrogen, (referred to as embrittlement) plays a fundamental role in materials physics [12]. Materials used in space technology are exposed to hazard and aggressive conditions like high-energetic radiation or rapid temperature changes. Bubble formation is one of four general processes of embrittlement [12]. When H_2-bubbles are formed on the surface of an irradiated solar sail, its reflectivity decreases with increasing surface density of molecular hydrogen bubbles. Since the reflectivity is directly proportional to the momentum transfer from solar photons to the sail, its decrease will reduce the propulsion performance of a sail-craft just linearly proportional to the reduction of the reflectivity.

For the bubble formation certain conditions with respect to sail temperature and dose of sail penetrating protons have to be fulfilled. It turns out, that bubble formation is very likely at least at distances from the Sun smaller than 1.5 AU, perhaps even for somewhat larger distances.

The here presented thermodynamic model describes the growth of H_2-bubbles. The model input parameters are the energy and flux of solar protons, properties of the irradiated metal, the dose of protons penetrated into the

metal, and its temperature. Another input parameter is the number of vacancies in the subsurface layers of the metal. Only at vacancy sites bubbles can be formed, i.e. their surface density is determined by the vacancy density which, in turn, is determined by the quality of the metal production process.

The model output is the velocity of bubble radius growth, the maximum possible bubble radius, and, for a given bubble density and average bubble radius, the reduction factor of the reflectivity with respect to its ideal value.

II. Temporal evolution of bubble formation

Incident protons, while penetrating the metallic target, recombine with its free electrons to neutral hydrogen atoms. There are four recombination processes of ions to neutral atoms: Auger- [6, 7, 8, 16, 17, 22], resonant- [6, 23], Oppenheimer-Brinkman-Kramers (OBK) - [1, 9], and Radiative Electron Capture (REC) [9] – recombination. Each one has its own probability of occurrence. It depends crucially on the energy of the incident ion. According to cross sections studies, the first two processes dominate the recombination of solar protons with energies lower than 100 keV. Above 100 keV the OBK process together with REC process (this even only above a few MeV) dominates recombination. The solar proton flux has its maximum at about 1 keV (see Fig. 1). At 100 keV it is already two orders of magnitude lower. Therefore, recombination studies of solar protons should be focused on Auger- and resonant – recombination processes.

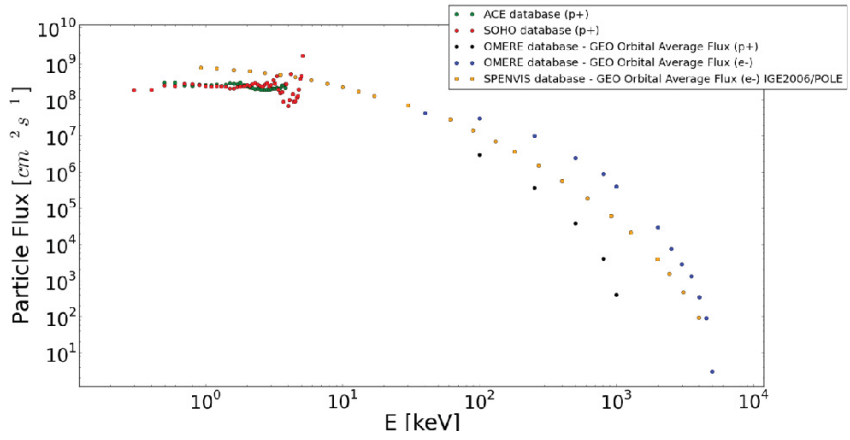

Fig. 1 Flux of solar protons and electrons as a function of energy. Data are taken from the databases of SOHO, ACE, OMERE, and SPENVIS.

Blistering occurs as irradiation damage. It changes the physical properties of the irradiated surface and increases the erosion rate [2]. Blistering is caused by formation of pockets of hydrogen molecular gas just below

the metal surface. There have been made many terrestrial irradiation experiments by use of Hydrogen ions as incident particles. Many types of target metallic materials have been investigated. An example of a population of bubbles is shown in the Fig. 2 [15]. Bubbles were formed on the 2 μm thick Aluminum sample after the critical dose of incident Hydrogen ions was reached. The Hydrogen ions energy was set to 1 keV; the sample was hold at a temperature of 300 K.

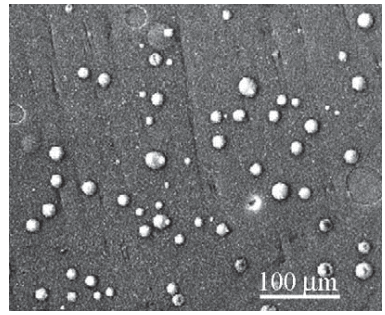

Fig. 2 Typical population of molecular hydrogen bubbles on an Aluminum sample. The diameter of the bubbles is in the range of a few μm [15].

The tendency to form bubbles depends on the proton energy, integrated proton flux (dose), temperature of the target, crystallographic orientation of the irradiated surface as well as on impurities and defects in the sample [5]. It is known from laboratory experiments that the minimum dose of protons above which the process occurs is $\sim 10^{16}$ H^+ cm^{-2} e.g. [14]. The temperature range in which bubbles were observed is between 288 and 573 K [5, 14].

Hydrogen atoms are much smaller than metal atoms, but they can introduce strain in metal lattices when being absorbed as interstitial atoms [21, 25]. They can also change the electronic structure of near neighbor metal atoms [21]. That causes an increase of the lattice energy. It may be decreased by the aggregation of the interstitial hydrogen atoms into hydrogen atom clusters, and then into molecular hydrogen bubbles [21]. Hydrogen could not agglomerate into H_2-clusters without the presence of vacancies. For instance a single vacancy in Aluminum can trap up to twelve H atoms. For comparison, a vacancy in Iron can trap only up to six H atoms [12].

Here, a thermodynamic model of bubble growth is proposed. The model is based on the assumption that the growth proceeds quasi-static i.e. during a j^{th} period of time Δt_j a small portion of H_2-molecules, $N_{H_2,i,j}$, is added to the i^{th} bubble. The thermodynamic equilibrium, however, is re-established very fast after each addition. For

simplicity it is assumed that a single bubble is a half of the sphere with a radius of r_i. The gas within a bubble behaves to a good approximation like an ideal gas:

$$p_i V_i = \sum_j^N N_{H_2,i,j} k_B T, \tag{1}$$

The total irradiation time of the sample, during which the bubble growth appears, is $N \times \Delta t_j$. The number $N_{H_2,i,j}$ is given by:

$$N_{H_2,i,j} = 0.5\left(N_B^T\right)^{-1} \Delta N_{p^+,j} \eta \xi (1 - BS), \quad N_B^T = N_B A, \quad \Delta N_{p^+,j} = I \Delta t_j A. \tag{2}$$

Here 0.5 reflects that a single H_2-molecule consists of two H atoms. While 100% of protons recombine into H atoms in the metal lattice, only a few percent of them recombine to H_2-molecules [3]. Hence, the coefficient η is the ratio between the number of H_2-molecules and the H-atoms in the lattice. Not all of the H_2 -molecules will merge into H_2-clusters and finally form H_2-bubbles. Thus, the coefficient ξ denotes the ratio of the number of H_2-molecules inside and outside the bubbles.

The first step to estimate the radius growth of the i[th] bubble is to calculate the Helmholtz free energy of the whole configuration F_{config}. Since the free energy is an additive quantity, the total free energy of bubble formation is a sum of following quantities: free energy of H_2 gas inside the i[th] bubble $\left(F_{gas,i}\right)$, of the metal surface deformation $\left(F_{md,i}\right)$ caused by the bubble growth itself, of the surface free energy $\left(F_{surf,i}\right)$ of the bubble cap, of the free energy of H_2-molecules $\left(F_{H_2}\right)$, and of H-atoms (F_H) placed outside the bubbles but within the metal lattice. The Helmholtz free energy of the whole configuration described above is then:

$$F_{config} = F_{gas,i} + F_{md,i} + F_{surf,i} + F_{H_2} + F_H. \tag{3}$$

The next step is to estimate the free energy of the i[th] bubble growth. It consists of the free energy of a gas filled in the bubble, the free energy of metal deformation, and of the bubble cap surface free energy. Using the thermodynamic relation between pressure of a gas and its Helmholtz free energy $p = \left(\frac{\partial F}{\partial V}\right)_T$ together with the ideal gas equation of state (Eq. (1)), the free energy of a gas within the i[th] bubble is

$$F_{gas,i} = -\sum_j^N N_{H_2,i,j} k_B T \ln\left(\frac{V_{max,i}}{V_{min}}\right), \tag{4}$$

The model assumes that two H_2 molecules form the smallest ("initial") possible bubble. The radius of such a bubble is approximately 3.2 Bohr radii [20]. Every bubble will crack if the pressure of the gas inside is higher than the pressure exerted by the metal deformation of the cap. The relation between the pressure of the gas, the strain σ of the metal, and the bubble radius corresponding to $V_{max,i}$ is [11]:

$$p_{gas,\ inside\ bubble} - p_{outside\ bubble} = \frac{2\sigma}{r_{max,i}}. \tag{5}$$

Since the sample is placed in the vacuum, the pressure outside the bubble is set to zero.

The free energy of metal deformation caused by the gas pressure inside the bubble with radius r_i can be found in [10], and is given by the following relation:

$$F_{\mathrm{md,i}} = \frac{4\pi}{3}\frac{r_\mathrm{i}^3(1+\gamma)}{E}p_\mathrm{i}^2. \tag{6}$$

The free energy of a surface of a cap of the i$^{\mathrm{th}}$ bubble is given by [13]:

$$F_{\mathrm{surf,i}} = 4\pi r_\mathrm{i}^2\sigma(T). \tag{7}$$

The Helmholtz free energy of the H_2-molecules located at certain positions in the metal lattice but outside the bubbles is calculated in the form $F = E_{\mathrm{int}} - TS$. Applying the statistical definition of the entropy, this free energy is (for a detailed derivation see M. Sznajder, thesis 2013 [24]):

$$F_{\mathrm{H_2}} = \left(N_{\mathrm{H_2}}^{\mathrm{T}} - \sum_{\mathrm{i}}^{\mathrm{N_B^T}}\sum_{\mathrm{j}}^{\mathrm{N}} N_{\mathrm{H_2,i,j}}\right)\left[\varepsilon + k_\mathrm{B}T\ln\left(\frac{N_{\mathrm{H_2}}^{\mathrm{T}} - \sum_{\mathrm{i}}^{\mathrm{N_B^T}}\sum_{\mathrm{j}}^{\mathrm{N}} N_{\mathrm{H_2,i,j}}}{N_0}\right)\right]. \tag{8}$$

The number of lattice sites N_0 can be expressed by:

$$N_0 = N_\mathrm{A} d_{\mathrm{csda}}\frac{A^2}{M_\mathrm{u}}. \tag{9}$$

The Helmholtz free energy of H atoms located at certain positions within the metal lattice is calculated in the same way as for Eq. (8):

$$F_\mathrm{H} = \left(N_\mathrm{H}^{\mathrm{T}} - 2N_{\mathrm{H_2}}^{\mathrm{T}}\right)\left[\varepsilon' + k_\mathrm{B}T\ln\left(\frac{N_\mathrm{H}^{\mathrm{T}} - 2N_{\mathrm{H_2}}^{\mathrm{T}}}{N_0}\right)\right]. \tag{10}$$

Since now each term of Eq. (3) is determined, the next step is to calculate the radius r_i of the i$^{\mathrm{th}}$ bubble at a given time *t*. This will be achieved by assuming that the process of bubble growth is quasi-static one, i.e. during each j$^{\mathrm{th}}$ time step Δt_j a small portion of H_2 molecules is merged to the i$^{\mathrm{th}}$ bubble and the thermodynamic equilibrium is rapidly re-established; the corresponding equilibrium condition is $\frac{\partial F_{\mathrm{config}}}{\partial N_{\mathrm{H_2,i,j}}} = 0$. This leads to the following fifth order equation for r_i:

$$8\pi\,\Xi_{\mathrm{i,j}}\,\sigma(T)r_\mathrm{i}^5 - H_\mathrm{i}r_\mathrm{i}^4 + \frac{3}{\pi}\frac{1+\gamma}{E}\left(\sum_{\mathrm{j}}^{\mathrm{N}} N_{\mathrm{H_2,i,j}}\right)k_\mathrm{B}^2T^2\left[2Nr_\mathrm{i} - 3\Xi_{\mathrm{i,j}}\sum_{\mathrm{j}}^{\mathrm{N}} N_{\mathrm{H_2,i,j}}\right] = 0. \tag{11}$$

$\Xi_{\mathrm{i,j}}$ is defined below in Eq. (13). H_i denotes the abbreviation:

$$H_\mathrm{i} = -\frac{\partial F_{\mathrm{gas,i}}}{\partial N_{\mathrm{H_2,i,j}}} - \frac{\partial F_\mathrm{H}}{\partial N_{\mathrm{H_2,i,j}}} - \frac{\partial F_{\mathrm{H_2}}}{\partial N_{\mathrm{H_2,i,j}}}. \tag{12}$$

III. Formation possibility of H_2 bubbles in the interplanetary medium, numerical analysis of bubble growth.

Growth of molecular hydrogen bubbles will be possible in the interplanetary space if the criterion of the minimum dose of protons is fulfilled. The temperature of the sample has to be high enough to start the bubble formation, but not too high to lose Hydrogen much too rapidly due to the high diffusivity of Hydrogen in metals.

To estimate, whether the conditions for bubble formation are fulfilled a thin $1\mathrm{m} \times 1\mathrm{m}$ non-rotating Aluminum foil (10 µm thick) is assumed to be located at 1 AU distance from the Sun. The foil surface normal is directed toward the Sun. The foil temperature will reach rapidly an equilibrium value of 520 K. Thus, the temperature criterion is fulfilled. Solar proton flux data, collected by the SOHO and ACE mission, return a typical flux at 5 keV of about 10^9 H^+ cm^{-2} s^{-1} (see Fig. 1). Under the simplifying assumption that the Sun generates only mono-energetic 5 keV protons, the criterion of minimum proton dose will be fulfilled after 116 days. In reality this happens even earlier (for a detailed analysis see M. Sznajder, thesis 2013 [24]). Therefore, both the temperature and the dose criterion for bubble formation are fulfilled if metallic surfaces are exposed to the solar wind in typical Earth distances from the Sun.

For the modeling of bubble growth, the i^{th} bubble located on the Aluminum surface (area $1cm^2$) consists at time t=0 only of two H_2-molecules, in accordance with an assumption about $V_{\min}$. At each time step Δt_j a small number of H_2 molecules are merged to the bubble. Therefore, Eq. (11) has to be solved at each time step. A realistic assumption of bubble growth dynamics is:

$$\Xi_{i,j} = \frac{\Delta r_i}{\Delta N_{H_2,i,j}} = j^{\frac{1}{3}} r_{i,0}. \tag{13}$$

The exponent $\frac{1}{3}$ is a model parameter of bubble growth; it is a good guess according to the experimentally found gas pressures inside the bubbles [18]. An increase of that exponent leads to a reduction of the pressure of the gas in the bubble. Six models are presented here. For each model it is assumed that the sample has been irradiated by a proton flux of 6.24×10^{12} H^+ cm^{-2} s^{-1} during more than 10^4 seconds. Hence the criterion of critical dose of protons above which the bubble formation can proceed is fulfilled. Two different temperatures of the sample are considered: 300, and 550 K. The *BS* coefficient is set to 10^{-2}. This value has been estimated by use of the SRIM software [26]. The coefficient η which determines the ratio of H_2-molecules to H-atoms is set to 10^{-2}, a value which is taken from the work of Canham et al. [3]. In order to study a wide range of the availability of vacancies within the lattice, the free parameter ξ, which prescribes the ratio of H_2-molecules inside and outside the bubbles, is set to: 10^{-6}, 10^{-4}, and 10^{-2}.

The results are presented in Fig. 3. While the left plot shows the temporal evolution of bubble growth at 300 K, the right plot presents it at 550 K. Blue-, green-, and red- lines denote models with ξ-factor of: 10^{-6}, 10^{-4}, and 10^{-2}, respectively. A significant influence on to the bubble growth has the ξ - factor. To illustrate this compare for a temperature of 300K the radii after an irradiation period of 10^4s. For $\xi = 10^{-4}$ the bubble radius is by a factor of about 5 larger than for $\xi = 10^{-6}$. In case of $\xi = 10^{-2}$ this factor increases to 30. Clearly, the larger the fraction of H_2-molecules that find their way into a bubble, the faster it grows. An increase in temperature has an effect in the like direction. After the same irradiation period and for $\xi = 10^{-2}$ is the radius at 550K about 33% larger than at 300K. Obviously, as long as the upper threshold is not exceeded, the bubble formation benefits from a higher temperature of the metal.

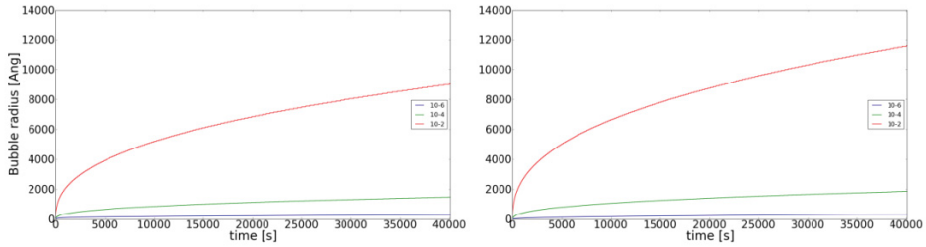

Fig. 3 Bubble radius as function of time. Left panel: T=300K, right panel: T=550K. The blue, green and red curves correspond to ξ –parameters of 10^{-6}, 10^{-4}, and 10^{-2}, respectively.

IV. The effect of bubble formation on to the reflectivity

The momentum transfer of a photon to an ideal reflecting surface is given by $\Delta q = 2q\cos\theta$, where the factor 2 is just in accordance with specular reflectivity. Certainly, the surface quality will suffer during the irradiation with protons from progressing bubble formation. At time $t = 0$ the foil has not been exposed to the electromagnetic radiation and/or charged particles, and is considered to be a perfect mirror with reflectivity $R = 1$. That means that all of the incident light rays are reflected perfectly, no light ray is absorbed or diffusively reflected by the target. Later, when the foil has been irradiated by a flux of protons and molecular hydrogen bubbles have been formed on its surface, the reflectivity of the degraded foil will be reduced. This deterioration is calculated in the following way: the foil is covered by a grid with a fixed single cell size of $\varepsilon_{\text{cell}} \times \varepsilon_{\text{cell}}$ (see Fig 4).

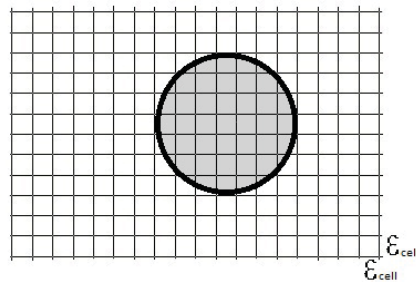

Fig. 4 Grid covered fraction of the foil with one spherical bubble. The cell area is $\varepsilon_{\mathrm{cell}} \times \varepsilon_{\mathrm{cell}}$.

The reflectivity of a single cell is by definition $\frac{\Delta q}{\Delta q_{\max}}$. Hence, taking into account all cells, one has:

$$R_{\mathrm{foil}} = \frac{\sum_{\mathrm{i}}^{\mathrm{N_{cell}}} \Delta q_{\mathrm{i}}}{\sum_{\mathrm{i}}^{\mathrm{N_{cell}}} \Delta q_{\mathrm{max,i}}} \tag{14}$$

The path of photons is directed parallel to the foil surface normal. Therefore, at time $t = 0$ the foil was a perfect mirror without surface imperfections and $\theta_{\mathrm{i}} = 0$. Later, when the surface is populated with bubbles, θ_{i} will vary between 0 and 90°. Thus, Eq. (14) reduces to:

$$R_{\mathrm{foil}} = \frac{\sum_{\mathrm{i}}^{\mathrm{N_{cell}}} 2q \cos\theta_{\mathrm{i}}}{N_{\mathrm{cell}} \times 2q} = \frac{\sum_{\mathrm{i}}^{\mathrm{N_{cell}}} \cos\theta_{\mathrm{i}}}{N_{\mathrm{cell}}}. \tag{15}$$

The reduction of reflectivity is here demonstrated for three different bubble number densities N_{B}: 500 cm^{-2}, 1000 cm^{-2}, and 1500 cm^{-2}. The ratio of the average radius of the bubble $\bar{r}$ to the size of the cell $\varepsilon_{\mathrm{cell}}$ is fixed and set to 100. Results are presented in Fig. 5. It turns out that the reflectivity is quite sensitive to the average radius and surface density of bubbles that cover the irradiated foil. For a small average radius of bubbles the change in the reflectivity is negligibly small. However, for bubbles with an average radius of 100 μm the change in the reflectivity with respect to the initial state of the foil is significant: 7%, 17%, and 27% for N_{B} = 500 cm^{-2}, 1000 cm^{-2}, and 1500 cm^{-2}, respectively.

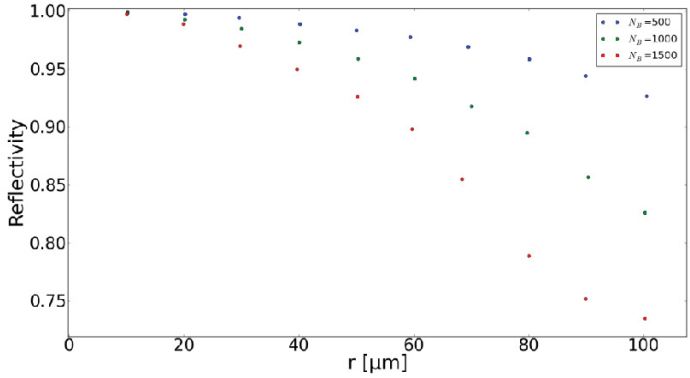

Fig. 5 Reflectivity of a metallic foil covered with hydrogen molecular bubbles as function of the average bubble radius for three bubbles surface densities N_B: 500 cm^{-2}, 1000 cm^{-2}, and 1500 cm^{-2}.

V. Conclusion

The main conclusion is that the bubble growth phenomenon and its consequence, the blistering, may have a serious influence on the propulsion efficiency of any sail-craft. The here presented model calculations have shown that blistering may appear on metallic surfaces exposed to the solar wind in the close vicinity of the Sun (detail calculations presented in Maciej Sznajder's thesis [24] has shown that the blistering process may occur in orbits between 0.5 AU and 1.5 AU). Under conditions that prevail there, high bubble surface densities with average bubble radius in the μm range can be expected. The corresponding reduction of the sail reflectivity must be considered in the planning of solar sail missions.

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Solar Sail: Materials and Space Environmental Effects

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Theoretical aspects of a solar sail material degradation are presented when the solar electromagnetic and corpuscular forms of radiation were considered as sources of degradation. The analysis of the interaction of two components of solar radiation, the electromagnetic radiation and radiation of low- and high-energy electrons, protons, and helium ions emitted by the Sun with the solar-sail materials is discussed. The physical processes of the interactions of photons, electrons, protons and α-particles with sail material atoms and nuclei, leading to the degradation and ionization of solar sail materials are analyzed. The dependence of reflectivity and absorption for solar sail materials on temperature and on wavelength of the electromagnetic spectrum of solar radiation is investigated. It is shown that the temperature of a solar sail increases approximately as $T \sim r^{-2/5}$, with the decrease of the heliocentric distance *r*, when are taking into account the temperature dependence of optical parameters of the solar sail material.

Nomenclature

ρ = density of the sail material, g/cm^3

d = thickness of solar sail material

$\sigma(\omega, T)$ = electrical conductivity, Ω^{-1}m^{-1}

σ_0 = electrical DC conductivity at temperature T_0, Ω^{-1}m^{-1}

$\rho(\lambda, T)$ = spectral reflectivity

$\alpha(\lambda, T)$ = spectral absorptivity

$\tau(\lambda, T)$ = spectral transmissivity

$\varsigma(\lambda, T)$ = spectral emissivity

$\varsigma(T)$ = emissivity

σ_{SB} = Stefan-Boltzman constant, Wm^{2}K^{-4}

μ = mass attenuation coefficient

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n = number of metal atoms in the unit volume, cm^{-3}

Z = atomic number

T = temperature, K

$\varepsilon(\omega)$ = dielectric function

I. Introduction

The solar sail is one of the very few proposed space-propulsion methods that has enormous potential, because it takes advantage of sunlight and does not require the chemical fuel that spacecraft currently relies on for propulsion. Solar sails accelerate slowly but surely, capable of eventually reaching tremendous speeds that may eventually be applied to interstellar exploration and travel. In fact, many scientists consider solar sailing the only reasonable way to make interstellar travel a reality. Usually a solar sail is considered as a thin membrane that uses the momentum carried by electromagnetic radiation originated from the sun to propel a spacecraft. However, the Sun space environment is a very dynamic place. As well as a high photon flux, there is a stream of electrically charged particles that is ejected from the Sun. This corpuscular part of the solar radiation is highly variable in terms of both velocity and density. The interactions of a solar-sail material with positive and negative Sun-generated particles must therefore be considered by interstellar mission planners because a solar sail will be a long term under an influence of different type space environmental effects. Solar energetic particle events can have a significant effect on both the operations and design of a solar-sail spacecraft. This was one of the reasons for formation of the Environmental Effects Group at NASA's Marshall Space Flight Center [1] that is tasked with characterizing the material properties of newly developed sail materials and further characterizing these materials in emulated space environments. Furthermore, the authors of Ref. [2] established in November 2004 the Solar Sail Degradation Model Working Group to take a step forward towards to study the general optical degradation behavior of solar sails and its impact on mission analysis. However, it is interesting to mention that more than four decades ago NASA experience has indicated a need for uniform criteria for the design of space vehicles related to effect of space environment on its material [3]. Present-day investigations [4-6] indicated that the space environmental effects degrade sail material thermo-optical properties and the sail material mechanical stability with increasing radiation exposure. Solar Sail Materials [7] is an on-going project for the European Space Agency relying on past and recent European solar sail design projects. It aims at developing and testing future technologies suitable for large, operational solar sailcrafts.

A solar-sail performance is significantly affected by four factors: the areal mass of the sail, the optical properties of the sail film, the mechanical properties of the sail films, and the sail geometry. The first three

factors are depending on each other that makes it difficult to study their influence on the performance of solar sail. The optical parameters such as a radiation absorption coefficient and emissivity are the major parameters governing the solar sail's surface temperature and its serviceability. This article focuses on theoretical aspects of a solar sail material degradation when the solar electromagnetic and corpuscular forms of radiation were considered as sources of degradation. The analysis of the interaction of two components of solar radiation, the electromagnetic radiation and radiation of low- and high-energy electrons, protons, and helium ions emitted by the Sun with the solar-sail material that lead to it degradation is addressed.

The remainder of this paper is organized in the following way. In Sec. II we discuss the structure and energy spectrum of the solar radiations. The temperature dependence of a solar sail parameters is discussed in Sec. III, where is presented the minimal thickness of the solar sail film that provides the maximum reflection of the solar radiation, as well as is given the dependence of the temperature of the solar sail material on the heliocentric distance when the optical parameters depend on temperature. The results of interaction of the ultraviolet radiation with the solar sail material are presented in Sec. IV A, while Sec. IV B has deal with the degradation of the solar sail material by solar electrons, protons and ions of helium. Finally, in Sec. V we summarize our studies and present the conclusions.

II. Solar Radiation

Solar radiation has two components: the electromagnetic radiation and the corpuscular radiation that consist from low- and high-energy elementary particles like electron, protons, neutrinos and ions of light nuclei emitted by the Sun. The solar corpuscular radiation mainly results from a solar wind, coronal mass ejections, solar flares, and solar prominences that are the main sources of electrons and protons. The first two produce electrons with energies from about 0.01 eV up to a few hundreds of eV and the flares are a source of high energy electrons with energies from 1 MeV up to hundreds of MeV. The energy spectrum of the solar protons extends from 0.2 keV to a few tens of keV for solar wind and coronal mass ejection protons and up to a few GeV for solar flare protons.

The spectrum of electromagnetic radiation of the sun is also diverse. Using the experimental data for the solar spectral irradiance in wide range of wavelengths from 119.5 nm (~ 10.4 eV) up to more than 5000 nm (~ 0.25 eV), with the maximum around 500 nm from Ref. [8] we present the dependence of the spectral irradiance of solar electromagnetic radiation on the energy of solar photon shown in Fig. 1. (The total area under the curve in Fig. 1 gives the value of the solar constant W_0=1366.1 W/m^2). Almost 7% of the solar electromagnetic radiation consisting from ultraviolet (UV), X - and γ -rays [9] takes part only in an ionization of the sail

producing the surface charge distribution. The visible wavelengths of the incident solar electromagnetic radiation are mostly reflected by the solar sail depending on the coefficient of reflection of the solar sail material. But a part of the visible radiation as well as the infrared portion of the spectrum with wavelengths are greater than 780 nm (~ 1.6 eV) will be mostly absorbed by the solar sail causing the heating of the sail and, therefore, the increase of its temperature. It is important to mention that about 47% of the solar radiation is in the range of infrared and microwave spectrum which is causing the heating of the sail and, as a result, its temperature increase.

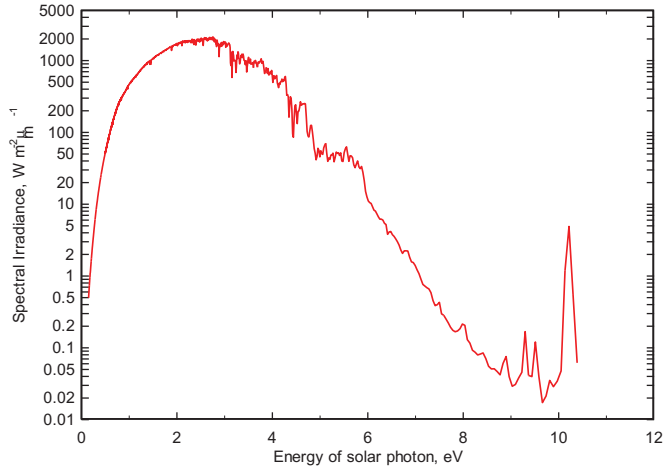

Fig. 1 Dependence of the Spectral Irradiance of solar electromagnetic radiation on the energy of a solar photon.

The interaction of the solar electromagnetic radiation with the solar sail not only accelerates the spacecraft but also induces the diversity of physical processes that change the physical properties of the sail. The interaction of the electromagnetic radiation with the solar sail material results in the photoelectric effect, the Compton effect and creation of electron-positron pairs that are the main processes that play a crucial role in the attenuation of the ultraviolet, X-ray and γ-ray by a solar sail. When the photons energy exceeds the threshold for nucleon knockout from solar sail material nuclei, a nuclear photoeffect occurs. All photons that produce the photoelectric effect or create the electron-positron pairs give up all their energy to the sail. For high-energy photons, which are X - and γ -rays portion of the spectrum, the sail is mostly transparent. However, metallic sails will be partially ionized by the solar UV radiation, as is shown in Ref. 10.

Consequently, we can conclude that basic components of the solar radiation that will interact with the solar sail are electromagnetic radiation with energy from a few tenths of eV to hundreds of MeV, and electrons and protons in the energy spectrum from a few tenths of eV to hundreds of MeV and up to GeV. Thus, the ionization of the sail, induced by the solar electrons, protons and helium ions, and its heating, resulted from the absorption

of the electromagnetic waves, are two main culprits in degradation of the sail material that required a consideration.

Below the degradation of solar sail material is considered under the following assumptions:

- The only sources of degradation of the solar sail material are the solar electromagnetic radiation and corpuscular part of the solar radiation: electrons, protons, helium ions (α-particles). The cosmic radiation that mostly includes electrons and protons can be considered as a part of the solar corpuscular radiation.
- The optical parameters such as a reflection coefficient, an absorption coefficient and a solar sail's surfaces emission coefficients depend on the sail temperature. However, the optical coefficients do not depend on the incidence angle of electromagnetic radiation. Also, we are not considering the solar events. Therefore, the solar electromagnetic and corpuscular fluxes do not depend on time.

III. Temperature Dependence of Solar Sail Parameters

A. Thickness of a solar sail material

The effect of ionizing radiations depends on both the radiation dose and the parameters of the solar sail material including the film thickness. It is already an established fact that the degradation is much more severe for the higher radiation doses and thinner films [11]. From the other hand one of the key design parameter, which determines the solar sail performance, is the solar sail areal mass, which depends on the thickness and the density of the sail material as follows:

$$s = \rho d \tag{1}$$

where d is the thickness of the sail and ρ is the density of the sail material. An optimal control of solar sails depends on the sail areal mass as well as the sail pitch angle. It is clear from Eq. (1) that to obtain a high performance sail we should select among the materials with the same optical properties, the material with low density and use a thin film of this material for the solar sail. On the other hand, the degradation of a solar sail through the ionization also depends on the thickness of a sail material. It is obvious that when thickness of the sail increases the penetration of the solar radiation decreases and, therefore, the attenuation increases that causes the degradation of the solar film. If the solar sail film is too thin its degradation by electrons and protons decreases, however, it may become transparent for a part of the electromagnetic spectrum of the solar electromagnetic radiation and, therefore, this part of solar radiation is lost for the acceleration of the sailcraft. *The question is how thin the film for a solar sail should be that it would produce the acceleration of the sailcraft based on the maximum reflection of solar electromagnetic radiation and its optical degradation by UV*

radiation, X , γ– rays and the corpuscular solar radiation would be minimal? For the electromagnetic part of the solar radiation electromagnetic fields inside a metallic conducting foil decay rapidly with depth. The distance it takes to reduce the amplitudes of the electromagnetic field by factor of $1/e$ (*e*-folding distance) is a skin depth and it is a measure of how far an electromagnetic wave penetrates into the conducting metallic foil. It is obvious that the foil thickness should be always larger than the skin depth; otherwise the solar sail material will be transparent to the electromagnetic radiation. Following the standard electrodynamics approach [12, 13] by applying the system of Maxwell's equations for linear conducting media in Refs. [14, 15] was found the minimum foil thickness that provides the maximum reflection and investigated dependence of this minimum thickness on wavelengths of solar electromagnetic radiation as well as on temperature. The condition for the minimum thickness of the solar sail that performs the acceleration of the sail based on the maximum reflection of the solar radiation should be at least the following [15]:

$$d = \omega^{-1}\left[\frac{\varepsilon(\omega)\mu_0}{2}\left(\sqrt{1+\left(\frac{\sigma(\omega,T)}{\varepsilon(\omega)\omega}\right)^2}-1\right)\right]^{-1/2}, \qquad (2)$$

where ω is a frequency of solar radiation, $\varepsilon(\omega)$ is the permittivity which is a function of the frequency and $\sigma(\omega,T)$ is the electrical conductivity of a material, which depends on the frequency (wavelength) of electromagnetic radiation and temperature. Using Eq. (2) we can estimate the minimal thickness of the solar sail required to achieve maximum reflection of the solar electromagnetic radiation for the given optical properties of the metallic foil. The required thickness of the metallic foil is governed by two properties of the solar sail material: the permittivity $\varepsilon(\omega)$ and the conductivity $\sigma(\omega,T)$ of the metallic foil. However, it is not the entire story. Actually, each of these parameters determines the optical properties of the solar sail material and depends to some extent on the frequency of the electromagnetic wave and the temperature. Indeed, Eq. (2) shows that the minimal thickness has the explicit dependence on the frequency through the factor $1/\sqrt{\omega}$ and it also has implicit dependence on frequency, because the conductivity $\sigma(\omega,T)$ and dielectric function $\varepsilon(\omega)$ are frequency dependent. Also the minimal thickness depends implicitly on the temperature through the temperature dependence of the conductivity.

The required thickness of the solar sail material is fluctuating depending on the temperature and frequency, therefore, we need to determine the minimal thickness of the sail material that would enable it to achieve the reflection of all solar spectrum frequencies at the temperatures that it got in the space environment. In studying the dependence of the sail material thickness on the temperature and on the solar electromagnetic spectrum

frequency (wavelength) we use Eq. (2) and have assumed the solar sail material to be nonmagnetic, therefore, the permeability $\mu_0 = 4\pi \times 10^{-7}$ T · m/A .

The calculations performed using for the beryllium and aluminum films to find the required thicknesses of the solar sail material providing the best reflection of electromagnetic radiation as a function of wavelength of the solar radiation and temperature show that the general behavior of this dependence is such that the increase in the temperature requires increases in the thickness of the sail foil to keep its best reflection ability [15]. Also the analysis shows that the thickness of the foil exhibits the negligible dependence on the wavelength at low temperature and weakly decreases for all wavelength ranges at high temperature. However, much stronger dependence of the minimal thickness on the temperature, especially in the range corresponding to the visual part of the solar radiation spectrum. Both solar sail materials exhibit strong dependence of the thickness on the wavelength in the wavelengths range $0.2\,\mu\text{m} < \lambda < 0.8\,\mu\text{m}$ and this dependence becomes stronger when temperature increases. The analysis of the results of calculations shows that the minimal thickness requirement that provides the best reflection and absorption of all solar radiation wavelengths for temperature range up to about 800-900 K for which the limit of the elastic deformation can be reached, is about 50 nm for beryllium and aluminum that corresponds to the areal mass 0.1 g/m^2 and 0.14 g/m^2 for beryllium and aluminum, respectively.

Above we mentioned the results of calculation of the temperature dependence of the minimal required thickness for the solar sail film for the constant value for the temperature coefficient of conductivity over the range of temperature. However, in reality, the temperature coefficient of conductivity is not a constant over the above range of temperature. It increases with a rise of temperature but not linearly [21]. To demonstrate the importance of this fact, we used the experimental values of the temperature coefficient of conductivity from Ref. [21] obtained for beryllium, extrapolated them, and made model calculations for the temperature dependence of the thickness of the beryllium film. The result of calculations shows that consideration of the temperature dependence of the temperature coefficient of conductivity leads to the increase of the required thickness by more than 35% at high temperatures [14]. Therefore, this factor must be also taken into consideration when designing the solar sail.

B. Optical Parameters

The treatment of the optical properties of material must be properly done by quantum theory. However, for many purposes, an adequate and insightful treatment can be given in purely classical electron theory. There are four radiative properties of the material surface: spectral reflectivity $\rho(\lambda, T)$, absorptivity $\alpha(\lambda, T)$, transmissivity $\tau(\lambda, T)$ and emissivity $\varsigma(\lambda, T)$. All four properties are functions of temperature as well as of

frequency of electromagnetic radiation. Absorptivity, reflectivity and emissivity can be evaluated in the framework of electromagnetic wave theory using the complex index of refraction, therefore, by using the electrical conductivity and the dielectric function of the material which must be known over the temperature and spectral range of interest. The electrical conductivity and dielectric function of the material are the functions of the wavelength of radiation and temperature. Such dependence for the case of the solar sail has been studied in detail in Refs. [14, 15]. Hence, the spectral absorptivity, reflectivity and emissivity of the surface of a solar sail should depend on the wavelength of radiation and temperature. However, the total absorptivity, reflectivity and emissivity of the surface are spectrally average value of the spectral absorptivity, reflectivity and emissivity and they depend on the temperature of the surface.

Let us consider the solar electromagnetic radiation traveling through the vacuum and hitting the sail surface of a conducting medium normally to the surface. Introducing the complex index of refraction as $n - im$ and taking into account that the solar electromagnetic radiation is unpolarized, the Fresnel's equation for the spectral reflectivity can be written in the following form [13]:

$$\rho_\lambda(\lambda,T)=\frac{[n(\lambda,T)-1]^2+m(\lambda,T)^2}{[n(\lambda,T)+1]^2+m(\lambda,T)^2}, \tag{3}$$

where $n(\lambda,T)$ and $m(\lambda,T)$ are the indices of refraction and absorption, respectively and they depend on the wavelength and temperature and related to the real and imaginary parts of the complex wave number given in Ref.[14] as

$$n(\lambda,T)=\frac{ck}{\omega}=\sqrt{\frac{\varepsilon(\omega)\mu}{2\varepsilon_0\mu_0}}\left[\sqrt{1+\left(\frac{\sigma(\omega,T)}{\varepsilon(\omega)\omega}\right)^2}+1\right]^{1/2}, \quad m(\lambda,T)=\frac{c\kappa}{\omega}=\sqrt{\frac{\varepsilon(\omega)\mu}{2\varepsilon_0\mu_0}}\left[\sqrt{1+\left(\frac{\sigma(\omega,T)}{\varepsilon(\omega)\omega}\right)^2}-1\right]^{1/2}. \tag{4}$$

For opaque media from Kirchhoff's law that states an object must emit at the same rate as it absorbs if equilibrium is to be maintained, follows that the spectral emissivity related to the spectral reflectivity as $\varsigma_\lambda(\lambda,T)=1-\rho_\lambda(\lambda,T)$. Therefore, from (3) it is easy to obtain that

$$\zeta_\lambda(\lambda,T)=\frac{4n(\lambda,T)}{[n(\lambda,T)+1]^2+m(\lambda,T)^2}. \tag{5}$$

For description of the optical properties of metals it is convenient to introduce the complex dielectric function

$$\varepsilon(\omega)=\varepsilon-\frac{i\sigma(\omega,T)}{\omega} \tag{6}$$

This function is related to the real and imaginary parts of the complex index of reflection in the following way

$$[n(\lambda,T)-im(\lambda,T)]^2=\varepsilon(\omega)-\frac{i\sigma(\omega,T)}{\omega}, \tag{7}$$

Therefore,

$$n(\lambda,T)^2 - m(\lambda,T)^2 = \varepsilon(\omega), \qquad n(\lambda,T)m(\lambda,T) = \frac{\sigma(\omega,T)}{2\omega}. \tag{8}$$

From (8) we can express $n(\lambda,T)^2$ and $m(\lambda,T)^2$ through the measurable electrical conductivity as

$$n(\lambda,T)^2 = \frac{1}{2}\left(\varepsilon(\omega) + \sqrt{\varepsilon(\omega)^2 + \left(\frac{\sigma(\omega,T)}{2\omega}\right)^2}\right), \quad m(\lambda,T)^2 = \frac{1}{2}\left(-\varepsilon(\omega) + \sqrt{\varepsilon(\omega)^2 + \left(\frac{\sigma(\omega,T)}{2\omega}\right)^2}\right). \tag{9}$$

The value of $n(\lambda,T)^2$ and $m(\lambda,T)^2$ can be predicted for all wavelengths and temperature or the corresponding values of $\varepsilon(\omega)$ and $\sigma(\omega,\mathrm{T})$ can be determined within certain theoretical model or measured experimentally.

Let us now consider the frequency dependence of the conductivity. The best way to do that is to use the classical Drude model for conductivity [16]. According to the Drude model, the frequency dependent conductivity is a complex function and given by

$$\sigma(\omega) = \frac{\sigma(T)}{1 - i\omega t}, \tag{10}$$

where $\sigma(T)$ is the DC Drude conductivity at temperature *T* and the relaxation time *t* for metal is about $t = 10^{-14}$ s. In Drude model we can consider

$$n^2 = m^2 \approx \frac{\sigma(T)}{\varepsilon_0 \omega} \tag{11}$$

and (3) and (5) become

$$\rho_\lambda(\lambda,T) = \frac{2n(\lambda,T)^2 - 2n(\lambda,T) + 1}{2n(\lambda,T)^2 + 2n(\lambda,T) + 1}, \tag{12}$$

$$\varsigma_\lambda(\lambda,T) = \frac{4n(\lambda,T)}{2n(\lambda,T)^2 + 2n(\lambda,T) + 1}. \tag{13}$$

The electrical conductivity depends on temperature and *approximately inversely* proportional to it: $\sigma(T) = \sigma_0 / [1 + \alpha(T - T_0)]$, where σ_0 is DC conductivity at temperature T_0 and α is the temperature coefficient of the conductivity. Hence, $n(\lambda,T)^2$ also depends on temperature, and therefore, spectral reflectivity and emissivity will show temperature dependence, however in different ways. Because $\rho_\lambda(\lambda,T)$ is the ratio of the same order polynomials the dependence on T in the denominator and the numerator in (12) cancels each other and it becomes negligible, but we cannot say the same about the emissivity. As it is seen from (13) the emissivity $\varsigma_\lambda(\lambda,T)$ is approximately directly proportional to $T^{1/2}$. It is easy to show that from the following expansion. Since $n > 1$ (12) and (13) may expanded as

$$\rho_\lambda(\lambda,T)=\frac{2n(\lambda,T)^2-2n(\lambda,T)+1}{2n(\lambda,T)^2+2n(\lambda,T)+1}\approx 1-\frac{2}{n(\lambda,T)}+\frac{2}{n(\lambda,T)^2}+...= \\ 1-2\sqrt{\frac{\varepsilon_0\omega[1+\alpha(T-T_0)]}{\sigma_0}}+2\frac{\varepsilon_0\omega[1+\alpha(T-T_0)]}{\sigma_0}+... \quad , \tag{14}$$

$$\varsigma_\lambda(\lambda,T)=\frac{4n(\lambda,T)}{2n(\lambda,T)^2+2n(\lambda,T)+1}\approx\frac{2}{n(\lambda,T)}-\frac{2}{n(\lambda,T)^2}+...= \\ 2\sqrt{\frac{\varepsilon_0\omega[1+\alpha(T-T_0)]}{\sigma_0}}+2\frac{\varepsilon_0\omega[1+\alpha(T-T_0)]}{\sigma_0}+.. \quad , \tag{15}$$

The consideration of the leading terms in (14) and (15) demonstrates that the spectral reflectivity weakly depends on temperature, emissivity depends on temperature and increases when temperature increases. The total hemispherical emissivity is can be approximated by following equation [17]

$$\varsigma(T)=7.66\left(\frac{T}{\sigma(T)}\right)^{1/2}. \tag{16}$$

For Al the coefficient is 7.52 $K^{-1/2}\Omega^{-1/2}m^{-1/2}$ [18]. Since the electrical conductivity σ is approximately inversely proportional to the temperature, therefore, the hemispherical emissivity, as well as the total normal emissivity is approximately linearly proportional to the temperature. Eq. (16) satisfactorily describes the temperature dependence of the experimental data for the hemispherical emissivity for various metals [17].

In general, the solar sail is the sandwich of films and front and back sides of the solar sail will have different surface emissivity. According to the Stefan-Boltzman's law the rate of energy emitted from a unit area of the surface material at temperature T is proportional to the fourth power of the temperature of the surface. The solar sail radiates heat into the space that has a temperature $2.725\,\mathrm{K}$ related to the cosmic micrrowave background radiation. For simplicity we consider that this temperature is 0 K. Taking this fact into consideration for the rate of energy emitted by both sides of a unit area of the solar sail we obtain

$$W_e=[(\varsigma_f(T)+\varsigma_b(T)]\sigma_{SB}T^4, \tag{17}$$

where $\varsigma_f(T)$ and $\varsigma_b(T)$ are the front and back sides surface emissivity of the sail at given temperature, respectively and $\sigma_{SB}=5.67\times10^{-8}\,\mathrm{Wm^2K^{-4}}$ is Stefan-Boltzman constant. Based on the law of conservation of energy the solar sail will be in thermal equilibrium if the total absorbed energy equals to total emitted energy. This condition allows finding the thermal equilibrium temperature [19, 20]

$$T=\left(\frac{[1-\rho(T)-\tau(T)]W_0R_0^2\cos\theta}{[\varsigma_f(T)+\varsigma_b(T)]\sigma_{SB}}\right)^{1/4}r^{-1/2}. \tag{18}$$

Eq. (18) is obtained under the assumption that solar energy flux has an inverse square variation with the distance from the Sun (in fact, this assumption is not valid when account is taken of the finite angular size of the solar sail [20] and considering that the solar energy flux with the angle of an incidence θ to the surface of the sail at the distance *r* from the Sun) and the solar irradiance W_0 at $R_0 = 1$ AU. From Eq. (18) it follows that the temperature of the solar sail material is increases when the solar sail approaches to the sun as $T \sim r^{-1/2}$. Taking into account the temperature dependence of the emissivity (17) (emissivity is approximately linearly proportional to temperature) in Eq. (18) we can conclude that the temperature increases approximately as $T \sim r^{-2/5}$, when the heliocentric distance *r* decreases. In other words, when we are taking into account the temperature dependence of the emissivity (17) and a conductivity of metals the temperature dependence on the distance is not the same as for the constant emissivity and conductivity. In this case the temperature of the solar sail material increases more slowly when sail approaches to the sun than in the case for the constant emissivity and conductivity. However, the increase of the temperature decreases the reflectivity of the solar sail which is a disadvantage for its dynamics (acceleration). Can the disadvantage related to the temperature be turned into the advantage for the solar sail dynamics? The hot sail may have one advantage related to the thermal desorption propulsion method proposed in Ref. [21]. The *thermal desorption* is the process for mass loss and dominates all other processes for mass loss above temperatures of 300–500 ^{0}C. If we can heat sails to temperatures > 1000K and the surface of solar sail has a coat of embedded atoms or paint, their *thermal desorption* can provide higher specific impulse than liquid rockets. In other words, atoms embedded in a substrate can be liberated by hear heating and provide an additional trust. Using the thermal desorption for thrust is not a new idea, but it is new to apply this idea to the solar sail that naturally gains temperature through the absorption of solar radiation. Heating of a sail by solar radiation to the temperature until its surface coat sublimes or desorbs and, therefore, can add far more thrust sounds attractive. However, it requires the search and surveys of promising new materials for thermal desorption that can be embedded on the surface of solar sail, study their optical properties and usage for thrusting applications.

IV. Degradation of Solar Sail Materials by Solar Radiations

The solar sail is exposed to different environmental factors that cause the degradation of the solar sail materials which can be a composite multilayer system of high reflective films on the polymeric substrates. The main factors of space environment that lead to the degradation of the solar sail material are vacuum UV radiation flux of electrons, protons and α-particles radiated by the sun.

A. UV Radiation

Vacuum ultraviolet radiation is one of the major environment factors causing the degradation of solar sail material. Although the UV energy percentage in the total spectrum of the Sun is very limited, its photon energy is high enough to significantly change reflectivity of the metallic films and break most chemical bonds in the polymers [23]. Under UV radiation, the mechanical properties as tensile properties of metallic film did not change significantly [24, 25], while the optical ones varied noticeably. With increasing the irradiation dose, the tensile fracture strength and elongation decreased slightly, and the spectral absorbance increased remarkably in the ultraviolet to visible regions.

The intensity of UV, X-ray, and γ-ray radiations penetrating a layer of a solar sail of material with thickness d is given by the exponential attenuation law $I = I_0 e^{-\mu \rho d}$, where I_0 is the incident photons intensity and μ is a mass attenuation coefficient. Using Eq. (1) the last equation can be rewritten in the following form:

$$I = I_0 e^{-\mu s}. \tag{19}$$

Eq. (19) gives the relationship between the key parameter s that determines the performance of the solar sail and the mass attenuation coefficient μ of the photon flux that related to the degradation of the surface of the solar sail. The mass attenuation coefficient μ is a basic quantity used in calculations of the penetration and the energy deposition by photons in materials that leads to degradation of optical properties of the solar sail. The mass attenuation coefficient is directly related to the total cross section of photon interaction with atoms of the solar sail material. The total cross section can be written as the sum of contributions from the principal photon interactions with a metal film: σ_{ape} is the atomic photoelectric effect cross section, σ_C is the Compton scattering cross section, σ_{pair} is the cross section for electron-positron production, and σ_{npe} is the photonuclear cross section. As a result for the mass attenuation coefficient we have

$$\mu = (n\sigma_{ape} + nZ\sigma_C + n\sigma_{pair} + n\sigma_{npe}), \tag{20}$$

where n is the number of metal atoms in the unit volume. The attenuation coefficient is a function of photon energy because each cross section depends on the photon energy. The first term in Eq. (20) is dominant for a low photon energy, the second term becomes important for photon energy up to hundreds of keV, the third term plays the major role for photons' energy higher than a few MeV, and the last term gives the dominating contribution for high-energy photons when they are interacted with nuclei of the solar sail material. Using the corresponding cross section for the considered processes we can estimate the reduction of the radiation through

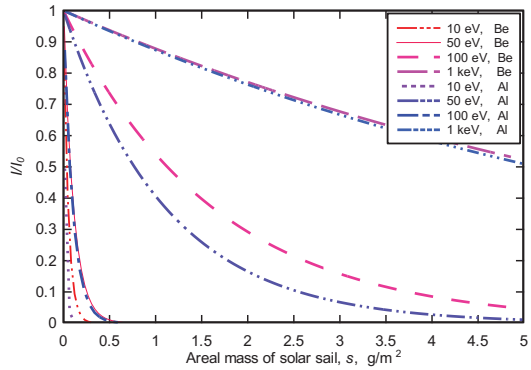

Fig. 2 Attenuation of photons by beryllium and aluminum films for different photon energies.

the solar sail. The results of the attenuation of the photons by beryllium and aluminum thin foils are presented in Fig. 2. It is important to mention that for the same areal mass of the solar sail the thickness of the aluminum film is 1.46 times larger than that for the beryllium film that is equal to the ratio of the densities of aluminum and beryllium. To make the most conservative prediction of the dependence of the ratio I/I_0 on the areal mass of the beryllium and aluminum films for the ionization processes, we use the maximum value of the attenuation coefficients for the corresponding processes. The results presented in Fig. 2 indicate that UV radiation with energy less that 10 eV are completely absorbed by Al and Be films when the areal mass more than 0.5 g/cm^2. It is interesting to mention that for the photon energy range between about 15 eV and 100 eV the ionization of aluminum is always less than that for beryllium. This is due to the fact that mass attenuation coefficient for Al at these photons energies is less than one for the beryllium. For photon energies more than 1keV ionization of the beryllium is relatively smaller than of the aluminum. However, at these energies less than 50% of photons penetrate the sail foil for both metals if the areal mass equals 5 g/m^2. At the energy of photons below 150 eV the major effect of the attenuation of the photons is the ionization of the metallic surface of the solar sail. Therefore, the solar sail is gaining the positive electric charge. Also our estimate based on the values of the mass attenuation coefficient at high energies shows that less than 1% of 1 MeV photons attenuated by each of films with the areal mass more than 1 g/cm^2. When the film areal mass increases the degree of ionization increases also as it was demonstrated in Ref. 26 for different candidates solar sail material.

B. Degradation by solar electrons, protons and α-particles

Electrons, protons and α-particles will also interact with the solar sail film. At low energy, electrons scatter on the sail material atoms and basic physical processes are the excitations and ionization of the atoms of solar

sail material. We can use Eq. (19) to determine the solar electron flux reduction; however, the mass attenuation coefficient for electrons is defined as

$$\mu_e = (n\sigma_{ex} + nZ\sigma_i), \tag{21}$$

where σ_{ex} is the sum of total cross sections for excitation processes in different states, and σ_i is the sum of total cross sections for the ionization with excitation. The energy dependence of the cross section results in the energy dependence of the mass attenuation coefficient for the electron. The scenario with the maximum value of the cross section for each process is conservatively assumed to better understand the processes' influence on the ionization of the solar sail surface. To make a meaningful estimate of all effects of the interaction of photons with the solar sail material leading to its ionization, we consider the upper limits of the experimental and theoretical values of the cross sections for all considered processes.

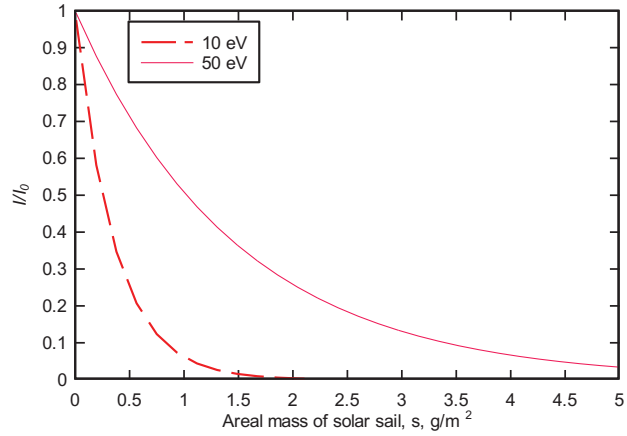

Fig. 3 The dependence of the ratio I/I_0 on the beryllium areal mass for different electron energies.

As an example we consider the beryllium and calculate the dependence of the ratio I/I_0 on the areal mass for different electron energies shown in Fig. 3. The results show that 10 eV electrons are stopped by the beryllium film with the areal mass 1.5 g/cm^2, while about 50% of 50 eV electrons penetrate the beryllium film with this areal mass. The stopped electrons mostly ionize the beryllium film and lead to the excitation of beryllium atoms that finally results their ionization through the different auto ionization channels.

When electrons, protons and α-particles pass through the sail material there is the energy loss for electrons, protons and α–particles. For electrons, this energy loss is the result of collisions with background atomic electrons and bremsstrahlung radiation when they are strongly deflected by the nuclei of solar-sail material

atoms. For protons and α–particles, it is caused by electronic and nuclear collisions. Figs. 4 and 5 present the dependence of the stopping range for protons and α-particles on the energy of proton and α-particle. We use the data from the ESTAR and ASTAR database [27] and the program, which calculates the stopping range for protons and α-particles in beryllium and aluminum. The comparison of Fig. 4 and 5 shows that the stopping range of the proton in beryllium is larger than that in aluminum, while there the situation is opposite for α-particle. The stopping range for α-particle in aluminum always exceeds that for the beryllium. From the presented result we can conclude that protons and α-particles with energies from 10 keV to 100 keV 1 MeV will be stopped by the beryllium or aluminum solar sail foil with the areal mass between 0.1 g/m^2 and 10 g/m^2. This results in ionization of the solar sail film and degradation its optical properties. Let us also mentioned that the stopping range much lower for the beryllium than for aluminum, scandium, titanium and niobium, which also are considered as candidate solar-sail materials [28].

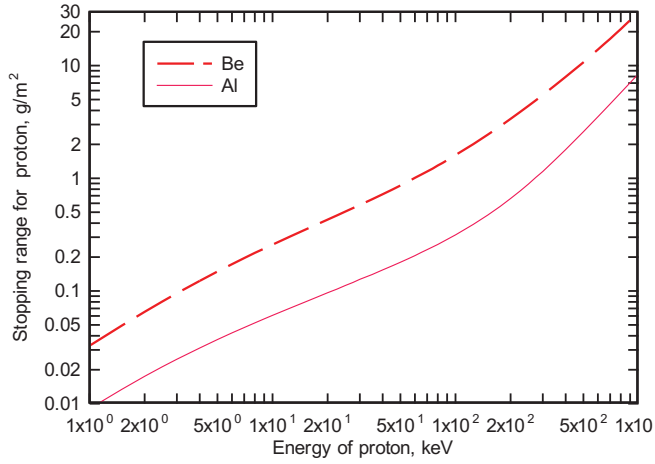

Fig. 4 The dependence of the stopping range of proton on the energy.

The interaction of photons, electrons, protons and α-particles with solar sail material produces atomic displacements, electronic excitations, or both. Atomic displacements result from the elastic scattering of the protons and α-particles by an atomic nucleus so that the kinetic energy transferred to the nucleus in the collision is sufficient to break the chemical bonds to neighboring atoms. Inelastic scatterings of these particles produce macroscopic modifications in the properties of solids that are dissimilar to those caused by the electrons. Electron-induced displacement damage in materials is qualitatively and quantitatively unlike that caused by the protons or α-particles.

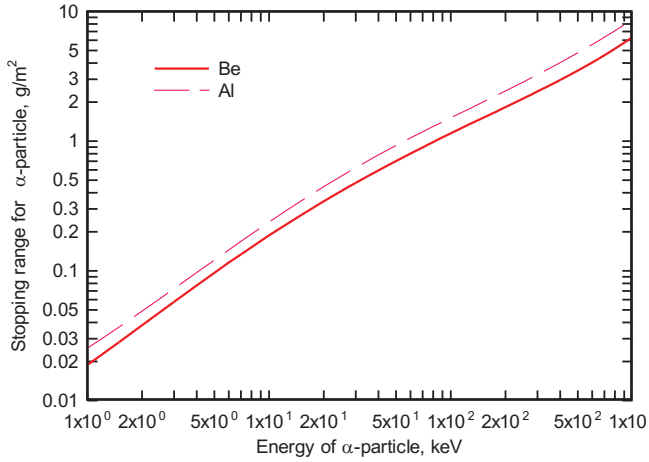

Fig. 5 The dependence of the stopping range of α–particle on the energy.

V. Discussion and Conclusions

Functional solar sails must be resistant to the effects of space environmental exposure. The influence of the effects of space environmental factors such as a vacuum UV radiation, electrons, protons and helium ions impact on materials were studied experimentally [1, 4-6, 29-30]. These experimental studies focused on the measurements of the hemispherical reflectance when samples under investigation have been exposed to electron, proton, and UV irradiation separately and combined were investigated. Peculiarities in changes of spectral reflectance of thermal control coatings under single and multiple influence of space environment were studied. Exposure of samples to environmental factors, separately and in combination, demonstrated different dependence of the spectral reflectance coefficient on the exposure that is associated with various processes of defect generation. The most significant irreversible changes of optical properties of materials are produced by protons. In this case the reflectance of materials changes appreciably. The most important conclusion that space environmental factors demonstrate the effect of nonadditivity: the influences of separate environmental factors and their combined action are not the same. One should take this fact into account in theoretical modeling of the space environmental effects.

We have discussed the influence the of space environment on the solar sail temperature that it gains as a result of absorption of solar radiation. The increase of temperature slightly change the reflectivity of the metals used in aerospace industry, and not significantly affect the propulsion efficiency of the sail. However it requires

having more thick material to reflect all wave lengths of electromagnetic radiation at high temperature than when solar sail is at low temperature. Of course this increases the areal mass affects significantly on the performance of solar sail. Therefore, the strategy of solar sail mission have to consider this fact up prior based on a projected maximum temperature that solar sail will reach in space depending on the distance from the sun. Results of our study show that the temperature of a solar sail increases approximately as $T \sim r^{-2/5}$, when the heliocentric distance r decreases, when are taking into account the temperature dependence of the emissivity and conductivity of the solar sail material. Thus, the temperature of the solar sail material increases more slowly than in the case for the constant emissivity and conductivity ($T \sim r^{-1/2}$).

In the present work is demonstrated that the space environment related to UV radiation, electrons, protons and helium ions influence:

1. Structural changes in the surface of solar sail that results in the radiation-enhanced adsorption, decrease of reflection and desorption. The radiation-enhanced adsorption leads to the increase the solar sail temperature.
2. Generation of the electric field due to accumulation of the volume charge at the materials surface under the action of ionizing radiation. The effect of ionizing radiations depends on the radiation dose.

Thus, the temperature increase and the accumulation of the positive electric charge are the most two disadvantages that causes by the space environmental factors. Let us address the issue how we can turn these disadvantages into the advantages. The ionized charged sail may have advantage related to the electrical propulsion by the solar wind proposed in Ref. [31]. As is shown above in our case as a result of ionization the solar sail naturally gains the positive charge (potential) which will deflect the proton component of the solar wind and therefore, extract momentum from the solar wind plasma. In Refs. [26, 32] was analyzed such opportunity for the beryllium hollow body solar sail. However, the proton component of the solar wind induced acceleration is considerably less that the solar radiation-pressure acceleration. When solar sail naturally gains the heat its temperature increases and can be so high that can be used the thermal desorption propulsion method proposed in Ref. [21]. Using coats of atoms, that liberate by hear heating at high temperature through thermal desorption, embedded in the reflecting surface of solar sail can provide an additional significant trust for a solar sail. After the coats sublime away, the sail can perform as a conventional solar sail, using a beryllium or aluminum coat beneath.

The determination of optical properties of the solar materials remains a challenging task, despite the advanced methods of data acquisition and analysis. It is required to develop mathematical model that allows

model the time dependence of optical parameters of the solar sail on dose of radiation exposed by the sun and use these parameter in orbital equation for a realistic mission analysis. In Ref. [33] are made some steps toward mathematical models describing the influence of electron radiation on outgassing of spacecraft materials. The theoretical works has been accomplished in the aria of an imperfectly reflecting solar sail dynamics modeling and considering optical degradations of the solar sail [2, 34, 35]. However, it was considered the static model assuming that the optical parameters are changed as a result of the space environment but not undergo the changes during the mission. Unique technology involving optical performance, environmental stability, thermal expansion and wrinkle management of the sail film, system dynamics and control strategies and many other issues must be addressed for realistic minimum mass solution [36]. This is problematic but a challenging task.

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Design and Sizing of the GOSSAMER Boom Deployment Concept

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Since 2011 DLR is preparing 3 orbit demonstrations of deployment and operation of a solar sail. For this mission bundle a new deployment control mechanism for DLR's tubular CFRP shell booms has been developed that is applicable to all 3 evolution steps of the GOSSAMER - Road Map. According to the mission goal of GOSSAMER-1 and its addressed deployed sail size of 5x5m², a down scaled configuration of the booms and an adapted deployment mechanism are currently under development.

The paper introduces the boom and its deployment mechanism concepts and describes and concludes tests and simulations that have been performed to proof the mechanical performance of the deployed booms.

I. Introduction

SOLAR sailing is a since several decades thought of and simple idea which in fact is extremely demanding to put into practice. There have been many European solar sail projects that either failed or were left unfinished [2, 3].

Based on previous successful work and recent developments, the enthusiasm of DLR researchers led to the internal agreement to start another effort to develop a reliable solar sail technology that should also drive the development of other deployable gossamer structures. The recently commenced Japanese IKAROS mission and

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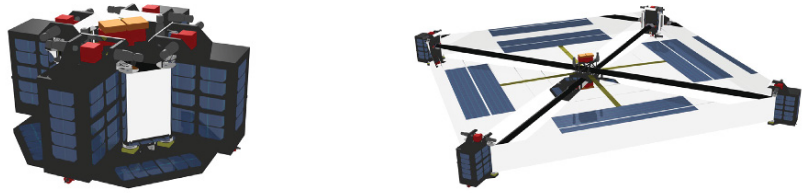


a) stowed configuration b) during parallel deployment of booms and sails

c) deployed configuration d) deployed sail with jettisoned BSDUs

Fig. 1 GOSSAMER-1 deployment process.

the successful deployment of the US sail NanoSail-D2 had additionally motivated management and engineers to pursue this topic.

One main difference to former European solar sail projects is the complete abandonment of any scientific payload. Scientific objectives would not only introduce a higher complexity and thereby introduce additional risk to a mission, which is solely focused on solar sail technology development, but also the linking of such a development to scientific objectives would increase mission cost to a non- justifiable level. Both aspects were the cause for previous failures of solar sail projects. Only if a conclusive proof of the reliable and robust operation of that alternative propulsion technique has been given, the scientific community will have confidence and will commit scientific payload to a sail craft. According to the "The 3-Step DLR-ESA Gossamer Road to Solar Sailing" [1] DLR is currently working on the first GOSSAMER-1 demo mission that should test and monitor the deployment of the 5x5m² Solar Sail in the next years. For this, the current launch date is set to mid-2015 and will be carried out in conjunction with the EU funded QB50 launch.

II. Basic Concept

Figure 1 shows the basic deployment concept of GOSSAMER-1. The booms and the sail segments are deployed simultaneously by electrically driven Boom & Sail Deployment Units (BSDUs). Thereby, a new sail

coiling concept is used, where each spool stores one half of a sail quadrant [5, 6]. Two of these spools are mounted to each of the four BSDUs. The advantage of this deployment concept is the well-defined position of sail and booms during deployment. The booms are under constant axial pressure load and the sails are mainly in tensioned condition. Consequently, the sails are always in defined positions during the deployment. However, the constant pressure on the booms is also a challenge for this concept as it demands a boom deployment mechanism that deploys the boom also under axial pressure or bending moments.

As indicated in Figure 1d) the BSDUs are foreseen to be jettisoned after full deployment of the sail to further minimize the mass of the sail craft and consequently optimize the resulting thrust.

The next sections introduce the booms and its deployment mechanisms with more details. Thereby the general design concepts as well as results of first tests are given.

III. Boom Design

Figure 2 shows the comparison of the booms cross section used for the former DLR/ESA Solar Sail Study that was conducted at the end of the 1990s, and the current boom design for GOSSAMER-1. To cope with the new requirements and parameters of GOSSAMER-1, the boom has been downscaled, and miniaturized to the smallest possible, but still coilable, design.

A. Key Parameters

For the GOSSAMER-1 project two booms are utilized and fixed in their center to the spacecraft in a cross like arrangement (see Figure 1). These two booms, each about 8 m in length, 6 cm of height and 7.4 cm of width, are crossing each other while maintaining an out of plane offset. One boom is fixed on top and one is fixed on the bottom side of a fixation plate. Thus building up the boom cross and holding the taut sails in cruise mode. This approach is considered to be beneficial for the transfer of bending and compression loads from one end of a boom to the other. The booms themselves are made of two CFRP half shells with an omega cross section, fabricated from a 0°/90° one layer fabric of 0.14 mm in thickness (0° is defined by the longitudinal axis of the

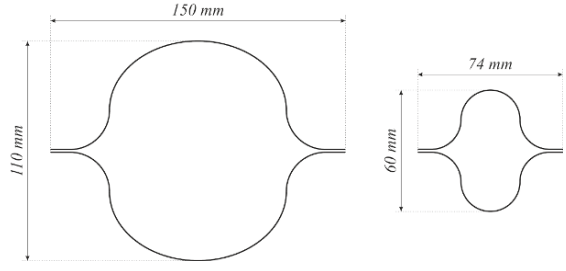


Fig. 2 Comparison of the former Solar Sail boom design (left) and the current GOSSAMER-1 design (right).

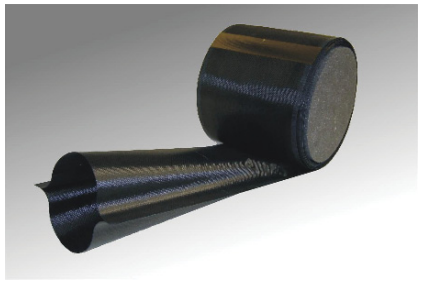


Fig. 3 Rolled up solar sail boom.

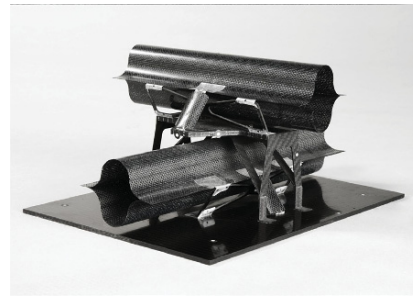


Fig. 4 Previous boom cross demonstrator.

booms). After bonding the half shells lengthwise to form the booms, the material's elasticity and the omega shaped cross section of the half shells enable the booms to be flattened out over the whole length and to be rolled up on a cylinder for stowage (see Figure 3). The lenticular shape and the main dimensions of the boom's cross section are depicted in Figure 2. Nevertheless an interface between boom and main module is required to handle non-symmetrical loads and to connect the sail to the sail craft. The interface for each boom comprises of two 3 mm thick solid CFRP panels, made of a standard +45°/-45°, 14 layer laminate cut as an X with closed frame, four CFRP tape springs with a thickness of 0.3 mm, supporting the interface in lateral direction, and elastic glass fibre fleece hinges connecting the parts. The booms are bonded with the elastic fleece hinges to the X-panels which are again bonded to the before mentioned fixation plate. Bonding is realized with a two component epoxy. The plate itself is fixed to a truss structure of the main module of the space craft. As supporting elements the tape springs connect the X-panels with the fixation plate as well. In this way the interface is flexible to react on the cross sectional change of the lenticular booms during folding and deployment, and locked in its final position facilitating the bi-stabile properties of the tape springs. In Figure 4 a previous, stacked and unsymmetrical, version of the boom cross is depicted, displaying the main components.

The used booms are per definition light weight profiles which are aiming to combine a very thin wall thickness, compared to the cross sectional dimensions, with a preferably very low mass and high bending and buckling stiffness. Under load these types of profiled structures tend to undergo certain types of deformations. On one hand the bonded flanges of the booms are providing a major part of stability, while under certain load levels of bending loads they exhibit spreading deformations. This reduces the occurring compression and relocates the load path thru the wall areas close to the curved cross section of the boom. On the other hand these walls are likely to buckle under compression and bending loads or a combination of both. This has a large negative influence on the overall stability of the booms. This and the fact that the booms will be coupled to the spacecraft via an elastic interface led to the need to verify the ability of the booms and the interface in finite

element analyses. Additionally this is realized in a vertical bending test stand. A large amount of the analysis and testing were supported by the work in [7].

B. Finite Element Analysis of Boom and Interface

1. Introduction to Finite Element Model

One half of the boom cross comprising one half of the boom including the interface area and the interface itself are modelled in order to perform a FE-Analysis in ANSYS. This was done to generate a reliable but simplified model being able to predict the structural behaviour under load. To comply with international standards within this project the ECSS (EUROPEAN COOPERATION FOR SPACE STANDARDIZATION) standard "Space engineering – Structural finite element models" [8] was regarded as guideline for units, model generation and validation area for comparable parameters. The model was composed in an APDL script, using variables for all parameters of the model, executed in key points, lines and areas. According to nature of the booms with their small wall thickness compared to the cross sectional dimensions the 4-Node structural shell element "SEHLL181" with six degrees of freedom at each node was chosen to model the boom. Regarding this it is assumed that tension over the wall thickness can be neglected. This choice of using shells makes a good compromise between very simple but rough calculations with beam elements and very demanding analyses, and because of the small wall thickness inappropriate, volume elements. This element type was also used to model all other parts of the interface such as the X-panels and tape springs. To start out with a fairly simple model the tape springs are considered as ideally stiff, since no folding will be simulated at this point. The elastic glass fibre fleece hinges are replaced with a merging of nodes in the according region for simplification. For the mesh generation rectangular elements are used supporting the approach with shell elements. This is considered to have a positive effect on the quality of results. The use of sub nodes is not necessary as these elements are being easily adjustable in size. Due to the orthotropic orientation of the fabric fibres (0°/90°) the Young´s modulus in x- and y-direction are the same while the one in the z-direction (out of plane of the laminate) is much smaller.

2. Boundary Condition, Loads and Solution Approach

The definition of a coordinate system is followed by dividing the load components in the 3 spatial directions. The external load is applied to one node at the boom tip in order to deflect the boom in lateral direction, in plane to the boom flanges. Nevertheless the load vector can be pointed also in other directions e.g. out of the flange plane. The X-panels, in reality bonded with the glass fibre fleece hinges to the fixation plate, are locked in all transversal directions at the nodes in the according region. The fixation plate is not modelled. Due to material

variations and the hand manufacturing of the booms, especially the thickness variation of the glue layer between two boom half shells are causing imperfections and lead to stress levels and deformations and finally to instability and buckling under load. Nevertheless this does not necessarily implicate the failure of material strength. These imperfections often show in geometrical imperfections, very small sized sine wave like shape deviation along the boom flanges, after bonding the half shells already in the unloaded state. When a load is applied these sine wave like deviations are amplified on the stressed side of the boom. The real influence of imperfections to the structural behaviour can only be estimated by a sufficient statistical evaluation in experiments as suggested by [9]. This can only be regarded in a simplified manner within the modelling. The implementation of these imperfections to the FE-model is pursued in a 3 step solution process. After a static linear computation with an extenuated load (1 % of the total load) a linear buckling analysis, again with an extenuated load of 1 % of the total load, is conducted. This results in a sine wave like deformation known from the real geometry imperfections of the unloaded booms. As third step a non-linear analysis with the applied total load is performed. At the beginning of this last step the values determined before are applied as a pre-deformation. In this way the influence of the geometric imperfection of a not perfectly planar flange area is applied to the simulation of the overall buckling behaviour. Alternatively the pre-deformation can be applied followed by a non-linear computation as demonstrated in [10] and [11]. Material imperfections such as deviations in the fibre alignment, corrugations of fibres or inhomogeneities like matrix clustering were not regarded at this point of FE-model development. Since the load is applied very slowly a quasi-static computation is performed and realized in a stepwise load application in the non-linear analysis. These load steps can be divided in sub load steps and are triggered by a time interval. Obtaining the point of time when the solution does not converge anymore from the non-linear computation the load of failure can be concluded by using an appropriate time interval. In this case 10 time units are chosen for the time interval to apply a load of 200 N. The solution is terminated when the computations do not converge anymore or in the case of stability failure a sudden and significant change in the course of events occurs.

3. Results

The applied load results in bending in x-direction (perpendicular to the boom axis, in plane to the boom flanges), in compression in the z-direction (longitudinal direction of the boom) and in a bending with a small deflection in the y-direction (out of plane relative to the plane of the boom flange). The FE-Analysis illustrates that due to the low stiffness in the y-direction a small angle of attack causes distinct deflections when a load is applied out of the flange plane. As a load at the boom tip is applied in the flange plane, the sine wave deviations triggered by imperfections along the compressed side of the boom start to form. Figure 5 shows these deviations

having their largest amplitude close to the interface fixation of the boom abating with an increasing distance from the interface. Although buckling occurs, the boom still shows a comparably high mechanical strength. Stability failure shows in a large and global buckling of the boom and can be illustrated in the plot of the sum of rotations. In this form of representation the occurring buckling can easily be observed as depicted in Figure 5. Here one can predict a possible location of stability failure at the regions of highest amplitudes (yellow regions) of the buckling pattern.

Within the FE-Simulation variations in boom length of 1 m, 2 m and 4 m are analysed. The cross sectional dimensions remain the same for all variations. Another parameter that is varied is the angle of attack α between the load vector and the y-z-plane (the plane perpendicular to the flange plane sharing the longitudinal boom axis as one dimension) of the model with 5°, 15° and 30°. The resulting final value of each simulation run is considered to be the one at the last load step before a stability failure occurs. In Figure 6 one can observe that with increasing values of angle of attack the bearable loads of the structure decrease. This behaviour can be expected when regarding the classical beam theory [12]. This leads to the logical consequence that increasing loads also increase the risk of buckling and stability failure of the structure. The analysis also implies that an increase in boom length goes hand in hand almost linearly with the increase of the bearable load since the main acting force is a bending moment. To investigate the influence of the interface compared to an ideally stiff clamping of the boom at the root several more simulation runs were performed. As one can see in Figure 6 the load values for the ideal clamping, without interface (suffix *o.I.* in diagram legend) were at least doubled compared to the values with the interface. Having this outcome in mind the analysis was aimed to identify this stiffness difference in order to characterize and tailor the interface coupled with a boom to comply with requirements of the structure resp. spacecraft. The abrupt change in slope for the 15° angle are interpreted as improvable deficiencies of the used FE-model resp. as a discordant value standing out. This will be improved by advancing the model and running more parameter variations to give a basis for a statement of the statistical significance.

C. Experimental testing in a vertical test stand

Before launching and deploying a spacecraft such as the GOSSAMER-1 demonstrator in space extensive testing and characterization of mechanical and material properties have to be performed. The CFRP-Boom concept of DLR for GOSSAMER-1 can be can be classified in a technical readiness level between TRL4 and

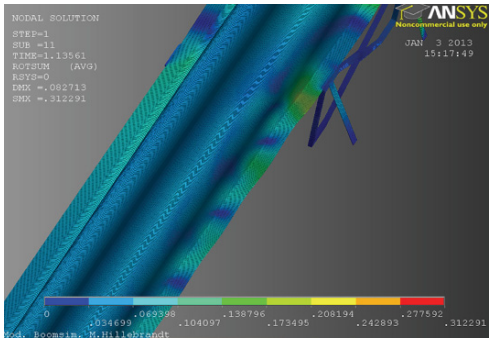


Fig. 5 FE-model showing flange deviations.

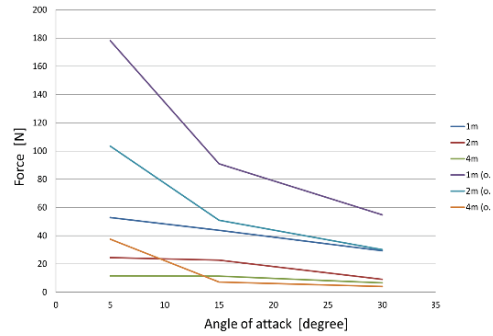
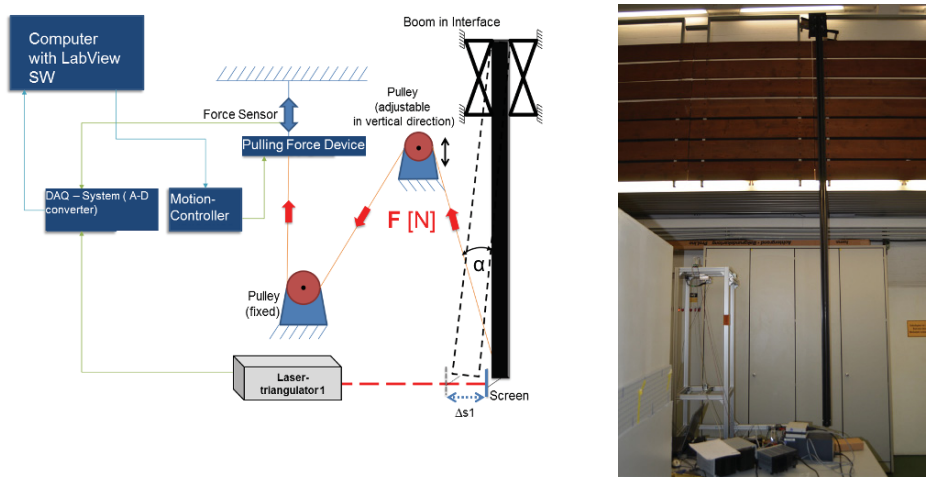


Fig. 6 FE-results of different boom lengths.

TRL5 as defined in [13]. Since experiments under relevant environment are the second step yet to be performed the first approach to obtain basic knowledge of the structure and the interaction of parts in an assembly resp. breadboard model are realized in a vertical test stand for combined lateral and compression loads. In previous experimental investigations [14] the booms were only tested as a stand-alone part fixed in a ideally stiff clamping device. In the tests presented here the booms are mounted to a somewhat elastic interface, providing elasticity at the fixation. This is done in a vertical hanging position resembling a realistic configuration of the hardware. In these experiments different boom lengths were tested at different angles of attack (α) to characterize the stability of the boom-interface structure. External loads are applied at the boom tip by a high strength line (rope), redirected with pulleys and winded up on a spool with a high precision electric motor. This test setup is also thought to feature a gravity compensating effect, cancelling out bending moments due to gravity force acting on the boom´s dead weight in a horizontal setup. In this way no complex counterweights, pulleys or gas balloons (see [14]) have to be implemented in the test stand. Although the gravity compensation might not have the same quality as achievable with the before mentioned mechanisms this setup is regarded as sufficient at this stage of development.

1. *Test Setup*

The force sensor is directly mounted to the pulling force device enabling an easy replacement. Figure 7a) shows the scheme of the test setup. The tip deflection resp. the load (force F in Figure 7a)) is applied by a pulling force device, pulling a rope which is tied to the boom tip. The rope is guided by pulleys and winded up on a motor driven spool which part of the pulling force device. This device is built as a hanger with the spool in the centre of gravity. Nevertheless the device has to be balanced with counterweights on the opposite side of the attached motor. A force sensor is mounted between the pulling force device and the frame of the test rig



a) Scheme of setup b) Real test setup with 4m boom

Fig. 7 Test setup for boom bending tests

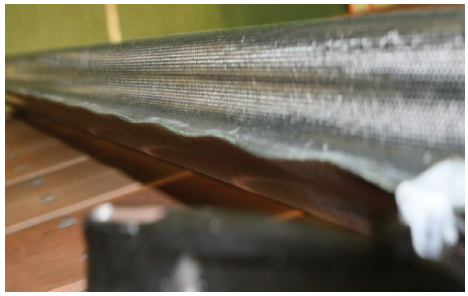
measuring the pulling force in regard to the dead weight of the device. A maximum torque of 2 Nm delivered by the motor can generate a maximum pulling force of 200 N. The angle of attack is varied by moving an adjustable pulley in the appropriate vertical position. The tip deflection Δs1 is measured as change in distance of a screen mounted to the boom tip relative to an optical sensor, a laser triangulator. The acquired analogue signals from the force and optical sensors are gathered and converted in a data acquisition system (LabJack U3) and fed into the data processing software (LabView) of a computer. From here the data can be analysed and exported to e.g. MS Excel or Matlab. A special software (Faulhaber Motion Manager 4.0) installed on the computer is used to control the motor of the pulling force device via a motion controller. The test rig comprises of a frame made of aluminium profiles, holding the laser triangulator, the motion controller (printed circuit board), a control board to calibrate the force sensor, the pulleys, and the pulling force device with the force sensor. Other necessary hardware are two DC power supply units, one for the force sensor and one for the motor to eliminate current disturbances, and a low pass filter ahead of the data acquisition system in order smoothen the signals. Whereas the test article has to be mounted appropriately in height relatively to the boom length on a fixation, the test rig itself can be placed freely while regarding the measurement length of 100 mm of the laser and the desired angle of attack. A larger distance Δs1 can be measured by halting the load application when 100 mm are reached and the laser triangulator is slid closer to the test rig on a cantilever beam. In Figure 7b) the boom and interface are clamped to a balustrade in a vertical orientation. The test rig, the adjustable pulley and the laser triangulator are accommodated in this manner that the desired angle of attack can be reached in the tip deflection signal is zero while no load is applied.

2. Testing and Test Results

At the start of every measurement the initial condition is re-established and all parameters are set to zero. At this point the rope is pulled by the motor iteratively in a state where it is not stretched too much but also not slack. In any case the boom must not be deflected yet. The measured values are obtained in discrete increments of a defined step width to reduce the influence of signal noise appearing during previous measurements. The force values as well as the values for the boom tip deflection are recorded with a rate of 100 Hz while a measurement lasts about 5 s. If the case of stability failure occurs, noticeable by a sudden drop in the load value at a growing value for the boom tip deflection, two more measurement steps are performed before the test is aborted. If the damage is not fatal to any part of the test article, the test is repeated to gain insight of the influence of pre-damage by stability failure to the residual strength and stability. Another criterion to abort the test is when the measurement limit of a sensor is reached.

Booms of 4 m and 2 m length with angles of attack of 5°, 15°, 30° and 45° were tested. From previous tests it is known that stability failure at the flanges usually occur only on the side of the boom where the load is applied. This justifies the testing of both sides of the boom, by just turning around, even when one side has already been buckled. For this kind of test setup a change in the angle of attack due to the change in geometric relation of the distance between boom tip and pulley and the height of the pulley when deflecting the boom occurs. This means that the angle of attack will not remain constant throughout a test run instead it will become smaller as the tip deflection grows. To reduce this effect the distance between the test rig and the load application point at the boom tip is increased.

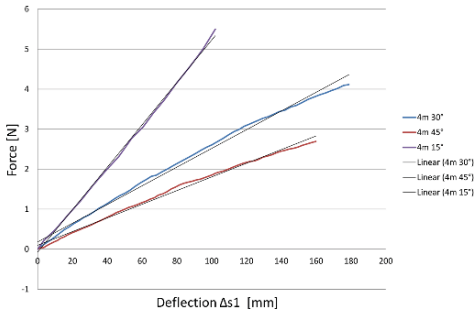
The when the applied load and tip deflection are high enough the sine wave like deviation pattern along the boom flange of the loaded side of the boom, starting at the interface, also show in the experiments as one can see in Figure 8a). The amplitude of the deviation increases with larger angles of attack and increasing loads. In this state of instability the structure is still able to carry loads. When the load is further increased the curved area of the boom starts to buckle all a sudden accompanied by a drop of the load value while the tip deflection value is still growing. This results in a stability failure as depicted in Figure 8b). During the experiment the cross section of the boom also deforms, especially at the connection to the X-panels of the interface. The X-panels themselves are being distorted. In general one can say that the boom flanges provide a large amount of stability and strength to the structure. The measurements as depicted in Figure 9 show a roughly linear behaviour under load. These results also illustrate that bearable loads for smaller angles of attack are higher than for larger angles. This is true



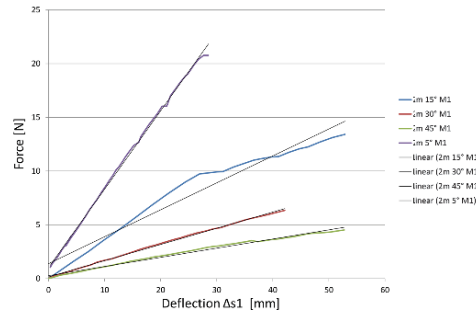
a) Sine wave like flange deviations

b) Buckling due to stability failure

Fig. 8 Boom deformations in experiments.



a) 4m booms.



b) 2m booms.

Fig. 9 Force over tip deflection.

for both lengths of tested booms and leads to the conclusion that this boom type tolerates compression loads very well while the ability to withstand bending loads decreases with increasing angles of attack.

D. Comparison of FE-Analysis and Experimental Results

The direct comparison of simulation results and experimental results makes clear that measured load values of boom failure are always under the values from simulation (see Figure 10). In the following paragraph possible reasons for these discrepancies are discussed. The FE-Simulations are based on an ideal fixation at the connection points of the interface, meaning the connections between boom and X-panels, X-panels and fixation plate (not modelled), tape springs and X-panels and tape springs and the fixation plate. Ideal fixation in this content is considered to have all degrees of freedom locked at the regarding nodes of the FE-model. The elasticity of the glass fibre hinges and the used epoxy are not regarded. Additionally the tape springs are modelled, unlike in reality, as ideally stiff, not regarding their bi-stable nature to keep this first FE-model rather simple. These unregarded properties of fixation and connection might explain a part of the difference in results of FE-simulation and experiments. Another factor is the change in angle of attack due to the change in geometric relation of distances when applying a load resp. displacing the boom tip in the test stand. For the FE-simulation a

constant angle of attack during the load application is realized. Especially for large angles of attack such as 30° and 45° this effect is distinct and shows in measurements. For the experiments an increase in distance between the test rig and the load application point at the boom tip would reduce this effect and could therefore reduce the discrepancy in results. The simplified modelling of imperfections such as material imperfections, manufacturing imperfections or inaccuracies in testing are considered influencing factors. Here an alternative modelling approach for imperfection according to [10] can help to bring the simulation closer to reality. In contrast to the numerical results the graphical analysis of simulations and experiments shows the same patterns of buckling and deformations as one compares Figure 5 with Figure 8a). This indicates that deformation during loading can be generated close to reality. More influencing factors such as the glue layer thickness between the boom half shells are difficult to resemble since they are varying and hard to characterize in reality. Consequently the FE-model needs to be improved and matched with the experimental conditions. This will be realized specifically in the scope of imperfections, fixation and boundary conditions and material properties.

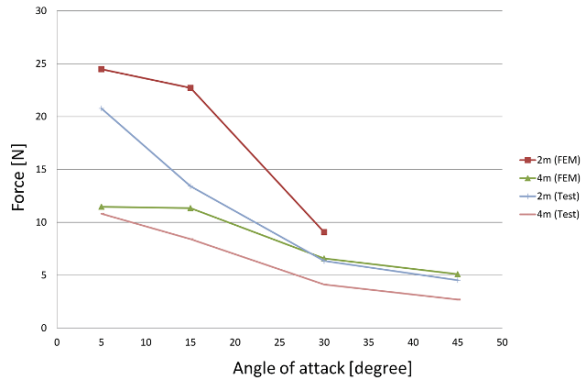


Fig. 10 Comparison between FE and test results

IV. Boom & Sail Deployment Unit Design

A. General Setup

Figure 11 shows a preliminary engineering model of the boom deployment unit with a visible motorization concept but without the attached sail deployment mechanisms. In this figure the following essential parts can be seen:

1. BSDU EM aluminium main structure (silver frame)
2. Deployable boom (black)
3. Boom hub for stowage of coiled boom (orange)
4. Hub braking device (aluminium gear wheel with a small gear wheel on a hysteresis brake)
5. Boom deployment belt (white)

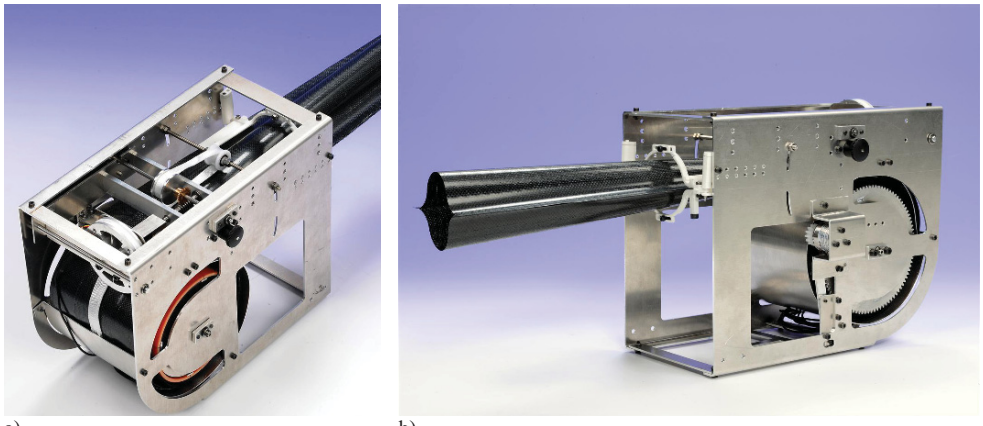
6. Boom deployment belt spool mechanism (silver spool with copper gear wheel visible in Figure 11 a))
7. Boom to Sail Fixation Ring (white ring with white brackets)

In addition, Figure 12 shows the current setup of the BSDU-EQM. The main differences to the EM are the nearly closed structure made of CFRP Honeycomb sandwich, the attached sail spools and the modified Boom to Sail Fixation Ring.

B. Deployment Drive

The mentioned belt is the essential part of the deployment principle and commonly used in the EM and EQM. It is of the same length as the boom and is co-coiled with it on the boom hub. The hub rotation is inhibited by a braking mechanism to suppress the self-deployment potential of the boom. Further details on the boom deployment behaviour can be found in [4].

First tests of the BSDU-EM had shown that about 10 N for the belt pulling force is required to drive the deployment mechanism against inner friction between BSDU-parts, boom and belt. Thereby, it has been observed, that bearing conditions in the BSDU where not optimal and that improvement in the boom guiding system will decrease this value. To deploy the complete sail the BSDUs will additionally have to provide the force necessary to unfurl the sails from the sail spools. For this reason a budget of 2 N has been reserved per spool resulting in a total belt force of 14 N.



a) b)

Fig. 11 BSDU engineering model (BSDU-EM).

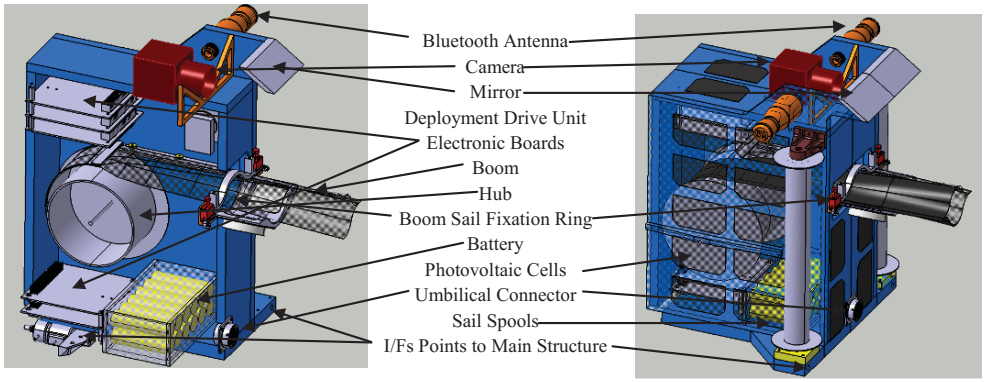


Fig. 12 BSDU engineering qualification model (BSDU-EQM).

C. Boom to Sail Fixation Ring

During the deployment, the belt is actively spooled by a motor driven drum. Consequently, the belt pulls the boom out of the stowage region. In this moment the boom exits the cylinder the boom cross section starts to deploy. The therefore partially deployed boom is passing the Boom to Sail Fixation Ring before it leaves the mechanism. This fixation ring is directly connected to the outer sail corners by ropes and contains two mechanisms: One locks the ring at the boom when the sail is fully deployed. The second mechanism separates the ring from the BSDU to enable jettisoning of the BSDU.

D. Interfaces to Main Structure

Figure 12 also indicates the interface points between the BSDU and GOSSAMER-1 main structure. The interface is designed iso-statically. A FRANGIBOLT®[§] provided by TiNi Aerospace Company is used as one out of three I/F points and as the release actuator for separation. The actuator is visible in the lower left corner of Figure 12a) as silver cylinder mounted to a stiffener. The other two I/F points are realized by two metallic husks that are laminated into the bottom plate of the BSDU structure and being the counter parts for the bolts (not depicted) that are firmly mounted to the octagonal baseplate of the main structure (Figure 1).

Electronically, the BSDUs and main module are connected by an umbilical connector that provides power and data lines. It will decouple automatically when the BSDU is moving out some millimeters.

E. Camera & Mirror System

As shown in Figure 12a) a camera is mounted to the top of each BSDU. The usage of these cameras is motivated by one main objective of GOSSAMER-1: It shall provide evidence of the successful deployment of the sail with this concept.

[§] http://www.tiniaerospace.com/frangibolt.html

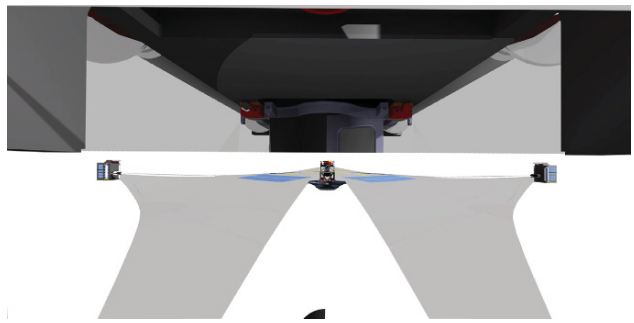


Fig. 13 View field study of BSDU camera including mirror (booms are set transparent)

The requirement for the view field of these cameras is to observe the whole deployment process and the deployed sail. Consequently, the horizontal view angle should be about 100° which results in a vertical view angle of 75° (for a 3:4 camera). As the vertical view angle is lavishly large for the observed flat sail, the decision has been made to reflect a part of the view by a mirror to other interesting points of the BSDU.

Figure 13 shows the result of a preliminary view field study of the combination of a 100° camera and a mirror. In this model, are not displayed. In a complete constellation they would be the field of view (see also Figure 1). This simulation shows that the entire sail could be observed by the camera. Moreover, the mirror reflects a part of the view field down along the inner plate of the BSDU which allows observing the Boom to Sail Fixation Ring and the Sail Spools during operation.

The study has been performed using the capabilities of Dassault's CATIA® software. For the later flight structure it is foreseen to use a CFRP honeycomb sandwich material with co-cured aluminized Kapton as efficient combination of mirror structure and reflecting mirror surface. At DLR the co-curing of those materials has been realized and proven as applicable before. It is expected that the mirror surface will not be perfectly even but it is considered to be sufficient for the short distance between camera and observed objects.

V. Conclusion

The achievements and gained knowledge of the previous pages could be concluded with the following bullets:

- The design for the boom and the BSDUs of GOSSAMER-1 has been presented and justified
- Tests on the boom had shown the potential of the GOSSAMER-1 booms and the boom interface but indicate that the FE models need to be further optimized
- The new boom bending test setup has proven it's functionality

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Bistable Over the Whole Length (BOWL) CFRP Booms for Solar Sails

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This paper presents novel ultra-light booms for solar sails and other large deployable space structures. These CFRP booms have a unique property: bistability over the whole length (BOWL), which enables simple and compact deployment mechanism designs that can reduce overall system mass. They were produced to solve some of the previously encountered problems with bistable composite tubular booms that reduced their optimal length and scalability due to local buckling phenomena when the diameter of the coil increased. A new low-cost manufacturing technique, which consists of using braids with a variable angle change over the boom length, was found to have a positive effect in reducing that tendency. An analytical model is used to explain this behavior and predict the secondary stable state properties and natural diameter of the coiled/packed boom. A 3.6m tape spring version of these bistable CFRP booms has been designed for a 25m^2 Gossamer Sail Deorbiter of future space assets and is being considered for an upcoming solar sail demonstration mission called CubeSail. Larger booms are being designed for a new scalable roll-up solar array concept.

I. Introduction

The majority of solar sail designs make use of deployable rigid structures or booms. These can not only extend and support the sail membrane, but can also provide enough rigidity in the deployed state for applications and missions which require relatively fast maneuvering, and that might need three-axis stabilization and control [1]. Mass minimization is of critical importance in making solar sailing a competitive and realistic

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form of propulsion. This minimization can be achieved through careful choice of sail film material, supporting structure, and operating mechanisms. The type of boom presented here is not only very low mass, but also aids in the minimization of 'parasitic mass' as the booms do not require a heavy hold-down mechanism to retain them in the stowed state.

Past solar sail designs have exploited different boom technologies such as coilable masts [2], inflatables [3], and metallic [4] and composite [5-7] thin shell structures. A number of low-cost solar sailing demonstration missions (NanoSail-D [8], LightSail-1 [9] or CubeSail [10]) making use of rigid sail support structures have been launched recently, or are scheduled for launch in the near future. One of the main considerations in the design of a supporting structure for a solar sail is the scalability of the concept. At present, most American researchers are focusing their efforts on the development of coilable masts, inflatable booms, and CFRP STEMs, while their European counterparts are developing composite thin shell structures. DLR's roadmap to solar sailing includes a series of upcoming launches with sails that utilize lenticular-shaped collapsible closed CFRP booms of increasing size [11].

This paper is divided into X sections. Section I introduces the different solar sail boom technologies developed to date. Section II gives a background to the study of bistability in thin shells and the analytical models developed to predict their behavior. Section III describes some of the problems previously encountered in trying to increase the scale of bistable cylindrical structures. Section IV introduces the model that determines the coiled curvature depending on the braid angle used . Section V explains the design of the Bistable Over the Whole Length (BOWL) boom. Section VI describes the new low-cost manufacturing process developed for them. Section VII describes the experiments carried out on small boom samples and a comparison of the results provided using an existing analytical model. Section VIII describes some of the current applications of the BOWL booms. Section IX discusses the benefits of using braids. Finally, Section X concludes the paper.

II. Bistability of thin composite shells

Tubular booms have been the most popular type of deployable structure for space applications since they were first used in 1961 on-board the Canadian Alouette I spacecraft as dipole antennas [12]. The STEM family of booms has been utilized on hundreds of occasions as antennas, gravity gradient booms, grasping devices, docking booms, and to extend payloads away from a spacecraft. Tape springs and STEMs are thin cylindrical shell structures that differ only on their subtended angle, α (see Fig. 1). They can be coiled for storage, yielding very high deployed-to-stored length ratios. Both structures are normally only stable in their deployed state (monostable). This necessitates the inclusion of a bulky mechanism to store the boom in the coiled

configuration. A second, coiled, stable state can be induced in isotropic shells through the introduction of an initial 'prestress' in the structure [13], and in composite shells by careful selection of the lay-up and fiber orientations [14]. Bistability can thus significantly simplify the way long slender structures may be stored and deployed.

Bistability in thin shells is an example of bio-mimicry, inspired by the closure-opening mechanism of the Venus Flytrap leaf [15]. Bistable composite slit tubes were discovered by Daton-Lovett under the patented name "Bistable Reeled Composites" (BRC) [16]. In order to achieve bistability in composite slit tubes, stiff fibers are placed at such an angle to the longitudinal axis that the shell exhibits anisotropic bending properties. The second stable state will then favor same-sense bending [17], meaning that the center of curvature of the shell in both stable states is on the same side of the surface. Each of the two stable states is a strain energy minimum, with the extended configuration existing in the lower of the two energy states. Bistable shell structures must undergo a transition between the two energy minima, requiring the input of energy. Once a certain transition point has been passed, the structure will follow the negative energy gradient to the other stable state. The shape of the energy gradient between the two stable states can be altered. Placing the transition point close to the stowed state energy minima leads to a structure which will self-deploy from the stowed state with little external input.

Several different concepts of deployable composite tubes with differing energy gradients between the stable states have been proposed in the last decade. For example, in semi-neutrally stable tubes both the extended and coiled configuration are zero energy points, meaning a partially unrolled tube will neither want to extend or coil up [18]. Neutrally stable tubes have a zero energy gradient between the deployed and stowed states, resulting in a resilient structure that is easy to control, can be fully or partially re-stowed, and which avoids deployment shocks [19]. It is also possible to construct a bistable composite shell with the coiled/stowed state representing the lower of the two energy minima.

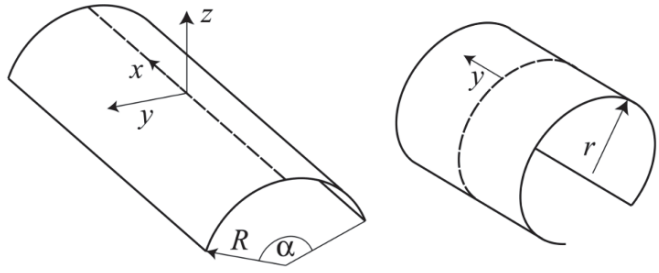


Fig. 1: A section of thin shell bistable composite in the initial, extended, stable configuration (left), and the second, coiled, stable state (right). *x*, *y* and z are the longitudinal, transverse and out-of-plane directions of the extended shell.

All the analytical models proposed to study the behavior of bistable tubes have been developed at the University of Cambridge. Iqbal and Pellegrino [17] first presented a simple analytical model that captures key features of bistable composite cylindrical shells. However, some of the assumptions made result in a model that does not distinguish between the bistable behaviors exhibited by symmetric and antisymmetric laminates. Galletly and Guest [20] extended the previous analytical model to include now all possible deformation modes, including twist and shear, and stretching-bending coupling effects. This was achieved by first assuming that the cross-section of the shell remains as an arc (constant transverse curvature), with a radius that is allowed to change. The model results do not correlate well with computational models, or the experiments undertaken by Iqbal and Pellegrino. This is mainly because of the simplifying assumption of constant transverse curvature. The model also fails to predict a second stable state for some symmetric lay-ups. In order to address this problem, a second shell model, which neglects the previous assumption of linear geometry, was produced [21]. It formulates a differential equation, based on Calladine's shell theory [22], to derive non-constant transverse curvature of the shell, which is used to calculate the second stable state. The findings are in good agreement with Iqbal's computational results. Nevertheless, there are still some disparities with the experimental results, mainly in failing to predict the second stable configuration of some anti-symmetric lay-ups. Galletly proposes that the result disparities are due to the fact that the polypropylene matrix had entered the plastic region, while the analytical models are based on linear-elastic material assumptions. Guest and Pellegrino [23] developed another inextensional bending model that is able to distinguish between symmetric and antisymmetric laminates. The analytical model assumes that the deformation of the shell is everywhere uniform, and that the Gaussian curvature (product of the two principal curvatures: coiled and extended) remains zero at all times. This model is discussed further in Section IV where it is used to analyze the experimental data produced by the authors.

III. Challenges in the design of long bistable composite tubes

Previous work showed that in order to have a bistable composite shell that tightly coils around itself, having the secondary stable curvature at a perpendicular axis to the extended curvature, the laminate needs to have no coupling between bending and twisting ($D_{16} = D_{26} = 0$) [23]. This initial work described that antisymmetric lay-ups made from unidirectional laminae with surface plies at sufficient angles from the shell longitudinal x-axis could be used for this. However, for ease of manufacturing of cylindrical shell tubes, fabrics like plain-weaves at ±45° to the longitudinal axis have been commonly used.

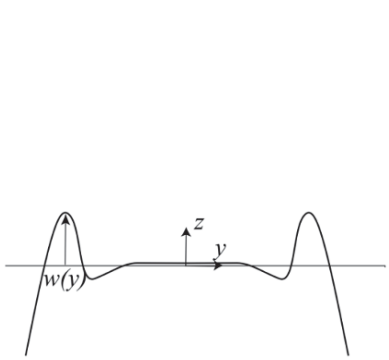
Murphey et al [24] presented 1 m long bistable CFRP tape springs that could be used as extensible booms and antennas. However, they are limited in length as the rolled-up stable configuration loses stability due to the

overlapping wraps that increase the diameter of the coil beyond the acceptable limit for bistability. The loss of bistability and buckled rolled shape is shown in Jeon and Murphey [25] where it is predicted that the tape springs manufactured have a maximum permissible length of 4.6 m. It is also shown that for thinner laminates and different lay-ups, slightly longer booms could be designed, but at the potential cost of reduced deployed structural stiffness and deployment energy.

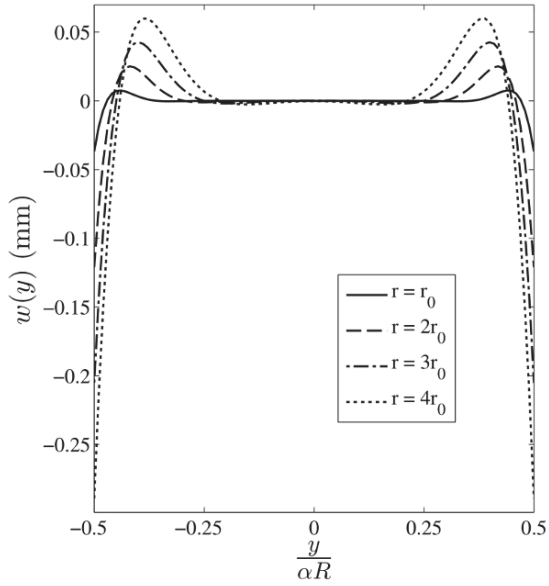
However, apart from short tape springs made for experimentation purposes, the commercial use of relatively long bistable composite slit tubes has only been exploited by RolaTube Ltd. The requirements for the different applications of BRCs have generally dictated tubes with relatively large radius of over 25 mm. This translates into a less challenging R/t (where t is the shell thickness) limit for elastic coiling of the tubes which is required to avoid permanent deformation of the matrix, fiber microbuckling or fiber and/or matrix fracture when rolled. Also, for the normal lengths utilized (< 15 m) the ratio of outer to inner coiled radii: r/r_0, is well outside the limits of the loss of bistability region. A drag sail/solar sail under development at the Surrey Space Centre involves the co-coiling of four 3.6 m long BRC booms (equivalent to a single 15m one) with a radius, R, of 16 mm. The booms have an inner coiled radius of 13 mm and an outer coiled radius of 43 mm, thus requiring a ratio of $r/r_0 \approx 3.3$. This large ratio exceeds the limit recommended by Murphey et al [24], whose tape springs were limited to a maximum ratio of $r/r_0 \approx 1.9$ in order to keep the bistable effect present in the whole coiled length. This large ratio was achieved by varying the fiber orientation along the boom length, using the method described in Section V.

Assumptions about uniform Gaussian curvature and inextensionality implicitly require the presence of non-zero bending moments along the edges. However, since these edge moments are certainly zero in the final stable states, the predicted stress distribution will be incorrect in a short boundary layer around the edges of the shell. Calladine [22] first demonstrated that the width of this boundary layer on isotropic shells can be approximated by the mean thickness of the shell times its radius of curvature. The boundary layer will affect the bistability of the tube and in reality needs to be accounted for as in Galletly's shell model [20]. Fig. 2 shows the transverse curvature profile of a tube calculated using Galletly's model for a [±45/0/±45] laminate with a subtended angle of α = 160º and an extended radius of curvature of R = 16 mm. It can be seen that, as one moves away from the initial stable radius of the coil, $r = r_0$, the width of the boundary layer grows as well. This reduces the flat central region of zero transverse curvature to only 40% of the total width of the tube for the largest radius, $r = 4r_0$. A similar effect occurs if the thickness of the tube is increased or the subtended angle is decreased. In

fact, for tubes that initially subtend a small cross-sectional angle (α), the relatively large size of the boundary layer significantly affects the position, and even the existence, of a stable second equilibrium point.



(a) Cross-section of coiled boom. w is the out of plane deflection of the shell when coiled.



(b) Predicted cross-section shapes based on Galletly's shell model [20] without considering the twist parameter. Here, r_0 is used to indicate the 'natural' coiled radius of the boom.

Fig. 2: Cross-section profiles of coiled composite bistable booms. As the boom coil radius is increased beyond its stable equilibrium value, undesirable edge effects begin to appear.

The loss of bistability of these thin shell structures is due to a combination of two effects that begin when the coil radius increases and the outer turns move away from the strain energy well of the second stable configuration. The first effect is the aforementioned increase in the boundary layer at the edge of the shell that produces local buckling in this area. As the radius of the coil and the stored strain energy grow, the width and length (coiled tube perimeter) of the boundary layer and the internal stresses will also increase. Hence, there will be an area in the edge where the local buckling load is surpassed and the area snaps back to the original extended curvature, relieving stress. However, at this point the rest of the edge and the shell still preserve the second state curvature. Consequently, the coiling diameter and thus the perimeter of the tube (edge length), can be further increased until another point of the edge locally buckles again. This process will occur repeatedly, and hence result in a wavy pattern located at the free edges of the shell (see Fig. 3(a)). The authors commonly name this edge phenomena '50 pence' effect in recognition of its semblance to the shape of the British coin.

It is important to eliminate the ‘50 pence’ effect, as it will hinder production of tight compact coils, and can produce non-circular rolled-up configurations that have a slight tendency to ‘blossom’ or self-deploy. A solution to this problem was found by suppressing bistability just at the free edges of the shell, which yields a less compliant boundary layer. For this, the two surface textile plies are removed near the edges of the shell, making them slightly narrower plies than the rest of the internal ones. The consequent gap is then filled with an extra unidirectional ply of the same thickness at 0° from the longitudinal x-axis, in order to keep the laminate thickness constant over the tube width. These extra surface 0º plies have a large effect in reducing local buckling of the free edge, allowing significant larger diameters before the ‘50 pence’ effect occurs (see Fig. 3, right). To delay the unstable behavior even longer, the surface fabric plies can be made tapered from root to tip of the tube, so that more resistant edges are created as the coil diameter increases. To prevent the border between the fabric and the 0º ply from failing along the boom longitudinal direction each of two boundary areas can be cut with pinking shears. This produces saw-tooth edges and eliminates crack propagation along the boundary between the two ply types.

(a) (b) (c)

Fig. 3: Bistable E-glass/ Polypropylene tape spring showing the ‘50 pence’ effect, or local buckling of the free edge (a), global buckling or warping of the boom (center), and the solution found for a compact coil (right).

The second cause of loss of bistability is more severe and involves the warping of the whole cross section of the shell, resulting in the total loss of bistability of the boom. This is a global effect on the coiled boom that results in an unstable roll, where the outer layers begin to prefer a doubly curved shape showing signs of warping (see Fig. 3(b)). This buckled rolled shape has a tendency to bloom, and is no longer self-containing. The radius at which the coil will start to show this type of failure can be approximately predicted by the strain energy equation if this point is taken as the radius of the coil where the two saddle equilibrium points lay. The curvature at which this occurs is indicated by the lighter colored line joining the two unstable regions (grey circles) in Fig. 5.

IV. Determining coiled curvature as a function of braid angle

A method for predicting the coiled radius of anti-symmetric bistable composite booms was developed by Guest [23]. This method assumes that the Gaussian curvature of the composite shell does not change; that is, the coiling-uncoiling process is inextensional. Composite shells which exhibit bistability can be constructed using unidirectional, woven or braided fabrics. In this case, braids were chosen because of the ease with which the 'braid angle' can be varied along the length of a composite boom. In order to better predict the way in which coiled radius changes with braid angle, Guest's model was augmented with experimentally derived material models for very thin carbon fiber braid shells.

Our goal is to find the braid angle δ required to cause a bistable laminate shell to match the coiled curvature, $C = 1/r$, of the cylindrical hub or spindle to which it is attached, and to determine how that braid angle should be varied along the shell length such that the shell stays in a minimum energy state as the radius of the coil increases from root to tip. Consider a situation in which the laminate shell is forced to conform to the surface of a cylindrical spindle with curvature C (see Fig. 4). An angle θ is defined as the angle between the longitudinal axes of the shell and the spindle. If the longitudinal axes of the shell and the spindle are aligned then $\theta = 0$, while if they are orthogonal, $\theta = \pi/2$. In the coiled state, it is necessary that $\theta = \pi/2$ for tight compact rolls.

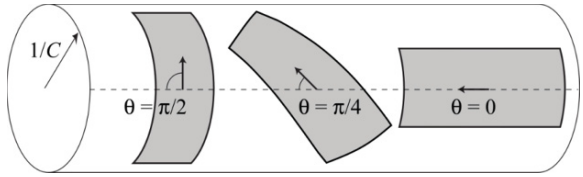


Fig. 4: Laminate shell conforming to the surface of a cylinder of curvature C. By rotating the shell longitudinal direction with respect to the cylinder's as shown, the shell's shape can be altered without changing its Gaussian curvature.

According to Guest [23], if a laminate's 'ABD' matrix has terms $D_{16} = D_{26} = 0$ (no coupling between bending and twisting), there will always be a stable equilibrium in the initial undeformed state ($\hat{C} = CR = 1$ at $\theta = 0$), as well as a second (not necessarily stable) equilibrium at $\hat{C} = CR = \hat{D}_{12} = D_{12}/D_{11}$ at $\theta = \pi/2$. A bending energy contour plot showing the location of this stable point is shown in Fig. 5.

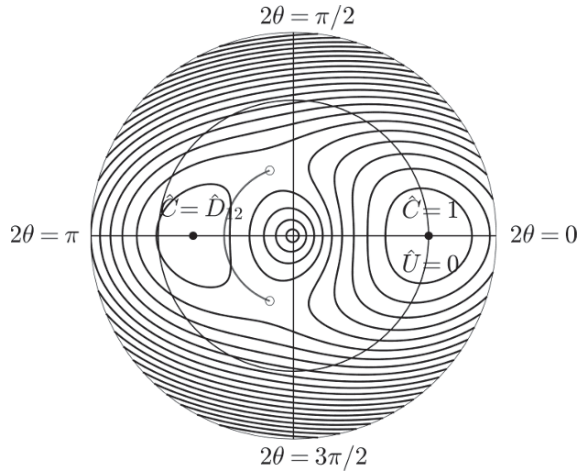


Fig. 5: Polar plot of the non-dimensional strain energy as a function of the coiled curvature and the angle of the shell relative to an underlying cylinder. The two stable locations are marked as dots, while two unstable saddle points are marked as circles. The contours shown are of the non-dimensional bending strain energy $\hat{U} = UR^2/D_{11}$ for a [±45/ 0°/ ±45°] braid lay-up. The contour lines are separated by a constant $\Delta\hat{U} = 0.05$.

V. Bistable Over the Whole Length (BOWL) booms

The main novelty of the booms presented herein resides in the introduction of a variable fiber orientation over the boom length that results in a tailored length-wise anisotropy of the bending stiffness of the laminate. A simple method to achieve this is to use braids as the surface fabric plies, in this case a 1x1 2D braided type. Unlike weave, the fiber angles of a braid can easily be varied by pre-forming the fabric on a mandrel. The length of the finished boom is also not limited by the width of the fabric roll, if a continuous boom without seams is the preferred choice. Also if variable angle braids are used in the laminate the previous strict conditions for bistability in composite shells can be relaxed. For example in the early version of CubeSail's solar sail booms, the laminate although not being strictly bistable produced booms that behaved in a bistable manner with structures that unravelled in a coherent way [7].

The increment in longitudinal bending stiffness (D_{11}) and reduction in transverse bending stiffness (D_{22}) along the length of the boom through the increase in braid angle from root to tip translates into a growth in the natural radius of the coil. The strain energy plots (see Fig. 12) of the laminate at each section of the boom reveal successively larger radii. With correct 'tuning,' that radius will coincide with the radius of the coil for every new overlapping tube section which can be estimated using Archimedes spiral approximation with the equation below.

$$r = \sqrt{\frac{Lt}{\pi} + r_0^2} \tag{1}$$

Consequently, when coiled, the tube is in a continuous stable equilibrium state over its whole length.

CubeSail's ultralight (15g/m) CFRP booms of R= 16mm, α = 160º, t = 0.22 mm, and L = 3.6 m have already faced a flight-qualification process. For this, resin systems that are less prone to creep, have acceptable outgassing properties, and are more dimensionally stable under harsh environments have been utilized. Both, space-qualified epoxies and cyanate ester were used, along with carbon as the reinforcing fibers. The booms are made from ultra-thin carbon fibers (T-300 1K and UTS-50) with an epoxy matrix (MTM44-1) from Advance Composites Group. The laminate [$\pm\delta$/ 0 / $\pm\delta$] consists of two surface braid plies sandwiching a unidirectional 0° ply that have a root braid angle of ±50º and a tip angle of ±35º. These structures have been named the Bistable Over the Whole Length (BOWL) boom, because every section of the coiled shell is now in a new stable equilibrium position that has a different second stable radius conveniently chosen to account for the thickness of each overlap.

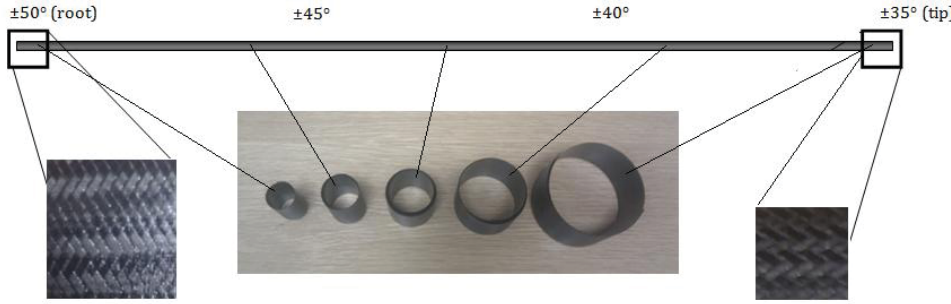


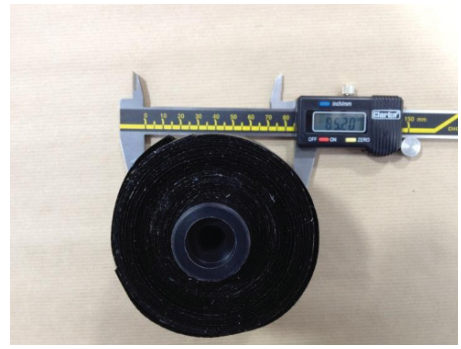
Fig. 6: Variable braid angle BOWL boom for CubeSail.

The unraveling/extension of the BOWL booms can be explained as a buckling instability spirally propagating along the length of the coiled boom through successive low-energy transitions from stables states, and thus guarantying a single plane (2D) coherent deployment. Fig. 7 shows the coiled state of a single BOWL boom with the designed gaps between the overlaps that make up for the thickness of the other three co-wrapped booms. The final diameter of the four co-coiled booms is 85 mm for a highly compact stored configuration.

Four of these booms have been coiled together for over a year at room temperature conditions showing no appreciable creep. No apparent loss of extension energy or significant change in cross-section shape and length-wise curvature due to creep effects has been found. It is considered that the fact that every section of the coiled

BOWL boom is in its lowest potential energy configuration helps reduce matrix creep by minimizing the internal stresses of the coiled state. For long term storage applications such as some solar sail missions or for a deorbiting system, matrix creep in the composite booms can be a serious problem that needs to be tackled not only through adequate material selection but with a proper design of the laminate.

(a) Single 3.6m BOWL boom showing the gaps between overlaps to account for the other three co-wrapped booms.

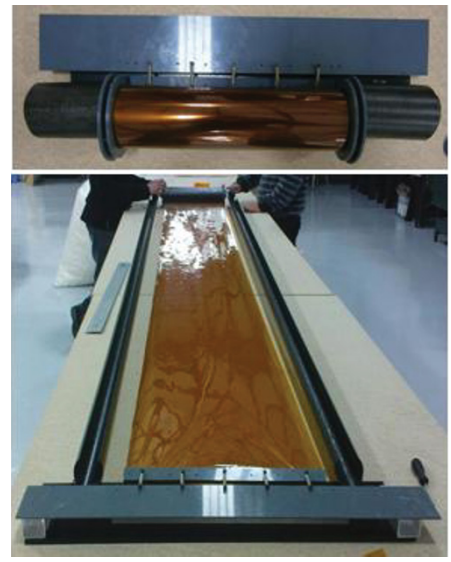


(b) Four co-coiled 3.6m BOWL booms showing a tight roll.

Fig. 7: BOWL CFRP booms coiled

Two larger booms of radius R = 25 mm and subtended angle α = 300º with a length of L = 5 m have been manufactured in a feasibility study of a new rollable solar array concept [26]. The lightweight (80g/m) booms are made of carbon/cyanate ester (T-700 12K/4.3P-CTD) with a laminate of [±45/ 0 /±45] that consist of a thicker 0º ply being sandwiched by two braid laminae at fix ±45 angles. The use of the thicker 0º ply was mainly established as the booms were intended to self-support under 1g condition during the ground testing of the 5m x 1m self-deployable solar array. As the boom design is completely scalable, if a BOWL approach is followed using variable braid angles the solar array could potentially be several tens of meters long. Through the FP7 project "Deploytech" [27] the scalability of the whole highly compact solar array concept is currently being studied. Fig. 8 shows the 5 m solar array prototype built with two bistable CFRP booms. The booms will deploy a drum that stores the dummy Kapton membrane used as the thin film photovoltaic (TFPV) roll, away from the satellite.

(a) 5m Bistable CFRP boom deploying.



(b) Rollable solar array prototype stored (up) and deployed (bottom).

Fig. 8: 5m Bistable CFRP boom for a new highly compact rollable solar array concept

VI. New low-cost manufacturing technique

The proposed tubular booms herein are made from low-density ultra-thin space qualified composite materials. A new cost-effective manufacturing technique for producing bistable CFRP masts with a thermosetting resin (epoxy or cyanate ester) has been produced. An innovative solution for the challenge of utilizing dry fabrics instead of more commonly used prepregs was found. Dry T-300 1K carbon fiber in the form of a biaxial braided sleeve was utilized in the boom laminate.

First, a thin tapered metal plate with the length of the required boom and the appropriate varying width required was produced. The metal tape was then coated with a PTFE glass fabric which was taped only to one of the long edges of the plate. The thin plate was then carefully passed inside the braid sleeve (see Fig. 9(a)). The metal plate was then hung vertically so as to aid in the stretching process of the fabric that settles the desired angles on the dry fabric. A small amount of spray mount from 3M was then used to fix the dry braid onto the PTFE glass fabric as if it was a backing paper (see Fig. 9(b)). Afterwards, impregnation with the appropriate space-qualified resin film was achieved with the use of a belt press at 100°C and 5 bar for 10 minutes. The same belt press was used for the impregnation of the unidirectional material too. The result of the process is a prepreg material that needs to be stored in a freezer below -20°C whenever not being used.

Afterwards, a wet lay-up process of the composite was carried out on a flat table. In order to achieve a fiber volume fraction of 50-60% in the final booms, an investigation into different stacking sequences of the consumable plies was carried out. Several release films (perforated or non-perforated) and different numbers of peel plies to bleed more or less residual resin out of the composite were considered. It was found that a PTFE glass fabric, two peel plies and a perforated release film at the intrados and extrados of the pre-processed composite part was the optimal sequence. The lay-up of the consumable plies and the composites was carried out directly on the cylindrical mandrel, (see Fig. 9(c)) . Given the low thickness of the composite, no substantial shear occurred between the layers of the laminate when laying the shell on the mandrel and therefore no wrinkles appeared on the final parts. The composite curing process was carried out on a temperature-controlled heated mandrel of the required size. A high-temperature polyimide shrinking tape provided the necessary pressure to cure the thermosetting resins.

(a) (b) (c) (d)

Fig. 9: Different steps of the new manufacturing process developed for the BOWL booms.

This cost-effective technique requires less time and power than traditional curing methods such as vacuum bagging, resin transfer moulding or autoclave curing. After the cure and postcure cycles the consolidated booms were released from their forming tubes. In order to minimize crack propagation on the edges due to the brittle nature of carbon fiber, the edges of the booms were trimmed using an oscillator and a bench specially designed for this purpose, (see Fig. 9(d)). Finally the trimmed edges are super-glued for reinforcement.

This low-cost manufacturing method can also be used for very long tubes. Such a process would entail manufacturing 5 m sections one at a time and then processing the joints. The laminate of the joint between two sections will have central interlaced plies and common surface plies.

VII. Experimental analysis of bistable composite booms

A number of short composite samples were produced to experimentally verify the coiled radius predictions. Six different fabric types were used to construct the samples; two different unidirectional fabrics, and a braided fabric stretched such that its fibers form four different angles, δ, to the longitudinal laminate direction. The individual ply details are given in Table 1.

The fabrics used here to construct the bistable shells are particularly thin. Due to the difficulties associated with accurately measuring the properties of very thin individual plies in their cured state, the six ply types were combined in various different lay-ups to produce two to four ply laminates, and the properties of the individual plies estimated via a regression analysis. The laminate details are given in Table 2. The areal density of the braided fabric is difficult to measure, and is a weak function of the braid angle itself. The braid areal density was measured directly to be 91.1 g at a braid angle of ± 45º. The areal densities at other braid angles were determined via the method described below, but were found not to vary greatly from the ± 45° value.

Table 1: Ply details

Ply fabric type	Ply name	Fiber angle to longitudinal (º)	Areal density, $W_{f,j}$ (g/m^2)	Fiber density, ρ_f (kg/m^3)	Typical ply thickness, t_j (mm)	Typical ply volume fraction, $V_{f,j}$	Typical matrix areal density, $W_{m,j}$ (g/m^2)
Braid	P_{35}	± 33.5	91.6	1790	0.100	0.51	49.1
Braid	P_{45}	± 43.7	91.1	1790	0.092	0.55	41.4
Braid	P_{50}	± 48.1	91.4	1790	0.089	0.57	37.9
Braid	P_{55}	± 53.5	91.1	1790	0.101	0.50	50.2
Unidirectional	P_{UD1}	0	32	1790	0.057	0.31	39.3
Unidirectional	P_{UD2}	0	139	1790	0.130	0.60	52.1

The same epoxy resin was used in the production of all samples. It has a density of ρ_m = 1180 kg/m^3, and comes in strips with approximately 60 g/m^2 pre-cure. It was determined that the cured resin contained a void fraction of ~ 15% on average, giving a total void fraction for the complete composite of ~ 6%, depending on the particular lay-up. To account for these voids, the parameter τ_m= ~ 0.85 was introduced to modify the matrix density as $\rho_{m+v} = \tau_m \rho_m$.

The fiber volume fraction for each sample has been calculated using:

$$V_{f,i} = \frac{\rho_{c,i} - \rho_{m+v}}{\rho_f - \rho_{m+v}} \tag{2}$$

Each fabric type/braid angle absorbs a different amount of resin, and produces a ply with a characteristic thickness. To estimate the ply thicknesses, it is assumed that each lay-up thickness is given by:

$$t_i = \sum_{j \in L_i} t_j \tag{3}$$

Table 2: Details of experimental composite samples

Sample number, i	Lay-up, L_i	Density, $\rho_{c,i}$ (kg/m^3)	Fiber volume fraction, $V_{f,i}$	Thickness, t_i (mm)
1	[P_{45}/ P_{UD1}/ P_{45}]	1388	0.49	0.240
2	[P_{45}/ P_{UD1}/ P_{45}]	1399	0.51	0.238
3	[P_{45}/ P_{UD1}/ P_{45}]	1420	0.53	0.235
4	[P_{45}/ P_{UD1}/ P_{45}]	1395	0.50	0.240
5	[P_{45}/ P_{UD1}/ P_{45}]	1401	0.51	0.240
6	[P_{50}/ P_{UD1}/ P_{50}]	1424	0.54	0.235
7	[P_{50}/ P_{UD1}/ P_{50}]	1384	0.49	0.242
8	[P_{50}/ P_{UD1}/ P_{50}]	1382	0.48	0.245
9	[P_{35}/ P_{35}]	1403	0.51	0.200
10	[P_{35}/ P_{35}/ P_{35}]	1406	0.51	0.305
11	[P_{45}/ P_{45}/ P_{45}]	1439	0.56	0.277
12	[P_{45}/ P_{45}/ P_{45}/ P_{45}]	1417	0.53	0.377
13	[P_{50}/ P_{50}]	1434	0.55	0.182
14	[P_{50}/ P_{50}/ P_{50}]	1499	0.63	0.253
15	[P_{55}/ P_{55}]	1439	0.56	0.198
16	[P_{55}/ P_{55}/ P_{55}]	1408	0.52	0.304
17	[P_{35}/ P_{45}/ P_{55}]	1417	0.53	0.293
18	[P_{35}/ P_{UD2}/ P_{35}]	1424	0.54	0.325
19	[P_{50}/ P_{UD2}/ P_{50}]	1437	0.55	0.310
20	[P_{55}/ P_{UD2}/ P_{55}]	1386	0.49	0.335

Performing a least squares regression over the thicknesses in Table 2 gives the approximate ply thicknesses given in Table 1. The thicknesses and densities shown in Table 2 can also be used to determine the approximate fiber volume fractions for each ply. The laminate sample densities in Table 2 can be used to construct the relationship:

$$V_{f,i} t_i = \frac{\rho_{c,i} - \rho_{m+v}}{\rho_f - \rho_{m+v}} t_i = \sum_{j \in L_i} t_j V_{f,j} = \sum_{j \in L_i} \frac{W_{f,j}}{\rho_f} \tag{4}$$

The areal densities for the two unidirectional fabrics are known accurately and were used to augment the estimation of the braid fabric areal densities. The resulting volume fractions and estimated cured matrix areal densities for each ply are shown in Table 1. A Kriging interpolation was used to fit the experimental data for the fiber volume fractions and ply thicknesses. The results are shown in Fig. 10.

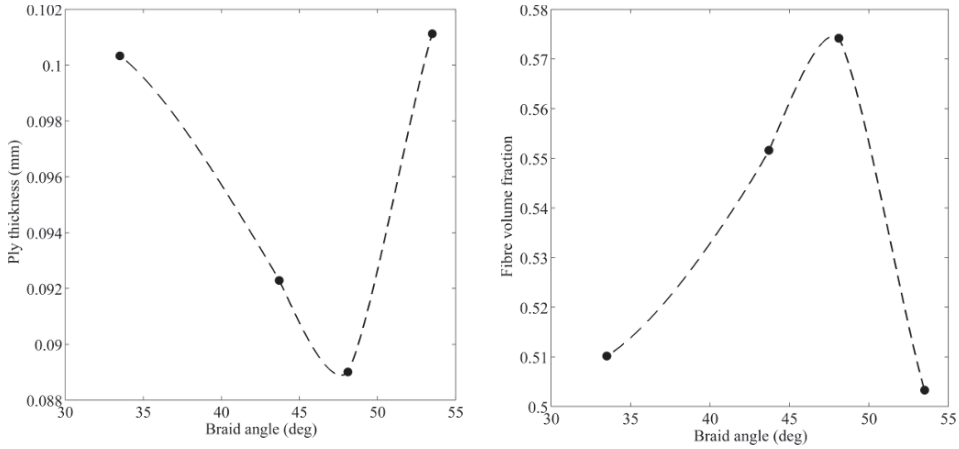


Fig. 10: Interpolation of experimental data. The experimental data points are indicated.

The braid angles examined above were used to predict the coiled radius for a three ply laminate booms with a central unidirectional P_{UD1} ply, and outer braid plies ([±δ°/ 0°/ ±δ°]) using Guest's model, described in Section IV. The fiber volume fraction and thickness of the central ply is assumed to be constant (value given in Table 1). The volume fraction and thickness of the braid plies vary with braid angle as shown in Fig. 10. The radius of curvature (R) of the undeformed laminate shells produced is 16 mm, subtending a uniform angle α = 160° along their length.

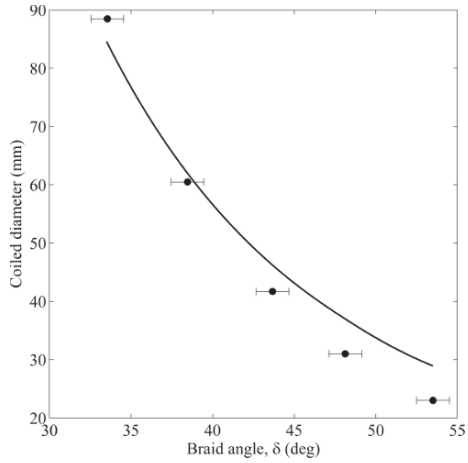
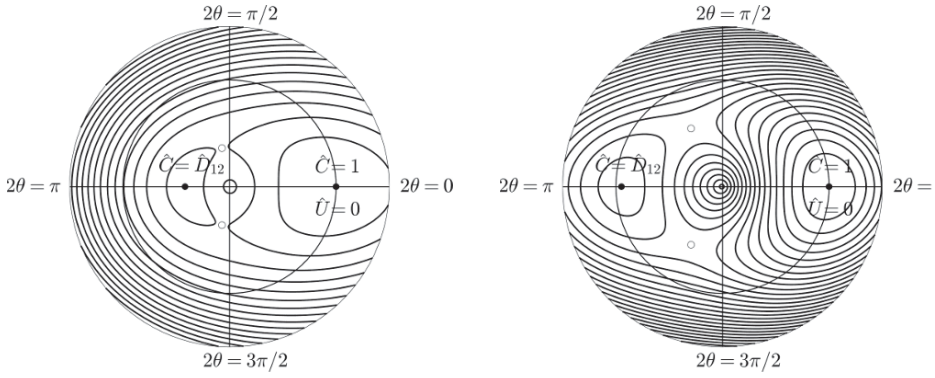


Fig. 11: Prediction of coiled laminate boom diameter with braid angle for a [±δ°/ 0°/ ±δ°] lay-up. Experimental results are shown as dots.

Fig. 11 shows the three-ply composite boom coiling diameter as a function of the braid angle in its outer plies. The inextensional model provides a good estimate of the coiling diameter at lower braid angles, but begins to deviate as the coiled curvature increases. It is likely that the purely elastic deformation assumptions are less valid at these higher curvatures.



(a) Braid angle $\delta = 35°$ (b) Braid angle $\delta = 50°$

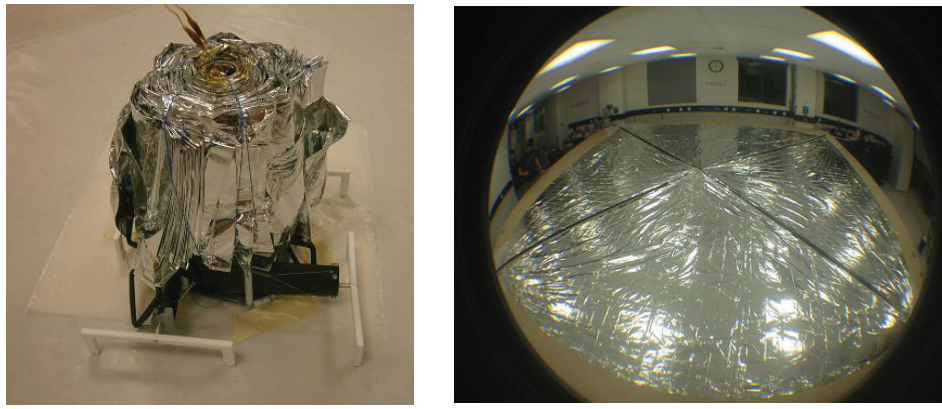
Fig. 12: Polar plots showing contours of the non-dimensional bending strain energy $\hat{U} = UR^2/D_{11}$ for a [±δ°/ 0°/ ±δ°] lay-up. The energy minima representing the stable coiled curvatures at $\hat{C} = \hat{D}_{12}$ are indicated as solid dots in both cases. The unstable equilibria are shown as circles. The contour lines are separated by a constant $\Delta\hat{U} = 0.05$.

Fig. 12 shows a contour plot of bending strain energy for two three-ply laminate booms. There are clear energy minima at $\hat{C} = \hat{D}_{12}$ and $\theta = \pi/2$ in both cases. The energy minimum is deeper for the case of δ =50°, and indeed, it was observed that laminates with higher braid angles did form more stable coils.

VIII. Current applications of BOWL CFRP booms in large gossamer structures

To date, two different BOWL CFRP booms have been manufactured. The first are 3.6m tape springs which are used as the booms for a Gossamer Sail Deorbiter for ESA and a 5m by 5m solar sail called CubeSail, Fig. 13. The tape springs have been used in deployments of a qualification model of the sail inside vacuum and thermal chambers. They have also been structurally characterized in bending, compression and torsion and their first natural frequencies have been found. In addition, a first assessment of the booms creep/relaxation effects after long term storage has been produced by coiling the booms for over a year under room temperature conditions and measuring the cross-section change and shape recovery process.

A second set of 5m long CFRP open-section STEMs of $\alpha = 300°$ have also been constructed (Fig. 8). These have been designed for a relatively small size roll-out solar array prototype. The scaled-up version of this concept would use a BOWL boom version that could potentially also be used in a medium size solar sail.



(a) Stored (b) Deployed

Fig. 13: 5m x 5m solar sail with BOWL CFRP booms

IX. Discussion

The advantages of using braids over other textile-based patterns or combination of angled unidirectional fabrics can be established from four points of view: bistability, ease of manufacturing, long term storage, and scalability.

As seen in Guest [23] the existence of a second stable equilibrium configuration at $\theta = \pi/2$ (compact coil) depends entirely on satisfying the condition shown below:

$$S = 4\widehat{D}_{66} + 2\widehat{D}_{12} - 2\frac{\widehat{D}_{22}}{\widehat{D}_{12}} > 0 \tag{5}$$

where $\widehat{D}_{ij} = D_{ij}/D_{11}$.

Therefore, in order to produce bistable booms increasing the laminate coupling between bending in the x and y directions (D_{12}), and twisting stiffness (D_{66}) is beneficial. Reducing the bending stiffness along either of the two principal directions of curvature of the shell (D_{11} and D_{22}) will also tend to make a shell bistable. All of these conditions are enhanced with the use of surface braid plies at sufficient angles to the longitudinal axis. Besides, braids can normally be found in thinner forms than other textile-based patterns and with a single ply they combine two symmetrical fiber angles when compared to unidirectional fabrics.

From a manufacturing point of view braids are easier to work with when long bistable boom structures are to be constructed. For unidirectional or other textile-based fabrics like weaves the width of the rolls available are relatively small. Therefore, given the large angles δ required for bistability, using these fabrics for long booms would entail having to butt together different sections in order to form a single lamina. However, commercially available biaxial braided sleeves come in hundreds of meters and can then produce continuous plies. The seamless final boom will then be less prone to cracks and will have more continuous properties over the length.

For long term storage applications using variable angle braid plies can benefit creep derived problems. This is due to the fact that the bistable deployable boom would then be at the lowest possible energy state when coiled, minimizing internal stresses on the matrix.

Finally, the scalability of the boom can be significantly improved with the use of variable angle braid plies. As previously seen tailoring the bending stiffness length-wise can further delay the onset of instabilities that hinder compact coils in long bistable booms.

X. Conclusion

The proposed tubular booms herein are made from low-density ultra-thin space qualified composite materials. A novel technique introduced in the composite laminate that consists of using surface braid plies with a fiber direction change along the boom length was found to have a positive effect against the scalability problems encountered on these structures. This way the bending properties of the shell structure are changed at every section, thus yielding a boom that wants to stably coil into a spiral as is imposed in reality. These special structures have been coined BOWL booms as they are now Bistable Over the Whole Length. The BOWL booms have been qualified for flight and are part of a near future solar sail demonstration mission called CubeSail.

An analytical model that produces the strain energy plots of these structures using the constitutive laminae properties has been utilized to explain this behavior and predict the characteristics of the packed configuration. When compared with the short booms manufactured, the model predicts well the secondary stable state for samples with low braid angles ($\delta < 45°$). The coiling diameter estimate for higher braid angles, that would correspond to the root area of a long BOWL boom, is less accurate. It is likely that the inextensional and purely elastic deformation assumptions used in the model becomes less valid at these higher curvatures.

Acknowledgments

Financial support from EADS Astrium as part of the CubeSail mission and the UK Technology Strategy Board through the "Feasibility Studies for Innovations in Space" program are gratefully acknowledged.

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Part IV

Dynamics

Heliogyro Solar Sail Research at NASA

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The recent successful flight of the JAXA IKAROS solar sail has renewed interest within NASA in spinning solar sail concepts for high-performance solar sailing. The heliogyro solar sail, in particular, is being re-examined as a potential game-changing architecture for future solar sailing missions. In this paper, we present an overview of ongoing heliogyro technology development and feasibility assessment activities within NASA. In particular, a small-scale heliogyro solar sail technology demonstration concept will be described. We will also discuss ongoing analytical and experimental heliogyro structural dynamics and controls investigations and provide an outline of future heliogyro development work directed toward enabling a low-cost heliogyro technology demonstration mission ca. 2020.

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I. Introduction

The heliogyro is a helicopter-like, spinning solar sail first introduced by MacNeal in the 1960s.[1] The heliogyro possesses the primary advantage of a spinning disk sail, namely low weight, while avoiding many of the difficulties associated with stowage and deployment of large membrane areas. The sail elements of the heliogyro are long – up to kilometers – high aspect ratio reflective membrane strips. These membrane strips, or "blades", spin about a central spacecraft hub and are stiffened by centrifugal forces only, thus making the design exceedingly light weight. Blades are stowed and deployed from reels; eliminating deployment and packaging problems associated with handling extremely large, and delicate, membrane sheets used with most traditional square-rigged or spinning disk solar sail designs. Attitude control is accomplished by pitching the blades at the root to change their orientation with respect to the sun. This may be performed collectively, to generate torques about the spin axis, for example to spin up or spin down the heliogyro, or cyclically, in a per-revolution fashion, to generate thrust components in the plane of rotation. Combinations of collective and cyclic pitch create overturning moments. Attitude control may thus be accomplished about all six axes using blade pitch actuation alone.

Despite these compelling advantages, very few heliogyro designs exist in the literature. The most in-depth heliogyro study, and one of the most detailed studies of any solar sail concept, was that conducted in 1977 by JPL for an ambitious Comet Halley rendezvous mission.[2, 3] The mission study considered both a heliogyro and a large square-rigged solar sail design (850 m x 850 m), with the heliogyro selected as the preferred approach. The final heliogyro design featured a dozen blades in two counter-rotating tiers, as shown in Fig. 1.

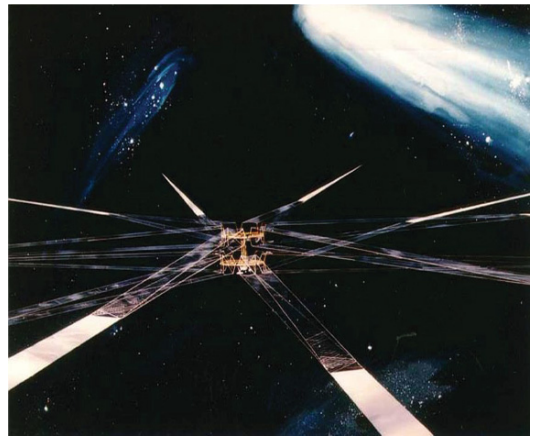

Fig. 1 JPL Comet Halley Rendezvous Mission heliogyro concept (1977).

Each blade was approximately 8 meters wide and 7500 meters long, and packaged would have required the full cargo bay of the Space Shuttle for launch; which, at that time, was still several years away from flight. Ultimately, a solar electric propulsion approach was selected over the heliogyro, due primarily to the perceived overall high risk associated with unproven solar sail technology.[3]

Heliogyro conceptual studies since 1977 have been very rare, with the most notable example a being a small heliogyro study in 1989 performed by an MIT design team as an entry into a solar sail race to Mars proposed by the Columbus Quincentenary Commission.[4] The MIT team proposed a very small, 8-bladed, 200 meter diameter heliogyro. Further heliogyro solar sail development since the MIT study has been almost nonexistent; the exception being some analytical heliogyro control and dynamics investigations by Blomquist; one of the original MIT design team members.[5]

The recent successful flight of the JAXA IKAROS solar sail [6] has renewed interest within NASA in spinning solar sail concepts for high performance solar sailing. The heliogyro solar sail, in particular, is being re-examined as a potential game-changing architecture for future solar sailing missions.[7] Demonstrating system-level feasibility of a large, spinning heliogyro on the ground is impossible; however, recent advances in microsatellite bus technologies for solar sails and other space missions, make an affordable, small-scale heliogyro technology flight demonstration potentially feasible.[8, 9] However, significant fundamental work in heliogyro membrane structural dynamics, guidance and control, and solarelastic stability and flutter [10] remains to be performed before the practicality of a heliogyro flight demonstrator can be determined. In this paper, we present an overview of recent heliogyro technology development and feasibility assessment activities within, and sponsored by, NASA. In particular, a small scale, approximately 440-m diameter, heliogyro technology demonstration concept, serving as a reference for our more detailed dynamics and controls studies, is discussed. We also give an overview of related ongoing analytical and experimental heliogyro investigations. Lastly, we provide some recommendations for future heliogyro development work to enable a low-cost, small heliogyro technology demonstration mission in the near-term, ca. 2020.

II. HELIOS: A Reference Design Concept for Heliogyro Technology Development

HELIOS (High-Performance, Enabling, Low-Cost, Innovative, Operational Solar Sail) is a conceptual, low-cost, CubeSat-based flight demonstration reference mission for our heliogyro technology development activities. HELIOS is based upon the 200-m diameter MIT heliogyro concept proposed in 1989 [4], and is designed to be capable of characteristic accelerations on the order of 0.5 mm/s^2 at 1.0 AU; placing it in the range of solar sail technologies required to enable a variety of science and human exploration support missions.[11] The HELIOS

spacecraft uses modern small-sat and CubeSat technology to minimize bus weight, and is packageable within a standard Evolved Expendable Launch Vehicle (EELV) Secondary Payload Adaptor (ESPA) rideshare payload envelope [12] in order to reduce launch costs. The HELIOS heliogyro vehicle, with blade reels, blades, and videogrammetry camera mast deployed, is shown in Fig. 2. Details of the current HELIOS working concept, including mission systems and mission design, are described below.

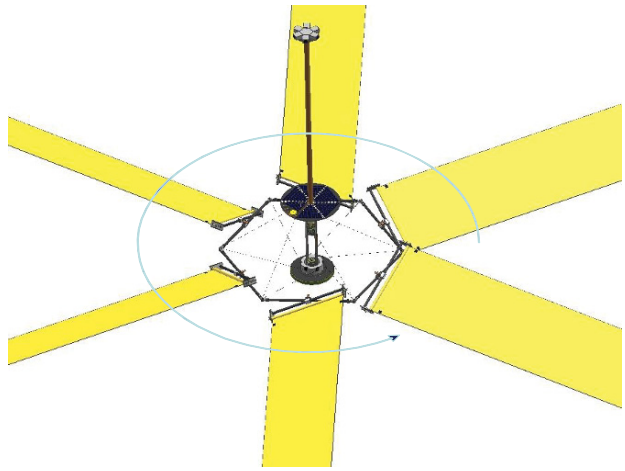

Fig. 2 HELIOS heliogyro technology flight demonstration reference concept. Blades at full deployment are 220 m long. Deployed blade area is 990 m^2. Total mass is 18.3 kg.

A. The HELIOS Heliogyro Concept

The primary objectives for the HELIOS reference mission are: 1) validation of heliogyro deployment technologies; 2) demonstration of controlled heliogyro solar sail flight at mission-enabling characteristic accelerations, a_c, defined here as an a_c on the order of 0.5 mm/s^2 or more; 3) validation of heliogyro structural dynamics behavior, including solarelastic stability, and thrust models; and 4) demonstration of orbit changing capabilities, specifically orbit raising and orbit lowering. These demonstration objectives must further be accomplished in an affordable manner, which constrains us to design a heliogyro technology demonstration spacecraft that can be packaged and flown as a low-cost rideshare payload. This also will, most likely, confine us to operations in a low Earth orbit (LEO), a difficult, but not impossible flight regime for solar sails. To minimize aerodynamics drag effects, we selected a 1200-1500 km altitude, initially dawn-dusk sun-synchronous orbit as our best option for a LEO mission. We also desire minimal eclipsing to avoid potential complications of thermal-

elastic transient dynamic effects. A nominal mission duration of approximately four months, with an intentional de-orbit at mission end-of-life, was also a requirement.

Packaging of the HELIOS heliogyro spacecraft as an ESPA auxiliary payload permits a number of rideshare launch options, depending on the intended orbit of the primary vehicle. The U.S. Department of Defense Space Test Program Mission S26 (STP-26) provides a particularly attractive launch scenario, well suited to HELIOS mission requirements.[13] STP-26 demonstrated a dual-orbit capability using the Hydrazine Auxiliary Propulsion System (HAPS), which achieved a secondary orbit of 1200 km altitude, which is at the edge of our acceptable minimum mission altitude. The stowed HELIOS spacecraft, with ESPA payload envelope indicated, is shown in a potential launch configuration on a Minotaur IV MPA atop HAPS in Fig. 3.

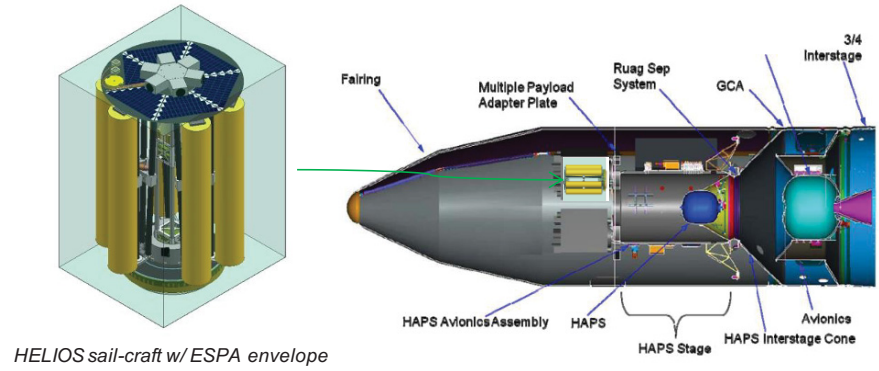

Fig. 3 HELIOS stowed payload, with potential Minotaur IV HAPS launch configuration shown.

After insertion into orbit, HELIOS separates from the launch vehicle with push-off springs, using a Light Band separation system, and will be tumbling. The HELIOS spacecraft must first acquire the sun, establish a sun-pointed attitude, deploy the blade reel hex truss, and spin-up prior to deployment of the sail blade membranes. The HELIOS spacecraft determines its attitude via sun sensors, providing 4 pi steradian coverage, and magnetometers. Attitude control of the core vehicle is accomplished using magnetic torque coils with large loop areas. The blade reel truss must be deployed prior to spin-up to establish the deployed axis of symmetry as the primary axis, with appropriate margin. The HELIOS deployment sequence is illustrated in Fig. 4.

Because the rotational speed will decrease as the sail deploys due to conservation of angular momentum, the spin-rate will initially be relatively high; on the order of 5 RPM. Achieving this initial high spin-rate without the assistance of thrusters could take some time. A mass-efficient approach using magnetic coils, with large loop

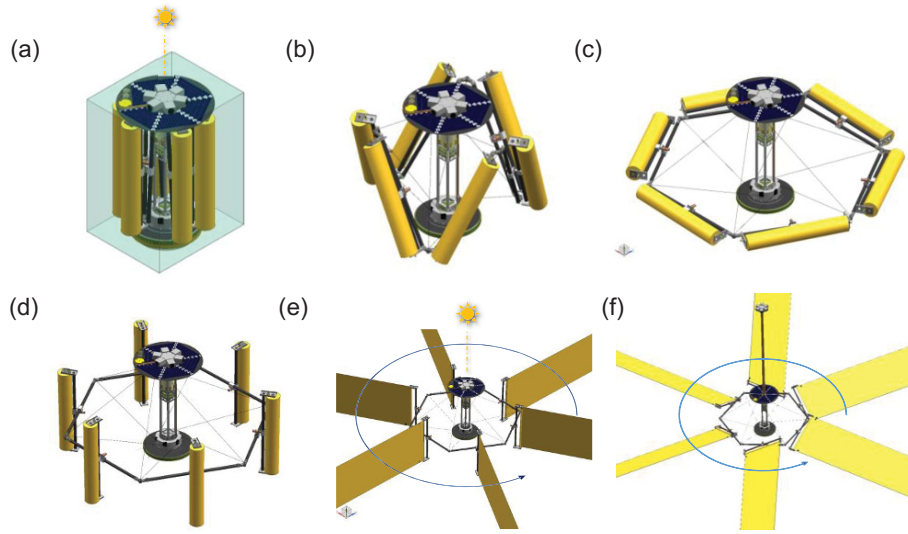

Fig. 4 HELIOS deployment sequence: After detumble, and acquisition and orientation of the +X axis toward the sun (a), blade reels are released via a burn wire (b). Blade reel hex ring deployment is actuated via strain energy stored in the vertex springs of the hex truss, with tensioned cables and dampers connecting the deployed truss to the central spacecraft bus. Once the blade truss is deployed (c), blade reels are pitched to +90 degrees (d) and magnetic coils are used to spin up the core spacecraft to the nominal pre-blade deployment spin rate. A synchronized, controlled partial blade deployment is initiated with the blades edge-on to the sun (e) while the spacecraft rotation rate slows to the nominal spin rate of 1 RPM via conservation of angular momentum. The blade camera mast is deployed and blades are pitched down to + 45 degrees (f). After a brief flight systems and control checkout, blades are then slowly deployed to their full radius using solar radiation pressure induced torque to manage the overall spin rate.

areas distributed on the HELIOS bus, is adopted here instead of heavier torque rods. For spin axis control the magnetic coil wraps around the circumference of the upper/lower deck. The precession coil/rod is mounted to the side of one of the cubes. Bus attitude determination and control functions are combined with sail pitch control functions in a dedicated CubeSat-like assembly located on the spin axis.

HELIOS blade pitch control is accomplished using conventional rotary stepper motor actuators. Blade root pitch angle is measured with analog output potentiometers. Blade bending and torsion deflection will be measured by estimating the local compound sun angle using small photovoltaic sensors located at intervals along the length of each blade. Blades will also be monitored using a dedicated camera system positioned above the plane of rotation by a deployable mast, with imagery stored and periodically downlinked for analysis on the ground. As noted by MacNeal in his original studies, the thin membrane blades of the heliogyro possess near-zero damping in both the out-of-plane (flapping) and torsion (pitch) directions[1, 14] Active damping will be provided using information from the distributed photovoltaic sensors along the blades, and closed-loop actuation of the pitch control motors. As rotation rates are very low; on the order of 1 RPM; blade dynamic time scales

will be very long and easily within the control bandwidth of the blade root pitch motors. Hub vibration will also be sensed using a centrally located MEMS 3-axis accelerometer. Blade deployment operations will take place at very slow rates; on the order of mm/sec. Controlled collective pitching of the blades, to adjust spin rate, will also take place during deployment. Deployment will be paused periodically to evaluate overall spin balance, structural dynamics, and flight control characteristics.

HELIOS science data consists of camera images from the six mast-mounted cameras, vibration data from bus-mounted accelerometers, and time series data from photovoltaic sensors on the sail blades. The data volume is much larger than most typical CubeSats and the higher orbit required by HELIOS also complicates communications. Our communications approach is to combine VHF/UHF for all uplink and engineering downlink and use S-Band for high rate downlink. As the HELIOS bus configuration is not favorable for omni-directional antennas, antennas will be mounted on both the +X and –X deck faces, with a 3 dB beam-splitter connecting the two. The S-Band antenna will be a patch antenna, which is commercially available and compatible with a 1U cube. The VHF/UHF antenna is a pop-up monopole antenna with the deck acting as a ground plane.

HELIOS power requirement is estimated to be 34 W, which exceeds the largest advertised single CubeSat power supply, which is capable of 30 W of power. Our approach is to place a separate 20 W supply in each cube. The solar cells on the +X deck will provide 35 W to 50 W. The –X deck has half the cells of the +X deck in order to support initial post-separation payload operations. A partial power budget for HELIOS systems is shown in Table 1.

The estimated nominal HELIOS spacecraft bus mass for all systems exclusive of the sail deployment mechanisms and sails is 4.5 kg, with a maximum expected mass, including mass contingency, of 5 kg. Unfortunately, the separation system scar mass, which must remain with the HELIOS sailcraft, is relatively heavy, at 785 gm. Mass of the sail deployment and pitch control system is also somewhat heavy, at 8.4 kg, although component masses have not yet been optimized and some mass savings may be achievable in a more detailed design.

A schematic of the sail membrane assembly is shown in Fig. 5. Each sail blade membrane consists of a 220-m x 0.75-m x 2.54-μm Mylar carrier film metallized on both sides with 0.1 μm of aluminum. Aluminized Mylar was deemed to have acceptable durability for the nominal HELIOS mission environment, based on previous materials testing performed under the NASA In-Space Propulsion Technology solar sail program.[15] Total mass allocated for all sail blades, including fittings, battens, tip mass and mass contingency, is 5.0 kg.

Table 1. HELIOS Estimated Power Budget

Element	*Power, W*
+ X Cube (bus systems):	14
OBC	*5*
VHF/UHF transceiver	*2*
S-band transmitter	*2*
Camera electronics	*4*
Power supply	*1*
-X Cube(sail systems):	20
ACS/Blade controller	*5*
Pitch motors and drivers	*6*
Mag torquers	*5*
I/O board	*2*
Power supply	*2*
Total	34

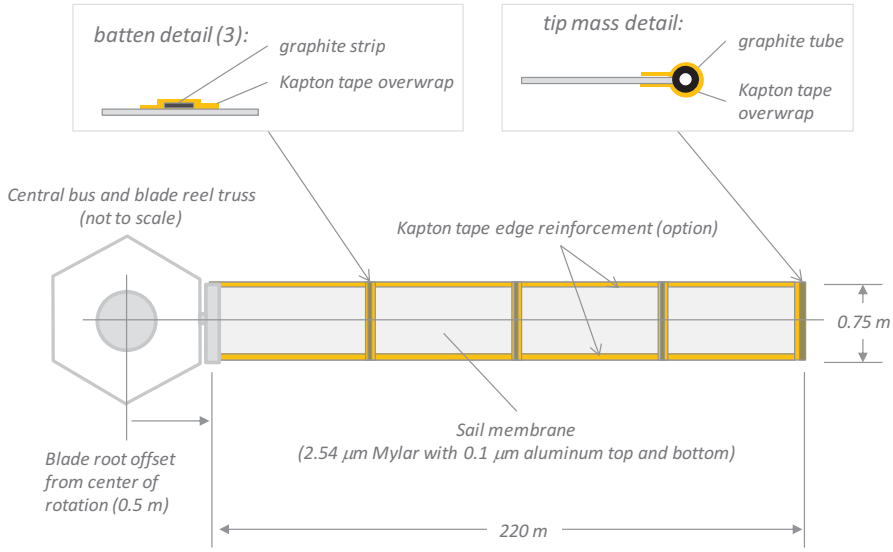

Fig. 5 HELIOS reference blade configuration.

Blade chord was limited by the maximum dimension of the ESPA payload envelope. Blade thickness was based upon the commercial availability of inexpensive aluminized Mylar films in appropriate widths and lengths.

Blade length was constrained by the available blade stowage volume, blade thickness, and packaging efficiency of the blades around the blade reels. Three thin battens composed of graphite strips are secured to the membrane film using Kapton tape at approximately 25% span intervals. The stiff battens provide resistance to chordwise curling of the blade membranes, which can introduce undesirable solarelastic dynamic responses.[14] Blade tip weights consist of graphite tubes attached with a 12.5 μm Kapton tape overwrap. Kapton tape edge reinforcements are applied along the blade leading and trailing edges. Edge reinforcements make the blades more robust to damage and tears in the membranes, and also act to carry the blade centrifugal loads, although, given the very low centrifugal stresses in the blades, blade designs without edge reinforcement may also be practical and are also under consideration. Eliminating edge reinforcements could reduce blade weight and increase packaging efficiency on the blade reels. This could permit longer blades to be stowed within the existing blade and reel stowage volume, and therefore larger sail areas to be deployed without significant modifications to the baseline HELIOS sail deployment mechanisms. At full deployment, the HELIOS sail area is 990 m^2. Assuming a reflective efficiency of 85%, HELIOS will have a conservative characteristic acceleration of 0.46 mm/s^2 at 1.0 AU, which is reasonably close to our nominal performance goal of 0.5 mm/s^2.

Final HELIOS design reference parameters are summarized in Table 2. These values are assumed for HELIOS studies described in this paper, unless otherwise noted.

Table 2 HELIOS Reference Parameters

Property	*Value*
Total sailcraft mass including contingency (kg)	18.3
Specular efficiency of blades (-)	0.85
Sail area (m^2)	990
Characteristic acceleration at 1 AU, a_c (mm/s^2)	0.46
Sail material	Mylar (2.54 μm), 2 x aluminum (0.1 μm)
Bus system mass (kg)	5
Deployment system mass (kg)	8.3
Sail mass (kg)	5
Number of blades	6
Blade chord (m)	0.75
Blade length (m)	220
Rotational period (minutes)	1
Blade root stress (Pa)	4.89×10^5
Blade root allowable stress (Pa)	55×10^6
Blade root tension load (N)	1.0

B. HELIOS Technology Demonstration Mission Reference Design

Some preliminary mission design of potential LEO missions has been accomplished using the baseline HELIOS design and an assumed characteristic acceleration of 0.5 mm/s^2.[16] Operating a solar sail in LEO has many disadvantages, such as gravity gradient and aero disturbance torques, a large gravity well, and a rapidly changing solar direction requires fast attitude maneuver slew rates. The economics and availability of secondary payload rides to LEO partially offsets these disadvantages given that low cost is an overriding concern for most near-term solar sail missions. Additionally, the heliogyro offers unique capabilities that give it an advantage over other solar sail architectures in LEO. The major advantage of the heliogyro over a more conventional square sail in LEO is the ability of the heliogyro to change thrust direction quickly using cyclic pitching of the sail blades. A secondary advantage is that the high characteristic acceleration performance possible with the low mass heliogyro offsets the deep gravity well to some extent. Finally, in this particular mission design study we have chosen to minimize the effects of the gravity gradient and aero torques by limiting the perigee to a minimum of 1500 km altitude.

We considered several potential LEO orbit-raising strategies and present two of particular interest herein. The first starts from a dawn-dusk sun-synchronous orbit that has long been a favorite for solar sail missions. This initially keeps the orbit plane constantly pointed towards the sun, greatly simplifying the pointing of the sail. Dawn-dusk sun-synchronous missions are also popular enough as a typical mission payload that secondary payload opportunities are available. The second mission of interest is in a geostationary transfer orbit (GTO) because secondary payload opportunities are readily available and because, as will be seen below, the heliogyro is well suited to a GTO mission.

We created the mission designs described below with the NASA software tool Copernicus.[17] Although Copernicus does not possess a solar sail model, per se, its solar electric propulsion (SEP) model can be used to emulate a solar sail closely by judicious selection of inputs. For example, it can account for changes in thrust due to shadowing and distance from the sun. The only significant restriction is that the trajectory must be broken into segments with a constant cone angle. The resulting mission is thus not fully optimized, but can be considered a conservative result of what is possible. The sail pointing guidance for the sun-synchronous mission uses a cyclic blade maneuver and was first proposed by MacNeal. [1] Since the heliogyro sail is always pointed at the sun, it has very low slew requirements. A Matlab [18] script was used to design a heliogyro pitch control strategy that would maximize the force along the velocity vector. This analysis determined that a 41° cyclic maneuver yields the largest lift force. In Fig. 6.a, the thick set of axes represent the Earth-fixed coordinates and the thin set of

axes are in the Earth-Sun, 2B Rotating frame. The sun lies along the thin x-axis. The spacecraft orbit is the cyan line and the red lines represent exaggerated thrust vectors. This orbit yields a change in semi-major axis (SMA) of 17 km per day. The orbit normal would stay within 15° of the sun line for about 160 days as the heliogyro gradually climbs out of the sun-synchronous altitude.

The other mission of interest is to GTO. For the purposes of this study, we assumed that the orbit plane was initially edge-on to the sun. Since this orbit is highly eccentric (e = 0.685), we included the argument of perigee (AOP) as an objective variable to allow Copernicus to optimize the orbit rotation within the orbit plane. Figure 6.b shows the final trajectory, and one can see that the optimizer converged to an AOP that lengthened the segment with the highest thrust (0° collective). The new trajectory raises the orbit by 159 km per day. McInnes [19] proves that the rate at which a solar sail can raise SMA ideally increases by the SMA cubed, so we expect this larger orbit to have a larger increase in SMA per day. Table 3 summarizes the results for both mission scenarios described in this section.

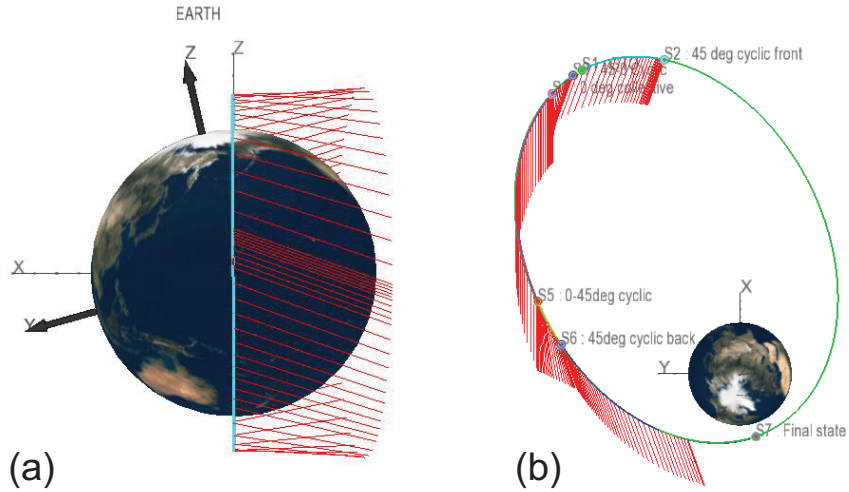

Fig. 6 HELIOS reference missions: a) Dawn-Dusk Sun-synchronous orbit-raising; b) Optimized GTO orbit raising

Table 3. Summary of orbit raising capability

Orbit	*Initial Semi-Major Axis, km*	*Orbit-Raising Capability, km/day*
Sun-synchronous	7878	17
GTO	25021	159

Both of these preliminary mission designs are viable candidates for a HELIOS heliogyro mission. The sun-synchronous orbit employs a much simpler control scheme using only one continuous maneuver. The GTO mission option would demonstrate orbit-raising more quickly and has the longest time-scales, which could also simplify flight validation operations.

III. Heliogyro Technology Development Activities

A. Heliogyro Structural Dynamics Analysis

As full-scale ground testing of the HELIOS heliogyro is impossible, validation of the HELIOS design will need to be performed analytically. High-fidelity, fully-coupled structural dynamic simulations of the HELIOS vehicle will need to be developed to accomplish this. These models will be used to simulate flight structural dynamics experiments and help validate simpler, more numerically efficient heliogyro dynamics analysis tools. For our high-fidelity heliogyro simulations, we are using the nonlinear modeling capabilities of the Abaqus commercial finite element code.[20] Examples of the coupled normal modes computed using Abaqus for the rotating HELIOS reference blade without edge stiffening and battens are shown in Fig. 7. Frequencies for the predominately out-of-plane, in-plane, and torsion modes compare well with the frequencies given by MacNeal's uncoupled analytical solution.[1] Note that the Abaqus model is capable of capturing chordwise "curling" modes on the blade (modes 3, 8, 10, and 12). These deflection modes would make control of a pure membrane heliogyro blade difficult, due to adverse couplings with solar radiation pressure, although the addition of battens should mitigate these issues.[14]

Fully-coupled HELIOS blade structural dynamics finite element based simulations including solar radiation pressure have also been performed using Abaqus. An example case is shown in Fig. 8. Here the HELIOS uniform reference blade is freely rotating about the hub at an initial speed of 1 RPM. Radiation pressure is subsequently applied, followed by a blade collective pitch input of -20 degrees. For this case the sun angle is at -180 degrees relative to the positive spin axis, i.e., normal to and below the spin plane. Solar radiation pressure has been adjusted using the nominal 0.85 specular efficiency of the HELIOS sail. The blade root pitch input profile is shown in Fig. 8.a with the resulting torsional deflection at the tip shown in Fig. 8.b. The average blade tip twist between 25 and 75 revolutions is approximately 6.3 degrees which is comparable to the theoretical tip twist of 7.1 degrees predicted by MacNeal.[14] A small difference in tip twist is expected because of the relatively coarse chord-wise finite element discretization and the fact that solar radiation pressure was not included in the theoretical solution. The center hub normalized angular velocity is shown in Fig. 8.c. The

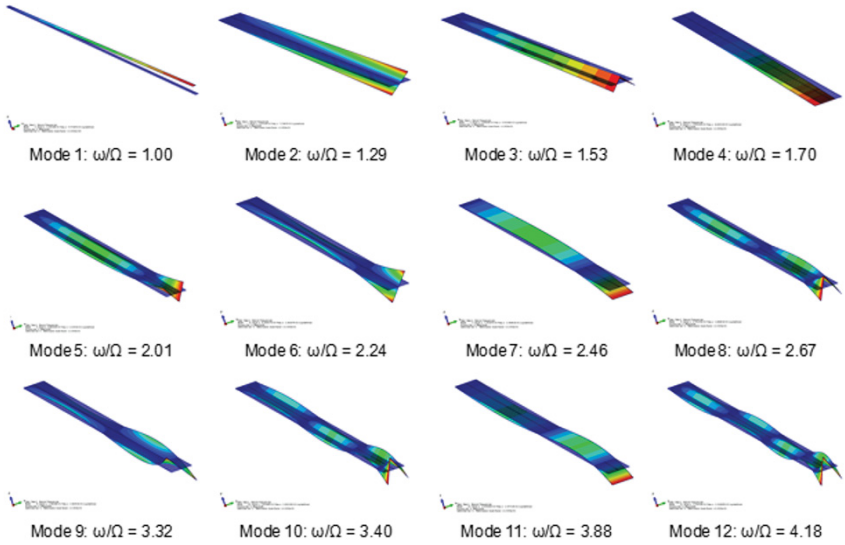

Fig. 7 Abaqus coupled finite element modal analysis of the uniform HELIOS reference blade.

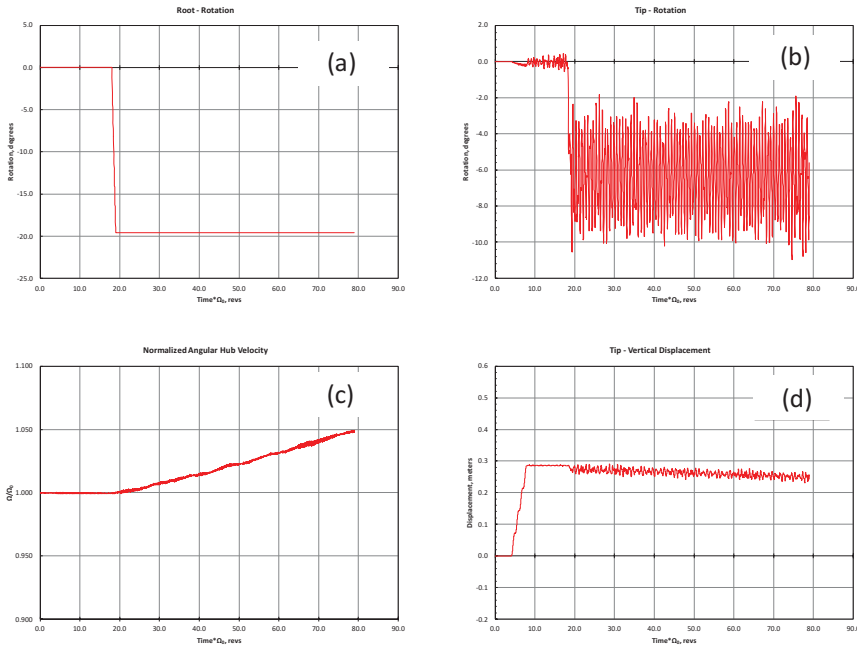

Fig. 8 Fully coupled Abaqus time domain simulation of HELIOS reference blade with solar radiation pressure at 1.0 AU. -20 degree collective pitch input; -180 degree sun angle, 1 RPM case shown. (a) root pitch input profile; (b) blade tip twist response; (c) normalized hub rotation rate; (d) blade tip vertical displacement response.

angular velocity increases as the blade is pitched due to the resulting solar radiation pressure induced torque about the spin axis. This is expected since the blade is freely spinning as it would be in flight. The average blade tip vertical displacement is shown in Fig. 8.d. The average blade tip vertical displacement before application of the root pitch input is approximately 0.286 meters which compares with the theoretical displacement of 0.281 meters.[1] Blade vertical deflection decreases over time as the blade and hub rotation rate, and resulting centrifugal forces, increase. Future studies will include nonuniform blade features and examine sensitivities of the finite element mesh discretization in more detail.

B. Heliogyro Flutter and Solarelastic Stability Studies

Heliogyro dynamics and solarelastic stability studies are possible using the fully-coupled, finite element analysis methods described above, although these simulations are computationally expensive and more efficient heliogyro analytical tools appropriate for rapid design and trade studies are needed. Our approach [21], previously explored by Natori [22], adapts rotating beam equations of motion developed for conventional helicopter rotor blade analysis [23] for use with the heliogyro. These equations can be numerically solved using MATLAB-based codes, and although not applicable to all heliogyro flight conditions, can be used to evaluate heliogyro structural dynamic and stability behavior for many restricted, yet still relevant, cases. One such case is for a heliogyro operating at zero blade pitch angle with the spin axis pointed toward the sun. For this case, the equations of motion can be transformed into a second-order set of ordinary differential equations with constant coefficients valid for small dynamic deflections. The eigenvalues of this system can be evaluated as a function of fixed rotational speed and incident solar radiation pressure to determine operational stability boundaries for a given heliogyro design. Example stability behavior trends for the HELIOS uniform reference blade, without battens and edge stiffening, are shown in Fig. 9. Here the imaginary parts of the blade eigenvalues; corresponding to the normalized, i.e., per revolution, blade modal frequencies; are tracked as a function of effective solar radiation pressure. A blade flutter instability, signified by coalescence of the fundamental in-plane and torsional vibration modes, is indicated near an effective solar radiation pressure of 41.0 x 10^{-6} Pa. This corresponds to a heliocentric distance of about 0.5 AU. Blade dynamics in the vicinity of 1.0 AU (~9.1 x 10^{-6} Pa) are stable; at least for the nominal HELIOS rotational speed of 1 RPM. Further instabilities at higher solar radiation pressures are also indicated, most notably, a divergence instability near 5.3 x 10^{-6} Pa and a higher frequency flutter condition at 5.7 x 10^{-6} Pa, although such post-critical instabilities are not physically significant.

Stability bounds for a given heliogyro blade design may also be mapped as function of rotor speed and effective solar radiation pressure. The stability boundary for the uniform HELIOS blade is shown in Fig. 10.

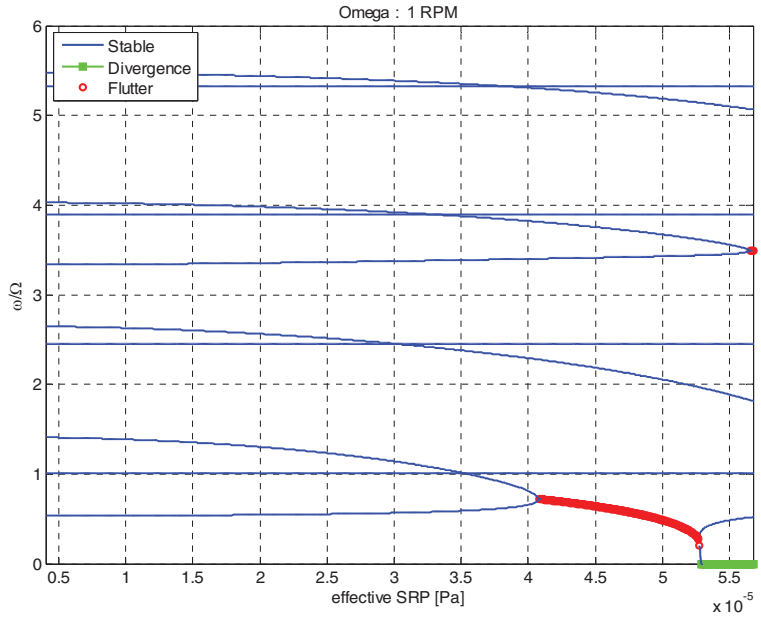

Fig. 9 Eigenvalue stability analysis of the HELIOS uniform reference blade at a nominal rotational speed of 1 RPM and zero sun angle.

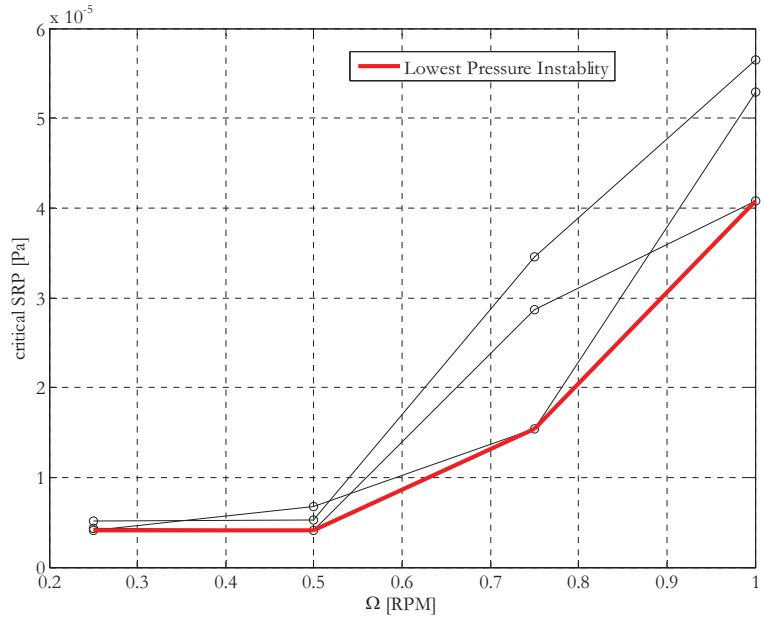

Fig. 10 Solarelastic instability boundary of HELIOS uniform reference blade.

Here, all unstable eigenvalues over the rotor speed range indicated have been tracked with the critical radiation pressure required to induce instability plotted against rotor speed. The lower bound in radiation pressure is indicated by the thick red line, with stable regions of rotational speed and radiation pressure lying below the line. Examination of this stability boundary plot indicates that at 1.0 AU solar radiation pressures (9.1×10^{-6} Pa) the HELIOS blade will encounter a solarelastic instability when rotational speed drops below approximately 0.6 RPM. Experimental verification of solarelastic stability characteristics would actually be an important goal of a heliogyro flight technology demonstration mission.[24] In practice, this could be accomplished by gradually reducing heliogyro rotation speed through the application of blade collective pitch; pausing periodically to experimentally evaluate the sub-critical changes in blade modal frequencies. As the dynamics of the heliogyro are relatively slow in real-time, stability boundaries may be approached slowly, and backed away from relatively quickly by re-application of blade collective pitch to increase the vehicle spin rate.

C. Ground Validation Experiments

Membrane structural damping will have a significant effect on all the dynamics of heliogyro blades, and reliable membrane damping estimates will be critical for designing the heliogyro blade control and damping augmentation systems. Although membrane damping is expected to be very small, even a small energy dissipation capability could be sufficiently stabilizing for many heliogyro blade vibration modes, particularly higher-order elastic modes which may cause undesirable narrow band instabilities at certain rotational speeds. To better understand and characterize damping mechanisms for very lightly loaded membranes, a series of modal tests on small-scale membrane blade test specimens were performed under 1 g in a vacuum chamber.[25] The experimental arrangement for our small-scale heliogyro blade dynamics and damping characterization studies is shown in Fig. 11. Out-of-plane and torsional dynamics of the vertically suspended model blade specimens were excited in the vacuum chamber by root piezoelectric actuators at the blade root. Dynamic responses were measured by laser Doppler vibrometers outside the chamber. Our small-scale test results are expected to be reasonably applicable to the full-scale HELIOS blade given that 1-g spanwise stress distributions in the 2.13-m model blades, although linearly varying, are on the same order as the very lightly loaded full-scale 220-m spinning HELIOS blade. Frequency correlation between analytical models of the vertical blades and experiment were generally very good, although residual atmospheric effects in the rough vacuum of the chamber prevented an accurate characterization of damping. Higher-vacuum testing is in work to further eliminate interference of chamber pressure on the experimentally observed damping. Dynamics and control experimental studies on larger-scale, “rope-ladder” membrane blade analogs are also being studied.[26] Use of the rope

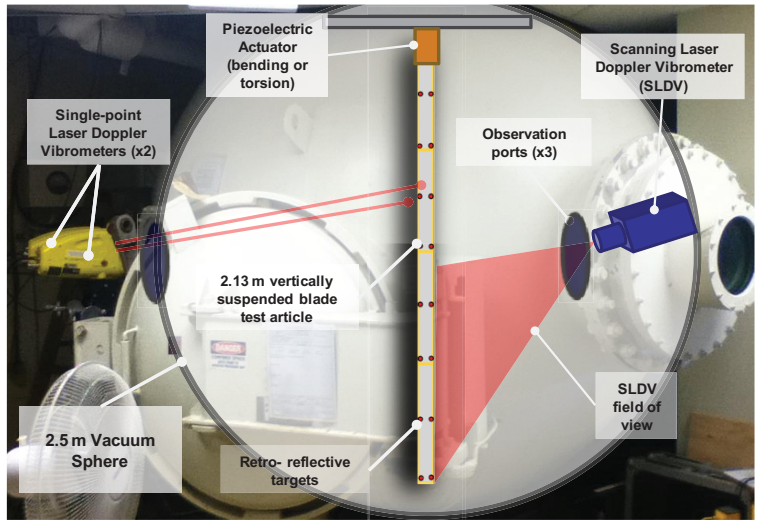

Fig. 11 1-g vertical membrane dynamics testing in 2.5 m vacuum sphere.

ladder models, which closely emulate the dynamics of membranes, permits us to conduct many structural dynamics and control experiments on blade-like structures in air, and at scales larger than those possible in our available vacuum chambers.

Although exact dynamic similitude with full-scale spinning heliogyros, including solar radiation pressure effects, is not possible on the ground due to the presence of gravity, analytical models of the sub-scale spinning membrane system at one-g can be validated and used to build confidence in analytical simulations of the full-scale vehicle. Such experiments were proposed during the JPL heliogyro study of the 1970s, although no experiments were performed,[27] and earlier heliogyro spin dynamics validation experiments attempted in air by MacNeal were severely complicated by atmospheric effects.[14] As an inexpensive pathfinder experiment, we have recently begun testing small-scale spinning membrane blades in our 2.5-m diameter vacuum sphere using a system constructed around low-cost, commercially available radio-controlled helicopter components. Fig. 12 shows the spin test apparatus in the vacuum chamber during initial testing.

The entire assembly is mounted upside-down to prohibit the blades from wrapping around the drive system during spin up. A hub-mounted videogrammetry camera system is used to recover displacement information from the spinning membrane blades. Post-test processing of the rotating system camera imagery using commercial videogrammetry software is now underway. Further calibration of the camera system should increase contrast and improve our ability to extract deflection time histories for the retroreflective targets located

on the membrane blades. Once preliminary testing in the 2.5 m vacuum sphere is completed, the entire spin test system will be moved to our 2.5 m x 5 m vacuum chamber for testing of blades up to one meter in length will be conducted under high vacuum conditions. This should eliminate unsteady aerodynamic effects more fully, allowing us to better evaluate blade transient dynamics and damping behavior.

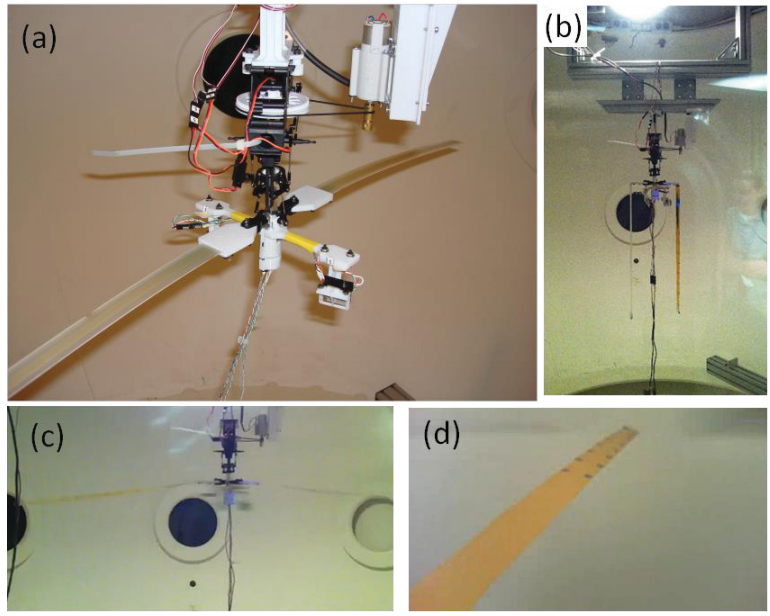

Fig. 12. Heliogyro membrane spin dynamics test apparatus in vacuum chamber: a) hub and camera assembly with rigid blades for checkout; b) spin test apparatus with 0.4 m Kapton membrane blades prior to spin up; c) fixed-frame camera view of spin test apparatus in vacuum at 120 RPM; d) rotating frame camera view of spinning membrane blade with videogrammetry targets visible.

IV. Conclusion

The HELIOS heliogyro concept described in this paper is being used as a near-term design reference mission for our NASA-sponsored heliogyro technology advancement activities. Several key examples of such activities were also presented. Our initial efforts have been encouraging and lead us to conclude that a credible, near-term heliogyro technology demonstration mission is possible at an affordable cost. Performance characteristics in the range needed for many science missions enabled by solar sail propulsion technology should also be achievable. This high performance is largely made possible by advances in small satellite and CubeSat technologies since the 1970s, which permit a small scale heliogyro solar sail to be flown with very lightweight bus system. Our renewed analytical investigations into the coupled structural dynamics of heliogyro membrane blades have also

revealed no intractable stability and control issues, although to what degree major damping augmentation systems will be needed to ensure solarelastic stability and damping of blade transient responses remains a subject for future work.

Also the subject of future work is the development of higher-fidelity blade deployment mechanisms. Development of these systems to high technology readiness levels should be possible with standard flight systems engineering practices and well designed ground tests. High altitude, balloon tests could also be used to experimentally validate deployment reel mechanisms at HELIOS-like full-scales, although without rotational dynamics. Ultimately, an actual spaceflight validation mission will be needed to prove feasibility of the heliogyro solar sail concept and retire risk. Fortunately, this appears possible at an affordable, non-flagship mission cost. Given a near term effort to advance the technology readiness of critical systems, most notably blade dynamics simulation capabilities, deployment mechanisms, solarelastic flutter dynamics and control systems including damping augmentation, and relevant ground test demonstrations, a HELIOS or HELIOS-like, low-cost heliogyro flight demonstration could be ready for launch in as few as five years.

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Solarelastic Stability of the Heliogyro

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This article presents a methodology for analyzing the solarelastic stability of a solar sail spacecraft blade. This paper couples linear and non-linear rotating structural models with an optical solar radiation pressure model for a completely reflective surface. The resulting time varying ordinary differential equations are solved in a quasi-static sense, where an instantaneous stability boundary is determined. The quasi-static analysis with the linear model predicts a divergence type instability and slow and non-uniform modal convergence using parameters for a representative heliogyro spacecraft blade. The non-linear model predicts a flutter instability at a lower radiation pressure and has improved modal convergence characteristics. The paper uses the non-linear model to evaluate the stability of a NASA heliogyro concept design and explore the dependence of the stability boundary on the spacecraft rotation rate for the case of the sun directly overhead. Increasing the spin rate of the spacecraft improves the solarelastic stability, but must be traded off with decreased spacecraft maneuverability.

I. Introduction

ON August 22nd 2011, NASA announced three technology demonstration missions. One of the missions was "Beyond the Plum Brook Chamber; An In-Space Demonstration of a Mission-Capable Solar Sail"[7]. The mission represents NASA's reinvestment in propellant-less space propulsion. Generally, solar sails are large membranes that are reflective on one side and emissive on the other. Photons impact these large membranes creating a solar pressure jump across the membrane and causing a thrust in the normal direction to the sail. Although a small force, the solar force is applied continuously, which when combined with the lack of energy dissipation in space, allows a spacecraft to accelerate indefinitely. Solar sail spacecraft are able to carry out certain missions that require sustained thrust, a feat that is unachievable by conventional spacecraft for long-duration missions. Proposed

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applications include orbital debris capture and removal, de-orbit of spent satellites, station keeping in unstable locations in space, and deep space propulsion [5].

One of the solar sail spacecraft architectures NASA is currently exploring is the heliogyro architecture originally proposed by Hedgepeth and MacNeal in the 1970's [4]. In this architecture, the solar sail spacecraft contains a set of helicopter-like blades as shown in Fig. 1. The whole system rotates at a fixed rate, centrifugally stiffening the blades. This configuration has the advantage of not requiring rigid support structures. This reduces the system mass and improves the spacecraft's performance. One of the potential problems faced by heliogyros is that the individual blades, which are thin membranes, remain only lightly tensioned and therefore may be susceptible to dynamic instabilities. Dowell [2] recently identified that instabilities may arise due to the coupling of the structural dynamics and the solar radiation pressure using an analogy to supersonic panel flutter. While references to solarelastic instabilities frequently appear in the literature [4], detailed modeling of the phenomenon is scarce.

Natori et.al. [6] provides an example of a solarelastic analysis in the literature. In their article, the authors develop a non-linear structural model based on the well-known Hodges-Dowell [3] equations for a rotating elastic beam. The model includes, in-plane bending, out-of-plane bending and torsion degrees of freedom. For the special case of the sun directly overhead, the authors present an eigenvalue analysis of the linear perturbation around a non-linear static equilibrium.

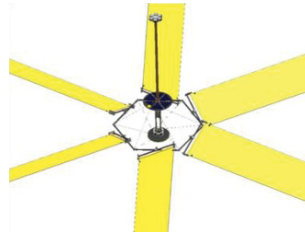


Fig. 1 HELIOS Concept

The present paper includes the development of a solarelastic model that can be used as a preliminary design tool during the development of a heliogyro spacecraft. Specifically, this research supports an effort lead by NASA's Langley Research Center (LaRC) to design, build and fly a cubesat sized heliogyro spacecraft. NASA calls the spacecraft the "High-Performance, Enabling, Low-Cost, Innovative, Operational Heliogyro Solar Sail" or HELIOS. The nominal design calls for a set of six blades, each with a span of 220 m, chord of 0.75 m and a nominal thickness of 2.54 microns that spin around the hub at a rate of 1/3 RPM. The analysis presented in this

paper is based on this configuration, with the goal being to develop the simplest model that can accurately predict the most critical dynamic instabilities.

The initial solarelastic model uses the linear model of the beam with bending and torsion degrees of freedom. We couple this structural model to a solar radiation force model derived from a classical optical model for solar radiation pressure across a reflective surface. The paper then presents the results of a structural analysis using the current model and compares these results with the published results for a rotating membrane. Next, the paper outlines the solarelastic stability boundary for the current HELIOS configuration. The analysis specifically focuses on the convergence behavior of the linear model. The paper then compares the initial linear model predictions with the results using the non-linear equations from Natori et.al. [6] for the case of the sun directly overhead. We show that the inclusion of the non-linear terms significantly changes the predicted solarelastic behavior of the system. Finally, the paper presents results that use the non-linear model to re-evaluate the stability boundary of the HELIOS spacecraft and conducts a parameter study using the HELIOS spin rate as a design variable.

II. Model Description

The initial model couples a classical linear structural model for a rotating beam with a optical model for the solar radiation pressure. Fig. 2 shows the configuration and coordinate system convention that used in this paper. As shown in the figure, the $x-z$ plane aligns with the incident radiation. γ is defined as the angle between the incident radiation and the z -axis. ψ is the angle between the x-axis and the elastic axis of the blade. Because the blade is rotating at a fixed rate Ω the angle ψ is linearly related to time.

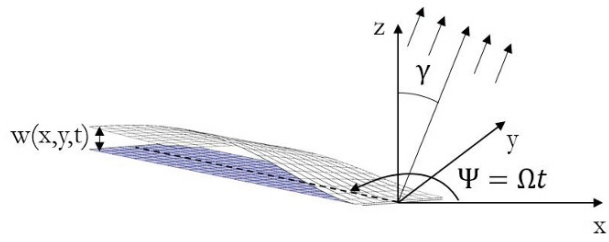


Fig. 2 Schematic of a Sample Heliogyro Blade

The derivation of the solar radiation pressure incident on a structure that has out of plane displacement that is defined by $w(x,y,t)$ can be found in the literature [2, 4]. Equation 1 gives the solar radiation pressure in terms of the angles defined for the current analysis. The radiation pressure acts normal to the surface of the structure.

$$P_r = P_o\left[\cos^2\gamma + \sin 2\gamma\left(\frac{\partial w}{\partial x}\sin\psi - \frac{\partial w}{\partial y}\cos\psi\right)\right] \tag{1}$$

The preliminary structural model is based on the MacNeal [4] model with the inclusion of terms to account for the bending and torsion rigidity of the structure. In general, the equations can be derived from the energies of the rotating structure. We transform the governing partial differential equations into ordinary differential equations using a separation of variables technique with spatial mode shapes that satisfy the appropriate boundary conditions.

$$\bar{w}(\tilde{t},\xi) = \sum_i W_i(\tilde{t}) w_i(\xi), \quad \bar{v}(\tilde{t},\xi) = \sum_i V_i(\tilde{t}) v_i(\xi), \quad \phi(\tilde{t},\xi) = \sum_i \Phi_i(\tilde{t}) g_i(\xi) \tag{2}$$

In the previous equation, $\tilde{t}$ is the non-dimensional time (t/Ω), ξ is the non-dimensional span (x/L), $w_i(\xi)$ is the i'th out-of-plane bending mode shape that satisfies the typical clamped-free beam in bending equation, $g_i(\xi)$ is the i'th torsion mode shape that satisfies the typical fixed-free beam torsion equation and $v_i(\xi)$ is is the i'th in-plane bending mode shape that satisfies the typical clamped-free beam bending equation. Equation 3 gives typical governing equation for each degree of freedom:

$$\begin{aligned}
\int_0^1 M_\phi g_i d\xi &= \sum_j \left[A_{ij}\Phi_j + B_{ij}\ddot{\Phi}_j\right] \\
\int_0^1 L_w w_i d\xi &= \sum_j \left[C_{ij}W_j + D_{ij}\ddot{W}_j\right] \\
\int_0^1 L_v v_i d\xi &= \sum_j \left[E_{ij}V_j + F_{ij}\ddot{V}_j\right]
\end{aligned} \tag{3}$$

with

$$\begin{aligned}
A_{ij} &= \int_0^1 \left[\overline{GJ} g_i' g_j' + \frac{1}{2}\bar{m}(1-\xi^2)\bar{k}_a^2 g_i' g_j' + \bar{m}\left(\bar{k}_{m2}^2 - \bar{k}_{m1}^2\right) g_i g_j\right] d\xi \\
B_{ij} &= \int_0^1 \bar{m}\bar{k}_m^2 g_i g_j \, d\xi \\
C_{ij} &= \int_0^1 \left[\overline{EI}_1 w_i'' w_j'' + \frac{1}{2}\bar{m}(1-\xi^2) w_i' w_j'\right] d\xi \\
D_{ij} &= \int_0^1 \bar{m} w_i w_j \, d\xi \\
E_{ij} &= \int_0^1 \left[\overline{EI}_2 v_i'' v_j'' + \frac{1}{2}\bar{m}(1-\xi^2) v_i' v_j' - \bar{m} v_i v_j\right] d\xi \\
F_{ij} &= \int_0^1 \bar{m} v_i v_j \, d\xi
\end{aligned} \tag{4}$$

The B, D, and F terms come from the inertia of the structure in each of the degrees of freedom. The A term accounts for the stiffness of the torsion modes. This term includes the normal $\overline{GJ}$ beam torsional stiffness term, the $\frac{1}{2}\bar{m}(1-\xi^2)\bar{k}_a^2 g_i' g_j'$ term captures the centripetal acceleration induced tension, and the $\bar{m}\left(\bar{k}_{m2}^2 - \bar{k}_{m1}^2\right) g_i g_j$ term comes from the so-called tennis-racket effect that acts to flatten the blade towards the axis of rotation. Similarly, the out-of-plane and in-plane bending terms C and E respectively contain beam bending and centripetal

acceleration terms. The in-plane bending mode also has the $\bar{m}v_i v_j$ tennis-racket effect term that is a softening influence due to the negative sign attached to it.

The non-dimensional parameters groups in Equation 4 are listed in Equation 5.

$$\begin{aligned} \bar{T} &= \frac{T}{m_r\Omega^2L^2}, \qquad \overline{GJ} = \frac{GJ}{m_r\Omega^2L^4}, \qquad \overline{EI}_1 = \frac{EI_1}{m_r\Omega^2L^4}, \qquad \overline{EI}_2 = \frac{EI_2}{m_r\Omega^2L^4} \\ \bar{K}_\phi &= \frac{K_\phi}{m_r\Omega^2L^3}, \qquad \bar{m} = \frac{m}{m_r} \end{aligned} \tag{5}$$

GJ is the torsional rigidity, EI_1 and EI_2 are the bending stiffness terms in the out-of-plane and in-plane directions respectively, m_r is the reference mass per unit length, Ω is the rotational speed of the structure, L is the structure span and $\bar{k}_{m1}^2$ and $\bar{k}_{m2}^2$ are the radii of gyration around the principal axes of the blade.

Using the solar radiation pressure from Equation 1 we can determine the forces on the structure as well.

$$\begin{aligned} M_\phi &= 0 \\ L_w &= \bar{p}\left[\cos^2\gamma + \sum_j \left(\sin 2\gamma \sin\psi\, \phi_j - \sin 2\gamma \cos\psi\, \bar{w}_j{}'\right)\right] \\ L_v &= -\bar{p}\sum_j \cos^2\gamma\, \phi_j \end{aligned} \tag{6}$$

$\bar{p}$ is the non-dimensional pressure that is equal $\bar{p} = (\mathrm{p_o c})/(\mathrm{m_r}\Omega^2\mathrm{L})$. Substituting Equation 6 into Equation 3 yields the following typical governing linear solarelastic equation for each degree of freedom.

$$\begin{aligned} 0 &= \sum_j \left[A_{ij}\Phi_j + B_{ij}\ddot{\Phi}_j\right] \\ \int_0^1 \bar{p}\cos^2\gamma\, d\xi &= \sum_j \left[C_{ij}W_j + D_{ij}\ddot{W}_j + C2_{ij}W_j + A2_{ij}\Phi_j\right] \\ 0 &= \sum_j \left[E_{ij}V_j + F_{ij}\ddot{V}_j + A3_{ij}\Phi_j\right] \end{aligned} \tag{7}$$

with

$$\begin{aligned} A2_{ij} &= -\bar{p}\sin 2\gamma \sin\psi \int_0^1 w_i g_j\, d\xi \\ A3_{ij} &= \bar{p}\sin 2\gamma \cos\psi \int_0^1 v_i g_j\, d\xi \\ C2_{ij} &= \bar{p}\cos^2\gamma \int_0^1 w_i w_j'\, d\xi \end{aligned} \tag{8}$$

The model incorporates the solar radiation pressure as a static term in the out-of-plane bending equation and additional stiffness terms in the out-of-plane and in-plane bending equations. For this model, the solar radiation pressure couples the in-plane and out-of-plane equations to the torsional degree of freedom, but the solar radiation pressure does not directly affect the torsion equation. Note, the solar radiation terms $A2_{ij}$ and $A3_{ij}$ are time varying when gamma is not equal to $n\pi/2$ because the ψ angle varies with time. To account for this we can either use a

time marching algorithm or do a snapshot analysis for a fixed ψ, the method of choice for this paper. In general we can construct a governing matrix equation and use MATLAB's eigenvalue solver to determine the solution to Equation 7.

Before moving onto a discussion of the results, this section will also outline the modifications to the linear model to include non-linear structural effects. As discussed in the introduction, we implemented the non-linear model outlined by Natori et.al. [6] for the specific case of no blade twist and no pre-coning. After following the steps outlined in the Natori et.al. [6] paper we arrive at the following characteristic perturbation equations around the non-linear static solution.

$$\begin{aligned} 0 &= \sum_j \left[A_{ij}\Phi_j + B_{ij}\ddot{\Phi}_j + \sum_k N1_{ijk} V_j W_{ok} \right] \\ \int_0^1 \bar{p}\cos^2\gamma\, w_i d\xi &= \sum_j \left[C_{ij}W_j + D_{ij}\ddot{W}_j + C2_{ij}W_j + A2_{ij}\Phi_j + \sum_k N2_{ijk}\dot{V}_j W_{ok} \right] \\ 0 &= \sum_j \left[E_{ij}V_j + F_{ij}\ddot{V}_j + A3_{ij}\Phi_j + \sum_k N3_{ijk}\Phi_j W_{ok} \right] \end{aligned} \tag{9}$$

with

$$\begin{aligned} N1_{ijk} &= \int_0^1 \Delta\, \bar{\mathrm{E}}\mathrm{I} g_i v_j'' w_k''\, d\xi \\ N2_{ijk} &= \int_0^1 2\bar{\mathrm{m}} w_i' \frac{v_j'''}{\omega_j^4} w_k'\, d\xi \\ N3_{ijk} &= \int_0^1 \Delta\, \bar{\mathrm{E}}\mathrm{I} v_i'' g_j w_k''\, d\xi \end{aligned} \tag{10}$$

The new equation have the additional terms $N1_{ijk}$, $N2_{ijk}$ and $N3_{ijk}$ that come from the solution to the non-linear static equation W_{ok}.The most significant additional term is the $N1_{ijk}$ term which couples the torsion equation to the in-plane bending degrees of freedom. Because there is solar radiation forces on the in-plane-bending modes, this coupling allows the torsion modes to interact with the radiation pressure. We will later show that this coupling is a critical component of the model.

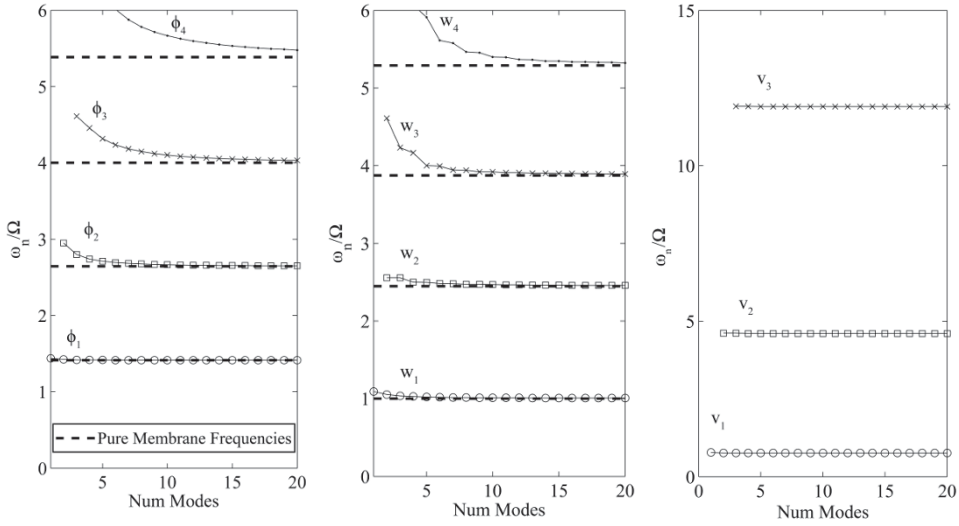
III. Structural Results

This section begins by exploring the structural dynamics, in the absence of solar radiation pressure. The structural analysis uses the parameters for the HELIOS concept listed in Table 1. By looking at the table, we can see that the terms associated with the bending stiffness terms are much smaller than one. Because the bending terms are associated with the highest order spatial derivative in the bending equations, our assumed modes, which are chosen to satisfy all of the boundary conditions, are no longer a good approximation of the leading order equation. This manifests itself in requiring a large number of modes to produce a converged solution.

Table 1. Baseline Non-Dimensional Parameters for HELIOS

Property	Value
$\overline{EI}_1$	3.8×10^{-13}
$\overline{EI}_2$	3.3×10^{-2}
$\overline{GJ}_1$	5.7×10^{-16}
$\bar{k}_{m1}$	3.2×10^{-10}
$\bar{k}_{m2}$	5.1×10^{-3}

Fig. 3 shows the natural frequencies as a function of the number of modes that are included in the modal expansion for each degree of freedom. Because the degrees of freedom are uncoupled in the linear structural model, we can look at each of the degrees of freedom independently. The figure also contains the closed form natural frequencies [4] for the pure membrane case, where the bending stiffness is identically equal to zero, for the torsion and out-of-plane bending modes. As we increase the number of modes, the natural frequencies predicted by the current model converge to the membrane solution because the bending stiffness is so small. The number of modes required to converge the natural frequencies is large because the assumed modes satisfy a fourth-order boundary value problem for the beam, while the dynamics of our spinning system are dominated by a second-order, tension-like term. While, it is possible to use modes that satisfy the second-order membrane equations, they would no

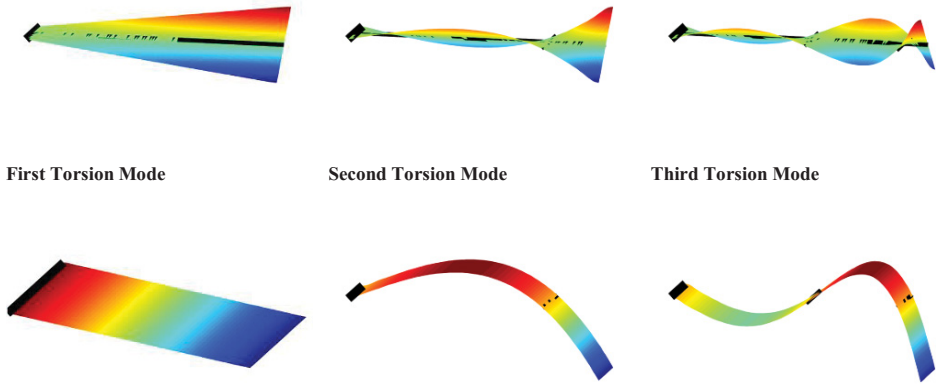


Twist Frequencies **Out-of-Plane Bending Frequencies** **In-Plane Bending Frequencies**

Fig. 3 Comparison of HELIOS Frequencies to Analytical Frequencies for a Pure Membrane

longer satisfy all of the boundary conditions required by the beam equation. An alternate approach is to conduct a boundary layer perturbation analysis that finds an outer solution that satisfies the beam equation in a small boundary layer at the root and the tip of the blade, and then an inner solution that satisfies the membrane equation over the remainder of the domain.

In addition to looking at the natural frequencies, we also looked at the mode shapes. Fig. 4 contains plots of the mode shapes for the first three torsion and out-of-plane bending degrees of freedom. These mode shapes are created using the eigenvectors of the structural equation with 20 modes in each degree of freedom. The bending mode shapes show the impact of a low beam bending stiffness. Although clamped-free beam modes are used, the mode shapes appear to have slopes all the way up to the root, more typical of a pinned beam or a string.



First Torsion Mode **Second Torsion Mode** **Third Torsion Mode**

First Out-of-Plane Bending Mode **Second Out-of-Plane Bending Mode** **Third Out-of-Plane Bending Mode**

Fig. 4 HELIOS Mode Shapes for Torsion and Out-of-Plane Bending Degrees of Freedom

IV. Solarelastic Results

This section explores both the linear and the linear perturbation around a non-linear static deflection solarelastic models. The section begins by using the linear model. For our initial analysis we choose the HELIOS parameters with the sun inclined at 45 degrees from directly overhead. As mentioned in the derivation for cases when the sun is not directly overhead the coefficients of solar radiation pressure are time varying as the heliogyro blade rotates through different angles of ψ. For this analysis, we conduct a quasi-steady analysis with the goal of solving for the linear stability boundary for a fixed rotation angle ψ. The linear stability of the system can be determined as a

function of the rotation angle. To determine completely the dynamic stability of the system a numerical time marching algorithm could be used.

The first analysis looks at the stability boundary as the parameter $\bar{p}$ is varied for different number of modes. This helps establish modal convergence and identifies the type of instability that may arise for the HELIOS system. For the initial analysis ψ is set to 0. Fig. 5 shows the results of the simulation. The figure shows the frequencies as a function of the non-dimensional solar radiation pressure. The figure identifies locations where the real part of the eigenvalue goes unstable with different color lines. For the current parameters the system transitions between a first bending mode divergence and a first and second bending coalescence flutter instability at a non-dimensional radiation pressure near one depending on if there are an even or odd number of modes in the modal expansion. For these parameters, the stability boundary does not appear to converge to a fixed value or instability type even as you increase the number of assumed modes over 100. We believe that this is a phenomenon similar to the membrane paradox found in the supersonic aeroelastic panel flutter literature [1]. Because the bending stiffness is such a small term for the current parameters, a more careful perturbation treatment of the system is required to determine the correct stability boundary.

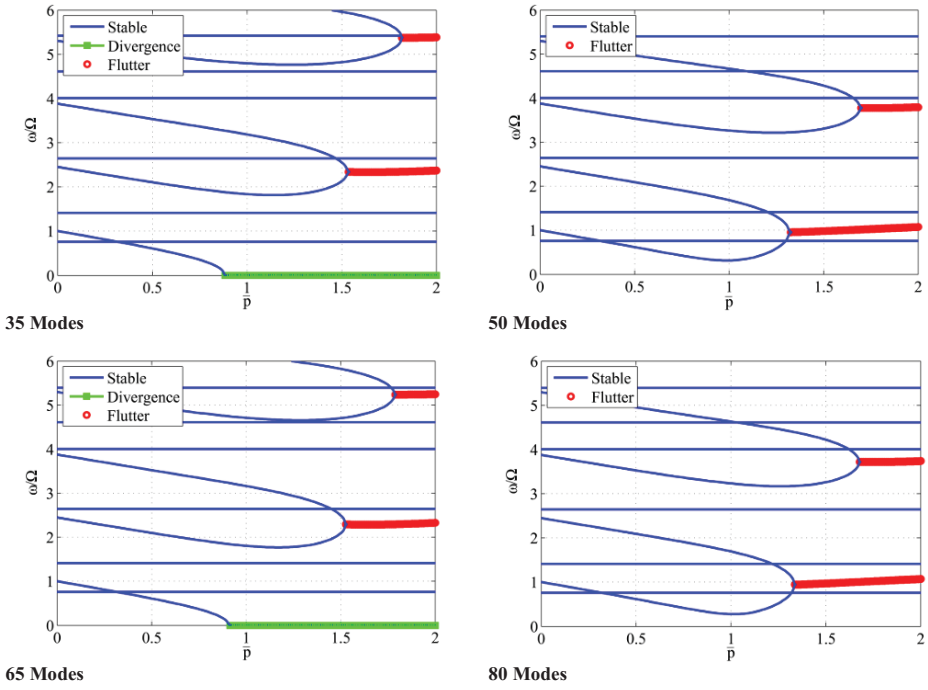


Fig. 5 Solar Radiation Pressure Stability Boundary for HELIOS with γ=45 deg

To estimate what the solution for our current system should be, we conducted a study of the stability boundary as a function of the non-dimensional bending stiffness $\overline{EI}_1$. For a larger, but still relatively small value of this parameter, we can increase the number of assumed modes until a converged solution is reached. For example, Fig. 6 shows the unstable solar radiation pressure as the number of modes is increased for $\overline{EI}_1$=3.8e-6. As the number of modes is increased above 40, both odd and even number of assumed modes predict a divergence instability at a radiation pressure $\bar{p}$ of 1.

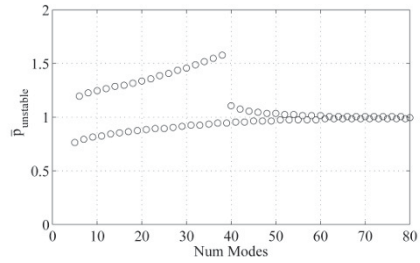


Fig. 6 Solar Radiation Pressure Boundary for HELIOS with $\overline{EI}_1$=3.8e-6

Fig. 7 shows the divergence boundary as the non-dimensional out-of-plane bending stiffness decreases. This plot suggests that as $\overline{EI}_1$ goes to zero the system will become unstable in a divergence mode at $\bar{p} = 1$. It is our belief, that if enough modes are used for the nominal HELIOS configurations parameters, the stability boundary will eventually converge to a static instability at $\bar{p} = 1$. Another method of confirming this hypothesis would be to conduct a careful perturbation analysis similar to the one conducted in Anderson et.al. [6] to confirm that a pure membrane will diverge at finite solar radiation pressure. Despite the modal convergence issues faced by the current model, the results are promising because they suggest that instabilities will not arise until a dimensional solar radiation pressure of 5e-4 N/m^2, which is much larger than the expected solar radiation pressure for the current HELIOS mission.

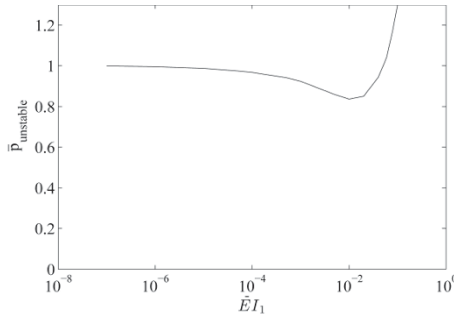


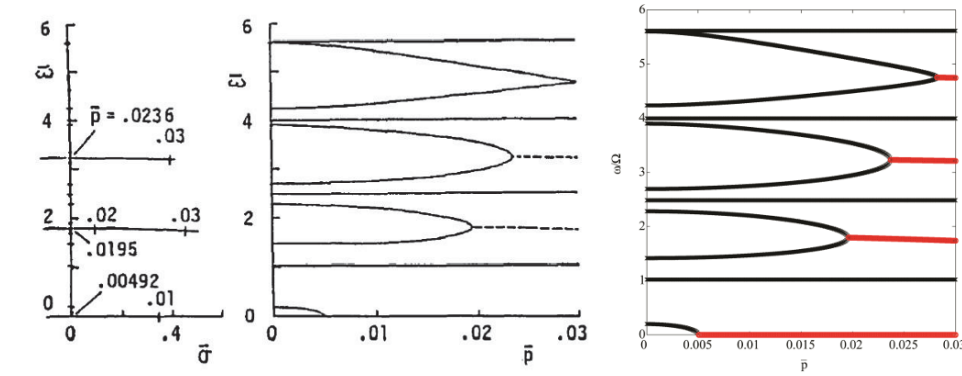
Fig. 7 Stability Boundary for HELIOS with Varying $\overline{EI}$

Because the initial analysis suggested a high solarelastic boundary, we undertook a validation effort by comparing our simulations to simulations using the *nonlinear* model and solution technique outlined by Natori et.al. [6]. We focused on the case of the sun directly overhead and used the previous author's parameters. These parameters are listed in Table 2 for reference

Table 2. Alternate Non-Dimensional Parameters

Property	Value
$\overline{EI}_1$	2×10^{-16}
$\overline{EI}_2$	4.1748×10^{-5}
$\overline{GJ}_1$	3×10^{-16}
$\bar{k}_{m1}$	6.01×10^{-10}
$\bar{k}_{m2}$	2.8868×10^{-4}

Before comparing the new nonlinear computational model to our fully linear model, we confirmed that our code could recreate the previous results. Fig. 8 includes the Natori et.al. [6] eigenvalue plot and an eigenvalue plot for the same configuration created using our new code. The two figures compare well with each other. The plot shows the natural frequency component of the eigenvalue as the solar radiation pressure strength increases. In Fig.



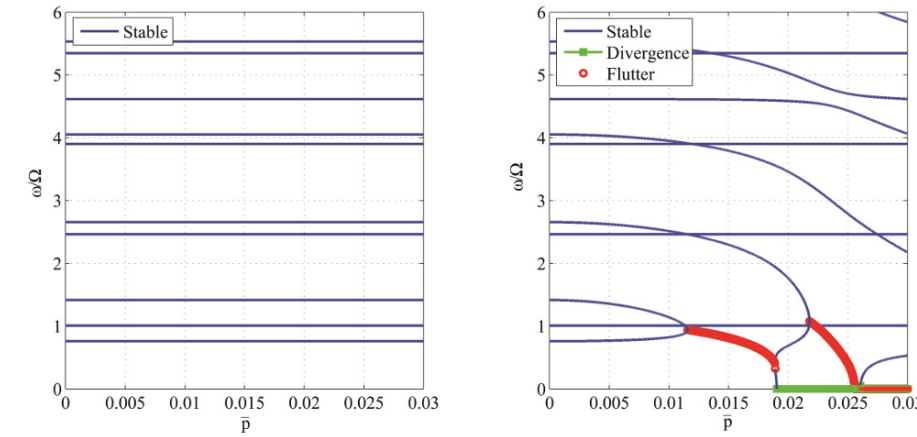
Natori et.al. Results **Current Model Results**

Fig. 8 Comparison of Eigenvalues for Natori et.al. [6] Parameters with Sun Overhead

8, the black lines corresponds to stable frequencies and the red line corresponds to unstable frequencies. The analyses in Fig. 8 include six modes in each degree of freedom.

By comparing the two models we confirmed that both models predict the same results. In both analyses the system loses stability in an in-plane divergence mode at a non-dimensional radiation pressure $\bar{p}$ of 0.005. Both models also predict that at a higher radiation pressure there is a flutter instability created by a coalescence between the first out-of-plane bending and second in-plane bending modes. The only visible difference between the published results and the current results is the radiation pressure of the coalescence between the fourth out-of-plane bending and fifth in-plane-bending modes. Because this is a post-critical coalescence between two of the highest modes that, according to Fig. 3, are most likely not converged, it is not a critical difference.

Once we confirmed that our implementation of the nonlinear perturbation method is accurate, we compared the stability boundaries from the nonlinear model to the linear model predictions. Fig. 9 shows the frequency component of the eigenvalue as a function of the non-dimensional radiation pressure for the two models using the HELIOS parameters with 15 modes for each of the degrees-of-freedom for the sun directly overhead. In the linear model, the dynamics do not depend on the solar radiation pressure and therefore the structure is predicted to be stable for all solar radiation pressures. On the other hand, for the non-linear model, there are instabilities predicted. The most critical instability, which occurs at a non-dimensional radiation pressure of 0.01, is a coalescence flutter between the first in-plane bending and first torsion degrees of freedom.



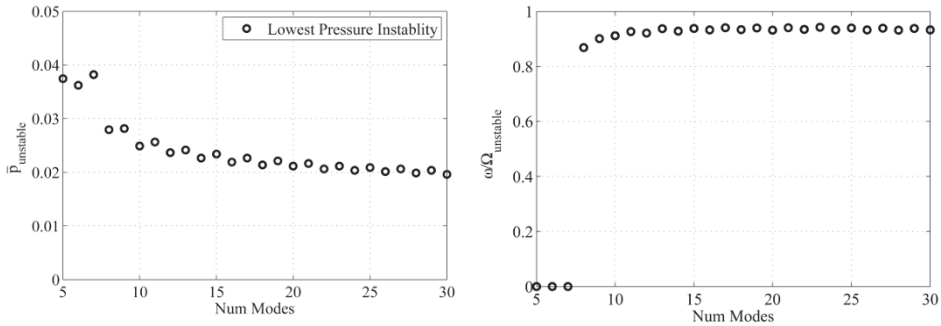
Linear Simulation **Nonlinear Simulation**

Fig. 9 Comparison of Stability Boundaries for HELIOS using Linear and Nonlinear models

The difference between the two results is significant. In this case, adding linear stiffness terms due to the static displacement of the structure creates an instability that does not arise for the purely linear model. Looking closely

at Equation 9 we can see that the critical terms are the $N1$ and $N3$ terms which fully couple the in-plane bending mode with the torsion mode and create the potential for a solarelastic instability between these two modes. This result implies that the simplest model to produce a conservative prediction of the solarelastic boundary requires these non-linear terms. This dictates that any tool used for designing a heliogyro spacecraft for solarelastic performance must at least include the non-linear terms that we identified in Equation 9.

Another benefit of the non-linear model is that, although the model still requires a relatively large number of modes to get a converged solarelastic stability boundary, the number of modes required is significantly less than for the purely linear model. Fig. 10 shows the stability boundary as the number of modes in the modal expansion changes for the HELIOS parameters with $\gamma = 45\text{deg}$ and $\psi = 0\text{deg}$, the same parameters used to create Fig. 5. For the improved model the solution has converged with less than 30 modes. By contrast, we were not able to get a converged prediction with the purely linear model. Additionally the stability boundary using the new nonlinear model is noticeably lower than the one predicted by the linear model even using 100 modes. The nonlinear model predicts a flutter instability at $\bar{p} = 0.02$ and a non-dimensional frequency near one. Because this frequency is the same order as the time scale of the rotating blade, the quasi-static approximation that we made to justify looking at the instantaneous stability for a fixed ψ is probably not appropriate. Thus future work will include using a time marching algorithm to capture fully the time varying strength of the solar radiation force on the heliogyro blades. Note that, the nonlinear stability boundary associated with this instability is the same order of magnitude of the solar radiation pressures that the HELIOS will encounter during its proposed mission and therefore the spacecraft designers must be prepared to design for solarelastic instabilities.



Unstable Solar Radiation Pressure **Unstable Frequency**

Fig. 10 Stability Boundary for HELIOS Configuration with $\gamma = 45\text{deg}$ and $\psi = 45\text{deg}$ Using Nonlinear Static Perturbation Method

Due to the potential of operating in a configuration when a solarelastic instability is possible, the next step is to evaluate what design variables can be changed to increase the stability boundary of the spacecraft. For this

paper, we choose to look at the impact that the spin rate of the heliogyro has on the stability of the structure. The analysis is conducted with the sun directly overhead so the forcing coefficients are time invariant and the eigenvalue technique used throughout this paper is valid. For this analysis we used the nominal HELIOS parameters and 20 modes in the modal expansion. Fig. 11 shows the stability boundary in terms of both the solar pressure and instability frequency as the spin rate varies. The figure shows that the solar pressure required to create an instability increases with the rotation rate. Interestingly at a rotation rate of 3/4 RPM, the instability changes from a flutter instability to a divergence instability. In general, the solarelastic stability of the heliogyro can be improved by increasing the spin rate, however there are trade-offs with the maneuverability and control system bandwidth that must be balanced when choosing the final rotation rate.

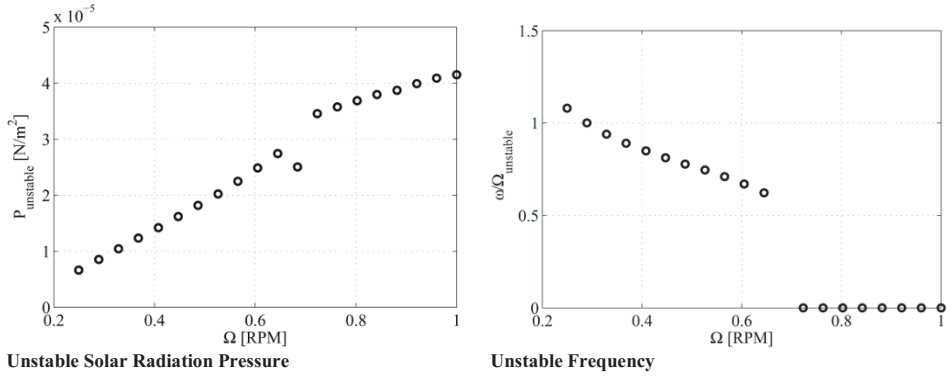


Fig. 11 Stability Boundary for HELIOS Configuration with Sun Overhead for Varying Spin Rates

V. Conclusion

In this paper we explored the solarelastic stability of a heliogyro blade using two different rotating beam structural models. The paper validated the linear structural modal by comparing it to the closed form solution for a rotating membrane. We then coupled the structural model to an optical solar radiation pressure model to analyze the solarelastic stability boundary. The initial simulations using the purely linear model predicted a high solar radiation pressure required to produce a divergence instability and required a large number of modes to get a converged solution. Next, a perturbation analysis of a non-linear structural model was compared with published results [6] to validate the model implementation. The paper then compared the non-linear and linear solarelastic models and showed the importance of the non-linear terms in producing a conservative estimate of the solarelastic stability boundary. Finally, we used the non-linear model to analyze the solarelastic stability of a proposed heliogyro spacecraft. A quasi-steady eigenvalue analysis of the HELIOS craft with the non-linear model predicted

a much lower solar radiation pressure instability boundary than the linear model. The stability boundary obtained when varying spacecraft spin rate was also presented suggesting that the faster the heliogyro spins, the more stable the system will be from a solarelastic perspective.

In addition to the insights presented in this paper, the current research also identified many avenues for future development. First, the structural model can be improved to include non-uniform properties and allow for a nonzero blade pre-twist angle. For these cases and for cases when the sun is not directly overhead the system has time varying forcing that can be captured using time marching algorithms. Additionally, numerical integration methods can be implemented to solve the non-linear solarelastic equations, instead of conducting a perturbation analysis around a static deflection. In addition to theoretical improvements, the current model can also be used to optimize the design of the HELIOS spacecraft with respect to its solarelastic stability. The future work will include using the model to evaluate the stability boundary in non-dimensional parameter space to identify trends that will be useful for the preliminary design of heliogyro solar sail spacecraft.

Acknowledgments

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Heliogyro Attitude Control Moment Authority via Blade Pitch Maneuvers

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Heliogyros generate attitude control moments by pitching their sail membrane blades collectively or cyclically, similar to a helicopter. Past work has focused on simple blade pitch profiles with the heliogyro normal to the sun; however, most solar sail missions will require sun angles of at least 35°. Furthermore, combination pitch profiles (e.g. cyclic plus collective) are needed for attitude control during all mission segments. The control moments for such situations vary in an unintuitive, nonlinear fashion. This paper explores heliogyro control moment authority with varying sun angle and combinations of pitch profiles, providing critical insight for future development of heliogyro attitude control schemes. Three strategies for generating control moments using various profile combinations are investigated for three-axis attitude control. These strategies indicate that the heliogyro can generate control moments while edge-on to the sun, allowing for recovery from any orientation. A control algorithm is presented that determines the required blade pitch profile combination to generate the desired attitude control torques. This algorithm could be employed for closed-loop control in attitude dynamics simulations.

Nomenclature

A	=	total sail area (m^2)
a_{xx}	=	collective (xx = *co*), half-p (xx = *hp*), or cyclic (xx = *cy*) profile amplitude (rad, deg)
a_0	=	characteristic acceleration (mm/s^2)
c	=	blade chord (m)
$\boldsymbol{F}$	=	force vector (N)
$\bar{\boldsymbol{F}}$	=	force vector normalized per total sail area (N/m^2)

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$\hat{\boldsymbol{h}}$ = $\hat{\boldsymbol{r}} \times \hat{\boldsymbol{v}}$, orbit momentum unit vector

$\boldsymbol{M}$ = moment vector (N-m)

$\bar{\boldsymbol{M}}$ = moment vector normalized per total sail area per blade radius (N/m^2)

N = number of blades

P = solar pressure force (4.563×10^{-6} Pa at 1 AU[1])

R = heliogyro blade radius (m)

$\hat{\boldsymbol{r}}$ = spacecraft orbital position unit vector

$\hat{\boldsymbol{s}}$ = sun-spacecraft unit vector that coincides with orbit position $\hat{\boldsymbol{r}}$ in heliocentric orbits

t = time (s)

$\hat{\boldsymbol{v}}$ = spacecraft orbital velocity unit vector

x = spanwise position (m)

y = chordwise position from centerline in the rotational plane (m)

z = position above rotational plane (m)

α = cone angle between the sun $\hat{\mathbf{s}}$ and the heliogyro spin axis $\hat{\boldsymbol{\Omega}}$ and limited to [0° 90°] (rad, deg)

δ = clock angle between $\hat{\boldsymbol{l}}$ and the projection of $\hat{\boldsymbol{d}}_2$ onto the $\hat{\boldsymbol{l}}$-$\hat{\boldsymbol{p}}$ plane (rad, deg)

θ_i = ith blade pitch (positive when pitching up in the direction of Ω)

ϕ_{xx} = half-p (xx = *hp*) or cyclic (xx = *cy*) phase angle (rad, deg)

χ_i = ith blade angle relative to blade 1 in the rotation plane, $\chi_1 = 0$ (rad, deg)

Ψ = in-plane moment M_{23} azimuth angle(rad, deg)

ψ_i = ith blade azimuth angle in the spin plane (rad, deg)

Ω = heliogyro inertial spin rate about primary spin axis (rad/s)

$\hat{\boldsymbol{\Omega}}$ = primary heliogyro spin axis

I. Introduction

THE heliogyro is a potentially high-performance solar sail architecture that divides the sail into strips or "blades" spun about a central hub. Pitching these blades either collectively or cyclically (sinusoidally)—similar to a helicopter—can provide attitude control torques without expending propellant. The literature suggests three blade pitch control profiles: collective, cyclic, and half-p [2,3,4,5,6,7,8]. Fig. 1 summarizes the effects of each of the three pitch profiles for the simple, sun-facing case. The first row is a visualization of a four-blade heliogyro showing the incident sunlight directions, spin direction, force components and moment components. Each

blade's pitch and the coordinate frame axes are also annotated. Section II defines these coordinate frames. Each profile is shown at the amplitude yielding the largest force or moment in the useful direction, assuming flat blades. Collective has the largest M_1 at 35°, half-p has the largest M_2 at 75°, and cyclic has the largest F_l at 45°. Note that, in this case, the cyclic profile (Fig. 1c) produces no moment but does generate thrust in the sail's rotation plane, a capability absent from traditional, square sails. As will be shown later however, the cyclic profile is useful for also generating moments in other situations.

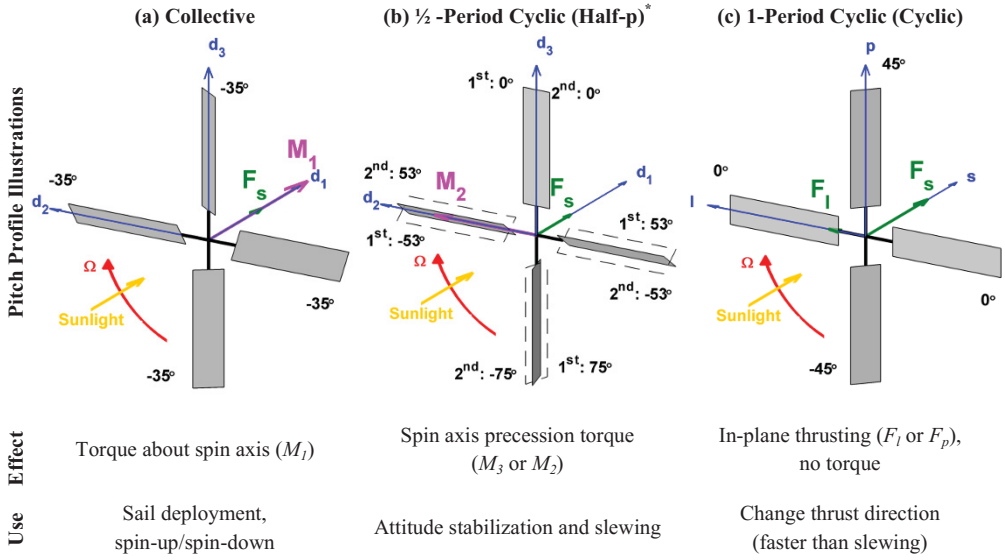


Fig. 1 Heliogyro blade pitch profile descriptions with zero cone angle α. *The half-p profile repeats after two revolutions. Dashed lines indicate the blade position for the 2nd revolution.

These three pitch profiles may be used in combination or independently for three-axis attitude control, although past work is predominantly limited to one pitch profile at a time with zero cone angle α. In reality, solar sail missions frequently require a cone angle of 35° since this angle yields the maximum thrust perpendicular to the sun-sail line [1]. The forces and moments resulting from blade pitch profiles vary dramatically and nonlinearly with cone angle. This can be problematic for maintaining attitude control at all orientations.

This paper explores the heliogyro's attitude control moment authority (CMA) with variation in cone angle and combinations of pitch profiles. We define CMA as the maximum possible moment in a given direction for a given cone angle and one of the attitude control strategies presented in section III. This investigation serves as a first step towards development of a comprehensive attitude control scheme. The algorithm we developed, `HGForce`, calculates resultant solar pressure force and moment given arbitrary heliogyro orientation and blade

pitches. Section II and the Appendix discuss the details of this calculation. We will use `HGForce` to design three different attitude control strategies in section III: collective, half-p, and cyclic. Each strategy finds the optimal control moment for different scenarios.

`HGForce` computes the heliogyro's forces and moments given blade pitch inputs; however, a blade pitch control law must compute the pitch control inputs from a desired moment vector given by the attitude control system. We must therefore find a way to invert `HGForce` and find the pitch control inputs to produce the desired force and moment vector. This is not intuitive since the resultant forces and moments for combination pitch profiles do not superimpose, and they change in direction and magnitude with cone angle. The mapping is nonlinear with many local minima, which makes inversion difficult. Section IV will present an inverse mapping for a limited case that optimizes only the attitude control moments.

II. Background

The `HGForce` algorithm finds the resultant solar pressure force and moment given an arbitrary heliogyro orientation and combination of pitch profiles. It returns the force in the sun frame, and the moment in the despun frame. The sun and despun reference frames are described in Table 1 and Table 2 and illustrated in Fig. 2. Two angles define the rotation between these frames: a clock angle δ rotation about the sun vector followed by a cone angle α pitch away from the sun vector. We define the cone and clock rotations such that +α produces a force along $+\hat{\boldsymbol{l}}$ and $+\alpha$ with $+\delta$ produces a force along $+\hat{\boldsymbol{p}}$. Consequently, the $\boldsymbol{F}$ and $\boldsymbol{M}$ of a given blade pitch profile are constant in the despun frame when averaged over two revolutions. The heliogyro's despun frame may be substituted for the flat sail frame used in most solar sail trajectory analyses. For preliminary insight, `HGForce` models the heliogyro as a set of rigid blades with freedom in pitch alone that are perfectly reflective on both sides. We normalize the forces and moments presented herein to make them independent of the heliogyro's dimensions and number of blades as follows:

$$\bar{\boldsymbol{F}} = \frac{\boldsymbol{F}}{A} \qquad \bar{\boldsymbol{M}} = \frac{\boldsymbol{M}}{A \cdot R} \tag{1}$$

Table 1 Sun coordinate frame $\boldsymbol{S}$: $\{\hat{\boldsymbol{s}}, \hat{\boldsymbol{l}}, \hat{\boldsymbol{p}}\}$

$\hat{\boldsymbol{s}}$	sun-spacecraft unit vector that coincides with $\hat{\boldsymbol{r}}$ in heliocentric orbits
$\hat{\boldsymbol{l}} = \hat{\boldsymbol{h}} \times \hat{\boldsymbol{s}}$	local horizontal w.r.t. the sun and along $\hat{\boldsymbol{v}}$
$\hat{\boldsymbol{p}} = \hat{\boldsymbol{s}} \times \hat{\boldsymbol{v}}$	the reference axis for clock angle δ, coincides with $\hat{\boldsymbol{h}}$ in heliocentric orbits
Uses:	establishing the direction of incident radiation the frame for net force used in trajectory analysis can be considered inertial in attitude control time scales

Table 2 Despun coordinate frame D: $\{\hat{\boldsymbol{d}}_1, \hat{\boldsymbol{d}}_2, \hat{\boldsymbol{d}}_3\}$

$[DS]$	direction cosine matrix (DCM) from the sun frame to the despun frame (see Appendix A for the equation definition)
$\hat{\boldsymbol{d}}_1$	overall sail normal coincident with $\hat{\boldsymbol{\Omega}}$
$\hat{\boldsymbol{d}}_2$	reference axis for clock and blade azimuth angles $\perp \hat{\boldsymbol{\Omega}}$
$\hat{\boldsymbol{d}}_3$	completes right hand system
Uses:	describing net solar pressure force and moment attitude dynamics

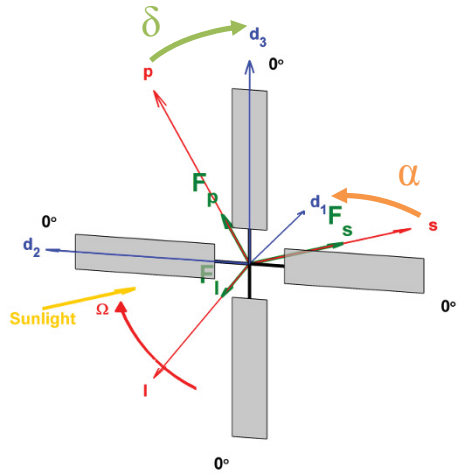


Fig. 2 Despun and Sun coordinate frames for $\alpha = 35°$ and $\delta = 45°$

`HGForce` has five blade pitch control inputs: the collective, half-p, and cyclic amplitudes and the half-p and cyclic phases. These five inputs are parameters in Eq. (2) that determine each blade's pitch over time. The "$-a_{co}$" and "$-\text{sign}(a_{hp})\frac{\pi}{2}$," terms ensure that positive amplitudes yield positive moments and negative amplitudes yield negative moments in the nominal case with zero cone angle. Without the extra term in half-p phase, both positive and negative amplitudes would yield a negative moment. This would lead to problems with perturbation-based optimization algorithms. Appendices A and B give the full derivation behind `HGForce`.

$$\theta_i = -a_{co} + a_{hp} \sin\left[\frac{1}{2}\left(\psi_i - \phi_{hp} - \text{sign}(a_{hp})\frac{\pi}{2}\right)\right] + a_{cy} \sin(\psi_i - \phi_{cy}) \tag{2}$$

$$\psi_i = \Omega t + \chi_i \qquad\qquad \chi_i = 2\pi\frac{i-1}{N} \tag{3}$$

While `HGForce` is an excellent tool to determine the forces and moments from a given set of pitch control inputs, an attitude control law would request a moment vector. The pitch control law would then have to determine the pitch control parameters required to produce the desired moment vector. We must therefore invert the problem to calculate the optimal pitch control inputs to achieve the desired force and moment outputs. This

inversion, illustrated in Fig. 3, is not straightforward because of the problem's nonlinearity; the resultant forces and moments for combination pitch profiles do not superimpose, and the resulting forces and moments depend significantly on cone angle. For example, note how the half-p profile's forces and moments vary with cone angle in Fig. 4.

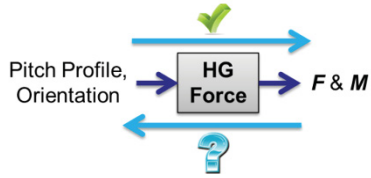


Fig. 3 HGForce control inversion problem

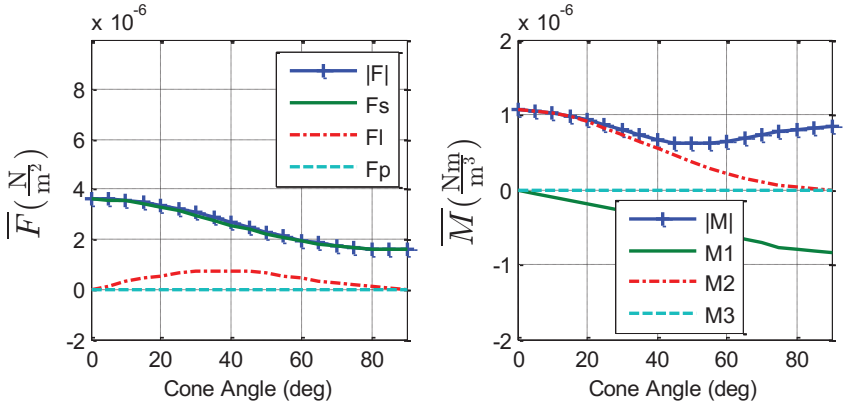


Fig. 4 Forces and moments using a 70° half-p profile with variation in cone angle.

As cone angle increases, an additional moment appears along M_1 and at the expense of M_2. This is undesirable if one is only trying to generate in-plane moments (M_2 or M_3). One might be able to counteract that effect with an additional pitch profile, but these profiles do not superimpose. Consider the superposition example in Fig. 5 using the sun-facing case. The collective profile only produces a moment about M_1 in isolation, and the cyclic profile only produces a force along F_1 in isolation. The combination of the two profiles, however, looks nothing like either one. The next section will investigate these effects in more detail and present a method of compensating for them to achieve the desired control moment.

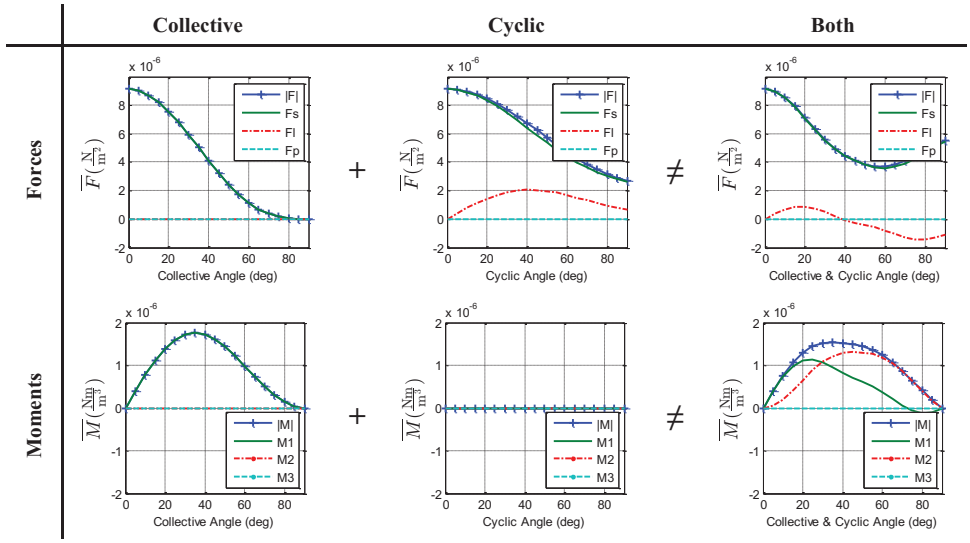


Fig. 5 Collective and cyclic superposition example for $\alpha = 0°$

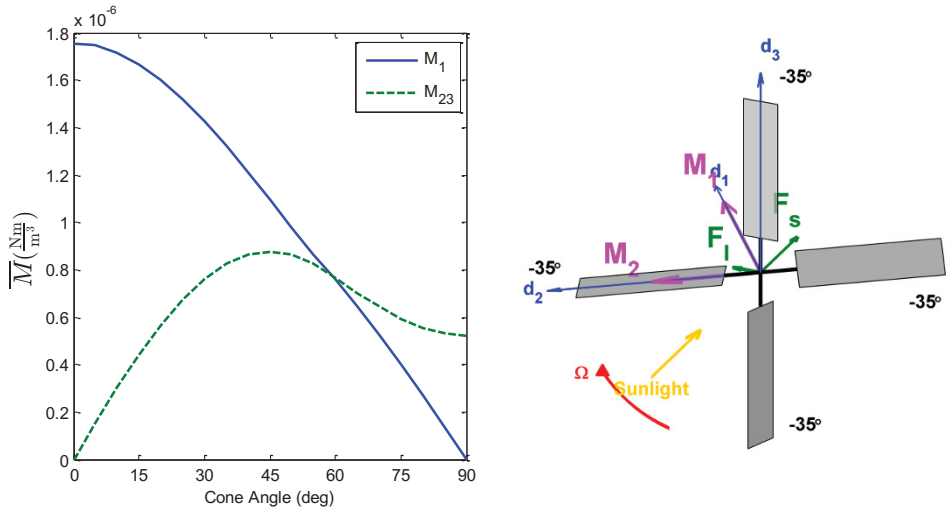
III. Discussion of Attitude Control Strategies

This section discusses the heliogyro's ability to generate control moments for several situations of interest. We use three strategies for generating these control moments, each of which combines two pitch profiles: a primary and a secondary. We name the strategies Collective, Half-p, and Cyclic after the strategy's primary pitch profile. We then find the CMA of each strategy and compare their results. This section represents the moments in cylindrical coordinates with the in-plane moment magnitude M_{23} and azimuth Ψ vice the in-plane components M_2 and M_3. This allows us to maximize the in-plane moment in a certain direction, and it separates the discussion of in-plane and out-of-plane moments. M_{23} would be used to precess the primary spin axis $\hat{\boldsymbol{\Omega}}$ for attitude control. M_1 would be used for spin-up during deployment and subsequent spin rate control.

A. CMA of the Collective Strategy

One of the first steps after launch in a heliogyro mission is to deploy the blades by slowly unreeling them from their stowage spools. The deploying blades' increasing mass moment of inertia would slow the spin rate to conserve angular momentum. A collective profile is the primary method of adding angular momentum and maintaining spin rate during blade deployment. Subsequently, collective can be used to maintain or change the heliogyro spin rate. Similar to the 35° cone angle yielding the maximum perpendicular force, a collective generates the maximum M_1 at 35° (assuming flat blades). Deployment would normally occur at $\alpha = 0°$ for maximum torque, but disturbance torques may cause the spacecraft attitude to drift. Furthermore, spin rate

control may be required when the heliogyro has a large cone angle for other mission segments. Fig. 6 shows the resulting moments as the cone angle increases.

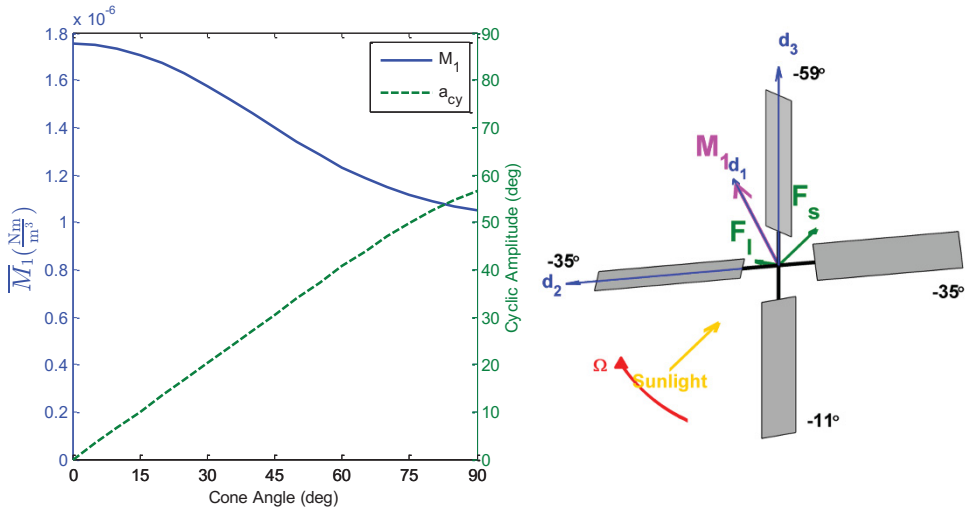


a) Moments **b) Example visualization at $\alpha = 35°$**

Fig. 6 Collective profile at 35° variation with cone angle

This profile generates an unwanted moment for $\alpha \neq 0°$ about $\hat{\boldsymbol{d}}_2$ ($\Psi = 0°$), and the M_1 CMA drops off to zero as the heliogyro points edge-on to the sun ($\alpha = 90°$). Fig. 6b illustrates this mechanism more clearly. As the sun falls away from the spin axis $\hat{\boldsymbol{d}}_1$, the top blade is more directly illuminated, and the bottom blade is angled away from the sun. The pressure differential causes a tip-over moment, which is always a $+M_2$ since we limit α to [0° 90°] by flipping the clock angle 180°. It is important to note that this moment is destabilizing in that it would increase the cone angle during spin-up. Therefore, active correction is required to maintain heliogyro attitude during deployment. We can zero out M_2 and restore the desired M_1 by overlaying a cyclic profile. This "α correction" is always provided by a $-a_{co}$ at $\phi_{cy} = 0°$ (equivalently a $+a_{co}$ at $\phi_{cy} = 180°$) regardless of the CMA strategy, because of how we define the despun frame. We determine the exact value of the α correction by iteration, but it is approximately $-2\alpha/3$ for the Collective strategy.

Fig. 7 shows the same 35° collective profile, but with the cyclic α correction applied. This successfully removes the unwanted moment (M_2) and largely restores the desired moment (M_1), even when the heliogyro is edge-on to the sun. Fig. 7a replaces M_{23} (set to zero at all cone angles) with the required cyclic correction a_{cy}. Fig. 7b shows that the upper blade is now pitched away from the sun and the lower blade towards the sun exactly enough to remove the tipping over moment.



a) Variation with cone angle **b) Example visualization at $\alpha = 35°$**

Fig. 7 CMA of 35° collective profile with α correction, $M_{23} = 0$

So far, this strategy has only produced in-plane moments about $\hat{\boldsymbol{d}}_2$, but attitude control during blade deployment will require CMA in all in-plane directions, preferably at all orientations. We therefore adjust the cyclic profile by adding two cyclic maneuvers together: the first to zero the unwanted in-plane moment, and the second to create a desired in-plane moment. The adjustment is given below and derived in Appendix C.

$$a_{cy} = \sqrt{\left(a^*_{cy}\right)^2 + 2a^*_{cy}a'_{cy}\cos\phi'_{cy} + \left(a'_{cy}\right)^2}$$
$$\phi_{cy} = \tan^{-1}\left(\frac{a'_{cy}\sin\phi'_{cy}}{a^*_{cy} + a'_{cy}\cos\phi'_{cy}}\right) \tag{4}$$

Star superscripts denote the α correction amplitude, and prime superscripts denote the additional cyclic maneuver to generate in-plane moments in the desired direction. Eq. (4) is the equation for a circle of radius a'_{cy} centered a distance a^*_{cy} away from the origin, as shown in Fig. 8 for an $a'_{cy} = 25°$. Fig. 9 shows the resulting in- and out-of-plane CMA. The curves represent the variation of cyclic phase ϕ_{cy} through 360°, and each curve is a different cone angle. We chose the 25° a'_{cy} to give an M_{23} similar to the Half-p strategy of section III.B while preserving most of the original M_1. In fact, M_1 is only about 20% lower than in Fig. 7 at all cone angles. Moreover, this strategy retains significant authority even edge-on to the sun at $\alpha = 90°$. If a larger M_{23} is desired, one could simply increase a'_{cy}. This would smoothly transition the Collective strategy into the third CMA strategy, Cyclic, discussed in section III.C.

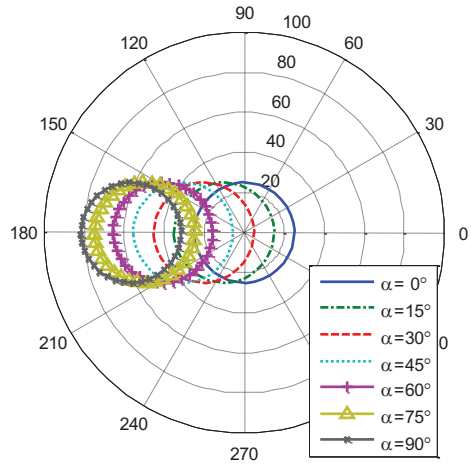
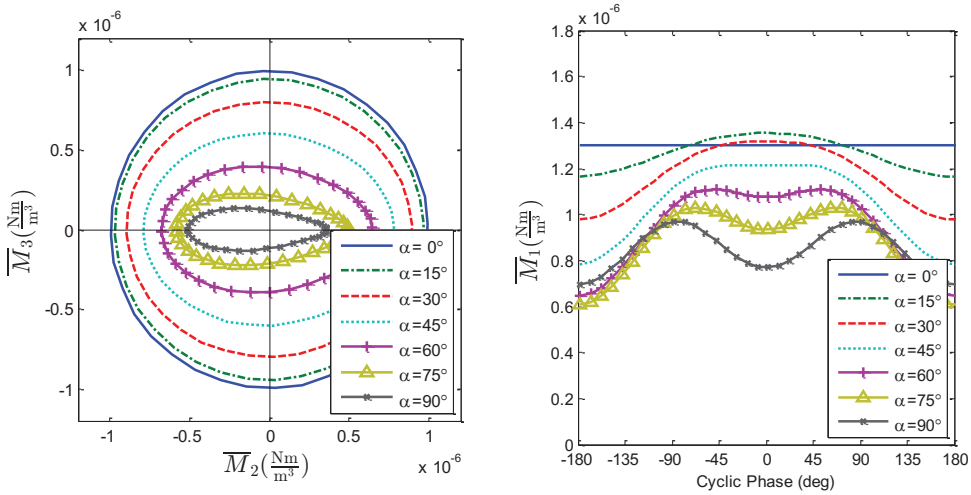


Fig. 8 Adjusted a_{cy} and ϕ_{cy} for 35° Collective strategy with 25° a'_{cy} and α correction

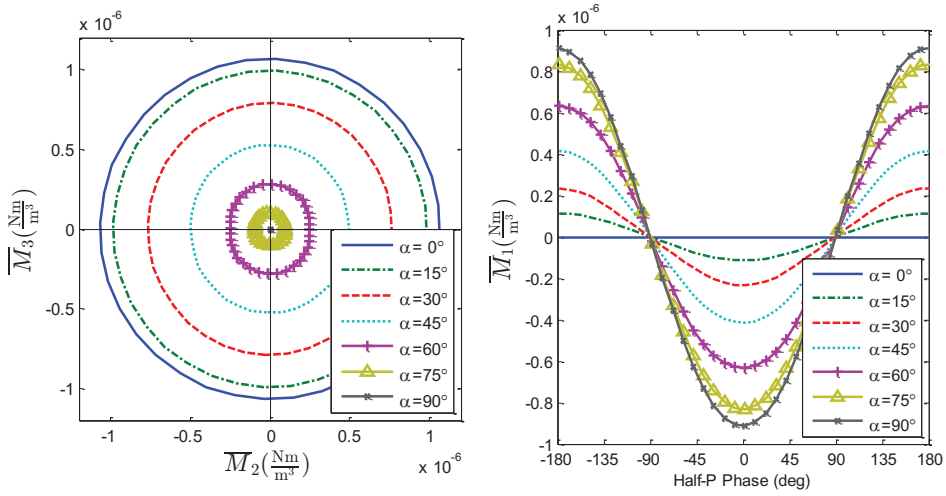


a) In-plane moment **b) Out-of-plane moment**

Fig. 9 Collective strategy CMA at 35° with 25° a'_{cy} and α correction

B. CMA of the Half-p Strategy

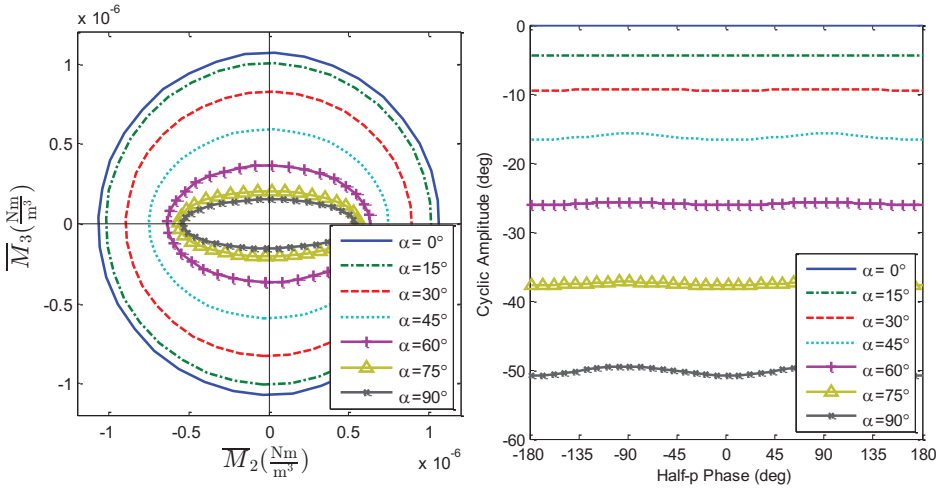
The half-p pitch profile is the primary method of controlling spacecraft attitude after blade deployment. As we will show, this profile smoothly transitions from zero to low to high torque and usually alters the solar thrust vector the least. Fig. 10a shows this profile's in-plane CMA using a 75° half-p maneuver, as this amplitude (±5°) generates the maximum M_{23}. Similar to uncompensated collective, the desired CMA decreases to zero as the sail moves edge-on to the sun. Furthermore, Fig. 10b shows an unwanted M_1 that arises with non-zero cone angles.



a) In-plane CMA **b) Residual out-of-plane moment**

Fig. 10 CMA of 75° half-p for varying cone angle

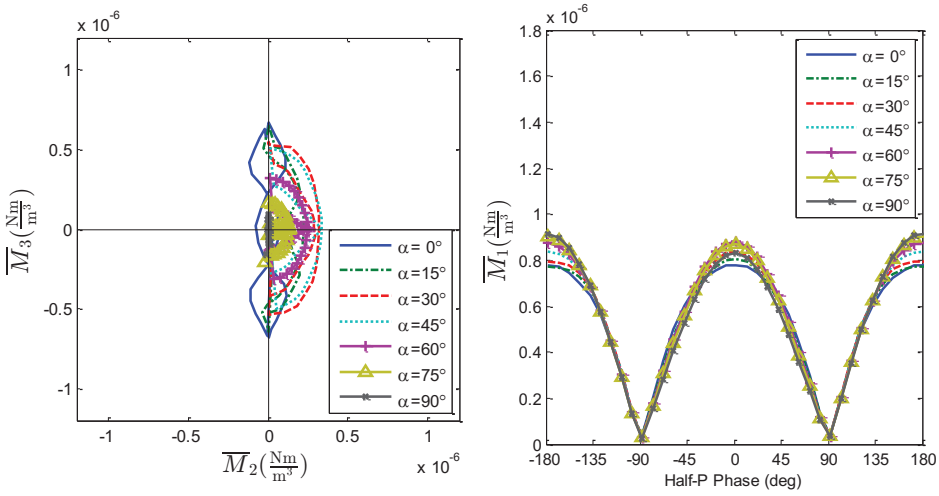
As before, we determine the exact cyclic α correction by iteration, but in this case $a_{cy}^{*} \cong -\alpha/2$. Fig. 11 shows the new CMA for the half-p profile with α correction. Fig. 11b shows the required α correction to set the out-of-plane moment to zero. Not only does the cyclic correction eliminate the unwanted M_1, but it also increases the CMA compared to Fig. 10a. Moreover, it even has CMA to recover from a worst-case orientation, α = 90°. Additionally, the cyclic compensation seen in Fig. 11b is nearly independent of the half-p phase.



a) In-plane CMA **b) Cyclic α correction amplitude**

Fig. 11 CMA of 75° half-p strategy for varying cone angle, applying α correction to set $M_1 = 0$

Cyclic compensation during a half-p profile can zero the out-of-plane moment, but it can also provide limited control authority on M_1. Fig. 12 uses the cyclic profile to maximize M_1 during a 75° half-p maneuver. This contrasts with Fig. 11, where we use cyclic to zero M_1 and maximize the in-plane moment M_{23}.



a) Residual in-plane moment **b) Out-of-plane CMA**

Fig. 12 Maximum M_1 by varying a_{cy} during a 75° half-p profile.

The CMA about M_1 is poor compared to the Collective strategy, and generating torques about M_1 significantly degrades the available M_{23}. There is no authority when half-p phase is at ±90° (i.e. the desired in-plane moment is about $\pm\hat{\boldsymbol{d}}_3$), and its maximum magnitude is only 0.91 μNm/m^3. Compare this to a 35° collective, which produces a maximum moment of 1.75 μNm/m^3. Clearly, one should use the Collective strategy when the primary goal is to spin-up or spin-down; however, the Half-p strategy could provide some limited control to maintain spin rate during attitude maneuvers. Fig. 13 shows the cyclic amplitudes required to achieve the moments shown in Fig. 12. As with Fig. 11b, the correct amplitude is weakly dependent on half-p phase and varies more strongly with cone angle. The two discontinuities in Fig. 13 at ±90° keep the in-plane moment of Fig. 12b positive.

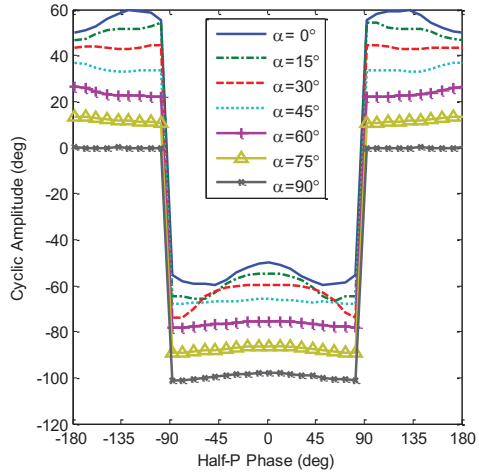
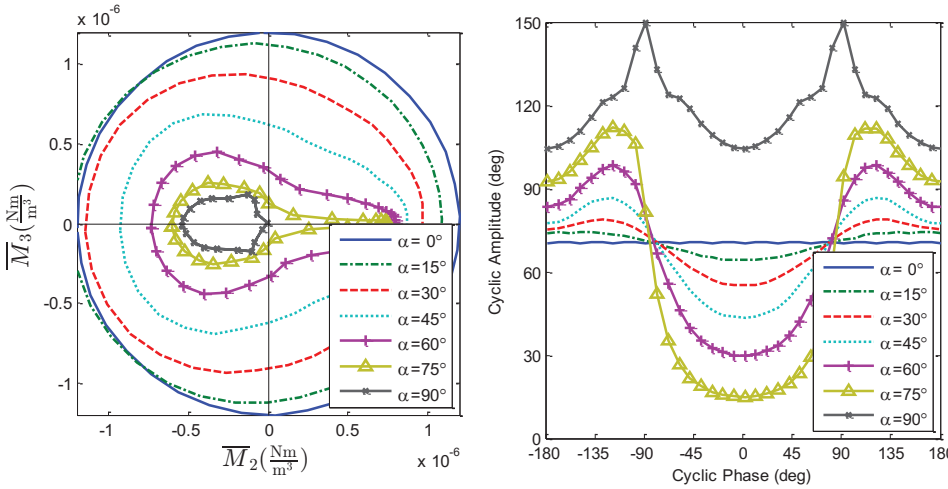


Fig. 13 Cyclic amplitude that maximizes M_1 during a 75° half-p profile

C. CMA of the Cyclic Strategy

The third and final strategy is similar to the Collective strategy. It starts with a baseline collective profile; however, the cyclic amplitudes are much larger to completely eliminate M_1. Fig. 14 shows the required cyclic correction to set M_1 to zero and the resulting M_{23}.



a) In-plane CMA **b) Cyclic compensation amplitude**

Fig. 14 Cyclic strategy CMA with 50° a_{co}, cyclic to set $M_1 = 0$, and α correction

Unlike previously, the cyclic correction is highly dependent on both the cone and control phase angles. The spikes at ±90° cyclic phase (i.e. when $M_3 = 0$) for α = 90° arise because the solution is singular at those points. With the heliogyro exactly edge-on to the sun, a pure collective profile yields the highest M_2, so the solution

tends towards a_{cy} = 0° or 180°. We avoided singularities throughout this paper by biasing ±2° around them, but that was not quite sufficient to avoid a spike in this case.

This strategy fills a similar role to the Half-p strategy, namely attitude slewing by precession of the spin axis, but there are several key differences. First, it has a larger in-plane CMA at most cone angles (18% larger at α = 0°), which means faster slewing. Second, it can smoothly transition to the collective strategy of section III.A., so it has excellent authority in M_1. On the other hand, it has lower in-plane CMA above about 60° cone angle in the $+\hat{\boldsymbol{d}}_2$ direction. This means that they Cyclic strategy could not torque to reach high cone angles if one desired such an orientation. Furthermore, it requires blade pitches of >50° even when the desired M_{23} is very low and >150° when the cone angle and desired M_{23} are high. Such high blade pitches have a number of disadvantages including a reduction in the solar thrust. Fig. 15 and Fig. 16 compare the resulting force and moment components for the Cyclic and Half-p strategies at α = 0°.

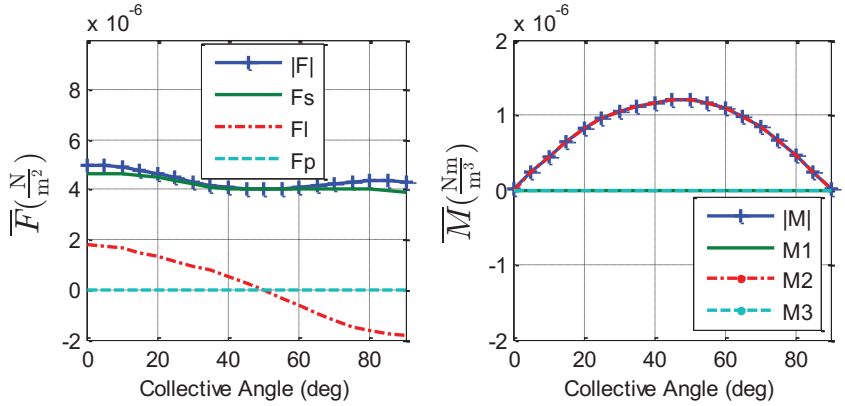


Fig. 15 Cyclic strategy force and moment components at α = 0°

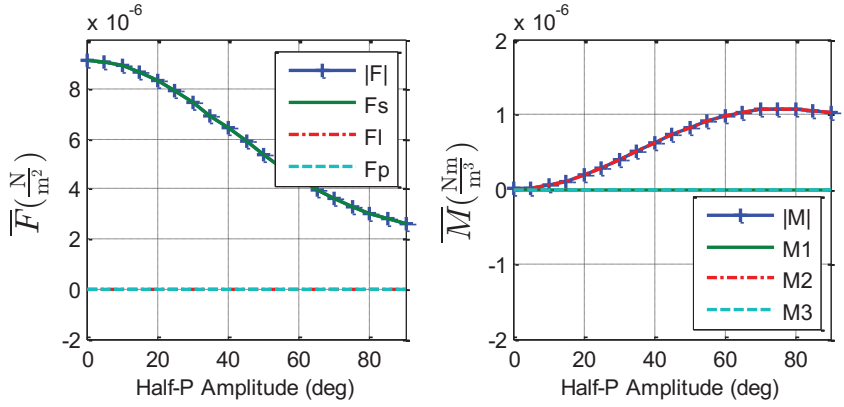


Fig. 16 Half-p strategy force and moment components at α = 0°

Both cases produce $\boldsymbol{M}$ only about $\widehat{\boldsymbol{d}}_{\boldsymbol{2}}$, but the forces are quite different. The cyclic strategy has an additional lateral force F_l, as expected from Fig. 1c. This force may or may not be desirable, but it is impossible to decouple the in-plane force and moment directions; they are both in the direction of the cyclic phase angle. At low desired moments, the half-p strategy has a much higher net solar thrust |**F**|. At around 1 μNm/m^3 and above, the cyclic strategy has a higher solar thrust. Therefore, the half-p strategy is better for routine attitude control and recovery from edge-on orientations, but the cyclic strategy is better when performing large slew maneuvers as quickly as possible. One example where rapid slewing would be useful is for missions with a change in orbit inclination. The optimal trajectory requires an instantaneous, 70° slew twice per orbit [1].

D. Section III summary

Table 1 summarizes the three CMA strategies developed in this section. It lists the two combined profiles of each strategy and some of the key advantages and disadvantages of each. Each strategy limits the inputs to three control parameters. From these investigations, we found that each of the three inputs primarily influence one of the three outputs (i.e. the moment cylindrical components). This makes it possible to construct an invertible Jacobian matrix with large diagonal terms. The Jacobian is the first step in a full inversion of `HGForce` and is the subject of the next section.

Table 3 Summary of attitude control strategies

CMA strategy	**Collective**	**Half-P**	**Cyclic**
Primary profile	Collective	Half-P	Cyclic
Secondary profile	Cyclic	Cyclic	Collective
	Parameters in 1st column primarily controlled by… (below)		
Ψ	ϕ_{cy}	ϕ_{hp}	ϕ_{cy}
M_{23}	a_{cy}	a_{hp}	a_{co}
M_1	a_{co}	a_{cy}	a_{cy}
α correction	$a_{cy}^* \cong -2\alpha/3$ $\phi_{cy}^* = 0°$	$a_{cy}^* \cong -\alpha/2$ $\phi_{cy}^* = 0°$	$a_{cy}^* \cong f(\alpha, \phi_{cy})$ $\phi_{cy}^* = 0°$
Pros:	Highest M_1 CMA Good M_{23} CMA @ α = 90°	Higher solar thrust Highest M_{23} CMA @ α = 90°	Highest M_{23} CMA @ α ≤ 60° Good M_1 CMA
Cons:	Lower M_{23} CMA No M_{23} without M_1	Worst M_1 CMA	Lower solar thrust Highest blade pitches Lowest M_{23} CMA @ α = 90°
Used for:	$\bar{M}_1 > \bar{M}_{23}$	Smaller $\bar{M}_{23}$ (<1 μNm/m^3)	Maximizing $\bar{M}_{23}$

IV. Inversion of `HGForce` with Half-p strategy

The investigations of section III led to the discovery of a limited inversion of the `HGForce` problem with the Half-p strategy. This algorithm, dubbed `ACHalf_p`, restricts the control inputs to a_{hp}, a_{co}, and ϕ_{hp} and the

outputs to $\boldsymbol{M}$ (ignoring the resulting forces). It does allow for small moments about M_1 in the limited cases discussed in section III.B., but its primary purpose is attitude control with M_{23}. First, it uses multivariate regression of the CMA data presented in Fig. 11 to make an initial estimate of the required control parameters. Then, it builds a Jacobian matrix by perturbing the initial estimate. Finally, it iterates to the true solution with the gradient descent optimization method. If the requested moment is unreachably large, it will generate a torque as large as possible in the intended direction.

Fig. 17 shows the results of `ACHalf_p` for two orientations of interest, $\alpha = \{0°, 35°\}$. These orientations appear frequently in solar sail trajectory studies. For example, the GEOSTORM mission to displace the Earth-Sun L1 point sunward primarily uses $\alpha \cong 0°$ [1], and the Solar Polar Imager mission stays near $\alpha \cong 35°$ for its entire trajectory [9]. The outer, darkest red areas in Fig. 17 are where the requested M_{23} is impossible to reach and `ACHalf_p` limits a_{hp} to 75°. Thus, it provides the largest possible moment in the desired direction Ψ.

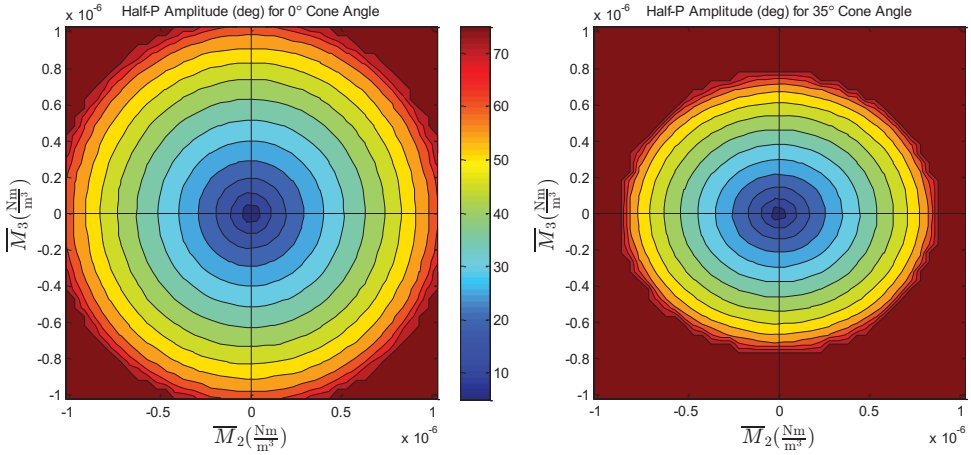


Fig. 17 Required a_{hp} for a range of requested in-plane moments

By comparing Fig. 17 to Fig. 11a, one can see that these are the inverse and forward mapping (respectively) of the same problem. The `ACHalf_p` algorithm would be suitable for attitude dynamics simulations of slewing maneuvers, but it would falter if significant changes in spin Ω were required. One could employ a similar approach on the other two CMA strategies, both of which have better control over spin rate.

V. Conclusions

The practicalities of actuating heliogyro blades to achieve desired attitude control moments have not been studied at any length until now. So far, we have found that the blade control problem is more complicated than previously imagined, but not intractable. Complex heliogyro operations (e.g. maintaining attitude control during

blade deployment) necessitate the use of combinations of collective, half-p, and cyclic blade pitch profiles. Unfortunately, these profiles do not superimpose intuitively due to the system's nonlinearity. Additionally, undesirable moments arise as the heliogyro cone angle increases, which many solar sail missions frequently require. We found that combining the baseline pitch profile with a cyclic profile was the best method to correct for these undesirable moments and restore the desired control moments.

This paper presented an algorithm that determined the resulting force and moment vectors for any given heliogyro orientation and combination of pitch profiles. It used that algorithm to maximize the heliogyro's attitude control moments via three different strategies. The Collective strategy with cyclic correction for cone angle was ideal for generating torques about the primary spin axis (e.g. during blade deployment). It also could generate attitude control torques to recover from any attitude. The Half-p strategy plus cyclic correction was useful for providing routine, low-magnitude, attitude control torques to precess the spin axis, and it generated these torques with the minimum reduction in solar thrust. It also could produce torque to recover from any attitude. The last strategy, Cyclic, was a variant on the Collective strategy that eliminated spin-axis torque and maximized in-plane torque, better for large slewing maneuvers. However, this strategy required very high blade pitches and yielded lateral forces that may not be desirable.

We also discussed a method to invert the heliogyro blade control problem and determine the required control input parameters to achieve a desired moment. This inversion was limited to a restricted set of inputs and only optimized the resulting moment, ignoring the force. Such a system is sufficient for attitude control simulations, but it does not necessarily arrive at the globally optimal solution. Future efforts will attempt to find a globally optimal, six degree-of-freedom force and moment control law. Lastly, this model made significant simplifying assumptions in order to gain a preliminary insight and make the problem manageable. Future work will transition to non-ideal reflection, a more realistic blade shape informed by our previous work [7,8], and permit blade bending out-of-plane.

Appendix

A. Additional coordinate frames and rotation matrices

Two other coordinate frames were used internally to `HGForce`: the body frame B and the ith blade frame L_i.

Table 4. Body coordinate frame B: $\{\hat{\mathbf{b}}_1, \hat{\mathbf{b}}_2, \hat{\mathbf{b}}_3\}$

[BD]	direction cosine matrix from the despun frame to the body frame
$\widehat{\boldsymbol{b}}_1$	axis to blade 1 midchord at the root
$\widehat{\boldsymbol{b}}_2$	completes RH system
$\widehat{\boldsymbol{b}}_3$	major body axis assumed coincident with $\widehat{\boldsymbol{d}}_1$ and $\widehat{\boldsymbol{\Omega}}$
Used for:	attitude dynamics

Table 5. Flat *i*th blade coordinate frame L_i: $\{\hat{\boldsymbol{x}}, \hat{\boldsymbol{y}}, \hat{\boldsymbol{z}}\}$

[L$_i$B]	DCM from the body frame to the *i*th blade frame
$\hat{\boldsymbol{x}}$	axis along blade span
$\hat{\boldsymbol{y}} = \hat{\boldsymbol{z}} \times \hat{\boldsymbol{x}}$	axis along blade chord
$\hat{\boldsymbol{z}}$	blade normal coincident with $\hat{\boldsymbol{n}}$
Used for:	solar pressure force calculation

The DCMs between frames are a combination of the following single-axis rotation matrices:

$$\mathrm{ROT1}(\theta) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & \sin\theta \\ 0 & -\sin\theta & \cos\theta \end{bmatrix} \tag{5}$$

$$\mathrm{ROT2}(\theta) = \begin{bmatrix} \cos\theta & 0 & -\sin\theta \\ 0 & 1 & 0 \\ \sin\theta & 0 & \cos\theta \end{bmatrix} \tag{6}$$

$$\mathrm{ROT3}(\theta) = \begin{bmatrix} \cos\theta & \sin\theta & 0 \\ -\sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \tag{7}$$

We define the DCMs with the notation ${}^{A}\hat{\boldsymbol{x}} = [AB]\,{}^{B}\hat{\boldsymbol{x}}$ where:

$$[DS] = \mathrm{ROT3}(\alpha)\mathrm{ROT1}(\delta) \tag{8}$$

$$[BD] = \mathrm{ROT3}(\phi_1)\mathrm{ROT1}\left(\frac{\pi}{2}\right)\mathrm{ROT3}\left(\frac{\pi}{2}\right) \tag{9}$$

$$[L_iB] = \mathrm{ROT1}(\theta_i)\mathrm{ROT3}(\chi_i) \tag{10}$$

B. Solar pressure force and moment calculation

Rotate $\hat{\boldsymbol{s}}$ into blade frame:

$${}^{L_i}\hat{\boldsymbol{s}} = [L_iB][BD][DS]\,{}^{S}\hat{\boldsymbol{s}} \tag{11}$$

Find blade's solar incidence angle (i.e. its cone angle):

$$\alpha_i = \cos^{-1}(\hat{\boldsymbol{s}} \bullet \hat{\boldsymbol{n}}_i) \tag{12}$$

The blade normal $\hat{\boldsymbol{n}}_i$ is:

$${}^{L_i}\hat{\boldsymbol{n}}_i = \left[0, 0, \mathrm{sign}\left({}^{L_i}\hat{\mathbf{s}}_3\right) \cdot 1\right]^T \tag{13}$$

The "sign$\left({}^{L_i}s_3\right)$" ensures the normal is pointing out the dark side of the blade regardless of which side is illuminated. Assuming perfect reflection, the solar pressure force on the blade is [1]:

$$\boldsymbol{F_i} = 2PA_i \cos^2 \alpha_i \, \hat{\boldsymbol{n}}_{\boldsymbol{i}}$$
$$A_i = R \cdot c \tag{14}$$

A_i is the area of one blade. The moment is:

$$\boldsymbol{M_i} = \boldsymbol{r_i^{cp}} \times \boldsymbol{F_i} \tag{15}$$

For a flat, uniform blade, the center of pressure is:

$$\boldsymbol{r_i^{cp}} = {}^{L_i}\left[\frac{R}{2}, 0, 0\right]^T \tag{16}$$

The net force and moment are found by summation of each blade's component in the despun frame.

$${}^{D}\boldsymbol{F} = \sum_{i=1}^{N} [BD]^T [L_i B]^T {}^{L_i}\boldsymbol{F_i} \tag{17}$$

$${}^{D}\boldsymbol{M} = \sum_{i=1}^{N} [BD]^T [L_i B]^T {}^{L_i}\boldsymbol{M_i} \tag{18}$$

C. Collective strategy adjustment of cyclic profile

The Collective strategy adds two cyclic maneuvers together. The first is an α correction of a_{cy}^* at $\phi_{cy}^* = 0°$ that zeros the unwanted in-plane moment, and the second is a given a'_{cy} and ϕ'_{cy} to generate the desired in-plane moment. The adjustment is calculated below, first by substituting $b'_{cy} = a'_{cy} \cos \phi'_{cy}$ and $c'_{cy} = a'_{cy} \sin \phi'_{cy}$.

$$\begin{aligned} a_{cy} \sin\left(\psi_i - \phi_{cy}\right) &= a'_{cy} \sin\left(\psi_i - \phi'_{cy}\right) + a_{cy}^* \sin(\psi_i) \\ &= b'_{cy} \sin \psi_i + c'_{cy} \cos \psi_i + a_{cy}^* \sin(\psi_i) \\ &= (b'_{cy} + a_{cy}^*) \sin \psi_i + c'_{cy} \cos \psi_i \end{aligned} \tag{19}$$

$$\begin{aligned} a_{cy} &= \sqrt{\left(b'_{cy} + a_{cy}^*\right)^2 + \left(c'_{cy}\right)^2} \\ &= \sqrt{\left(b'_{cy}\right)^2 + 2b'_{cy} a_{cy}^* + \left(a_{cy}^*\right)^2 + \left(c'_{cy}\right)^2} \\ &= \sqrt{\left(a_{cy}^*\right)^2 + 2a_{cy}^* a'_{cy} \cos \phi'_{cy} + \left(a'_{cy}\right)^2} \end{aligned} \tag{20}$$

$$\begin{aligned} \phi_{cy} &= \tan^{-1}\left(\frac{c'_{cy}}{a_{cy}^* + b'_{cy}}\right) \\ &= \tan^{-1}\left(\frac{a'_{cy} \sin \phi'_{cy}}{a_{cy}^* + a'_{cy} \cos \phi'_{cy}}\right) \end{aligned} \tag{21}$$

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Dynamics of a Coupled Pendulum Model of a Heliogyro Membrane Blade

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As a necessary precursor to a heliogyro solar sail flight demonstration, meaningful ground test experiments are necessary for predicting the linear and nonlinear structural dynamics of the heliogyro membrane blades in flight. This paper describes analytical comparisons of linear and nonlinear behavior of a multi-link discrete model of a heliogyro blade under 1-g gravitational and centrifugal loads, and one setup for experimental validation of 1-g out-of-plane motion. Linear system-identification is performed on the multi-link experimental data to validate the 1-g multi-link model of a heliogyro membrane blade.

Nomenclature

A	=	discrete-time state matrix
B	=	discrete-time input matrix
C	=	output matrix
D	=	direct transmission matrix
$\hat{e}_x(t)$	=	unit vector pointing to the direction of the gravitation field
$\hat{e}_y(t)$	=	unit vector perpendicular to $\hat{e}_x(t)$ and $\hat{e}_x(t)$ obeying the usual ordering of the axes
$\hat{e}_z(t)$	=	unit vector perpendicular to the plane of the chain in still motion
ℓ	=	total length of the membrane blade, meter
m	=	total mass of the membrane blade, gram
m_{ij}	=	discretized mass of the membrane blade; $i = 1,2,\ldots n$; $j = 1,2$

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n = degree of freedom (number of discretized sectors of the membrane blade)

s = half width of the membrane, meter

θ_i = ith angular displacement of in-plane motion, radian; i = 1,2,…n

ϕ_i = ith angular displacement of out-of-plane/bending motion, radian; i = 1,2,…n

φ_i = ith angular displacement of twisting motion, radian; i = 1,2,…n

τ_{θ_i} = torque for the ith angular displacement of in-plane motion; i = 1,2,…n

τ_{ϕ_i} = torque for the ith angular displacement of out-of-plane/bending motion; i = 1,2,…n

τ_{ϕ_i} = torque for the ith angular displacement of twisting motion; i = 1,2,…n

I. Introduction

A HELIOS system called *heliogyro* is a high-performance, spinning solar sail architecture that uses long (order of kilometers) reflective membrane strips to produce thrust from solar radiation pressure. The heliogyro, was first proposed by MacNeal [1,2] and studied extensively by JPL in the 1970s. The membrane "blades" are joined to a central hub and stiffened by only centrifugal forces, making the design exceedingly light-weight. The heliogyro concepts have existed for decades, but their implementation has been elusive. Without a more rigorous analytical and empirical investigation, the reliability and robustness of heliogyro control remains unknown. Typical sail materials are aluminized polymer films with only a few micrometers thick [3]. The extremely flexible membrane blades can develop significant twisting (radial pitch variation), bending in the rotor plane (lead/lag) and out of plane (flap) directions, which makes dynamic analysis and experimental validation a considerable challenging task. MacNeal [2] in 1970s used the techniques of helicopter rotor to study the structural dynamics of the heliogyro. The uncoupled beam equations derived in [1] to characterize in-plane, out-of-plan, and twisting motions have been used as a foundation for engineering development and research studies [4-7]. Recently, Juang, Hung and Wilkie [8] proposed another approach to characterize linear and nonlinear behavior of the heliogyro with a simpler discrete-mass and multi-link model technique. Despite analytical studies, no one has ever experimentally validated the structural dynamics and control issues. Guerrant, Lawrence, and Wilkie [9-13] have examined the stability and effectiveness of a torque-source motor at the blade root to control blade pitch. They have conducted tests on a hanging heliogyro specimen in a vacuum chamber to validate their membrane-ladder finite-element model, and the studies [13] give an encouraging signal that the blade twist control for the heliogyro is feasible.

Here we consider the twisting and bending motions of a coupled pendulum model of a heliogyro blade. We will use the approach developed in [8] for describing a slender spinning membrane with a simpler multi-link model. The nonlinear models in [8] can be linearized to produce identical frequencies for any arbitrary degree of freedoms with those computed from a continuum model described by partial differential equations [1]. We would conduct analytical studies and compare the discrete-mass/multi-link and continuum models under the gravitational and centrifugal loads. We hope to replace some experiments in the spinning setting by experiments in the gravitational setting, which is more plausible to perform on earth. Computer simulations and experimental results would be performed to study the out-of-plane motion of a 1-g multi-link model. System identification [14] would be used to identify modal parameters from the multi-link experimental data to validate the 1-g multi-link model.

II. Discrete Configuration of a Membrane

In a previous work by Juang, Hung, and Wilkie [8], a discretized model of a slender membrane was considered. This discretized model differs from the usual finite element discretization. We reuse the model in [1] and consider motions under both centrifugal and gravitational loads. We assume throughout this article that our membrane has fixed length ℓ, fixed width 2*s*, and fixed total mass *m*. The model looks like a ladder, and we formulate the model of *n* sectors as follows. The model consists of pairs of masses (m_{i1}, m_{i2}), each mass with mass $m/2n$. The quantities m_{i1} and m_{i2} are connected by an inextensible weightless rod of length 2*s*. In addition, m_{ij} and $m_{(i+1)j}$ are connected by an inextensible weightless rod of length ℓ/n for $i = 1,\ldots, n\text{-}1$ and $j = 1,2$. Finally, m_{11} is connected to the point (ℓ/n, *s*, 0) and m_{12} is connected to the point (ℓ/n, -*s*, 0) each by an inextensible weightless rod of length ℓ/n. For notational convenience, we consider m_{01} and m_{02} as weightless masses at positions (0, *s*, 0) and (0, -*s*, 0), respectively.

For the spinning model, we use a spinning frame with basis $\hat{e}_x(t)$, $\hat{e}_y(t)$ and $\hat{e}_z(t)$ spinning at a constant rate ω about $\hat{e}_z(t)$ as shown in Fig. 1. For the gravitational model, we use a fixed reference frame with unit vector $\hat{e}_x(t)$ pointing in the direction of the gravitational field, $\hat{e}_z(t)$ as the direction perpendicular to the plane of the chain in still motion and $\hat{e}_y(t)$ as the third unit vector perpendicular to $\hat{e}_x(t)$ and $\hat{e}_x(t)$ obeying the usual ordering of the axes.

The deformation of the slender membrane as well as our discrete model can be characterized into three types: in-plane deformation, out-of-plane deformation (or we may refer to it as the bending deformation), and twisting deformation. We give formal definitions below in terms of angles by which the degree of deformation is measured. We assume that all motions are decoupled.

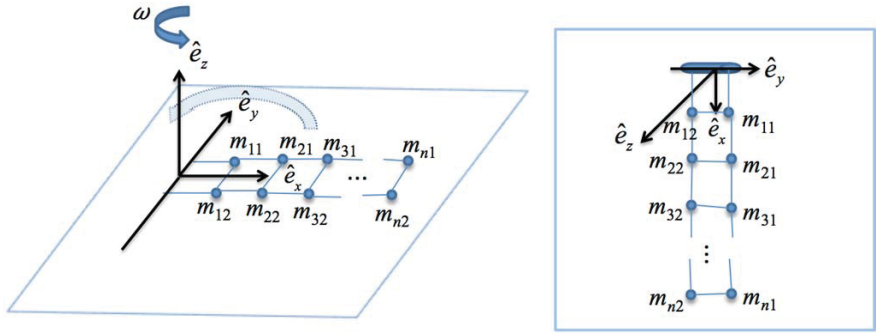

Fig. 1 The chain model under the centrifugal and the gravitational loads

A. In-plane deformation

In-plane deformation can be interpreted as side-ways movement. A property of such deformation is that movement remains in the same plane. We measure the degree of deformation at each sector by the angle θ_i, which is the angle between $\hat{e}_z(t)$ and the link connecting m_{i1} and $m_{(i-1)1}$ (or the link connecting m_{i2} and $m_{(i-1)2}$).

B. Out-of-plane deformation

Out-plane deformation is measured by the angle ϕ_i between the plane containing $\hat{e}_x(t)$, $\hat{e}_y(t)$ and the plane containing m_{i1}, $m_{(i-1)1}$, m_{i2}, and $m_{(i-1)2}$. The angle between between two planes is the angle between their normal vectors.

C. Twisting deformation

Twisting deformation is measured by the angle φ_i between the link connecting m_{i1} and m_{i2}, and its projection on the rotating *x*-*y* plane.

III. Dynamic Equations of Motion

The dynamic equations of motion for the spinning condition are listed in the first subsection. The formulations for the gravitational condition are derived in the second subsection.

A. Spinning condition

The equations of three different types of motions (uncoupled) in the spinning condition were derived in [1] through the Lagrangian equations. We list the equations and their linearized equations below. Equations below are taken from [1] with masses replaced by $m/2n$ and lengths replaced by ℓ/n. Dots on variables represent time derivatives.

1. *In-plane motion*

$$\frac{d}{dt}\left(\frac{\partial \mathcal{L}_{\text{in-plane,spin}}}{\partial \dot{\theta}_p}\right)-\left(\frac{\partial \mathcal{L}_{\text{in-plane,spin}}}{\partial \theta_p}\right)=\frac{m}{n}\left(\frac{\ell}{n}\right)^2\sum_{k=p}^{n}\sum_{j=1}^{k}\left(\cos(\theta_p-\theta_j)\ddot{\theta}_j+\sin(\theta_p-\theta_j)(\dot{\theta}_j+\omega)^2\right)=\tau_{\theta_p}. \tag{1}$$

2. *Out-of-plane motion*

$$\frac{d}{dt}\left(\frac{\partial \mathcal{L}_{\text{bending,spin}}}{\partial \dot{\phi}_p}\right)-\left(\frac{\partial \mathcal{L}_{\text{bending,spin}}}{\partial \phi_p}\right)=\frac{m}{n}\left(\frac{\ell}{n}\right)^2\sum_{k=p}^{n}\sum_{j=1}^{k}\left(\cos(\phi_p-\phi_j)\ddot{\phi}_j-\frac{1}{2}\sin(\phi_p-\phi_j)\dot{\phi}_j^2+\omega^2\cos\phi_j\sin\phi_p\right)=\tau_{\phi_p}. \tag{2}$$

3. *Twisting motion*

$$\begin{aligned}
&\frac{d}{dt}\left(\frac{\partial \mathcal{L}_{\text{twist,spin}}}{\partial \dot{\phi}_p}\right)-\left(\frac{\partial \mathcal{L}_{\text{twist,spin}}}{\partial \varphi_p}\right)\\
&=\frac{m}{n}s^2\ddot{\varphi}_p+\frac{m}{n}\omega^2 s^2\sin 2\varphi_p\\
&+\frac{m\ell}{n^2}s\sum_{k=p}^{n}\sum_{j=1}^{k}\left(\ddot{\psi}_j\sin\psi_j\tan\psi_p\cos\left(\frac{\varphi_p-\varphi_{p-1}}{2}\right)+\dot{\psi}_p\dot{\psi}_j\left(\sec^2\psi_p-1\right)\sin\psi_j\cos\left(\frac{\varphi_p-\varphi_{p-1}}{2}\right)\right)\\
&-\frac{m\ell}{n^2}s\sum_{k=p+1}^{n}\sum_{j=1}^{k}\left(\ddot{\psi}_j\sin\psi_j\tan\psi_p\cos\left(\frac{\varphi_{p+1}-\varphi_p}{2}\right)+\dot{\psi}_{p+1}\dot{\psi}_j\left(\sec^2\psi_{p+1}-1\right)\sin\psi_j\cos\left(\frac{\varphi_{p+1}-\varphi_p}{2}\right)\right)\\
&-\frac{m}{n}s^2\sum_{k=p}^{n}\sum_{j=1}^{k}\dot{\psi}_j\frac{\sin^2\psi_p}{\cos^3\psi_p}\sin\psi_j\cos^2\left(\frac{\varphi_p-\varphi_{p-1}}{2}\right)\left(\dot{\varphi}_p-\dot{\varphi}_{p-1}\right)\\
&+\frac{m}{n}s^2\sum_{k=p+1}^{n}\sum_{j=1}^{k}\dot{\psi}_j\frac{\sin^2\psi_{p+1}}{\cos^3\psi_{p+1}}\sin\psi_j\cos^2\left(\frac{\varphi_{p+1}-\varphi_p}{2}\right)\left(\dot{\varphi}_{p+1}-\dot{\varphi}_p\right)\\
&+\frac{m\ell}{n^2}s\sum_{k=p}^{n}\sum_{j=1}^{k}\left(\dot{\psi}_j^2+\omega^2\right)\tan\psi_p\cos\psi_j\cos\left(\frac{\varphi_p-\varphi_{p-1}}{2}\right)\\
&-\frac{m\ell}{n^2}s\sum_{k=p+1}^{n}\sum_{j=1}^{k}\left(\dot{\psi}_j^2+\omega^2\right)\tan\psi_{p+1}\cos\psi_j\cos\left(\frac{\varphi_{p+1}-\varphi_p}{2}\right)=\tau_{\varphi_p}
\end{aligned} \tag{3}$$

where $\frac{\ell}{n}\sin\psi_i=2s\sin\left(\frac{\varphi_i-\varphi_{i-1}}{2}\right)$ and p = 1, 2,…, n. The quantities τ_{θ_p}, τ_{ϕ_p}, and τ_{φ_p} represent the torques associated with the angles θ_p, ϕ_p, and φ_p, respectively.

4. *Linearized in-plane motion*

$$\begin{aligned}
&\frac{m\ell^2}{n^3}\begin{bmatrix} n & n-1 & \cdots & 1\\ n-1 & n-1 & \cdots & 1\\ \vdots & \vdots & \ddots & \vdots\\ 1 & 1 & \cdots & 1\end{bmatrix}\begin{bmatrix}\ddot{\theta}_1\\ \ddot{\theta}_2\\ \vdots\\ \ddot{\theta}_n\end{bmatrix}\\
&-\omega^2\frac{m\ell^2}{n^3}\left(\begin{bmatrix} n & n-1 & \cdots & 1\\ n-1 & n-1 & \cdots & 1\\ \vdots & \vdots & \ddots & \vdots\\ 1 & 1 & \cdots & 1\end{bmatrix}-\begin{bmatrix}\sum_{k=1}^{n}k & 0 & \cdots & 0\\ 0 & \sum_{k=2}^{n}k & \cdots & 0\\ \vdots & \vdots & \ddots & \vdots\\ 0 & 0 & \cdots & \sum_{k=n}^{n}k\end{bmatrix}\right)\begin{bmatrix}\theta_1\\ \theta_2\\ \vdots\\ \theta_n\end{bmatrix}=\begin{bmatrix}\tau_{\theta_1}\\ \tau_{\theta_2}\\ \vdots\\ \tau_{\theta_n}\end{bmatrix}.
\end{aligned} \tag{4}$$

5. Linearized out-of-plane motion

$$\frac{m\ell^2}{n^3}\begin{bmatrix} n & n-1 & \cdots & 1 \\ n-1 & n-1 & \cdots & 1 \\ \vdots & \vdots & \ddots & \vdots \\ 1 & 1 & \cdots & 1 \end{bmatrix}\begin{bmatrix} \ddot{\phi}_1 \\ \ddot{\phi}_2 \\ \vdots \\ \ddot{\phi}_n \end{bmatrix}+\omega^2\frac{m\ell^2}{n^3}\begin{bmatrix} \sum_{k=1}^{n}k & 0 & \cdots & 0 \\ 0 & \sum_{k=2}^{n}k & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \sum_{k=n}^{n}k \end{bmatrix}\begin{bmatrix} \phi_1 \\ \phi_2 \\ \vdots \\ \phi_n \end{bmatrix}=\begin{bmatrix} \tau_{\phi_1} \\ \tau_{\phi_2} \\ \vdots \\ \tau_{\phi_n} \end{bmatrix}. \tag{5}$$

6. Linearized twisting motion

$$\begin{aligned}&\frac{ms^2}{n}\mathbf{I}_{n\times n}\begin{bmatrix} \ddot{\varphi}_1 \\ \ddot{\varphi}_2 \\ \vdots \\ \ddot{\varphi}_n \end{bmatrix}\\ &+\frac{m\omega^2 s^2}{n}\begin{bmatrix} 1+\sum_{k=1}^{n}k+\sum_{k=2}^{n}k & -\sum_{k=2}^{n}k & 0 & \cdots & 0 \\ -\sum_{k=2}^{n}k & 1+\sum_{k=2}^{n}k+\sum_{k=3}^{n}k & -\sum_{k=3}^{n}k & \cdots & 0 \\ 0 & -\sum_{k=3}^{n}k & 1+\sum_{k=3}^{n}k+\sum_{k=4}^{n}k & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 1+\sum_{k=n}^{n}k \end{bmatrix}\begin{bmatrix} \varphi_1 \\ \varphi_2 \\ \vdots \\ \varphi_n \end{bmatrix}=\begin{bmatrix} \tau_{\varphi_1} \\ \tau_{\varphi_2} \\ \vdots \\ \tau_{\varphi_n} \end{bmatrix}.\end{aligned} \tag{6}$$

B. Gravitational condition

With the same method as in [1], we can derive equations of motions in the gravitational condition. For example, for the out-of-plan motion, the position vectors would be

$$\begin{aligned}\vec{\mathbf{R}}_{k1} &= \left[\sum_{i=1}^{k}\frac{\ell}{n}\cos\phi_i(t)\right]\hat{\mathbf{e}}_x+s\hat{\mathbf{e}}_y+\left[\sum_{i=1}^{k}\frac{\ell}{n}\sin\phi_i(t)\right]\hat{\mathbf{e}}_z \\ \vec{\mathbf{R}}_{k2} &= \left[\sum_{i=1}^{k}\frac{\ell}{n}\cos\phi_i(t)\right]\hat{\mathbf{e}}_x-s\hat{\mathbf{e}}_y+\left[\sum_{i=1}^{k}\frac{\ell}{n}\sin\phi_i(t)\right]\hat{\mathbf{e}}_z\end{aligned} \tag{7}$$

for $k=1,2,\cdots,n$. The time derivatives of the position vectors are

$$\begin{aligned}\dot{\vec{\mathbf{R}}}_{k1} &= \left[-\sum_{i=1}^{k}\frac{\ell}{n}\sin\phi_i(t)\dot{\phi}_i(t)\right]\hat{\mathbf{e}}_x+\left[\sum_{i=1}^{k}\frac{\ell}{n}\cos\phi_i(t)\dot{\phi}_i(t)\right]\hat{\mathbf{e}}_z \\ \dot{\vec{\mathbf{R}}}_{k2} &= \left[-\sum_{i=1}^{k}\frac{\ell}{n}\sin\phi_i(t)\dot{\phi}_i(t)\right]\hat{\mathbf{e}}_x+\left[\sum_{i=1}^{k}\frac{\ell}{n}\cos\phi_i(t)\dot{\phi}_i(t)\right]\hat{\mathbf{e}}_z.\end{aligned} \tag{8}$$

The Lagrangian $\mathcal{L}_{\text{bend,grav}}$ which is defined as the kinetic energy minus the potential energy is then

$$\mathcal{L}_{\text{bend,grav}} = \sum_{k=1}^{n} \frac{m}{2n} \left[\sum_{i=1}^{k} \sum_{j=1}^{k} \left(\frac{\ell}{n} \right)^2 \cos(\phi_i - \phi_j) \dot{\phi}_i \dot{\phi}_j \right] - 2 \sum_{k=1}^{n} \frac{m}{2n} g \sum_{i=1}^{k} \frac{\ell}{n} \cos\phi_i \tag{9}$$

We list equations of different motions under the gravitational load below.

1. *Gravitational In-plane motion*

$$\begin{aligned} &\frac{d}{dt}\left(\frac{\partial \mathcal{L}_{\text{in-plane,grav}}}{\partial \dot{\theta}_p} \right) - \left(\frac{\partial \mathcal{L}_{\text{in-plane,grav}}}{\partial \theta_p} \right) \\ &= \frac{m\ell^2}{n^3} \sum_{k=p}^{n} \sum_{j=1}^{k} \left(\ddot{\theta}_j \cos\left(\theta_j - \theta_p\right) - \dot{\theta}_j^2 \sin\left(\theta_j - \theta_p\right) \right) + g \frac{m\ell}{n^2} (n - p + 1) \sin\theta_p = \tau_{\theta p} \end{aligned} \tag{10}$$

2. *Gravitational Out-of-plane motion*

$$\begin{aligned} &\frac{d}{dt}\left(\frac{\partial \mathcal{L}_{\text{bend,grav}}}{\partial \dot{\phi}_p} \right) - \left(\frac{\partial \mathcal{L}_{\text{bend,grav}}}{\partial \phi_p} \right) \\ &= \frac{m\ell^2}{n^3} \sum_{k=p}^{n} \sum_{j=1}^{k} \left(\ddot{\phi}_j \cos\left(\phi_j - \phi_p\right) - \dot{\phi}_j^2 \sin\left(\phi_j - \phi_p\right) \right) + g \frac{m\ell}{n^2} (n - p + 1) \sin\phi_p = \tau_{\phi p} \end{aligned} \tag{11}$$

   Note that Eq. (11) is identical to Eq. (10) with the replacement of θ by ϕ. It occurs only in the gravitational condition but not in the spinning condition.

3. *Gravitational Twisting Motion*

$$\begin{aligned} &\frac{d}{dt}\frac{\partial \mathcal{L}_{\text{twist,grav}}}{\partial \dot{\varphi}_p} + \frac{\partial \mathcal{L}_{\text{twist,grav}}}{\partial \varphi_p} \\ &= \frac{m}{n} s^2 \ddot{\varphi}_p + \frac{m}{n} s^4 \sum_{k=p}^{n} \sum_{j=1}^{k} \frac{\sin(\Delta\varphi_p)\sin(\Delta\varphi_j)(\ddot{\varphi}_j - \ddot{\varphi}_{j-1})}{\sqrt{\alpha_p}\sqrt{\alpha_j}} \\ &+ \frac{m}{n} s^4 \sum_{k=p}^{n} \sum_{j=1}^{k} \frac{\sin(\Delta\varphi_p)\cos(\Delta\varphi_j)(\Delta\dot{\varphi}_j)^2}{\sqrt{\alpha_p}\sqrt{\alpha_j}} + \frac{m}{n} s^6 \sum_{k=p}^{n} \sum_{j=1}^{k} \frac{\sin(\Delta\varphi_p)\sin^2(\Delta\varphi_j)(\Delta\dot{\varphi}_j)^2}{\sqrt{\alpha_{p+1}}\left(\alpha_j\right)^{3/2}} \\ &- \frac{m}{n} s^4 \sum_{k=p+1}^{n} \sum_{j=1}^{k} \frac{\sin(\Delta\varphi_{p+1})\sin(\Delta\varphi_j)(\Delta\ddot{\varphi}_j)}{\sqrt{\alpha_{p+1}}\sqrt{\alpha_j}} - \frac{m}{n} s^4 \sum_{k=p+1}^{n} \sum_{j=1}^{k} \frac{\sin(\Delta\varphi_{p+1})\cos(\Delta\varphi_j)(\Delta\dot{\varphi}_j)^2}{\sqrt{\alpha_{p+1}}\sqrt{\alpha_j}} \\ &- \frac{m}{n} s^6 \sum_{k=p+1}^{n} \sum_{j=1}^{k} \frac{\sin(\Delta\varphi_{p+1})\sin^2(\Delta\varphi_j)(\Delta\dot{\varphi}_j)^2}{\sqrt{\alpha_{p+1}}\left(\alpha_j\right)^{3/2}} + \frac{m}{n} g s^2 (n - p + 1) \frac{\sin(\Delta\varphi_p)}{\sqrt{\alpha_p}} \\ &- \frac{m}{n} g s^2 (n - p) \frac{\sin(\Delta\varphi_{p+1})}{\sqrt{\alpha_{p+1}}} = \tau_{\varphi_p} \end{aligned} \tag{12}$$

where

$$\Delta\varphi_k = \varphi_k - \varphi_{k-1}; \quad k = p,\ p+1,\ j$$

$$\alpha_k = \left(\frac{\ell}{n} \right)^2 + 2s^2 \left(\cos(\Delta\varphi_k) - 1 \right)$$

4. *Linearized Gravitational In-plane Motion*

$$\frac{m\ell^2}{n^3} \begin{bmatrix} n & (n-1) & \cdots & 1 \\ (n-1) & (n-1) & \cdots & 1 \\ \vdots & \vdots & \ddots & \vdots \\ 1 & 1 & \cdots & 1 \end{bmatrix} \begin{bmatrix} \ddot{\theta}_1 \\ \ddot{\theta}_2 \\ \vdots \\ \ddot{\theta}_n \end{bmatrix} + \frac{m\ell}{n^2} g \begin{bmatrix} n & 0 & \cdots & 0 \\ 0 & (n-1) & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{bmatrix} \begin{bmatrix} \theta_1 \\ \theta_2 \\ \vdots \\ \theta_n \end{bmatrix} = \begin{bmatrix} \tau_{\theta_1} \\ \tau_{\theta_2} \\ \vdots \\ \tau_{\theta_n} \end{bmatrix} \tag{13}$$

5. Linearized Gravitational Out-of-plane Motion

$$\frac{m\ell^2}{n^3}\begin{bmatrix} n & (n-1) & \cdots & 1 \\ (n-1) & (n-1) & \cdots & 1 \\ \vdots & \vdots & \ddots & \vdots \\ 1 & 1 & \cdots & 1 \end{bmatrix}\begin{bmatrix} \ddot{\phi}_1 \\ \ddot{\phi}_2 \\ \vdots \\ \ddot{\phi}_n \end{bmatrix}+\frac{m\ell}{n^2}g\begin{bmatrix} n & 0 & \cdots & 0 \\ 0 & (n-1) & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{bmatrix}\begin{bmatrix} \phi_1 \\ \phi_2 \\ \vdots \\ \phi_n \end{bmatrix}=\begin{bmatrix} \tau_{\phi_1} \\ \tau_{\phi_2} \\ \vdots \\ \tau_{\phi_n} \end{bmatrix} \tag{14}$$

6. Linearized Gravitational Twisting Motion

$$\begin{aligned}&\frac{m}{n}s^2\begin{bmatrix} 1 & 0 & 0 & \cdots & 0 \\ 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 1 \end{bmatrix}\begin{bmatrix} \ddot{\varphi}_1 \\ \ddot{\varphi}_2 \\ \ddot{\varphi}_3 \\ \vdots \\ \ddot{\varphi}_n \end{bmatrix} \\ &+\frac{m}{\ell}gs^2\begin{bmatrix} n+(n-1) & -(n-1) & 0 & 0 & \cdots & 0 \\ -(n-1) & (n-1)+(n-2) & -(n-2) & 0 & \cdots & 0 \\ 0 & -(n-2) & (n-2)+(n-3) & -(n-3) & \cdots & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & 0 \\ 0 & \cdots & \cdots & -2 & 2+1 & -1 \\ 0 & 0 & \cdots & 0 & -1 & 1 \end{bmatrix}\begin{bmatrix} \varphi_1 \\ \varphi_2 \\ \varphi_3 \\ \vdots \\ \varphi_n \end{bmatrix}=\begin{bmatrix} \tau_{\varphi_1} \\ \tau_{\varphi_2} \\ \tau_{\varphi_3} \\ \vdots \\ \tau_{\varphi_n} \end{bmatrix}\end{aligned} \tag{15}$$

IV. Continuous Equations

Equations for the continuum of the discrete model can be derived via limiting methods that are commonly used in vibrational problems. We show here the derivation of continuous equations for the linearized gravitational bending equations.

According to equation (14), the linearized bending equation under the gravitational load is

$$M_\phi\ddot{\phi}+K_\phi\phi=\tau_\phi \tag{16}$$

where

$$M_\phi=\frac{m\ell^2}{n^3}\begin{bmatrix} n & (n-1) & \cdots & 1 \\ (n-1) & (n-1) & \cdots & 1 \\ \vdots & \vdots & \ddots & \vdots \\ 1 & 1 & \cdots & 1 \end{bmatrix},\ K_\phi=\frac{m\ell}{n^2}g\begin{bmatrix} n & 0 & \cdots & 0 \\ 0 & (n-1) & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{bmatrix},\ \tau=\begin{bmatrix} \tau_{\phi_1} \\ \tau_{\phi_2} \\ \vdots \\ \tau_{\phi_n} \end{bmatrix}. \tag{17}$$

Rewriting the equation as

$$\ddot{\phi}+M_\phi^{-1}K_\phi\phi=M^{-1}\tau_\phi \tag{18}$$

we have that the *p*th equation of motion is

$$\ddot{\phi}_p-\left(\frac{n}{\ell}g\right)(n+2-p)\phi_{p-1}+2\left(\frac{n}{\ell}g\right)(n+1-p)\phi_p-\left(\frac{n}{\ell}g\right)(n-p)\phi_{p+1}=\frac{-n^3}{\ell^2m}\tau_{\phi_{p-1}}+\frac{2n^3}{\ell^2m}\tau_{\phi_p}-\frac{n^3}{\ell^2m}\tau_{\phi_{p+1}}. \tag{19}$$

Reorganizing the equation yields

$$\ddot{\phi}_p + 2\left(\frac{n}{\ell}g\right)\left(\phi_p - \phi_{p-1}\right) - \left(\frac{n}{\ell}g\right)(n-p)\left(\phi_{p+1} - 2\phi_p + \phi_{p-1}\right) = \frac{-n^3}{\ell^2 m}\left(\tau_{\phi_{p+1}} - 2\tau_{\phi_p} + \tau_{\phi_{p-1}}\right). \tag{20}$$

Now consider each small increment of x, as $\Delta x = \frac{\ell}{n}$ and $x_p = \frac{\ell p}{n}$, then

$$\ddot{\phi}_p + 2g\left(\frac{\phi_p - \phi_{p-1}}{\Delta x}\right) - g(\ell - x_p)\left(\frac{\frac{\phi_{p+1} - \phi_p}{\Delta x} - \frac{\phi_p - \phi_{p-1}}{\Delta x}}{\Delta x}\right) = \frac{-n}{m}\left(\frac{\frac{\tau_{\phi_{p+1}} - \tau_{\phi_p}}{\Delta x} - \frac{\tau_{\phi_p} - \tau_{\phi_{p-1}}}{\Delta x}}{\Delta x}\right). \tag{21}$$

Let us now consider angle ϕ and torque τ_ϕ as functions of x

$$\begin{aligned}\ddot{\phi}(x_p) + 2g\left(\frac{\phi(x_p) - \phi(x_p - \Delta x)}{\Delta x}\right) - g(\ell - x_p)\left(\frac{\frac{\phi(x_p + \Delta x) - \phi(x_p)}{\Delta x} - \frac{\phi(x_p) - \phi(x_p - \Delta x)}{\Delta x}}{\Delta x}\right) \\ = \frac{-n}{m}\left(\frac{\frac{\tau_\phi(x_p + \Delta x) - \tau_\phi(x_p)}{\Delta x} - \frac{\tau_\phi(x_p) - \tau_\phi(x_p)}{\Delta x}}{\Delta x}\right).\end{aligned} \tag{22}$$

Consider x_p as any point $x \in [0, \ell]$, and let $\Delta x \to 0$, we get

$$\frac{\partial^2 \phi(x,t)}{\partial t^2} + 2g\frac{\partial \phi(x,t)}{\partial x} - g(\ell - x)\frac{\partial^2 \phi(x,t)}{\partial x^2} = -\frac{1}{\rho}\frac{\partial^2 \overline{\tau}_\phi(x)}{\partial x^2}. \tag{23}$$

We compare the continuous equations in both spinning and gravitational conditions in **Table 1**. Note that partial differential equations for the in-plane and out-of-plane motions of the hanging model are identical but not true for the spinning model. Thus, we suggest that the oscillation between in-plane and out-of-plane motions may take place in the hanging model but may not occur in the spinning model.

Table 1 Comparison of continuous equations in the spinning and gravitational conditions

In-plane	Spinning	$\frac{\partial^2 \theta(x,t)}{\partial t^2} + 2\omega^2 x\frac{\partial \theta(x,t)}{\partial x} - \omega^2\left(\frac{\ell^2 - x^2}{2}\right)\frac{\partial^2 \theta(x,t)}{\partial x^2} = -\frac{1}{\rho}\frac{\partial^2 \overline{\tau}_\theta}{\partial x^2}$
	Hanging	$\frac{\partial^2 \theta(x,t)}{\partial t^2} - g(\ell - x)\frac{\partial^2 \theta(x,t)}{\partial x^2} + 2g\frac{\partial \theta(x,t)}{\partial x} = -\frac{1}{\rho}\frac{\partial^2 \overline{\tau}_\theta}{\partial x^2}$
Out-of-plane	Spinning	$\frac{\partial^2 \phi(x,t)}{\partial t^2} + \omega^2\phi(x,t) + 2\omega^2 x\frac{\partial \phi(x,t)}{\partial x} - \omega^2\left(\frac{\ell^2 - x^2}{2}\right)\frac{\partial^2 \phi(x,t)}{\partial x^2} = -\frac{1}{\rho}\frac{\partial^2 \overline{\tau}_\phi}{\partial x^2}$
	Hanging	$\frac{\partial^2 \phi(x,t)}{\partial t^2} - g(\ell - x)\frac{\partial^2 \phi(x,t)}{\partial x^2} + 2g\frac{\partial \phi(x,t)}{\partial x} = -\frac{1}{\rho}\frac{\partial^2 \overline{\tau}_\phi}{\partial x^2}$
Twisting	Spinning	$\frac{\partial^2 \varphi(x,t)}{\partial t^2} + \omega^2\varphi(x,t) - \omega^2\left(\frac{\ell^2 - x^2}{2}\right)\frac{\partial^2 \varphi(x,t)}{\partial x^2} = \frac{1}{s^2\rho}\overline{\tau}_\varphi$
	Hanging	$\frac{\partial^2 \varphi(x,t)}{\partial t^2} - g(\ell - x)\frac{\partial^2 \varphi(x,t)}{\partial x^2} + g\frac{\partial \varphi(x,t)}{\partial x} = \frac{1}{s^2\rho}\overline{\tau}_\varphi$

V. Natural Frequencies Observed from the Linearized Equations

The frequencies of the linearized equations for three different types of spinning motions were calculated in [3] and we list the results here. An interesting observation is that all frequencies shown in

Table 2 Frequencies of the membrane model in the spinning coincide with the frequencies in MacNeals equations [2] in each degree of freedom. This spurred our further interest in investigating the frequencies of motion under the gravitational load. In Table 3, we list the frequencies obtained from the linearized equations of the gravitational equations. We observe that the natural frequencies of the in-plane, bending, and twisting motions for the gravitational model are identical in each mode but not true for the spinning model.

Table 2 Frequencies of the membrane model in the spinning condition

Sets of masses (degrees of freedom)	Type of motion	Frequencies
1 set of masses	In-plane	0
	Out-of-plane	ω
	Twisting	$\omega\sqrt{2}$
2 sets of masses	In-plane	$0,\ \omega\sqrt{5}$
	Out-of-plane	$\omega,\ \omega\sqrt{6}$
	Twisting	$\omega\sqrt{2},\ \omega\sqrt{7}$
3 sets of masses	In-plane	$0,\ \omega\sqrt{5},\ \omega\sqrt{14}$
	Out-of-plane	$\omega,\ \omega\sqrt{6},\ \omega\sqrt{15}$
	Twisting	$\omega\sqrt{2},\ \omega\sqrt{7},\ \omega\sqrt{16}$
MacNeal's equations	In-plane	$\omega\sqrt{\frac{n(n+1)}{2}-1},\ n=1,3,5,\rightleftharpoons$
	Out-of-plane	$\omega\sqrt{\frac{n(n+1)}{2}},\ n=1,3,5,\rightleftharpoons$
	Twisting	$\omega\sqrt{\frac{n(n+1)}{2}+1},\ n=1,3,5,\rightleftharpoons$

Table 3 Frequencies of the membrane model in the gravitational condition

Sets of masses (degrees of freedom)	Type of motion	Frequencies
1 set of masses	In-plane	$\sqrt{g/\ell}$
	Out-of-plane	$\sqrt{g/\ell}$
	Twisting	$\sqrt{g/\ell}$
2 sets of masses	In-plane	$1.08239\sqrt{g/\ell},\ 2.61313\sqrt{g/\ell}$
	Out-of-plane	$1.08239\sqrt{g/\ell},\ 2.61313\sqrt{g/\ell}$
	Twisting	$1.08239\sqrt{g/\ell},\ 2.61313\sqrt{g/\ell}$
3 sets of masses	In-plane	$1.11684\sqrt{g/\ell},\ 2.62352\sqrt{g/\ell},\ 4.34394\sqrt{g/\ell}$
	Out-of-plane	$1.11684\sqrt{g/\ell},\ 2.62352\sqrt{g/\ell},\ 4.34394\sqrt{g/\ell}$
	Twisting	$1.11684\sqrt{g/\ell},\ 2.62352\sqrt{g/\ell},\ 4.34394\sqrt{g/\ell}$
Continuous equations	In-plane	To be determined
	Out-of-plane	To be determined
	Twisting	To be determined

VI. Simulation Results for a 3-dof Hanging Model

In Fig. 2, we show a schematic diagram of the three-degree-of-freedom (3-dof) along with some notations defined on the right-hand side. The quantity ϕ_1 is the angle between the plane containing $\hat{\mathbf{e}}_x$ and $\hat{\mathbf{e}}_y$, and the plane containing m_{01}, m_{02}, m_{11} and m_{12} where m_{01}, m_{02} are virtual masses implying the positions of the root of the model. The angle between two planes is the angle between two normal vectors. Similarly, ϕ_2 is the angle between the plane containing $\hat{\mathbf{e}}_x$ and $\hat{\mathbf{e}}_y$, and the plane containing m_{11}, m_{12}, m_{21} and m_{22}. The angle ϕ_3 is similar to ϕ_2 with m_{11}, m_{12}, m_{21} and m_{22} replaced by m_{21}, m_{22}, m_{31} and m_{32}.

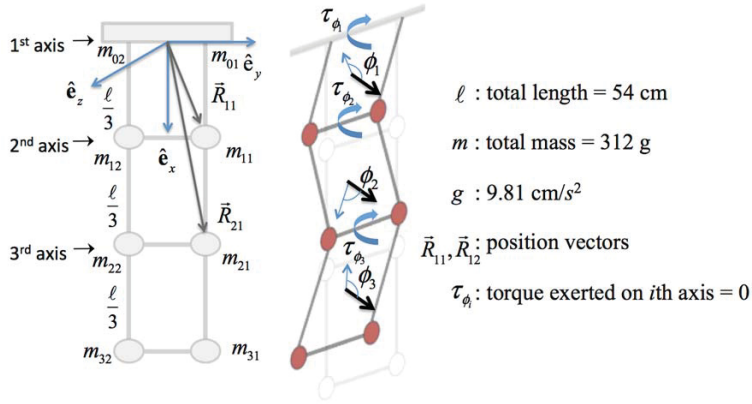

Fig. 2 Schematic diagram of the out-of-plane motion for the 3-dof hanging system

Let the initial angular displacements and velocities be

$$\left[\ \phi_1(0)\ \ \phi_2(0)\ \ \phi_3(0)\ \ \dot{\phi}_1(0)\ \ \dot{\phi}_2(0)\ \ \dot{\phi}_2(0)\ \right]=\left[\ 0.2\ \ 0.05\ \ 0.1\ \ 0\ \ 0\ \ 0\ \right]$$

Free-decay responses of theoretically derived linear and nonlinear models for the 3-dof hanging system are shown in Fig. 3 where the solid line is the time history of the m_{11} displacement from the linear model, Eq. (14), and the dashed line from the nonlinear model, Eq. (11). The nonlinear free-decay responses for the angles ϕ_1, ϕ_2, and ϕ_3 were first generated by integrating Eq. (11) with the MATLAB function file "ODE45". Equation (7) was then used to compute the out-of-plane positions of the masses m_{11}, m_{21}, m_{31} or m_{12}, m_{22}, m_{32}. No significant discrepancy in amplitude and frequency is observed, which implies that the frequencies computed from the linear model may be useful for studying/comparing the experiment and simulation results. Similar time responses (not shown) with larger displacement were obtained for the out-of-plane positions of m_{21} and m_{31}.

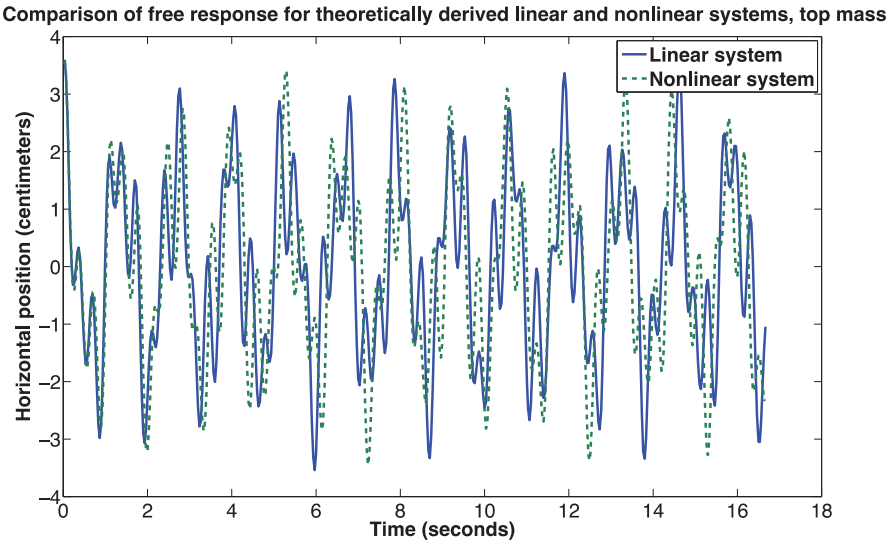

Fig. 3 Free-decay responses for the linear and nonlinear models.

VII. Experiment Results for a 3-dof Model

The laboratory test structure (Fig. 4) for validation of our analytical studies consists of a vertical 3-dof hanging model and a digital camcorder for deflection measurements. The overall dimensions are 54 cm in height and 15 cm in width, and the total mass is approximately 312 g. The structure is hinged at its top to a board.

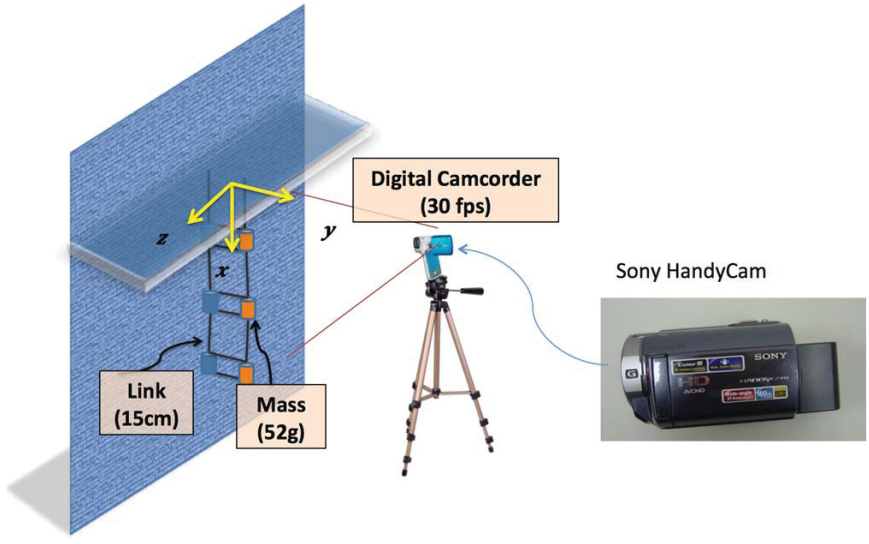

Fig. 4 Hanging mass experiment setup

A flow diagram of building the three-mass-set hang model is shown in Fig. 5. Each horizontal pair required two straws to prevent masses from additional swaying. Each discrete mass was made by stacking 13 coins (New Taiwan Dollar), which were wrapped with a tape, and then tied up with a string. Putting a double-string through a straw to link one coin-mass to another completed the model. Colors of the masses and links were adjusted for an easy capture by a digital camcorder to make mass tracking more accurate. A typical image of three masses captured by the digital camcorder of 1075 by 1111 pixels is shown in Fig. 6.

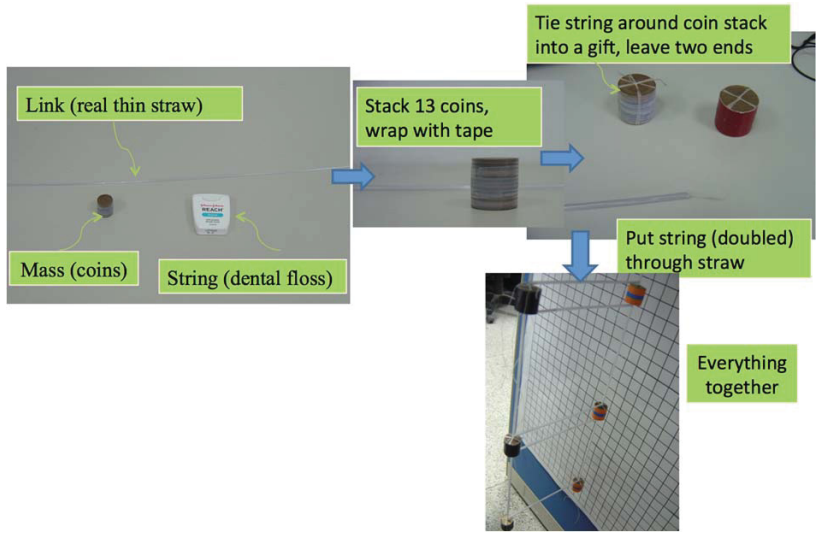

Fig. 5 A flow diagram for making the coupled pendulum model.

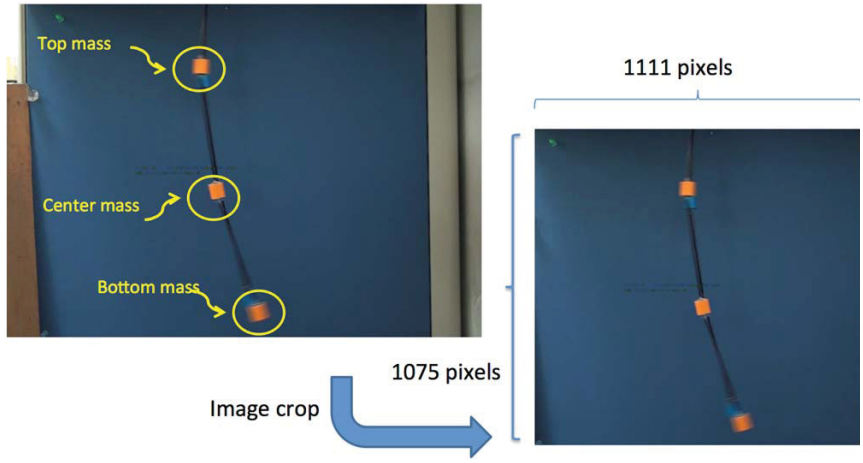

Fig. 6 Image capture by a digital camcorder

In Fig. 7, we demonstrate a total of 4810 data points of the experimental free-decay responses for the top mass m_{11}, the center mass m_{21}, and the bottom mass m_{31} with the initial displacements given by a hand and the sample interval 1/30 sec. The bottom-mass displacement is approximately twice larger than the top-mass displacement. Note that the initial conditions are different from the theoretical simulation for analyzing linear and nonlinear behavior.

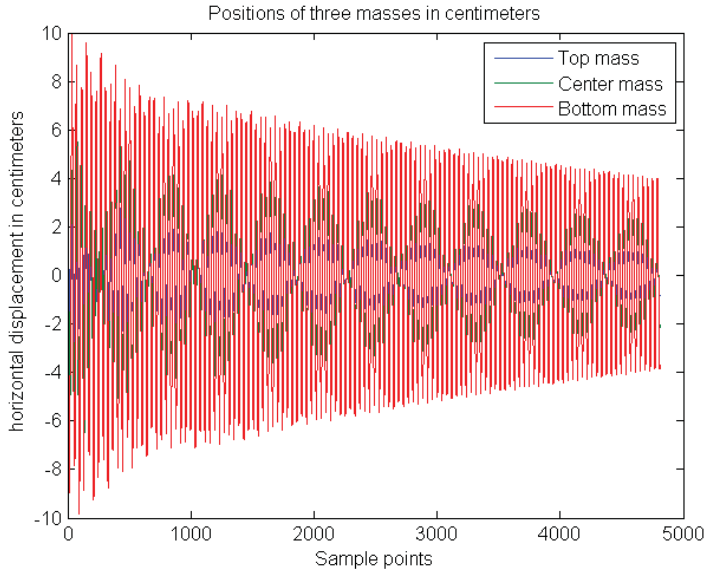

Fig. 7 Experimental data rescaled to actual size (cm), center adjusted to zero

In Fig. 8, we give the flow diagram of using the Eigensystem Realization Algorithm (ERA) [14] on the experimental data shown in Fig. 7. The ERA method developed at NASA Langley Research Center is commonly used for producing a state-space model from experimental data for structural analysis and controller design. Each ERA was performed using a single matrix (4810 x 3) of data from the three response measurements and one initial condition at a time. The first step is to convert the positions of three masses in pixels to actual displacement response $y(k)$ where k is the time index. The second step is to form two Hankel matrices $H(0)$ and $H(1)$ consisting of the response $y(k)$, and perform singular value decomposition on Hankel matrix $H(0)$ to determine the system order by truncating relatively small singular values. The third step is to determine the discrete-time state-space matrices A (state matrix), B (input matrix), C (output matrix), and D (direct transmission matrix). For the free-decay response, the state matrices A, C, and $x(0)$ (initial condition) are identified instead of A, B, C, and D. The last step is to transform A, C, and $x(0)$ into modal coordinates for identification of system damping ratio, frequencies, and mode shape, and for model reduction if necessary for

controller designs. The identified matrices/vectors A, C, and $x(0)$ describe a linear model that produces an "optimal-linear" map from the initial condition to the measured free-decay response.

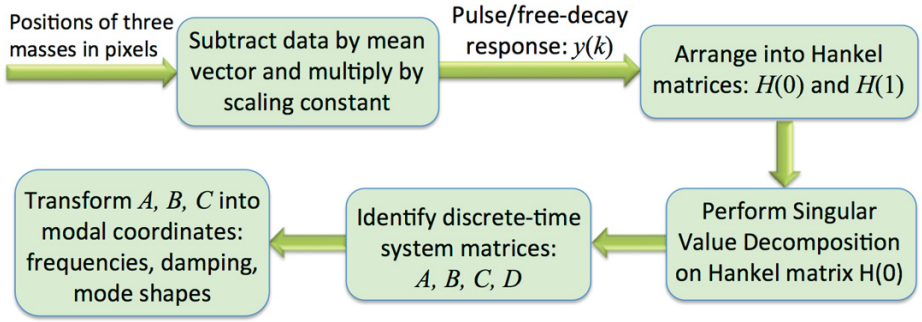

Fig. 8 Flow diagram of using ERA on experimental data

We performed system identification with two different sets of data in length (2811 and 4810 points) for the purpose of comparison. The theoretical and ERA-identified system frequencies and damping ratios are compared in Table 4. The identified results agree with the theoretical results in frequencies. The final aspect of the ERA method available for assessing identification accuracy is the process of data reconstruction. This procedure consists of comparing the original free-decay time histories with those calculated using the ERA-reduced model, which contains 3 modes. If the ERA modal decomposition process is performed accurately, the reconstruction results should closely match the original data. A typical top-mass actual time history is compared with a 3-mode reconstruction in Figs. 9 and 10, and the reconstruction result is seen closely following the original data in both amplitude and phase.

Table 4 Analytical and identified system damping ratios and frequencies

Theoretical frequencies	Identified frequency using 2811 points		Identified frequency using 4810 points	
Freq. (Hz)	Freq. (Hz)	Damping (%)	Freq. (Hz)	Damping (%)
0.7576	0.763	0.117	0.763	0.111
1.7797	1.798	0.342	1.798	0.340
2.9467	3.088	0.342	3.088	0.342

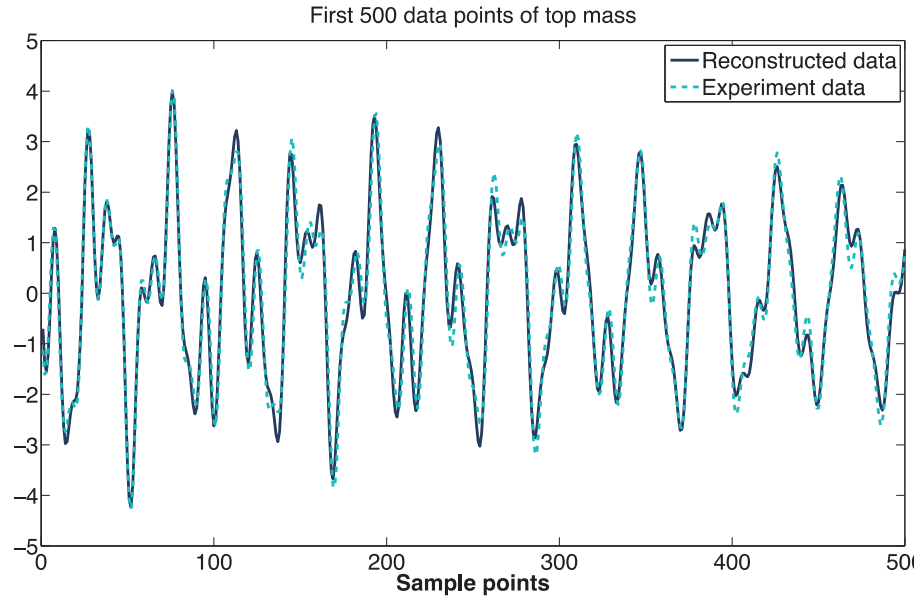

Fig. 9 Comparison of the top-mass first 500 measurements with a 3-mode reconstruction.

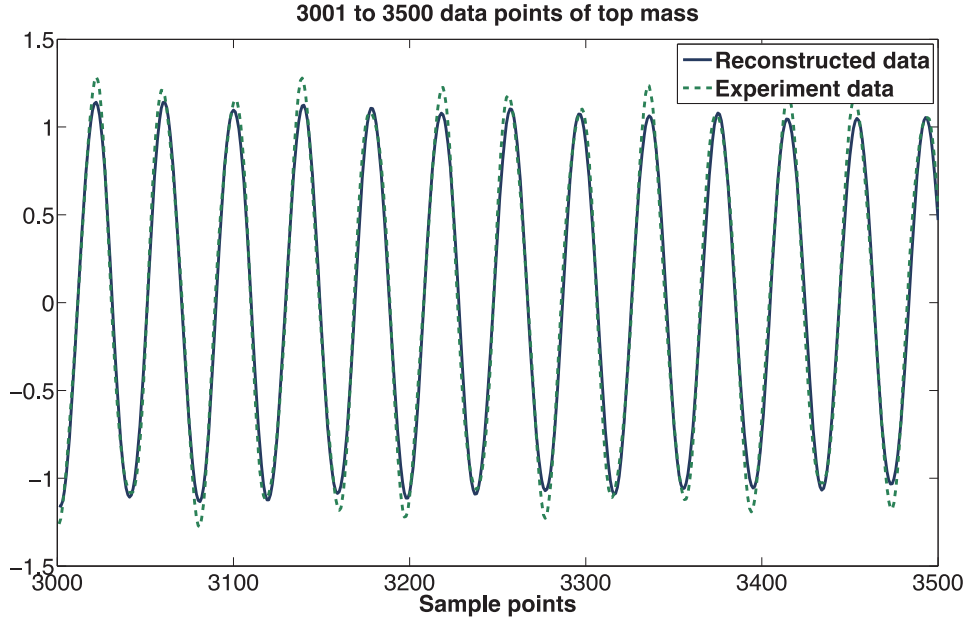

Fig. 10 Comparison of the top-mass 3001-3500 measurements with a 3-mode reconstruction

VIII. Conclusion

We were motivated by hoping to replace some experiments in the spinning setting by experiments in the gravitational setting, which is more plausible to perform on earth. The bending (out-of-plane) motions of a coupled pendulum model of a heliogyro blade were studied analytically and experimentally. Although our efforts to make this conversion were not as expected, we discovered some interesting properties on the motion of the multi-link coupled pendulum, namely, the linearized (motion in small angles) system of the twisting and bending motions give identical frequencies. This means that resonance is likely to occur in small angle movements once either type of motion is present in conjunction with offset centers of structural geometry and external force. We derived the equations of motion by using the Lagrangian equations, and provided computer simulations and experimental results to support our studies.

Acknowledgments

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Challenges Associated with System Identification and Control of a Heliogyro Membrane Blade

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Initial studies of a Heliogyro membrane blade have shown significant nonlinearities in the equations of motion. The objective of this paper is to demonstrate a system identification method applicable for describing the dynamic behavior of a Heliogyro membrane blade and to use it for control design. The approach is to use simulation data generated from linear and nonlinear models of a spinning membrane to identify linear models using time and frequency domain techniques. A controller design is introduced and its stability robustness is described. Results from the implementation of the controller shows that the achievable level of damping is diminished by the system nonlinearities.

Nomenclature

A_c = continuous-time state matrix

B_c = continuous-time input matrix

C = output matrix

D = direct transmission matrix

$D(s)$ = distance from the uncertainty circle to the critical point using a controlled plant in Nyquist plot

F_c = control gain matrix

G_c = observer gain matrix

ℓ = total length of the membrane blade, meter

m = lump mass of the membrane blade, kilogram

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n = degree of freedom (number of repeated cells for the discretized membrane)

r = span wise length between two mass sets, meter

S_m = stability margin

s = half membrane chord width, meter

T = kinetic energy, J

V = potential energy, J

$\hat{x}$ = observer state

Ξ = proportional damping matrix

$\Delta\lambda_R$ = distance from the uncertainty circle to the critical point using a controlled plant in Nyquist plot

ϕ_i = ith angular displacement of out-of-plane/bending motion, radian; $i = 1,2,\ldots n$

φ_i = ith angular displacement of twisting motion, radian; $i = 1,2,\ldots n$

ω = spinning rate along the z axis, radian/second

I. Introduction

The basic concept of solar sails is to use the direct momentum transfer of photons to generate propellant-free spacecraft propulsion. A classical/representative solar sail architecture is the heliogyro first proposed by MacNeal [1]. The heliogyro consists of several extremely high aspect-ratio reflective membrane 'blades' spinning about a central spacecraft hub. The membrane blades are extremely flexible so it can develop considerable coupled in-plane (spinning plane), out-of-plane (flap/bending), and twisting (pitch angle) motions. The dynamic behavior of such a lightweight thin membrane is a considerable challenge for system identification, and structural and control analyses. To fully understand the dynamic behaviors of coupled motions, decoupled models [2,3] are developed to help us easily analyze the modal frequencies of the system individually. Stability and effectiveness of a torque-source motor at the blade root to control blade pitch have been examined in [4-6], giving a promising picture that the blade twist control for the heliogyro is feasible.

System identification is a way to build a mathematical model from experimental data for dynamic analysis and controller designs of an engineering system. In this paper, system identification techniques [7-8] are used to identify models while the system is driven with inputs using various amplitudes. Because the system is nonlinear, changing input amplitudes results in identified models which are functions of the input amplitude. Principal gains of the transfer function are then calculated to determine an uncertainty envelope for the identified nominal model due to input amplitude variations. The envelope gives information on how much amplitude

variation occurs in the model, for the input range chosen. Also shown is the variation in modal frequencies as the system becomes more nonlinear. This is important for controlling purposes because one can use the uncertainty envelope along with the Nyquist criterion to conduct stability assessments of the nominal plant. To demonstrate the process, a full state feedback controller is then designed on the basis of the nominal linear model to increase the damping ratio for all the system modes. In this example, uncertainty bounds along with controllers designed using the nominal linear plant are used to add damping to all flexible modes of a nonlinear plant while maintaining stability. At last, the stability of the controlled nonlinear plants is described by numerically calculating the Nyquist.

II. Model Description

A spinning thin membrane is modeled by using a set of discrete lumped mass connected with inextensible links. The total mass of the membrane is assumed to be evenly distributed among the lumped masses. The governing equations for three different types of motions (uncoupled in-plane, out-of-plane bending, and twisting) for a spinning membrane were derived in [3] using the Lagrangian equations.

As shown in Fig. 1, coupled bending-twisting deformation is measured by the bending (out-of-plane) angle ϕ_1, and the twisting angle φ_1 along with its associated angle ψ_1. A spinning frame is used to describe the dynamic behavior with basis $\vec{e}_x(t)$, $\vec{e}_y(t)$ and $\vec{e}_z(t)$ spinning at a constant rate ω about $\vec{e}_z(t)$. The Lagrangian for the coupled bending and twisting motion was derived in [9]. In this approximation, each discrete lumped mass is defined as $m = \frac{M}{2n}$ (M is the total mass of the membrane blade), the span wise length between two mass sets is $r = \frac{\ell}{n}$, (ℓ is its total length), s is half width of the membrane, and n is the number of repeated cells for the discretized model. Dots on variables correspond to differentiation with respect to time.

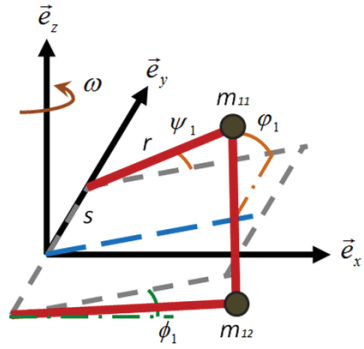

Fig. 1 Coupled out-of-plane and twisting motion for the spinning model

The Lagrangian of the system [9] is

$$\begin{aligned} L &= T - V \\ &= m[r^2\dot{\phi}_1^2\cos^2(\frac{s\varphi_1}{r})\sin^2\phi_1 + s^2\dot{\phi}_1^2\sin^2\varphi_1\sin^2\phi_1 + r^2\dot{\phi}_1^2\cos^2\phi_1 \\ &+2rs\dot{\varphi}_1\dot{\phi}_1\cos(\frac{s\varphi_1}{r})\sin(\frac{s\varphi_1}{r})\cos\phi_1\sin\phi_1 - 2s^2\dot{\varphi}_1\dot{\phi}_1\cos\varphi_1\sin\varphi_1\cos\phi_1\sin\phi_1 \\ &+s^2\dot{\varphi}_1^2\sin^2(\frac{s\varphi_1}{r})\cos^2\phi_1 + w^2r^2\cos^2(\frac{s\varphi_1}{r})\cos^2\phi_1 \\ &+s^2\dot{\varphi}_1^2\cos^2\varphi_1\cos^2\phi_1 + s^2\dot{\varphi}_1^2\sin^2\varphi_1 + w^2s^2\cos^2\varphi_1] \end{aligned} \tag{1}$$

Application of Lagrange's equations of motion

$$\frac{d}{dt}\frac{\partial L}{\partial\dot{\phi}_1} - \frac{\partial L}{\partial\phi_1} = \tau_{\phi_1} \tag{2}$$

$$\frac{d}{dt}\frac{\partial L}{\partial\dot{\varphi}_1} - \frac{\partial L}{\partial\varphi_1} = \tau_{\varphi_1} \tag{3}$$

yields the nonlinear equations of motion in the form of

$$M(\theta_1)\ddot{\theta}_1 + \mathbf{k}(\theta_1,\dot{\theta}_1) = \tau_1 \tag{4}$$

where the state vector θ_1 and the external torque vector τ_1 are

$$\theta_1 = \begin{bmatrix}\phi_1\\ \varphi_1\end{bmatrix};\quad \tau = \begin{bmatrix}\tau_{\phi_1}\\ \tau_{\varphi_1}\end{bmatrix} \tag{5}$$

and the positive-definite mass matrix $M(\theta_1)$ of 2×2 is a nonlinear function of θ_1 and the vector $\mathbf{k}(\theta_1,\dot{\theta}_1)$ of 2×1 is a nonlinear function of θ_1 and $\dot{\theta}_1$. The coupled bending-twisting model can be linearized to be in the form of

$$M\ddot{\theta}_1 + K\theta_1 = \tau_1 \tag{6}$$

where M and K are constant matrices

$$M = 2m\begin{bmatrix}r^2 & 0\\ 0 & s^2\end{bmatrix};\quad K = 2m\omega^2\begin{bmatrix}r^2 & 0\\ 0 & 2s^2\end{bmatrix} \tag{7}$$

Equations (6) and (7) imply that the bending and twisting motions are decoupled in the linearized model, and the bending frequency is ω and the twisting frequency is $\sqrt{\omega}$. In theory, this discrete-mass/multi-link modeling approach [2,3] can be extended to any arbitrary degrees of freedom. Table 1 lists the frequencies of the bending and twisting motions of the linearized model [3]. No repeated frequencies for any degree of freedom exist, indicating that no oscillation is expected to take place between bending and twisting motions.

Table 1 Frequencies of the membrane model up to 3 sets of masses

Sets of masses (degrees of freedom)	Type of motion	Frequency
1 set of masses	Bending	ω
	Twisting	$\omega\sqrt{2}$
2 set of masses	Bending	$\omega, \omega\sqrt{6}$
	Twisting	$\omega\sqrt{2}, \omega\sqrt{7}$
3 set of masses	Bending	$\omega, \omega\sqrt{6}, \omega\sqrt{15}$
	Twisting	$\omega\sqrt{2}, \omega\sqrt{7}, \omega\sqrt{16}$

III. Simulation of Spinning Membrane Model

Table 2 lists the basic properties of the Heliogyro membrane used for simulation. To simulate the response of the membrane, input torques in bending and twisting were applied to the model. The excitation signal was a burst random with 2048 points sampled at a sampling rate of 10 sec with the input on for 10% of the time. Input torques magnitudes were limited to maintain response angles for bending within $-30° \le \phi_i \le 30°$, and for twisting within $-30° \le \varphi_i \le 30°$, which corresponded to an input torque smaller than 3.25 $\mathrm{N \cdot M}$ for bending and 0.000125 $\mathrm{N \cdot M}$ for twisting. The input and output signals were then used to compute the linear frequency response functions to approximate the nonlinear system. In our simulation model, coupled bending-twisting motions were examined separately on the bending and twisting responses. For simplicity in the following example, we use only the bending case.

Table 2 Summary of simulation parameters

Feature	
Spinning Rate, ω	0.5 rpm
Number of mass sets	3
Membrane length, ℓ	220 m
Membrane chord width, 2s	0.75 m
Mass of each blade, M	0.8 kg
Revolution period,	120 sec
Number of points in time history	4096 points
Sample interval	10 sec
Nyquist Frequency	0.05 Hz
Proportional damping ratio	3 %

The frequency response function (FRF) (using 99% overlap and 900 data blocks for average) of a single-input single-output system is shown in Fig. 2. The FRF were computed from the time history data that were generated from the linearized out-of-plane model with n = 3, i.e., three degrees of freedom. A 3% proportional damping ratio was added to the linear/nonlinear model in the form of

$$M(\theta)\ddot{\theta} + \Xi(\theta)\dot{\theta} + \mathbf{k}(\theta, \dot{\theta}) = \tau; \;\; M = M^T > 0; \;\; \Xi = \Xi^T > 0 \tag{8}$$

where

$$\theta = \begin{bmatrix} \phi_1 & \phi_2 & \phi_3 & \varphi_1 & \varphi_2 & \varphi_3 \end{bmatrix}^T , \tag{9}$$

$$\tau = \begin{bmatrix} \tau_{\phi_1} & \tau_{\phi_2} & \tau_{\phi_3} & \tau_{\varphi_1} & \tau_{\varphi_2} & \tau_{\varphi_3} \end{bmatrix}^T , \tag{10}$$

and

$$\Xi = 2\rho M^{1/2}(M^{-1/2}KM^{-1/2})^{1/2}M^{1/2} \tag{11}$$

The constant matrix K is the stiffness matrix of the linearized model as shown in Eq. (7), and ρ is the proportional damping ratio.

In Fig. 2, we demonstrate that the identified FRFs for linearized model are independent of input excitation magnitude and the approach used to compute FRFs is leakage free. The frequencies (i.e., the peaks of the curve) as seen in Fig. 2 match the bending frequencies given in Table 2. The same input signals were then used to perform simulations for the nonlinear model.

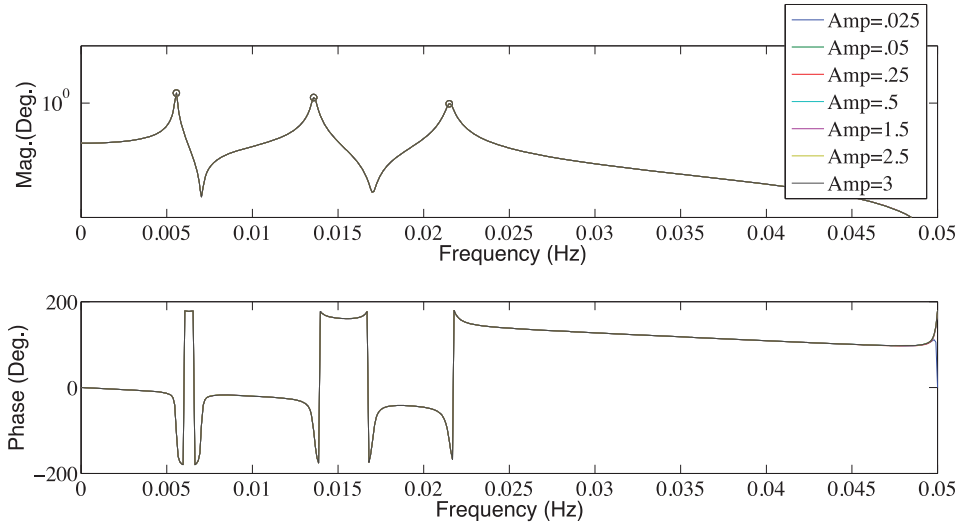

Fig. 2 FRF with various input amplitudes for linearized out-of-plane model

In Fig. 3, we show the FRFs from the simulation data with different amplitudes for the burst random signal. Nonlinear effects start appearing when large input signal (maximum torque amplitude > 0.5) is used to excite the nonlinear model. The FRF amplitudes are a function of input amplitudes, which reflects the nonlinearities in the system. From the frequency response function (FRF), a shift in frequencies is apparent in Fig. 3.

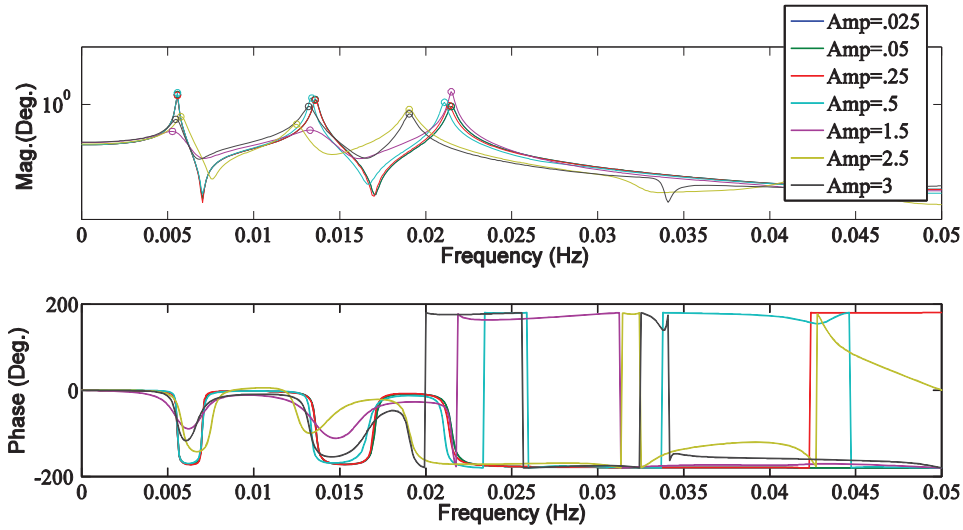

Fig. 3 FRF with various input amplitudes for nonlinear out-of-plane model

For controller designs, we chose the linear plane as the nominal plant. By calculating the principal gain [10], computed using singular value decomposition of the frequency response, an uncertainty envelope of the FRF could be created. Multiple simulation runs using burst random inputs to excite the model, even with the same amplitude level, may result in a small variation in the envelope of the FRF response. The FRF envelope is shown in Fig. 4 for a single-input multi-output uncoupled bending case. Ten simulation runs were conducted using burst random input signal, each run with 50 different input amplitudes ranging from 0.065 to 3.25 to excite the system to generate 50 identified models. In other words, the probability P having a FRF outside the envelope is lower than 1/500, which is sufficiently small. The plotted Upper/Lower bound were determined by taking the maximum/minimum bound value over the 10 runs. The envelope shows the variation for the responses of different amplitude simulation runs. Using the linearized model as the nominal plant along with the bound information, one can anticipate that any input in the chosen input bound will have a response that will not exceed the envelope. Hence the range of variation with respect to the nominal plant can be used to assess stability.

IV. Design of Controller

The goal of our controller design is to increase the damping level of all flexible modes while maintaining the system stability. Initially, the nominal plant had 3% of proportional damping, which was based on the experimental results of a small blade sample tested inside a vacuum chamber [6]. Assuming full state feedback, a pole placement technique [11] is implemented to increase damping to a user-defined value.

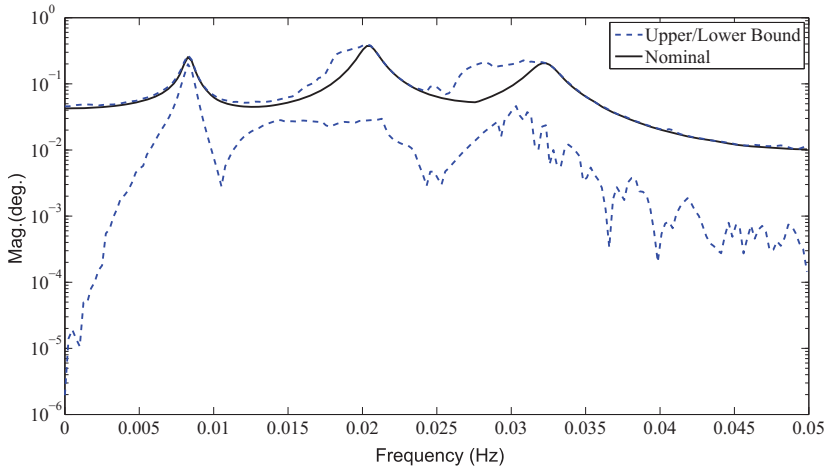

Fig. 4 Envelope of the out-of-plane bending modes using 50 identified models

To begin the controller design process, a nominal linear model was identified with the OKID frequency domain algorithm [7-8] from FRF data created by using an input level of 0.217 $\mathrm{N}\cdot\mathrm{M}$. From the identification process a nominal plant of order 6 that was identified and used to implement a full state feedback controller using a state observer. Because the true state is unknown in the state feedback law, an estimated state of the true state $x(t)$ is needed for feedback,

$$\frac{dx(t)}{dt} = A_c + B_c u(t) = A_c x(t) - B_c F_c \hat{x}(t) = (A_c - B_c F_c)\hat{x}(t) \tag{12}$$

$$\frac{d\hat{x}(t)}{dt} = (A_c + G_c C)\hat{x}(t) + (B_c + G_c D)u(t) - G_c y(t) \tag{13}$$

where $\hat{x}(t)$ is the observer state. The system and observer gain matrices A_c, B_c, C, D and G_c could be identified from experimental data. The feedback gain F_c is to be determined from the pole placement technique. A flow diagram for the state feedback with an estimator is shown in Fig. 5.

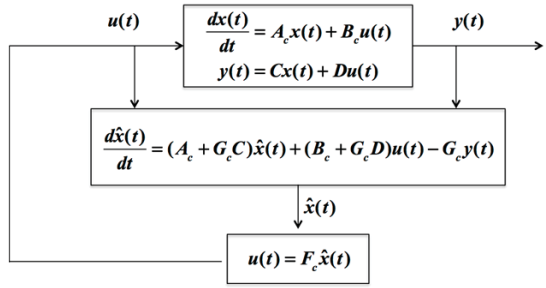

Fig. 5 State feedback with an estimator

Two types of pole placements approaches were applied to move the poles on the complex plane. Assuming the original real part of a eigenvalue is λ_{oR}, one approach, as shown in Fig. 6(a), moves the real part of all the poles to the left at the same distance $\Delta\lambda_R$ to a desired position λ_{dR};

$$\Delta\lambda_R = \lambda_{dR} - \lambda_{oR} \tag{14}$$

Another approach moves the poles to produce a desired level of proportional damping ζ_d, as shown in Fig. 6 (b). Results are listed in Table 3 for both approaches. Ideally the level of damping should be a prescribed parameter, however practically speaking the stability of the system for all input levels restricts the level of achievable damping to those shown in Table 3. In fact, the minimum values shown are from the models that have the smallest adjustment tolerance. For all cases, 10 runs were conducted with 50 identified models in each run. Note that conducting more runs may cause additional changes in the minimum values. As poles are moved farther away from the imaginary axis, the system becomes unstable more easily. Table 3 shows that the twisting motion has smallest minimum tolerance for both methods.

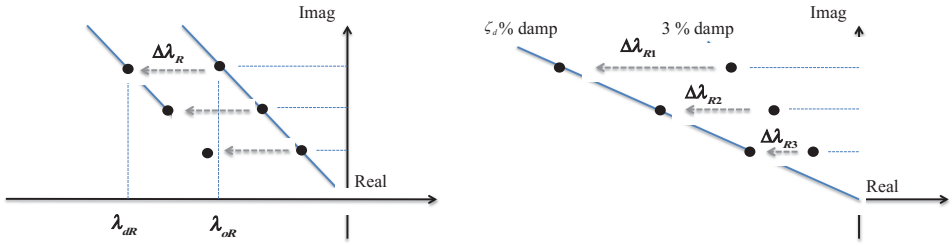

Fig. 6 (a) Pole placement according to $\Delta\lambda_R$ **(b) Pole placement according to ζ_d**

Table 3 Results of pole placement

Method for pole placement (for 10 runs)	Minimum	
	Out-of-plane	Twisting
$\Delta\lambda_R$	0.0006	0.0003
ζ_d	5.625%	3.375%

V. Stability robustness

Robustness of the plant is determined by numerically calculating the stability margin using the Nyquist criterion. Assuming that the original plant is $G(s)$ and the controller is $H(s)$, then the open loop transfer function is $O(s) = G(s)H(s)$. The stability margin defined in [12] is the shortest distance from any point $O(s = jw)$ to the critical point -1. Written explicitly, the shortest distance is

$$S_m = \min_{\forall s} \left|1 + O(s)\right| \tag{15}$$

In Fig. 7, we show the stability margin for the nominal and nonlinear plant as a function of the desired damping level. By applying the same controller to different nonlinear plants, a range of stability margin can be calculated. The top red line in Fig. 7 corresponds to controllers applied to the linear plant (no change in margin) and the lower blue line corresponds to the nonlinear plant. Examining the stability margin for the nonlinear controlled model (blue line), we observe that as the desired damping ratio becomes larger, the smaller the stability margin becomes.

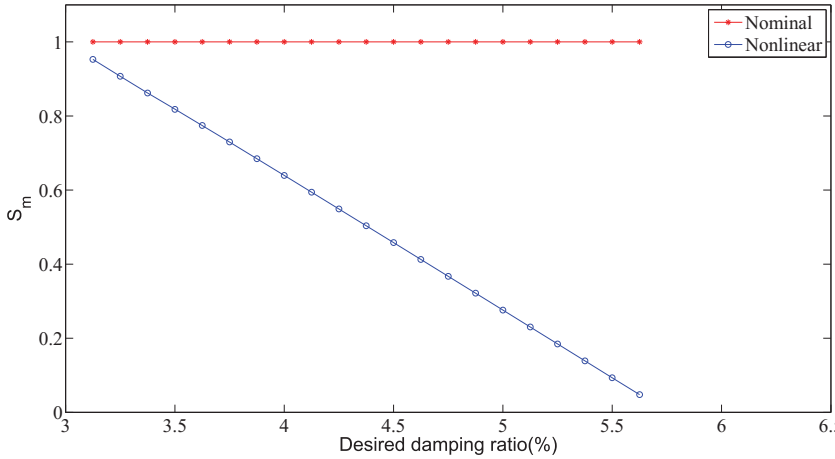

Fig. 7 Trend of stability margin related with damping ratio

With computed uncertainty envelope $\Delta(s)$ at each frequency point, a control design must ensure that the system is stable for all possible variations of the open loop plant. That is

$$\left|\Delta(s)\right| < \left|1 + O(s)\right| \tag{16}$$

After checking the stability of the controlled plant, one can calculate the distance $D(s)$, which is the distance from the critical -1 point to the Nyquist curve with uncertain distance shown in Fig. 8.

$$D(s) = \left|1 + O(s)\right| - \left|\Delta(s)\right| \tag{17}$$

When the distance is larger than 0,

$$D(s) > 0, \forall s \tag{18}$$

the stability of the controlled plant can be ensured.

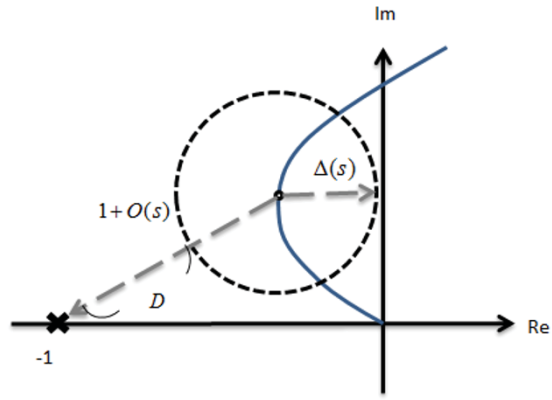

Fig. 8 Nyquist curve with uncertainty circle and stable distance D

VI. Conclusion

A simplified model of a spinning membrane using discrete lump masses has been used to create simulated time history data for system identification and control studies. Because the system is nonlinear, a frequency response approach has been implemented to envelope nonlinear effects for stability assessments of the control system. An example was presented where simulated time history data from the out-of-plane/bending response was used to identify a nominal model, to design a controller, and quantify the nonlinearities in terms of system uncertainty. A full state feedback control system was implemented to increase damping while maintaining stability under nonlinearities. With this control system, it is demonstrated that the achievable level of damping is diminished by the system nonlinearities.

Acknowledgments

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Two-blade solar sail dynamics

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This paper is about research of dynamics of spatial motion of two-blade heliogyro solar sail considering different types of perturbing effects with their responsiveness and range. The main purpose of this work is determination of list of design objectives and their range to provide sustainable movement of picosatellite with two-blade solar sail in experiment BMSTU-Sail.

I. Introduction

The accuracy of satellite manufacturing and assembling is limited, this causes displacements of different structural elements of satellite from their nominal (theoretical) position. For spinning bodies these displacements are responsible for additional parasite momenta causing precession.

BMSTU-Sail project [1] proposes space experiment with spinning picosatellite with two deployable non-rigid solar sail blades of considerable length up to 5 m each. Fig. I.1 represents the design of picosatellite from BMSTU-Sail experiment. Precession caused by irregularity of solar sail blades frame, imperfection of picosatellite geometry, can be the reason of unpredictable and undesirable effects. To exclude these effects dynamics of spatial motion of imperfect satellite with heliogyro-like solar sail should be considered.

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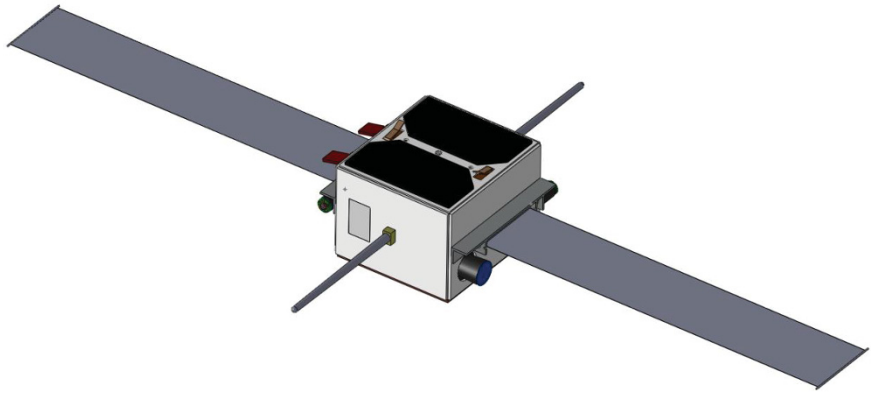

Fig. I.1 Picosatellite from BMSTU-Sail space experiment

II. Problem formulation

In this work we considered several imperfection factors affecting satellite motion which are described below:

- accuracy of reaction wheel mounting;
- parasitic initial angular velocity vector;
- non-simultaneity of solar sail blades uncoiling;
- accuracy of central body balancing;
- etc. (see section III “Mathematical model”).

Mathematical formulation of these factors and constrains is provided in section III “Mathematical model”.

Depending on BMSTU-Sail experiment sequence there are two possible stabilization modes:

1) unstabilized picosatellite (total rotational momentum of reaction wheel and picosatellite is approximately equals to zero);
2) spin-stabilized picosatellite (total rotational momentum is much greater than zero).

In first case any small parasitic initial angular velocity of picosatellite non-collinear to main rotation axis will cause significant change of satellite attitude during experiment. This will create additional difficulties for experiment's imaging and will influence to solar sail blades during deployment. By this reason, we selected spin-stabilized picosatellite with experiment sequence described below:

1) Mounting of picosatellite on launch device.
2) Switching-on and spinning picosatellite's reaction wheel to maximum speed (rotational momentum is compensated by cosmonaut attached to space station).

3) Separation of picosatellite from launch device.

4) Braking of reaction wheel.

5) Solar sail deployment.

There were two different models utilized for satellite dynamic simulation: one with stowed solar sail and the second with solar sail deployed to defined length. Direct modeling of solar sail deployment with imperfection is not considered, assuming that deployment process is long and in each moment the whole system is in quasi steady state equilibrium between centrifugal force and Coriolis forces, which are effecting to solar sail blades during their unrollment.

Therefore, the modeling of dynamics of picosatellite comes down to modeling of steady state rotation of picosatellite with solar sail blades with different length.

III. Mathematical model

In this section we describe mathematical statement of problem of picosatellite dynamics. Utilizing this model some numerical experiments were performed determining dependence of precession angle of picosatellite from modeled imperfections. The MSC Adams was used for calculations.

Imperfection vector $\vec{\Delta}$ from now on we mean the set of parameters which are modeling imperfection factors. Elements of this vector, their nomenclature, dimension and limits of variation are given in table III.1.

Table III.1 Elements of imperfection vector $\vec{\Delta}$

Element	Dimension	Tolerance	Physical meaning
$\overrightarrow{\Delta_{cm}^{rw}}=$ $(\Delta_{cm}^{rw})_X$ $(\Delta_{cm}^{rw})_Y$ $(\Delta_{cm}^{rw})_Z$	m	 -0.003…0.003 -0.003…0.003 -0.025…0	Displacement of mass center of the reaction wheel from its nominal position that should match with nominal position of mass center of picosatellite – initial point of base coordinate system marked “BCS” (fig. III.1)
$\overrightarrow{\Delta_{cm}^{SC}}=$ $(\Delta_{cm}^{SC})_X$ $(\Delta_{cm}^{SC})_Y$ $(\Delta_{cm}^{SC})_Z$	m	 -0.003…0.003 -0.003…0.003 -0.003…0.003	Displacement of picosatellite mass center from its nominal position that matches initial point of base coordinate system of MSC Adams (BCS on fig. III.1)
$\Delta\alpha$	deg	-5…5	Angle between rotation wheel spinning axis and nominal spinning axis of picosatellite – Z axis of BCS (fig. III.1)
ω_0^X	deg/s	-5…5	Initial angular velocity of picosatellite after separation from launcher around X axis of BCS (fig. III.1)
ω_0^Y	deg/s	-5…5	Initial angular velocity of picosatellite after separation from launcher around Y axis of BCS (fig. III.1)
$\Delta\varphi$	deg	-5…5	Difference between angular deviation of blades #2 and #1 from XZ plane of BCS during deployment (fig. III.5)

Table III.1 Elements of imperfection vector $\vec{\Delta}$

Element	Dimension	Tolerance	Physical meaning
ΔL	m	-0.1·L…0.1·L	Difference between lengths of partially deployed blades #2 and #1 during deployment (fig. III.5)

A. Position with stowed blades

In stowed position picosatellite can be considered as rigid body with rotating wheel inside. Base coordinate system matches with theoretical position of picosatellite mass center. Z axis of BCS matches with theoretical spinning axis of picosatellite, X axis of BCS directed in the line of solar sail blade #1.

Real mass center of picosatellite is displaced from BCS on vector $\overrightarrow{\Delta_{cm}^{sc}}$.

Real mass center of rotating wheel is displaced from BCS on vector $\overrightarrow{\Delta_{cm}^{rw}}$.

Real axis of rotating wheel inclined from Z axis of BCS on angle $\Delta\alpha$ (fig. III.1).

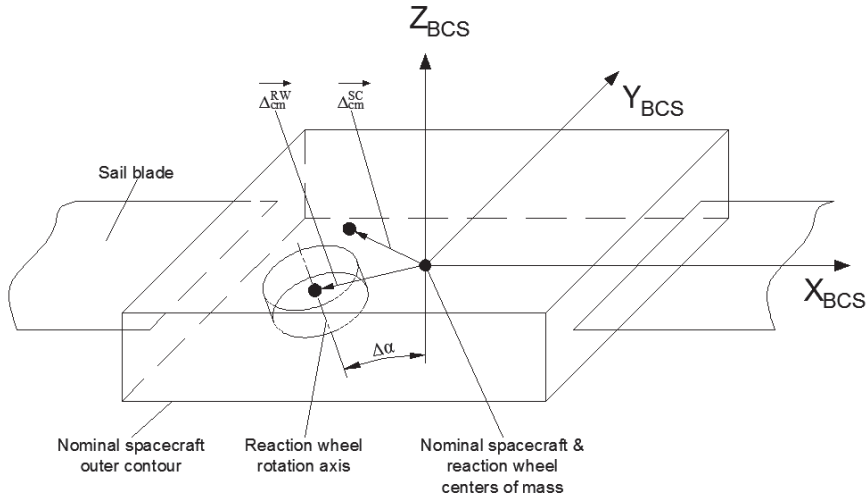

Fig. III.1 Calculation scheme of picosatellite with stowed solar sail

Initially, the spinning velocity of reaction wheel is maximum. Initial angular velocity of picosatellite corresponds to disturbances from separation process. Angular velocity of reaction wheel decreased from initial value to zero linearly during first 10 seconds of simulation translating its kinematic momentum to picosatellite. During next 10 seconds of simulation (approximately 10 turns of picosatellite) the attitude of picosatellite is changing providing an information of precession angle with given value of imperfection vector. An overview of model in MSC Adams View is shown on fig. III.2.

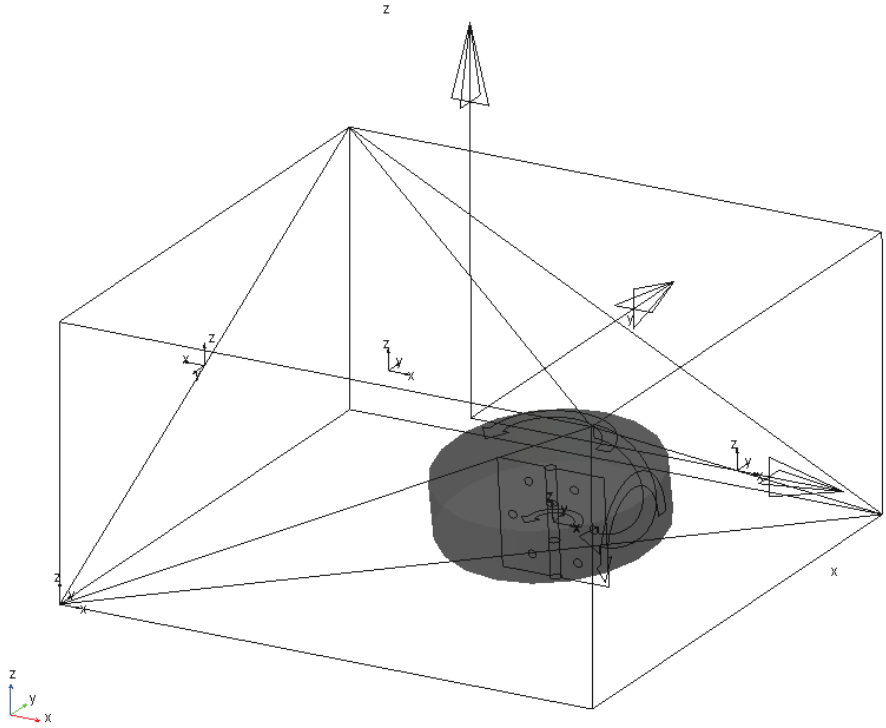

Fig. III.2 Overview of picosatellite model in MSC Adams View

MSC Adams model consists of two rigid bodies:

- rigid part of picosatellite «SC» (primary structure, PCB, bobbins with solar sail, stator of reaction wheel);
- reaction wheel «RW».

On the rigid part of picosatellite there is a marker «BCS» which initially matches with origin point of Adams base coordinate system, axes of marker are initially directed parallel to Adams «BSC» axes. Other elements are positioned relative to «BCS» marker.

Rigid part of picosatellite is represented as the rectangular prism. Mass center of this prism is displaced by vector $\overrightarrow{\Delta_{cm}^{sc}}$ from marker «BCS». Prism faces are perpendicular to the axes of marker «BCS».

Reaction wheel is represented as the cylinder. Mass center of cylinder is displaced by vector $\overrightarrow{\Delta_{cm}^{rw}}$ from «BCS» marker. The axis of cylinder is inclined by angle $\Delta\alpha$ relative to Z axis of «BCS» marker (see fig. III.1 and III.2).). If necessary, reaction wheel imbalance can be simulated e.g. as third point-like body attached to the edge of

reaction wheel by rigid link. But in this study we assumed that reaction wheel can be accurately balanced during manufacturing and testing so the reaction wheel imbalance is not considered.

The constrained between reaction wheel and rigid picosatellite is the cylindrical joint. Deceleration of reaction wheel and acceleration of picosatellite is provided by three-dimensional torque with Z axis directed along the axis of cylindrical joint. The torque is defined between two bodies (SC and RW). Initial angular velocity of RW body is equals to maximum angular velocity of reaction wheel's electric motor according to its technical specifications. Deceleration momentum is selected assuming that total deceleration time of reaction wheel should be 10 s. To simulate the end of deceleration 10 seconds after beginning the deceleration momentum is set to zero by STEP() function. Total duration of simulation is 20 s.

Precession angle is measured with MSC Adams Solver «THETA» function applied to SC body. In addition to precession angle we measure angular velocity vector of SC body projected to axis of BCS marker. These measurements are used in calculation of initial conditions for model with deployed blades.

To automate the study of dependence of precession from imperfection vector the preprocessor script in Mathematica which generates Adams model file (.adm) and Adams Solver Script (.acf) for a given imperfection vector was written. After finishing it launches Adams Solver which automatically performs all computations. Then Mathematica's script loads solver output (.res file), processed it and plotting the dependence of precession angle from time for given imperfection vector, and, as a final result, dependence of precession angle from varied imperfection vector.

Figure III.3 is representing the output result examples: plot of precession angle with time for $\Delta\boldsymbol{\alpha} = \mathbf{5°}$.

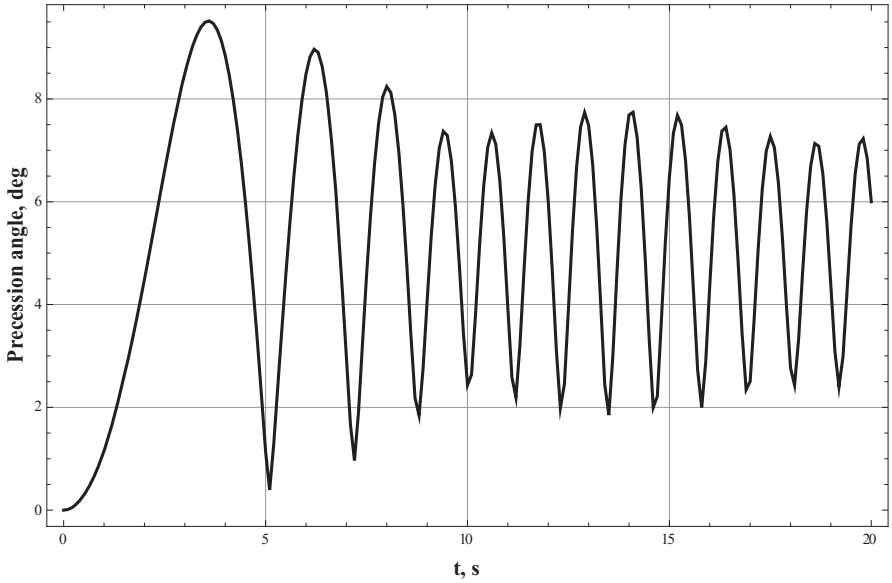

Fig. III.3 Example plot of precession angle depending on time

The maximum precession angle on simulation time interval accepted as the output value.

B. Position with partially deployed solar sail blades

Mathematical model of picosatellite with partially deployed solar sail blades differs from model of picosatellite with stowed solar sail by following points:

- reaction wheel is stopped (relative to rigid part of picosatellite) with «FIXED» constrained.
- Blades are modeled as a set of rigid elements connected with spherical joints (fig. III.4). Length of blade's finite elements is selected from numerical experiments. The masses of tip blade's finite elements are increased on tip masses weight.

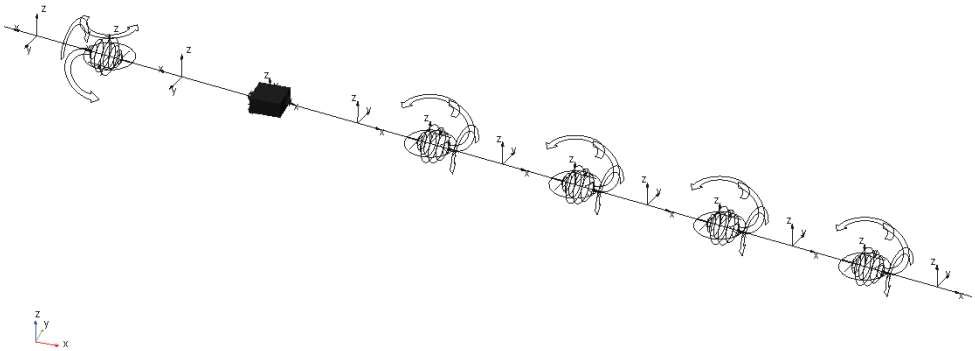

Fig. III.4 Overview of picosatellite with partially deployed solar sail blades in MSC Adams View

Coriolis forces are modeled as additional momentum in every spherical joint. This momentum is calculated by formula [1]:

$$M = (2 \cdot \omega \cdot v) \cdot \left(\rho b h \frac{l^2}{2} + m_{tip} l \right), \tag{III.1}$$

$$\omega = \omega_{RW} \frac{(Y_{RW}^0)_Z}{(Y_{SC}^0)_Z + (Y_{RW}^0)_Z + \frac{2}{3}\rho b h L^3 + 2 \cdot m_{tip} L^2},$$

where: ω – nominal angular velocity of picosatellite with solar sail blades deployed to current length, rad/s, v – nominal linear velocity of solar sail blade deployment, m/s; ρ – density of solar sail material, kg/m^3; b – width of solar sail blades, m; h – thickness of solar sail film, m; l – distance from this joint to blade tip, m; $(Y_{RW}^0)_Z$ – nominal inertia momentum of rotation wheel around rotation axis, kg·m^2; $(Y_{SC}^0)_Z$ – nominal inertia momentum of rigid part of picosatellite relative to nominal position of rotation axis, kg·m^2; L - length of deployed blades, m; m_{tip} - tip mass's weight, kg; ω_{RW} - nominal angular velocity of reaction wheel, rad/s.

Initially, solar sail blades are based in XY plane of «BCS» and displaced from XZ plane of «BCS» on angle corresponding to theoretical equilibrium of Coriolis force and centrifugal force. This angle can be calculated by formula below [1]:

$$\varphi = \operatorname{asin} \frac{2\,v}{\omega\, L_0}, \tag{III.2}$$

$$v = \frac{\omega\, L_0}{2} \sin \varphi\,,$$

where: L_0 – distance from nominal axis of rotation to attachment point of solar sail blade to picosatellite – center of root spherical joint between solar sail blade and rigid picosatellite structure (fig. III.5).

Considering that the problem is quasi-static there is a problem of setting of initial condition – initial angular velocity vector. This vector cannot be calculated in this statement however considering some assumptions it is possible to estimate initial condition by using the law of variation of rotational momentum. These assumptions are:

- there are no external torques for picosatellite with solar sail during deployment;
- there are no energy dissipation in system;
- orbital movement of picosatellite's mass center is not considered during analysis of angular motion.

The first assumption is correct firstly because external momenta (magnetic and aerodynamic couple) are small and secondly that the duration of space experiment is only several minutes.

The second assumption is correct because of fact that deceleration of rotating wheel is forced by electric motor and not by friction and that energy dissipation in solar sail film is small.

The third assumption is correct firstly because of small gravitation momentum, secondary because of small orbital angular velocity relative to spinning velocity of picosatellite (~10^{-3} rad/s against ~0.2 rad/s after full deployment of each solar sail blade 5 m length each) considering that duration of experiment is short.

Assuming that there are no external torques, we can write the law of variation of rotational momentum:

$$\frac{d\vec{K}}{dt} = 0 \;\Rightarrow \vec{K} = const, \tag{III.3}$$

where: $\vec{K}$ – rotational momentum of whole system.

In this case the angular velocities before and after deployment are linked with the following formula:

$$[Y]_0\{\Omega\}_0 = [Y]_1\{\Omega\}_1, \tag{III.4}$$

where: $[Y]_0$, $[Y]_1$ – picosatellite inertia tensor before and after deployment; $\{\Omega\}_0$, $\{\Omega\}_1$ - picosatellite angular velocity vector before and after deployment.

In expression (III.4) all elements excluding $\{\Omega\}_1$ can be calculated or roughly estimated. $\{\Omega\}_1$ is unknown initial angular velocity vector of picosatellite. This vector $\{\Omega\}_1$ can be calculated as:

$$\{\Omega\}_1 = [Y]_1^{-1}([Y]_0\{\Omega\}_0) = ([Y]_1^{-1}\,[Y]_0)\,\{\Omega\}_0\,. \tag{III.5}$$

To simplify the calculation model there is one more assumption: in statement (III.5) all vectors and tensors are calculated in base coordinate system (marker «BCS») which is not the coordinate system of mass center. The correctness of this assumption in design study should be investigated in further works.

Vector $\{\Omega\}_0$ is given as picosatellite angular velocity vector in body-fixed coordinates (marker «BCS») in the end of integration of picosatellite angular movement with stowed solar sail blades.

Components of inertia tensor in «BCS» can be estimated by following statements (see fig. III.1 and III.2):

$$
\begin{aligned}
Y^0_{xx} &= (Y^0_{SC})_X + m_{SC}\,([(\Delta^{SC}_{cm})_Y]^2 + [(\Delta^{SC}_{cm})_Z]^2) + (Y^0_{RW})_X + m_{RW}\,([(\Delta^{RW}_{cm})_Y]^2 + [(\Delta^{RW}_{cm})_Z]^2)\,;\\
Y^0_{yy} &= (Y^0_{SC})_Y + m_{SC}\,([(\Delta^{SC}_{cm})_X]^2 + [(\Delta^{SC}_{cm})_Z]^2) + (Y^0_{RW})_Y + m_{RW}\,([(\Delta^{RW}_{cm})_X]^2 + [(\Delta^{RW}_{cm})_Z]^2)\,;\\
Y^0_{zz} &= (Y^0_{SC})_Z + m_{SC}\,([(\Delta^{SC}_{cm})_X]^2 + [(\Delta^{SC}_{cm})_Y]^2) + (Y^0_{RW})_Z + m_{RW}\,([(\Delta^{RW}_{cm})_X]^2 + [(\Delta^{RW}_{cm})_Y]^2)\,;\\
Y^0_{xy} &= 0.0 + m_{SC}(\Delta^{SC}_{cm})_X\,(\Delta^{SC}_{cm})_Y + 0.0 + m_{RW}(\Delta^{RW}_{cm})_X\,(\Delta^{RW}_{cm})_Y\,;\\
Y^0_{xz} &= 0.0 + m_{SC}(\Delta^{SC}_{cm})_X\,(\Delta^{SC}_{cm})_Z + 0.0 + m_{RW}(\Delta^{RW}_{cm})_X\,(\Delta^{RW}_{cm})_Z\,;\\
Y^0_{yz} &= 0.0 + m_{SC}(\Delta^{SC}_{cm})_Y\,(\Delta^{SC}_{cm})_Z + 0.0 + m_{RW}(\Delta^{RW}_{cm})_Y\,(\Delta^{RW}_{cm})_Z\,;
\end{aligned}
\qquad \text{(III.6)}
$$

where: m_{SC} - weight of picosatellite with blades, kg; m_{RW} - weight of reaction wheel, kg; $(Y^0_{SC})_{X,Y,Z}$. – nominal (theoretical) inertia momenta of picosatellite with solar sail blades.

Expressions (III.6) are derived using the following assumptions:

- blades moved only in XY plane of base coordinate system;
- the blades shape– straight line.

Exclusion of first assumption is beyond the scope of this work. The second assumption is argued in [1] – in equilibrium state blade's shape under Coriolis and centrifugal forces is straight line (blades was represented as a flexible threads).

Calculation scheme for determination of components of inertia tensor after blade's deployment on specific length is shown on fig. III.5.

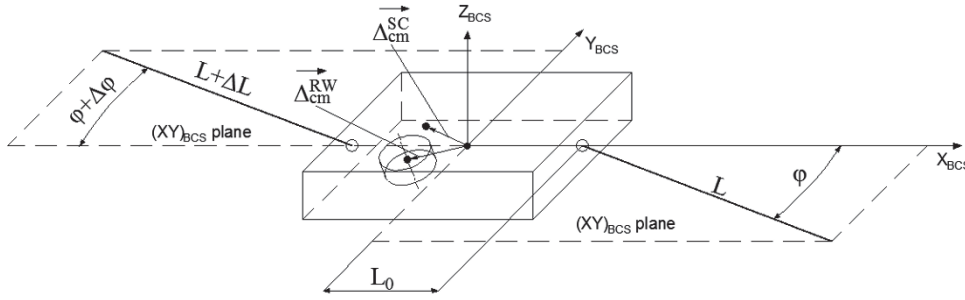

Fig. III.5 Calculation scheme for determination of components of inertia tensor after blades deployment

Inertia tensor components in base coordinate system after deployment are estimated by following statements:

$$\boldsymbol{Y^1_{xx}} = Y^0_{xx} + \int_0^L \rho bh \cdot (L \cdot \sin\varphi)^2\, dL + \int_0^{L+\Delta L} \rho bh \cdot (L \cdot \sin(\varphi + \Delta\varphi))^2\, dL$$

$$+ m_{tip}((L \cdot \sin\varphi)^2 + (L \cdot \sin(\varphi + \Delta\varphi))^2) =$$

$$= Y^0_{xx} + \rho bh \cdot \frac{L^3}{3}\sin^2\varphi + \rho bh \cdot \frac{(L+\Delta L)^3}{3}\sin^2(\varphi + \Delta\varphi)$$

$$+ m_{tip}\left((L \cdot \sin\varphi)^2 + \left((L+\Delta L) \cdot \sin(\varphi + \Delta\varphi)\right)^2\right);$$

$$\boldsymbol{Y^1_{yy}} = Y^0_{yy} - 2L_0^2 \cdot \left(\rho bhL + m_{tip}\right) + \int_0^L \rho bh \cdot (L \cdot \cos\varphi)^2\, dL + \int_0^{L+\Delta L} \rho bh \cdot (L \cdot \cos(\varphi + \Delta\varphi))^2\, dL$$

$$+ \rho bh(2L + \Delta L)L_0^2$$

$$+ m_{tip}\left((L \cdot \cos\varphi + L_0)^2 + \left((L+\Delta L) \cdot \cos(\varphi + \Delta\varphi) + L_0\right)^2\right)$$

$$= Y^0_{yy} + \rho bh \cdot \Delta L \cdot L_0^2 - 2m_{tip}L_0^2 + \rho bh \cdot \frac{L^3}{3}\cos^2\varphi + \rho bh$$

$$\cdot \frac{(L+\Delta L)^3}{3}\cos^2(\varphi + \Delta\varphi) +$$

$$+ m_{tip}\left(\left((L + L_0) \cdot \cos\varphi\right)^2 + \left((L + \Delta L + L_0) \cdot \cos(\varphi + \Delta\varphi)\right)^2\right);$$

$$\boldsymbol{Y^1_{zz}} = Y^0_{zz} + \rho bh \cdot \Delta L \cdot L_0^2 + \rho bh \cdot \frac{L^3}{3} + \rho bh \cdot \frac{(L+\Delta L)^3}{3} + m_{tip}(L^2 + (L + \Delta L)^2); \qquad \text{(III.7)}$$

$$\boldsymbol{Y^1_{xy}} = Y^0_{xy} + \int_0^L \rho bh \cdot (L \cdot \sin\varphi) \cdot (L \cdot \cos\varphi + L_0)\, dL$$

$$+ \int_0^{L+\Delta L} \rho bh \cdot (L \cdot \sin(\varphi + \Delta\varphi)) \cdot (L \cdot \cos(\varphi + \Delta\varphi) + L_0)\, dL + m_{tip} \cdot (L \cdot \sin\varphi)$$

$$\cdot (L \cdot \cos\varphi + L_0) + m_{tip} \cdot \left((L + \Delta L) \cdot \sin(\varphi + \Delta\varphi)\right)$$

$$\cdot \left((L + \Delta L) \cdot \cos(\varphi + \Delta\varphi) + L_0\right)$$

$$= Y^0_{xy} + \rho bh \cdot$$

$$\cdot \left(\frac{L^3}{3}\sin\varphi\cos\varphi + \frac{L^2}{2}L_0\sin\varphi + \frac{(L+\Delta L)^3}{3}\sin(\varphi + \Delta\varphi)\cos(\varphi + \Delta\varphi)\right.$$

$$\left. + \frac{(L+\Delta L)^2}{2}L_0\sin(\varphi + \Delta\varphi)\right) + m_{tip}$$

$$\cdot \left((L \cdot \sin\varphi) \cdot (L \cdot \cos\varphi + L_0) + (L + \Delta L) \cdot \sin(\varphi + \Delta\varphi)\right.$$

$$\left. \cdot \left((L + \Delta L) \cdot \cos(\varphi + \Delta\varphi) + L_0\right)\right);$$

$$\boldsymbol{Y^1_{xz}} = \boldsymbol{Y^0_{xz}};\ \boldsymbol{Y^1_{yz}} = \boldsymbol{Y^0_{yz}};$$

where: L – nominal length of deployed blades, m; ΔL – blade #2 length excess relative to blade #1, m; φ – nominal angle of deviation of blades from XZ plane (marker BCS), rad; $\Delta\varphi$ - excess of blade #2 angle of deviation relative to blade's #1 angle of deviation, rad.

The special script was written in Mathematica similar to previous model.

IV. Numerical experiments

Values of picosatellite's design parameters which was used in numerical experiments are given in table IV.1.

Table IV.1 Design parameters

Parameter	Dimension	Value
m_{SC}	kg	0.5
m_{RW}	kg	0.05
$(Y_{SC}^0)_X$	kg·m^2	0.0035
$(Y_{SC}^0)_Y$	kg·m^2	0.0035
$(Y_{SC}^0)_Z$	kg·m^2	0.004
$(Y_{RW}^0)_X$	kg·m^2	0.00003
$(Y_{RW}^0)_Y$	kg·m^2	0.00003
$(Y_{RW}^0)_Z$	kg·m^2	0.00003
ρ	kg/m^3	1470
b	m	0.05
h	m	$10\cdot10^{-6}$
φ	deg	10
L	m	0…5
L_0	m	0.05
ω_{RW}	rad/s	650
m_{tip}	kg	0.0005

The WSTIFF solver was used for simulations with error parameter = 10^{-7} (fortran executable).

A. Stowed position

Calculation showed that the maximum influence to precession angle is provided by following components of imperfection vector:

- $\Delta\alpha$;
- ω_0^X, ω_0^Y (symmetrically).

Plots of precession angle depending to these components of imperfection vector are shown in fig. IV.1…IV.2.

As it was expected, elements $\overrightarrow{\Delta_{cm}^{SC}}$, $\overrightarrow{\Delta_{cm}^{rw}}$ without any other elements not affect the precession angle within the accuracy of the calculations (fig. IV.3, IV.4).

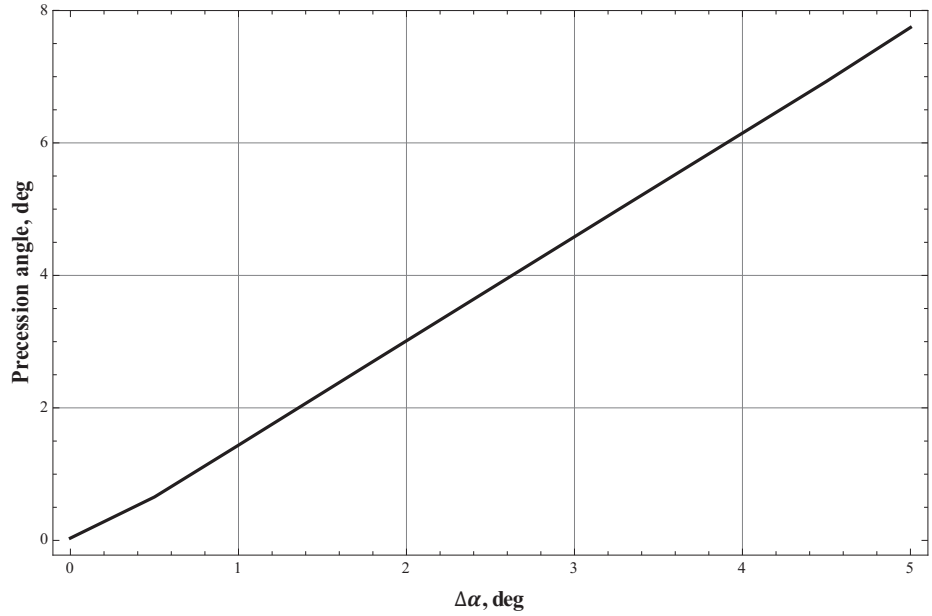

Fig. IV.1 Dependence of precession angle from $\Delta\alpha$

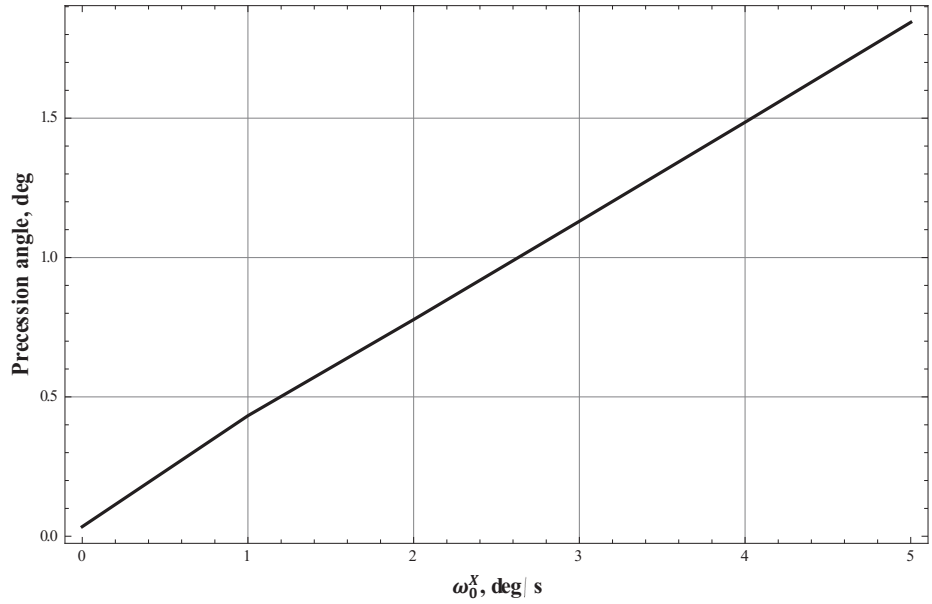

Fig. IV.2 Dependence of precession angle from ω_0^X (ω_0^Y)

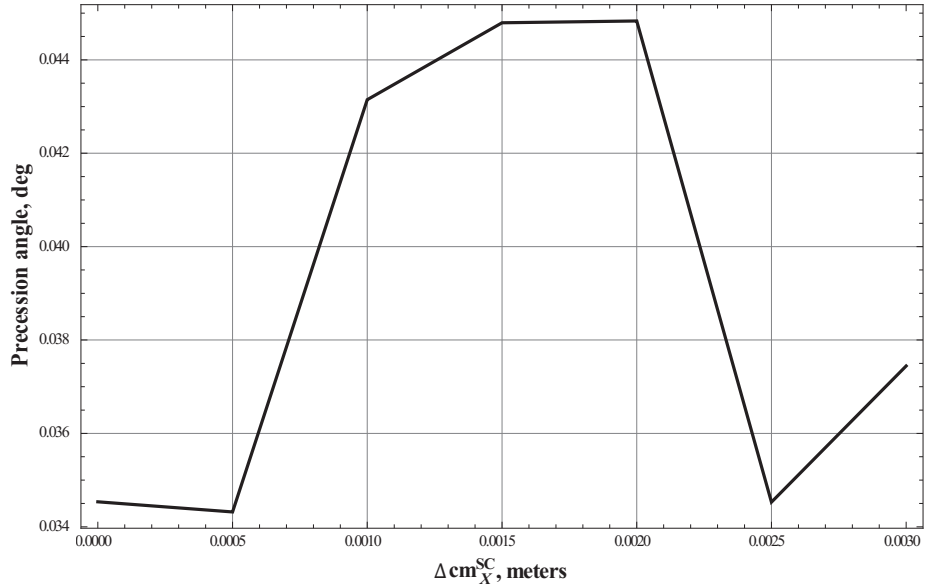

Fig. IV.3 Dependence of precession angle from $(\Delta_{cm}^{SC})_X$

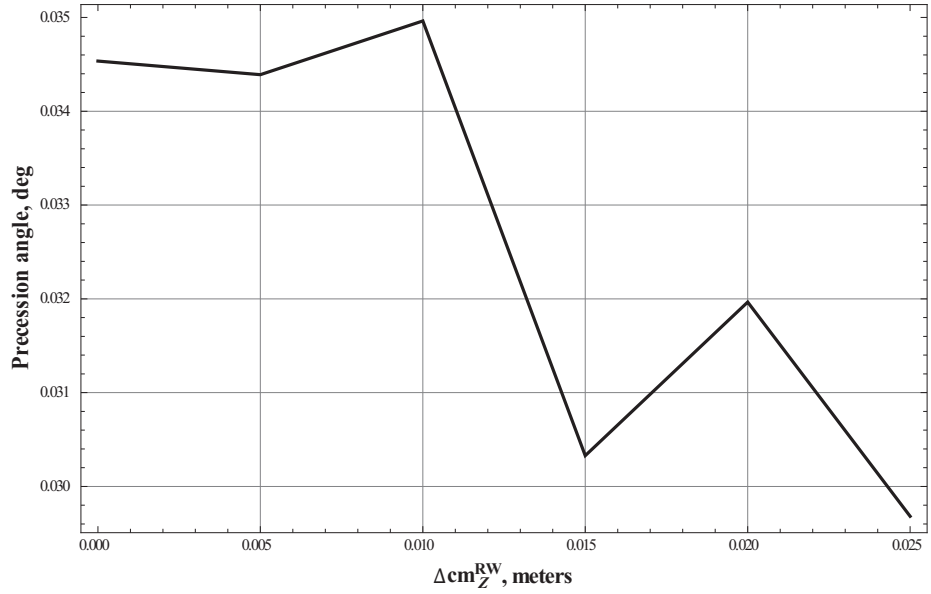

Fig. IV.4 Dependence of precession angle from $(\Delta_{cm}^{rw})_Z$

B. Deployed position

Calculation showed that the maximum influence to precession angle is provided by following components of imperfection vector:

– $\Delta\alpha$;

– ω_0^X;

– ω_0^Y.

Plots of precession angle depending to components of imperfection vector are shown in fig. IV.5…IV.7.

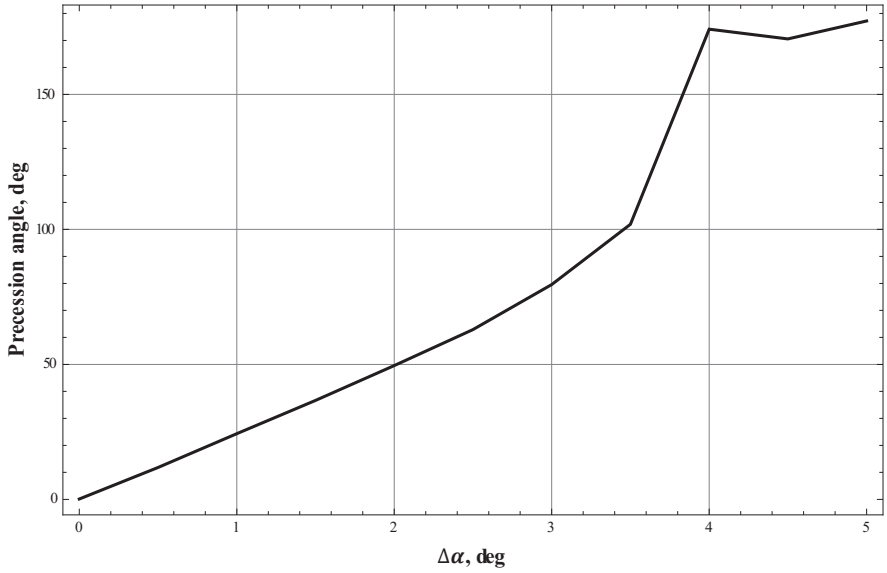

Fig. IV.5 Dependence of precession angle from $\Delta\alpha$

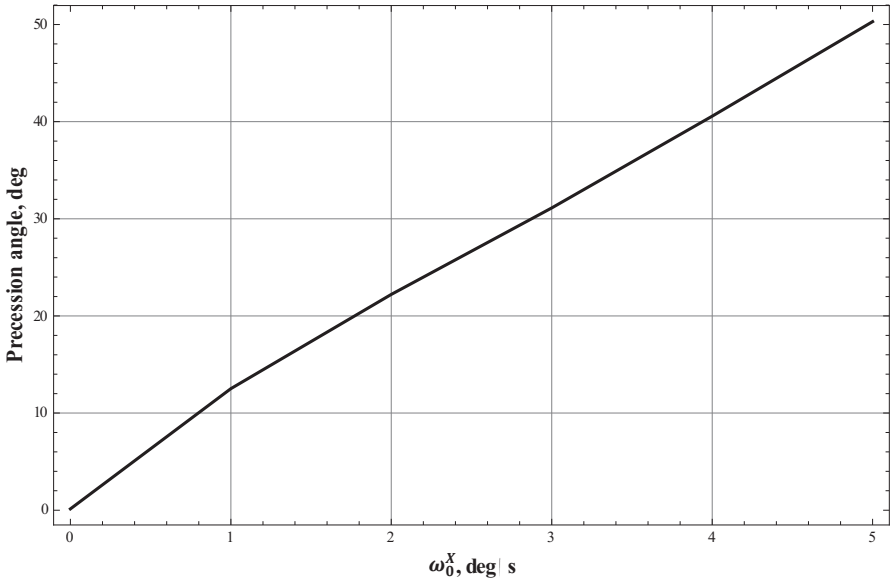

Fig. IV.6 Dependence of precession angle from ω_0^X

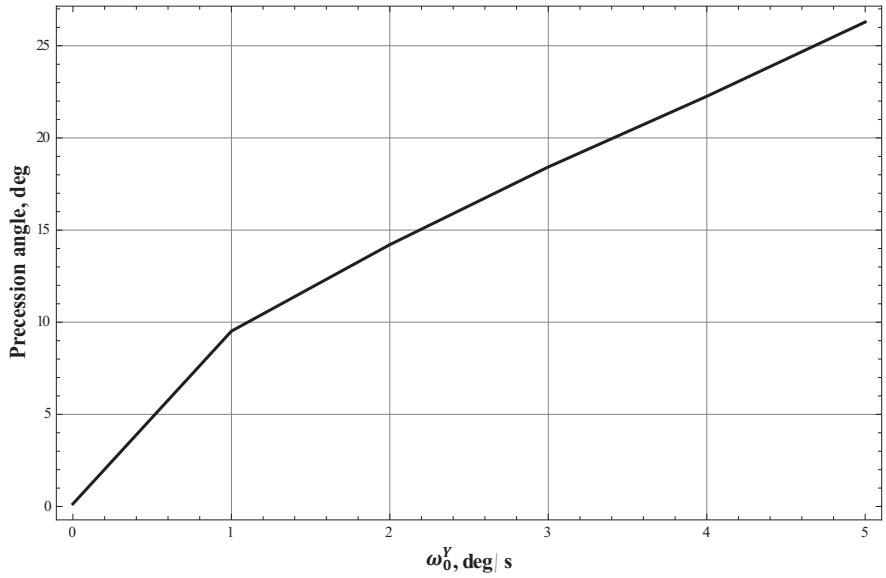

Fig. IV.7 Dependence of precession angle from ω_0^Y

V. Estimation of design parameters

Considering that the number of components of imperfection vector is high, it is practically difficult to create full-scale multidimensional problem to determine the dependence of precession angle from imperfection vector. Thus we use the following algorithm to calculate constraints of accuracy of picosatellite manufacturing and assembling:

- calculate this dependence for every component of imperfection vector assuming that all other components are equals to zero;
- calculate response level of precession relative to selected component of imperfection vector by formula below:

$$c_i = \frac{max\left(\theta\left(\overrightarrow{\Delta_i}\right)\right)}{max(\theta(\vec{\Delta}))}, \tag{V.1}$$

where: c_i – coefficient of response level to component number i of imperfection vector $\vec{\Delta}$; $\overrightarrow{\Delta_i}$ – component number i from imperfection vector $\vec{\Delta}$; $\theta\left(\overrightarrow{\Delta_i}\right)$ – dependence of precession angle from component number i from imperfection vector $\vec{\Delta}$; $max\left(\theta\left(\overrightarrow{\Delta_i}\right)\right)$ – maximum value of precession angle

in given limits of variation for element i of imperfection vector $\vec{\Delta}$; $max(\theta(\vec{\Delta}))$ - maximum value of precession angle for all elements of imperfection vector $\vec{\Delta}$.

Coefficient c_i corresponds to sensitivity to this element relative to other elements of imperfection vector.

Then the weighted vector of imperfections is calculated:

$$\tilde{\vec{\Delta}}_i = \mu \cdot \frac{\vec{\Delta}_i}{\max(c_i, 0.3)}, \qquad \text{(V.2)}$$

where: μ - "degree" of imperfection – generalized design parameter.

Condition "not less than 0.3" is given in formula (V.2) to limit the growth of maximum value of imperfection vector element to be not greater than 3 times relative to nominal value shown in table III.1.

Using this model we calculated the dependence of precession angle form parameter μ. Originating from design restriction we can chose maximum precession angle, get corresponding value of μ, and using formula (V.2), can estimate requirements to accuracy of picosatellite manufacturing and assembling.

Dependence of precession angle from generalized degree of imperfection is shown on fig. V.1.

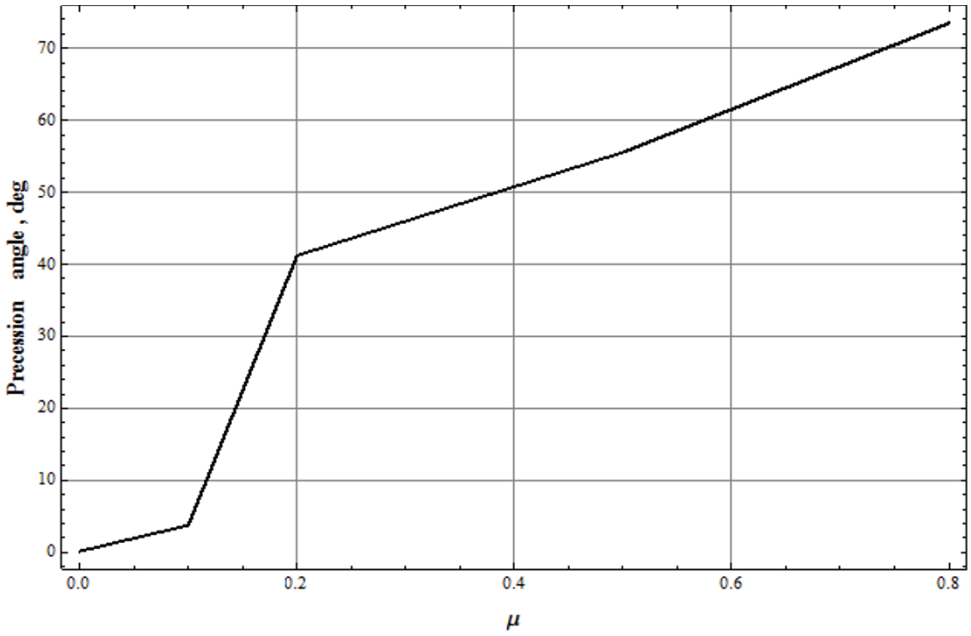

Fig. V.1 Dependence of precession angle from generalized degree of imperfection μ

The value of **0.1** for μ asserted for future analysis.

This algorithm allows the determination of design requirements to picosatellite only in first approximation. Using this algorithm the worst combinations of imperfection vector components easily can be missed.

Thus for improving these estimations, for vector of imperfections weighted by $\boldsymbol{\mu = 0.1}$ we carry out some calculations using Monte-Carlo method [2]. The uniform distribution inside the tolerance ranges of weighted imperfection vector assumed for all elements. For every calculation the weighted imperfection vector was corrected manually. As the result, the final design requirements vector was determined.

Table V.1 is representing the final requirements to accuracy of picosatellite manufacturing and assembling.

Table V.1 Requirements to accuracy of picosatellite manufacturing and assembling

Element of imperfection vector $\vec{\Delta}$	Dimension	Tolerance
$\overrightarrow{\Delta_{cm}^{rw}}=$ $(\Delta_{cm}^{rw})_X$ $(\Delta_{cm}^{rw})_Y$ $(\Delta_{cm}^{rw})_Z$	m	 ±0.001 ±0.001 -0.025…0.0
$\overrightarrow{\Delta_{cm}^{SC}}=$ $(\Delta_{cm}^{SC})_X$ $(\Delta_{cm}^{SC})_Y$ $(\Delta_{cm}^{SC})_Z$	m	 ±0.001 ±0.001 ±0.001
$\Delta\alpha$	deg	0…0.5
ω_0^X	deg/s	±0.1
ω_0^Y	deg/s	±0.1
$\Delta\varphi$	deg	±1.0
ΔL	m	0.05·L

The histogram of Monte-Carlo calculations is presented on fig. V.2.

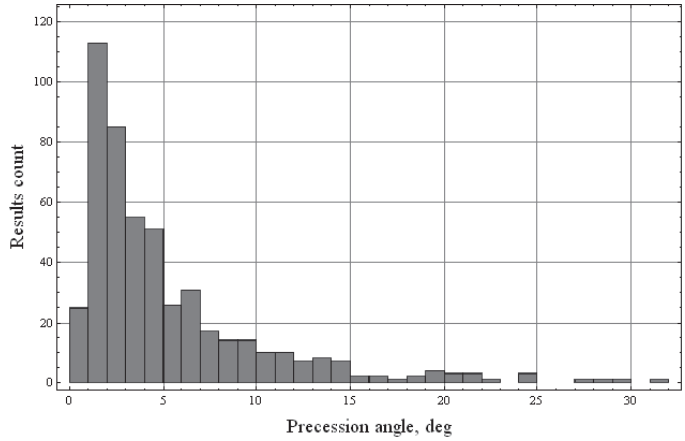

Fig. V.2 Monte-Carlo calculations histogram

The expected value of precession angle is **5.2º**. The standard deviation is **5.0º**. The **90%** of all results are inside region where precession angle is less than **13º**.

The number of calculations in one Monte-Carlo method launch was 500 (selected according to the convergence analysis).

VI. Conclusion

The mathematical model for estimation of precession angle for picosatellite BMSTU-Sail is proposed. The design requirements for satellite manufacturing, assembling and launching identified.

In future works should be paid attention to simulation deploying process directly, without assumptions used in this work. Such simulations allows improving the accuracy of precession angle prediction.

References

[1] Mayorova, V. I., Dmitriev A., Nerovnyy N., Rachkin D., Tenenbaum S., “ISS - test bed for the future solar sail systems”, *63rd International Astronautical Congress,* IAC-12-B3.3.10, Naples, Italy, 2012.

[2] Weisstein, Eric W. "*Monte Carlo Method*." From MathWorld-A Wolfram Web Resource. URL: http://mathworld.wolfram.com/MonteCarloMethod.html [cited 10 May 2013].

Solar Sail Coning Control to Induce Orbital Effects in Spinning Versus Non-Spinning Sails

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Sail attitude equilibria exist under the influence of aerodynamic, gravity gradient and solar torques. Precession of the sail normal from these equilibria causes sail normal coning about that equilibrium attitude. If the coning happens at orbit rate, wide variety of orbital effects can be induced. Past work developed a coning controller that enables orbit rate circular coning of the sail normal about the attitude equilibrium. This controller enables the sail normal vector to trace a desired coning trajectory at orbit rate. In this analysis, the performance of spinning vs. non-spinning solar sails in inducing orbital effects is studied using the coning controller. Results show that control torques required are on the order of 10^{-6} Nm, and it is advised to operate a non-spinning sail to avoid complex mechanisms and to maintain system robustness.

Nomenclature

β	=	cone angle [deg]
$\delta\beta$	=	angular measure for cone perturbation from the equilibrium point [deg]
β_c	=	desired cone angle [deg]
β_0	=	equilibrium cone angle [deg]
C_T	=	coning trajectory
$\vec{h}$	=	angular momentum vector [kgm^2/s]
$\vec{h}_c$	=	desired angular momentum vector [kgm^2/s]
$\vec{h}_S$	=	angular momentum vector from the sail simulation [kgm^2/s]
$\hat{n}$	=	sail normal unit vector
$\hat{n}_c$	=	desired sail normal unit vector (on the desired cone)
$\hat{n}_0$	=	sail equilibrium attitude in the L-frame
$\Delta\hat{n}$	=	sail normal error between $\hat{n}$ and $\hat{n}_c$
ϕ	=	clock angle [deg]
ϕ_0	=	equilibrium clock angle [deg]
ϕ_c	=	desired clock angle [deg]

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$\vec{\tau}$	=	net external torque acting on the sail [Nm]
$\tau_{control}$	=	control torque [Nm]
t	=	time [s]
ω_{no}	=	equilibrium sail inertial spin rate [rad/s]
ω_{nc}	=	desired inertial sail spin rate [rad/s]

I. Solar Sail Introduction

Solar sails are an attractive solution for expensive and massive space missions. Traditional spacecraft must carry in-space propulsion fuel that increases both launch mass and cost. A sail exposed to solar radiation offers free and continuous propulsion by manipulating the sail thrust vector direction relative to the Sun. Fig. 1 illustrates a simple solar sail configuration and Fig. 2 shows how the solar radiation thrust force is used for propulsion.

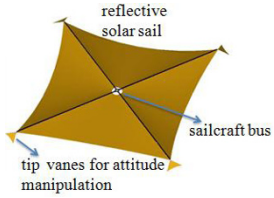

Fig. 1 Solar sail configuration

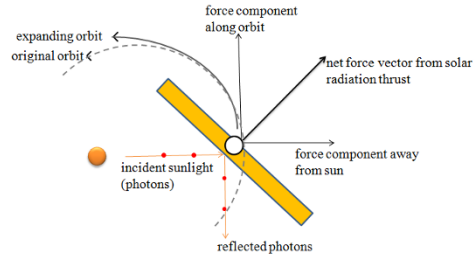

Fig. 2 Solar radiation thrust force

A solar sail consists of a sailcraft bus that houses the necessary electronics and hardware, a large, reflective, gossamer sail and an attitude manipulation component (tip vanes, thrusters, reaction wheels). The integrated effect of the reflected photons provides the propulsive thrust force. A large sail area is required in order to interrupt the photon radiation and produce an appreciable amount of thrust force. Since acceleration is inversely proportional to mass for a given thrust force, the mass of the sailcraft must be kept to a minimum. Thus, the sail and bus are designed to provide a large area-to-mass ratio and maximize the propulsion acceleration. Incident rays of sunlight reflect off of the sail (assume specular reflection from a perfectly flat sail) and produce two force components: one in the direction of the incident sunlight and another in the opposite direction of the reflected rays. In the net force vector, the components tangent to the sail surface cancel and the components normal to the surface add to produce the thrust force approximately in the sail normal direction. NASA's CubeSail project used a perfectly reflective 40 m^2 square sail in simulation to show that at 1 AU from the Sun, 0.03 N of solar radiation thrust force can be produced.[1] Although this force is relatively small compared with other propulsion methods, it is available continuously and hence can still be used to propel spacecraft for long distances without carrying any propellant.

The free propulsion from solar radiation makes extended mission durations feasible with reduced spacecraft mass and cost. A typical science mission is expected to cost on the order of one million dollars per kilogram of spacecraft mass.[1] This leaves high potential for low-cost science missions that have reduced spacecraft mass. As an example, compare results from a comet rendezvous mission study to reach Comet 88P/Howell using chemical propulsion versus solar sailing. Via traditional chemical propulsion, a Hohmann transfer from the Earth to the comet required 400 kg of propellant mass, whereas the solar sail completed the same mission with a total spacecraft mass of 3 kg (a factor of ~135 reduction in spacecraft mass). This translates into lower launch costs to Earth escape velocity and reduced development costs for the spacecraft.[2] The transfer time for the high thrust approach using chemical propulsion was 0.7 years as compared with 10 years for the solar sail. Therefore, for missions where transfer time is not a critical parameter, solar sail propulsion can result in significant fuel mass savings.

The significant solar thrust propulsion enables non-Keplerian orbits for solar sails. With this unique capability of solar sails, many space science missions can be achieved which are difficult to implement using conventional (chemical) propulsion techniques. Conventional propulsion can only produce Keplerian orbits such as ellipses, parabola and hyperbola (parts of a conic section). However, non-Keplerian orbits can be produced with constant sail thrusting through which orbit raising and precession can be achieved. One such mission is the study of the Earth magnetotail for which non-Keplerian orbits are desired. This requires the spacecraft orbit to continuously rotate to follow a Sun-synchronous path and also raise its orbit to explore the entire magnetotail. With this approach at least two years of scientific data could be returned.[3] Solar sail propulsion may provide an optimum propulsion system over conventional chemical propulsion, at least in some missions. It may also have advantages over electric propulsion due to the ease in attaining non-Keplerian orbits with reduced mission mass and cost.[4]

II. Solar Sail Attitude Dynamics and Control

Solar sail research on orbit analysis focuses on producing orbit raising/lowering and inducing orbital effects using the sails. These effects can yield otherwise expensive orbits such as Sun-synchronous and halo types. Orbit changes result from orienting the thrust vector, and in order to achieve the desired orbital effects, the sail must operate at the required attitude to modulate the solar thrust. Thus, orbit analysis has also motivated the research work on sail attitude manipulation. Sails typically contain a small spacecraft bus in the midst of a large, gossamer structure. Most sail work has addressed feasibility concerns related to orbit analysis and structural sail dynamics. Due to this configuration, significant solar, aerodynamic and gravity gradient torques act on the sail

and can disturb the sail orientation (attitude) relative to the sun. Accordingly, large control torques are needed to counteract these attitude disturbances, and an understanding of sail attitude dynamics is required in order to design appropriate control algorithms.

Recent studies have analyzed natural sail dynamics in order to maintain the desired thrust vector pointing.[5&6] Generally, large external torques are required to maintain the desired thrust vector pointing relative to the Sun. Lawrence, et. al. have shown that specific kinds of torques can be generated naturally under the influence of solar, aerodynamic and gravity gradient torques.[5] This reduces the need for expensive and massive traditional attitude control techniques (attitude jets or reaction wheels). The basic idea is to operate at the attitude equilibria of the sail normal vector in the local vertical local horizontal (LVLH) frame. An extension of this idea is to utilize a slight deviation of the sail normal from these equilibria, which results in sail normal coning about that equilibrium. McMahon, et. al. have shown that any desired orbit changes can be obtained with sail normal coning at orbit rate (circular cones).[6]

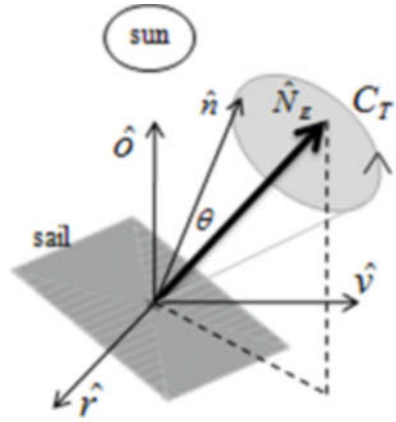

Fig. 3 Solar normal coning in local vertical local horizontal frame

In Fig. 3, the sail normal cones about the LVLH equilibrium attitude, $\hat{N}_E$. With a slight deviation, θ from the equilibrium attitude, the normal vector, $\hat{n}$ traces a coning trajectory, C_T in the L-frame (defines natural sail coning). The cone tracing should occur at orbit rate to attain desired sail orbit changes. The sail normal coning results show that any desired sail orbit changes can be obtained with sail normal coning at orbit rate (circular cones). In past work on the subject[7], a controller is designed that can achieve sail normal coning at orbit rate to attain the desired orbital effects. The natural dynamics of a rigid, uniform, flat and square solar sail under the influence of aerodynamic, solar and gravity gradient torques is used as the plant. It is found that even though sail normal cones about the equilibrium attitude, the coning is not of the required shape or at the required orbit

rate. Thus, a lypunov approach is used to develop the sail normal coning controller to enable the correct coning by studying the motion of the sail angular momentum vector. This controller enables the sail normal to track the desired coning trajectory at orbit rate, and hence induce the orbital effects. In this analysis, the performance of the coning controller on a spinning and non-spinning sails is studied.

III. Solar Sail Coning Dynamics

In this section, the dynamics of the solar sail are presented. With the sail dynamics, the concept of sail attitude equilibrium in the local vertical local horizontal frame (L-frame) is explained. The L-frame sail attitude equilibria enable sail normal coning in the inertial frame (A-frame). A small perturbation from the sail attitude equilibria induces L-frame coning of the sail normal about those sail equilibria. This L-frame coning of the sail is discussed. Natural environmental torques can cause sail precession and enable L-frame sail coning.

Many orbital effects can be obtained due to the L-frame coning of the sail normal about the sail equilibria. However, the natural rate and shape of coning of the sail normal about the sail equilibrium point does not yield the desired orbital effects. Control torques can be used to enforce the desired rate and coning shape (circular cones). Results showing the coning of the sail normal in the L-frame about the sail equilibrium are presented.

A. Reference Frames

In order to describe the sail dynamics in a circular orbit, the reference frames used are given in Fig. 4.

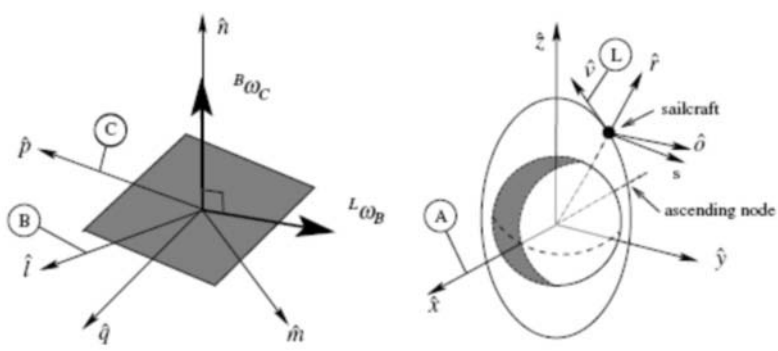

Fig. 4 Reference frames used to develop sail dynamics[5]

The A-frame $\{\hat{x}, \hat{y}, \hat{z}\}$ is the inertial (fixed) frame. The $\hat{x}$ and $\hat{y}$ define the Earth equatorial frame with $\hat{x}$ pointing along the vernal equinox. The $\hat{z}$ is normal to the equatorial frame. The local vertical local horizontal or L-frame $\{\hat{r}, \hat{v}, \hat{o}\}$ rotates along with the orbit at orbit rate. The $\hat{r}$ points along the orbit radial direction, $\hat{v}$ is in the direction of the sail velocity vector and $\hat{o}$ is aligned with the orbit angular momentum. The C-frame $\{\hat{n}, \hat{p}, \hat{q}\}$ is the sail body-fixed frame. The $\hat{n}$ points along the sail normal vector (normal to the plane of the sail). The $\hat{p}$ and

$\hat{q}$ remain in the plane of the sail. The B-frame $\{\hat{n}, \hat{l}, \hat{m}\}$ is also a body frame except that it does not rotate with the sail in the rotation about $\hat{n}$. The $\hat{l}$ and $\hat{m}$ lie in the plane of the sail. Thus, the only difference between the B-frame and C-frame is the rotation about $\hat{n}$. The B-frame is taken to be aligned with the C-frame at the initial time epoch.

B. Description of Sail Equilibria in L-frame and Inertial Sail Normal Coning

Any fixed sail normal $\hat{n}$ in the L-frame describes the sail equilibrium attitude in the L-frame. The sail attitude can be defined by the sail normal vector, $\hat{n}$ because the sail is symmetric about $\hat{n}$ and the rotation about $\hat{n}$ does not alter the forces on the sail. When the sail normal, $\hat{n}$ is fixed in the L-frame, but not aligned with $\hat{o}$, the sail rotates with the orbit and produces inertial coning of $\hat{n}$ at orbit rate.

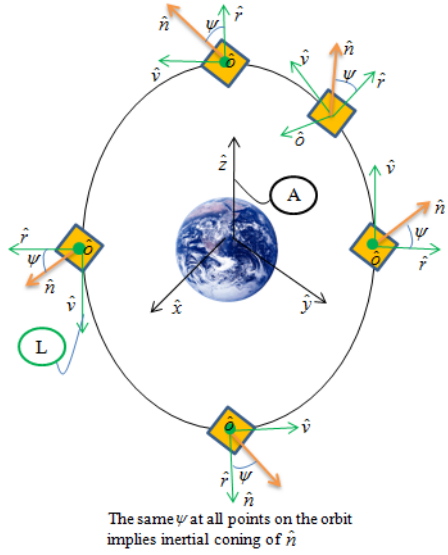

Fig. 5 Illustration of inertial coning of the L-frame fixed sail normal

In Fig. 5, the sail normal remains fixed in the L-frame (indicated by the constant angle ψ) and causes inertial coning of $\hat{n}$ at orbit rate. By choosing ψ appropriately, the sail angular momentum precesses to provide inertial coning with desirable orbit change effects. The angular momentum precession, in turn, is caused by torques acting on the sail. For $\hat{n}$ to remain fixed in the L-frame and enable inertial coning, the sail angular momentum must precess at a desired rate.

$$^{A}\left[\dot{\vec{h}}_{desired}\right] = \tau_{required} \tag{1}$$

where $\dot{\vec{h}}_{desired}$ and $\tau_{required}$ are the desired precession of the angular momentum and required torque to attain this desired motion, respectively. Lawrence, et. al. have shown that this $\tau_{required}$ can be achieved with naturally induced environmental torques (atmospheric, gravity gradient and solar) acting on the sail.

C. L-frame Sail Normal Coning

The sail normal $\hat{n}$ can be described in the L-frame via a cone angle β and a clock angle ϕ.

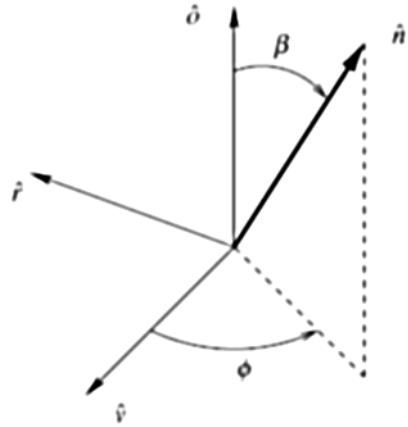

Fig. 6 Cone and clock angles of the sail normal relative to the L-frame[5]

From Fig. 6, the sail normal, $\hat{n}$ becomes

$$\hat{n} = -(\sin\beta\sin\phi)\hat{r} + (\sin\beta\cos\phi)\hat{v} + (\cos\beta)\hat{o} \tag{2}$$

Lawrence, et. al. have shown that for the developed sail dynamics, a small sail normal perturbation from the L-frame attitude equilibrium enables the sail normal to cone about that equilibrium point.[5] This constitutes sail normal coning in the L-frame illustrated in Fig. 7.

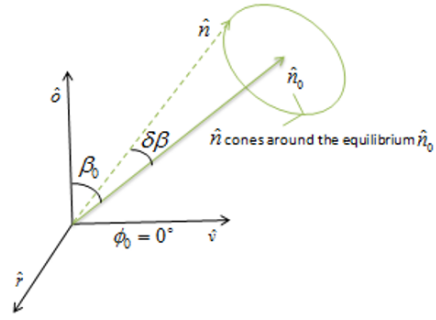

Fig. 7 Illustration of the sail normal coning in the L-frame

The $\hat{n}_0$ exhibits the sail equilibrium point and $\hat{n}$ is the coning sail normal vector. The perturbation is given by a small deviation, $\delta\beta$ in the cone angle, β. The coning is defined such that the cone need not be circular or have a

fixed coning rate. McMahon, et. al. have shown that desired orbital effects can be induced when the L-frame coning occurs at orbit rate such that one rotation of $\hat{n}$ around $\hat{n}_0$ is completed in one orbital period and has a circular coning shape.[6] They state that the greatest orbital effects can be induced when the sail normal is operated at $\beta = 35^\circ$ and $\phi^\circ = 0^\circ$ for 1° cones. This study is performed using these parameters.

D. Natural Dynamics of Sail Normal Coning in L-frame using CubeSail Simulation

Lawrence, et. al. have created a MATLAB simulation for a CubeSail (small solar sail satellite) that propagates the sailcraft attitude under the influence of aerodynamic, solar pressure and gravity gradient torques over a circular, Low Earth Orbit.[5] The sailcraft studied consists of a solar sail and sailcraft bus, which is located out of the sail plane. The sailcraft model used in this study is shown in Fig. 8.

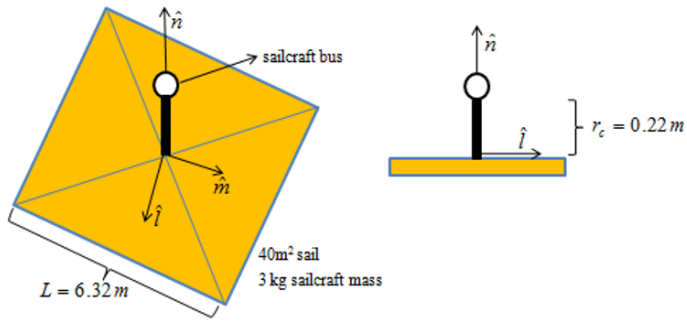

Fig. 8 Physical components and dimensions of the sailcraft model

The sail is assumed to be flat, rigid, uniform, square with a sailcraft bus located out of the sail plane at a distance r_c along the sail normal, $\hat{n}$. Since the sail is symmetric and rotation about $\hat{n}$ does not alter any solar, aerodynamic, or gravity gradient forces, the sail attitude can be described only by the sail normal vector when considering these external effects. The bus is located near the plane of the sail (0.22 m) as compared with the sail size (40 m^2), which enables the sailcraft moment of inertia to be similar to that of a flat plate.[8] A large, gossamer sail will be non-rigid in space. It is argued that the orbital element control applications using coning motions produce smooth, low frequency environmental torques, on the order of orbit frequency and require closed loop settling times on the order of several orbits. The disturbance frequencies and control system bandwidths are on the order of 10^{-4} Hz (for 700 km Low Earth Orbit), whereas the lowest structural modes of the sail are in the range of 10^{-1} to 10^{-2} Hz.[4] This suggests that the torque applications do not excite the sail structural modes, thus justifying the rigid body assumption from a control-structure interaction viewpoint. Non-uniform material properties within the sail will cause imperfect solar reflections and variations in pressure-loading on the sail. The varying pressure-loading issue will deform the sail and hence it will no longer remain flat in orbit. A deformed

sail will experience different torques as a function of attitude. However, the varying pressure-loading issue is a secondary effect and offers more insight on the sailcraft torque as a function of attitude by deviating from the ideal case. This study focuses on the ideal (flat) case. The case of a non-spinning sail is analyzed in this work. The orbit and sailcraft parameters used are shown in Table 1.

Table 1. Orbit and Sailcraft Parameters

Orbit Parameters	
Altitude, a	700 km
Inclination, i	90°
Right ascension of ascending node, Ω	180°
Orbit rate, ω_o	1.06×10^{-3} rad/s
Right ascension of sun, r_s	90°
Declination of sun, d_s	0°
Solar flux at 1 AU, F_s	1358 W/m^2
Atmospheric density, ρ	5×10^{-14} kg/m^3
Aerodynamic moment coefficient, τ_{am}	1.18×10^{-5} Nm
Solar moment coefficient, τ_{sm}	9.84×10^{-6} Nm
Gravity gradient moment coefficient, τ_{gm}	6.33×10^{-6} Nm
Sailcraft Parameters	
Sail side length, L	6.325 m
Sail mass, m_s	1.7 kg
Sailcraft bus mass, m_b	1.3 kg
Distance to bus from sail plane in $\hat{n}$ direction, r_c	0.22 m
Axial moment of inertia, I_n	11.3 kgm^2
Transverse moment of inertia, I_t	5.7 kgm^2
Sail reflectivity, s	0.9
Aerodynamic coefficient of drag, C_d	2.2
Specular reflectance fraction, f_s	0.7

E. Solar Sail Spin Rate

To attain sail equilibria in the L-frame and enable inertial sail coning, the sail angular momentum must achieve the desired precession rate. Besides control torques, environmental torques also affect sail dynamics. These environmental torques emerge from gravity gradient, atmospheric and aerodynamic drag torques. In order to precess the sail, the net combination of all three environmental torques must provide the required torque for sail precession. The details of this torque equation are found in Ref. 5. For the desired precession rate, the sail must spin about its axis at a required rate. The sail spin about its axis is described in Fig. 9.

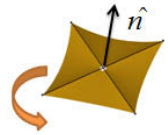

Fig. 9 Sail spin about the sail normal vector

The desired sail spin rate is the inertial spin rate referred to as the equilibrium sail spin rate (includes the sail spin rate about its axis). The sail spin rate is treated as a variable in order to balance the torque equation, and the spin rate that balances the equation is called the equilibrium spin rate. Sail normal coning about an orbit lowering equilibrium point (case $\phi = 0^\circ$) is considered in this analysis because this is a typical application for solar sails, precession about which produces the maximum orbital effects. For a given β orientation of $\hat{n}$ in the L-frame (and $\phi = 0^\circ$), given orbital parameters and sail design parameters, the required equilibrium spin rate is

$$\omega_{no} = \frac{\cos\beta}{I_n \omega_o}\left(\omega_o{}^2 I_T + \tau_{am} - \tau_{sm} sign(\cos\beta)\right) \tag{3}$$

where τ_{am}, τ_{sm} are environmental torque parameters, ω_o is the orbit rate, and I_n, I_T are the sailcraft design parameters for moment of inertia (see Table 1).

IV. Natural Dynamics of Spinning Versus Non-Spinning Solar Sails

As the sail normal precesses from the equilibrium point and cones in the L-frame, the sail need not spin at the equilibrium rate. The control torques required to remain on the desired coning trajectory differ based upon the sail spin rate used. In order to minimize costs, the sail spin rate requiring the least control torque magnitude needed to trace the desired orbit rate circular coning trajectory should be used. The control torque induces the desired coning and enforces orbit rate coning. The part of the control torque used to induce the desired coning (referred to as coning control torque) allows the sail normal to trace the desired shape of the coning trajectory, whereas the other part of the control torque enforcing orbit rate coning (called as the rate control torque) allows the sail normal to trace the trajectory at the desired rate. The total control torque required to attain the desired orbit rate circular coning is a combination of these two control torques. When the sail normal is precessed from the attitude equilibrium and has the equilibrium sail spin rate, Lawrence, et. al. have shown that the sail normal exhibits natural coning about that sail equilibrium in the L-frame.[5] Due to the already coning behavior of the sail normal, the coning control torque magnitude is reduced. The natural coning motion does not occur at the desired orbit rate.[5] In order to enforce the coning at orbit rate, the magnitude of the rate control torque increases for the equilibrium sail spin case. This is because the spinning sail adds to the total angular momentum of the sailcraft

and thus increases the rate control torque required to provide the desired momentum precession. As the sail spin rate magnitude is reduced however, the sail does not carry as much momentum due to which the momentum can be precessed more easily with reduced rate control torque. On the other hand, deviations from the equilibrium spin rate can cause loss of natural coning behavior. This increases the need to require more coning control torque such that the desired coning motion can be attained. The rate control torque magnitude is the least for the zero sail spin rate because a non-spinning sail adds no additional momentum to the sailcraft momentum. This makes the desired momentum precession possible with the least rate control torque. Conversely, the coning control torque magnitude is the least for the equilibrium sail spin rate because the already coning behavior (natural coning) is best attained when the sail spins at the equilibrium rate. Due to this trade-off between the rate control torque and coning control torque (combination of both yields the total required control torque), it is interesting to study which sail spin rate can result in the desired orbit rate circular coning with the least control torque magnitude. In this thesis, the L-frame sail coning is analyzed for two sail spin rates: equilibrium spin rate and zero spin rate. Before determining the required control torques needed to enforce the desired coning, it is important to study the natural motion of the sail with the equilibrium and zero sail spin rates under the influence of the environmental (natural) torques. In the CubeSail simulation, a small perturbation ($\delta\beta = 1^\circ$) is added to a sail equilibrium having a cone angle, $\beta = 35^\circ$ and clock angle, $\phi = 0^\circ$. The simulation results showing the coning of $\hat{n}$ in the L-frame for these two spin rates are presented.

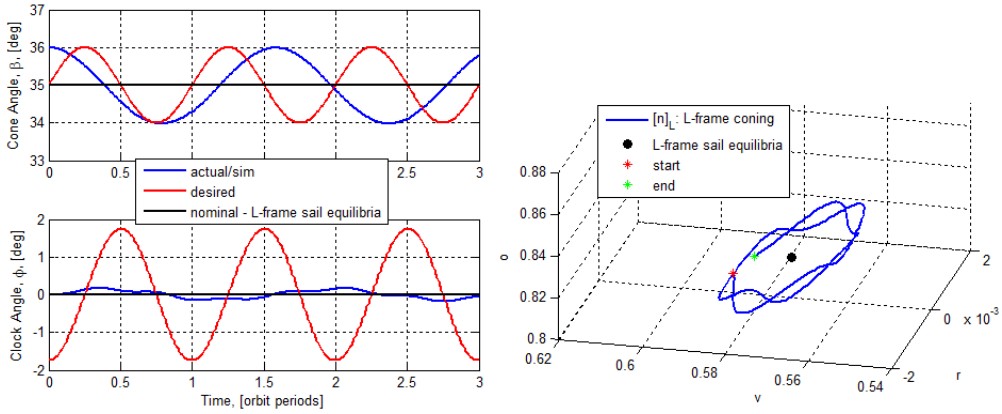

Fig. 10 Sail normal coning in the L-frame about the equilibrium point, [β=35°, ϕ=0°] using the equilibrium sail spin rate

The three-dimensional plot shows the motion of $\hat{n}$ in the L-frame. The two-dimensional plot gives a time history of the cone and clock angles over the course of several orbital periods. The $\hat{n}$ cones about the equilibrium point in the L-frame. However, the coning does not occur at orbit rate (one cone is traced in one and a half orbits).

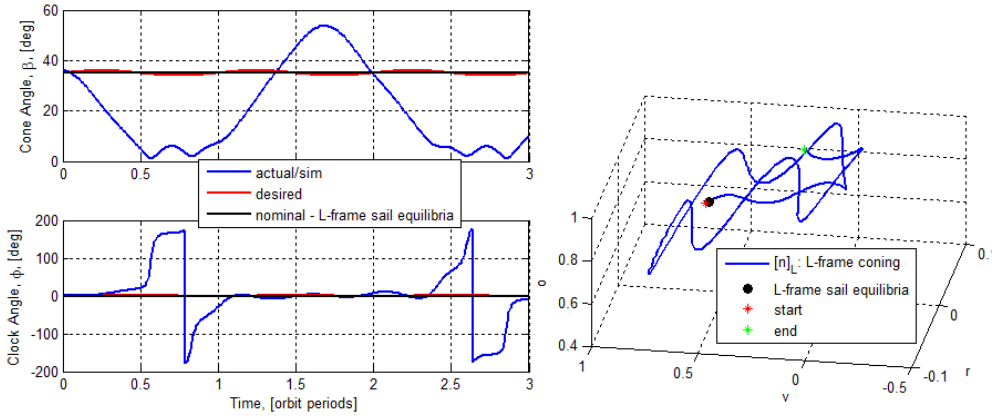

Fig. 11 Sail normal coning in the L-frame about the equilibrium point, [β=35°, ϕ=0°] using zero sail spin rate

Even though $\hat{n}$ stays in the vicinity of the equilibrium point, the coning is not well-defined in Fig. 11. Since the sail is not spinning about $\hat{n}$ at the equilibrium spin rate (has zero sail spin), the dynamics are different, and required L-frame coning is not observed. Although quasi-periodic, the motion of $\hat{n}$ in the L-frame does not occur at orbit rate (one cone is traced every two orbits in this case).

Since the natural motion is not orbit rate circular coning (desired for useful orbital effects) control torques on the sailcraft are necessary.

V. Dynamics of Spinning Versus Non-Spinning Solar Sails using Sail Coning Controller

The control method developed in Ref. 7 is used in this analysis. This control method enables the sail normal $\hat{n}$ to track the desired sail normal $\hat{n}_c$ on the circular coning trajectory and provide orbit rate coning in the L-frame. The relationship between the sail angular momentum, $\vec{h}$ and $\tau_{control}$ ($\left[\dot{\vec{h}}\right]^{inertial} = \tau$) is used to construct the control law. The control method is developed such that $\vec{h}$ tracks the desired angular momentum, $\vec{h}_c$ on the coning trajectory at orbit rate. The control law is used to reduce the error between $\vec{h}$ and $\vec{h}_c$. Since sail angular momentum is a function of sail normal/angular position or sail attitude, the control law is created with the hope

that error reduction in angular momentum and thus tracking the desired angular momentum at orbit rate will also lead to tracking the desired sail angular position at orbit rate (enable $\hat{n}$ to track $\hat{n}_c$). Please see Ref. 7 for details on this angular momentum error reduction control method. This control method also includes the sail spin rate controller that maintains the equilibrium spin rate of the sail. Results for the coning dynamics of non-spinning and spinning sails are presented below.

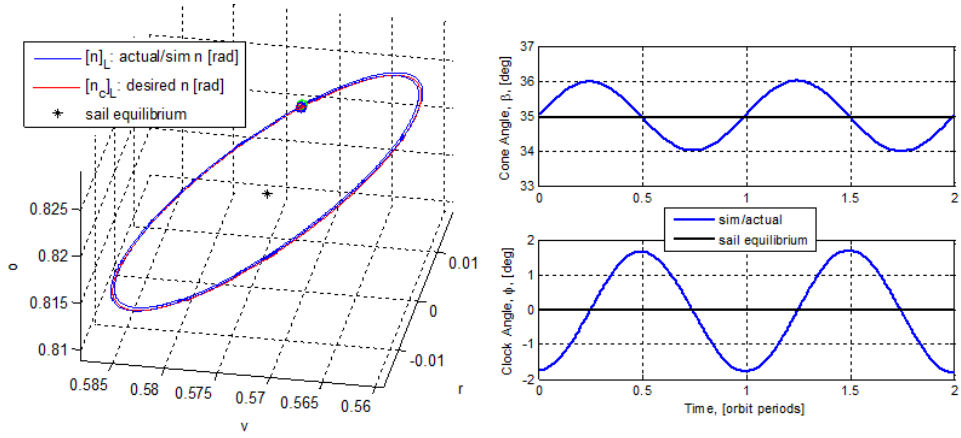

Fig. 12 Non-spinning sail angular positions for coning trajectory about nominal sail normal of β_0=35°, Φ_0=0° in the L-frame

The three-dimensional plot indicates that the control method enables the simulation sail normal $\hat{n}$ to trace the desired circular cone. In the cone and clock angles plot, both β and ϕ complete one cycle in one orbital period and repeat the same pattern in the next orbital period. Thus, the control method also enables orbit rate coning. The errors in the individual components of sail normal in the coning tracing are shown in Fig. 13.

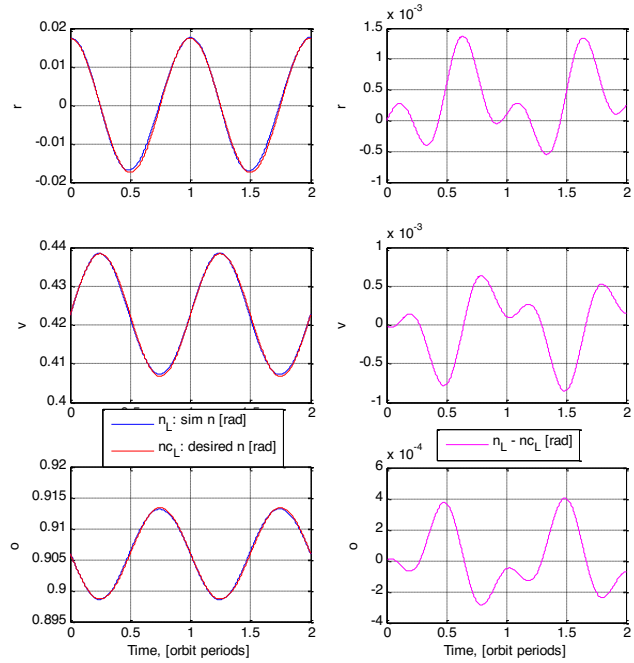

Fig. 13 Individual L-frame components of simulated and desired sail normal with errors

As the cone is traced, Fig. 13 shows that the error magnitudes of the $\hat{r}$, $\hat{v}$ and $\hat{o}$ sail normal components oscillate. Even though there are errors in the cone tracing, they appear to be bounded. In addition, the error magnitudes are significantly lower than the absolute component magnitudes (10^1-10^3 times lower). McMahon, et. al. assert that the shape of the coning need not be accurate as long as orbit rate coning is achieved in order to yield the orbital effects.[6] Thus, slight deviations from the circular coning are not expected to have significant deviations in the averaged orbital effects induced over an entire orbit. The control torque required to enable this orbit rate coning is on the order of 10^{-6} Nm.[7]

For the spinning sail case with the given L-frame equilibrium point and from Eq. 3, the sail spin rate is calculated to be -2.97 x 10^{-4} rad/s (component of the equilibrium sail spin rate). The coning dynamics for this spinning sail are presented next.

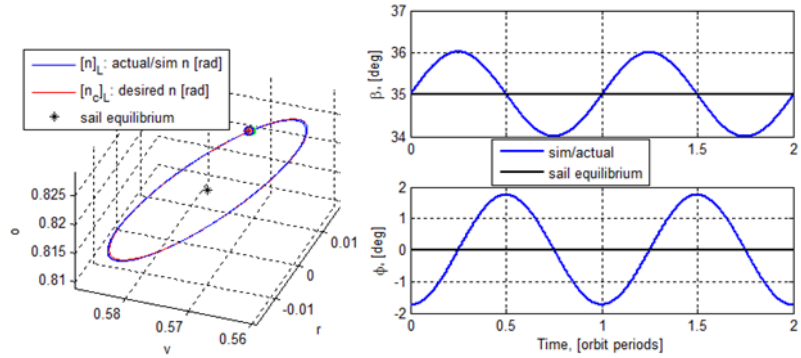

Fig. 14 Spinning sail angular positions for coning trajectory about nominal sail normal of β_0=35°, Φ_0=0° in the L-frame

The simulation sail normal, $\hat{n}$ traces the desired circular cone. In addition, the cone and clock angles repeatedly complete one cycle in one orbital period. This indicates orbit rate coning. As compared with the zero sail spin (Fig. 12), the cone tracing appears more accurate. To gain more confidence in this, the errors in the individual components of sail normal while tracing the desired cone are shown in Fig. 15.

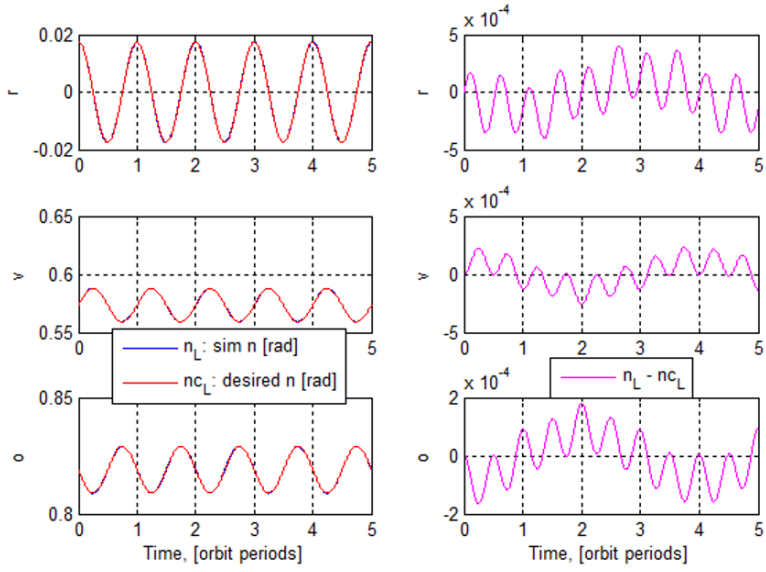

Fig. 15 Individual L-frame components of simulated and desired sail normal with errors

As the cone is traced, the oscillating nature of the error magnitudes is similar to the ones in the zero sail spin. However, the magnitudes are 10 times lower in the equilibrium spin case than the zero spin case (Fig. 13). This means that the control method was able to provide better sail normal tracking at the same cost (same control

torque magnitude of 10^{-6} Nm is required[7]) with the equilibrium spin as compared with the zero spin for this L-frame sail equilibrium point ($\beta = 35^{\circ}, \phi = 0^{\circ}$).

VI. Conclusion

Past work has shown that orbit rate coning of the sail normal vector about an LVLH attitude equilibrium point induces desired orbital effects. Sail attitude equilibria exist in the LVLH frame under the influence of gravity gradient, aerodynamic and solar torques. When the sail normal is precessed from some of these equilibria, the sail normal naturally cones about that equilibrium point. However, the sail normal coning has to follow a circular coning trajectory at orbit rate to induce the desired orbital effects. Previous work developed a control method that enables sail normal coning (circular cones) about the LVLH attitude equilibria at orbit rate. In this analysis, the dynamics of spinning versus non-spinning sails under the influence of natural and control torques was explored for one of the LVLH equilibria points (point which induces maximum orbital effects).

The control method allowed the sail normal to trace the desired sail normal on the circular coning trajectory at orbit rate. The coning was at orbit rate (accurate coning rate), but there were errors in the shape of the coning (inaccurate circular coning). Even though there were errors in the circular cone tracing (10 to 10^3 times lower than the absolute sail attitude), they were bounded. In addition, past work asserted that the shape of coning need not be as accurate as long as orbit rate coning is achieved in order to attain the desired orbital effects. Thus, slight deviations from circular coning that cause these error magnitudes in the cone tracing are within the acceptable range.

The required control torque is a combination of two control torque components: rate control torque and coning control torque. The rate control torque defines that piece of the total control torque which enforces orbit rate coning and allows the sail normal to trace the trajectory at the desired rate. The coning control torque is that piece which is used to induce the desired coning and allows the sail normal to trace the desired shape of the coning trajectory. It was argued that rate control torque is lower in the non-spinning sail case (total angular momentum is lower for a non-spinning sail and thus the momentum vector is easier to precess), whereas the coning control torque is lower for the equilibrium sail spin case (equilibrium sail spin causes natural coning that reduces the required control authority). It was interesting to determine which sail spin rate could result in the desired orbit rate circular coning with a lower cost (lower control torque magnitude). Even though, the required control torque was on the order of 10^{-6} Nm for both a spinning and non-spinning sail (10^{-6} Nm was found to be a reasonable torque magnitude from other work on small sailcraft with similar dimensions and mass properties),

the equilibrium spin case provided a more effective control method performance in tracking the desired angular momentum and sail normal as compared with the zero sail spin case (10 times more effective). However, since slight deviations from the actual coning trajectory are allowed and given the same required control torque, it is advisable to operate a non-spinning sail to avoid additional mechanisms and making the system more susceptible to structural and mechanical failures.

Acknowledgments

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The Attitude Control of a Tri-Spin Solar Sail Satellite

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The aim is to develop a conceptual satellite that combines the scalability and stability of a spinning sail with the agility of a 3-axis stabilized sail. A tri-spin satellite is proposed making use of a spinning sail and a momentum countering system (MCS) rotating relative to the satellite body. The MCS reduces the angular momentum bias increasing the maneuverability of the satellite. Induced oscillations in the non-rigid elements due to attitude changes are investigated. A complete attitude determination and control system (ADCS) for such a satellite in an earth-centered orbit is proposed and simulated. The simulation reveals that the tri-spin concept is viable.

Nomenclature

$\bar{x}_I, \bar{y}_I, \bar{z}_I$	=	Inertial earth-centered coordinates
$\bar{x}_O, \bar{y}_O, \bar{z}_O$	=	Orbit referenced coordinates
$\bar{x}_B, \bar{y}_B, \bar{z}_B$	=	Satellite body coordinates
θ, ϕ, ψ	=	Euler 213 pitch, roll, yaw attitude angles (rad)
$\boldsymbol{q}$	=	Attitude quaternion set
$\boldsymbol{\omega}_{B/O}$	=	Orbit referenced body rates (rad/s)
$\boldsymbol{\omega}_{B/I}$	=	Inertial referenced body rates (rad/s)
$\hat{\boldsymbol{\omega}}_{B/O}$	=	Estimated orbit referenced body rates (rad/s)
$\boldsymbol{q}_e$	=	Error quaternion set
$\boldsymbol{A}_O^B$	=	Orbit to body reference frame direction cosine matrix
$\boldsymbol{\omega}_{B/I}$	=	Inertial referenced body rate vector (rad/s)
$\boldsymbol{\omega}_{B/O}$	=	Orbit referenced body rate vector (rad/s)
$\hat{\boldsymbol{\omega}}_{B/O}$	=	Estimated orbit referenced body rate vector (rad/s)
ω_{sy}	=	Angular rate of sail relative to satellite body (rad/s)

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ω_{cy} = Angular rate of momentum counter system relative to satellite body (rad/s)

$\boldsymbol{I}$ = Moment of inertia of entire spacecraft (kg·m^2)

$\boldsymbol{I}_s$ = Moment of inertia of sail and sail mechanism (kg·m^2)

$\boldsymbol{I}_c$ = Moment of inertia of momentum counter system (kg·m^2)

$\boldsymbol{I}_h$ = Moment of inertia of stabilized satellite bus (kg·m^2)

$\boldsymbol{N}$ = External torque vector (N·m)

$\boldsymbol{h}_w$ = Angular momentum of reaction wheels (N·m·s)

$\boldsymbol{h}_s$ = Angular momentum of spinning sail relative to satellite body (N·m·s)

$\boldsymbol{h}_c$ = Angular momentum of momentum counter system relative to satellite body (N·m·s)

$\boldsymbol{e}_{sun}$ = Sun unit vector in body coordinates

$\boldsymbol{e}_u$ = The control reference vector

I. Introduction

THIS paper introduces a conceptual design of a spinning solar sail satellite in an earth-centered orbit. The satellite performs attitude maneuvers to control the solar thrust to change its orbit. This thrust can increase the energy of the orbit to create an escape trajectory or reduce the altitude to deorbit. The IKAROS and NanoSail-D2 missions have demonstrated solar sails successfully in orbit [1, 2]. CubeSats are a popular test bed for solar sail systems proving its worth in the NanoSail-D2 mission and future missions of Surrey's CubeSail and DeOrbitSail [3], Planetary Society's LightSail-1 and Illinois' CubeSail [4] that are all planned to be launched in the near future [5].

IKAROS was a spinning solar satellite using wire booms with tip masses as tensioners to support the sail. The rotation produced a continuous centrifugal force, which kept the sail and wire booms stiff [1]. Nanosail-D2, Surrey CubeSail, and LightSail-1 deploy semi-rigid booms used as sail masts [2, 3, 5, 6]. These semi-rigid booms are curved strips that are rolled up and kept under pressure. When released, the strips straighten and produce an almost rigid boom on which the sail is deployed. Other mechanisms to deploy and supply structure to a solar sail include deployable rigid trusses [7] and inflatable structures.

Spinning solar sails provide greater scalability and stability when compared to 3-axis stabilized sails. The continual rotation creates a centrifugal force, which keeps the sail stiff and makes it possible to use wire booms with tip masses to supply further structure. The spinning sail produces a large angular momentum, which makes the satellite more resistant to disturbances, but also increases the control energy and maneuver time required to

change its attitude. Stabilized sails require deployable semi-rigid booms to provide structure to the sail. A non-rotating satellite body can make use of conventional reaction wheels to perform fast attitude maneuvers.

This paper will continue to introduce a conceptual satellite, which aims to combine the maneuverability of a 3-axis stabilized sail with the disturbance immunity and scalability of a spinning sail. Section 2 will present the conceptual design of a tri-spin solar sail satellite and introduce the new subsystems required. Section 3 introduces a basic analysis of the dynamics of the satellite system. Section 4 discusses the attitude control modes required and introduces a simulation of the conceptual satellite performing the required maneuvers to control the solar thrust. The results of the simulation are discussed.

II. Spinning Sail Concept

The energy of a satellite's orbit energy changes when an acceleration (Δv) is experienced in the direction of the satellite's linear velocity. The orbit trajectory will correlate more to a parabolic escape trajectory as the orbit energy increases. A decrease in orbit energy will shrink the elliptic orbit until the satellite reaches earth's atmosphere. A solar sail in an earth-centered orbit must change its angle relative to the sun to control the Δv from the solar thrust. The aerodynamic drag force per unit area of a satellite at altitudes below 600km ($P_a > 6.488\times10^{-6}$ N/m^2) is larger than the solar pressure force per unit area ($P_s = 4.563\times10^{-6}$ N/m^2). The deorbit lifetime of satellite greatly increases if it is stationed beyond this altitude. Solar thrust can be used to decrease the altitude to obtain denser atmosphere and so greatly reduce the deorbit duration. The solar thrust can also be used to perform station keeping and increase the satellite's lifetime in its orbit without the need for chemical/electrical propulsion.

The orbital period of a satellite in a low earth orbit (LEO) is between 85 and 130 minutes. Attitude maneuvers must be completed in a fraction of the satellite's orbital period to maximize the orbital changing thrust. This requirement prevents the use of low torque actuators to change the attitude of the satellite. For example, actuator mechanisms using solar thrust to create a controlled torque, have much longer settling times for a change in orientation [8, 9, 10]. Conventional actuators, like reaction wheels and thrusters, produce larger torques that will result in faster maneuvering.

A spinning solar sail satellite can use wire booms and tip masses to supply a mast structure for the sail but the rotating satellite body prevents the use of reaction wheels. However, a despun satellite body can still use reaction wheels. The rotating sail becomes a large external momentum wheel. This rotating structure supplies gyroscopic stiffness making the satellite more resistant to external disturbance torques. This also increases the control energy required to change the attitude. The system's maneuverability is improved by counteracting the

angular momentum bias through an internal momentum wheel. This solution is not scalable, for the counter momentum wheel can become large when the sail size is increased.

A scalable momentum counter system (MCS) solution consists of an external momentum wheel which deploys wire booms and tip masses. The MCS can scale by adjusting the wire boom length or tip mass as the size and nominal rate of the spinning sail changes. The resultant conceptual satellite is a tri-spin solar sail satellite with fast maneuvering capabilities with the solar sail and MCS rotating relative to the satellite bus. Fig. 1 shows the conceptual satellite with the satellite bus (blue), MCS deployment mechanism (green), sail deployment mechanism (red), solar panels (yellow) and sensors (cyan).

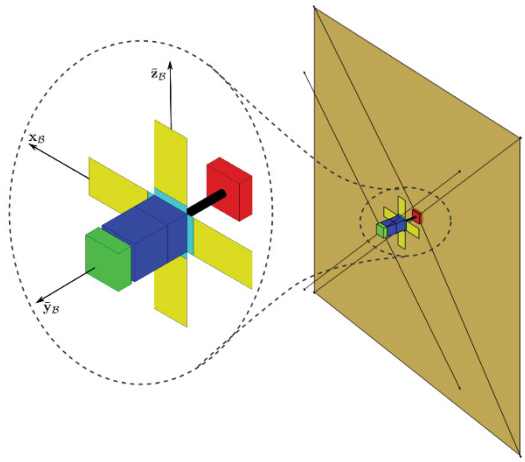

Fig. 1 Conceptual satellite

III. Dynamic Model

The satellite has three rotating parts and non-rigid components. Attitude changes of the satellite bus will result in oscillations in the non-rigid components, which will influence the dynamics of the attitude changes. Deploying large structures will have a large effect on the satellite's attitude. The feasibility of the attitude control is dependent on the output of a model that describes the dynamics of the entire satellite system.

A. Reference Frame

The attitude of the satellite is described in three reference frames. The inertial reference frame, I: $\{\bar{x}_I, \bar{y}_I, \bar{z}_I\}$, is fixed to the center of the earth (geocentric system) with one axis in the same direction as the spin vector of the earth. The second axis is in direction of the Vernal equinox (♈) the third axis is to create a valid right-handed

frame. The orbit fixed frame, $O:\{\bar{x}_O, \bar{y}_O, \bar{z}_O\}$, is defined with the origin at the center of mass (CoM) of the satellite. The first axis is in the direction of the velocity vector at apogee and perigee of the satellite, the second axis is in the orbit anti-normal direction and the third axis is nadir pointing. The body-fixed frame, $B:\{\bar{x}_B, \bar{y}_B, \bar{z}_B\}$, is defined at the CoM and points in the directions of the principle moment of inertia.

B. Kinematics

The attitude relative to the orbit frame is described by a quaternion set ($\boldsymbol{q}=[q_1\ q_2\ q_3\ q_4]^T$) or by Euler angles. The relation of the satellite's body rates relative to the orbit frame and the quaternion set is:

$$\omega_{B/O}=\begin{bmatrix}\omega_{bx}\\ \omega_{by}\\ \omega_{bz}\end{bmatrix}=2\begin{bmatrix}q_4 & q_3 & -q_2 & -q_1\\ -q_3 & q_4 & q_1 & -q_2\\ q_2 & -q_1 & q_4 & -q_3\end{bmatrix}\begin{bmatrix}\dot{q}_1\\ \dot{q}_2\\ \dot{q}_3\\ \dot{q}_4\end{bmatrix} \tag{1}$$

The direction cosine matrix (DCM) from the orbit frame to the body fixed frame is described in quaternions

$$A_O^B=\begin{bmatrix}q_1^2-q_2^2-q_3^2+q_4^2 & 2(q_1q_2+q_3q_4) & 2(q_1q_3-q_2q_4)\\ 2(q_1q_2-q_3q_4) & -q_1^2+q_2^2-q_3^2+q_4^2 & 2(q_2q_3+q_1q_4)\\ 2(q_1q_3+q_2q_4) & 2(q_2q_3-q_1q_4) & -q_1^2-q_2^2+q_3^2+q_4^2\end{bmatrix} \tag{2}$$

and is described in an Euler 2-1-3 sequence by

$$A_O^B=\begin{bmatrix}C\psi C\theta+S\psi S\phi S\theta & S\psi C\phi & -C\psi S\theta+S\psi S\phi C\theta\\ -S\psi C\theta+C\psi S\phi S\theta & C\psi C\phi & S\psi S\theta+C\psi S\phi C\theta\\ C\phi S\theta & -S\phi & C\phi C\theta\end{bmatrix} \tag{3}$$

with C the cosine function and S the sine function. The inertial referenced body rates are calculated from, $\boldsymbol{\omega}_{B/I}=\boldsymbol{\omega}_{B/O}+\boldsymbol{A}_O^B[0\quad -\omega_o\quad 0]^T$, with ω_o the instantaneous orbital rate. The orbital rate is constant when assuming a circular orbit.

C. Dynamics

The attitude dynamics of the satellite is described by extending the standard Newton-Euler equations. The extended Newton-Euler equation is

$$\boldsymbol{I}\dot{\boldsymbol{\omega}}_{B/I}=\boldsymbol{N}-(\dot{\boldsymbol{I}}_s+\dot{\boldsymbol{I}}_c)\boldsymbol{\omega}_{B/I}-\dot{\boldsymbol{h}}_s-\dot{\boldsymbol{h}}_c-\dot{\boldsymbol{h}}_w+\boldsymbol{\omega}_{B/I}\times(\boldsymbol{I}\boldsymbol{\omega}_{B/I}+\boldsymbol{h}_s+\boldsymbol{h}_c+\boldsymbol{h}_w). \tag{4}$$

The external torque includes the controlled magnetic torques ($\boldsymbol{N}_m$), controlled thruster torques ($\boldsymbol{N}_t$) and the sum of all the disturbance torques ($\boldsymbol{N}_D$). The moment of inertia of the sail, MCS and satellite bus contributes to the total moment of inertia of the satellite ($\boldsymbol{I}=\boldsymbol{I}_s+\boldsymbol{I}_c+\boldsymbol{I}_h$). The angular momentum relative to the satellite body of the sail and the MCS is defined as $\boldsymbol{h}_s=[0\ \boldsymbol{I}_s\omega_{sy}\ 0]^T$ and $\boldsymbol{h}_c=[0\ \boldsymbol{I}_c\omega_{cy}\ 0]^T$ respectively.

D. Non-Rigid Dynamics

The rotating sail and MCS consist of non-rigid elements. Attitude changes of the satellite bus will result in in-plane and out-plane angle offsets of the tip masses. This angular offsets will result in an oscillation in the tip mass positions and therefore influence the moment of inertia of the entire satellite. The effect of the changing moment of inertia on the attitude due to the deployment of structures as well as the tip mass oscillations is captured in (4) by the $\dot{\boldsymbol{I}}_s$ and $\dot{\boldsymbol{I}}_c$ terms.

Investigating the effects of the attitude changes on the non-rigid elements require another approach. Methods like Finite Element Method (FEM) and elasticity theory supplies very accurate approximations of the effects of angular accelerations. These methods produce very large and complex models that are more difficult to integrate into an attitude simulation. An attitude simulation containing the non-rigid dynamics will confirm whether the non-rigid elements remain in a stable state throughout the maneuver sets. A model using Lagrangian mechanics and defining two degrees of freedom for each wire boom produce a good lower order approximation.

This model was used to investigate the dynamics of the tip mass under different attitude maneuvers and physical parameters. A satellite with a sail with a constant spin rate ω_{sy} performs an 90° attitude maneuver around the $\bar{z}_B$ axis by means of a controller of the form $K(K_d\omega_{bz}+K_p\psi_e)$. The maximum out-plane angle due to an attitude step command is investigated by varying the spin rate of the sail (ω_{sy}) and the overall controller gain (K). These attitude steps were conducted at $t=t_1$, with $t_1>0$. Figure 2 shows the result of this investigation and reveals that the tip mass offset grows as the controller gain increases, until settling at a maximum offset value. This indicates, as one would expect, that the rate at which ω_{bz} change ($\dot{\omega}_{bz}$) cause the out-plane offset. By limiting $\dot{\omega}_{bz}$ the amplitude of the tip mass oscillations will be reduced. This results in a time specification that attitude controllers should have high damping ratio and little to no overshoot to limit $\dot{\omega}_{bz}$. It is also clear from Fig. 2 that the spin rate of the sail has a more dominant effect, i.e. decreasing the tip mass offset at an exponential rate as the spin rate increases.

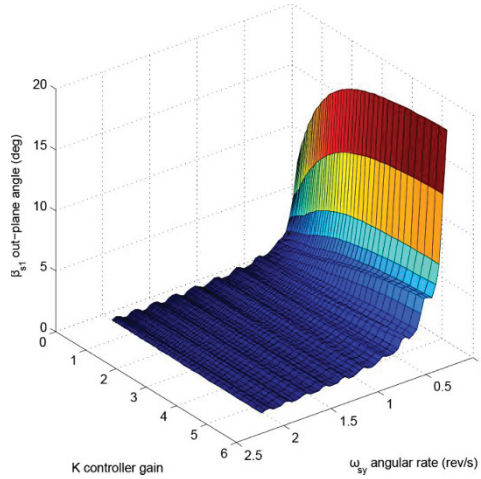

Fig. 2 Maximum out-plane angle as function of spin rate and controller gain

IV. Attitude Control

An attitude determination and control system (ADCS) for a tri-spin solar sail satellite is proposed. The performance of the ADCS is investigated with an extensive attitude simulation. It is assumed that the satellite is in a low earth orbit (LEO) and makes use of magnetic control and two reaction wheels. The ADCS will consist of a number of control modes including: detumbling, deployment, solar-tracking and maximum aerodynamic drag.

The ADCS design will be based on a satellite containing, which contains torquer rods in all body axes, and two reaction wheels in the $\bar{x}_B$ and $\bar{z}_B$ axes. The control torques created by magnetic torquer rods can be substituted by torques from electric/chemical thrusters for satellites at higher altitudes. A fine sun sensor, nadir sensor, magnetometer and coarse sun sensors are used to determine the satellite's current attitude.

A. Control Modes

The tri-spin solar sail satellite will have several control modes (see Fig. 3). The detumbling phase implements magnetic controllers which removes the initial angular momentum present in the satellite system This is done to ensure a near to zero angular momentum starting condition when deploying the sail and the MCS. The deployment phase consists firstly of deploying the MCS and then the sail. The satellite will change its attitude relative to the sun to generate maximum or minimum solar thrust. The order of the sun-following and sun-avoidance phases will depend on the satellite's orbit and whether positive (increasing orbit energy) or negative (decreasing orbit energy) thrust is required. Further control modes include a controller for keeping the

satellite stable during eclipse when no sun vector is present, a controller to desaturate the reaction wheels and a controller that points the sail normal in the velocity vector. The last controller will be used in a very low orbit and the sail is then used to maximize the aerodynamic drag for fast orbit decay.

The detumbling phase will make use of external torques to remove the angular momentum present in the satellite system. This includes the angular momentum of a rotating satellite body and the internal momentum exchange devices. This is to ensure that the disturbances during the deployment of the sail and the MCS are within controllable ranges. The angular rates of a rotating satellite body can be reduced by either using a simple magnetic Bdot and Y-spin controller [8] or by more complex magnetic cross-product controllers.

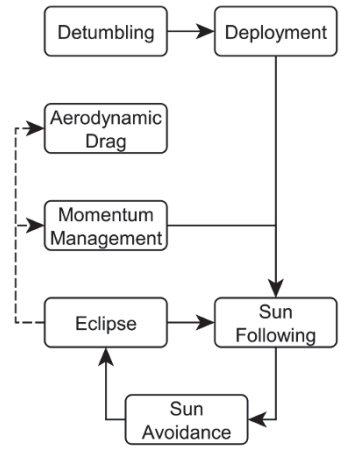

Fig. 3 Control modes for decreasing the altitude

The deployment of the sail and MCS contributes to disturbance torques around the $\bar{\boldsymbol{y}}_B$ axis. The MCS is deployed first with the stowed sail counteracting the effect from the deployment of the MCS. The disturbances caused by the sail deployment are controlled by the deployed MCS. In the case of the detumbling and the deployment phase, the body rates of the satellite are controlled. Rate sensors or a Rate Kalman Filter (RKF) [9] using magnetometer measurements can be used to supply the required body rate information.

The satellite is required to change from a maximum solar thrust orientation to a minimum solar thrust orientation to ensure that the maximum altitude changing thrust is generated. The maximum solar thrust orientation is when the solar rays are perpendicular to the sail surface. The minimum solar thrust is generated when maintaining an orientation with the minimum projected area to the sun. The quaternion error is calculated directly by determining the quaternion rotation of the $\bar{\boldsymbol{y}}_B$ axis towards the required sun pointing attitude. The quaternion rotation for the $\bar{\boldsymbol{y}}_B$ axis to point towards the sun is calculated from,

$$\boldsymbol{e}=\boldsymbol{e}_{sun}\times\boldsymbol{e}_{u} \tag{5}$$

and

$$\Phi=\cos^{-1}\left(\frac{\boldsymbol{e}_{sun}\cdot\boldsymbol{e}_{u}}{|\boldsymbol{e}_{sun}|\cdot|\boldsymbol{e}_{u}|}\right) \tag{6}$$

with the control reference vector defined in this case as the $\bar{\boldsymbol{y}}_B$ axis (the axis perpendicular to the sail surface) and $\boldsymbol{e}=[e_x\ e_y\ e_z]^T$, see Fig. 4.

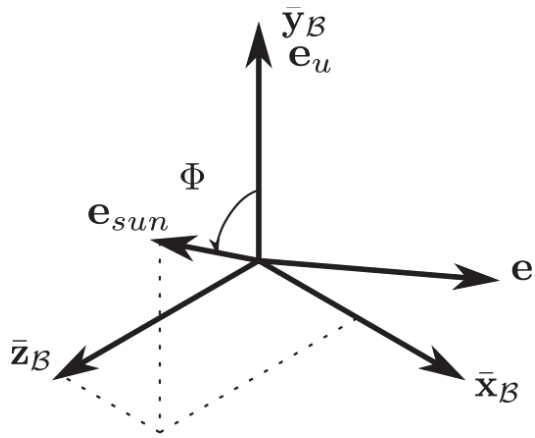

Fig. 4 The quaternion definition for tracking the sun vector

The error quaternion is determined by

$$q_{e1}=\frac{e_x}{|\boldsymbol{e}|}\sin\left(\frac{\Phi}{2}\right) \tag{7}$$

$$q_{e2}=\frac{e_y}{|\boldsymbol{e}|}\sin\left(\frac{\Phi}{2}\right) \tag{8}$$

$$q_{e3}=\frac{e_z}{|\boldsymbol{e}|}\sin\left(\frac{\Phi}{2}\right) \tag{9}$$

$$q_{e4}=\cos\left(\frac{\Phi}{2}\right) \tag{10}$$

The sun-avoidance controller uses a similar method to determine the error quaternion set except for an additional relative offset angle of 90° to ensure the sail normal vector will be perpendicular to the sun vector. The sun-avoidance rotation angle is calculated as,

$$\Phi=\cos^{-1}\left(\frac{\boldsymbol{e}_{sun}\cdot\boldsymbol{e}_{u}}{|\boldsymbol{e}_{sun}|\cdot|\boldsymbol{e}_{u}|}\right)\text{-}90°. \tag{11}$$

When the sun is not visible to the satellite's sun sensors, the $\boldsymbol{e}_{\text{sun}}$ is estimated using a sun orbit model, which propagates the modeled orbit referenced sun vectors and transform these to body coordinates using the estimated DCM. The estimated DCM can be obtained from an Extended Kalman Filter [9] utilizing that uses the magnetic, sun and horizon vectors to determine the satellite's current attitude. The quaternion error is used in a quaternion

feedback reaction wheel controller, which uses the reaction wheels in the $\bar{x}_B$ and $\bar{z}_B$ axes and the MCS in the $\bar{y}_B$ axis to reduce the attitude error and track the sun

$$\boldsymbol{N}_w = -\boldsymbol{K}_{dw}\,\hat{\boldsymbol{\omega}}_{B/O} - \boldsymbol{K}_{qw}\boldsymbol{q}_e \tag{12}$$

with

$$\boldsymbol{N}_w = \dot{\boldsymbol{h}}_w + \dot{\boldsymbol{h}}_c \tag{13}$$

The feedback gains ($\boldsymbol{K}_{dw}$ and $\boldsymbol{K}_{qw}$) are functions of the time specifications and the principle moment of inertia terms of the satellite as described in [10]. Referring back to Fig. 2 these gain values do have an effect on the stability of the sail and supporting wire booms. The specifications should be adjusted to obtain a step response with higher damping and no overshoot to prevent the introduction of unnecessary wire boom tip mass oscillations.

Propagating a sun model that supplies the modeled orbit referenced sun vector the conversion from a sun-following to a sun-avoidance controller can be automated. The sign of the $\bar{x}_O$ component of the modeled sun vector indicates the type of (positive or negative) thrust the sail can generate when pointing to the sun vector.

B. Simulation

The performance of the ADCS and the effect of the solar thrust are investigated within a simulation. The simulation implements the ADCS algorithms on a CubeSat sized satellite with the mission objective to illustrate the solar thrust from a tri-spin solar sail satellite. The satellite is a 4U CubeSat with 1.6U dedicated for the satellite bus, 0.4U for the MCS deployment mechanism and 2U for the sail storage and deployment mechanism (refer to Figure 1). The simulated satellite will deploy a 25m^2 sail, which is similar to the sail size that the Surrey CubeSail and LightSail-1 are planning to deploy. The simulated satellite is released from the launcher in its stowed configuration with initial orbit-referenced body rates. After successfully detumbling using a Bdot and Y-spin magnetic controller with a magnetic RKF, the satellite will deploy its solar panels. The final configuration is achieved after the successful deployment of the MCS and the sail. The principal moment of inertia of the different configurations and release rates are summarized in Table 1. The simulated satellite will then enter the solar tracking phase of operation. The simulation is repeated showing both orbit increasing and orbit decreasing capabilities. Table 2 contains the orbital elements of the simulation. The satellite is outfitted with a magnetometer, coarse sun sensors, fine sun sensor and horizon sensor. Three magnet torquers and two reaction wheels, in the $\bar{x}_B$ and $\bar{z}_B$ axes, is used as actuators. The specifications of the actuators and sensors are

seen in Table 3 and 4. These specifications were extracted from CubeSat components that are commercially available.

Table 1 Simulation parameters

Parameter	Value
Sail area	$A_s = 25m^2$
MCS wire boom length	$l_{MCS} = 1.5m$
Principal moment of inertia in stowed configuration	$\boldsymbol{I}(t=0) = \begin{bmatrix} 0.0814 \\ 0.0078 \\ 0.0815 \end{bmatrix} kg{\cdot}m^2$
Principal moment of inertia in final stage configuration	$\boldsymbol{I}(t=0) \approx \begin{bmatrix} 0.707 \\ 1.260 \\ 0.708 \end{bmatrix} kg{\cdot}m^2$
Initial release body rates assumed	$\boldsymbol{\omega}_{B/O}(t=0) = \begin{bmatrix} 1.4 \\ -5 \\ -1.4 \end{bmatrix}$ °/s

Table 2 Simulated orbit parameters

Parameter	Value
Semi-major axis a	7169.65km
Eccentricity e	0.001
Inclination i	98.24°
Period P	6041.7s

Table 3 Sensor noise

Sensor	Noise level (1σ)
Magnetometer	12nT
Fine sun sensor	0.4°
Coarse sun sensor	5°
Nadir/Horizon sensor	0.1°

Table 4 Actuator specifications

Actuator	Restriction
Torquer rod	$0.2Am^2$
Internal reaction wheels	40mNms
	2.1mNm
Sail and MCS driving motor	10mNm

The satellite's orbit is simulated using a Simplified General Perturbation No.4 (SGP4) model. The geomagnetic field vector is modeled using a 10^{th} order International Geomagnetic Reference Field (IGRF). The theoretical solar thrust generated by the sail uses the model presented in [11, 12].

The results of the ADCS simulation for both decreasing and increasing the orbit energy are seen in Fig. 5 to 9. The shaded areas reveal the periods where the satellite is in the eclipse period of its orbit. Figure 5 shows the true orbit referenced body rates of the satellite bus during the detumbling and deployment of the sail and MCS.

The initial body rates were reduced with a BdotBDot and Y-spin magnetic controller until the satellite has an almost zero angular momentum. The deployment of the MCS resulted in larger disturbances than those duringthe sail deployment. The disturbances were increased to investigate the ability of the satellite to correct these faults. The deployment phase controlled the disturbances, effectively stabilizing the satellite's body rates at zero body rates.

The effectiveness of the sun-following and sun-avoidance controllers can be seen in Fig. 6 and 7. The sun-following controller was activated and the major component of the sun vector is in the $\bar{y}_B$ axis direction, indicating that the solar rays are almost perpendicular to the sail surface. The sun-avoidance controller almost zeroed the $\bar{y}_B$ sun vector component. Figures 8 and 9 reveal the orbit referenced solar thrust force generated during this period. The solar thrust component in the $\bar{x}_O$ axis direction will change the energy of the orbit. In the orbit decrease mode (see Fig. 6) the $\bar{x}_O$ component of the thrust is always negative and in the orbit increase mode (see Fig. 6) it is always positive. This indicates the successful switching between the sun-following and the sun-avoidance controllers producing a maximum orbit changing thrust force. The 90° attitude maneuver is completed within 4 to 5 minutes. The solar-avoidance controller also successfully zeroed the solar thrust.

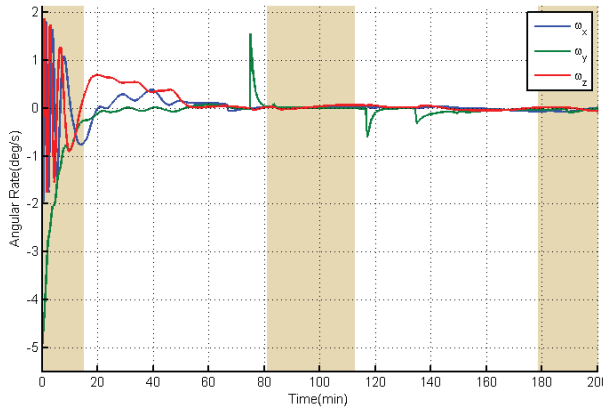

Fig. 5 ω_{by} body rate during detumbling and deployment

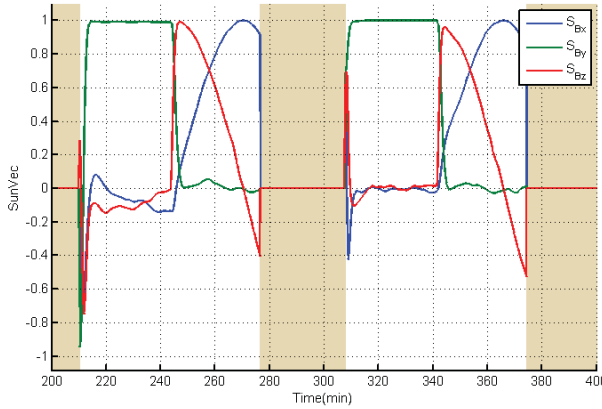

Fig. 6 Body referenced sun vectors for orbit decrease

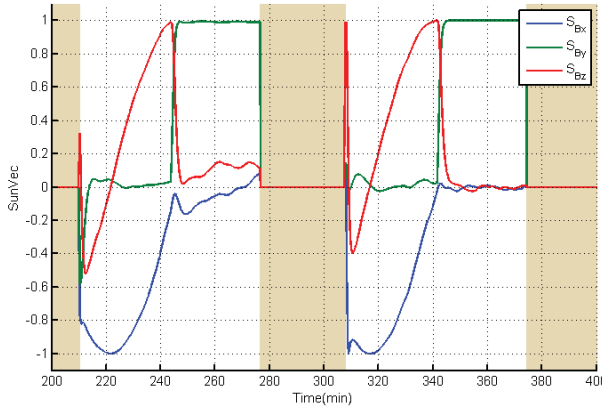

Fig. 7 Body referenced sun vectors for orbit increase

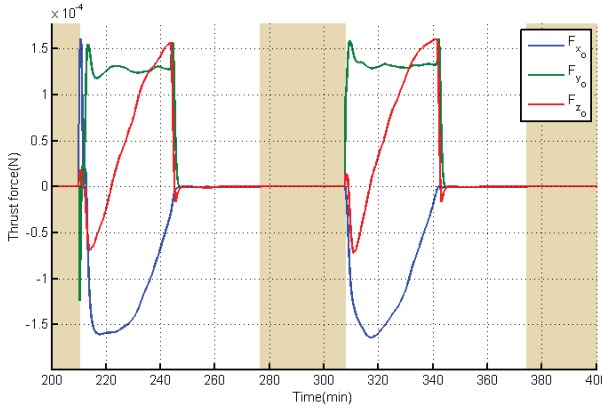

Fig. 8 Orbit referenced solar thrust force for orbit decrease

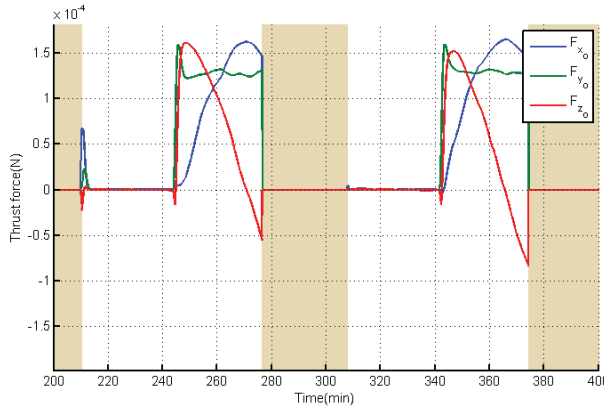

Fig. 9 Orbit referenced solar thrust force for orbit increase

V. Conclusion

A concept for a highly maneuverable spinning solar sail was presented. The satellite consists of three parts, with the sail and the momentum counter system (MCS) rotating relative to the satellite body. The MCS is a deployable momentum wheel, which counter the large angular momentum created by the rotating sail.

The high maneuverability of the satellite can induce oscillations in the non-rigid components of the sail and MCS. A simple and low order model revealed that the angular acceleration of the satellite contributes to tip mass offsets. By limiting the angular acceleration with highly damped, zero overshoot attitude controllers, the amplitude of these oscillations will be small.

The ADCS for a conceptual tri-spin solar sail satellite in a LEO was proposed. The ADCS includes controllers to detumble, deploy the sail and MCS and to perform solar tracking or avoidance to control the solar thrust. Simulation tests were conducted to investigate the performance of these controllers and estimators. The simulation results showed that the satellite would be able to reduce its initial release rates and handle the disturbances caused by the sail and MCS deployment. Alternating between a sun-following and sun-avoidance controller proved to effectively use solar thrust to increase or decrease the orbital energy by raising or lowering of the satellite's altitude. By changing the order of these controllers during the sunlit part of each orbit either an increase or decrease in orbital energy can be obtained.

It is therefore feasible to construct a maneuverable spinning solar sail satellite, which can effectively control its orbit altitude using solar thrust. This approach is suitable for satellites required to change their altitude without the need of a propulsion system consuming propellant. This concept is ideally suited for satellites in an earth-centered orbit, aiming for either an escape trajectory, a deorbiting trajectory or orbit altitude maintenance.

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Influence of Attitude Control on Orbital Plane Change for Flexible Solar Sail

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The solar sail is large and highly flexible, and its motion involves a coupling of the orbit, the attitude and structural vibration. This paper establishes the reduced dynamic model for a flexible solar sail with foreshortening deformation and coupling of its attitude and vibration. In the process of attitude control with the large-angle maneuvers, large orbital deviations and structural vibrations are generated. When initial deviations and solar pressure disturbance torques are considered, the process of attitude control leads to greater accumulated error in the transfer trajectory, which demonstrates that the process of attitude control is important to the solar sail mission.

Nomenclature

$\boldsymbol{AU}$ = astronomical unit, 1.4959787×10^{11} m

$\boldsymbol{C}_t$ = coupling matrix of transfer and structural vibration

$\boldsymbol{C}_r$ = coupling matrix of rotation and structural vibration

f = true anomaly, deg

$\boldsymbol{F}_g$ = gravitational force, N

$\boldsymbol{F}_s$ = solar radiation pressure vector, N

$\boldsymbol{F}_t$, $\boldsymbol{F}_r$, $\boldsymbol{F}_f$ = generalized forces caused by solar radiation pressure

$\boldsymbol{F}_{gt}$, $\boldsymbol{F}_{gr}$, $\boldsymbol{F}_{gf}$ = generalized forces caused by gravity

$\boldsymbol{J}_f$ = mass moments of inertia tensor coupled with the attitude and the structural vibration, kg•m^2

$\boldsymbol{K}_f$ = modal stiffness matrix

$\boldsymbol{K}_P$, $\boldsymbol{K}_D$ = control coefficient of proportion and differential

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$\boldsymbol{L}_S$ = offset between the mass center and pressure center, m

m_{tot} = total mass of the solar sail, kg

$\boldsymbol{M}_S$ = solar pressure disturbance torque, N•m

$\boldsymbol{q}$ = modal coordinate

$\boldsymbol{Q}_f$, $\boldsymbol{Q}_f$, $\boldsymbol{Q}_f$ = generalized forces

$\boldsymbol{Q}_{ft}$, $\boldsymbol{Q}_{fr}$, $\boldsymbol{Q}_{ff}$ = generalized forces caused by coupling

$\boldsymbol{R}_0$ = displacement vector of the hub of the solar sail in the global inertial frame, m

$\boldsymbol{R}$ = displacement vector from the sail to the sun, m

$d\boldsymbol{R}$ = initial orbital deviation, m

$\boldsymbol{r}_f$ = local position vector in the undeformed state, m

$\boldsymbol{S}_f$ = static moment coupled with the attitude and the structural vibration, kg•m

$\boldsymbol{U}$ = control torque vector, N•m

$\boldsymbol{u}_f$ = local displacement vector, m

$\boldsymbol{V}$ = velocity vector from the sail to the sun, m/s

w = transverse deflection from an arbitrary point on the sail, m

(x_0, y_0) = a reference point on the middle surface with zero longitudinal displacement, m

α = cone angle, deg

δ = clock angle, deg

$\boldsymbol{\Phi}$ = mode shape matrix that can be obtained using a finite element program

$\boldsymbol{\Phi}_{NL}$ = first derivatives of local displacement vector with respect to modal coordinate

φ, θ, ψ = Euler angles between the orbital frame and the body frame, deg

μ = gravity constant of the sun, 1.3271×10^{20} N•m^2/ kg

$\boldsymbol{\Omega}$ = vector of Euler angles of the local frame with respect to the orbital frame of the solar sail, deg

$d\boldsymbol{\Omega}$ = initial deviation of the vector of Euler angles, deg

ω = argument of periapsis, deg

$\boldsymbol{\omega}$ = angular velocity vector of the floating frame with respect to the inertial frame, deg/ s

$\boldsymbol{\omega}_{bo}$ = angular velocity with respect to the orbital frame, deg/ s

$\boldsymbol{\omega}_r$ = orbital angular velocity with respect to the orbital frame, deg/ s

$d\boldsymbol{\omega}$ = angular acceleration increment, deg/ s^2

$\dot{(\,)}, \ddot{(\,)}$ = first and the second time derivatives, respectively, with respect to the inertial frame

(′), (″) = first and the second time derivatives, respectively, with respect to the floating frame

(˜) = cross product matrix of the vector

($_{ref}$) = desired state variables

I. Introduction

A solar sail is a new type of spacecraft for deep-space exploration that is propelled by the pressure of sunlight. To reduce the launch cost and achieve the highest thrust, the solar sail must employ large-scale, lightweight structures, which cause the solar sail to be highly flexible. When the flexible structures are in motion, the vibration of the structure influences the dynamic characteristics of the entire structure and the control, and the influence is significant, especially when the structure undergoes large-angle maneuvering [1].

Given the coupled effect of orbit and attitude, some special orbits that are difficult for ordinary spacecraft can be designed, such as observation of the sun polar, displaced orbits, exploration of planets and the outer solar system, and so on. The coupled attitude–orbit control of solar sails has been the subject of much research, but the research models used to date have been models of either a rigid sail or a point mass, and due to the control of the attitude, the thrust was changed to achieve orbit control [2,3].

The Solar Polar Imager (SPI) mission is one of several Sun–Earth Connection solar sail roadmap missions envisioned by NASA. The objectives of the SPI mission are to investigate the global structure and dynamics of the solar corona and reveal the secrets of the solar cycle and the origins of solar activity [4]. It is also defined that the required sail slew rate as 10^0 per day during the cranking phase with the examination of locally optimal inclination control law [4].

Wie and Murphy [5], for example, studied a 160-m, 450-kg solar sail spacecraft, considering an uncertain center-of-mass/center-of-pressure offset of ±0.4 m. The SPI sailcraft first spirals inwards from 1 AU to a heliocentric circular orbit at 0.48 AU, followed by a cranking orbit phase, to achieve a science mission orbit at a 75^0 inclination over a total sailing time of 6.6 years. A reference AOCS consisting of a propellantless primary ACS and a microthruster-based secondary (optional) ACS is used. A relatively fast, 70^0 pitch reorientation within 3 hours of every half orbit period occurs during the 5-year cranking orbit phase in this study, and it is also validated by a high-fidelity dynamic model of flexible sail. The orbit and the attitude of the solar sail were examined in this study, but the influence of the attitude on the orbit was not studied [5].

Using a conservative solar sail film temperature limit, a “hot” mission scenario was designed and compared with a “cold” mission scenario in [6]. Based on this “hot” mission scenario, different sail temperature limits, different characteristic accelerations, different interplanetary insertion energies, and different sail degradation

behaviors were investigated to gain deeper insight into the trade space of the SPI mission and to help the designer of such a mission to estimate the required transfer time [6].

Based on a transfer mission with 2 preceding phases, a transfer process of 3 phases was researched in [7]. In the first phase (the approach phase), the sailcraft follows a trajectory coplanar to the heliocentric circular intermediate orbit, whose orbital radius depends on the sail's characteristic acceleration. The second phase (the cranking phase) is used to vary the orbital plane position alone by maintaining a nearly constant distance from the sun while driving the orbital inclination to the desired final value. In the third phase (raising phase), the orbital radius is increased up to the achievement of the desired value while maintaining the orbital inclination unchanged. The choice of 2 or 3 phases was selected to depend on the value of the inclination change between the initial and final orbits [7].

There have been many studies on the transfer mission of heliocentric polar orbit, and the available organs and control methods for attitude control are represented in these studies. However, the coupled effect of the orbit and the attitude on the transfer mission has not been studied. This paper establishes a reduced dynamic model for a flexible solar sail, coupled with its orbit, attitude and structural vibration. Based on this coupled dynamic model, the phase of orbital plane change in the transfer mission is analyzed, and the influences of attitude changing and flexibility on the transfer trajectory are studied.

II. Coupled Dynamic Model of a Flexible Solar Sail

A. Coordinate system definition

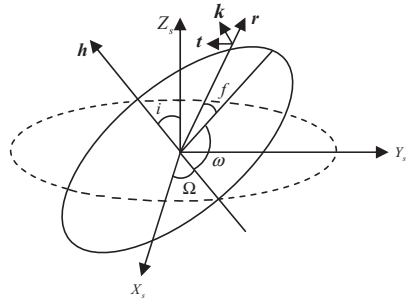


Fig. 1 Representation of a coordinate systems definition

Before describing the orbital and attitude dynamics of the solar sail, three coordination systems are defined: the solar inertial frame $OX_sY_sZ_s$, the orbital frame $oxyz$ and the body frame $o_bx_by_bz_b$. The definitions of the coordinate systems are shown in Fig. 1 and Table 1.

Table 1 Coordinate system definitions

Name	Origin coordinate	Axes
Solar inertial frame $OX_sY_sZ_s$	Center of mass of the sun	OX_s is in the solar ecliptic plane and points to the vernal equinox, OZ_s is perpendicular to the ecliptic plane, and OY_s forms a right-handed triad.
Orbital frame $oxyz$	The orbital displacement of the solar sail	$ox(\boldsymbol{r})$ lies along the orbital displacement vector of the sail, $oz(\boldsymbol{k})$ is perpendicular to the orbital plane, and $oy(\boldsymbol{t})$ forms a right-handed triad.
Body frame $o_bx_by_bz_b$	The center of the square sail	o_bx_b and o_by_b are in the plane of the sail, and o_bz_b lies along the normal of the sail.

The transformation matrix between each frame are represented as

$$\boldsymbol{A}_{s2o}=\left(\frac{\boldsymbol{R}}{\|\boldsymbol{R}\|}\quad\frac{\boldsymbol{R}\times(\boldsymbol{R}\times\boldsymbol{V})}{\|\boldsymbol{R}\times(\boldsymbol{R}\times\boldsymbol{V})\|}\quad\frac{\boldsymbol{R}\times\boldsymbol{V}}{\|\boldsymbol{R}\times\boldsymbol{V}\|}\right)^T \tag{1}$$

$$\boldsymbol{A}_{o2b}=R_y(\theta)R_z(\psi)R_x(\varphi)=\begin{pmatrix}\cos(\theta)&0&-\sin(\theta)\\0&1&0\\\sin(\theta)&0&\cos(\theta)\end{pmatrix}\begin{pmatrix}\cos(\psi)&\sin(\psi)&0\\-\sin(\psi)&\cos(\psi)&0\\0&0&1\end{pmatrix}\begin{pmatrix}1&0&0\\0&\cos(\varphi)&\sin(\varphi)\\0&-\sin(\varphi)&\cos(\varphi)\end{pmatrix} \tag{2}$$

where $\boldsymbol{A}_{s2o}$ is the transformation matrix from the solar inertial frame to the solar orbital frame, $\boldsymbol{A}_{o2b}$ is the transformation matrix from the orbital frame to the body frame, the order of rotation is 1-3-2, and the Euler angles are defined as $\boldsymbol{\Omega}=(\varphi\quad\theta\quad\psi)^T$. The transformation matrix from the solar inertial frame to the body frame is written as

$$\boldsymbol{A}_{s2b}=\boldsymbol{A}_{o2b}\cdot\boldsymbol{A}_{s2o} \tag{3}$$

The attitude of the solar sail is described by 2 angles, the cone angle α and the clock angle δ, as illustrated in Fig. 2. α is the angle between the normal of the sail $\boldsymbol{n}$ and the sunlight, and δ is the angle between the normal of the sail $\boldsymbol{n}$ and $\boldsymbol{k}$ in the plane formed by $\boldsymbol{k}$ and $\boldsymbol{t}$ [5].

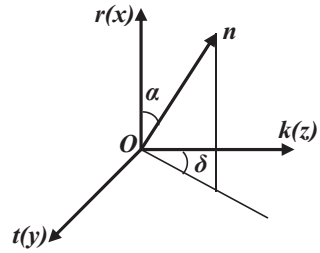


Fig. 2 Representation of the sail normal in the orbital frame

The relationship between the 2 attitude angles and the Euler angles is obtained from Fig. 2

$$\begin{aligned}\varphi &= -\delta \\ \theta &= \pi/2 - \alpha \\ \psi &= 0\end{aligned} \tag{4}$$

The angular velocity and the angular acceleration of the solar sail in the body frame are written as

$$\boldsymbol{\omega}^{b} = \boldsymbol{\omega}_{bo} + \boldsymbol{A}_{o2b} \cdot \boldsymbol{\omega}_{r} \tag{5}$$

$$\dot{\boldsymbol{\omega}}^{b} = \dot{\boldsymbol{\omega}}_{bo} + \dot{\boldsymbol{A}}_{o2b} \cdot \boldsymbol{\omega}_{r} = \dot{\boldsymbol{\omega}}_{bo} + d\boldsymbol{\omega}_{2} \tag{6}$$

where the angular velocity and the angular acceleration with respect to the orbital frame are described by Euler angles as

$$\boldsymbol{\omega}_{bo} = \begin{pmatrix} -\dot{\psi}\sin\theta + \dot{\varphi}\cos\theta\cos\psi \\ -\dot{\varphi}\sin\psi + \dot{\theta} \\ \dot{\psi}\cos\theta + \dot{\varphi}\sin\theta\cos\psi \end{pmatrix} = \begin{pmatrix} \cos\theta\cos\psi & 0 & -\sin\theta \\ -\sin\psi & 1 & 0 \\ \sin\theta\cos\psi & 0 & \cos\theta \end{pmatrix} \begin{pmatrix} \dot{\varphi} \\ \dot{\theta} \\ \dot{\psi} \end{pmatrix} = \boldsymbol{A}_{\Omega} \cdot \dot{\boldsymbol{\Omega}} \tag{7}$$

$$\dot{\boldsymbol{\omega}}_{bo} = \begin{pmatrix} \cos\theta\cos\psi & 0 & -\sin\theta \\ -\sin\psi & 1 & 0 \\ \sin\theta\cos\psi & 0 & \cos\theta \end{pmatrix} \begin{pmatrix} \ddot{\varphi} \\ \ddot{\theta} \\ \ddot{\psi} \end{pmatrix} + \begin{pmatrix} -\dot{\varphi}\dot{\psi}\cos\theta\sin\psi - \dot{\theta}\dot{\psi}\cos\theta - \dot{\theta}\dot{\varphi}\sin\theta\cos\psi \\ -\dot{\varphi}\dot{\psi}\cos\psi \\ -\dot{\varphi}\dot{\psi}\sin\theta\sin\psi - \dot{\theta}\dot{\psi}\sin\theta + \dot{\theta}\dot{\varphi}\cos\theta\cos\psi \end{pmatrix} = \boldsymbol{A}_{\Omega} \cdot \ddot{\boldsymbol{\Omega}} + d\boldsymbol{\omega}_{1} \tag{8}$$

Substituting Eq. (8) into Eq. (6), the angular acceleration is expressed in terms of the Euler angles as

$$\dot{\boldsymbol{\omega}}_{b} = \boldsymbol{A}_{\Omega} \cdot \ddot{\boldsymbol{\Omega}} + d\boldsymbol{\omega}_{1} + d\boldsymbol{\omega}_{2} = \boldsymbol{A}_{\Omega} \cdot \ddot{\boldsymbol{\Omega}} + d\boldsymbol{\omega} \tag{9}$$

B. Dynamic model of a flexible solar sail

As Fig. 3 shows, $OX_sY_sZ_s$ represents the global inertial frame of reference, and $o_bx_by_bz_b$ represents the local floating frame of reference fixed to the center of the solar sail.

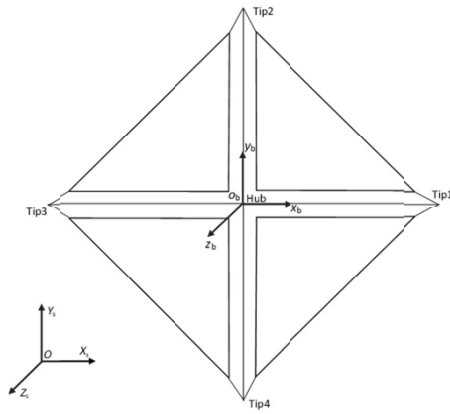


Fig. 3 Representation of a five-point-connected square solar sail

The main flexible structures of a solar sail are the booms and the membranes. The booms are modeled as Euler–Bernoulli beams, based on the assumptions that the cross section is rigid and remains perpendicular to the

vertical axis after deformation. The sails are modeled as thin plates. Because they interact, they should be modeled as parts of a single structure. The vector from an arbitrary point on the structure to O is written as

$$\boldsymbol{R} = \boldsymbol{R}_O + \boldsymbol{r}_f + \boldsymbol{u}_f \tag{10}$$

Only the out-of-plane deformation and the foreshortening deformation are of concern. The local elastic displacement of an arbitrary point on the booms or the sails with respect to the floating frame is written as

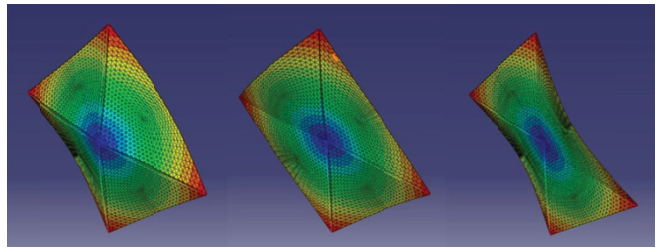
$$\boldsymbol{u}_f = \begin{pmatrix} -z\dfrac{\partial w(x,y,t)}{\partial x} - \dfrac{1}{2}\displaystyle\int_{x_0}^{x}\left(\frac{\partial w(x,y,t)}{\partial x}\right)^2 \mathrm{d}x \\ -z\dfrac{\partial w(x,y,t)}{\partial y} - \dfrac{1}{2}\displaystyle\int_{y_0}^{y}\left(\frac{\partial w(x,y,t)}{\partial y}\right)^2 \mathrm{d}y \\ w(x,y,t) \end{pmatrix} \tag{11}$$

where (x_0, y_0) is taken as the origin of the floating frame [8].

The transverse deflection is represented in terms of the modal coordinates as

$$w(x,y,t) = \boldsymbol{\Phi q} \tag{12}$$

and the 3 preceding modes are used for the modal superposition. The 3 preceding frequencies and modes of the sail can be obtained using the software ABAQUS/Standard. Finite element models of the 3 preceding frequencies and modes are shown in Fig. 4.



a) Mode 1 at 0.00932759 Hz b) Mode 2 at 0.0109624 Hz c) Mode 3 at 0.0109624 Hz

Fig. 4 3 preceding frequencies and modes of the solar sail

Considering the gravitational force, the solar radiation pressure and the control torque, and ignoring dissipative forces, application of the variational principles gives the virtual power in the form

$$-\int_V \delta\dot{\boldsymbol{R}}\cdot\ddot{\boldsymbol{R}}\,dm + \int_V \delta\dot{\boldsymbol{R}}\cdot d(\boldsymbol{F}_s) + \int_V \delta\dot{\boldsymbol{R}}\cdot d(\boldsymbol{F}_g) + \delta\boldsymbol{\omega}\cdot\boldsymbol{U} - \delta\dot{\boldsymbol{q}}^T\boldsymbol{K}_f\boldsymbol{q} = 0 \tag{13}$$

This equation indicates that the summation of each element power about its arbitrary virtual displacement is equal to 0, including the power caused by active forces, inertial forces, and vibration.

Substituting the first and second time derivatives of $\boldsymbol{R}$ into Eq. (13) yields

$$\begin{cases} m_{tot}\ddot{\boldsymbol{R}}_O - \boldsymbol{S}_f \times \dot{\boldsymbol{\omega}} + \int_V \boldsymbol{u}_f'' dm + \boldsymbol{Q}_{ft} = \boldsymbol{F}_t + \boldsymbol{F}_{gt} \\ \boldsymbol{S}_f \times \ddot{\boldsymbol{R}}_O + \boldsymbol{J}_f \cdot \dot{\boldsymbol{\omega}} + \int_V (\boldsymbol{r}_f + \boldsymbol{u}_f) \times \boldsymbol{u}_f'' dm + \boldsymbol{Q}_{fr} = \boldsymbol{F}_r + \boldsymbol{F}_{gr} + \boldsymbol{U} \\ \boldsymbol{C}_t^T \cdot \ddot{\boldsymbol{R}}_O + \boldsymbol{C}_r^T \cdot \dot{\boldsymbol{\omega}} + \int_V \boldsymbol{\Phi}_{NL}^T \cdot \boldsymbol{u}_f'' dm + \boldsymbol{K}_f \boldsymbol{q} + \boldsymbol{Q}_{ff} = \boldsymbol{F}_f + \boldsymbol{F}_{gf} \end{cases} \tag{14}$$

To simplify the numerical analysis, Eq. (14) can be written in matrix form as follows:

$$\begin{pmatrix} m_{tot}\boldsymbol{I}_3 & -\tilde{\boldsymbol{S}}_f & \boldsymbol{C}_t \\ \tilde{\boldsymbol{S}}_f & \boldsymbol{J}_f^b & \boldsymbol{C}_r \\ (\boldsymbol{C}_t)^T & (\boldsymbol{C}_r)^T & \boldsymbol{M}_f \end{pmatrix} \begin{pmatrix} \ddot{\boldsymbol{R}}_O \\ \dot{\boldsymbol{\omega}} \\ \ddot{\boldsymbol{q}} \end{pmatrix} + \begin{pmatrix} \boldsymbol{0}_{3\times3} & \boldsymbol{0}_{3\times3} & \boldsymbol{0}_{3\times n} \\ \boldsymbol{0}_{3\times3} & \boldsymbol{0}_{3\times3} & \boldsymbol{0}_{3\times n} \\ \boldsymbol{0}_{n\times3} & \boldsymbol{0}_{n\times3} & \boldsymbol{K}_f \end{pmatrix} \begin{pmatrix} \boldsymbol{R}_O \\ \boldsymbol{\Omega} \\ \boldsymbol{q} \end{pmatrix} = \begin{pmatrix} \boldsymbol{Q}_t \\ \boldsymbol{Q}_r \\ \boldsymbol{Q}_f \end{pmatrix} \tag{15}$$

Substituting Eq. (9) into Eq. (15), the dynamic equations described by the Euler angles can be written as

$$\begin{pmatrix} m_{tot}\boldsymbol{I}_3 & -\tilde{\boldsymbol{S}}_f \cdot \boldsymbol{A}_\Omega & \boldsymbol{C}_t \\ \tilde{\boldsymbol{S}}_f & \boldsymbol{J}_f \cdot \boldsymbol{A}_\Omega & \boldsymbol{C}_r \\ (\boldsymbol{C}_t)^T & (\boldsymbol{C}_r)^T \cdot \boldsymbol{A}_\Omega & \boldsymbol{M}_f \end{pmatrix} \begin{pmatrix} \ddot{\boldsymbol{R}}_O \\ \ddot{\boldsymbol{\Omega}} \\ \ddot{\boldsymbol{q}} \end{pmatrix} + \begin{pmatrix} \boldsymbol{0}_{3\times3} & \boldsymbol{0}_{3\times3} & \boldsymbol{0}_{3\times n} \\ \boldsymbol{0}_{3\times3} & \boldsymbol{0}_{3\times3} & \boldsymbol{0}_{3\times n} \\ \boldsymbol{0}_{n\times3} & \boldsymbol{0}_{n\times3} & \boldsymbol{K}_f \end{pmatrix} \begin{pmatrix} \boldsymbol{R}_O \\ \boldsymbol{\Omega} \\ \boldsymbol{q} \end{pmatrix} = \begin{pmatrix} \boldsymbol{Q}_t \\ \boldsymbol{Q}_r \\ \boldsymbol{Q}_f \end{pmatrix} - \begin{pmatrix} -\tilde{\boldsymbol{S}}_f \cdot d\boldsymbol{\omega} \\ \boldsymbol{J}_f \cdot d\boldsymbol{\omega} \\ (\boldsymbol{C}_r)^T \cdot d\boldsymbol{\omega} \end{pmatrix} \tag{16}$$

C. Model parameters of the solar sail

As Fig. 3 shows, the sail consists of four booms and four triangular sail membranes, and each sail membrane is connected to the booms at the three corners through twelve cables, an out-of-plane control mast, and several point masses, representing the sail stowage masses and the payload. The details of the structural parameters are given in Table 2 and are taken from [9].

Table 2 Structural parameters of the solar sail

Structure name	Size	Density, kg•m^{-3}	Young's Modulus, GPa	Poisson's ratio
boom	106-m thin-walled pipes, each with a circular cross section with a radius of 0.229 m and a thickness of 7.5 μm.	1908	124	0.30
membrane	Triangular pieces of film, 105 m on a right-angle side and 2.50 μm in thickness, with a metallic coating.	1572	2.48	0.34
cable	Four 0.42-m-long cables at the center of the solar sail and eight 0.67-m-long cables at the ends of the booms, each with a solid circular cross section with a radius of 0.5 mm.	1440	62	0.36
control mast	2-m thin-walled pipe, directed along the z-axis from the hub. The circular cross section has a radius of 0.01705 m and a thickness of 0.005 m.	7660	124	0.30

A point mass representing a 0.58-kg tip plate is located at the end of each boom. Another point mass representing a 228-kg payload mass is located at the end of the control mast, and a concentrated mass representing a 54.84-kg stowage mass is located at the hub.

III. Optimal trajectory design for the transfer mission

With the coupling effect of the orbit and attitude, the thrust for the orbital transfer of the solar sail is generated only by changing the attitude, so the transfer trajectory to the heliocentric polar orbit is designed according to the theory of time-optimal control.

The transfer target is a heliocentric polar orbit at an orbital radius of 0.48 AU and an orbital inclination of 90°. There are 2 phases for the transfer mission. The first is a continuous-attitude control phase from an earth-centric orbit to a heliocentric interim orbit, which reaches an orbital radius of 0.48 AU and an orbital inclination of 15°. The second is a step-attitude control phase from the heliocentric interim orbit to a heliocentric polar orbit, which increases the orbital inclination to 90° with an orbital radius of 0.48 AU.

A. Time-optimal control for the transfer mission of the solar sail

Based on the time-optimal control theory, the transfer mission from the earth-centric orbit to the heliocentric orbit is described as a time-optimal problem with initial and terminal states. The state variables are the orbital position $\boldsymbol{R}$ and velocity $\boldsymbol{V}$ of the solar sail, the optimal variables are the cone angle α and the clock angle δ, and the optimal target is the transfer time $t_f - t_0$.

The assistant state variables of displacement and velocity are chosen to form the Hamiltonian function. In the time-optimal problem of low thrust, the Hamiltonian function generates its maximum value when the direction of low thrust is in the direction of the assistant state variables of velocity $\boldsymbol{\lambda}_v$. However, the direction of the thrust for the solar sail cannot change randomly as it can with ordinary spacecraft, so locally optimal control is used to obtain the maximum value of the Hamiltonian function, which maximizes the projection of the solar pressure to $\boldsymbol{\lambda}_v$ [10].

B. The fastest increase in orbital inclination for the solar sail

The orbital dynamic equations show that the orbital inclination of the solar sail is only related to the normal component of solar pressure on its orbit. Based on the locally optimal control, to make the orbital inclination increase as quickly as possible, the projection of the solar pressure to the direction of sunlight should be as large as possible [2]. The transfer trajectory from the heliocentric interim orbit to the heliocentric polar orbit is then obtained, as shown in Fig. 5. Without the process of attitude changing, the orbital radius and orbital inclination of the heliocentric polar orbit are 0.4826 AU and 89.9995°, respectively, and the time of flight is 1336.3 days. The sequences of the optimal cone angle α and clock angle δ are shown in Fig. 6.

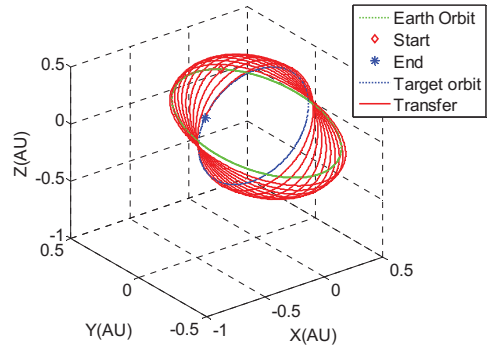


Fig. 5 The transfer trajectory from the heliocentric interim orbit to the heliocentric polar orbit

Fig. 8 shows that for sustained acceleration, the optimal cone angle α is a constant and the optimal clock angle δ inverts twice per orbital period. The time of inversion can be determined by the true anomaly f and the argument of periapsis ω, which means that the optimal attitude cone angle α and clock angle δ are not directly related to the time. It also means that the attitude control for the second phase is a problem of large-angle maneuver, which has a greater influence on the flexible solar sail mission.

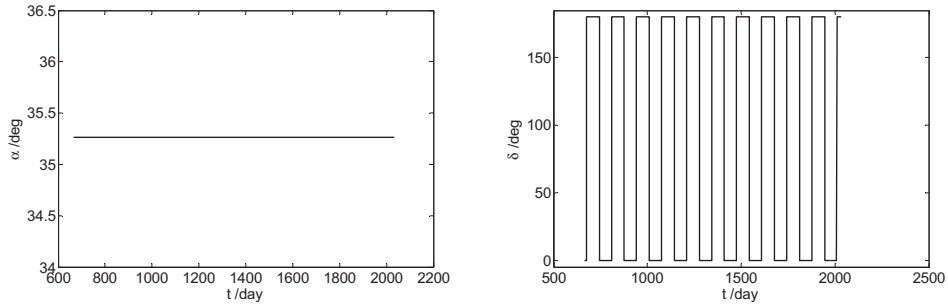


Fig. 6 The sequences of the optimal cone angle α and clock angle δ

IV. Analysis of the transfer mission of the flexible solar sail with attitude control

As shown before, the transfer mission from an earth-centric orbit to a heliocentric polar orbit can be achieved by the design of the attitude angles (the cone angle α and clock angle δ), but it is based on an assumption of the attitude changing transiently. It is obvious that the solar sail experiences the coupled effect of attitude and orbit and that the flexible structure also has some influence on the attitude change, especially for large-angle maneuvers. Therefore, an analysis of the transfer mission with the attitude-changing process is necessary.

The sequences of attitude angles are generated by time-optimal control, so PD control (with the feedback of angular displacement and angular velocity) is used to achieve attitude tracking control. The dynamic responses

are generated using Eq. (16). The terminal orbit with the attitude-changing process is considered first, and the orbital deviation with respect to the designed orbit (without the attitude-changing process) is given to describe it. The elastic deformation and vibration of the flexible structure are considered next, and the influence of structural vibration on the attitude control is analyzed at the same time.

The attitude tracking control torque with PD control is shown as

$$\boldsymbol{U} = \boldsymbol{J}_f \cdot \boldsymbol{A}_\Omega \cdot \left(\boldsymbol{K}_P \cdot \left(\boldsymbol{\Omega} - \boldsymbol{\Omega}^{ref} \right) + \boldsymbol{K}_D \cdot \dot{\boldsymbol{\Omega}} \right) \tag{17}$$

where $\boldsymbol{\Omega}^{ref}$ are the desired Euler angles, which can be generated using Eq. (4).

For the desired Euler angles, the sequence is obtained from the optimal attitude angle sequence. Take the orbital plane changing phase for example, the changes in optimal attitude angles over time depend on the orbital position, so the mathematical expressions of the desired attitude angles varying with orbital position can be given for real-time control.

There are usually initial deviations and disturbance forces in space missions. The initial attitude and orbital deviations are considered for solar sails as they would be for ordinary spacecraft. Because of the special structure of the solar sail, the initial vibration is also considered. The solar pressure disturbance torque has been shown to have a significant effect because of the special structure and force model of a flexible solar sail, so it was considered as well.

The analysis of the orbital plane change for the flexible solar sail with the attitude-changing process is described in this section. The effect of the flexibility on the transfer mission is examined first in section A. Large-angle maneuvers with initial attitude and orbital deviation, along with the solar pressure disturbance torque and the influence of flexibility on the transfer mission, are examined in section B. The influence of initial vibration is studied in section C.

A. Attitude tracking control of the orbital plane change

The optimal cone angle α and clock angle δ for real-time control in this phase are represented as

$$\begin{aligned} \alpha &= \arctan\left(1/\sqrt{2}\right) \\ \delta &= \begin{cases} 0, \cos\left(f+\omega\right) \geq 0 \\ \pi, \cos\left(f+\omega\right) < 0 \end{cases} \end{aligned} \tag{18}$$

Because a larger coefficient of proportion results in failure of the numerical computation during large-angle maneuvers, the smaller control parameters are chosen for this phase to be

$$\begin{cases} \boldsymbol{K}_P = -0.0001 I_3 \\ \boldsymbol{K}_D = -0.1 I_3 \end{cases} \tag{19}$$

Assuming there is no initial deviation of orbit and attitude, the results of the process of attitude changing for rigid sails and flexible sails are shown in Figs. 7–10.

The orbital deviation with respect to the designed orbit is shown in Fig. 7. The solar sail reaches the heliocentric polar orbit, where the orbital radius is 0.4832 AU and the orbit inclination is 89.8896°, which is different from the designed orbit, and the orbital deviation is 0.0006 AU (nearly 10^7 m). There is also some difference for rigid sails versus flexible sails. The absolute orbital deviations between rigid sails and flexible sails are 10^{-6} AU, 10^{-7} AU, and 10^{-6} AU.

The Euler angles for rigid and flexible sails are shown in Fig. 8, and the Euler angles are almost coincident with the desired angles. When the Euler angle along the x axis has a large-angle maneuver, the Euler angles along the y and z axes have some deviation because of the coupling of attitude equations. The deviation for flexible sails is greater than for rigid sails.

The control torques for rigid and flexible sails are shown in Fig. 9. There are sudden changes in the control torque when the attitude along the x axis involves a large-angle maneuver. The largest control torque is 81.64 Nm for the rigid sail along the z axis. The largest control torque for the flexible sail is 31.26 Nm, also along the z axis.

The stretching and deflection deformations of the flexible sail at tip 1 are shown in Fig. 10. The largest stretching and deflection deformations occur when the sail makes a large-angle maneuver. The largest deflection deformation is 0.0525 m. When the attitude is stable, there is sustained high-frequency vibration of the flexible structure, and the amplitude is 10^{-3} m.

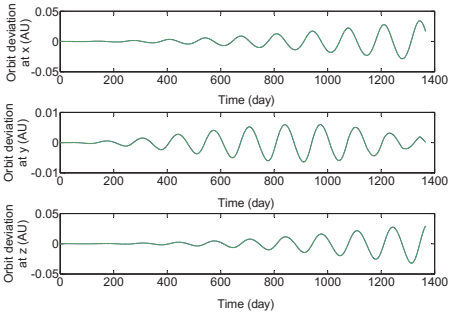


Fig. 7 Comparison of orbital deviation with respect to the designed orbit for rigid and flexible sails

(‘—’: rigid sail, ‘- -’: flexible sail)

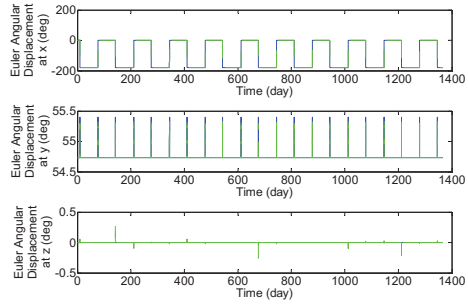


Fig. 8 Comparison of Euler angular displacement for rigid and flexible sails

(‘—’: rigid sail, ‘- -’: flexible sail)

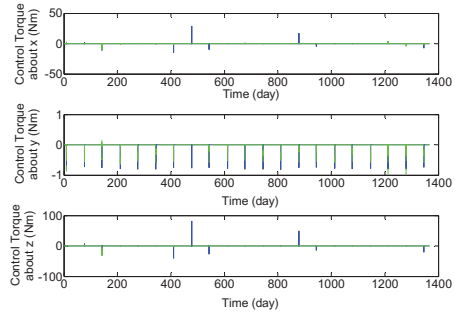


Fig. 9 Comparison of control torque for rigid and flexible sails

(‘—’: rigid sail, ‘- -’: flexible sail)

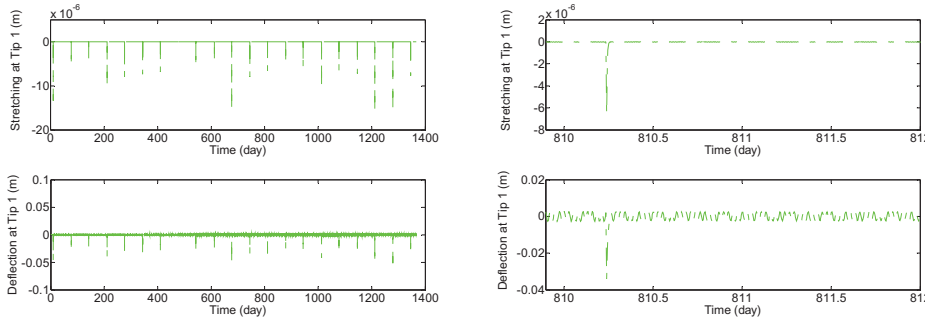


Fig. 10 Stretching and deflection deformation at tip 1

These results show that the orbital plane changing phase of the solar sail can be achieved using attitude tracking control, and the sail will reach the target orbit within the margin of error. Thus, the process of attitude changing has a significant influence on the transfer trajectory. The deformation of the flexible structure is very small because the control torque of the tracking control is small, so there is no significant effect of flexibility for this solar sail.

B. Attitude tracking control with initial deviation and disturbance torque

Assuming that, there are both attitude and orbital deviations for the step-attitude-control phase at the end of the first transfer phase, which are approximately given as

$$\begin{aligned} d\boldsymbol{\Omega} &= \begin{pmatrix} 65.89 & 5.16 & 0 \end{pmatrix}^T \circ \\ d\boldsymbol{R} &= \begin{pmatrix} 10^7 & 10^7 & 10^6 \end{pmatrix}^T m \end{aligned} \tag{20}$$

And there is an offset between the mass center and pressure center along the x axis of the sail, the vector of offset is represented in the body frame as

$$\boldsymbol{L}_S = \begin{pmatrix} 0.4 & 0 & 0 \end{pmatrix}^T m \tag{21}$$

and the solar pressure disturbance torque is written as

$$\boldsymbol{M}_S = \boldsymbol{F}_S \times \boldsymbol{L}_S \tag{22}$$

taking $\boldsymbol{Q}_r$ in Eq. (16) instead of $\boldsymbol{Q}_r + \boldsymbol{M}_S$. The dynamic responses of the flexible sail are shown in Figs. 11–14.

The orbital deviation with respect to the designed orbit is shown in Fig. 11. The solar sail reached an orbital radius of 0.4830 AU and an orbit inclination of 89.575°, which is different from the designed orbit. With the effect of the solar pressure disturbance torque considered, the orbital deviation of the sail is larger.

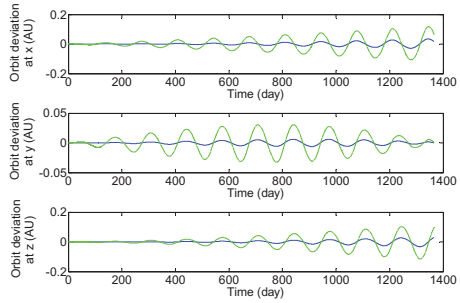


Fig. 11 Orbital deviation with respect to the designed orbit with initial deviation and solar pressure disturbance torque

(‘—’: none, ‘- -’: with initial deviation and solar pressure disturbance torque)

The Euler angles of the flexible sail with initial attitude and orbital deviation and solar pressure disturbance torque are shown in Fig. 12. Because of the initial deviation and disturbance torque, the time of diversion has a delay of 2 days, as shown in Fig. 12, and the delay increases with the orbital time, which also explains the decrease in the orbital inclination. Because of the effect of the disturbance torque, the Euler angle along the y axis has a deviation of 0.63°, which explains the orbital period delay.

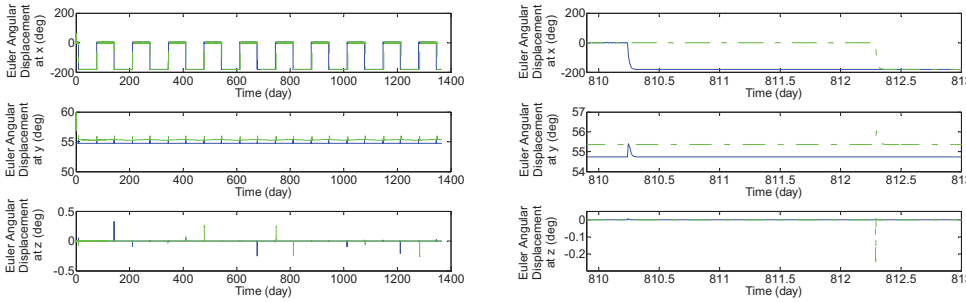


Fig. 12 Euler angular displacement with initial deviation and solar pressure disturbance torque

(‘—’: none, ‘- -’: with initial deviation and solar pressure disturbance torque)

The control torques for the flexible sail are shown in Fig. 13. There are sudden changes of the control torque when the attitude along the x axis involves a large-angle maneuver. The largest control torque for the flexible sail is 95.76 Nm, which is along the z axis. There is also a constant torque of 0.2134 Nm to counteract the solar pressure disturbance torque.

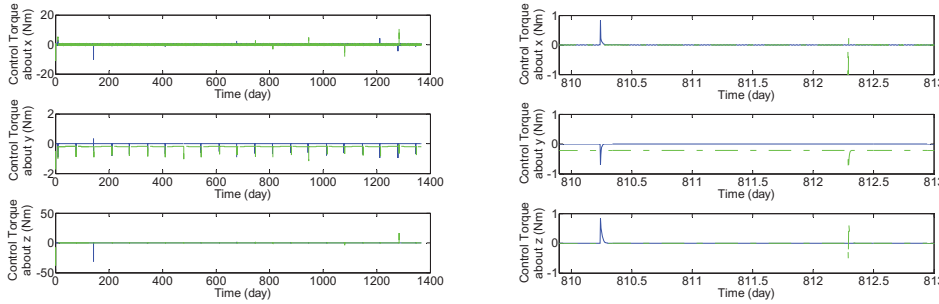


Fig. 13 Control torque with initial deviation and solar pressure disturbance torque

(‘—’: none, ‘- -’: with initial deviation and solar pressure disturbance torque)

The stretching and deflection deformations of the flexible sail at tip 1 are shown in Fig. 14. The largest deflection deformation with initial deviation and disturbance torque is 0.0536 m, which is almost the same as with no initial deviation and disturbance torque. When the attitude is stable, there is also sustained and high-frequency vibration of the flexible structure, and the amplitude is 10^{-3} m.

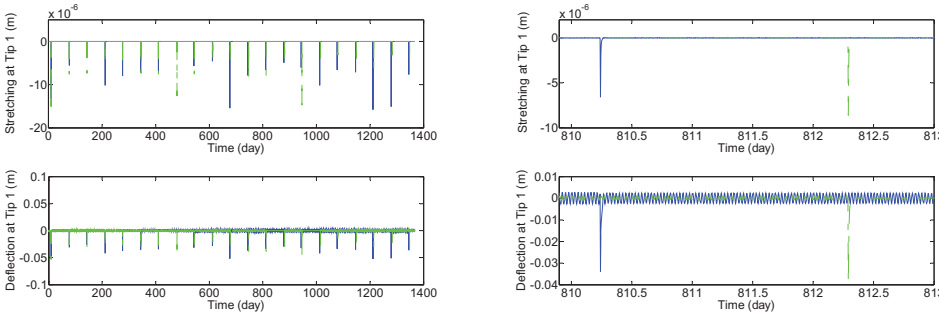


Fig. 14 Stretching and deflection deformation at tip 1 with initial deviation and solar pressure disturbance torque

(‘—’: none, ‘- -’: with initial deviation and solar pressure disturbance torque)

Because the coefficient of proportion is very small in this phase, the solar pressure disturbance torque will lead to deviation of the Euler angles, which leads to a larger deviation of the orbit and also explains the delay of the orbital period. The initial deviations of attitude and orbit also lead to some delay of the orbital period.

C. Attitude tracking control with initial vibration

There is a slight deformation in the flexible sail when only small attitude and orbit deviations are given, and the slight deformation is not dissipated without a damping term. Next, the initial vibration condition is considered, because the deployment will cause the structural vibration of the solar sail. Assuming the initial modal coordinates are

$$\boldsymbol{q}_0 = \begin{pmatrix} 10 & 0.5 & 0.5 \end{pmatrix}^T \tag{23}$$

and the initial stretching and deflection deformations are 5.222×10^{-3} m and 0.9527 m, respectively.

The offset between the mass center and pressure center and the solar pressure disturbance torque are given in Eq. (21) and Eq. (22), and the dynamic responses of the flexible sail are shown in Figs. 15–18.

The orbital deviation with respect to the designed orbit is shown in Fig. 15. The solar sail reached an orbital radius of 0.4830 AU and an orbit inclination of 89.575°, which is same as the orbit without initial vibration.

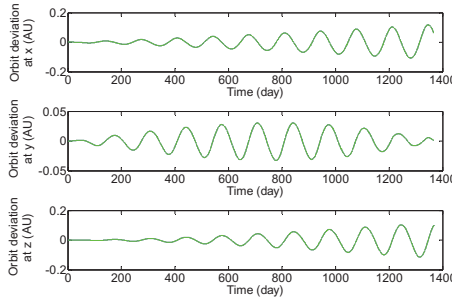
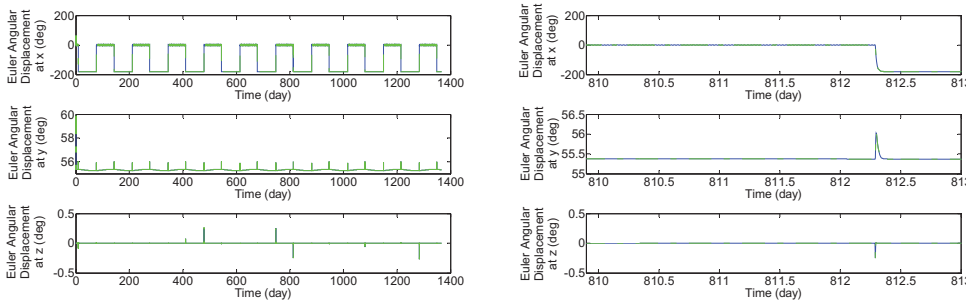


Fig. 15 Orbital deviation with respect to the designed orbit with initial vibration
(‘—’: none, ‘- -’: with initial vibration)

The Euler angles of the flexible sail are shown in Fig. 16. There is no obvious effect of initial vibration to the attitude while the large-angle maneuvering. But when the attitude is stable, the attitude has slight and high-frequency vibration with the effect of structural vibration.



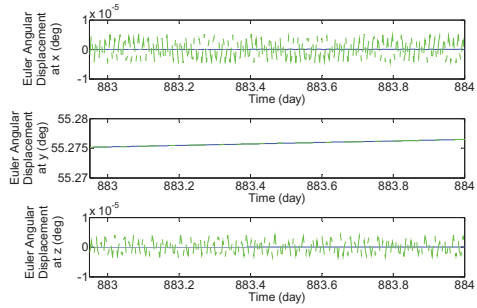


Fig. 16 Euler angular displacement with initial vibration

(‘—’: none, ‘- -’: with initial vibration)

The control torques are shown in Fig. 17. Similar to the attitude of the solar sail, with the effect of structural vibration, the control torques have slight and high-frequency vibration when the attitude stable.

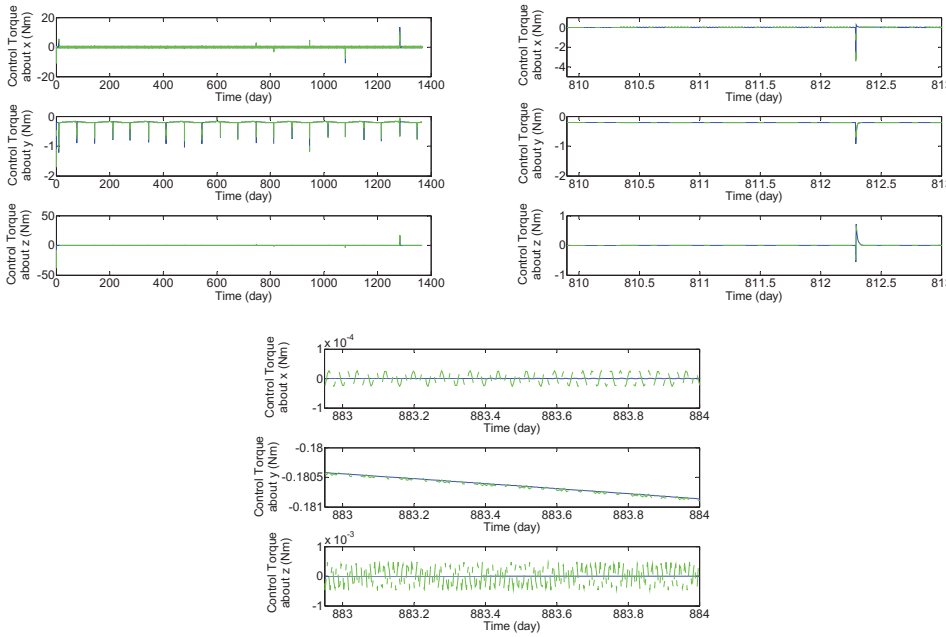


Fig. 17 Control torque with initial vibration

(‘—’: none, ‘- -’: with initial vibration)

The stretching and deflection deformations of the flexible sail at tip 1 are shown in Fig. 18. The flexible structure has sustained, high-frequent vibrations without the damping term, and the amplitude is the same as that of the initial vibration condition.

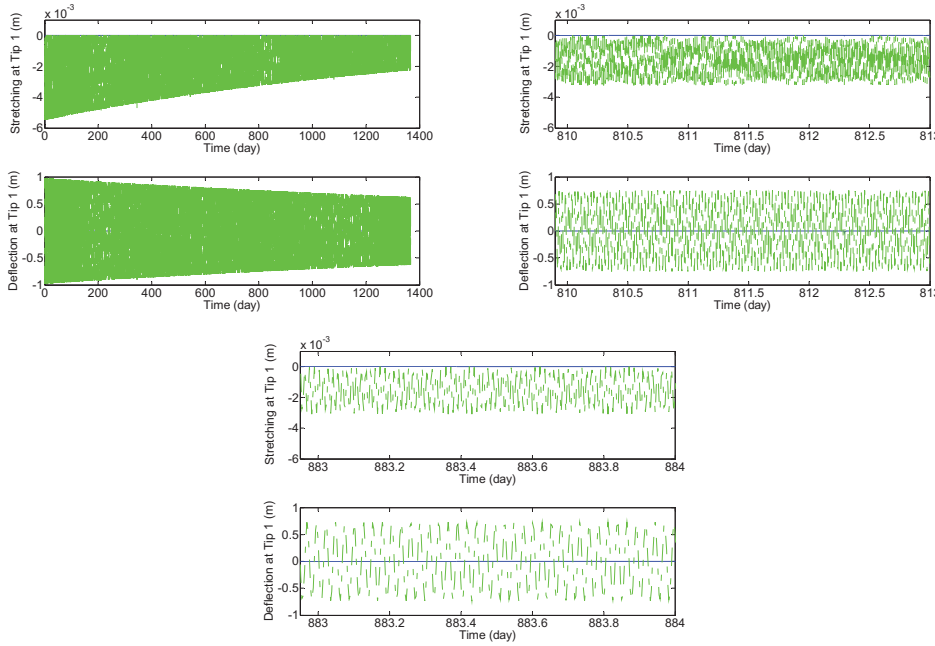


Fig. 18 Stretching and deflection deformation at tip 1 with initial vibration

(‘—’: none, ‘- -’: with initial vibration)

This finding indicates that when the flexible sail has a large initial structural vibration, as a result of the sustained vibration, the attitude and the control torques have slight and high-frequency vibration, but its amplitude is very small.

V. Conclusion

Based on the reduced dynamic model of the flexible solar sail, the process of orbital plane changing phase of SPI for the solar sail is analyzed. It is demonstrated that with tracking control of attitude, the transfer phase of the solar sail can be accomplished within the margin of error, and the flexibility of the sail has little influence on this mission. When the solar sail makes a large-angle maneuver, it will generate some orbital deviation. Furthermore, the flexible structure of the sail will generate some deformation and elastic vibration, which also influence the attitude control. The initial deviation of attitude will lead to greater deformation and elastic vibration of the solar sail, and the solar pressure disturbance torque will lead to greater deviation of the orbit. The initial structural vibration will lead to the slight and high-frequency vibration of the solar sail, but the influence is very small.

The coupled effect of the flexible solar sail is proven in this transfer mission of deep-space exploration. The process of attitude control will generate some orbital deviation and even larger deviations for some special

conditions, which could have significant influences on some missions. In particular, the solar pressure disturbance torque has a great influence on the transfer mission, so it needs to be considered in the analysis of missions for the solar sail. If the flight time is not strict, the orbital inclination could be gotten by extending the flight time.

Acknowledgments

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Coupled Attitude–orbit Dynamics and Control of Reflectivity Modulated Solar Sail for GeoSail Formation Flying

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Abstract:

Electrochromic device has been proposed for the attitude control of spacecraft for that the optical characteristics of the device can be altered by applying electrical excitation to generate a torque on the spacecraft. The technology was recently employed and verified successfully on IKAROS, the world's first interplanetary solar sail. It is found that the reflection control device can not only be used to generate desired torque to control the attitude of a sail, but also it can be used to change the total force on the sail instantaneously. The property is proved to be very helpful for the active control of a sailcraft. In this study, the force and torque models of a reflectivity modulated spinning disk sail experienced by solar radiation pressure are discussed. A reflectivity modulation scheme, which was used in IKAROS mission, is investigated for the active control in a GeoSail formation flying application. The main objective of the study is to construct a method for the active control of solar sail in some situation where an additional control input is required in addition to two sail attitude angles. The coupled attitude and orbit dynamics of the sail on a GeoSail orbit is discussed and the equation of relative orbit and attitude motion is given. Finally, the performance of the reflectivity modulation control for GeoSail formation flying is numerically demonstrated.

I. Introduction

Solar sailing has long been envisioned as a promising technology since the early part of last century. In the last twenty years, solar sailing has been extensively researched and proposed for a

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wide range of mission applications [1, 2]. The typical examples of mission applications which could be enabled by the technology include GeoSail mission [3-5], the mission around Lagrange points [6], etc.

In solar sail research, the attitude dynamics and control is another important topic beside orbital dynamics. Many researchers investigated the passive and active control for solar sail. The gimbaled control boom and sail control vanes are discussed in detail by Wie [8, 9]. The passive control of solar sail on displaced solar orbits is discussed by Gong [10]. In 2010, IKAROS(Interplanetary Kite-craft Accelerated by Radiation Of the Sun) [11] was launched by JAXA(Japan Aerospace Exploration Agency) and became the first interplanetary sailcraft. Among the verified techniques, one attitude control technique was tested. An electrochromic device called RCD (reflection control device) was designed to control the attitude of the sail. The devices are a thin layer of liquid crystal films stuck near the edges of the membrane and they can be switched specular and diffuse reflection by being powered on and off according to the spin rate and generate a torque by changing the induced force on each small element's surface through switching between two states. The performance of the electrochromic attitude control was successfully verified [12]. The technology will have a great potential for future solar sail missions. The technology that can control the attitude by the variation of reflectivity was referred as reflectivity modulation in Ref. [8].

For many solar sail mission applications, an active control is often necessary to stabilize equilibria or orbits. For a spacecraft with fixed area of sail membrane and constant characterization acceleration, it is often difficult to implement the controllability only relying on the attitude variation. Generally two solutions were proposed to overcome the difficulty: one is to assume that the area of sail can be varied and used as a control variable [13], the other one proposes to install additional solar electric propulsion (SEP) device to implement a hybrid SEP/sail technology [14, 15]. Both solutions will inevitably lead to more complicated design and higher mission costs.

In most published literatures, attitude dynamics is often discussed separately from orbit dynamics. When orbit dynamics is discussed, attitude is assumed to be agile enough to achieve instantaneous changes. However, for reflectivity modulation control of a solar sail, things are quite different from other control methods. Firstly, the control torque is highly constrained and dependent on reflectivity modulation magnitude, its distribution and the Sun angle. Secondly, the variation of

reflectivity may not only lead to a torque variation on the sailcraft, but also generate a change in total force experienced by the sail. In the past studies about the reflectivity modulation techniques [11, 12, 16], only the attitude and its induced effect on orbit are discussed, the orbit change directly caused by the force change incurred by reflectivity modulation is largely neglected. Recently, the application of electrochromic device for the orbit control of smart-dust device was proposed by Lücking [17]. However, the orbit control is discussed without the consideration of the attitude control for the smart-dust device in the study.

This paper discusses an extension of the reflectivity modulation technology to the active control of solar sail and tackles both the attitude and orbit control together through the coupled attitude and orbit dynamics under constrained reflectivity modulation control. Its application in GeoSail formation flying is investigated. GeoSail was envisioned as a near term solar sail mission application [5, 18] since it has lower demand for sail size. GeoSail formation flying can further reduce the weight of payload and the size of each sail by coordinating to carry out the designated tasks.

The paper is outlined as follows: In section II, the torque and force models of a reflectivity modulated spinning disk sail are discussed and constrained reflectivity control inputs are described. Section III describes the coupled attitude and orbit dynamics of a reflectivity modulated spinning disk sail in an elliptical GeoSail Earth orbit. The equation of the relative attitude and orbit motion for GeoSail formation flying is presented. A typical controller is employed to track an artificially designed periodic relative orbit. Numerical results are given to demonstrate the performance of the active control for the relative GeoSail orbit tracking problem using the reflectivity modulation.

II. Force and Torque Models of Reflectivity Modulated Solar Sail

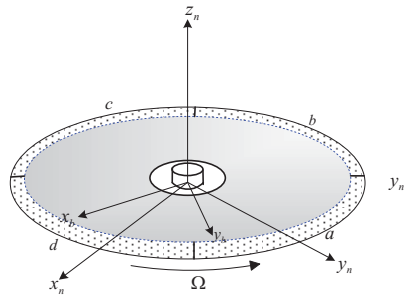


Fig. 1 Configuration of spinning disk sail.

Assume that the solar sail considered here is a spinning disk sail with a total area A_s and radius r_s. The basic configuration of the sail is depicted in Fig. 1. Two reference coordinate frames are used to describe the configuration: One is a spin-free body frame $o_n x_n y_n z_n$ and another is a body frame $o_b x_b y_b z_b$. The origins of both frames are the center of mass of the sail and the z-axes of both frames are along the normal vector of the sail. For a normal part of the sail with an area Δs, the force experienced by SRP [2, 19] is

$$F = P\Delta s(\mathbf{n}_s \cdot \mathbf{n})[(C_a + C_d)\mathbf{n}_s + (\frac{2}{3}C_d + 2C_s(\mathbf{n}_s \cdot \mathbf{n}))\mathbf{n}] \tag{1}$$

where P is the solar radiation pressure exerted on the sail surface, $\mathbf{n}_s$ and $\mathbf{n}$ are the unit Sun vector and the unit normal vector of the sail, respectively. C_a, C_d and C_s denote the coefficients of absorption, diffusive and specular reflections, respectively.

Another layer of electrochromic devices with total area A_f are evenly distributed and adhered around the edges of the sail. The ratio of the total area of the electrochromic surface to the total area of the sail is denoted as $u_{max} = \frac{A_f}{A_s}$. The devices are assumed to be divided into a number of cells which can be switched on and off independently and each device is small enough to make the area of switched-off surface to change smoothly. When a part of RCDs with an area Δs is switched on, the SRP force can be expressed as

$$F_{on} = P\Delta s(\mathbf{n}_s \cdot \mathbf{n})[(C_{a_on} + C_{d_on})\mathbf{n}_s + (\frac{2}{3}C_{d_on} + 2C_{s_on}(\mathbf{n}_s \cdot \mathbf{n}))\mathbf{n}] \tag{2}$$

And when the part of RCDs is switched off, the SRP force can be expressed as

$$F_{off} = P\Delta s(\mathbf{n}_s \cdot \mathbf{n})[(C_{a_off} + C_{d_off})\mathbf{n}_s + (\frac{2}{3}C_{d_off} + 2C_{s_off}(\mathbf{n}_s \cdot \mathbf{n}))\mathbf{n}] \tag{3}$$

The difference of force between two states is then obtained as

$$\Delta F = P\Delta s(\mathbf{n}_s \cdot \mathbf{n})[(\Delta C_a + \Delta C_d)\mathbf{n}_s + (\frac{2}{3}\Delta C_d + 2\Delta C_s(\mathbf{n}_s \cdot \mathbf{n}))\mathbf{n}] \tag{4}$$

where $\Delta C_a = C_{a_on} - C_{a_off}$, $\Delta C_d = C_{d_on} - C_{d_off}$, $\Delta C_a = C_{s_on} - C_{s_off}$. It is assumed that the surface where the electrochromic devices are switched on has the same optical properties as a normal sail surface. Thus, the three coefficients $C_{a_on} = C_a$, $C_{d_on} = C_d$ and $C_{s_on} = C_s$

respectively.

At a given moment t_0, the total area of RCDs which are switched off is A_o, and the ratio of the area to the total sail area is denoted as the reflection modulation rate $u = \frac{A_o}{A_s}$. The SRP force on the part of solar sail can be obtained

$$F_1 = uPA_s(\mathbf{n}_s \cdot \mathbf{n})[(C_{a_off} + C_{d_off})\mathbf{n}_s + (\frac{2}{3}C_{d_off} + 2C_{s_off}(\mathbf{n}_s \cdot \mathbf{n}))\mathbf{n}] \tag{5}$$

Then, the SRP force on the remaining part of the sail can be got

$$F_2 = (1-u)PA_s(\mathbf{n}_s \cdot \mathbf{n})[(C_a + C_d)\mathbf{n}_s + (\frac{2}{3}C_d + 2C_s(\mathbf{n}_s \cdot \mathbf{n}))\mathbf{n}] \tag{6}$$

The total force on the sail can be obtained

$$F = F_1 + F_2 \tag{7}$$

When two pieces of RCDs, which are symmetrically located beside two sides of a certain axis of the sail with the same distance, stay in opposite states and the position vector to the axis for the piece with RCDs switched on is $\vec{r_d}$, then the torque on the axis generated by the state difference can be expressed as

$$\vec{T} = (\Delta F \cdot \mathbf{n})(\vec{r_d} \times \mathbf{n}) = P\Delta s \cos\alpha_s[\cos\alpha_s(\Delta C_a + \Delta C_d + 2\Delta C_s) + \frac{2}{3}\Delta C_d)](\vec{r_d} \times \mathbf{n}) \tag{8}$$

where α_s is the Sun angle.

It is assumed that two reference frames $o_n x_n y_n z_n$ and $o_b x_b y_b z_b$ coincide with each other at the moment t_0. The ratio of the area of the RCDs switched off, which are located at the first and fourth quadrants of x-y plane of the frame $o_n x_n y_n z_n$ to the total area of the sail, is u_{x+}, and the ratio of those in the second and third quadrants is u_{x-}, the torque exerted on y-axis of the frame can be written as

$$T_y = -PA_s \cos\alpha_s[\cos\alpha_s(\Delta C_a + \Delta C_d + 2\Delta C_s) + \frac{2}{3}\Delta C_d)][\int_0^{u_{x+}} r_{dx+}(u)du - \int_0^{u_{x-}} r_{dx-}(u)du] \tag{9}$$

where r_{dx+} and r_{dx-} are the distance of each cell of RCDs to the axis. If the RCDs switched off are evenly distributed in the stripe of the related half plane. The equation above can be

approximated as

$$T_y \approx -u_x P A_s \cos\alpha_s [\cos\alpha_s (\Delta C_a + \Delta C_d + 2\Delta C_s) + \frac{2}{3}\Delta C_d)\overline{r_d} \tag{10}$$

where $u_x = u_{x+} - u_{x-}$ and $\overline{r_d}$ is the average distance of the two stripes to the y-axis. Similarly, the torque on x-axis can be approximated as

$$T_x \approx u_y P A_s \cos\alpha_s [\cos\alpha_s (\Delta C_a + \Delta C_d + 2\Delta C_s) + \frac{2}{3}\Delta C_d)\overline{r_d} \tag{11}$$

where $u_y = u_{y+} - u_{y-}$, and u_{y+}, u_{y-} are the distribution of RCDs turned off on the upper and lower half planes besides x-axis.

It can be seen from the configuration of the sail that the torque exerted on the sail only relies directly on the distribution of the RCDs switched off on the sail, since that if all RCDs switched off are distributed evenly around the edges of the sail, there would be no torque exerted on the sail. However, the maximum torque is constrained by the reflection modulation rate u. When u reaches zero or its maximum, there is no more capacity for the attitude control of the sail. On the other hand, the two variables u_x and u_y are also constrained to each other and have to compete for the resource when u is constant. A rigid analysis of the constraint model is cumbersome. To decouple the use of control variables u_x and u_y, the ring of RCDs can be divided into four arcs with equal length in the frame $o_n x_n y_n z_n$ as shown in Fig. 1. The parameter u_y is decided by the distribution of RCDs in the arc a and c. The RCDs switched off in the arc a and c can be evenly distributed relative to y-axis, so that u_y is independent from the torque exerted on y-axis. Similarly, the parameter u_x is only decided by the distribution of RCDs in the arc b and d. For the configuration discussed above, the average distance $\overline{r_d}$ in Eqs. (10) and (11), same as the average distance of the arc a to x-axis, is $0.91 \times r_s$. The three parameters u, u_x and u_y are independent variables and satisfy the following constraints

$$\begin{cases} 0 \le u \le u_{max} \\ |u_x| + |u_y| \le \min(\frac{1}{2}u_{max}, u, u_{max} - u) \end{cases} \tag{12}$$

For a spinning sail, a control logic can be designed to make the change of the three parameters

independent from the spin rate of the sail [19]. In order to generate a control torque for a direction in the frame $o_n x_n y_n z_n$, RCDs should be switched on and off in synchronization with the spin motion. The torque generated in frame $o_b x_b y_b z_b$ can be expressed as

$$\begin{bmatrix} T_{bx} \\ T_{by} \end{bmatrix} = \begin{bmatrix} \cos(\Omega(t-t_0)) & \sin(\Omega(t-t_0)) \\ -\sin(\Omega(t-t_0)) & \cos(\Omega(t-t_0)) \end{bmatrix} \begin{bmatrix} T_x \\ T_y \end{bmatrix} \tag{13}$$

where T_{bx} and T_{by} is the torque on the two axes of the body frame, Ω is the spin rate of the sail.

Two types of reflectivity modulation technology can be used to control solar sails: One is that the electrochromic devices can be switched between specular and diffuse states; another one is that the devices can be switched between specular and absorptive states. The paper only discusses the first model which was already implemented and verified in IKAROS mission. In the idealized model of the first scheme, the three coefficients are got $\Delta C_a = 0$, $\Delta C_d = -1$, $\Delta C_s = 1$. The Eqs. (7) and (11) can be rewritten as

$$F = PA_s \cos\alpha_s \{u\mathbf{n}_s + [\frac{2}{3}u + 2(1-u)\cos\alpha_s]\mathbf{n}\} \tag{14}$$

$$T_x \approx u_y PA_s \cos\alpha_s (\cos\alpha_s - \frac{2}{3})\overline{r_d} \tag{15}$$

III. Attitude-orbit Coupled Dynamics of Reflectivity Modulated Solar Sail on a GeoSail Orbit

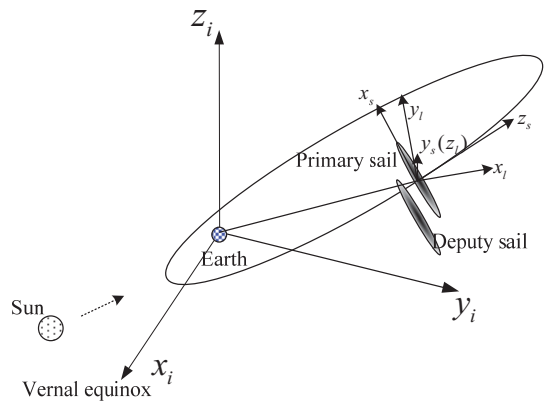


Fig. 2 Formation of reflectivity modulated spinning disk sails on a GeoSail orbit.

Two coordinate frames are used to describe the attitude and orbit motion of a sail in a GeoSail

orbit. The first one is an ecliptic inertial frame $o_i x_i y_i z_i$ used to describe the motion of solar sail in an elliptical orbit as shown in Fig. 2. The origin of the frame is the center of the Earth. The x-axis is directed towards the vernal equinox, and z-axis is directed towards the north ecliptic pole, the y-axis completes a right-hand orthogonal frame; The second one is a sunlight pointing frame $o_s x_s y_s z_s$. The origin of the frame is the center of mass of the sail; z-axis is directed towards the sail from the Sun. The x-axis stays in the orbit plane of the sail, and the y-axis completes a right hand orthogonal frame. For a sail in a GeoSail orbit, the transit matrix from the inertial frame $o_i x_i y_i z_i$ to the frame $o_s x_s y_s z_s$ is $R_x(\frac{\pi}{2})R_y(\lambda_s+\pi)$ and the transit matrix from the frame $o_s x_s y_s z_s$ to the body frame $o_b x_b y_b z_b$ is $R_y(\beta)R_x(\alpha)R_z(\psi)$. where λ_s is the angle of the Sun within the ecliptic plane from the vernal equinox, and β, α and ψ are denoted here as yaw, pitch and roll angles of the sail, respectively.

The dynamics of the reflectivity controlled solar sail on an Earth orbit can be expressed as perturbed two body problem. Its equation of motion in the inertial frame $o_i x_i y_i z_i$ can be written as

$$\ddot{\mathbf{r}} = -\frac{\mu}{r^3}\mathbf{r} + \mathbf{a}_s \tag{16}$$

The diffuse reflectivity modulation model, which was implemented in IKAROS, is used to investigate the active control of solar sails formation in GeoSail mission. Because the mission orbit considered here is an elliptical Earth orbit with zero inclination, the distances between the solar sails and the Sun can be assumed as the distance between the Earth and the Sun. The perturbed acceleration exerted by SRP in Eq. (16) can be written as

$$\mathbf{a}_s = \frac{1}{2}\kappa\cos\alpha_s\{u\mathbf{n}_s + [\frac{2}{3}u + 2(1-u)\cos\alpha_s]\mathbf{n}\} \tag{17}$$

where κ is the characteristic acceleration of the solar sail which is defined as the acceleration experienced by the solar sail at a heliocentric distance of 1 astronomical unit (AU) while the sail normal is directed along the sun line in the anti-Sun direction [2]. The Sun vector and the normal vector of the sail on the inertial frame can be written as follows

$$\mathbf{n}_s = [-\cos\lambda_s, -\sin\lambda_s, 0]^T \tag{18}$$

$$\mathbf{n} = [-\cos\alpha\cos(\beta+\lambda_s), -\cos\alpha\sin(\beta+\lambda_s), -\sin\alpha]^T \tag{19}$$

Th dot product of two vectors is obtained as

$$\mathbf{n}\cdot\mathbf{n}_s = \cos\alpha_s = \cos\alpha\cos\beta \tag{20}$$

To achieve a GeoSail obit, the rotation of the apse line of the orbit should be synchronous with the annual rotation of Sun and the argument of perigee of the orbit $\omega = \lambda_s$ must always hold. When the inclination of the GeoSail orbit is zero, and under the simple steering law that the Sun angle of the sail is always zero, the perturbed acceleration $\mathbf{a}_s$ on the sail is found to satisfy the condition of being synchronous with the Sun [5]

$$\mathbf{a}_s = \frac{2}{3}\dot{\lambda}_s e\sqrt{\frac{\mu(1-e^2)}{a}} \tag{21}$$

where $\dot{\lambda}$ is the rotation rate of the Sun, a and e are the semi-major axis and eccentricity of the sail respectively, μ is the gravitational constant of the Earth. The characteristic acceleration of the sail can be obtained

$$\kappa = \frac{4}{-u+6}\dot{\lambda}_s\frac{e}{\sqrt{1-e^2}}\sqrt{\frac{\mu}{a}} \tag{22}$$

The sail considered here is a spinning disk sail, and the angular velocity along its spinning axis is far larger than other two. Even though the sail does not have any rigid structure to support its membrane, under the continuous small torque generated by the reflectivity modulation, the attitude dynamics can be approximated as that of a rigid sail. The yaw and pitch angles β and α are assumed small, the kinematic equation for the spinning disk sail can be approximated as

$$\begin{aligned}
&\dot{\theta} = \Omega \\
&\dot{\beta} = \omega_y\cos(\Omega t) + \omega_x\sin(\Omega t) - \dot{\lambda} \\
&\dot{\alpha} = \omega_x\cos(\Omega t) - \omega_y\sin(\Omega t)
\end{aligned} \tag{23}$$

where Ω is the spin rate of the sail about its roll axis, ω_x, ω_y and ω_z are the angular velocity of the sail in its body frame $o_b x_b y_b z_b$. The solution for ω_x, ω_y can be obtained

$$\begin{aligned}
\omega_y &= (\dot{\beta}+\dot{\lambda})\cos(\Omega t) - \dot{\alpha}\sin(\Omega t) \\
\omega_x &= (\dot{\beta}+\dot{\lambda})\sin(\Omega t) + \dot{\alpha}\cos(\Omega t)
\end{aligned} \tag{24}$$

The general attitude equation of motion for a rigid structure is given by

$$\dot{\mathbf{H}} - \vec{\omega}_b \times \mathbf{H} = T_g + T_b \tag{25}$$

where $\vec{\omega}_b$ is the angular velocity vector of the sail in body frame $o_b x_b y_b z_b$ and $\mathbf{H}$ is the angular momentum of the sail in the frame. T_g is the gravity gradient torque and T_b is the control torque. Since T_g in the GeoSail orbit is much smaller than the control torque T_b, it can be neglected in the equation. For an axis-symmetric circular spinning sail, its rotational equation of motion can be obtained

$$\begin{aligned} J\dot{\omega}_x - (J - J_z)\omega_y\omega_z &= T_{bx} \\ J\dot{\omega}_y + (J - J_z)\omega_x\omega_z &= T_{by} \\ \dot{\omega}_z &= \Omega \end{aligned} \tag{26}$$

where J_z and J are the moments of inertia about the spinning axis and other two axis of the sail, respectively. T_{bx} and T_{by} are the control torque about the x-axis and the y-axis of the body frame of the sail as shown in Eq. (13). With Eq. (24), the above equation can be simplified as

$$\begin{bmatrix} \ddot{\alpha} \\ \ddot{\beta} \end{bmatrix} + \mathbf{M_1} \begin{bmatrix} \dot{\beta} + \dot{\lambda} \\ \dot{\alpha} \end{bmatrix} = \mathbf{M}_2 \begin{bmatrix} u_x \\ u_y \end{bmatrix} \tag{27}$$

where $\mathbf{M_1} = \frac{J_z}{J}\Omega \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$, $\mathbf{M}_2 = \frac{T}{J}\begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$ and T can be written as

$$T = \frac{1}{2}\kappa m \bar{r}_d \cos\alpha \cos\beta (\cos\alpha \cos\beta - \frac{2}{3}) \tag{28}$$

where m is the total weight of the solar sail.

IV. Coupled Dynamics and Control of Relative Motion of Two Sails

To describe the relative motion of the solar sail formation, a LVLH Cartesian frame $o_l x_l y_l z_l$ as shown in Fig. 2 is used. The origin of the frame is the center of mass of the primary solar sail, and the relative motion of the deputy solar sail along the x-axis, the y-axis and the z-axis are also referred to as the radial, the along-track and the out-of-plane motion respectively. The relative motion of solar sail formation can be written in the following equation

$$\begin{cases} \ddot{x} - 2\dot{f}\dot{y} - \ddot{f}y - \dot{f}^2 x = -\dfrac{\mu(r_1 + x)}{((r_1 + x)^2 + y^2 + z^2)^{\frac{3}{2}}} + \dfrac{\mu}{r_1^2} + S_r \\ \ddot{y} + 2\dot{f}\dot{x} + \ddot{f}x - \dot{f}^2 y = -\dfrac{\mu y}{((r_1 + x)^2 + y^2 + z^2)^{\frac{3}{2}}} + T_r \\ \ddot{z} = -\dfrac{\mu z}{((r_1 + x)^2 + y^2 + z^2)^{\frac{3}{2}}} + W_r \end{cases} \tag{29}$$

where $\dot{f}$ and $\ddot{f}$ are the angular velocity and angular acceleration of the rotating frame. S_r, T_r, and W_r are the relative SRP acceleration on the primary and the deputy sails in the LVLH frame $o_l x_l y_l z_l$ of the primary sail

$$\begin{bmatrix} S_r \\ T_r \\ W_r \end{bmatrix} = R_z(\lambda + f)(\mathbf{a}_{s2} - \mathbf{a}_{s1}) \tag{30}$$

where f is the true anomaly of the primary sail. $\mathbf{a}_{s1}$ and $\mathbf{a}_{s2}$ are the SRP acceleration experienced by the primary and the deputy sails in the inertial frame $o_i x_i y_i z_i$ of the primary sail. The SRP acceleration of both sails is close to each other so that the relative SRP acceleration can be linearized with three independent variables u, α and β

$$\begin{bmatrix} S_r \\ T_r \\ W_r \end{bmatrix} \approx \begin{bmatrix} \dfrac{\partial S}{\partial \alpha} & \dfrac{\partial S}{\partial \beta} \\ \dfrac{\partial T}{\partial \alpha} & \dfrac{\partial T}{\partial \beta} \\ \dfrac{\partial W}{\partial \alpha} & \dfrac{\partial W}{\partial \beta} \end{bmatrix}\Big|_{\alpha=\alpha_1;\beta=\beta_1} \begin{bmatrix} \delta\alpha \\ \delta\beta \end{bmatrix} + \begin{bmatrix} \dfrac{\partial S}{\partial u} \\ \dfrac{\partial T}{\partial u} \\ \dfrac{\partial W}{\partial u} \end{bmatrix}\Big|_{u=u_1} \delta u = \mathbf{B}_{3\times 2} \begin{bmatrix} \delta\alpha \\ \delta\beta \end{bmatrix} + \mathbf{B}_{3\times 1}\delta u \tag{31}$$

where $\delta u = u_2 - u_1$, $\delta\alpha = \alpha_2 - \alpha_1$ and $\delta\beta = \beta_2 - \beta_1$. The subscripts 1 and 2 denote the primary and the deputy solar sails, respectively.

The Eq. (29) can be linearized into the following form

$$\ddot{\mathbf{x}} = -\mathbf{C}\dot{\mathbf{x}} - \mathbf{D}\mathbf{x} - n(r_1, \mathbf{x}) + \mathbf{B}_{3\times 2} \begin{bmatrix} \delta\alpha \\ \delta\beta \end{bmatrix} + \mathbf{B}_{3\times 1}\delta u \tag{32}$$

where $\mathbf{x} = [x, y, z]^T$. The relative dynamics equation of attitude motion can be obtained from Eq. (27)

$$\begin{bmatrix} \delta\ddot{\alpha} \\ \delta\ddot{\beta} \end{bmatrix} + \mathbf{M}_1 \begin{bmatrix} \delta\dot{\alpha} \\ \delta\dot{\beta} \end{bmatrix} = \mathbf{M}_2 \begin{bmatrix} \delta u_x \\ \delta u_y \end{bmatrix} \tag{33}$$

where $\delta u_x = u_{x2} - u_{x1}$ and $\delta u_y = u_{y2} - u_{y1}$. The relative attitude and orbit equations of motion are incorporated and linearized into the following coupled equation

$$\dot{\mathbf{X}} = \mathbf{A}\mathbf{X} + \mathbf{B}\mathbf{u} \tag{34}$$

where $\mathbf{X} = \begin{bmatrix} \delta\alpha & \delta\beta & x & y & z & \delta\dot{\alpha} & \delta\dot{\beta} & \dot{x} & \dot{y} & \dot{z} \end{bmatrix}^T$ and $\mathbf{u} = \begin{bmatrix} \delta u_x & \delta u_y & \delta u \end{bmatrix}^T$. The matrices $\mathbf{A}$ and $\mathbf{B}$ are expressed in the following forms

$$\mathbf{A} = \begin{bmatrix} \mathbf{0} & \mathbf{0} & \mathbf{I} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & 0 & \mathbf{I} \\ \mathbf{0} & \mathbf{0} & -\mathbf{M}_1 & \mathbf{0} \\ \mathbf{B}_{3\times2} & -\mathbf{D} & 0 & -\mathbf{C} \end{bmatrix}, \mathbf{B} = \begin{bmatrix} \mathbf{M}_2 & \mathbf{0} \\ \mathbf{0} & \mathbf{B}_{3\times1} \end{bmatrix} \tag{35}$$

where $\mathbf{0}$ and $\mathbf{I}$ denote the zero matrix and the identity matrix with compatible dimension, respectively.

To track an artificially designated relative orbit, a control law based on Linear Quadrature Regulator (LQR) is designed to minimize the following cost function

$$L = \frac{1}{2}\int_{t_0}^{\infty} \left[\mathbf{X}^T\mathbf{Q}\mathbf{X} + \mathbf{u}^T\mathbf{R}\mathbf{u} \right] \mathrm{d}t \tag{36}$$

where $\mathbf{Q}$ and $\mathbf{R}$ are symmetric positive semidefinite and definite weighting matrices, respectively.

The linear state feedback control law is obtained by solving the following Ricatti equation

$$\begin{aligned} &\dot{\mathbf{P}}(t) = -\mathbf{P}(t)\mathbf{A}(t) - \mathbf{A}^T(t)\mathbf{P}(t) + \mathbf{P}(t)\mathbf{B}\mathbf{R}^{-1}\mathbf{B}^T\mathbf{P}(t) - \mathbf{Q} \\ &\mathbf{P}(t_f) = 0 \end{aligned} \tag{37}$$

Since $\mathbf{P}(t)$ is obtained by backward integration in time, Eq.(37) can not be integrated with the control equation. The time series $\mathbf{P}(t)$ should be calculated and stored before hand. The control input without constraints can be obtained by seeking a gain matrix $\mathbf{K}$ such that the controlled system is asymptotically stable

$$\mathbf{u} = -\mathbf{K}(\mathbf{X} - \mathbf{X}_{ref}) \tag{38}$$

where the feedback matrix is solved by

$$\mathbf{K}(t) = \mathbf{R}^{-1}\mathbf{B}^T\mathbf{P}(t) \tag{39}$$

An important issue of the control design is that the three control inputs are highly constrained according to the Eq. (12) . The reflectivity control rate u of each sail is constrained by the maximum value allowed for the sail u_{max} and the sum of the absolute value of parameter u_x and u_y are constrained by the minimum value among $\frac{1}{4}u_{max}$, u and $u_{max}-u$. Furthermore, when the sum of the absolute value of parameter u_x and u_y obtained by the LQR exceeds the admissible scope, a strategy has to be devised to decide the share of each variable. There are rigorous methods designed for constrained control problem in which the admissible region is fixed in most cases. However, the admissible region discussed here is changing with the variation of u. To simplify the complicated constraints, a saturator is imposed directly upon the control inputs generated by the LQR. For the GeoSail formation flying, the three variables of the primary sail are used as control inputs to maintain a sun-synchronous orbit under the simple steering law and those of the deputy sail are employed to track a reference orbit relative to the primary. If the primary sail is flying in the sun-synchronous GeoSail orbit without error, its three control variables can be kept unchanged namely, $u_1 = U_c$ and $u_{x1} = u_{y1} = 0$. One part of saturator imposed the control variable δu is expressed as follows

$$S_i(\delta u) = \begin{cases} \delta u_{max}, & \delta u > \delta u_{max} \\ \delta u, & \delta u_{min} < \delta u < \delta u_{max} \\ \delta u_{min}, & \delta u \le \delta u_{min} \end{cases} \tag{40}$$

where $\delta u_{min} = -U_c$ and $\delta u_{max} = u_{max} - U_c$ denote the relative lower and the upper bounds, respectively, for the control variable δu. The part of the saturator imposed on the control variable δu_x is expressed as

$$S_i(\delta u_x) = \begin{cases} \delta u_x = \mathrm{sgn}(\delta u_x)\dfrac{u_{max}}{4}, & |\delta u_x| > \dfrac{u_{max}}{4} \\ \delta u_x = \mathrm{sgn}(\delta u_x)\min\left(\dfrac{|\delta u_x| U_b}{(|\delta u_y| + |\delta u_x|)}, \dfrac{u_{max}}{4}\right), & |\delta u_x| + |\delta u_y| > U_b \\ \delta u_x = \delta u_x, \quad \text{else} \end{cases} \tag{41}$$

where $\mathrm{sgn}(\cdot)$ is the signum function and $U_b = \min(U_c + \delta u, u_{max} - U_c - \delta u)$. The part of the

saturator imposed on the control variable δu_y is the same as that of δu_x. When $|\delta u_x|+|\delta u_y|$ exceeds the bound U_b, two variables will share the constrained resource based on their unconstrained ratio.

V. Numerical Results

In the simulation, the primary solar sail is assumed to orbit the Earth Sun-synchronously based on the simple steering law. Its classical orbital elements in the initial epoch is a_0 = 127,000 km, e_0 = 0.5, $i_0 = 0^\circ$, $\Omega_0 = 0^\circ$, $\omega_0 = 0^\circ$ and $f_0 = 0^\circ$. The maximum values u_{max} of both sails are set to be 0.2. The characteristic acceleration κ required for the orbit is 1.38172*10^{-7} mm/s^2 when the primary's reflectivity modulation rate is kept at 0.1. Its corresponding sail loading is approximately 66.0 g/m^2. The deputy solar sail and the primary sail are set to have the same characteristic acceleration and configuration. The other parameters used in the simulation are set in the Table 1.

Table 1 Main specifications of the sails

Parameter	Specification
Mass(m)	80kg
Radius(r_s)	19.64m
Moments of inertia(J_x,J_y,J_z)	(4000,4000,8000)kg.m^2
Spin rate(Ω)	0.5 rpm
Radius of relative reference orbit	1km
Orbit period of primary sail(T_s)	5.4425 days

When three control inputs of the primary sail are kept at the initial value with $u_{x1} = 0$, $u_{y1} = 0$ and $u = 0.1$, The values of the control variables δu_x, δu_y and δu are decided by the LQR with the saturator discussed above and bounded in the following range

$$\begin{cases} -0.1 \le \delta u \le 0.1 \\ -0.05 \le \delta u_x \le 0.05 \\ -0.05 \le \delta u_y \le 0.05 \end{cases} \tag{42}$$

The deputy sail is commanded to track the reference relative orbit and attitude around the primary sail. The reference relative orbit and attitude are given by

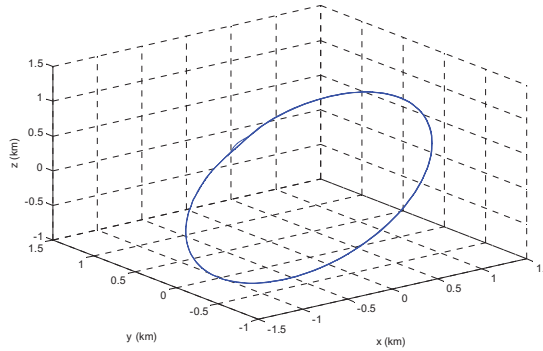
$$\left[\begin{array}{cccc} 0 & 0 & r_0 \sin(\frac{2\pi}{T_s}t) & r_0 \cos(\frac{2\pi}{T_s}t) \quad r_0 \sin(\frac{2\pi}{T_s}t) \end{array}\right]^T .$$

The simulation is conducted as follows: Firstly, the reference relative orbit is calculated, and the terminal state is got. Then the time series $\mathbf{P}(t)$ in Eq. (37) are calculated and stored for further interpolation. At a given epoch, the sail states and the matrix $\mathbf{P}(t)$ are obtained through interpolation, and the difference between the true and the reference states $\delta\mathbf{X}$ can be obtained. Then, the control variables are calculated according to the control law in Eq. (38). Finally, the saturator discussed above is imposed directly on the control inputs and a controlled system is decided.

For the simulation, the initial values of the relative state are $[\mathbf{x}_{ref}(0) \quad \dot{\mathbf{x}}_{ref}(0)]^T$ without initial errors with respect to the reference orbit. It is a tricky task to select proper values for the two weight matrices $\mathbf{Q}$ and $\mathbf{R}$ to achieve a better performance. If the eigen values of $\mathbf{Q}$ increase relative to those of $\mathbf{R}$, tracking errors will decrease, but with the risk of exceeding the bounds in Eq. (42). That could possibly lead to the instability for the controller. To balance the performance and stability, the two matrices are finally selected to be $\mathbf{Q} = 2 \times \mathrm{diag}([1,\ 1,\ 1000,\ 1000,\ 1000,\ 1,\ 1,\ 1000,\ 1000,\ 1000]^T)$ and $\mathbf{R} = 10^3 \times \mathrm{diag}([1,\ 1,\ 1]^T)$, and both are diagonal positive-definitive matrices. The simulation results generated by the LQR are shown in the following figures. Figure 3a shows the controlled relative orbit. The relative position errors with respect to the reference orbit are given in Fig. 3b to Fig. 3d. The corresponding constrained control inputs by the LQR are shown in Fig.4. The dynamic admissible region in Eq. (41) is shown in Fig. 5. Fig. 6 shows the controlled relative attitude of the deputy sail which is actually the attitude motion of the spin-free body frame $o_n x_n y_n z_n$ of the deputy sail in the spin-free body frame $o_n x_n y_n z_n$ of the primary sail.

The errors of controlled relative orbit are bounded in small ranges, and maximum errors in x-axis and y-axis are below 30 meters and that of z-axis below one meter. Except for the initial stage, the demand for control inputs doesn't reach the boundary of the dynamic admissible region any more afterward. It should be mentioned that the errors increase acutely nearby the perigee. Since the reference relative orbit is an artificially designated orbit which is evenly distributed with time. The

orbit of the primary sail is highly elliptical orbit which has shorter duration and there is greater difference in the relative orbital state between the reference relative orbit and their natural relative orbit nearby the perigee. Thus, the artificial relative orbit will demand a higher control torque to "brake" the rapider change of relative position at the arc. The maneuver is reflected in the attitude variation. Especially for angle β, it varies about 1.5° in less than one day.



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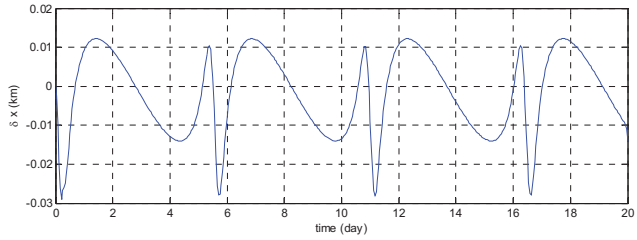

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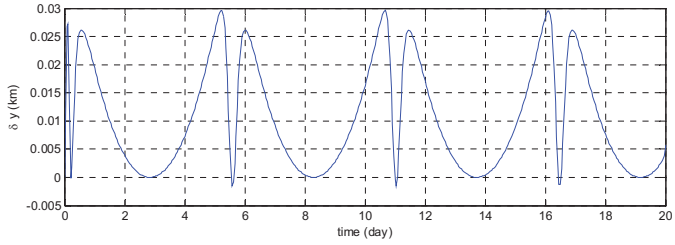

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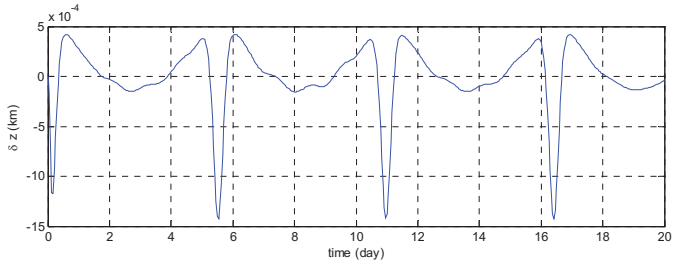
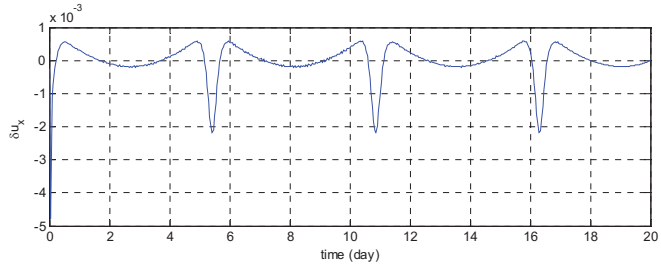


d)

Fig. 3 Controlled relative orbit and orbit errors of the controlled dynamics



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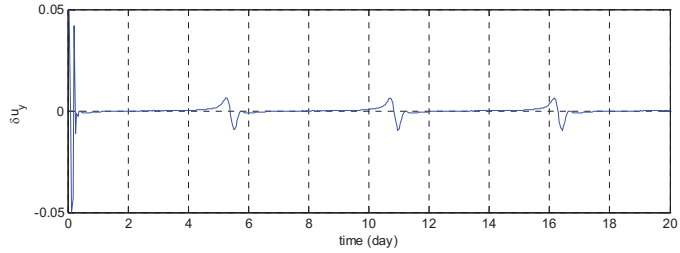

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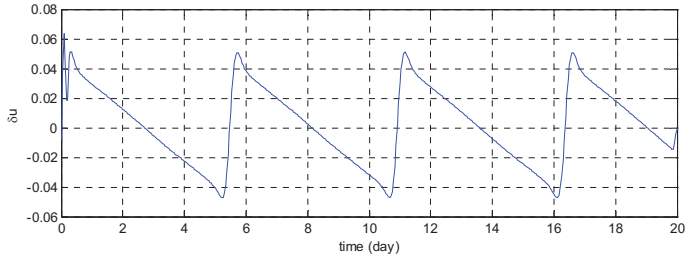


c)

Fig. 4 Three control variables for relative reference trajectory tracking for the LQR

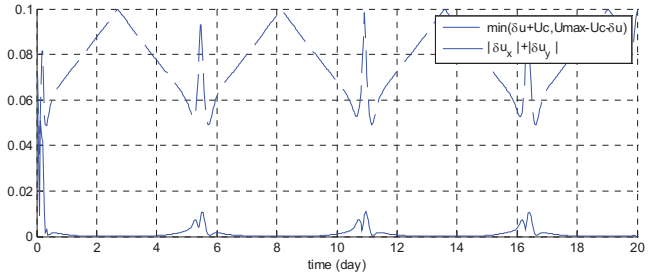
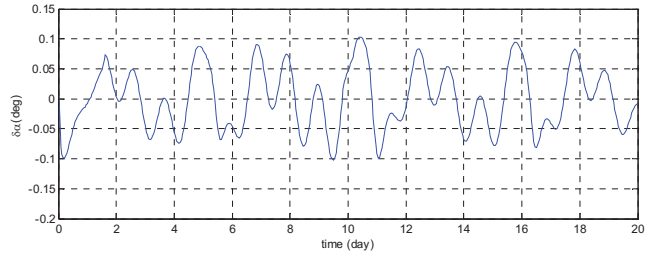
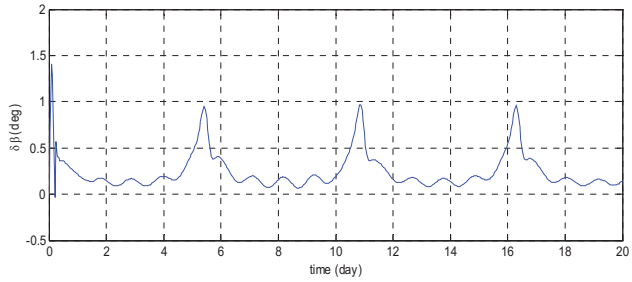


Fig. 5 The dynamic admissible region



a)



b)

Fig. 6 Controlled relative attitude

VI. Conclusion

The approach discussed in the paper is an extension to the solar sail reflectivity technology which was already implemented. It proposed to use the reflectivity modulation technology for the active orbit and attitude control of solar sails. Coupled orbit and attitude dynamics were considered together. It has been shown that the reflectivity modulation rate and its distribution can be used to track an artificially designed relative reference orbit in GeoSail formation flying mission. In addition to the three control variables discussed in the paper, the spin rate of the sail could be used to further control attitude motion of the sail. In contrast to traditional methods, the proposed

approach does not complicate the solar sail design. The method can be further used in many other situations where an active control is needed.

Acknowledgements

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Dynamical Analysis of The Deployment for A Reduced Spinning Solar Sail Model

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The deployment stability of the spinning solar sail spacecraft has been discussed in this paper. First, two-step deployment method is discussed. To get the solar sail model in every deployment stage, we study a three-dimensional reduced model including a rigid body and four cables with corner masses, consider the cables noncoplanar and the mass of the cables and derive the analytical model. After that, the characteristics of every stage are analysed. Finally, with the given controller, we study the deployment stability of the system and discuss how to keep the successful deployment with small pitch angle and none re-wrapping.

Nomenclature

(*Nomenclature entries should have the units identified*)

$\mathcal{L}_0$	=	angular momentum of the hub
M	=	external torque
ω	=	angular velocity of the solar sail under the inertial frame
n	=	numbers of the tip mass or the membrane arms
S	=	length of a side of the square solar sail
r	=	radius of the central hub
h	=	height of the central hub
L	=	length of the membrane arms unfolded
L_f	=	maximum unfolded length of the membrane arms in the first step
L_s	=	maximum unfolded length of the membrane arms in the second step
N/N_0	=	force at the tip/bottom of the membrane arms
dm/dl	=	mass/length infinitesimal of the membrane arms
α	=	pitch angle of the membrane arms
ϕ	=	wrapping angle of the membrane arms

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ψ = swing angle of the membrane arms

r_{cm} = position of center of mass of entire system

m_a = mass of entire system

m_b = mass of entire membrane

m_h = mass of hub

m_c = total mass of all corner masses

m_ω = total mass of all expandable membrane arms

m_s = total mass of all non-expandable membrane arms

ρ = linear density of the membrane arms

I. Introduction

THE trend in space industry is to build successively larger structures. The thin membrane is one of the potential candidates used for such structures [1,2] , because it is easy to fold into small volume, and enables a larger space structures with high area-to-mass ratio. However, it is so difficult to deploy the gossamer appendage properly in space. So far, a lot of research has been made on this subject, and some methods for the deployment of the gossamer membrane have been proposed. Spin deployment method is the most potential one in which the membrane is deployed and maintained flat by the centrifugal force [3]. This method is expected to be realized with simpler and lighter-weight mechanism than other ways, because it does not require rigid structural elements and has the ability to deploy far large membrane. Japan Exploration Agency (JAXA) lunched a spin solar sail IKAROS on May 2010, which successfully spinning deployed the sail [4-7].

One of the simplest deployment methods of gossamer membrane in spinning soar sail is the one-step deployment method [8]. In this method, membrane/cable elements are initially coiled around the center hub, and their tips, for instance the four corners of a square membrane of IKAROS, are fixed on the hub. These fixed points are then released simultaneously, starting the deployment by the centrifugal forces. Another deployment method which has been successfully used in IKAROS is the two-step deployment method [7]. In the first step whose initial condition is equal to that of the one-step deployment method, four bundles of the folded membrane are slowly released starting from the tip. Then four vertical stopper bars rotates around the hub, rolling petals are extracted like a Yo-Yo despinner, and form a cross shape maintained by stopper bars. In the second step, these bars are released, and four extended petals form a square shape. However, membrane dynamic analysis needs to be done to make it into use for both of the deployment methods [9,10]. As the deployment sequence must be as simple as possible and can avoid the re-wrapping phenomenon after the deployment as well as the suppression of the nutation dynamics.

Due to the great efforts devoted by researchers, a number of valuable results on spin deployment technique have been reported in literature. Among various deployable structures on spinning spacecraft, Yo-Yo despinner, a reduced analytical model, is one of the most primitive analytical ones. Ferdor [11,12] carried out a two-dimensional analysis of Yo-Yo despinner dividing the motion into two phases. The qualitative cables are initially coined around the hub. In phase 1, the cables are released, and remain tangent to the wrapping circle. In phase 2, the cables have fixed length, and attached to the hub, swing away from the tangent. Based on this analytical model, Gärdsback [8] designed optimal control criteria to stabilize the deploying solar sail, but his model is only simple two-dimensional and nutation dynamics hasn't been considered. Collins [13] extended the analytical model to three-dimensional case. Based on this model, Haraguchi [9] studied the problem of selecting structural parameters and designing control laws, but the mass of membrane band was ignored, and the assumption of symmetric distribution membrane band is inconsistent with the actual situation.

The purpose of the present research is to study two-step deployment method and built a three-dimensional analytical model considering the nutation dynamics of the system and the mass of the membrane band which is more accuracy than Yo-Yo models mentioned above. On the basis of the model, we will study the deployment performance of the spinning solar sail with the given controller. Finally, the proper structural parameters are proposed to keep the successful deployment with small nutation angular and none re-wrap around the hub.

II. Two Steps Deployment Method

In this section, a two steps deployment method is discussed for a three-dimensional reduced spinning solar sail model. Based on some assumptions, one gives the initial and restrictive conditions for all the deployment stages.

A. Coordinate Definition

The inertial frame $\left\{e^{(0)}\right\}=\left\{e_x^{(0)},e_y^{(0)},e_z^{(0)}\right\}$, the body rotation frame of the center hub $\left\{e^{(1)}\right\}=\left\{e_x^{(1)},e_y^{(1)},e_z^{(1)}\right\}$, and the body rotation frame of the membrane $\left\{e^{(2)}\right\}=\left\{e_x^{(2)},e_y^{(2)},e_z^{(2)}\right\}$ are defined, as shown in Fig. 1. In the problem, we assume that no external translational force is applied; thus, the center of mass (CM) of the entire system does not move throughout the analysis. The CM position is defined as the origin of the inertial frame $\left\{e^{(0)}\right\}$. The origin of the body-fixed frame $\left\{e^{(1)}\right\}$ is $-\boldsymbol{r}_{cm}$ in $\left\{e^{(0)}\right\}$. In this model, the external torque M can be applied on the hub around its $e_z^{(0)}$ axis.

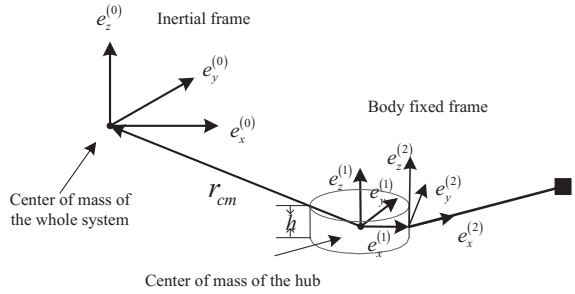


Fig. 1 Reference frames

B. Some Assumptions

Simple analytical models can be used to describe the deployment dynamics qualitatively. The development of our analytical model follows the same principles that Daisuke Haraguchi [9] use to describe the deployment of split and solid reflectors and tether systems from a rotating central satellite. In order to analyse the deployment of the three-dimensional rotational solar sail, the following assumptions were made:

1) Two-step deployment method is considered to use in this study.

2) The size of the corner mass and the membrane arm were neglected.

3) The membrane arms were supposed to be straight and deployed symmetrically relative to $e_z^{(1)}$.

4) The gravity gradient, environmental effects and the elasticity in the membrane arm were neglected.

5) Pitch angle of the membrane arms is included. Thus, the problem was three-dimensional.

A. The First Step of Deployment

The first step of deployment can be called as pre-deployment, and divided into two phases.

1) Phase 1 (unwrap phase): The cables with corner masses are initially wrapped in a circle, as in Fig. 2, and in phase 1, the cables are unwrapping, remain tangent to the wrapping circle shown in Fig. 3. When the unfixed cables L_f have been released totally, the unwrap phase is finished.

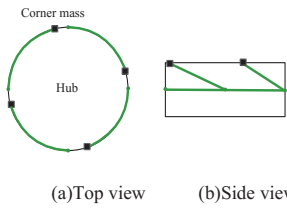


Fig. 2 The initial state of pre-deployment

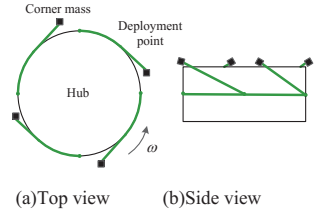
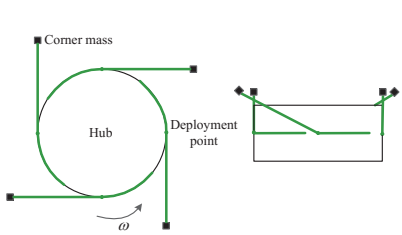


Fig. 3 The unwrap phase of the pre-deployment

2) Phase 2 (swing phase): the released cables have a fixed length and are completely unwrapped. In this phase, the corner masses swing away from the tangent, as in Fig. 4. The attachment points of the cables are fixed to the hub. If the controller is designed properly, in the final of phase 2 is the swing angle can converge to zero, shown as Fig. 5.



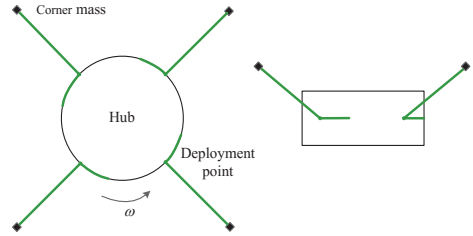


Fig. 4 The initial state of the swing phase **Fig. 5 The fina state of swing phase**

B. The Second Step of Deployment

In the second step, these stopper bars are slowly rotated around the hub to deploy the whole cables, and the released cables L_s are extracted like a Yo-Yo despinner, as in Fig. 6. The deployment solar sail maintained cross shape by four rotated stopper bars. The step didn't finish until the whole cables are released.

We consider the phase of the release of tip mass and define the phase as pre-deployment. Note that this phase contains unwrap sub-phase and swing sub-phase. The second phase of deployment is an active deployment phase. We assume the attachment point move slowly and the movement is controllable. The characteristic condition of every deployment phase is listed in Table 1.

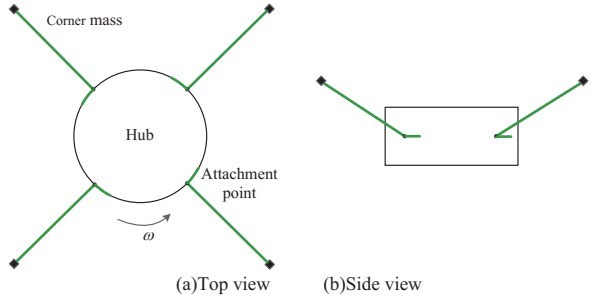


Fig. 6 The equilibrium state of the quasi static deployment

III. Modeling of Three Dimensional Yo-Yo Booms

In this section, a reduced spinning solar sail model will be derived based on some fundamental laws. All of these make the dynamic performance analysis of the deployment for a reduced spinning solar sail model become possible.

Table 1 Key conditions of every deployment phase

	pre-deployment		quasi static
	unwrap phase $\left(t\in[t_0,t_1]\right)$	swing phase $\left(t\in[t_1,t_2]\right)$	deployment $\left(t\in[t_2,t_3]\right)$
Restrained conditions	$\phi>0,\dot{\phi}<0,\psi=\pi/2,$ $\dot{\psi}=0,N_0\geq 0,N\geq 0$	$\phi=\phi(t_1),\dot{\phi}=0,N_0\geq 0,$ $-\pi/2\leq\psi\leq\pi/2,N\geq 0$	$\dot{\phi}\leq 0,N_0\geq 0,N\geq 0,$ $-\pi/2\leq\psi\leq\pi/2$
Initial conditions	$\omega(0),\phi(0),\alpha(0)$	$\phi(t_1)=L_s/r$	$\psi(t_2)=0,\alpha(t_2)=0$
Finish conditions	$\phi(t_1)=L_s/r$	$\psi(t_2)=0,\alpha(t_2)=0$	$\phi(t_3)=0$

A. Equations of Motion

For the purpose of obtaining the exact deployment information of the deployment solar sail, the following variables α,ϕ,ψ,L must be known. Since the length of the membrane arms already unwrapped L directly depends on ϕ, therefore, we only need to solve for three unknown variables. As a result, three equations are required to solve. Here, the change of angular momentum for the hub and the equations of motion for the corner masses and the membrane arms are available.

First consider the change in angular momentum, due to the applied external torque and the torque exerted by the pulling arms, for the central hub in Fig. 7 is.

$$\dot{\boldsymbol{\mathcal{L}}}_0=\boldsymbol{M}-n\left(\boldsymbol{r}\times\boldsymbol{F}_0\right) \tag{1}$$

where $\boldsymbol{F}_0$ is the tension force of the hub on the membrane arms at the bottom.

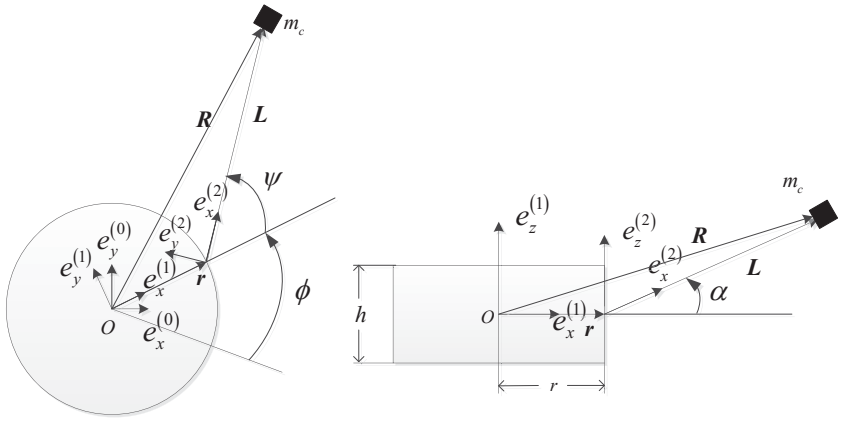


Fig.7 The analytical model for a point mass

Projected along the axis of rotation it becomes

$$J\dot{\omega}=M+nN_0r\sin\psi\cos\alpha \tag{2}$$

Note that all the membrane arms are deployed symmetrically. When the pitch angle $\alpha\neq 0$, the whole system will not in one plane. J and $\boldsymbol{r}_{cm}$ will change with the deployment, and for the case shown in Fig. 2, we have

$$J=\frac{1}{4}m_h r^2+\frac{1}{12}m_h h^2+m_s r^2+m_\omega r^2-\frac{L^2}{\left(L_f+L_s\right)^2}m_\omega r^2 \tag{3}$$

$$\boldsymbol{r}_{cm}=\frac{n\int_0^L \rho(l)\,l\sin\alpha\,\mathrm{d}l+m_c\boldsymbol{r}_c}{m_a}=\frac{m_\omega\boldsymbol{r}_\omega+m_c\boldsymbol{r}_c}{m_a} \tag{4}$$

where, $\boldsymbol{r}_\omega=\left[0\quad 0\quad L^3\sin\alpha\Big/3\left(L_f+L_s\right)^2\right]^{\mathrm{T}}$, $\boldsymbol{r}_c=\left[0\quad 0\quad L\sin\alpha\right]$.

Second, we consider the motion equations of the corner masses

$$\boldsymbol{F}=\frac{m_c}{n}\ddot{\boldsymbol{R}}(L) \tag{5}$$

Its position is obtained from Fig. 8 as

$$\boldsymbol{R}(L)=\boldsymbol{r}-\boldsymbol{r}_{cm}+\boldsymbol{L} \tag{6}$$

and the derivatives of $\boldsymbol{R}$ become

$$\dot{\boldsymbol{R}}(L)=\left(\dot{\boldsymbol{r}}-\dot{\boldsymbol{r}}_{cm}+\dot{\boldsymbol{L}}\right)+\left(\boldsymbol{\omega}+\dot{\phi}\boldsymbol{e}_z^{(2)}+\dot{\psi}\boldsymbol{e}_z^{(2)}\right)\times\left(\boldsymbol{r}-\boldsymbol{r}_{cm}+\boldsymbol{L}\right) \tag{7}$$

$$\begin{aligned}\ddot{\boldsymbol{R}}(L)=&\left(\ddot{\boldsymbol{r}}-\ddot{\boldsymbol{r}}_{cm}+\ddot{\boldsymbol{L}}\right)+\left(\dot{\boldsymbol{\omega}}+\ddot{\phi}\boldsymbol{e}_z^{(2)}+\ddot{\psi}\boldsymbol{e}_z^{(2)}\right)\times\left(\boldsymbol{r}-\boldsymbol{r}_{cm}+\boldsymbol{L}\right)+2\left(\boldsymbol{\omega}+\dot{\phi}\boldsymbol{e}_z^{(2)}+\dot{\psi}\boldsymbol{e}_z^{(2)}\right)\times\left(\dot{\boldsymbol{r}}-\dot{\boldsymbol{r}}_{cm}+\dot{\boldsymbol{L}}\right)\\&+\left(\boldsymbol{\omega}+\dot{\phi}\boldsymbol{e}_z^{(2)}+\dot{\psi}\boldsymbol{e}_z^{(2)}\right)\times\left[\left(\boldsymbol{\omega}+\dot{\phi}\boldsymbol{e}_z^{(2)}+\dot{\psi}\boldsymbol{e}_z^{(2)}\right)\times\left(\boldsymbol{r}-\boldsymbol{r}_{cm}\right)\right]\end{aligned} \tag{8}$$

where $\boldsymbol{\omega}=\begin{bmatrix}0\\0\\\omega\end{bmatrix}$, $\boldsymbol{r}=\begin{bmatrix}r\cos\psi\\-r\sin\psi\\0\end{bmatrix}$, $\boldsymbol{L}=\begin{bmatrix}L\cos\alpha\\0\\L\sin\alpha\end{bmatrix}$, $\boldsymbol{e}_z^{(2)}=\begin{bmatrix}0\\0\\1\end{bmatrix}$, $\boldsymbol{F}=\begin{bmatrix}-N\cos\alpha\\0\\-N\sin\alpha\end{bmatrix}$.

Expand equation (5) into scalar form, and simplify them available, then we can obtain

$$\begin{aligned}-N\cos\alpha=&\frac{m_c}{n}\left(r\dot{\omega}\sin\psi+r\ddot{\phi}\sin\psi-r\omega^2\cos\psi-r\dot{\phi}^2\cos\psi-2r\omega\dot{\phi}\cos\psi\right)\\&+\frac{m_c}{n}\left[-2\dot{L}\dot{\alpha}\sin\alpha+\ddot{L}\cos\alpha-L\dot{\alpha}^2\cos\alpha-L\ddot{\alpha}\sin\alpha-L\left(\omega+\dot{\phi}+\dot{\psi}\right)^2\cos\alpha\right]\end{aligned} \tag{9}$$

$$\begin{aligned}0=&\frac{m_c}{n}\left[r\dot{\omega}\cos\psi+r\ddot{\phi}\cos\psi+r\omega^2\sin\psi+r\dot{\phi}^2\sin\psi+2r\omega\dot{\phi}\sin\psi+2\dot{L}\left(\omega+\dot{\phi}+\dot{\psi}\right)\cos\alpha\right]\\&+\frac{m_c}{n}\left[+L\left(\dot{\omega}+\ddot{\phi}+\ddot{\psi}\right)\cos\alpha-2L\dot{\alpha}\left(\omega+\dot{\phi}+\dot{\psi}\right)\sin\alpha\right]\end{aligned} \tag{10}$$

$$\begin{aligned}-N\sin\alpha=&\frac{m_c}{n}\left[2p\dot{L}\dot{\alpha}\cos\alpha+p\ddot{L}\sin\alpha-pL\dot{\alpha}^2\sin\alpha+pL\ddot{\alpha}\cos\alpha-\frac{2m_\omega L\dot{L}^2\sin\alpha}{m_a\left(L_f+L_s\right)^2}\right]\\&+\frac{m_c m_\omega}{3nm_a\left(L_f+L_s\right)^2}\left(-3L^2\ddot{L}\sin\alpha-6L^2\dot{L}\dot{\alpha}\cos\alpha+L^3\dot{\alpha}^2\sin\alpha-L^3\ddot{\alpha}\cos\alpha\right)\end{aligned} \tag{11}$$

where $p=1-\dfrac{m_c}{m_a}$.

Finally, we consider the motion equations of the membrane arms shown in Fig. 8.

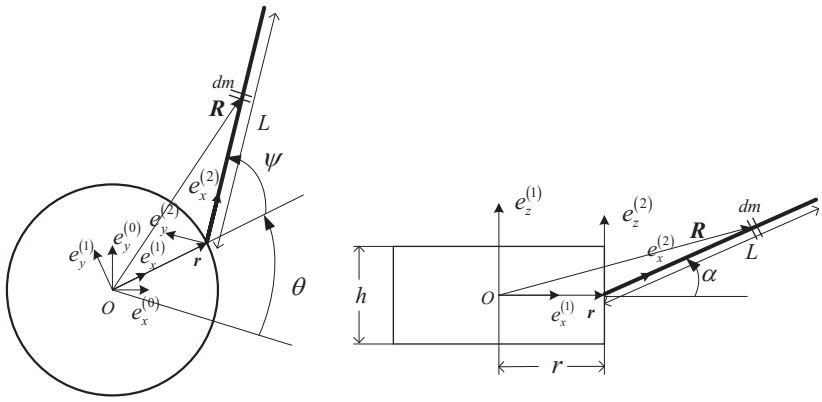


Fig. 8 The analytical model for a distributed mass

Since the membrane arms are always remain straight, we have the sum acceleration of all the mass infinitesimal $\mathrm{d}m=\rho(l)\mathrm{d}l$ just the same as the corner masses

$$\boldsymbol{F}_0-\boldsymbol{F}=\int_0^L\ddot{\boldsymbol{R}}(l)\rho(l)\mathrm{d}l \tag{12}$$

and the position of $\mathrm{d}m$ is

$$\boldsymbol{R}(l)=\boldsymbol{r}-\boldsymbol{r}_{cm}+\boldsymbol{l} \tag{13}$$

where $\boldsymbol{F}_0=\left[-N_0\cos\alpha \quad 0 \quad -N_0\sin\alpha\right]^{\mathrm{T}}$, $\boldsymbol{l}=\left[l\cos\alpha \quad 0 \quad l\sin\alpha\right]^{\mathrm{T}}$.

For the square solar sail which is folded in 'Z' patterns, the mass per length $\rho(l)$ varies linearly, starting from zero at the tip of the arm.

$$\rho(l)=\frac{2m_\omega}{n\left(L_f+L_s\right)^2}(L-l),(0\le l\le L) \tag{14}$$

the maximum length of the membrane arms unfolded can be described as

$$L_f=\frac{S}{2}-\frac{\pi}{4}r-L_s \tag{15}$$

and the mass of m_b, m_ω and m_s have the following relationship

$$m_b=m_s+m_\omega \tag{16}$$

$$m_\omega=n\int_0^{L_f+L_s}\frac{2m_\omega}{n\left(L_f+L_s\right)^2}\left(L_f+L_s-l\right)\mathrm{d}l \tag{17}$$

$$\frac{m_\omega}{m_b}=\frac{4L_s^2}{S^2} \tag{18}$$

In view of $\dot{l}$ and $\ddot{l}$ are function relate to time t not directly to l, then we have

$$\dot{l} = \dot{L}\left(0 \le l \le L\right) \tag{19}$$

$$\ddot{l} = \ddot{L}\left(0 \le l \le L\right) \tag{20}$$

so we can obtain further

$$\int_0^L \left(L-l\right) l^k \mathrm{d}l = \frac{L^{k+2}}{\left(k+1\right)\left(k+2\right)}, k \in \square \tag{21}$$

$$\int_0^L \dot{l}\mathrm{d}l = L\dot{L} \tag{22}$$

$$\int_0^L \ddot{l}\mathrm{d}l = L\ddot{L} \tag{23}$$

Based on the above conclusions, equation (12) can be simplified as

$$\begin{aligned}\left(N-N_0\right)\cos\alpha = {} & \frac{2m_\omega}{n\left(L_f+L_s\right)^2}\frac{L^2}{2}\left(r\dot{\omega}\sin\psi + r\ddot{\phi}\sin\psi - r\omega^2\cos\psi - r\dot{\phi}^2\cos\psi - 2r\omega\dot{\phi}\cos\psi\right) \\ & + \frac{2m_\omega}{n\left(L_f+L_s\right)^2}\left\{\frac{L^2}{2}\left(-2\dot{L}\dot{\alpha}\sin\alpha + \ddot{L}\cos\alpha\right) - \frac{L^3}{6}\left[\dot{\alpha}^2\cos\alpha + \ddot{\alpha}\sin\alpha + \left(\omega+\dot{\phi}+\dot{\psi}\right)^2\cos\alpha\right]\right\}\end{aligned} \tag{24}$$

$$\begin{aligned}0 = {} & \frac{2m_\omega}{n\left(L_f+L_s\right)^2}\frac{L^2}{2}\left[r\dot{\omega}\cos\psi + r\ddot{\phi}\cos\psi + r\omega^2\sin\psi + r\dot{\phi}^2\sin\psi + 2r\omega\dot{\phi}\sin\psi + 2\dot{L}\left(\omega+\dot{\phi}+\dot{\psi}\right)\cos\alpha\right] \\ & + \frac{2m_\omega}{n\left(L_f+L_s\right)^2}\frac{L^3}{6}\left[\left(\dot{\omega}+\ddot{\phi}+\ddot{\psi}\right)\cos\alpha - \dot{\alpha}\left(\omega+\dot{\phi}+\dot{\psi}\right)\sin\alpha\right]\end{aligned} \tag{25}$$

$$\begin{aligned}\left(N-N_0\right)\sin\alpha = {} & \frac{2m_\omega}{n\left(L_f+L_s\right)^2}\left\{\frac{L^2}{2}\left(2p\dot{L}\dot{\alpha}\cos\alpha + p\ddot{L}\sin\alpha\right) + \frac{L^3}{6}\left[p\ddot{\alpha}\cos\alpha - p\dot{\alpha}^2\sin\alpha - \frac{2m_\omega\dot{L}^2\sin\alpha}{m_a\left(L_f+L_s\right)^2}\right]\right\} \\ & - \frac{m_\omega^2 L^4}{6nm_a\left(L_f+L_s\right)^4}\left(\ddot{L}\sin\alpha + 2\dot{L}\dot{\alpha}\cos\alpha\right) + \frac{m_\omega^2 L^5}{30nm_a\left(L_f+L_s\right)^4}\left(\dot{\alpha}^2\sin\alpha - \ddot{\alpha}\cos\alpha\right)\end{aligned} \tag{26}$$

In order to calculate easily, we combine the corner mass and its corresponding membrane arm, the equations of motion are added together, so that

$$\begin{aligned}-nN_0\cos\alpha = {} & A\left(r\dot{\omega}\sin\psi + r\ddot{\phi}\sin\psi - r\omega^2\cos\psi - r\dot{\phi}^2\cos\psi - 2r\omega\dot{\phi}\cos\psi - 2\dot{L}\dot{\alpha}\sin\alpha + \ddot{L}\cos\alpha\right) \\ & - B\left[\dot{\alpha}^2\cos\alpha + \ddot{\alpha}\sin\alpha + \left(\omega+\dot{\phi}+\dot{\psi}\right)^2\cos\alpha\right]\end{aligned} \tag{27}$$

$$\begin{aligned}0 = {} & A\left[r\dot{\omega}\cos\psi + r\ddot{\phi}\cos\psi + r\omega^2\sin\psi + r\dot{\phi}^2\sin\psi + 2r\omega\dot{\phi}\sin\psi + 2\dot{L}\left(\omega+\dot{\phi}+\dot{\psi}\right)\cos\alpha\right] \\ & + B\left[\left(\dot{\omega}+\ddot{\phi}+\ddot{\psi}\right)\cos\alpha - \dot{\alpha}\left(\omega+\dot{\phi}+\dot{\psi}\right)\sin\alpha\right]\end{aligned} \tag{28}$$

$$-nN_0\sin\alpha = A\left(2p\dot{L}\dot{\alpha}\cos\alpha + p\ddot{L}\sin\alpha\right) - B\left[p\dot{\alpha}^2\sin\alpha - p\ddot{\alpha}\cos\alpha + \frac{2m_\omega\dot{L}^2\sin\alpha}{m_a\left(L_f+L_s\right)^2}\right] - C\left(\ddot{L}\sin\alpha + 2\dot{L}\dot{\alpha}\cos\alpha\right) + D\left(\dot{\alpha}^2\sin\alpha - \ddot{\alpha}\cos\alpha\right) \tag{29}$$

where

$$A = m_c + m_\omega\frac{L^2}{\left(L_f+L_s\right)^2} \tag{30}$$

$$B = m_c L + m_\omega\frac{L^3}{3\left(L_f+L_s\right)^2} \tag{31}$$

$$C = \frac{m_\omega m_c L^2}{m_a\left(L_f+L_s\right)^2} + \frac{m_\omega^2 L^4}{6m_a\left(L_f+L_s\right)^4} \tag{32}$$

$$D = \frac{m_\omega m_c L^3}{3m_a\left(L_f+L_s\right)^2} + \frac{m_\omega^2 L^5}{30m_a\left(L_f+L_s\right)^4} \tag{33}$$

B. Equations for The Unwrap Phase in First Stage

As we known, in the unwrap phase of the first stage, $\dot{\phi}<0$, $\psi=\pi/2$, $\dot{\psi}=0$ and $\ddot{\psi}=0$. To simulate space webs that are coiled around the center hub, first set the initial length of the coiled off part

$$L(0) = L_{ini} > 0 \tag{34}$$

then the initial wrapping angle can be derived from

$$\phi(0) = \left(L_f + L_s - L_{ini}\right)/r \tag{35}$$

Because the current length of the coiled off part

$$L = L_f + L_s - r|\phi| \tag{36}$$

thus we have the derivatives of L as

$$\dot{L} = -r\dot{\phi} \tag{37}$$

$$\ddot{L} = -r\ddot{\phi} \tag{38}$$

Clearly, if we substitute those into (2), (27), (28) and (29), then we have

$$J\dot{\omega} = M + nN_0 r\cos\alpha \tag{39}$$

$$-nN_0\cos\alpha = A\left(r\dot{\omega} + r\ddot{\phi} + 2r\dot{\alpha}\dot{\phi}\sin\alpha - r\ddot{\phi}\cos\alpha\right) - B\left[\dot{\alpha}^2\cos\alpha + \ddot{\alpha}\sin\alpha + \left(\omega+\dot{\phi}\right)^2\cos\alpha\right] \tag{40}$$

$$0 = A\left[r\omega^2 + r\dot{\phi}^2 + 2r\omega\dot{\phi} - 2r\left(\omega+\dot{\phi}\right)\dot{\phi}\cos\alpha\right] + B\left[\left(\dot{\omega}+\ddot{\phi}\right)\cos\alpha - \dot{\alpha}\left(\omega+\dot{\phi}\right)\sin\alpha\right] \tag{41}$$

$$-nN_0\sin\alpha = -A\left(2pr\dot{\alpha}\dot{\phi}\cos\alpha + pr\ddot{\phi}\sin\alpha\right) - B\left[p\dot{\alpha}^2\sin\alpha - p\ddot{\alpha}\cos\alpha + \frac{2m_\omega r^2\dot{\phi}^2\sin\alpha}{m_a\left(L_f + L_s\right)^2}\right] + C\left(r\ddot{\phi}\sin\alpha + 2r\dot{\alpha}\dot{\phi}\cos\alpha\right) + D\left(\dot{\alpha}^2\sin\alpha - \ddot{\alpha}\cos\alpha\right) \tag{42}$$

where L in A , B , C and D are replaced with equation (36).

Substitute (40) and (42) into (39), respectively, eliminate the intermediate variable nN_0 , and define the state variable

$$x = \begin{bmatrix} x_1 & x_2 & x_3 & x_4 & x_5 \end{bmatrix}^{\mathrm{T}} = \begin{bmatrix} \omega & \alpha & \dot{\alpha} & \phi & \dot{\phi} \end{bmatrix}^{\mathrm{T}} \tag{43}$$

the control input variable

$$u = M \tag{44}$$

then the governing equations for the unwrap can be obtain.

A sign of the end of the stage is the wrapping angle ϕ equal to L_s/r , at the same time, all the expandable membrane arms deployed complete and $L = L_f$.

C. Equations for The Swing Phase in First Stage

With the end of the unwrap phase, the swing phase begins. In this phase the membrane arm swings like a pendulum respect to its attachment point. In the following, we will discuss the dynamic constraints for arms with three cases.

1) Normal Swing Case $\psi + \phi \le \pi/2$

As we known, in this case, $\phi = L_s/r$, $\dot{\phi} = 0$ and $\ddot{\phi} = 0$. Substitute those into (2), (27), (28) and (29), then we have

$$J\dot{\omega} = M + nN_0 r\sin\psi\cos\alpha \tag{45}$$

$$-nN_0\cos\alpha = A\left(r\dot{\omega}\sin\psi + r\ddot{\phi}\sin\psi - r\omega^2\cos\psi\right) - B\left[\dot{\alpha}^2\cos\alpha + \ddot{\alpha}\sin\alpha + \cos\alpha\left(\omega + \dot{\psi}\right)^2\right] \tag{46}$$

$$0 = A\left(r\dot{\omega}\cos\psi + r\omega^2\sin\psi\right) + B\left[\left(\dot{\omega} + \ddot{\psi}\right)\cos\alpha - \dot{\alpha}\left(\omega + \dot{\psi}\right)\sin\alpha\right] \tag{47}$$

$$-nN_0\sin\alpha = -pB\left(\dot{\alpha}^2\sin\alpha - \ddot{\alpha}\cos\alpha\right) + D\left(\dot{\alpha}^2\sin\alpha - \ddot{\alpha}\cos\alpha\right) \tag{48}$$

Now, we can define the state variable

$$x = \begin{bmatrix} x_1 & x_2 & x_3 & x_4 & x_5 \end{bmatrix}^{\mathrm{T}} = \begin{bmatrix} \omega & \alpha & \dot{\alpha} & \psi & \dot{\psi} \end{bmatrix}^{\mathrm{T}} \tag{49}$$

substitute (46) and (48) into (45), respectively, and eliminate the intermediate variable nN_0 , then we will obtain the system dynamic equations.

2) Counter-clockwise Coiled Case $\psi+\phi>\pi/2+L_s/r$

In this case, $\phi>L_s/r$, $\psi=\pi/2$, $\dot{\psi}=0$ and $\ddot{\psi}=0$. Substitute those into (2), (27), (28) and (29), then we have

$$J\dot{\omega}=M+nN_0r\cos\alpha \tag{50}$$

$$-nN_0\cos\alpha=A\left(r\dot{\omega}+r\ddot{\phi}+2r\dot{\alpha}\dot{\phi}\sin\alpha-r\ddot{\phi}\cos\alpha\right)-B\left[\dot{\alpha}^2\cos\alpha+\ddot{\alpha}\sin\alpha+\left(\omega+\dot{\phi}\right)^2\cos\alpha\right] \tag{51}$$

$$0=A\left[r\omega^2+r\dot{\phi}^2+2r\omega\dot{\phi}-2r\left(\omega+\dot{\phi}\right)\dot{\phi}\cos\alpha\right]+B\left[\left(\dot{\omega}+\ddot{\phi}\right)\cos\alpha-\dot{\alpha}\left(\omega+\dot{\phi}\right)\sin\alpha\right] \tag{52}$$

$$\begin{aligned}-nN_0\sin\alpha=&-A\left(2pr\dot{\alpha}\dot{\phi}\cos\alpha+pr\ddot{\phi}\sin\alpha\right)-B\left[p\dot{\alpha}^2\sin\alpha-p\ddot{\alpha}\cos\alpha+\frac{2m_\omega r^2\dot{\phi}^2\sin\alpha}{m_a\left(L_f+L_s\right)^2}\right]\\&+C\left(r\ddot{\phi}\sin\alpha+2r\dot{\alpha}\dot{\phi}\cos\alpha\right)+D\left(\dot{\alpha}^2\sin\alpha-\ddot{\alpha}\cos\alpha\right)\end{aligned} \tag{53}$$

3) Clockwise Coiled Case $\psi+\phi<-\pi/2+L_s/r$

In this case, $\phi<L_s/r$, $\psi=-\pi/2$, $\dot{\psi}=0$ and $\ddot{\psi}=0$. Substitute those into (2), (27), (28) and (29), then we have

$$J\dot{\omega}=M-nN_0r\cos\alpha \tag{54}$$

$$-nN_0\cos\alpha=-A\left(r\dot{\omega}+r\ddot{\phi}+2\dot{L}\dot{\alpha}\sin\alpha-\ddot{L}\cos\alpha\right)-B\left[\dot{\alpha}^2\cos\alpha+\ddot{\alpha}\sin\alpha+\left(\omega+\dot{\phi}\right)^2\cos\alpha\right] \tag{55}$$

$$0=-A\left[r\omega^2+r\dot{\phi}^2+2r\omega\dot{\phi}-2\dot{L}\left(\omega+\dot{\phi}\right)\cos\alpha\right]+B\left[\left(\dot{\omega}+\ddot{\phi}\right)\cos\alpha-\dot{\alpha}\left(\omega+\dot{\phi}\right)\sin\alpha\right] \tag{56}$$

$$\begin{aligned}-nN_0\sin\alpha=&-A\left(2pr\dot{\alpha}\dot{\phi}\cos\alpha+pr\ddot{\phi}\sin\alpha\right)-B\left[p\dot{\alpha}^2\sin\alpha-p\ddot{\alpha}\cos\alpha+\frac{2m_\omega r^2\dot{\phi}^2\sin\alpha}{m_a\left(L_f+L_s\right)^2}\right]\\&+C\left(r\ddot{\phi}\sin\alpha+2r\dot{\alpha}\dot{\phi}\cos\alpha\right)+D\left(\dot{\alpha}^2\sin\alpha-\ddot{\alpha}\cos\alpha\right)\end{aligned} \tag{57}$$

This case has the same state variable and control variable with counter-clockwise coiled case, and We can use the same method to solve the system response.

In all three cases, we expect the first one. Only in this case, can the expandable membrane arms deployed successfully in the first step without re-wrapping. In addition, the target we design controller is to deploy successfully with small pitch angle and none re-wrapping.

D. Equations for The Second Stage

After the successfully deployment in the first step, the pitch angle, the swing angle and the unwrap angle satisfy

$$\alpha=0 \tag{58}$$

$$\psi = 0 \tag{59}$$

$$\phi = L_s / r \tag{60}$$

then the second stage can be start smoothly.

Assume the speed of membrane arms draw out can be control as (61).When T is very small, then the unwrap angle can keep zeros always.

$$\dot{\phi} = T \tag{61}$$

$$J\dot{\omega} = M + nN_0 r \tag{62}$$

$$-nN_0 = A\left(r\dot{\omega}\sin\psi - r\omega^2\cos\psi - r\dot{\phi}^2\cos\psi - 2r\omega\dot{\phi}\cos\psi\right) - B\left(\omega + \dot{\phi} + \dot{\psi}\right)^2 \tag{63}$$

$$0 = A\left[r\dot{\omega}\cos\psi + r\omega^2\sin\psi + r\dot{\phi}^2\sin\psi + 2r\omega\dot{\phi}\sin\psi - 2r\dot{\phi}\left(\omega + \dot{\phi} + \dot{\psi}\right)\right] + B\left(\dot{\omega} + \ddot{\psi}\right) \tag{64}$$

Define the state variable

$$x = \begin{bmatrix} x_1 & x_2 & x_3 & x_4 \end{bmatrix}^{\mathrm{T}} = \begin{bmatrix} \omega & \psi & \dot{\psi} & \phi \end{bmatrix}^{\mathrm{T}} \tag{65}$$

and the control input variable

$$u = \begin{bmatrix} u_1 & u_2 \end{bmatrix}^{\mathrm{T}} = \begin{bmatrix} M & T \end{bmatrix}^{\mathrm{T}} \tag{66}$$

substitute (63) into (62), respectively, and eliminate the intermediate variable , then we will obtain the dynamic equations of the system. When the length of membrane arms $L = L_f + L_s$ without re-wrapping, the whole deployment completed.

IV. Numerical Simulations

Based on the reduced model derived above, deployments under different controllers are executed. Furthermore, the deployment performance is analysed after simulation, and the effective of different controllers is discussed finally.

In the following simulation study, the simple spin-rate feedback laws are used

$$M = \begin{cases} M_{\max} & ,\omega < 0 \\ M_{\max}\left(1 - \dfrac{\omega}{\omega_0}\right)^{\gamma} & ,0 \le \omega \le \omega_0 \\ 0 & ,\omega > \omega_0 \end{cases} \tag{67}$$

A. Controlled Deployment Response with γ = 1

First, the uncontrolled deployment is simulated with the structure parameters and the initial conditions of $S=14\text{m}$, $r=0.76\text{m}$, $m_h=291\text{kg}$, $m_b=15\text{kg}$, $m_c=2\text{kg}$, $\omega_0=3\text{rpm}$, $L_f=2\text{m}$, $\omega(0)=1.5\text{rpm}$, $\psi(0)=\pi/2\text{rad}$, $\phi(0)=8.412\text{rad}$, $L(0)=0.01\text{m}$, $\alpha(0)=\pi/6\text{rad}$, $M_{\max}=5\text{Nm}$, $T=-0.05\text{rad/s}$. Moreover, select $\gamma=1$, then we obtain the closed-loop transient responses shown in Fig. 9.

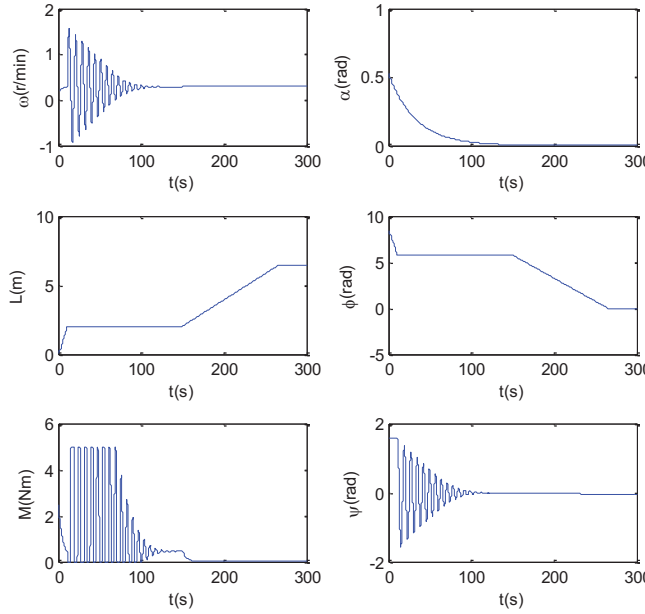


Fig.9 Closed-loop transient responses with $\gamma=1$

From the results, we can observe that the unwrap angle ϕ, the swing angle ψ and the pitch angle α are all converge to zero, finally, and $t_1=10.86\text{s},t_2=140\text{s},t_3=115.87\text{s}$. In other word, the deployment is successful.

B. Controlled Deployment Response with $\gamma \neq 1$

We use the same structure parameters and the initial conditions of solar sail as the previous simulation. Meanwhile, we consider power control law. Set $\gamma=5$, then we obtain the related system responses shown in Fig. 10.

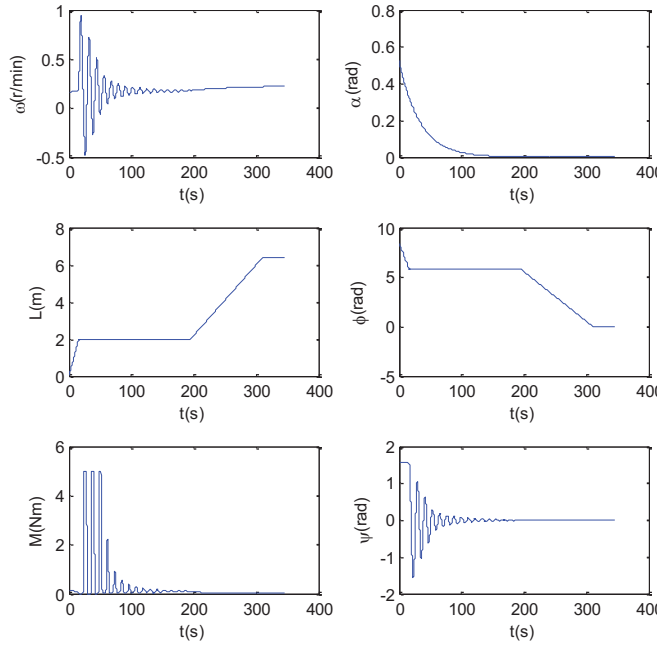


Fig.10 Closed-loop transient responses with $\gamma = 5$

Fig. 9 and Fig. 10 show that the controller can realize the successful deployment of the rotation square solar sail. The swing angle ψ can converges to zero equilibrium. The unwrap angle ϕ decreases to zero monotonically and remains unchanged. There isn't increasing trend of the pitch angle caused by three-dimensional motion.

V. Conclusion

In this paper, a model for spinning solar sail is derived, which first takes account of the three-dimensional dynamic of the system, the mass of the cables and the deployment performance in pre-deployment phase. Based on this model, the deployment stability is analyzed with different control. Simultaneously, the issue of re-wrapping is discussed and the corresponding model is derived as well. In addition, the detail of the two-step deployment method is described and the related restrictive condition is proposed. Theoretical analysis and numerical simulation show that the reduced model fully allows for three-dimensional dynamics and the mass of membrane bands. Due to the famous control strategies, the validity is verified that the solar sail can deploy successfully and avoid re-wrapping and large pitch angle. Not only for spinning solar sail spacecraft but also many other large structures in space can greatly benefit from the research in system design, analysis and control.

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Part V

Advanced Concepts

Optimal Solar Sail Interplanetary Trajectories with Constant Cone Angle

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A simplified solar sail steering law is discussed in which the sail clock angle is used as the only control variable, while the sail cone angle is maintained constant along the whole spacecraft heliocentric trajectory. Within this control strategy, the steering law that minimizes the flight time is obtained with a closed form solution using a classical variational approach. Sub-optimal performances, that is, optimal constrained solutions, have been analyzed for different mission scenarios assuming three-dimensional (ephemeris free) problems. Numerical simulations show that for suitable combinations of cone angle and spacecraft characteristic acceleration, the percentage increase of flight time of the simplified steering law is moderate when compared to a minimum time solution, in which the sail cone and clock angles are both allowed to vary during the whole transfer phase.

Nomenclature

$\mathbb{A}$	= matrix $\in \mathbb{R}^{6\times 3}$, see Eq. (6)
A	= sail reflective area [m^2]
a	= semimajor axis [AU]
a_c	= characteristic acceleration [mm/s^2]
$\boldsymbol{a}_s$	= propulsive acceleration vector [mm/s^2]
b_1, b_2, b_3	= force coefficients
$\boldsymbol{c}$	= vector $\in \mathbb{R}^{6\times 1}$, see Eq. (7)
e	= orbital eccentricity

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f, g, h, k = modified equinoctial elements

$\mathcal{H}$ = Hamiltonian

$\hat{\boldsymbol{i}}_R, \hat{\boldsymbol{i}}_T, \hat{\boldsymbol{i}}_N$ = unit vectors of RTN reference frame, see Fig. (1)

J = performance index

L = true longitude [deg]

m = spacecraft total mass [kg]

$\hat{\boldsymbol{n}}$ = sail normal unit vector

p = semilatus rectum [AU]

r = Sun-spacecraft distance [AU] (with $r_{\oplus} \triangleq 1$ AU)

t = time [days] (t_f is the flight time)

$\boldsymbol{x}$ = state vector

α = sail cone angle [deg]

β_σ = dimensionless sail loading [$\mathrm{g/m^2}$]

λ_i = adjoint to variable i

$\mu_\odot$ = Sun's gravitational parameter [$\mathrm{km^3/s^2}$]

ω = argument of perihelion [deg]

Ω = right ascension of the ascending node [deg]

Subscripts

0 = initial

des = design value

f = final

Superscripts

$\star$ = optimal

$\bullet$ = time derivative

$\wedge$ = unit vector

I. Introduction

THE study of a solar sail interplanetary transfer is a classical problem, especially for rendezvous missions toward an inner planet of the Solar System [1,2]. This kind of problem is usually studied within an optimal framework by calculating the time histories of the two control angles (that is, the cone angle $\alpha \in [0,90]\text{deg}$ and the clock angle $\delta \in [-180,180]\text{deg}$, see Fig. 1), necessary to minimize the total flight time, and the results are expressed, in a parametric way, as a function of the spacecraft characteristic acceleration a_c.

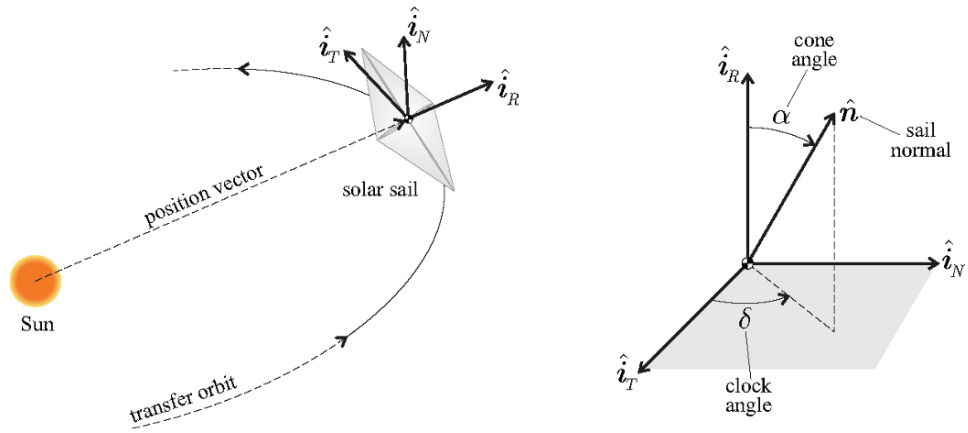


Fig. 1 Control angles (adapted from Ref. [3]).

The optimal steering law requires, in general, a continuous variation of the two control angles within the set of their admissible values. The main consequence is that the solar sail attitude should be continuously adjusted to track the optimal sail orientation with respect to a given reference frame (for example a Radial-Transverse-Normal (RTN) frame). However, such a steering law may be complex to implement, especially for large size solar sails, and this difficulty could jeopardize the effectiveness of the attitude control system. In most cases sub-optimal solutions are therefore advisable. For example, because the sail cone angle can, in principle, be maintained at a constant value with a suitable choice of the sail's shape, a possible answer to the previous problem is offered by a piecewise-constant steering law [4]. In particular, the solution discussed in Ref. [4] involves a two-dimensional problem in which the sail cone angle is optimally chosen within a finite set of admissible values.

In this paper the problem of sub-optimal trajectories is addressed using a viewpoint different from that of Ref. [4]. Assuming a three-dimensional rendezvous mission of a flat solar sail, the whole trajectory is calculated

using a single (fixed) value of the sail cone angle, while the clock angle δ is adjusted with continuity to minimize the total flight time. The resulting trajectory is therefore characterized by the fact that the axis perpendicular to the mean sail's plane rotates following a conical motion when compared to the direction of the local (solar) gravity vector. Therefore, the sail clock angle is the only control variable and the corresponding optimal constrained steering law is found by solving a variational problem through an indirect method. In this case, the mathematical approach to the optimization problem is derived from the model discussed in Refs. [3,5].

The advantage of using an approach in which the sail cone angle is maintained constant along the whole mission, turns into a substantial simplification of the spacecraft attitude control history. A second favorable point is in terms of reduction of the computational complexity of the optimal constrained trajectory. Indeed, assuming a flat solar sail model, the optimal constrained steering law for the sail clock angle δ can be expressed in a closed-form solution as a function of both the adjoint variables and the optical characteristics of the sail reflecting film [6,7], as is shown in the next section.

On the other hand, when the cone angle is not constrained to a fixed value, its optimal value is obtained, at each time instant, by minimizing the Hamiltonian function, with an increase of computational time of about an order of magnitude when compared to the sub-optimal model discussed in this paper.

Of course, the exclusion of the sail cone angle from the optimization process implies a reduction of the thrust effectiveness, that is, an increase of the total flight time.

To avoid ambiguities, in what follows a steering law in which both α and δ are varied to minimize the total flight time will be referred to as optimal or minimum time solution. A steering law in which α is maintained constant and δ is the only control variable will be referred to as optimal constrained or sub-optimal solution.

II. Mathematical Model

The equations of motion [8] of a (flat) solar sail-based spacecraft of mass m and area S, in a heliocentric inertial reference frame, can be expressed in terms of Modified Equinoctial Orbital Elements [9, 10] (MEOE) p, f, g, h, k and L as:

$$\dot{\boldsymbol{x}} = \mathbb{A}\boldsymbol{a}_s + \boldsymbol{c} \tag{1}$$

where $\boldsymbol{x} \triangleq \left[p, f, g, h, k, L\right]^{\mathrm{T}}$ is the state vector, and $\boldsymbol{a}_s$ is the solar sail propulsive acceleration vector. Assuming a non-perfect reflection model, the propulsive acceleration $\boldsymbol{a}_s$ at a distance r from the Sun is given by [11]

$$\boldsymbol{a}_s = \frac{\beta_\sigma \mu_\odot}{2r^2}\left(\hat{\boldsymbol{i}}_R \cdot \hat{\boldsymbol{n}}\right)\left[b_1\hat{\boldsymbol{i}}_R + \left(b_2\hat{\boldsymbol{i}}_R \cdot \hat{\boldsymbol{n}} + b_3\right)\hat{\boldsymbol{n}}\right] \tag{2}$$

where $\mu_\odot = 132712439935.5\,\mathrm{km}^2/\mathrm{s}^2$ is the Sun's gravitational parameter, $\beta_\sigma \triangleq \tilde{\sigma}/(m/S)$ is the dimensionless sail loading [12] with $\tilde{\sigma} \triangleq 1.539\,\mathrm{g/m}^2$, and r is the Sun-spacecraft distance. Also, b_1, b_2 and b_3 are the dimensionless force coefficients related to the optical properties of the reflective film [6,7], $\hat{\boldsymbol{i}}_R$ is the radial unit vector of an RTN reference frame (see Fig. 1), and $\hat{\boldsymbol{n}}$ is the unit vector normal to the sail plane in a direction such that $\hat{\boldsymbol{i}}_R \cdot \hat{\boldsymbol{n}} > 0$. For a sail with a reflective aluminum-coated front side and an emissive chromium-coated back side the force coefficients [11] are $b_1 = 0.1728$, $b_2 = 1.6544$ and $b_3 = -0.0109$. If the characteristic acceleration a_c (that is, the maximum sail propulsive acceleration at a Sun-spacecraft distance $r_\oplus \triangleq 1$ AU) is used as a performance metric [7], Eq. (2) becomes

$$\boldsymbol{a}_s = \frac{a_c(r_\oplus/r)^2}{b_1+b_2+b_3}\left(\hat{\boldsymbol{i}}_R \cdot \hat{\boldsymbol{n}}\right)\left[b_1\hat{\boldsymbol{i}}_R + \left(b_2\hat{\boldsymbol{i}}_R \cdot \hat{\boldsymbol{n}} + b_3\right)\hat{\boldsymbol{n}}\right] \tag{3}$$

Note that the sail cone angle $\alpha \in [0,90]\mathrm{deg}$ is given by

$$\alpha = \arccos\left(\hat{\boldsymbol{i}}_R \cdot \hat{\boldsymbol{n}}\right) \tag{4}$$

whereas the Sun-spacecraft distance r, required for the calculation of the solar sail propulsive acceleration (see Eq. (2)), can be written in terms of MEOE as

$$r = \frac{p}{1 + f\cos L + g\sin L} \tag{5}$$

In Eq. (1), $\mathbb{A} \in \mathbb{R}^{6\times 3}$ is a matrix in the form

$$\mathbb{A} \triangleq \sqrt{\frac{p}{\mu_\odot}}\begin{bmatrix} 0 & \dfrac{2p}{1+f\cos L+g\sin L} & 0 \\ \sin L & \dfrac{(2+f\cos L+g\sin L)\cos L+f}{1+f\cos L+g\sin L} & -\dfrac{g(h\sin L-k\cos L)}{1+f\cos L+g\sin L} \\ -\cos L & \dfrac{(2+f\cos L+g\sin L)\sin L+g}{1+f\cos L+g\sin L} & \dfrac{f(h\sin L-k\cos L)}{1+f\cos L+g\sin L} \\ 0 & 0 & \dfrac{(1+h^2+k^2)\cos L}{2(1+f\cos L+g\sin L)} \\ 0 & 0 & \dfrac{(1+h^2+k^2)\sin L}{2(1+f\cos L+g\sin L)} \\ 0 & 0 & \dfrac{h\sin L-k\cos L}{1+f\cos L+g\sin L} \end{bmatrix} \tag{6}$$

and vector $\boldsymbol{c} \in \mathbb{R}^{6\times 1}$ is defined as

$$\boldsymbol{c} \triangleq \left[0,0,0,0,0,\sqrt{\mu_{\odot} p}\left(\frac{1+f\cos L+g\sin L}{p} \right)^2 \right]^{\mathrm{T}} \tag{7}$$

where p is the semilatus rectum of the spacecraft osculating orbit. The transformations from MEOE to the classical orbital elements are

$$a = \frac{p}{1-f^2-g^2} \tag{8}$$

$$e = \sqrt{f^2+g^2} \tag{9}$$

$$i = 2\arctan\sqrt{h^2+k^2} \tag{10}$$

$$\sin\omega = gh - fk \;\;,\;\; \cos\omega = fh + gk \tag{11}$$

$$\sin\Omega = k \;\;,\;\; \cos\Omega = h \tag{12}$$

$$\nu = L - \Omega - \omega \tag{13}$$

where a is the semimajor axis, e is the eccentricity, i the orbital inclination, ω is the argument of perihelion, Ω is the longitude of the ascending node, and ν is the spacecraft's true anomaly.

A. Optimal control law with a constant cone angle

Assume that the initial (corresponding to $t_0 \triangleq 0$) spacecraft osculating orbit coincides with the Earth's (Keplerian) heliocentric orbit, viz.

$$\begin{aligned} &p(t_0) = 9.9878\times10^{-1}\,\mathrm{AU} \;\;,\;\; f(t_0) = -3.5778\times10^{-3} \;\;,\;\; g(t_0) = 1.5344\times10^{-2} \\ &h(t_0) = -1.5181\times10^{-5} \;\;,\;\; k(t_0) = 2.1250\times10^{-5} \end{aligned} \tag{14}$$

This scenario is representative of a solar sail deployment on a parabolic Earth escape trajectory with zero hyperbolic excess energy with respect to the planet.

The optimization problem consists of finding the minimum time trajectory (and the corresponding optimal control law) that transfers the spacecraft from the initial orbit to a final (target) planet's osculating orbit. This amounts to maximizing the performance index $J \triangleq -t_f$, where t_f is the total flight time, subject to the mission constraint

$$\hat{\boldsymbol{i}}_R \cdot \hat{\boldsymbol{n}} = \cos\alpha_{\mathrm{des}} \tag{15}$$

where $\alpha_{\mathrm{des}} \in [0,90)\,\mathrm{deg}$ is the fixed (that is, the design) value of the sail cone angle.

Using an indirect approach [13], from Eq. (1) the Hamiltonian of the system is

$$\mathcal{H} \triangleq (\mathbb{A}\boldsymbol{a}_s)\cdot\boldsymbol{\lambda} + \boldsymbol{c}\cdot\boldsymbol{\lambda} \tag{16}$$

where $\boldsymbol{\lambda} \triangleq [\lambda_p, \lambda_f, \lambda_g, \lambda_h, \lambda_k, \lambda_L]^{\mathrm{T}}$ is the adjoint vector whose time derivative is given by the Euler-Lagrange equation:

$$\dot{\boldsymbol{\lambda}} = -\frac{\partial \mathcal{H}}{\partial \boldsymbol{x}} \tag{17}$$

Recalling Eqs. (2) and (15), the optimal constrained sail attitude, that is, the value of the sail clock angle δ (see Fig. 1), is obtained from Pontryagin's maximum principle by maximizing, at any time, the function

$$\mathcal{H}' \triangleq (\mathbb{A}\hat{\boldsymbol{n}}) \cdot \boldsymbol{\lambda} \tag{18}$$

which coincides with that portion of the Hamiltonian $\mathcal{H}$ that explicitly depends on the control angle δ. According to Ref. [3], the optimal constrained value of the sail clock angle δ is given by

$$\sin\delta = A_{23}\lambda_f + A_{33}\lambda_g + A_{43}\lambda_h + A_{53}\lambda_k + A_{63}\lambda_L \tag{19}$$

$$\cos\delta = A_{32}\lambda_g + A_{12}\lambda_p + A_{22}\lambda_f \tag{20}$$

where A_{ij} is the entry (i, j) of the matrix $\mathbb{A}$, see Eq. (6). The optimal constrained direction of the unit vector $\hat{\boldsymbol{n}}$ is therefore

$$\hat{\boldsymbol{n}} = (\cos\alpha_{\mathrm{des}})\hat{\boldsymbol{i}}_R + (\sin\alpha_{\mathrm{des}}\cos\delta)\hat{\boldsymbol{i}}_T + (\sin\alpha_{\mathrm{des}}\sin\delta)\hat{\boldsymbol{i}}_N \tag{21}$$

where $\sin\delta$ and $\cos\delta$ are given by Eqs. (19) and (20), respectively, as a function of MEOE and adjoint variables.

The spacecraft motion is described by the six equations of motion (1) and the six Euler-Lagrange equations (17). This differential system must be completed with twelve suitable boundary conditions, five of these are shown in Eq.(14). Other five boundary conditions refer to the value of MEOE p, f, g, h and k at the (given) target orbit. Because the initial and final spacecraft angular position is left free, both the initial and final true longitude ($L(t_0)$ and $L(t_f)$) are outputs of the optimization process. The two last conditions, along with the equation required to evaluate the minimum flight time t_f, are obtained by enforcing the transversality conditions [14], viz.

$$\lambda_L(t_0) = 0 \quad , \quad \lambda_L(t_f) = 0 \quad , \quad \mathcal{H}(t_f) = 1 \tag{22}$$

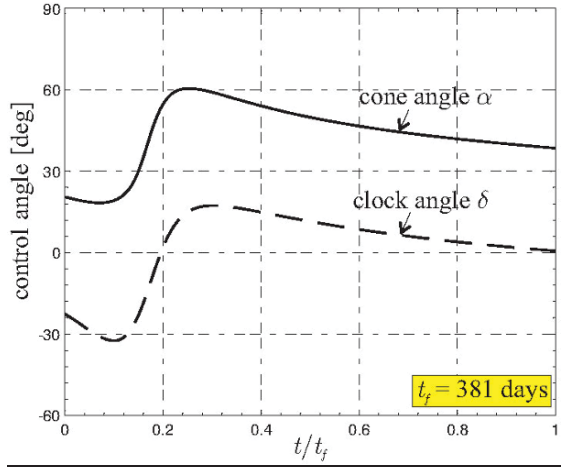
A set of heliocentric canonical units [15] has been used in the integration of the differential equations to reduce their numerical sensitivity. The equations of motion and the Euler-Lagrange equations have been integrated in double precision using a variable order Adams-Bashforth-Moulton solver scheme [16, 17] with absolute and relative errors of 10^{-12}.

III. Mission Applications

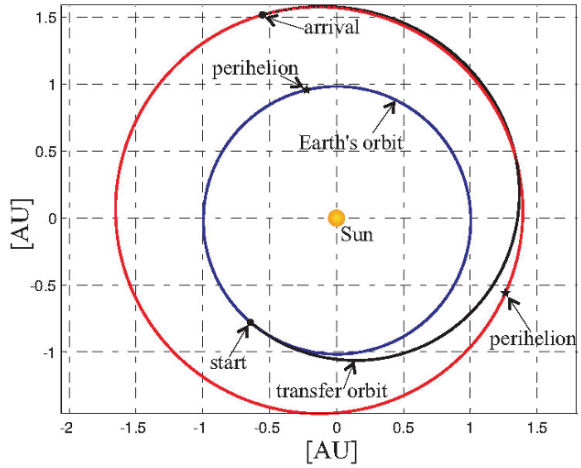
Two classical mission scenarios are now investigated, that is, an interplanetary transfer and a mission toward a near-Earth asteroid. Three-dimensional planetary orbits are assumed in all of the following simulations. A parametric approach, involving different values of sail cone angle $\alpha_{\rm des}$ and spacecraft characteristic acceleration a_c, is used to detect those mission scenarios in which the percentage increase of flight time is tolerable, and justifies the advantages introduced by the simplification of the attitude control law.

A. Interplanetary mission case

An Earth-Mars three-dimensional rendezvous mission is first discussed. Simulations show that an Earth-Mars transfer (without ephemeris constraints) for a flat solar sail, with a canonical value of characteristic acceleration $a_c = 1$ mm/s^2 and an optical force model [7], requires a minimum flight time of about $t_f \simeq 381$ days. Assuming a constant cone angle equal to $\alpha_{\rm des} = 45$ deg, the optimal constrained time history of the sail clock angle can be calculated using the approach discussed above. The resulting flight time is about 407 days, that is, only a 6% larger that the minimum value. Figures 2 and 3 show the ecliptic projection of the transfer trajectories and the time histories of the control variables for the two cases.

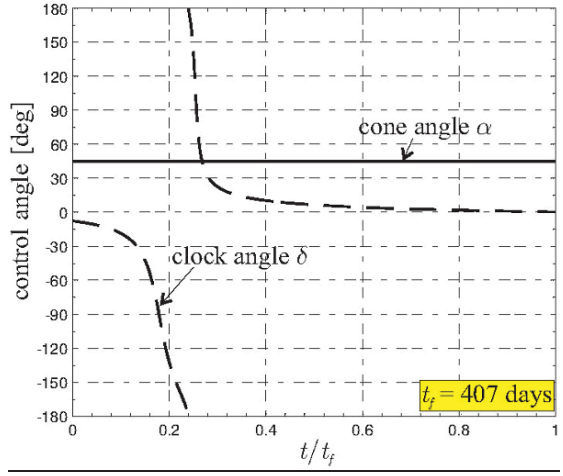


(a) Control angles.

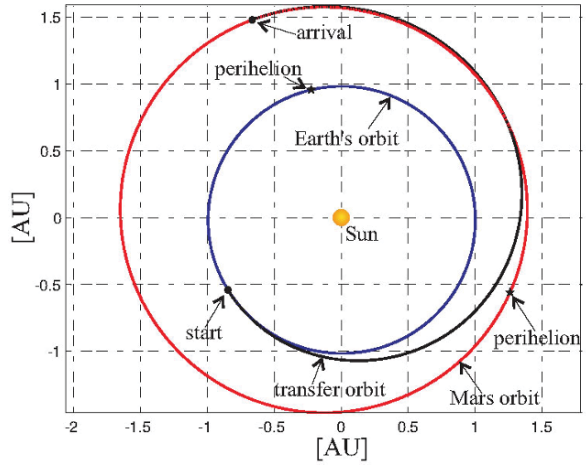


(b) Transfer trajectory (ecliptic projection).

Fig. 2 Optimal Earth-Mars transfer trajectory for a solar sail with a characteristic acceleration of 1 mm/s^2 **(optical force model).**



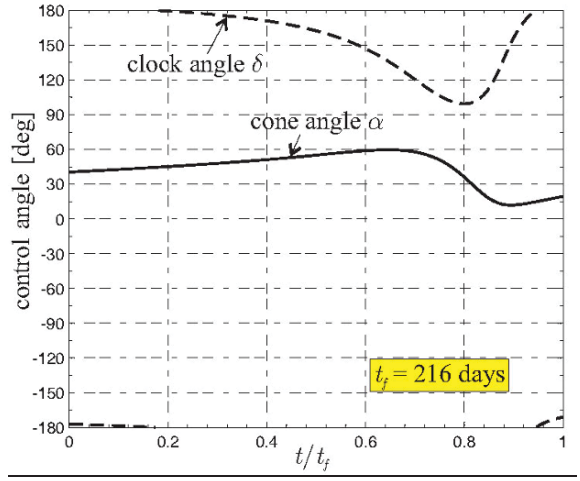
(a) Control angles.



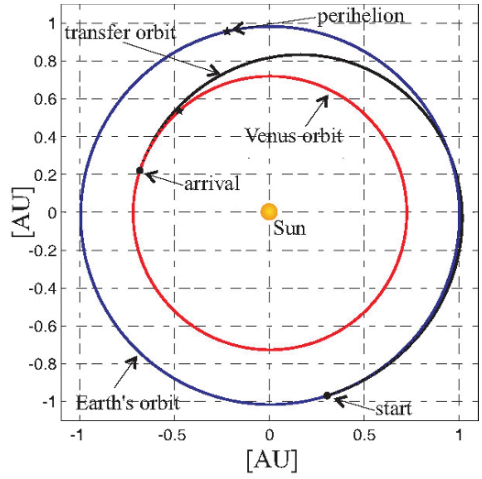
(b) Transfer trajectory (ecliptic projection).

Fig. 3 Optimal constrained Earth-Mars transfer trajectory for a solar sail with $\alpha = 45\,\text{deg}$ and a characteristic acceleration of $1\ \text{mm/s}^2$ (optical force model).

As a second example, consider a transfer trajectory toward Venus, using a flat solar sail with a characteristic acceleration $a_c = 1\ \text{mm/s}^2$, where the minimum time solution gives a transfer time of about 216 days. In this case, the characteristic acceleration being the same, the optimal constrained flight time with $\alpha_{\text{des}} = 45$ deg becomes 243 days, with an increase of 12.5% with respect to the global minimum. The corresponding control law and transfer trajectory (ecliptic projection) are illustrated in Figs. 4 and 5.

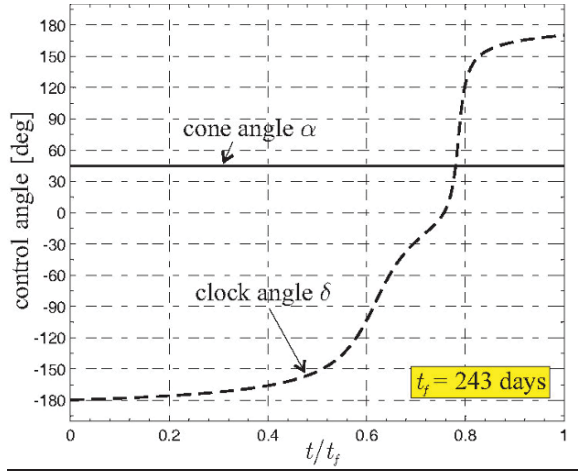


(a) Control angles.

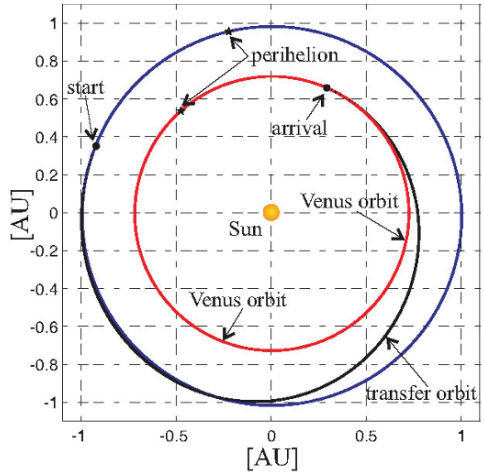


(b) Transfer trajectory (ecliptic projection).

Fig. 4 Optimal Earth-Venus transfer trajectory for a solar sail with a characteristic acceleration of $1\ \mathrm{mm/s^2}$ **(optical force model).**



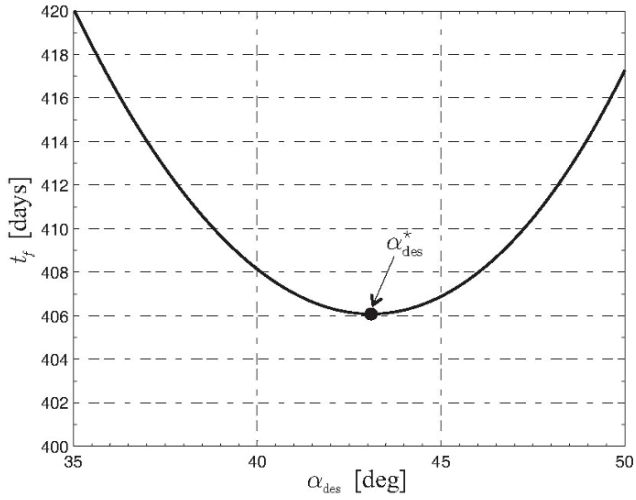
(a) Control angles.



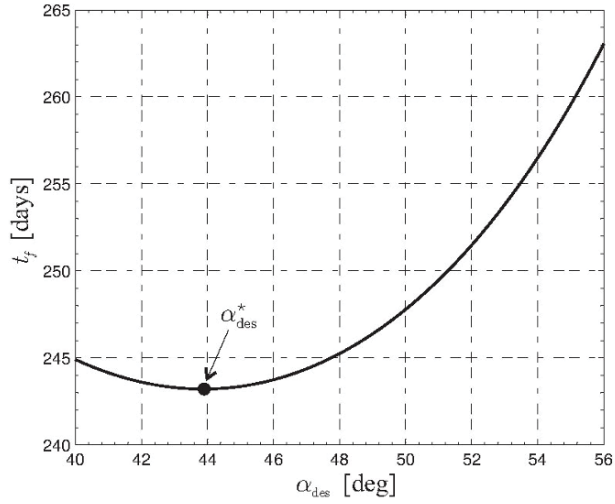
(b) Transfer trajectory (ecliptic projection).

Fig. 5 Optimal constrained Earth-Venus transfer trajectory for a solar sail with $\alpha = 45\,\mathrm{deg}$ and a characteristic acceleration of $1\ \mathrm{mm/s^2}$ (optical force model).

For a given value of a_c, the sail performance (in terms of flight time) is a function of the selected value of α_{des}. It is therefore interesting to investigate the sensitivity of t_f with respect to α_{des}. The results of this analysis are summarized in Fig. 6 for the two previous missions scenarios. In both cases it is seen that there exists an optimal cone angle (indicated with the symbol $\alpha^{\star}_{\mathrm{des}}$) that minimizes the flight time for a given spacecraft characteristic acceleration. Such an optimal constrained value of cone angle, however, changes when a_c is varied. This behavior is illustrated in Fig. 7, which shows that for both mission scenarios $\alpha^{\star}_{\mathrm{des}}$ increases when a_c is increased.

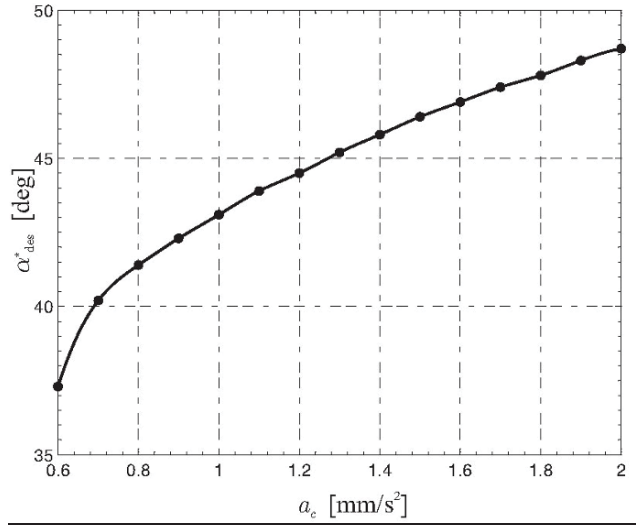


(a) Earth-Mars transfer.

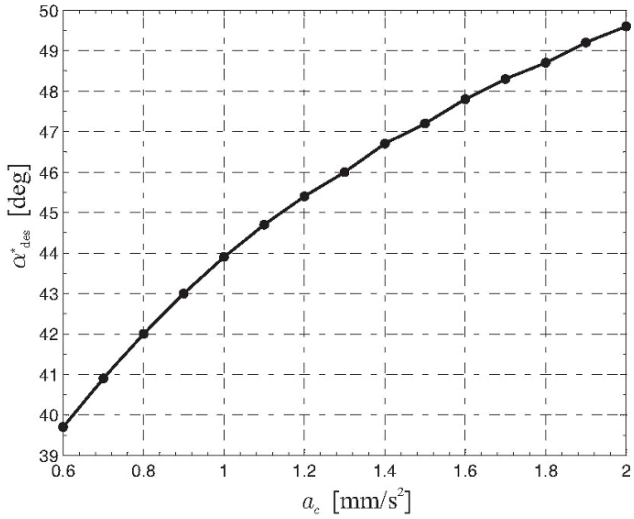


(b) Earth-Venus transfer.

Fig. 6 Optimal constrained transfer time as a function of the fixed cone angle α_{des} for a solar sail with $a_c = 1$ mm/s^2.



(a) Earth-Mars transfer.

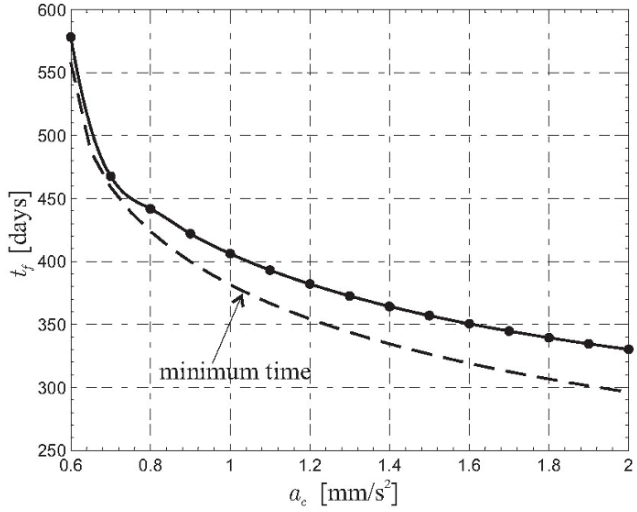


(b) Earth-Venus transfer.

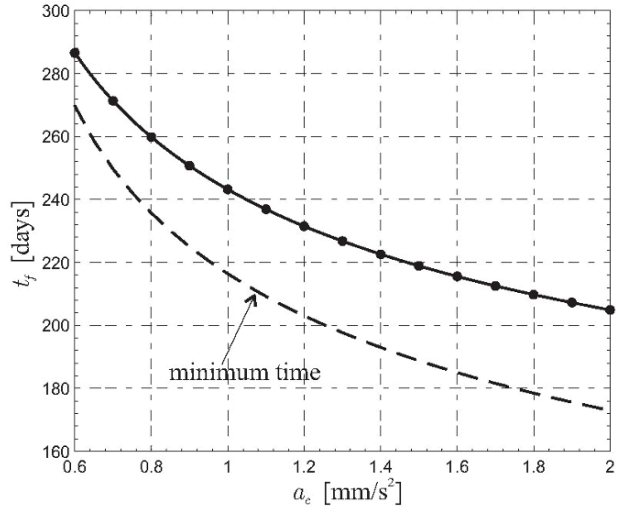
Fig. 7 Optimal constrained cone angle $\alpha^*_{\rm des}$ as a function of the characteristic acceleration a_c.

It is now possible to compare the performance of the optimal constrained solution with $\alpha = \alpha^*_{\rm des}$ and the minimum time case. In this analysis, for a given value of a_c, the value of the cone angle is taken from Fig. 7. The simulation results are shown in Fig. 8, which summarizes the performance of the proposed model. It can be verified that the solution with a constant value of cone angle guarantees a reasonable compromise between

complexity and performance of the resulting control law. In particular, the flight time increase associated to the use of the simplified control law is moderate in percentage terms for low-medium values of a_c.



(a) Earth-Mars transfer.



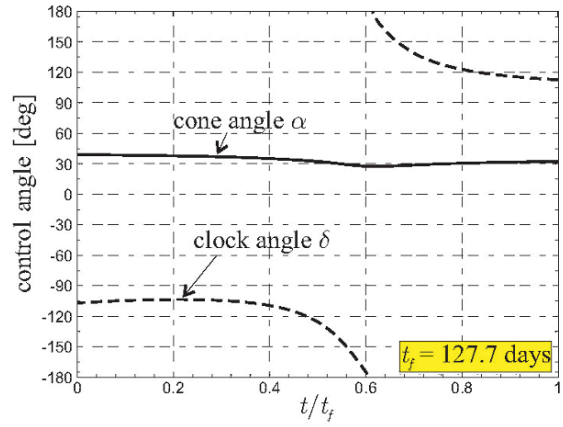
(b) Earth-Venus transfer.

Fig. 8 Transfer time when $\alpha = \alpha^{\star}_{\mathrm{des}}$ (solid line) as a function of the characteristic acceleration a_c.

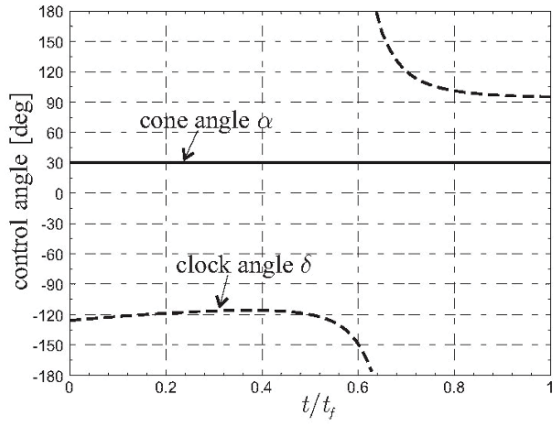
B. Mission to 99942 Apophis

Near-Earth asteroid 99942 Apophis is now used as an example to enable design of a mission to a potentially dangerous asteroid. In particular, rapid transfer trajectories are investigated, that is, missions whose transfer times are less than one terrestrial year. With this choice the results can be compared to the minimum time solutions discussed in Ref. [3].

Using a spacecraft characteristic acceleration $a_c = 1$ mm/s^2 and an optical force model [7], a minimum time rendezvous mission to asteroid Apophis requires a flight time of $t_f \simeq 127.7$ days [3]. Assuming a constant cone angle equal to $\alpha_{\text{des}} = 30$ deg, the resulting flight time is about 134 days, that is, only 5% larger that the minimum value. Figure 9 shows the time histories of the control variables for the two cases. The main reason for this small difference between minimum time and optimal constrained solutions is due to the fact that the minimum time trajectory requires a modest variation of the cone angle, see Fig. 9(a).



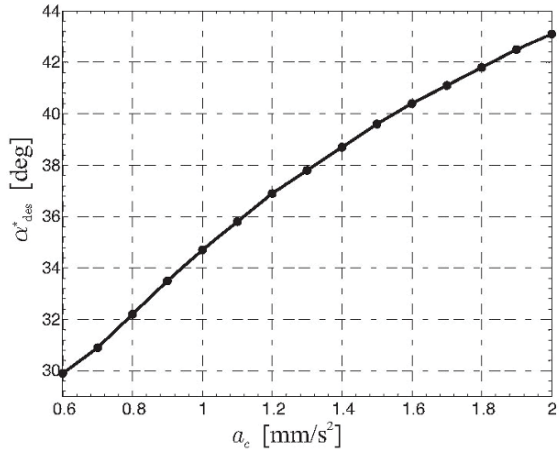
(a) Optimal case.



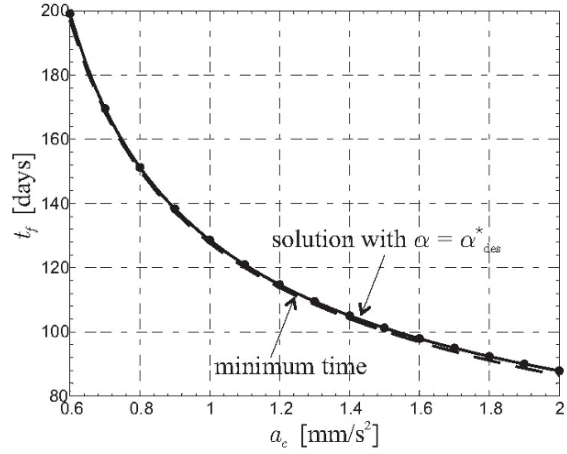
(b) Optimal constrained case.

Fig. 9 Comparison between optimal and optimal constrained ($\alpha = 30\,\mathrm{deg}$) transfer trajectory to Apophis for a solar sail with a characteristic acceleration of $a_c = 1$ mm/s^2 (optical force model).

Figure 10(a) shows the optimal value of cone angle as a function of a_c, and Fig. 10(b) summarizes the performance of the optimal constrained model (with $\alpha = \alpha^*_{\mathrm{des}}$). Notably, the optimal constrained case is nearly coincident with the minimum time solution for all values of a_c, with a maximum time increase of flight time always less than 2% only.



(a) Optimal constrained cone angle.



(b) Minimum flight time.

Fig. 10 Comparison of optimal and optimal constrained solar sail performance for a mission to Apophis as a function of a_c .

IV. Conclusions

The attitude control law of a solar sail can be substantially simplified by maintaining constant the sail cone angle along the whole spacecraft trajectory. Assuming a flat solar sail and an optical force model, the total flight time is minimized by varying the sail clock angle according to a closed-form expression that depends on the adjoint variables of the variational problem. The proposed solution is able to reduce remarkably the computational complexity of the optimal control law. Different mission scenarios have been discussed, and the simulation results show that the simplified steering law is able to guarantee moderate differences, in flight times, with respect to an optimal control law in which the sail cone angle varies with continuity. In particular, for a three-dimensional rendezvous mission (without ephemeris constraints) toward asteroid Apophis, the simplified steering law provides a solution that approximates the minimum time trajectory with a maximum error less than 2% in terms of total flight time.

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New Families of non-Keplerian Orbits: Solar Sail Motion over Cylinders and Spheres

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This paper presents new families of Sun-centered non-Keplerian orbits (NKOs), where the motion of a high-performance solar sail is confined to either a cylindrical or spherical surface. These orbits are found by investigating the geometrically constrained sail dynamics and imposing further constraints on the angular velocity and lightness number to generate pure solar sail trajectories. By considering the sail motion in the phase space of the problem, families of new NKOs are identified, and by investigating the oscillating behavior of the orbits, true periodic orbits are found. As extension to the well-known families of displaced NKOs, these three-dimensional NKOs generate a wealth of new solar sail orbits and novel sail applications.

I. Introduction

THE dynamics of two-body solar sail displaced non-Keplerian orbits (NKOs) is well-known in the literature [1, 2]: by considering the equations of motion of a solar sail in a rotating frame of reference, equilibrium solutions can be found that correspond to a circular orbit displaced away from the natural Keplerian orbit in an inertial frame. Through the choice for the angular velocity of the rotating frame, families of displaced NKOs can be identified. The dynamics, stability and control of these families have been investigated [3] as well as their abundant applications: to guarantee future geostationary slots for telecommunication, Earth observation and weather satellites the use of NKOs in the form of displaced geostationary orbits has been investigated [4, 5]; NKOs have been suggested to hover above Saturn's ring for high resolution in-situ observations [6]; and highly

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elliptic geocentric NKOs that lie in the ecliptic plane and rotate such that apogee is always located behind the Earth have been proposed for geomagnetic tail investigations [7].

However, as this paper will demonstrate, by moving away from these two-dimensional non-Keplerian orbits, and instead constraining the motion of the solar sail to be over a three-dimensional surface such as a cylinder or sphere, a wealth of new orbits is created that enable a range of novel space applications, including interplanetary communication, astronomical observations, solar physics, and many more.

The required solar sail technology to enable these orbits ranges from near-term to far-term, depending on the out-of-plane displacements considered. However, with advances in solar sail technology through JAXA's successful IKAROS mission [8] and NASA's NanoSail-D2 sail [9] and with more missions scheduled for the next few years, including NASA's Sunjammer mission [10] and The Planetary Society's LightSail mission [11], the far-term solutions get more and more within reach. Furthermore, similar far-term technology has been proposed in previous studies such as those for the Heliopause mission [12] and additional technologies such as perforated sails [13] may in the future enable the far-term solutions presented in this paper.

To introduce these novel cylindrical and spherical orbits, the structure of the paper will be as follows. The paper will start with the cylindrical case: the adopted reference frame is introduced, the equations of motion are derived and the control law required to keep the solar sail on a cylindrical surface are obtained. This control law is a function of the acceleration and for the cylindrical case, two types of solar sail accelerations will be considered: the first type of acceleration assumes only small out-of-plane displacements such that the pitch and cone angles can be assumed to be equal. This will allow for analytical investigations, while the second, true solar sail acceleration only allows for numerical investigations. For both types of acceleration, constraints on the in-plane angular velocity and the sail lightness number are introduced which result into a feasibility map of the cylindrically constrained orbits. The phase spaces of the orbits within this feasibility map allow to structure the obtained orbits into families and Poincaré maps are used to show the periodicity of the orbits. True periodic orbits are subsequently obtained and a set of typical, periodic orbits is presented. Very similar analyses are subsequently performed for the spherical case and at the end of the paper the conclusions will be drawn.

II. Cylindrically constrained NKOs

A. Reference frames and equations of motion

In order to derive the cylindrically constrained NKOs, a cylindrical coordinate frame is adopted with ρ the projected radius, z the out-of-plane displacement and θ the angle measured from the reference x-axis, see Fig.

1. Furthermore, an acceleration of magnitude a is assumed which lies in the (ρ, z)-plane only and is pitched with respect to the (x, y)-plane at an angle α. Under the assumption of a central gravitational force field with a gravitational potential $V = -\frac{\mu}{r}$, the equations of motion are given by:

$$\ddot{\rho} - \rho\dot{\theta}^2 = -\frac{\mu}{r^3}\rho + a\cos\alpha$$
$$\rho^2\ddot{\theta} + 2\rho\dot{\rho}\dot{\theta} = 0$$
$$\ddot{z} = -\frac{\mu}{r^3}z + a\sin\alpha$$

In order for the orbit to be constrained to a cylindrical surface, the following constraint is enforced:

$$\rho = \text{constant} \quad \rightarrow \quad \dot{\rho} = \ddot{\rho} = 0$$

which reduces the equations of motion to:

$$-\rho\dot{\theta}^2 = -\frac{\mu}{r^3}\rho + a\cos\alpha$$
$$\rho^2\ddot{\theta} = 0$$
$$\ddot{z} = -\frac{\mu}{r^3}z + a\sin\alpha$$

From these equations it becomes clear that the in-plane angular velocity $\dot{\theta} = \omega$ is constant. Furthermore, using $\bar{\omega}$ to denote the angular velocity of a Keplerian orbit with radius ρ:

$$\bar{\omega} = \sqrt{\frac{\mu}{\rho^3}}$$

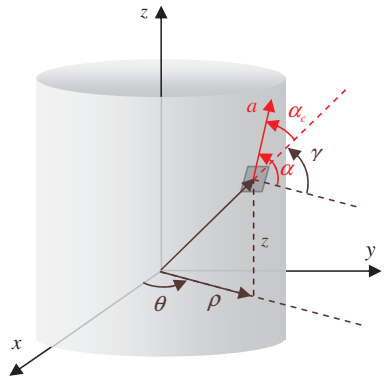

Fig. 1 Cylindrical coordinate frame definition.

and using $r = \sqrt{\rho^2 + z^2} = \rho\sqrt{1+\left(\frac{z}{\rho}\right)^2}$, the equations of motion can be further rewritten as:

$$\begin{aligned} \cos\alpha &= \frac{\rho\bar{\omega}^2}{a}\left(\left(1+\left(\frac{z}{\rho}\right)^2\right)^{-\frac{3}{2}} - \left(\frac{\omega}{\bar{\omega}}\right)^2\right) \\ \ddot{z} &= -z\bar{\omega}^2\left(1+\left(\frac{z}{\rho}\right)^2\right)^{-\frac{3}{2}} + \text{sign}(z)\, a\sqrt{1-\cos^2\alpha} \end{aligned} \tag{1}$$

The first equation in Eq. (1) gives the control law required to stay on a cylindrical surface. Note that, due to the symmetry of the problem, the $\text{sign}(z)$-term is included to mirror the acceleration in the (x, y)-plane upon crossing.

As mentioned in the introduction of this paper, two different solar sail accelerations will be considered:

1. A true solar sail acceleration law:

$$a = \beta\frac{\mu}{r^2}\cos^2\left(\alpha - \tan^{-1}\left(\frac{z}{\rho}\right)\right)$$

with β the solar sail lightness number [13] and the term $\alpha - \tan^{-1}\left(\frac{z}{\rho}\right)$ equal to the cone angle, α_c, see Fig. 1.

2. A solar sail law where the cone angle is approximated by the pitch angle under the assumption that the out-of-plane displacement is small: $\alpha_c = \alpha - \tan^{-1}\left(\frac{z}{\rho}\right) \approx \alpha$.

$$a = \beta\frac{\mu}{r^2}\cos^2\alpha$$

The second type of acceleration, for small-displacements, which allows for a range of analytical investigations, will be investigated first.

B. Small displacements solar sail acceleration law

The small-displacements solar sail acceleration law can be written as:

$$a = \beta\frac{\mu}{r^2}\cos^2\alpha = \frac{\beta\bar{\omega}^2\rho}{1+\left(\frac{z}{\rho}\right)^2}\cos^2\alpha$$

Substitution into Eq. (1) then gives:

$$\cos\alpha = \sqrt[3]{\frac{1}{\beta}\left(\left(1+\left(\frac{z}{\rho}\right)^2\right)^{-\frac{3}{2}} - \left(\frac{\omega}{\bar{\omega}}\right)^2\right)\left(1+\left(\frac{z}{\rho}\right)^2\right)}$$

$$\ddot{z} = -z\bar{\omega}^2\left(1+\left(\frac{z}{\rho}\right)^2\right)^{-\frac{3}{2}} + \mathrm{sign}(z)\frac{\beta\bar{\omega}^2\rho}{1+\left(\frac{z}{\rho}\right)^2}\cos^2\alpha\sqrt{1-\cos^2\alpha} \tag{2}$$

The most well-known cylindrically constrained orbit is the two-dimensional (2D) NKO as described in Ref. [1, 3, 13]. The sail lightness number required to maintain this 2D NKO can be obtained by setting $\ddot{z}=0$ and substituting $\cos\alpha$:

$$\beta_{2D} = \left(\frac{\left(\frac{z}{\rho}\right)^2\left(1+\left(\frac{z}{\rho}\right)^2\right)^{-1} + \left(\left(\left(1+\left(\frac{z}{\rho}\right)^2\right)^{-\frac{3}{2}} - \left(\frac{\omega}{\bar{\omega}}\right)^2\right)\left(1+\left(\frac{z}{\rho}\right)^2\right)\right)^2}{\left(\left(\left(1+\left(\frac{z}{\rho}\right)^2\right)^{-\frac{3}{2}} - \left(\frac{\omega}{\bar{\omega}}\right)^2\right)\left(1+\left(\frac{z}{\rho}\right)^2\right)\right)^{\frac{4}{3}}}\right)^{\frac{3}{2}} \tag{3}$$

Contours of equal β_{2D} as well as the required direction of the acceleration is provided in Fig. 2a for $\omega = 1$. Note that the results in Fig. 2, as well as all other results for the cylindrical case, are made dimensionless by setting $\rho=1$ and $\bar{\omega}=1$. Furthermore, note that the contours differ from the well-known results in Ref. [2] due to the difference in definition for the acceleration: Ref. [2] assumes a true solar sail law, while here the cone angle is approximated by the pitch angle. The correctness of the results in Fig. 2a can thus be verified by calculating the required sail lightness number for a true solar sail, $\beta_{2D,s}$, starting from β_{2D} in Eq. (3):

$$\beta_{2D,s} = \frac{\beta_{2D}\cos^2\alpha}{\cos^2\alpha_c} \tag{4}$$

Contours for equal $\beta_{2D,s}$ are provided in Fig. 2b which correspond to the results in Ref. [2] and therefore confirm the correctness of Eq. (3).

When using a solar sail, it is well-known that the following constraint on the solar sail attitude has to be taken into account since the sail cannot generate a solar sail component in the direction of the Sun:

$$\cos\alpha \approx \cos\alpha_c \geq 0 \tag{5}$$

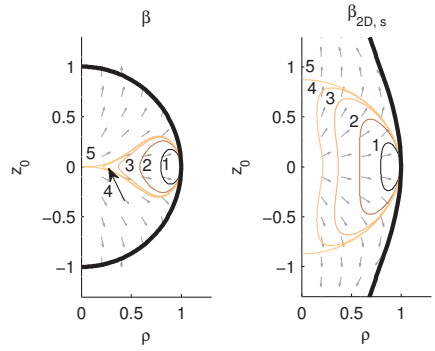

Fig. 2 Contours of equal lightness number to maintain a 2D NKO for $\bar{\omega}$= 1. 1) 0.5; 2) 0.8; 3) 0.95; 4) 0.99; 5) 1.0. Left: sail lightness number for a small displacements solar sail acceleration law. Right: sail lightness number for a true solar sail acceleration law.

This constraint is represented in Fig. 2 by thick black lines. The figure shows that, for small displacements, this line is similar for both types of acceleration, but unnecessarily restricts the feasibility region for the small displacements solar sail acceleration law for larger displacements.

By deviating from β_{2D}, either in positive or negative direction, and keeping the control law as given in Eq. (2), additional, and three dimensional, cylindrical orbits can be found. To obtain *feasible* orbits, not only the lower limit on $\cos\alpha$ in Eq. (5) needs to be imposed, but also an upper limit: $0 \le \cos\alpha \le 1$. The lower limit, i.e. $\cos\alpha \ge 0$ can be satisfied by defining the following maximum value for the in-plane angular velocity:

$$\left(1+\left(\frac{z}{\rho}\right)^2\right)^{-\frac{3}{2}} \ge \left(\frac{\omega}{\bar{\omega}}\right)^2 \quad \rightarrow \quad \omega \le \omega_{\max} = \bar{\omega}\left(1+\left(\frac{z}{\rho}\right)^2\right)^{-\frac{3}{4}} = f(z) \tag{6}$$

Since $\omega_{\max}$ depends on the value for z, which changes during the orbit, the minimum value for $f(z)$ needs to be found to ensure that the value chosen for ω satisfies Eq. (6) throughout the orbit. The minimum value for $f(z)$ occurs for the maximum value for z, $z_{\max}$, which results in the following final constraint on ω to satisfy $\cos\alpha \ge 0$:

$$\omega \le \omega_{\max} = \bar{\omega}\left(1+\left(\frac{z_{\max}}{\rho}\right)^2\right)^{-\frac{3}{4}} \tag{7}$$

Note that, for a 2D NKO, where $z_{\max} = z_0$, Eq. (7) gives the thick black line in Fig. 2a and can also be used to determine the maximum allowable value for ω for a particular value for z_0. As will be seen later, $z_{\max}$ is

unknown for orbits that lie completely above the 2D NKO and can only be determined through an integration of the equations of motion. However, in case the orbits lies entirely below the 2D NKO, $z_{\max}$ is again equal to z_0.

The second inequality constraint, $\cos\alpha \le 1$ can be satisfied through a constraint on β:

$$\beta \ge \beta_{\min} = \left(\left(1 + \left(\frac{z}{\rho} \right)^2 \right)^{-\frac{3}{2}} - \left(\frac{\omega}{\bar{\omega}} \right)^2 \right) \left(1 + \left(\frac{z}{\rho} \right)^2 \right) = g(z) \tag{8}$$

A similar reasoning as for the constraint on ω can be applied: since the minimum required value for β depends on the value for z, which changes during the orbit, the maximum value for $g(z)$ needs to be found to ensure that the value chosen for β satisfies Eq. (8) throughout the orbit. The maximum value for $g(z)$ occurs for the minimum value for z, $z_{\min}$. The final constraint to satisfy $\cos\alpha \le 1$ thus equals:

$$\beta \ge \beta_{\min} = \left(\left(1 + \left(\frac{z_{\min}}{\rho} \right)^2 \right)^{-\frac{3}{2}} - \left(\frac{\omega}{\bar{\omega}} \right)^2 \right) \left(1 + \left(\frac{z_{\min}}{\rho} \right)^2 \right) \tag{9}$$

Again, as will be seen later, $z_{\min}$ is unknown for orbits that lie completely below the 2D NKO and can only be determined through an integration of the equations of motion. However, in case the orbits lies entirely above the 2D NKO, $z_{\min}$ is given by z_0. Note that β_{2D} is always larger than $\beta_{\min}$ as the latter can be written as:

$$\beta_{2D} = \left(\frac{\left(\frac{z_0}{\rho} \right)^2 \left(1 + \left(\frac{z_0}{\rho} \right)^2 \right)^{-1} + \beta_{\min}^2}{\beta_{\min}^{\frac{4}{3}}} \right)^{\frac{3}{2}} = \beta_{\min}^{-2} \left(\frac{z}{\rho} \right)^3 \left(1 + \left(\frac{z}{\rho} \right)^2 \right)^{-\frac{3}{2}} + \beta_{\min} \ge \beta_{\min}$$

An overview of the achievable orbits for a particular value of the initial out-of-plane displacement, z_0, is provided in Fig. 3, which shows the phase spaces for a range of values for ω (along the z-axis) and percentage deviation from β_{2D}, $\Delta\beta$ (through the used color scheme), for the case of $\rho = 1$ and $z_0 = 0.5$. The step size in $\Delta\beta$ between two phase spaces belonging to the same ω-value is 5 percent.

In Fig. 3, black dots are used on a gray transparent surface to represent the 2D NKO from which the cylindrically constrained orbits bifurcate. As indicated below Eq. (7), a maximum allowable value for ω exists for the 2D NKO. For $z_0 =$ 0.5, this value is $\omega_{\max} = 0.846$ and is indicated in Fig. 3 with a red dot.

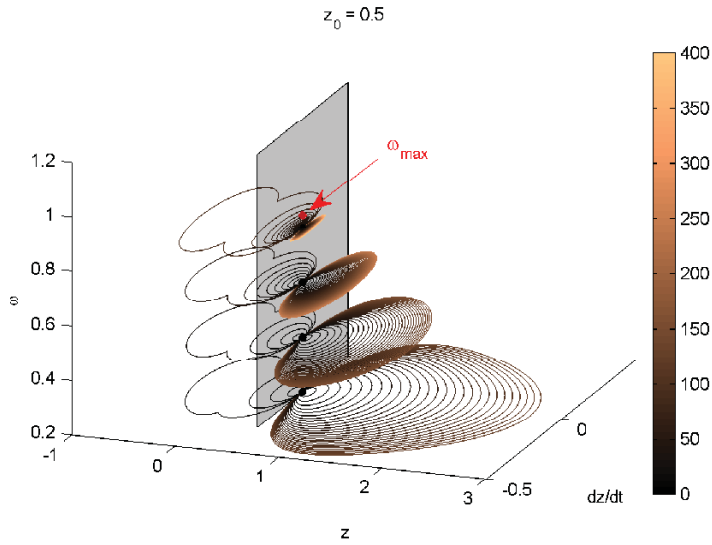

Fig. 3 Phase spaces for cylindrically constrained NKOs with a small-displacements solar sail acceleration law and for ρ = 1 and z_0 = 0.5. The colors indicate the percentage deviation from β_{2D} with a step size of 5% (negative deviation to the left of the gray transparent plane; positive deviation to the right of the gray transparent plane).

The phase spaces to the right of the transparent surface are created by *increasing* β with respect to β_{2D}, while phase spaces to the left of this surface are generated by *decreasing* β with respect to β_{2D}. Inspecting the phase spaces it becomes clear that, for $\beta > \beta_{2D}$, the spacecraft will move in a confined band *above* z_0 ("north orbits") such that $z_0 \le z \le z_{\max}$, while for $\beta < \beta_{2D}$, the spacecraft will move in a confined band *below* z_0 ("south orbits") such that $z_{\min} \le z \le z_0$. For some cases, β can be decreased to such extent that the phase space crosses the $z = 0$ plane, after which the $\text{sign}(z)$-term in Eq. (2) mirrors the acceleration vector in the (x, y)-plane and the orbit immediately changes into an orbit that oscillates around the (x, y)-plane, i.e. $-z_0 \le z \le z_0$. Decreasing β further does not change this behaviour, i.e. the orbit will always oscillate around the (x, y)-plane and will not go beyond $-z_0$.

Additional observations that can be made from Fig. 3 are as follows: firstly, the limitations introduced by the constraints in Eqs. (7) and (9) are clear both in terms of the possible deviations from β_{2D} and the attainable values for ω. This is further illustrated in Fig. 4, which shows the feasibility region in terms of achievable

values for ω and allowable percentage deviation from β_{2D} for the same case as presented in Fig. 3, i.e. $\rho = 1$ and $z_0 = 0.5$. Furthermore, for a particular value for ω, the larger $\Delta\beta$, the wider the band covered on the cylinder. Comparing the phase spaces between different values for ω shows that, the larger ω, the narrower is the band for a particular value for β. This means that for large values for ω combined with large deviations from β_{2D}, the sail does not move away much from the 2D NKO. For south orbits (i.e. $\beta < \beta_{2D}$), this can result in significant savings in the lightness number required compared to a 2D NKO while only introducing a slight penalty on the constancy in $z-$direction, providing a potential alternative to the 2D NKO.

Since Fig. 3 and Fig. 4 only present the cylindrically orbits in an abstract way, Fig. 5 is included to show some of the types of trajectories that are possible. In counterclockwise direction and starting from the top left plot in Fig. 5a, the following cases are represented: an orbit oscillating around the (x, y)-plane, an orbit that creates a band south of the corresponding 2D NKO, and finally two orbits that create a band north of the corresponding 2D NKO. From the plots, especially the last one, it becomes clear that some of the trajectories are (quasi-) periodic, which is confirmed by the Poincaré maps of these trajectories in Fig. 5b, which shows the state-vector after each full revolution. Although the required lightness numbers for the orbits in Fig. 5 are significant, it must be noted that these can be reduced substantially by decreasing the out-of-plane displacement. However, for illustration purposes and clear interpretation of the results, an out-of-plane displacement of 0.5 will be used throughout the paper. Furthermore, the introduction of the paper already hinted at ways to achieve these high solar sail performances, e.g. through the use of perforated sail.

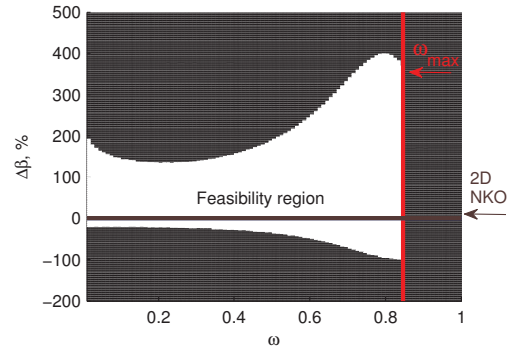

Fig. 4 Feasibility region (in white) in terms of in-plane angular velocity, ω, and percentage deviation from β_{2D}, $\Delta\beta$, for cylindrically constrained NKOs with a small-displacements solar sail acceleration law and for $\rho = 1$ and $z_0 = 0.5$.

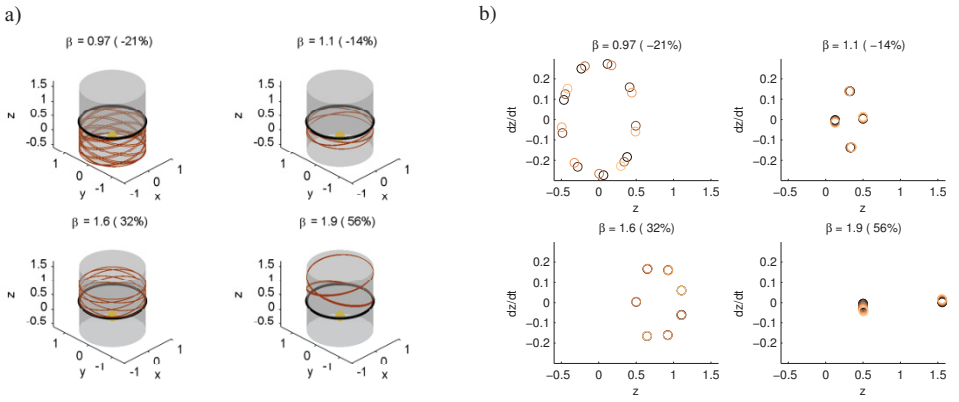

Fig. 5 Examples of cylindrically constrained NKOs with a small-displacement solar sail acceleration law for $\rho = 1$, $z_0 = 0.5$ and $\omega = 0.2$ and 20 orbital revolutions. a) Trajectories. b) Poincaré maps with the revolution number increasing from dark to light colors.

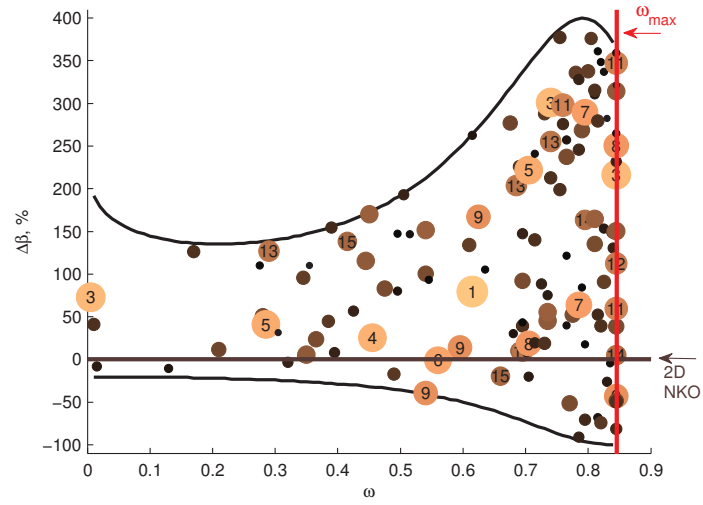

Fig. 6 Periodicity of the set of feasible cylindrically constrained NKOs with a small displacements solar sail acceleration law and for $\rho = 1$ and $z_0 = 0.5$, where orbits with a periodicity of up to 30 orbital revolutions are included and orbits with periods of up to 15 are indicated with a numerical value. The smaller the marker and the lighter the color, the shorter the period.

True periodic orbits can be found numerically by integrating the equations of motion and truncating the integration when the orbit returns back to the initial conditions in terms of z - and $\dot{z}$ -values. The time at which this occurs is expressed as a fraction of a full revolution and as such gives an insight in the periodicity. The results are presented in Fig. 6, which are obtained through a fine grid search within the feasibility region. Only orbits with a period of less than 30 revolutions are provided and those with a period of up to 15 revolutions are indicated with the actual period. Furthermore, the smaller and darker the marker, the larger the period. The figure shows that a combination of ω and $\Delta\beta$ exists for which the orbital period is one revolution. This orbit, together with some additional periodic orbits are illustrated in Fig. 7 together with their corresponding Poincaré maps.

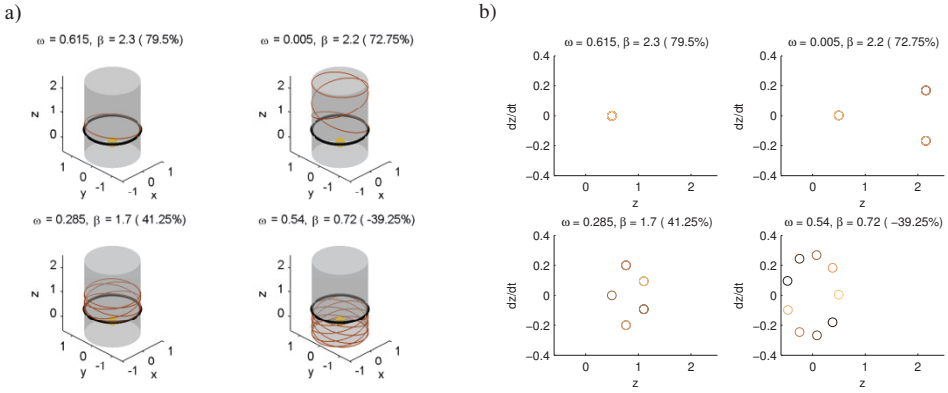

Fig. 7 Periodic cylindrically constrained NKOs for a small-displacement solar sail acceleration law for $\rho = 1$ and $z_0 = 0.5$. a) Trajectories. b) Poincaré maps with the revolution number increasing from dark to light colors.

C. Solar sail acceleration law

When replacing the small displacements solar sail acceleration law with the true solar sail acceleration law, the acceleration to be used in Eq. (1) is:

$$a = \beta \frac{\mu}{r^2} \cos^2\left(\alpha - \tan^{-1}\left(\frac{\rho}{z}\right)\right) = \beta \frac{\bar{\omega}^2 \rho}{1 + \left(\frac{z}{\rho}\right)^2} \cos^2\left(\alpha - \tan^{-1}\left(\frac{z}{\rho}\right)\right)$$

The required control law for the pitch angle α to remain on a cylindrical surface can then be derived from the first equation in Eq. (1) and is implicitly given through:

$$\cos^2\left(\alpha-\tan^{-1}\left(\frac{z}{\rho}\right)\right)\cos\alpha=\frac{1}{\beta}\left(1+\left(\frac{z}{\rho}\right)^2\right)\left(\left(1+\left(\frac{z}{\rho}\right)^2\right)^{-\frac{3}{2}}-\left(\frac{\omega}{\bar{\omega}}\right)^2\right) \tag{10}$$

To solve Eq. (10) for $\cos\alpha$, the roots of the following sixth order polynomial need to be found:

$$\begin{aligned}\left(1+2c_1^{\,2}+c_1^{\,4}\right)x^6+\left(-2c_1^{\,2}-2c_1^{\,4}\right)x^4+\left(-2c_2+2c_1^{\,4}c_2\right)x^3+c_1^{\,4}x^2+\\ \left(-2c_1^{\,2}c_2-2c_1^{\,4}c_2\right)x+\left(c_2^{\,2}+2c_1^{\,2}c_2^{\,2}+c_1^{\,4}c_2^{\,2}\right)=0\end{aligned} \tag{11}$$

with

$$c_1=\frac{z}{\rho},\qquad c_2=\frac{1}{\beta}\left(\left(1+\left(\frac{z}{\rho}\right)^2\right)^{-\frac{3}{2}}-\left(\frac{\omega}{\bar{\omega}}\right)^2\right)\left(1+\left(\frac{z}{\rho}\right)^2\right)$$

and

$$x=\cos\alpha$$

Although this polynomial has six roots, it appeared that only one of those six roots is a real solution as well as a true solution to Eq. (10). Since no explicit expression exists for $\cos\alpha$, also no explicit expression can be found for β_{2D} from Eq. (1). However, from the literature it is known that β_{2D} is given through [13]:

$$\beta_{2D}=\left(1+\left(\frac{z}{\rho}\right)^2\right)^{\frac{1}{2}}\frac{\left(\left(\frac{z}{\rho}\right)^2+\left(1-\left(\frac{\omega}{\bar{\omega}}\right)^2\right)^2\right)^{\frac{3}{2}}}{\left(\left(\frac{z}{\rho}\right)^2+\left(1-\left(\frac{\omega}{\bar{\omega}}\right)^2\right)\right)^2} \tag{12}$$

The lack in an expression for $\cos\alpha$ furthermore implies that also the constraints on the in-plane angular velocity and allowed deviation from β_{2D} cannot be derived. They can only be enforced within the integration of the equations of motion as soon as the constraint $0\le\cos\alpha_s\le 1$ is violated. When doing so, the phase spaces as depicted in Fig. 8 can be obtained for $\rho=1$ and $z_0=0.5$. Comparing Fig. 8 with Fig. 3 immediately shows the effect of the real solar sail acceleration law as it significantly limits the orbits that are achievable both in terms of the in-plane angular velocity and the deviation from β_{2D}. This reduction in the feasibility region becomes even clearer from Fig. 9. However, the behavior of the orbits is still the same, i.e. in terms of north- and south orbits for increasing and decreasing β with respect to β_{2D}, respectively, and the switch to orbits that oscillate around the (x,y)-plane for a large enough reduction in β. Furthermore, again a fairly large region for negative values of $\Delta\beta$ exists, which again suggests the potential of the cylindrical orbits for decreasing the sail technology

requirements with respect to a 2D NKO at the cost of a slight penalty on the constancy in z-direction. This can among others be seen from the results for the periodicity analysis which are presented in Fig. 10 and Fig. 11. For $\omega =$ 0.725 an almost 10 percent reduction in the sail lightness number provides an orbit that has a period of 4 revolutions while almost coinciding with the original 2D NKO, see the top right figure in Fig. 11a.

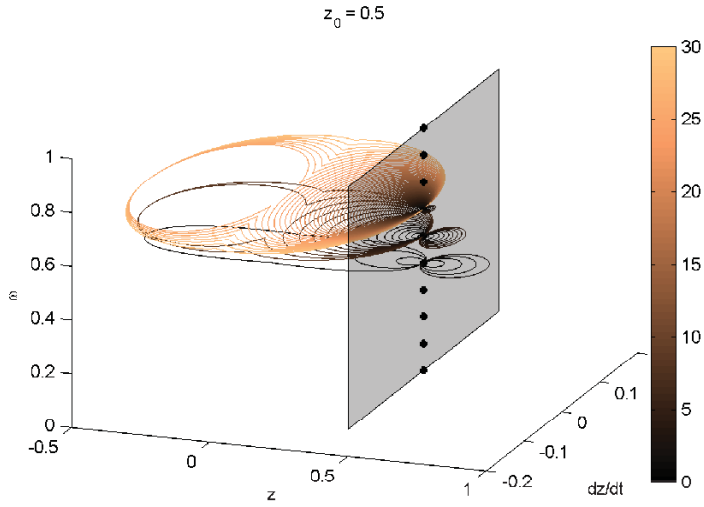

Fig. 8 Phase spaces for cylindrically constrained NKOs with a solar sail acceleration law and for ρ = 1 and z_0 = 0.5. The colors indicate the percentage deviation from β_{2D} with a step size of 1% (negative deviation to the left of the gray transparent plane; positive deviation to the right of the gray transparent plane).

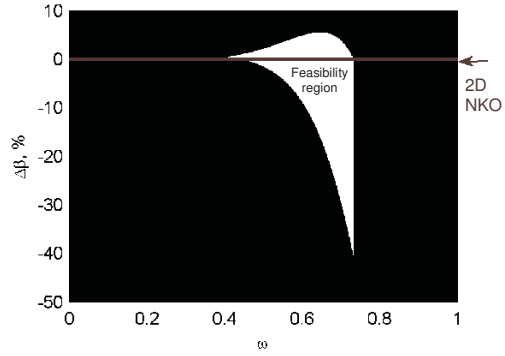

Fig. 9 Feasibility region (in white) in terms of in-plane angular velocity, ω, and deviation from β_{2D} for cylindrically constrained NKOs with a solar sail acceleration law and for ρ = 1 and z_0 = 0.5.

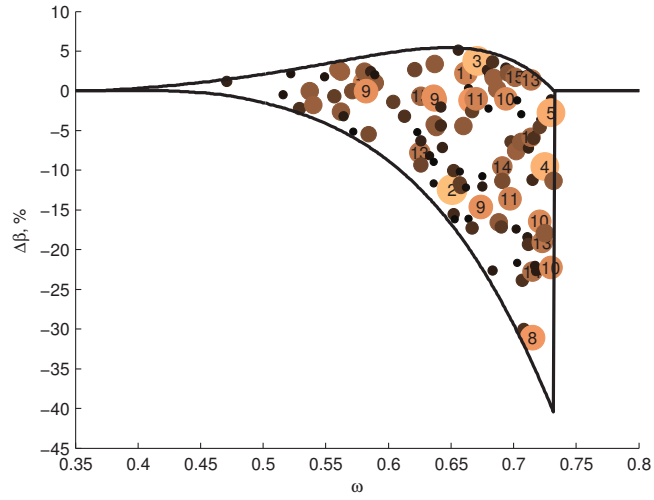

Fig. 10 Periodicity of the set of feasible cylindrically constrained NKOs with a solar sail acceleration law and for $\rho = 1$ and $z_0 = 0.5$, where orbits with a periodicity of up to 30 orbital revolutions are included and orbits with periods of up to 15 are indicated with a numerical value. The smaller the marker and the lighter the color, the shorter the period.

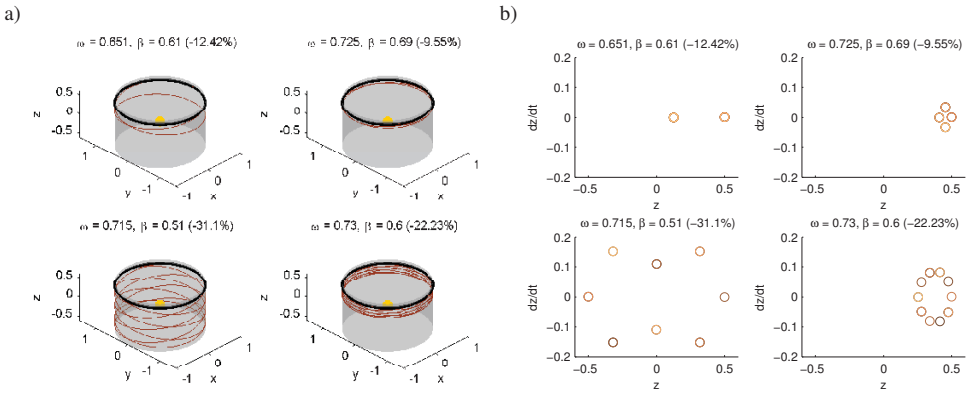

Fig. 11 Periodic cylindrically constrained NKOs for a solar sail acceleration law for $\rho = 1$ and $z_0 = 0.5$. a) Trajectories. b) Poincaré maps with the revolution number increasing from dark to light colors.

III. Spherically constrained NKOs

In order to obtain spherically constrained NKOs, a spherical cylindrical coordinate frame is adopted with r the projected radius, θ the in-plane angle between the x-axis and the projected radius and ϕ the out-of-plane angle, see Fig. 12. Again, assuming an acceleration in the (r,ϕ)-plane only, pitched at an angle α (which for the spherical orbits is defined with respect to the radius rather than with respect to the projected radius), the equations of motion for a central gravitational force field are given by:

$$\begin{aligned} \ddot{r} - r\dot{\theta}^2\cos^2\phi - r\dot{\phi}^2 &= -\frac{\mu}{r^2} + a\cos\alpha \\ r\cos\phi\ddot{\theta} + 2\dot{r}\dot{\theta}\cos\phi - 2r\dot{\theta}\dot{\phi}\sin\phi &= 0 \\ r\ddot{\phi} + 2\dot{r}\dot{\phi} + r\dot{\theta}^2\sin\phi\cos\phi &= a\sin\alpha \end{aligned} \tag{13}$$

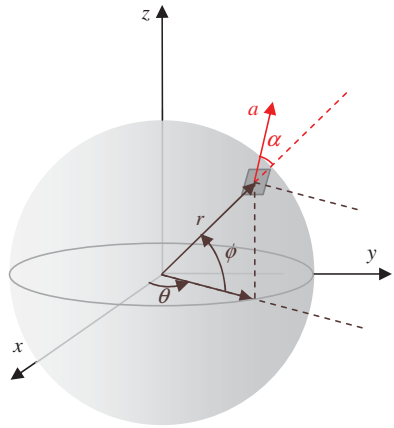

Fig. 12 Spherical coordinate frame definition.

The geometrical constraint required to restrict the motion to a spherical surface is:

$$r = \text{constant} \quad \rightarrow \quad \dot{r} = \ddot{r} = 0$$

Substituting these conditions into the equations of motion, Eq. (13) reduces to:

$$\begin{aligned} -\dot{\theta}^2\cos^2\phi - \dot{\phi}^2 &= -\frac{\mu}{r^3} + \frac{a}{r}\cos\alpha \\ \ddot{\theta} &= 2\dot{\theta}\dot{\phi}\tan\phi \\ \ddot{\phi} + \dot{\theta}^2\sin\phi\cos\phi &= \frac{a}{r}\sin\alpha \end{aligned} \tag{14}$$

Note that, contrary to the cylindrical case, the in-plane angular velocity, $\dot{\theta}$, is not constant and can therefore not be replaced by the notation ω as done for the cylindrical case. Furthermore, the parameter $\bar{\omega}$ will now denote the angular velocity of a Keplerian orbit with radius r:

$$\bar{\omega} = \sqrt{\frac{\mu}{r^3}}$$

Finally, since for the spherical case the pitch angle α is identical to the cone angle, only a true solar sail acceleration law will be considered as this will immediately allow for some analytical investigations:

$$a = \beta \frac{\mu}{r^2} \cos^2 \alpha = \beta \bar{\omega}^2 r \cos^2 \alpha \tag{15}$$

Substituting the above, the equations of motion in Eq. (14) can be further reduced to:

$$\begin{aligned} \cos\alpha &= \sqrt[3]{\frac{1}{\beta}\left(1 - \frac{1}{\bar{\omega}^2}\left(\dot{\theta}^2 \cos^2\phi + \dot{\phi}^2\right)\right)} \\ \ddot{\theta} &= 2\dot{\theta}\dot{\phi}\tan\phi \\ \ddot{\phi} &= -\dot{\theta}^2 \sin\phi\cos\phi + \text{sign}(\phi)\beta\bar{\omega}^2\cos^2\alpha\sqrt{1-\cos^2\alpha} \end{aligned} \tag{16}$$

As for the cylindrical case, the analysis for the spherically constrained NKOs will start from the 2D NKO, which can be derived from Eq. (16) by setting $\phi = \text{constant} \rightarrow \dot{\phi} = \ddot{\phi} = 0$, which gives:

$$\beta_{2D} = \left(\frac{\left(\sin\phi\cos\phi\left(\frac{\omega}{\bar{\omega}}\right)^2\right)^2 + \left(1 - \cos^2\phi\left(\frac{\omega}{\bar{\omega}}\right)^2\right)^2}{\left(1 - \cos^2\phi\left(\frac{\omega}{\bar{\omega}}\right)^2\right)^{\frac{4}{3}}} \right)^{\frac{3}{2}} \tag{17}$$

Since in this case the definition for the acceleration is the same as used in Ref. [2], the contours of equal β_{2D} are exactly the same as those in Ref. [2] and as in the right figure of Fig. 2.

In order for the control law to be feasible, the constraint $0 \le \cos\alpha \le 1$ needs to be imposed on the first equation in Eq. (16). Since it was found that the minimum and maximum values for the term $1 - \frac{1}{\bar{\omega}^2}\left(\dot{\theta}^2\cos^2\phi - \dot{\phi}^2\right)$ occur at the maximum and minimum value for z (or equivalently for ϕ), respectively, the following constraints on the in-plane angular velocity and sail lightness number can be derived:

$$\dot{\theta}\Big|_{\phi=\phi_{\max}} \le \dot{\theta}_{\max}\Big|_{\phi=\phi_{\max}} = \sqrt{\frac{\bar{\omega}^2 - \dot{\phi}^2}{\cos^2\phi}}\Bigg|_{\phi=\phi_{\max}} \tag{18}$$

$$\beta \ge \beta_{\min} = 1 - \frac{1}{\bar{\omega}^2}\left(\dot{\theta}^2\cos^2\phi + \dot{\phi}^2\right)\Big|_{\phi=\phi_{\min}} \tag{19}$$

As for the cylindrically constrained orbits, spherically constrained orbits can be found by deviating from β_{2D}, either in positive or negative direction, while keeping the control law as given in Eq. (16). By increasing

β with respect to β_{2D}, the spacecraft will move in a confined band above ϕ_0 ("north orbits") such that $\phi_0 = \phi_{\min}$ while $\phi_{\max}$ is unknown. The constraint in Eq. (18) can therefore only be determined through an integration of the equations of motion, while the constraint in Eq. (19) can be rewritten as:

$$\beta \geq \beta_{\min} = 1 - \left(\frac{\dot{\theta}_0}{\bar{\omega}} \right)^2 \cos^2 \phi_0 \tag{20}$$

Contrary, for the case of decreasing β with respect to β_{2D}, the spacecraft will move in a confined band below ϕ_0 ("south orbits") such that $\phi_0 = \phi_{\max}$, while $\phi_{\min}$ is unknown. The constraint in Eq. (19) can therefore only be determined through an integration of the equations of motion, while the constraint in Eq. (18) can be rewritten as:

$$\dot{\theta}_0 \leq \dot{\theta}_{0,\max} = \frac{\bar{\omega}}{\cos \phi_0} \tag{21}$$

A set of feasible orbits for the case of $r = 1$ and $z_0 = 0.5$ (i.e. $\phi_0 = 0.5236$ rad) is shown in Fig. 13. Note that, for the spherically constrained orbits, the results in Fig. 13 and all subsequent figures are made dimensionless by setting $r = 1$ and $\bar{\omega} = 1$. The behavior of the orbits in the phase space is very similar to the behavior seen in Fig. 3 and Fig. 8 for the cylindrical case: the bands above and below the 2D NKO are clearly visible, where a larger value for the in-plane angular velocity leads to tighter bands, i.e. a smaller deviation from ϕ_0 for a particular percentage deviation from β_{2D}. Furthermore, also the orbits oscillating around the (x, y)-axis are once again clear for large enough negative values of $\Delta\beta$.

The full feasibility region in terms of $\dot{\theta}_0$ and $\Delta\beta$ is shown in Fig. 14. Although the figure suggests that β can be increased unlimitedly, this is not the case. For large enough values for $\Delta\beta$ the constraint in Eq. (18) will eventually be violated. However, this only occurs for extremely large values for $\Delta\beta$, which are not considered reasonable and are therefore not included in Fig. 14.

As for the cylindrically constrained orbits, a periodicity analysis has been performed for the spherically constrained orbits. The results are shown in Fig. 15 with some characteristic orbits provided in Fig. 16. The orbits in Fig. 16a show, in clockwise direction and starting from the top left figure, a north orbit that covers a narrow band on the spherical surface; a north orbit that covers a much wider band and closely passes to the spherical poles; a south orbit; and a south orbit that oscillates around the (x, y)-plane. When considering applications for these types of orbits, one can imagine that the second orbit would be very suitable for solar polar observation, especially if the out-of-plane displacement is increased further such that the solar sail spends even

more time directly above the solar poles, while the fourth orbit, which covers a significant part of the spherical surface would be suitable for three-dimensional imaging of the Sun.

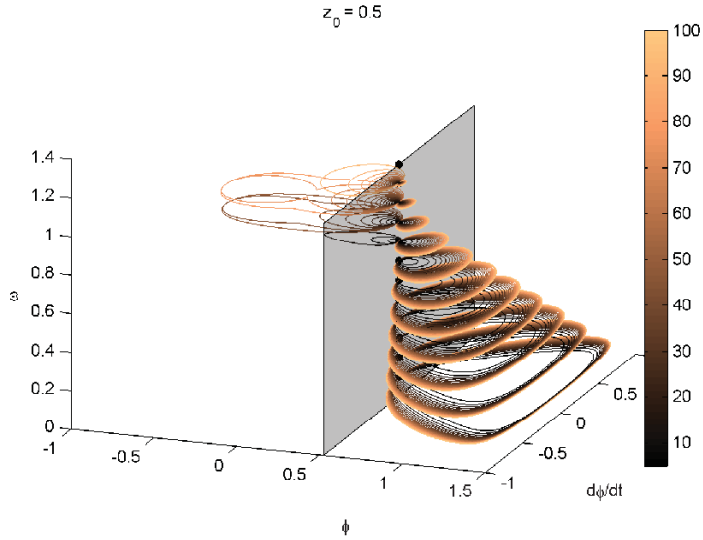

Fig. 13 Phase spaces for spherically constrained NKOs with a solar sail acceleration law and for $r = 1$ and $z_0 = 0.5$ (i.e. $\phi_0 = 0.5236$ rad). The colors indicate the percentage deviation from β_{2D} with a step size of 5% (negative deviation to the left of the gray transparent plane; positive deviation to the right of the gray transparent plane).

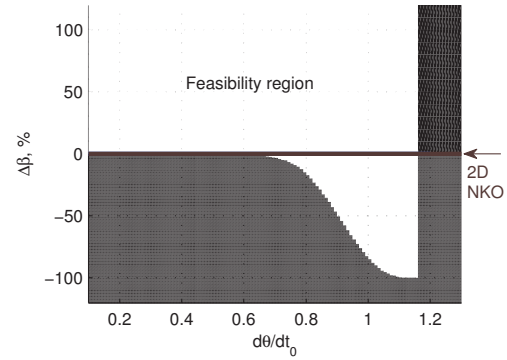

Fig. 14 Feasibility region (in white) in terms of the initial in-plane angular velocity, $\dot{\theta}_0$, and deviation from β_{2D} for spherically constrained NKOs with a solar sail law and for $r = 1$ and $z_0 = 0.5$ (i.e. $\phi_0 = 0.5236$ rad).

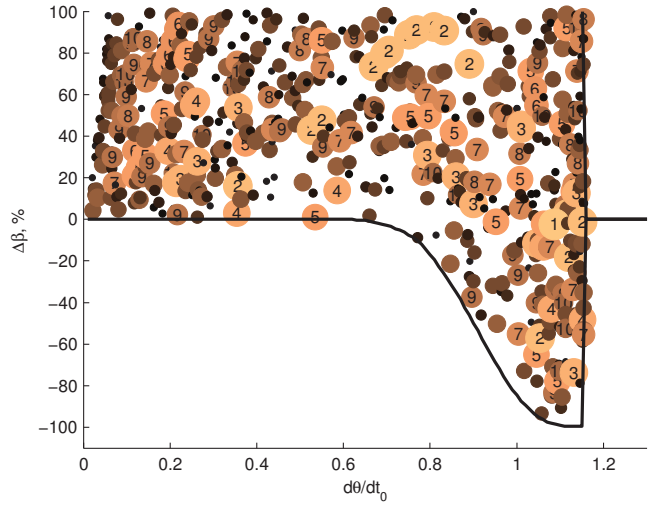

Fig. 15 Periodicity of the set of feasible spherically constrained NKOs with a solar sail acceleration law and for $\rho = 1$ and $z_0 = 0.5$, where orbits with a periodicity of up to 20 orbital revolutions are included and orbits with periods of up to 10 are indicated with a numerical value. The smaller the marker and the lighter the color, the shorter the period.

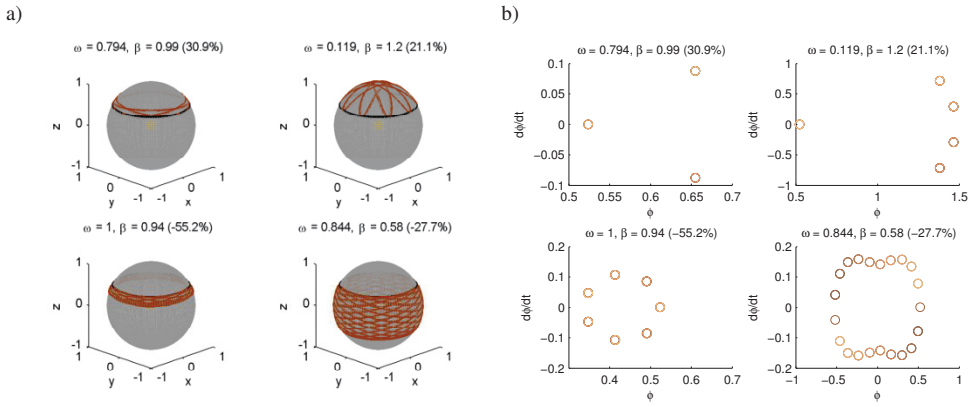

Fig. 16 Periodic spherically constrained NKOs for a solar sail acceleration law for $\rho = 1$ and $z_0 = 0.5$. a) Trajectories. b) Poincaré maps with the revolution number increasing from dark to light colors.

IV. Conclusions

In this paper, new families of non-Keplerian solar sail orbits have been introduced where the sail motion is restricted to a cylindrical or spherical surface. The feasibility region in terms of acceleration magnitude and in-plane angular velocity has been established and true periodic orbits have been found. The range of possible orbits include both those that cover only a narrow band of the cylindrical or spherical surface and those that cover a significant portion of that surface. With applications in planetary observation, interplanetary communication, astronomical observations and solar physics, the potential of these novel orbits is clear. For example, the latter can significantly benefit from the spherically constrained orbits as they enable orbits high above the solar poles for solar polar observations as well as orbits that cover a significant part of the spherical surface for obtaining a full three dimensional picture of the features and structures of the Sun.

Finally, the results indicate that a reduction in the required propulsion system performance can be achieved through the use of these 3D rather than 2D NKOs, at the cost of only a minor deviation from the original 2D NKO, increasing the potential of these cylindrically and spherically constrained orbits even further.

Acknowledgements

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Families of periodic orbits for solar sails in the CRBTP

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We use the numerical continuation package AUTO to investigate families of periodic orbits in the solar sail circular restricted three-body problem. For a sail orientated perpendicular to the Sun-line we find significant differences to the classical case for some families near the Earth, including the L_1 Halo family and retrograde satellite family. Specifically, we expand on existing results and find that the change in the Halo family H1 is associated with a bifurcation of a branch point in the retrograde satellite family, which splits H1 in half. We also track regions of stability within the family, and find some large amplitude stable orbits. For a sail tilted relative to the Earth-Sun line only we find large amplitude families with some stable orbits. Interestingly there is also a small range of parameters for which L_1 bifurcates into three separate points in this system.

I. Introduction

FAMILIES of periodic orbits are well known in the context of the circular restricted three-body problem (CRTBP) and have found numerous applications to space craft trajectory design. In this work we consider a modified CRTBP that includes radiation pressure, so that it is applicable to solar sails. We use the simple model of radiation pressure acceleration given by [1] for a perfect sail. This model uses the sail's lightness number β and normal vector $\boldsymbol{n}$ to parameterize the problem.

The classical CRTBP is conservative and as a consequence it is well known that periodic orbits form one-parameter families in the energy level for a given mass ratio. The solar sail CRTBP is only conservative if the sail's normal vector $\boldsymbol{n}$ is directed radially along the Sun-line. For a generic choice of $\boldsymbol{n}$ the system is non-conservative, and families can only be formed in β and the angles used to define the sail orientation. However,

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there are some specific sail orientations that result in a non-conservative but reversible system, in which time reversible periodic orbits will also form one-parameter families [2, 3].

Numerical continuation methods provide a powerful tool for investigating such families of periodic orbits. For example, [4] demonstrate how the AUTO package [5] can be used to study elementary families emanating from the Lagrangian points in the classical CRTBP. Here one-parameter families can be continued by introducing artificial dissipation controlled by an unfolding parameter which will be zero for periodic solutions [6, 7, 8, 9].

Using AUTO we investigate families of periodic orbits for two specific choices of the solar sail orientation. The first case is the conservative system with the sail orientated perpendicular to the Sun-line. The second is the non-conservative but reversible system with the sail tilted relative to the Earth-Sun line. The following two sections describe the mathematical model and numerical continuation methods for generating the periodic orbits. The results for the two different sail angles are then presented and finally conclusions are given.

II. The Solar Sail CRTBP

We consider the motion of a solar sail in the context of the Earth-Sun CRTBP i.e. the Sun is the primary mass m_1, the Earth the secondary mass m_2 and the sail the massless test particle. In the rotating reference frame the Sun is located at $(-\mu,0,0)^T$ and the Earth at $(1-\mu,0,0)^T$, where $\mu = \frac{m_2}{m_1+m_2} = 3 \times 10^{-6}$ is the mass ratio. Units are used such that the distance between the Sun and the Earth, their angular velocity in the non-rotating frame and $G(m_1+m_2)$ are all unity, where G is the gravitational constant. The sail is located at $\boldsymbol{r} = (x, y, z)^T$ in this frame and is assumed to have a fixed orientation defined by its normal vector $\boldsymbol{n}$. The sail's position relative the Sun and Earth are

$$\boldsymbol{r}_1 = (x + \mu, y, z)^T,$$

$$\boldsymbol{r}_2 = (x - 1 + \mu, y, z)^T$$

respectively, as shown in Fig. 1. Following [1] the additional acceleration on a perfectly reflecting solar sail in this system is

$$\boldsymbol{a} = \beta \frac{(1-\mu)}{r_1^2} (\hat{\boldsymbol{r}}_1 \cdot \boldsymbol{n})^2 \boldsymbol{n}$$

where $\hat{\boldsymbol{r}}_1$ is the unit vector in the direction of $\boldsymbol{r}_1$ and $r_1 = |\boldsymbol{r}_1|$. The first case we consider has the sail orientated along the Sun-line so the acceleration is in the radial direction only i.e. $\boldsymbol{n} = \hat{\boldsymbol{r}}_1$. This system is conservative and Hamiltonian, and has equations of motion given by

$$\ddot{\boldsymbol{r}} + 2\boldsymbol{\omega} \times \dot{\boldsymbol{r}} = \nabla V$$

where a dot indicates differentiation with respect to time, $\boldsymbol{\omega} = (0,0,1)^T$ and the potential V is

$$V = \frac{(1-\mu)(1-\beta)}{r_1} + \frac{\mu}{r_2} + \frac{1}{2}|\boldsymbol{\omega} \times \boldsymbol{r}|^2$$

where $r_2 = |\boldsymbol{r_2}|$.

The second case we consider has the sail orientated at a constant angle γ to the Sun-Earth line i.e. $\boldsymbol{n} = (\cos\gamma\,, 0, \sin\gamma)^T$. The equations of motion in this case are

$$\ddot{\boldsymbol{r}} + 2\boldsymbol{\omega} \times \dot{\boldsymbol{r}} = \nabla U + \boldsymbol{a}$$

where the potential U is now

$$U = \frac{(1-\mu)}{r_1} + \frac{\mu}{r_2} + \frac{1}{2}|\boldsymbol{\omega} \times \boldsymbol{r}|^2.$$

This system is no longer Hamiltonian but remains reversible as the equations of motion are invariant under $(x, y, z, \dot{x}, \dot{y}, \dot{z}, t) \rightarrow (x, -y, z, -\dot{x}, \dot{y}, -\dot{z}, -t)$. Both systems are known to possess one-parameter families of orbits emanating from the equilibria equivalent to the Lagrangian points [10, 11, 12].

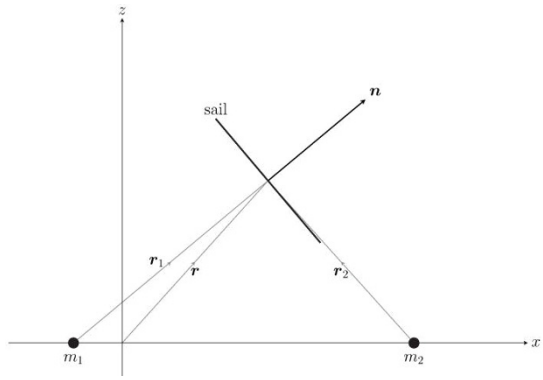

Fig. 1. The solar sail CRTBP, shown in the *x*-*z* plane.

III. Numerical Continuation

One-parameter families of periodic orbits can be continued using boundary value techniques as implemented in the package AUTO [5]. Natural families in a conservative system can be continued through use of an 'unfolding parameter' λ which introduces artificial dissipation into the system [6, 7, 8, 9]. The equations of motion of the system are then given by

$$\ddot{\boldsymbol{r}} + 2\boldsymbol{\omega} \times \dot{\boldsymbol{r}} = \nabla V + \lambda\, \dot{\boldsymbol{r}}.$$

Families are continued in the unfolding parameter, but periodic solutions will always have $\lambda = 0$ to numerical precision. Justification of this approach and its implementation in AUTO can be found in [4, 6, 7, 8, 9], where it is used to follow families in the classical CRTBP as well as detect branch points and continue these in a system parameter. Alternatively, the equations of motion can be reformulated to explicitly include the energy in the equations of motion through the addition of a term of the form

$$\nabla h(\boldsymbol{r}, \mu, \beta)^T (h(\boldsymbol{r}, \mu, \beta) - h_0)$$

where h is the energy, which takes the constant value h_0 along a trajectory. This method permits folds in the energy level to be detected and continued in the system parameter h_0.

Preliminary work has found that the numerical continuation of families of periodic orbits in a reversible system can be implemented in AUTO in a similar way to the unfolding method used for the conservative system.

IV. The Radial Case

The first case investigated is the conservative system with the sail's normal aligned with the Sun-line i.e. $\boldsymbol{n} = \hat{\boldsymbol{r}}_1$. In this modified version of the CRTBP the L_1 Lagrangian equilibrium point remains on the x-axis and moves closer to the Sun as the sail's lightness number increases. This system has been investigated by numerous authors, for example the planar Lyapunov family, or L1 following the notation of [4], and the Halo family, or H1, associated with the L_1 point are known to exist in this case [10, 11, 12, 13]. In particular, [12] found a significant change in the shape of H1 occurs for values of β around 0.04 to 0.05.

We extend on previous results by continuing the L_1 Halo family H1 out to large amplitudes for a wide range of β. Our aim is to provide a full picture of the system's evolution and as such we consider values of the lightness number up to about 0.5, although we recognize that these are not realistic for current solar sail technology.

In the classical case ($\beta = 0$) the H1 family branches from L1 in a pitchfork bifurcation to a north and south branch, symmetric about the x-y plane [4, 10]. We adopt the notation $\mathrm{B}_{\mathrm{L1}}^{\mathrm{H1}}$ for this branch point between families, were in general the lower label indicates the family with the higher symmetry. As the two branches of H1 are identical we will concentrate on the North branch. This family ends in a collision with m_2. At higher mass ratios the collision does not occur and the family connects to a planar family of retrograde circular orbits about both masses, called C2 by [4]. This family bifurcating from C2 still exists in the classical Earth-Sun case, and also ends in a collision with m_2 as would be expected. At higher values of β we find that the increasing amplitudes and changing position of the family means that the collision disappears. Thus we follow H1 from both the L1 and C2 branch points. A bifurcation diagram for the classical case is shown in the first panel of Fig. 2.

As mentioned above, [12] found a marked change in H1 for values of β greater than about 0.04. We also observe this, but find it is due to the appearance of an additional branch point in the family at a critical value of the lightness parameter $\beta_1 \approx 0.0387$. As $\beta \to \beta_1$ the family starts to fold back on itself down to the ecliptic plane. At $\beta = \beta_1$ the family intersects the plane and a branch point appears to a planar family of orbits about m_2, as shown in panel 2 of Fig. 2. This planar family is often referred to in the classical case as the retrograde satellite family and corresponds to Strömgren's family *f* [14, 15], and hence we will label it RS. In the classical case this family is found to contain one branch point to a non-planar family we will call HR, as also shown in panel 1 of Fig. 2.

As β increases above β_1 this branching orbit bifurcates into two and gradually moves apart, essentially splitting the Halo family in half. This is shown in panel 3 of Fig. 2. The second half still ends in collision at this point, but as β increases to about 0.16 this disappears and it reconnects to the C2 half of the family. We call the first 'new' family, the one from the L1 branch point to the first RS branch point, H1B and the second from the other RS branch point to the C2 branch point H1C. These are labeled in Fig. 2.

For completeness we follow these families to higher values of β. Here we find another critical value $\beta_2 \approx 0.2894$ at which the reverse happens: the H1B branch point in RS moves towards and annihilates with the HR branch point (panel 4 of Fig. 2), and the H1B and HR families merge to become a family we designate H1R (panel 5 of Fig. 2). We note that the additional branch points in RS are a result of two folds in β in the locus of the original HR family branch point. In fact as $\beta \to 1$ the branch points in the RS and C2 families merge, as shown in Fig. 3.

The linear stability along the families is determined by the Floquet multipliers (see for example [16]), which are provided by AUTO. As the system is Hamiltonian the multipliers are always reciprocal pairs, and two of the six are unity as they represent perturbations along the periodic orbit. Information about the linear stability of a periodic orbit is therefore determined by the remaining four multipliers. [19] and [12] define order-0 instability as the case with all multipliers on the unit circle, order-1 as one pair only on the unit circle and one pair on the real axis, and order-2 as no multipliers on the unit circle. Generically, changes in the stability can occur at branch points and folds (a pair of multipliers collide at +1), period doubling bifurcations (a pair of multipliers collide at -1), and Krein collisions (two pairs of multipliers collide simultaneously on the unit circle, or for an inverse Krein collision, on the real axis).

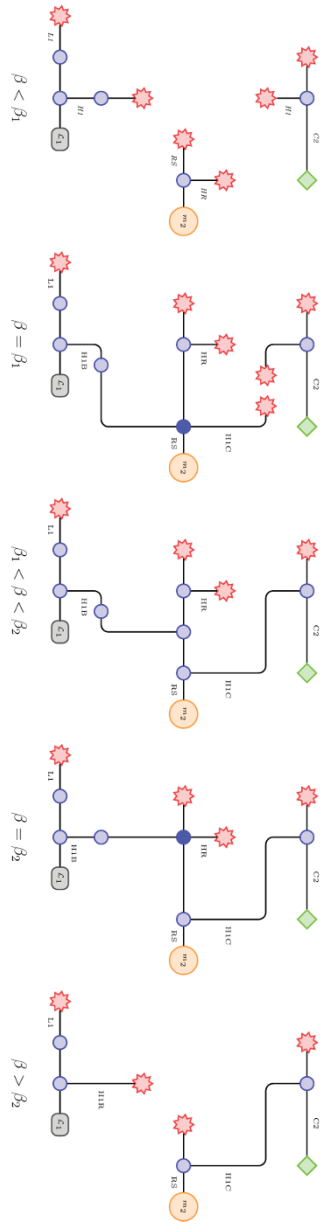

Fig. 2. Schematic bifurcation diagrams of H1 and related families for increasing values of β. Blue circles represent branch points and red stars collisions. The L_1 equilibrium point is shown as a grey rectangle and the secondary mass as a yellow circle. The green diamond indicates that the C2 family appears to continue out to infinity. The darker blue circles represent the two special cases where two branch points in the RS family either appear or collide.

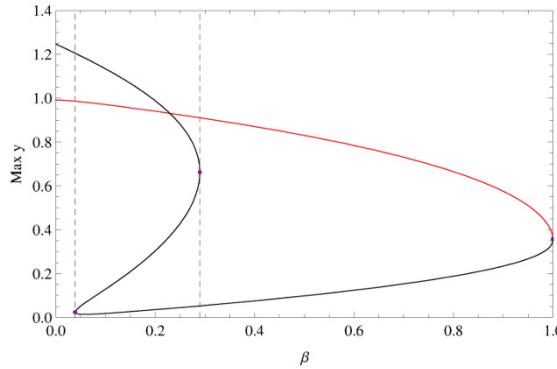

Fig. 3. The evolution of the branch points on the C2 and RS families in β. The C2 branch point is shown in red, the RS branch point in black. The C2 and RS branch points merge at $\beta = 1$. Folds in β are marked as a purple dot. Gray dashed lines mark the region in β for which three branch points exist along the RS family. This region occurs between $\beta_1 \approx 0.0387$ and $\beta_2 \approx 0.2894$.

All these can be detected and continued in AUTO, with the exception of Krein collisions which can be located through examination of the Floquet multipliers. Using these methods we investigate how the linear stability of each family changes as β increases. Fig. 4 shows schematically this stability for the H1, H1B, H1C, HR and H1R families for the entire range of β considered.

For $\beta = 0$ the branch of H1 from L1 is well known to have a small region of order-0 instability defined by a fold and period doubling bifurcation [17], as shown on the left-side of Fig. 4. At small values of β the fold disappears, as does the order-0 region. However, another order-0 region appears, defined by a branch point to an asymmetric family called W5 by [4] and a Krein collision. (This region exists for higher mass ratios in the classical case but is lost in the collision with m_2 here.) Although it appears fairly large on the diagrams, the region is in fact a fairly small portion of each family. As β increases further this region is split by the appearance of two folds. As β approaches the critical value β_1 it becomes part of the H1C family. However, near β_1 a new order-0 region appears in-between a period doubling bifurcation and fold and this continues into H1B. As β approaches 0.041 and 0.045 respectively the fold and period doubling bifurcations disappear and the majority of the region is order-0, defined by the W5 branch point and the branch point to the RS family. This is the large region of order-0 instability seen by [12] at $\beta = 0.05$.

The HR family appears to have order-0 instability for its entirety, and this persists as it merges with the stable part of H1B. These orbits are all large amplitude, and are in marked contrast to orbits obtainable in the classical CRTBP. Examples of the H1, H1B, H1C and H1R families are shown in Fig. 5 to 8, and in each case the order-0 instability portion of the family has been marked. Note that the families are shown on different scales.

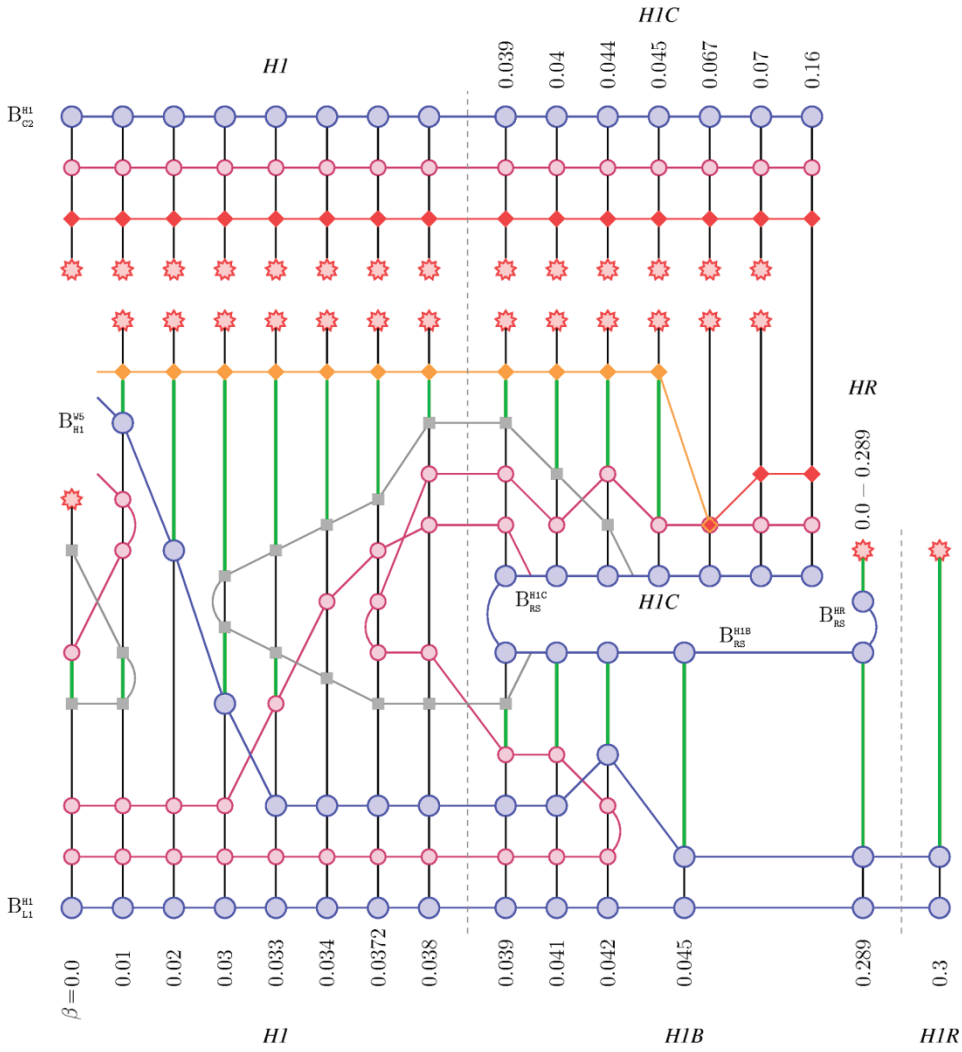

Fig. 4. Stability of H1 and associated families. Vertical lines represent families, and β increases left to right as labeled. Blue circles: branch points to other families, purple circles: period doubling bifurcations, gray squares: folds, orange diamonds: Krein collisions, red diamonds: inverse Krein collisions, red stars: collisions. The thick green line marks the portion of each family that has order-0 instability. The H1 family always starts from the L1 branch point with order-1 instability, and the linear stability of the rest of the family can be deduced from the diagram. The gray dashed lines indicate the points at which the H1 family splits in half or merges with another family. Branch points, period doubling bifurcations and folds have been continued in AUTO and the lines joining these points reflects this. The Krein collisions have been located through examination of each family's Floquet multipliers.

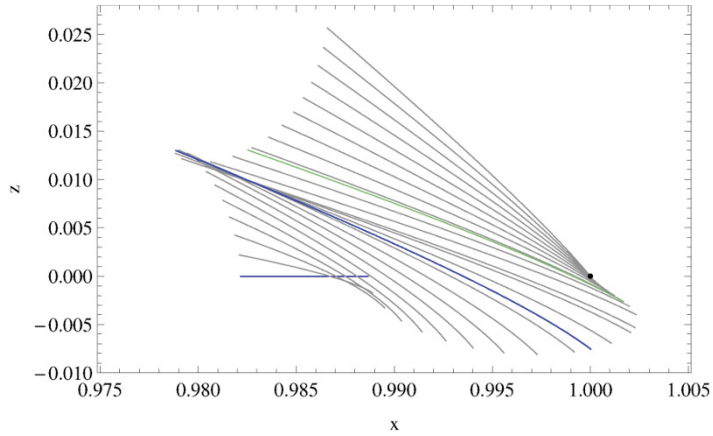

Fig. 5. The H1 family for $\beta = 0.03$, shown in the *x*-*z* plane. Order-0 instability orbits are shown in green, others in gray. The branching orbits to L1 and W5 are shown in blue, and the position of the Earth marked as a black dot.

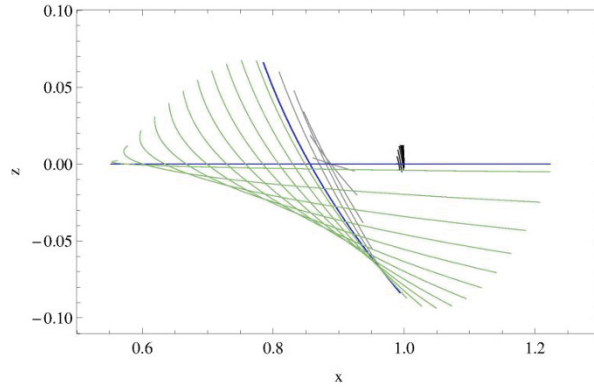

Fig. 6. The H1B family for $\beta = 0.289$, shown in the *x*-*z* plane. Order-0 instability orbits are shown in green, others in gray. The classical H1 family is shown in black for comparison. The branching orbits to L1 and W5 are shown in blue.

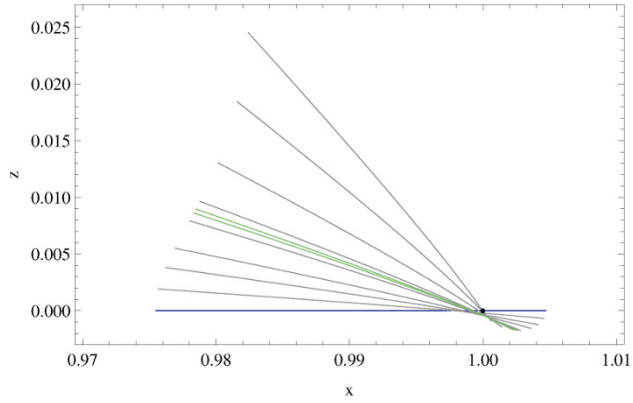

Fig. 7. The H1C family for $\beta = 0.04$, shown in the *x*-*z* plane. Order-0 instability orbits are shown in green, others in gray. The branching orbit to RS is shown in blue, and the position of the Earth marked as a black dot.

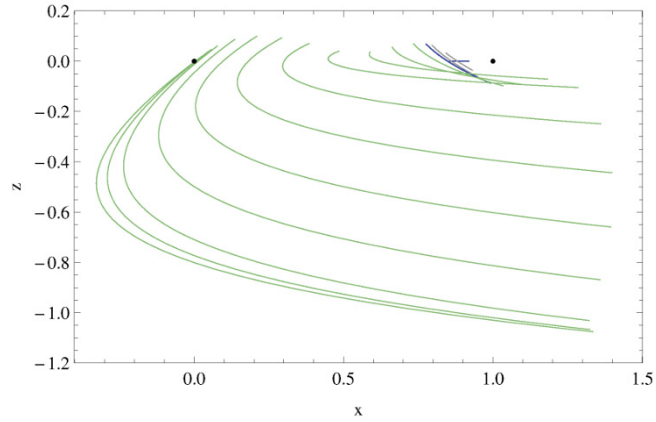

Fig. 8. The H1R family for $\beta = 0.3$, shown in the *x*-*z* plane. Order-0 instability orbits are shown in green, others in gray. The branching orbits to L1 and W5 are shown in blue, and the positions of the Earth and Sun marked as black dots.

V. The Non-Radial Case

The second case we look at is a sail orientated at a constant angle γ to the x-axis, i.e. $\boldsymbol{n} = (\cos\gamma, 0, \sin\gamma)^T$. As mentioned this system is not Hamiltonian, but is reversible, so one-parameter families of periodic orbits exist for fixed values of γ and β. This has previously been considered by [18] and [10]. [18] look at families near L_1 continued in γ, suitable as pole sitter orbits. [10] look at one-parameter families for $\beta = 0.051689$ and show how the L1 to H1 pitchfork bifurcation is broken and the L1 and H1 families merge as the sail angle changes.

In general, five equilibria exist in this system for any fixed values of γ and β, corresponding to the Lagrangian points in the classical case [1, 10, 18]. However, for values of β greater than about 0.15 there is a fold in the curve of equilibria continued from L_1 in γ. That is, there is a small range of angles where three equilibria exist near L_1 instead of one. It can be shown that this coincides with a change in linear stability of one point from center-center-saddle to center-saddle-saddle.

The reversible version of the Lyapunov center theorem [2, 3] implies two families emanate from the points that are center-center-saddle (corresponding to L1 and the vertical family V1 in the classical CRTBP), and one from the center-saddle-saddle point (corresponding to the L1 family). These three points are shown in Fig. 9. We label the center-center-saddle points as L_1^l and L_1^u for the lower and upper point respectively, and the middle center-saddle-saddle point as L_1^m.

Preliminary work has shown that the family emanating from the L_1^m point connects to the lower of the other points, L_1^l. The other family from L_1^l and the two from L_1^u continue out to large amplitudes and are terminated by collisions. These families are also shown in Fig. 9. At large amplitudes the 'vertical' family is similar to large amplitude vertical family orbits in the classical case. However, in contrast to the classical case, some possess order-0 instability.

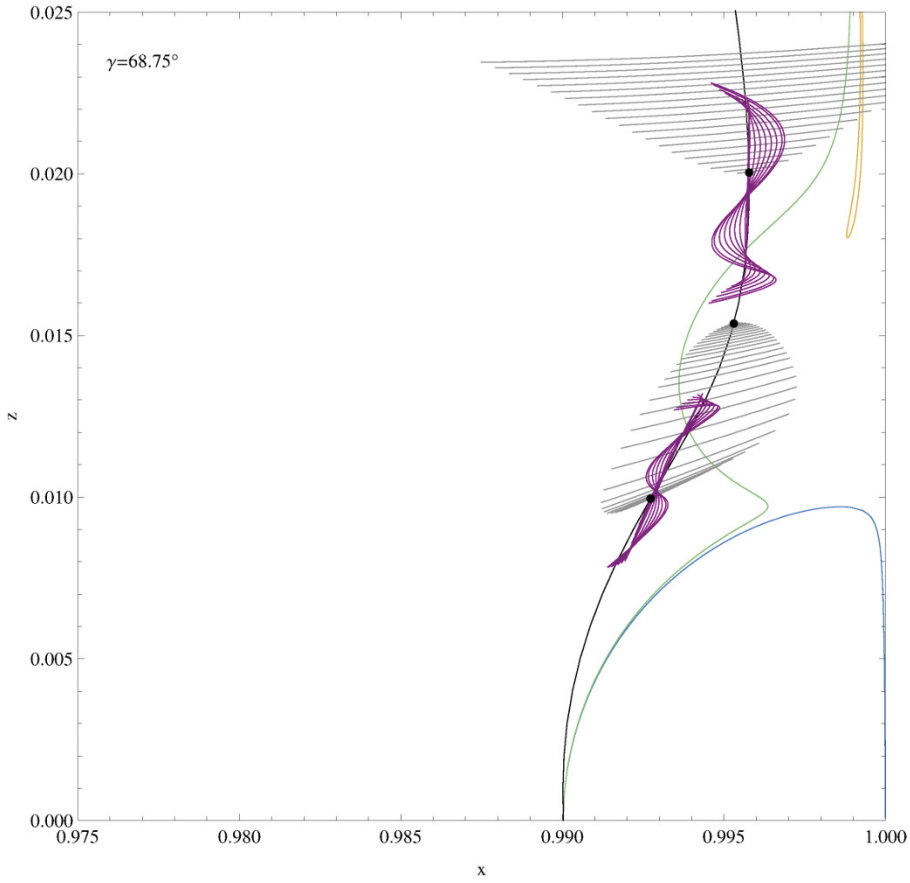

Fig. 9. The families of periodic orbits associated with the three L_1 equilibrium points for $\beta = 0.2$ and $\gamma = 68.75^{\circ}$, shown in the *x*-*z* plane. The equilibria L_1^l, L_1^m and L_1^u are shown as black points, the 'L1' families in gray and the 'V1' families in purple. Note that the families continue out to larger amplitudes than shown here. The black curve is the locus of L_1 in γ for this value of β. The green line marks the location of the fold in γ (for all β) that results in the bifurcation of L_1 into three. The region marked in blue indicates an inaccessible region for L_1, and the region marked in orange a region where the linear stability of L_1^m changes form center-saddle-saddle to center-focus-focus.

VI. Conclusion

Numerical continuation techniques such as those implemented in AUTO provide a means of obtaining a broad overview of periodic orbits obtainable by solar sails, as well as the connections between them and their linear stability.

For a sail orientated perpendicular to the Sun-line the Halo family undergoes significant evolution as the sail's lightness number increases. A fold in the locus of branch point in the retrograde satellite family results in the appearance of two new families of periodic orbits. Even at relatively low lightness numbers these families have fairly large amplitude orbits with order-0 linear instability. Preliminary investigation for a sail orientated instead relative to the Sun-Earth line has shown complex changes occur near the Earth to the L_1 equilibrium point and associated families.

The appearance of large amplitude order-0 instability orbits is a significant difference between solar sails and the classical CRTBP. This provides new opportunities for mission design.

Acknowledgments

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Periodic Motion for an Imperfect Solar Sail near an Asteroid

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In this paper we consider the Hill 3-body problem with the extra effect of the solar radiation pressure as a model for the motion of a solar sail close to an asteroid. To model the sail's acceleration, we include both reflectivity and absorption of the sail's material. We describe the most relevant dynamical properties of the system for different reflectivity and absorption coefficients as well as different fixed sail orientations. We show families of periodic orbits, describe how they relate to the parameters of the sail and discuss their stability.

Introduction

DUE to the small gravitational field around asteroids, solar radiation pressure (SRP) plays an important role on the dynamics around them. Hence, the systematic use of SRP to propel a satellite via specialized reflecting areas, such as solar sails, can give rise to new and interesting mission concepts. When we look at the dynamics of a satellite close to an asteroid there are three effects that must be taken into account: gravitational field, solar attraction and solar radiation pressure [1]. The first effect includes the shape and spin states of the asteroid, and it is relevant within a few radii from its surface, in what is known as the gravity regime. Far from the gravity regime, a good approximation of the asteroid's contribution to the real dynamics is obtained by considering it as a point mass. Here effects like SRP and the Sun's gravity must be taken into account. For asteroids following very eccentric orbits around the Sun, the variation of the distance to the Sun also plays an important role.

The goal of this paper is to study the dynamics of a solar sail near an asteroid. A flat solar sail will be considered, for which we will take into account not only the reflectivity but also the absorption of the sail's material [2, 3]. Our reference model will be the Augmented Hill 3-body problem, in which the asteroid is considered as a

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point mass and the effect of SRP due to the solar sail is added. We will give a description of some of the most relevant dynamical properties of the system (fixed points and periodic orbits) and describe some of their possible applications to asteroid mission analysis.

It is well known that the Hill 3-body problem has two equilibrium points L_1 and L_2. When we add the effect of SRP, these two fixed points move towards the Sun. L_1 quickly moves away from the small body, while L_2 comes reasonably close to it [4, 5, 6]. For this reason, in this paper we will focus on the dynamics close to L_2 as it is more relevant for future mission applications. It is true that L_2 is in the shadow side of asteroids, and this could compromise observational missions. We will show, however, that there are many orbits on the family of orbits emanating from L_2 that spend most of their orbital period on the Sun side, making them very interesting for this kind of missions.

As we know, the linear dynamics around the displaced L_2 point is of the type center$\times$center$\times$saddle. Hence, two families of periodic orbits are born from L_2: a planar and a vertical family, each one related to the one of the two center directions. We also find two families of Halo orbits at the 1:1 resonance between the planar and vertical frequencies as we move along the planar family of periodic orbits. We will see that, as we move along the family of planar orbits, they acquire a boomerang shape that leaves the asteroid in the middle. On the other hand, the orbits on the vertical family start with an eight shape, with a large vertical amplitude. As we move along the family, they loose part of their vertical amplitude to become almost planar, and spend half of their period on the Sun side of the asteroid. Both the planar and vertical families of periodic orbits are unstable, needing a control strategy to remain on them for a long period. Finally, as we move along the family of Halo orbits, these ones come closer to the asteroid, but stay in the shadow side all the time. Nevertheless, the terminating orbits of the family are linearly stable.

When we change the orientation of the solar sail, we can displace the position of these fixed points and periodic orbits. Tilting the solar sail vertically with respect to the Sun-line will displace these invariant objects above and bellow the orbital plane, whereas tilting the solar sail horizontally with respect to the Sun-line will shift these objects to one side or another of the Sun-asteroid line. We will compute these families of equilibrium points and periodic orbits for different fixed sail orientations by means of continuation methods, and we will discuss how different reflectivity parameters for the solar sail affect their shape and properties. We will pay special attention to the terminating orbits of the families, since they come close to the small body and present interesting features for asteroid mission analysis.

The Augmented Hill for a Non-Perfectly Reflecting Solar Sail

As it has been mentioned before, to capture the dynamics of a solar sail around an asteroid we use the Augmented Hill 3-body problem (H3BP), that models the motion of two small masses (asteroid and spacecraft) which interact due to their mutual gravitational attraction and is perturbed by a distant larger body (the Sun). The Augmented H3BP also includes the effect of SRP as our satellite is propelled by a solar sail [7, 8]. Hence, in this model

the effect of the Sun is included as a uniform gravity vector field and uniform SRP, both in the horizontal direction.

We consider a rotating reference frame centered on the small body, defined such that: the x direction points from the Sun to the asteroid; the z direction is aligned with the Sun–asteroid angular velocity; and the y direction completes a positive coordinate system. We normalize the units of distance and time such that: $L = (\mu_{sb}/\mu_{sun})^{1/3}R$ and $T = 1/\omega$ respectively, where μ_{sun} is the gravitational parameter of the Sun ($\approx 1.327 \times 10^{11}\text{km}^3/\text{s}^2$), μ_{sb} is the gravitational parameter of the smaller body, R is the mean distance distance between the Sun and the asteroid, and $\omega = \sqrt{\mu_{sun}/R^3}$ is the angular velocity. Using these normalized units, the equations for the Augmented Hill Three Body Problem (AH3BP) for a solar sail [5, 6] are:

$$\begin{aligned} \ddot{X} - 2\dot{Y} &= -\frac{X}{r^3} + 3X + a_x, \\ \ddot{Y} + 2\dot{X} &= -\frac{Y}{r^3} + a_y, \\ \ddot{Z} &= -\frac{Z}{r^3} - Z + a_z, \end{aligned} \tag{1}$$

where (X, Y, Z) denotes the position of the solar sail in the rotating frame, $r = \sqrt{X^2 + Y^2 + Z^2}$ is the distance between the solar sail and the center of the asteroid, and $\mathbf{a_{sail}} = (a_x, a_y, a_z)$ is the acceleration given by the solar sail.

To model the acceleration of the solar sail we use the simplified model for a non-perfectly reflecting (SNPR) solar sail from [3], which only takes into account absorption and reflection of the sail material: if we denote the absorption coefficient as a and the reflectivity coefficient as ρ, we have $a + \rho = 1$. Besides, the force due to absorption and reflection is given by $\mathbf{F_a} = PA\langle \mathbf{r_s}, \mathbf{n}\rangle \mathbf{r_s}$ and $\mathbf{F_r} = 2PA\langle \mathbf{r_s}, \mathbf{n}\rangle^2 \mathbf{n}$ respectively. In the previous expressions, $P = P_0(R_0/R)^2$ is the SRP magnitude at a distance R from the Sun ($P_0 = 4.563\mu\text{N/m}^2$, the SRP magnitude at 1AU and $R_0 = 1$AU), A is the area of the solar sail, $\mathbf{r_s}$ is the direction of SRP and $\mathbf{n}$ is the normal direction to the surface of the sail. Hence, the acceleration for the SNPR solar sail is given by:

$$\mathbf{a_{sail}} = \frac{2PA}{m}\langle \mathbf{r_s}, \mathbf{n}\rangle \left(\rho\langle \mathbf{r_s}, \mathbf{n}\rangle \mathbf{n} + \frac{1}{2}(1-\rho)\mathbf{r_s} \right). \tag{2}$$

Notice that $\rho = 1$ corresponds to a perfectly reflecting solar sail, and $\rho = 0$ to a perfect solar array where the absorption of the solar panels is the only effect. According to [3], a solar sail with a highly reflective aluminum-coated side has an estimated value of $\rho \approx 0.88$. Nevertheless, in this paper we present a comparative result of the dynamical properties of a solar sail for different values for ρ.

Observe that, in the rotating reference frame, $\mathbf{r_s} = (1, 0, 0)$. We will parametrize the normal direction to the surface of the sail using two angles α, δ, representing the horizontal and vertical displacement with respect to $\mathbf{r_s}$: $\mathbf{n} = (\cos\alpha\cos\delta, \sin\alpha\cos\delta, \sin\delta)$. As the sail cannot point towards the Sun, we have that $\alpha, \delta \in [-\pi/2, \pi/2]$. Finally, the efficiency of our solar sail will be measured in terms of $\beta = 2PA/m$, also know as the sail lightness

number. One can check that, with these definitions, the acceleration for the solar sail in Eq.(2) is written as

$$\begin{aligned} a_x &= \bar{\beta}\cos^2\alpha\cos^2\delta(\rho\cos\alpha\cos\delta + 0.5(1-\rho)), \\ a_y &= \bar{\beta}\cos^2\alpha\cos^2\delta(\rho\sin\alpha\cos\delta), \\ a_z &= \bar{\beta}\cos^2\alpha\cos^2\delta(\rho\sin\delta), \end{aligned} \tag{3}$$

where $\bar{\beta} = \beta(\mu_{sb}\omega^4)^{-1/3}$ is the sail lightness number in normalized units [5].

Some comments on the normalized units

To derive the equations of motion for the AH3BP, we have considered a set of normalized units. Recall that the unit of length is defined as $L = (\mu_{sb}/\mu_{sun})^{1/3}R$, and the unit of time is defined as $T = 1/\omega$. Notice that they both depend on the target asteroid for our mission. In Table 1 we summarize the physical data for different asteroids, as well as the normalized units of distance and time in the normalized Hill problem [5]. We also include the maximum radius of the asteroids in the normalized unit of distance, in order to provide an estimate of the gravity regime for each asteroid (gravity regime is around 3 to 4 times the asteroids' radii). Notice that, for all the asteroids considered, the radius distance varies between 0.001 and 0.005.

Table 1: Relevant parameters for different asteroids

asteroid	μ_{sb} (km^3/s^2)	R (AU)	RADII (km / UD)	UD (km)	UT (day)
Eros	4.463×10^{-4}	1.45	8.42 (2.5×10^{-3})	3265.4	102.2
Vesta	17.8	2.36	265 (1.4×10^{-3})	180782	210
Itokawa	2.1×10^{-9}	1.32	0.165 (3.3×10^{-3})	49.57	88.15
Golevka	$\sim 1.4\times10^{-8}$	2.49	0.265 (1.5×10^{-3})	175.96	229.1
Apophis	$\sim 1.8\times10^{-9}$	0.92	0.163 (4.9×10^{-3})	32.89	51.4

The normalized sail lightness number is $\bar{\beta} = \beta(\mu_{sb}\omega^4)^{-1/3}$. We recall that $\beta = 2PA/m$, $\omega = \sqrt{\mu_{sun}/R^3}$ and $P = P_0(R_0/R)^2$. Taking this into account, we can rewrite $\bar{\beta}$ as

$$\bar{\beta} = \frac{2P_0R_0^2(A/m)}{\mu_{sun}^{2/3}\mu_{sb}^{1/3}} = K_1(A/m)\mu_{sb}^{-1/3}, \tag{4}$$

where $K_1 \approx 7.8502$ if A is given in m^2 and m in kg. Hence, the value of $\bar{\beta}$ depends on both: $\sigma = (A/m)$, the area-to-mass ratio of the satellite, and μ_{sb}, the gravitational parameter of the small body. We could have the same $\bar{\beta}$ for two completely different missions scenarios. Notice that $\bar{\beta}$ is inversely proportional to $\mu_{sb}^{1/3}$. Hence, for a fixed area-to-mass ratio, the smaller the asteroid the larger the effect of SRP will be.

In Fig. 1 we show the contour levels of the normalized sail lightness number $\bar{\beta}$ as a function of σ and μ_{sb}. As we can see, for small asteroids ($\mu_{sb} \approx 10^{-9}$) and an area-to-mass ratio of a regular spacecraft ($\sigma \approx 0.01$), $\bar{\beta}$ takes values between 50 and 100. In Table 2 we summarize the values of $\bar{\beta}$ for the 5 asteroids in Table 1 and different

[5] The physical parameters have been taken from `http://ssd.jpl.nasa.gov/sbdb.cgi`

area-to-mass rations. As examples, we have taken an estimated area-to-mass ratio for Ikaros ($\sigma \approx 0.63\text{m}^2/\text{kg}$), NanoSail-D2 ($\sigma \approx 2.5\text{m}^2/\text{kg}$), Hayabussa ($\sigma \approx 0.032\text{m}^2/\text{kg}$) and Dawn ($\sigma \approx 0.17\text{m}^2/\text{kg}$). Although Hayabussa and Dawn are not solar sails, they have very large solar panels that could be considered for orbital control [6]. In Table 2 we see again that, around very small asteroids like Itokawa, Golevka and Apophis, SRP is very important, even for a spacecraft like Hayabussa and Dawn.

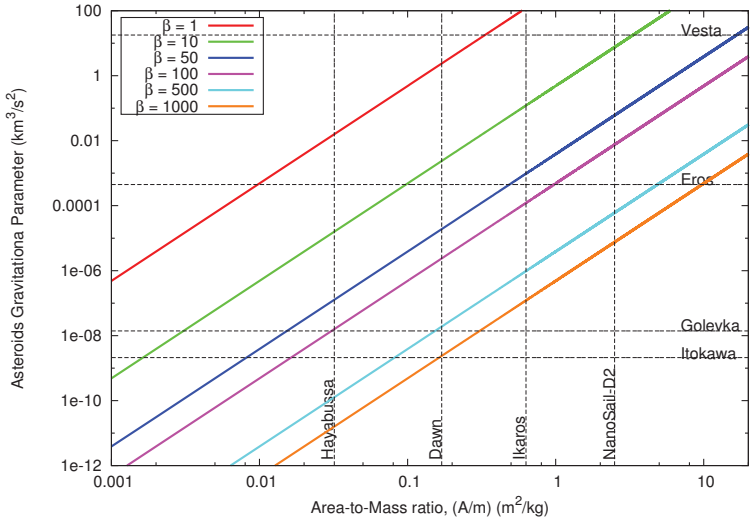

Fig. 1: Contours for $\bar{\beta}$ as a function of (A/m), area-to-mass ratio, and μ_{sb}, the gravitational parameter of the asteroid.

Table 2: Normalized values for $\bar{\beta}$ for different mission configurations

	Ikaros	NanoSail-D2	Hayabussa	Dawn
Eros	64.71	256.809	3.287	17.463
Vesta	1.894	7.51647	0.0962	0.511
Itokawa	3862.02	15325.48	196.166	1042.13
Golevka	2052.0	8142.88	104.228	553.716
Apophis	4065.65	16133.548	206.509	1097.081

Equilibrium Points

The description of the equilibrium points in the AH3BP for a solar sail has already been studied by many authors [7, 8]. Here we give a brief summary of the most important aspects.

It is well known that (x_0, y_0, z_0) is an equilibrium point of the AH3BP for the set of sail parameters $(\bar{\beta}_0, \alpha_0, \delta_0, \rho_0)$

if it satisfies

$$\begin{aligned} 0 &= -\frac{X}{r^3} + 3X + a_x + \bar{\beta}\cos^2\alpha\cos^2\delta(\rho\cos\alpha\cos\delta + 0.5(1-\rho)), \\ 0 &= -\frac{Y}{r^3} + \bar{\beta}\cos^2\alpha\cos^2\delta(\rho\sin\alpha\cos\delta), \\ 0 &= -\frac{Z}{r^3} - Z + \bar{\beta}\cos^2\alpha\cos^2\delta(\rho\sin\delta). \end{aligned} \tag{5}$$

If we neglect the effect of the solar sail ($\bar{\beta} = 0$), we have only two equilibrium points L_1, L_2, symmetrically located around the origin, with coordinates $(\pm 3^{-1/3}, 0, 0)$. When we include SRP and keep the solar sail perpendicular to the Sun-line ($\alpha = \delta = 0$), the position of L_1 and L_2 moves towards the Sun as we increase the value of $\bar{\beta}$. As it can be seen in [9], for $\bar{\beta} > 50$, L_1 is already at more than 16 unit distances from the small body. On the other hand, L_2 gets close to the small body, but still far away from the gravity regime where the shape and spin-rate of the asteroid must be considered. For this reason, from now on we will not consider the dynamics around the displaced L_1 point

Notice that, if $\rho = 0$ (i.e. the only acceleration for the solar sail is given by the absorption), even if we change the sail orientation the equilibrium points will always lie in the x axis, as the effect of changing the sail orientation implies a change on SRP magnitude only, moving the position of equilibria back and forth along the x axis. If $\rho \neq 0$, changing the sail orientation will displace the equilibrium points away from the Sun–asteroid line. For instance, if we tilt the sail vertically w.r.t. the Sun–line (i.e. $\alpha = 0, \delta \neq 0$), we displace the equilibria outside the orbital plane, while if we tilt the sail horizontally w.r.t. the Sun–line (i.e. $\alpha \neq 0, \delta = 0$), we displace the equilibria away from the x axis inside the orbital plane. On the right-hand side of Fig. 2, we have the equilibria on the orbital plane ($Z = 0$) for $\rho = 0.85$, $\bar{\beta} = 10, 20, 30, 50, 75, 100$, $\alpha \in [-\pi/2, \pi/2]$ and $\delta = 0$, where the dashed lines represent values of α between $\pm\pi/4$. We note that: for each $\bar{\beta}$, the equilibrium point closer to the origin corresponds to $\alpha = 0$, and the furthest equilibrium point corresponds to $\alpha = \pm\pi/2$; and the position of the equilibrium points is symmetric with respect to $Y = 0$, hence, if (x_0, y_0, z_0) is a equilibrium point for α_0, then $(x_0, -y_0, z_0)$ is also an equilibrium point for $-\alpha_0$.

Finally, changes on the reflectivity parameter ρ will change the maximal displacement of the "artificial" equilibria with respect to the x-axis. The larger the reflectivity coefficient the more we will be able to displace the equilibrium point from the x axis. This phenomenon can be seen on the left-hand side of Fig. 2, where we have the equilibria on the orbital plane ($Z = 0$) for $\bar{\beta} = 75$, $\alpha \in [-\pi/2, \pi/2]$, $\delta = 0$ and $\rho = 0, 0.15, 0.3, 0.5, 0.75, 0.85$ and 1. Although we just plot the behavior of the equilibrium points for $\delta = 0$, the same effects, for a slightly different scale, are observed when $\alpha = 0$ and $\delta \neq 0$ where the equilibria lie on the $Y = 0$ plane [8].

When we study the stability of these equilibrium points we see that they are all unstable, and that their instability increases as the position of the fixed point comes close to the small body (i.e. as $\bar{\beta}$ increases). This can be seen in Fig. 3, where we plot the maximum absolute value of the eigenvalues $|\lambda|$ versus the x coordinate of the equilibrium points. The equilibria represented in this figure are the same as the ones on Fig. 2.

An important remark is that even for large SRP accelerations, the linear dynamics for all the equilibrium points

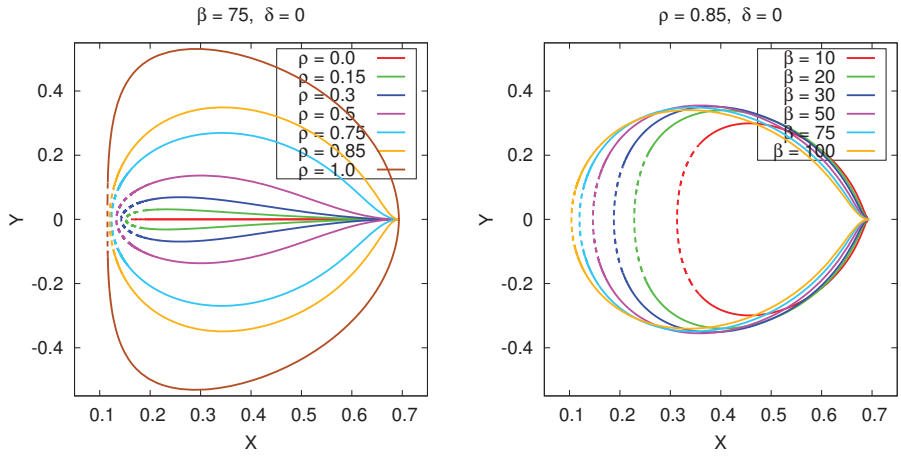

Fig. 2: Equilibrium points on the orbital plane $Z = 0$ (i.e. $\alpha \in [-\pi/2, \pi/2], \delta = 0$) for $\bar{\beta} = 75$ and $\rho = 0, 015, 0.3, 0.5, 0.75, 0.85, 1$ (left), and for $\rho = 0.85$ and $\bar{\beta} = 10, 20, 30, 50, 75, 100$ (right).

has the same structure: saddle×center×center, so the typical families of periodic orbits present in the AH3BP for smaller values of SRP are still present. Namely, the planar and vertical Lyapunov families, as well as the two Halo families, which we will describe in the next section.

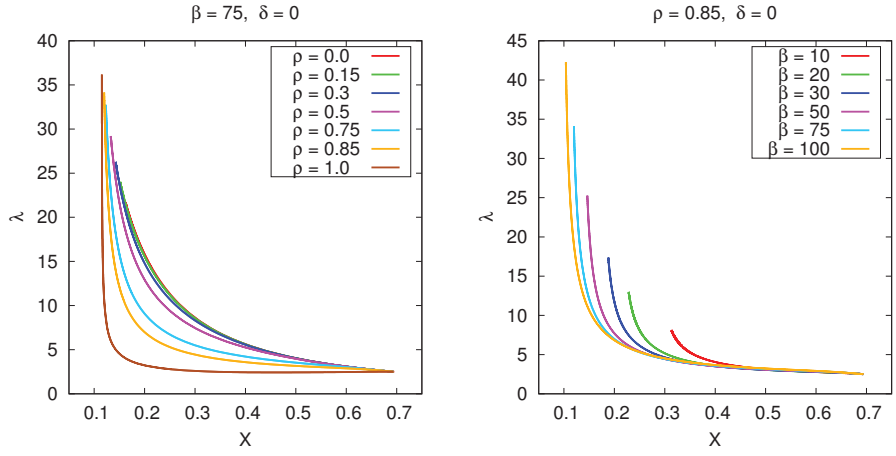

Fig. 3: Maximum absolute value of the eigenvalues, $|\lambda|$, vs the x coordinate of the equilibrium point for: $\bar{\beta} = 75$ and $\rho = 0, 015, 0.3, 0.5, 0.75, 0.85, 1$ (left) and for $\rho = 0.85$ and $\bar{\beta} = 10, 20, 30, 50, 75, 100$ (right).

Periodic Orbits

As mentioned in the previous section, the equilibrium points described are of the type saddle×center×center. Moreover, the system is Hamiltonian for any set of parameters for the sail. Hence, Lyapunov's Center Theorem [10] ensures that, under generic non-resonant conditions, two families of periodic orbits emanate from the fixed point.

Each family is related to one of the two complex pairs of eigenvalues $(\pm \mathrm{i}\omega_1, \pm \mathrm{i}\omega_2)$, and the period of the orbits tends to $2\pi/\omega_i$ when approaching the equilibrium point.

In Fig. 4 we plot for different values of β, ρ, α and δ the difference between the two complex eigenvalues $(\omega_1 - \omega_2)$. When ω_1 and ω_2 coincide, we cannot guarantee the existence of periodic orbits and a more detailed study of this case should be done. Notice in Fig. 4 that this is the case for $\alpha \neq 0, \delta = 0$ (i.e. equilibrium points on the $Z = 0$ plane) where for $\rho \geq 0.5$ we find at least four values for α where this is true. It is not the case for $\alpha = 0, \delta \neq 0$ (i.e. equilibrium points on the $Y = 0$ plane), where $|\omega_1 - \omega_2|$ has a minimum for $\alpha = \delta = 0$, whose value tends to zero as $\bar{\beta}$ increases, but is positive always positive. Still in the $\alpha = 0$ case, the difference between the two center oscillations increases as we move along the family of equilibria.

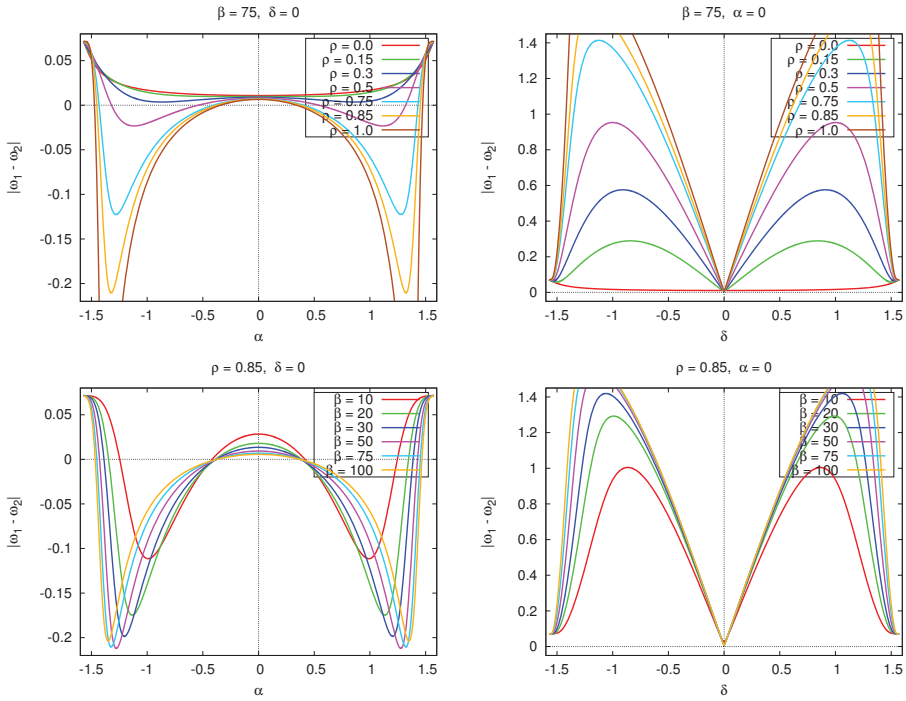

Fig. 4: Difference between the two complex eigenvalues $\omega_1 - \omega_2$ vs α (left) and δ (right) for: $\bar{\beta} = 75$ and $\rho = 0, 015, 0.3, 0.5, 0.75, 0.85, 1$ (top) and for $\rho = 0.85$ and $\bar{\beta} = 10, 20, 30, 50, 75, 100$ (bottom).

Let us describe the families of periodic orbits for $\bar{\beta} = 75$ and their dependence with respect to the sail orientation α, δ and the reflectivity ρ. Taking $\bar{\beta} = 75$ corresponds to a solar sail with an area-to-mass ratio similar to Ikaros on a mission to a middle size asteroid like Eros, or a smaller spacecraft with an area-to-mass ratio like Hayabussa close to smaller asteroids like Golevka (Fig. 1).

To compute the families of periodic orbits for different sail orientations, we will consider appropriate Poicaré sections to ensure the transversality of the flow. For all the computations we have taken as Poincaré section a plane perpendicular to $\{Z = 0\}$ and contains the origin and the equilibrium point for the given sail parameters. Hence, if

(x_0, y_0, z_0) is an equilibrium point for $(\alpha_0, \delta_0, \rho_0)$, then we consider as Poincaré section: $\Gamma = \{x_0 Y - y_0 X = 0\}$. For each family of periodic orbits, we use the linear approximation of the flow around (x_0, y_0, z_0) to compute the first orbit. Then we use a continuation method to compute the rest of orbits of the family, taking the x coordinate of the point on the Poincaré section, Γ, as the continuation parameter.

On the left-hand side of Fig. 5 we plot the continuation diagram of the two families of periodic orbits for $\alpha = \delta = 0$ and $\rho = 0.85$, i.e. the displaced L_2. Each point corresponds to the x and z components of a periodic orbit on the Poincaré section. We have assigned a color to each orbit depending on the linear dynamics around the periodic orbit: blue stands for saddle×center, red stands for saddle×saddle, and green stands for center×center. As we can see, when the planar family crosses a 1:1 resonance, two families of Halo-type orbits are born. As the two center librations of the displaced L_2 point are close a 1:1 resonance ($\mathtt{i}\omega_1 = \mathtt{i}24.1591043, \mathtt{i}\omega_2 = \mathtt{i}24.1522099$) the two Halo orbits bifurcate from the planar family very close to the equilibrium point. Their bifurcation takes place at the rightmost point of Fig. 5 left. At the scale of this Figure, it is indistinguishable from the birth of the planar and vertical Lyapunov families. On the right-hand side of Fig. 5 we plot the variation of the *stability parameters*[6], $s_{1,2}$, along the family.

Both the planar and vertical families of periodic orbits are linearly unstable. The planar family is center×saddle close the equilibrium point, but as it crosses the 1:1 resonance, a bifurcation occurs and the orbits become saddle×saddle. The Halo orbits that are born inherit the center×saddle dynamics at first, but as we move along the family they become stable. Nevertheless, notice that, for each family, instability decreases as we reach the terminating orbits.

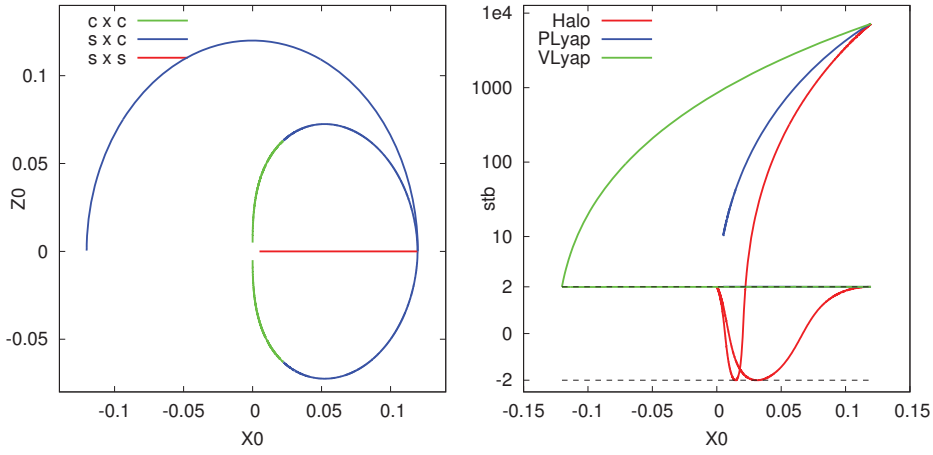

Fig. 5: For $\bar{\beta} = 75$, $\alpha = \delta = 0$ and $\rho = 0.85$. Left: continuation diagram of the families of periodic orbits, Right: variation of the stability parameters for the orbits on the family.

In Fig. 6 we show the X, Y, Z projection for these three type of orbits. Form left to right we have the Planar,

[6]The *stability parameters* of a periodic orbit are defined as $s_i = \lambda_i + \lambda_i^{-1}$, for $i = 1, 2$, where λ_i belongs to the spectra of the monodromy matrix associated to a periodic orbit. As the system is autonomous and Hamiltonian with two degrees of freedom, if M is the monodromy matrix of a periodic orbit, $\mathrm{Spec}(M) = \{1, 1, \lambda_1, \lambda_1^{-1}, \lambda_2, \lambda_2^{-1}\}$.

Vertical and Halo type orbits. We observe that the orbits on the planar family are on the orbital plane ($Z = 0$), and have a boomerang shape, that gets more elongated as we move along the family, and leaves the asteroid in the middle. The orbits on the vertical family have the familiar eight shape, spending half of their orbital period above the orbital plane, and as we move along the family the orbits loose vertical displacement, to become almost planar close to their terminator. Finally, the two families of Halo orbits are almost circular and remain on the shadow side of the asteroid.

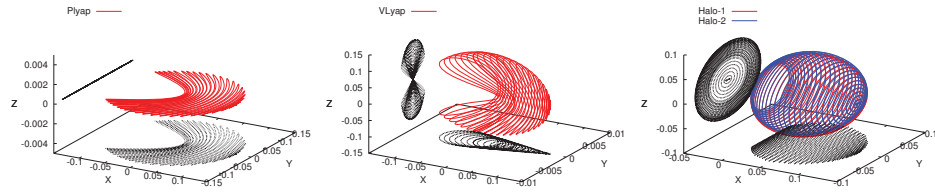

Fig. 6: Family of periodic orbits for $\bar{\beta} = 75$, $\alpha = \delta = 0$ and $\rho = 0.85$. From left to right: Planar Lyapunov orbits, Vertical Lyapunov orbits and Halo-type orbits

As it happened with the equilibrium points, changing the sail orientation allows us to displace these families of periodic orbits. Small changes on the sail orientation will not change the general structure of the families, and we will still find the three kind of periodic orbits (Planar, Vertical and Halo). If we take $\alpha = 0$ and $\delta \neq 0$, we will displace the families of periodic orbits above the orbital plane. As it happened in [11], there is a symmetry breaking on the equations for $\delta \neq 0$, which changes the way in which these orbits relate. The planar family of orbits emanating from the fixed point starts almost planar, but as we move along the family, these orbits start to gain vertical amplitude in such a way that they can be considered Halo orbits. Nevertheless, the planar orbits still exist, and can be found by continuing planar orbits for $\delta = 0$ with respect to δ. These orbits still have the boomerang shape and are slightly inclined with respect to the orbital plane. The orbits in the vertical family still have an eight shape, but the loops are slightly asymmetric. If we take $\delta = 0$ and $\alpha \neq 0$, we displace the families of periodic orbits sideways with respect to the Sun–asteroid line. The overall structure is the same as for $\alpha = \delta = 0$, i.e. we have the planar and vertical families of orbits, and at the 1:1 resonance between the planar and vertical frequencies the Halo orbits appear. Planar orbits still have the boomerang shape, orbits in the vertical family have an eight shape and Halo orbits are almost circular and perpendicular to the orbital plane. The main difference is that they are tilted, but still approach the small body as we move along the family.

We must mention that, for large changes of the sail orientation, the nature of the families periodic orbits that we have just described can drastically change. For instance, for $|\alpha| > 0.5$, we have observed that, if we continue along the planar family of orbits, we do not cross a 1:1 resonance and the Halo-type orbits do not appear. A study of these phenomena is in progress.

In Fig. 7 we show the X, Y, Z projection of the three type of orbits for $\alpha = 0.2$rad, $\delta = 0$rad and $\rho = 0.85$. We can see how the orbits have the same shape as in Fig. 6 but they are tilted with respect to the asteroid–equilibrium

point line. In Fig. 8 we show the X, Y, Z projection of the three type of orbits for $\alpha = 0$rad, $\delta = 0.02$rad and $\rho = 0.85$. Observe that here the orbits are tilted above the orbital plane.

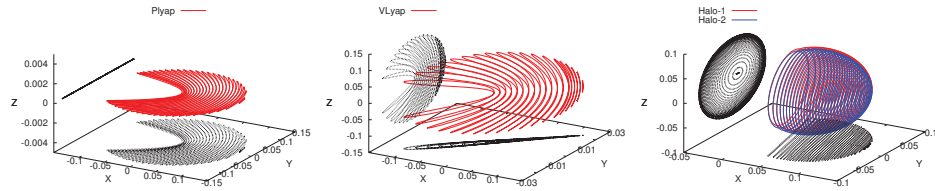

Fig. 7: Family of periodic orbits for $\bar{\beta} = 75$, $\alpha = 0.2, \delta = 0$ and $\rho = 0.85$. From left to right: Displaced planar Lyapunov orbits, displaced vertical Lyapunov orbits and displaced Halo-type orbits

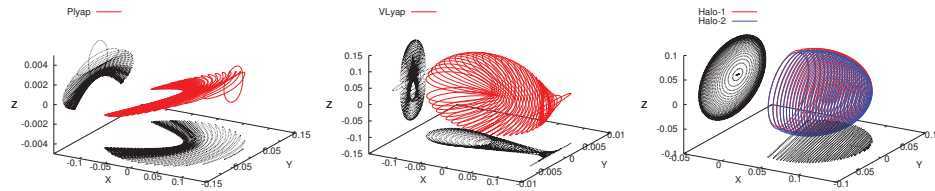

Fig. 8: Family of periodic orbits for $\bar{\beta} = 75$, $\alpha 0, \delta = 0.02$ and $\rho = 0.85$. From left to right: Displaced planar Lyapunov orbits, displaced vertical Lyapunov orbits and displaced Halo-type orbits

Finally, a change on the reflectivity parameter ρ, will also displace these families of periodic orbits. If we look at Eq. (3), we notice that if we change ρ we are changing the direction of a_{SRP}. Hence, we can think of a change in ρ as a change in the sail orientation. In Fig. 9 we plot several planar and vertical orbits from $\bar{\beta} = 75, \alpha = 0.2, \delta = 0$ and different values of ρ. As we can see, the effect of varying the reflectivity is similar to the one obtained varying the sail orientation. The periodic orbits are tilted towards the Sun–asteroid line as ρ tends to 0.

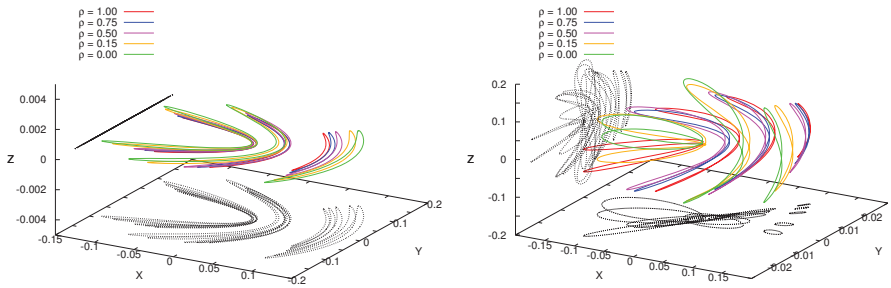

Fig. 9: Periodic orbits for $\bar{\beta} = 75$, $\alpha 0.2, \delta = 0$ and $\rho = 0, 0.15, 0.5, 0.75$ and 1. Left: displaced planar Lyapunov orbits and right: displaced vertical Lyapunov orbits

Some comments on the Terminator Orbits

If we were to plan a mission to an asteroid with a solar sail, there are a several restrictions that must be taken into account. For instance: orbiting close to the asteroid to allow more detailed scientific observations; keeping the solar sail almost perpendicular to the SRP to magnify the performance of the sail; and being able to observe the day side of the asteroid.

As we know, L_2 and "artificially" displaced equilibrium points are on the shadow side of the asteroid, which could compromise an observational mission [8]. On the other hand, some of the periodic orbits that we have just described spend a part of their orbital period on the day side of the asteroid, and could be used for observational missions at asteroids. This is the case for the terminator orbits from the planar Lyapunov and vertical Lyapunov families. In Fig. 10 we show one orbit of each family that spends some time on the day side of the asteroids.

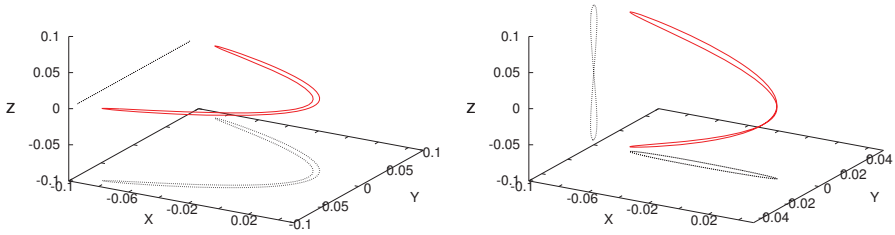

Fig. 10: A planar Lyapunov (left) and a vertical Lyapunov (right) periodic orbit on the terminator family that spends more that half of its period in the day side of the asteroid

We have done some preliminary tests on the feasibility of these orbits, checking the maximum and minimum distance to the center of the asteroid, as well as the amount of time each orbit spends on the day side of the asteroid. We have seen that, for $\bar{\beta} = 75, \rho = 0.85$ and $\alpha = \delta = 0$, we can find terminator orbits in the planar and vertical Lyapunov families that have the closest approach with the small body at a distance of 0.01 or less normalized units (that correspond to 4 radii in the case of Eros), and a maximum distance from the asteroid of ≈ 0.12 normalized units ($\approx$48 radii for Eros). Finally, the terminator planar Lyapunov orbits spend more than 90% of their orbital period on the day side, and their average period is of ≈ 0.15 (around 15 days for Eros). The terminator vertical Lyapunov orbits spend around 70% of their orbital period on the day side, and have an average period of ≈ 0.11 (around 11 days for Eros).

Conclusions and Future Work

In this paper we have considered the Augmented H3BP as a model for the dynamics of a solar sail close to an asteroid, and we have described some aspects of the natural dynamics of the system for the different parameters defining the performance of the solar sail and its orientation. We have computed fixed points and families of periodic orbits. Some of these periodic orbits, such as the terminator planar and vertical Lyapunov orbits, spend most of their orbital period on the day side of the asteroid and could be interesting for observational missions to

asteroids. Nevertheless, these orbits are unstable and require control strategies. A study on the controllability of these orbits is progress.

There are many aspects on the dynamics of the Augmented H3BP that still need to be addressed. In the near future, we plan to do a complete study on the non-linear dynamics around the equilibrium points. We are currently performing a reduction to the center manifold, to be able to have a complete description of the periodic and quasi-periodic motion. Moreover, the stable and unstable manifolds of the equilibrium points and periodic orbits, can be interesting for low-energy transfers or approximation maneuvers to the surface of the asteroid. Finally, it is known that some asteroids follow very eccentric orbits around the Sun. The variation of the distance to the Sun also plays an important role in the dynamics of the system [9], and should be taken into account.

Acknowledgments

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Variable-geometry solar sailing: the possibilities of the quasi-rhombic pyramid

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Variable geometry solar sailing potentially offers enhanced delta-V capabilities and new orbital solutions. We propose a device with such capabilities, based upon an adjustable quasi-rhombic pyramid sail geometry, and examine the benefits that can be derived from this additional flexibility. The enabling technology for this concept is the bevel crux drive, which can maintain tension in the solar sail across a wide range of apex angles. This paper explores the concept of such a device, discussing both the capabilities of the architecture and the possibilities opened up in terms of orbital and attitude dynamics.

I. Introduction

SOLAR sails have long been proposed as a mechanism for interplanetary travel [1], using sunlight to accelerate across space unconstrained by propellant reserves. In more recent years the opportunities that solar sails provide in terms of highly non-Keplerian behavior, with applications such as displaced orbits [2] and polar loitering [3], have come to be recognized, and attractive concepts such as orbit raising from low Earth orbit [4] and inclination change [5] have been proposed.

To achieve these objectives, some form of control over the direction and magnitude of the thrust produced by the solar sail has usually been required [6]. It has been suggested that the solar sail can be tilted, using masses or propellant, as part of an active attitude and orbit control system [7] and that the thrust vector could be modified by changing the reflectivity of the membrane [8].

However it is now proposed that, in some cases, management of both the direction and magnitude of the thrust vector could be achieved without resorting to the use of propellant, control masses and exotic membranes equipped with liquid crystals or e-ink. This could be done by combining a "heliostable" solar sail, which is to say

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a shape that, when deflected, produces a restoring moment back towards he sun, with a geometry that can vary the apparent size and inclination of the solar sails without rotation of the bus.

Such a geometry might be realized in a quasi-rhombic pyramid (QRP), in which the spacecraft bus would lie at the apex and deploy booms along the slant edges, with the membranes filling the slant faces. By considering the four booms to be arranged in two opposing pairs, and starting in the degenerate case of a square-based pyramid, it is apparent from Fig. 1 that increasing the flare angle of one opening boom pair (orange arrows) and simultaneously reducing the flare angle of the other closing pair (blue arrows) will serve to reduce the area presented to the sun without creating any net torque about the apex. Provided that the angular positions of the opening and closing pairs are carefully matched, the membrane will remain taut and the heliostable behavior will be maintained throughout, albeit to diverging extents in the two orthogonal planes.

This paper seeks to outline the capability that such variable-geometry solar sailing could provide to nanosat-sized spacecraft operating in near-Earth space. Deploying sail-like structures from a nanosats can be done [9], but most architectures offer only a simple deploy-and-forget behavior similar to that provided by the commercial AEOLDOS module. In these cases, multiple booms are often spooled around a shared hub to reduce complexity and provide some measure of mutual support.

We therefore propose a new mechanical arrangement, the bevel crux drive (BCD), that builds on traditional deployment concepts but which provides the flexibility required to vary the boom angles independently after deployment. This is achieved by deploying each boom from its own dedicated spool, the spool rotating when released due to strain energy stored in the deflected boom itself. Each boom is deployed tangentially to the spool and is constrained by a cage of roller bearings, as has been proposed elsewhere [10], with mutual support between the spools now being provided by linking each to its neighbors through a simple arrangement of bevel gears. After the deployment is complete the spools become fixed but each circumferential cage of roller bearings is permitted to rotate about its host spool over a short angular distance. This permits independent pointing of each deployed boom.

The BCD arrangement has an additional advantage in that, in contrast to some similar deployment architectures [11], the deployed sails exhibit perfect radial symmetry. This means that fold patterns can be simplified, which facilitates packing, and the potential for significant out-of-plane forces that could tend to buckle the booms during deployment is reduced.

This paper investigates these factors in some detail and goes on to consider how a QRP-sail might be used to raise the orbit of a small satellite in LEO. We suggest that the solar sail could be opened such that the solar

radiation pressure may provide an accelerating force as the satellite moves away from the sun and then closed to minimize the braking force as it moves towards the sun in the second half of the orbit, the heliostable behavior and onboard dampers keeping the apex pointed approximately towards the SRP vector throughout.

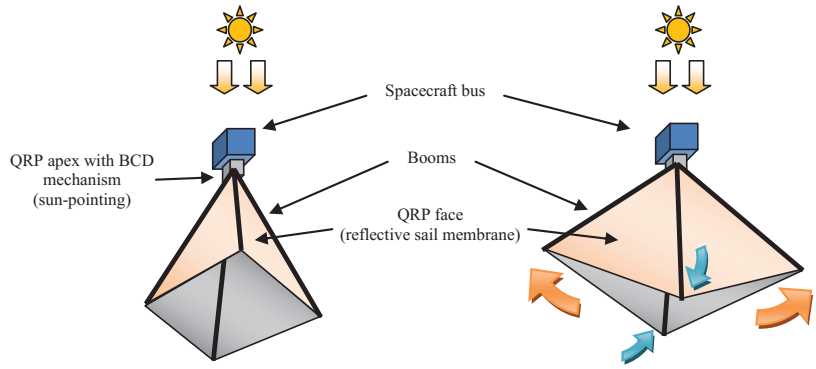


Fig. 1 The quasi-rhombic pyramid concept.

II. Geometry and mass properties

This paper will consider that the booms have already fully deployed, and all the analyses will be done in this configuration. We assume the (extended) booms have the same length, which is set to l. Since the sail membrane must be always tensioned, the size of each triangular face of the QRP cannot be altered. This means that the distance between any two adjacent boom tips (e.g. A and B) is constant and set to b, see Fig. 2 (which also shows the body axes).

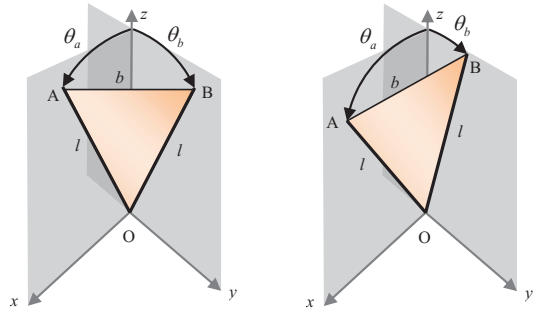


Fig. 2 One face of the QRP, in two different configurations. Highlighted are the body axes and the planes *x-z* and *y-z* in which the two booms move.

The area of each triangular face, which can be computed through Heron's formula, is:

$$S = \frac{b}{4}\sqrt{4l^2 - b^2}$$

The control of the sail is done by varying the flare angle of the booms with respect to the spacecraft body axis *z*. Because of symmetry, opposite booms must have the same flare angle at all times, hence we can consider the angles of the two adjacent booms, θ_A, θ_B. Referring to Fig. 2, the Cartesian coordinates in body frame of their tips A and B are functions of these angles:

$$\mathbf{r}_A = l\begin{bmatrix}\sin\theta_A \\ 0 \\ \cos\theta_A\end{bmatrix}; \quad \mathbf{r}_B = l\begin{bmatrix}0 \\ \sin\theta_B \\ \cos\theta_B\end{bmatrix} \tag{1}$$

By writing the expression for the Cartesian distance between *A* and *B*, setting it equal to *b* and squaring:

$$(b/l)^2 = \sin^2\theta_A + \sin^2\theta_B + (\cos\theta_A - \cos\theta_B)^2$$

Assuming boom A is the driver, and boom B is driven, we can derive an expression that defines the flare angle of B, for any flare angle of A, in order to maintain the sail tension:

$$\theta_B = \arccos\frac{2-(b/l)^2}{2\cos\theta_A} \tag{2}$$

The maximum value for *b/l* is clearly $\sqrt{2}$, such that both θ_A and θ_B are acute. In such case, each face is an isosceles right triangle. All four booms have the same flare angle when $\theta_A = \theta_B$, or:

$$\cos\theta_{A,B} = \sqrt{\frac{2-(b/l)^2}{2}} \tag{3}$$

Taking the (time) derivative of Eq. (2), allows to find the gear ratio to connect the driver booms to the driven booms:

$$\dot{\theta}_B = \frac{\sin(\theta_A)\left(\frac{b^2}{l^2}-2\right)}{2\cos(\theta_A)^2\sqrt{1-\frac{\left(\frac{b^2}{l^2}-2\right)^2}{4\cos(\theta_A)^2}}}\dot{\theta}_A \tag{4}$$

Equation (2) also gives a maximum value for the angle of the driver boom A, which is obtained when the driven boom B is fully closed, i.e. $\theta_B = 0$. With this condition, we can write (for both boom *A* and *B*):

$$\theta_{A,B} \le \arccos\frac{2-(b/l)^2}{2}$$

In the particular case of equilateral face ($l = b$), the maximum angle of the booms is $\pi/3$, while in the (limit) case where $b/l = \sqrt{2}$, then from Eq. (2), $\theta_B = \pi/2$, so the driven boom B is always fully open, regardless of the position of the driver boom A.This corresponds to a square sail that folds along the diagonal.

Fig. 3 shows the relations between angular positions [Eq. (2)] and angular velocities [Eq. (4)] between two adjacent booms, for different values of b/l. Note that when $b/l = \sqrt{2}$, one set of booms is not moving, but it is constantly deployed at 90 deg, and the sail folds along the diagonal of a square.

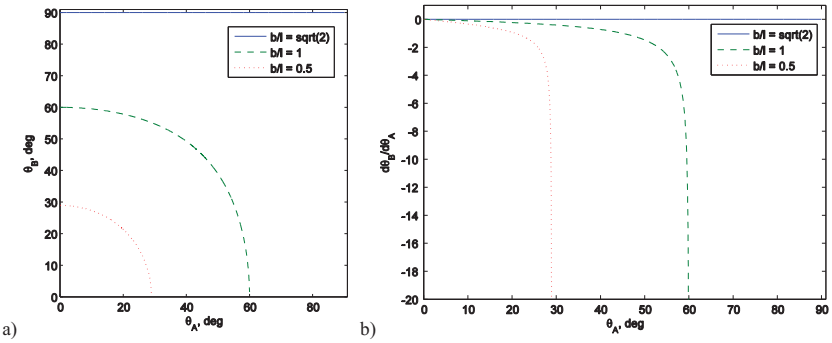


Fig. 3 a) θ_B as function of θ_A for different values of *b/l*. b) Derivative (angular velocity of one boom with respect to the other one).

With no loss in generality, let us consider face AOB. The normal to the face, pointing outwards, is:

$$\hat{\mathbf{n}}_{AOB} = \frac{\mathbf{r}_B \times \mathbf{r}_A}{|\mathbf{r}_B \times \mathbf{r}_A|}$$

The centroid of the face, which is also the geometric barycenter, is simply the average of the three vertices:

$$\mathbf{r}_{CM,AOB} = \frac{1}{3}(\mathbf{r}_A + \mathbf{r}_B + \mathbf{r}_O) \tag{5}$$

Three different sail configurations, varying the flare angles of the booms, are represented in Fig. 4.

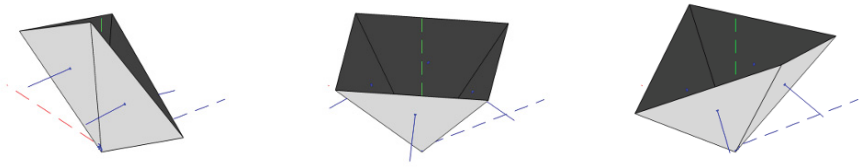


Fig. 4. Sail configurations for $b = l = 1$: a) $\theta_a = 5$ deg; a) $\theta_a = 55$ deg (maximum effective area); c) $\theta_a = 55$ deg. Face normal and principal axes of inertia are also represented.

A. Mass properties

If we consider a uniform areal density σ of the sail material, then this point coincides with the center of mass of the face. The value of σ however shall take into account an average density that considers the mass of the booms. So for a square sail with X-type booms, the equivalent σ, considering a membrane of mass per unit area $\sigma_{membane}$ and booms of mass per unit length ρ_{boom}, is:

$$\sigma = \frac{4S\sigma_{membane} + 4l\rho_{boom}}{4S}$$

The mass of the spacecraft has two contributions: one due to the QRP sail assembly and one due to the spacecraft bus. The bus is considered to be a point mass at (0,0,0) with mass m_{bus} and moments of inertia $\mathbf{I}_{bus}$, with respect to its own center of mass:

$$m = 4S\sigma + m_{bus}$$

The center of mass of the whole spacecraft is then:

$$\mathbf{r}_{CM} = \frac{\sum_{i=1}^{4} \mathbf{r}_{CM,i}\sigma S}{m}$$

It can easily be shown that $r_{CM,x} = r_{CM,y} = 0$, and this will be used for computing the moments of inertia.

B. Moments of inertia

1. Sail face

Due to symmetries, the principal axes of inertia are aligned with the body axes *x*, *y*, *z* in any configuration of the spacecraft. The moment of inertia of one sail face (e.g. AOB), is:

$$I_{AOB} = \int_S \sigma d(\mathbf{r})^2 \, dS \tag{6}$$

where $d(\mathbf{r})$ is the distance of the point to the considered axis of inertia. The integration domain *S* is the triangular sail surface.

Let us introduce a parameterization of the surface (s,t) where *s* is along the edge and *t* is parallel to the base (see Fig. 5). The infinitesimal area of each surface element then becomes:

$$dS = dtds = dt\cos\varphi ds$$

where the angle $\varphi = \arcsin(b/2l)$. Equation (6) can be re-written as:

$$I_{AOB} = \int_0^l \int_0^{\frac{b}{l}s} \sigma d^2 \cos\varphi dtds \tag{7}$$

where $\cos\varphi = \sqrt{1-\frac{b^2}{4l^2}}$. To compute the distance of each point on the face to one of the axes, we shall compute the coordinates of the generic point *R* on the triangle:

$$\mathbf{r}_R = \mathbf{r}_P + \frac{\mathbf{r}_Q - \mathbf{r}_P}{\overline{PQ}} t = \begin{bmatrix} \sin\theta_A \\ 0 \\ \cos\theta_A \end{bmatrix} s + \begin{bmatrix} -\sin\theta_A \\ \sin\theta_B \\ \cos\theta_B - \cos\theta_A \end{bmatrix} \frac{l}{b} t$$

The distance of *R* to the three principal axes of inertia is then:

$$\begin{aligned} d_x^{\,2} &= r_{R,y}^2 - \left(r_{R,z} - r_{CM,z}\right)^2 \\ d_y^{\,2} &= r_{R,x}^2 - \left(r_{R,z} - r_{CM,z}\right)^2 \\ d_z^{\,2} &= r_{R,y}^2 - r_{R,x}^2 \end{aligned}$$

The integral in Eq. (7) is then polynomial in *s*, *t* and can be computed analytically, for given flare angles of the booms.

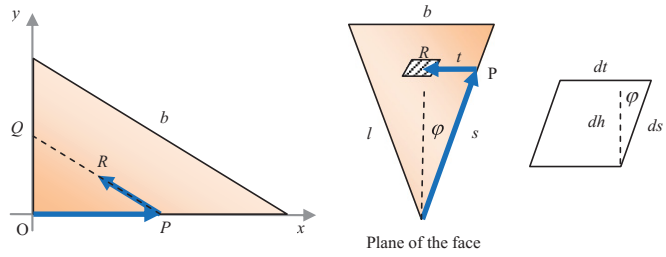


Fig. 5 Parameterization of the sail face.

2. Overall spacecraft

Due to symmetry, each face contributes in the same way to the moment of inertia of the spacecraft about each principal axis. The principal moments of inertia for the whole spacecraft are then:

$$\begin{aligned} I_x &= 4I_{x,AOB} + m_{bus} r_{CM,z}^2 + I_{bus,x} \\ I_y &= 4I_{y,AOB} + m_{bus} r_{CM,z}^2 + I_{bus,y} \\ I_z &= 4I_{z,AOB} + I_{bus,z} \end{aligned} \tag{8}$$

where $\mathbf{I}_{bus}$ are the components of the principal moments of inertia of the spacecraft bus with respect to its own center of mass [assumed in (0, 0, 0)]. Considering a uniform cube of side l_{bus}:

$$I_{bus,x} = I_{bus,y} = I_{bus,z} = \frac{m_{bus} l_{bus}^2}{6}$$

C. Forces and torques

The net force generated by solar radiation pressure (SRP) in ideal conditions, on each face *i*, is [12]:

$$\mathbf{f}_i = -2\eta P_{sun} S \left(\hat{\mathbf{n}}_i \cdot \hat{\mathbf{r}}_s\right)^2 \hat{\mathbf{n}}_i \tag{9}$$

in which $\mathbf{r}_s$ is the sun vector, representing the direction of the sun in the same reference frame (body axes), and P_{sun} is the SRP of the sun, which varies as $1/r_s^2$ and at 1 AU is approximately 4.56 x 10^{-6} N/m^2. η is the efficiency of the solar sail material in terms of reflectivity, ranging from 0.5 (full absorption) to 1 (full specular reflection). Note that diffraction and other effects are not explicitly considered here, but embedded in the efficiency η. The force $\mathbf{f}_i$ is only experienced on a face, if the face is lit by the sun, i.e. $\hat{\mathbf{n}}_i \cdot \hat{\mathbf{r}}_s > 0$. If not, then the face does not provide force and it is not taken into account. The total acceleration experienced by the spacecraft, considering all four faces, is simply the sum of the forces on each face:

$$\mathbf{a}_{sail} = \frac{1}{m} \sum_{i=1}^{4} \mathbf{f}_i \tag{10}$$

Similarly, given the position *C* of the center of mass of the spacecraft, the total torque about that point is:

$$\mathbf{t} = \sum_{i=1}^{4} \left(\mathbf{r}_{CM,i} - \mathbf{r}_{CM}\right) \times \mathbf{f}_i \tag{11}$$

III. Mechanical layout

The solar sail is extended in a largely-passive deployment phase, during which each section of the sail is extended from its stowage volume by the booms on either side; and subsequently articulated, for the remainder of the mission, in an active control phase during which the booms are pivoted about their inboard end.

The deployment phase is a mechanical process that uses a four-boom BCD assembly powered by strain stored in the booms themselves. Subsequently, in the active control phase, the flare angle of each boom may be varied either independently or collectively using one of three proposed electromechanical techniques to position the cage of roller bearings that surrounds each spool. The complexity of these electromechanical processes varies, but each presents its own advantages and disadvantages.

A. Deployment Phase

The BCD is a novel gearing architecture that communicates torques generated by tape spring booms through a series of bevel gear subassemblies arranged in a closed loop, thus providing mutual support, shared momentum and mechanical synchronization between the four spool assemblies.

Although the present work considers a four-boom case, due to the specific requirements of the quasi-rhombic pyramid solar sail, the BCD is a highly flexible architecture that can accommodate multiple booms and thus approximate a cone if required. Fig. 6 illustrates that, provided that the pitch cone angles and subassembly

spacings remain equal around the loop, the relationship between the number of booms and the pitch cone angle is expressed by $N_{boom} = 360°/2\alpha_{cone}$, where N_{boom} is the number of booms arrayed radially and α_{cone} is the pitch cone angle of each bevel gear.

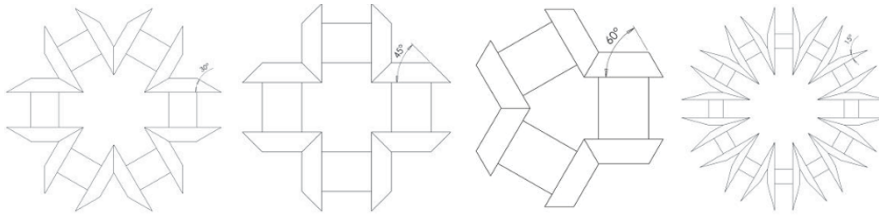


Fig. 6 BCD configurations showing the number of bevel subassemblies with associated pitch cone angles required.

B. Active Control Phase

During deployment, the booms extend tangentially away from their spools and a circumferential cage of roller bearings around each spool is used to control the direction of the tangent. These cages remain fixed during deployment to ensure that the booms extend with a common flare angle, but in the active control phase the cages may be rotated about their spools. As is shown in Fig. 7, this means that the flare angle of each boom can be adjusted independently.

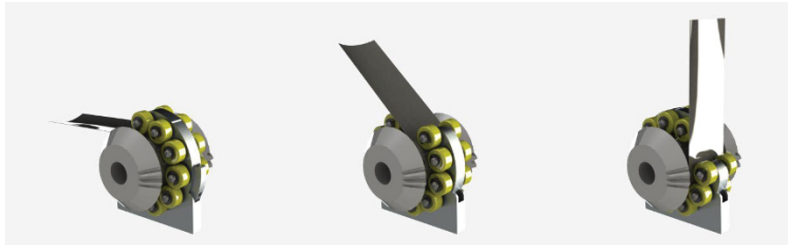
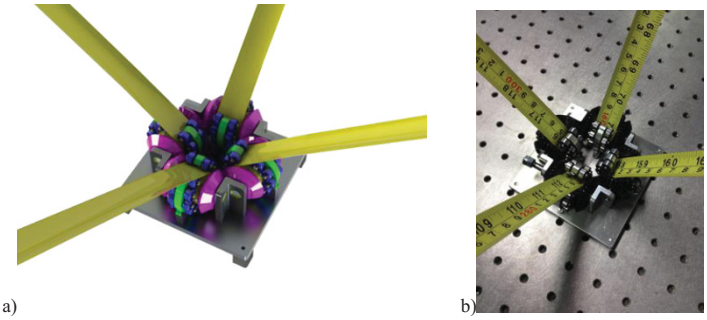


Fig. 7 Variable pitch capability provided by rotating a cage structure about a single spool of the BCD.

Assembling four spool-and-cage units into a BCD layout provides the mechanical basis for the deployment of the quasi-rhombic pyramid from a realistic CubeSat module. Such a device is represented in Fig. 8a, where the yellow booms (in this case, tape-springs) are deployed from purple spools and directed by blue roller bearings in green cages. Fig. 8b shows a physical prototype of this concept, which was approximately 0.6U and which successfully deployed a Mylar membrane against gravity in laboratory conditions.



a) b)

Fig. 8 a) Preliminary concept of the BCD in a 4-boom configuration. b) First technology demonstrator of the BCD in a 4-boom configuration.

C. Articulation

Three specimen articulation methods are explored, namely independent stepper motors for each cage, two stepper motors each linked to a pair of cages, and a single stepper motor mechanically linked to all four cages.

1. Independent Stepper Motors

By employing the approach depicted in Fig. 9a, the flare angle of each boom may be controlled independently via a stepper motor and worm gear engaging with teeth fabricated into the roller cage. Algorithms will be necessary to control each stepper motor and thus ensure that all four booms move in a manner that maintains the desired global shape. This approach has the advantage that, at the end of the deployment phase, the sail need not come under any undue stress because the booms may be arranged such that all the membrane quadrants are initially slack. This could reduce the risk of tearing, with tension subsequently being applied in a controlled fashion via the stepper motors.

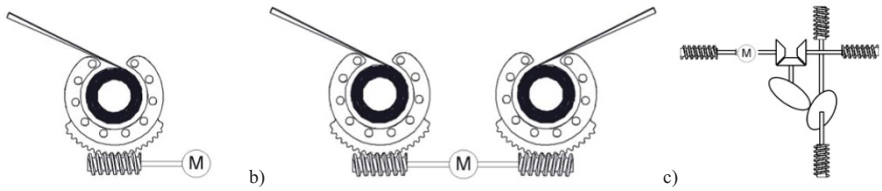
2. Paired Stepper Motors

The number of stepper motors may be halved by appropriately linking each opposing pair of cages, in compliance with the global behaviour of the quasi-rhombic pyramid, through a shaft driven by a single motor. The control algorithm is still required to match the speed of the two shafts, and a stress-relieving procedure at end-of-deployment is still possible. To ensure that both booms in each set rotate in the correct direction from one another when the controlling motor turns either clockwise or anticlockwise it would be necessary to use both a lefthand and a righthand driven worm gear in each set, as indicated by Fig. 9b.

3. Single Mechanically Linked Stepper Motor

It may be possible to eliminate all but one motor by performing the matching function of the control algorithm with a mechanical device such as an elliptical gear train [13]. Fig. 9c shows a schematic of one such

approach, using a combination of bevel and elliptical gears to transmit motion from a single motor to all four worm gears. It should be noted that full rotations are not required in the current architecture and so the elliptical gears do not have to achieve full rotations in this case. The elliptical gear ratio is given in Eq. (4).



a) b) c)

Fig. 9 a) one motor per cage; b) one motor per pair of cages; and c) a single motor for all four cages.

IV. Spacecraft data

For the simulations shown later in this paper we will use three spacecraft, which differ in dimensions and masses, corresponding to three different levels of technology. The data presented in Table 1 is common to the three, but Table 2 presents dimensions, masses and other quantities that change between the spacecraft models. By using the formulas in Section II, it is possible to compute the values in Table 3, which essentially characterize each individual spacecraft.

As will be explained in detail in a later section, the control law requires two different configurations: one in which the driver booms are fully closed and one in which the booms are fully open (the transition between the two configurations is not considered). In the "closed" configuration, we consider $\theta_A = 5°$, while in the "open" configuration, the booms are such that $\theta_A = \theta_B$, according to Eq. (3), and the frontal sail area is maximized.

Table 1 Common spacecraft data

$\sigma_{membrane}$, g/m^2	13.2
ρ_{boom}, g/m	16.3
Efficiency of the sail, η	0.85

Table 2 Specific spacecraft data

Spacecraft	1	2	3
Boom length, *l*, m	1	2	3
QRP base length, *b*, m	1	2	3
Bus mass, m_{bus}, kg	1	2	3
Bus size, l_{bus}, cm	10	12.6	14.4
Rotational damping coefficient, *c*, Nm/rad s	$1\cdot10^{-6}$	$4\cdot10^{-6}$	$4\cdot10^{-8}$
Sail assembly mass per unit area, σ, kg/m^2	0.050	0.032	0.026

Table 3 Computed data

Spacecraft	1	2	3
Total mass, m, kg	1.088	2.221	3.401
Equivalent flat sail area-to-mass ratio,* AMR_{eq}, m^2/kg (booms open)	0.001	0.002	0.003
Equivalent flat sail area-to-mass ratio,* AMR_{eq}, m^2/kg (booms closed)	0.30	0.60	0.88
Moment of inertia of the damping fluid,* I_f, kg m^2	$1.25 \cdot 10^{-4}$	$3.97 \cdot 10^{-4}$	$7.80 \cdot 10^{-4}$
Principal moments of inertia, **I**, kg m^2 (booms open)	0.0294 0.0294 0.0163	0.281 0.281 0.153	1.120 1.120 0.612

* See following Sections V and VI for definition.

V. Attitude motion

In this section and in the following, simulations of the attitude motion and orbital motion will be presented. Although the two types of motion are in fact simultaneous, for the sake of this simulation we assume that the spacecraft deploys the sail, stabilizes itself in terms of attitude, and then begins the analyzed orbit. In this way, we can consider the orbital motion as being not affected by initial attitude changes.

One of the advantages of the QRP sail is its equilibrium position, under SRP, with the tip pointing towards the sun. An angular displacement from this position results in a heliostable torque in the opposite direction (see Fig. 10) which, if undamped, essentially provides an elastic oscillation about the equilibrium position.

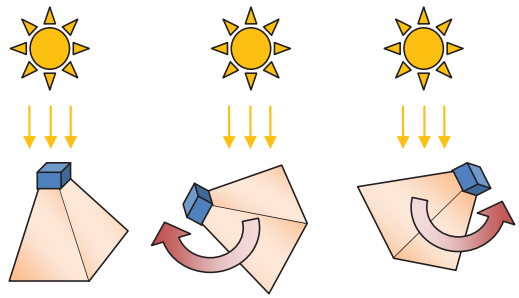


Fig. 10 Sun-pointing effect.

Damping can be obtained with a pair of fluid dampers, one for each body axis x and y, of the type used on spun spacecraft for nutation damping. They usually consist of a sealed ring attached to the spacecraft bus, filled with a viscous fluid. When the spacecraft experiences an angular acceleration in the direction perpendicular to the ring, a difference in angular velocity between the fluid inside the ring and the ring walls is created, which causes viscous friction and therefore dissipates rotational energy as heat.

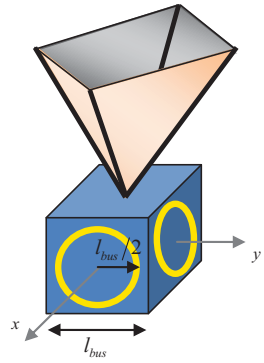


Fig. 11 Annular damping fluid rings (yellow).

The differential equations of motion of the fluid in each ring can be written considering that the fluid is accelerated by viscous effects which depend on the relative velocity between the fluid itself and the ring. Named $\omega_{f,x}, \omega_{f,y}$ the (average) velocity of the fluid in each ring, and $\boldsymbol{\omega}$ the angular velocity of the spacecraft in body axes, we can write:

$$\dot{\omega}_{fx} = \frac{-c\left(\omega_{fx} - \omega_x\right)}{I_f}, \qquad \dot{\omega}_{fy} = \frac{-c\left(\omega_{fy} - \omega_y\right)}{I_f} \tag{12}$$

where c is a coefficient that depends on the fluid viscosity and I_f is the moment of inertia of the fluid in the ring. Considering a ring whose radius is half of the spacecraft bus side (see again Fig. 11), and which has a mass 1/20 of the spacecraft bus mass, we have $I_f = m_{bus}/20 \cdot \left(l_{bus}/2\right)^2$.

The dynamical equation for the angular velocity in body axes $\boldsymbol{\omega}$ of the spacecraft can now be derived. They are essentially the classic Euler equations, combining the fluid rings and the rest of the spacecraft:

$$\begin{aligned}
&I_x\dot{\omega}_x + \left(I_z - I_y\right)\omega_z\omega_y + I_f\dot{\omega}_{fx} - I_f\omega_z\omega_{fy} + t_x \\
&I_y\dot{\omega}_y + \left(I_x - I_z\right)\omega_x\omega_z + I_f\dot{\omega}_{fy} + I_f\omega_{fx}\omega_z + t_y \\
&I_z\dot{\omega}_z + \left(I_z - I_y\right)\omega_z\omega_y + I_f\omega_x\omega_{fy} - I_f\omega_{fx}\omega_y + t_z
\end{aligned} \tag{13}$$

The SRP torque **t** is computed in Eq. (11) and matrix of inertia **I**, in principal inertia axes, in Eq. (8). Equations (12) can be solved for $\dot{\omega}_{fx}, \dot{\omega}_{fy}$, and these values substituted in Eqs. (13). Equations (12) and (13) can therefore be integrated in time to determine the attitude motion of the spacecraft under SRP.

It is possible to show that there is an optimal value for the fluid coefficient, c, to maximize the damping action, and that this value depends on the moments of inertia of the spacecraft. In essence, a small value of c will result in a negligible acceleration of the fluid within one period of the oscillation, while a great value of c will

result in a fluid that is almost rigidly connected with the ring walls. In either case, the damping action is diminished.

Attitude parameterization is performed with quaternions [14]. A quaternion $\mathbf{q}$ defines the attitude of the body reference frame with respect to the inertial one. The dynamics of the quaternions can be expressed as a function of the angular velocity $\boldsymbol{\omega}$ in body axes $\dot{\mathbf{q}} = \dot{\mathbf{q}}(\boldsymbol{\omega}, \mathbf{q})$ (see Ref. [14], pp. 104-110), and this equation can therefore be integrated over time, together with the Euler equations (13). The attitude parameterization is used to compute the direction of the sun $\hat{\mathbf{r}}_s$ in body axes, as required in Eq. (9). In general, from the quaternions, it is possible to compute the direction cosine matrix as $\mathbf{A} = \mathbf{A}(\mathbf{q})$, and therefore we obtain the sun vector in body axes, $\hat{\mathbf{r}}_s = \mathbf{A}\hat{\mathbf{r}}_s^{(ECI)}$. The superscript *(ECI)* denotes an inertial frame (and we assume the direction of the sun is known and fixed in this frame), e.g. Earth-Centered Inertial. For the sake of the attitude simulation, we can arbitrarily choose $\hat{\mathbf{r}}_s^{(ECI)} = \begin{bmatrix} 0 & 0 & -1 \end{bmatrix}^T$.

A. Results

Results are presented here for each one of the three spacecraft in the open configuration. The spacecraft initial position is rotated of an angle of 30 deg around its *y* axis, and the initial angular velocity of the spacecraft (and the fluid dampers) is initially zero. The damping effect is different on the three spacecraft due to their various moments of inertia, which differ by about one order of magnitude each time (see Table 3). For spacecraft 1, after about 60 hours the amplitude of the oscillations is reduced to about 3 deg, and this is considered a sufficient pointing accuracy (see Fig. 12a, which represents the angular velocity of the spacecraft and the damping fluid). A closer inspection to the first periods of the motion (see Fig. 12b) highlights the velocity of the spacecraft with respect to the fluid: their difference is related to the instantaneous dissipation. For spacecraft 2 and 3, the dissipation becomes less and less effective: after 6 days, the amplitude of the oscillations is about 5 deg and 15 deg respectively. For reference, the torque experienced by each spacecraft at 30 deg tilt are: -1.50×10^{-6}, -1.18×10^{-5}, -3.94×10^{-5} Nm^2 respectively.

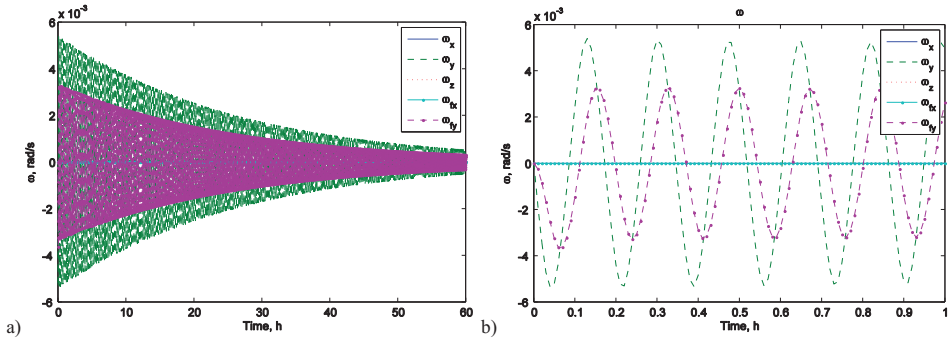


Fig. 12 Angular velocity of spacecraft and damping fluid for spacecraft 1 on initial displacement of 30 deg in *y*. a) 6 hours; b) a magnification of the first hour.

With the selected initial conditions, and since the *y* axis is a principal axis, no motion around *x* is involved. A coupled movement can be obtained by initiating the simulation with the spacecraft rotated around an arbitrary axis, for example $[0.5, 0.866, 0]^T$. This case, for spacecraft 1, is represented in Fig. 13, which shows how both annular dampers are effective. Although it is not visible from the figure, the damping time is very similar to the previous case.

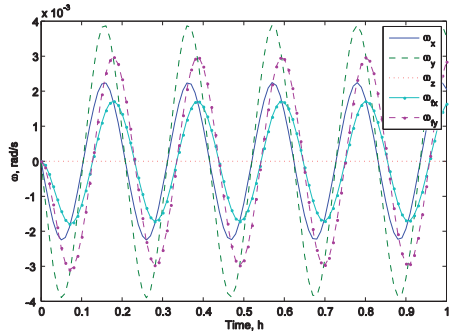


Fig. 13 Angular velocity of spacecraft and damping fluid for spacecraft 1 on initial displacement of 30 deg around axis [0.5, 0.866, 0].

VI. Orbital motion

For the orbital motion of the spacecraft we consider the action of the gravity of the Earth (considered a point mass with planetary constant $\mu = 3.986 \cdot 10^5$ km^{3}s^{-2}) and the effect of SRP.

Given the heliostable behavior of the spacecraft explained above, we consider the orbital simulation when the attitude transient phase is concluded and the spacecraft is in an equilibrium attitude with the tip of the sail towards the sun. This attitude is maintained in an inertial, Earth-centered reference frame, while the sun direction

rotates around the Earth, so with a period of one year the equilibrium attitude of the spacecraft will execute a complete revolution.

Under this scenario, the net force experienced by the spacecraft due to SRP can be thought as one given by a flat sail, facing the sun, of the same efficiency η and equivalent area $AMR_{eq} = \sum_{i=1}^{4} \mathbf{f}_i \Big/ 2\eta P_{sun} m$ (see again Table 2).

The differential equation of motion of the spacecraft is:

$$\ddot{\mathbf{r}}^{(ECI)} = -\frac{\mu}{r^2}\hat{\mathbf{r}}^{(ECI)} - a_{sail}\hat{\mathbf{r}}_s^{(ECI)} \tag{14}$$

where $\mathbf{r}$ is the position vector in an Earth-centered, equatorial inertial (ECI) reference frame and a_{sail} is the magnitude of the acceleration due to SRP computed in Eq. (10). The direction is opposite to the direction of the sun, as explained.

The direction of the Sun, in ECI and as function of time, assuming a circular Earth orbit around the sun, is given by [15]:

$$\hat{\mathbf{r}}_s^{(ECI)} = \begin{bmatrix} \cos\lambda & \sin\lambda\cos\varepsilon & \sin\lambda\sin\varepsilon \end{bmatrix}^T, \qquad \lambda = \lambda_0 + t\dot{\lambda}$$

where λ is the longitude of the sun, $\dot{\lambda}$ is the angular velocity of the Earth around the sun (assumed constant at one revolution per year), and $\varepsilon = 23.5°$ is the obliquity of the equator on the ecliptic plane. By choosing $\lambda_0 = 0$, then at time $t = 0$ the sun is at the vernal equinox in the equatorial plane.

Due to the small perturbing acceleration with respect to the gravitational acceleration, the Keplerian elements of the orbit a, e, i, Ω, ω (apart from the true anomaly f) change very slowly. For this reason, the integration is performed using Gauss' variational equations (Ref. [16], pp. 488-489). In this formulation, the perturbing acceleration $\mathbf{a}_{sail}$ is decomposed in tangential $\hat{\mathbf{t}} \equiv \hat{\mathbf{v}}$, normal $\hat{\mathbf{n}}$ and out-of-plane direction $\hat{\mathbf{h}}$, with respect to the orbital plane (*tnh*). This can be obtained through the rotation matrix $\mathbf{B} = \begin{bmatrix} \hat{\mathbf{v}} & \hat{\mathbf{n}} & \mathbf{h} \end{bmatrix}$ from ECI system to *tnh*, as $\mathbf{a}_{sail}^{(tnh)} = \mathbf{B}^T \mathbf{a}_{sail}^{(ECI)} = -\mathbf{B}^T a_{sail} \hat{\mathbf{r}}_s^{(ECI)}$.

A. Control law

When the sun direction is aligned with the pyramidal axis, the sail acceleration naturally depends on the length of the booms *l* and *b,* but also depends on the instantaneous angle of the booms. Thus, varying the angle of the booms, according to the law in Eq. (2), can be used to change the acceleration experienced by the spacecraft. Indeed, we consider that the spacecraft can control the driver boom angle to toggle between two

values, one in with the boom is fully closed, and the other where the four booms are open to the same angle, according to the value found in Eq. (3). In the former case, the effective area exposed to the sun (and hence the acceleration) is minimized, in the latter maximized (see Fig. 4).

We now consider control laws that can either increase or decrease the semi-major axis. By inspecting the Gauss' equation for semi-major axis it can be seen that, for a maximum change, the sail acceleration shall be tangential. However, the direction of the acceleration in an ECI frame cannot be arbitrarily decided, being always pointing away from the sun. Therefore the switching timing between one configuration or the other can be used as control law.

To increase the semi-major axis, the sail shall be open when the spacecraft is travelling away from the sun, i.e. $\hat{\mathbf{r}}_s \cdot \hat{\mathbf{v}} < 0$. If the sun is in the same plane of the orbit, then there will be one true anomaly in which $\hat{\mathbf{r}}_s, \hat{\mathbf{v}}$ are antiparallel, and thus the only acceleration is tangential. In the other parts of the orbit, it will have a normal component. If the sun is out of the orbital plane, then there will always be an out-of-plane acceleration component. When $\hat{\mathbf{r}}_s \cdot \hat{\mathbf{v}} \approx 0$, the open sail can still provide a (small) contribution to the tangential acceleration, but there will be major normal and out-of-plane components that will not contribute to the semi-major axis change. For this reason, it is decided to open the sail when the following condition is verified:

$$\arccos\left(-\hat{\mathbf{r}}_s \cdot \hat{\mathbf{v}}\right) < \theta_{control}$$

where the angle $\theta_{control} \in \left[0, \pi/2\right]$ can be arbitrarily selected. On the lower end of the interval, the spacecraft receives a quasi-tangential acceleration, but for a limited fraction of the orbit, if any depending on the position of the sun; on the upper end, the spacecraft maximizes the tangential acceleration, but it experiences a considerable amount of other components as well. Research has also investigated optimal values for $\theta_{control}$ [17], however their application is beyond the scope of this paper. The value selected is $\theta_{control} = 80°$, which offers a substantial increase in semi-major axis and very small changes of the other elements. This will result in a control law similar to what is proposed in Ref. [18] for deorbit, by varying the effective area exposed to SRP by changing the pitch of the solar panels.

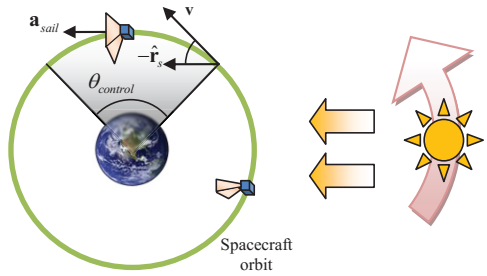


Fig. 14 Scheme of the control law.

By inspecting the Gauss' equation for eccentricity it is clear that a tangential acceleration changes the eccentricity of the orbit. However the change depends on the true anomaly *f*, and in particular for a circular orbit ($e = 0$), the change is positive for half of the orbit, and negative for the other half. If we consider an equatorial or quasi-equatorial circular orbit, where the sun rotates around the orbit and it is in average in the orbit plane over one year, then the contribution to the increase in eccentricity vanishes during a full year, because the tangential acceleration is experienced approximately to the same amount at every anomaly. This allows us to produce a net increase (or decrease) of semi-major axis without substantial change to the other orbital elements, i.e. maintaining the orbit in a circular shape.

B. Orbit simulation

The selected initial orbit is ideally circular and equatorial (zero inclination), but in the simulation an orbit with very small eccentricity and inclination is used, to avoid singularities in the Gauss' equations. These are integrated for one year, using the control law presented above, to produce the following results. Fig. 15 shows the change in semi-major axis, eccentricity and inclination for initial altitude of 600 km. The increase in semi-major axis is almost constant in time and the eccentricity is kept to a very low value, but there is a net increase of inclination of a fraction of a degree. This is due to a seasonal out-of-plane component of the sun vector.

Because atmospheric effects are not considered the increment of the semi-major axis has a very weak relationship to the initial orbital altitude, and Table 4 illustrates that spacecraft 1 climbs at approximately 20 km/year, spacecraft 2 at approximately 40 km/year, and spacecraft 3 at approximately 60 km/year irrespective of initial altitude (see Table 4). It is however expected that, at lower altitudes, atmospheric effects would have a significant effect on the behavior of a practical device.

To estimate the magnitude of this effect we consider an aggressive scenario in which the atmosphere is near its maximum density in LEO, namely 1400h on a day during the solar maximum of 2001. Evaluating the

dynamic pressure in a circular equatorial at 600 km, 800 km and 1000 km, and comparing these values to the solar radiation pressure, we find that the aerodynamic pressure is over 12 times higher at 600 km, falling to 1.3 times higher at 800 km, and only 0.2 times as much at 1000 km. We therefore expect that actual altitude gains would tend towards the values presented in Table 4 above approximately 1000 km, but are aware that a much more complex behavior must be expected at lower altitudes. This is due not only to the competing forces but also to the destabilizing effects of the competing aerodynamic and SRP torques [19].

Finally, we note that altitude reduction, as a part of an end-of-life deorbit scheme, may also be an attractive application. Under these circumstances a different control law would be required to reduce the semi-major axis, and it can be envisioned that the heliostable behavior might be exploited above 1000 km and the aerostable behavior below approximately 600 km, depending on the solar activity. The behavior in the altitude band where aerodynamic and solar radiation pressures are comparable would naturally be more complex but it is worth noting that, if sufficient pointing capability is available, relatively early compliance with the aerodynamic vector may be attractive. This would enable best use of the solar radiation pressure to be made around 0600h, when it is most effective, and best use to be made of the dynamic pressure around 1400–1600h, when that effect is strongest. This approach would be further supported by silvering the front surface of the sail to maximize the solar radiation force in the morning, and blackening the rear surface to minimize the force in the afternoon.

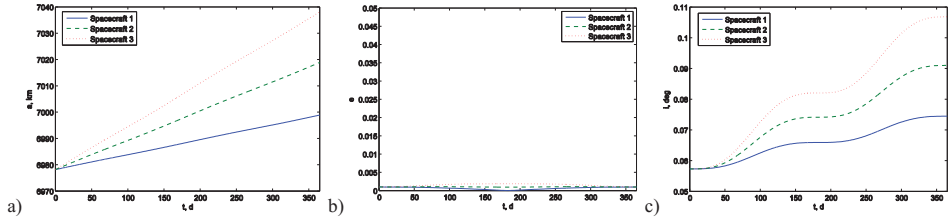


Fig. 15 Change in a) semi/major axis, b) eccentricity, and c) inclination during one year, starting from circular equatorial orbit at 600 km altitude.

Table 4 Altitude after one year (km).

Initial altitude, km	1	2	3
600	620.7216	640.7249	659.9745
800	821.6338	842.4915	862.5775
1000	1022.551	1044.2801	1065.216
1200	1223.4786	1246.098	1267.896

VII. Conclusions

A new concept of quasi-rhombic pyramid (QRP) solar sail for nanosatellites was explored. The spacecraft bus, at the apex of the pyramid, deploys booms along the slant edges, with the membranes filling the slant faces.

The enabling technology of the QRP is the bevel crux drive (BCD) mechanism, which allows the deployment of the quasi-rhombic pyramid sail from a realistic CubeSat-class spacecraft, and to vary the boom angles independently after deployment.

The QRP shape provides a passive, self-stabilizing effect under solar radiation pressure, such that the apex of the pyramid will always point to the sun. In addition, by varying the boom angles, it is possible to change the effective area-to-mass ratio of the spacecraft for orbit control. Although aerodynamic effects may be significant below approximately 1000 km, realistic architectures appear likely to have the capability to raise the orbit of CubeSat-class spacecraft above this altitude by several tens of kilometers per year.

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Real Solar Sails Are Not Ideal, And Yes It Matters

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Ideal solar sails are perfectly flat, rigid, and reflective, can hold arbitrary sun angles perfectly, etc. Real sails are very different. Yet much preliminary mission analysis is done with ideal sails, on the assumption that the only important effect of non-ideal behavior is slightly reduced performance. This assumption is often false. Real sails simply cannot fly some mission plans that look plausible for ideal sails, and in other cases, realistic mission design and realistic sailcraft design much consider some aspects of non-ideal sail behavior from the start.

I. Introduction

The ideal solar sail is a rigid flat surface set at an angle to incoming sunlight, which experiences perfect specular reflection on hitting the sail. Real sails deviate from this simple model in many ways.

There is nevertheless a great temptation to assume an ideal solar sail when doing preliminary mission analysis. The reason is obvious: the ideal sail is so much simpler to analyze. Real sails are complicated and messy. A detailed model of a real sail's properties requires considerable effort and a number of assumptions, is typically too complicated for analytical use, and adds significant overhead to computer simulations.

The usual (often implicit) assumption for preliminary mission analysis is that the various non-ideal aspects of a real sail can be treated as merely reducing its performance somewhat, effectively giving a slightly higher sail loading (slightly less sail area or slightly more mass). This yields a very simple model that is easy to work with.

Unfortunately, this model is *too* simple, and frequently yields over-optimistic results – in some cases, to the point of dangerously misleading the analyst about what is possible and what isn't. This can require significant changes to mission and system designs late in development, which is very undesirable: the later a change is made in a spacecraft program, the more expensive it is, often greatly so.

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We will first examine the details of non-ideal behavior and look at what simplifying assumptions can (and more importantly, *cannot*) be made, and at how simple a model can be and still be reasonably faithful to reality. We will then look at some important consequences that follow from the differences between such a model and the ideal sail, illustrating these with examples from several recent solar-sail design studies carried out in conjunction with the graduate Microsatellite Design Project course at the Space Flight Laboratory (SFL) of the University of Toronto Institute for Aerospace Studies (UTIAS). These include a polesitter mission and a near-Earth asteroid rendezvous mission, in which we have been exploring the potential for applying nanosatellite technology to deep-space solar sail missions.

II. Non-Ideal Sail Behavior

Non-ideal solar-sail behavior broadly falls into three categories: optical, shape, and attitude control.

All sail interactions with light are proportional to what might be called the *illumination* of the sail: $P(x) \cos \alpha$, where $P(x)$ is the pressure (as a function of distance from the Sun) on a perfect absorber facing the Sun, and α is the *pitch angle*, the angle between the sunlight and the sail normal (taken to point outward from the *back* of the sail). For simplicity, formulae given for interaction forces in this section assume unit illumination and unit area, i.e. they must be multiplied by illumination and sail area to give total force.

The ideal sail's only interaction with light is perfect specular reflection, yielding a force of magnitude $2 \cos \alpha$, directed along the sail normal. The non-ideal aspects of a sail membrane's interaction with light are moderately well-known and are discussed in standard references, e.g. [1]. However, some examination of them is needed with an eye to possible simplifications.

A. Transmission

Realistic sail membranes are optimized for low mass and compact packing, and their metallizations typically are thin enough that they are not completely opaque: some light passes completely through them. However, for near-term membranes, even aggressive designs, transmission at most 1-2% [2]. (Note also that transmission is normally measured at a pitch angle of zero, and could be expected to diminish with increasing pitch angle.)

Transmission is one form of non-ideal behavior that *can* be modeled as a small increase in sail loading, since it amounts to slightly less light *interacting* with the sail per unit mass. Alternatively, given the small size of the effect, it can be completely neglected for first-approximation analysis.

B. Imperfect Reflection I: Non-Constant Reflectivity

The reflectivity of a sail membrane is typically quoted as a single number. This is unrealistic. Real sail metallizations, like aluminum, have an approximately-constant reflectivity only at low pitch angles. The actual angle-dependent reflectivity of aluminum is approximately as in Fig. 1 [2]:

Reflectivity is nearly constant out to about 45°, but not thereafter. In particular, at pitch angles around 83°, the absorbed fraction of incoming light is nearly double what it is at low angles. The discrepancy is not large at the pitch angles used in realistic maneuvering, but can be serious when an attempt is made to put the sail nearly edge-on to approximately zero out thrust; see later.

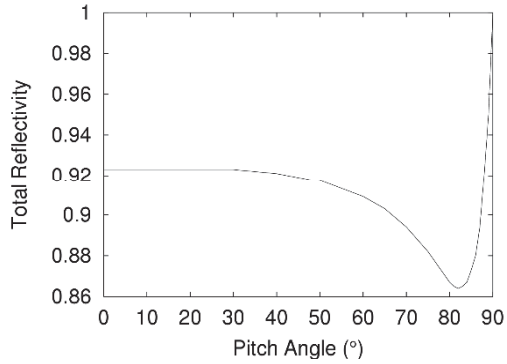

Fig. 1 Aluminum reflectivity as function of angle

C. Absorption and Re-emission

Realistic sail metallizations are not perfectly reflective: some of the light hitting them is absorbed rather than reflected, warming the sail, and is then re-radiated as heat. For near-term sail membranes, typically 8-12% of incident light is absorbed at low pitch angles. (As noted in the previous section, this number can nearly double at high pitch angles.)

The absorption itself transfers the momentum of the light to the sail, producing a force in the same direction as the light, i.e. generally *not* along the surface normal. This force is directly proportional to illumination, and hence at high pitch angles, where $\cos\alpha$ becomes small, the absorption force can become quite important compared to forces that scale with $\cos\alpha$. *This force is not equivalent to an adjustment of sail loading, is typically quite significant, and must be explicitly included in a realistic model.*

The re-emission as heat carries the same amount of momentum, since the momentum of W watts of any form of electromagnetic radiation is just W/c, where c is the speed of light, but is not in the same direction. Both sides of the sail radiate, and not in a single direction or uniformly.

A sail with the same metallization on both sides would, to a good approximation, see no net re-emission thrust. In practice, most real sail membranes have a front and a back, with the front optimized for reflecting sunlight and the back optimized for radiating heat. Much of the re-emission thus is from the back, *opposing* the specular-reflection thrust. Back re-emission is spread over a full hemisphere, although not uniformly, and there is some re-emission from the front as well.

When the details are accounted for, for near-term sail membranes, the re-emission force typically has a magnitude of 0.4-0.6 that of the absorption force (again, directly proportional to illumination). Note, however, that it is *not* in the same direction. For uniform surfaces on a symmetrical sail, re-emission is symmetrical around the sail axis, and so the re-emission force is directed opposite to the sail normal.

The re-emission force could be modeled as a small increase in sail loading (to reduce the specular-reflection force slightly) combined with a small reduction in reflectivity (to maintain the same absorption force), although the exact numbers would be pitch-angle dependent. It's also small enough (and further diminished by diffuse reflection, to be discussed next) that for first-approximation analysis, ignoring it completely might be defensible, *except* at high pitch angles. (That case will be discussed later.)

D. Imperfect Reflection II: Diffuse Reflection

Reflectivity is normally quoted as *total* reflectivity, including both specular reflection and diffuse reflection. For sail-membrane reflective surfaces, diffuse reflection is quite small: the *specularity* (fraction reflected specularly) is typically 0.94 or higher.

Classical diffuse reflection is light scattered over approximately the full hemisphere, although not entirely uniformly, much like the front-side component of the re-emission force. For modeling purposes, this is effectively a form of *absorption*, not reflection; in particular, it has the same direct proportionality to illumination. It can be modeled with adjustments to the absorption/re-emission constants. Diffuse reflection typically opposes the re-emission force and is somewhat smaller, further reducing re-emission‘s significance.

There is a temptation to use a relatively strict definition of “specular”, inherited from optics, allowing only a narrow cone around the nominal specular-reflection direction, and declare any reflection outside that cone to be “diffuse”. *This is a mistake.* Macroscopic surface irregularities, to be discussed shortly, cause modest amounts of divergence of the specular beam, *not* hemispherical scattering, and must be analyzed separately.

E. Shape I: Small-scale Irregularities

The ideal sail is flat. Real sails are not, even on a small scale. Local departures from flatness can occur for several reasons, most notably unbalanced strains ("wrinkles") and residual deformations resulting from handling ("crinkles") and folding ("creases").

All these forms can be modeled, to a first approximation, as pairs of equal-sized facets, departing from the theoretical flat surface by equal angles in opposite directions. If the angles are small and the angle of incidence of incoming light not too large, the primary effect of this is *divergence*: the reflected beam is split into two, symmetrically around the nominal specular-reflection direction. With many pairs of facets at many random angles, the result is a reflected beam which diverges somewhat, with a power distribution (and hence a momentum distribution) matching the distribution of facet angles.

While the exact result depends on the power distribution in the reflected beam, a first approximation can be had by assuming that the beam is a cone with a uniform power density within the cone and none outside it. In this case, the force loss due to divergence is the same as the divergence loss in a conical rocket nozzle: momentum is reduced by a factor of $\frac{1}{2}(1 + \cos \beta)$, where β is the half-angle of the cone.

Broadly similar results hold so long as the power distribution is symmetrical around the specular-reflection direction: the reflected beam is equivalent to somewhat-weaker perfect specular reflection. When symmetry is not present, then the effective direction of the reflected beam may depart from the specular direction, but not by a large angle unless there is large-scale asymmetry in the surface irregularities.

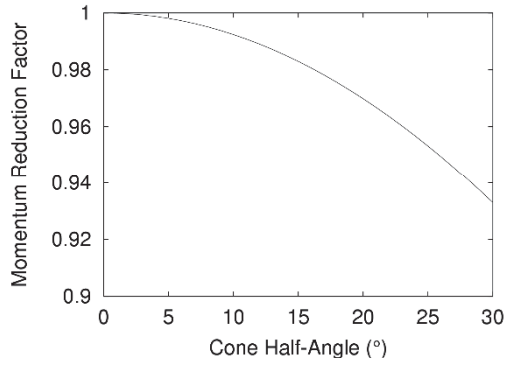

Fig. 2 Divergence Loss

Divergence losses actually are surprisingly small. See Fig. 2 for the result using the uniform-cone approximation. For a beam that is 60° wide (!), the momentum loss is less than 7%. Such a modest loss in the momentum of the reflected beam is trivial to model: it is equivalent to a small reduction in reflectivity, i.e. a small increase in absorption thrust, except that it does not increase re-emission thrust. For more reasonable beam

widths like 30°, the loss is under 2%, which a first-approximation mission analysis could reasonably even ignore. As noted in the discussion of diffuse reflection, treating beam divergence as classical diffuse reflection *is grossly inaccurate and greatly exaggerates its impact*.

When both the facet angles and the pitch angle become sizable, this analysis is complicated by a "cosine effect": facets that are closer to perpendicular to the incoming light see more light than other facets. At the extreme, if the sum of incidence angle and facet angle starts to exceed 90°, this effect starts to diminish because favored facets start being shadowed by their neighbors. Cosine effect skews the reflected-beam power distribution somewhat toward the sail axis, increasing specular-reflection thrust and moving it away from the sail normal toward the incident-light direction. For reasonable surfaces, cosine effect is small except at extreme pitch angles. If it becomes necessary to model this effect, this can generally be done as a small reduction in sail loading (to supply the higher thrust) combined with an increase in absorption (to shift its direction).

F. Shape II: Large-Scale Departures from Flatness

The most obvious sail-scale departure from flatness is *billow*: deformation of the surface due to light-pressure loads; this can be reduced by higher membrane tension or sail designs that reduce unsupported membrane spans, but cannot be eliminated. For a square sail, the support booms may also deform slightly under light pressure, unless boom shape is pre-curved to counteract this, or perhaps actively controlled; this has been called *droop*. Finally, net-supported slack sails [3] also experience local billow between net strands.

If these various effects are asymmetrical, torques obviously can result, potentially complicating attitude control of the sail. Symmetrical billow and droop can also cause a "shuttlecock" torque via a large-scale cosine effect. However, provided attitude control is maintained, these torques do not affect performance.

The *performance* effect of large-scale departure from flatness is quite simple to analyze: it's just bigger facets! Again, the result can be some more divergence of the reflected beam, but as noted earlier, even quite sizable beam divergence has only quite small effects on performance, and this is easily modeled.

G. Pitch-Angle Limits

Ideal sails can hold any chosen pitch angle exactly. Real sails depart from this in two important ways: attitude-control errors, and pitch-angle limits. Attitude-control errors generally do not have much effect on overall performance, but pitch-angle limits are more significant.

An ideal sail can operate at any chosen pitch angle, including 90°, i.e., edge-on to the light. Real sails may not be able to do that. Apart from questions of whether attitude-control actuators remain effective at very high

pitch angles (tip vanes will, given adequate range of motion; movable masses probably won't), there is also the problem of the effect of very high pitch angles on the sail itself.

For one thing, a nearly-edge-on sail gets very cold. The theoretical membrane temperature is easy to evaluate with a one-node thermal model; see Fig. 3 for an example. Thin polymer films are more flexible than bulk polymer at low temperatures, but still there is a potential problem. The effects on support structures like booms (and the net of a net-supported sail) are more difficult to evaluate but also more worrisome.

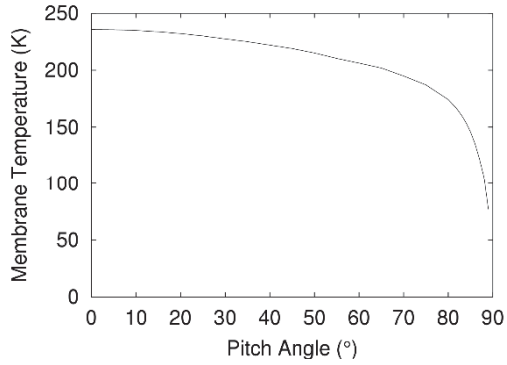

Fig. 3 Membrane Temperature vs. Pitch Angle

Possibly more problematic yet is the effect of high pitch angles on the *shape* of the sail. A truly edge-on sail will have ill-defined billow, possibly including jibing (snapping from one extreme to the other as attitude varies slightly). Net-supported slack sails will be less prone to large-scale jibing, but will have very poorly-controlled local shape, especially since the sail actually rests against the net on only one side. (Typically it is secured to the net at intervals to ensure that it can't depart far, but this still gives less control.)

In practice it will probably be necessary to set an upper limit on pitch angle at significantly less than 90°. Just how close to edge-on is achievable depends on issues like attitude-control precision, but nominal pitch angles higher than perhaps 85° seem likely to be impractical, and even that value may be optimistic.

H. Simple (But Not Too Simple) Sail Models

Figure 4 compares a “full” sail model with an ideal sail, showing the force components in orbital coordinates, with leeward force along the light direction and transverse force at right angles to it. (For assessing sail capabilities and performance, this is a more useful reference frame than sail-fixed coordinates.) The full model includes all of the optical non-ideal effects, with parameters estimated for a net-supported sail [3], plus beam divergence with 20° half-angle to account for shape effects. Clearly they don't match very well, and no simple scaling (e.g. adjustment of sail loading) will fix this.

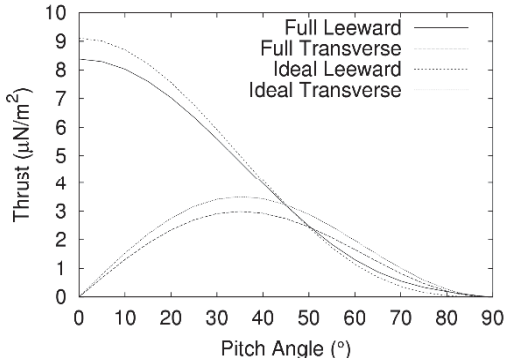

Fig. 4 Full Sail Model vs. Ideal Sail

However, almost all the effects discussed above can be modeled, as discussed, with simple combinations of specular reflection and absorption. Figure 5 compares the same full sail model with a "simple" model that adds only one complication to the ideal sail: only 84% of the light is reflected specularly, while the rest is absorbed (with no re-emission thrust). The actual reflectivity in the full model is rather higher, but setting the simple model's to 84% achieves, empirically, a good simulation of the full model's more complex effects. These two models are almost indistinguishable: the differences are at most 1-2% until pitch angle approaches 70°. (The situation at very high pitch angles will be discussed in the next section.)

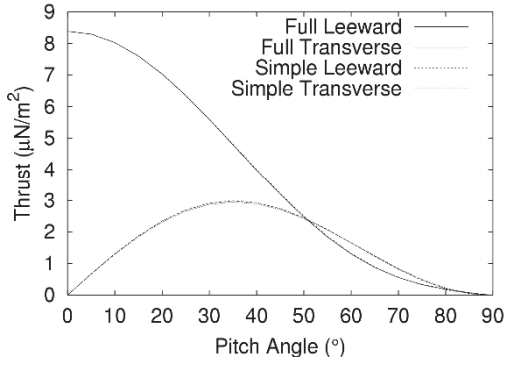

Fig. 5 Simple Sail Model vs. Full Sail Model

In short, it is possible to model most real-sail effects quite well, at least for the purposes of first-approximation mission analysis, by complicating the ideal sail only slightly, *but that complication is necessary.* The following sections will discuss some other reasons why that is true.

III. Leeward Force

For certain types of solar-sail missions (examples will be discussed later), at times during the mission there is a desire to minimize sail thrust. The classic solution to this is to put the sail edge-on to the light. As noted earlier, in practice a 90° pitch angle is unlikely to be very practical, with perhaps 85° being more reasonable. This obviously raises the question of just how well sail thrust approximates zero at such angles.

Using the terminology of the previous section, the key issue is the leeward thrust, along the light direction. At least in principle, transverse thrust can be averaged out to zero by tactics such as *coning*: slowly rotating the sail around the Sun axis, so that the direction of the transverse thrust sweeps out a great circle on the sky. There is no equivalent for leeward thrust: it always points outward from the Sun.

Here again there are large differences between the ideal sail and a more detailed model. Figure 6 is an enlargement of the lower right corner of Fig. 4, showing only the leeward components. Clearly, modeling a real sail as an ideal sail *grossly* underestimates the magnitude of leeward thrust at high pitch angles.

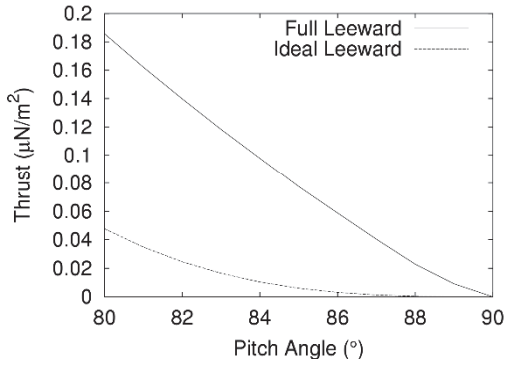

Fig. 6 Full vs. Ideal Sail Models at High Pitch Angle

Figure 7 is a similar enlargement of Fig. 5, showing that the simple model, while it has some error in this region, is much closer to being realistic.

Given that the whole point of operating at very high pitch angles is to come as close as practical to zeroing out thrust, there seems little need to operate over a *range* of such angles. So probably the best approach is to pick a specific low-thrust angle, typically the highest angle that the particular sail can hold without practical difficulties, and evaluate the sail's properties there using a detailed model.

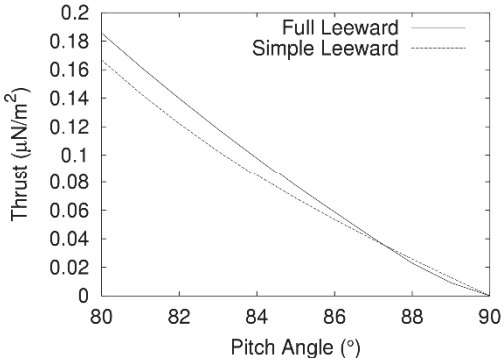

Fig. 7 Full vs. Simple Models at High Pitch Angle

Regardless of the exact model used, clearly any realistic model evaluated at a realistic low-thrust angle will still produce some leeward thrust. The amount is not large, typically perhaps 1-2% of full face-on thrust (depending on angle and sail properties), but it is *not* entirely insignificant, *and it cannot be switched off*.

Residual leeward thrust is especially important in cases where sail navigation involves careful balancing of forces. Examples include maneuvering to and near Lagrange points (natural or artificial), and flying formation with asteroids. Subsequent sections go into detail on these particular cases.

IV. Leeward Force Consequences 1: L1 Maneuvering

The publication in the early 1990s by Forward and McInnes of several papers [4], [5] on the topic of "statite," "halo orbit" and "polesitter" equilibria generalized the concept of the Euler's and Lagrange's five Sun/Earth equilibrium point solutions to the restricted three-body problem, into the concept of equilibrium *surfaces* achievable using solar sail propulsion, these surfaces (under the same restricted three-body problem assumptions) being stationary in an Earth-fixed (non-inertial) reference frame that co-rotates with the Earth around the Sun. A spacecraft stationed at a location on one of these surfaces may be said to be located in a "non-Keplerian orbit," located at an "artificial Lagrange point" (ALP).

Those early papers investigated the nature of ALPs for perfectly-reflecting solar sails. In [6], one of us extended Forward's analysis to include the effect of imperfect reflection, in particular for an 85% reflective sail. The result was a very significant change in the shape of the equilibrium surfaces, generally pushing the "upwind" ALPs for a given sail mass/area ratio to a higher altitude and latitude. McInnes also developed a vector-equation version of this analysis, in [7], also discussed briefly in section 5.5 of [1].

The ALP idea has led to several attractive concepts for solar-sail-propelled missions, including weather and/or communications satellites "hovering" at an ALP that has a continual view of Earth's polar regions, and

the Geostorm mission concept for keeping station at an ALP at or near the Earth-Sun line (like the L1 point), but higher above the Earth than L1 (a "super-L1" point). Many of the studies that have been carried out of such missions have focused on operations *after* the spacecraft has been maneuvered to its ALP station, frequently glossing over the question of how to reach the final ALP (e.g., as in the description of this phase of the Geostorm mission in section 6.2.3 of [1]). An early common assumption about such missions was that, since all the "upwind" equilibrium surfaces pass through L1, once in the vicinity of L1 it should be possible to move from near L1 to anywhere on any of these surfaces, by "sliding" so slowly along an equilibrium surface that dynamic effects are small, so that the spacecraft remains near equilibrium the whole time

A bit more than a decade after this type of mission first started to be contemplated, Chen-wan [8] published a what seems to be the first sophisticated study of this type of maneuvering in the vicinity of the Earth/Sun L1 point. As part of a JPL study of the Geostorm mission's orbital dynamics and control, he designed a numerically-optimized trajectory from an L1 halo orbit to a super-L1 location (which he calls "Sub-L1"), using Sauer's SAIL trajectory optimization software as a starting point. Chen-wan assumed a perfectly reflecting solar sail, with a mass/area ratio of 0.029 kg/m^2, with a characteristic acceleration of 0.31 mm/s^2 when face-on to the Sun at 1 AU distance. Figure 8, copied from Chen-wan's paper, illustrates some of his results. The trajectory labeled "Halo to Sub-L1 Sailing, cone <90°, FT=206d" is the result of allowing the pitch angle (which Chen-wan refers to as "cone angle") to vary through the range -90° to +90°, with pitch angles as large at 70° near the beginning of the trajectory. He found the duration of this trajectory to be 206 days.

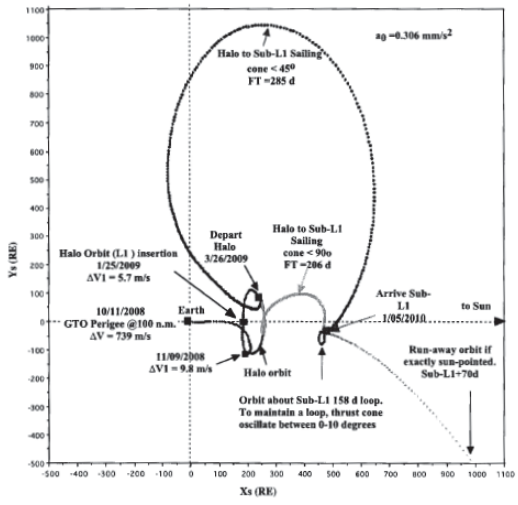

Fig. 8 Trajectories from Near L1 to Geostorm ALP Station (from [Chen-wan, 2004])

One particular type of sail "non-ideal" behavior was investigated in [8]. Noting that "current sail technology predicts uncertain performance when the sail is tilted more than 45° away from Sun," a second trajectory was designed in which the pitch angle was constrained to stay under 45°, resulting in the trajectory labeled "Halo to Sub-L1 Sailing, cone <45°, FT=285d". This trajectory takes 285 days to complete, and is notable in that it includes a very large, looping deviation in the +y direction (in his graph, the direction opposite to the Earth's orbital velocity vector about the Sun), taking the spacecraft as far as ~ 1100 Earth radii (~ 7 million km) away from Earth.

It was with this background that we started a study at Space Flight Laboratory for the ASPEN mission, which aimed to design a minimal polesitter demonstration mission based on SFL's nanosatellite technology --- essentially a nanosat version of the earlier POLARES mission concept. The result was a 21 kg spacecraft, of which 7 kg was a 35 m square solar sail of area 1225 m^2, with a mass/area ratio of 0.017 kg/m^2, carrying a 0.83 kg payload of imagers and magnetometers. (It is not a coincidence that this is the same size as L'Garde's Sunjammer sail, as we based the ASPEN design on L'Garde's solar sail technology.) Figure 9 shows the bus portion of the ASPEN spacecraft, along with the central portion of the solar sail and its support booms. Launch was assumed to be as a secondary payload on a mission to an L1 halo orbit, with solar sailing from near L1 to the ALP location. ASPEN was designed assuming an 85% reflective sail. Its target ALP location was chosen to always be able to see the entire Earth's north polar region above latitude 76°. Figure 10 shows the chosen ALP location in the Earth/Sun-fixed rotating reference frame, 1.5 million km both above the ecliptic and Sunwards of Earth, at a range of 2.25 million km (the units there actually being hundredths of an AU, with Earth at (0,0)).

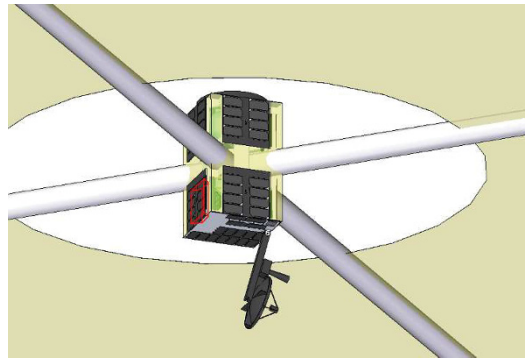

Fig. 9 ASPEN Spacecraft Bus

The maximum range to Earth emerged as a design driver for ASPEN, due to the difficulty of achieving an adequate downlink communications data rate at near-interplanetary distances --- to date, virtually all nanosat and

microsat missions have operated in low Earth orbit, and significant changes to equipment designs clearly are needed to operate at the chosen ALP distance. The most stringent requirement was to achieve a (bare minimum) 2 bps downlink data rate to a medium-sized (7m) receive antenna on Earth, through omnidirectional transmit antennas on the spacecraft, to support recovery of the spacecraft from safe/hold mode. On top of that, Chen-wan's paper raised the question of whether the range to Earth might grow to be much larger than 2.25 million km during the period of transit to the ALP location. The ASPEN designers assumed that this distance might grow to be as large as 5 million km; with that as the maximum design range, the S-band transmitter used in safe/hold mode needs 10 W of RF output power to close the communications link. This in turn requires 30 W of DC power to be provided by the power subsystem to this transmitter, which drove the size of the solar arrays and batteries, and hence the spacecraft mass and its solar sailing performance.

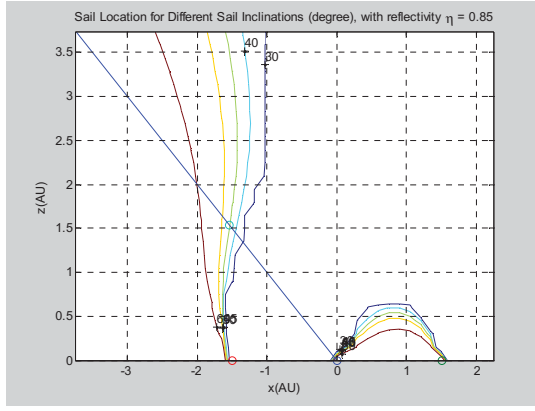

Fig. 10 ASPEN ALP Station Location

A particular concern was the question of whether imperfect reflectivity might drive the transit trajectory to be more like the second case in [8], rather than the (fairly benign) first case. The rationale for this concern is that to leave L1 in a controlled fashion, the sail obviously must be oriented to generate some thrust; for a real sail, that thrust vector will have a leeward component, which will push the sail towards Earth, away from the "upwind" ALP equilibrium surfaces. Once in that zone, the sail could be forced to fall uncontrollably to leeward. This tendency can be reduced by thrusting aftward (in the direction opposite the spacecraft's orbital direction around the Sun), to reduce the sail's angular velocity about the Sun; the resulting reduction in leeward centrifugal force will create "headroom" for accepting some leeward thrust from the solar sail, without forcing a fall to leeward. The question is whether enough reduction in centrifugal force can be developed quickly enough to avoid that fate; clearly the larger leeward thrust due to less-than-100% sail reflectivity will have a bearing on this question.

To investigate this, we have simulated the initial stages of a hypothetical transit trajectory from L1 towards the vicinity of a Geostorm-like "super-L1" ALP location (keeping the analysis in the ecliptic plane for the purposes of clarity, with similar results expected for cases of sailing from L1 to out-of-the-ecliptic polesitter ALP locations), for a range of sail reflectivity values. Figure 11 shows the results of one such set of 200-day simulations, which assume Earth in a circular orbit around the Sun as the only gravitating bodies, for a 0.02 kg/m^2 solar sail with reflectivity values of 100%, 84% and 80%, shown as the 3 trajectories from left to right in the graph. These trajectories result from angling the sail at its best-lateral-thrust pitch angle of $\alpha=35^o$ for 30 days, to accelerate initially in the aft (-y) direction, then coasting with the sail edge-on to the Sun for the remaining 170 days; the red dots along each trajectory are spaced 2 days apart. The coordinate frame here is Earth-fixed, rotating once per year. The blue circle at (0,0) represents the Earth's location. The Sun is located at x=-1 AU (about -149.6 million km). The blue circle at x=-1.5 million km is the Earth/Sun L1 point; all three trajectories start there, with zero velocity in this reference frame. The blue circle at about x=-3.6 million km is the "super-L1" ALP location for the 84% reflective sail, for a 0^o pitch angle.

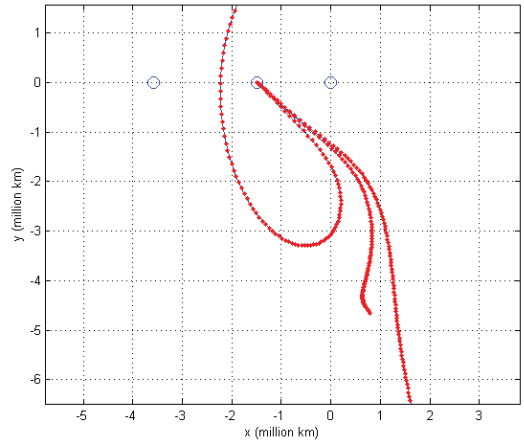

Fig. 11 Thrust Profile: Pitch = 35^o for 30 days, Then Coast With Pitch = 0^o for 170 Days

The only difference between the trajectories is the reflectivity of the sail, which is seen to have a significant effect on the path followed by the sail. The initial leeward tendency of the sail is apparent, even for the ideal sail, with about as much initial motion in the leeward (+x) direction as in the aft (-y) direction. The 100% reflective sail recovers from this leeward motion after about 100 days, while the 84% reflective sail perhaps barely recovers after 200 days, and the 80% reflective sail falls somewhat to leeward (this particular thrust profile being chosen to highlight this difference).

This shows that the effect of sail absorptivity will definitely affect trajectories in sailing from near L1 to an ALP. It does not yet confirm or refute the question raised by the ASPEN designers, regarding maximum distance to Earth during that maneuver. Detailed design of such a trajectory is needed, which must be done using numerical trajectory optimization software; we are not yet to that point.

Figure 12 shows one further step towards that. For the same three reflectivity values, it shows the trajectories resulting from a thrust strategy that aims to reduce the amount of leeward displacement, by gradually ramping up the aftward thrust rather than starting out with full aftward thrust. Here the sail's pitch angle starts at 90° (edge-on to the Sun), and over the course of 50 days is ramped to 65°; after that the pitch angle is held at 65° for the remaining 150 days of the simulation. From left to right the trajectories are shown for reflectivity values of 100%, 84% and 80%; the full trajectory is shown for the latter two cases, while only the first 140 days is shown for the 100% reflectivity case (with red dots again spaced 2 days apart for each).

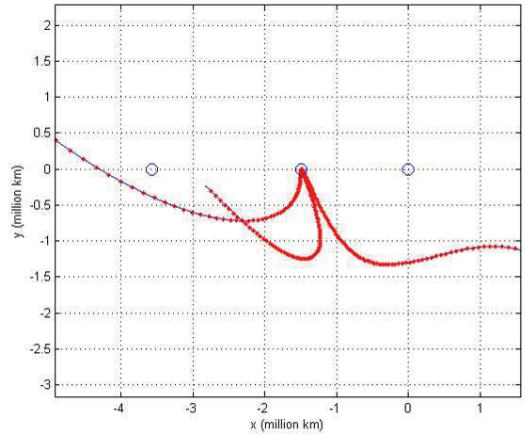

Fig. 12 Thrust Profile: Ramp Pitch from 90° to 65° Over 50 Days, Then Hold for 150 Days

The first notable result is that the intent of the thrust strategy worked as hoped-for: all 3 trajectories depart from the initial L1 point at a shallower angle, with less velocity in the leeward (+x) direction. Also notable is the greatly magnified difference between the three trajectories in this case, with the 100% reflective sail overshooting super-L1 to end up falling rapidly Sunward, the 84% reflective sail closing in on super-L1, and the 80% reflective sail falling catastrophically to leeward. While the question raised by the ASPEN designers remains to be answered, this analysis illuminates aspects of the tools and approach needed to answer it properly.

V. Leeward Force Consequences 2: Asteroid Formation Flying

We have recently carried out near-Earth asteroid rendezvous mission studies, again based on combining SFL's deep expertise in microspace missions with L'Garde's solar sail technology. One of these started just after L'Garde's Sunjammer technology demonstration mission was approved by NASA; our CARAVEL mission aims to investigate what sort of a useful asteroid exploration mission could be accomplished by an SFL-type nanosat, propelled by a Sunjammer-sized sail. In developing an initial preliminary design for CARAVEL, the effects of leeward thrust once again proved to be a mission and system design driver, this time for a different reason.

ASPEN made abundantly clear the challenges of communicating with a very small spacecraft at long range from Earth. CARAVEL was posed as an as-low-cost-as-possible asteroid mission, and to minimize the maximum range to Earth range (and hence the cost of the equipment needed to close comms links), we chose CARAVEL's target is asteroid to be (164207) 2004-GU9. This Apollo-class asteroid is one of the very few "quasi-satellites" of Earth discovered to date, with orbit parameters that will keep it within a range of 50 million km of Earth for more than a decade from now. The relative ease of communicating at that distance (as compared to typical near-Earth asteroid maximum ranges of >300 million km) makes 2004-GU9 an intriguing target for a rendezvous mission. Its orbital inclination of 13.65° makes it very expensive to reach from Earth via chemical rocket, requiring a plane-change ΔV of about 7.1 km/s --- rather higher than the delta-V of most rocket-propelled space missions. However, that amount of ΔV can easily be achieved using a modest-performance solar sail. CARAVEL is designed to use a square L'Garde solar sail, 35 m on a side, with a total area of 1164 m^2, to propel a spacecraft whose total mass is 17 kg, 5.7 kg of which is the solar sail plus its support structure, and 1.2 kg of which is a payload comprising a camera and a pair of science-grade magnetometers. The mass/area ratio for this solar sail is 0.015 kg/m^2; with that sail loading, we found a trajectory concept that seems able to bring the spacecraft from an Earth C3=0 escape trajectory to rendezvous with the asteroid within 22 months. Figure 13 shows the overall layout of CARAVEL on the left, and on the right shows the CARAVEL bus and its ejectable sail support module along with the inner portions of the sail and its inflatable booms.

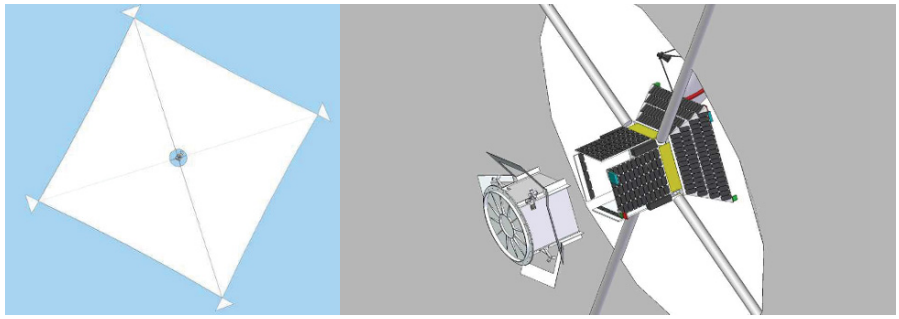

Fig. 13 CARAVEL

Our original mission concept was to sail to the asteroid, rendezvous with it, then carrying out an extended period (several months, at least) of science imaging while flying in close formation with it. We understood from the outset that the formation-flying phase would be complicated by the solar sail --- a device which is so useful in reaching this otherwise nearly-inaccessible target, but which is something of an encumbrance after arrival. Our initial concept was to enter a non-Keplerian orbit, keeping station with the asteroid at a Sun/asteroid ALP; in [9], Morrow *et al.*, provide a first look at what that involves. However, when we began working on the details of just *how* to operate the spacecraft in the vicinity of 2004-GU9, we found that the details contained a devil.

In [9] a maximum sail acceleration is determined as a function of asteroid mass, above which orbital operations around an asteroid are not feasible, due to solar radiation force on the sail being too large relative to the asteroid's gravitational force. Trajectories around several relatively large asteroids are studied; operations at those appear feasible for reasonable-performance solar sails. For the smallest of these, 433 Eros, whose radius is 8.9 km and whose mass is estimated to be 6.69×10^{15} kg, any sail with a maximum characteristic sail acceleration of <0.79 mm/s^2 was found to be acceptable; this corresponds to a sail mass/area ratio >0.0115 kg/m^2, indicating that the CARAVEL sail design could operate about Eros (although perhaps just barely). [9] notes that, "smaller asteroids will require low values of sail acceleration for orbital options to be feasible about them."

Our problem was that 2004-GU9 is much smaller than Eros --- as are a great many other asteroids that are now considered to be attractive near-term exploration targets --- with an estimated diameter of 150 m, and mass of $5.7\text{x}10^9$ kg [10]. Since the face-on-to-the-Sun acceleration of CARAVEL's solar sail ($6\text{x}10^{-4}$ m/s^2) is even higher than the estimated gravitational acceleration at the asteroid's surface ($6.8\text{x}10^{-5}$ m/s^2), it is clear that the artificial Lagrange point concept is not meaningful for that solar sail at that asteroid. Some approach other than that explored in [9] would be needed, for CARAVEL to keep station with this asteroid.

An alternate concept which we investigated was to "switch off" the solar sail on arrival at the asteroid, by turning it edge-on to the Sun. If the sail is assumed to be perfectly reflecting, then the magnitude of the thrust varies with $\cos^2 \alpha$, which drops to zero at $\alpha = 90^\circ$. A real attitude control system will not be able to keep the sail's orientation *exactly* at $\alpha = 90^\circ$; in practice α would wander about 90° by some amount, likely over a range of a few degrees. However, near $\alpha = 90^\circ$, for an ideal sail the thrust direction is almost entirely transverse, as the transverse thrust component goes with $\cos^2 \alpha \sin \alpha$, while the leeward thrust goes with $\cos^3 \alpha$; the "Ideal Leeward" line in Fig. 6 illustrates how small the leeward thrust is for several degrees out from $\alpha = 90^\circ$. Even if the transverse thrust component is relatively large, the "coning about the Sun-line" strategy discussed in Section III can be used to cause it to average out to zero over some reasonable period of time.

Assuming that staying 3 km from the asteroid is sufficient to avoid a collision despite the practical difficulties of remote-control navigation, and with the use of coning, calculations show that an ideal sail could plausibly keep station with 2004 GU9 at $\alpha=88.5^\circ$ --- rather close to 90°, but perhaps not unachievably so. Thus, with an ideal sail, flying formation with even such a small asteroid looks challenging but perhaps feasible.

However, when a realistic sail optical model is used, the situation is significantly different, remembering Fig. 6. While the transverse thrust at $\alpha=88.5^\circ$ is about the same for both models, the leeward thrust of the full sail model is nearly ten times the gravitational acceleration at 3 km from the asteroid --- and, unlike transverse thrust, coning about the Sun-line doesn't average leeward thrust to zero. Neither going much closer to the asteroid nor going to an even higher pitch angle is likely to help enough to solve this problem.

Thus, while the ideal sail assumption made it appear that keeping station with this small asteroid could be plausible, it is totally implausible with a realistic sail model. Trying to do so is analogous to attempting to dock a sailboat in a small harbor while carrying all sail, in a stiff breeze --- a practical impossibility. For CARAVEL this led us to drop the original mission requirement for keeping station for an extended period of time, instead switching to a mission design in which a very slow flyby is executed, with the spacecraft being within (say) 100 km of the asteroid for a period (say) a day or so. This has strong implications for achieving low mission cost, as it requires the mission operations team to get the proximity-operations software and procedures worked out in advance; also, this is strongly different from SFL's tried-and-true "microspace" approach that has worked so well in keeping costs down in microsat and nanosat missions in LEO, where typically weeks or months are spent in commissioning all of a satellite's equipment. It also drove payload design, to an imager that could take sufficient-resolution images from a rather larger distance than if station-keeping with the asteroid. (The

alternative, of having the sailcraft drop off an autonomous probe at the asteroid, is more promising but dodges the problem rather than solving it.)

VI. Conclusion

Real sails are not ideal, and yes, it matters. Mission analysis done assuming ideal sails, even with somewhat derated performance, can yield misleading results. For some purposes it may be good enough; for others, it definitely is not. Some missions that look plausible with ideal sails cannot be done with real sails; others can be, but only with significant changes to plans, schedules, and even spacecraft designs.

The good news is that for most mission-design work, complex real-sail models considering a wide variety of effects are unnecessary. Merely adding absorption to the sail model suffices (given some adjustment of parameters) to model real behavior well enough for preliminary mission planning.

One particular consequence of real-sail effects is the inescapable presence of a certain amount of leeward thrust, even in mission phases when sail thrust is to be minimized. Two mission types that are particularly affected by this are maneuvering at and near Lagrange points (natural or artificial), and flying formation with small asteroids. Lagrange points involve delicate balances of multiple forces, and even a small added force can easily make maneuvering plans much more complex, requiring longer transit times, greater communications ranges, and other complications. Flying formation with a small asteroid is simply *impossible* for a real sail, because there is no natural force that can be used to balance out the leeward force; a sail mission to such an asteroid must either jettison its sail on arrival, or drop a probe in passing and proceed elsewhere.

Acknowledgments

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Dr. Robert E. Zee, the director of SFL, has allowed us to mercilessly exploit his students while we educate them, and has encouraged us to present them with design problems that go well beyond SFL's more normal projects.

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Inspection of a co-orbital solar sail using a microthruster attitude control system

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The use of a solar sail allows a spacecraft to generate thrust at no expense of propellant. However, there is little data to quantify the effect of sail structure on thrust performance. The InspectorSat mission will use a microsatellite to autonomously perform close proximity manoeuvres to inspect a co-orbital solar sail. Using a novel microelectromechanical thruster for precision orbital and attitude control and a simple monocular imaging sensor for inspection, the InspectorSat will observe sail deployment, surface features and record measurements of solar radiation pressure. This work, therefore, reflects the mission analysis inputs to the PRECISE micro-chemical propulsion system development, demonstrating the utility of a microscopic, low impulse thruster for the actuation micro- and nanosatellite platforms. Simulation results are used to determine thruster performance and Δv requirements for deployment, rendezvous and inspection. The proposed inspection schemes are designed to allow the inspector to take advantage of multiple observation opportunities. The results of this study provide a comprehensive analysis of the manoeuvring requirements of a small inspector satellite in close proximity to a passive observation target. As such, the analysis demonstrates the practicality of the micro-chemical propulsion system for inspection manoeuvres.

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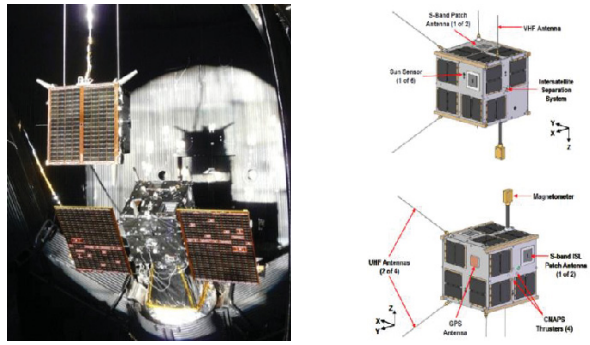
I. Introduction

Although solar sails have now accumulated a modest degree of flight heritage [1-3], little data exists to illustrate either the geometry of a sail during and after deployment or the dynamics of the sail membrane during flight. To tackle this problem, the InspectorSat mission concept proposes the use of a tracking microsatellite flying in formation with a solar sail to autonomously observe and inspect surface features during flight. Manoeuvres suitable for the inspection of the sail during deployment, observation of sail surface features such as creasing and billow and measurement of Solar Radiation Pressure (SRP) have been conceptually validated for a 50 kg or smaller microsatellite using a single imaging sensor [4]. Previous work has also aimed to develop sail reflectance models that provide light-maps highlighting observation opportunities and so-called "keep-out" zones, where specular reflections may either damage an imaging sensor or saturate the image [5].

Close proximity on-orbit operations, however, such as collision avoidance and formation flying, also rely on the estimation of keep-out zones, defined by ellipsoidal volumes of position uncertainty in spacecraft trajectories. The magnitudes of these volumes are often based on statistical modelling of the error margins in trajectory estimates and are typically of the order of several kilometres [6, 7]. Precise sensing is required to provide relative motion predictions of sufficient accuracy that allow autonomous operations at much closer proximities, of the order of hundreds, tens or even fewer metres. In doing so, applications including mission lengthening operations, such as autonomous on-orbit servicing, repair, construction and reconfiguration, become possible.

Autonomous co-orbital motion and formation flying has been demonstrated on several small satellite missions to date, including the PRISMA and CanX-4 & 5 missions, shown in Fig. 2. Three Guidance, Navigation and Control (GNC) methods based on combinations of Global Positioning System (GPS), Radio Frequency (RF) and visual measurements were utilised in the PRISMA mission [8]. New methods enabling formation flying by means of GPS measurements alone were demonstrated on the CanX-2 nanosatellite and are also planned for the CanX-4/5 mission [9]. While these methods have shown promising results, all of the above examples depend upon some level of cooperation from the target spacecraft. Furthermore, the continuous use of subsystems such as GPS is typically limited on nanosatellite platforms by constraints in the available power budget. Performing observations without the aid of GPS or other cooperative methods while in orbit requires a high performance and robust Attitude and Orbital Control System (AOCS), coupled with passive target detection sensors and a precise spacecraft actuation system. The purpose of the InspectorSat mission is to demonstrate



a) PRISMA Mango and **b) CanX-4 & 5**

Fig. 2. Examples of cooperative relative motion missions (courtesy SSC Space [11] (a) and after Orr et al. [9] (b))

such a system in the form of a "Visual Inspection Payload" (VIP), thus providing a platform for the direct and controlled on-orbit observation of a target spacecraft for extended periods of time.

The VIP will provide GNC telecommands based on predictions of the target spacecraft trajectory. The methods employed by this system are based on the monocular imaging of the target to provide estimates of the relative position and pose at close proximities. To enable the InspectorSat to maintain its planned trajectory relative to the target a highly responsive and precise actuation system is required. The use of a micro-chemical propulsion system (μCPS) – a mono-propellant hydrazine thruster based on the use of microelectromechanical systems (MEMS) – would provide such control that close-proximity inspection manoeuvres may be executed. A suitable system is under development as part of the "chemical-μPRopulsion for an Efficient and accurate Control of Satellites for Space Exploration" (PRECISE) project consortium, with the objective of producing a MEMS hydrazine μCPS, similar to that shown in Fig. 1, capable of providing impulses of the order of 10mN [10].

The InspectorSat mission, therefore, aims to implement a high-precision AOCS that utilises such a thruster to

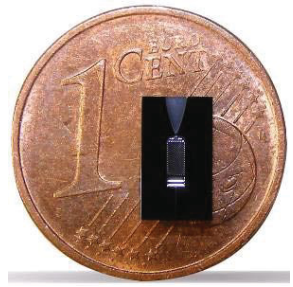


Fig. 1. A prototype MEMS μCPS, shown with a 1 cent coin for scale (Image © 2012 EADS Astrium [10])

allow the execution of close-proximity, autonomous manoeuvres around a passive target. In this case, the target will be a solar sail spacecraft (SSSC), currently under development at the Russian aerospace company NPO Mashinostroyenia. In addition to acting as the target in the demonstration mission, it will also carry a PRECISE μCPS and perform its own manoeuvres and experiments during the InspectorSat observation campaign. Using the targeted performance characteristics of the μCPS, the thruster's utility for the execution of the inspection manoeuvres can be validated. To do so, it is necessary to define attitude and orbital manoeuvre requirements which will allow the InspectorSat to complete its observation objectives.

The purpose of this paper is to study the performance of such a thruster to demonstrate its suitability as an AOCS actuator for the InspectorSat. Section II will provide a description of the mission concept; including the μCPS, VIP and AOCS elements and the solar sail target. Section III will discuss the manoeuvre requirements and modelling methodology used to estimate thruster utilisation. Section IV presents a nominal set of thruster performance specifications, based on those targeted in the PRECISE μCPS, against which the suitability of the system is validated for the actuation requirements of an inspection spacecraft.

II. The InspectorSat mission concept

A. Mission overview

The InspectorSat mission statement reads as follows:

> *To demonstrate efficient and accurate control of a satellite using a micro-chemical propulsion thruster and visual inspection payload, enabling the development and execution of formation flying manoeuvres around a target solar sail spacecraft for the purposes of visual inspection and measurement of co-orbital relative motion.*

The initial stage in characterising this mission has been addressed in the analysis of light reflectance patterns and the identification of focal points where light intensity is greatest. Modelling billow sizes, folding patterns and material properties allows the estimation of where such points may occur relative to the sail surface [5]. In addition to the inspection of sail surface features, the proposed mission also involves the observation of the SSSC to characterise the relative motion of the two spacecraft. In total, this will require the use of μCPS for manoeuvres including launch vehicle separation, detumbling, observation of sail deployment and unfolding, rendezvous and orbit (re-)phasing and compensation for SRP. Further complexity is added by considering the resolution and field of view provided by a monoscopic imaging sensor. These data provide inputs for the

estimation of observation opportunities for a given co-orbital configuration and thus the utilisation requirements of a thruster system for spacecraft actuation.

These requirements, therefore, provide a baseline μCPS requirement to maintain position in formation, compensate for perturbing forces and perform inspection manoeuvres. Considering these motivations, two levels of objectives are proposed for the InspectorSat mission. The primary objectives aim to provide a technical demonstration of the novel hardware:

- Demonstrate the μCPS thruster capability for AOCS actuation
- Optically capture the solar sail spacecraft for relative motion estimation using the VIP

The secondary objectives aim to demonstrate the scientific and mission-specific applications provided by the μCPS and VIP. That is, to:

- Perform orbit maintenance, perturbation mitigation, formation flying and attitude control manoeuvres
- Visually capture the solar sail deployment and surface features, including billow, creasing and folding patterns
- Characterise SRP on the solar sail spacecraft by analysis of spacecraft relative motion estimates from the VIP and μCPS requirements to mitigate these effects

B. InspectorSat payloads

1. *The PRECISE μCPS*

The InspectorSat mission will demonstrate the use of the μCPS being developed by the PRECISE consortium. The thruster provides a low mass, low volume monopropellant propulsion system with thrust performances suitable for orbital maintenance and attitude control of small spacecraft of up to 50 kg mass. Orbital maintenance manoeuvres typically require low-frequency high-Δv burns, necessary for co-orbital rendezvous, formation flying position holding and atmospheric drag compensation. Conversely, high-frequency low-Δv burns are used for manoeuvres necessary for SRP compensation, orbital maintenance and attitude control.

Historically, such manoeuvres have been performed by various macroscopic thruster technologies, including electrostatics and electromagnetics, arcjets, resistojets, monopropellants, liquid and solid chemical propellants and cold gas jets [12]. The relative performance of these technologies is shown in Fig. 3, illustrating how monopropellant thrusters fill a performance gap between cold gas and liquid/solid chemical thrusters. Such a

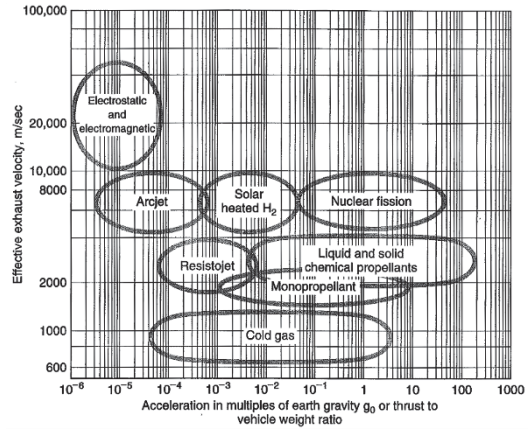


Fig. 3. Comparison of thruster technology performance (after Sutton and Biblarz [12])

thruster provides the mechanical simplicity of a cold gas system while allowing the utilisation of the propellant chemical energy through a catalytic breakdown mechanism.

The PRECISE µCPS is a hydrazine monopropellant microthruster which incorporates new technologies constructed using MEMS processes. Monolithic hardware components found in typical monopropellant thrusters, such as valves, heaters and sensors, are substituted for equivalent micron-scale MEMS parts. The benefit of such technology is a significant reduction in mass and volume requirements – at the expense of the magnitude of the thrust – making a MEMS µCPS an ideal solution for implementation on small spacecraft platforms. MEMS thruster technology was flown on the PRISMA mission as a technology demonstrator experiment. Each module, shown in Fig. 4, incorporates four thrusters, including MEMS nozzles, valves, filters and heaters [13]. The PRECISE µCPS will further develop this concept to incorporate a micron-scale catalyst bed and improved component performance, to meet the thermal and pressure requirements a monopropellant thruster demands. A comprehensive discussion of the novel technologies employed by the PRECISE µCPS can be found in Reference [10].

The performance characteristics of the µCPS tend themselves to allowing very small impulses, either through discrete “impulse bits” by using a fast acting valve or a continuous varying impulse with a proportional valve. One objective of the PRECISE project is to determine which approach is best suited to the MEMS manufacturing process. For the purposes of the analysis presented in this paper it is assumed that a proportional valve is used, providing a continuous, varying thrust when in operation. Such a thruster is beneficial on a small spacecraft as it provides impulses suitable for extremely precise orbital and attitude control, making the spacecraft an ideal on-orbit observation and inspection platform.

Fig. 4. Cold gas MEMS micro-thruster (after Rangsten et al [13])

2. *Visual inspection payload*

The InspectorSat GNC system is incorporated into the VIP, where a closed feedback loop provides an estimation routine for the relative position and pose of a target spacecraft. These data are used to estimate the relative motion of the target spacecraft and, therefore, predict its motion on orbit. The precise estimation of position and pose parameters is required for the safe execution of close proximity inspection manoeuvres. The implementation of the μCPS provides an additional actuator for the control of the spacecraft motion. It both compliments more traditional attitude actuation systems, such as magnetorquers or reaction wheels, and provides a means for manipulation of the InspectorSat's orbital motion.

The GNC system architecture includes inputs from systems normally distributed around a spacecraft system and centralised through the on-board computer (OBC). Therefore, the VIP resides on a dedicated computing board and includes the functions relating to inspection manoeuvres, i.e. planning, control and frame transformation routines. Sensing of the target spacecraft motion is based on a monocular imaging sensor. The image frames are filtered to provide state parameters describing the relative motion of the target spacecraft. Real-time processing provided by an FPGA soft-processor allows relative motion estimates to be incorporated into the InspectorSat GNC loop, complementing the orbit and attitude sensory data provided by GPS, sun sensors, star cameras or magnetometers. This data is fused to provide control inputs to the navigation computer path planner which is responsible for firing the AOCS actuators.

The imaging sensor is a standard CMOS part equipped with a novel sphere lens. Initial ground experiments have focused on the use of a conceptual imager comprised of a 1.3 megapixel CMOS sensor paired with a uniform refractive index lens, 8 mm in diameter, constructed from S-LAH79 glass, with a refractive index of 2.00. To mitigate the typical issue of interfacing such lenses to a commercial-off-the-shelf (COTS) planar sensor,

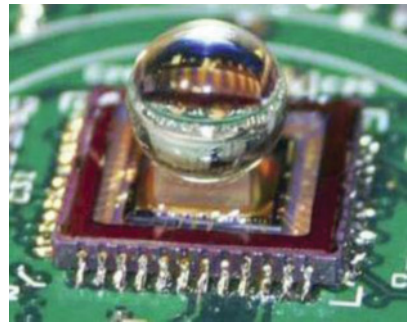


Fig. 5. Sphere lens camera comprised of sensor, FOFP and sphere lens (after Laycock and Handerek [14])

a faceted FOFP is used to mount the lens to the sensor [14]. The nature of the optics allows a wide field of view of 120 degrees, while the FOFP allows the total package volume and mass to remain low. The complete imager, shown in Fig. 5, is less than 2 cm in height and has a mass of approximately 46 grams. The compact nature of this hardware naturally aligns with the low mass, low volume design requirements for the InspectorSat platform.

C. Solar Sail Spacecraft

The SSSC ("Vympel") will demonstrate two elements of attitude and orbital control using solar sails; firstly, apogee adjustment using a solar sail and, secondly, attitude manipulation using a pair of "ailerons". Launched in a stowed configuration, the 5 kg spacecraft is contained within a "corset", permanently mounted to the launch vehicle upper stage. Upon separation, the supporting booms are opened in a similar fashion to an umbrella, providing an 8 m square surface area. The primary spacecraft systems are mounted to a 3 m boom attached to the centre of the sail. The μCPS thruster is mounted to the primary spacecraft, providing two degrees of rotational freedom (pitch and yaw). The boom also supports the pair of solar-ailerons, aligned with their longest dimension parallel to the face of the sail. The purpose of these secondary sails is to act as a control surface for the roll-stabilisation of the sail structure.

The μCPS will be used to introduce controlled perturbations under microgravity conditions that disturb the structure of the soil sail, using the assumption that the system acts in an elastic manner. Repeated yaw or pitch rotations of 90° about the SSSC centre of mass will instigate these disturbances. Each manoeuvre will comprise of a simple burn-coast-burn series to initiate and cancel the rotational motion of the SSSC. The burn times are limited by the structural strength of the spacecraft and the maximum rotational acceleration produced by the μCPS. Assuming a maximum thrust of 10 mN it has been estimated that 50 mL of propellant will allow approximately 20 rotational manoeuvres, with a total burn time of nearly 2000 seconds.

III. Inspection and co-orbital motion manoeuvres

D. Formation flying configurations

While the SSSC is performing its experimental manoeuvres the InspectorSat will be responsible for observing these changes in attitude and maintaining a relative position that avoids undesirable reflections. Furthermore, the solar sail orbital trajectory will also vary due to the incident SRP. The role of the InspectorSat must, therefore, include manoeuvres that allow the safe observation of the SSSC while maintaining the co-orbital formation. The difference in spacecraft mass-to-area ratios will result in a differential in SRP and atmospheric drag pressure between the InspectorSat and the SSSC. To do so, the comparative effects can be reviewed. The magnitude of SRP and drag forces as a function of altitude is shown in Fig. 6. At approximately 800 km, atmospheric drag is shown to be an order of magnitude greater than SRP. For SRP to be the dominant force, a minimum altitude of 1400 km is needed. Currently, the proposed mission orbit is at a sun-synchronous inclination and altitude of between 1500 and 2000 km.

Characterisation of perturbations on orbit due to SRP and inspection of other sail features will be accomplished by management of the relative motion between the two spacecraft. Three such manoeuvres are considered for inspection of a co-orbital target [4].

Assuming the SSSC orbit is circular or nearly circular, a 2-1 safety ellipse trajectory relative to the SSSC the InspectorSat will remain within a range defined by its orbit. The motion is illustrated in Fig. 7a) and demonstrates how a complete relative orbit of the target spacecraft is achieved for every complete orbit of the Earth. The 2-1 ellipse is a stable, natural motion but will drift slowly relative to the target over the course of several orbits due to SRP and other perturbation effects. Such disturbances can be mitigated by the μCPS based on observations made by the VIP, forming the basis for the measurement of the relative drift of the InspectorSat and SSSC orbits.

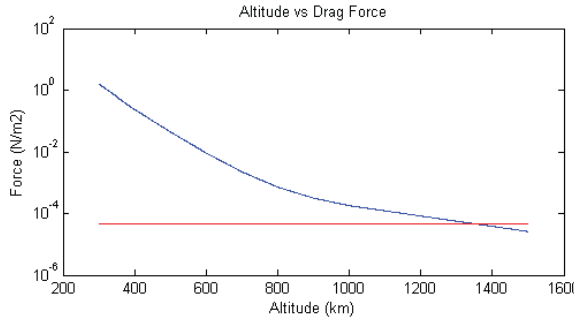
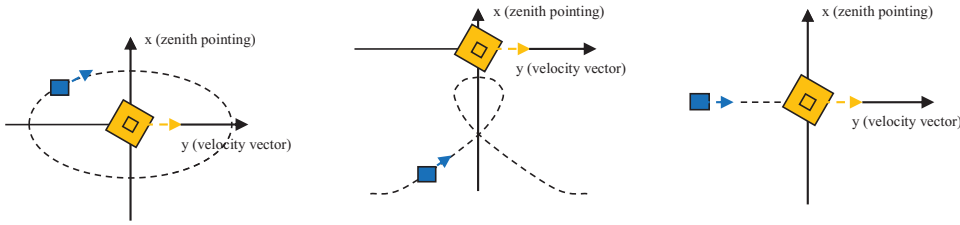


Fig. 6. Atmospheric (blue) and solar radiation (red) pressures as a function of altitude



a) 2-1 "safety ellipse" **b) "Pogo" motion** **c) "V-bar" motion**

Fig. 7. Co-orbital inspection trajectories

Observation of the target spacecraft can be achieved either by maintaining a fixed orientation of the InspectorSat in either the Earth-Centric Inertial (ECI) or Local-Vertical Local-Horizontal (LVLH) frames. In the former case the maximum range of observation is limited by the Field-of-View (FoV) of the inspection camera optics. The latter case represents the more typical nadir-pointing mode of satellite operation and allows for the use of narrow-FoV or telephoto optics. However, active control is required to maintain pointing toward a target.

Forced motion modes may also be used to manoeuvre the InspectorSat into a specific orientation relative to the target, e.g. to align the principal axis of the camera orthogonal to the sail surface. To achieve such manoeuvres, a "pogo" orbit can be used to maintain position above or below the target spacecraft with respect to the Earth. The pogo orbit is artificial and not naturally stable. When compared to the 2-1 safety ellipse, co-orbital motion by this method would require a more intensive regime of periodic thrusting, however, it would allow longer linger times around a specific position relative to the target spacecraft. The relative motion of the inspection platform takes the distinctive "tear drop" form shown in Fig. 7b).

Finally, a "V-bar perch" configuration, as illustrated in Fig. 7c), would position the InspectorSat in an identical orbit to the SSSC in either a lagging or leading position along-track. Unlike the pogo orbit, the V-bar perch would require very little fuel to maintain its relative position but would require attitude manoeuvres by the SSSC in order to perform observation and inspection tasks. This simple co-orbital configuration does, however, provide a naturally fixed position relative to the SSSC in order to perform SRP drag characterisation measurements.

E. Visual inspection of a solar sail

For the InspectorSat to take images of the solar sail, an investigation into the solar sail billowing effect has previously been undertaken to find potential focal lengths and the resultant ideal point at which to take images [5]. At this position the light is sufficient to detect differences between the target satellite's distance and attitude and avoid high intensity light areas. The general shape of the solar sail with solar radiation pressure on the sail is

analogous to a catenary described typically by Earth's gravity on the curve where both shapes are defined by a uniform force exerted across the shape. A 3-dimensional model can be built to observe a generalised square solar sail shape. Adding stowage fold-marks and material effects, physically represented by wrinkles and creases, is achieved by using a number of directed Fourier-series saw tooth waves.

To investigate the inspection opportunity of the sail, ray tracing of reflected sunlight is performed on the new solar sail model. Using an average light energy of 1370 W/m^2, each ray can be divided into representative quantities and their reflected signature can be traced and followed to investigate light intensities at varying distances, as shown in Fig. 8 for a 4-quadrant sail (artificially heavily billowed for effect).

This technique can then be used to investigate differing light reflection effects of:

- Billow sizes – fully deployed fixed-boom sail, partial deployment of sail or booms
- Folding patterns – sine, tan, sawtooth functions at varying heights and frequencies
- Material properties – aluminium, mylar, teflon light coefficients and elasticity characteristics

The new fold line reflections shown highlight the imaging opportunities in the direct sunlight focal point of the reflections, which can be used to estimate shape and attitude. Side lobe reflections can also be investigated where the macro-structure of the sail is observable at lower light intensities. At the focal points, the light intensity can be high and manoeuvres will be necessary to move to a safer viewing area where there is enough light to take images but also prevent saturation and damage to the imager.

Considering such constraints, it is necessary to estimate the resolution requirements for inspection of the SSSC sail surface to determine the magnitude of formation separations. For surface feature and detail inspection the imager instantaneous field of view (IFoV), or angular resolution, can be calculated as a function of range,

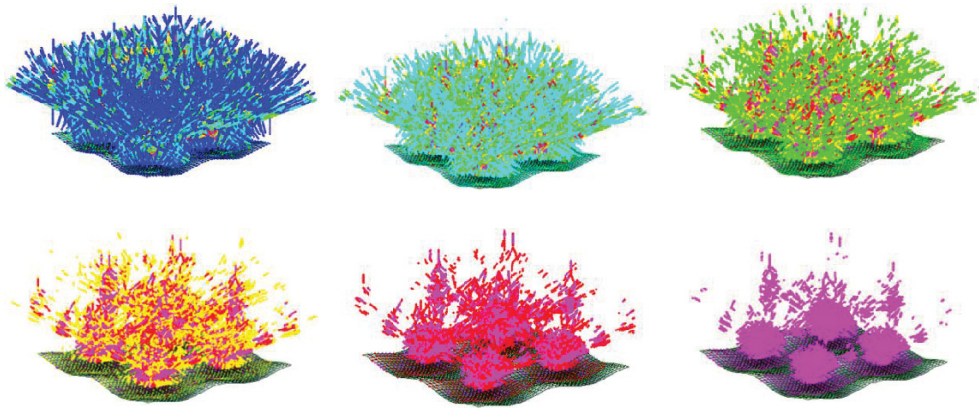


Fig. 8. Light energy levels for a solar sail (divided into 20 W bands up to 120 m) (after Bridges and Palmer [5])

considering the angular FoV and candidate image sensor resolutions. Fig. 9 illustrates the effect of range on IFoV assuming a wide angle lens with a FOV of 120 degrees.

Adding a further constraint of an ideal sail surface resolution of 0.01 m^2/px, very close inspection proximities of under 50 m are required. In Fig. 10 the absolute minimum range required to meet the desired IFoV is shown with respect to sensor resolution. Inspection using a wide angle lens would require extremely close co-orbital manoeuvres at ranges of under 10 m, even with a high resolution sensor. Furthermore, higher resolution sensors typically trade resolution improvements for pixel sensitivity, due to limitations in the physical size of the sensor. Selection of an appropriate sensor will result from a trade-off study using sensor parts suitable for use in the space environment.

F. Relative motion at deployment

If the SSSC and InspectorSat spacecraft are mounted independently on the launch vehicle, any differential in the separation characteristics will result in a cumulative increase in relative distance during the early commissioning stages of the mission. Before rendezvous and phasing of the orbits can be considered, the spacecraft must be checked out for normal operations and brought into a stable attitude regime. The typical duration of this "launch and early operations" (LEOP) stage is assumed to lie between two and seven days. Therefore, after detumbling, a primary objective of the μCPS will be to approach the SSSC. To analyse the motion of the spacecraft orbits, two tools are used:

- High precision orbital propagation using a Bulirsch-Stoer integrator with varying solar conditions and NOAA NRLMSISE-00 atmospheric drag and solar radiation models (locally called BSAT) [15]

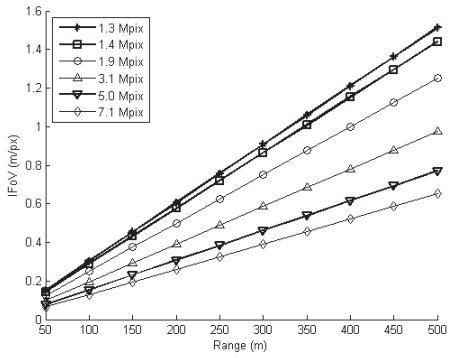


Fig. 9. IFoV as inspector-target range increases for a range of sensor resolutions (120 degree FoV)

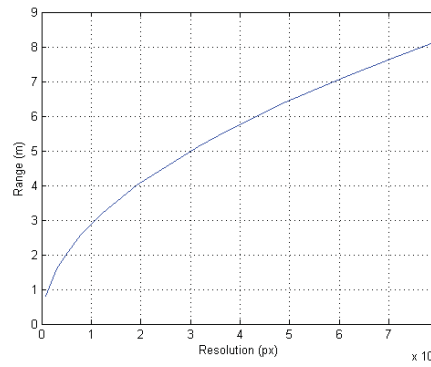


Fig. 10. Minimum range for 0.01m^2/px

- Further analysis using SatLauncher, a custom formation flying visualisation tool developed at the SSC.

The orbital motion of the respective spacecraft was propagated from identical initial conditions for a number of deployment scenarios, where relative deployment angle (α) and Δv differences were considered. The orbit injection scenarios which were considered were based on the following variables:

- Altitude of 1500 km (fixed to ensure drag is not a key parameter in observed measurements).
- $\alpha = 0$ to 180° in steps of 10°. This relates to a rotation about the along-track axis.
- $\Delta v = 1$ to 5 m/s in steps of 0.5 m/s. From the initial conditions, this initial acceleration is added to simulate ejection from the launch vehicle.

The simulation variables were considered during a solar average environment where the solar flux at 10.7 cm (f107a) is 120 (electrons/cm^2/s/sr) and geomagnetic kp-index is 14. For the purposes of this analysis solar average conditions were chosen.

If the separation of the spacecraft is identical in terms of angular offset and velocity, the development of the relative range is due to differences in the SRP and other drag effects incident on the respective spacecraft. The resulting relative range of the two spacecraft over a period of seven days due to SRP only is shown in Fig. 11. Over the period of one week the natural increase in range reaches an estimated distance of 25.42 km.

When considering the angular deployment difference between spacecraft ejection from the launch vehicle, a coupling in motion between the radial and along-track motion is seen. At ~90° (inclination dependant), there is minimal radial and along-track motion – leaving only cross-track motion. Despite this minimisation, a drift range of 200 km was estimated over the period of one week. When deployed at 180° or in opposite directions, a

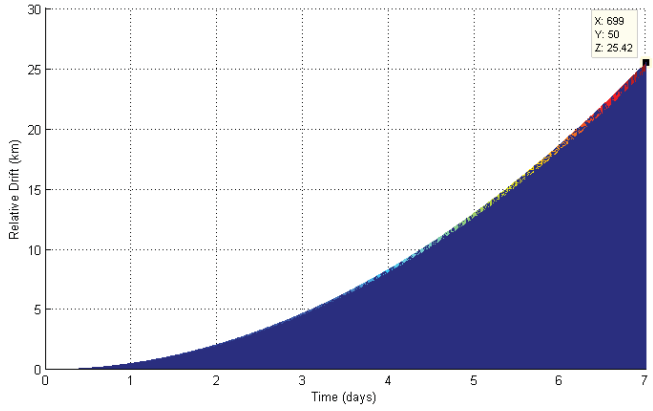


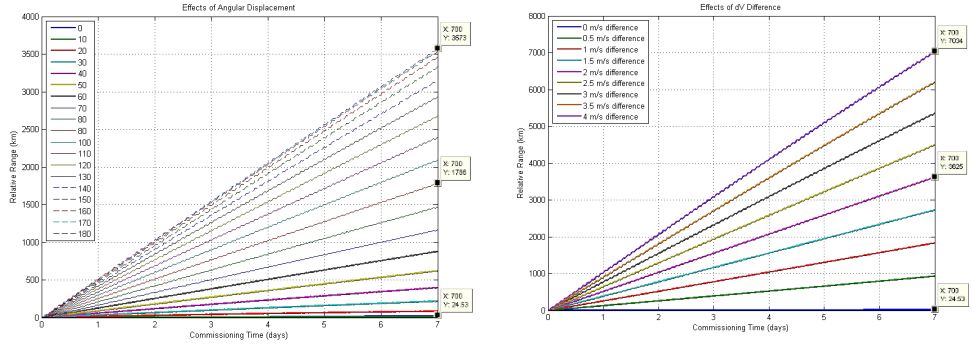
Fig. 11. Relative range with matched injection vector & magnitude

maximum drift range of approximately 3,500 km occurs in the same period of time. The development of the separation of the spacecraft due to angular differences at separation is shown in Fig. 12a). For rendezvous, a classic circular coplanar phasing manoeuvre is required to match semi major axis, *a*, and notably the true anomaly, *v*.

As with angular differences, the relative difference in ejection velocity from the launch vehicle ultimately leads to large relative ranges between the SSSC and InspectorSat. As shown in Fig. 12b), with 1 m/s difference in Δv, the relative range is estimated to reach approximately 1,800 km, increasing to over 7,000 km at 5 m/s Δv difference. Again, these large displacements would require a circular coplanar phasing manoeuvre to return back to the SSSC target.

To perform cost optimal continuous thrust manoeuvres, the methodologies developed by Palmer and Sauter were used to determine the Δv rendezvous requirements for numerous trajectories [16, 17]. Table 1 and Table 2 show the estimated Δv requirements to allow rendezvous of the spacecraft after one week of natural motion. Management of the total Δv can be achieved by performing low-thrust long-term μCPS burns or vice versa. Clearly, co-axial deployment of equal impulses is most desirable in minimising the total thruster utilisation and fuel quantity requirement. The results also highlight the diminishing returns of extending the burn duration. For example, in Table 2 the reduction factor of the integrated Δv between a 1 and 5 orbit burn is 55, which reduces to only 1.3 by extending the burn to 11 orbits.

To visualise these relative motions, the SSC tool “SatLauncher” is used. SatLauncher is a Java-based orbit propagation and visualization tool designed for investigating deployment of satellite formations or clusters. The



a) Effect of angular displacement **b) Effect of Δv difference**

Fig. 12. Relative ranges with differing separation injection vector & magnitude

Table 1: Local Frame Δv Requirements (angle difference)

Solar Sail to InspectorSat Deployment Conditions	Solar Sail to InspectorSat Range (km)	Optimal (integrated) Δv Requirement (m/s)		
		1 orbit burn	*5 orbit burn*	*11 orbit burn*
20° Angle, 0 Δv Difference	84.9192	41.5919	1.0390	0.4828
40° Angle, 0 Δv Difference	400.2576	186.4709	4.2112	3.5774
60° Angle, 0 Δv Difference	883.1932	388.6965	25.5587	21.0631

Table 2: Local Frame Δv Requirements (Δv difference)

Solar Sail to InspectorSat Deployment Conditions	Solar Sail to InspectorSat Range (km)	Optimal (integrated) Δv Requirement (m/s)		
		1 orbit burn	*5 orbit burn*	*11 orbit burn*
0° Angle, 0 Δv Difference	24.6243	11.1148	0.2092	0.1526
0° Angle, 1 Δv Difference	1838.40	1157.60	158.7388	112.2188

simulator allows the input launcher settings: altitude, radius, rotation rates, and satellite and deployment settings such as initial size, position on the launcher, deployment time, and velocity direction (in Cartesian or Epicycle coordinates). SatLauncher allows the user to visualise motion based on the Hills-Clohesy Wiltshire (HCW) equations, where classical orbital motion is rederived into a linearised Cartesian frame. The final derived equations for a safety ellipse, for example, are given in both *x* and *y* coupling, whereby any motion in the *x*-axis (radial) can be driven by the *y*-axis (along track) and vice versa. This is described in the following equations:

$$\dot{x}_0 = \frac{\omega y_0}{2} \quad \text{Eq. 1}$$

$$\dot{y}_0 = -2\omega x_0 \quad \text{Eq. 2}$$

Utilising these equations, SatLauncher can visualise and control the relative drift between satellites ejected from a launcher or between the launch vehicle and its payload spacecraft. The variables tested in the previous high precision propagator are also used for input with SatLauncher. In Fig. 13 and Fig. 14, the effect of changes in Δv injection and angle of deployment, respectively, are shown. The Figures illustrate the immediate imaging opportunities and collision risks after launch vehicle separation. In particular, Fig. 13 shows the effect of varying Δv on the ellipse size; where Fig. 14 shows the effect of varying x-velocities on the final along-track positions.

IV. μCPS thrust requirements

G. Fuel and performance estimates

A typical mission will require further manoeuvres in addition to those necessary for rendezvous. These manoeuvres would make use of a microthruster over the course of its mission lifetime. Definition of a nominal mission comprised of such manoeuvres allows for the development of a Δv budget. The operational requirements

would use the steady state firing (SSF) thrust modes planned for the proposed μCPS thruster. The following analyses are based on the targeted performance parameters for the μCPS thruster:

- Thrust: 1-10 mN
- I_{sp} ~180s
- Operational mode: Steady-State Firing (SSF)

The InspectorSat and SSSC missions represent a formation flying configuration. In practice, it would be desirable to utilise a combination of high-frequency short-duration and low-frequency long-duration burns. The final requirements are dependent on the manoeuvre classifications summarised in Table 3. These include detumbling, orbital adjustments for co-orbital motion such as natural motion formation flying, SRP drag compensation, momentum dumping and attitude control manoeuvres. A given mission will comprise of a number of these manoeuvres. Identification of the mission parameters, for example number of observations or total lifetime, allows a Δv budget to be compiled based on the total number of manoeuvres required for the mission.

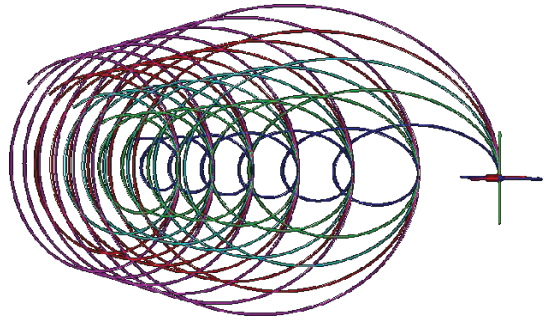


Fig. 13. SatLauncher output trajectories for deployment at varying velocities (1-5 m/s)

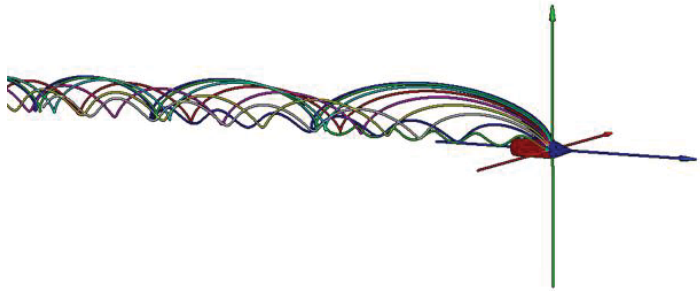


Fig. 14. SatLauncher output trajectories for radially-spaced deployment (0-90°)

Table 3: Manoeuvre descriptions and nominal Δv requirements

Manoeuvre	Description	Required Δv and burn duration
De-tumble	The mechanism of separation from the launch vehicle will induce unpredictable torques on the spacecraft body, causing it to tumble randomly. The AOCS will utilise reaction wheels and the thrusters to bring the spacecraft attitude into a controlled state.	3 m/s for 18000 s
Drift and rendezvous	After detumbling, the orbital control system will utilise the thrusters to implement a trajectory to bring the InspectorSat to a position where co-orbital motion can be initiated.	1.4 m/s for 8100 s
Circumnavigation entry	The thrusters are used to insert the InspectorSat into the desired inspection orbit in formation with the SSSC target.	0.3 m/s for 1800 s
SRP compensation	To obtain accurate SRP measurements from SSSC observations, it is necessary to consider the thruster requirements to compensate for SRP on the InspectorSat	0.1 m/s for 600 s
Momentum wheel unloading	Utilisation of the thrusters to compensate for momentum wheel spin-down	0.05 m/s for 290 s
Circumnavigation maintenance	Maintenance of the co-orbital formation	0.05 m/s for 290 s
Station keeping	Compensation for other perturbations and orbital corrections	0.05 m/s for 290 s

The calculation of a Δv budget is based on the ideal rocket equation. Two approaches are taken, one to define the mission utility in terms of the total Δv for a series of manoeuvres and a second in terms of the I_{sp} and spacecraft mass. When implemented, longer burn durations of the μCPS are desirable to obtain a constant mass flow rate, targeted at 6 mg/s. As such, a conceptual mission profile has been defined in terms of Δv and burn duration, summarised in the right hand column in Table 3. The manoeuvres presented in this example have been selected to indicate a mission duration of approximately 6 months. The burn durations are otherwise arbitrary, but indicative of the manoeuvres that would be required for an inspection mission and provide a baseline analysis for revision as further mission parameters are defined. Using the stated assumptions the total Δv and required propellant mass can be estimated. The results are summarised in Table 4 for spacecraft with masses of 5, 10 and 50 kg at the maximum mission altitude of 2000 km.

A low-frequency high-duration burn requirement for an attitude control manoeuvre, such as momentum dumping, for spacecraft masses within the 5 to 50 kg range is shown in Fig. 15. The axes show Δv requirements and propellant mass for relieving wheel saturation. In this example the use of an SSTL 10SP-M micro-reaction wheel is assumed. This wheel is suitable for the attitude manoeuvres required from a small satellite, such as InspectorSat. The results illustrate, therefore, that the μCPS thruster would be suitable for such attitude control.

After rendezvous and in addition to attitude control, the remaining orbital control manoeuvres include SRP and atmospheric drag compensation, natural motion maintenance and station keeping. Due to SRP, station keeping is a necessary manoeuvre if a relative position to the target SSSC is required for observation purposes. Such a manoeuvre would be similar to the V-bar perch, in the following analysis an alternative perches of equal

Table 4: µCPS Δv and fuel requirements for inspection manoeuvres

Satellite mass (kg)	Estimated total Δv (m/s) (inc. 15% error margin)	Total fuel required (g)
5	27.42	77
10		154
50		771

distances along-track and in radial separation are considered, but assumes in-plane relative motion. The formation flying relative motion assumes maintaining perch distances of 1 m to 500 m for spacecraft with masses within the assumed range of 5 to 50 kg. Altitude is again fixed at 2000 km and a burn is required every 60 seconds. In Fig. 16, propellant flow rate is presented for these spacecraft parameters. In Fig. 17, the fuel mass requirement to perform this manoeuvre in 10 minutes is presented.

The assumed I_{sp} of 180 s is used in all of the above calculations. However, as the final µCPS thruster I_{sp} will not be known until it is characterised experimentally, it is sensible to study the effect of variation in I_{sp} for the proposed spacecraft masses and propellant mass estimates. This can be achieved through the use of a Dummkopf plot. The purpose of such analysis is to determine how I_{sp} is affected by the inert mass of the spacecraft. This allows the assessment of sensitivity to minor differences in the final mass budget and thruster performance.

In Fig. 18, a series of curves is presented for the three suggested spacecraft masses, representing a range of inert mass fractions assuming thruster dry mass to lie between 0.1 and 1 kg. This is based on a ΔV of 27.4 m/s and inert mass is equal to the assumed spacecraft masses, after subtracting the initial propellant estimates and thruster mass (i.e. the thruster wet mass). The three Dummkopf charts illustrate the magnitude of the inert mass fraction. At the nominal I_{sp} of 180 s the slope of the plots are of a shallow gradient. This would suggest that small variations in thruster or spacecraft mass would have a lower risk of affecting thruster performance.

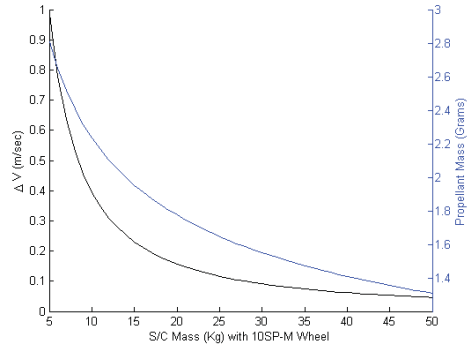


Fig. 15. Δv and propellant requirements for momentum dumping on a single axis for a range of microsatellite masses

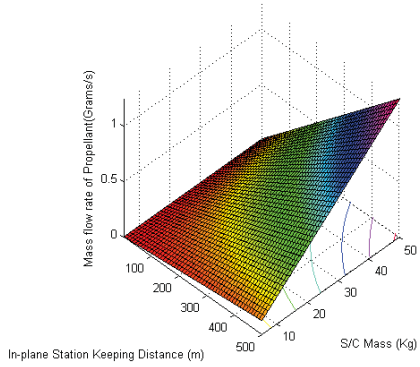


Fig. 16. Mass flow rate (g/s) for station keeping at 2000 km altitude

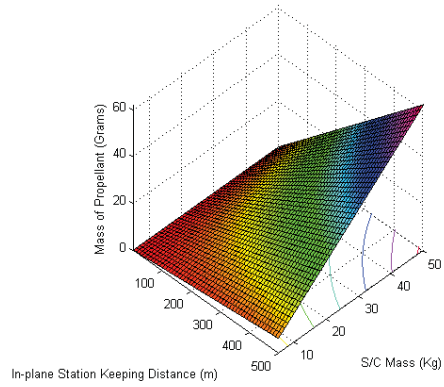
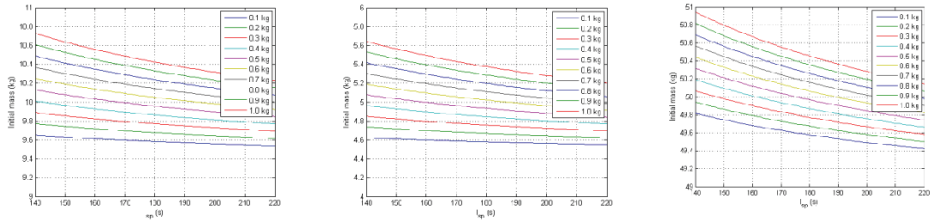


Fig. 17. Manoeuvre fuel requirement for station keeping at 2000 km altitude



a) 5 kg spacecraft **b) 10 kg spacecraft** **c) 50 kg spacecraft**

Fig. 18. Dummkopf charts for spacecraft with thruster masses of between 0.1 and 1.0 kg

V. Conclusions

The analysis presented above has taken a nominal series of manoeuvres for which a microthruster with the performance expected of the PRECISE µCPS can be used. The results demonstrate the µCPS is capable of providing the actuation requirements necessary for on-orbit inspection of a target such as the SSSC. The activities expected of such a thruster include the technical capability to manoeuvre the InspectorSat around a target and also to enable new methods for the collection of scientific data, such as the measurement of SRP. Furthermore, the system design is also seen to be robust against variations in final thruster efficiency. Longevity of the thruster is dependent upon minimising total on time to extend the mission life time. As such. smaller spacecraft have been shown to be more suited to the PRECISE system as better agility provides a lower demand on the total fuel capacity required. Combining the launch vehicle separation to ensure deployment at similar

velocities and angular displacements have also demonstrated how the total fuel and on-time requirements of the μCPS can be minimised to better aid in the longevity of the thruster lifetime.

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Deployment Simulations of the Space Tow Solar Sail

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The Space Tow is a modular solar sail design, uniting versatility of size with ease of manufacturing in an easily stowable package. All stowed structures must be deployed to their mission configuration, which is challenging for such a light and flexible design. The present study explores the "leave behind" and "drag along" deployment strategies in simulation and mathematical analysis. The conclusions are that the "drag along scheme" is sensitive to perturbations and that the "leave behind" scheme needs careful consideration of its parameters or risk that the undeployed stack is overrun by the deployed structure. The present study also discusses the energy dissipation needed for a robust deployment and proposes both a frictional and acceleration-rate proportional damping such as occurs in deformations in an accelerating frame.

Nomenclature

c_j = Jerk damping coefficient, Ns^2

d_b = Braking distance, m

E_k = Kinetic Energy, J

F_f = Frictional force, N

F_n = Normal force, N

F_s = Solar pressure force on panel, N

l_0 = Longeron length, m

m_{bay} = Sail and longeron (sail bay) mass, kg

m_i = Deploying mass, kg

m_s = Mass of sail panel, kg

M_i = Deployed mass

t = Time, s

t_s = Sail thickness, m

U_f = Frictional work, J

v_0 = Initial deployment velocity, m/s

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v_l = Velocity at full length, m/s

$v_{m,i}$ = Velocity of deploying stack, m/s

$v_{M,i}$ = Velocity of deployed structure, m/s

x_M = Position of payload, m

x_m = Position of deploying stack, m

μ_f = Friction coefficient

I. Introduction

Solar sailing for space missions is just around the corner, with the first demonstrations of flight-worthy solar sails successfully launched[a] [1, 2].

A. Why solar sailing?

Solar sailing uses the momentum of ambient photons to continuously accelerate a spacecraft, which is attractive not only as an elegant notion, but from a variety of viewpoints including allowing open ended missions and unique mission physics. Solar sails seem like an elegant solution to space propulsion, in no small part due to their light weight and compact size, as very little fuel is needed to propel the craft and the designs require a light craft for effective propulsion, enabling launch into orbit with a comparatively small footprint in mass as well as size. When no fuel is needed to propel the craft, the limitation of impulse force on the mission is dropped as well, as provided enough time, the accumulated photon thrust would provide all Δv needed for any mission, effectively enabling long-distance planetary missions and missions with sample retrieval that otherwise would have a prohibitive fuel cost. This would significantly enhance long-range missions such as an interstellar heliopause probe or a Kuiper belt fly-through [3]. Among the most interesting aspects is the sustained acceleration available through the solar sails, a continuous acceleration without affecting the craft enables access to highly non-keplerian or non-inertial orbits [4, p. 11–24], allowing for the Geostorm mission or a solar polesitter mission [3, 5].

B. The space tow

One interesting design solving many scale and fabrication difficulties is Greschik's Space Tow, a scalable solar sailing system composed of a series of smaller sails connected together in a train, the main advantage of the Space Tow would be that it is easy to tailor total sail area by adjusting the number of small sails, Fig 1. The smaller dimension of the sails would also facilitate design, manufacture and testing [6–10]. Compactness, lightness and manufacturability are all attractive qualities for the Space-Tow concept, facilitating a launch in the foreseeable future. The thin sails and filament like cords give very little stiffness to use for deployment, and the sudden slack-taut transition induces dynamic shock that the lack of stiffness cannot alleviate. Tibert and Lennon recommend an

[a] "NASA Nanosail-D Home Page" http://www.nasa.gov/mission_pages/smallsats/nanosaild.html (accessed 2013-03-15)

inertial deployment system [11], proposed by Greschik [12], where the stack of sails and payload are separated by a "kick" that deploys the sails one by one from the payload and gradually accelerate the payload and craft to mission specifications. In Tibert and Lennon's investigation, one main cause for concern is the bungee effect at the slack-taut transition, with potential to collapse the whole structure. They propose that this could be resolved by frictional damping. Frictional damping is not the only damping present in such a structure, as due to the deformation of sail and filament, heat is dissipated in a way that could be conveniently described in the accelerated frame of the Space Tow, with said damping being acceleration-rate, or "jerk", proportional damping [9].

This study shows initial simulations of the deployment strategies in a commercial finite element software as well as discusses the energies the frictional damping need to dissipate for the deployment of a Space Tow structure, and how this would relate to a model of jerk-based damping. The focus is to shed light on the behavior and magnitude of the involved quantities.

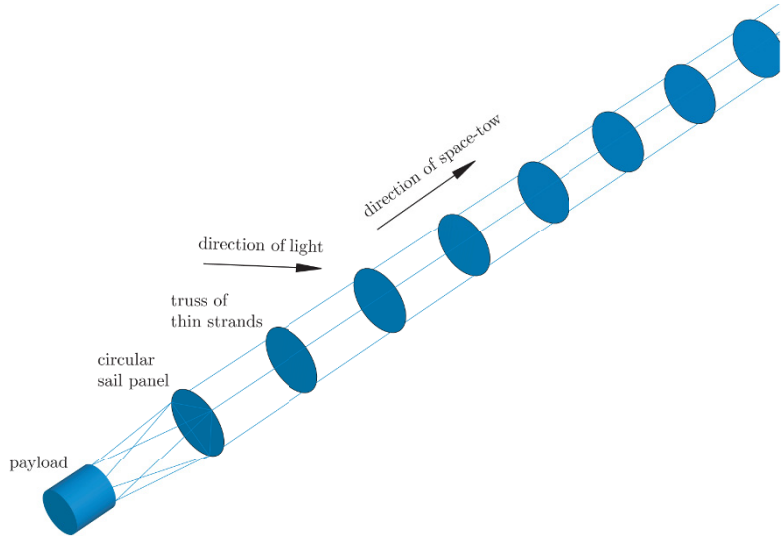


Figure 1. Fully deployed Space Tow, reproduced from [11].

II. Deployment Simulations

A. Model setup

The model studied consists of a five panel version of a Space Tow with each panel consisting of a sail sheet and a reinforcing ring, the panels are interconnected by three longerons. The ring is modeled with a diameter of 4 m, consisting of carbon fibre with density ρ= 1800 kg/m^3, Young's modulus $E = 560$ GPa and Poisson's ratio ν = 0.35 [10]. The ring has a rectangular cross-section 1.65 mm by 0.60 mm with the wider direction in the ring plane and having a total mass of 22.4 $\cdot 10^{-3}$ kg. In this ring is a sheet of 0.90 μm Mylar with density 1390 kg/m^3 and Young's modulus 5 GPa, Poisson's ratio 0.38 resulting in a sail mass $m_s = 17.5 \cdot 10^{-3}$ kg, the ring is connected to

the rest of the structure by three longerons of carbon fibre filament evenly spaced around the ring. Each filament is modeled with a circular cross section with a radius of 91 μm, corresponding to a 60-filament cable like the one proposed by Tibert and Lennon [10], the material parameters are the same as the ring. This gives a longeron mass of $0.33 \cdot 10^{-3}$ kg, or about $0.98 \cdot 10^{-3}$ kg per set of three longerons. This corresponds to a total mass of about 40.9 $\cdot 10^{-3}$ kg per bay of three longerons, a ring and sail sheet, the model's total mass adds up to about 0.21 kg. In a full scale Space Tow, this would correspond to a total tow mass of 32.51 kg plus the 50 kg payload.

The model was simulated in ABAQUS 6.11, with the 7 m longerons meshed as 16 truss elements of equal lengths. The rings were meshed as beam elements with an element length of 0.2 m, for a total of 63 elements, the sheets as triangular shell elements with a side length of 0.5 m, total 117 elements per sheet, Fig 2.

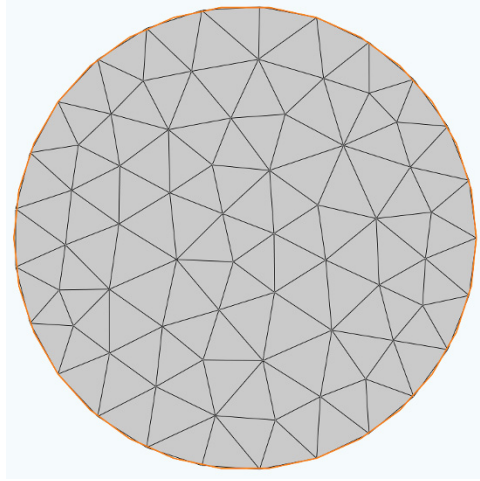


Figure 2. The meshed sail sheet, 117 elements total.

The model was constructed with each of the rings having an attached sail sheet by tie constraint along the circumference of the ring, forcing all translational degrees of freedom to be equal on the nodes constrained [13, sect. 33.3.1]. The rings are each separated by 1 mm of "packing height", for a total model height of 5 mm from the plane of the boundary conditions. Each ring is fastened with a tie to three longeron ends, evenly spaced around the circumference of the ring. The longerons are each folded in four even lengths, in a slight zig-zag pattern. Each length ends 0.25 mm higher relative the panel plane so that the total height is the same as the 1 mm packing spacing between each panel. Fig 3. The deployment simulations are set up with the first set of longerons pinned at one end to a rigid boundary spaced as on the panels, the other to the first panel.

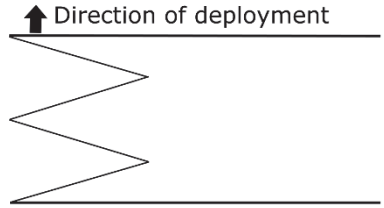


Figure 3. Schematic over the stowed longeron geometry, detail depicts only one of three stowed longerons.

B. Leave behind deployment

The leave behind deployment simulation models a stack of panels separated from the payload with an initial kick, where individual sail panels separate from the stack as their longerons unfold and are stretched taut [12]. When the longerons of a sail bay, here defined as panel with longerons, are stretched a solar force is applied to the film sheet. An initial velocity $v_0 = 0.03$ m/s is applied along the axis of deployment to all films, rings and longerons except the set fastened to the boundary points. The model then sequentially adds a uniform pressure of $1.01 \cdot 10^{-4}$ Pa normal to the film on the side facing the rigid boundary, the pressure is added instantaneously when the preceding set of longerons are fully unfolded, Fig 4.

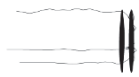
C. Drag along deployment

The drag along deployment simulation models a stack of panels where only the furthermost panel is separated from the payload, by initial kick, when its longerons become stretched, it lifts the next panel from the stack and so on. As the tow deploys the increasing solar pressure accelerates the deployment [12]. An initial velocity of $v_0 = 0.3$ m/s is applied only to the outermost ring and film. The model then adds a uniform pressure $1.01 \cdot 10^{-4}$ Pa normal to the film on the side facing the stack and/or rigid boundary, the pressure is added instantaneously when the preceding set of longerons are fully unfolded, Fig 5

D. Discussion

The initial velocity of the outermost panel in the drag along setup produces an accordion effect where the kinetic energy of the outermost panel is transferred almost entirely to the next panel when the longerons become taut. This effect propagates throughout the structure causing chaotic perturbations of the panel orientation relative to the axis of motion, when the perpendicular solar pressure drives the panel forward, these perturbations are exaggerated further with uncontrolled deployment as a result, Figs. 5(d)–(j).

One could consider damping the bungee-effect from the initial kick and acceleration, alleviating such a problem, but if perturbations arise from other sources, the same situation applies. Control would then need to be applied on each panel, costing mass, or with high precision, possibly over long time spans, to counter not only



(a) Stack deployed by initial kick, first panel illuminated and deployed.

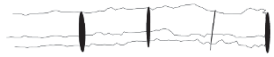
(b)

(c) Second panel illuminated.

(d) Note the slight perturbation on the first panel (on the far left).

(e) Third Panel illuminated and perturbations spread.

(f) Perturbations are slowly ebbing out.



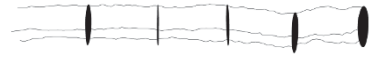
(g)

(h) Fourth panel illuminated.

(i)

(j) All five panels are fully deployed and illuminated. Structure is settling.

(k)



(l)

Figure 4. Figures from Space Tow "leave behind" deployment simulation.

(a) First panel deployed by initial kick and illuminated.

(b) Second panel deployed and illuminated, most of the initial kick is transferred to the second panel.

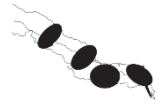
(c) Third panel deployed, most of the initial kick is transferred to the third panel.

(d) Fourth panel deployed, most of the initial kick is transferred to the fourth panel. Perturbations from uneven longeron stretching start skewing illuminated panels.

(e) Fifth panel deployed, most of the initial kick is transferred to the fifth panel.

(f) All panels illuminated and deployed. Panels 3–5 are skewed so that the normal pressure produces a moment.

(g) The moment is starting to turn the deployed structure.



(h)

(i)

(j)

Figure 5. Figures from Space Tow "drag along" deployment simulation.

chaotic perturbations but possible skew and translational accelerations. The constant acceleration will also yield increasingly larger shocks on the panel lifted from the stack, ending in a dimensioning shock on the payload itself.

In the leave-behind deployment, the mass of the stack provides inertia during deployment, making it more resistant to small perturbations, something Grechik, Mikulas & Freeland presented in 1999 [12], Figs. 4(d)–(h). Intuition indicates that if a few panels are perturbed, they are connected to the mass-bearing portion of the Tow, which not only provides a slight restoring effect due to longeron stretching against the inertia of the payload, but also gives access to centralized control systems as in [8, 9]. One can see that the bungee effect is mitigated by this effect in Fig. 4. In contrast to the drag along system, the payload is accelerated in a stepped fashion, and damping will be needed as each panel decelerates to the speed of the payload and deployed tow. The panels will accelerate over the deployment so that the last panel will have the largest kinetic energy and thus be dimensioning for the shock. In the leave behind setup, the initial stack velocity is limited by the overall deployment time span of the Space Tow but also lower bound by the acceleration of the payload, the initial panel velocity must be large enough not to be overtaken by the payload, this is covered in more detail in a later section.

A consideration of the effect of the boundary conditions should be made, for the drag along deployment, the force of the omitted panels is not accounted for, such that the panels deploy as if the structure were made of only 5 panels. Couple this with the pinned boundary of the payload, the simulation is expected to show a reasonable sequence for the deployment of the first panels until the longerons that connect to the clamped boundary are stretched. When the structure is deployed it will represent a slightly milder (low-energy) acceleration of the payload although without the effects of non-simultaneous stretching of the longerons; in reality the payload will be expected to be pulled by one of the longerons before the others due to the perturbed deployment.

In the simulations of the leave behind deployment, no acceleration of the deploying stack occurs, as well as no acceleration of the clamped boundary, the numerical calculations shown in a later section indicate that the effect of these are of the same order of magnitude, and four orders lower than the velocities in the finite element simulation. As the perturbations are smaller, the effects of uneven payload acceleration are expected to be negligible.

III. Deployment Analytical Solutions

To better understand the underlying principles of the differences of the deployment process, one could examine the mechanics of a simple case. Here is considered a one-dimensional Space Tow accelerated by increasing photon pressure modelling shading effects of the stowed panel stack. Fig 6. The model considers a force linearly proportional to the distance between a deploying panel and the one behind, such as might be expected from a square sail illuminated perfectly from one of its sides. A circular panel will be illuminated in a much less linear fashion, as Tibert & Lennon proposed [11].

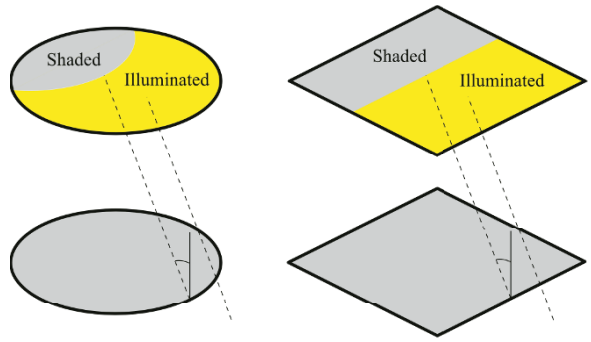


Figure 6. Shading on circular and square Space Tow sail panels, image from [11].

The one-dimensional equation $m\ddot{x} = F(x)$ is solved exactly for each deployment step and corresponding force function $F(x)$.

Leave behind deployment

Assuming that the payload starts in relative rest, and the stack is moving as an accelerated unit until the longerons are stretched. When longerons are fully stretched, the movement of the panel is damped so that it stops instantaneously and then starts to accelerate the payload. Due to the velocity of the panel stack, it continues to deploy and as the distance to the deployed panel increases, the stack accelerates again. Solving for an arbitrary step i, where i panels have been deployed and are accelerating the payload and deployed bay masses. The deployed part has initial conditions as:

$$\begin{cases} \text{From start of deployment} & \text{step i} \\ t = \Sigma_0^i t_i & t = 0 \\ x_M = \Sigma_0^i x_{M,i} & x_M = 0 \\ v_M = v_{M,i} & v_M = v_{m,i} \\ \ddot{x}_M = iF_s/M_i & \ddot{x}_M = iF_s/M_i \\ M_i = m_{\text{payload}} + im_{\text{bay}} & \end{cases} \tag{1}$$

where x_M corresponds to the position of the last deployed sail. Then the differential equation gives that

$$x_M = \frac{iF_s t^2}{2M_i} + v_{M,i}t \tag{2}$$

Simultaneously, the stack is deployed under acceleration with initial conditions such as:

$$\begin{cases} \text{step i} \\ t = 0 \\ x_m = 0 \\ v_m = v_{m,i} \\ F = F_s(x_m - x_M)/l_0 \\ \ddot{x}_m = F_s(x_m - x_M)/(m_i l_0) \\ m_i = N_{\text{panels}} m_{\text{bay}} - i m_{\text{bay}} \end{cases} \tag{3}$$

where N_{panels} is the total number of panels. And the inhomogeneous nonlinear differential equation is solved with the boundary conditions as:

$$\begin{cases} t = 0 \\ x = 0 \\ v = \dot{x} = v_{m,i} \end{cases} \tag{4}$$

$$\begin{cases} t = t_{i+1} \\ x = l_0 + x_M \\ v = v_{i+1} \end{cases} \tag{5}$$

so that the end time of the ith deployment step is described by

$$t_{i+1} = \frac{1}{2\lambda_i} \ln\left(\frac{c \pm \sqrt{c^2 - b^2 + a^2}}{a + b} \right) \tag{6}$$

With the constants given as

$$\begin{aligned} a &= \mp \left(\frac{v_{m,i} - v_{M,i}}{\lambda_i} \right) \left(\frac{i m_i l_0}{M_i} \right) \\ b &= \frac{1}{2} \left(\frac{v_{m,i} - v_{M,i}}{\lambda_i} \right)^2 + \frac{1}{2} \left(\frac{i m_i l_0}{M_i} \right)^2 \\ c &= l_0^2 \left(1 - \frac{i m_i}{M_i} \right)^2 + \frac{1}{2} \left(\frac{v_{m,i} - v_{M,i}}{\lambda_i} \right)^2 - \frac{1}{2} \left(\frac{i m_i l_0}{M_i} \right)^2 \\ \lambda_i &= \sqrt{\frac{F_s}{m_i l_0}} \end{aligned}$$

For the data output, the lowest positive time of the sign-combinations was used. Finally, the stack velocity

$v_{m,i+1} = v_m(t = t_{i+1})$

$$v_{m,i+1} = \frac{v_{m,i} - v_{M,i}}{2}(e^{\lambda_i t_{i+1}} + e^{-\lambda_i t_{i+1}}) - \frac{im_i l_0 \lambda_i}{2M_i}(e^{\lambda_i t_{i+1}} - e^{-\lambda_i t_{i+1}}) + \frac{iF_s t_{i+1}}{M_i} + v_{M,i} \tag{7}$$

where $v_{m,0} = v_0$ is the initial kick velocity. The payload velocity at time $t = t_{i+1}$ is described as

$$v_{M,i+1} = v_M(t_{i+1}) = \frac{iF_s t_{i+1}}{M_i} + v_{M,i} \tag{8}$$

where the initial payload velocity, as well as the payload end velocity at step 1 $v_{M,0} = v_{M,1} = 0$ as no longerons have yet stretched to transfer the force from the solar panels as well as assuming that the kinetic energy of the deployed panel is damped out. The subindex 0 corresponds to the initial deployment problem where the stack is accelerated as a whole from the payload and no longerons are stretched.

For deployment to continue, $\Delta v_{i+1} = v_{m,i+1} - v_{M,i+1} \geq 0$, it can be shown that this will be the case as long as the relation of the previous step fulfills

$$v_{m,i} - v_{M,i} \geq \frac{im_i l_0 \lambda_i}{M_i} \tanh(\lambda_i t_{i+1}) \tag{9}$$

This indicates that as the number of deployed panels grow, the stack will have a slower relative velocity increase $v_{m,i+1} - v_{M,i+1}$. This is the problem of the payload being accelerated by more force than the deploying stack as mentioned earlier. The payload will quite simply start to catch up to the stack as more panels are deployed. The inequality is also dependent on the relative masses of payload and tow, as well as the force and distance affecting the panels, many of which will be set mission criteria and are thus not further discussed here. A parameter study of the starting velocity of the simulated structure with full 796 panels and $N_{\text{panels}} F_s = 7.27$ N reveal that the starting velocity $v_0 \geq 2.111$ m/s, with equality resulting in the velocity difference between the deployed last panel and tow is $\Delta v_{796} \geq 6.9 \cdot 10^{-5}$ m/s. With this velocity the structure is deployed in about 1 h 14 min with a final velocity of 2.3 m/s.

Drag along deployment

For the drag along mode of deployment, we separate only the outermost sail panel and let the solar thrust accelerate it until the longerons stretch enough to pull the next panel from the stack and so on. It is assumed that the payload and stack of undeployed sails starts in relative rest, and the deployed sails are moving as a unit accelerated by the sum of deployed sails until the longerons are stretched. When stretched, the next panel is instantaneously accelerated to the deployed sail velocity and continues to be accelerated both by the deployed sails as well as the increasing partial illumination of itself. For an arbitrary step i, where i panels have been deployed and are

accelerating the deploying bay and previously deployed bay masses. The deployed part has initial conditions as:

$$\begin{cases} \text{relative} \\ t = 0 \\ x = 0 \\ v = v_i \\ \ddot{x} = F_s(x + il_0)/(m_i l_0 \\ m_i = m_{\text{bay}}(1 + i) \end{cases} \tag{10}$$

The inhomogeneous DE is formulated as

$$\ddot{x} = F_s \frac{x + il_0}{m_i l_0}\,; \quad x = x_h + x_p$$

For each step, the boundary conditions

$$\begin{cases} t = 0 \\ x = 0 \\ v = \dot{x} = v_i \end{cases} \quad \begin{cases} t = t_{i+1} \\ x = l_0 \\ v = v_{i+1} \end{cases} \tag{11}$$

This then solves so that

$$t_{i+1} = \frac{1}{\lambda_i} \ln \left(\frac{\zeta \pm \sqrt{\zeta^2 - \beta^2 + \alpha^2}}{\alpha + \beta} \right) \tag{12}$$

With

$$\alpha = \frac{v_i}{\lambda_i}$$
$$\beta = il_0$$
$$\zeta = l_0(1 + i)$$

and the end velocity given as

$$v_{i+1} = \frac{v_i}{2}(e^{\lambda_i t_{i+1}} + e^{-\lambda_i t_{i+1}}) + \frac{il_0\lambda_i}{2}(e^{\lambda_i t_{i+1}} - e^{-\lambda_i t_{i+1}}) \tag{13}$$

This would mean that the panels instantaneously accelerate from rest to tow speed without energy loss, which of course is impossible, but by handling each step individually, one could interject a transition stage between them where acceleration could be handled, then changing the step-initial velocity accordingly.

It is clear that this form of deployment accelerates with number of panels, with a quick decrease of deployment

time per step. For the same structure and parameters as in the "leave behind" analysis, the structure deploys in about 40 min with an end velocity $v_{796} = 4.4$ m/s

If we change the initial velocity to a gentler $v_0 = 0.03$ m/s, the structure reaches full deployment in about 43 minutes, with the end velocity 10 mm/s lower than the case above.

A. Damping the jerk

One pervasive problem of both these deployment schemes is that they will cause panels to spring back, due to tensional elasticity coupled with lack of compressive stiffness in the longerons. To avoid this, the kinetic energy of the deploying panel needs to be dissipated somehow, either via dry friction sliding along the longerons, or dissipation due to flexural waves in longerons or sail sheet, or some other mechanism.

Simulated damping in the longerons

A few simulations were done with ABAQUS's mass-damping on the longerons, which translates to velocity-proportional damping. The observed effect was that of the longerons lagging behind the deploying structure, in a very unnatural way, clearly illustrating Greschik's point that for continuously accelerated structures, such as solar sails, an acceleration-rate ("jerk") proportional damping would be a better description for models of such structures [9].

Damping with dry friction

Exploring the energies needed to be removed in a classic dry friction example, a sliding length on the longerons, ended by either a knot or thickening of the longeron. Allowing a clamped panel fastening to slide a short distance along the longeron whilst affected by frictional force.

Classic approach

At some time the longeron will have stretched to its maximum length and the panel will have reached l_0 under continuous acceleration from the Sun, the panel will then have accelerated so that the kinetic energy is

$$E_k = \frac{m_s v_i^2}{2}$$

When the filament is stretched, the panel starts to glide until stopped by a frictional force. We assume that the solar panel is fastened onto the cable with a shoe of the same carbon material, resulting in a frictional force $F_f = \mu_f F_n$ from the normal force F_n of the clamp acting on the longeron. Classically, to make this damping halt the panel, it would have to do work

$$E_k \leq U_f = \int_0^{d_b} F_f \, \mathrm{d}x$$

which for the equality yields

$$F_n = \frac{E_k}{d_b \mu_f} \tag{14}$$

The work is done over a period $\Delta t = t_b - t_0$, t_b being the braking time, and is one-dimensionally described as

$$\dot{x} = \int_{t_0}^{t_b} \frac{F_s - F_f}{m_s} \, \mathrm{d}t = \{t_0 = 0\} = (F_s - F_f)\frac{t_b}{m_s} + v_i \tag{15}$$

as $\dot{x}(t = 0) = v_i$. This can be solved by integrating once again and applying our boundary conditions for the position.

$$u|_{t=t_b} = d_b = (F_s - F_f)\frac{t_b^2}{m_s} + v_i t_b \tag{16}$$

as $u(t = 0) = 0$. Solving for $t_b > t_0$ yields

$$t_b = \frac{-v_i m_s + \sqrt{v_i^2 m_s^2 + 2 m_s d_b (F_s - F_f)}}{F_s - F_f} \tag{17}$$

When the work is done, the kinetic energy has been lowered so that the panel has come to the same velocity as the deployed structure. It then adds its solar force F_s to the thrust upon the deployed structure, this remaining force will play a role in the the "jerk"-proportional damping in the next section. It is clear that a large friction coefficient $\mu_f \gg 1$ is required to achieve a reasonable braking distance $d_b \ll l_0$.

Jerk-proportional approach

Greschik [9] proposes that jerk damping, can be physically rationalized by energy dissipated in movements of the sail sheet, cords or other components changing shape or just increasing and decreasing their deformations. Greschik also shows that the mechanical energy removed from the system is described as

$$E_c = \pm \int_{\dddot{x}_0}^{\dddot{x}_1} c_j(\ddot{x}) \, \mathrm{d}\ddot{x} = \int_{t_0}^{t_b} c_j(\ddot{x}) |\dddot{x}| \, \mathrm{d}t; \tag{18}$$

where $\ddot{x}(t)$ monotonous in $[t_0, t_b]$, c_j is a proportional damping coefficient.

For a constant $c_j(\ddot{x})$ we write

$$\Delta E_c = c_j |\Delta \ddot{x}| \, ; \quad F_c = -c_j |\Delta \ddot{x}| / \Delta x \tag{19}$$

and

$$\Delta v_c = F_c \Delta t / m = -\frac{c_j |\Delta \ddot{x}|}{m_s \Delta x} \Delta t \tag{20}$$

which can be restructured to

$$c_j = -\frac{\Delta v_c m_s \Delta x}{|\Delta \ddot{x}| \Delta t} \tag{21}$$

with all else being equal to the situation in the classical solution and the model in the simulations,

$$\begin{cases} \Delta\ddot{x} = -F_f/m_s \\ \Delta x = d_b \\ \Delta t = t_b - t_0 = t_b \\ \Delta v_c = v_i - 0 = v_i \end{cases} \tag{22}$$

which yields

$$c_j = -\frac{v_i m_s d_b}{t_b F_f/m_s} \tag{23}$$

For the last longeron set in the "drag along" deployment system, this would correspond to $E_k = m_s v_{796}^2/2 \approx 322.31$ J and the jerk damping $\Delta E_c = c_j|\Delta\ddot{x}| = -322.31$ J.

IV. Conclusions

The present work has explored two different strategies for deployment of a Space Tow solar sail structure, the leave behind and drag along strategies. Simulations as well as mathematical analysis illuminate obstacles with both of these strategies.

The simulated model presented here shows the first five panels deploying in a Space Tow design. The simulations illustrate a bungee effect from the slack-taut dynamics of the longerons in both strategies, and also illustrate the "drag along" strategy's weakness to perturbations. The mathematical analysis shows the "leave behind" strategy's complex criteria for successful deployment as well as the dimensioning energies to be damped out in the respective strategies.

The "drag along" deployment, due to perturbations quickly multiplying, would demand much of a control system to deploy with only an initial kick, introducing cost in control mechanisms and mass. A more practical solution would then be to drag deploy the tow not by a pilot sail but by a controlled rocket or other steered craft. As each panel is lifted from the stack a sharp increase in acceleration would demand sturdier materials or necessitate damping of each sail bay.

The "leave behind" approach, due to the increasing and larger acceleration of the payload, demand a high enough initial speed for the Space Tow to be fully deployed. This initial speed is dependent of several different factors that limit design specifications.

The work also discusses how these energies could be damped. The model presented proposes a frictional damping that could work over a wide range of tolerances, as the proposed friction stretch needs little special preparation and as the masses and velocities are low, the frictional stretch could be kept short on each sail bay, keeping the stowed dimensions small. The energy loss has also been compared to a coefficient for acceleration-rate proportional damping, which could be a property of interest for material selection of both sail and truss.

Control and navigation are important issues that will very much be affected by the choice of damping, but were outside the scope of this study.

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Electric Solar Wind Sail: Deployment, Long-Term Dynamics, and Control Hardware Requirements

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The deployment, dynamics, and control of the electric solar wind sail is addressed in terms of the single tether motion. Based on a simple model of a rotating rigid tether as a spherical pendulum, estimates for the goodness of the control system are shown for strictly planar tether tip orbits. It is concluded that the control system is rather inefficient for a slowly rotating sail with a large tether coning angle. The results are conservative as they do not take into account the actual shape of the tether. The long and short-term effects on the sail rotation rate arising from the single tether dynamics are addressed. We also present preliminary results on a new control scheme assuming non-planar tether tip orbits. It is argued that the scheme will improve the electric sail control system efficiency.

Nomenclature

(*Nomenclature entries should have the units identified*)

r,Θ,φ	=	spherical coordinates
X,Y,Z	=	cartesian coordinates
α	=	sail angle
φ_0	=	solar zenith angle
λ	=	electric sail force constant
l	=	tether length
t	=	time
ω	=	tether rotation rate
Λ	=	tether coning angle
g	=	tether voltage modulation
X	=	$\tan(\alpha)\tan(\Lambda)$
ρ	=	electric force ratio to the centrifugal force
Ω	=	Earth orbital angular velocity
Ψ	=	tether rotation plane tilt angle

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Introduction

Electric sail space propulsion system is an electro-mechanical system with strawman components of conducting lightweight micro-meteoroid tolerant tethers, electron guns, and solar panels [1,2]. The tethers are charged to a positive high voltage of a few tens of kV to deviate the solar wind proton flow to obtain the wind momentum as spacecraft thrust. The thrust has been estimated to be several hundreds of nN per tether unit length based on particle-in-cell plasma simulations [2] and analytical analysis [3]. The positive tethers collect electrons and the positive charge state is maintained by the electron gun. The resulting electron current loop connects the tethers, electron gun, and solar wind plasma and using only a moderate amount of solar panel power.

Presently, there are two principal designs of the electric sail (Fig. 1). One is such that the tether tips are mechanically connected with perforated auxiliary tether. Each of the tips are equipped with a remote unit that includes the auxiliary tether reels and cold gas or FEEP thrusters. The remote unit with a mass about 500 grams was manufactured and space quality tested under the ESAIL project of EU/FP7 program. As the electric sail is deployed from a spinning spacecraft by unreeling the long tethers, the remote unit thrust sources have to be added to obtain the required angular momentum of the fully opened sail. Recently, it was suggested that relatively small solar blades can be attached to the tether tips for the deployment thrust and flight time control of the tether rotation rates individually [4]. The other sail design is then such that the tether tips are mechanically free and the flight time control of the tether rotation rates is done individually by using Freely Guided Photonic Blades (FGPB).

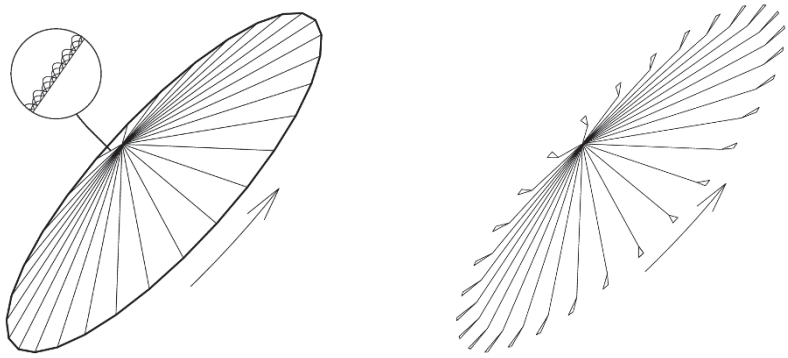

Fig. 1 Electric sail designs use either auxiliary tethers and remote units (left) or Freely Guided Photonic Blades (right) for the sail flight time stability.

The electric sail control algorithms depend on the sail design. Most notably, for the design with the mechanically connected tether tips, the tethers rotate in unison whereas for the case of free tethers, the tether rotation rate is typically a function of the rotation phase [5]. In this paper, we consider the controlled planar motion of free tethers. We address this control scheme in terms of both the sail power system mass efficiency and the ratio of the radial and azimuthal thrust components directed perpendicular and parallel to the sail orbital motion. It is shown that the control routine becomes inefficient for large coning angles corresponding to slow sail rotation rates. Motivated by these findings, we suggest a new control scheme that does not assume strict planar motion. Our preliminary numerical calculation indicates that the suggested types of tether tip orbit exist, and the control scheme based on these orbits seems to improve the control system efficiency.

Tether Planar Orbit Control Efficiency

Assuming the tether being a thin rigid beam, the rotational dynamics of the tether can be described by a spherical pendulum. The equation of motion is written spherical coordinates as

$$\begin{aligned} \ddot{\theta} - \sin\theta\cos\theta\,\dot{\phi}^2 &= \frac{\lambda}{l} g\left(\sin\alpha\cos\theta\cos(\phi-\phi_0) - \cos\alpha\sin\theta\right) \\ \frac{d}{dt}\left(\sin^2\theta\dot{\phi}\right) &= -\frac{\lambda}{l} g\sin\alpha\sin\theta\sin\phi, \end{aligned} \tag{1}$$

where α and φ are the sail inclination angle and the sun azimuth angle [5]. The tether motion is considered in the SE (Solar Ecliptic) coordinate system shown in Fig. 1. In this system, the apparent controlled tether motion is planar with a constant coning angle ($\Lambda = \Theta - \pi/2$). It can be shown that for a voltage modulation of

$$g(\phi) = \frac{(1-\chi)^3}{(1+\chi\cos\phi)^3} \tag{2}$$

the solution with ($d\Theta/dt = 0$) exist for any given realistic sail angle ($\chi = tan(\alpha)tan(\Lambda)$).

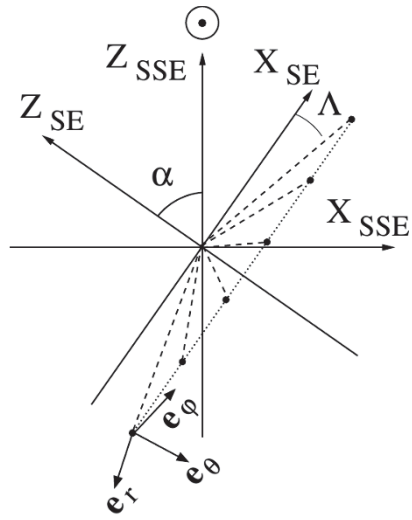

Fig. 2 Two coordinate systems of SE (Sail Eccliptic) and (Sail-centric Solar Ecliptic).

The tether motion can be parametrized by the ratio (ρ) of the electric sail and centrifugal forces. Fig. 3a shows the sail coning angle as a contour plot with the equicontour values in degrees. As expected the coning angle increases for decreasing sail rotation rate (increasing ρ), but it depends also strongly on the sail inclination angle. Another central feature of this control modulation is that the tether rotation rate shows considerable dependence on the rotation phase when the sail is inclined (Fig. 3b).

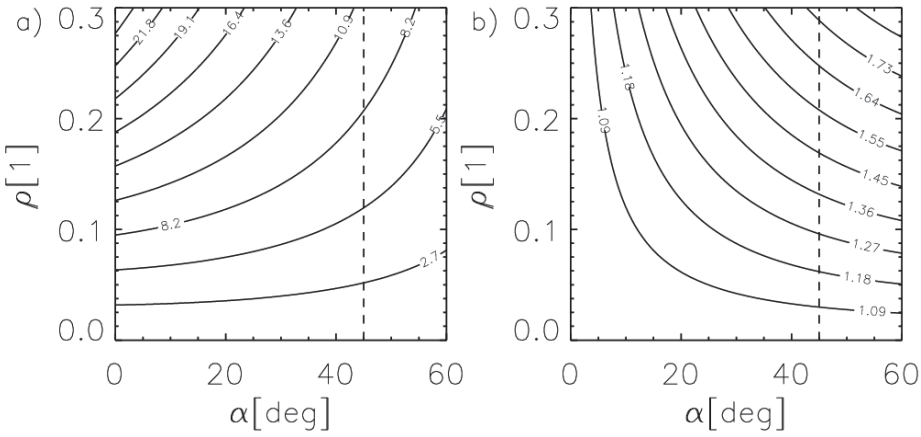

Fig. 3 Equicontours of the coning angle (a) and the amplitude ratio of the tether rotation rate (b).

Figure 4 shows the thrust integrated over the tether rotation phase. The azimuthal component aligned with the sail orbital motion along the X_{SSE} axis (Fig. 4a) has its maximum at $\rho = 0$ and $\alpha = 45$ deg. For larger rho values, the azimuthal thrust decreases clearly and the maximum is reached with sail angles below 45 deg. For example, when ρ equals to about 0.2, the maximum is at around α equals to 40 deg. Note that the same level of azimuthal thrust is gained with smaller sail inclination with a sail spinning faster. The radial component aligned with the Z_{SSE} axis (Fig. 4b) decreases with the decreasing sail spin rate and increasing sail inclination as expected.

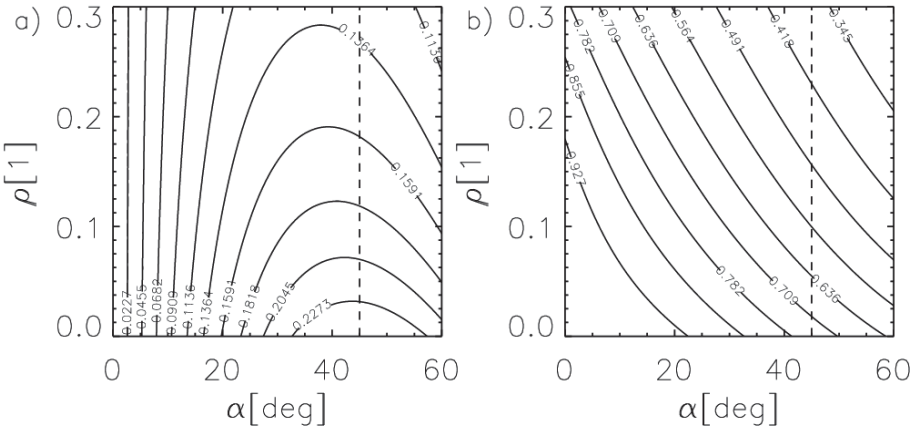

Fig. 4 Equicontours of the total thrust in azimuthal and radial directions. The contour levels are scaled to maximum at (ρ = α = 0).

The inefficieny of the sail control for slowly rotating sails (large ρ) is also apparent when the mass of the power system running the sail current system is considered. The power system mass scales as $V^{3/2}$ to the maximum voltage V of the power system design [6]. Hence the tether voltage averaged over the tether rotation gives the power system mass efficiency implied the applied control scheme (Fig. 5). It can be seen that the effciency is significantly reduced as a function of ρ.

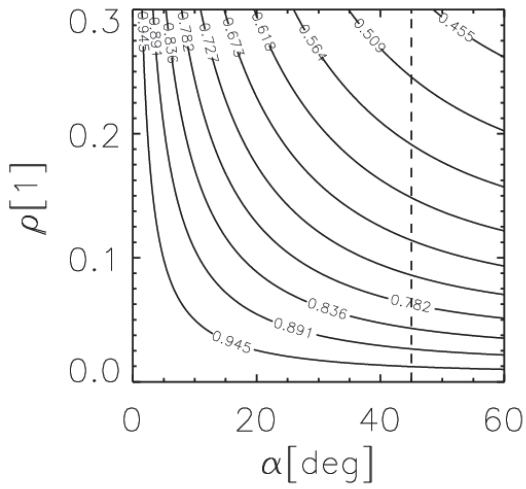

Fig. 5 Equicontours of the power system mass efficiency propotional to the averaged modulation voltage over the maximum voltage to the power of 3/2.

Ultimately the inefficiency of the control scheme arises from the fact that the electric sail force is directed along the solar wind component perpendicular to the tether. Thus depending on the tether rotation phase the instantaneous thrust has a significant azimuthal component only when it is in the sectors near the X_{SSE} axis. In sectors around the Y_{SSE} axis, the instantaneous thrust is mostly radial. Furthermore, maintaining the tether planar motion requires large voltage modulation amplitude relative to the maximum tether voltage. This leads to large variations of the tether angular speed as a function of the rotation phase. Thus a considerable amount of the available thrust is lost by this control scheme for slowly rotating sails. This can be avoided by increasing the sail spin rate in the limits of the tether material tensile strength. Note that these conclusions are strictly based on the assumption the tether being a rigid beam and on the adopted control scheme. These topics are further addressed below. However, the simple model is adequate to get insight into the short and long-term evolution of the sail key parameters, the coning angle and spin rate.

Sail Spin Rate Evolution

There are several factors that affect the single tether or collectively the sail spin rate. These can be attributed to the mechanics of the rotating tethers, solar wind variations, and celestial mechanics [Toivanen2013, Toivanen2012].

Tether Deployment

During the tether deployment, the tether rotation slows down while reeled out. This mundane effect arises from the conservation of the angular momentum, and the rotation rate scales as $1/l^3$ to the tether length l. Although the tethers are lightweight, any realistic initial spin of the spacecraft is far from being adequate for the deployment of a full-scale electric sail and already from this perspective the tether deployment is untrivial. To reduce the required delta-v for the sail deployment the additional thrust is applied to the tether tips either by remote units or FGPB depending on the sail design.

Sail Turning

The electric sail pointing with respect to the sun direction can be controlled with the tether voltage modulation. As the sail is turned, the modulated thrust that turns the tether spin plane also has a component along the tether angular speed enhancing the sail spin rate. Based on the results of reviewed above, the spin rate is proportional to $1/\cos(\alpha)$ (Ref. [5]). If the initial deployment is realized with the sail point to the sun, turning the sail to the flight inclination, increases the spin rate by a factor of about 1.4, an effect that can be used to reduce the amount of propellant of the remote units reserved for the sail deployment.

Solar wind Variation

The solar wind has non-radial components significant in terms of the individual tether rotation. These components induces drift in the tether rotation phase leading ultimately to tether collisions and collapse of the sail in matter of days. As the angular speed of the tether depends on the inclination of its spin plane, it can be controlled by inclining the tether spin planes individually according. Each tether tip is then kept in cone around the nominal phase. This way the solar wind variations can be tolerated and the time scale to the tether collisions can be extended [7]. However, additional thrust by FGPB are required. For the sail design using the auxiliary tethers the solar wind variations play no role.

Coriolis Effect

With a low thrust system, the spacecraft orbits are typically spirals around the sun. When orbiting around the sun, the sail spin plane conserves the orientation to the distant stars as depicted in Figure 6, the SSE coordinate system is not inertial, and the sail appears to be slowly rotating in the SSE system. This corresponds to a small Coriolis force affecting the sail in the SSE system (the term is not included in (1), see Ref. [5]). Thus this force has to be controlled by an additional modulation embedded in the voltage modulation (2) in order to maintain a sail inclination constant with respect to the sun direction. Such an additional modulation induces also a thrust component along the tether rotation accumulating slowly in the sail rotation rate. The sail rotation rate is slowed down when spiraling inward and enhanced when spiraling outward. The temporal change in the rotation rate (ω)is expressed as

$$\omega = \omega_0 e^{\Omega \tan \alpha (t - t_0)} \tag{3}$$

where Ω is rotation rate of the SSE around the sun. This effect has to be taken into account in mission analysis when designing the additional thrust source both in terms of the auxiliary tether remote units and FGPB.

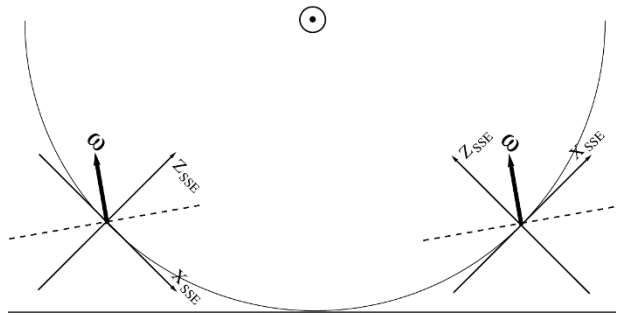

Fig. 6 Slow rotation of the sail in the non-inertial SSE coordinate system while orbiting around the sun.

Non-Planar Tether Orbit Control

As concluded above, the apparent control scheme that maintains the sail tether tips rotating in a plane with a constant coning angle is inefficient in many respects, and there is a need for more efficient control schemes. Here, we sketch an alternative orbit for the tether tip and show that such an orbit exist based on preliminary numerical analysis. Figure 7 shows the proposed orbit of the tether tip (thick solid line) in the SE coordinates. If the tether voltage is turned off, the tether rotates freely on any given plane. Two such planes tilted by some angle

(Ψ) around the X_{SE} axis are considered as shown in Fig. 7. When approaching the rotation phase sector near the X_{SE} axis, the tether voltage is turned on and kept constant at maximum. The voltage is turned off when the tether reaches the plane of free motion with the opposite tilt. The preliminary numerical integration of the equation of motion suggests that orbits of the proposed type exist as shown in Fig. 8. The sail inclination angle is 45 deg.

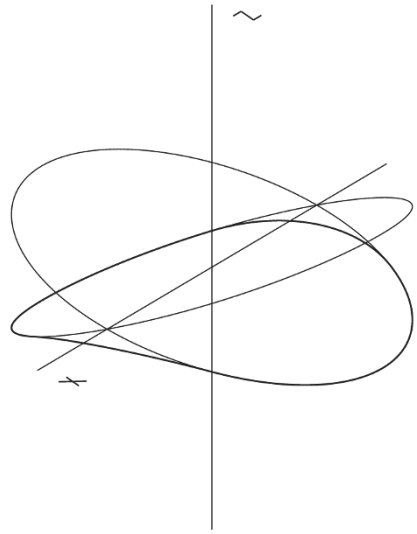

Fig. 7 Alternative orbit of the tether tip for a control scheme with an enhanced efficiency.

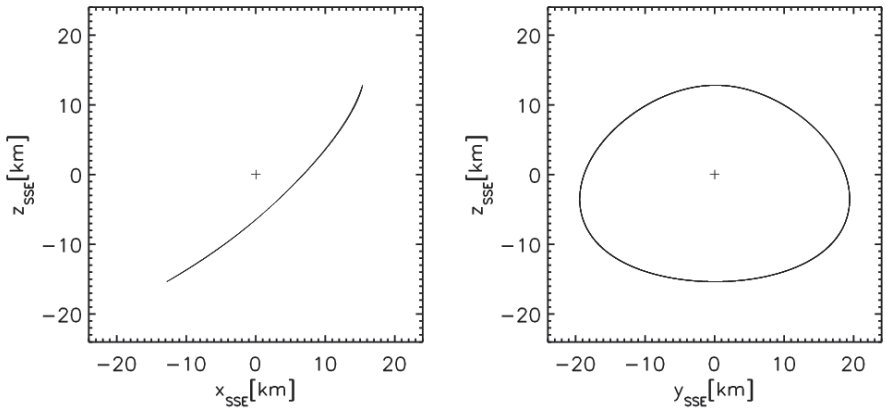

Fig. 8 Numerical solution for the orbit of the tether tip.

There are several promising aspects in the proposed orbit type, and improvement to the tether control efficiency can be expected. The voltage is used at or near the maximum (Fig. 9). The tether rotation rate varies less over the rotation phase (Fig. 9; during the free motion, the rate is constant). The voltage is turned on when the tether is producing azimuthal thrust. When the voltage is on the tether is located near the plane of Z_{SE} equals to zero implying a small coning angle and reduced power consumption and power system mass.

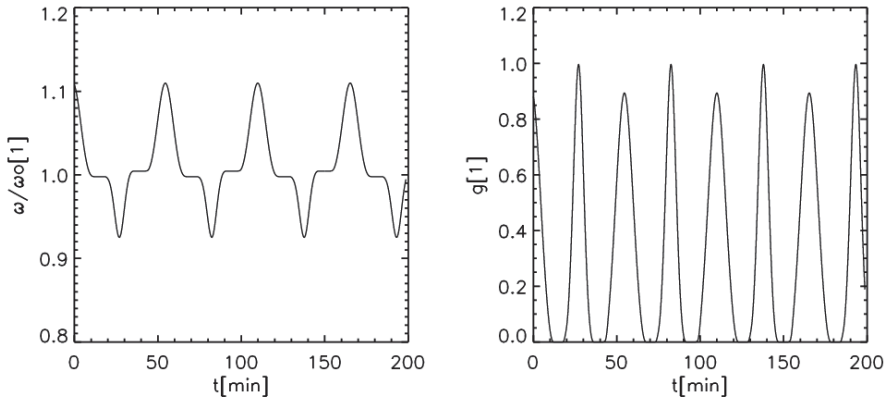

Fig. 9 Relative angular frequency of the tether rotation (left) and voltage modulation (right).

Conclusions and Discussion

The results of this paper concerns the electric sail design with mechanically free tethers. Such design is realized by using the Freely Guided Photonic Blades [4]. Based on our previous analysis on the controlled motion of a free tether [5], we addressed the efficiency of this control scheme that assumes strictly planar tether tip motion. The total thrust integrated over the tether rotation rate shows that this scheme is rather inefficient in producing significant azimuthal thrust component along the sail orbital motion. It also implies low power system mass efficiency for slowly rotating sails with large coning angle.

We also sketched a tether tip orbit alternative to the strictly planar motion. The control scheme based on such orbits seems to be more efficient than that based on the strictly planar orbits. The tether voltage is turn to its maximum in sectors where the tether produces most of the azimuthal thrust while the rest of the rotation phase the voltage is zero and the tether undergoes free motion. The estimates for the efficiency is a topic of a future study.

The results assume a rigid thin beam to model the tether. While the model allows us to identify several effects affecting the sail rotation rate, the estimates on the control system efficiency presented in this paper are rather

conservative. The actual shape of the tether rotating under the influence of the solar wind flow is such that the tether coning angle decreases towards the tip of the tether, and for a considerable portion of the tether, the coning angle approaches to zero. While this is a matter of further study, it can be expected that there is an effective coning angle that is smaller than that derived under the rigid tether assumption, and the control system efficiency is underestimated for a slowly rotating sail (the sail rotation rate is a free parameter only to the limit of the tether material tensile strength). Furthermore, similar analysis for the sail designs based on auxiliary tethers are also left for a future study.

Acknowledgments

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